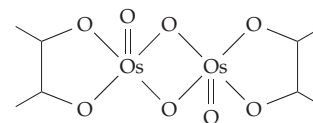


for electron microscopy, their membranes show a characteristic three layered structure similar to that observed for biological membranes. Two darkly stained lines ~2–2.5 nm thick are separated by a clear space ~2.5–3.5 nm wide in the center. Both myelin and the retinal rod outer segments show closely spaced pairs of such membranes with a combined width of 18 nm. These results seemed to support the original Davsen–Danielli model. However, many questions must be raised about the interpretation of these results. Why does OsO_4 stain only the outer protein layer when it is known to react also with double bonds of hydrocarbon side chains of lipids to form osmate esters which are readily reduced to a diol and osmium?^{77a-c,78} Membranes from which most of the lipid has been extracted still stain with OsO_4 to give three-layered electron micrographs. Perhaps little can be concluded from the three-layered appearance. We have learned that it is difficult to determine even the thickness, let alone the complete structure of an object that is only 6–10 nm thick.

Strong support for the lipid bilayer model comes from the preparation of another type of artificial mem-



An osmate ester formed from two unsaturated groups

brane which can be made from a solution of phosphatidylcholine or of a mixture of phospholipids plus cholesterol in a hydrocarbon solvent. A droplet of solution is placed on a small orifice in a plastic sheet, separating two compartments filled with an aqueous medium (Fig. 8-8). The solution in the orifice quickly drains, just as does a soap bubble, and the resulting film eventually becomes so thin that the bright colors disappear and a “**black membrane**” is formed. Similar membranes, but without a residual content of hydrocarbon solvent, have been created by apposition of two lipid monolayers formed at an air–water interface.^{79,80} The thickness of such artificial membranes is thought to be only 6–9 nm. Resilient and self-sealing, the membranes can be stained with OsO_4 to give a typical three-layered pattern.

TABLE 8-3
Estimated Chemical Compositions of Some Membranes

Compound	Percentage of total dry weight of membrane ^a					
	Myelin (bovine)	Retinal rod	Plasma membrane (human erythrocyte)	Mitochondrial membranes	<i>E. coli</i> ^{b,c,d} (inner and outer membranes)	Chloroplasts ^e
Protein	22	59	60	76	75	48
Total lipid	78	41	40	24	25	52 ^f
Phosphatidylcholine	7.5	13	6.9	8.8		
Phosphatidylethanolamine	11.7	6.5	6.5	8.4	18	
Phosphatidylserine	7.1	2.5	3.1			
Phosphatidylinositol	0.6	0.4	0.3	0.75		
Phosphatidylglycerol					4	
Cardiolipin ^g		0.4		4.3	3	
Sphingomyelin	6.4	0.5	6.5			
Glycolipid	22.0	9.5	Trace	Trace		23
Cholesterol	17.0	2.0	9.2	0.24		
Total phospholipid	33	27	24	22.5	25	4.7
Phospholipid as a percentage of total lipid	42	66	60	94	>90%	9

^a Dewey, M. M. and Barr, L. (1970) *Curr. Top. Membr. Transp.* **1**, 6.

^b Kaback, H. R. (1970) *Curr. Top. Membr. Transp.* **1**, 35–99.

^c Mizushima, S. and Yamada, H. (1975) *Biochim. Biophys. Acta.* **375**, 44–53.

^d Yamato, I. Anraku, Y. and Hirose, H. (1975) *J. Biochem. (Tokyo)* **77**, 705–718. These investigators found 67% protein, 21% lipids, 10% carbohydrate, and 2% RNA.

^e Lichtenthaler, H. K. and Park, R. B. (1963) *Nature (London)* **198**, 1070–1072.

^f About 14% is accounted for by chlorophyll, carotenoids, and quinones.^e

^g Diphosphatidylglycerol (Fig. 8-2).

The study of monolayers formed on a water surface has also provided important information. A thin film of an amphiphilic (containing both polar and non-polar groups) compound such as a fatty acid is prepared. This is done by depositing a small quantity of the compound dissolved in a volatile solvent on a clean aqueous surface between the barriers of a **Langmuir trough** (Fig. 8-8).^{81,82} The difference in surface tension (π) across the barriers is measured with a suitable device⁸¹ for different areas of the monolayer, i.e., for different positions of the moveable barrier. The value of π is low for expanded monolayers and falls to nearly zero when the surface is no longer completely covered. The pressure reaches a plateau when a compact monolayer is formed, after which it rises again (Fig. 8-8B). At very high values of π the monolayer collapses (buckles). Both the cross-sectional area per molecule in the monolayer and the collapse pressure can be determined. For typical fatty acids, regardless of chain length, the area covered is only $\sim 0.2 \text{ nm}^2$ per molecule indicating that the fatty acid chains are stacked vertically to the surface in the monolayer. The collapse

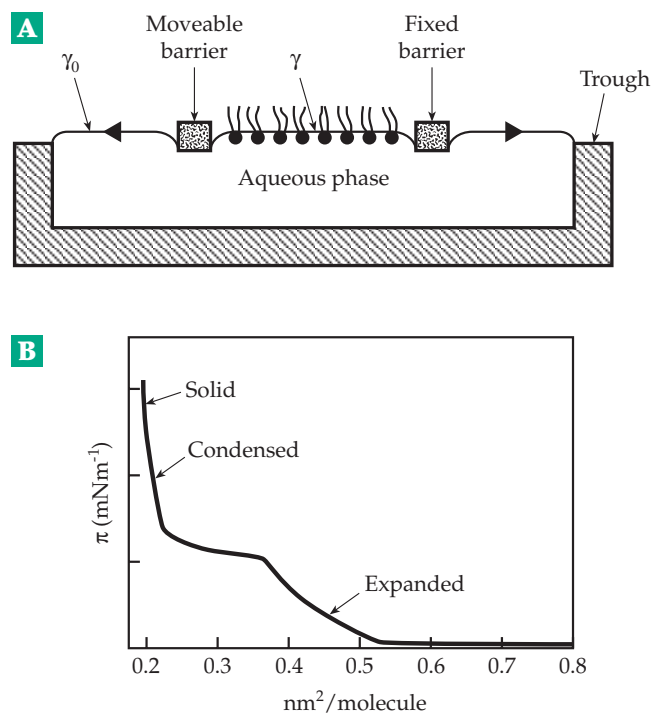


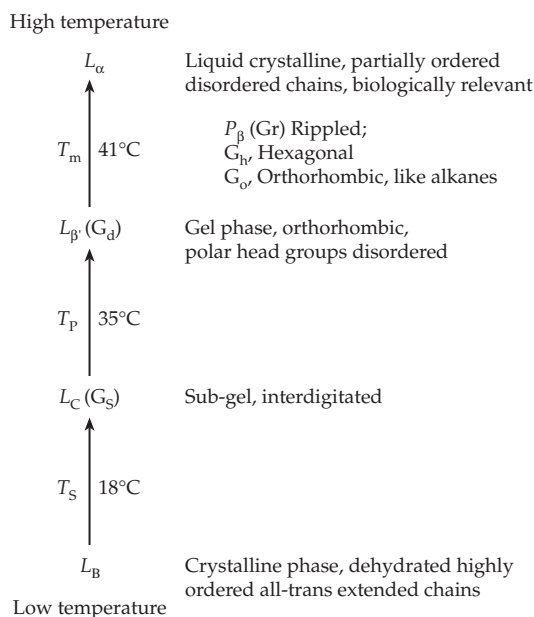
Figure 8-8 (A) The Langmuir–Adam film balance. Tension on the moveable barrier is recorded for different areas of the surface between the barriers. This gives the surface pressure π , which is the difference between the surface tension (γ_0) of a clean aqueous surface and that of a spread monolayer (γ): $\pi = \gamma_0 - \gamma$. Courtesy of Jones and Chapman.⁸¹ (B) Surface pressure (π)–area per molecule isotherm for a typical fatty acid (e.g., pentadecanoic acid $\text{C}_{14}\text{H}_{29}\text{CO}_2\text{H}$) at the aqueous–air interface. From Knobler.^{81a}

pressure is higher for longer molecules as a result of the greater number of van der Waals interaction between the chains. **Langmuir–Blodgett layers** are prepared by transferring one or more monolayers onto a smooth solid surface (Fig. 8-7).^{82,83}

Physical properties of membrane lipids. A completely extended C_{18} fatty acid chain as shown in Fig. 8-4 has a length of $\sim 2.0 \text{ nm}$ and occupies, either in crystals or in monolayers, when viewed “end-on,” an area of $\sim 0.2 \text{ nm}^2$. The hydrocarbon layer in a lipid bilayer containing such chains would have a thickness of about 4.0 nm ; that determined by X-ray diffraction for myelin is $\sim 3.5 \text{ nm}$. However, for artificial black membranes the thickness of the hydrocarbon layer can be as little as 3.1 nm when all solvent is removed.⁸⁴ These and many other results⁸⁵ indicate that the hydrocarbon chains are to some extent folded and that the membrane is expanded over that expected according to the simplest model.

Structure determinations on crystalline alkanes confirm that the chains exist in a completely extended conformation and that adjacent chains often pack together in the orthorhombic arrangement shown in Fig. 8-6B. As the temperature of such crystals is raised a series of solid–solid phase transitions is observed below the melting point of the crystals.⁸⁶ These can be detected by changes in the infrared absorption spectrum or by small amounts of heat absorption revealed by **differential scanning calorimetry** (Fig. 8-9). Each new phase permits a greater degree of mobility for the hydrocarbon chains. Thus, at a high enough temperature but below the melting point, the chains are able to rotate freely about their own axes in a so-called **hexagonal phase**. Now the chains are packed in a hexagonal array instead of the orthorhombic array of Fig. 8-6B. At intermediate temperatures, some of the chains may assume nonplanar conformations and changes in the tilt of the hydrocarbon chains (Fig. 8-6) may occur.

Similar phase transitions are observed for bilayers.^{88–90} For dipalmitoyl phosphatidylcholine the first detectable **subtransition**⁹¹ is centered at a temperature T_s of 18°C . The second, known as the **pretransition**, occurs at 35°C (T_p). The structure below T_s may be described as rigid or crystalline and that above T_s as a **gel** in which the hydrocarbon side chains twist and turn much more freely but in which the orthorhombic packing is maintained.⁸⁶ Above T_p the head groups become disordered. Although the orthorhombic packing of the tails may be maintained, there are several distinct phases,^{92,93} including one or more in which the gel is thought to assume a structure analogous to that in the hexagonal phase of hydrocarbons. At 41°C the **main transition** occurs.



This is a sharper transition with a well-defined melting temperature designated T_m . Above T_m the lipid is in the lamellar liquid crystalline or L_α state. The bilayer continues to hold together, but the fatty acid chains have melted and are now free to rotate and undergo twisting movements more freely than at lower temperatures (Fig. 8-11). The main transition is highly, but not completely, cooperative. Thus, the melting of the membrane occurs over a range of several degrees. The presence in biological membranes of a variety of

different components containing a variety of fatty acid chains leads to a broadening of the melting range.

The behavior of bilayers is strongly influenced by the lipid composition. Phospholipids containing saturated, long-chain fatty acids have high transition temperatures. The presence of unsaturated fatty acyl groups with *cis* double bonds in membrane lipids encourages folding of the hydrocarbon chains and lowers T_m . Even a single double bond lowers T_m , the decrease being greatest when the double bond is near the center of the chain.⁹⁴⁻⁹⁶ While T_m for dipalmitoyl phosphatidylcholine is 41°C, that of 1,2-dipalmitoyl phosphatidyl-*sn*-glycerol, which lacks the phosphocholine head group, is 70°C. This falls to 11.6°C for the polyunsaturated 1-stearoyl-2-linoleoyl-*sn*-glycerol, whose melting curve is shown in Fig. 8-9.⁸⁷ This lipid also shows a complex phase behavior and a melting point for the stable, crystalline β' phase higher than that of the α phase.

Inclusion of other molecules of irregular shape within membranes also lowers T_m . However, a molecule of cholesterol can pack into a bilayer with a cross-sectional area of 0.39 nm², just equal to that of two hydrocarbon chains.⁴⁹ It tends to harden membranes above T_m but increases mobility of hydrocarbon chains below T_m .⁹⁷⁻¹⁰⁰ A complex of cholesterol and phosphatidylcholine may form a separate phase within the membrane.^{101,102} The ether-linked plasmalogens may account for over 30% of the phosphoglycerides of the white matter of the brain and of heart and ether linked phospholipids are the major lipids of many anaerobic bacteria.¹⁰³ Their T_m values are a few degrees higher than those of the corresponding acyl phospholipids.¹⁰⁴

Between the pretransition temperature and T_m solid and liquid regions may coexist within a bilayer.¹⁰¹ The term **lateral phase separation** has been applied to this phenomenon.^{105,106} Since changes in the equilibrium between solid and liquid can be induced readily, e.g., by changes in the ionic environment surrounding the bilayer, lateral phase separation may be of significance in such phenomena as nerve conduction.¹⁰⁷

The phase transitions in bilayers can be recognized in many ways. Differential scanning calorimetry has already been mentioned. Another approach is to measure the spacing between molecules by X-ray diffraction. The cross-sectional area occupied by a phospholipid in a bilayer is always greater than the 0.40 nm² expected for closest packing of a pair of extended hydrocarbon chains.^{39,85}

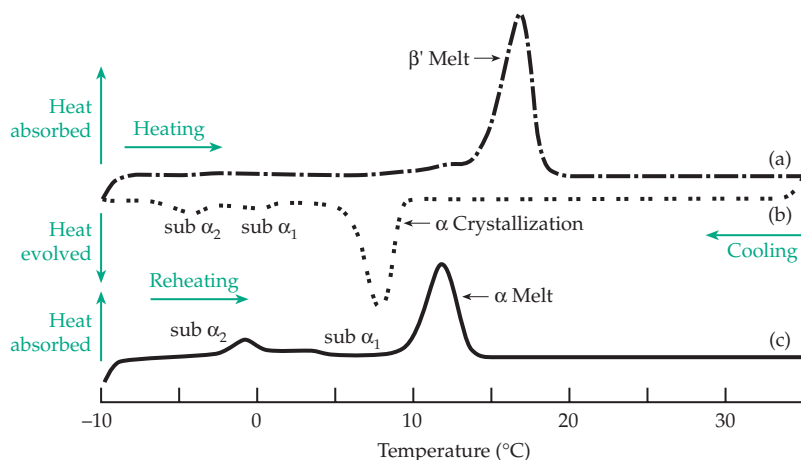


Figure 8-9 Differential scanning calorimetric curves for 1-stearoyl-2-linoleoyl-*sn*-glycerol. (A) Crystals of the compound grown from a hexane solution were heated from -10° to 35°C at a rate of 5°C per minute and the heat absorbed by the sample was recorded. (B) The molten lipid was cooled from 35° to -10°C at a rate of 5° per minute and the heat evolved was recorded as the lipid crystallized in the α phase and was then transformed through two sub- α phases. (C) The solid was reheated. From Di and Small.⁸⁷ Courtesy of Donald M. Small.

Below T_m the spacing between chains is about 0.42 nm corresponding to close packing of the fatty acid chains in a hexagonal array with an area per phospholipid of 0.41 nm². As the temperature is raised above T_m the spacing increases⁸⁵ to give an average area per phospholipid of 0.64–0.73 nm². Another technique (Box 8-C) is to study a **spin label** by EPR while yet another is to observe the fluorescence of a **polarity-dependent fluorescence probe** such as *N*-phenylnaphthylamine or other fluorescent probes¹⁰⁸ (see Chapter 23). The compound is incorporated into the membrane and undergoes changes in the intensity of its fluorescence when the state of the membrane is altered.

A variety of NMR techniques are being applied^{109–113} both to liposomes and to natural membranes.^{111,114} Incorporation of ¹³C or ²H into various positions in the hydrocarbon chains has allowed measurements of the relative degree of mobility of the chains at different depths in the bilayer (Fig. 8-10).^{109,115–117} The results are in agreement with statistical mechanical predictions that configurational freedom increases with depth toward the midplane of the bilayer. Separation of a

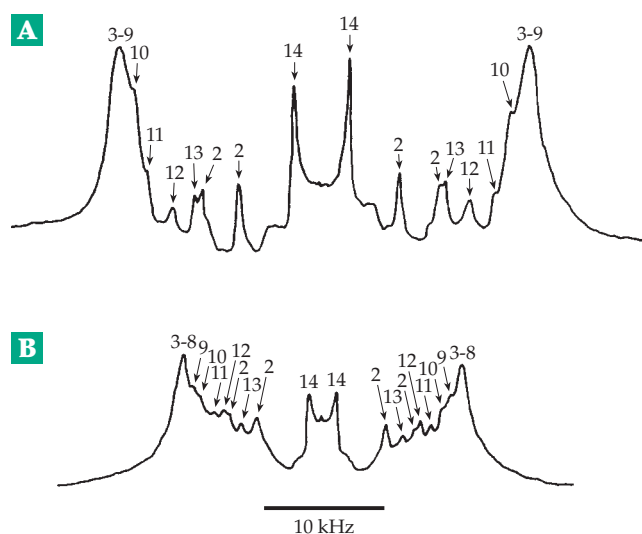


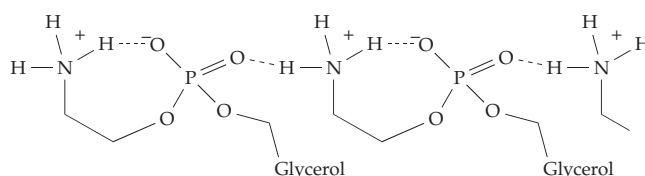
Figure 8-10 ²H NMR spectra of dimyristoyl phosphatidylcholine-*d*₂₇/water in lamellar phases at 40°C. One chain of the phosphatidylcholine is fully deuterated, containing 27 atoms of ²H. The mole ratios of water to lipid were 5.0 in (A) and 25.0 in (B). The average interfacial areas per alkyl chain as measured by X-ray diffraction were 0.252 nm² for (A) and 0.313 nm² for (B). ²H NMR spectra are presented as “powder patterns” because the lipid molecules are randomly oriented in the magnetic field of the spectrometer as if in a powder. This gives rise to pairs of peaks symmetrically located on both sides of the origin. The separation distances are a measure of the quadrupole splitting of the NMR absorption line caused by the ²H nucleus. The various splittings of the resonances of the 13 –CH₂– and one –CH₃ groups reflect differences in mobility.¹⁰⁹ The peaks have been assigned tentatively as indicated. From Boden, Jones, and Sixl.¹¹⁵ Courtesy of N. Boden.

bilayer into two or more phases can be observed using ²H- or ³¹P- NMR.^{118–120} The orientation and dynamic behavior of various head groups has been explored,^{110,121} as have effects of mixing into the bilayer other lipids such as glycosphingolipids¹²² and cholesterol.^{123,124} Crystalline phospholipids are being investigated by solid-state NMR.¹²⁵

Fourier transform infrared spectroscopy^{126,127} also provides information about conformation of both hydrocarbon chains and head groups. EPR spectroscopy (Box 8-C) with doxyl probes on carbon atoms at different depths within the bilayer has also been employed.¹²⁸

In recent years **molecular dynamics simulations** have been used to predict behavior of membranes. As is indicated in Fig. 8-11, the molten interior of the liquid crystalline *L*_α state is portrayed clearly.^{129–131} In the gel state the hydrocarbon chains maintain a closer packing and undergo coordinated movement.⁸⁸ It is difficult to know how realistic the simulations are. To calibrate the method efforts are made to correctly predict a series of known properties such as density and area per lipid (0.61 nm).¹³⁰

Functions of phospholipid head groups. The dipolar ionic head groups of phosphatidylcholine and phosphatidylethanolamine occupy about the same cross-sectional area as the two hydrocarbon tails. Thus, they are in rather close contact with each other. In crystals chains of hydrogen-bonded atoms may be formed. In phosphatidylethanolamine the phosphate and –NH₃⁺ ions may alternate in these chains.¹³²



In phosphatidylcholine, in which the nitrogen is surrounded by methyl groups and cannot form this kind of chain, water molecules bridge between the phosphates but the positive charges still interact with the adjacent negative charges.

The chains of hydrogen bonds between the head groups of phosphatidylethanolamine help to stabilize the bilayer and are apparently responsible for the elevation of T_m by 10–30° above that observed for phosphatidylcholine.¹³² In contrast, the negatively charged carboxyl groups of phosphatidylserine make the membrane less stable. The melting point is increased if the pH is lowered, protonating these groups. Their presence also makes the membrane sensitive to the concentration of cations.¹³³ The same is true of phosphatidylglycerol, whose head group contains a negatively charged phosphate without an attached

counterion. Addition of calcium ions increases T_m greatly and causes either phosphatidylglycerol or phosphatidylserine to form a separate phase with a more crystalline-like packing of the hydrocarbon side chains.¹³⁴ Hydrogen bonding between head groups also occurs with glycolipids.¹³⁵

Non-bilayer structures of phospholipids.

Under appropriate conditions some aqueous phospholipids can exist in non-bilayer phases, a fact that may be of considerable biological importance.^{119,136,137} In the presence of Ca^{2+} some pure phospholipids can be converted to the **inverted hexagonal** or H_{II} phase (Fig. 8-12).^{136,138-140} In this phase the phospholipid heads are clustered together in cylindrical "inverted" micelles which pack in a hexagonal array. The ease with which this transition can occur is increased by the presence of small amounts of diacylglycerols or lysolecithins.¹⁴¹ Some lipids, such as the galactosyldiacylglycerol of chloroplasts, do not form bilayers but prefer the hexagonal phase structure.^{29,32a} This is

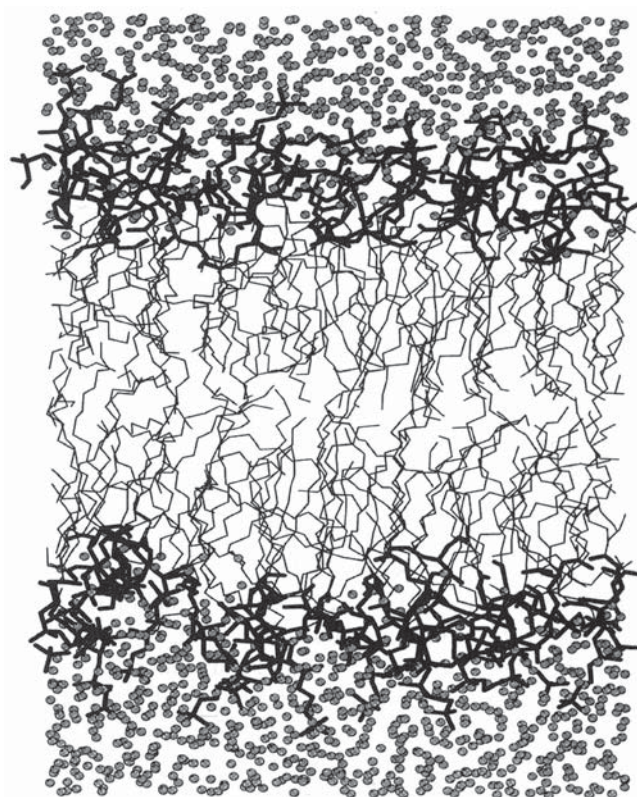


Figure 8-11 Results of simulated motion in a lipid bilayer consisting of 64 molecules of dipalmitoylphosphatidylcholine and 23 water molecules per lipid at a pressure of 2 atm and 50°C. The view is that observed after 500 ps of simulation. Bold lines represent the head group and glycerol parts of the structures and the thin lines the hydrocarbon chains. The gray spheres represent water molecules. From Berger, Edholm, and Jähnig.¹³⁰ Courtesy of Dr. Olle Edholm.

thought to be a result of very high curvature of a bilayer that arises from the sizes and packing of their head groups. Another phase, even though it is liquid, has a three-dimensional cubic symmetry.^{142-144a} It apparently consists of a complex arrangement of polyhedral bilayer surfaces with interpenetrating water channels between them.¹⁴³

Membrane fluidity and life. In agreement with the known behavior of bilayers, the lipids of most membranes in all organisms are partially liquid at those temperatures suitable for life. Organisms have developed at least three distinct means of ensuring that membrane lipids remain liquid.¹⁴⁵ (1) In our bodies (as well as in *E. coli*) the unsaturated fatty acids that are present lower the melting point. Mutants of *E. coli* that are unable to synthesize unsaturated fatty acids cannot live unless these materials are supplied in the medium.¹⁴⁶ (2) In *Bacillus subtilis*, which contains no unsaturated fatty acids when grown at 37°C, and in other gram-positive bacteria, more than 70% of membrane

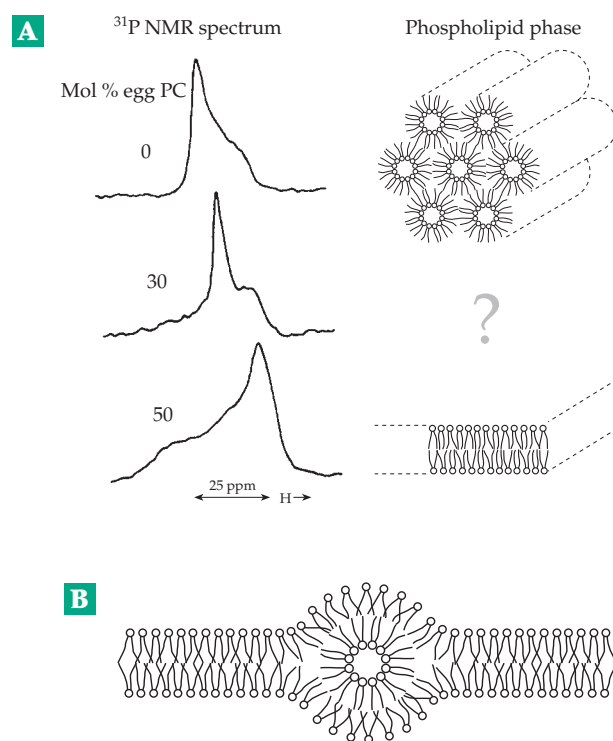


Figure 8-12 (A) ^{31}P NMR spectra of different phospholipid phases. Hydrated soya phosphatidylethanolamine adopts the hexagonal H_{II} phase at 30°C. In the presence of 50 mol% of egg phosphatidylcholine only the bilayer phase is observed. At intermediate (30%) phosphatidylcholine concentrations an isotropic component appears in the spectrum. (B) Inverted micelles proposed to explain "lipidic particles" seen in freeze fracture micrographs of bilayer mixture of phospholipids, e.g., of phosphatidylethanolamine + phosphatidylcholine + cholesterol. From de Kruijft *et al.*¹¹⁹ Courtesy of B. de Kruijft.

BOX 8-C ELECTRON PARAMAGNETIC RESONANCE (EPR) SPECTRA AND "SPIN LABELS"

Unpaired electrons have magnetic moments and are therefore suitable objects for magnetic resonance spectroscopy. The technique is similar to NMR spectroscopy, but microwave frequencies of $\sim 10^{10}$ Hz are employed, the energies being ~ 100 times greater than those used in NMR.^{a,b} Unpaired electrons are found in organic free radicals and in certain transition metal ions, both of which are important to many enzymatic processes. Furthermore, **spin labels** in the form of stable organic free radicals, can be attached to macromolecules at many different points.^{a,c-e} Coupling of such artificially introduced unpaired electrons with the magnetic moments of other unpaired electrons or of magnetic nuclei can often be observed by EPR techniques.

The conditions for absorption of energy in the EPR spectrometer are given by the equation

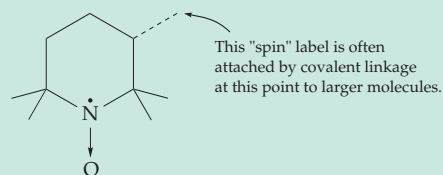
$$h\nu = g\beta H_0$$

which is identical in form to that for NMR spectroscopy. Here H_0 is the external magnetic field strength and β is a constant called the Bohr magneton. The value of g , the **spectroscopic splitting factor**, is one of the major characteristics needed to describe an EPR spectrum. The value of g is exactly 2.000 for a free electron but may be somewhat different in radicals and substantially different in transition metals. One factor that causes g values to vary with environment is **spin-orbit coupling** which arises because the p and d orbitals of atoms have directional character. For the same reason g sometimes has three discrete values for the three different directions (g value anisotropy). The g value parallel to the direction of H_0 ($g_{\parallel}$) often differs from that in the perpendicular direction ($g_{\perp}$). Both values can be ascertained experimentally.

A second feature of an EPR spectrum is **hyperfine structure** which results from coupling of the magnetic moment of the unpaired electron with nuclear spins. The coupling is analogous to the spin-spin coupling of NMR (Chapter 3). The hyperfine splitting constant A , like the coupling constant J of NMR spectroscopy, is given in Hertz. Splitting may be caused by a magnetic atomic nucleus about which the electron is moving or by some adjacent nucleus or other unpaired electron. Sometimes important chemical conclusions can be drawn from the presence or absence of splitting. Thus, the EPR spectrum of a metal ion in a complex will be split by nuclei in the ligand only if covalent bonding takes place.

It is customary in EPR spectroscopy to plot the first derivative of the absorption rather than the

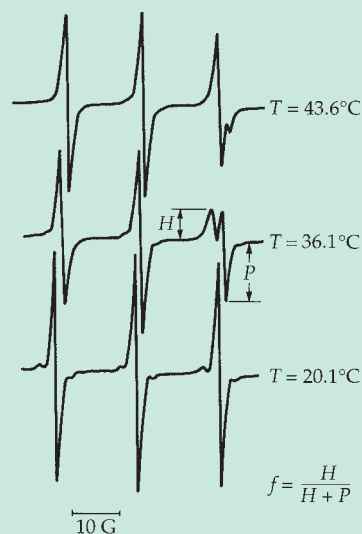
absorption itself. Thus, for the paramagnetic nitroxide 2,2,6,6-tetramethylpiperidine-1-oxyl the EPR spectrum consists of three equally spaced bands whose peaks are marked at the points where the steep



2,2,6,6-Tetramethylpiperidine-1-oxyl

lines in the middle of the first derivative plots cross the horizontal axis.

Coupling with the ^{14}N nuclear spin causes splitting into three lines as shown in the accompanying figure.

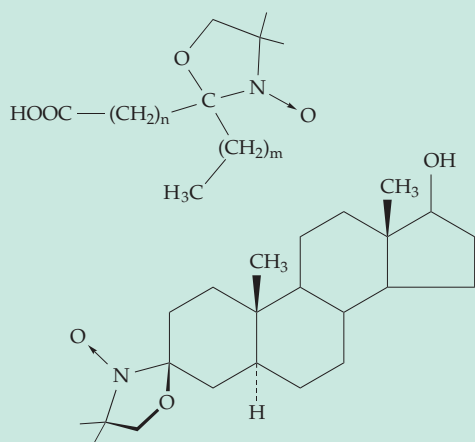


EPR spectrum of tetramethylpiperidine-1-oxyl dissolved in an aqueous dispersion of phospholipids. (Top) above the main bilayer transition temperature T_t ; (center) between T_t and pretransition temperature; (bottom) below pretransition temperature. From Shimshick and McConnell.^f

This nitroxide is more soluble in liquid regions of bilayers than it is in solid regions. As bilayers are warmed in the EPR spectrometer, the solubility of this spin-labeled compound in the lipid can be followed (see figure). The lower of the three spectra approximates that of the spin label in water alone, while the others are composite spectra for which part of the spin label has dissolved in the phospholipid bilayers.

BOX 8-C (continued)

Since frequencies for EPR spectroscopy are ~ 100 times higher than those for NMR spectroscopy, correlation times (Chapter 3) must be less than $\sim 10^{-9}$ s if sharp spectra are to be obtained. Sharp bands may sometimes be obtained for solutions, but samples are often frozen to eliminate molecular motion; spectra are taken at very low temperatures. For spin labels in lipid bilayers, both the bandwidth and shape are sensitively dependent upon molecular motion, which may be either random or restricted. Computer simulations are often used to match observed band shapes under varying conditions with those predicted by theories of motional broadening of lines. Among the many spin-labeled compounds that have been incorporated into lipid bilayers are the following:



Much of the interpretation of the observed changes in EPR spectra of spin labels is empirical. For example, the spectra in the accompanying figure can be interpreted to indicate that the spin label dissolves in the lipid to a greater extent at higher temperatures. The ratio f (defined in the figure) is an empirical quantity whose change can be monitored as a function of temperature. Plots of f vs T have been used to identify transition and pretransition temperatures in bilayers.^f

EPR spectroscopy is used widely in the study of proteins and of lipid-protein interactions.^c It has often been used to estimate distances between spin labels and bound paramagnetic metal ions.^g A high-resolution EPR technique that detects NMR transitions by a simultaneously irradiated EPR transition is known as electron-nuclear double resonance (ENDOR).^h

^a Berliner, L. J., and Reuben, J., eds. (1989) *Spinlabeling. Theory and Applications*, Vol. 8, Plenum, New York ((Biological Magnetic Resonance Series)

^b Cantor, C. R., and Schimmel, P. R. (1980) *Biophysical Chemistry*, Freeman, San Francisco, California (pp. 525–536, 1352–1362)

^c Marsh, D. (1983) *Trends Biochem. Sci.* **8**, 330–333

^d Esmann, M., Hideg, K., and Marsh, D. (1988) *Biochemistry* **27**, 3913–3917

^e Millhauser, G. L. (1992) *Trends Biochem. Sci.* **17**, 448–452

^f Shimshick, E. J., and McConnell, H. M. (1973) *Biochemistry* **12**, 2351–2360

^g Voss, J., Salwinski, L., Kaback, H. R., and Hubbell, W. L. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 12295–12299

^h Lubitz, W., and Babcock, G. T. (1987) *Trends Biochem. Sci.* **12**, 96–100

fatty acids contain methyl branches (Chapter 21)^{147,148} which can decrease the melting point and increase the monolayer surface area by a factor of as much as 1.5. (3) Yet another mechanism for lowering the melting point of fats is the incorporation of **cyclopropane-containing fatty acids** (Chapter 21).

On the other hand, as we have already seen, cholesterol tends to reduce the mobility of molecules in membranes and causes phospholipid molecules to occupy a smaller area than they would otherwise. Myelin is especially rich in long-chain sphingolipids and cholesterol, both of which tend to stabilize artificial bilayers. Within our bodies, the bilayers of myelin tend to be almost solid. Bilayers of some gram-positive bacteria growing at elevated temperatures are stiffened by biosynthesis of bifunctional fatty acids with covalently joined “tails” that link the opposite sides of a bilayer.¹⁴⁹

Why must membrane lipids be mobile? One reason is probably to be found in the participation of

membranes in many vital transport processes. Biological membranes have a relatively high permeability to neutral molecules (including H_2O),^{64,150} and it has been suggested that above T_m fatty acid chains are free to rotate by 120° around single bonds from trans to gauche conformations. When such rotation occurs about adjacent, or nearly adjacent single bonds, **kinks** are formed. If a kink originates near the bilayer surface, as will usually be the case, a small molecule may jump into the void created. Since the kink can easily migrate through the bilayer, a small molecule may be carried through with it.^{151,152} The same factors may assist larger protein molecules which function in membrane transport. They probably also account for the substantial degree of **hydration** of bilayers which involves both the polar head groups and water diffusing through the nonpolar interior.¹⁵³

Not only can molecules diffuse through membranes but also membrane lipids and proteins can move with respect to neighboring molecules. The rates of **lateral**

diffusion of lipids in bilayers and of antigenic proteins on cell surfaces are rapid.^{80,154} If diffusion of phospholipids is assumed to occur by a pairwise exchange of neighboring molecules, the frequency of such exchanges can be estimated¹⁵⁵ as $\sim 10^7 \text{ s}^{-1}$. However proteins may meet many obstacles to free diffusion.¹⁵⁶ Lateral diffusion is often measured by the technique of **fluorescence recovery after photobleaching**. One small spot in a bilayer that contains a dye attached to a lipid or a protein is bleached by a laser beam. Lateral diffusion of nearby unbleached molecules into the bleached spot can then be observed.⁸⁰ Lateral diffusion can also be observed by NMR spectroscopy¹⁵⁷ and by single-particle tracking.^{158,159} In addition to diffusion there may often be a flow of membrane constituents in directions dictated by metabolism.¹⁶⁰ Although lateral diffusion is fast a “flip-flop” transfer of a phospholipid from one side of the bilayer to the other may require many seconds.¹⁶¹ However, a sudden increase in the calcium ion concentration, an important intramolecular signal (Chapter 11), activates a “scramblase” protein which promotes a rapid transbilayer movement of phospholipids.¹⁶²

Electrical properties of membranes. Biological membranes serve as barriers to the passage of ions and polar molecules, a fact that is reflected in their high electrical resistance and capacitance. The electrical resistance is usually $10^3 \text{ ohms cm}^{-2}$, while the capacitance is $0.5\text{--}1.5 \text{ microfarad } (\mu\text{F}) \text{ cm}^{-2}$. The corresponding values for artificial membranes are $\sim 10^7 \text{ ohms cm}^{-2}$ and $0.6\text{--}0.9 \mu\text{F cm}^{-2}$. The lower resistance of biological membranes must result from the presence of proteins and other ion-carrying substances or of pores in the membranes. The capacitance values for the two types of membrane are very close to those expected for a bilayer with a thickness of $\sim 2.5 \text{ nm}$ and a dielectric constant of 2.^{54,84,163} The electrical potential gradient is steep.

Outer cell surfaces usually carry a net negative charge, the result of phosphate groups of phospholipids, of carboxylate groups on proteins, and of sialic acids attached to glycoproteins. This negatively charged surface layer attracts ions of the opposite charge (counterions), including protons, and repels those of the same charge. The result is development of a diffuse **electrical double layer** consisting of the fixed negative charges on the surface and a positive **ionic atmosphere** extending into the solution for a distance that depends upon the ionic strength.^{164–166} This ionic atmosphere is analogous to that postulated by the Debye–Hückel theory (Chapter 6). At the physiological ionic strength of 0.145 M the thickness of the double layer, taken as the distance at which the electrical potential falls to a certain fraction of that at the cell surface,^{164,165} is about 0.8 nm . However, the double-layer thickness increases to about three times this

value at an ionic strength of 10^{-3} M and to still greater distances at lower ionic strengths.

The net surface charge of a cell and the associated electrical double layer are important in interactions between cells and may influence the development of extracellular structure such as basement membranes. The net negative charge on cells also gives rise to an experimentally measurable electrophoretic mobility.

A characteristic of living cells is the maintenance of **ionic gradients** across the plasma membrane. Thus almost all cells accumulate K^+ , even “pumping” it from very dilute external solutions. Cells also exclude sodium, pumping it out from the cytoplasm by mechanisms considered in Section C,2. If a microelectrode is inserted through a cell membrane and the potential difference is measured between the inside and outside of the cell, a **resting potential** which, in nerve cells, may be as high as 90 mV is observed. The origin of the potential appears to lie in the concentration differences of ions. From the value of ΔG for dilution of an ion (Eq. 6-25) and the relationship between ΔG and electrode potential (Eq. 6-63), the Nernst equation (Eq. 8-2) can be derived. According to this equation, which applies to a single ion for which the membrane is permeable,

$$E_m = \frac{RT}{nF} \ln \left(\frac{c_1}{c_2} \right) = \frac{0.059}{n} \log \left(\frac{c_1}{c_2} \right) \quad \text{at } 25^\circ\text{C} \quad (8-2)$$

a 10-fold concentration difference across the membrane for a monovalent ion ($n = 1$) would lead to a 59-mV membrane potential, E_m . Since membranes are relatively impermeable to sodium ions, it is generally conceded that for many membranes the origin of the membrane potential lies mainly with the potassium ion concentration difference which is maintained by the Na^+ , K^+ -ATPase (Section C). A more complete equation takes account of K^+ , Na^+ , and Cl^- together with their respective permeabilities.^{167–169} Note also that Eq. 6-64 is also often called the Nernst equation.¹⁷⁰

Protons are also pumped across cytoplasmic and inner mitochondrial membranes, a topic of Chapter 18. The flow of protons from inside to outside also contributes to the membrane potential. The positive charges of H^+ , K^+ , and other cations associated with the external membrane surface are balanced by the negative charges of protein molecules as well as Cl^- and phosphate anions that are in or near to the inner surface of the membranes.

Another possibility for proton flow has intrigued biophysicists for years. Membranes often display a substantial electrical conductivity in a lateral direction along the membrane surface.^{171–173} Electrical conduction may involve movement of protons along hydrogen-bonded lines, e.g., involving ethanolamine head groups or phosphate groups and bridging water as

previously discussed (see also Eq. 9-96). Alternatively, conduction may depend upon membrane-associated proteins.¹⁷⁴ This lateral proton conduction may be important to many proton-driven membrane processes, such as rotation of bacterial flagella, ATP synthesis, and pumping of ions (Chapter 18).

The two sides of a membrane. Many observations indicate great differences between the inside and outside of the membranes that surround cells.^{51,175,176} Bretscher and Raff⁶² observed that, among the phospholipids of the erythrocyte membrane, phosphatidylcholine predominates in many mammals but is replaced by sphingomyelin in ruminants. Sheep erythrocytes are resistant to cobra venom phospholipase A, which is known to remove the fatty acid from the central position on the glycerol of phosphatidylcholine, causing lysis of the cells. The resistance of sheep erythrocytes suggested that the sphingomyelin is on the outside of the membrane while the phosphatidylethanolamine and other phospholipids are inside. By inference, phosphatidylcholine is also largely on the outside of plasma membranes. Supporting this conclusion is the observation that most of the reactive amino groups of phosphatidylethanolamine and phosphatidylserine are found on the inside (cytoplasmic) surfaces.¹⁷⁷ Since the total content of phosphatidylcholine and sphingomyelin often exceeds that of phosphatidylethanolamine and phosphatidylserine, the bilayer would be incomplete on the inside of the membrane were it not for the presence of proteins, which contribute more to the inside than to the outside surface.

Glycolipids are usually on the *outside* of plasma membranes with the attached sugar chains projecting into the surrounding water. An important generalization is that *sugar groups attached either to lipids or to proteins tend to be on outer cell surfaces or on materials that are being exported from cells.* An exception is found in the abundant galactolipids of chloroplasts.

2. Membrane Proteins

The many proteins present within or attached to membranes have a variety of functions. Some are obviously structural, tying other proteins to a membrane or providing a base for projecting fimbriae, flagella, and other appendages. Some proteins of the outer surface act as anchoring points for macromolecules that lie between cells. The inner surfaces of membranes are attached to the cytoskeleton (Chapter 7). Many of the proteins embedded in membranes control the passage of materials across membranes. Others serve as receptors that sense the presence of specific compounds or of light. Membranes may also contain foreign proteins such as subunits of virus coats. Proteins that are deeply embedded in membranes are

referred to as **intrinsic** or **integral membrane proteins**. Proteins that are more loosely associated with the membrane, principally at the inner surface, are called **peripheral**.⁶¹

Integral membrane proteins. Membrane proteins are hard to crystallize¹⁷⁸ and precise structures are known for only a few of them.^{179–181} A large fraction of all of the integral membrane proteins contain one or more **membrane-spanning helices** with loops of peptide chain between them. Folded domains in the cytoplasm or on the external membrane surface may also be present. The best-known structure of a transmembrane protein is that of the 248-residue bacteriorhodopsin. It consists of seven helical segments that span the plasma membrane (Fig. 23-45) and serves as a light-activated proton pump. Other proteins with similar structures act as hormone receptors in eukaryotic membranes. A seven-helix protein embedded in a membrane is depicted in Fig. 8-5 and also, in more detail, in Fig. 11-6.

The most hydrophobic integral membrane proteins can be extracted into organic solvents such as mixtures of chloroform and methanol. One such **proteolipid protein**, the 23.5-kDa lipophilin, accounts for over half the protein of myelin.^{57,182} The purified protein from rat brain contains 66% of nonpolar amino acids and six molecules of covalently bound palmitic acid and other fatty acids per peptide chain in thioester linkage to cysteine side chains. This protein evidently has four transmembrane helical segments with the six fatty acid chains incorporated into the membrane bilayer. It also has cytoplasmic and extracellular loops, one of which binds inositol hexakisphosphate (Ins P-6). (Fig. 11-9).¹⁸³ The myelin proteolipid is an essential component of the myelin sheath and defects in this protein are associated with some demyelinating diseases⁵⁷ which are discussed in Chapter 30.

There are many known topologies for helix-bundle membrane proteins with the number of membrane-spanning helices ranging from 1 to 14 or more e.g., see Fig. 8-23.^{184–186} A topology can often be predicted using suitable computer programs.^{187–191} A first step is to identify all sequences of 20 or more residues that could form a helix sufficiently hydrophobic to allow good nonpolar interactions with the bilayer core. Sequences that can form **amphipathic** (amphiphilic) helices must also be considered because two or more of these can pack together in a membrane with their hydrophilic sides together, sometimes forming pores (see Section C,1).¹⁹²

Predictions of membrane protein structure are in part based on the **positive-inside rule** which states that positively charged lysine and arginine residues will not pass through a membrane but will remain on the negatively charged cytoplasmic surface. Often, N-terminal parts of a protein will pass through a

membrane and will contain glutamate or aspartate residues whose side chains carry negative charges. These tend to remain outside of the membrane where they are attracted to the positive charges on the membrane outer surface. Positively charged residues may interact with phosphate groups of phospholipids and tyrosine and tryptophan side chains may interact with the carbonyl groups of the ester linkages.^{192a,b}

After probable transmembrane helical regions have been identified a residue-by-residue attempt can be made to identify cytosolic and extracellular loops. Chemical reactivities of the naturally occurring side chains can also be examined. Residues within the helix bundle will be protected. Mutations can be prepared systematically by “**scanning mutagenesis**.” For example, presumed extracellular loops in the erythrocyte band 3 protein (next section) have been mutated by introduction of *N*-glycosylation acceptor sites (Asn-X-Ser / Thr). If the sequence is at least 12–14 residues away from transmembrane sequences it will probably be glycosylated in a suitable laboratory test system.¹⁹³ Within a transmembrane domain amino acid residues can be systematically replaced with alanine (“alanine scanning mutagenesis”) or with cysteine and effects on the protein can be observed. Substitution with cysteine allows another possibility: Two cysteines on adjacent transmembrane helices may be linked as disulfides if they are close enough.^{194,195} Discovery of such neighboring pairs can be very valuable in attempting to establish relationships of one helix to another. A cloned gene can also be split into two pieces prior to crosslinking of cysteine side chains. This may facilitate mapping of tertiary interactions within transmembrane proteins.¹⁹⁶ Molecular dynamics methods for modeling helix bundle proteins are also being developed.¹⁹⁷

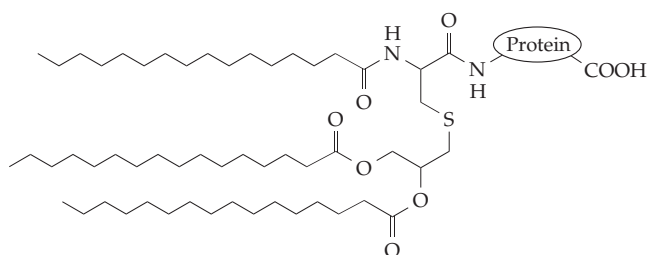
Not all integral membrane proteins have a helix bundle structure. Some of the simplest transmembrane proteins are the subunits of bacterial viruses and pili (Figs. 7-7 and 7-9). The 7-nm α -helical rods of phage M13 have a 20-residue hydrophobic section (residues 25-46) which is preceded by a negatively charged sequence that appears to form a short *amphipathic* helix that lies on the external surface of the membrane and helps to anchor the subunit. A positively charged cluster at the opposite end of the hydrophobic helical region remains in the cytoplasm.^{180,186,198} The **porins**, which form pores in outer membranes of bacteria and ribosomes, are large 16-strand β cylinders (Section C,1).^{179,180} It has been suggested that this different basic architecture may be related to the fact that the porin polypeptides must be exported through the inner membrane before being refolded in the periplasm.¹⁹⁹ Keep in mind that a β strand of nine residues can span the 33-nm bilayer core just as well as can a 22-residue helix. Beta structures as well as shorter helices may well be present in proteins that also contain membrane-spanning helices.

A third important structural pattern involves extensive use of amphipathic helices that lie partially embedded in a membrane surface. For example, the blood lipoproteins are lipid particles partially coated by amphipathic helices (Chapter 21).^{200,201}

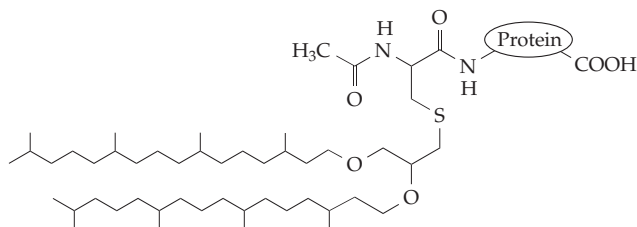
Anchors for proteins. Proteins with membrane-spanning sequences are usually anchored into the membranes with the help of many polar side chains, often including glycosylated residues, in the ends and loops that protrude on the two sides of the membranes. Other proteins are anchored by insertion into membranes of nonpolar groups. These include fatty acyl groups at N termini, fatty acid ester groups on serine, threonine, or cysteine side chains, and polyprenyl groups attached to cysteine side chains as thioethers.²⁰² Many membrane-anchored proteins carry the saturated 14-carbon **myristoyl** group in amide linkage with an N-terminal glycine of the protein.^{203–206} The fatty acid chain is added at the time of the protein synthesis, i.e., **cotranslationally**. It can intercalate into the membrane bilayer but provides a relatively weak anchor.²⁰⁷ The 16-carbon **palmitoyl** group is typically added to a cysteine thiol in recognition sequences at various positions in a protein.^{182,202,208,209} The modification occurs after protein synthesis, i.e., **posttranslationally**. Because thioesters are relatively unstable, palmitoylation is regarded as a reversible modification.²¹⁰

A third lipid anchor is provided by the polyprenyl **farnesyl** (15-carbon) and **geranylgeranyl** (20-carbon) groups in thioether linkage to cysteine residues. These must be present in specific recognition sequences at the C termini of proteins, most often with the sequence CAAX.^{211–215} The prenylation (also called isoprenylation) reaction is followed by proteolytic removal of the last three residues (AAX) and methylation of the new C-terminal carboxyl group as is discussed in Chapter 11, Section D,3. See also Chapter 22, Section A,4.

A lipoprotein present in the periplasmic space of *E. coli* is anchored to the outer bacterial membrane by a triacylated modified N-terminal cysteine containing a glyceryl group in thioether linkage as shown in the following structure (see also Section E,1).



A related anchor that uses diphytanylglyceroylation is found in certain proteins of archaeobacteria.²¹⁶



A series of **glycosylphosphatidylinositol** (GPI) anchors that are covalently linked to a variety of proteins utilize diacylglycerol or alkylacylglycerol for attachment to a bilayer. The proteins are joined through their C-terminal carboxyl groups to the diacylglycerol by a chain of covalently linked ethanolamine, phosphate, mannose, glucosamine, and *myo*-inositol as shown in Fig. 8-13.^{217–223} The proteins are linked to the diacylglycerol through the conserved structure: $\text{H}_2\text{N-protein} \rightarrow \text{ethanolamine} \rightarrow \text{P} \rightarrow 6\text{Man}\alpha 1 \rightarrow 2\text{Man}\alpha 1 \rightarrow 6\text{Man}\alpha 1 \rightarrow 4\text{GlcNAc} \rightarrow \text{Ins} \rightarrow \text{diacylglycerol}$.

The structure in Fig. 8-13, which anchors the small Thy-1 antigen to surfaces of rat thymocytes,²²⁴ contains additional mannose, *N*-acetylgalactosamine, and ethanolamine phosphate. These groups may be missing or substituted by other groups in other anchors.²²⁵ For example, that of human erythrocyte acetylcholinesterase lacks the extra mannose and GalNAc of the Thy-1 anchor but contains a palmitoyl group attached to an oxygen atom of the inositol. This provides an additional hydrocarbon tail that can enter the bilayer.²¹⁸ Other PI-anchored proteins include enzymes, such as alkaline phosphatase and lipoprotein lipase, and surface proteins of the parasites *Trypanosoma*,²²⁵ *Leishmania*, and *Toxoplasma*.²²⁰ Some adhesion molecules and a variety of other outer surface proteins are similarly attached to membranes. Analysis of the genome of *Caenorhabditis elegans* suggests that the nematode contains over 40 GPI-tailed proteins and perhaps more than 120.^{225a}

Analyzing erythrocyte membranes. The proteins of red blood cell membranes were among the first to be studied. Because membrane proteins are present in small amounts and tend to be hard to dissolve without denaturation, they have been difficult to study.

They usually do not dissolve readily in water, but red cell membranes can be almost completely solubilized in water using a 5×10^{-3} M solution of the chelating agent EDTA (Table 6-9) or by 0.1 M tetramethyl ammonium bromide.²²⁶ These observations suggested that ionic linkages between proteins, or between proteins and phospholipids, are important to membrane stability. Nonionic detergents such as those of the Triton X series or β -octylglucopyranoside also solubilize most membrane proteins,^{178,227–229} whereas ionic detergents such as sodium dodecyl sulfate (SDS) often cause unfolding and denaturation of peptide chains.

Gel electrophoresis of plasma membrane proteins in SDS solution yields ~ 10 prominent bands and at

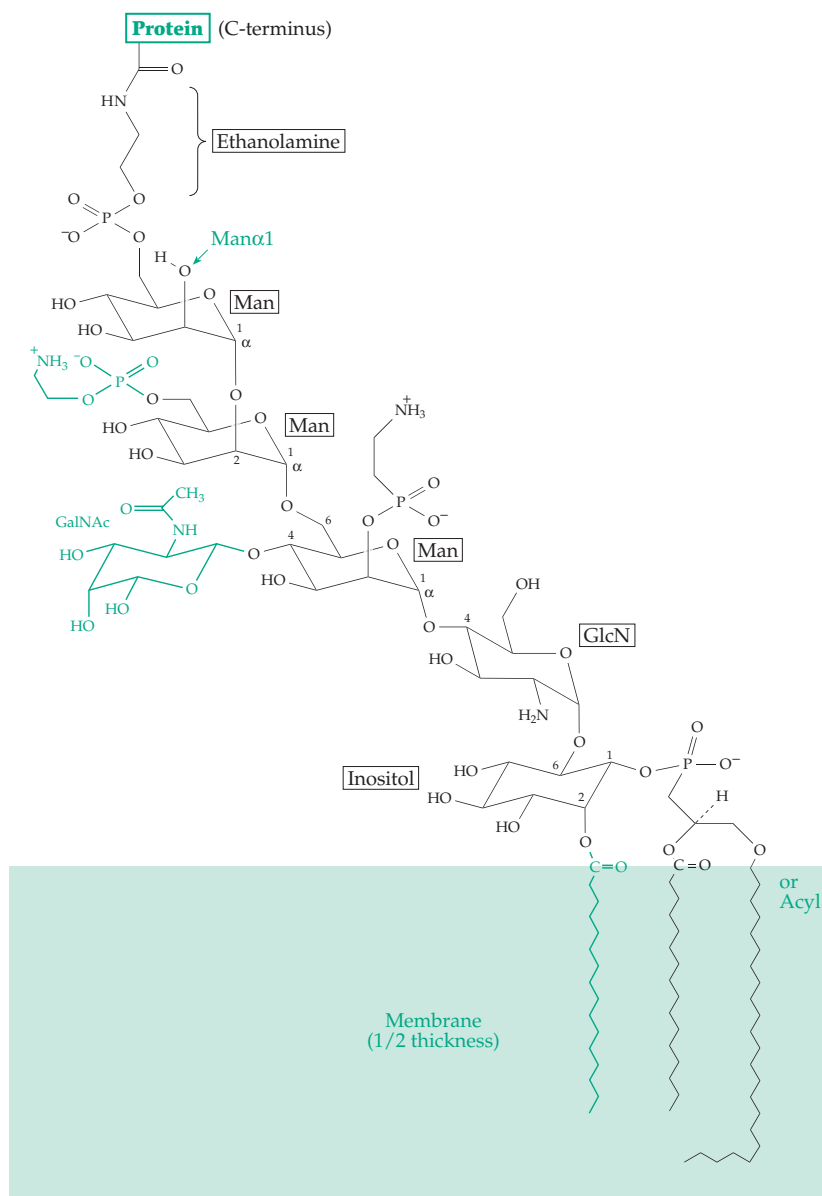
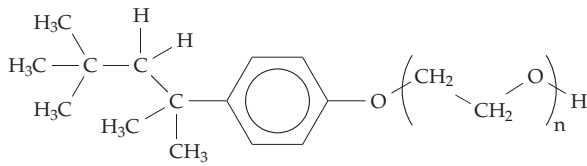


Figure 8-13 Structure of glycosylphosphatidylinositol (also called phosphatidylinositol-glycan) membrane anchors. The core structure is shown in black. The green parts are found in the Thy-1 protein and / or in other anchors.



Polyoxyethylene p-t-octyl phenol: the Triton X series of detergents.
n = 9–10 for Triton X-100 and 7–8 for Triton X-114.

least 30 less intense bands ranging in molecular mass from 10 to 360 kDa.^{230,231} Proteins of erythrocyte ghosts, and the usual system for numbering them, are shown in Fig. 8-14. Some very important proteins known to be present in this membrane, such as (Na⁺ + K⁺)-activated ATPase (Section C,2), are found in such low quantities (e.g., a few hundred molecules in a single red blood cell)⁶² that they do not show up in electropherograms. Mitochondrial membranes appear to be more complex than plasma membranes, but myelin has a somewhat simpler composition.

Glycoproteins. Many of the integral proteins of membranes are glycoproteins.^{234–236} These may sometimes be recognized in electropherograms because they are stained by the periodic acid-Schiff (PAS) procedure. At least 20 glycoproteins are present in erythrocyte ghosts, and glycoproteins appear to be prominent protein components of the plasma membranes of all eukaryotes and of primitive archaeobacteria such as *Halobacterium*.

The most abundant glycoprotein of red blood cell membranes is the 95-kDa PAS-reactive **band 3 protein** (Fig. 8-14) which makes up ~25% of the total membrane protein.^{236–240} A variety of asparagine-linked oligosaccharides based on the core hexasaccharide structure shown in Chapter 4, Section D,2 are present and apparently project into the surrounding medium. Electron micrographs of freeze-fractured surfaces through the membrane bilayer (Fig. 8-15) show ~4200 particles of 8 nm diameter per square micrometer, randomly distributed and apparently embedded in the membrane. These probably represent dimers of the glycoprotein. The amino acid sequence of the 911-residue protein suggests 13 membrane-spanning helices in the 550-residue C-terminal domain and that the 41-kDa N-terminal domain projects inward into the cytoplasm.^{236,240} Among the first 31 residues are 16 of aspartate or glutamate. These provide a highly negatively charged tail that is able to interact electrostatically with other proteins.²³⁶ Among these are components of the cytoskeleton, which appears to be anchored to the membrane via the band 3 protein. The band 3 protein is also a substrate for transglutaminase (Eq. 2-23) which creates covalent crosslinks to other proteins.²³⁷ Another major function of the band 3 glycoprotein is to form channels for the transport of anions (Section C,2).

Another integral glycoprotein of erythrocytes, the 31-kDa **glycophorin A** (PAS-1),^{235,241–244} is ~60% by

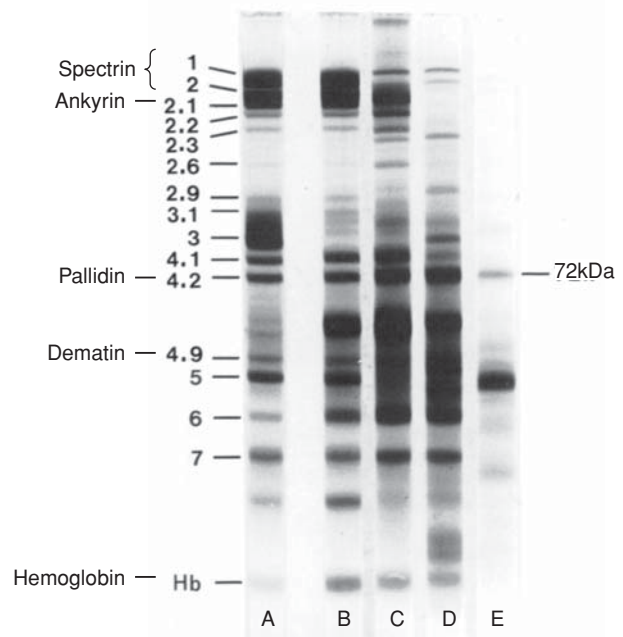


Figure 8-14 SDS-polyacrylamide gel electrophoresis of human erythrocyte ghosts. (A) From untreated cells. (B) From cells digested externally with chymotrypsin. (C) Inside-out vesicles prepared from cells pretreated with chymotrypsin. (D) The same inside-out vesicles after further treatment with chymotrypsin. (E) Polypeptides released by the chymotryptic treatment of the inside-out vesicles. The peptides are numbered according to the system of Steck²³²; Hb, hemoglobin. From Luna *et al.*²³³

weight carbohydrate. Its 131-residue chain has a single ~23-residue membrane-spanning helix. The first 50 residues at the N terminus, which project from the outside surface of the membrane, include many serines and threonines. Their side chains carry 15 O-linked tetrasaccharides and one complex N-linked oligosaccharide. There are a total of ~160 sugar residues per peptide chain, largely *N*-acetylgalactosamine, galactose, and sialic acid. Some of the oligosaccharides contain MNO blood group determinants^{234,241} (Box 4-C). Because of a high content of sialic acid, glycophorin also carries a large negative charge. The 35-residue C-terminal domain is hydrophilic and rich in proline, glutamate, and aspartate. It probably extends into the cytoplasm and may bind calcium ions or interact with $-\text{NH}_3^+$ groups on phospholipid heads.

If all the sugar residues of the glycophorin molecules in an erythrocyte were spread over the surface of the cell they could cover approximately one-fifth of its surface in a loose network. However, it is more likely that they form bushy projections of a more localized sort. These oligosaccharides not only act as immunological determinants but also serve as receptors for influenza viruses. Other glycoproteins related to glycophorin A occur in smaller amounts.²⁴⁴

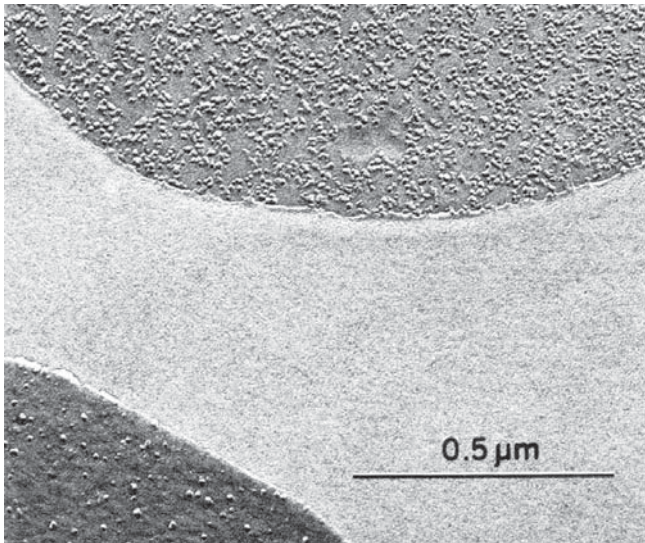


Figure 8-15 Freeze-fractured membranes of two erythrocyte “ghosts.” The upper fracture face (PF) shows the interior of the membrane “half” closest to the cytoplasm. The smooth region is lipid and contains numerous particles. The lower face, the extracellular half (EF), possesses fewer particles. The space between the two is nonetched ice. See Figs. 1-4 and 1-15 for electron micrographs of sections through biological membranes. Courtesy of Knute A. Fisher.

Connections to the cytoskeleton. About one-third of the protein of the red blood cell membrane is accounted for by a pair of larger hydrophobic peptides called **spectrin** with molecular masses of 280 kDa (α chain) and 246 kDa (β chain).^{245–250a} These are found in bands 1 and 2 of Fig. 8-14. The spectrin monomers consist largely of 106- to 119- residue repeat sequences each of which folds into a short triple-helical bundle (Fig. 8-16B). The beaded-chain monomers associate readily to $\alpha\beta$ dimers, long ~ 100 nm thin flexible rods which associate further to $(\alpha\beta)_2$ tetramers. The latter, in turn, bind to monomers or to small oligomers of actin. In red blood cells the actin crosslinks the spectrin tetramers into a two-dimensional “fishnet” (Fig. 8-16A), the $\sim 85,000$ spectrin tetramers uniformly covering the entire inner surface ($130 \mu\text{m}^2$) of the erythrocyte.^{245,251} The inner location of spectrin was established by the fact that chemical treatments²⁵² that covalently label groups on proteins exposed on the outer surface of erythrocytes (e.g., iodination with lactoperoxidase; Chapter 16) did not label spectrin.²⁵⁷

Spectrin also binds to one domain of another large 215-kDa peripheral protein called **ankyrin** (band 2.1, in Fig. 8-14) which anchors the spectrin network to the membrane.^{258,259} Ankyrin is actually a multigene family of related proteins that are present in many metazoan tissues.^{258,260} These are modular proteins

with separate binding domains for spectrin and band 3 protein. The latter domain contains 24 **ankyrin repeats**, 33-residue modules, also found in a variety of other proteins.^{258,259}

Ankyrin binds firmly to the band 3 glycoprotein which is embedded in the membrane. The 78 kDa band 4.1 protein is another major component of the erythrocyte **membrane skeleton**.^{261,262} Protein 4.1 binds both to ankyrin and also to **glycophorin C**, providing another anchor to an integral membrane protein (Fig. 8-16C). Spectrin:actin:protein 4.1 in a 1:2:1 ratio are the major components of the membrane skeleton.²⁶² Other less abundant proteins include **adducin**, protein 4.2 (**pallidin**, which interacts with band 3 protein),²⁶³ protein 4.9 (**dematin**, an actin bundling protein),²⁶⁴ and the muscle proteins tropomyosin and tropomodulin (Chapter 19). While spectrin and ankyrin of erythrocyte membranes have been studied most intensively, related proteins occur in other cells.²⁶⁵ Spectrin of brain and other tissues is also known as **fodrin**.²⁶⁶ **Dystrophin** and **α -actinin**, actin-crosslinking proteins of muscle, are also members of the spectrin superfamily.²⁴⁸ Protein 4.1 also occurs in various organisms,²⁶⁷ in various tissues, and in various locations in cells.²⁶⁸

What is the function of the membrane skeleton? There is a group of hereditary diseases including **spherocytosis** in which erythrocytes do not maintain their biconcave disc shape but become spherical or have other abnormal shapes and are extremely fragile.^{269–272} Causes of spherocytosis include defective formation of spectrin tetramers and defective association of spectrin with ankyrin or the band 4.1 protein.^{265,273} Thus, the principal functions of these proteins in erythrocytes may be to strengthen the membrane and to preserve the characteristic shape of erythrocytes during their 120-day lifetime in the bloodstream. In other cells the spectrins are able to interact with microtubules, which are absent from erythrocytes, and to microtubule-associated proteins of the cytoskeleton (Chapter 7, Section F).²⁷⁰ In nerve terminals a protein similar to erythrocyte protein 4.1 may be involved in transmitter release.²⁷⁴ The cytoskeleton is also actively involved in transmembrane signaling.

Integrins and focal adhesions. Mature erythrocytes have no nucleus and lack the microtubules and actin filaments that span other cells. In nonerythroid cells the major connections of the cytoskeleton to the membrane are through large $\alpha\beta$ heterodimeric membrane-spanning proteins called integrins (Fig. 8-17). These proteins, as well as the ends of the actin filaments, tend to be concentrated in regions long observed and described by microscopists as focal adhesions.^{275–279} These are also sites of interaction with the external proteins that form the **extracellular matrix (ECM)**. There are at least 16 different integrin

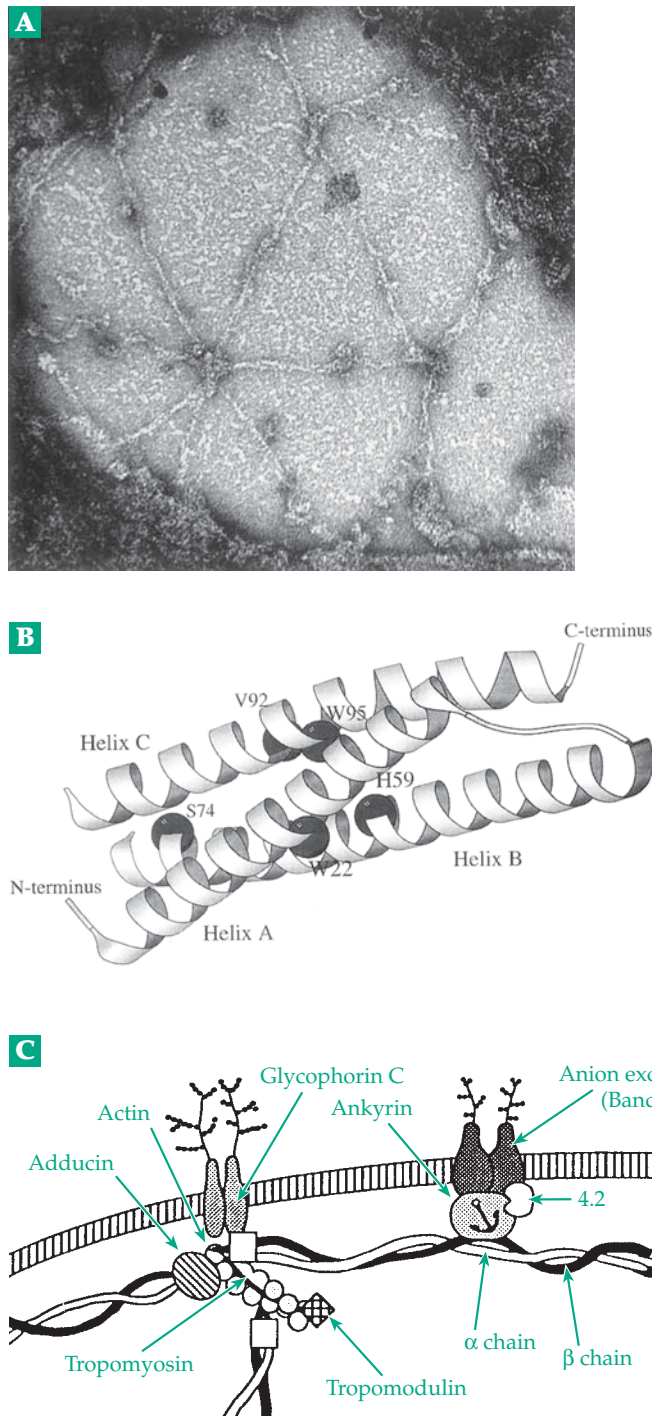


Figure 8-16 The erythrocyte membrane skeleton. (A) Electron micrograph showing a region of the membrane skeleton (negatively stained, X 200,000) and artificially spread to a surface area nine to ten times as great as in the native membrane. Spreading makes it possible to obtain clear images of the skeleton whose protein components are so densely packed and so subject to thermal flexing on the native, unspread membrane that it is difficult to visualize the individual molecules and the remarkably regular way that they are connected. The predominantly hexagonal and pentagonal network is composed of spectrin tetramers cross-linked by junctions containing actin oligomers and band 4.1 protein. Band 4.9 protein and tropomyosin are probably also bound to the oligomers, whose length (13 actin monomers long) corresponds to the length of a tropomyosin molecule. From Byers and Branton.^{253,254} Courtesy of Daniel Branton. (B) Proposed triple α -helical structure of a single spectrin repeat unit.²⁵⁵ Courtesy of Ruby I. MacDonald and Alfonso Mondragón. Each α spectrin chain consists of 20 and each β spectrin chain of 17 such repeats, which have only partially conserved sequences. The α and β chains are thought to associate in a side-by-side fashion and the $\alpha\beta$ heterodimers in an end-to-end fashion to give tetramers. These are the rod-like structures seen in (A) and (C). (C) Cross section of the unspread membrane and cytoskeleton as pictured by Luna and Hitt.²⁵⁶ Major interactions among components of the cytoskeleton are shown. Apparent sizes of the protein subunits, based on migration positions in SDS-polyacrylamide gel electrophoresis are: spectrin (260 and 225 kDa), adducin (105 and 100 kDa), band 3 (90 to 100 kDa), protein 4.1 (78 kDa), protein 4.2 (pallidin, 72 kDa), dematin (protein 4.9, 48 kDa), glycophorin C (~25 kDa), actin (43 kDa), and tropomyosin (29 and 27 kDa). Courtesy of Elizabeth J. Luna.

α subunits, whose external domains consist of up to 1114 residues, at least 8 β subunits with external domains of up to 678 residues, and at least 22 distinctly different $\alpha\beta$ heterodimers.^{280–282} The N-terminal part of each α subunit contains seven repeats of ~60 residues each, probably arranged as a β -propeller (see Fig. 11-7D or 15-23).²⁸¹ The integrins are structurally complex. Some contain a nucleotide-binding domain (see Figs. 2-13 and 2-27C).²⁸³ Integrins have differing and quite exacting specificities toward the proteins of the

external matrix to which they bind. They span the cell membrane and appear to be actively involved in communication between the cytoskeleton and external proteins.^{275,277,284} They are often described as **receptors** for the proteins that bind to them.

Other cell adhesion molecules. Long before the discovery of integrins another class of transmembrane adhesion molecules were recognized as members of the immunoglobulin family. These were called cell

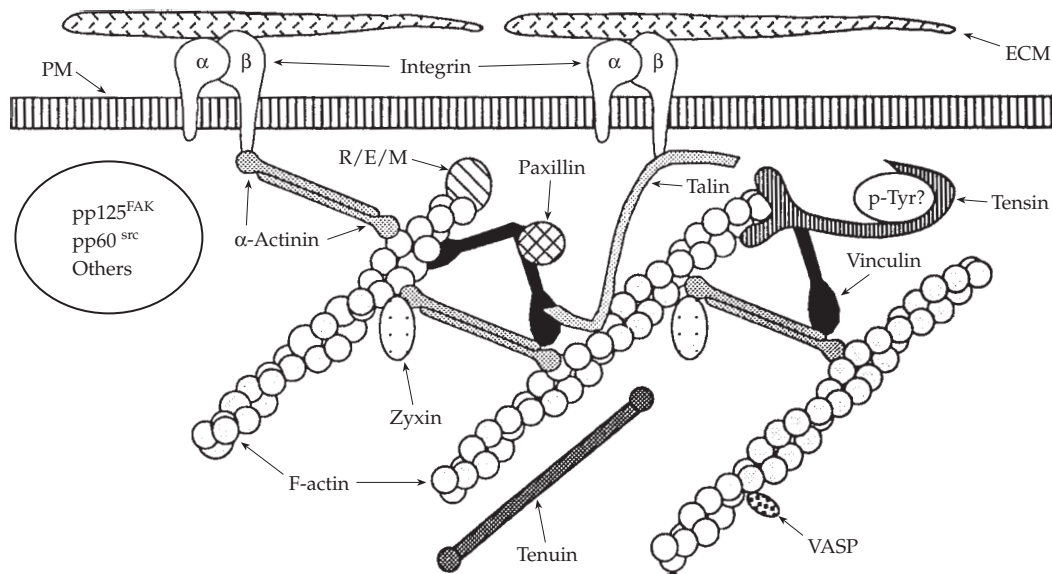


Figure 8-17 Working model of the protein–protein interactions in focal adhesions determined by *in vitro* binding experiments and immunolocalization. In addition, several interactions are of relatively low affinity in solution but may be enhanced at the membrane surface. Abbreviations are: ECM, extracellular matrix; PM, plasma membrane; p-Tyr-?, unknown phosphotyrosine-containing protein; R/E/M, member of the radixin/ezrin/moesin family; VASP, vasodilator-stimulated phosphoprotein. Diagram is modified from Simon *et al.*²⁸⁵

adhesion molecules (**CAMs**) by Edelman.^{286,287} These molecules go by names such as intercellular adhesion molecule-1 (ICAM-1), neural cell adhesion molecule (NCAM),^{288,289} liver cell adhesion molecule (LCAM), vascular cell adhesion molecule (VCAM), and platelet endothelial cell adhesion molecule (PECAM).^{290,291}

Many of these proteins, including the **T-cell antigen CD2**,^{291a} were first recognized on leukocyte surfaces as differentiation antigens and are often designated by the “Cluster of Differentiation” names (Chapter 31). For example, ICAM-1 is also called CD54 and PECAM CD31.^{290,292} The genes for these proteins are often expressed differentially in various tissues and the mRNA molecules formed undergo alternative splicing.^{287,289} The extracellular domains of these adhesion molecules consist largely of Ig domains and most have a single transmembrane helix and a small cytoplasmic C-terminal domain. Some of the alternatively spliced forms are attached to the membrane by PGI tails.^{287,289} ICAM-1 (Fig. 8-18) has five Ig domains and VCAM-1 has seven,²⁹³ but some CAMs have only two.²⁹⁴ The CAMs are glycoproteins, often with large N-linked oligosaccharides attached. The widely distributed NCAM contains long $\alpha 2 \rightarrow 8$ linked polysialic acid chains on two of the three N-glycosylation sites in the fifth Ig domain.²⁸⁹ The CAMs are often referred to as **receptors**. Their ligands include surface proteins such as fibronectin (next section) but also the integrins, which are also called **coreceptors** for

receptors of the Ig superfamily. Each of the coreceptors in a pair has binding sites for other ligands as well (Fig. 8-18). The CAMs and many other adhesion molecules are most abundantly expressed in embryonic tissues in which cells often migrate to new locations and for which the communication with neighboring cells is especially active. Many adhesion molecules bind only weakly and reversibly to their ligands, allowing cells to move.

The **cadherins** are calcium-dependent adhesion proteins that mediate direct cell–cell interactions.^{295,296} The external parts of the cadherins also have repeated structural domains with the Ig fold.^{297–298b} They have high affinity for each other, allowing cadherins from two different cells to interact and tie the cells together with a zipper-like interaction that is stabilized by the bound Ca^{2+} ions,^{297,300} and may be relatively long-lived. The gene for cadherin E is often mutated in breast cancers and may be an important **tumor suppressor gene** (Box 11-D).³⁰¹

Peripheral proteins of the outer membrane surface. Many integral membrane glycoproteins have their sugar-bearing portions exposed on the outer surface of the plasma membrane. Among these are receptors, ion pumps, and biochemical markers of individuality. In addition to these proteins, which are actually embedded in the bilayer, there are external peripheral proteins. One of the best known of these is

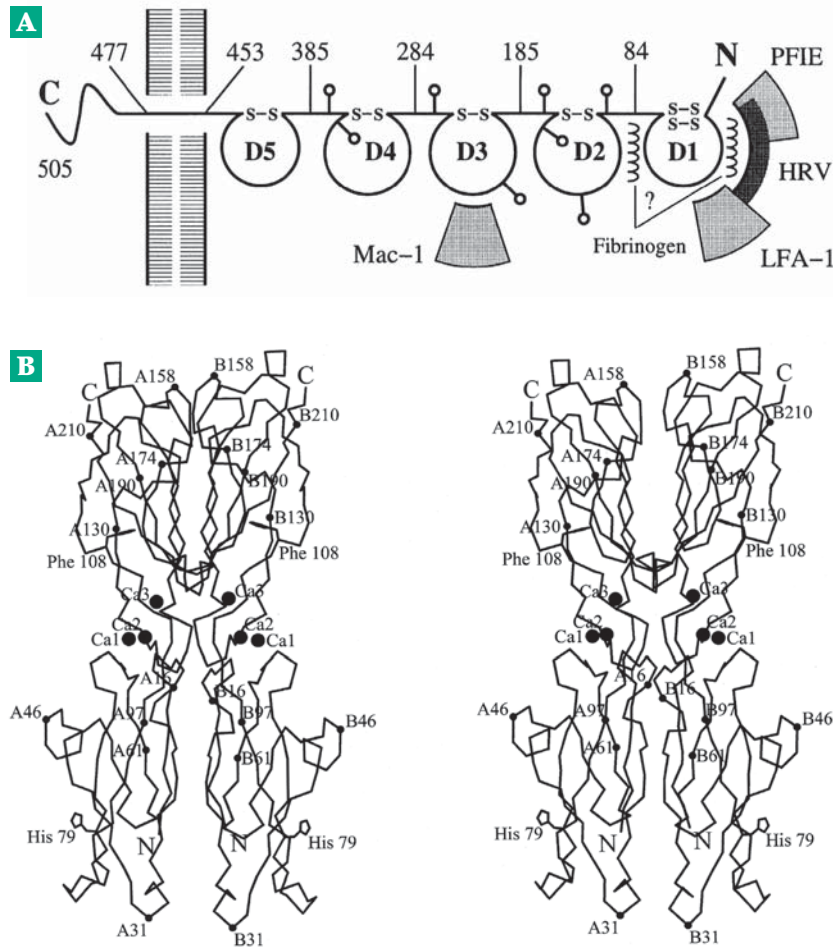


Figure 8-18 (A) Diagram of the ICAM-1 molecule. The structures labeled D1–D5 are the Ig domains. The glycosylation sites are labeled with small lollipop and approximate sites for binding of the α chain (CD11a) of the integrin called LFA-1, for the macrophage antigen Mac-1, for fibrinogen, for a ligand from erythrocytes infected by the malaria organism *Plasmodium falciparum* (PFIE), and for human rhinoviruses (HRV) are labeled. The binding sites are indicated schematically but each one is a complex interacting surface complementary to its ligand. From Bella *et al.*²⁹⁹ Courtesy of Michael Rossman. (B) Structure of two N-terminal domains of E-cadherin. The molecules of the dimer are related by a noncrystallographic twofold symmetry axis running vertically in the plane of the page. Clusters of three calcium ions are bound in the linker regions, connecting the N- and C-terminal domains of each molecule and, in this view, are separated by the twofold axis. The N- and C-terminal domains are composed of seven-stranded β -barrels showing the same topology and similar three-dimensional structures. From Nagar *et al.*²⁹⁶

fibronectin (from *fibra*, “fiber”, and *nectare*, “to bind”).^{302–309} This very large 470-kDa glycoprotein is a disulfide-linked dimer. Appearing under the electron microscope as having two 60-nm arms,³⁰⁵ fibronectin molecules join together and surround animal cells, anchoring other proteins and carbohydrates of the ECM to the cells. Fibronectin binds tightly to several different cell surface integrins and also to collagen and to glycosaminoglycans of the matrix. It binds to the blood-clotting protein fibrinogen, to actin, and also to staphylococci and other bacteria. Each peptide chain of the fibronectin dimer is organized as several domains. Fibrin and staphylococci bind to the N-termi-

nal domain. A second domain binds collagen and a domain near the C-terminus binds heparin. Several different integrins of cell surfaces bind to the region of the 8th, 9th, and 10th type III repeats.³⁰⁸ The specificity of the binding depends to a large extent on the presence of the specific tripeptide sequence Arg-Gly-Asp (**RGD**) in a type III repeat (Fig. 8-19B). Peptide sequences such as the PHSRN (Fig. 8-19B) and others³¹⁰ also participate in binding to specific integrins. Initial interactions are noncovalent but fibrin and collagen gradually become covalently attached to fibronectin through isopeptide linkages formed by transglutaminase (Eq. 2-23). This enzyme is abundant

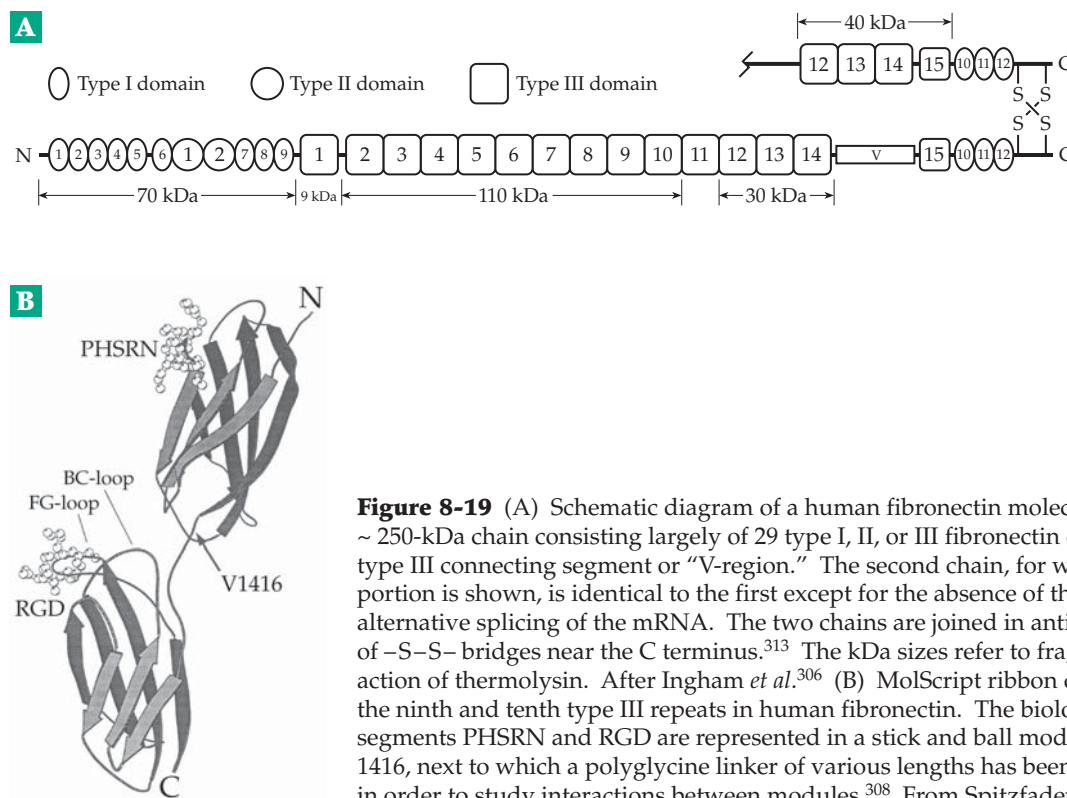


Figure 8-19 (A) Schematic diagram of a human fibronectin molecule showing one complete ~ 250-kDa chain consisting largely of 29 type I, II, or III fibronectin domains with a 12-kDa type III connecting segment or “V-region.” The second chain, for which only the C-terminal portion is shown, is identical to the first except for the absence of the V-region as a result of alternative splicing of the mRNA. The two chains are joined in antiparallel fashion by a pair of –S–S– bridges near the C terminus.³¹³ The kDa sizes refer to fragments generated by the action of thermolysin. After Ingham *et al.*³⁰⁶ (B) MolScript ribbon drawing of the structure of the ninth and tenth type III repeats in human fibronectin. The biologically active peptide segments PHSRN and RGD are represented in a stick and ball mode. The arrow marks valine 1416, next to which a polyglycine linker of various lengths has been inserted experimentally in order to study interactions between modules.³⁰⁸ From Spitzfaden *et al.*³¹⁴

in many developing tissues but often disappears as the tissues mature.^{311,312} Fibronectin was formerly known as the “large external transformation-sensitive protein,” a name derived from the fact that its quantity is greatly reduced in many virus-transformed cells. This might explain the loss of adhesiveness and of “contact inhibition” observed for cancer cells (Box 11-D).

There are 20 isoforms of human fibronectin. These arise by alternative splicing of the primary gene transcript. Their formation is developmentally regulated.^{308,315} One of the isoforms is present as a soluble protein (cold-insoluble globulin) in blood plasma.³⁰² The fibronectins belong to a larger family of **cytoadhesins**, among which are the blood-clotting proteins **fibrinogen** and **von Willebrand factor** (Chapter 12), and also **thrombospondin**, **vitronectin**,³¹⁶ **tenascin**,^{317–320} **laminin**,^{321,322} **osteopontin**,³²³ and **collagens**. These proteins all have a modular construction with repeated domains, often of several different types.³²⁴ For example, tenascin contains 14 1/2 EGF domains (Table 7-3), 16 fibronectin type III repeats, and a C-terminal segment homologous to fibrinogens.³²⁵ Laminin (see Fig. 8-33) also has EGF-like modules. Most of these proteins also have in common the presence of the RGD sequence, which binds the cytoadhesins to those integrins and other molecules that carry RGD receptor sites.^{304,326–328} The extracellular matrix is discussed further in Section E,2.

3. Enzymes and Membrane Metabolism

Many of the proteins of membranes are enzymes. For example, the entire electron transport system of mitochondria (Chapter 18) is embedded in membranes and a number of highly lipid-soluble enzymes have been isolated. Examples are **phosphatidylserine decarboxylase**, which converts phosphatidylserine to phosphatidylethanolamine in biosynthesis of the latter, and **isoprenoid alcohol phosphokinase**, which participates in bacterial cell wall synthesis (Chapter 20). A number of **ectoenzymes** are present predominantly on the outsides of cell membranes.³²⁹ Enzymes such as phospholipases (Chapter 12), which are present on membrane surfaces, often are relatively inactive when removed from the lipid environment but are active in the presence of phospholipid bilayers.^{330,331} The distribution of lipid chain lengths as well as the cholesterol content of the membrane can affect enzymatic activities.³³²

Why are membranes so important to cells? Besides their obvious importance in enclosing and defining the limits of cells, membranes are the result of a natural aggregation of amphipathic molecules. They also represent a natural arrangement for boundaries between different aqueous phases within a cell. In addition, membranes are the “habitat” for many relatively nonpolar molecules formed by metabolism. These include proteins with hydrophobic surfaces and

those with membrane anchors. The semiliquid interior of the membrane permits distortion of the bilayer and the addition or subtraction of proteins (and low-molecular-mass materials) in response to metabolic processes in the adjacent cytoplasm.

The principal factor providing stability to macromolecules and membranes is the hydrophobic nature of reduced organic compounds. This characteristic leads to the separation of lipids, proteins, and other molecules from the aqueous cytoplasm into oligomeric aggregates and membranes. However, the best catalysts, including most enzymes, are soluble in water. Thus, membranes represent thin regions of relative stability adjacent to aqueous regions in which chemical reactions occur readily and which tend to contain the more polar, the smaller, and the more water-soluble materials. The stability of membrane surfaces provides a means of bringing together reactants and of promoting complex sequences of biochemical reactions. For example, membranes contain both oxidative enzymes and reactive, dissolved quinones. The membrane–cytoplasmic interfaces may often be the metabolically most active regions of cells.

Despite their stability, membrane components have a metabolism of their own which is related to the high concentrations of oxidizing enzymes located in or on membranes. Oxidative reactions provide a mechanism for modification of hydrophobic membrane constituents. For example, sterols, prostaglandins, and other regulatory molecules are initially synthesized as hydrophobic chains attached to water-soluble “head groups” (Chapter 21). The hydrophobic products of these synthetic reactions tend to be deposited in membranes. However, attack by oxygen leads to introduction of hydroxyl groups and to a gradual increase in water solubility. As the hydrophilic nature of the compound is increased through successive enzymatic hydroxylation reactions, the hydrophobic membrane constituents eventually redissolve in the water and are completely metabolized. Another process that actively degrades membrane lipids is attack by hydrolytic enzymes such as the phospholipases.

C. The Transport of Molecules through Membranes

Small neutral molecules, such as water or ethanol, can penetrate membranes by **simple diffusion**.^{64,150} The rate is determined by the solubility of a substance in the membrane, by its diffusion coefficient (see Eq. 9-24) in the membrane, and by the difference in its concentration between the outside and the inside of the cell. This concentration difference is commonly referred to as the **concentration gradient** across the membrane. The ease of diffusion through a membrane is described quantitatively by a **permeability**

coefficient P which is related to the diffusion coefficient D (Eq. 8-3).

$$J = -D_m K c / \Delta x = -P \Delta c \quad (8-3)$$

Here J is the **flux** of molecules across the membrane, i.e., the number of molecules crossing one cm^2 per second. D_m is the diffusion coefficient in the bilayer, while K is a **partition coefficient**, the ratio of the concentration of the diffusing solute in the bilayer to that in water. The concentration gradient of the solute across the membrane is Δc , while Δx is the membrane thickness in centimeters. The permeability coefficient P for H_2O through biological membranes³³³ is about $1 - 10 \mu\text{m s}^{-1}$. For H^+ and OH^- P is $0.1 \mu\text{m s}^{-1}$. For halide ions diffusing across liposome bilayers P ranges from 10^{-5} to $10^{-3} \mu\text{m s}^{-1}$,³³⁴ fast enough to be of some biological significance. However, for most other ions P is less than $10^{-6} \mu\text{m s}^{-1}$. Because of their high lipid content membranes are quite permeable to nonpolar materials. For example, anesthetics usually have a high solubility in lipids, enabling them to penetrate nerve membranes.

1. Facilitated Diffusion and Active Transport

While simple diffusion may account for the entrance of water, carbon dioxide, oxygen, and anesthetic molecules into cells, movement of most substances is facilitated by protein **channels** and **transporters**.³³⁵ Genes of 76 families of such proteins have been located in the genomes of 18 prokaryotes.^{335a} Some of these provide for **facilitated diffusion**.^{169,335} Like simple diffusion, it depends upon a concentration gradient and molecules always flow from a higher to a lower concentration. A distinguishing feature of facilitated diffusion is a **saturation effect**, i.e., a tendency to reach a maximum rate of flow through the membrane as the concentration of the diffusing substance, on the high concentration side, is increased. In this characteristic it is similar to enzymatic action (Chapter 9).

In **active transport** a material is carried across a membrane against a concentration gradient, i.e., from a lower concentration to a higher concentration. This process necessarily has a positive Gibbs energy change (as given by Eq. 6-25) of approximately $5.71 \log c_2 / c_1 \text{ kJ mol}^{-1}$, where c_2 and c_1 are the higher and lower concentrations, respectively. The transport process must be coupled with a spontaneous exergonic reaction. In **primary active transport** there is a direct coupling to a reaction such as the hydrolysis of ATP to “pump” the solute across the membrane. **Secondary active transport** utilizes the energy of an **electrochemical gradient** established for a second solute; that is, a second solute is pumped against a concentration gradient and the first solute is then allowed to

cross the membrane through an exchange process with the second solute (**antiport** or **exchange diffusion**).

Alternatively, both the first and the second solutes may pass through the membrane bound to the same carrier (**cotransport** or **symport**). Another form of active transport is **group translocation**, a process in which the substance to be transported undergoes covalent modification, e.g., by phosphorylation. The modified product enters the cell and within the cell may be converted back to the unmodified substance. Transport processes, whether facilitated or active, often require the participation of more than one membrane protein. Sometimes the name **permease** is used to describe the protein complexes utilized.

Like facilitated diffusion, active transport depends upon conformational changes in carrier or pore proteins, the equilibrium between the two conformations depending upon the coupled energy-yielding process. Thus, if ATP provides the energy a phosphorylated carrier will probably have a different conformation than the unphosphorylated protein. A carrier with Na^+ bound at one site may have a different affinity for glucose than the same carrier lacking Na^+ . A hypothetical example of the kind of cycle that can function in active transport is provided by the picture of the “sodium pump” given in Fig. 8-25.

2. Pores, Channels, and Carriers

To accommodate the rapid diffusion that is often needed to supply food, water, and inorganic ions to cells, membranes contain a variety of small pores and channels. The pores may be nonspecific or they may be selective for anions or cations or for some other chemical characteristics. They may be permanently open or sometimes closed and referred to as **gated**. The gating may be controlled by the membrane electrical potential, by a hormone, by the specific ligand, or by other means. Some pores may be small enough to allow only small molecules such as H_2O to pass through. Others may be large enough to allow for nonspecific simple diffusion of molecules of low molecular mass. Structures are known for only a few.

Large pores tend to be nonspecific, but when the solute approaches the pore diameter in size the specificity increases. Furthermore, diffusion of ions through pores is influenced strongly by any charged groups in or near the pore. Thus, a cation will not enter a pore containing a net positive charge in its surface. Any electrical potential difference across the membrane, resulting from accumulation of excess negative ions within the cell, will also affect the diffusion of ions.^{336,337}

Porins. The outer membranes of gram-negative bacteria contain several 34- to 38-kDa proteins known

as porins. They form a large number of trimeric pores which allow molecules and ions with molecular masses <600 Da to enter. However, even small proteins are excluded. This appears to be a means for protecting the bacteria against enzymes such as lysozyme (Chapter 12). Four distinct porins are among the most abundant proteins of the outer membrane of *E. coli*.³³⁸ They are designated according to the names of their genes.

OmpF (a nonspecific, open channel) and **OmpC** (**osmoporin**) are encoded by *OmpF* (outer membrane protein F) and *OmpC* genes, respectively. **Maltoporin** (LamB) is selective for maltodextrins but also allows other small molecules and ions to pass.^{338–341a} Its name comes from its original discovery as a receptor for bacteriophage lambda. **PhoE** (**phosphoporin**) is a porin with a preference for anions such as sugar phosphates while OmpF prefers cations.³⁴² Porin **FepA** is ligand gated, opening to take up the chelated iron from ferric enterochelin (Chapter 16). OmpF, OmpC, and PhoE all have 16-stranded β -barrel structures (Fig. 8-20).^{199,343} Maltoporin forms a quite similar 18-stranded barrel. Similar porins are present in many bacteria.^{199,344}

The porin monomers associate to form trimeric channels as is shown in Fig. 8-20B. They all have a central water-filled, elliptical channel that is constricted in the center to an “eye” $\sim 0.8 \times 1.1$ nm in size. In this restriction zone the channel is lined with polar residues that provide the substrate discrimination and gating. For example, in OmpF and PhoE there are many positively and negatively charged side chains that form the edge of the eye (Fig. 8-20C). The electrostatic potential difference across the outer membrane is small, but apparently determines whether the porins are in an open or a closed state.^{344a} The voltage difference has opposite effects on OmpF and PhoE, apparently as a result of the differing distribution of charged groups.^{342,345,346} A key role in determining the voltage dependence may be played by Lys 18 (Fig. 8-20C).³⁴² Polyamines (Chapter 24), which are present in the outer membrane, induce closing of porin channels by binding to specific aspartate and tyrosine side chains.³⁴⁷

The most *abundant* protein in the *E. coli* outer membrane is OmpA. It appears to form a transmembrane helical bundle. Although it is regarded primarily as a structural protein it too acts, in monomeric form, as an inefficient diffusion pore.³⁵⁰ Mitochondrial outer membranes contain nonspecific pores (mitochondrial porins) that allow passage of sucrose and other saccharides of molecular mass up to 2 to 8 kDa.^{351,352} Similar pore-forming proteins have been found in plant peroxisomes.³⁵³

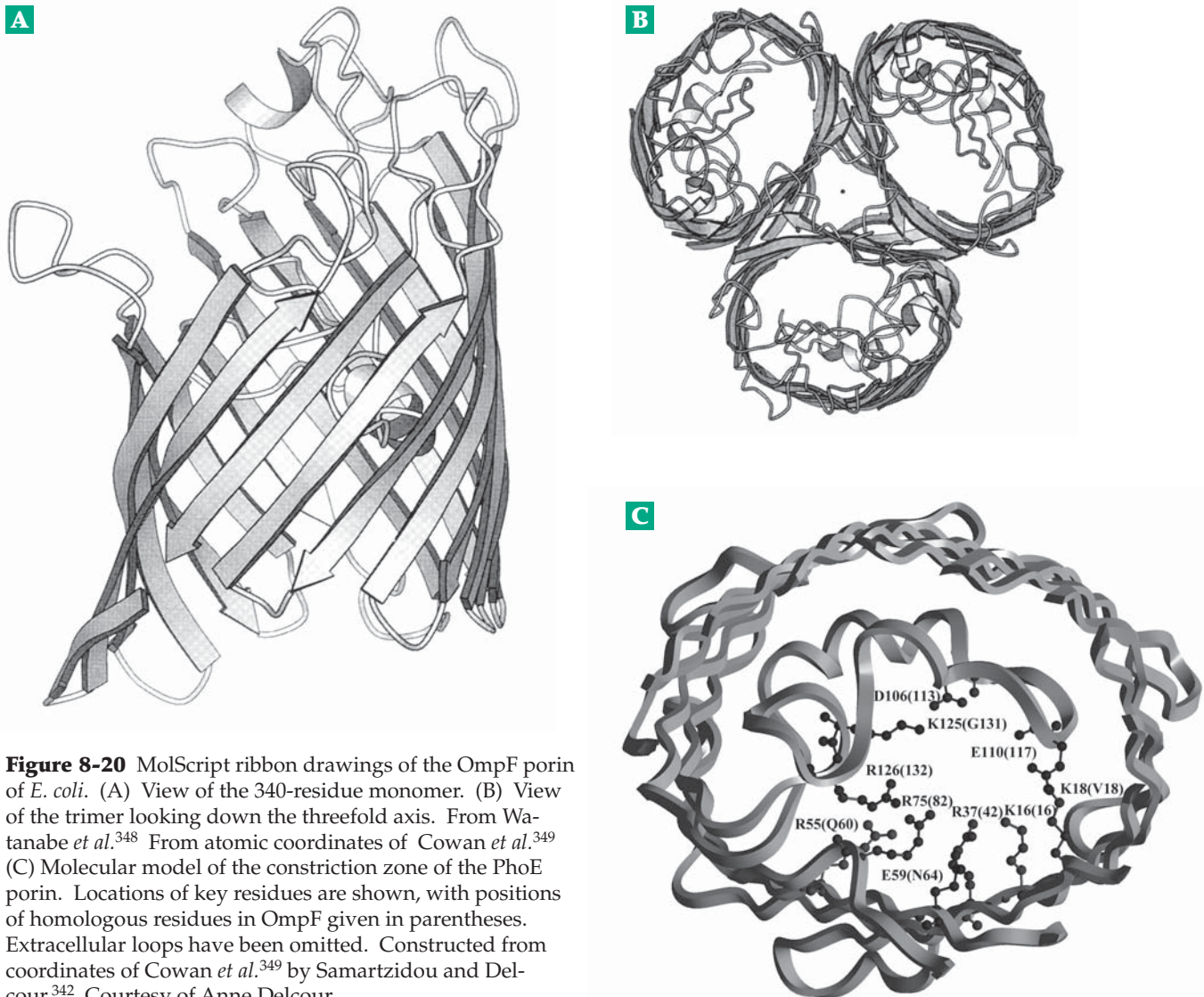
Aquaporins. Many biological membranes are not sufficiently permeable to water to allow for rapid osmotic flow. For example, the kidney membranes in

portions of Henle's loop have permeabilities as high as $2500 \mu\text{m s}^{-1}$ (compared to $10\text{--}20 \mu\text{m s}^{-1}$ in a 1:1 cholesterol / phospholipid bilayer).^{354-355a} This high permeability is provided by aquaporin-1 (**AQP-1**), formerly called CHIP (channel-forming integral membrane protein) or AQP-CHIP. Aquaporin-1 was first identified in erythrocyte membranes and is present in many tissues. The 28-kDa subunits of the protein form six-helix bundles, each with a pore in the center. These are associated as tetramers in the membrane.³⁵⁶⁻³⁵⁸

Other aquaporins with related sequences occur broadly. There are at least ten in mammals.^{359,359a} Plants, which must accommodate to heavy loss of water in hot dry weather, have aquaporins in both plasma membranes and tonoplasts.³⁶⁰ Bacteria also have aquaporins.^{356,361} A defect in aquaporin-2 of the kidney collecting duct leads to **nephrogenic diabetes insipidus**, in which the kidneys fail to concentrate urine in response to secretion of the hormone vasopressin.^{355a,362,363}

Ion channels. Most organisms contain a large number of ion channels. One of these, which plays a key role in nerve conduction, is the **voltage-gated K^+ channel**. It is closed most of the time but opens when a nerve impulse arrives, dropping the membrane potential from its resting 50 - to 70-mV (negative inside) value to below zero. This voltage change opens the channel, allowing a very rapid outflow of K^+ ions.³⁶⁴⁻³⁶⁷ The channels then close spontaneously. There are many different K^+ channels but most have a similar architecture.³⁶⁸⁻³⁷⁰ The three-dimensional structure has been determined for the membrane-spanning part of the K^+ channel of *Streptomyces lividans* (Fig. 8-21). The funnel-shaped tetrameric molecule has a narrow **conduction channel** which contains the **selectivity filter**.

The conduction channel is lined largely with hydrophobic groups. The selectivity filter, which discriminates between K^+ and Na^+ , is a short ($\sim 1.2\text{-nm}$ -long) narrow ($\sim 1.0\text{-nm}$ -diameter) portion of the



pore that is aligned roughly with the center of the bilayer. It is formed by four extended peptide chains, one from each subunit, each having the “**signature sequence**” for K^+ channels, TVGYG. In this sequence the peptide carbonyl groups all point into the channel. The consecutive groups of four carbonyls along the channel form binding sites for K^+ , whose ionic diameter (Table 6-10) is 0.27 nm. The K^+ must lose its hydration sphere to fit into the 1.0-nm channel. The site of strongest binding, occupied by K^+ in Fig. 8-21C, lies just at the C-terminal ends of four helices, and the partial negative charges of the helix dipoles probably contribute to the binding. The pore may constrict to strengthen the bonds. When a second K^+ ion enters the channel it appears to bind ~ 0.75 nm from the central K^+ , repelling it and weakening its interaction with the filter, allowing it to pass through the pore. Rb^+ (0.30 nm diameter) and Cs (0.34 nm diameter) also pass through.

How are the smaller Na^+ (0.19 nm diameter) and Li^+ (0.12 nm diameter) excluded from the pore? The pore is too small for the hydrated ions and perhaps too large to bind the dehydrated ion well enough to let it escape from its hydration sphere. Four negatively charged side chains in the cytoplasmic mouth of the pore presumably discourage anions from entering. At the other end, the extracellular entryway is a site that can be blocked specifically by 35- to 40-residue scorpion toxins.³⁶⁸

The *S. lividans* K^+ channel is *not* voltage gated. Voltage-gating mechanisms must be learned from study of other channels! The voltage-dependent K^+ channels from the rat have an $\alpha_4\beta_4$ composition. The α and β subunits coassemble in the endoplasmic reticulum and remain as a permanent complex,^{370a-c} After insertion into the plasma membrane the α subunits form a channel as in Fig. 8-21. However, an additional intracellular domain of

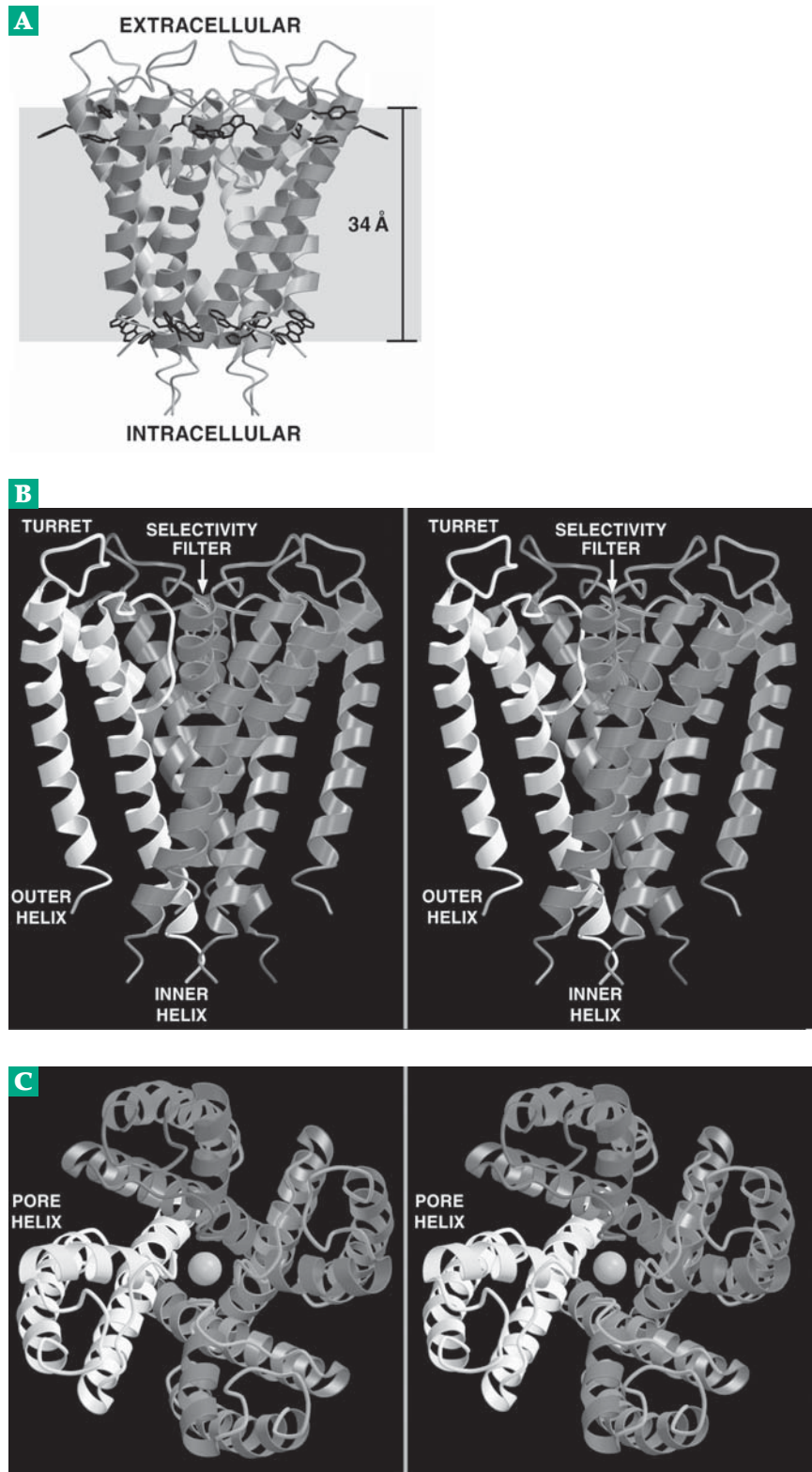


Figure 8-21 Views of the tetrameric K^+ channel from *Streptococcus lividans*. (A) Ribbon representation as an integral membrane protein. Aromatic amino acids on the membrane-facing surface are also shown. (B) Stereoscopic view. (C) Stereoscopic view perpendicular to that in (B) with a K^+ ion in the center. From Doyle *et al.*³⁶⁶

the α subunits, together with the β_4 complex, provide an elaborate and as yet poorly understood internal structure. The β subunits are oxidoreductases containing bound NADH, whose function is also uncertain.

There are many more pores and carriers of various kinds in biological membranes. Some, like the *S. lividans* K^+ channel, facilitate diffusion of a single ion or compound. They are **uniporters**. Others promote cotransport in which an ion such as H^+ or Na^+ also passes through the carrier. In some cases a pore is formed by a single molecule. Other pores follow a twofold, threefold, or fourfold axis (Fig. 8-21) of an oligomeric protein.³⁷¹ Two different conformations for the protein, one in which a “gate” opens to one side of the membrane and one in which it opens to the other, are usually involved. Interconversion between the two conformations is spontaneous but the equilibrium between them may be influenced by the binding of the solute, by the membrane potential, or by binding of inhibitors or activators. The latter may bind differently to the parts of the carrier (pore) protein exposed on the two sides of the membrane.

Channel-forming toxins and antibiotics. Some of the bacterial toxins known as **colicins** (Box 8-D) kill susceptible bacteria by creating pores that allow K^+ to leak out of the cells. One part of the **complement system** of blood (Chapter 31) uses specific proteins to literally punch holes in foreign cell membranes. **Mel-litin**, a 26-residue peptide of bee venom,^{372,373} as well as other hemolytic toxins and antibiotic peptides of insects, amphibians, and mammals (Chapter 31) form amphipathic helices which associate to form voltage-dependent anion-selective channels in membranes.^{374–377}

The polypeptide antibiotic **gramicidin A** consists of 15 nonpolar residues of alternating D- and L-configuration; two molecules can form a channel with a right-handed β helix structure.^{378,379} The central 0.48-nm hole in the β helix is large enough to allow unhydrated cations such as Na^+ or K^+ to pass through. The same peptide, under other conditions, forms left-handed helical channels whose structures are known.³⁷⁹ Aggregates of β_{10} helices may form channels between helices of **suzukacillin**.³⁸⁰ For this antibiotic as well as for **alamethicin** (Chapter 30) the conductance depends upon the membrane potential, a characteristic shared with the pores of nerve membranes. A variety of synthetic channel-forming peptides have been prepared. Some of these form α -helical bundles with a central pore. A five helix bundle of this type³⁸¹ has a structure reminiscent of the acetylcholine receptor channel of neurons.³⁸²

Ionophores and other mobile carriers. Facilitated diffusion of a molecule or ion is sometimes accomplished by binding to a **mobile carrier**. An example is the diffusion of a complex of K^+ with the

low-molecular-mass lipid-soluble carrier, or ionophore **valinomycin** (Fig. 8-22). The K^+ –valinomycin complex diffuses the short distance to the other side of the membrane and discharges the bound ion. If the rates of binding to a carrier and of release from the carrier are greater than those of the diffusion process, Michaelis–Menten kinetics are observed. The maximum velocity $V_{\max}$ and Michaelis constant K_m can be defined as in Eq. 9-15 for enzymatic catalysis.

Valinomycin is a **depsipeptide** which contains ester linkages as well as amide linkages. The antibiotic is made up of D- and L-valine, L-lactic acid, and D-hydroxyisovaleric acid. When incorporated into an artificial membrane bathed in a K^+ -containing medium, valinomycin increases the conductance greatly and when it is added to a suspension of *Streptococcus faecalis* cells the high ratio of $[K^+]_i / [K^+]_o$ falls rapidly.³⁸³ The loss of K^+ from cells probably explains the antibiotic activity. However, under suitable conditions, with a high external $[K^+]$, the bacteria will continue to grow and reproduce in the presence of the antibiotic.³⁸⁴

Uncomplexed valinomycin has a more extended conformation than it does in the potassium complex.^{385,386} The conformational change results in the breaking of a pair of hydrogen bonds and formation of new hydrogen bonds as the molecule folds around the potassium ion. Valinomycin facilitates potassium transport in a passive manner. However, there are cyclic changes between two conformations as the carrier complexes with ions, diffuses across the membrane, and releases ions on the other side. The rate of transport is rapid, with each valinomycin molecule being able to carry $\sim 10^4$ potassium ions per second across a membrane. Thus, a very small amount of this ionophore is sufficient to alter the permeability and the conductance of a membrane.

Because the stability constant of its complex with potassium is much greater than that with sodium, valinomycin is a relatively specific potassium ionophore. In contrast, the mushroom peptide **antamanide** has a binding cavity of a different geometry and shows a strong preference for *sodium* ions.^{388,390} The structure of the Na^+ –antamanide complex is also shown in Fig. 8-22B. The *Streptomyces* polyether antibiotic **monensin** (Fig. 8-22D),^{389,391} a popular additive to animal feeds, is also an ionophore. However, its mode of action, which involves disruption of Golgi functions, is uncertain.³⁹²

Anions of lipid-soluble phenols such as 2,4-dinitrophenol can serve as effective carriers of *protons* (Chapter 18). However, proteins usually serve as the natural carriers, both of protons and of other ions. A protein is sometimes pictured as rotating to present the solute-binding surface first to one side, then to the other side of a membrane. However, gated pores or channels are probable for most biological transport.

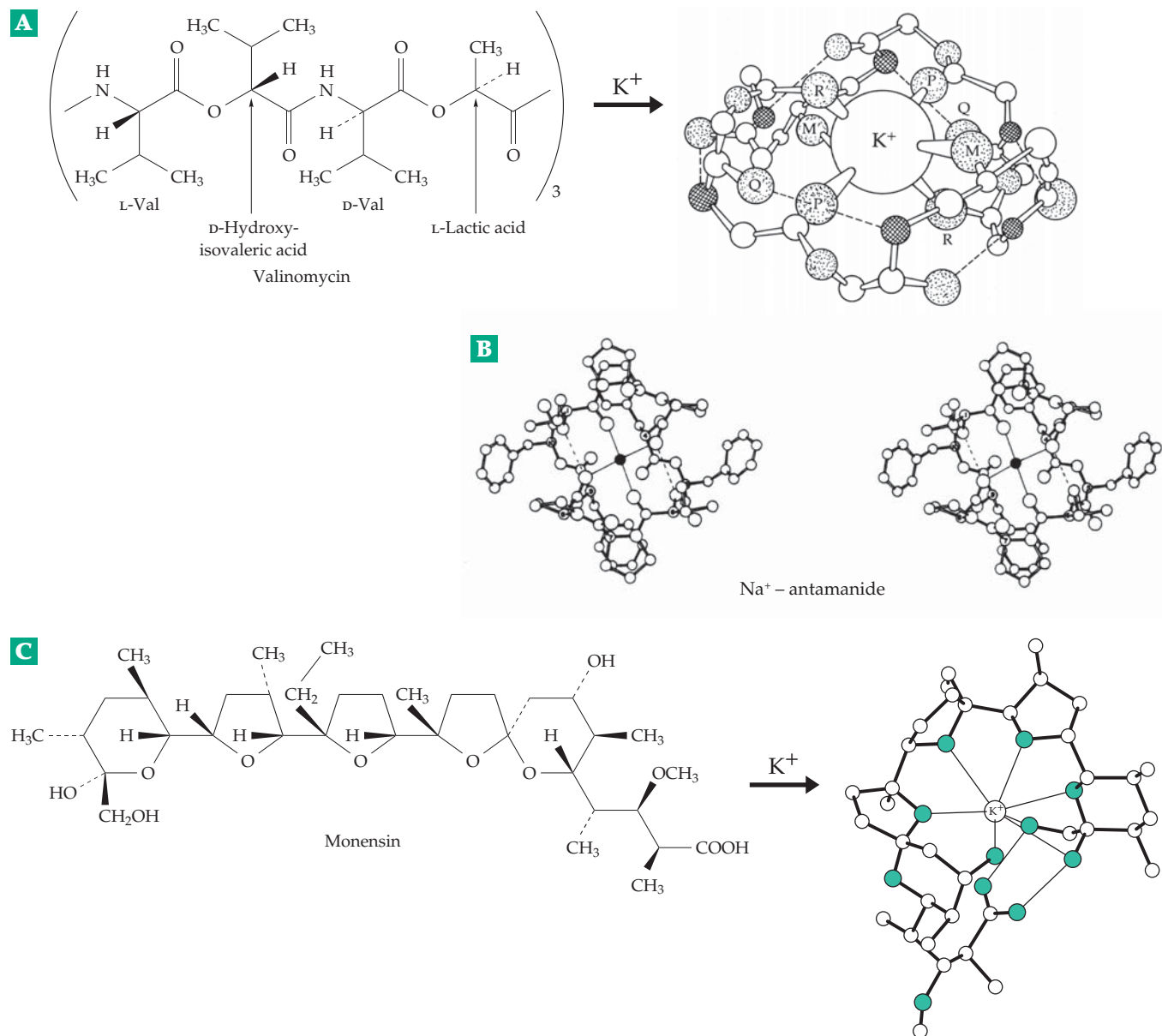


Figure 8-22 (A) Uncomplexed valinomycin and its complex with K⁺ (from Duax *et al.*³⁸⁷). (B) Stereodiametric of the Na⁺-[Phe⁴, Val⁶]antamanide complex. A molecule of C₂H₅OH, which forms the fifth ligand to the Na⁺, is omitted for clarity. From Karle *et al.*³⁸⁸ (C) Monensin and its complex with K⁺. From Pangborn *et al.*³⁸⁹

3. The 12-Helix Major Facilitator Superfamily

A large family of transmembrane facilitators from bacteria and eukaryotes appear to consist largely of 12 transmembrane helices with intervening cytosolic and extracellular loops. Some of these transporters facilitate simple uniport diffusion, but others participate in active transport of the symport or antiport type.^{393,394} Several hundred members of the family are known.³⁹⁵

Entrance of sugars into cells. It is important that sugars be able to enter cells rapidly. However, the

permeability coefficient P for D-glucose across a lipid bilayer is only 10^{-6} to 10^{-5} $\mu\text{m s}^{-1}$. For an intact erythrocyte, P is much greater: ~ 1 $\mu\text{m s}^{-1}$. This is the result of facilitated diffusion by a transport protein with a high specificity for hexose and pentose sugars having a pyranose ring in a C1 chain conformation.³⁹⁶ This human erythrocyte glucose transporter, now known as **GLUT1**, has a K_m of 1.6 mM for D-glucose but of >3 M for L-glucose. It is a 55-kDa intrinsic membrane glycoprotein migrating in band 4.5 of Fig. 8-14. From the sequence of its cloned gene the unglycosylated carrier was deduced to be a 54-kDa peptide of 492 residues.

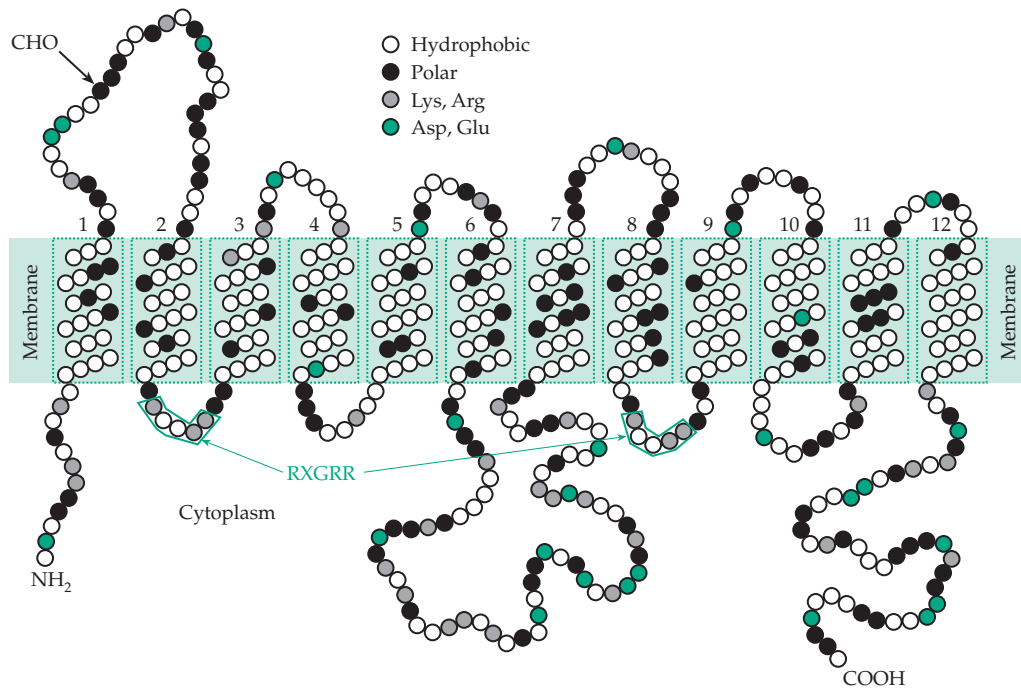
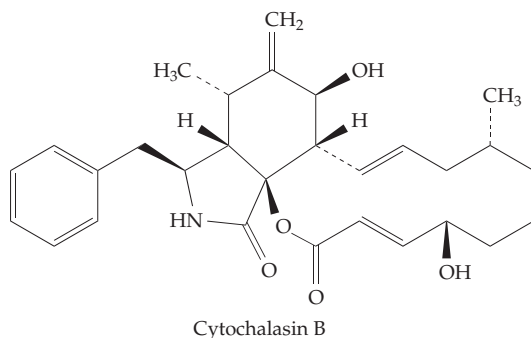


Figure 8-23 Predicted topology of the human glucose transporter GLUT1. The 12 predicted helices are numbered and the single external N-linked oligosaccharide is marked CHO. The sequence RXGRR (marked) is found in many 12-helix transporters. Its occurrence in these two positions suggests that the transporters may have evolved by duplication of a 6-helix motif. However, the human glucose transporters otherwise show no sequence similarity to other 12-helix transporters. After Bell *et al.*³⁹⁸ See also Muekler.⁴⁰⁰

The glucose transporter is specifically inhibited by the fungal metabolite **cytochalasin B** which can also be used for photoaffinity labeling of the transporter.³⁹⁷



There are at least six closely related facilitative sugar transporters in the human body.^{398–400} GLUT1 is found not only in red cells but also in brain⁴⁰¹ and other tissues. GLUT2 is the principal liver transporter and GLUT3 is found along with GLUT1 and GLUT5 in the brain.⁴⁰¹ GLUT5 is primarily a fructose transporter.^{402,403} The latter is present in spermatozoa, in which fructose transport is especially important (Box 20-A), and also in the small intestine. Both the GLUT1 and GLUT5 genes are overexpressed in breast cancer.⁴⁰³

GLUT6 is an unexpressed pseudogene (see Chapter 27) and GLUT7 has been found only in liver microsomes. GLUT4 of skeletal and cardiac muscle and adipose tissue has received a great deal of attention because of its response to insulin^{400,404–405a} and to exercise⁴⁰⁶ (see Chapter 17).

All of the GLUT family appear to be 12-helix transmembrane-regulated, gated-pore proteins^{407,408} with relatively short cytosolic N- and C-terminal ends. The proposed topology of GLUT1, which has been supported by much experimental data,^{400,406} is shown in Fig. 8-23. However, the helices are thought to be bundled with the glucose channel centered within a helix bundle. The structure is unknown, but it could resemble the α, α -barrel of glucoamylase (Fig. 2-29), which has 12 helices, a glucose-binding site in the center, and a structure that could easily be modified to form a central pore.

Dehydroascorbate, the oxidized form of vitamin C (Box 18-D) is also transported into cells by GLUT1 and GLUT3.⁴⁰⁹ A related transporter carries L-fucose into mammalian cells.⁴¹⁰ Another facilitates the uptake of galactose in yeast.⁴¹¹

Cotransport of sugars and other nutrients with H^+ or Na^+ . Epithelial cells of the small intestine or of kidney tubules must take up glucose at low concentra-

tions and discharge it into the bloodstream at a higher concentration.⁴¹² This active transport is accomplished by cotransport of glucose with Na^+ in a 1:2 ratio, with the sodium ion concentration gradient across the membrane providing a usable source of energy amounting to 5.8 kJ / equivalent of Na^+ .⁴¹³ The 662-residue human, sodium-dependent transporter **SGLT1** may have 14 transmembrane helices, 5 of which have been proposed to provide the sugar pathway.⁴¹⁴

Cotransport with Na^+ is also observed for transport of many other sugars, amino acids, neurotransmitters, and cofactors.⁴¹⁵ A confusing variety of transporter molecules have been identified and are now being classified into families based on gene sequences as well as function.^{416,417} Transporters from intestinal mucosal cells, kidney membranes, and synaptic endings of neurons have been studied most and have been the source for many of the cloned genes.⁴¹⁸ The Na^+ -dependent transporters of neutral amino acids from these tissues have long been classified as system A, system B, and system ASC (alanine, serine, cysteine) according to substrate specificities.^{417,419–420a} Both cationic amino acids and cystine are taken up by kidney tubules and intestinal epithelial cells by another Na^+ -dependent transporter which is defective in human **cystinuria**, a common metabolic genetic problem.^{421,422}

The brain contains several transporters specialized for rapid uptake of neurotransmitters glutamate and aspartate,^{423–425} glycine,⁴²⁶ γ -aminobutyrate (Gaba),^{427,428} and catecholamines and also for taurine, L-proline,⁴²⁹ serotonin,⁴³⁰ and other substances. Many of these are not only Na^+ dependent but also require cotransport of Cl^- .^{428–430} There are several different glutamate transporter genes with specialized distribution in the brain and other tissues.^{423,424}

Cotransport of sugars with H^+ is especially common in bacteria⁴³¹ but also occurs in eukaryotes. For example, the alga **chlorella** employs hexose / H^+ symporters.⁴³² The most investigated H^+ cotransporter is probably the **lactose (lac) permease** from *E. coli* which enables *E. coli* to take up lactose and other β -galactosides from very dilute solutions.^{433–436} From a variety of measurements it has been possible to propose a stacking arrangement for the 12 helices⁴³⁴ and to identify groups that are essential for function of the 417-residue protein. Glutamates as well as an arginine, and a histidine, all in transmembrane regions, are essential and may be involved in the gating and transport functions.⁴³⁶ The 469-residue **melibiose permease** of *E. coli* transports α -D-galactopyranosides, including melibiose ($\text{Galp}\alpha 1 \rightarrow 6 \text{ Glc}$) and raffinose ($\text{Galp}\alpha 1 \rightarrow 6 \text{ Glcp}\alpha 1 \rightarrow 2 \text{ Fruf}$).^{437,438} This permease will couple sugar uptake to either the H^+ or Na^+ gradient (or to a gradient of Li^+ but *not* of K^+).⁴³⁷ Cells of *E. coli* also contain H^+ symporters for D-galactose, D-xylose, and L-

arabinose that are homologous to the mammalian GLUT proteins.⁴³⁹

A series of specific H^+ -linked cotransporters are found in the brush border membranes of small intestine and in kidney epithelial cells.^{440,441} In green plants H^+ -linked cotransport of amino acids is used in the distribution of amino acids synthesized in the roots and leaves to other parts of the plants.^{442,443}

Antiporter or ion exchange transporters are also common. For example, *E. coli* uses a metal ion–tetracycline / H^+ transporter to carry the antibiotic tetracycline out of cells. This protein, when present, provides a high level of antibiotic resistance to the bacteria.⁴⁴⁴

4. Active Transport Systems

Both bacteria and eukaryotes also possess complex active transport systems for uptake of sugars, amino acids, and other nutrients and for pumping out toxic xenobiotics. One of the most important groups are the high affinity **ABC** (ATP-binding cassette) **transporters**, also called **traffic ATPases**.^{445–448a} The *E. coli* genome contains genes for 80 ABC transporters, 54 of which had been identified before the genome sequence was completed.⁴⁴⁷ In the human body an ABC transporter enables eukaryotic cells to pump out a large number of different drugs and other foreign compounds. This **multidrug resistance protein** (or P-glycoprotein) not only protects cells but also can seriously interfere with drug treatment. For example, cancer cells with increased amounts of this transporter often arise during chemotherapy. The transporter protein is a single 1280-residue, 170-kDa chain which probably has 12 transmembrane helices and two ATP-binding domains.^{448,449} Other human ABC transporters include the antigen processing transporter **TAP** (Chapter 31), the 1480-residue anion transporter **CFTR**, which is defective in cystic fibrosis (Box 26-C),⁴⁵⁰ the erythrocyte glutathione-conjugate exporter (Box 11-B), and a long-chain fatty acid transporter of peroxisomes. ABC transporters usually consist of four domains. Two are hydrophobic intrinsic membrane domains, each with six membrane-spanning helices and two are peripheral membrane ATP binding domains. All four domains may be in a single peptide chain, as in CFTR, or they may be separate smaller proteins as in bacterial periplasmic permeases.^{431,446}

Periplasmic permeases. Gram-negative bacteria contain numerous ABC transporters with components located on the periplasmic surfaces of their plasma membranes. Many of these can be dissociated from the surfaces by **osmotic shock**, i.e., by sudden changes in the osmotic pressure of the medium.^{451,452} For example, cells of *E. coli* suspended in 0.5 M sucrose, treated with 10^{-4} M EDTA for 10 min, and then diluted

with cold water release ~50 **binding proteins** that hold sugars, amino acids, ions, and other substances tightly with $K_f = 10^6$ – 10^8 M⁻¹.^{453,454} An example is the **L-arabinose binding protein** of *E. coli*, a 306-residue peptide organized into two large α/β units with a deep cleft between them. The sugar is bound into this cleft^{455,456} by an extensive network of hydrogen-bonding interactions between polar groups in the sugar and in the protein similar to the network that binds either D-galactose (Fig. 4-18) or D-glucose to another of the periplasmic binding proteins.^{457,458} Proteins with the same general architecture bind D-ribose, L-arabinose,⁴⁵⁸ and maltodextrins.⁴⁵⁹ The $\alpha 1,4$ -linked maltodextrins, as well as maltose, are major nutrients for *E. coli* and enter the periplasmic space via the previously men-

tioned LamB porin. Other periplasmic proteins bind histidine,⁴⁶⁰ basic amino acids,⁴⁶¹ branched chain amino acids, only leucine, oligopeptides, polyamines,⁴⁶² and the tetrahedral anions phosphate⁴⁵⁴ and sulfate.⁴⁶³ The sulfate²⁻ anion is held to its binding protein by seven hydrogen bonds—one from a serine –OH, one from an indole ring NH, and five from peptide NH groups, three of which are at the positive ends of α helices. No permanently charged groups nor cations nor water molecules come into contact with the SO_4^{2-} .⁴⁶³ The phosphate-binding protein also binds a tetrahedral dianion HPO_4^{2-} but it doesn't bind sulfate. It forms numerous hydrogen bonds with its ligand and also an ion pair with a guanidinium group. An aspartate carboxylate hydrogen bonds to the –OH

BOX 8-D COLICINS: ANTIBIOTIC PROTEINS

Certain strains of *E. coli* and related bacteria synthesize proteins known as colicins that kill cells of other susceptible strains.^{a–c} Three kinds of colicins, each encoded in its own small DNA plasmid (colicinogenic factor), are known. **Colicin E3** is a 58-kDa ribonuclease (RNase) that attacks 26S ribosomal RNA of susceptible bacteria;^d **colicin E2** is a deoxyribonuclease (DNase) that cleaves the bacterial chromosome.^e **Colicin E1** and its relatives, colicins A, B and Ia, Ib and N, attack the bacterial inner membranes and form lethal pores which allow K⁺ and other ions to flow out of the cell. The presence of a single channel will kill the bacterium. The effect is similar to that of valinomycin.^{f,g} The small colicin V is an 88-residue peptide antibiotic that is secreted by a dedicated ABC export system.^h Colicins are members of a larger group of **bacteriocins**. One of these proteins, megacin Cx from *Bacillus megaterium*, kills bacteria of sensitive strains by blocking protein synthesis.ⁱ

The channel-forming colicins bind to bacterial surface molecules that serve as their receptors. For example, colicin N binds to the abundant *E. coli* surface protein OmpF. Interaction with a complex of membrane proteins known as tol Q, R, A, and B then leads to translocation across both outer and inner membranes and refolding of the colicin in a pore-forming conformation.^g This mechanism is also used by colicins A, E, and K. The single-stranded bacteriophages M13, fd, and f1 “parasitize” the same transport system. Colicins B, D, I, and M enter bacteria with the aid of a second transport system consisting of proteins TonB, ExbB, and ExbD which participate in uptake of chelated iron (Chapter 16) and of vitamin B₁₂. This system is also parasitized by bacteriophages T1, T5, and $\phi 80$.^j

The N-terminal portion of the 522-residue polypeptide chain of colicin E1 appears to be required for transport into the membrane and the central part for binding to the receptor; the channel-forming property is characteristic of the C-terminal region.^k A similar organization has been established for the smaller colicin N:translocation domain, (residues 1–66), receptor domain, (residues 67–182), and pore-forming domain (residues 183–387).

The colicin E1 plasmid is a 4.43 MDa circular double stranded DNA molecule consisting of 6646 base pairs.^l Only one site is susceptible to cleavage by the restriction endonuclease ECoR1 (Chapter 26). This feature has led to its widespread use in cloning of genes.

^a Luria, S. E. (1975) *Sci. Am.* **233**(Dec), 30–37

^b Parker, M. W., Tucker, A. D., Tsernoglou, D., and Pattus, F. (1990) *Trends Biochem. Sci.* **15**, 126–129

^c Parker, M. W., and Pattus, F. (1993) *Trends Biochem. Sci.* **18**, 391–395

^d Escuyer, V., Boquet, P., Perrin, D., Montecucco, C., and Mock, M. (1986) *J. Biol. Chem.* **261**, 10891–10898

^e Dvhsllrt, K., and Nomura, M. (1976) *Proc. Natl. Acad. Sci. U.S.A.* **73**, 3989–3993

^f Wiener, M., Freymann, D., Ghosh, P., and Stroud, R. M. (1997) *Nature (London)* **385**, 461–464

^g Evans, L. J. A., Labeit, S., Cooper, A., Bond, L. H., and Lakey, J. H. (1996) *Biochemistry* **35**, 15143–15148

^h Fath, M. J., Zhang, L. H., Rush, J., and Kolter, R. (1994) *Biochemistry* **33**, 6911–6917

ⁱ Brusilow, W. S. A., and Nelson, D. L. (1981) *J. Biol. Chem.* **256**, 159–164

^j Derouiche, R., Bénédicti, H., Lazzaroni, J.-C., Lazdunski, C., and Lloubès, R. (1995) *J. Biol. Chem.* **270**, 11078–11084

^k Griko, Y. V., Zakharov, S. D., and Cramer, W. A. (2000) *J. Mol. Biol.* **302**, 941–953

^l Chan, P. T., Ohmori, H., Tomizawa, J.-I., and Lebowitz, J. (1985) *J. Biol. Chem.* **260**, 8925–8935

of the phosphate. Since the sulfate ion lacks the necessary H to form this bond it is excluded from the binding site.⁴⁵⁴

All of these binding proteins have similar architectures. The ligands fit into a groove between two domains as is seen in Fig. 8-24 for the histidine binding protein. The histidine is held by formation of carboxylate–arginine and $-\text{NH}_3^+$ –aspartate ion pairs and an additional hydrogen bond to the imidazole. The proteins are able to bend in the hinge region between the two domains to give a better fit to their ligands.

The periplasmic binding proteins function together with the other subunits of the ABC transporter system. One of the best understood systems is encoded by the histidine transport operon of *Salmonella typhimurium*.^{445,453} There are four genes: *hisJ* (encoding the histidine-binding protein), *hisQ*, *hisM*, and *hisP*. The Q and M proteins are hydrophobic integral membrane proteins which interact with two copies of the P protein, which contains the ATP-binding motif, to form a His QMP₂ membrane complex. The soluble HisJ transfers its bound histidine to this complex, which with the hydrolysis of ATP supplying the driving force transfers the histidine across the membrane, presumably via a channel.⁴⁴⁵ In *E. coli* the gene *malE* encodes the periplasmic maltose binding protein, while *malF* and *malG* are genes for the integral plasma membrane components. The *malK* gene encodes an ATP-binding protein homologous to the *hisP* protein. As with the histidine permease, a multiprotein complex MalFGK₂ is formed. It accepts maltose or a maltodextrin and ATP is hydrolyzed to drive the transport.⁴⁵⁹ *MalT* encodes a positive regulatory protein that stimulates transcription of the *mal* genes.⁴⁶⁴ Membrane

transport systems are made more complex by the fact that several of the binding proteins serve also as receptors for stimulation of **chemotaxis** (Chapter 19).

The bacterial phosphotransferase system.

A third system for uptake of sugars is utilized by *E. coli* and by many other bacteria. This phosphoenolpyruvate-dependent phosphotransferase system converts glucose, mannose, fructose, other sugars, or mannitol into their 6-phosphate esters, at the same time transporting the latter across the membrane (a group translocation).^{431,465} Four proteins form a cascade (Eq. 8-4). Phosphoenolpyruvate, an intermediate in sugar metabolism whose high energy of hydrolysis provides the

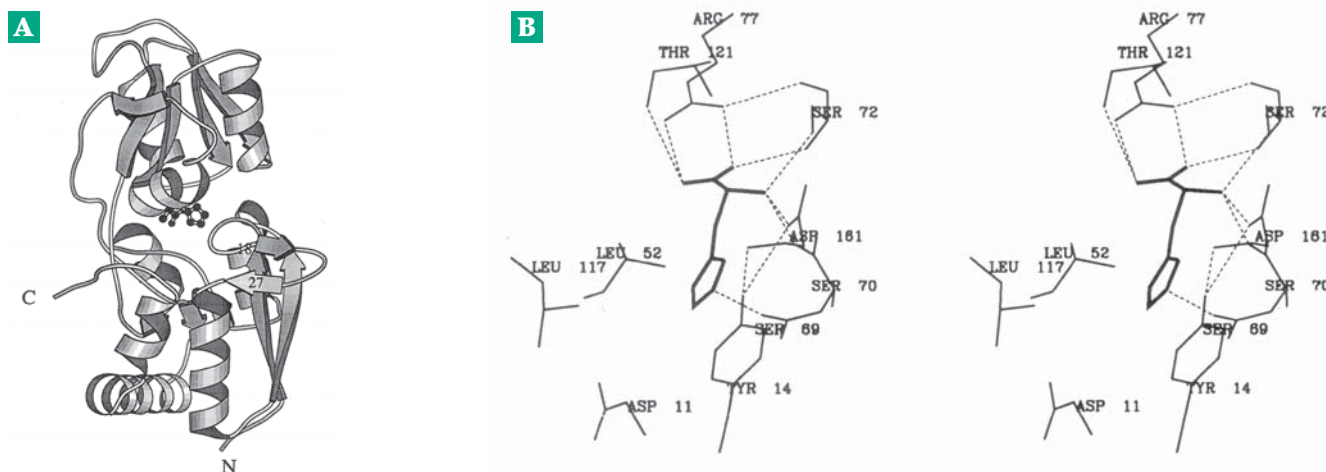
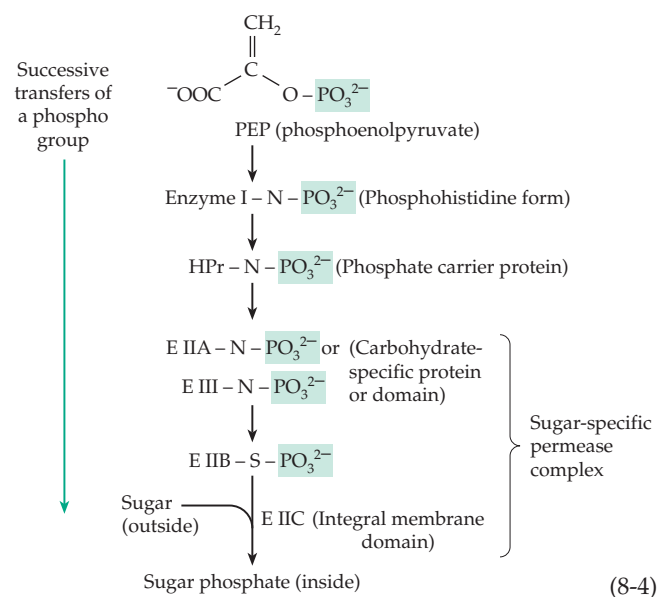


Figure 8-24 (A) MolScript ribbon drawing of the periplasmic histidine-binding protein HisJ, a component of an ABC transporter system of *Salmonella*. The bound L-histidine is shown as a ball-and-stick model. (B) Stereoscopic view of the histidine-binding site showing hydrogen-bonding interactions of protein side chains with the histidine. From Oh *et al.*⁴⁶⁰ Courtesy of Giovanna Ferro-Luzzi Ames.

driving force, phosphorylates $N^{\epsilon 2}$ of a histidine side chain on **enzyme I**, a large 64-kDa soluble membrane-associated protein.^{466-467a}

The phospho group is then transferred sequentially to the small 88-residue dimeric phosphate carrier protein **HPr**, to the carbohydrate-specific membrane proteins **IIA** and **IIB**, and to the sugar being transported. A histidine side chain at position 15 of HPr is phosphorylated by PEP to form $N^{\delta 1}$ -phosphohistidine.⁴⁶⁸ The $NH^{\epsilon 2}$ of the same histidine forms a hydrogen bond to C-terminal glutamate 85, which is also hydrogen bonded to an arginine side chain. This chain of interacting groups may function in the phospho transfer reactions.⁴⁶⁹ Both enzyme I and HPr function in the transport of many ligands but there are specific **enzymes II** (EIIs) for each sugar or other ligand.⁴⁷⁰ In *E. coli* there are at least 13 different PTS transporters.⁴⁷¹ A complete EII usually consists of three domains—EIIA (or EIII), EIIB, and EIIC. In some cases all three domains are in a single polypeptide chain, but in other cases they are individual proteins or some combination of individual and bifunctional proteins. The 637-residue mannitol-specific EII contains all three domains. Domains A and B are cytoplasmic while the N-terminal segments form the integral membrane C domain. The two phosphorylation sites are His 554 (EIIA) and Cys 384 (EIIB).⁴⁷² The glucose-specific EII from *E. coli* consists of two subunits, IIA and IICB.^{471,473-475} The mannose transporter has three subunits representing domains IIAB, IIC, and an additional integral membrane subunit IID.⁴⁷⁶ In addition to their direct transport functions, components of the PTS system play regulatory roles in chemotaxis, transcription, and control of other transporters.^{477,478}

5. Transport of Ions

Cell membranes are impermeable to most ions. Only a small number of ions can enter cells readily and these usually do so with the assistance of protein channels or pores. The principal anion of plasma (Box 5A) is Cl^- , which passes through membranes readily by virtue of the presence of channel-forming proteins. Chloride ions are often distributed across membranes passively according to Eq. 8-5, which describes the **Donnan equilibrium**.^{167,479,480}

$$[K^+]_i[Cl^-]_i = [K^+]_o[Cl^-]_o \quad (8-5)$$

Here the subscripts *i* and *o* refer to the inside and the outside of the cell, respectively. The potassium ion concentration within a cell is maintained at a high value by the operation of the $Na^+ + K^+$ pump and by the presence of nondiffusible anions within the cell. According to Eq. 8-5, the internal chloride concentration must be low, with the product of $[K]_i[Cl]_i$ equaling

that of the low exterior $[K^+]$ and high exterior $[Cl^-]$. The internal $[Na^+]$ and $[Ca^{2+}]$ are both low, while the internal $[K^+]$ is high. These differences are also linked to the membrane potential (Eq. 8-2), which is ordinarily expressed as a negative voltage of the interior of a cell, mitochondrion, plastid, etc. with respect to a reference electrode in the external medium.

The maintenance of both the membrane potential and the steep gradients of ionic concentrations is essential to cells, both as a means of coupling metabolic energy to transport and other processes and for electrical signaling. The effects are most pronounced for mitochondrial membranes for which E_m may attain -140 to -170 mV and for plasma membranes of excitable cells such as neurons ($E_m = -70$ to -90 mV). For liver and kidney cells E_m of plasma membranes may be approximately -35 mV and for erythrocytes only -9 mV.⁴⁸⁰

In excitable cells electrical impulses are initiated by opening or closing ion channels. They are propagated along an axon by a complex sequence of opening and closing of voltage-gated channels,⁴⁸¹ a process that is described in Chapter 30. All cells appear to also contain ATP-driven ion pumps as well as simple channels, cotransporter proteins, and ion exchangers.

Anions. Cell membranes have long been known to be relatively permeable to Cl^- and other small anions. However, the molecular basis of this permeability is quite complex.⁴⁸² Voltage-gated selective anion channels, often called **chloride channels**, are important in electrically excitable membranes, where they ensure a high resting chloride conductance and stability.⁴⁸³ The gene for one of these channel proteins was first cloned from the electric ray *Torpedo*^{484,485} and is designated *Clc-0*. Similar channels have been found in organisms ranging from bacteria to yeast, green plants, and vertebrate animals. The yeast genome contains just one *Clc* gene⁴⁸⁶ but mammals have at least nine. One of these, *Clc-2*, is defective in **myotonia congenita**, a human disease of impaired muscle relaxation; and in similar diseases of mice, goats, and horses.^{483,487} A mutation in the chloride channel *Clc-5* causes kidney problems, including proteinuria, hypercalciuria, and kidney stones.⁴⁸³

Erythrocyte membranes permit rapid transport of anions to allow for the exchange of Cl^- within the red cells for HCO_3^- generated by tissue metabolism. The HCO_3^- binds to deoxyhemoglobin (Eq. 7-47) and is carried to the lungs, where it is released upon oxygenation of the hemoglobin. Then the reverse exchange of internal HCO_3^- for Cl^- occurs. This electroneutral ion exchange is mediated by the **band 3 protein** (Fig. 8-14) which contains a channel that allows anions but not cations to pass.^{238,239,488} The band 3 protein, also known as **AE1** (anion exchanger 1), is found principally in red blood cells but is also present in kidney tubules.⁴⁸⁹

Related proteins occur in other tissues.⁴⁸⁸ The 911-residue band 3 protein consists of two distinct parts of nearly equal size. The N-terminal portion is attached to the membrane skeleton (Fig. 8-16). The C-terminal part, which is embedded in the membrane, is thought to form 14 transmembrane helices and to contain the ion exchange channel or channels.^{489a} As previously mentioned, defects in the N-terminal portion cause spherocytosis. The mutation Arg 589 His in the C-terminal half causes **renal tubular acidosis** in which the kidneys do not adequately remove acids from the body.^{238,489} Band 3 proteins can also exchange phosphate, sulfate, and phosphoenolpyruvate for Cl^- or bicarbonate.

Another chloride channel, which is regulated by cyclic AMP (Chapter 11), functions in secretory epithelia. Its regulation is faulty in **cystic fibrosis** (Box 26-A), one of the most common human genetic defects, especially among persons of European descent.⁴⁹⁰ As was previously mentioned, this **cystic fibrosis transmembrane conductance regulator** (CFTR) is a member of the ABC superfamily of transporters. The large 1480-residue protein apparently has two 6-helix membrane-spanning domains, two cytoplasmic nucleotide-binding domains,⁴⁹¹ and another large cytoplasmic regulatory domain.

In addition to AE1 (band 3 protein), kidneys depend upon other modes of reabsorption of HCO_3^- from the proximal tubules. These include a $\text{Na}^+/\text{HCO}_3^-$ cotransporter, which seems to be related to the AE family of ion exchangers. However, it transfers three HCO_3^- ions per Na^+ and is therefore highly **electrogenic**.^{492,493} Transporters for phosphate, sulfate, and small organic anions are found in many organisms. Bacterial periplasmic transporters have already been described. Plants employ $\text{H}^+/\text{phosphate}$,⁴⁹⁴ $\text{H}^+/\text{sulfate}$,⁴⁹⁵ and $\text{H}^+/\text{nitrate}$ ⁴⁹⁶ cotransporters. Phosphate transporters are probably essential to all organisms. One of the best known is the mitochondrial P_i/H^+ cotransporter which carries phosphate ions originating from hydrolysis of ATP to ADP + P_i back into the mitochondria.^{497,498} See also Table 18-8. A human Na^+/P_i cotransporter in the kidney is also essential. An X-linked trait leading to inadequate synthesis of the transporter causes hypophosphatemic vitamin D-resistant **rickets**.⁴⁹⁹

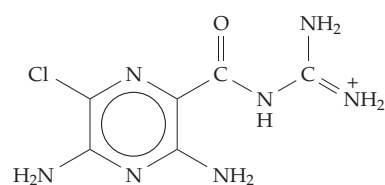
Monocarboxylates such as lactate and pyruvate enter animal cells with the aid of monocarboxylate/ H^+ cotransporters of low specificity.⁵⁰⁰ A $\text{Cl}^-/\text{oxalate}$ transporter is one of several ion exchange proteins in the kidney.⁵⁰¹ Transport systems for ADP, phosphate, dicarboxylates, and other anions are very active in mitochondrial membranes (Chapter 18).

Cation channels. When a nerve impulse passes along an axon gated pores or channels specifically permeable to Na^+ and K^+ open for short periods of

time as a result of changes in membrane potential induced by the advancing wave of the action potential (Chapter 30). We have already considered the structure of a potassium ion channel (Fig. 8-21). However, there are at least 30 types of K^+ channels that can be distinguished.⁵⁰² Many of these appear to have similar channel structures but to serve a variety of purposes. While the K^+ channels of neurons are voltage gated many others are controlled by hormones, neurotransmitters, or mechanical stimuli.⁵⁰³ One of the most investigated K^+ channels, known as K_{ATP} , sets the resting potential in the insulin-secreting β cells of the pancreas by facilitating a flow of K^+ into cells. Such channels, which help equilibrate intracellular and extracellular K^+ at near equilibrium, are found in cells with low negative values of E_m . They are called *inwardly rectifying*. When the internal glucose concentration in the β cells rises it initiates a complex signaling sequence involving blockage of the K_{ATP} channels by ATP, opening of voltage-sensitive Ca^{2+} channels, and insulin secretion.⁵⁰⁴⁻⁵⁰⁶ The K_{ATP} channel is also blocked by **sulfonylureas**. As a result, these compounds induce insulin release and are useful in treatment of diabetes mellitus (Box 17-G). When its gene was cloned the **sulfonylurea receptor** was found to be a transmembrane protein of the ABC transporter family.⁵⁰⁵ The potassium channel protein is another subunit of the transporter. Similar K_{ATP} channels are found in the kidneys⁵⁰⁷ and also in embryonic cells in which they may participate in regulation of the cell cycle.⁵⁰⁸

Voltage-regulated **sodium channels** are the major participants in propagation of nerve impulses. The large 260-kDa α subunit of the sodium channel of nerve membranes contains four homologous repeat sequences, each of which may form transmembrane helices and also contain a loop that may participate in forming a pore similar to the K^+ pore of Fig. 8-21.^{509-510a} However, the structure is uncertain.⁵¹¹ The channel complex also contains 36- or 33-kDa β_1 and β_2 subunits that appear to be members of the Ig superfamily.

Epithelial cells contain a quite different Na^+ channel that participates in reabsorption of urinary Na^+ and in control of blood pressure (Box 22-D). The channel consists of at least three structurally similar subunits, each with two predicted transmembrane helices and a large extracellular domain.⁵¹² These channels are blocked specifically by the diuretic compound **amiloride**.⁵¹³



Amiloride, a diuretic

The best known sodium channel is, in fact, a general cation channel which is part of the “nicotinic” **acetylcholine receptor**. Acetylcholine is the principal excitatory transmitter in the peripheral nervous system and upon release from synaptic endings or neuromuscular junctions occupies receptors on the “postsynaptic” membranes of one or more adjacent neurons. When acetylcholine binds in the ion pore, the acetylcholine receptor opens and cations flow out, depolarizing the membrane. Under favorable circumstances, this initiates a nerve impulse (action potential) in the postsynaptic neuron. The receptor gene was first cloned from *Torpedo* and the receptor protein has been studied extensively.³⁸² The 290-kDa protein consists of five similar-sized subunits with an $\alpha_2\beta\gamma$ structure. These form a nearly symmetric fivefold oligomer with a pore in the center. Images of both the open and the closed states, obtained by electron microscopy at a resolution of 0.9 nm, suggest that the inner pore of ~2.6 nm diameter is formed by five α helices, one from each subunit. It is open when acetylcholine binds to the two α subunits and closes to a much smaller diameter when the acetylcholine leaves (and is destroyed by hydrolysis).

Calcium channels are a third major group of cation-selective channels.⁵¹⁴ As pointed out in Box 6-D, calcium ions are involved in a very wide range of signaling functions. These are discussed in several places in this book. Several of these functions depend upon voltage-gated Ca^{2+} channels. Muscle is rich in **L-type** or DHP-sensitive channels (Box 6-D) which play a role in transmission of nerve impulses to muscles by allowing rapid flow of calcium ions into cells from outside.⁵¹⁵

The structure appears to be homologous to that of voltage-gated Na^+ channels with a large 170-kDa subunit with a fourfold repeat plus smaller subunits. The **ryanodine receptors** of muscle control the release of Ca^{2+} from stores in the endoplasmic reticulum^{516,517} (see also Chapter 19). These receptors are ligand gated, being activated by cyclic ADP-ribose (Chapter 11). Two additional types of voltage-sensitive Ca^{2+} channels, **N** and **P**, are found in the central nervous system.^{514,518} Another ligand gated calcium channel has been found in endothelial cells.⁵¹⁹ It is activated by **sphingosylphosphocholine**⁵¹⁹ rather than by cAMP-ribose or inositol triphosphate.⁵²⁰ A sodium / calcium ion exchanger and cotransporter⁵²¹ utilizes the Na^+ gradient to exchange three external Na^+ ions for one internal Ca^{2+} .

Many aspects of calcium function are poorly understood. Among these is the role of a group of proteins known as **annexins** (formerly lipocortins, calpactins, endonexins, etc.).⁵²² The ten or more members of the annexin family^{523–527} share the property of binding to phospholipid membranes in the presence of Ca^{2+} . One of the several proposed functions of annex-

ins is formation of Ca^{2+} channels, a function that is suggested by the modular three-dimensional structures.^{525,526} Other suggested functions include roles in membrane fusion, exocytosis, and adhesion.

Active transport of cations. Most organisms take up ions from their surroundings by active transport. Green plants extract essential nutrients from the extremely dilute solutions in contact with their roots. Microorganisms such as yeast and bacteria have the same ability, and specific concentrating systems for many ions such as K^+ , Ca^{2+} , sulfate, and phosphate have been identified. The skin of a frog can take up Na^+ from a 10^{-5} M solution of NaCl and extrude it into the internal fluids whose Na^+ concentration may be greater than 0.1 M. Ions can also be concentrated from internal fluids and excreted at higher concentrations. Some seabirds and marine animals rid their bodies of excess salt by secretion from salt glands. The lining of the human stomach is able to concentrate hydrogen ions in gastric juice to ~0.16 M.

Organelles within cells have their own ion-concentrating mechanisms. Thus, mitochondria can concentrate K^+ , Ca^{2+} , Mg^{2+} , and other divalent metal ions as well as dicarboxylic acids (Chapter 18). The entrance and exit of many substances from mitochondria appear to occur by exchange diffusion, i.e., by secondary active transport. Such ion exchange processes may also occur in other membranes.

ATP-driven ion pumps. Within virtually all cells the sodium concentration is relatively low, while that of potassium is high (Box 5-A). One theory⁵²⁸ regards the cytoplasm as analogous to an ion exchange resin with fixed charges in a lattice. Highly crosslinked ion exchange resins exhibit specificity toward binding of certain ions; e.g., sulfonic acid resins tend to bind K^+ preferentially, while phosphonic acid resins tend to bind Na^+ . Do proteins also prefer K^+ to Na^+ ?

In contrast to the ion exchange theory, much evidence indicates that cells have an active **ion pump** that removes Na^+ from cells and introduces K^+ . For example, the cytoplasm of the giant axons of nerves of squid can be squeezed out and replaced by ionic solutions. Erythrocyte ghosts can be allowed to reseal with various materials inside. Ion transport into or out of cells has been demonstrated with such preparations and also with intact cells of many types. Such transport is blocked by such inhibitors as cyanide ion, which prevents nearly all oxidative metabolism. However, the cyanide block can be relieved by introduction into the cells of ATP and other phosphate compounds of high group-transfer potential.

Uptake of K^+ by cells and extrusion of Na^+ from cells are also specifically blocked by “cardiac glycosides” such as **ouabain** (Fig. 22-12). Ouabain labeled with ^3H binds to the outer surface of cells, and from

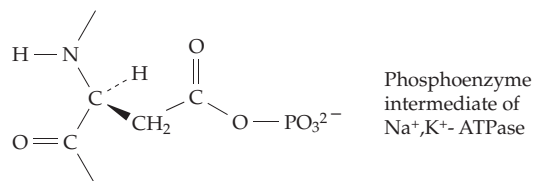
this binding it was estimated that erythrocytes possess 100–200 ion pumping sites per cell (~ 1 site / μm^2).⁵²⁹ For the HeLa cell (a widely studied strain of human cancer cells) 10^5 to 10^6 sites / cell (~ 10 / μm^2) were found. Further experiments showed that in the presence of Na^+ within the cell and K^+ on the outside of the cell, ATP is hydrolyzed. The rate of hydrolysis was directly related to the concentrations of these two ions and to the number of ouabain binding sites and also required the presence of Mg^{2+} . These observations led to the concept of an **(Na^+ + K^+)- activated ATPase** (often abbreviated **Na^+ , K^+ -ATPase**) as synonymous with the membrane-bound ion pump. Within the cell Na^+ must be located on one side of the membrane and K^+ on the other to activate this enzyme. However, the purified enzyme would be expected to hydrolyze ATP in the test tube in the presence of Na^+ + K^+ + Mg^{2+} . Such a protein was isolated from several sources and has been studied intensively. It is an $\alpha\beta$ mixed dimer with molecular masses of ~ 113 kDa for the α chains and ~ 55 kDa for the glycoprotein β chains.^{530,531} The proteins may associate to $\alpha_2\beta_2$ tetramers in membranes. The genes for various isoforms of the proteins from several sources have been cloned and sequenced.^{531,532} The large α subunit may span the bilayer of the membrane as many as ten times; the glycoprotein β subunit is thought to be largely on the outer surface and may have only one membrane-spanning helix.^{533,534} A small 68-residue γ subunit copurifies with the pump protein. It may be involved in control of the ATPase, which has complex regulatory properties.^{535,536} Sulfatides (ceramide galactose-3-sulfate) may also play a role in the enzymatic activity.⁵³⁵

The sodium–potassium pump displays a curious stoichiometry. *Three sodium ions are pumped from the inside and two potassium ions from the outside of a cell for each molecule of ATP cleaved.* Thus, an excess of positive ions is pumped out with the result that a negative charge develops inside the cell and a positive charge accumulates on the outside. This action of the Na^+ , K^+ -ATPase is the primary source of the membrane potential for most eukaryotic cells and is said to be **electrogenic**. Because the cell membrane is somewhat permeable to K^+ , outward diffusion of K^+ through the “leaky” membrane along its concentration gradient helps to maintain the membrane potential as does inward leakage of Cl^- . At the same time, Na^+ diffuses inward, aided by the membrane potential. Even though the permeability of Na^+ is low, a steady state is reached at which the rate of passive inward diffusion of cations just balances the membrane potential set up by the active transport.

The energy for transport of Na^+ and K^+ by the ion pump is supplied by ATP. The Na^+ , K^+ -ATPase does not merely catalyze the hydrolysis of ATP but also couples its cleavage to the pumping of the ions. The

pumping of sodium and potassium ions is one of the most important energy-requiring activities of cells. It is said to account for 23% of the ATP utilization in a resting human. *Thus, it constitutes an important fraction of the basal metabolic activity.*

The Na^+ , K^+ -ATPase is one of a family of over 50 ion pumps that are characterized by transfer of a phospho group from ATP to an aspartate side chain carboxylate in the invariant sequence DKTG to give an intermediate phosphoenzyme + ADP.^{530,537}



These **P-type ATPases** are characterized by phosphoenzyme intermediates, by a conserved consensus sequence, and through inhibition by vanadate ion.^{537–539} The structures are poorly known. Some consist of single chains (perhaps dimerized) and some have more than one chain. However, the major subunit always appears to have about ten transmembrane helices with a large ~ 430 -residue cytoplasmic domain between the fourth and fifth helices. This domain contains the ATP binding site and the phosphoaspartyl group of the phosphoenzyme.⁵³⁴ This is Asp 369 for the Na^+ , K^+ -ATPase.

In addition to the Na^+ , K^+ -ATPases there is a very active **Ca^{2+} -ATPase** which transports two Ca^{2+} from the inside of cells to the outside while returning two H^+ from outside per ATP.^{540–543a} This is the primary transporter by which cells maintain a low internal $[\text{Ca}^{2+}]$. During its action it becomes phosphorylated on Asp 351. However, in neurons, in which the membrane potential is maintained at a high negative value by the sodium pump, an Na^+ / Ca^{2+} ion exchange plays an even more important role.⁵⁴⁰

Other P-type ATPases include the gastric H^+ , K^+ -ATPase, which acidifies the stomach and has a high degree of sequence homology with the Na^+ , K^+ -ATPase.⁵⁴⁴ Secretion of HCl into the stomach apparently involves diffusion of K^+ together with Cl^- from the bloodstream through the cells lining the stomach. The K^+ is then pumped back into these cells in exchange for H^+ by H^+ , K^+ -ATPase.⁵⁴⁵ The chloride channel may be in the same protein as the (K^+ + H^+) pump.⁵⁴⁶ The kidney is the principle acid excretory organ of the body and as such also contains proton pumps. An electrogenic H^+ -ATPase pumps H^+ alone outward through the plasma membranes of fungi and of green plants.^{547,548} The resulting proton gradient may be used to provide energy for transport of other materials into cells. A group of metal ion P-type transporters carry copper and other nutrient ions into cells and

extrude Cd^{2+} and other toxic ions.⁵³⁹ Some **alkaliphilic bacteria** pump Na^+ to create a sodium ion gradient.^{539a} All of these ion pumping systems require MgATP as the source of energy and function via phosphoenzyme intermediates. The ionic gradients generated can be used to move other ions or nonionic compounds into or out of cells by exchange or cotransport processes. For example, internal H^+ may be exchanged for external Na^+ in an exchanger-mediated process that assists in control of cytoplasmic pH.^{549,549a} The reverse process in *E. coli*^{549b} and many other bacteria provides the principal mechanism by which those cells export Na^+ . This exchange is driven by the electrochemical gradient of the H^+ ion created by oxidative phosphorylation (Chapter 18). The Na^+ ion, in turn, can be used by bacterial cells to drive other uptake processes, e.g., sugar or amino acid- Na^+ cotransport.

What is the mechanism by which ATPase transporters function? We still do not know.⁵⁵⁰ The pumping cycles for the Na^+, K^+ -ATPase and the Ca^{2+} -ATPase are similar although different in details. The ATPases are reversible and with suitable ionic gradients will work as ATP synthases.⁵⁵¹ A strictly hypothetical model for the Na^+, K^+ -ATPase is shown in Fig. 8-25. There are at least two conformations of the ion pump proteins.^{552,552a} In one conformation the protein binds three sodium ions tightly, while in the other conformation it binds two potassium ions. The ATP operates the “motor” that carries out the conformational changes. In Fig. 8-25 the ion pump, in conformation A, is shown embedded in a membrane. In the center, perhaps between three or more transmembrane helices, there is a narrow cavity, perhaps resembling that of the K^+ channel (Fig. 8-21), into which chelating groups (e.g., $\text{C}=\text{O}$ groups of the peptide chain) protrude. These groups form the three binding sites for the 0.19-nm-diameter Na^+ ion. The spontaneous binding of the sodium ions triggers a phosphorylation reaction by which a phospho group from the MgATP^{2-} complex is transferred to the side chain carboxyl of the active site aspartate. This phosphorylation in turn triggers a change to the second conformation in which the channel to the outside is open and that to the inside is closed. At the same time the affinity

for Na^+ is decreased over 100-fold and the sodium ions dissociate on the outside. The affinity for Na^+ may decrease because the diameter of the pore is increased to accommodate the larger 0.27-nm diameter of K^+ ions, perhaps by a twisting motion of the peptide chains that form the channel.

The next step is loading with two K^+ ions. The affinity for K^+ in the second conformation is high. The return to conformation 1 with release of K^+ to the inside is triggered by hydrolytic removal of the phospho group as inorganic phosphate (P_i). It may seem surprising that a channel could be opened and closed so readily with synchronous changes in the number and specificity of ion binding sites. However, recall the type of structural alteration occurring upon oxygenation of hemoglobin (Fig. 7-25). Rotation of the hemoglobin subunits with respect to one another causes small changes in the geometrical relationships of groups protruding into the central channel. This strongly

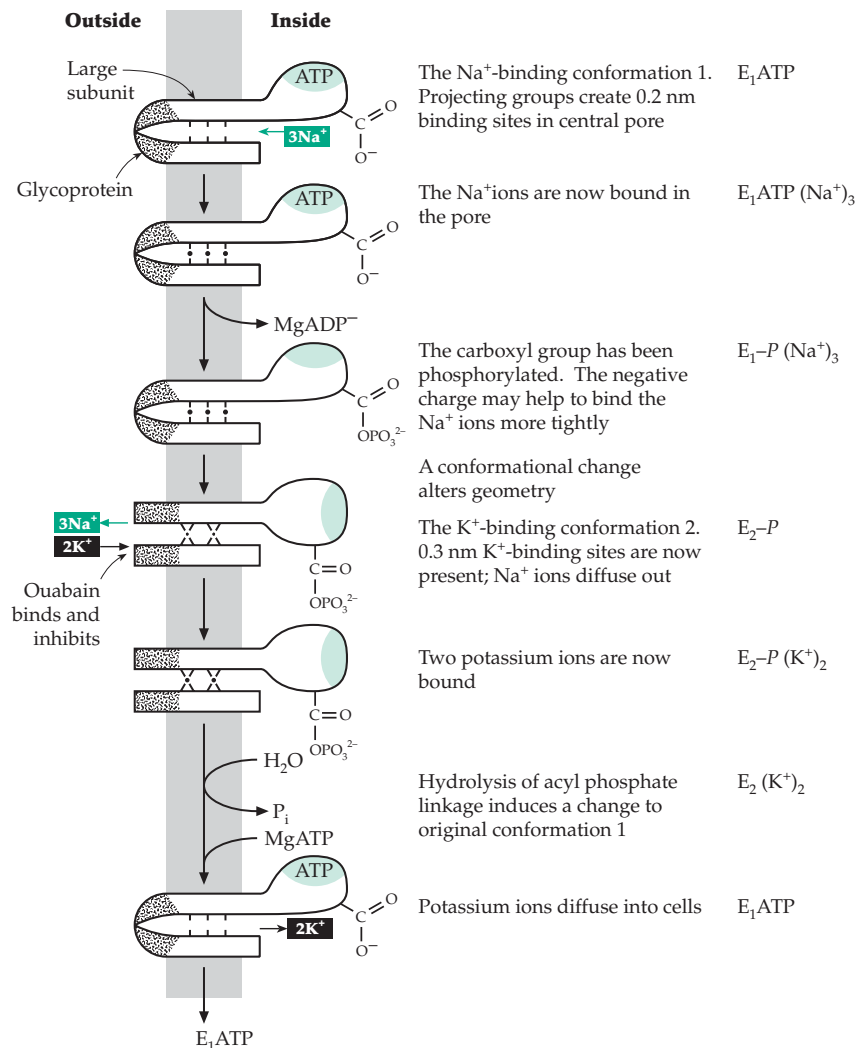


Figure 8-25 A strictly hypothetical model of a $\text{Na}^+ + \text{K}^+$ pump which operates by ATP-driven opening and closing of a channel at opposite ends and with alternate tight binding of Na^+ and K^+ .

affects the binding of 2,3-bisphosphoglycerate. Very small movements could open up the Na^+ binding groups and create new binding sites for the larger K^+ ion, using in part the same chelating groups.

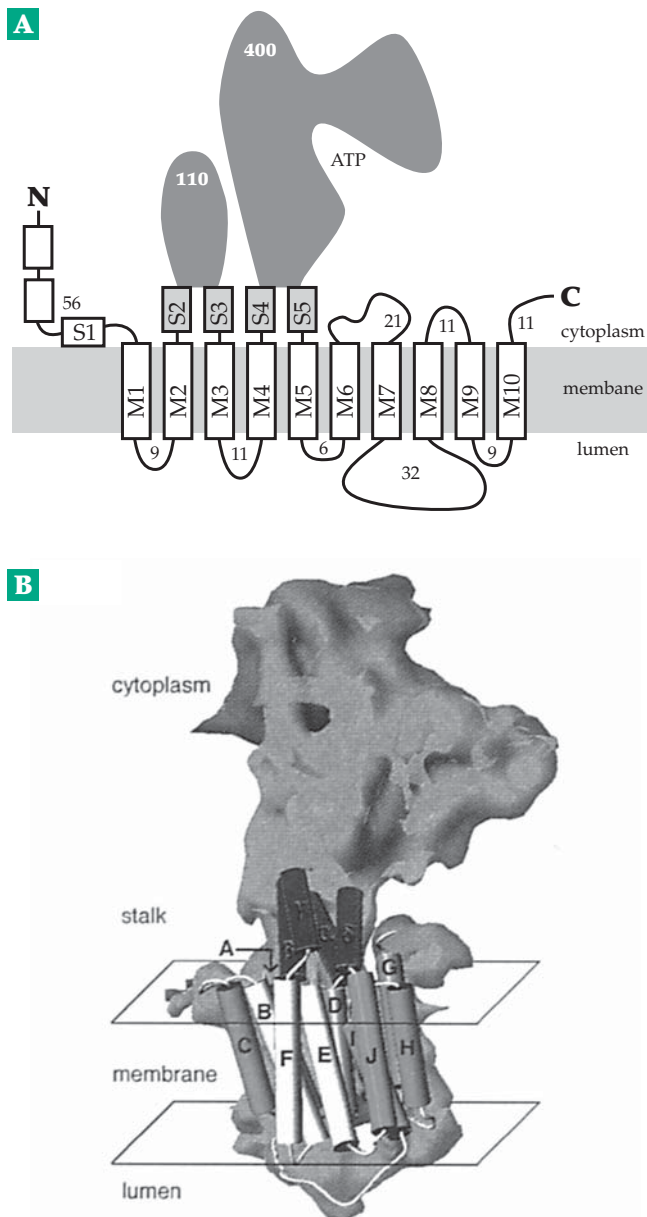


Figure 8-26 The structure of the 994-residue Ca^{2+} -ATPase of the endoplasmic reticulum of rabbit muscle at 0.8-nm resolution. (A) Predicted topology diagram organized to correspond to the electron density map prepared by electron crystallography of frozen-hydrated tubular crystals. The number of amino acid residues in each connecting loop is marked. (B) The electron density map with the predicted structure embedded. The relationships of the helices in (B) to those in (A) are not unambiguous. The helices marked B, D, E, and F in (B) may form the Ca^{2+} channel. The large cytoplasmic loops, which are black in (A), were not fitted. From Zhang *et al.*⁵⁵³ Courtesy of David L. Stokes.

Images of both the Ca^{2+} -ATPase (Fig. 8-26)⁵⁵³ and the H^+ -ATPase of *Neurospora* plasma membranes⁵⁴⁸ at 0.8 nm resolution reveal similar transmembrane regions and large cytoplasmic domains which are somewhat differently organized. The picture in Fig. 8-26 has been greatly clarified by determination of the structure by X-ray diffraction to a resolution of 0.26 nm.^{553a,b,c} Two calcium-binding sites have been located in the transmembrane domain between the helices marked M4, M5, M6, and M8. The Ca^{2+} ions are apparently coordinated by side chains of Asp, Glu, Gln, and Thr. There are three cytosolic domains. The site of phosphorylation, Asp 351, lies within a large ~ 27 kDa **P** (phosphorylation) domain adjacent to the membrane. The ATP is held by a nucleotide-binding **N** domain which must at some point in the cycle move close to Asp 351 for phosphorylation to occur. The third cytosolic (**A**, actuator) domain is thought to be involved in control of the conformational alterations. The nucleotide-binding domain lacks the “P loop” characteristic of many ATPases and GTPases (see p. 648), but is homologous in its sequence to L-2-haloacid dehalogenase (Eq. 12-2)

Two other types of proton-pumping ATPases are considered in Chapter 18. One is the mitochondrial F_1F_0 ATPase, which ordinarily operates in the reverse direction as the body’s principal **ATP synthase**. The other type, which in some ways resembles the mitochondrial F_1F_0 ATPase, is the **vacuolar ATPase** (V-ATPase). These are true proton pumps which acidify vacuoles of plants and also lysosomes and phagocytic vacuoles.^{554,555} They are also considered in Chapter 18.

6. Exocytosis, Endocytosis, and the Flow of Membrane Constituents

Observation of cells under the microscope with time-lapse photography reveals that the plasma membrane as well as the mitochondria and other organelles are in a constant state of motion. Mitochondria twist and turn and the surface membrane undulates continuously. Vesicles empty their contents to the outside of the cells, while materials are taken into cells through endocytosis. In addition, chemical evidence indicates a directed flow of the materials of which membranes are constructed from the endoplasmic reticulum (ER) to the Golgi vesicles, excretion granules, and plasma membrane (see Fig. 10-8). Along this route new materials are inserted from the cytoplasmic side of the membrane, while enzymes within the vesicles add glycosyl units and make other modifications. The plasma membrane surface area grows quite rapidly. In secretory cells fusion of secretion granules with the plasma membrane also adds additional material to the membrane.

Counterbalancing this expansion of the plasma

membrane is active endocytosis of fluids and of solid materials from outside the cell. This not only brings new materials into the cell but also accomplishes removal of material from the plasma membrane and partial recycling of its components. One form of endocytosis is seen with the amoeba.^{555a} The cytoplasm flows around a smaller organism or other particle of food enclosing it in an internal membrane-bound compartment (**endocytic vacuole** or **endosome**). This vacuole then fuses with lysosomes which supply the necessary enzymes to digest the food. In a similar way phagocytic cells of our bodies engulf microorganisms or other particles to form **phagosomes** which, over a period of more than 24 hours, undergo extensive biochemical changes.⁵⁵⁶ They acquire digestive enzymes, vacuolar ATPase,⁵⁵⁷ other proteins needed to kill bacteria, and other parasites.

Uptake of smaller particles including protein molecules occurs by **micropinocytosis**, a process that can be seen only by electron microscopy. This often takes place via **coated pits**, indentations of $\sim 0.3 \mu\text{m}$ diameter underlain by a thickened membrane.^{558–559b} The pit membrane is also coated with protein molecules and appears to have many short bristles or spikes protruding into the cytoplasm (Fig. 8-27A). After endocytosis the coated pits become **coated vesicles** of $0.15\text{--}0.25 \mu\text{m}$ (Fig. 8-27B). Within a few seconds, however, these vesicles lose their coat and become endosomes.

The major protein making up the coat is the 180-kDa **clathrin**, but smaller 33- to 36-kDa peptides of several types also contribute.⁵⁶¹ The coat forms a “basket” with pentagonal and hexagonal faces surrounding the lipid bilayer of the vesicle. At each vertex of the basket is a “triskelion,” a trimer of clathrin together with an equal number of the smaller chains. The smallest baskets consist of 12 pentagons plus 4, 8 or more hexagons, a relationship that allows formation of a variety of larger baskets.^{562,563} Additional 50- and 100-kDa accessory proteins form a shell around the clathrin cage. That clathrin is essential for normal cell growth has been established by the observation that deletion of its structural gene from yeast is lethal.⁵⁶⁴ The addition of more trimer units from a reserve of soluble clathrin in the cytoplasm allows the vesicles to develop and break off from the membrane. From studies with inhibitors it is evident that metabolic energy is required to drive the process.

Other vesicles are surrounded by nonclathrin membrane coats. Some of these originate from **caveolae** (little caves), which act in endocytosis, exocytosis, and transmembrane signaling.^{564a,b,c} A **coatamer** complex of eight subunits with molecular masses from 20 to 60 kDa coats vesicles involved in transport between compartments of the Golgi.^{565–567}

What is inside a coated vesicle? Cells take up a variety of peptide hormones and proteins. This usual-

ly occurs with the aid of specific receptor proteins located in or on the outside of the plasma membrane. Some of these, e.g., receptors for the low-density lipoprotein of plasma (Chapter 22), are clustered in coated pits. Other receptors, such as those for insulin or epidermal growth factor, are spread more evenly across the membrane but collect in coated pits when the hormone binds. Endocytosis provides a means for the cell to take up and in some cases destroy the hormone or the receptor or both.

Transmembrane proteins, including hormone receptors, are incorporated into coated vesicles with the help of **clathrin adapter proteins** (APs). These

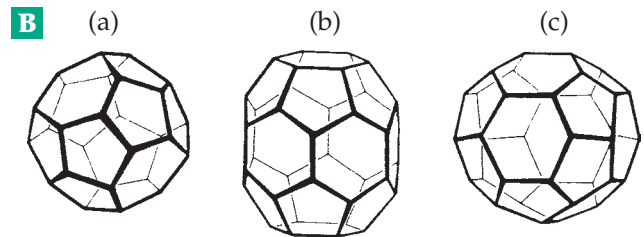
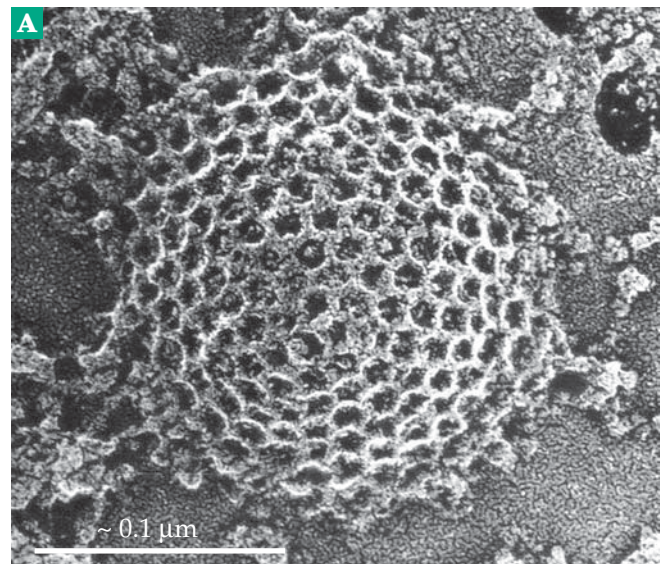


Figure 8-27 (A) Region of a coated membrane from fibroblasts at an intermediate stage of the budding process, demonstrated by deep etching and rotary replication (by J. E. Heuser^{559b}). From Pearse and Bretscher.⁵⁶⁰ Courtesy of Barbara Pearse. (B) Three structures identified among the smallest coated vesicles. Structure (a) contains 12 pentagons and four hexagons, the latter lying at the vertices of a tetrahedron; (b) has a barrel shape built of 12 pentagons and eight hexagons; structure (c) also has twelve pentagons and eight hexagons, but the latter are arranged in two arcs of four, related in the same way as the two parts of a tennis ball. Larger coats seem to be constructed on similar principles, with the addition of further hexagons. From Pearse and Bretscher.⁵⁶⁰

complex oligomeric proteins bind to recognition or “sorting” sequences such as dileucine on YXX ϕ (Y = Tyr, X = any amino acid, ϕ = bulky hydrophobic).^{567a–d} The adapter proteins also bind to clathrin, the N-terminal β -propeller domain associating with the sequence L ϕ X ϕ D / E of some AP adapters.^{567c} Completion of a coated vesicle requires membrane fusion as the vesicle is pinched off from the membrane surface. A GTPase (Chapter 11) called **dynamin** as well as another protein **endophilin I** are required. Endophilin I is an acyltransferase able to transfer the fatty acyl group of arachidonoyl-coenzyme A to lysophosphatidic acid in the cytosolic surface of the membrane. This may change the curvature of the membrane, assisting in vesicle formation.^{567e}

Once inside a cell the vesicles lose their coats to become endosomes which may then fuse with lysosomes or with Golgi membranes. The removal of a clathrin coat requires ATP as well as the chaperonin Hsp 70 (Chapter 10) and a coat protein called **auxilin**.⁵⁶⁸ Triskelion is distorted and displaced from the clathrin cage. The interior of the newly formed endosome is quickly acidified by the action of a proton pump in the vesicle walls.^{554,569} This sometimes leads to dissociation of enclosed receptors from their ligands and permits recycling of receptors and lipids of the vesicle membranes to the cell surface. This is the case for the low-density lipoprotein receptor.^{570,571}

Exocytosis, by which the content of a secretion vesicle is released to the outside of a cell, is just as important as endocytosis. The process is sometimes very specialized. For example, the release of a nematocyst from *Hydra* (Fig. 1-13) can occur in about 3 ms.⁵⁷² Exocytosis involves **fusion** of membranes,^{562,573} a process also occurring during the movement of endosomes along the endocytic pathway,⁵⁷⁴ during vesicular transport between Golgi compartments, and in many other biological processes. Exocytosis is often triggered by the binding of Ca²⁺ to specific proteins of the vesicle wall and of the cytoskeleton (Chapter 7).⁵⁷⁵ The fusion of membranes at several stages in the vesicle-mediated transport of materials between Golgi compartments requires a specific protein known as the N-ethylmaleimide-sensitive fusion protein (**NSF**).^{576,577} A host of other specialized proteins are also involved^{562,578} and are discussed in Chapter 10. See also Chapters 20 and 29.

D. Communication

External coats and cell walls help to control the access of materials to a cell. However, it is the outer surface of the plasma membrane that makes the cell's first contact with nutrients, hormones, and other important chemicals. The membrane must often not only detect these materials but also send signals to the

interior of the cell and sometimes to adjacent cells. These signals may be about changes in pH or nutrient concentration or the presence of hormones, neurotransmitters, or harmful materials. For these reasons cell membranes contain many embedded receptors and signaling complexes. These are discussed in Chapter 11 and later sections of the book.

The plasma membrane also contains many “markers” of the individuality of the species or of an individual. These are chemical groupings that, in higher animals, can be recognized by the immune system as “self” rather than as a foreign invader. These surface markers may also be used by parasites as camouflage to evade the immune response of the host. Such chemical groupings on cell surfaces are often described as **antigenic determinants** and the molecules that carry them as **antigens**. Each antigenic determinant elicits the production of antibodies that will bind specifically to it. Over 250 different antigenic groups have already been described for the surface of the red blood cell. They determine the blood type. Groups on the surfaces of other cells determine whether a transplanted tissue will be rejected. Various proteins from plant and other sources act as **agglutinins** by binding to surface groups much as do antibodies. Viruses that attack cells may also become adsorbed onto specific surface molecules, which act as receptors.

Immunoglobulins can also be receptors. For example, molecules of IgE bound to basophils and the related **mast cells** of tissues serve as receptors for allergens. Binding of an allergen to the IgE molecules stimulates the release of granules containing histamine and other substances (Chapter 31).

E. The Extracellular Matrix and Cell Walls

The surroundings of cells are extremely complex and vary from one organism to another and from one tissue to another. The principal function of cell walls and other surface coats is to protect cells against attack by organisms and against physical disruption.

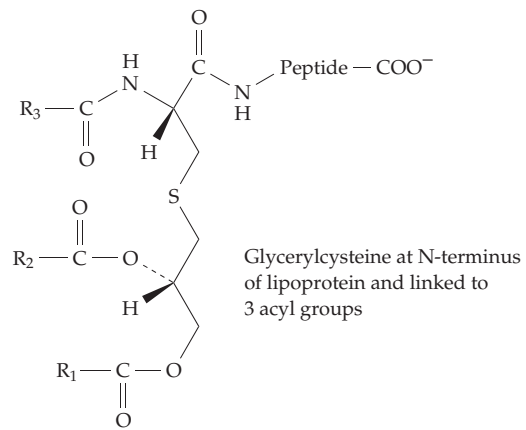
1. The Structure of Bacterial Cell Walls

The plasma membrane of bacterial cells, other than the wall-less mycoplasmas and some archaeobacteria, is surrounded by a multilayered wall which may be separated from the membrane by a thin **periplasm** (or periplasmic space). This can be seen most clearly in suitably prepared thin sections of cells of *E. coli* or other gram-negative bacteria as a relatively empty space of 11- to 25-nm thickness (Fig. 8-28).^{579–581} The volume of this space (which may be filled with gelled material) depends upon the osmotic pressure of the medium. In *E. coli* it contains 20–40% of the total

cell water. In gram-negative bacteria the innermost layer of the walls lies within the periplasm (Fig. 8-28). It is a porous network of a highly crosslinked material known as **peptidoglycan** or **murein**. The backbone of the peptidoglycan is a β -1,4-linked polymer of alternating *N*-acetylglucosamine and *N*-acetylmuramic acid residues. Alternate units of the resulting chitin-like molecule carry unusual peptides attached to the lactyl groups of the *N*-acetylmuramic acid units (Fig. 8-29). The peptide side chains are crosslinked as indicated in the figure. The peptides vary considerably in structure and crosslinkages.^{582,583} In *E. coli* and other gram-negative bacteria the peptidoglycan forms a thin (2-nm) continuous network around the cell, but in gram-positive bacteria the highly crosslinked peptidoglycan forms a layer as much as 10 nm thick.⁵⁸⁴ The peptidoglycan layer is surrounded by other layers whose structures vary from one organism to another, with the outermost antigenic layers being the most variable.

The outer membrane of gram-negative bacteria.

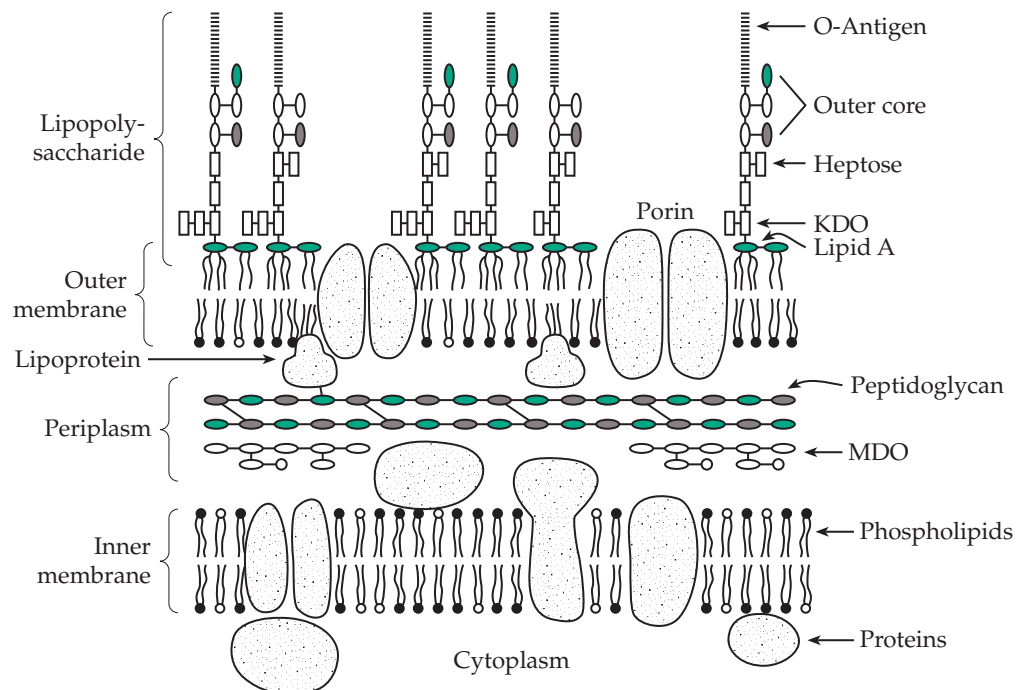
Outside the murein layer of *E. coli* and other gram-negative organisms is a phospholipid-containing **outer membrane** which has the thickness and something of the structure of a typical biological membrane (Fig. 8-28).^{585,586} This membrane is attached to the peptidoglycan layer with the aid of a small hydrophobic 58-residue **lipoprotein** whose N terminus contains a **glycerylcysteine** which carries three fatty acids connected by ester and amide linkages:



The fatty acid chains are evidently embedded in the outer membrane as an anchor. About one-third of the lipoprotein molecules are attached covalently to the peptidoglycan through an amide linkage between the side chain amino group of the C-terminal lysine of the protein and a diaminopimelic acid residue of the peptidoglycan (Fig. 8-29). Thus, the protein replaces one of the terminal D-alanine residues of about one in ten of the murein peptides. There are $\sim 2.5 \times 10^5$ molecules of the bound form of the lipoprotein per cell spread over a surface area of peptidoglycan of $\sim 3 \mu\text{m}^2$. They appear to be associated as trimers located primarily in the periplasmic space.⁵⁸⁹

About 10^5 copies per cell of the previously mentioned (Section C,2) larger 325-residue structural protein, **OmpA protein**,³⁵⁰ after its gene symbol *ompA*

Figure 8-28 Schematic molecular structure of the *E. coli* envelope. Sugar residues are represented by ovals and rectangles. Circles represent polar head groups of phospholipids. MDO, membrane-derived oligosaccharides; KDO, 3-deoxy-manno-octulosonic acid (structures for KDO and heptose are in Fig. 4-15). From Raetz and Dowhan.⁵⁸⁷



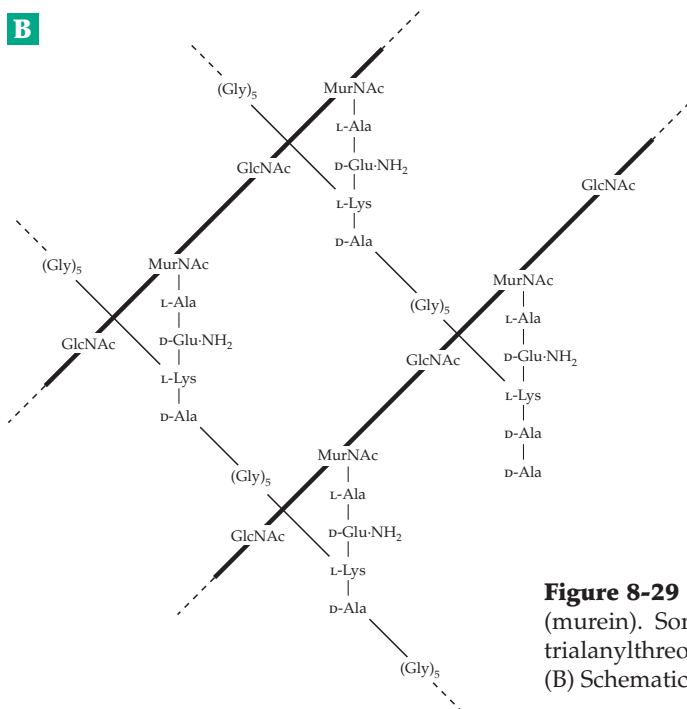
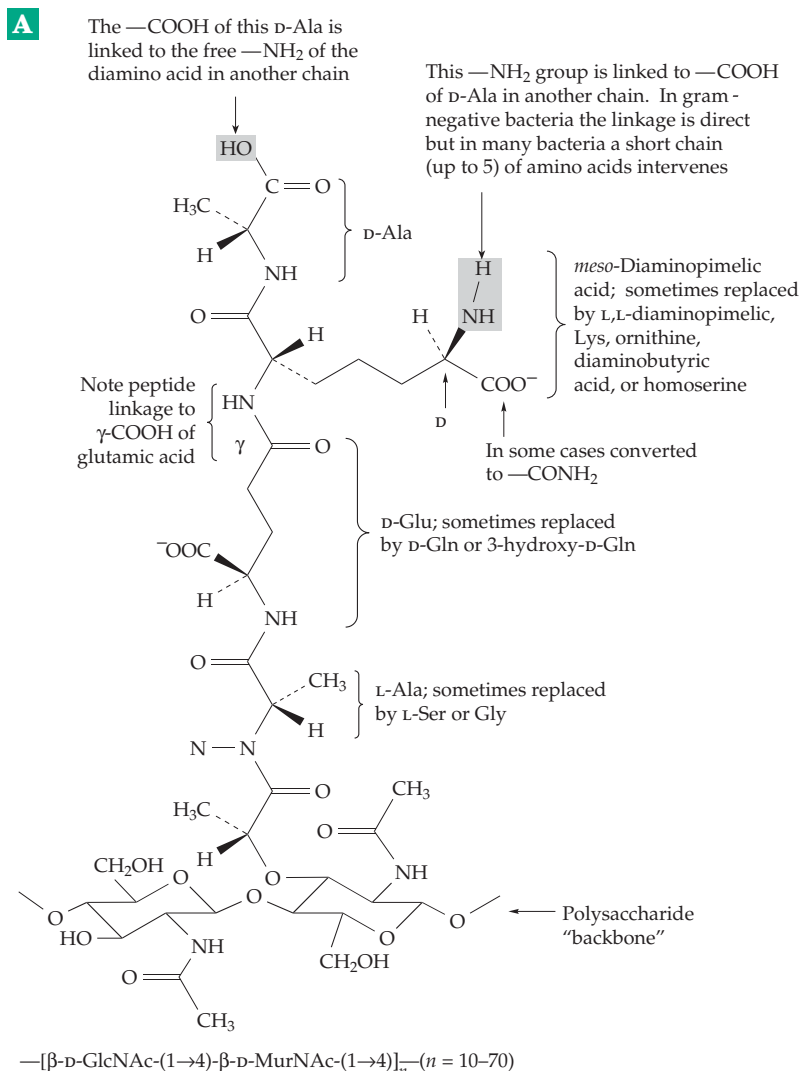


Figure 8-29 (A) Repeating unit of structure of a bacterial peptidoglycan (murein). Some connecting bridges are pentaglycine (*Staphylococcus aureus*), triallanylthreonine (*Micrococcus roseum*), and polyserine (*S. epidermis*). (B) Schematic drawing of the peptidoglycan of *S. aureus*. From Osborn.⁵⁸⁸

(outer membrane protein A), are also present. The outer membrane contains almost the same number of molecules of two porins. Together with phospholipids or with the lipopolysaccharide discussed in the next paragraphs, the porins and OmpA protein associate in hexagonal arrays which provide the basic framework structure of the outer membrane.⁵⁸⁵ Two of the outer membrane proteins have been shown to contain some of the modified lysine α-amino acid 5-semialdehyde (**allysine**).⁵⁹⁰ The aldehyde groups of allysine are able to form crosslinks to other proteins as has been well established for collagen and elastin in the human body.

A characteristic feature of the outer surface of gram-negative bacteria is a **lipopolysaccharide**⁵⁸⁶ that is anchored in the outer membrane. It was discussed briefly in Chapter 4 where the structures of the repeating oligosaccharides known as O-antigens are given. Figures 8-28 and 8-30 show the manner in which the oligosaccharide bearing the O antigen is attached to a lipophilic anchoring group that is embedded in the outer membrane of the bacteria.⁵⁹¹ The anchor, which is called **lipid A**, is a β-1,6-linked disaccharide of N-acetylglucosamine. It is also linked both to phosphate groups and to the fatty acyl groups that fit into the lipid bilayer of the membrane. In *E. coli* and *Salmonella typhimurium* four molecules of **3-D-hydroxymyristic** acid are joined by ester and amide linkages to the two GlcN units (Fig. 8-30). Other fatty acids, including lauric, myristic, and palmitic acids, are esterified to the hydroxyl groups of two or three residues of hydroxymyristic acid (Fig. 8-30). Lipid A from other gram-negative bacteria is similar but with variations in the fatty acyl group composition and linkages to the carbohydrate.^{592–594}

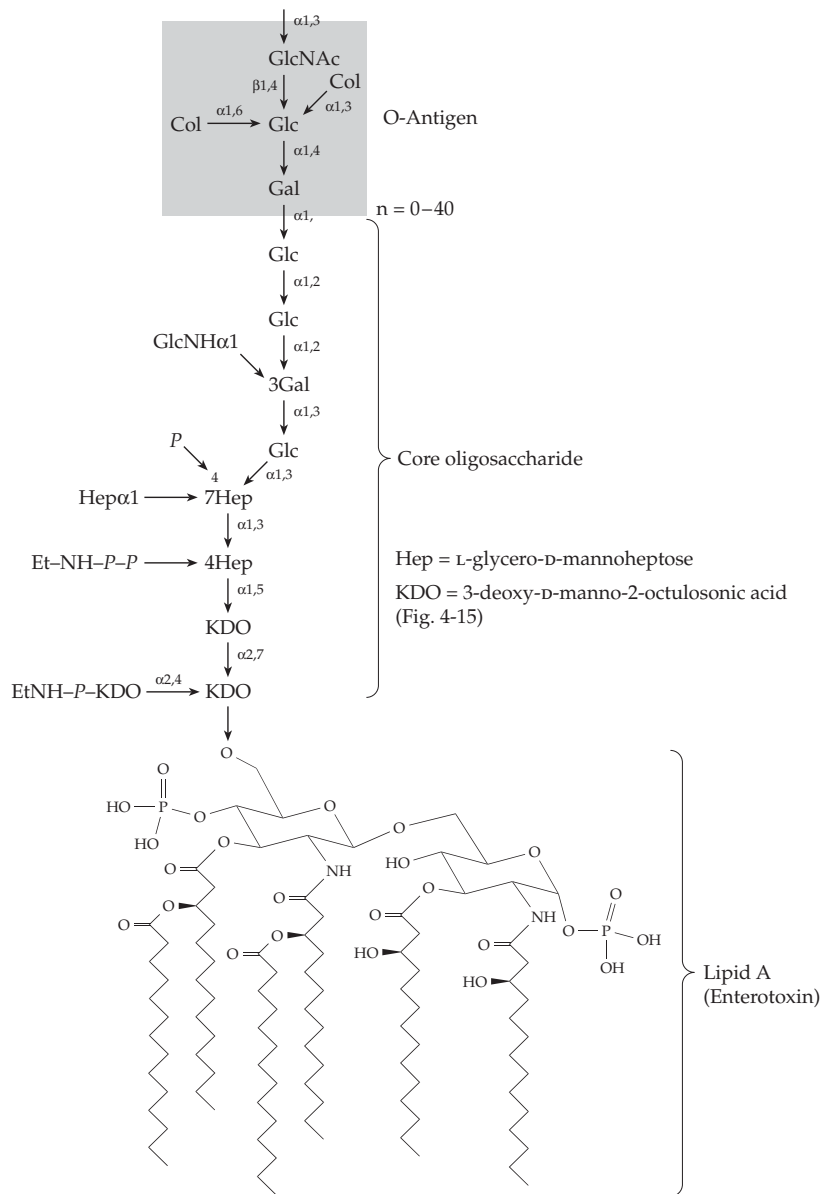
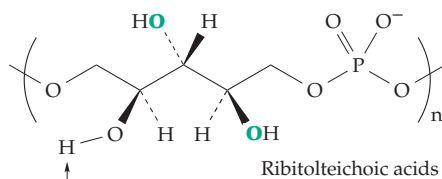
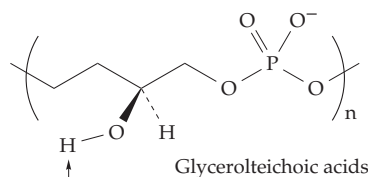


Figure 8-30 Structures of the lipopolysaccharides of the outer membrane of *E. coli* and *S. typhimurium* including the bilayer anchor lipid A. For structures of L-glycero-D-mannoheptose, KDO, and colitose, see Fig. 4-15.

In *Bacillus subtilis* D-alanine is attached to one of the green oxygen atoms in at least half of the units



In *B. subtilis* β -D-glucose is attached here



In *Lactobacillus arabinosus* D-alanine in ester linkage occupies this position but is replaced by β -D-glucose on about one unit in nine

Many differences are also found in the "core" oligosaccharide (Fig. 8-30) and in the O-antigens.⁵⁹⁴⁻⁵⁹⁶

Why do cells of *Salmonella* have a thousand distinguishable surface antigens, many based on differences in the O-antigens? The ends of these carbohydrate clusters are the groups to which the antibodies in animals clamp themselves if the bacteria enter the bloodstream. Mutants known as R forms (because of the growth as rough colonies on agar plates) completely lack the outer O-antigen. The R mutants of *Salmonella* are nonpathogenic, whereas the smooth strains with intact O-antigen often cause illness. Perhaps, if the O-antigen has the right cluster of sugar rings at the ends, the host organism does not recognize it as dangerous. This is only part of a continuous battle between the immune system of the body and camouflaged surfaces of attacking pathogens.

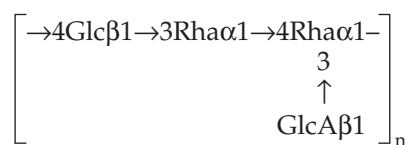
The lipopolysaccharides of bacterial surfaces have been identified as the “**endotoxins**” that cause many of the worst effects of gram-negative bacterial infections and that are often lethal.⁵⁹⁷ Although the name endotoxin came from the assumption that there was an internal heat-resistant toxin that was released from the bacteria, the toxin is part of the bacterial cell wall. While the O-antigens provide for immunological recognition, it is the *lipid A* part that is responsible for the toxicity and unusually strong fever-inducing properties of the lipopolysaccharide.^{592–594,597}

Teichoic and teichuronic acids. The cell walls of gram-positive bacteria are composed of a thick peptidoglycan layer which also contains proteins and additional polymers known as **teichoic acids** and **teichuronic acids**. In some species these account for 50% of the dry weight of the cells.^{598,599} Teichoic acids are high polymers of the following general types: They are often attached through phosphodiester linkages to *N*-acetylglucosamine or a disaccharide which, in turn, is attached to muramic acid residues of the peptidoglycan. Since the teichoic acid is uniformly distributed in its attachment to the peptidoglycan, it must be intimately associated with peptidoglycan throughout the wall.

Teichoic acids are often covalently attached to glycolipids which are part of the plasma membrane. For example, the glycerolteichoic acid of *Streptococcus faecium* contains about 28 monomer units of glycerol phosphate, approximately 60% of which carry residues of kojibiose (Gluα1 → 2Glu) as a phosphatidylkojibiosyl diacylglycerol membrane anchor.⁶⁰⁰ Teichuronic acids contain uronic acids:



From *Micrococcus luteus*⁶⁰¹



From *Bacillus megaterium*⁶⁰²

Nearly 800 of the foregoing tetrasaccharide units are joined to give a large, densely packed, ~500-kDa polymer. The reducing end of the polymer is covalently attached to the peptidoglycan layer.

Mycobacteria. Most mycobacterial species seem to be harmless saprophytic soil bacteria of the gram-positive group. However, tuberculosis (caused by *Mycobacterium tuberculosis*) may infect one-third of the inhabitants of the earth⁶⁰³ and kills about three million

people per year.⁶⁰⁴ Leprosy, caused by *M. leprae*, affects 10 million or more.⁶⁰⁵ Most antibiotics are not effective against mycobacteria because of the unusual nature of their multilayered cell walls. The muramic acid of their peptidoglycans contains glycolyl groups instead of acetyl groups, perhaps providing extra hydrogen bonding within the wall.⁶⁰⁴ Most characteristically, they are rich in lipid materials. Among these are a unique polysaccharide, a highly branched **arabinogalactan**, that is covalently attached to muramic acid residues of the peptidoglycan. Clusters of arabinofuranosyl units at the nonreducing ends of the chains are esterified with **mycolic acids** (Section A,1).^{10,604,605} Mycobacteria also contain **lipoarabinomannans** that act as major antigens.^{603,606} Some mycobacterial species synthesize small glycopeptidolipids that may disrupt cell membranes of hosts.⁶⁰⁷ Mycobacteria don't produce toxins of usual types but cause slow, long-lasting infections.

Other bacterial coats. Archaeobacteria not only have unusual plasma membranes that contain phytanyl and diphytanyl groups (Section A,3)⁶⁰⁸ but also have special surface layers (**S-layers**) that may consist of many copies of a single protein that is anchored in the cell membrane.⁶⁰⁹ The surface protein of the hypothermic *Staphylothermus marinus* consists of a complex structure formed from a tetramer of 92-kDa rods with an equal number of 85-kDa “arms.”^{610,611} S-layers are often formed not only by archaeobacteria but also by eubacteria of several types and with quite varied structures.^{612–614} While many bacteria carry adhesins on pili, in others these adhesive proteins are also components of surface layers.⁶¹⁵ Additional sheaths, capsules, or slime layers, often composed of dextrans (Chapter 4) and other carbohydrates, surround some bacteria.

2. The Surroundings of Animal Cells

Cells in the external epithelial layers are always surrounded by a protective covering of some kind. Our own skin is made up of specialized cells which become filled with microfibrils of keratin as they move outward and become the relatively dry nonliving external surface (Box 8-F). Internal epithelial cells secrete protein and carbohydrate materials that form a thin **basement membrane** around the exposed parts of the cells. The **connective tissue** that lies between organs and which also includes tendons, cartilage, and bone consists of a relatively small number of cells surrounded by a “matrix” consisting of the protein fibers collagen and elastin in a “ground substance” rich in proteoglycans (Chapter 4).^{616–618} In bone, the calcium phosphate is deposited within this matrix.

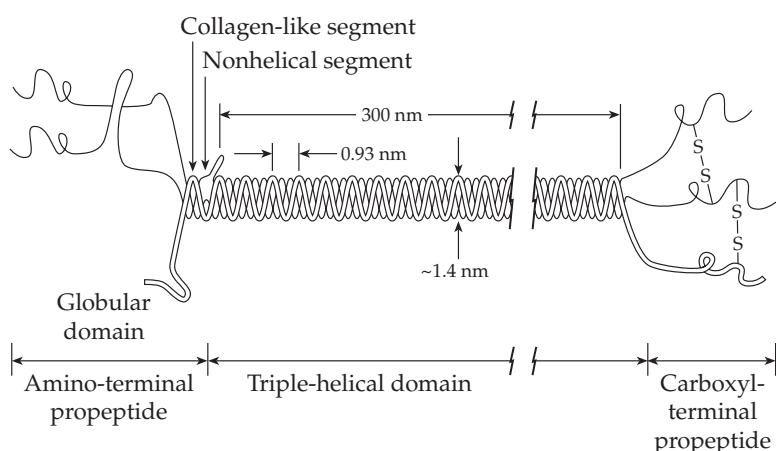


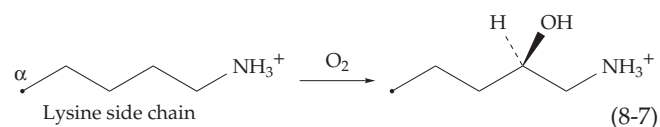
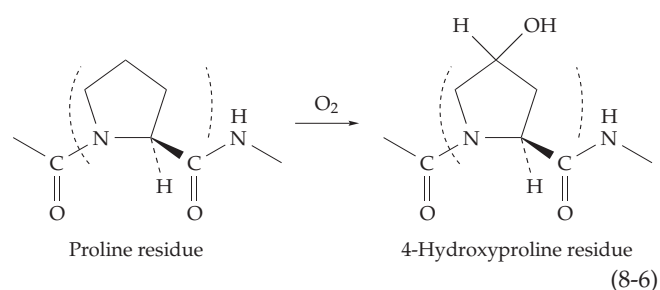
Figure 8-31 Schematic representation of type I procollagen. The molecule is composed of two identical pro α 1 chains (solid lines) and one pro α 2 chain (dashed line). In addition to the central triple-helical region that gives rise to the collagen molecule, as portrayed in Fig. 2-23(C), the precursor contains amino- and carboxyl-terminal non-triple-helical domains. The amino-terminal domain is composed of a presumably globular region, a short collagen-like segment, and a non-triple-helical region in which cleavages by the amino-terminal protease occur. Interchain disulfide bonds are limited to the carboxyl-terminal domain. The short telopeptides at the ends of collagen α chains represent the residual sequences of the linkage regions between the collagen helix and the terminal domains. After Prockop⁶²⁸ and Byers.⁶²⁵

The collagens. The most abundant proteins in the body are the collagens,^{619–624} a family of closely related materials that account for 20% of the total protein in higher animals. Collagens make up much of the organic mass of skin, tendons, blood vessels, bone, the cornea, vitreous humor of the eye, and basement membranes. They are found in every metazoan phylum studied. There are at least 16 types in the human body.⁶²⁵ Type I collagen, which accounts for 90% of the total, is the major form occurring in skin, tendon, and bone. It is synthesized by the fibroblasts and is excreted into the extracellular space where it is polymerized into a durable long-lived material. Collagen II is found exclusively in cartilage and the vitreous humor of the eyes. Form III is located in blood vessels and intestines and is prominent in embryonic tissues. Collagen IV is the major form found in basement membranes. Some other collagens and characteristic features are listed in Table 8-4.

All collagens contain the triple-helical structure shown in Fig. 2-23. Collagen I consists of two chains of one kind (α 1) and one of another (α 2), while most other collagens have three identical chains. In these chains over 1000 residues have the characteristic GlyXY sequence.^{624,626} At each end of the rodlike molecules short segments of the peptides fold into globular domains. For type I collagen 16 residues at the N termini and 25 at the C termini form these domains. Collagens are synthesized as intracellular precursors known as **procollagens**. The three chains of procollagens are much longer than in the mature proteins and have larger non-triple-helical ends (Fig. 8-31). The C-terminal extensions are crosslinked by S–S bridges.^{625,627} Synthesis of the collagen chains requires at least six minutes after which they are released into the cisternal space of the ER, associate, and become crosslinked (Fig. 8-32). This crosslinking

ensures that the three chains remain in proper register in the triple helix while the procollagen is converted into collagen, a process involving additional crosslinking.

Before this “maturation” can occur there must be other modifications to procollagen. These begin while the peptide chains are still attached to ribosomes of the rough ER. Hydroxylases (Chapter 18) localized in the membranous vesicles of the ER convert some of the proline and lysine residues of the procollagen chains into **4-hydroxyproline**^{625,629,630} and **hydroxylysine** (Eqs. 8-6 and 8-7). Lesser amounts of 3-hydroxyproline are formed. About 100 molecules of 4-hydroxyproline and 50 of 5-hydroxylysine are created in each α 1 chain.



Galactosyl units are then transferred onto some of the hydroxyl groups of the hydroxylysine side chains, and glucosyl groups are transferred onto some of the galactosyl groups. This glycosylation may prevent incorrect association of the procollagen molecules.

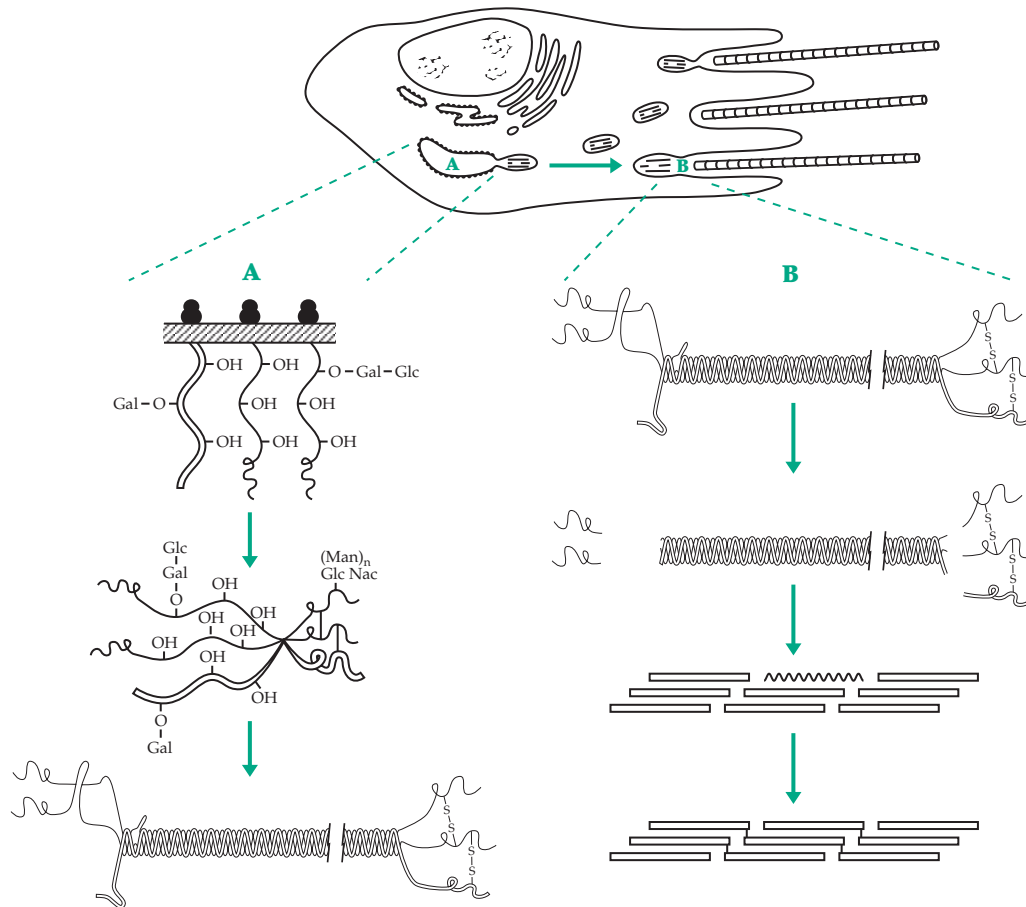


Figure 8-32 Scheme summarizing the biosynthesis of a fibrillar collagen by a fibroblast. (A) Assembly of pro- α chains in cisternae of the rough endoplasmic reticulum; posttranslational hydroxylations and glycosylations; association and disulfide bonding of C-propeptides; and zipper-like folding of the triple helix by nucleated growth. (B) Proteolytic processing of procollagen to collagen; self-assembly of fibrils by nucleated growth; and covalent crosslinking of the fibrils. The collagen molecule is first shown as a triple helix and then either as a *wavy line* to depict a molecule assembling on the surface of a fibril or as a *rectangle* to depict the quarter-staggered assay of monomers in a fibril. The proteolytic processing of procollagen and assembly of fibrils may occur within crypts of fibrils as shown here or perhaps at some distance from the cell. After D. J. Prockop.⁶²⁸

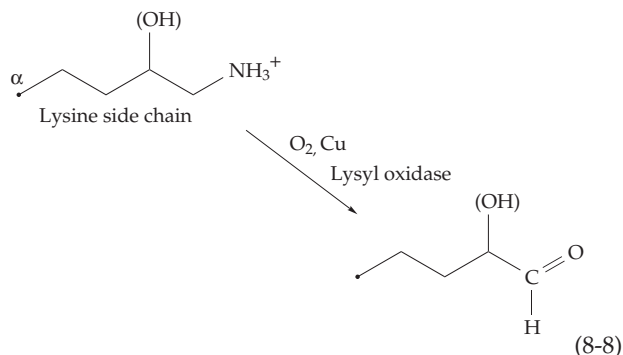
Within the extracellular space two **procollagen peptidases** act to cleave a 35-kDa peptide from the C terminus⁶³¹ and a 20-kDa peptide from the N-terminal end of each of the three chains of the secreted procollagen. The amino acid composition of the peptides removed is quite unlike that of the remaining collagen monomer (also called tropocollagen) which contains one-third glycine and much proline.

The three-stranded monomers of collagens I–III are rods of dimensions $\sim 1.4 \times 300$ nm (Fig. 8-23). When they reach their final location they associate and become crosslinked to form strong fibers with diameters ranging from 8 nm to 0.5 m. Tendons tend to contain large fibrils, while those in bone are small. The smallest 8-nm fibrils must contain about 20 triple helices in a single cross section but the successive monomers are staggered by 6.4–6.7 nm (234 residues for type I collagen) in such a way that 3-nm gaps are

left between the ends (Fig. 8-32). These gaps give rise to the characteristic banding pattern seen in the fibrils in Fig. 2-23D. Finer analysis of the bands together with the known sequences shows that the bands also reflect the locations of residues with charged side chains.⁶³² These ionized groups are thought to provide electrostatic stabilization to the fibrils.⁶³³ The exact packing of the collagen molecules into sheetlike or microfibrillar crystalline arrays is still uncertain.⁶³⁴ However, there is agreement on the staggered arrangement^{634a} and upon the fact that precisely formed crosslinks with neighboring rods prevent the gaps from weakening the fibrils.

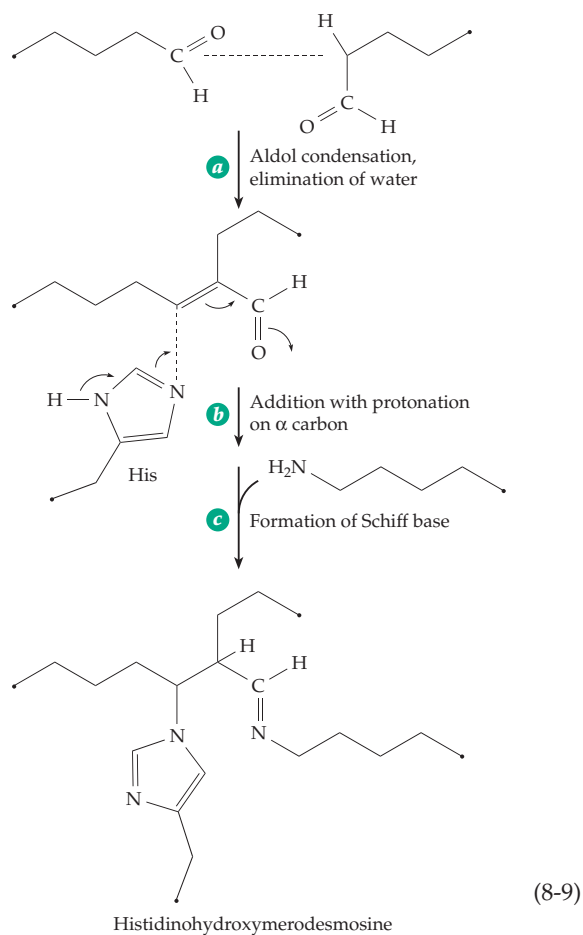
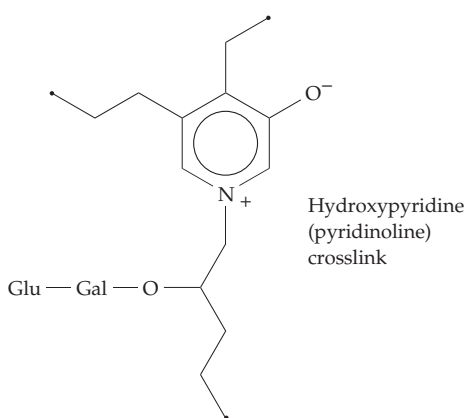
Crosslinking of collagen is initiated by oxidation of some of the lysyl and hydroxylysyl side chains from amino groups to aldehyde groups under the action of a copper-containing oxidase (Eq. 8-8, Chapter 18). The aldehyde groups enter into a variety of reactions that

lead to crosslinking of the collagen monomers and to



the formation of insoluble fibers.⁶³⁵ One reaction is an aldol condensation followed by elimination of water (Eq. 8-9, step *a*). If one of the two aldehydes involved in the condensation is derived from hydroxylysine and the other from lysine, two isomeric condensation products are formed. The aldol condensation product can react further: An imidazole group from a histidine side chain can add to the carbon-carbon double bond and (either before or after this reaction) another lysine side chain can form a Schiff base with the free aldehyde. The results of these two processes are summarized in Eq. 8-9, step *b*. The final product **histidinohydroxymerodesmosine** links four different side chain groups. In other instances, simple Schiff bases (**aldimines**) are formed between aldehyde and ϵ -amino groups. If there is an adjacent hydroxyl group these can isomerize to ketoamines (Eq. 8-10). Two residues of hydroxylysine, one of which is glycosylated as shown in Eq. 8-10, are often involved. Borohydride reduction (Eq. 4-2) and hydrolysis leads to isolation of **dihydroxylysinonorleucine**, the predominant product of such treatment of bone or cartilage.

Crosslinkages reducible with borohydride are characteristic of newly formed collagen but these disappear and are replaced by more stable crosslinks as collagen matures. For example, a **3-hydroxypyridine** that joins three triple helices may be formed from the reaction of two ketoamine groupings (with elimination of one glycosylated hydroxylysine residue).⁶³⁶⁻⁶³⁹ Similar chemistry can also produce pyrrole crosslinks.⁶³⁹



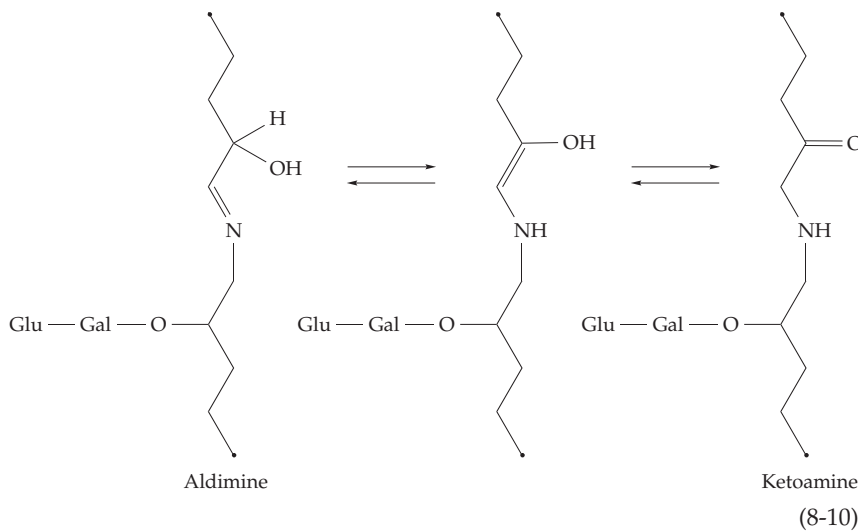
The crosslinkages in collagen are not located at random but are found in certain positions, often toward the ends of the collagen monomers. Thus, histidine residues are found only at positions 89, 929, and 1034 in the $\alpha 1(I)$ peptide chains. Residue 9 in the N-terminal globular portion of one chain is linked to residue 946 of another while residue 103 is linked to 1047 in the globular C-terminal peptide.⁶⁴⁰ The variety and number of crosslinkages vary among different species. As collagen ages through a lifetime, glycation (Eq. 4-8) leads to crosslinkages in which two ketoamines are formed via glycation cyclize.⁶⁴¹

Collagens I, II, and III form fibrils with similar structures. However, other collagens are longer or shorter and aggregate in different ways. A pepsin-resistant part of the collagen V molecules resembles collagens I, II, and III but it may contain an additional non-collagenous segment at the N terminus. Collagens V and XI are quantitatively minor components of the extracellular matrix but are thought to provide a core around which the fibrils of collagens I and III may form.⁶⁴² Types XII, XIV, IX, and XVI collagens contain interruptions in the helix which create bends, flexible sites, and sites of increased proteolytic susceptibility. They may link the fibrils to other components of the surrounding matrix.⁶⁴³

TABLE 8-4
Types of Vertebrate Collagen

Type number	Location	Characteristics	Gene location: human chromosome
Forming quarter-staggered fibrils			
I ^{a,b}	Skin, bone, tendon, dentin	Most abundant, banded quarter-staggered fibrils	7, 17
III ^{a,c}	Skin, blood vessels	Abundant, small banded fibrils	2
V ^{a,d}	Most interstitial tissues; cartilage, bone	Abundant, small fibrils	2
Predominant in cartilage and bone			
II ^a	Hyaline cartilage, vitreous humor	Very abundant, small banded fibrils	12
XI ^a	Hyaline cartilage		
With interrupted triple helices			
XII ^{a,e}	Embryonic tendon, periodontal ligaments	Fibril associated	
XIV ^{a,e}	Fetal skin, tendon	Fibril associated	
IX ^{a,e}	Cartilage, vitreous humor	Minor, contains attached glycosaminoglycan, fibril associated	
XVI ^f	Cartilage	Fibril associated	
Forming sheets and networks			
IV ^{a,g}	All basement membranes	Nonfibrillar network	13
X ^a	Mineralizing cartilage, growth plate	Short chain	
VIII ^{a,h}	Endothelial cells; Descemet's membrane of the cornea	Small helices linked in hexagonal arrays	
Forming beaded filaments			
VI ^a	Most interstitial tissues, intervertebral discs	Beaded microfilaments	
Forming anchoring fibrils			
VII ^{a,i}	Basement membranes	Long-chain, antiparallel dimers, anchoring fibrils	

^a Martin, G. R., Timpl, R., Müller, P. K., and Kühn, K. (1985) *Trends Biochem. Sci.* **10**, 285-287^b van der Rest, M., and Garrone, R. (1991) *FASEB J.* **5**, 2814-2823
^c Prockop, D. J., and Kivirikko, K. I. (1995) *Ann. Rev. Biochem.* **64**, 403-434^d Prockop, D. J. (1990) *J. Biol. Chem.* **265**, 15349-15352^e Nah, H.-D., Niu, Z., and Adams, S. L. (1994) *J. Biol. Chem.* **269**, 16443-16448^f Myers, J. C., Loidl, H. R., Stolle, C. A., and Seyer, J. M. (1985) *J. Biol. Chem.* **260**, 5533-5541^g Shaw, L. M., and Olsen, B. R. (1991) *Trends Biochem. Sci.* **16**, 191-194^h Myers, J. C., Yang, H., D'Ippolito, J. A., Presente, A., Miller, M. K., and Dion, A. S. (1994) *J. Biol. Chem.* **269**, 18549-18557ⁱ Hudson, B. G., Reenders, S. T., and Tryggvason, K. (1993) *J. Biol. Chem.* **268**, 26033-26036^j Benya, P. D., and Radilla, S. R. (1986) *J. Biol. Chem.* **261**, 4160-4169^k Lunstrum, G. P., Sakai, L. Y., Keene, D. R., Morris, N. P., and Burgeson, R. E. (1986) *J. Biol. Chem.* **261**, 9042-9048



The basement membrane collagen IV forms a non-fibrillar network.^{644,645} The 400-nm-long molecules aggregate via their identical ends. Four molecules are held together through their triple-helical N termini, while the C-terminal globular domains connect pairs of molecules. Type IV collagen is also a proteoglycan with a glycoaminoglycan chain attached to one of its nonhelical domains. It may become covalently attached to type II collagen via a hydroxypyridine linkage.⁶⁴⁶ Dimers of the microfibrillar collagen VI are formed from 105-nm-long monomers by antiparallel and staggered alignment, with the 75-nm overlapping helical segments twisting around each other to form coiled dimers. After a symmetrical association of dimers to tetramers, fibrillar structures are formed by end-to-end aggregation. The connection of monomers to dimers, tetramers, and polymers occurs by disulfide bonds between triple-helical segments and globular domains. The 450-nm-long collagen VII molecules associate into antiparallel dimeric structures which show a 60-nm overlap and which subsequently assemble laterally with their ends in register.⁶²²

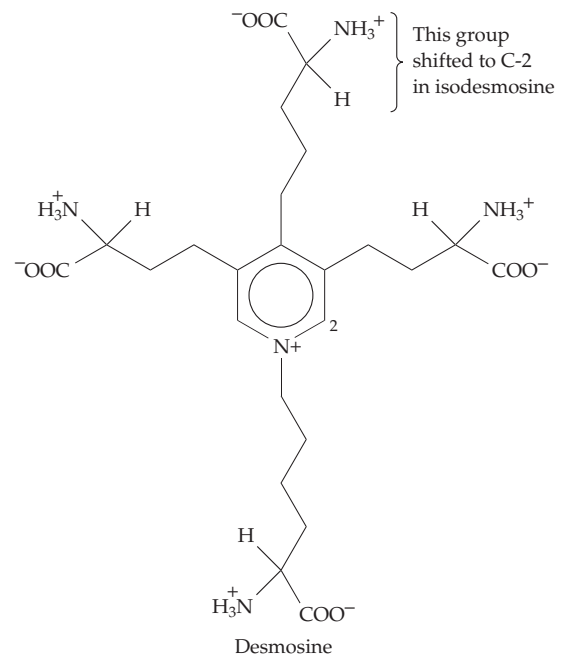
At least 32 genes encode the α peptide chains of vertebrate collagens.^{647,648} These chains are assembled into the 19 known types of collagen. Alternative splicing of the mRNAs provides additional isoforms.⁶⁴⁹ The collagen $\alpha 2(I)$ gene from both the chick and human DNA is ~38 kb in length and consists of 52 exons separated by introns ranging in length from 80 to 2000 base pairs.^{647,650} At least nine of the exons that encode the triple-helical regions have exactly 54 bp. All are multiples of 9 bp, i.e., the length needed to encode one Gly-X-Y triplet (Chapter 2, Section D.4). The significance of these observations is unclear but the presence of so many introns does suggest ways in which collagen sequences could have been transferred into the genes for such proteins as acetylcholinesterase and the C1q component of complement.⁶⁴⁷ The human $\alpha 1(I)$ collagen gene also consists of 51 segments but they lie

within a shorter 18-kb length of DNA.⁶⁵¹ A collagen gene from *Drosophila* is much less fragmented.

Collagens are found in all metazoan organisms.⁶⁵² Invertebrate collagens play a variety of specialized roles.⁶⁵³ For example, minicollagens from *Hydra* strengthen the walls of their nematocysts.⁵⁷²

Elastic fibers. The elastic properties of lung, skin, and large blood vessels are provided by elastic fibers in the extracellular matrix.⁶⁵⁴ The fibers consist of amorphous material together with the insoluble protein **elastin**, which is rich in glycine, proline, and hydrophobic

amino acids. Its special structure (Chapter 2) provides elasticity to the fibers. A 72-kDa precursor **tropoelastin** is secreted into the extracellular space where it is acted upon by lysyl oxidase (Eq. 8-7) and crosslinked into a rubber-like network.⁶⁵⁴ Remarkable crosslinkages are formed. Three aldehyde groups derived by oxidation of lysine side chains combine with one lysine amino group through aldol condensations, dehydration, and oxidation to form residues of **desmosine** and **isodesmosine**.



Cartilage and basement membranes. Tendons consist largely of collagen, but in most tissues the collagen fibrils are embedded in a matrix of proteoglycans and various other proteins.⁶⁵⁵⁻⁶⁵⁷ Both the core proteins of the proteoglycans and the attached

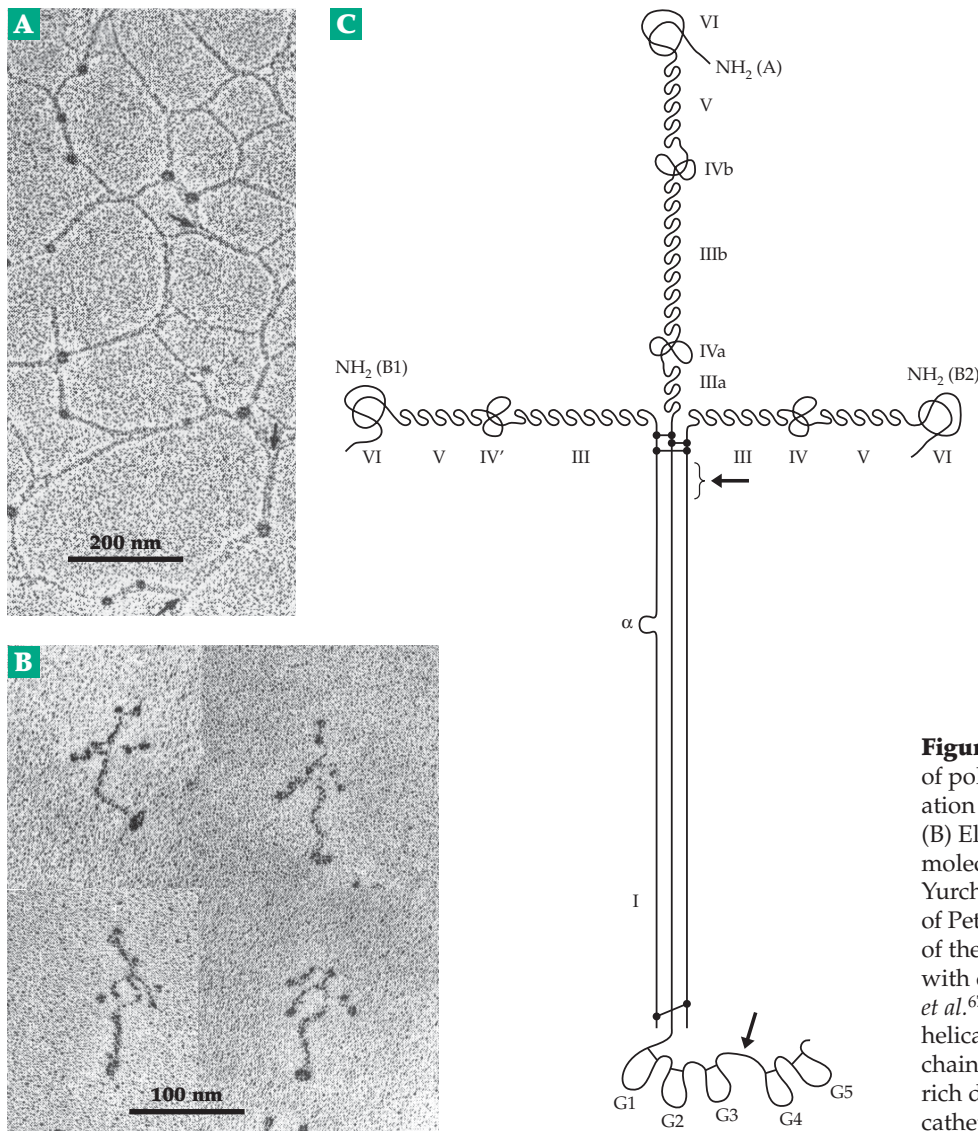


Figure 8-33 (A) Electron micrograph of polygonal network formed by association of collagen type IV monomers. (B) Electron micrographs of single molecules of laminin. (A) and (B) from Yurchenko and Schittny.^{663,663a} Courtesy of Peter Yurchenko (C) Structural model of the three-chain laminin molecule with domain designations. From Beck *et al.*⁶⁷⁰ Domains I and II are a triple-helical coiled coil rod which, in the B1 chain, is interrupted by a small cysteine-rich domain α . Sites of cleavage by cathepsin are marked by arrows.

polysaccharides interact with collagen fibrils and with fibronectin and other (previously discussed) cell surface proteins. The cartilage matrix^{658,659} consists largely of proteoglycans (Fig. 4-16)⁶⁶⁰ and of several difficult to study, insoluble proteins. One of these is the 148-kDa **cartilage matrix protein**, which yields 52-kDa subunits upon reduction.^{661,662} It interacts with both proteoglycans and collagen and may help to integrate the cartilage matrix.

Basement membranes (Fig. 1-6)⁶⁶³ function in part as an exoskeleton that helps keep cells positioned. However, the thick basement membranes of the capillary walls of the glomeruli of the kidney provide the ultrafilters that prevent most proteins from entering the urine. Basement membranes contain large amounts of collagen IV, which forms a polygonal network (Fig. 8-33A). A second macromolecular network is formed by the very large 950-kDa cross-shaped multisubunit protein called **laminin** (Fig.

8-33B).^{664–666} Laminin is one of a series of extracellular proteins which appear to have arisen by shuffling of structural modules during evolution.⁶⁶⁷ It contains sites for binding to heparin, to integrins,⁶⁶⁸ to the heparin sulfate proteoglycan **agrin** (see also Fig. 4-11), and to the 150-kDa sulfated glycoprotein **nidogen** (entactin).⁶⁶⁹ At least seven isoforms of laminin, with varying tissue distributions, are formed, in part as a result of alternative splicing of the mRNA transcripts.⁶⁶⁶ Laminin is rich in EGF-like modules.⁶⁶⁵ An X-ray structure of three of them shows that they form a continuous rod of complex structure.³²¹ A smaller 100 kDa basement membrane **fibulin** also contains multiple EGF-like repeats.⁶⁷¹ As with other extracellular structures, crosslinking of laminin and other components of basement membranes by transglutaminase provides additional stability.⁶⁷²

BOX 8-E GENETIC DEFECTS OF COLLAGEN STRUCTURE

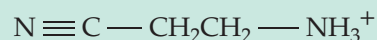
Any major protein of the body is likely to be associated with a number of genetic problems. In the case of collagen, the possibility for harmful mutations is enhanced by the existence of a large number of genes that encode the more than 16 types of collagen which are expressed differently in different tissues. There are known human disorders resulting from defects in synthesis, secretion, or structure of types I, II, III, IV, and VII collagens. Other defects involve lysyl hydroxylase and procollagen N-proteinase.^{a-d}

In the severe lethal form of **osteogenesis imperfecta** (brittle bone syndrome) the victims' collagen I may contain an $\alpha 1$ chain lacking as many as 100 residues or a shortened $\alpha 2$ chain. In other cases a cysteine, arginine, or other amino acid has been substituted for glycine in the triple-helical region of an $\alpha 1$ chain.^{e,f} The cloning of collagen genes has permitted an exact description and precise location of the defects in these genes. Alterations toward the N-terminus of the $\alpha 1$ chain or in the $\alpha 2$ chain often cause a milder type of osteogenesis imperfecta.^g Some patients have deletions in the pro- $\alpha 2$ chains of collagen I causing the chain to be incorporated into the collagen without removal of the N-terminal or C-terminal peptide to give a protein with poor stability.^h In other cases amino acid substitutions in the $\alpha 2$ chain cause formation of chains with excessive posttranslational modification. Sometimes the $\alpha 2$ chain is not incorporated into the triple helix and the collagen I formed contains three $\alpha 1$ chains.

Another well established abnormality of collagen is found in cattle suffering from **dermatosparaxis**, a disease in which the skin is extremely brittle. The collagen chains are disorganized and have poor fiber-forming properties. The procollagen peptidase that cleaves a peptide from the N termini of the chains of procollagen is apparently defective. A similar human collagen disease is the **Ehlers-Danlos syndrome**, which in some instances is accompanied with recurrent joint dislocations and curvature of the spine. At least ten different types of the disorder are known.^d The procollagen peptidase is sometimes lacking.ⁱ In other cases a person synthesizes an abnormal pro- $\alpha 2$ chain that is resistant to cleavage by the peptidase because of deletion of the normal cleavage site. In others collagen is formed in only small amounts or is degraded rapidly. Some individuals lack lysyl hydroxylase and others have a defect in mRNA splicing which causes loss of an exon from the mRNA and synthesis of shortened pro- $\alpha 2$ chains.^j

Somewhat similar symptoms are observed in **lathyrism**, a disease which arises when animals ingest seeds of *Lathyrus odoratus*, the common sweetpea. Since lathyrus peas form part of the diet of some peoples, the condition is also known in humans and often causes curvature of the spine and rupture of the aorta.

The biochemical problem has been traced to the presence in the seeds of **β -cyanoalanine** and of its decarboxylation product **β -aminopropionitrile**.



Although the mode of action is not certain, this compound is an inhibitor of lysyl oxidase essential to the crosslinking of both collagen and elastin. A hereditary defect with a similar effect in the mouse involves a defect in lysyl oxidase.^{k,l}

Collagen defects account for a variety of other skeletal problems^m including some cases of the common **osteoarthritis**.ⁿ Mice lacking the $\alpha 1$ chain of collagen IX develop a degenerative joint disease resembling human osteoarthritis.^o An inherited defect in the basement membrane collagen IV is responsible for the inherited **Alport disease** in which kidney filtration is defective.^{p,q} Similar symptoms are observed with the acute autoimmune disease **Goodpasture syndrome** and in **diabetic nephropathy**.^p

^a Prockop, D. J. (1990) *J. Biol. Chem.* **265**, 15349–15352

^b Kuivaniemi, H., Tromp, G., and Prockop, D. J. (1991) *FASEB J.* **5**, 2052–2060

^c Prockop, D. J., and Kivirikko, K. I. (1995) *Ann. Rev. Biochem.* **64**, 403–434

^d Byers, P. H. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 3 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 4029–4077, McGraw-Hill, New York

^e Cohen-Solal, L., Zylberberg, L., Sangalli, A., Gomez Lira, M., and Mottes, M. (1994) *J. Biol. Chem.* **269**, 14751–14758

^f Lightfoot, S. J., Atkinson, M. S., Murphy, G., Byers, P. H., and Kadler, K. E. (1994) *J. Biol. Chem.* **269**, 30352–30357

^g Marini, J. C., Lewis, M. B., Wang, Q., Chen, K. J., and Orrison, B. M. (1993) *J. Biol. Chem.* **268**, 2667–2673

^h Mundlos, S., Chan, D., Weng, Y. M., Silience, D. O., Cole, W. G., and Bateman, J. F. (1996) *J. Biol. Chem.* **271**, 21068–21074

ⁱ Holmes, D. F., Watson, R. B., Steinmann, B., and Kadler, K. E. (1993) *J. Biol. Chem.* **268**, 15758–15765

^j Weil, D., D'Alessio, M., Ramirez, F., Steinmann, B., Wirtz, M. K., Glanville, R. W., and Hollister, D. W. (1989) *J. Biol. Chem.* **264**, 16804–16809

^k Pope, F. M., Martin, G. R., Lichtenstein, J. R., Penttinen, R., Gerson, B., Rowe, D. W., and McKusick, V. A. (1975) *Proc. Natl. Acad. Sci. U.S.A.* **72**, 1314–1316

^l Rowe, D. W., McGoodwin, E. B., Martin, G. R., and Grahn, D. (1977) *J. Biol. Chem.* **252**, 939–942

^m Freisinger, P., Ala-Kokko, L., LeGuellec, D., Franc, S., Bouvier, R., Ritvaniemi, P., Prockop, D. J., and Bonaventure, J. (1994) *J. Biol. Chem.* **269**, 13663–13669

ⁿ Ala-Kokko, L., Baldwin, C. T., Moskowitz, R. W., and Prockop, D. J. (1990) *Proc. Natl. Acad. Sci. U.S.A.* **87**, 6565–6568

^o Fässler, R., Schnegelsberg, P. N. J., Dausman, J., Shinya, T., Muragaki, Y., McCarthy, M. T., Olsen, B. R., and Jaenisch, R. (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 5070–5074

^p Zhou, J., Hertz, J. M., Leinonen, A., and Tryggvason, K. (1992) *J. Biol. Chem.* **267**, 12475–12481

^q Gunwar, S., Ballester, F., Noelken, M. E., Sado, Y., Ninomiya, Y., and Hudson, B. G. (1998) *J. Biol. Chem.* **273**, 8767–8775

BOX 8-F SKIN

Mammalian skin must be tough, water-resistant, self-renewing, and rapidly healing. The outer layers of cells or **epidermis** consist principally of **keratinocytes**, epithelial cells specialized for formation of keratin (Fig. 7-31). In the inner layer of the epidermis the **basal stem cells** divide, providing a constant outward flow of cells which become progressively flattened, dehydrated, and filled with keratin fibrils.^a The outer layers contain only dead cells which are finally sloughed or abraded from the surface. Human epidermis is completely renewed in about 28 days!

About 25 different human genes encode the keratins of skin and other soft tissues. Others specify the keratins of hair and nails.^b Both of these hard tissues as well as claws, hoofs, beaks, horns, scales, quills, and feathers are largely keratin. However, there are additional constituents. During the final stages of keratinocyte differentiation a 15-nm-thick crosslinked sheath of protein, the **cornified cell envelope** (CE), forms beneath the plasma membrane.^c Crosslinkages between keratin and other proteins are formed by the action of transglutaminases.^{d-f} A specialized protein **involucrin**, which contains glutamine-rich repeating sequences, provides many of the side chain amide groups for the crosslinking reaction (Eq. 2-23).^g **Loricin**, a protein containing glycine-rich flexible loops,^h is also a major partner in these crosslinkages.^{c,h} The histidine-rich **filaggrin** undergoes complex processing before binding to keratin fibrils to provide another form of crosslinkage.^{i,j} Small proline-rich proteins, desmosomal proteins, and others are also present in the CE.^c

As the final outer **stratum corneum** is formed the phospholipid bilayer deteriorates and intercellular lipid layers are formed.^{k,l} These contain principally ceramides, cholesterol, and free fatty acids. Some sphingolipids are covalently attached to proteins.^a

Also present in the epidermis are embedded macrophage-like Langerhans cells as well as pigmented **melanocytes**, cells with highly branched dendrites, which lie just above the basal stem cell layer. Each melanocyte contains hundreds of pigmented organelles called **melanosomes**. They contain not only the black or reddish **melanin** pigments but also the enzymes needed to form them (Chapter 25).^{n,o}

The dendrites of a melanocyte contact about 36 keratinocytes and are able to transfer melanosomes to these adjacent cells. The numbers and sizes of the melanosomes as well as melanin structure determine differences in skin color.^m Similar cells in amphibians, the **melanophores**, also contain light receptors.^p Their melanosomes are not transferred to other cells but may be either clustered near the center of the cell or dispersed. The location can be changed quickly by transport of the melanosomes along a network of microtubules allowing the animals to change in response to changes in light color.^q Various stimuli, including ultraviolet irradiation of melanocytes, cause increased synthesis of melanin with a resultant tanning^o and added protection against sunburn.

Beneath the basement membrane of the epidermis is the **dermis**, a thick, tough, collagen-rich connective tissue. Blood vessels and nerve endings are found in this layer, as are roots of hairs and oil and sweat glands.^r

Skin suffers from a variety of ailments including serious hereditary diseases.^a One group of **keratinization disorders**, known as **ichthyoses**, are characterized by thickened, scaly skin. Some hereditary ichthyoses result from defects in type II

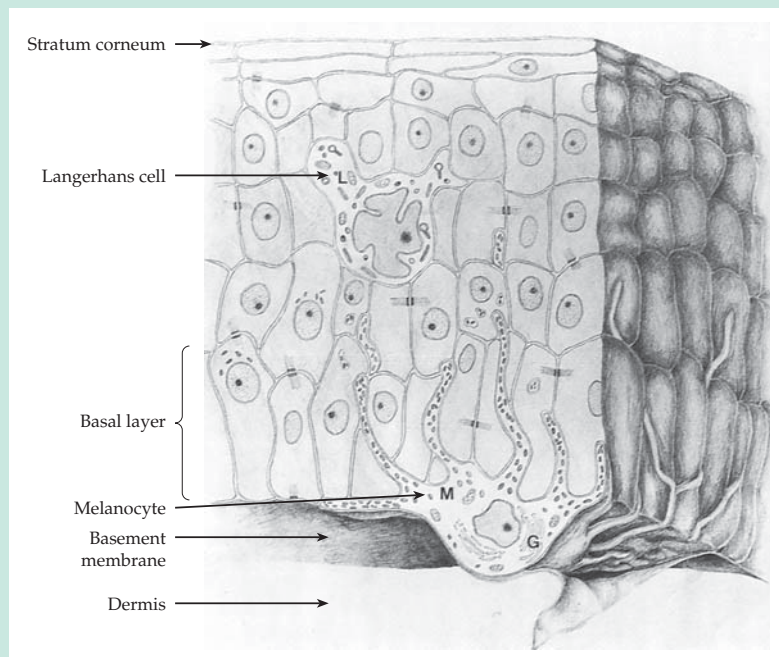


Diagram of a dendritic melanocyte surrounded by satellite keratinocytes. The Golgi area (G), where the melanosomes are synthesized, is shown around the nucleus. The other branched cell, higher in the epidermis, is a Langerhans cell with its tennis racquet-shaped granules. Courtesy of Dr. W. Quevedo, Jr. From Montagna *et al.*^m

BOX 8-F SKIN (continued)

keratin.^b **Lamellar ichthyosis** reflects defects in the crosslinking enzyme transglutaminase.^{e,f} **Epidermolysis bullosa** is a heterogeneous group of disorders characterized by easy formation of blisters. One form has been traced to a defect in the anchoring fibrils of type VII collagen, which tie cells of the basal layer to the basement membrane.^{s,t} Others are defects in keratins of the basal or intermediate layers.^a Yet others involve the lipid metabolism of skin, e.g. a steroid sulfatase deficiency.^a

The most frequent skin disorder, which affects about 2% of the world's population is **psoriasis**. The thickened, scaly patches can cover much of the skin and become disabling. The inflammation and excessive epidermal growth are usually a T-cell mediated immunologic response to antigenic stimuli.^{p,u,v} However, there is a hereditary form.^u Other common skin disorders include actinic keratosis induced by light and cancer.

^a Roop, D. (1995) *Science* **267**, 474–475

^b Takahashi, K., Paladini, R. D., and Coulombe, P. A. (1995) *J. Biol. Chem.* **270**, 18581–18592

^c Robinson, N. A., Lopic, S., Welter, J. F., and Eckert, R. L. (1997) *J. Biol. Chem.* **272**, 12035–12046

^d Kim, S.-Y., Chung, S.-I., and Steinert, P. M. (1995) *J. Biol. Chem.* **270**, 18026–18035

^e Huber, M., Rettler, I., Bernasconi, K., Frenk, E., Lavrijsen, S. P. M., Ponc, M., Bon, A., Lautenschlager, S., Schorderet, D. F., and Hohl, D. (1995) *Science* **267**, 525–528

^f Candi, E., Melino, G., Lahm, A., Ceci, R., Rossi, A., Kim, I. G., Ciani, B., and Steinert, P. M. (1998) *J. Biol. Chem.* **273**, 13693–13702

^g Yaffe, M. B., Beegen, H., and Eckert, R. L. (1992) *J. Biol. Chem.* **267**, 12233–12238

^h Hohl, D., Mehrel, T., Lichti, U., Turner, M. L., Roop, D. R., and Steinert, P. M. (1991) *J. Biol. Chem.* **266**, 6626–6636

ⁱ Resing, K. A., Walsh, K. A., Haugen-Scofield, J., and Dale, B. A. (1989) *J. Biol. Chem.* **264**, 1837–1845

^j Mack, J. W., Steven, A. C., and Steinert, P. M. (1993) *J. Mol. Biol.* **232**, 50–66

^k ten Grotenhuis, E., Demel, R. A., Ponc, M., Boer, D. R., van Miltenburg, J. C., and Bouwstra, J. A. (1996) *Biophys. J.* **71**, 1389–1399

^l Bouwstra, J. A., Thewalt, J., Gooris, G. S., and Kitson, N. (1997) *Biochemistry* **36**, 7717–7725

^m Montagna, W., Prota, G., and Kenney, J. A., Jr. (1993) *Black Skin Structure and Function*, Academic Press, San Diego, California

ⁿ Potterf, S. B., Muller, J., Bernardini, I., Tietze, F., Kobayashi, T., Hearing, V. J., and Gahl, W. A. (1996) *J. Biol. Chem.* **271**, 4002–4008

^o Roméro-Graillet, C., Aberdam, E., Biagoli, N., Massabni, W., Ortonne, J.-P., and Ballotti, R. (1996) *J. Biol. Chem.* **271**, 28052–28056

^p Greaves, M. W., and Weinstein, G. D. (1995) *N. Engl. J. Med.* **332**, 581–588

^q Rogers, S. L., Tint, I. S., Fanapour, P. C., and Gelfand, V. I. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 3720–3725

^r Martin, P. (1997) *Science* **276**, 75–81

^s Byers, P. H. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 3 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 4029–4077, McGraw-Hill, New York

^t Christiano, A. M., Ryyänen, M., and Uitto, J. (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 3549–3553

^u Tomfohrde, J., Silverman, A., Barnes, R., Fernandez-Vina, M. A., Young, M., Lory, D., Morris, L., Wuepper, K. D., Stastny, P., Menter, A., and Bowcock, A. (1994) *Science* **264**, 1141–1145

^v Boehncke, W.-H., Dressel, D., Zollner, T. M., and Kaufmann, R. (1996) *Nature (London)* **379**, 777

Fibrillin and Marfan's syndrome. Most connective tissues contain insoluble, beaded microfibrils 10–12 nm in diameter. A component of some of these microfibrils, which are often found in elastic tissue, was purified from media used to culture human fibroblasts in 1986. This protein, called fibrillin, is a single-chain 350-kDa glycoprotein which contains ~14% cysteine.^{673–674} Using a DNA probe based on the partially cloned fibrillin gene, the location of the gene was established on the long arm of chromosome 15 at a site previously identified as that of a gene defective in Marfan's syndrome. This disorder often causes dislocation of lenses of the eyes and aortic aneurysm as well as elongated limbs and fingers. Point mutations in the fibrillin gene have been identified in both Marfan's patients and family members that carry the defective gene.^{675,676}

The cuticles of invertebrates. The tough elastic cuticle of the nematode *Caenorhabditis* is largely collagen. However, the molecules are smaller than in

vertebrates and there are ~100 different genes whose transcription gives rise to a large variety of similar proteins.⁶⁵² Cuticles of some annelids have unusually long collagens.^{653,677} In contrast, the epithelial cells of insects and other arthropods secrete chitin which serves as the framework for development of a thick and often hard cuticle or exoskeleton. The cuticle also contains a variety of proteins.^{678,679} In some instances mineralization by calcium carbonate occurs. During the later phases of the cuticle development extensive crosslinking of the proteins takes place. This is largely by reactions between modified aromatic side chains and resembles the chemistry of formation of melanin and lignin (Chapter 25).⁶⁸⁰

Bones, teeth, and shells. Living organisms are able to induce the formation of over 60 inorganic compounds.⁶⁸¹ Most of these are formed by animals. Two forms of calcium carbonate, **calcite** and **aragonite**, predominate.⁶⁸² These minerals form shells, exoskeleton bones, bones, teeth, and other specialized structures.

While some organisms promote mineral deposition completely outside of their cell coats, the mineralization is usually controlled by the proteins and polysaccharides lying around and between cells.^{683–687}

Bone deposition begins in the proteoglycan and collagen II matrix of cartilage. Later these polymers are largely replaced by collagen I which accounts for 90% of the organic material in mature bone. Bone collagen has a distinctive pattern of crosslinking.^{639,688,689} Some borohydride-reducible crosslinks remain throughout adult life.⁶³⁹ However, complex hydroxypyridine and pyrrole linkages are more characteristic of bone collagen. Embedded in the bone matrix are spidery cells, the **osteocytes**. Among these are **osteoblasts**,^{689a} which secrete the collagen and other proteins that promote the laying down of calcium phosphate. Also present are large multinucleate **osteoclasts**^{689b} which dissolve bone and reabsorb calcium and phosphate. Cells of both types remain active in mature bone, which is both a structural material and a store of calcium and phosphorus.

How do osteoblasts induce calcium phosphate deposition? The first crystals laid down within the cartilage matrix are of carbonate apatite only 2–3 nm thick and tens of nanometers in length. They are intimately associated with the collagen and other components of the matrix⁶⁹⁰ and may be formed within matrix vesicles.⁶⁹¹ From observations with large biologically formed calcium carbonate crystals it is known that proteins or other organic initiators of crystallization can be found embedded in mature crystals. If the same is true in bone what are the initiators? The answer is uncertain but it is known that without osteoblasts the partially mineralized cartilage will not become bone.⁶⁹²

A key to the development of osteoblasts appears to be an **osteoblast-specific transcription factor OS/2** or **Cbfa1**.^{693–695} Mutations in human Cbfa1 are linked to a series of skeletal defects.⁶⁹⁶ A unique change accompanying conversion of a precursor cell into an osteoblast is the formation of a 49-residue γ -carboxyglutamate (Gla)-containing protein called **osteocalcin**.⁶⁹⁷ (See also Box 15-F). It is the most abundant noncollagenous protein of bone. Its three Gla residues doubtless help to bind calcium ions and osteocalcin may be an initiator of crystallization. Osteocalcin has also been found in fish scales.⁶⁹⁸ Also present in bone is a 74-residue **matrix Gla protein** which has 5 Gla residues.⁶⁹⁹

Other phosphoproteins, glycoproteins, and proteoglycans may also be required for mineralization. For example, the 286-residue glycoprotein **osteonection**^{323,700,701} accounts for 3% of total bone protein. It contains two Ca^{2+} -binding motifs and inhibits growth of hydroxylapatite crystals, but its role in bone development is not clear. One of the phosphoproteins of developing bone has been identified as

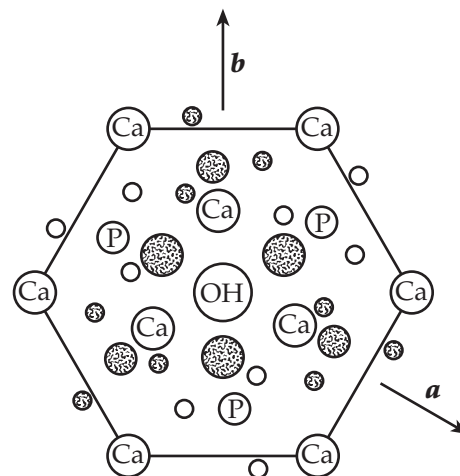


Figure 8-34 An end-on view of a crystallite of hydroxylapatite. The shaded atoms of Ca, P, and O represent an underlying layer. The OH^- groups form a longitudinal H-bonded array in the center. From J. A. Weatherell and C. Robinson,⁷⁰⁵ p. 55. A small amount of Mg^{2+} is present in place of Ca^{2+} and a very small fraction of the OH^- is replaced by fluoride ion which has a bone strengthening effect.

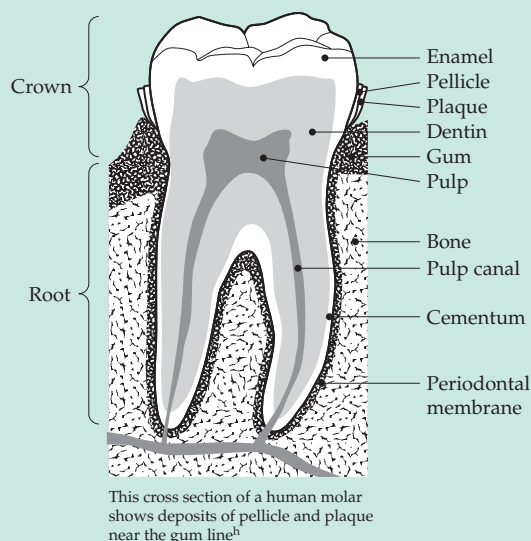
the 24 kDa propeptide cut from the N-terminal end of the $\alpha 1$ chain of type I procollagen.⁷⁰² Another bone protein, the sialic acid-containing **osteopontin** has the cell-binding sequence GRGDS identical to that in fibronectin.^{703,704} Since it also binds to hydroxylapatite this protein may form a bridge between cell surfaces and the mineral matrix.

The mineral phase of bone is largely hydroxylapatite, $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$ (Fig. 8-34), which is essentially in chemical equilibrium with the calcium and phosphate ions present in the blood serum. Thus, bone cells can easily promote either the deposition or dissolution of the mineral phase by localized change in pH, concentrations of Ca^{2+} or HPO_4^{2-} , or of chelating compounds such as ATP or inorganic pyrophosphate. Small 100 nm vesicles rich in acidic phospholipids and containing both Ca^{2+} and the enzymes alkaline phosphatase and pyrophosphatase, may play an essential role in calcification. Perhaps they release the calcium and enzymes that generate inorganic phosphate.⁶⁹¹

Bone is noted for its continual “remodeling” caused by the action of both osteoblasts and osteoclasts.^{704a,b} The latter are large multinuclear cells derived from the same precursors as give rise to macrophages.^{689b,704} Osteoclasts generate a sealed, acidic compartment on a bone surface, using a vacuolar ATPase to pump protons from the cytosol of the osteoclast, using H_2CO_3 as a source of protons.⁷⁰⁶ The HCO_3^- then leaves cells in exchange for Cl^- which then also enters the acidified compartment. Not only is the calcium phosphate dissolved but lysosomal enzymes digest the organic materials. Each osteoclast creates a long tun-

BOX 8-G THE BIOCHEMISTRY OF TEETH

Like other bones, mammalian teeth are composed largely of collagen and hydroxylapatite. The much studied rat incisor is 65% mineral; about 85% of the organic material is type I collagen. A tooth consists of three mineralized tissues together with the internal blood vessels and nerves of the pulp.^a The outer layer of **enamel** is formed by secretion from a thin layer of epithelial cells, the **ameloblasts**. The enamel matrix is devoid of collagen but contains two characteristic groups of proteins, the **amelogenins** and the **enamelin**s.



The amelogenins are hydrophobic but are also rich in proline, histidine, and glutamic acid.^{a,b} They account for 90% of the matrix proteins but are replaced by the initially less abundant enamelin^c in fully mineralized teeth.^{a,d} Another protein present in developing enamel is **ameloblastin**, which appears to be unique to ameloblasts.^a As mineralization of the enamel progresses the matrix is lost and 98% of the enamel is hydroxylapatite.^a

Dentin is of epithelial-mesenchymal origin. The **odontoblasts** secrete an extracellular matrix that is rich in collagen I and which also contains all of the other major bone proteins as well as a **dentin sialoprotein**^e and poorly characterized phosphate carriers, the **phosphophoryns**.

The third hard tissue of teeth is **cementum**, which binds teeth to the periodontal ligaments. The ligaments contain fibrillar collagen which inserts into the cementum.^{f,g}

Except for the common cold, tooth decay (**caries**) is the most prevalent disease in the United States.^{h,i} Caries is initiated by attack of acids produced by bacterial fermentations on the enamel. The saliva contains calcium and phosphate and is supersatu-

rated with respect to these ions. As a result the enamel surface is continuously recalcified. The 43-residue salivary protein **statherin** retards precipitation of calcium phosphates from the saliva, preventing excessive build-up of calcified deposits on the teeth.^j Other proline-rich proteins may play a role in recalcification repair.^k

A freshly cleaned tooth surface quickly becomes coated with a thin **pellicle** of salivary proteins. This provides a surface for growth of **dental plaque**, which contains many bacteria and adhesive polysaccharides such as dextrans.^l The latter are generated from dietary sucrose by such bacteria as *Streptococcus mutans*. (Chapter 20) and others.^m Many factors affect the probability of tooth decay. A high sucrose diet promotes decay.ⁿ While most people have some trouble with tooth decay, 1 or 2 per thousand remain totally free of caries and seem to be immune. Many factors must affect resistance to caries. For example, individuals vary in the kinds and numbers of bacteria present on teeth and in the structure of tooth enamel.^o Addition of fluoride ion to water supplies at a level of 1ppm (0.05 mM) is generally believed to reduce the incidence of tooth decay. However, caries has been declining in many developed countries at rates that are the same for water with or without fluoride.^{p,q} If teeth escape caries **periodontal disease**, caused by bacteria, is often a major problem for older people.^r

^a Krebsbach, P. H., Lee, S. K., Matsuki, Y., Kozak, C. A., Yamada, K. M., and Yamada, Y. (1996) *J. Biol. Chem.* **271**, 4431–4435

^b Renugopalakrishnan, V., Strawich, E. S., Horowitz, P. M., and Glimcher, M. J. (1986) *Biochemistry* **25**, 4879–4887

^c Deutsch, D., Palmon, A., Fisher, L. W., Kolodny, N., Termine, J. D., and Young, M. F. (1991) *J. Biol. Chem.* **266**, 16021–16028

^d Robinson, C., Weatherell, J. A., and Höhling, H. J. (1983) *Trends Biochem. Sci.* **8**, 284–287

^e Ritchie, H. H., Hou, H., Veis, A., and Butler, W. T. (1994) *J. Biol. Chem.* **269**, 3698–3702

^f Arzate, H., Olson, S. W., Page, R. C., Gown, A. M., and Narayanan, A. S. (1992) *FASEB J.* **6**, 2990–2995

^g Yamauchi, M., Katz, E. P., and Mechanic, G. L. (1986) *Biochemistry* **25**, 4907–4913

^h Sanders, H. J. (1980) *Chem. Eng. News* (Feb. 25) **58**(8) 30–39

ⁱ Shaw, J. H. (1987) *N. Engl. J. Med.* **317**, 996–1004

^j Schlesinger, D. H., and Hoy, D. I. (1977) *J. Biol. Chem.* **252**, 1689–1695

^k Clements, S., Mehansho, H., and Carlson, D. M. (1985) *J. Biol. Chem.* **260**, 13471–13477

^l Kolenbrander, P. E., Ganeshkumar, N., Cassels, F. J., and Hughes, C. V. (1993) *FASEB J.* **7**, 406–408

^m Abeygunawardana, C., and Bush, C. A. (1990) *Biochemistry* **29**, 234–248

ⁿ Newbrun, E. (1982) *Science* **217**, 418–423

^o Cevc, G., Cevc, P., Schara, M., and Skaleric, U. (1980) *Nature (London)* **286**, 425–426

^p Diesendorf, M. (1986) *Nature (London)* **322**, 125–129

^q Hileman, B. (1988) *Chem. Eng. News* **Aug.** **1**, 26–42

^r Williams, R. C. (1990) *N. Engl. J. Med.* **322**, 373–382

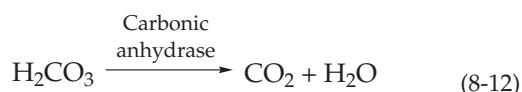
nel into the interior of the bone. However, the tunnel is soon lined with new osteoblasts and new bone is laid down.⁷⁰⁷ In **osteoporosis**, a very common disease of older persons, the rate of resorption of bone by osteoclasts exceeds that of bone formation by osteoblasts. Women typically lose 50% and men ~30% of their calcium phosphate from vertebrae and ends of long bones as they age.⁷⁰⁸ In osteopetrosis (marble bone disease) excessive calcification of bone and other tissues occurs as a result of a deficiency of carbonic anhydrase.⁷⁰⁹ See also Chapter 13. A disease of progressive calcification of soft tissues⁷¹⁰ results from an excess of **bone morphogenic factor-4**, one of a group of protein factors acting on bone development (Chapter 32).⁶³¹ **Paget's disease** is another disorder of bone remodeling that leads to overproduction of bone of poor quality.⁷¹¹ As discussed in Box 22-C, the circulating level of Ca^{2+} as well as the cellular uptake of this ion are controlled by vitamin D and its metabolites and by calcitonin and the parathyroid hormones.

Lack of a pyrophosphate ion pump in cartilage cells may cause a deficit in pyrophosphate in the surroundings of the forming bone. In mice with a defect in this pump bony spurs, similar to those in human osteoarthritis, are formed.^{711a} Over half of the world's population of persons over 65 years of age are afflicted by arthritis, over 100 types being known.^{711b}

Calcium carbonate in several different crystalline forms arises biologically.^{705,712} Sometimes the mineral is deposited within vesicles inside the cell. Thus some species of the protozoan *Hymenomonas* use Golgi vesicles as sites for construction of segments of plantlike cell walls complete with crystalline calcium carbonate plates attached to the outer surface.⁷¹³ These wall segments are then transported to their final locations. The intricately sculptured spicules of sea urchins and sponges are single crystals of calcium carbonate which have grown within intracellular vesicles.^{687,714} Diatoms also form their shells of nearly pure, hydrated SiO_2 (See Box 4-B) entirely within membrane-bound vesicles.⁷¹⁵ On the other hand, shells of molluscs are usually deposited outside the cells of the organism but again under the influence of a protein matrix.^{716,717} The animals apparently actively pump bicarbonate outward where it reacts with Ca^{2+} (Eq. 8-11).⁷¹⁸



The protons released then react with more bicarbonate to form carbonic acid which is converted to CO_2 and water (Eq. 8-12) by the enzyme carbonic anhydrase.



A problem that arises within organisms is avoiding

crystallization under supersaturating conditions. For example, normal urine is supersaturated with calcium oxalate. To prevent formation of renal calculi (stones)⁷¹⁹ an inhibitory glycoprotein is present and slows the formation and growth of crystals.⁷²⁰ Under some disease conditions calcium carbonate stones may form in pancreatic ducts. A 17 kDa lectinlike glycoprotein called **lithostatine** has been proposed to inhibit stone formation by binding to certain planes on CaCO_3 microcrystals just as antifreeze proteins (Box 4-D) inhibit ice formation.⁷²¹ However, this proposed function for lithostatine is doubtful.^{722,723} Pathological deposits of crystalline calcium pyrophosphate and basic calcium phosphates are sometimes present in joints,⁷²⁴ even in Neanderthal skeletons.⁷²⁵

3. Cell Walls of Fungi and Green Plants

The cell walls of yeasts and other fungi are made up largely of glucans, chitin, and a mannan-protein complex. Yeasts contain predominately glucans but chitin is the major polysaccharide in many other fungi. The most abundant glucan is a $\beta 1,3$ linked polymer with about 3% $\beta 1,6$ linkages to the branches and a molecular mass of ~ 240 kDa. In addition there is about 15% of a highly branched 1,6 linked glucan containing 1,3 linked branches. At the outer surface of the wall are mannoproteins⁷²⁶⁻⁷²⁸ which carry small serine- or threonine-linked oligomannans as well as large highly branched mannans linked to asparagine through the usual *N*-acetylglucosamine-containing core structure. Some of these highly branched mannan chains serve as species-specific antigens.⁷²⁷ Like those of the bacterial and animal cell surfaces, the antigens vary in structure, a fact with important medical implications. Curiously, many fungi have surface fimbriae which are composed of **collagen**, which is usually regarded as exclusively an animal protein.⁷²⁹

Cell walls of higher plants (Fig. 1-7, 4-14) are composed largely of polysaccharides. They are discussed briefly in Chapter 20.

References

- Christie, W. W. (1973) *Lipid Analysis; Isolation, Preparation, Identification and Structural Analysis of Lipids*, Pergamon, Oxford
- Hitchcock, C., and Nichols, B. W. (1971) *Plant Lipid Biochemistry*, Academic Press, New York
- Kates, M. (1972) in *Laboratory Techniques in Biochemistry and Molecular Biology*, Vol. 3, Part II (Work, T. S., and Work, E., eds), North-Holland Publ., Amsterdam
- Weete, J. W. (1980) *Lipid Biochemistry of Fungi and Other Organisms*, Plenum, New York
- Gurr, M. I., and James, A. T. (1975) *Lipid Biochemistry, an Introduction*, 2nd ed., Cornell Univ. Press, Ithaca, New York
- Porter, K. R., and Bonneville, M. A. (1973) *Fine Structure of Cells and Tissues*, 4th ed., Lea & Febiger, Philadelphia, Pennsylvania
- Avelaño, M. I., and Sprecher, H. (1987) *J. Biol. Chem.* **262**, 1180–1186
- Gunstone, F. D. (1967) *An Introduction to the Chemistry and Biochemistry of Fatty Acids and Their Glycerides*, 2nd ed., Chapman and Hall, London (Chapter 6)
- Steck, P. A., Schwartz, B. A., Rosendahl, M. S., and Gray, G. R. (1978) *J. Biol. Chem.* **253**, 5625–5629
- McNeil, M., Daffe, M., and Brennan, P. J. (1991) *J. Biol. Chem.* **266**, 13217–13223
- Qureshi, N., Takayama, K., and Schnoes, H. K. (1980) *J. Biol. Chem.* **255**, 182–189
- Adlof, R. O., Duval, S., and Emken, E. A. (2000) *Lipids* **35**, 131–135
- Cravatt, B. F., Prospero-Garcia, O., Siuzdak, G., Gilula, N. B., Henriksen, S. J., Boger, D. L., and Lerner, R. A. (1995) *Science* **268**, 1506–1509
- Hummel, H. E., Gaston, L. K., Shorey, H. H., Kaae, R. S., Byrne, K. J., and Silverstein, R. M. (1973) *Science* **181**, 873–874
- Klun, J. A., Chapman, O. L., Mattes, K. C., Wojtkowski, P. W., Beroza, M., and Sonnet, P. E. (1973) *Science* **181**, 661–663
- Moolenaar, W. H. (1995) *J. Biol. Chem.* **270**, 12949–12952
- Wells, W. W., and Eisenberg, F., Jr., eds. (1978) *Cyclitols and Phosphoinositides*, Academic Press, New York
- DeBach, H. (1978) *Trends Biochem. Sci.* **3**, 44–45
- Chilton, F. H., and Connell, T. R. (1988) *J. Biol. Chem.* **263**, 5260–5265
- Ford, D. A., and Gross, R. W. (1994) *Biochemistry* **33**, 1216–1222
- Friedman, S. M., ed. (1978) *Biochemistry of Thermophily*, Academic Press, New York
- Kushwaha, S. C., Kates, M., Sprott, G. D., and Smith, I. C. P. (1981) *Science* **211**, 1163–1164
- Luzzati, V., Gambacorta, A., DeRosa, M., and Gulik, A. (1987) *Ann. Rev. Biophys. Biophys. Chem.* **16**, 25–47
- Goldfine, H., and Langworthy, T. A. (1988) *Trends Biochem. Sci.* **13**, 217–221
- Ferrante, G., Ekiel, I., and Sprott, G. D. (1986) *J. Biol. Chem.* **261**, 17062–17066
- Pond, J. L., Langworthy, T. A., and Holzer, G. (1986) *Science* **231**, 1134–1136
- Florin-Christensen, J., Florin-Christensen, M., Knudson, J., and Rasmussen, L. (1986) *Trends Biochem. Sci.* **11**, 354–355
- Cooney, R. V., Mumma, R. O., and Benson, A. A. (1978) *Proc. Natl. Acad. Sci. U.S.A.* **75**, 4262–4264
- Renou, J.-P., Giziewicz, J. B., Smith, I. C. P., and Jarrell, H. C. (1989) *Biochemistry* **28**, 1804–1814
- Gounaris, K., and Barber, J. (1983) *Trends Biochem. Sci.* **8**, 378–381
- Knowles, F. C., and Benson, A. A. (1983) *Trends Biochem. Sci.* **8**, 178–180
- Tupper, S., Wong, P. T. T., Kates, M., and Tanphaichitr, N. (1994) *Biochemistry* **33**, 13250–13258
- Andersson, A.-S., Rilfors, L., Bergqvist, M., Persson, S., and Lindblom, G. (1996) *Biochemistry* **35**, 11119–11130
- Vikström, S., Li, L., and Wieslander, Å. (2000) *J. Biol. Chem.* **275**, 9296–9302
- Nishihara, M., Utagawa, M., Akutsu, H., and Koga, Y. (1992) *J. Biol. Chem.* **267**, 12432–12435
- Powles, T. J., Easty, D. M., Easty, G. C., Bondy, P. K., and Munro-Neville, A. (1973) *Nature (London)*, *New Biol.* **245**, 83
- Lynch, D. V. (1993) in *Lipid Metabolism in Plants* (Moore, T. S., Jr., ed), pp. 285–308, CRC Press, Boca Raton, Florida
- Araki, S., Satake, M., Ando, S., Hayashi, A., and Fujii, N. (1986) *J. Biol. Chem.* **261**, 5138–5144
- Araki, S., Abe, S., Odani, S., Ando, S., Fujii, N., and Satake, M. (1987) *J. Biol. Chem.* **262**, 14141–14145
- Song, W., and Rintoul, D. A. (1989) *Biochemistry* **28**, 4194–4200
- Hsieh, T. C.-Y., Lester, R. L., and Laine, R. A. (1981) *J. Biol. Chem.* **256**, 7747–7755
- Barr, K., and Lester, R. L. (1984) *Biochemistry* **23**, 5581–5588
- Dickson, R. C., Nagiec, E. E., Wells, G. B., Nagiec, M. M., and Lester, R. L. (1997) *J. Biol. Chem.* **272**, 29620–29625
- Yamakawa, T., and Nagai, Y. (1978) *Trends Biochem. Sci.* **3**, 128–131
- Hansson, G. C. (1983) *J. Biol. Chem.* **258**, 9612–9615
- Clausen, H., Holmes, E., and Hakomori, S.-I. (1986) *J. Biol. Chem.* **261**, 1388–1392
- Farooqui, A. A. (1981) *Adv. Lipid Res.* **18**, 159–202
- Godchaux, W., III, and Leadbetter, E. R. (1984) *J. Biol. Chem.* **259**, 2982–2990
- Abbanat, D. R., Leadbetter, E. R., Godchaux, W., III, and Escher, A. (1986) *Nature (London)* **324**, 367–369
- Jones, M. N., and Chapman, D., eds. (1994) *Micelles, Monolayers and Biomembranes*, Wiley, New York
- Jain, M. (1988) *Introduction to Biological Membranes*, 2nd ed., Wiley, New York
- Lofan, R., and Nicolson, G. L. (1981) in *Advanced Cell Biology* (Schwartz, L. M., and Azar, M. M., eds), Van Nostrand Reinhold Co., New York
- Storch, J., and Kleinfeld, A. M. (1985) *Trends Biochem. Sci.* **10**, 418–421
- Robertson, R. N. (1983) *The Lively Membranes*, Cambridge Univ. Press, Cambridge
- Starzak, M. (1984) *The Physical Chemistry of Membranes*, Academic Press, San Diego, California
- Gennis, R. B. (1989) *Biomembranes: Molecular Structure and Function*, Springer-Verlag, New York
- Morell, P., and Norton, W. T. (1980) *Sci. Am.* **242**(May), 88–116
- Martenson, R. E., ed. (1992) *Myelin: Biology and Chemistry*, CRC Press, Boca Raton, Florida
- Weimbs, T., and Stoffel, W. (1994) *Biochemistry* **33**, 10408–10415
- Korn, E. D. (1966) *Science* **153**, 1491–1498
- Tanford, C. (1980) *The Hydrophobic Effect: Formation of Micelles and Biological Membranes*, 2nd ed., Wiley, New York
- Danielli, J. F., and Davsen, H. (1985) *J. Cell. Comp. Physiol.* **5**, 495–508
- Singer, S. J., and Nicolson, G. L. (1972) *Science* **175**, 720–731
- Bretscher, M. S., and Raff, M. C. (1975) *Nature (London)* **258**, 43–49
- Quinn, P. J., and Cherry, R. J., eds. (1992) *Structural and Dynamic Properties of Lipids and Membranes*, Portland Press, London
- Disalvo, E. A., and Simon, S. A., eds. (1995) *Permeability and Stability of Lipid Bilayers*, CRC Press, Boca Raton, Florida
- Jacobson, K., Sheets, E. D., and Simson, R. (1995) *Science* **268**, 1441–1442
- Pascher, I., Lundmark, M., Nyholm, P.-G., and Sundell, S. (1992) *Biochim. Biophys. Acta.* **1113**, 339–373
- Cameron, D. G., Gudgin, E. F., and Mantsch, H. H. (1981) *Biochemistry* **20**, 4496–4500
- Zasadzinski, J. A. N., Helm, C. A., Longo, M. L., Weisenhorn, A. L., Gould, S. A. C., and Hansma, P. K. (1991) *Biophys. J.* **59**, 755–760
- Wiener, M. C., and White, S. H. (1992) *Biophys. J.* **61**, 428–433
- Smith, D. P. E., Bryant, A., Quate, C. F., Rabe, J. P., Gerber, C. H., and Swalen, J. D. (1987) *Proc. Natl. Acad. Sci. U.S.A.* **84**, 969–972
- Yang, J., Tamm, L. K., Tillack, T. W., and Shao, Z. (1993) *J. Mol. Biol.* **229**, 286–290
- Nagle, J. F., Zhang, R., Tristram-Nagle, S., Sun, W., Petrache, H. I., and Suter, R. M. (1996) *Biophys. J.* **70**, 1419–1431
- Mueller, P., Rudin, D. O., Tien, H. T., and Wescott, W. C. (1962) *Nature (London)* **194**, 979–980
- Howard, K. P., and Prestegard, J. H. (1995) *J. Am. Chem. Soc.* **117**, 5031–5040
- Ostro, M. J. (1987) *Sci. Am.* **256**(Jan), 103–111
- Philippot, J. R., and Schubert, F., eds. (1995) *Liposomes as Tools in Basic Research and Industry*, CRC Press, Boca Raton, Florida
- Harwood, J. L. (1992) *Trends Biochem. Sci.* **17**, 203–204
- Schröder, M. (1980) *Chem. Rev.* **80**, 187–213
- Mohapatra, S. K., and Behrman, E. J. (1982) *J. Inorg. Biochem.* **16**, 85–89
- Deetz, J. S., and Behrman, E. J. (1981) *Intl. J. Peptide Prot. Res.* **17**, 495–500
- Norby, P.-O., Becker, H., and Sharpless, K. B. (1996) *J. Am. Chem. Soc.* **118**, 35–42
- Montal, M., and Mueller, P. (1972) *Proc. Natl. Acad. Sci. U.S.A.* **69**, 3561–3566
- Ladha, S., Mackie, A. R., Harvey, L. J., Clark, D. C., Lea, E. J. A., Brullemans, M., and Duclouhier, H. (1996) *Biophys. J.* **71**, 1364–1373
- Jones, M. N., and Chapman, D. (1995) *Micelles, Monolayers, and Biomembranes*, Wiley-Liss, New York
- Knobler, C. M. (1990) *Adv. Chem. Phys.* **77**, 397–449
- Bryce, M. R., and Petty, M. C. (1995) *Nature (London)* **374**, 771–776
- Zasadzinski, J. A., Viswanathan, R., Madsen, L., Garnaes, J., and Schwartz, D. K. (1994) *Science* **263**, 1726–1733
- Fettiplace, R., Andrews, D. M., and Haydon, D. A. (1971) *J. Membr. Biol.* **5**, 277–296
- White, S. W., and King, G. I. (1985) *Proc. Natl. Acad. Sci. U.S.A.* **82**, 6532–6536
- Snyder, R. G., Maroncelli, M., Qi, S. P., and Strauss, H. L. (1981) *Science* **214**, 188–190
- Di, L., and Small, D. M. (1995) *Biochemistry* **34**, 16672–16677
- Tu, K., Tobias, D. J., Blasie, J. K., and Klein, M. L. (1996) *Biophys. J.* **70**, 595–608
- Katsaras, J., Raghunathan, V. A., Dufourcq, E. J., and Dufourcq, J. (1995) *Biochemistry* **34**, 4684–4688
- Snyder, R. G., Liang, G. L., Strauss, H. L., and Mendelsohn, R. (1996) *Biophys. J.* **71**, 3186–3198
- Chen, S. C., Sturtevant, J. M., and Gaffney, B. J. (1980) *Proc. Natl. Acad. Sci. U.S.A.* **77**, 5060–5063

References

92. Sun, W.-J., Tristram-Nagle, S., Suter, R. M., and Nagle, J. F. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 7008–7012
93. Mou, J., Yang, J., and Shao, Z. (1994) *Biochemistry* **33**, 4439–4443
94. Wang, Z.-q, Lin, H.-n, Li, S., and Huang, C.-h. (1995) *J. Biol. Chem.* **270**, 2014–2023
95. Wang, G., Lin, H.-n, Li, S., and Huang, C.-h. (1995) *J. Biol. Chem.* **270**, 22738–22746
96. Wang, Z.-q, Lin, H.-n, Li, S., and Huang, C.-h. (1994) *J. Biol. Chem.* **269**, 23491–23499
97. Dufourc, E. J., Parish, E. J., Chitrakorn, S., and Smith, I. C. P. (1984) *Biochemistry* **23**, 6062–6071
98. Maulik, P. R., and Shipley, G. G. (1996) *Biochemistry* **35**, 8025–8034
99. McMullen, T. P. W., and McElhaney, R. N. (1997) *Biochemistry* **36**, 4979–4986
100. Finegold, L., ed. (1993) *Cholesterol in Membrane Models*, CRC Press, Boca Raton, Florida
101. Recktenwald, D. J., and McConnell, H. M. (1981) *Biochemistry* **20**, 4505–4510
102. Needham, D., McIntosh, T. J., and Evans, E. E. (1988) *Biochemistry* **27**, 4668–4673
103. Goldfine, H., Johnston, N. C., and Phillips, M. C. (1981) *Biochemistry* **20**, 2908–2916
104. Lewis, R. V. A. H., Pohle, W., and McElhaney, R. N. (1996) *Biophys. J.* **70**, 2736–2746
105. Shimshick, E. J., and McConnell, H. M. (1973) *Biochemistry* **12**, 2351–2360
106. Almeida, P. F. F., Vaz, W. L. C., and Thompson, T. E. (1992) *Biochemistry* **31**, 7198–7210
107. Träuble, H., and Eibl, H. (1974) *Proc. Natl. Acad. Sci. U.S.A.* **71**, 214–219
108. Basáñez, G., Nieva, J. L., Rivas, E., Alonso, A., and Goni, F. M. (1996) *Biophys. J.* **70**, 2299–2306
109. Smith, R. L., and Oldfield, E. (1984) *Science* **225**, 280–288
- 109a. Smith, I. C. P. (1983) in *NMR of Newly Accessible Nuclei*, Vol. 2 (Laslo, P., ed), pp. 1–26, Academic Press, New York
110. Marassi, F. M., and Macdonald, P. M. (1992) *Biochemistry* **31**, 10031–10036
111. Oldfield, E., Bowers, J. L., and Forbes, J. (1987) *Biochemistry* **26**, 6919–6923
112. Bruzik, K. S., and Nyholm, P.-G. (1997) *Biochemistry* **36**, 566–575
113. Hong, M., Schmidt-Rohr, K., and Nanz, D. (1995) *Biophys. J.* **69**, 1939–1950
114. Thurmond, R. L., Niemi, A. R., Lindblom, G., Wieslander, Å., and Rilfors, L. (1994) *Biochemistry* **33**, 13178–13188
115. Boden, N., Jones, S. A., and Sixl, F. (1991) *Biochemistry* **30**, 2146–2155
116. Davis, J. H. (1993) in *Cholesterol in Membrane Models* (Finegold, L., ed), pp. 67–157, CRC Press, Boca Raton, Florida
117. Baenziger, J. E., Jarrell, H. C., and Smith, I. C. P. (1992) *Biochemistry* **31**, 3377–3385
118. De Boeck, H., and Zidovetzki, R. (1992) *Biochemistry* **31**, 623–630
119. de Kruijff, B., Cullis, P. R., and Verkleij, A. J. (1980) *Trends Biochem. Sci.* **5**, 79–81
120. Peng, X., and Jonas, J. (1992) *Biochemistry* **31**, 6383–6390
121. Hong, M., Schmidt-Rohr, K., and Zimmermann, H. (1996) *Biochemistry* **35**, 8335–8341
122. Lu, D., Singh, D., Morrow, M. R., and Grant, C. W. M. (1993) *Biochemistry* **32**, 290–297
123. Salmon, A., and Hamilton, J. A. (1995) *Biochemistry* **34**, 16065–16073
124. Ruocco, M. J., Siminovich, D. J., Long, J. R., Das Gupta, S. K., and Griffin, R. G. (1996) *Biophys. J.* **71**, 1776–1788
125. Bruzik, K. S., and Harwood, J. S. (1997) *J. Am. Chem. Soc.* **119**, 6629–6637
126. Casal, H. L., and McElhaney, R. N. (1990) *Biochemistry* **29**, 5423–5427
127. Hübner, W., Mantsch, H. H., Paltauf, F., and Hauser, H. (1994) *Biochemistry* **33**, 320–326
128. Léger, C. L., Daveloose, D., Christon, R., and Viret, J. (1990) *Biochemistry* **29**, 7269–7275
129. Jakobsson, E. (1997) *Trends Biochem. Sci.* **22**, 339–344
130. Berger, O., Edholm, O., and Jähnig, F. (1997) *Biophys. J.* **72**, 2002–2013
131. Chiu, S.-W., Clark, M., Balaji, V., Subramaniam, S., Scott, H. L., and Jakobsson, E. (1995) *Biophys. J.* **69**, 1230–1245
132. Browning, J. L. (1981) *Biochemistry* **20**, 7144–7151
133. Cevc, G., Watts, A., and Marsh, D. (1981) *Biochemistry* **20**, 4955–4965
134. Harlos, K., and Eibl, H. (1980) *Biochemistry* **19**, 895–899
135. Koyanova, R. D., Kuttentreich, H. L., Tenchov, B. G., and Hinz, H.-J. (1988) *Biochemistry* **27**, 4612–4619
136. Hayakawa, E., Naganuma, M., Mukasa, K., Shimozaawa, T., and Arais, T. (1998) *Biophys. J.* **74**, 892–898
137. Katsaras, J., Jeffrey, K. R., Yang, D. S.-C., and Eppard, R. M. (1993) *Biochemistry* **32**, 10700–10707
138. Srisiri, W., Sisson, T. M., O'Brien, D. F., McGrath, K. M., Han, Y., and Gruner, S. M. (1997) *J. Am. Chem. Soc.* **119**, 4866–4873
139. Siegel, D. P., and Eppard, R. M. (1997) *Biophys. J.* **73**, 3089–3111
140. Thurmond, R. L., Lindblom, G., and Brown, M. F. (1993) *Biochemistry* **32**, 5394–5410
141. Siegel, D. P., Banschbach, J., and Yeagle, P. L. (1989) *Biochemistry* **28**, 5010–5019
142. Siegel, D. P., and Banschbach, J. L. (1990) *Biochemistry* **29**, 5975–5981
143. Landau, E. M., and Rosenbusch, J. P. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 14532–14535
144. Delacroix, H., Gulik-Krzywicki, T., and Seddon, J. M. (1996) *J. Mol. Biol.* **258**, 88–103
- 144a. Tenchov, B., Koyanova, R., and Rapp, G. (1998) *Biophys. J.* **75**, 853–866
145. Russell, N. J. (1984) *Trends Biochem. Sci.* **9**, 108–112
146. deMendoza, D., and Cronan, J. E., Jr. (1983) *Trends Biochem. Sci.* **8**, 49–52
147. Mantsch, H. H., Madec, C., Lewis, R. N. A. H., and McElhaney, R. N. (1987) *Biochemistry* **26**, 4045–4049
148. Bechtel, D. B., Mueller, D. D., Whaley, T. W., and Bulla, L. A., Jr. (1985) *J. Biol. Chem.* **260**, 9784–9792
149. Bérubé, L. R., and Hollingsworth, R. I. (1995) *Biochemistry* **34**, 12005–12011
150. Huster, D., Jin, A. J., Arnold, K., and Gawrisch, K. (1997) *Biophys. J.* **73**, 855–864
151. Träuble, H. (1971) *J. Membr. Biol.* **4**, 193–208
152. Wilson, M. A., and Pohorille, A. (1996) *J. Am. Chem. Soc.* **118**, 6580–6587
153. Ho, C., and Stubbs, C. D. (1997) *Biochemistry* **36**, 10630–10637
154. Bretscher, M. S. (1980) *Trends Biochem. Sci.* **5**, VI–VII
155. Devaux, P., and McConnell, H. M. (1972) *J. Am. Chem. Soc.* **94**, 4475–4481
156. Kell, D. B. (1984) *Trends Biochem. Sci.* **9**, 86–88
157. Picard, F., Paquet, M.-J., Dufourc, E. J., and Auger, M. (1998) *Biophys. J.* **74**, 857–868
158. Schmidt, T., Schütz, G. J., Baumgartner, W., Gruber, H. J., and Schindler, H. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 2926–2929
159. Sheets, E. D., Lee, G. M., Simson, R., and Jacobson, K. (1997) *Biochemistry* **36**, 12449–12458
160. Hopkins, C. R. (1992) *Trends Biochem. Sci.* **17**, 27–32
161. Kleinfeld, A. M., Chu, P., and Storch, J. (1997) *Biochemistry* **36**, 5702–5711
162. Zhao, J., Zhou, Q., Wiedmer, T., and Sims, P. J. (1998) *Biochemistry* **37**, 6361–6366
163. Wolff, D., Canessa-Fischer, M., Vargas, F., and Diaz, G. (1971) *J. Membr. Biol.* **6**, 304–314
164. Sherbet, G. V. (1978) *The Biophysical Characterisation of the Cell Surface*, Academic Press, London
165. Jones, M. N. (1975) *Biological Interfaces*, Elsevier, Amsterdam
166. McLaughlin, S. (1989) *Ann. Rev. Biophys. Biophys. Chem.* **18**, 113–136
167. Aidley, D. J. (1971) *The Physiology of Excitable Cells*, Cambridge Univ. Press, London and New York
168. Nystrom, R. A. (1973) *Membrane Physiology*, Prentice-Hall, Englewood Cliffs, New Jersey
169. Nicholls, D. G., and Ferguson, S. J. (1992) *Bioenergetics 2*, Academic Press, London
170. Wood, P. M. (1985) *Trends Biochem. Sci.* **10**, 106–107
171. Teissie, J., Prats, M., LeMassu, A., Stewart, L. C., and Kates, M. (1990) *Biochemistry* **29**, 59–65
172. Teissie, J., Gabriel, B., and Prats, M. (1993) *Trends Biochem. Sci.* **18**, 243–246
173. Heberle, J., Riesle, J., Thiedemann, G., Oesterheld, D., and Dencher, N. A. (1994) *Nature (London)* **370**, 379–382
174. Gabriel, B., and Teissie, J. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 14521–14525
175. Devaux, P. F. (1991) *Biochemistry* **30**, 1163–1173
176. Victorov, A. V., Janes, N., Taraschi, T. F., and Hoek, J. B. (1997) *Biophys. J.* **72**, 2588–2598
177. Shiffer, K. A., Rood, L., Emerson, R. K., and Kuypers, F. A. (1998) *Biochemistry* **37**, 3449–3458
178. Michel, H. (1983) *Trends Biochem. Sci.* **8**, 56–59
179. Cowan, S. W., and Rosenbusch, J. P. (1994) *Science* **264**, 914–916
180. von Heijne, G. (1994) *Annu. Rev. Biophys. Biomol. Struct.* **23**, 167–192
181. Bloom, M. (1995) *Biophys. J.* **69**, 1631–1632
182. Weimbs, T., and Stoffel, W. (1992) *Biochemistry* **31**, 12289–12296
183. Yamaguchi, Y., Ikenaka, K., Niinobe, M., Yamada, H., and Mikoshiba, K. (1996) *J. Biol. Chem.* **271**, 27838–27846
184. Jennings, M. L. (1989) *Ann. Rev. Biochem.* **58**, 999–1027
185. Bowie, J. U. (1997) *J. Mol. Biol.* **272**, 780–789
186. Galvelin, G., Sakaguchi, M., Andersson, H., and von Heijne, G. (1997) *J. Biol. Chem.* **272**, 6119–6127
187. Jähnig, F. (1990) *Trends Biochem. Sci.* **15**, 93–95
188. Jones, D. T., Taylor, W. R., and Thornton, J. M. (1994) *Biochemistry* **33**, 3038–3049
189. Persson, B., and Argos, P. (1996) *Protein Sci.* **5**, 363–371
190. Fasman, G. D., and Gilbert, W. A. (1990) *Trends Biochem. Sci.* **15**, 89–92
191. Milpetz, F., Argos, P., and Persson, B. (1995) *Trends Biochem. Sci.* **20**, 204–205
192. Shai, Y. (1995) *Trends Biochem. Sci.* **20**, 460–464
- 192a. Killian, J. A., and von Heijne, G. (2000) *Trends Biochem. Sci.* **25**, 429–434
- 192b. Lew, S., Ren, J., and London, E. (2000) *Biochemistry* **39**, 9632–9640
193. Popov, M., Tam, L. Y., Li, J., and Reithmeier, R. A. F. (1997) *J. Biol. Chem.* **272**, 18325–18332
194. Lee, G. F., Burrows, G. G., Lebert, M. R., Dutton, D. P., and Hazelbauer, G. L. (1994) *J. Biol. Chem.* **269**, 29920–29927
195. Lee, G. F., Dutton, D. P., and Hazelbauer, G. L. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 5416–5420
196. Yu, H., Kono, M., McKee, T. D., and Oprian, D. D. (1995) *Biochemistry* **34**, 14963–14969
197. Sansom, M. S. P., Son, H. S., Sankaramakrishnan, R., Kerr, I. D., and Breed, J. (1995) *Biophys. J.* **68**, 1295–1310

References

198. Stopar, D., Jansen, K. A. J., Páli, T., Marsh, D., and Hemminga, M. A. (1997) *Biochemistry* **36**, 8261–8268
199. Nikaido, H. (1994) *J. Biol. Chem.* **269**, 3905–3908
200. Mishra, V. K., and Palgunachari, M. N. (1996) *Biochemistry* **35**, 11210–11220
201. Mishra, V. K., Palgunachari, M. N., Lund-Katz, S., Phillips, M. C., Segrest, J. P., and Anantharamaiah, G. M. (1995) *J. Biol. Chem.* **270**, 1602–1611
202. Casey, P. J. (1995) *Science* **268**, 221–225
203. McIlhinney, R. A. J. (1990) *Trends Biochem. Sci.* **15**, 387–391
204. Buss, J. E., Mumby, S. M., Casey, P. J., Gilman, A. G., and Sefton, B. M. (1987) *Proc. Natl. Acad. Sci. U.S.A.* **84**, 7493–7497
205. Johnson, D. R., Bhatnagar, R. S., Knoll, L. J., and Gordon, J. I. (1994) *Ann. Rev. Biochem.* **63**, 869–914
206. Pesceckis, S. M., and Resh, M. D. (1994) *J. Biol. Chem.* **269**, 30888–30892
207. McLaughlin, S., and Aderem, A. (1995) *Trends Biochem. Sci.* **20**, 272–276
208. Milligan, G., Parenti, M., and Magee, A. I. (1995) *Trends Biochem. Sci.* **20**, 181–186
209. Wilson, P. T., and Bourne, H. R. (1995) *J. Biol. Chem.* **270**, 9667–9675
210. Duncan, J. A., and Gilman, A. G. (1996) *J. Biol. Chem.* **271**, 23594–23600
211. Marshall, C. J. (1993) *Science* **259**, 1865–1866
212. Glomset, J. A., Gelb, M. H., and Farnsworth, C. C. (1990) *Trends Biochem. Sci.* **15**, 139–142
213. Casey, P. J., and Seabara, M. C. (1996) *J. Biol. Chem.* **271**, 5289–5292
214. Shipton, C. A., Parmryd, I., Swiezewska, E., Andersson, B., and Dallner, G. (1995) *J. Biol. Chem.* **270**, 566–572
215. Wedegaertner, P. B., Wilson, P. T., and Bourne, H. R. (1995) *J. Biol. Chem.* **270**, 503–506
216. Kikuchi, A., Sagami, H., and Ogura, K. (1999) *J. Biol. Chem.* **274**, 18011–18016
217. Turner, A. J. (1994) in *Essays in Biochemistry*, Vol. 28 (Tipton, K. F., ed), pp. 113–128, Portland Press, London
218. Low, M. G. (1987) *Biochem. J.* **244**, 1–13
219. Lisanti, M. P., and Rodriguez-Boulant, E. (1990) *Trends Biochem. Sci.* **15**, 113–118
220. Thomas, J. R., Dwek, R. A., and Rademacher, T. W. (1990) *Biochemistry* **29**, 5413–5422
221. Brewis, I. A., Ferguson, M. A. J., Mehler, A., Turner, A. J., and Hooper, N. M. (1995) *J. Biol. Chem.* **270**, 22946–22956
222. Previato, J. O., Jones, C., Xavier, M. T., Wait, R., Travassos, L. R., Parodi, A. J., and Mendonça-Previato, L. (1995) *J. Biol. Chem.* **270**, No 13, 7241–7250
223. Heinz, D. W., Ryan, M., Smith, M. P., Weaver, L. H., Keana, J. F. W., and Griffith, O. H. (1996) *Biochemistry* **35**, 9496–9504
224. Williams, A. F., and Gagnon, J. (1982) *Science* **216**, 696–703
225. Mehler, A., Richardson, J. M., and Ferguson, M. A. J. (1998) *J. Mol. Biol.* **277**, 379–392
- 225a. Eisenhaber, B., Bork, P., Yuan, Y., Löffler, G., and Eisenhaber, F. (2000) *Trends Biochem. Sci.* **25**, 340–341
226. Reynolds, J. A., and Trayer, H. (1971) *J. Biol. Chem.* **246**, 7337–7342
227. Pryde, J. G. (1986) *Trends Biochem. Sci.* **11**, 160–163
228. Bennett, V. (1985) *Ann. Rev. Biochem.* **54**, 273–304
229. Justice, J. M., Murtagh, J. J., Jr., Moss, J., and Vaughan, M. (1995) *J. Biol. Chem.* **270**, 17970–17976
230. Neville, D. M., Jr., and Glossmann, H. (1971) *J. Biol. Chem.* **246**, 6335–6338
231. Inaba, M., and Maede, Y. (1988) *J. Biol. Chem.* **263**, 17763–17771
232. Steck, T. L. (1974) *J. Cell Biol.* **62**, 1–19
233. Luna, E. J., Kidd, G. H., and Branton, D. (1979) *J. Biol. Chem.* **254**, 2526–2532
234. Viitala, J., and Järnefelt, J. (1985) *Trends Biochem. Sci.* **10**, 392–395
235. Marchesi, V. T., Furthmayr, H., and Tomita, M. (1976) *Ann. Rev. Biochem.* **45**, 667–698
236. Jay, D., and Cantley, L. (1986) *Ann. Rev. Biochem.* **55**, 511–538
237. Murthy, S. N. P., Wilson, J., Zhang, Y., and Lorand, L. (1994) *J. Biol. Chem.* **269**, 22907–22911
238. Tanner, M. J. A. (1996) *Nature (London)* **382**, 209–210
239. Müller-Berger, S., Karbach, D., König, J., Lepke, S., Wood, P. G., Appelhans, H., and Passow, H. (1995) *Biochemistry* **34**, 9315–9324
240. Erickson, H. K. (1997) *Biochemistry* **36**, 9958–9967
241. Fukuda, M., Lauffenburger, M., Sasaki, H., Rogers, M. E., and Dell, A. (1987) *J. Biol. Chem.* **262**, 11952–11957
242. MacKenzie, K. R., Prestegard, J. H., and Engelman, D. M. (1997) *Science* **276**, 131–133
243. Challou, N., Goormaghtigh, E., Cabiaux, V., Conrath, K., and Ruysschaert, J.-M. (1994) *Biochemistry* **33**, 6902–6910
244. Huang, C.-H., Reid, M., Daniels, G., and Blumenfeld, O. O. (1993) *J. Biol. Chem.* **268**, 25902–25908
245. Marchesi, V. T. (1985) *Ann. Rev. Cell Biol.* **1**, 531–561
246. Fujita, T., Ralston, G. B., and Morris, M. B. (1998) *Biochemistry* **37**, 264–271
247. Yan, Y., Winograd, E., Viel, A., Cronin, T., Harrison, S. C., and Branton, D. (1993) *Science* **262**, 2027–2030
248. Cherry, L., Menhart, N., and Fung, L. W.-M. (1999) *J. Biol. Chem.* **274**, 2077–2084
249. Begg, G. E., Harper, S. L., Morris, M. B., and Speicher, D. W. (2000) *J. Biol. Chem.* **275**, 3279–3287
250. Fujita, T., Ralston, G. B., and Morris, M. B. (1998) *Biochemistry* **37**, 272–280
- 250a. Rief, M., Pascual, J., Saraste, M., and Gaub, H. E. (1999) *J. Mol. Biol.* **286**, 553–561
251. Fowler, V. M. (1986) *Nature (London)* **322**, 777–778
252. Carraway, K. L. (1975) *Biochim. Biophys. Acta.* **415**, 379–410
253. Byers, T. J., and Branton, D. (1985) *Proc. Natl. Acad. Sci. U.S.A.* **82**, 6153–6157
254. Elgsaeter, A., Stokke, B. T., Mikkelsen, A., and Branton, D. (1986) *Science* **234**, 1217–1223
255. Pantazatos, D. P., and MacDonald, R. I. (1997) *J. Biol. Chem.* **272**, 21052–21059
256. Luna, E. J., and Hitt, A. L. (1992) *Science* **258**, 955–964
257. Reichstein, E., and Blostein, R. (1975) *J. Biol. Chem.* **250**, 6256–6263
258. Bennett, V. (1992) *J. Biol. Chem.* **267**, 8703–8706
259. Michaely, P., and Bennett, V. (1993) *J. Biol. Chem.* **268**, 22703–22709
260. Kordeli, E., Lambert, S., and Bennett, V. (1995) *J. Biol. Chem.* **270**, 2352–2359
261. An, X.-L., Takakuwa, Y., Nunomura, W., Manno, S., and Mohandas, N. (1996) *J. Biol. Chem.* **271**, 33187–33191
262. Workman, R. F., and Low, P. S. (1998) *J. Biol. Chem.* **273**, 6171–6176
263. Malik, S., Sami, M., and Watts, A. (1993) *Biochemistry* **32**, 10078–10084
264. Azim, A. C., Knoll, J. H. M., Beggs, A. H., and Chishti, A. H. (1995) *J. Biol. Chem.* **270**, 17407–17413
265. Wolfe, L. C., John, K. M., Falcone, J. C., Byrne, A. M., and Lux, S. E. (1982) *N. Engl. J. Med.* **307**, 1367–1374
266. Diakowski, W., and Sikorski, A. F. (1995) *Biochemistry* **34**, 13252–13258
267. Marfatia, S. M., Lue, R. A., Branton, D., and Chishti, A. H. (1995) *J. Biol. Chem.* **270**, 715–719
268. Krauss, S. W., Chasis, J. A., Rogers, C., Mohandas, N., Krockmalnic, G., and Penman, S. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 7297–7302
269. Khodadad, J. K., Waugh, R. E., Podolski, J. L., Josephs, R., and Steck, T. L. (1996) *Biophys. J.* **70**, 1036–1044
270. Hansen, J. C., Skalak, R., Chien, S., and Hoyer, A. (1996) *Biophys. J.* **70**, 146–166
271. Rice-Evans, C. A., and Dunn, M. J. (1982) *Trends Biochem. Sci.* **7**, 282–286
272. Mohandas, N., and Evans, E. (1994) *Annu. Rev. Biophys. Biomol. Struct.* **23**, 787–818
273. Agre, P., Casella, J. F., Zinkham, W. H., McMillan, C., and Bennett, V. (1985) *Nature (London)* **314**, 380–383
274. Baines, A. J., and Bennett, V. (1985) *Nature (London)* **315**, 410–413
275. Clark, E. A., and Brugge, J. S. (1995) *Science* **268**, 233–239
276. Dejana, E., Corada, M., and Lampugnani, M. G. (1995) *FASEB J.* **9**, 910–918
277. Malik, R. K., and Parsons, J. T. (1996) *J. Biol. Chem.* **271**, 29785–29791
278. Lo, S. H., An, Q., Bao, S., Wong, W.-K., Liu, Y., Janmey, P. A., Hartwig, J. H., and Chen, L. B. (1994) *J. Biol. Chem.* **269**, 22310–22319
279. Horwitz, A. F. (1997) *Sci. Am.* **276**(May), 68–75
280. de Pereda, J. M., Wiche, G., and Liddington, R. C. (1999) *EMBO J.* **18**, 4087–4095
281. Springer, T. A. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 65–72
282. Bazzoni, G., and Hemler, M. E. (1998) *Trends Biochem. Sci.* **23**, 30–34
283. Emsley, J., King, S. L., Bergelson, J. M., and Liddington, R. C. (1997) *J. Biol. Chem.* **272**, 28512–28517
284. Lusinskas, F. W., and Lawler, J. (1994) *FASEB J.* **8**, 929–938
285. Simon, K. O., Otey, C. A., Pavelko, F. M., and Burridge, K. (1991) *Curr. Top. Membr. Transp.* **38**, 57–64
286. Edelman, G. M. (1983) *Science* **219**, 450–457
287. Becker, J. W., Erickson, H. P., Hoffman, S., Cunningham, B. A., and Edelman, G. M. (1989) *Proc. Natl. Acad. Sci. U.S.A.* **86**, 1088–1092
288. Cunningham, B. A., Hemperly, J. J., Murray, B. A., Prediger, E. A., Brackenbury, R., and Edelman, G. M. (1987) *Science* **236**, 799–806
289. Kudo, M., Kitajima, K., Inoue, S., Shiokawa, K., Morris, H. R., Dell, A., and Inoue, Y. (1996) *J. Biol. Chem.* **271**, 32667–32677
290. Newman, P. J., Berndt, M. C., Gorski, J., White, G. C., II, Lyman, S., Paddock, C., and Muller, W. A. (1990) *Science* **247**, 1219–1222
291. DeLisser, H. M., Yan, H. C., Newman, P. J., Muller, W. A., Buck, C. A., and Albelda, S. M. (1993) *J. Biol. Chem.* **268**, 16037–16046
- 291a. Chen, H. A., Pfuhl, M., McAlister, M. S. B., and Driscoll, P. C. (2000) *Biochemistry* **39**, 6814–6824
292. Casanovas, J. M., Stehle, T., Liu, J.-h, Wang, J.-h, and Springer, T. A. (1998) *Proc. Natl. Acad. Sci. U.S.A.* **95**, 4134–4139
293. Wang, J.-H., Pepinsky, R. B., Stehle, T., Liu, J.-h, Karpus, M., Browning, B., and Osborn, L. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 5714–5718
294. Springer, T. A. (1990) *Nature (London)* **346**, 425–434

References

295. Wagner, G. (1995) *Science* **267**, 342
296. Nagar, B., Overduin, M., Ikura, M., and Rini, J. M. (1996) *Nature (London)* **380**, 360–364
297. Shapiro, L., Fannon, A. M., Kwong, P. D., Thompson, A., Lehmann, M. S., Grubel, G., Legrand, J.-F., Als-Nielsen, J., Colman, D. R., and Hendrickson, W. A. (1995) *Nature (London)* **374**, 327–336
298. Overduin, M., Harvey, T. S., Bagby, S., Tong, K. I., Yau, P., Takeichi, M., and Ikura, M. (1995) *Science* **267**, 386–389
- 298a. Nollet, F., Kools, P., and van Roy, F. (2000) *J. Mol. Biol.* **299**, 551–572
- 298b. Pertz, O., Bozic, D., Koch, A. W., Fauser, C., Brancaccio, A., and Engel, J. (1999) *EMBO J.* **18**, 1738–1747
299. Bella, J., Kolatkar, P. R., Marlor, C. W., Greve, J. M., and Rossmann, M. G. (1998) *Proc. Natl. Acad. Sci. U.S.A.* **95**, 4140–4145
300. Patel, D. J., and Gumbiner, B. M. (1995) *Nature (London)* **374**, 306–307
301. Berx, G., Cleton-Jansen, A.-M., Nollet, F., de Leeuw, W. J. F., van de Vijver, M. J., Cornelisse, C., and van Roy, F. (1995) *EMBO J.* **14**, 6107–6115
302. Vartio, T., and Vaheri, A. (1983) *Trends Biochem. Sci.* **8**, 442–444
303. Hynes, R. O. (1986) *J. Am. Chem. Soc.* **108**, 42–51
304. Ruoslahti, E. (1988) *Ann. Rev. Biochem.* **57**, 375–413
305. Engel, J., Odermatt, E., Engel, A., Madri, J. A., Furthmayr, H., Rohde, H., and Timpl, R. (1981) *J. Mol. Biol.* **150**, 97–120
306. Ingham, K. C., Brew, S. A., Huff, S., and Litvinovich, S. V. (1997) *J. Biol. Chem.* **272**, 1718–1724
307. Johnson, K. J., Sage, H., Briscoe, G., and Erickson, H. P. (1999) *J. Biol. Chem.* **274**, 15473–15479
308. Grant, R. P., Spitzfaden, C., Altmoff, H., Campbell, I. D., and Mardon, H. J. (1997) *J. Biol. Chem.* **272**, 6159–6166
309. Hynes, R. O. (1990) *Fibronectins*, Springer-Verlag, New York
310. Moyano, J. V., Carnemolla, B., Dominguez-Jiménez, C., García-Gila, M., Albar, J. P., Sánchez-Aparicio, P., Leprini, A., Querzé, G., Zardi, L., and García-Pardo, A. (1997) *J. Biol. Chem.* **272**, 24832–24836
311. Greenberg, C. S., Birckbichler, P. J., and Rice, R. H. (1991) *FASEB J.* **5**, 3071–3077
312. Jeong, J.-M., Murthy, S. N. P., Radek, J. T., and Lorand, L. (1995) *J. Biol. Chem.* **270**, 5654–5658
313. An, S. S. A., Jiménez-Barbero, J., Petersen, T. E., and Llinás, M. (1992) *Biochemistry* **31**, 9927–9933
314. Spitzfaden, C., Grant, R. P., Mardon, H. J., and Campbell, L. D. (1997) *J. Mol. Biol.* **265**, 565–579
315. Kornblihtt, A. R., Pesce, C. G., Alonso, C. R., Cramer, P., Srebrow, A., Werbach, S., and Muro, A. F. (1996) *FASEB J.* **10**, 248–257
316. Yoneda, A., Ogawa, H., Kojima, K., and Matsumoto, I. (1998) *Biochemistry* **37**, 6351–6360
317. Fischer, D., Chiquet-Ehrismann, R., Bernasconi, C., and Chiquet, M. (1995) *J. Biol. Chem.* **270**, 3378–3384
318. Yokosaki, Y., Matsuura, N., Higashiyama, S., Murakami, I., Obara, M., Yamakido, M., Shiget, N., Chen, J., and Sheppard, D. (1998) *J. Biol. Chem.* **273**, 11423–11428
319. Denda, S., Müller, U., Crossin, K. L., Erickson, H. P., and Reichardt, L. F. (1998) *Biochemistry* **37**, 5464–5474
320. Oberhauser, A. F., Marszalek, P. E., Erickson, H. P., and Fernandez, J. M. (1998) *Nature (London)* **393**, 181–185
321. Stetefeld, J., Mayer, U., Timpl, R., and Huber, R. (1996) *J. Mol. Biol.* **257**, 644–657
322. Colognato, H., MacCarrick, M., O'Rear, J. J., and Yurchenco, P. D. (1997) *J. Biol. Chem.* **272**, 29330–29336
323. Lane, T. F., and Sage, E. H. (1994) *FASEB J.* **8**, 163–173
324. Venstrom, K. A., and Reichardt, L. F. (1993) *FASEB J.* **7**, 996–1003
325. Gherzi, R., Carnemolla, B., Siri, A., Ponassi, M., Balza, E., and Zardi, L. (1995) *J. Biol. Chem.* **270**, 3429–3434
326. Ruoslahti, E., and Pierschbacher, M. D. (1987) *Science* **238**, 491–497
327. D'Souza, S. E., Ginsberg, M. H., and Plow, E. F. (1991) *Trends Biochem. Sci.* **16**, 246–250
328. Leahy, D. J., Hendrickson, W. A., Aukhil, I., and Erickson, H. P. (1992) *Science* **258**, 987–991
329. Stanley, K. K., Newby, A. C., and Luzio, J. P. (1982) *Trends Biochem. Sci.* **7**, 145–147
330. Roberts, M. F. (1996) *FASEB J.* **10**, 1159–1172
331. Han, S. K., Yoon, E. T., Scott, D. L., Sigler, P. B., and Cho, W. (1997) *J. Biol. Chem.* **272**, 3573–3582
332. Caruthers, A., and Melchior, D. L. (1986) *Trends Biochem. Sci.* **11**, 331–335
333. Gutknecht, J. (1987) *Proc. Natl. Acad. Sci. U.S.A.* **84**, 6443–6446
334. Paula, S., Volkov, A. G., and Deamer, D. W. (1998) *Biophys. J.* **74**, 319–327
335. Stein, W. D. (1990) *Channels, Carriers and Pumps: an Introduction to Membrane Transport*, Academic Press, San Diego, California
- 335a. Paulsen, I. T., Nguyen, L., Sliwinski, M. K., Rabus, R., and Saier, M. H., Jr. (2000) *J. Mol. Biol.* **301**, 75–100
336. Fox, R. O., Jr., and Richards, F. M. (1982) *Nature (London)* **300**, 325–330
337. Kempf, C., Klausner, R. D., Weinstein, J. N., Van Renswoude, J., Pincus, M., and Blumenthal, R. (1982) *J. Biol. Chem.* **257**, 2469–2476
338. Klebba, P. E., Hofnung, M., and Charbit, A. (1994) *EMBO J.* **13**, 4670–4675
339. Schirmer, T., Keller, T. A., Wang, Y.-F., and Rosenbusch, J. P. (1995) *Science* **267**, 512–514
340. Meyer, J. E. W., Hofnung, M., and Schulz, G. E. (1997) *J. Mol. Biol.* **266**, 761–775
341. Jordy, M., Andersen, C., Schüle, K., Ferenci, T., and Benz, R. (1996) *J. Mol. Biol.* **259**, 666–678
- 341a. Dumas, F., Koebnik, R., Winterhalter, M., and Van Gelder, P. (2000) *J. Biol. Chem.* **275**, 19747–19751
342. Samartzidou, H., and Delcour, A. H. (1998) *EMBO J.* **17**, 93–100
343. Paupit, R. A., Schirmer, T., Jansonius, J. N., Rosenbusch, J. P., Parker, M. W., Tucker, A. D., Tsernoglou, D., Weiss, M. S., and Schulz, G. E. (1991) *J. Structural Biol.* **107**, 136–145
344. Kreusch, A., Neubüser, A., Schiltz, E., Weckesser, J., and Schulz, G. E. (1994) *Protein Sci.* **3**, 58–63
- 344a. Müller, D. J., and Engel, A. (1999) *J. Mol. Biol.* **285**, 1347–1351
345. Bainbridge, G., Mobasher, H., Armstrong, G. A., Lea, E. J. A., and Lakey, J. H. (1998) *J. Mol. Biol.* **275**, 171–176
346. Van Gelder, P., Saint, N., Phale, P., Eppens, E. F., Prilipov, A., van Bostel, R., Rosenbusch, J. P., and Tommassen, J. (1997) *J. Mol. Biol.* **269**, 468–472
347. Lyer, R., Wu, Z., Woster, P. M., and Delcour, A. H. (2000) *J. Mol. Biol.* **297**, 933–945
348. Watanabe, M., Rosenbusch, J., Schirmer, T., and Karplus, M. (1997) *Biophys. J.* **72**, 2094–2102
349. Cowan, S. W., Schirmer, T., Rummel, G., Steiert, M., Ghosh, R., Paupit, R. A., Jansonius, J. N., and Rosenbusch, J. P. (1992) *Nature (London)* **358**, 727–733
350. Stathopoulos, C. (1996) *Protein Sci.* **5**, 170–173
351. Mannella, C. A. (1992) *Trends Biochem. Sci.* **17**, 315–320
352. Li, A. Z., Huang, H., Re, X., Qi, L. J., and Marx, K. A. (1998) *Biophys. J.* **74**, 964–973
353. Reumann, S., Maier, E., Benz, R., and Heldt, H. W. (1995) *J. Biol. Chem.* **270**, 17559–17565
354. Chrispeels, M. J., and Agre, P. (1994) *Trends Biochem. Sci.* **19**, 421–425
355. Knepper, M. A. (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 6255–6258
- 355a. Borgnia, M., Nielsen, S., Engel, A., and Agre, P. (1999) *Ann. Rev. Biochem.* **68**, 425–458
356. Ringler, P., Borgnia, M. J., Stahlberg, H., Maloney, P. C., Agre, P., and Engel, A. (1999) *J. Mol. Biol.* **291**, 1181–1190
357. de Groot, B. L., Heymann, J. B., Engel, A., Mitsuoka, K., Fujiyoshi, Y., and Grubmüller, H. (2000) *J. Mol. Biol.* **300**, 987–994
358. Cheng, A., van Hoek, A. N., Yeager, M., Verkman, A. S., and Mitra, A. K. (1997) *Nature (London)* **387**, 627–630
359. Ma, T., Song, Y., Gillespie, A., Carlson, E. J., Epstein, C. J., and Verkman, A. S. (1999) *J. Biol. Chem.* **274**, 20071–20074
- 359a. Zeuthen, T., and Klaerke, D. A. (1999) *J. Biol. Chem.* **274**, 21631–21636
360. Barone, L. M., Shih, C., and Wasserman, B. P. (1997) *J. Biol. Chem.* **272**, 30672–30677
361. Calamita, G., Bishai, W. R., Preston, G. M., Guggino, W. B., and Agre, P. (1995) *J. Biol. Chem.* **270**, 29063–29066
362. Deen, P. M. T., Verdijk, M. A. J., Knoers, N. V. A. M., Wieringa, B., Monnens, L. A. H., van Oost, C. H., and van Oost, B. A. (1994) *Science* **264**, 92–95
363. Bai, L., Fushimi, K., Sasaki, S., and Marumo, F. (1996) *J. Biol. Chem.* **271**, 5171–5176
364. Jan, L. Y., and Jan, Y. N. (1994) *Nature (London)* **371**, 119–122
365. Armstrong, C. (1998) *Science* **280**, 56–57
366. Doyle, D. A., Cabral, J. M., Pfuetzner, R. A., Kuo, A., Gulbis, J. M., Cohen, S. L., Chait, B. T., and MacKinnon, R. (1998) *Science* **280**, 69–77
367. Heginbotham, L., Odessey, E., and Miller, C. (1997) *Biochemistry* **36**, 10335–10342
368. MacKinnon, R., Cohen, S. L., Kuo, A., Lee, A., and Chait, B. T. (1998) *Science* **280**, 106–109
369. Aidley, D. J., and Stanfield, P. R. (1996) *Ion Channels, Molecules in Action*, Cambridge Univ. Press, New York
370. Kreusch, A., Pfaffinger, P. J., Stevens, C. F., and Choe, S. (1998) *Nature (London)* **392**, 945–948
- 370a. Gulbis, J. M., Zhou, M., Mann, S., and MacKinnon, R. (2000) *Science* **289**, 123–127
- 370b. Kobertz, W. R., Williams, C., and Miller, C. (2000) *Biochemistry* **39**, 10347–10352
- 370c. Roux, B., Bernèche, S., and Im, W. (2000) *Biochemistry* **39**, 10347–10352
371. Klingenberg, M. (1981) *Nature (London)* **290**, 449–454
372. Prince, R. C., Gunson, D. E., and Scarpa, A. (1985) *Trends Biochem. Sci.* **10**, 99
373. Smith, R., Separovic, F., Milne, T. J., Whittaker, A., Bennett, F. M., Cornell, B. A., and Makriyannis, A. (1994) *J. Mol. Biol.* **241**, 456–466
374. Dathe, M., Schümann, M., Wieprecht, T., Winkler, A., Beyermann, M., Krause, E., Matsuzaki, K., Murase, O., and Bienert, M. (1996) *Biochemistry* **35**, 12612–12622
375. Kiyota, T., Lee, S., and Sugihara, G. (1996) *Biochemistry* **35**, 13196–13204
376. Monette, M., and Lafleur, M. (1996) *Biophys. J.* **70**, 2195–2202
377. Ojcius, D. M., and Young, J. D.-E. (1991) *Trends Biochem. Sci.* **16**, 225–229
378. Langs, D. A. (1988) *Science* **241**, 188–191

References

379. Doyle, D. A., and Wallace, B. A. (1997) *J. Mol. Biol.* **266**, 963–977
380. Iqbal, M., and Balaram, P. (1981) *Biochemistry* **20**, 7278–7284
381. Malashkevich, V. N., Kammerer, R. A., Efimov, V. P., Schulthess, T., and Engel, J. (1996) *Science* **274**, 761–765
382. Unwin, N. (1995) *Nature (London)* **373**, 37–43
383. Harrold, F. M., and Baarda, J. R. (1967) *J. Bacteriol.* **94**, 53–60
384. Harold, F. M., and Van Brunt, J. (1977) *Science* **197**, 372–373
385. Pressman, B. C. (1976) *Ann. Rev. Biochem.* **45**, 501–530
386. Hamilton, J. A., Sabesan, M. N., and Steinrauf, L. K. (1981) *J. Am. Chem. Soc.* **103**, 5880–5885
387. Duax, W. L., Hauptman, M., Weeks, C. M., and Norton, D. A. (1972) *Science* **176**, 911–914
388. Karle, I. L., Karle, J., Wieland, T., Burgemeister, W., Faulstich, H., and Witkop, B. (1973) *Proc. Natl. Acad. Sci. U.S.A.* **70**, 1836–1840
389. Pangborn, W., Duax, W., and Langa, D. (1987) *J. Am. Chem. Soc.* **109**, 2163–2165
390. Karle, I. L. (1985) *Proc. Natl. Acad. Sci. U.S.A.* **82**, 7155–7159
391. Inabayashi, M., Miyauchi, S., Kamo, N., and Jin, T. (1995) *Biochemistry* **34**, 3455–3460
392. Ledger, P. W., and Tanzer, M. L. (1984) *Trends Biochem. Sci.* **9**, 313–314
393. Marger, M. D., and Saier, M. H., Jr. (1993) *Trends Biochem. Sci.* **18**, 13–20
394. Goswitz, V. C., and Brooker, R. J. (1995) *Protein Sci.* **4**, 534–537
395. Seidel, H. M., and Knowles, J. R. (1994) *Biochemistry* **33**, 5641–5646
396. Cloherty, E. K., Sulzman, L. A., Zottola, R. J., and Carruthers, A. (1995) *Biochemistry* **34**, 15395–15406
397. Shanahan, M. F., and D'Artel-Ellis, J. (1984) *J. Biol. Chem.* **259**, 13878–13884
398. Bell, G. I., Burant, C. F., Takeda, J., and Gould, G. W. (1993) *J. Biol. Chem.* **268**, 19161–19164
399. Silverman, M. (1991) *Ann. Rev. Biochem.* **60**, 757–794
400. Mueckler, M. (1994) *Eur. J. Biochem.* **219**, 713–725
401. Maher, F., Vannucci, S. J., and Simpson, I. A. (1994) *FASEB J.* **8**, 1003–1011
402. Burant, C. F., Takeda, J., Brot-Laroche, E., Bell, G. I., and Davidson, N. O. (1992) *J. Biol. Chem.* **267**, 14523–14526
403. Zamora-León, S. P., Golde, D. W., Concha, I. I., Rivas, C. I., Delgado-López, F., Baselga, J., Nualart, F., and Vera, J. C. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 1847–1852
404. Coderre, L., Kandror, K. V., Vallega, G., and Pilch, P. F. (1995) *J. Biol. Chem.* **270**, 27584–27588
405. Fischer, Y., Thomas, J., Sevilla, L., Munoz, P., Becker, C., Holman, G., Kozka, I. J., Palacín, M., Testar, X., Kammermeier, H., and Zorzano, A. (1997) *J. Biol. Chem.* **272**, 7085–7092
- 405a. Shepherd, P. R., and Kahn, B. B. (1999) *N. Engl. J. Med.* **341**, 248–257
406. Mueckler, M., and Makepeace, C. (1997) *J. Biol. Chem.* **272**, 30141–30146
407. Cloherty, E. K., Diamond, D. L., Heard, K. S., and Carruthers, A. (1996) *Biochemistry* **35**, 13231–13239
408. Seatter, M. J., De La Rue, S. A., Porter, L. M., and Gould, G. W. (1998) *Biochemistry* **37**, 1322–1326
409. Rumsey, S. C., Kwon, O., Xu, G. W., Burant, C. F., Simpson, I., and Levine, M. (1997) *J. Biol. Chem.* **272**, 18982–18989
410. Wiese, T. J., Dunlap, J. A., and Yorek, M. A. (1994) *J. Biol. Chem.* **269**, 22705–22711
411. Kasahara, M., Shimoda, E., and Maeda, M. (1997) *J. Biol. Chem.* **272**, 16721–16724
412. Turk, E., Kerner, C. J., Losta, M. P., and Wright, E. M. (1996) *J. Biol. Chem.* **271**, 1925–1934
413. Chen, X.-Z., Coady, M. J., Jackson, F., Berteloot, A., and Lapointe, J.-Y. (1995) *Biophys. J.* **69**, 2405–2414
414. Panayotova-Heiermann, M., Eskandari, S., Turk, E., Zampighi, G. A., and Wright, E. M. (1997) *J. Biol. Chem.* **272**, 20324–20327
415. Prasad, P. D., Wang, H., Kekuda, R., Fujita, T., Fei, Y.-J., Devoe, L. D., Leibach, F. H., and Ganapathy, V. (1998) *J. Biol. Chem.* **273**, 7501–7506
416. Malandro, M. S., and Kilberg, M. S. (1996) *Ann. Rev. Biochem.* **65**, 305–336
417. Utsunomiya-Tate, N., Endou, H., and Kanai, Y. (1996) *J. Biol. Chem.* **271**, 14883–14890
418. Kong, C.-T., Yet, S.-F., and Lever, J. E. (1993) *J. Biol. Chem.* **268**, 1509–1512
419. Su, T.-Z., Wang, M., Syu, L.-J., Saltiel, A. R., and Oxender, D. L. (1998) *J. Biol. Chem.* **273**, 3173–3179
420. Yao, D., Mackenzie, B., Ming, H., Varoqui, H., Zhu, H., Hediger, M. A., and Erickson, J. D. (2000) *J. Biol. Chem.* **275**, 22790–22797
- 420a. Reimer, R. J., Chaudhry, F. A., Gray, A. T., and Edwards, R. H. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 7715–7720
421. Moskovitz, R., Udenfriend, S., Felix, A., Heimer, E., and Tate, S. S. (1994) *FASEB J.* **8**, 1069–1074
422. Chillarón, J., Estévez, R., Mora, C., Wagner, C. A., Suessbrich, H., Lang, F., Gelpi, J. L., Testar, X., Busch, A. E., Zorzano, A., and Palacín, M. (1996) *J. Biol. Chem.* **271**, 17761–17770
423. Trotti, D., Rossi, D., Gjesdal, O., Levy, L. M., Racagni, G., Danbolt, N. C., and Volterra, A. (1996) *J. Biol. Chem.* **271**, 5976–5979
424. Peghini, P., Janzen, J., and Stoffel, W. (1997) *EMBO J.* **16**, 3822–3832
425. Lebrun, B., Sakaitani, M., Shimamoto, K., Yasuda-Kamatani, Y., and Nakajima, T. (1997) *J. Biol. Chem.* **272**, 20336–20339
426. Ponce, J., Biton, B., Benavides, J., Avenet, P., and Aragón, C. (2000) *J. Biol. Chem.* **275**, 13856–13862
427. Pantanowitz, S., Bendahan, A., and Kanner, B. I. (1993) *J. Biol. Chem.* **268**, 3222–3225
428. Bennett, E. R., and Kanner, B. I. (1997) *J. Biol. Chem.* **272**, 1203–1210
429. Velaz-Faircloth, M., Guadano-Ferraz, A., Henzi, V. A., and Fremeau, R. T., Jr. (1995) *J. Biol. Chem.* **270**, 15755–15761
430. Chen, J.-G., Liu-Chen, S., and Rudnick, G. (1998) *J. Biol. Chem.* **273**, 12675–12681
431. Nikaido, H., and Saier, M. H., Jr. (1992) *Science* **258**, 936–942
432. Will, A., Grassl, R., Erdmenger, J., Caspari, T., and Tanner, W. (1998) *J. Biol. Chem.* **273**, 11456–11462
433. Kaback, H. R., Bibi, E., and Roepe, P. D. (1990) *Trends Biochem. Sci.* **15**, 309–314
434. Frillingos, S., Wu, J., Venkatesan, P., and Kaback, H. R. (1997) *Biochemistry* **36**, 6408–6414
435. Green, A. L., Anderson, E. J., and Brooker, R. J. (2000) *J. Biol. Chem.* **275**, 23240–23246
436. Zhao, M., Zen, K.-C., Hubbell, W. L., and Kaback, H. R. (1999) *Biochemistry* **38**, 7407–7412
437. Zani, M. L., Pourcher, T., and Leblanc, G. (1994) *J. Biol. Chem.* **269**, 24883–24889
438. Pourcher, T., Bibi, E., Kaback, H. R., and Leblanc, G. (1996) *Biochemistry* **35**, 4161–4168
439. McDonald, T. P., Walmsley, A. R., and Henderson, P. J. F. (1997) *J. Biol. Chem.* **272**, 15189–15199
440. Fei, Y.-J., Liu, W., Prasad, P. D., Kekuda, R., Oblak, T. G., Ganapathy, V., and Leibach, F. H. (1997) *Biochemistry* **36**, 452–460
441. Brandsch, M., Thuncke, F., Küllertz, G., Schutkowski, M., Fischer, G., and Neubert, K. (1998) *J. Biol. Chem.* **273**, 3861–3864
442. Boorer, K. J., Frommer, W. B., Bush, D. R., Kremann, M., Loo, D. D. F., and Wright, E. M. (1996) *J. Biol. Chem.* **271**, 2213–2220
443. Boorer, K. J., and Fischer, W.-N. (1997) *J. Biol. Chem.* **272**, 13040–13046
444. Someya, Y., Niwa, A., Sawai, T., and Yamaguchi, A. (1995) *Biochemistry* **34**, 7–12
445. Nikaido, K., Liu, P.-Q., and Ames, G. F.-L. (1997) *J. Biol. Chem.* **272**, 27745–27752
446. Létoffé, S., Deleplaire, P., and Wandersman, C. (1996) *EMBO J.* **15**, 5804–5811
447. Blattner, F. R., Plunkett, G., III, Bloch, C. A., Perna, N. T., Burland, V., Riley, M., Collado-Vides, J., Glasner, J. D., Rode, C. K., Mayhew, G. F., Gregor, J., Davis, N. W., Kirkpatrick, H. A., Goeden, M. A., Rose, D. J., Mau, B., and Shao, Y. (1997) *Science* **277**, 1453–1462
448. Decottignies, A., Lambert, L., Catty, P., Degand, H., Epping, E. A., Moye-Rowley, W. S., Balzi, E., and Goffeau, A. (1995) *J. Biol. Chem.* **270**, 18150–18157
- 448a. Driessen, A. J. M., Rosen, B. P., and Konings, W. N. (2000) *Trends Biochem. Sci.* **25**, 397–401
449. Sharom, F. J., DiDiodato, G., Yu, X., and Ashbourne, K. J. D. (1995) *J. Biol. Chem.* **270**, 10334–10341
450. Ko, Y. H., and Pedersen, P. L. (1995) *J. Biol. Chem.* **270**, 22093–22096
451. Nossal, N. G., and Heppel, L. A. (1966) *J. Biol. Chem.* **241**, 3055–3062
452. Ames, G. F.-L. (1986) *Ann. Rev. Biochem.* **55**, 397–425
453. Ames, G. F.-L., and Higgins, C. F. (1983) *Trends Biochem. Sci.* **8**, 97–100
454. Yao, N., Ledvina, P. S., Choudhary, A., and Quirocho, F. A. (1996) *Biochemistry* **35**, 2079–2085
455. Quirocho, F. A., and Vyas, N. K. (1984) *Nature (London)* **310**, 381–386
456. Gilliland, G. L., and Quirocho, F. A. (1981) *J. Mol. Biol.* **146**, 341–362
457. Vyas, N. K., Vyas, M. N., and Quirocho, F. A. (1988) *Science* **242**, 1290–1295
458. Vyas, N. K., Vyas, M. N., and Quirocho, F. A. (1991) *J. Biol. Chem.* **266**, 5226–5237
459. Shilton, B. H., Shuman, H. A., and Mowbray, S. L. (1996) *J. Mol. Biol.* **264**, 364–376
460. Oh, B.-H., Kang, C.-H., De Bondt, H., Kim, S.-H., Nikaido, K., Joshi, A. K., and Ames, G. F.-L. (1994) *J. Biol. Chem.* **269**, 4135–4143
461. Oh, B.-H., Ames, G. F.-L., and Kim, S.-H. (1994) *J. Biol. Chem.* **269**, 26323–26330
462. Sugiyama, S., Vassilyev, D. G., Matsushima, M., Kashiwagi, K., Igarashi, K., and Morikawa, K. (1996) *J. Biol. Chem.* **271**, 9519–9525
463. Pflugrath, J. W., and Quirocho, F. A. (1988) *J. Mol. Biol.* **200**, 163–180
464. Manson, M. D., Boos, W., Bassford, P. J., Jr., and Rasmussen, B. A. (1985) *J. Biol. Chem.* **260**, 9727–9733
465. Reizer, J., Michotey, V., Reizer, A., and Saier, M. H., Jr. (1994) *Protein Sci.* **3**, 440–450
466. Garrett, D. S., Seok, Y.-J., Liao, D.-I., Peterkofsky, A., Gronenborn, A. M., and Clore, G. M. (1997) *Biochemistry* **36**, 2517–2530
467. Chauvin, F., Brand, L., and Roseman, S. (1994) *J. Biol. Chem.* **269**, 20263–20269
- 467a. Zhu, P.-P., Szczepanowski, R. H., Nosworthy, N. J., Ginsburg, A., and Peterkofsky, A. (1999) *Biochemistry* **38**, 15470–15479
468. Jones, B. E., Rajagopal, P., and Klevit, R. E. (1997) *Protein Sci.* **6**, 2107–2119
469. El-Kabbani, O. A. L., Waygood, E. B., and Delbaera, L. T. J. (1987) *J. Biol. Chem.* **262**, 12926–12929

References

470. Reizer, J., Hoischen, C., Reizer, A., Pham, T. N., and Saier, M. H., Jr. (1993) *Protein Sci.* **2**, 506–521
471. Eberstadt, M., Grdadolnik, S. G., Gemmecker, G., Kessler, H., Buhr, A., and Erni, B. (1996) *Biochemistry* **35**, 11286–11292
472. Van Dijk, A. A., Scheek, R. M., Dijkstra, K., Wolters, G. K., and Robillard, G. T. (1992) *Biochemistry* **31**, 9063–9072
473. Gemmecker, G., Eberstadt, M., Buhr, A., Lanz, R., Golc Grdadolnik, S., Kessler, H., and Erni, B. (1997) *Biochemistry* **36**, 7408–7417
474. Pelton, J. G., Torchia, D. A., Remington, S. J., Murphy, K. P., Meadow, N. D., and Roseman, S. (1996) *J. Biol. Chem.* **271**, 33446–33456
475. Lanz, R., and Erni, B. (1998) *J. Biol. Chem.* **273**, 12239–12243
476. Nunn, R. S., Markovic-Housley, Z., Génovésio-Taverne, J.-C., Flükiger, K., Rizkallah, P. J., Jansonius, J. N., Schirmer, T., and Erni, B. (1996) *J. Mol. Biol.* **259**, 502–511
477. Roseman, S., and Meadow, N. D. (1990) *J. Biol. Chem.* **265**, 2993–2996
478. Hurley, J. H., Faber, H. R., Worthylake, D., Meadow, N. D., Roseman, S., Pettigrew, D. W., and Remington, S. J. (1993) *Science* **259**, 673–677
479. Cantor, C. R., and Schimmel, P. R. (1980) *Biophysical Chemistry*, Freeman, San Francisco, California
480. Masuda, T., Dobson, G. P., and Veech, R. L. (1990) *J. Biol. Chem.* **265**, 20321–20334
481. Catterall, W. A. (1995) *Ann. Rev. Biochem.* **64**, 493–531
482. Valverde, M. A., Hardy, S. P., and Sepúlveda, F. V. (1995) *FASEB J.* **9**, 509–515
483. Schmidt-Rose, T., and Jentsch, T. J. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 7633–7638
484. Jentsch, T. J., Steinmeyer, K., and Schwarz, G. (1990) *Nature (London)* **348**, 510–514
485. Middleton, R. E., Pheasant, D. J., and Miller, C. (1994) *Biochemistry* **33**, 13189–13198
486. Schwappach, B., Stobrawa, S., Hechenberger, M., Steinmeyer, K., and Jentsch, T. J. (1998) *J. Biol. Chem.* **273**, 15110–15118
487. Gronemeier, M., Condie, A., Prosser, J., Steinmeyer, K., Jentsch, T. J., and Jockusch, H. (1994) *J. Biol. Chem.* **269**, 5963–5967
488. Pucéat, M., Korichneva, I., Cassoly, R., and Vassort, G. (1995) *J. Biol. Chem.* **270**, 1315–1322
489. Jarolim, P., Shayakul, C., Prabakaran, D., Jiang, L., Stuart-Tilley, A., Rubin, H. L., Simova, S., Zavadil, J., Herrin, J. T., Brouillette, J., Somers, M. J. G., Seemanova, E., Brugnara, C., Guay-Woodford, L. M., and Alper, S. L. (1998) *J. Biol. Chem.* **273**, 6380–6388
- 489a. Fujinaga, J., Tang, X.-B., and Casey, J. R. (1999) *J. Biol. Chem.* **274**, 6626–6633
490. Welsh, M. J., and Smith, A. E. (1995) *Sci. Am.* **273**(Dec), 52–59
491. Seibert, F. S., Linsdell, P., Loo, T. W., Hanrahan, J. W., Riordan, J. R., and Clarke, D. M. (1996) *J. Biol. Chem.* **271**, 27493–27499
492. Romero, M. F., Hediger, M. A., Boulpaep, E. L., and Boron, W. F. (1997) *Nature (London)* **387**, 409–413
493. Burnham, C. E., Amlal, H., Wang, Z., Shull, G. E., and Soleimani, M. (1997) *J. Biol. Chem.* **272**, 19111–19114
494. Muchhal, U. S., Pardo, J. M., and Raghothama, K. G. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 10519–10523
495. Smith, F. W., Ealing, P. M., Hawkesford, M. J., and Clarkson, D. T. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 9373–9377
496. Zhou, J.-J., Theodoulou, F. L., Muldin, I., Ingemarsson, B., and Miller, A. J. (1998) *J. Biol. Chem.* **273**, 12017–12023
497. Schroers, A., Burkovski, A., Wohrlab, H., and Krämer, R. (1998) *J. Biol. Chem.* **273**, 14269–14276
498. Phelps, A., Briggs, C., Mincone, L., and Wohrlab, H. (1996) *Biochemistry* **35**, 10757–10762
499. Collins, J. F., and Ghishan, F. K. (1996) *FASEB J.* **10**, 751–759
500. Garcia, C. K., Brown, M. S., Pathak, R. K., and Goldstein, J. L. (1995) *J. Biol. Chem.* **270**, 1843–1849
501. Kuo, S.-M., and Aronson, P. S. (1996) *J. Biol. Chem.* **271**, 15491–15497
502. Yellen, G. (1987) *Ann. Rev. Biophys. Biophys. Chem.* **16**, 227–246
503. Miller, C. (1991) *Science* **252**, 1092–1096
504. Philipson, L. H., and Steiner, D. F. (1995) *Science* **268**, 372–373
505. Inagaki, N., Gono, T., Clement, J. P., IV, Namba, N., Inazawa, J., Gonzalez, G., Aguilar-Bryan, L., Seino, S., and Bryan, J. (1995) *Science* **270**, 1166–1170
506. Bränström, R., Efendic, S., Berggren, P.-O., and Larsson, O. (1998) *J. Biol. Chem.* **273**, 14113–14118
507. Ruknudin, A., Schulze, D. H., Sullivan, S. K., Lederer, W. J., and Welling, P. A. (1998) *J. Biol. Chem.* **273**, 14165–14171
508. Day, M. L., Johnson, M. H., and Cook, D. I. (1998) *EMBO J.* **17**, 1952–1960
509. McCormick, K. A., Isom, L. L., Ragsdale, D., Smith, D., Scheuer, T., and Catterall, W. A. (1998) *J. Biol. Chem.* **273**, 3954–3962
510. Favre, L., Moczydlowski, E., and Schild, L. (1996) *Biophys. J.* **71**, 3110–3125
- 510a. Lipkind, G. M., and Fozzard, H. A. (2000) *Biochemistry* **39**, 8161–8170
511. Pérez-García, M. T., Chiamvimonvat, N., Marban, E., and Tomaselli, G. F. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 300–304
512. Kosari, F., Sheng, S., Li, J., Mak, D.-O. D., Foskett, J. K., and Kleyman, T. R. (1998) *J. Biol. Chem.* **273**, 13469–13474
513. Garty, H. (1994) *FASEB J.* **8**, 522–528
514. Miller, R. J. (1992) *J. Biol. Chem.* **267**, 1403–1406
515. Mitterdorfer, J., Sinnegger, M. J., Grabner, M., Striessnig, J., and Glossmann, H. (1995) *Biochemistry* **34**, 9350–9355
516. McPherson, P. S., and Campbell, K. P. (1993) *J. Biol. Chem.* **268**, 13765–13768
517. Jayakumar, L. H., Copello, J. A., O'Malley, A. M., Wu, G.-M., Grassucci, R., Wagenknecht, T., and Fleischer, S. (1998) *J. Biol. Chem.* **273**, 16011–16020
518. Kraus, R. L., Sinnegger, M. J., Glossmann, H., Hering, S., and Striessnig, J. (1998) *J. Biol. Chem.* **273**, 5586–5590
519. Kim, S., Lakhani, V., Costa, D. J., Sharara, A. I., Fitz, J. G., Huang, L.-W., Peters, K. G., and Kindman, L. A. (1995) *J. Biol. Chem.* **270**, 5266–5269
520. Mezna, M., and Michelangeli, F. (1996) *J. Biol. Chem.* **271**, 31818–31823
521. Nicoll, D. A., Hryshko, L. V., Matsuoka, S., Frank, J. S., and Philipson, K. D. (1996) *J. Biol. Chem.* **271**, 13385–13391
522. Crumpton, M. J., and Dedman, J. R. (1990) *Nature (London)* **345**, 212
523. Swairjo, M. A., and Seaton, B. A. (1994) *Annu. Rev. Biophys. Biomol. Struct.* **23**, 193–213
524. Benz, J., Bergner, A., Hofmann, A., Demange, P., Göttig, P., Liemann, S., Huber, R., and Voges, D. (1996) *J. Mol. Biol.* **260**, 638–643
525. Burger, A., Berendes, R., Liemann, S., Benz, J., Hofmann, A., Göttig, P., Huber, R., Gerke, V., Thiel, C., Römisch, J., and Weber, K. (1996) *J. Mol. Biol.* **257**, 839–847
526. Demange, P., Voges, D., Benz, J., Liemann, S., Göttig, P., Berendes, R., Burger, A., and Huber, R. (1994) *Trends Biochem. Sci.* **19**, 272–276
527. Campos, B., Mo, Y. D., Mealy, T. R., Li, C. W., Swairjo, M. A., Balch, C., Head, J. F., Retzinger, G., Dedman, J. R., and Seaton, B. A. (1998) *Biochemistry* **37**, 8004–8010
528. Ling, G. N. (1984) *In Search of the Physical Basis of Life*, Plenum, New York
529. Baker, P. F., and Willis, J. S. (1970) *Nature (London)* **226**, 521–523
530. Kyte, J. (1972) *J. Biol. Chem.* **247**, 7642–7649
531. Lingrel, J. B., and Kuntzweiler, T. (1994) *J. Biol. Chem.* **269**, 19659–19662
532. Lutsenko, S., Daoud, S., and Kaplan, J. H. (1997) *J. Biol. Chem.* **272**, 5249–5255
533. Scheiner-Bobis, G., and Schreiber, S. (1999) *Biochemistry* **38**, 9198–9208
534. Gatto, C., Wang, A. X., and Kaplan, J. H. (1998) *J. Biol. Chem.* **273**, 10578–10585
535. Rossier, B. C., Geering, K., and Kraehenbuhl, J. P. (1987) *Trends Biochem. Sci.* **12**, 483–487
536. Minor, N. T., Sha, Q., Nichols, C. G., and Mercer, R. W. (1998) *Proc. Natl. Acad. Sci. U.S.A.* **95**, 6521–6525
537. Lutsenko, S., and Kaplan, J. H. (1995) *Biochemistry* **34**, 15607–15613
538. Nelson, N., and Taiz, L. (1989) *Trends Biochem. Sci.* **14**, 113–116
539. Solioz, M., and Vulpe, C. (1996) *Trends Biochem. Sci.* **21**, 237–241
- 539a. Ueno, S., Kaieda, N., Koyama, N. (2000) *J. Biol. Chem.* **275**, 14537–14540
540. Carafoli, E. (1994) *FASEB J.* **8**, 993–1002
541. MacLennan, D. H., Rice, W. J., and Green, N. M. (1997) *J. Biol. Chem.* **272**, 28815–28818
542. Myung, J., and Jencks, W. P. (1995) *Biochemistry* **34**, 3077–3083
543. Dode, L., De Greef, C., Mountian, I., Attard, M., Town, M. M., Casteels, R., and Wuytack, F. (1998) *J. Biol. Chem.* **273**, 13982–13994
- 543a. Menguy, T., Corre, F., Bouneau, L., Deschamps, S., Vuust Moller, J., Champeil, P., le Maire, M., and Falsion, P. (1998) *J. Biol. Chem.* **273**, 20134–20143
544. Melle-Milovanovic, D., Milovanovic, M., Nagpal, S., Sachs, G., and Shin, J. M. (1998) *J. Biol. Chem.* **273**, 11075–11081
545. Rulli, S. J., Horiba, M. N., Skripnikova, E., and Rabon, E. C. (1999) *J. Biol. Chem.* **274**, 15245–15250
546. Asano, S., Inoue, M., and Takeguchi, N. (1987) *J. Biol. Chem.* **262**, 13263–13268
547. Seto-Young, D., Hall, M. J., Na, S., Haber, J. E., and Perlin, D. S. (1996) *J. Biol. Chem.* **271**, 581–587
548. Auer, M., Scarborough, G. A., and Kühlbrandt, W. (1998) *Nature (London)* **392**, 840–843
549. Orlowski, J., and Grinstein, S. (1997) *J. Biol. Chem.* **272**, 22373–22376
549. Orlowski, J., and Grinstein, S. (1997) *J. Biol. Chem.* **272**, 22373–22376
- 549a. Wakabayashi, S., Pang, T., Su, X., and Shigekawa, M. (2000) *J. Biol. Chem.* **275**, 7942–7949
- 549b. Williams, K. A. (2000) *Nature (London)* **403**, 112–115
550. Kasho, V. N., Stengelin, M., Smirnova, I. N., and Faller, L. D. (1997) *Biochemistry* **36**, 8045–8052
551. Campos, M., and Beaugé, L. (1997) *Biochemistry* **36**, 14228–14237
552. Vilsen, B. (1997) *Biochemistry* **36**, 13312–13324
- 552a. Boxenbaum, N., Daly, S. E., Javaid, Z. Z., Lane, L. K., and Blostein, R. (1998) *J. Biol. Chem.* **273**, 23086–23092
553. Zhang, P., Toyoshima, C., Yonekura, K., Green, N. M., and Stokes, D. L. (1998) *Nature (London)* **392**, 835–839
- 553a. Toyoshima, C., Nakasako, M., Nomura, H., and Ogawa, H. (2000) *Nature (London)* **405**, 647–655

References

- 553b. MacLennan, D. H., and Green, N. M. (2000) *Nature (London)* **405**, 633–634
- 553c. Zhang, Z., Lewis, D., Strock, C., and Inesi, G. (2000) *Biochemistry* **39**, 8758–8767
554. Crider, B. P., Andersen, P., White, A. E., Zhou, Z., Li, X., Mattsson, J. P., Lundberg, L., Keeling, D. J., Xie, X.-S., Stone, D. K., and Peng, S.-B. (1997) *J. Biol. Chem.* **272**, 10721–10728
555. Nanda, A., Brumell, J. H., Nordström, T., Kjeldsen, L., Sengelov, H., Borregaard, N., Rotstein, O. D., and Grinstein, S. (1996) *J. Biol. Chem.* **271**, 15963–15970
- 555a. Kelly, R. B. (1999) *Trends Biochem. Sci.* **24**, M29–M33
556. Desjardins, M., Celis, J. E., van Meer, G., Dieplinger, H., Jahraus, A., Griffiths, G., and Huber, L. A. (1994) *J. Biol. Chem.* **269**, 32194–32200
557. Peng, S.-B., Crider, B. P., Tsai, S. J., Xie, X.-S., and Stone, D. K. (1996) *J. Biol. Chem.* **271**, 3324–3327
558. Pearse, B. M. F., and Crowther, R. A. (1987) *Ann. Rev. Biophys. Biophys. Chem.* **16**, 49–68
559. Schmid, S. L., and Damke, H. (1995) *FASEB J.* **9**, 1445–1453
- 559a. Marsh, M., and McMahon, H. T. (1999) *Science* **285**, 215–220
- 559b. Heuser, J. (1981) *Trends Biochem. Sci.* **6**, 64–68
560. Pearse, B. M. F., and Bretscher, M. S. (1981) *Ann. Rev. Biochem.* **50**, 85–101
561. Pishvae, B., Munn, A., and Payne, G. (1997) *EMBO J.* **16**, 2227–2239
562. Rothman, J. E. (1996) *Protein Sci.* **5**, 185–194
563. Brodsky, F. M., Hill, B. L., Acton, S. L., Näthke, I., Wong, D. H., Ponnambalam, S., and Parham, P. (1991) *Trends Biochem. Sci.* **16**, 208–213
564. Lemmon, S. K., and Jones, E. W. (1987) *Science* **238**, 504–509
- 564a. Anderson, R. G. W. (1998) *Ann. Rev. Biochem.* **67**, 199–225
- 564b. Shin, J.-S., Gao, Z., and Abraham, S. N. (2000) *Science* **289**, 785–788
- 564c. Mulvey, M. A., and Hultgren, S. J. (2000) *Science* **289**, 732–733
565. Lowe, M., and Kreis, T. E. (1995) *J. Biol. Chem.* **270**, 31364–31371
566. Schekman, R., and Orci, L. (1996) *Science* **271**, 1526–1533
- 566a. Eugster, A., Frigerio, G., Dale, M., and Duden, R. (2000) *EMBO J.* **19**, 3905–3917
567. Bloomer, A. C., Champness, J. N., Bricogne, G., Staden, R., and Klug, A. (1978) *Nature (London)* **276**, 362–368
- 567a. Rapoport, I., Chen, Y. C., Cupers, P., Shoelson, S. E., and Kirchhausen, T. (1998) *EMBO J.* **17**, 2148–2155
- 567b. Huang, K. M., D'Hondt, K., Riezman, H., and Lemmon, S. K. (1999) *EMBO J.* **18**, 3897–3908
- 567c. Owen, D. J., Vallis, Y., Pearse, B. M. F., McMahon, H. T., and Evans, P. R. (2000) *EMBO J.* **19**, 4216–4227
- 567d. Wendland, B., Steece, K. E., and Emr, S. D. (1999) *EMBO J.* **18**, 4383–4393
- 567e. Scales, S. J., and Scheller, R. H. (1999) *Nature (London)* **401**, 123–124
568. Ungewickell, E., Ungewickell, H., Holstein, S. E., Lindner, R., Prasad, K., Barouch, W., Martin, B., Greene, L. E., and Eisenberg, E. (1995) *Nature (London)* **378**, 632–635
569. Mellman, I., Fuchs, R., and Helenius, A. (1986) *Ann. Rev. Biochem.* **55**, 663–700
570. Brown, M. S., and Goldstein, J. L. (1974) *Sci. Am.* **251**(Nov), 58–66
571. Brown, M. S., and Goldstein, J. L. (1986) *Science* **232**, 34–47
572. Holstein, T. W., Benoit, M., Herder, G., Wanner, G., David, C. N., and Gaub, H. E. (1994) *Science* **265**, 402–404
573. Ferro-Novick, S., and Jahn, R. (1994) *Nature (London)* **370**, 191–193
574. Mayorga, L. S., Berón, W., Sarrouf, M. N., Colombo, M. I., Creutz, C., and Stahl, P. D. (1994) *J. Biol. Chem.* **269**, 30927–30934
575. Burgoyne, R. D. (1988) *Nature (London)* **331**, 20
576. Wilson, D. W., Wilcox, C. A., Flynn, G. C., Chen, E., Kuang, W.-J., Henzel, W. J., Block, M. R., Ullrich, A., and Rothman, J. E. (1989) *Nature (London)* **339**, 355–359
577. Colombo, M. I., Taddese, M., Whiteheart, S. W., and Stahl, P. D. (1996) *J. Biol. Chem.* **271**, 18810–18816
578. Goda, Y. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 769–772
579. Lugtenberg, B. (1981) *Trends Biochem. Sci.* **6**, 262–266
580. Graham, L. L., Beveridge, T. J., and Nanninga, N. (1991) *Trends Biochem. Sci.* **16**, 328–329
581. Oliver, D. B. (1987) in *Escherichia coli and Salmonella typhi-murium* (Neidhardt, F. C., ed), pp. 56–70, Am. Soc. for Micro-biology, Washington, DC
582. Glauner, B., Höltje, J.-V., and Schwarz, U. (1988) *J. Biol. Chem.* **263**, 10088–10095
583. Park, J. T. (1987) in *Escherichia coli and Salmonella typhimurium* (Neidhardt, F. C., ed), pp. 23–32, Am. Soc. for Microbiology, Washington, DC
584. Scherrer, R. (1984) *Trends Biochem. Sci.* **9**, 242–245
585. Nikaido, H., and Vaara, M. (1987) in *Escherichia coli and Salmonella typhimurium* (Neidhardt, F. C., ed), pp. 7–22, Am. Soc. for Microbiology, Washington, DC
586. Hancock, R. E. W., Karunaratne, D. N., and Bernegger-Egli, C. (1994) in *Bacterial Cell Wall (New Comprehensive Biochemistry)*, Vol. 27 (Ghuysen, J.-M., and Hakenbeck, R., eds), pp. 263–279, Elsevier, Amsterdam
587. Raetz, C. R. H., and Dowhan, W. (1990) *J. Biol. Chem.* **265**, 1235–1238
588. Osborn, M. J. (1969) *Ann. Rev. Biochem.* **38**, 501–538
589. Choi, D.-S., Yamada, H., Mizuno, T., and Mizushima, S. (1986) *J. Biol. Chem.* **261**, 8953–8957
590. Diedrich, D. L., and Schnaitman, C. A. (1978) *Proc. Natl. Acad. Sci. U.S.A.* **75**, 3708–3712
591. Raetz, C. R. H., Ulevitch, R. J., Wright, S. D., Sibley, C. H., Ding, A., and Nathan, C. F. (1991) *FASEB J.* **5**, 2652–2660
592. Hollingsworth, R. I., and Carlson, R. W. (1989) *J. Biol. Chem.* **264**, 9300–9303
593. Ogawa, T. (1994) *Eur. J. Biochem.* **219**, 737–742
594. Rietschel, E. T., Kirikae, T., Schade, F. U., Mamat, U., Schmidt, G., Loppnow, H., Ulmer, A. J., Zähringer, U., Seydel, U., Di Padova, F., Schreier, M., and Brade, H. (1994) *FASEB J.* **8**, 217–225
595. Masoud, H., Moxon, E. R., Martin, A., Krajcarski, D., and Richards, J. C. (1997) *Biochemistry* **36**, 2091–2103
596. Aspinall, G. O., and Monteiro, M. A. (1996) *Biochemistry* **35**, 2498–2504
597. Rietschel, E. T., and Brade, H. (1992) *Sci. Am.* **267**(Aug), 54–61
598. Costerton, J. W., Irwin, R. T., and Cheng, K. J. (1981) *Crit. Rev. Microbiol.* **8**, 303–338
599. Fischer, W. (1994) in *Bacterial Cell Wall (New Comprehensive Biochemistry)*, Vol. 27 (Ghuysen, J.-M., and Hakenbeck, R., eds), pp. 199–215, Elsevier, Amsterdam
600. Ganfield, M.-C. W., and Pieringer, R. A. (1980) *J. Biol. Chem.* **255**, 5164–5169
601. Johnson, S. D., Lacher, K. P., and Anderson, J. S. (1981) *Biochemistry* **20**, 4781–4785
602. Ivatt, R. J., and Gilvarg, C. (1979) *J. Biol. Chem.* **254**, 2759–2765
603. Venisse, A., Rivière, M., Vercauteren, J., and Puzo, G. (1995) *J. Biol. Chem.* **270**, 15012–15021
604. Brennan, P. J., and Nikaido, H. (1995) *Ann. Rev. Biochem.* **64**, 29–63
605. Liu, J., Barry, C. E., III, Besra, G. S., and Nikaido, H. (1996) *J. Biol. Chem.* **271**, 29545–29551
606. Mikusová, K., Mikus, M., Besra, G. S., Hancock, I., and Brennan, P. J. (1996) *J. Biol. Chem.* **271**, 7820–7828
607. Lopez-Marin, L. M., Quesada, D., Lakhdar-Ghazal, F., Tocanne, J.-F., and Lanée, G. (1994) *Biochemistry* **33**, 7056–7061
608. Sprott, G. D., Ekiel, I., and Dicaire, C. (1990) *J. Biol. Chem.* **265**, 13735–13740
609. Peters, J., Nitsch, M., Kühlmorgen, B., Golbik, R., Lupas, A., Kellermann, J., Engelhardt, H., Pfander, J.-P., Müller, S., Goldie, K., Engel, A., Stetter, K.-O., and Baumeister, W. (1995) *J. Mol. Biol.* **245**, 385–401
610. Thomas, S. R., and Trust, T. J. (1995) *J. Mol. Biol.* **245**, 568–581
611. Peters, J., Baumeister, W., and Lupas, A. (1996) *J. Mol. Biol.* **257**, 1031–1041
612. Fischetti, V. A. (1991) *Sci. Am.* **264**(Jun), 58–65
613. Goward, C. R., Scawen, M. D., Murphy, J. P., and Atkinson, T. (1993) *Trends Biochem. Sci.* **18**, 136–139
614. Schneewind, O., Fowler, A., and Faull, K. F. (1995) *Science* **268**, 103–106
615. Knörle, R., and Hübner, W. (1995) *Biochemistry* **34**, 10970–10975
616. Shaper, N. L., Hollis, G. F., Douglas, J. G., Kirsch, I. R., and Shaper, J. H. (1988) *J. Biol. Chem.* **263**, 10420–10428
617. Lin, C. Q., and Bissell, M. J. (1993) *FASEB J.* **7**, 737–743
618. Har-El, R., and Tanzer, M. L. (1993) *FASEB J.* **7**, 1115–1123
619. Mayne, R., and Burgeson, R. E., eds. (1987) *Structure and Function of Collagen Types*, Academic Press, New York, Orlando, San Diego
620. Nimni, M. E., ed. (1988) *Collagen: Biochemistry, Biomechanics, Biotechnology*, CRC Press, Boca Raton, Florida
621. Ramachandran, G. N. (1988) *Intl. J. Peptide Prot. Res.* **31**, 1–16
622. Martin, G. R., Timpl, R., Müller, P. K., and Kühn, K. (1985) *Trends Biochem. Sci.* **10**, 285–287
623. Prockop, D. J., and Kivirikko, K. I. (1995) *Ann. Rev. Biochem.* **64**, 403–434
624. van der Rest, M., and Garrone, R. (1991) *FASEB J.* **5**, 2814–2823
625. Byers, P. H. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 3 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 4029–4077, McGraw-Hill, New York
626. Brodsky, B., and Shah, N. K. (1995) *FASEB J.* **9**, 1537–1546
627. Koivu, J., and Myllylä, R. (1987) *J. Biol. Chem.* **262**, 6159–6164
628. Prockop, D. J. (1990) *J. Biol. Chem.* **265**, 15349–15352
629. Annunen, P., Autio-Harmainen, H., and Kivirikko, K. I. (1998) *J. Biol. Chem.* **273**, 5989–5992
630. Reiser, K., McCormick, R. J., and Rucker, R. B. (1992) *FASEB J.* **6**, 2439–2449
631. Li, S.-W., Sieron, A. L., Fertala, A., Hojima, Y., Arnold, W. V., and Prockop, D. J. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 5127–5130
632. Meek, K. M., Chapman, J. A., and Hardcastle, R. A. (1979) *J. Biol. Chem.* **254**, 10710–10714
633. Chan, V. C., Ramshaw, J. A. M., Kirkpatrick, A., Beck, K., and Brodsky, B. (1997) *J. Biol. Chem.* **272**, 31441–31446

References

634. Wess, T. J., Hammersley, A. P., Wess, L., and Miller, A. (1998) *J. Mol. Biol.* **275**, 255–267
- 634a. Kramer, R. Z., Venugopal, M. G., Bella, J., Mayville, P., Brodsky, B., and Berman, H. M. (2000) *J. Mol. Biol.* **301**, 1191–1205
635. Eyre, D. R., Paz, M. A., and Gallop, P. M. (1984) *Ann. Rev. Biochem.* **53**, 717–748
636. Miller, A. (1982) *Trends Biochem. Sci.* **7**, 13–18
637. Eyre, D. R. (1980) *Science* **207**, 1315–1322
638. Wu, J.-J., and Eyre, D. R. (1984) *Biochemistry* **23**, 1850–1857
639. Hanson, D. A., and Eyre, D. R. (1996) *J. Biol. Chem.* **271**, 26508–26516
640. Kiss, I., Deák, F., Holloway, R. G., Jr., Delius, H., Mebust, K. A., Frinberger, E., Argraves, W. W., Tsonis, P. A., Winterbottom, N., and Goetink, P. F. (1989) *J. Biol. Chem.* **264**, 8126–8134
641. Tanaka, S., Avigad, G., Eikenberry, E. F., and Brodsky, B. (1988) *J. Biol. Chem.* **263**, 17650–17657
642. Kleman, J.-P., Aeschlimann, D., Paulsson, M., and van der Rest, M. (1995) *Biochemistry* **34**, 13768–13775
643. Mazzorana, M., Gruffat, H., Sergeant, A., and van der Rest, M. (1993) *J. Biol. Chem.* **268**, 3029–3032
644. Martin, G. R., Timpl, R., Miller, P. K., and Kuhn, K. (1985) *Trends Biochem. Sci.* **10**, 285–287
645. Gunwar, S., Ballester, F., Noelken, M. E., Sado, Y., Ninomiya, Y., and Hudson, B. G. (1998) *J. Biol. Chem.* **273**, 8767–8775
646. van der Rest, M., and Mayne, R. (1988) *J. Biol. Chem.* **263**, 1615–1618
647. de Crombrughe, B., and Pastan, I. (1982) *Trends Biochem. Sci.* **7**, 11–13
648. Myers, J. C., Yang, H., D'Ippolito, J. A., Presente, A., Miller, M. K., and Dion, A. S. (1994) *J. Biol. Chem.* **269**, 18549–18557
649. Lui, V. C. H., Ng, L. J., Sat, E. W. Y., Nicholls, J., and Cheah, K. S. E. (1996) *J. Biol. Chem.* **271**, 16945–16951
650. de Wet, W., Bernard, M., Benson-Chanda, V., Chu, M.-L., Dickson, L., Weil, D., and Ramirez, F. (1987) *J. Biol. Chem.* **262**, 16032–16036
651. Chu, M.-L., de Wet, W., Bernhard, M., and Ramirez, F. (1985) *J. Biol. Chem.* **260**, 2315–2370
652. Kramer, J. M. (1994) *FASEB J.* **8**, 329–336
653. Engel, J. (1997) *Science* **277**, 1785–1786
654. Rosenbloom, J., Abrams, W. R., and Mecham, R. (1993) *FASEB J.* **7**, 1208–1218
655. Hukins, D. W. L. (1984) *Connective Tissue Matrix*, MacMillan, New York
656. Fransson, L.-Å. (1987) *Trends Biochem. Sci.* **12**, 406–411
657. Burg, M. A., Tillet, E., Timpl, R., and Stallcup, W. B. (1996) *J. Biol. Chem.* **271**, 26110–26116
658. Caplan, A. I. (1984) *Sci. Am.* **251**(Oct), 84–94
659. Heinegård, D., and Oldberg, Å. (1989) *FASEB J.* **3**, 2042–2051
660. Cheng, F., Heinegård, D., Fransson, L.-Å., Bayliss, M., Bielicki, J., Hopwood, J., and Yoshida, K. (1996) *J. Biol. Chem.* **271**, 28572–28580
661. Hauser, N., and Paulsson, M. (1994) *J. Biol. Chem.* **269**, 25747–25753
662. Beck, K., Gambee, J. E., Bohan, C. A., and Bächinger, H. P. (1996) *J. Mol. Biol.* **256**, 909–923
663. Yurchenco, P. D., and Schittny, J. C. (1990) *FASEB J.* **4**, 1577–1590
- 663a. Yurchenco, P. D., Birk, D. E., and Mecham, R. P., eds. (1994) *Extracellular Matrix Assembly and Structure*, Academic Press, San Diego, California
664. Utani, A., Nomizu, M., Timpl, R., Roller, P. P., and Yamada, Y. (1994) *J. Biol. Chem.* **269**, 19167–19175
665. Pöschl, E., Mayer, U., Stetefeld, J., Baumgartner, R., Holak, T. A., Huber, R., and Timpl, R. (1996) *EMBO J.* **15**, 5154–5159
666. Durkin, M. E., Gautam, M., Loebel, F., Sanes, J. R., Merlie, J. P., Albrechtsen, R., and Wewer, U. M. (1996) *J. Biol. Chem.* **271**, 13407–13416
667. Beckmann, G., Hanke, J., Bork, P., and Reich, J. G. (1998) *J. Mol. Biol.* **275**, 725–730
668. Colognato-Pyke, H., O'Rear, J. J., Yamada, Y., Carbonetto, S., Cheng, Y.-S., and Yurchenco, P. D. (1995) *J. Biol. Chem.* **270**, 9398–9406
669. Kramer, J. M., Cox, G. N., and Hirsch, D. (1985) *J. Biol. Chem.* **260**, 1945–1951
670. Beck, K., Hunter, I., and Engel, J. (1990) *FASEB J.* **4**, 148–160
671. Sasaki, T., Kostka, G., Göhring, W., Wiedemann, H., Mann, K., Chu, M.-L., and Timpl, R. (1995) *J. Mol. Biol.* **245**, 241–250
672. Aeschlimann, D., and Paulsson, M. (1991) *J. Biol. Chem.* **266**, 15308–15317
673. Reinhardt, D. P., Ono, R. N., and Sakai, L. Y. (1997) *J. Biol. Chem.* **272**, 1231–1236
- 673a. Kettle, S., Yuan, X., Grundy, G., Knott, V., Downing, A. K., and Handford, P. A. (1999) *J. Mol. Biol.* **285**, 1277–1287
- 673b. Ritty, T. M., Broekelmann, T., Tisdale, C., Milewicz, D. M., and Mecham, R. P. (1999) *J. Biol. Chem.* **274**, 8933–8940
674. Reinhardt, D. P., Keene, D. R., Corson, G. M., Pöschl, E., Bächinger, H. P., Gambee, J. E., and Sakai, L. Y. (1996) *J. Mol. Biol.* **258**, 104–116
675. McKusick, V. A. (1991) *Nature (London)* **352**, 279–281
676. Francke, U., and Furthmayr, H. (1994) *N. Engl. J. Med.* **330**, 1384–1385
677. Mann, K., Mechling, D. E., Bächinger, H. P., Eckerskorn, C., Gail, F., and Timpl, R. (1996) *J. Mol. Biol.* **261**, 255–266
678. Durkin, M. E., Carlin, B. E., Vergnes, J., Bartos, B., and Merlie, J. (1987) *Proc. Natl. Acad. Sci. U.S.A.* **84**, 1570–1574
679. Richards, A. G. (1978) in *Biochemistry of Insects* (Rockstein, M., ed), Academic Press, New York (p. 205)
680. Schaefer, J., Kramer, K. J., Garbow, J. R., Jacob, G. S., Stejskal, E. O., Hopkins, T. L., and Speirs, R. D. (1987) *Science* **235**, 1200–1204
681. Simkiss, K., and Wilbur, K. M. (1989) *Biomineralization*, Academic Press, San Diego, California
682. DeOliveira, D. B., and Laursen, R. A. (1997) *J. Am. Chem. Soc.* **119**, 10627–10631
683. Weiner, S., and Addadi, L. (1991) *Trends Biochem. Sci.* **16**, 252–256
684. Albeck, S., Aizenberg, J., Addadi, L., and Weiner, S. (1993) *J. Am. Chem. Soc.* **115**, 11691–11697
685. Bonucci, E., ed. (1992) *Calcification in Biological Systems*, CRC Press, Boca Raton, Florida
686. Mann, S., Archibald, D. D., Didymus, J. M., Douglas, T., Heywood, B. R., Meldrum, F. C., and Reeves, N. J. (1993) *Science* **261**, 1286–1292
687. Aizenberg, J., Hanson, J., Ilan, M., Leiserowitz, L., Koetzie, T. F., Addadi, L., and Weiner, S. (1995) *FASEB J.* **9**, 262–268
688. Otsubo, K., Katz, E. P., Mechanic, G. L., and Yamauchi, M. (1992) *Biochemistry* **31**, 396–402
689. Fledelius, C., Johnsen, A. H., Cloos, P. A. C., Bonde, M., and Qvist, P. (1997) *J. Biol. Chem.* **272**, 9755–9763
- 689a. Ducey, P., Schinke, T., and Karsenty, G. (2000) *Science* **289**, 1501–1504
- 689b. Teitelbaum, S. L. (2000) *Science* **289**, 1504–1508
690. Weiner, S., and Traub, W. (1992) *FASEB J.* **6**, 879–885
691. McLean, F. M., Keller, P. J., Genge, B. R., Walters, S. A., and Wuthier, R. E. (1987) *J. Biol. Chem.* **262**, 10481–10488
692. Komori, T., Yagi, H., Nomura, S., Yamaguchi, A., Sasaki, K., Deguchi, K., Shimizu, Y., Bronson, R. T., Gao, Y.-H., Inada, M., Sato, M., Okamoto, R., Kitamura, Y., Yoshiki, S., and Kishimoto, T. (1997) *Cell* **89**, 755–764
693. Ducey, P., Zhang, R., Geoffroy, V., Ridall, A. L., and Karsenty, G. (1997) *Cell* **89**, 747–754
694. Dickman, S. (1997) *Science* **276**, 1502
695. Rodan, G. A., and Harada, S.-i. (1997) *Cell* **89**, 677–680
696. Mundlos, S., Otto, F., Mundlos, C., Mulliken, J. B., Aylsworth, A. S., Albright, S., Lindhout, D., Cole, W. G., Henn, W., Knoll, J. H. M., Owen, M. J., Mertelsmann, R., Zabel, B. U., and Olsen, B. R. (1997) *Cell* **89**, 773–779
697. Price, P. A., and Williamson, M. K. (1985) *J. Biol. Chem.* **260**, 14971–14975
698. Nishimoto, S. K., Araki, N., Robinson, F. D., and Waite, J. H. (1992) *J. Biol. Chem.* **267**, 11600–11605
699. Price, P. A., Rice, J. S., and Williamson, M. K. (1994) *Protein Sci.* **3**, 822–830
700. Hohenester, E., Maurer, P., and Timpl, R. (1997) *EMBO J.* **16**, 3778–3786
701. Kelm, R. J., Jr., Swords, N. A., Orfeo, T., and Mann, K. G. (1994) *J. Biol. Chem.* **269**, 30147–30153
702. Fisher, L. W., Robey, P. G., Turoso, N., Otsuka, A. S., Tepen, D. A., Esch, F. S., Shimasaki, S., and Termine, J. D. (1987) *J. Biol. Chem.* **262**, 13457–13463
703. Denhardt, D. T., and Guo, X. (1993) *FASEB J.* **7**, 1475–1482
704. Shanmugam, V., Chackalaparampil, I., Kundu, G. C., Mukherjee, A. B., and Mukherjee, B. B. (1997) *Biochemistry* **36**, 5729–5738
- 704a. Rodan, G. A., and Martin, T. J. (2000) *Science* **289**, 1508–1514
- 704b. Service, R. F. (2000) *Science* **289**, 1498–1500
705. Zipkin, I., ed. (1973) *Biological Mineralization*, Wiley, New York
706. Schlesinger, P. H., Blair, H. C., Teitelbaum, S. L., and Edwards, J. C. (1997) *J. Biol. Chem.* **272**, 18636–18643
707. Alberts, B., Bray, D., Lewis, J., Raff, M., Roberts, K., and Watson, J. D. (1994) *Molecular Biology of the Cell*, 3rd ed., Garland, New York (p. 1182–1186)
708. Riggs, B. L., and Melton, L. J. (1992) *N. Engl. J. Med.* **327**, 620–627
709. Sly, W. S., and Hu, P. Y. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 3 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 4113–4124, McGraw-Hill, New York
710. Roush, W. (1996) *Science* **273**, 1170
711. Delmas, P. D., and Meunier, P. J. (1997) *N. Engl. J. Med.* **336**, 558–566
- 711a. Ho, A. M., Johnson, M. D., and Kingsley, D. M. (2000) *Science* **289**, 265–270
- 711b. Hagmann, M. (2000) *Science* **289**, 225–226
712. Hohling, H. J., Barckhaus, R. H., and Kreftin, E. R. (1980) *Trends Biochem. Sci.* **5**, 8–11
713. Outka, D. E., and Williams, D. C. (1971) *Journal of Protozoology* **18**, 285–297
714. Inoue, S., and Okazaki, K. (1977) *Sci. Am.* **236**(Apr), 83–92
715. Volcani, B. E. (1981) in *Silicon and Silicious Structures in Biological Systems* (Simpson, T. L., and Volcani, B. E., eds), Springer-Verlag, New York (p. 157)
716. Mann, S. (1988) *Nature (London)* **332**, 119–124
717. Calvert, P. (1988) *Nature (London)* **334**, 651–652
718. Watanabe, N., and Wilbur, K. M., eds. (1976) *The Mechanisms of Mineralization in Invertebrates and Plants*, Univ. N. Carolina Press, Columbia, South Carolina

References

719. Lemann, J., Jr. (1993) *N. Engl. J. Med.* **328**, 880–881
720. Nakagawa, Y., Abram, V., Kézdy, F. J., Kaiser, E. T., and Coe, F. L. (1983) *J. Biol. Chem.* **258**, 12594–12600
721. Bertrand, J. A., Pignol, D., Bernard, J.-P., Verdier, J.-M., Dagorn, J.-C., and Fontecilla-Camps, J. C. (1996) *EMBO J.* **15**, 2678–2684
722. Bimmler, D., Graf, R., Scheele, G. A., and Frick, T. W. (1997) *J. Biol. Chem.* **272**, 3073–3082
723. De Reggi, M., Gharib, B., Patard, L., and Stoven, V. (1998) *J. Biol. Chem.* **273**, 4967–4971
724. Cheung, H. S., Kurup, I. V., Sallis, J. D., and Ryan, L. M. (1996) *J. Biol. Chem.* **271**, 28082–28085
725. Rothschild, B. M., and Thillaud, P. L. (1991) *Nature (London)* **349**, 288
726. Cabib, E., Roberts, R., and Bowers, B. (1982) *Ann. Rev. Biochem.* **51**, 763–793
727. Ballou, C. E., and Raschke, W. C. (1974) *Science* **184**, 127–134
728. Frevert, J., and Ballou, C. E. (1985) *Biochemistry* **24**, 753–759
729. Celerin, M., Ray, J. M., Schisler, N. J., Day, A. W., Stetler-Stevenson, W. G., and Laudenbach, D. E. (1996) *EMBO J.* **15**, 4445–4453

Study Questions

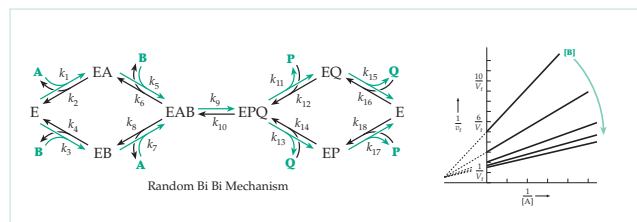
- Compare the chemical makeup of the extracellular “coat” or “matrix” materials secreted by the following cells: bacteria, fibroblasts, osteoblasts, plant cells, fungi.
- The iodine number of a compound is defined as the number of grams of I_2 absorbed (through addition to $C=C$ bonds to give a diiodo derivative) per 100 g of fat. NOTE: Iodine monochloride or iodine monobromide are the usual halogenating reagents but the iodine number is expressed in terms of grams of I_2 . The saponification number is the number of milligrams of KOH needed to completely saponify (hydrolyze and neutralize the resulting fatty acids) 1 g of fat. A pure triglyceride has a saponification number of 198 and an iodine number of 59.7.
 - What is the relative molecular mass?
 - What is the average chain length of the fatty acids?
 - What is the number of double bonds in the molecule?
- Spermaceti (a wax from the head of the sperm whale) resembles high molecular mass hydrocarbons in physical properties and inertness toward $Br_2/CHCl_3$ and $KMnO_4$; on qualitative analysis, it gives positive tests only for carbon and hydrogen. However, its IR spectrum shows the presence of an ester linkage, and quantitative analysis gives the empirical formula $C_{16}H_{32}O$. A solution of the wax in alcoholic KOH is refluxed for a long time. Titration of an aliquot shows that one equivalent of base is consumed for every 475 g of wax. Water and ether are added to the cooled reflux mixture, and the aqueous and ethereal layers are separated. Acidification of the aqueous layer yields a solid A with a neutralization equivalent of 260 ± 5 . Evaporation of the ether layer gives solid B, which could not be titrated. Reduction of either spermaceti or A by lithium aluminum hydride gave B as the only product.
- Stearic acid (1.16 g) was dissolved in 100 ml of ethanol. A 10 μ l portion of the resulting solution was pipetted onto a clean surface of a dilute HCl solution (in a shallow tray) where it spread to form a monolayer of stearic acid. The layer was compressed (by moving a Teflon barrier across the tray) until the surface pressure π started to rise sharply and reached ~ 20 dyn/cm. Note that $\pi = \gamma_0 - \gamma$ where γ is the measured surface tension with the film present and γ_0 is the higher surface tension of water alone. The compressed film occupied a 20 x 24 cm area. Calculate the cross-sectional area of an alkyl chain in stearic acid. [See J. B. Davenport, in *Biochemistry and Methodology of Lipids* (A. R. Johnson and J. B. Davenport, eds.), pp. 47–83. Wiley-Interscience, New York, 1971; and M. C. Phillips, in *Progress in Surface and Membrane Science* (J. F. Danielli, D. M. Rosenberg, and D. A. Cadenhead, eds.) Vol. 5, pp. 139–221. Academic Press, New York, 1972.]
- In 1925, E. Gorter and F. Grendel (*J. Exp. Med.* **41**, 439) reported measurements in which they extracted lipid from red blood cell membranes with acetone, spread the lipids as a monolayer, and measured the area of the compressed monolayer. They then estimated the surface area of an erythrocyte and calculated that the ratio of the lipids (as a monolayer) to the surface area of the red blood cell was 1.9–2.0. More modern experiments gave the following: each erythrocyte membrane contains 4.5×10^{-16} mol of phospholipid and 3.1×10^{-16} mol of cholesterol.
 - If the cross-sectional areas of phospholipid and cholesterol molecules in a membrane are taken as 0.70 and 0.38 nm², respectively, what surface area would be occupied in a monolayer?
 - If the measured surface area of an erythrocyte is 167 μ m², what is the ratio of the area calculated in (a) to the area of the cell surface?
 - How might you explain the difference between this answer and that of Gorter and Grendel? See E. D. Korn (1966) *Science* **153**, 1491–1498.

What is the likely structure of spermaceti?

Study Questions

6. The following experimental observations are related to biological membrane structure and function. Discuss the implications of each observation with respect to membrane structure.
 - a) Many macrocyclic antibiotics (nonactin, valinomycin, and others) form 1:1 complexes with alkali metal cations in a highly selective manner. The complexes are readily soluble in nonpolar organic solvents. These antibiotics increase the electrical conductance and permeability to alkali metal cations of synthetic phospholipid membranes. Valinomycin increases the electrical conductance of thylakoid membranes of chloroplasts in the presence of K^+ but not in the presence of Na^+ ; it also uncouples oxidative phosphorylation in mitochondria. (See Chapter 18.)
 - b) Treatment of intact chloroplasts with a galactolipase releases galactose from galactosyl diglycerides. Treatment of red blood cell ghosts with phospholipase C releases about 75% of the lipid P in water-soluble form. In neither case is the structural integrity of the membrane destroyed.
 - c) Using sphingomyelin as a hapten, antibodies specific for this lipid can be produced. When red blood cell ghosts are exposed to these antibodies, it can be shown that the antibodies react, but only on one side of the membrane.
7. Describe the structure of biological membranes and the characteristic functions of lipid-, protein-, and carbohydrate-containing components. Describe the differences between inner and outer membrane surfaces.
8. Compare the distribution of triglycerides, phosphatidylcholine, phosphatidylethanolamine, sphingomyelins, glycolipids, and cholesterol within cells. Consider differences between the two sides of membranes.
9. Consider the chemistry underlying the labeling of cell surfaces with each of the following:
 - a) Lactoperoxidase,
 - b) galactose oxidase,
 - c) formylmethionylsulfone methyl phosphate,
 - d) the diazonium salt of diiodosulfanilic acid,
 - e) fluorescent antibodies,
 - f) antibodies conjugated with ferritin.
10. Which would be the more effective detergent in the pH range 2 to 3, sodium lauryl sulfonate or sodium laurate? Why?
11. Suppose that a cell contains 10 mM Na^+ and 100 mM K^+ and that it is bathed in extracellular fluid containing 100 mM Na^+ and 5 mM K^+ . How much energy will be required to transport three equivalents of Na^+ out and two equivalents of K^+ in? Compare this with $\Delta G'$ for hydrolysis of ATP at pH 7. Assume that the membrane is permeable to Cl^- .
12. An *E. coli* cell is said to contain about 10^5 molecules of an envelope protein of MW = 36,500. If the latter is spherical and the spheres are closely packed in a hexagonal lattice, how much of the surface area of the bacterium would be covered? What would the diameter of the protein be? What spacing would be required if 10^5 molecules covered the surface completely? Suggest a shape for the protein molecule that is consistent with the requirement.

Write the equations for the chemical reactions involved. State what surface groups will be labeled by each reagent. List special advantages of each of these reagents.



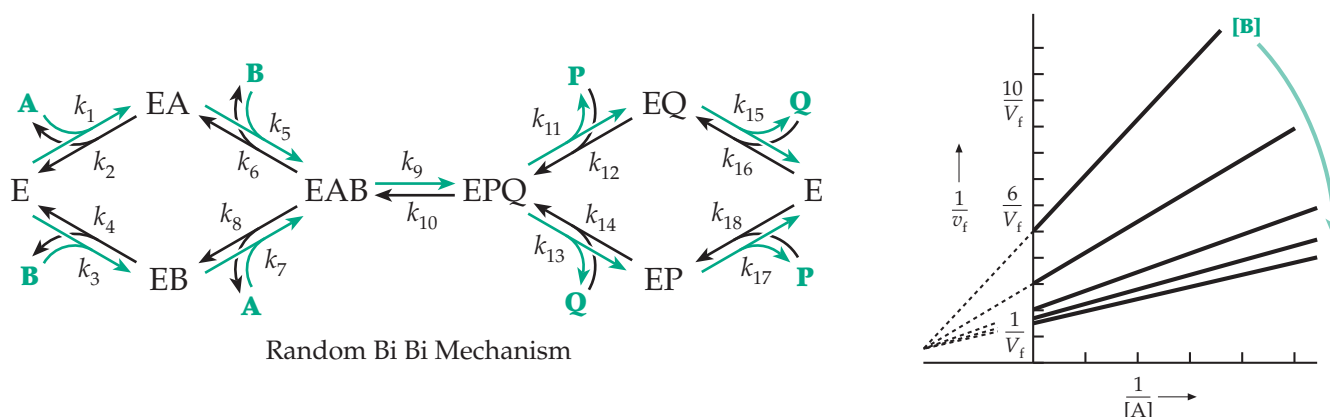
Bimolecular reactions of two molecules, A and B, to give two products, P and Q, are catalyzed by many enzymes. For some enzymes the substrates A and B bind into the active site in an ordered sequence while for others, binding may be in a random order. The scheme shown here is described as random Bi Bi in a classification introduced by Cleland. Eighteen rate constants, some second order and some first order, describe the reversible system. Determination of these kinetic parameters is often accomplished using a series of double reciprocal plots (Lineweaver-Burk plots), such as those at the right.

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Enzymes: The Catalysts of Cells

9



Most of the machinery of living cells is made of enzymes. Thousands of them have been extracted from cells and have been purified and crystallized. Many others are recognized only by their catalytic action and have not yet been isolated in pure form. Most enzymes are soluble globular proteins but an increasing number of RNA molecules are also being recognized as enzymes. Many structural proteins of the cell also act as catalysts. For example, the muscle proteins actin and myosin together catalyze the hydrolysis of ATP and link the hydrolysis to movement (Chapter 19). Catalysis is one of the most fundamental characteristics of life.

How do we recognize that a protein or RNA molecule is an enzyme? The answer is that *enzymes are recognized primarily by their ability to catalyze a chemical reaction*. For this reason, an everyday operation for many biochemists is the measurement of the catalytic activity of enzymes. Only by measuring the rates of the catalyzed reactions carefully and quantitatively has it been possible to isolate and purify these remarkable molecules.

Since the beginning of biochemical investigation enzymes have held a special fascination for chemists and biologists. How can these easily destroyed substances catalyze reactions with such speed and without formation of significant quantities of side products? Some enzymes increase the velocity of a single chemical reaction of a specific compound by a factor of as much as 10^{10} . How can a protein do this? In this chapter we'll consider both ways of measuring enzymatic activity and basic mechanisms of catalysis.

A. Information from Kinetics

The quantitative study of catalysis by enzymes, i.e., the study of enzyme kinetics, is a highly developed branch of biochemistry. It is one of our most important means of learning about the mechanisms of catalysis at the active sites of enzymes.^{1-13a} By determining **rate constants k** under a variety of conditions we can learn just how fast an enzyme can act, how tightly it binds its substrates to form the enzyme-substrate (ES) complexes essential to catalysis, how specific it is with respect to substrate structure, and how it is affected by compounds that inhibit or activate the catalysis.

1. Measuring the Speed of an Enzymatic Reaction

A major goal in kinetic studies is to establish a **rate equation** which describes the velocity of a reaction in terms of **kinetic constants** and other experimentally measurable parameters. To measure the velocity of any chemical reaction precisely we must start the reaction at a definite time by rapidly mixing together the two or more reactants. Then, while keeping the mixture at an accurately constant temperature and pH, we must measure the concentration of a reactant or product after a fixed time interval, or at various times. No end of ingenuity has gone into devising ways of doing this for particular enzymes. Whatever the procedure, the information we must obtain is the **rate** at which some concentration changes with time. We can then construct a **progress curve** showing the decrease in concentration [S] of the reactant (**substrate**) or the increase in the concentration of **product** [P] with time (Fig. 9-1).

BOX 9-A A HISTORICAL NOTE ON ENZYMES

While the earliest physiologists postulated a “vital force” to explain the chemical reactions of cells, the existence of biochemical substances promoting reactions outside of the body was recognized at least by the early 1600s. However, the role of yeast in fermentation was still unknown and it was thought that both alcoholic fermentation and animal digestion were caused by unknown substances called **ferments**. In 1752, Reamur demonstrated the solvent power of the gastric juice of birds and by 1783 Spallanzani had extended the studies to humans and other species.^a In 1836 Schwann isolated the enzyme **pepsin**^b from gastric juice.

In the same year Jacob Berzelius introduced the concept of catalysis, which he developed as a result of studies of the effects of acids and bases in promoting the hydrolysis of starch and of the effects of metals on the decomposition of hydrogen peroxide. Berzelius proposed the term **catalyst** from the Greek “katalysis,” meaning “dissolution.” Although he had been concerned primarily with inorganic catalysts, Berzelius recognized that a natural catalyst,

an amylase that causes the hydrolysis of starch, had already been isolated from germinating barley in 1833 and that “in living plants and animals thousands of catalytic processes take place.”^c Some chemists and physiologists accepted Berzelius’ concept of biochemical catalysis immediately, but many did not. The matter was complicated by Pasteur’s discovery that yeast cells were the causative agent of alcoholic fermentation. Only after Edward Buchner’s reports in 1897 that a juice formed by grinding yeast with sand and filtering could still ferment sugar (see Chapters 15 and 17) was the reality of enzymes in metabolism^d generally accepted. The word **enzyme** (from the Greek “in yeast”) was introduced earlier by F. W. Kuhne, professor of physiology in Heidelberg and a person who had accepted Berzelius’ concept.^c

^a Richmond, C. (1986) *Trends Biochem. Sci.* **11**, 528–530

^b Schwann, T. (1836) *Arch. Anat. Physiol.*, 90–138

^c Hoffmann-Ostenhof, O. (1978) *Trends Biochem. Sci.* **3**, 186–187

^d Buchner, E. (1897) *Ber. Deut. Chem. Ges.* **30**, 117–124

The velocity v of an enzymatic reaction is defined as the rate at which a substrate disappears or at which a product is formed, the two being identical:

$$v = -d[S] / dt = d[P] / dt \quad (9-1)$$

Under the steady-state conditions that usually apply (see p. 449), the rate of increase of product will be the same as the rate of decrease of substrate. The units of velocity are moles per liter per second ($M s^{-1}$) or more traditionally in enzymology moles per liter per minute. We are interested in the **instantaneous velocity**, which at

any time is given by the slope of the progress curve (Fig. 9-1). We usually want to measure the velocity immediately after the reaction is started to avoid the decrease in rate that comes from the depletion of substrate or accumulation of products. In some cases, as in the example given in Fig. 9-1, this is difficult to do with accuracy.

In some chemical reactions, which involve first-order processes, a logarithmic plot of the progress curve ($\log[S]$ vs t) gives a straight line so that the initial slope need not be determined. However, most **enzyme assays** give progress curves that remain nearly linear for only short periods of time. In these cases we need very sensitive methods for detecting products. These may involve colorimetric or fluorimetric measurements or the use of radioactive substrates. One of the most sensitive approaches is to arrange the assay so that a product of the reaction serves as a catalyst for another enzymatic process, thus amplifying the amount of final product to be measured.^{14,15} Another approach is to measure velocities of reactions in a very small volume, e.g., 200 nanoliters, and to continuously separate and measure products using capillary electrophoresis and fluorescence detection.¹⁶

If the progress curve is not a straight line at the beginning of the reaction (0 time) and if the amount of compound that reacts in a fixed

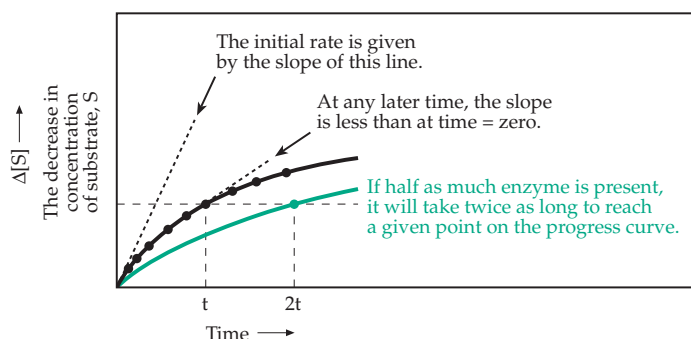


Figure 9-1 The progress curve for an enzymatic reaction in which the substrate S is converted into products.

interval of time is taken as the rate, an erroneous answer will be obtained. Sometimes an **integrated rate expression** that describes the time course of product formation can be used (see Eq. 9-22). Even when the progress curve is nonlinear as shown in Fig. 9-1, it is possible to estimate relative rates by noting that in most cases if one unit of enzyme yields a certain amount of product in time t_1 , the same amount of product will be formed by n units of enzyme in time t_1/n .

First-order reactions. In many chemical reactions the rate of decrease of the concentration of a given reactant [A] is found experimentally to be directly proportional to the concentration of that reactant at any given time:

$$v = -d[A] / dt = k[A] \quad (9-2)$$

Such a reaction is described as **first order** and the proportionality constant k is known as the **rate constant**. Such first-order kinetics is observed for **unimolecular processes** in which a molecule of A is converted into product P in a given time interval with a probability that does not depend on interaction with another molecule. An example is radioactive decay. Enzyme–substrate complexes often react by unimolecular processes. In other cases, a reaction is **pseudo-first order**; compound A actually reacts with a second molecule such as water, which is present in such excess that its concentration does not change during the experiment. Consequently, the velocity is apparently proportional only to [A].

A first-order rate constant k has units of s^{-1} . When $[A] = 1$, $v = k$. Thus, k is a measure of the speed in $\text{mol l}^{-1} \text{s}^{-1}$ of the reaction of a substance at unit activity. As a first-order reaction proceeds, [A] decreases and at time t is given by any one of the following three equivalent expressions (Eq. 9-3). These are obtained by integration of Eq. 9-2 in which t_0 is the time at which the reaction was started.

$$\begin{aligned} [A] &= A_0 e^{-kt} \\ \ln([A_0] / [A]) &= kt \\ \log[A_0] - \log[A] &= kt / 2.303 \end{aligned} \quad (9-3)$$

Equation 9-3 is the equation of exponential decay. A characteristic of exponential decay is that [A] is halved in a time that is independent of concentration. The **half-life** is $t_{1/2}$:

$$t_{1/2} = \ln 2 / k = 0.693 / k \quad (9-4)$$

The **relaxation time** τ for A is defined by Eq. 9-5 and represents the time required for the concentration [A] to fall to $1/e$ (or ~ 0.37) of its initial value.

$$\tau = 1 / k = t_{1/2} / \ln 2 \quad (9-5)$$

Turnover numbers and units of activity. When an enzyme is catalyzing product formation at the maximum possible rate, $V_{\max}$, we can usually assume that the active site of every molecule of enzyme contains a substrate in the form of an intermediate enzyme–substrate complex ES and that this complex is being converted to products according to Eq. 9-6:

$$\begin{aligned} d[P] / dt &= V_{\max} = k[ES] = k[E]_t \\ \text{or} \quad V_{\max} &= k_{\text{cat}}[E]_t \end{aligned} \quad (9-6)$$

Here E_t is the total enzyme, namely, the free enzyme E plus enzyme–substrate complex ES. The equation holds only at **substrate saturation**, that is, when the substrate concentration is high enough that essentially all of the enzyme has been converted into the intermediate ES. The process is first order in enzyme but is zero order in substrate. The rate constant k is a measure of the speed at which the enzyme operates. When the concentration $[E]_t$ is given in moles per liter of *active sites* (actual molar concentration multiplied by the number of active sites per mole) the constant k is known as the **turnover number**, the **molecular activity**, or k_{cat} . The symbol k_{cat} is also used in place of k in Eq. 9-6 for complex rate expressions in which k_{cat} cannot represent a single rate constant but is an algebraic expression that contains a number of different constants.

Turnover numbers can be measured only when the concentration of the enzyme is known. Partly for this reason the activity of an enzyme is usually given as **specific activity**, the *units of activity per milligram of protein*. One **international unit** is the amount of enzyme that produces $1 \mu\text{mol}$ of product per minute under standard (usually optimal) conditions. The International Union of Biochemistry¹⁷ has recommended a larger unit, the **katal** (kat), the amount of enzyme that converts one mol s^{-1} of substrate to product.

$$\begin{aligned} 1 \text{ kat} &= 6 \times 10^7 \text{ international units} \\ 1 \text{ international unit} &= 16.67 \text{ nkat (nanokatals)} \end{aligned}$$

If the enzyme is pure and is saturated with substrate under the standard assay conditions, the following relationships hold.

$$\begin{aligned} \text{Turnover No.} &= \text{katal} / \text{mol of active sites} \\ &= [\text{nkat} / \text{mg}] \times M_r \times 10^{-6} / n \\ &= [\text{international units} / \text{mg}] \times M_r \times 10^{-3} / 60n \end{aligned}$$

Here M_r is the relative molecular mass of the enzyme and n is the number of active sites per molecule. Since the activity of an enzyme is dependent on both temperature and pH, these variables must be specified.

Turnover numbers of enzymes vary from <1 to $\sim 10^6 \text{ s}^{-1}$. Trypsin, chymotrypsin, and many intracellular

enzymes have turnover numbers of $\sim 10^2 \text{ s}^{-1}$. The fastest enzymes, which include **catalase** (Chapter 16), **carbonic anhydrase** (Chapter 13), and **Δ^5 -3-oxosteroid isomerase** (Chapter 13), have maximum turnover numbers of $2 \times 10^5 \text{ s}^{-1}$ or more. Compare these reaction rates with those of a typical organic synthesis in the laboratory. A reaction mixture must often be heated for hours ($k < 10^{-3} \text{ s}^{-1}$). Many enzymes accelerate rates by factors of greater than 10^6 over those observed in the absence of an enzyme at a comparable temperature and pH. Enzymes often bring two or more substrates together, binding them at a specific location in their active sites. Because of this, rapid reactions can be catalyzed even when the reactants are present in low concentrations.

Second-order reactions. For a chemical reaction to occur between two molecules, A and B (Eq. 9-7), they must meet and collide.



The velocity of such a **second-order** process is characterized by a **bimolecular rate constant** k_2 and is proportional to the product of the concentrations of A and B:

$$v = k_2[\text{A}][\text{B}] \quad (9-8)$$

The units of k_2 are $\text{M}^{-1} \text{ s}^{-1}$. If [B] is present at unit activity, the rate is $k_2[\text{A}]$, a quantity with units of s^{-1} . We can see that the bimolecular, or second-order, rate constant for reaction of A with B may be compared with first-order constants when the second reactant B is present at unit activity. In many real situations, reactant B is present in large excess and in a virtually constant concentration. The reaction is pseudo-first order and the experimentally observed rate constant $k_2[\text{B}]$ is an *apparent first-order rate constant*. The bimolecular rate constant k_2 can be obtained by dividing the apparent constant by [B].

Reversible chemical reactions. In any reversible process, we must consider rate constants for both the forward and the reverse reactions. At equilibrium a reaction proceeds in the forward direction at exactly the same velocity as in the reverse reaction so that no change occurs. For this reason there is always a relationship between the equilibrium constant and the rate constants. For Eq. 9-9, k_1 is the bimolecular rate constant

$$\text{A} + \text{B} \xrightleftharpoons[k_2]{k_1} \text{P}$$

$$\text{Equilibrium constant} = K = \frac{[\text{P}]}{[\text{A}][\text{B}]} = \frac{k_1}{k_2} \quad (9-9)$$

for the forward reaction and k_2 the unimolecular rate constant for the reverse reaction. The equilibrium constant K can easily be shown, from Eq. 9-2 and 9-8, to equal k_1/k_2 for the reaction of Eq. 9-9.

The student should be aware that in kinetic equations rate constants are usually numbered consecutively via subscripts and that the subscripts do not imply anything about the molecularity. The system which is used here employs odd-numbered constants for steps in the forward direction and even-numbered constants for steps in the reverse direction. However, *many authors number the steps in the forward direction consecutively and those in the reverse direction with corresponding negative subscripts*.

What relationships exist between experimentally observable rates and k_1 and k_2 for a reversible reaction? Consider first the simplest case (Eq. 9-10):



If pure A is placed in a solution, its concentration will decrease until it reaches an equilibrium with the B which has been formed. It is easy to show that in this case [A] does not decay exponentially but $[\text{A}] - [\text{A}]_{\text{equil}}$ does. If $\log([\text{A}] - [\text{A}]_{\text{equil}})$ is plotted against time a first-order rate constant k , characteristic of *the rate of approach to equilibrium*, will be obtained. Its relationship to k_1 and k_2 is given by Eq. 9-11.

$$k = k_1 + k_2 \quad (9-11)$$

The relaxation time τ for approach to equilibrium can be expressed as follows:

$$\tau^{-1} = k = \tau_1^{-1} + \tau_2^{-1} = k_1 + k_2 \quad (9-12)$$

and for the more complex case of Eq. 9-9 as

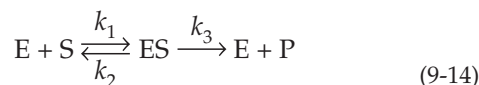
$$k = \tau^{-1} = k_1([\text{A}]_e + [\text{B}]_e) + k_2 \quad (9-13)$$

where $[\text{A}]_e$ and $[\text{B}]_e$ are the equilibrium concentrations of A and B.¹⁸

2. Formation and Reaction of Enzyme-Substrate Complexes

An abundance of evidence indicates that the first step in enzymatic catalysis is the combining of the enzyme and substrate reversibly to form a complex, ES (Eq. 9-14). Formation of the complex is normally reversible. ES can either break up to form enzyme and substrate again or it can undergo conversion to a product or products, often by a unimolecular process. Three rate constants are needed to describe this system for a reaction that is irreversible overall. A complete

description of the kinetic behavior is fairly involved.



For example, the kinetics may be different within cells, where molar concentrations of enzymes often exceed those of substrate, than in the laboratory. In most laboratory experiments the enzyme is present at an extremely low concentration (e.g., 10^{-8} M) while the substrate is present in large excess. Under these circumstances the **steady-state approximation** can be used. For this approximation *the rate of formation of ES from free enzyme and substrate is assumed to be exactly balanced by the rate of conversion of ES on to P*. That is, for a relatively short time during the duration of the experimental measurement of velocity, the concentration of ES remains essentially constant. To be more precise, the steady-state criterion is met if the absolute rate of change of a concentration of a transient intermediate is very small compared to that of the reactants and products.¹⁹

The **Michaelis–Menten equation** (Eq. 9-15) describes the initial reaction rate of a single substrate with an enzyme under steady-state conditions.

$$v = \frac{V_{\max}}{1 + K_m/[S]} = \frac{V_{\max}[S]}{K_m + [S]} \quad (9-15)$$

where $K_m = (k_2 + k_3)/k_1$

This can be rearranged as follows:

$$\frac{V_{\max}}{v} = 1 + K_m/[S] \quad (9-16)$$

Equation 9-15 provides a relationship between the velocity observed at a particular substrate concentration and the maximum velocity that would be achieved at infinite substrate concentration. The quantities $V_{\max}$ and K_m are often referred to as the **kinetic parameters** of an enzyme and their determination is an important part of the characterization of an enzyme. Equation 9-15 can be derived by setting the rate of formation of the ES complex ($k_1[E][S]$) in the steady state equal to its rate of breakdown, ($[k_2 + k_3][ES]$). Rearranging and substituting K_m , as defined in Eq. 9-15, we obtain Eq. 9-17.

$$[E][S] = \frac{(k_2 + k_3)}{k_1} [ES] = K_m [ES] \quad (9-17)$$

Using this equation, together with a mass balance relationship ($[E] = [E]_t - [ES]$), we can solve for $[ES]/[E]_t$, the fraction of enzyme combined as enzyme–substrate complex (Eq. 9-18).

$$\frac{[ES]}{[E]_t} = \frac{[S]}{K_m + [S]} \quad (9-18)$$

The maximum velocity, $V_{\max} = k_3[E]_t$, is attained only when all of the enzyme is converted into ES. Under other conditions $v = k_3[ES]$ and Eq. 9-19 holds.

$$[ES]/[E]_t = v/V_{\max} \quad (9-19)$$

Substituting from Eq. 9-19 into Eq. 9-18 gives the Michaelis–Menten equation (Eq. 9-15).

In many cases, the rate at which ES is converted back to free E and S is much greater than the rate of conversion of ES to products ($k_2 \gg k_3$). In such cases K_m equals k_2/k_1 , the dissociation constant for breakdown of ES to free enzyme and substrate (sometimes called K_s). Thus, K_m sometimes has a close inverse relationship to the strength of binding of substrate to enzyme. In such cases, $1/K_m$ is a measure of the affinity of the substrate for its binding site on the enzyme and for a series of different substrates acted on by the same enzyme. The more tightly bound substrates have the lower values of K_m . But beware! The condition that k_3 is negligible compared to k_2 may be met with some (poorer) substrates but may not always be met by others. From Eq. 9-15 we see that K_m is always greater than or equal to K_s , but it may be less than K_s for more complex mechanisms.

Figure 9-2 shows a plot of velocity against substrate concentration as given by Eq. 9-15. The position of $V_{\max}$ on the ordinate is marked, but it should be clear that the experimental velocity (v) can never attain $V_{\max}$ unless $[S]$ is very high relative to K_m . The value of v approaches $V_{\max}$ asymptotically. Since K_m is defined as the value of $[S]$ at which $v = V_{\max}/2$, its value can be estimated from Fig. 9-2. However, K_m cannot be determined reliably because of the difficulty of establishing the value of $V_{\max}$ from a plot of this type. Notice that the curve of Figure 9-2 is identical in form to the saturation curve for reversible binding shown in Fig. 7-1.

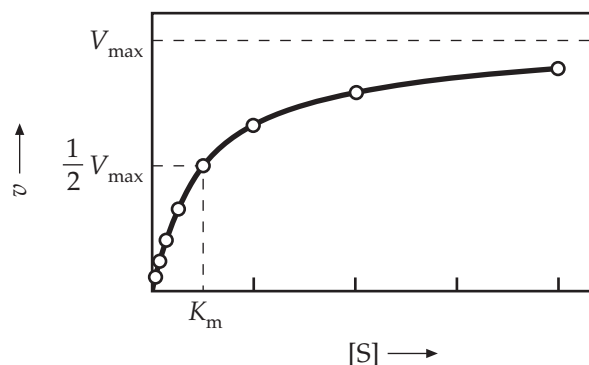


Figure 9-2 Plot of observed velocity v vs substrate concentration $[S]$ for an enzyme-catalyzed reaction.

Linear forms for rate equations. To obtain K_m and $V_{\max}$ from experimental rate data, Eq. 9-15 can be transformed by algebraic rearrangement into one of several linear forms. The popular **double-reciprocal** or **Lineweaver–Burk** plot of $1/v$ against $1/[S]$ (Fig. 9-3) is described by Eq. 9-20. The values of $K_m/V_{\max}$ and $1/V_{\max}$ can be evaluated from the slope and intercept, respectively, of this straight line plot.

$$\frac{1}{v} = \frac{1}{V_{\max}} + \frac{K_m}{V_{\max}} \frac{1}{[S]} \quad (9-20)$$

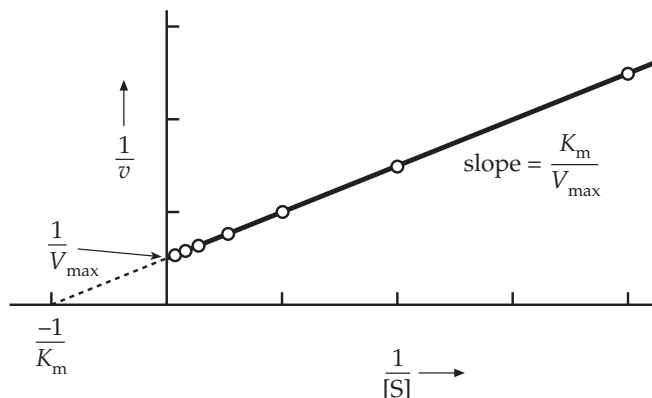


Figure 9-3 Double-reciprocal or Lineweaver–Burk plot of $1/v$ vs $1/[S]$. The intercept on the vertical axis gives $1/V_{\max}$ and the slope gives $K_m/V_{\max}$. The intercept on the horizontal axis equals $-1/K_m$.

Another linear plot, the **Eadie–Hofstee plot**, is that of $v/[S]$ vs v (Fig. 9-4). It is related to the Scatchard plot (Fig. 7-3) and is fitted by Eq. 9-21.

$$\frac{v}{[S]} = \frac{V_{\max}}{K_m} - v \frac{1}{K_m} \quad (9-21)$$

While the point for $[S] = 0$ and $v = 0$ cannot be plotted, the ratio $v/[S]$ approaches $V_{\max}/K_m$ as v approaches zero. Notice the distribution of the points in Fig. 9-4. Substrate concentrations were chosen such that the increase in velocity from point to point is more or less constant, a desirable experimental situation. The points on the Eadie–Hofstee plot are also nearly evenly distributed, but those of the Lineweaver–Burk plot are compressed at one end. (However, if the substrate concentrations for successive points are selected in the ratios 1, 1/3, 1/5, 1/7, and 1/9, the spacing will be uniform on the Lineweaver–Burk plot.) A second advantage of the Eadie–Hofstee plot is that the entire range of possible substrate concentration from near zero to infinity can be fitted onto a single plot.

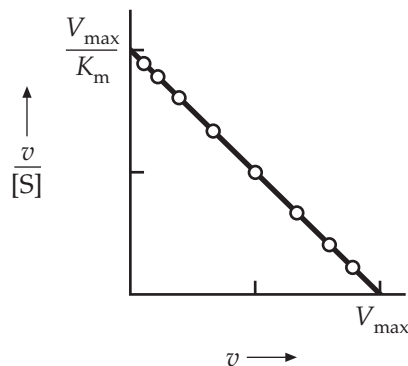


Figure 9-4 The Eadie–Hofstee plot of $v/[S]$ against v . The slope is $-1/K_m$; the intercept on the vertical axis is $V_{\max}/K_m$ and that on the horizontal axis is $V_{\max}$.

As pointed out at the beginning of Section A, depletion of substrate with time lowers $[S]$ from its initial value to some extent. This can have an especially adverse effect on points obtained at low $[S]$, e.g., points on the right side of Fig. 9-3 or on the left side of Fig. 9-4. Equations 9-20 and 9-21 will be more precise if the average substrate concentration over the time period of the assay, rather than that at zero time,²⁰ is used.

Nonlinear equations and integrated rate equations. In discussions of the *control* of metabolism through regulation of enzymatic activity, it is often better to plot v against $\log [S]$ as in Fig. 9-5. This plot also has the virtue that the entire range of attainable substrate concentrations can be plotted on one piece of paper if the point for $[S] = 0$ (at minus infinity) is omitted. The same scale can be used for all enzymes. The plot is S-shaped, both for simple cases that are represented by Eq. 9-15 and for enzymes that bind substrate cooperatively (Section B.5). Thus, the classification or “hyperbolic” vs “sigmoidal” is lost. However, the degree of cooperativity can be directly measured from the midpoint slope of the curve. A disadvantage is that it is awkward to measure $V_{\max}$ from a plot of this type, and it may be preferable to obtain it from a linear plot (Figs. 9-3 or 9-4). Alternatively, computer-assisted methods can be used to obtain both K_m and $V_{\max}$ and to fit a curve to the experimental points as in Fig. 9-5.

An attractive alternative to the use of initial velocities and linear plots is to measure the kinetic parameters $V_{\max}$ and K_m using points all along the progress curve (Fig. 9-1). Various procedures for doing this have been devised.^{21,22} For example, the integrated form of Eq. 9-15 can be given as Eq. 9-22 or 9-23.

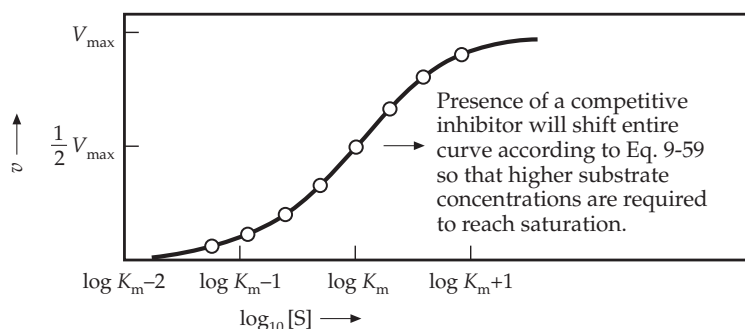


Figure 9-5 Plot of v against $\log [S]$ for an enzyme-catalyzed reaction.

$$V_{\max} t = [P] + K_m \ln \frac{[S]_0}{[S]_0 - [P]} \quad (9-22)$$

$$V_{\max} t = [S]_0 - [S] + K_m \ln ([S]_0 / [S]) \quad (9-23)$$

Here $[S]_0$ is the initial substrate concentration at $t = 0$ and $[S]$ and $[P]$ are the substrate and product concentrations at any later time. In one type of linear plot,²¹ based on Eq. 9-23, $([S]_0 - [S]) / t$ is plotted against $1/t \ln ([S]_0 / [S])$, the slope of the plot being $-K_m$ and the intercepts being $V_{\max}$ and $V_{\max} / K_m$. Computer programs are available for analysis of enzymatic progress curves, even for complex mechanisms.^{23,24}

Kinetics with high enzyme concentrations.

Laboratory studies of the kinetics of purified enzymes are usually conducted with enzyme concentrations of 10^{-7} to 10^{-10} M, but within cells enzyme concentrations are probably often in the range 10^{-6} to 10^{-5} M,²⁵ which may be higher than the concentrations of the substrates upon which the enzymes act. Be cautious in drawing conclusions about kinetics under such circumstances! Methods have been devised for handling kinetic data when the concentration of enzyme is greater than K_m , a condition that can lead to intolerably high errors if the usual equations are applied.^{26,27}

3. Diffusion and the Rate of Encounter of an Enzyme with Substrate

What determines the value of k_1 in Eq. 9-14? This rate constant represents the process by which the substrate and enzyme find each other, become mutually oriented, and bind to form ES. If orientation and binding are rapid enough, the rate will be determined by the speed with which the molecules can come together by diffusion. Large molecules in solution are free to travel for only a tiny fraction of their diameter as a result of their frequent collision with solvent molecules.

The result is visible in the **Brownian movement** of microscopic particles suspended in a fluid. If an individual particle is followed, it is seen to undergo a "random walk," moving in first one direction then another. Albert Einstein showed that if the distances transversed by such particles in a given time Δt are measured, the mean square of these Δx values Δ^2 can be related by Eq. 9-24 to the diffusion constant D (which is usually given in units of $\text{cm}^2 \text{s}^{-1}$).

$$\Delta^2 = 2D \Delta t \quad (9-24)$$

For molecules, the Brownian movement cannot be observed directly but the diffusion constant can be measured, for example, by observing the rate of spreading of a boundary between two different concentrations of the substance.²⁸ The diffusion constant for $^1\text{H}_2\text{HO}$ (HDO) in H_2O at 25°C is $2.27 \times 10^{-5} \text{ cm}^2 \text{s}^{-1}$, and the values for the ions K^+ and Cl^- are about the same.²⁹ For many small molecules D is approximately $10^{-5} \text{ cm}^2 \text{s}^{-1}$, and the value decreases as the size of the molecule increases. Thus, for the 13.7-kDa ribonuclease, $D = 1.1 \times 10^{-6} \text{ cm}^2 \text{s}^{-1}$, and for the 500-kDa myosin, D is $\sim 1 \times 10^{-7} \text{ cm}^2 \text{s}^{-1}$. For the spherical particle the **Stokes-Einstein equation** (Eq. 9-25) can be used to relate the diffusion coefficient to the radius, the coefficient of viscosity η , the Boltzmann constant k_B , and the temperature (Kelvin) T .

$$D = \frac{k_B T}{6\pi\eta r} \quad (9-25)$$

To estimate the rate constant for a reaction that is controlled strictly by the frequency of collisions of particles, we must ask how many times per second one of a number n of particles will be hit by another of the particles as a result of Brownian movement. The problem was analyzed in 1917 by Smoluchowski,^{30,31} who considered the rate at which a particle B diffuses toward a second particle A and disappears when the two collide. Using Fick's law of diffusion, he concluded that the number of **encounters** per milliliter per second was given by Eq. 9-26.

$$\text{Number of encounters/ml s}^{-1} = 4\pi (D_A + D_B) (r_A + r_B) n_A n_B \quad (9-26)$$

Here D_A and D_B are the two diffusion coefficients, r_A and r_B are the radii (in Å) of the two particles, and n_A and n_B are the numbers of particles per milliliter. The number of encounters per liter per second is $(4\pi / 1000) (D_A + D_B) (r_A + r_B) N^2 [A][B]$, where N is Avogadro's number. Dividing this frequency by N gives the rate of collision v in M s^{-1} , a velocity that can be equated (Eq. 9-27) with $k_D [A][B]$, where k_D is a second-order rate constant

which determines the **diffusion controlled limit**, that is, the maximum possible rate of a reaction.

$$v = \frac{4\pi}{1000} (D_A + D_B)(r_A + r_B)[A][B]N = k_D[A][B] \quad (9-27)$$

This equation can be rearranged to Eq. 9-28.

$$k_D = \frac{4\pi N}{1000} (D_A + D_B)(r_A + r_B) \text{M}^{-1} \text{s}^{-1} \quad (9-28)$$

While this equation is thought to overestimate the diffusion-limited rate constant slightly, it is a good approximation. If the diffusing particles are approximately spherical, diffusion constants D_A and D_B can be calculated from Eq. 9-25, and Eq. 9-28 becomes Eq. 9-29.

$$k_D = \frac{2RT}{3000\eta} \left(2 + \frac{r_A}{r_B} + \frac{r_B}{r_A} \right) \quad (9-29)$$

The value of k_D does not vary greatly as the ratio of radii r_A/r_B is changed, and in most cases it may be assumed that $r_A/r_B \approx 1$, in which case the equation simplifies (Eq. 9-30).

$$k_D \approx \frac{8RT}{3000\eta} \text{M}^{-1} \text{s}^{-1} \quad (9-30)$$

For water at 25°C, the coefficient of viscosity η is ~ 0.01 poise (1 poise = 10^{-5} newton cm^{-2}) which leads, according to Eq. 9-30, to $k_D \sim 0.7 \times 10^{10} \text{M}^{-1} \text{s}^{-1}$. It was shown by Debye³² that this rate constant must be multiplied by a correction factor when charged particles rather than uncharged spheres diffuse together. This factor may be of the order of 5–10 for a substrate and enzyme carrying two or three charges and may act to either increase or decrease reaction rates. Since the viscosity of cytoplasm is 0.02–0.03 poise,³³ k_D will be reduced correspondingly.

A simple alternative derivation of Eq. 9-30 was developed by Dexter French,³⁴ who also provided the author with most of this discussion of encounter theory. Consider a small element of volume ΔV swept out by a particle as it moves through the solution for a distance equal to its own radius. This element of volume will equal πr^3 :

$$\Delta V = \pi r^2 \times r = \pi r^3 \quad (9-31)$$

It will be swept out in a time Δt which can be calculated from Eq. 9-24 as $r^2 / 2D$. Substituting the value of D given by Eq. 9-25, we obtain for Δt :

$$\Delta t = \frac{3\pi\eta r^3}{k_B T} \quad (9-32)$$

Division of Eq. 9-31 by Eq. 9-32 gives the volume swept out per second by one particle:

$$\Delta V / \Delta t = k_B T / 3\eta \text{cm}^3 \text{s}^{-1} \quad (9-33)$$

Since the collision radius for two particles of equal size is two times the particle radius, the effective volume swept out will be four times that given by Eq. 9-33. Since both particles are diffusing, the effective diffusion constant will be twice that used in obtaining Eq. 9-28. Thus, the effective volume swept out by the particle in a second will be eight times that given by Eq. 9-33. The volume swept out by one mole of particles is equal to k_D (recall that the second-order rate constant has dimensions of liter $\text{mol}^{-1} \text{s}^{-1}$). Thus, when converted to a moles per liter basis and multiplied by 8, Eq. 9-33 should (and does) become identical with the Smoluchowski equation (Eq. 9-30).

The volume given by Eq. 9-33 is about $1.4 \times 10^{-11} \text{cm}^3$, which could be represented approximately by a cube 2.4 μm on a side. If we compare this volume with that of a cell (Table 1-2) or of an organelle, we see that in one second an enzyme molecule will sweep out a large fraction of the volume of a small cell, mitochondrion, chloroplast, etc.

The “cage effect” and rotation of molecules. It is of interest to compare the bimolecular rate constant for encounters calculated by the Smoluchowski theory (Eq. 9-29) with the corresponding bimolecular rate constant for molecular collisions given by the kinetic theory of gases (Eq. 9-34).

$$k(\text{collision}) = \frac{N(r_A + r_B)^2}{1000} \left[8\pi k_B T \left(\frac{1}{m_A} + \frac{1}{m_B} \right) \right]^{1/2} \quad (9-34)$$

Here m_A and m_B are the masses of the two particles. This rate constant is also relatively independent of molecular size and for spheres varies from $(4 \text{ to } 11) \times 10^{11} \text{M}^{-1} \text{s}^{-1}$, over an order of magnitude greater than the rate constant for encounters. In a solution, molecules still collide at about the same rate as in a gas so that *100 to 200 collisions occur between two particles for each encounter*. However, during the time of the single encounter, the particles are together in a solvent “cage.” While in this solvent cage, both substrate and enzyme molecules undergo random rotational motions. Successive collisions bring them together in different orientations, one of which is likely to lead to a sufficiently close match of complementary surfaces (of substrate and binding site) so that formation of a “productive” ES complex takes place.

Molecular rotation in a solution is described quantitatively by diffusion laws (analogous to Fick’s laws)

for which a **rotary diffusion constant** θ is defined.^{35,36} Consider a group of molecules all oriented the same way initially, then undergoing rotary diffusion until their orientations become random. If we measure the orientation of each molecule by an angle we see that initially the value of $\cos \alpha$ is 1 but that when the angles become random the mean value of $\cos \alpha$ averaged over all molecules is zero. The rotary relaxation time τ is the time required for the mean value of $\cos \alpha$ to fall to



$1/e$ (which occurs when $\alpha = 68.5^\circ$). For a sphere θ is given by Eq. 9-35.

$$\theta = \frac{1}{2}\tau = \frac{k_B T}{8\pi\eta r^3} \quad (9-35)$$

Ellipsoidal or rod-shaped molecules have two different rotary diffusion constants while, if the dimensions of the molecules are different along all three axes, three constants must be specified.³⁶

From Eq. 9-35 we can calculate (if $\eta = 0.01$ poise) the following values for a small spherical molecule (substrate) of ~ 1 nm length and for a spherical enzyme of 5 nm diameter:

$$\begin{aligned} r = 0.5 \text{ nm} & \quad \theta \approx 1.3 \times 10^9 \text{ s}^{-1} \\ r = 2.5 \text{ nm} & \quad \theta \approx 1.0 \times 10^7 \text{ s}^{-1} \end{aligned}$$

We see that smaller molecules rotate much faster than large ones and that rotational relaxation times for small proteins are of the same order of magnitude as k_D for diffusion-limited encounter. However, for very large molecules, especially long rods, the rotary relaxation time about a short axis may be a large fraction of a second.

The rate of substrate binding. At very low substrate concentrations the Michaelis–Menten equation (Eq. 9-15) simplifies as follows:

$$v = (V_{\max}/K_m)[S] \quad (9-36)$$

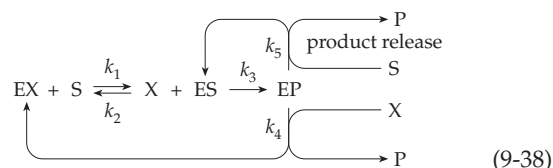
Since $V_{\max} = k_{\text{cat}}[E]_t$ (Eq. 9-6) and at low $[S]$ most of $[E]_t$ is free E, we obtain the following equation:²

$$v = (k_{\text{cat}}/K_m)[E][S] \quad (9-37)$$

From this we see that k_{cat}/K_m is the *apparent second-order rate constant for the reaction of free enzyme with substrate*. As such it cannot exceed the diffusion controlled limit k_D of Eqs. 9-28 to 9-30 which falls in the range of $10^9 - 10^{11} \text{ M}^{-1} \text{ s}^{-1}$. Experimentally observed

values of k_{cat}/K_m are always less than this limit, indicating that a certain time is required for a substrate molecule to become oriented and seated in the active site.³⁷ However, for several real enzymes values of k_{cat}/K_m of 10^7 to $3 \times 10^8 \text{ M}^{-1} \text{ s}^{-1}$ have been observed. For triose phosphate isomerase Albery and Knowles obtained a value of $4 \times 10^8 \text{ M}^{-1} \text{ s}^{-1}$, so close to the diffusion controlled limit that these authors regard this enzyme as a *nearly perfect catalyst*, one that could not have evolved further because it is already catalyzing the reaction with substrate at almost the maximum velocity that is possible.^{37–39}

The displacement of bound ligands by substrate. Jenkins pointed out that in many instances a substrate must displace another ligand from the active site to form the ES complex.⁴⁰ For example, a binding site for an ionic substrate often already contains an ion, either of the product P or some other ion X which Jenkins calls a **substrate surrogate**. For this situation Eq. 9-14 must be replaced by the following set of equations:



Here k_2 and also k_4 and k_5 , are second-order rate constants. The release of product, as determined by k_4 and k_5 , may be rate-limiting. At zero time the reverse reactions may be ignored, and steady-state analysis shows that the Michaelis–Menten equation (Eq. 9-16b) will be replaced by Eq. 9-39. Here, D is a constant and A is also constant if X is present at a fixed concentration.

$$\begin{aligned} \frac{V_{\max}}{v} &= 1 + \frac{k_2[X] + k_3}{k_1[S]} + \frac{k_3(1/k_5 - 1/k_1)}{k_4[X]/k_5 + [S]} \\ &= 1 + K_m/[S] + D/(A + [S]) \end{aligned} \quad (9-39)$$

When k_5 , the rate constant for displacement of product by substrate, is very small, this equation simplifies to Eq. 9-40.

$$\begin{aligned} \frac{V_{\max}}{v} &= 1 + k_3/k_4[X] + (k_2[X] + k_3)/k_1[S] \\ &= 1 + k_3/k_4[X] + K_m'/[S] \end{aligned} \quad (9-40)$$

At high concentrations of X the second term becomes negligible and the equation becomes identical to the Michaelis–Menten equation (Eq. 9-16) except that K_m' , the apparent Michaelis constant, now increases as $[X]$

increases (Eq. 9-41). This is exactly what is observed for competitive inhibitors where

$$K_m' = \frac{k_2[X] + k_3}{k_1} \quad (9-41)$$

Thus substrate surrogates at high concentrations are often competitive inhibitors. However, at low $[X]$ the second term in Eq. 9-41 may be important and X then serves as an activator. Under many circumstances, Eq. 9-39 does not simplify further and a nonlinear relationship between $1/v$ and $1/[S]$ is observed, with the shape of the curves being influenced by the values of constants A and D (see also Section B,4).⁴⁰

4. Reversible Enzymatic Reactions

For some enzyme-catalyzed reactions the equilibrium lies far to one side. However, many other reactions are freely reversible. Since a catalyst promotes reactions in both directions, we must consider the action of an enzyme on the reverse reaction. Let us designate the maximum velocity in the forward direction as V_f and that in the reverse direction as V_r . There will be a Michaelis constant for reaction of enzyme with product K_{mp} , while K_{ms} will refer to the reaction with substrate.

As in any other chemical reaction, there is a relationship between the rate constants for forward and reverse enzyme-catalyzed reactions and the equilibrium constant. This relationship, first derived by the British kineticist J. B. S. Haldane and proposed in his book *Enzymes*⁴¹ in 1930, is known as the **Haldane relationship**. It is obtained by setting $v_f = v_r$ for the condition that product and substrate concentrations are those at equilibrium. For a single substrate–single product system it is given by Eq. 9-42.

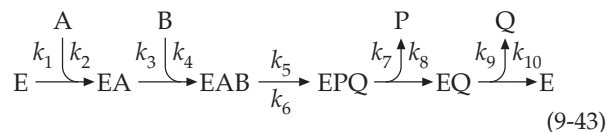
$$K_{eq} = V_f K_{mp} / V_r K_{ms} \quad (9-42)$$

Because the Haldane relationship imposes constraints on the values of the velocity constants and Michaelis constants, it is of some value in understanding regulation of metabolism. Consider the case that the maximum forward velocity V_f is high and that K_{ms} has a moderately low value (fairly strong binding of substrate). If the reaction is freely reversible ($K_{eq} \sim 1$) and the velocity of the reverse reaction V_r is about the same as that of the forward reaction, it will necessarily be true (see Eq. 9-42) that the product P will also be fairly tightly bound. If $V_r \ll V_f$, the value of K_{mp} will have to be much lower than that of K_{ms} . In such a situation P will remain tightly bound to the enzyme and since V_r is low, it will tend to clog the enzyme. Such **product inhibition** may sometimes slow down a whole pathway of metabolism. In such a case, the

only way that an enzymatic sequence can keep going in the forward direction is for product P to be removed rapidly by a subsequent reaction with a second enzyme. The first enzyme may be thought of as possessing a kind of “one-way valve” that turns off the flow in a metabolic pathway when the concentration of its product rises.

Reactions of two or more substrates. Enzymes frequently catalyze the reaction of two, three, or even more different molecules to give one, two, three, or more products. Sometimes all of the substrate molecules must be bound to an active site at the same time and are presumably lined up on the enzyme molecule in such a way that they can react in proper sequence. In other cases, the enzyme may transform molecule A to a product, and then cause the product to react with molecule B . The number of variations is enormous.^{1,10,12}

The order in which two molecules A and B bind to an enzyme to form a complex EAB may be completely *random* or it may be *obligatorily ordered*. Both situations occur with real enzymes. Cleland introduced a widely used method of depicting the possibilities.⁴² For example, Eq. 9-43 shows the reaction of A and B in an *ordered sequence* to form the complex EAB which is then isomerized to EPQ , the complex formed by binding the two products P and Q to the enzyme. The rate constant to the left of each vertical arrow or above each horizontal arrow refers to the reaction in the



forward direction as indicated by the arrow while the other constants (to the right or below the arrows) refer to the reverse reactions. The velocity in the forward direction for an enzyme with ordered binding is given by Eq. 9-44a,

$$v_f = \frac{V_f [A][B]}{K_{eqA} K_{mB} + K_{mB} [A] + K_{mA} [B] + [A][B]} \quad (9-44a)$$

which may also be written in the reciprocal form (Eq. 9-44b):

$$\frac{1}{v_f} = \frac{1}{V_f} \left(1 + \frac{K_{mA}}{[A]} + \frac{K_{mB}}{[B]} + \frac{K_{eqA} K_{mB}}{[A][B]} \right) \quad (9-44b)$$

An alternative form of this equation (Eq. 9-45), proposed by Dalziel,⁴³ is sometimes used.

$$\frac{[E]_t}{v_f} = \phi_0 + \frac{\phi_1}{[A]} + \frac{\phi_2}{[B]} + \frac{\phi_{12}}{[A][B]} \quad (9-45)$$

The kinetic parameters of Eq. 9-44 are V_f , the maximum velocity in the forward direction, the two Michaelis constants, K_{mB} and K_{mA} , and the equilibrium constant K_{eqA} for reversible dissociation of the complex EA and which is equal to k_2/k_1 . The relationship between the parameters of Eq. 9-44 (K_m 's, V 's, and K_{eqA} 's) and the rate constants k_1 – k_{10} is not obvious. However, remember that the parameters are experimental quantities determined by measurements on the enzyme. Sometimes, but not always, it is possible to deduce some of the values of individual rate constants from the experimental parameters.

An equation similar to 9-44a can be written for the velocity v_r of the reaction of P and Q. Also an equation can be written in similar form for $v_f - v_r$, i.e., the instantaneous velocity of the reaction in any mixture of all four components A, B, P, and Q.

The kinetic parameters of Eq. 9-44b are often obtained from experimental data through the use of reciprocal plots (Fig. 9-6). However, Eq. 9-44b is linear only if the concentration of one or the other of the substrates A and B is kept constant. For this reason a series of experiments is usually performed in which [A] is varied while [B] is held constant. Then [A] is held constant and [B] is varied. Each of these experiments leads to a family of lines (Fig. 9-6A) whose slopes and intercepts are measured. The slopes and intercepts of this family of curves are then plotted against the reciprocal of the second concentration, i.e.,

the one that was held fixed.

From a set of these **secondary plots**, V_f and one of the Michaelis constants can be determined (Fig. 9-6B and C). Using two sets of secondary plots, all of the constants of Eq. 9-44 may be established. Alternatively, a computer can be used to examine all of the data at once and to obtain the best values of the parameters. The latter approach is desirable because estimates of the standard deviations of the parameters can be obtained. However, the user must take care to ensure that the experimental errors are correctly estimated and are not simply estimates of how well the computer has fitted the points on the assumption that they contain no error.²³

The meaning of the kinetic parameters may be slightly difficult to grasp. V_f is the velocity that would be obtained if both [A] and [B] were high enough to saturate the enzyme. Each K_m corresponds to that for a simple system in which the concentration of the second substrate is at a high, saturating value.

For the bimolecular reaction that we have considered, there are two Haldane relationships:

$$K_{eq} = \frac{V_f K_{mP} K_{dQ}}{V_r K_{dA} K_{mB}} = \left(\frac{V_f}{V_r} \right)^2 \frac{K_{dP} K_{mQ}}{K_{mA} K_{dB}} \quad (9-46)$$

Of these, only the first is ordinarily used.

“Ping-pong” mechanisms. A common type of mechanism that is especially prevalent for enzymes with tightly bound cofactors has been dubbed **ping-pong** because the enzyme alternates between

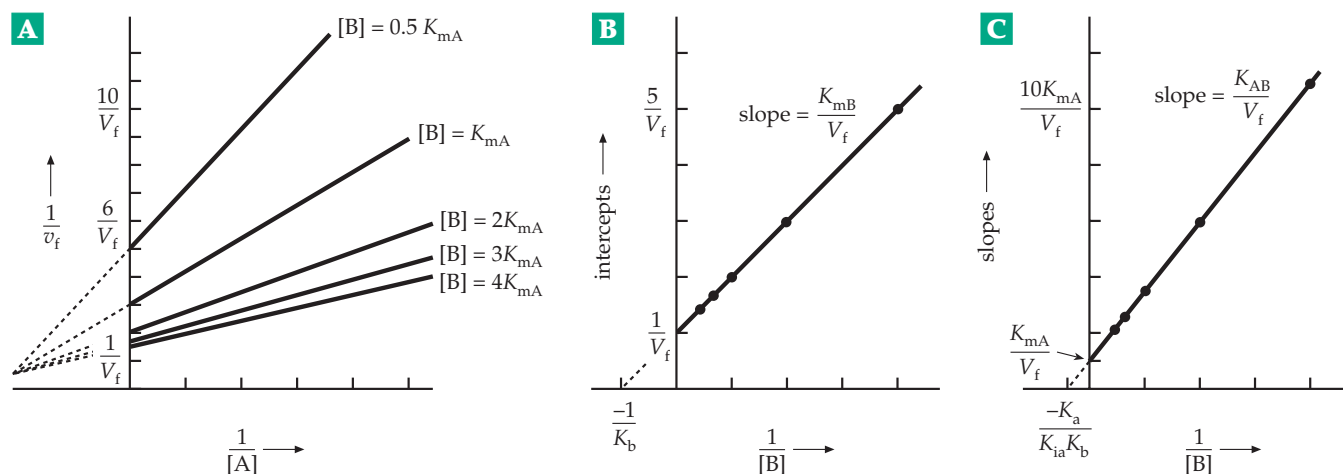
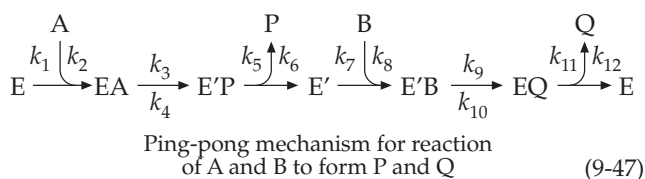


Figure 9-6 Reciprocal plots used to analyze kinetics of two-substrate enzymes. (A) Plot of $1/v_f$ against $1/[A]$ for a series of different concentrations of the second substrate B. (B) A secondary plot in which the intercepts from graph A are plotted against $1/[B]$. (C) Secondary plot in which the slopes from graph A have been plotted against $1/[B]$. The figures have been drawn for the case that $K_{mA} = 10^{-3}$ M, $K_{mB} = 2 K_{mA}$, and $K_{AB} = K_{eqA} K_{mB}$ (Eq. 9-46) = $K_{mA}/200$ and [A] and [B] are in units of moles per liter. Eadie–Hofstee plots of $v_f/[A]$ vs v_f at constant [B] can also be used as the primary plots. The student can easily convert Eq. 9-44 to the proper form analogous to Eq. 9-21.

two forms E and E' (Eq. 9-47). Substrate A reacts via complex EA to form E', a modified enzyme that often



contains an altered coenzyme. At the same time A is changed to product P still bound to the enzyme. P dissociates leaving E' which is then able to react with the second substrate B and to go through the second half of the cycle during which E' is converted back to E. An example is provided by the aminotransferases (Eq. 14-25) in which the coenzyme pyridoxal phosphate is interconverted with pyridoxamine phosphate.

The rate equations of the ping-pong mechanism resemble that for the ordered bimolecular reaction (Eq. 9-44), but each has one less term (Eq. 9-48):

$$v_f = \frac{V_f[A][B]}{K_{mB}[A] + K_{mA}[B] + [A][B]} \quad (9-48a)$$

or

$$\frac{1}{v_f} = \frac{1}{V_f} \left(1 + \frac{K_{mA}}{[A]} + \frac{K_{mB}}{[B]} \right) \quad (9-48b)$$

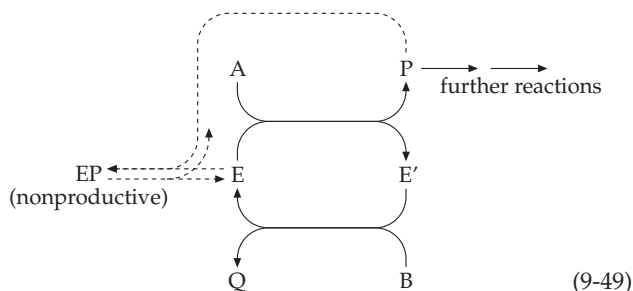
A diagnostic feature of the ping-pong mechanism is that the families of lines (in double reciprocal plots) obtained when one substrate is held constant while the other is varied no longer intersect as they do in Fig. 9-6A but are parallel. They must be truly parallel and the experimentalist must be aware that nearly parallel lines may sometimes be observed for sequential reactions. Thus, if K_{eqA} of Eq. 9-44b is small enough the last term of that equation will be small compared to the other terms and the equation will be approximated by Eq. 9-48b. The reaction will appear to be ping-pong even though it is sequential and the reaction proceeds through the ternary complex EAB.

One less kinetic parameter can be obtained from an analysis of the data for a ping-pong mechanism than can be obtained for ordered reactions. Nevertheless, in Eq. 9-47, twelve rate constants are indicated. At least this many steps must be considered to describe the behavior of the enzyme. Not all of these constants can be determined from a study of steady-state kinetics, but they may be obtained in other ways.

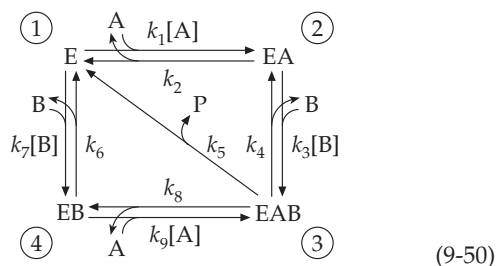
Isomechanisms. A catalyst functions over and over again without being altered. However, during a single turnover it is likely to undergo a temporary change. For example, if an enzyme assists in removing a proton from a substrate, some functional group of

the enzyme will be protonated during part of the catalytic cycle. This proton must dissociate before another catalytic cycle begins, and in some instances this dissociation may be a rate-limiting step. Northrup and coworkers developed the use of product inhibition to identify such isomechanisms.⁴⁴

Dead-end complexes. In the steady state of action of an enzyme with ping-pong kinetics, part of the enzyme is in form E and part in form E'. Ideally, E would have affinity only for A and Q, while E' would have affinity only for B and P. However, in many real situations E also binds B and P weakly; similarly, E' binds A and Q. This is because the products and reactants usually have structural features in common. The propensity of enzymes with ping-pong kinetics to form **dead-end complexes** (also called abortive complexes) may sometimes have a regulatory function. In Eq. 9-49 the reactions of Eq. 9-47 have been rearranged to depict a situation in which product P normally undergoes a sequence of further reactions. However, if P accumulates to a high enough concentration it can react reversibly to form a dead-end complex EP. This is an effective form of product inhibition which can be relieved only by a lowering of the concentration of P through its further metabolism.



Handling rate equations for complex mechanisms. While steady-state rate equations can be derived easily for the simple cases discussed in the preceding sections, enzymes are often considerably more complex and the derivation of the correct rate equations can be extremely tedious. The **topological theory of graphs**, widely used in analysis of electrical networks, has been applied to both steady-state and nonsteady-state enzyme kinetics.⁴⁵⁻⁵⁰ The method employs diagrams of the type shown in Eq. 9-50. Here



the reaction of an enzyme with two substrates A and B with a **random** order of binding is depicted. (In contrast, Eq. 9-43 shows the case of ordered binding of two substrates.) When complex EAB is formed, it can decompose to free enzyme and to the single product P. Each one of the **nodes**, which are numbered 1–4 in the diagram, corresponds to a single form of the enzyme. The appropriate first-order rate constant or apparent first-order constant is placed by each arrow. The methods provide easy rules for deriving from such a scheme the steady-state rate equation.

The importance of the simplified schematic methods is apparent when one considers that the steady-state rate equation for Eq. 9-50 would have 6 terms in the numerator and 12 terms in the denominator.⁵¹ In the more complex case in which EAB breaks down to two products P and Q with a random order of release, the rate equation contains 672 terms in the denominator. In such cases it is worthwhile to enlist the help of a computer in deriving the equation.^{24,52–54}

The rapid equilibrium assumption. Rate equations for enzymes are often simplified if a single step, e.g., that of reaction of complex EAB to product in Eq. 9-50, is **rate limiting**.^{54a} If it is assumed that all reaction steps preceding or following the rate-limiting step are at equilibrium, the equation for random binding with a two-substrate and two-product reaction simplifies to one whose form is similar to that obtained for ordered binding (Eq. 9-44). In the absence of products P and Q Eq. 9-44 will correctly represent the steady-state rate equation corresponding to Eq. 9-50. However, this simplification may not be valid for a very rapidly acting enzyme.

Isotope exchange at equilibrium. Consider the reaction of substrates A and B to form P and Q (Eq. 9-51). If both reactants and both products are present with the enzyme and in the ratio found at equilibrium no net reaction will take place. However, the reactants and products will be continually interconverted under the action of the enzyme. Now if a small amount of



highly labeled reactant (A^* or B^*) is added, the rate at which isotope is transferred from the labeled reactant into one or the other of the products can be measured. In general, a label in one of the substrates will appear in only one of the products.

Figure 9-7A shows the rate of exchange of isotopically labeled glucose (glucose*) with glucose 6-phosphate as catalyzed by the enzyme hexokinase (Chapter 12). The exchange rate is plotted against the concentration of glucose 6-phosphate with the ratio $[\text{glucose}] / [\text{glucose 6-phosphate}]$ constant at 1/19, such that an equilibrium ratio for reactants and products is always

maintained. As can be seen from the graph, this exchange rate increases monotonically as substrate concentrations are increased. This is also true for the rate of ATP–ADP exchange. The fact that both exchange rates increase continuously indicates random binding of substrates.⁵⁵ The inequality of the two maximal exchange rates suggests that release of glucose 6-phosphate may be slower than that of ADP.

Figure 9-7B shows similar plots for lactate dehydrogenase.⁵³ In this case, after an initial rise (that is not regarded as significant), the pyruvate*–lactate exchange reaches a high constant value as the amount of pyruvate is increased (with a constant $[\text{pyruvate}] / [\text{lactate}]$ ratio of 1/35). However, the NAD^* –NADH exchange increases rapidly at first but then drops abruptly as the pyruvate and lactate concentrations continue to increase. This suggests an ordered mechanism (Eq. 9-43) in which NAD^+ and NADH represent A and Q, respectively, and pyruvate and lactate represent B and P. As the concentrations of B and P become very high, the

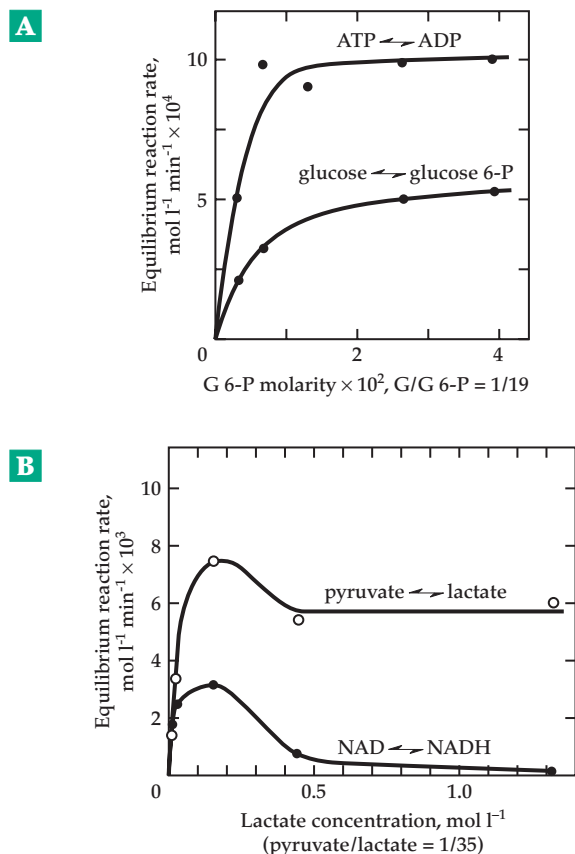


Figure 9-7 (A) Effect of glucose and glucose 6-phosphate concentrations on reaction rate of yeast hexokinase at equilibrium. Reaction mixtures contain 1–2.2 mM ATP, and 25.6 mM ADP at pH 6.5. From Fromm *et al.*⁵¹ (B) Effect of lactate and pyruvate concentrations on equilibrium reaction rates of rabbit muscle lactate dehydrogenase. Reaction mixtures contained 1.7 mM NAD^+ , and 30–46 μM NADH in Tris-nitrate buffer, pH 7.9, 25°C. From Silverstein and Boyer.⁵³

enzyme shuttles back and forth between EA and EQ, but these two complexes rarely dissociate to give free enzyme and A or Q. Hence, the A^*-Q exchange rate drops.

In other cases a label may be transferred from A into P or from B into Q. Information on such exchanges has provided a valuable criterion of mechanism which is considered in Chapter 12, Section B,4.

5. Kinetics of Rapid Reactions

The fastest steps in an enzymatic process cannot be observed by conventional steady-state kinetic methods because the latter cannot be applied to reactions with half-times of less than about 10 s. Consequently, a variety of methods have been developed^{18,56–59a} to measure rates in the range of 1 to 10^{13} s^{-1} .

Flowing substrates together. One of the first rapid kinetic methods to be devised consists of rapidly mixing two flowing solutions together in a special mixing device and allowing the resulting reaction mixture to move at a rate of several meters per second down a straight tube. At a flow velocity of 10 m s^{-1} a solution will move 1 cm in 10^{-3} s . Observations of the mixture are made at a suitable distance, e.g., 1 cm, and with various flow rates. Using spectrophotometry or other observation techniques, the formation or disappearance of a product or reactant can be followed. The special advantage of this technique is that observation can be made slowly. However, it may require large amounts of precious reactant solutions, e.g., those of purified enzymes.

In the **stopped flow** technique two solutions are mixed rapidly by the flow technique during a period of only 1–2 (or a few) milliseconds. A ram drives the solutions from syringes through a mixing chamber into an observation chamber. After the flow stops light absorption, fluorescence, conductivity, or other property, is measured. A means of rapid observation of changes during the reaction is essential. For example, light absorption may be measured by a photomultiplier with data being collected by a computer. Relaxation times as short as a few milliseconds or less can be observed in this way.^{59a,b}

Observing relaxation. Kinetic measurements over periods of tens of microseconds or less can be made by rapidly inducing a small displacement from the equilibrium position of a reaction (or series of reactions) and observing the rate of return (relaxation) of the system to equilibrium. Best known is the **temperature jump** method devised by Eigen and associates. Over a period of about 10^{-6} s a potential difference of $\sim 100 \text{ kV}$ is applied across the experimental solution. A rapid electrical discharge from a bank of condensers passes

through the solution (without any sparking) raising the temperature 2–10 degrees. All the chemical equilibria for which $\Delta H \neq 0$ are perturbed. If some property, such as the absorbance at a particular wavelength or the conductivity of the solution, is measured, very small relaxation times can be determined.

While it may not be intuitively obvious, if the displacement from equilibrium is small, the rate of return to equilibrium can always be expressed as a first-order process (e.g., see Eq. 9-13). In the event that there is more than one chemical reaction required to reequilibrate the system, each reaction has its own characteristic relaxation time. If these relaxation times are close together, it is difficult to distinguish them; however, they often differ by an order of magnitude or more. Therefore, two or more relaxation times can often be evaluated for a given solution. In favorable circumstances these relaxation times can be related directly to rate constants for particular steps. For example, Eigen measured the conductivity of water following a temperature jump¹⁸ and observed the rate of combination of H^+ and OH^- for which τ at 23°C equals $37 \times 10^{-6} \text{ s}$. From this, the rate constant for combination of OH^- and H^+ (Eq. 9-52) was calculated as follows (Eq. 9-53):



$$k = 1 / \{\tau([\text{OH}^-] + [\text{H}^+])\} = 1.3 \times 10^{11} \text{ M}^{-1} \text{ s}^{-1} \quad (9-53)$$

Pressure jump and electric field jump methods have also been used, as have methods depending upon periodic changes in some property. For example, absorption of ultrasonic sound causes a periodic change in the pressure of the system.

Rapid photometric methods. Another useful method has been to discharge a condenser through a flash tube over a period of 10^{-12} to 10^{-4} s , causing a rapid light absorption in a sample in an adjacent parallel tube. Following the flash, changes in absorption spectrum or fluorescence of the sample can be followed. The availability of intense lasers as light sources has made it possible to follow the results of light flashes of 5–10 picosecond duration and to measure extremely short relaxation times (Chapter 23).^{58,59}

Some results. Rapid kinetic methods have revealed that enzymes often combine with substrates extremely quickly,⁶⁰ with values of k_1 in Eq. 9-14 falling in the range of 10^6 to $10^8 \text{ M}^{-1} \text{ s}^{-1}$. Helix–coil transitions of polypeptides have relaxation times of about 10^{-8} s , but renaturation of a denatured protein may be much slower. The first detectable structural change in the vitamin A-based chromophore of the light-operated proton pump bacteriorhodopsin occurs in $\sim 5 \times 10^{-8} \text{ s}$, while a proton is pumped through the membrane in

$\sim 10^{-4}$ s.⁶¹ Interconversion between chair and boat forms of cyclohexane derivatives may have $\tau \sim 10^{-5}$ s at room temperature, while rotation about a C–N bond in an amide linkage may be very slow with $\tau \sim 0.1$ s. The nonenzymatic hydration of the aldehyde pyridoxal phosphate via Eq. 13-1 occurs with $\tau = 0.01 - 0.1$ s, depending upon the pH.⁵⁹

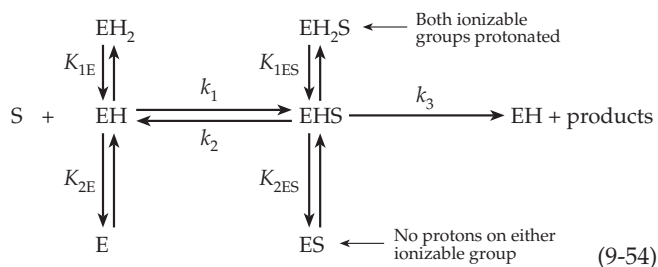
6. Cryoenzymology

An alternative to studying rapid reactions is to cool enzymes to a subzero temperature (down to -100°C) where reactions proceed more slowly.^{60,62} The enzyme must be dissolved in a suitable “cryosolvent,” often containing 50–80% by volume of an organic solvent or solvents such as methanol, ethanol, dimethyl sulfoxide, dimethyl formamide, or ethylene glycol. Those containing methanol are especially desirable because of their low viscosities. Kinetics can be studied by various spectroscopic methods and stopped-flow, temperature-jump, and other rapid-reaction techniques can be applied. One goal of cryoenzymology is to stabilize otherwise unstable intermediates. X-ray crystallographic measurements can also be made at these low temperatures and it should be possible to observe structures of intermediate ES complexes. A problem is that the forms of the complexes stabilized may be side-products rather than true intermediates. However, as discussed in Chapter 3, a combination of low-temperature Laue X-ray diffraction and a laser-induced temperature jump may be feasible.

7. The Effect of pH on Enzymatic Action

Because proteins contain many acidic and basic groups it is not surprising that the activity of enzymes often varies strongly with pH. However, it is usually found that the state of protonation of only a few groups has a strong effect on activity. This is understandable because most ionized groups are on the outer surface of protein molecules and most are not close to an active site. Protonation or deprotonation of those groups will hardly ever have a major influence on events in the active site. Often only one or two ionizable groups have highly significant effects.

For many enzymes a plot of $V_{\max}$ against pH is a bell-shaped curve (Fig. 9-8). The **optimum** rate is observed at some intermediate pH, which is often, but not always, in the range of pH 6–9. This type of curve can be interpreted most simply by assuming the presence in the active site of two ionizable groups and three forms of the enzyme with different degrees of protonation: E, EH, and EH_2 .



Let us designate the acid dissociation constants (K_a) for the two groups in the enzyme as $K_{1\text{E}}$ and $K_{2\text{E}}$ and those of the ES complex $K_{1\text{ES}}$ and $K_{2\text{ES}}$. However, as discussed in Chapter 6, Section E,2 these consecutive K_a values cannot necessarily be assigned to single groups. They may belong to a system of interacting groups. Each may be the sum of more than one microscopic K_a and there may be extensive tautomerism within the active site. The rate constants k_1 , k_2 , and k_3 define the rates of formation and breakdown of the ES complex.

If it is assumed that only form EHS reacts to give products, the bell-shaped curves of Fig. 9-8 are obtained. The frequent observation of such curves supports the model of Eq. 9-54. It also suggests that the two ionizable groups may be intimately involved in catalysis, one as an acid and the other as a conjugate base (see Section E,5).

For the simple case illustrated in Eq. 9-54, the pH dependence of the initial maximum velocity, the apparent Michaelis constant, and $V_{\max}/K_m$ are given by Eqs. 9-55 to 9-57.

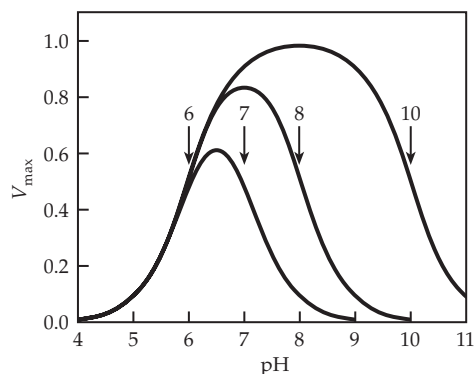


Figure 9-8 Expected dependence of $V_{\max}$ on the pH according to Eq. 9-54 with $k_3[\text{E}_t] = 1$, $\text{p}K_{1\text{ES}} = 6$, and $\text{p}K_{2\text{ES}}$ as given on the graph. After Alberty.⁶³ Computer-drawn graph courtesy of Carol M. Metzler.

$$V_{\max} = \frac{k_3[E]_t}{(1 + [H^+]/K_{1ES} + K_{2ES}/[H^+])} = \frac{k_3[E]_t}{F_{2ES}} \quad (9-55)$$

$$K_m = \frac{(k_2 + k_3)}{k_1} \cdot \frac{(1 + [H^+]/K_{1E} + K_{2E}/[H^+])}{(1 + [H^+]/K_{1ES} + K_{2ES}/[H^+])} = K_m F_{2E}/F_{2ES} \quad (9-56)$$

$$\frac{V_{\max}}{K_m} = \frac{k_1 k_3}{(k_2 + k_3)} \cdot \frac{1}{(1 + [H^+]/K_{1E} + K_{2E}/[H^+])} = \frac{k_3[E]_t}{K_m F_{2E}} \quad (9-57)$$

The denominators (which are the Michaelis pH functions given by the first three terms of Eq. 3-6) represent the fraction of enzyme or of ES complex in the monoprotonated state. The pH dependence of enzymatic action is often more complex than that shown in Fig. 9-8 and given by the foregoing equations. However, it is easy to write Michaelis pH functions (see Chapter 3) for enzymes with any number of dissociable groups in both E and ES and to write appropriate equations of the type of

BOX 9-B GROWTH RATES OF CELLS

How do we correctly describe the rate at which a cell grows? Consider bacteria in their rapidly growing “log phase.”^a Each cell divides after a fixed length of time, the **doubling time**. For *E. coli* this may be as short as 17 min in the early stages of growth but becomes somewhat longer as time goes on. A mean value of about 26 min. for *E. coli* at 37°C is typical. In contrast, the doubling time for mammalian cells in tissue culture is often about one day.

If a given volume of culture contains N_0 cells initially, the number N_n after n cell divisions will be:

$$N_n = 2^n N_0$$

From this equation we can calculate that a single bacterium with a generation time of 20 min can produce 2^{144} cells in 48 h of exponential growth. The **exponential growth rate constant k** is equal to the number of doublings per unit time. Thus, k is the reciprocal of the doubling time. It is easy to show that the number of bacteria present at time t will be given by the following equation.

$$N_t = 2^{kt} N_0$$

This can be rewritten in a form that can be used to determine k by counting the number of bacteria at zero time and at time t .

$$kt = \log_2(N_t/N_0) = \log_{10}(N_t/N_0) / 0.301$$

Another way of expressing the growth is to equate the rate of increase of the number of bacteria with a growth rate constant μ multiplied by the number of bacteria present at that time.

$$dN/dt = \mu N$$

This is a general equation for an autocatalytic reaction and N could be replaced with a concentration, for example, the total content of cellular matter per

liter in the medium. From the two preceding equations the following can be derived.^a

$$\mu = k \ln 2 = 0.69k$$

When bacteria are transferred to new medium there is usually a lag before exponential growth begins. Exponential growth eventually stops and the culture enters **stationary phase**, which is usually followed by relatively rapid death of cells in the culture. It is often desirable to study cell growth under conditions of **continuous cultivation** in which a constant generation time is maintained but the density of cells in the medium does not increase. This can be done with a simple device known as the **chemostat**.^{a,b} A culture vessel containing bacteria is stirred to ensure homogeneity. Fresh culture medium continuously enters the vessel from a reservoir and part of the content of the vessel, suspended bacteria included, is continuously removed through another tube. The bacterial population in the vessel builds up to a constant level and can be maintained at the same level for relatively long periods of time.

Here are some other statistics on cell growth. Bacteria growing exponentially expand their linear dimensions by 1.5 nm and synthesize 1.6×10^7 Da of new cell material in one second. This is equivalent to ~1000 small proteins and includes 23 ribosomes and 3000 base pairs of DNA at each growing point. The much larger HeLa human tumor cell grows by 0.13 nm/s but makes 4.6×10^8 Da/s of material.^c

^a Stanier, R. Y., Douderoff, M., and Adelberg, E. A. (1970) *The Microbial World*, 3rd ed., Prentice-Hall, Englewood Cliffs, New Jersey (pp. 298–324)

^b Smith, H. L., and Waltman, P., eds. (1995) *The Theory of the Chemostat; Dynamics of Microbial Competition*, Cambridge University Press, London and New York

^c Pollard, E. C. (1973) in *Cell Biology in Medicine* (Bittar, E. E., ed), pp. 357–377, Wiley, New York

Eqs. 9-55 to 9-57. Bear in mind that if the **free substrate** contains groups dissociating in the pH range of interest, a Michaelis pH function for the free substrates will also appear in the numerator of Eq. 9-56. If the pH dependence of the enzyme is regulated by a conformational change in the protein, there may be a cooperative gain or loss of more than one proton and the Michaelis pH function must reflect this fact. This can sometimes be accomplished by addition of a term related to Eq. 7-45. For more information see Dixon and Webb,⁶⁴ Cleland,^{65,66} or Kyte.⁶⁷

Plots of $\log V_{\max}$ (or log-specific activity) and $-\log K_m$ or $\log (V_{\max} / K_m)$ versus pH yield graphs of the type shown in Fig. 9-9. The curved segments of the graphs that extend for about 1.5 pH units on either side of each pK_a are asymptotic to straight lines of slope 1 when a single proton is involved or of a higher slope for multiple cooperative proton dissociation. The straight lines can be extrapolated and intersect at the pK_a values. (However, it is better to fit a complete curve of theoretically correct shape to the points.) Note that the curved line always passes below or above the intersection point at the value of $\log 2 = 0.30$ except in the case of cooperative proton dissociation when it is closer.

Upward turns in the curve of $\log K_m$ vs pH correspond to pK_a values in the free enzyme or substrate and downward turns to pK_a values in ES. This approach to analysis of pH dependence has been adopted widely but often incorrectly. For example, many published curves have very sharp bends in which the curved portion covers less than 3 pH units and the curve is much closer than 0.30 to the extrapolated point. This suggests cooperative proton binding and an *apparent* pK_a that is related to $\bar{K}$ of Eq. 7-21. Cooperativity is always a possibility if a conformational change in the protein is involved.⁶⁷

The simple treatment given above is based on the assumption that all proton dissociations are rapid compared to k_{cat} that enzyme in only one state of protonation binds substrate, and that ES in only one state of protonation yields products. These assumptions are not always valid. It also assumes that both binding and dissociation of substrate are rapid, that is, to use Cleland's terminology the substrate is not "sticky." For a sticky substrate that dissociates more slowly than it reacts to form products ($k_3 > k_2$; Eq. 9-54), the values of pK_{1E} will be lowered and pK_{1E} of Eq. 9-53 will be raised by $\log (1 + k_3 / k_2)$.^{65,66} In addition to the articles by Cleland, other detailed treatments of pH effects have been prepared by Brocklehurst and Dixon⁶⁹ and Tipton and Dixon.⁷⁰

Fumarate hydratase (fumarase), which is discussed in Chapter 13, catalyzes the reversible hydration of fumaric acid to malic acid (Eq. 13-11). It was one of the first enzymes whose pH dependence was studied intensively. A bell-shaped pH dependence

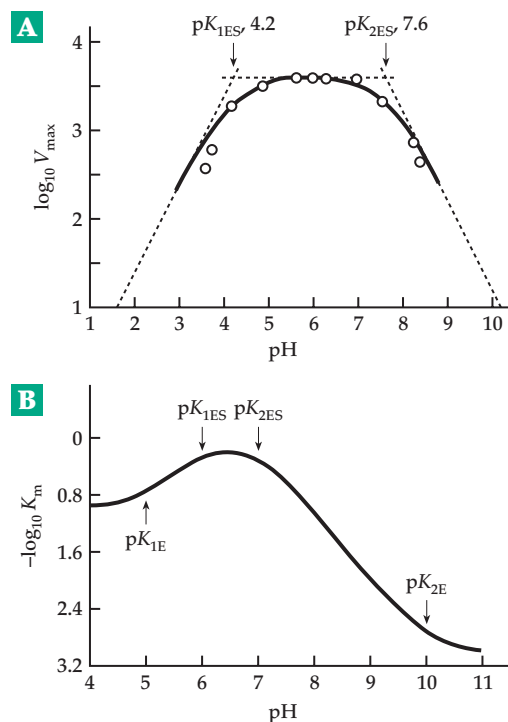


Figure 9-9 (A) Plot of $\log V_{\max}$ vs pH for a crystalline bacterial α -amylase. From Ono, et al.⁶⁸ (B) Theoretical curve of $\log K_m$ vs pH for Eq. 9-56 with $pK_{1E} = 5$, $pK_{2E} = 10$, $pK_{1ES} = 6$, and $pK_{2ES} = 7$. Courtesy of C. Metzler.

for both forward and reverse reactions was observed by Alberty and coworkers⁶³ and, using Eqs. 9-54 and 9-55, the two apparent pK_a values were measured.

B. Inhibition and Activation of Enzymes

The action of most enzymes is inhibited by many substances. Inhibition is often specific, and studies of the relationship between inhibitor structure and activity have been important to the development of our concepts of active sites and of complementarity of surfaces of biomolecules. Inhibition of enzymes is also the basis of the action of a very large fraction of important drugs. Inhibition may be **reversible** or **irreversible**, the latter leading to permanent inactivation of the enzyme. Often, but not always, irreversible inhibition is preceded by reversible binding of the inhibitor at a complementary site on the enzyme surface.

1. Competitive Inhibitors

Inhibitors with close structural similarities to a substrate tend to bind to the substrate site. In truly competitive inhibition, substrate and inhibitor not only

compete for the same site but also their binding is reversible and mutually exclusive. The affinity of the inhibitor for the enzyme is expressed quantitatively through the **inhibition constant** K_i which is the *dissociation constant of the enzyme inhibitor complex EI*:

$$K_i = [E][I] / [EI] \quad (9-58)$$

Using the steady state assumption for the mechanism shown in Eq. 9-14, and writing a mass balance equation that includes not only free enzyme and ES but also EI we obtain an equation relating rate to substrate concentration. It is entirely analogous to Eq. 9-15 but K_m is replaced by an apparent Michaelis constant, K'_m :

$$K'_m = K_m \left(1 + \frac{[I]}{K_i} \right) \quad (9-59)$$

The relationships between v and $[S]$ and between $1/v$ and $1/[S]$ are as follows:

$$\frac{v}{[S]} = \frac{V_{\max}}{K'_m} - \frac{v}{K'_m} \quad (9-60)$$

$$\frac{1}{v} = \frac{1}{V_{\max}} + \frac{K_m}{V_{\max}[S]} \left(1 + \frac{[I]}{K_i} \right) \quad (9-61)$$

A commonly used test for competitive inhibition is to plot $1/v$ vs $1/[S]$ (Eq. 9-61), both in the absence of inhibitor and in the presence of one or more fixed concentrations of I. The result, in each case, is a family of lines of varying slope (Fig. 9-10) that converge on one of the axes at the value $1/V_{\max}$. We see that the maximum velocity is unchanged by the presence of inhibitor. If sufficient substrate is added, the enzyme will be saturated with substrate and the inhibitor cannot bind. The value of K_i can be calculated using Eq. 9-61 from the change in slope caused by addition of inhibitor.

The effect of a fixed concentration of a competitive inhibitor on a plot of v against $\log [S]$ (Fig. 9-11) is to shift the curve to the right, i.e., toward higher values of $[S]$, but without any change in shape (or in the value of $V_{\max}$).

Another plot, introduced by Dixon,⁷¹ is that of $1/v$ versus $[I]$ at two or more fixed substrate concentrations. The student should be able to demonstrate that this plot contains a family of straight lines that intersect at a point to the left of the origin with coordinates $[I] = -K_i$ and $1/v = 1/V_{\max}$. This plot may fail to distinguish certain types of inhibition discussed in the next section.⁷²

Competitive inhibition is extremely common and has great significance for metabolic control and for the effects of drugs and of poisons. Simple ions are often competitive inhibitors. Since many biochemical sub-

stances carry negative charges, anions such as Cl^- , HCO_3^- , HPO_4^{2-} , and acetate⁻ frequently act as competitive inhibitors.

Two special classes of competitive inhibitors are characterized by **slow binding** and **slow, tight binding** to active sites.⁷³⁻⁷⁷ Among the very tight-binding inhibitors, which may also be slow to dissociate from active sites, are transition state inhibitors discussed in Section D,1.

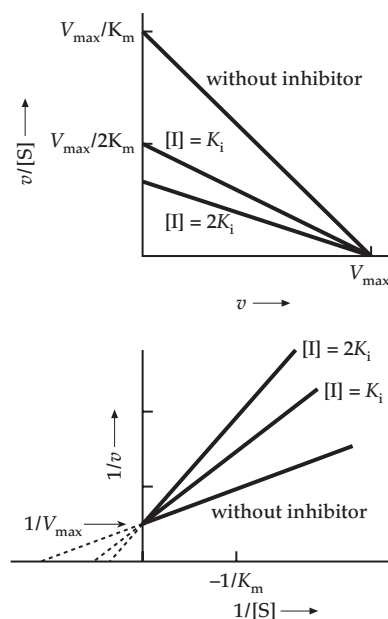


Figure 9-10 Effect of a competitive inhibitor on the Eadie-Hofstee plot (top) and on a double reciprocal plot (bottom). The apparent K_m (Eq. 9-59) is increased by increasing $[I]$, but $V_{\max}$ is unchanged.

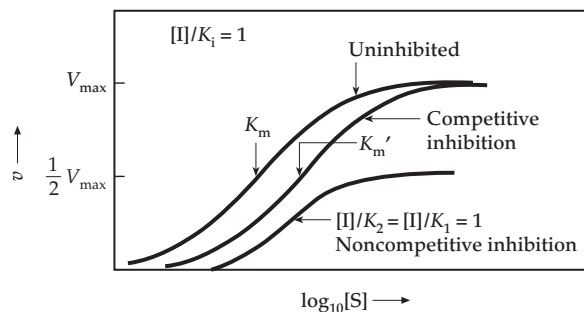


Figure 9-11 Plots of v vs. $\log [S]$ for competitive and noncompetitive inhibition.

2. Noncompetitive Inhibition and Activation; Allosteric Sites

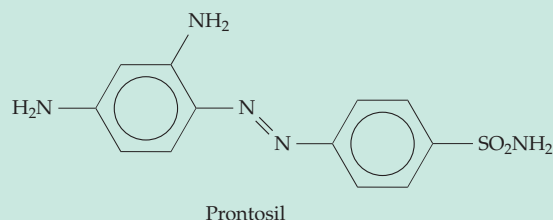
If an inhibitor binds not only to free enzyme but also to the enzyme substrate complex ES, inhibition is **noncompetitive**. In this case, S and I do not mutually exclude each other and both can be bound to the enzyme at the same time. Why does such an inhibitor slow an enzymatic reaction? In most instances, the structure of the inhibitor does not show a close similarity to that of substrate, which suggests that the binding of inhibitors is at an **allosteric site**, that is, at a site other than that of the substrate. The inhibition of the enzyme may result from a distortion of the three-dimensional structure of the enzyme which is caused by the binding of the inhibitor. This distortion may be

transmitted to the active site even though the inhibitor binds far from that site. In some cases two distinctly different conformers of the protein may exist, one binding substrate well and the other binding inhibitor well (Fig. 9-12). In other instances the bound inhibitor may interfere with the catalytic action by partially overlapping the active site. In either case the ES complex reacts to give product in a normal way, but the ESI complex reacts more slowly or not at all.

Binding of a substance to an allosteric site sometimes has the effect of *increasing* the activity of an enzyme rather than inhibiting it. This may occur because the **activator** stabilizes the conformation that binds substrate best (Fig. 9-12). The quantitative treatment of such activation is similar to that of inhibition; allosteric inhibitors and activators are often considered together and are referred to as **modifiers** or

BOX 9-C THE SULFONAMIDES AS ANTIMETABOLITES

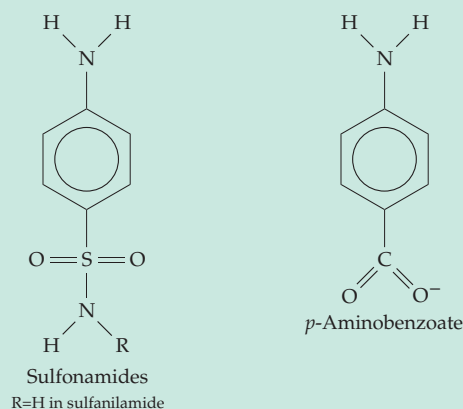
The development of the “sulfa drugs,”^{a-c} derivatives of **sulfanilamide**, originated with studies of the staining of protozoal parasites by Paul Ehrlich. In 1932 it was shown that the red dye 2,4-diaminoazobenzene-4'-sulfonamide (Prontosil) dramatically cured systemic infections by gram-positive bacteria. Subsequent studies revealed that bacteria converted



the azo dye to sulfanilamide, a compound with strong bacteriostatic activity; that is, it inhibited bacterial growth without killing the bacteria. Although sulfanilamide had been used in large quantities since 1908 as an intermediate in synthesis of dyes, its potential as an antibacterial agent had not been recognized.

In 1935, D. D. Woods found that the growth inhibition of sulfanilamide was reversed by yeast extract.^d From this source, in 1940 he isolated **p-aminobenzoic acid** and demonstrated that the inhibitory effect of 3×10^{-4} M sulfanilamide was overcome by 6×10^{-8} M p-aminobenzoate. The relationship between the two compounds was shown to be strictly competitive. If the sulfanilamide concentration was doubled, twice as much p-aminobenzoate was required to reverse the inhibition as before. These facts led to the formulation by Woods and by P. Fildes (in 1940) of the

antimetabolite theory.^{d,e} It was proposed that p-aminobenzoate was needed by bacteria and that sulfanilamide competed for a site on an enzyme designed to act on p-aminobenzoate. We now know that the idea was correct and that the enzyme on which the competition occurs catalyzes the synthesis of dihydropteroic acid (Fig. 25-19), a precursor to folic acid.



While sulfanilamide itself is somewhat toxic, a variety of related drugs of outstanding value have been developed. Over 10,000 sulfonamides and related compounds have been tested for antibacterial action.

^a Bardos, T. J. (1974) *Top. Curr. Chem.* **52**, 63–98

^b Gale, E. F., Cundliffe, E., Reynolds, P. E., Richmond, M. H., and Waring, M. J. (1972) *The Molecular Basis of Antibiotic Action*, Wiley, New York

^c Shepherd, R. G. (1970) in *Medicinal Chemistry*, 3rd ed. (Burger, A., ed), pp. 255–304, Wiley (Interscience), New York

^d Woods, D. D. (1940) *Brit. J. Exp. Pathol.* **21**, 74–90

^e Fildes, P. (1940) *Lancet* **1**, 955–957

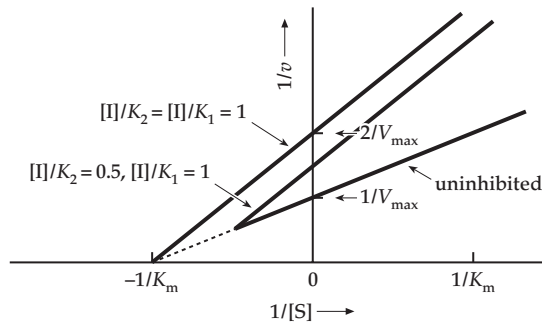
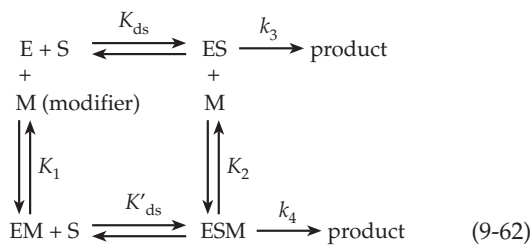


Figure 9-12 Double reciprocal plots for two cases of non-competitive inhibition.

effectors. A general scheme^{78,79} is given by Eq. 9-62.



Here K_1 and K_2 are equilibrium constants for dissociation of M from EM and ESM, respectively, while K_{ds} and K'_{ds} are the dissociation constant of ES and of ESM (to S and EM), respectively. Notice that, K'_{ds} is not independent of the others (Eq. 9-63):

$$K'_{ds} = K_{ds} K_2 / K_1 \quad (9-63)$$

We have already considered competitive inhibition which is obtained when $K_2 = 0$ (and therefore $K'_{ds} = 0$). For this case, M is always an inhibitor and no activation is possible. Notice that the inhibition will appear competitive even if M binds at an allosteric site as in Fig. 9-13 or if the inhibited form does not react at all with substrate. Noncompetitive inhibition will be observed if ESM is formed but does not react, i.e., if $k_4 = 0$. Then the rate equation in reciprocal form will be given by Eq. 9-64.

$$\frac{1}{v} = \frac{1}{V_{\max}} \left(1 + \frac{[I]}{K_2} \right) + \frac{K_m}{V_{\max}[S]} \left(1 + \frac{[I]}{K_1} \right) \quad (9-64)$$

We see that $1/V_{\max}$ is multiplied by a term containing $[I]$ and K_2 . It is a characteristic of noncompetitive inhibition that the maximum velocity is decreased from that observed in the absence of the inhibitor. Also, no matter how high the substrate concentration,

inhibition cannot be reversed completely.

Figure 9-12 shows a plot of $1/v$ against $1/[S]$ at a series of fixed values of $[I]$. For the case that $K_1 = K_2$ (**classical noncompetitive inhibition**), a family of reciprocal plots that intersect on the horizontal axis at a value of $-1/K_m$ is obtained. On the other hand, if K_1 and K_2 differ (the general case of noncompetitive inhibition), the family of curves intersect at some other point to the left of the vertical axis and, depending upon the relative values of K_1 and K_2 , either above or below the horizontal axis. The example illustrated is for $K_2 = 0.5K_1$; that is, for the binding of M to ES being twice as strong as that to E.

Figure 9-11 shows inhibition data for both the noncompetitive and the competitive cases plotted vs $\log[S]$. The shift of the midpoint to the right in each case reflects the tendency of the inhibitor to exclude the substrate from binding, while the lowered value of the maximum velocity in the case of noncompetitive inhibition results from the failure of the substrate to completely displace the inhibitor from the enzyme

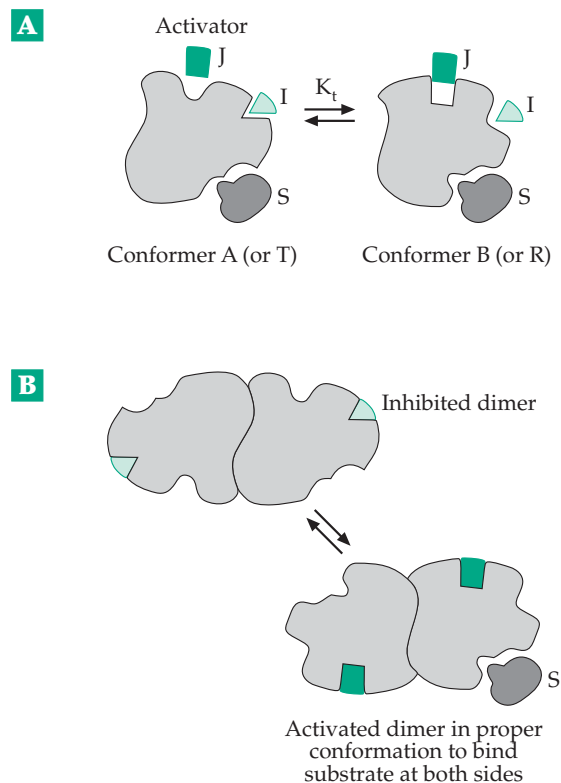


Figure 9-13 (A) An enzyme with binding sites for allosteric inhibitor I and activator J. Conformer A binds inhibitor I strongly but has little affinity for activator J or for substrate S. Conformer B binds S and catalyzes its reaction. It also binds activator J whose presence tends to lock the enzyme in the "on" conformation B. Conformers A and B are designated T and R in the MWC model of Monod, Wyman, and Changeux.⁸⁰ (B) Inhibited and activated dimeric enzymes.

even as $[S]$ becomes very high. If an inhibitor binds only to ES and not to E, i.e., $K_1 = 0$, a family of parallel double-reciprocal plots of $1/v$ vs $1/[S]$ will be obtained. This case is referred to as **uncompetitive inhibition**. Multisubstrate enzymes with either ordered sequential or ping-pong mechanisms also often give parallel line plots with inhibitors.

Uncompetitive inhibitors of liver alcohol dehydrogenase (Chapter 15) could be used to treat cases of poisoning by methanol or ethylene glycol.^{81–83} The aim is to prevent rapid oxidation to the toxic acids HCOOH and HOCH₂COOH, which lower blood pH, while the alcohols are excreted. Uncompetitive inhibitors have an advantage over competitive inhibitors as therapeutic agents in that the inhibition is not overcome when the substrate concentration is saturating.⁸⁴

If rate constant k_4 of Eq. 9-62 is not zero we may observe either inhibition or activation. If $k_3 = k_4$ the effect of a modifier will be to alter the apparent K_m by either increasing it (inhibition) or decreasing it (activation). The maximum velocity will be unchanged. Monod *et al.*⁸⁰ referred to enzymes showing such behavior as **K systems**. On the other hand, if $K_1 = K_2$ and k_3 differs from k_4 we have a purely **V system**. In the general case a modifier affects both the apparent K_m and $V_{\max}$. Although in the foregoing discussion it has been assumed that activators bind at allosteric sites, as was pointed out in Section A,11, ions and other small molecules (substrate surrogates) that act as competitive inhibitors at high concentrations may be activators at low concentrations.⁴⁰

Activation of enzymes by specific metallic ions is often observed. In many instances the metallic ion is properly regarded as a *second substrate* which must bind along with the first substrate before reaction can occur. Alternatively, the complex of the organic substrate with the metal ion can be considered the “true substrate.” Thus, many enzymes act upon the magnesium complex of ATP (Chapter 12). The enzymes can either be regarded as three-substrate enzymes requiring $Mg^{2+} + ATP^{4-}$ and another substrate or as two-substrate enzymes acting upon $ATPMg^{2-}$ and a second substrate.

3. Inhibitors in the Study of Mechanisms

A substrate analog will frequently inhibit only one of the two forms of a multisubstrate enzyme with a ping-pong mechanism.^{1,72} Reciprocal plots made for various inhibitor concentrations consist of a family of parallel lines reminiscent of uncompetitive inhibition. Observation of such parallel line plots can support a ping-pong mechanism for an enzyme but cannot prove it because in some cases parallel lines are observed for inhibition of enzymes acting by an ordered sequential mechanism. The following question arises naturally for any ordered bimolecular reaction (Eq. 9-43): Of the

two substrates required by the enzymes, which one binds to the enzyme first? If the concentration of one substrate is kept constant while varying concentrations of an inhibitory analog of that substrate are added and $1/v$ is plotted against the reciprocal of the concentration of the other substrate, parallel lines are obtained if, and only if, the substrate of fixed concentration is B, the substrate is added second in the binding sequence, and if I is its analog. The substrate binding first (A) is the one whose concentration was varied in the experiment.

Product inhibition (Section A,12) can also provide information about mechanisms. For example, if $1/v$ is plotted against $1/[A]$ in the presence and absence of the product Q, the product will be found to compete with A and to give a typical family of lines for competitive inhibition. On the other hand, a plot of $1/v$ vs $1/[B]$ in the presence and absence of Q will indicate noncompetitive inhibition if the binding of substrates is ordered (Eq. 9-43). In other words, only the A–Q pair of substrates are competitive. Product inhibition is also observed with enzymes having ping-pong kinetics (Eq. 9-47) as a result of formation of nonproductive complexes.

4. Allosteric Effectors in the Regulation of Enzyme Activity

The binding of a substance at an allosteric site with the induction of a conformational change forms the basis for many aspects of regulation. The term **allostery** (allosterism) usually refers to the effects of allosteric modifiers, which may be either inhibitors or activators, on oligomeric enzymes. However, as we have already seen (Eq. 9-62), monomeric enzymes may also be subject to allosteric regulation by modifiers. Consider a monomer that contains binding sites for substrate, inhibitor, and activator and which exists in conformations A and B as in Fig. 9-13. Let us assume (see Eq. 7-31) that conformer B binds both substrate and activator well but that it binds inhibitor poorly or not at all. On the other hand, A binds inhibitor well but binds substrate and activator poorly. This simple combination of two conformers with different binding properties provides a means by which enzymes can be turned “on” or “off” in response to changing conditions.

If an inhibitory substance builds up to a high concentration within a cell, it binds to conformer A; if the inhibitor concentration is high enough, virtually all of the enzyme will be locked in the inactive conformation A. The enzyme will be turned off or at least reduced to a low activity. On the other hand, in the presence of a high concentration of activator the enzyme will be turned on because it is locked in the B conformation. The relative concentrations of inhibitor, activator, and substrate within a cell at any given time

will determine what fraction of the enzyme is in active conformation B. It is this interplay of inhibitory and activating effects that provides the basis for much of the regulation of cell chemistry.^{80,85}

The effects of inhibitors or activators on the kinetics of the monomeric enzyme of Fig. 9-13 can be described by Eq. 9-62 to 9-64. Separate terms for both inhibition and activation can be included. The equilibrium between the two conformers can also be indicated explicitly according to Eq. 7-30. However, for monomeric enzymes it is usually not profitable to try to separate the two constants K_t and K_{bx} which describe the conformational change and binding of substrate or activator, respectively, in Eq. 7-30.

Most intracellular enzymes are oligomeric, and the binding of allosteric effectors leads to additional interesting effects. Binding constants or dissociation constants must be defined for both inhibitor and activator to both conformers A and B. Since all species must be taken into account in the mass balance, the equations are complex. However, the Monod–Wyman–Changeux (MWC) model (Chapter 7) gives a relatively simple picture. The saturation curve for an oligomeric enzyme following this model may be derived from Eq. 7-39 and is given by the following expression:

$$Y = \frac{Lc\alpha(1 + c\alpha)^{n-1} + \alpha(1 + \alpha)^{n-1}}{L(1 + c\alpha)^n + (1 + \alpha)^n} \quad (9-65)$$

Here, L is the **allosteric constant** which is given (for a dimer) by Eq. 7-36. The constant c is the ratio of dissociation constants K_{BS} and K_{AS} for the two conformers:

$$c = K_{BS} / K_{AS} \quad (9-66)$$

Notice that in this chapter *dissociation* constants (K_d) of ES complexes are being used, whereas the equations of Chapter 7 are all written in terms of association constant (K_f). The parameter α is defined as follows, where K_{BS} equals $1/K_{bx}$ of Chapter 7.

$$\alpha = [S] / K_{BS} \quad (9-67)$$

To take account of effects of inhibitor and activator the *ratios* of dissociation constants of I from BI and AI and of activator J from BJ and AJ are defined as in Eq. 9-68. Likewise, “normalized concentrations” of I and J are defined (Eq. 9-69) as β and γ , respectively.

$$d = \frac{K_{BI}}{K_{AI}} > 1 \quad e = \frac{K_{BJ}}{K_{AJ}} < 1 \quad (9-68)$$

$$\beta = [I]/K_{AI} \quad \gamma = [J]K_{BJ} \quad (9-69)$$

According to the MWC model, in the presence of inhibitor and activator at normalized concentrations β and γ an enzyme will still follow Eq. 9-65, but the allosteric constant L will be replaced by an apparent allosteric constant L' (Eq. 9-70).⁸⁶ Figure 9-14 shows plots of Y vs. $\log \alpha$ for two different values of L' for a tetramer with a specific value assumed for c . In both

$$L' = L \left[\frac{(1 + \beta d)(1 + \gamma e)}{(1 + \beta)(1 + \gamma)} \right]^n \quad (9-70)$$

cases, Y approaches 1 as $\log \alpha$ increases, but since we are dealing with noncompetitive inhibition at high values of L' , much of the enzyme will be in the T(A) conformation at saturation.

Noncompetitive inhibition cannot be completely reversed by very high substrate concentrations. Monod *et al.* defined for an allosteric enzyme a **function of state** $\bar{R}$ (Eq. 9-71) which is the fraction of total enzyme in the $\bar{R}$ (B) conformation:

$$\begin{aligned} \bar{R} &= \frac{(1 + \alpha)^n}{L(1 + c\alpha)^n + (1 + \alpha)^n} \\ &= \text{function of state} \end{aligned} \quad (9-71)$$

In a K system (Section B,2) it is the value of $\bar{R}$ that determines the velocity with which an enzyme reacts. Figure 9-14 also shows $\bar{R}$ as a function of $\log [\alpha]$. Note that when L' is low $\bar{R}$ does not approach zero even when $[S] \rightarrow 0$. In other words, the enzyme is never completely turned off, just as when L' is high the enzyme is never completely turned on.

Figure 9-14 may be compared with Fig. 9-11 which shows similar curves for noncompetitive inhibition of a monomeric enzyme. The significant difference between the two figures is that saturation of the oligomeric enzyme occurs over a narrower concentration range than does that of the monomer, i.e., saturation of the oligomeric enzyme, especially in the presence of inhibitor, is *cooperative*. Note that cooperative binding of substrate requires that the free enzyme be largely in conformation T (A), as it is in the presence of an inhibitor. Allosteric interactions between two identical molecules, whether of substrate or of effector, are described by Monod *et al.*⁸⁰ as **homotropic interactions**. Such interactions lead to cooperativity or anticooperativity in binding. Allosteric interactions between two different molecules, e.g., a substrate and an activator are designated **heterotropic**.

For many enzymes the MWC model is unrealistically simple. The more general treatment of binding equilibria given in Chapter 7 may be applicable. However, in addition to K systems there are V systems in which a conformational change alters the maximum velocity (see Eq. 9-62)⁸⁷ and sometimes both substrate

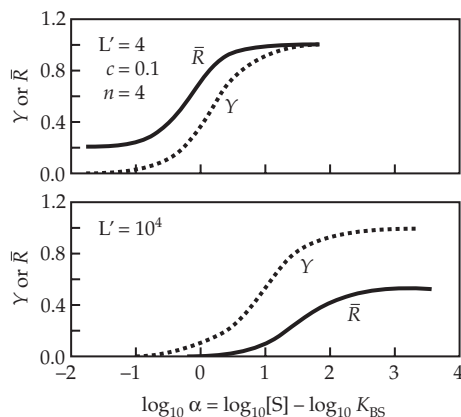
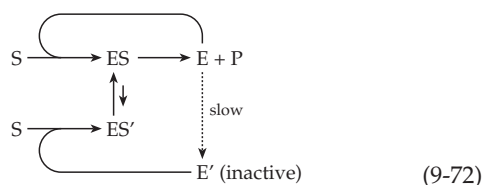


Figure 9-14 Fractional saturation Y and “function of state” $\bar{R}$ for hypothetical tetrameric enzymes following the MWC model. Curves are calculated for two different values of the apparent allosteric constant L' (Eq. 9-70) and for $c = 0.1$ (Eq. 9-66). After Rubin and Changeux.⁸⁶

affinity and maximum velocity.

The fact that data can be fitted to an equation is not proof that a mathematical model is correct; other models may predict the results just as well. For example, Jenkins has shown that the presence of ions that act as substrate surrogates (Section A,11) can produce cooperative or anticooperative binding curves for substrates.^{40,88} Rabin⁸⁹ suggested the following explanation of cooperativity for a monomeric enzyme with a single substrate-binding site. In its active conformation E the enzyme reacts with substrate rapidly and the ES complex yields product rapidly as in the upper loop of Eq. 9-72. However, a slow conformational change interconverts E and E' , a less active form with much lower affinity for substrate.



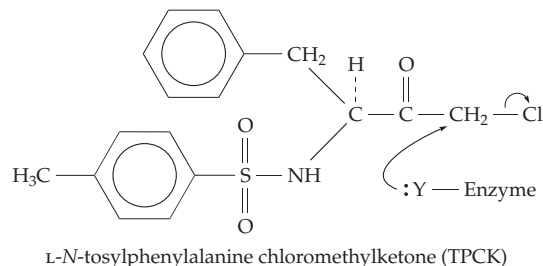
At the same time, the complex ES' , if formed, can equilibrate with ES and thereby alter the conformational state of the protein. At low substrate concentrations E' will predominate and the enzymatic activity will be low. At high substrate concentrations most E will exist as ES and after release of product will tend to remain in the active conformation long enough to bind and act on another substrate molecule. This and other kinetic models of cooperativity are discussed by Newsholme and Start.⁹⁰

The physiological significance of cooperative binding of substrates to enzymes is analogous to that of cooperative binding of oxygen by hemoglobin which provides for more efficient release of oxygen to tissues. However, in the presence of excess activator an enzyme is locked in the R (B) conformation and no cooperativity is seen in the binding of substrate. In this case, each binding site behaves independently. On the other hand, *there will be strong cooperativity in the binding of the activator*. The result is that control of the enzyme is sensitive to a higher power than the first of the activator concentration. Likewise, the turning off of the enzyme is more sensitive to inhibitor concentration as a result of cooperative binding of the latter. It seems likely that the evolution of oligomeric enzymes is at least partly a result of the greater efficiency of control mechanisms based on cooperative binding of effectors.

5. Irreversible Inhibition of Enzymes

Some of our most effective drugs, such as penicillin (Box 20-G) are irreversible inhibitors of specific enzymes. Such inhibitors are also of practical importance to biochemists who wish to inhibit specific enzymes such as proteases that might otherwise destroy proteins they are studying. Transition state analogs (Section E,1) and slow-binding inhibitors (Section B,1) may appear to inactivate an enzyme irreversibly because they bind tightly and dissociate slowly. However, true irreversible inhibitors, such as oxidizing agents or alkylating agents, cause a more permanent chemical modification, usually at or near the active site. Two groups of irreversible inhibitors are of special interest: affinity labeling agents and enzyme-activated inhibitors.

Affinity labeling agents (active-site directed inactivating reagents) have two essential properties: *a high affinity for the active site* of the specific enzyme to be inhibited and *the presence of a chemically highly reactive group* which can attack a functional group in a protein. A good example is provided by derivatives of chloromethylketones, which are reactive alkylating agents. One of these, *N*-tosylphenylalanine chloromethyl ketone (TPCK), is a potent inhibitor of chymotrypsin. In addition to the chloromethyl ketone group it contains

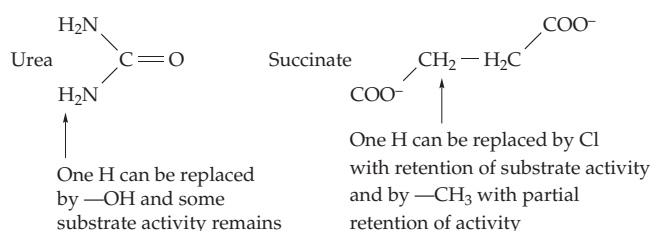


the phenylalanine side chain, which helps direct it into the substrate binding site of chymotrypsin, and a sulfonamide linkage that mimics the normal amide linkages of the substrate. The group Y, which becomes alkylated in an S_N2 -like displacement of a chloride ion (see Eq. 9-76) is thought to be His 57 of the active site (see Fig. 12-10). For additional discussion see Walsh,⁹¹ Kyte,⁶⁷ and Plapp.^{91a}

Enzyme-activated inhibitors (also called suicide substrates, k_{cat} inhibitors, or mechanism-based inhibitors) are chemically inert until they enter the active site of their target enzymes. Then, by passing through at least some of the normal stages of the catalytic action of the enzyme, they are converted to reactive intermediates that can become irreversibly bound to an enzyme.^{92,93} For example, a halogen atom (F, Cl, Br) together with a proton may be eliminated from an intermediate to give an unsaturated compound to which a nucleophilic side chain of the protein adds. Several of these inhibitors are discussed in later chapters. Because of their high specificity many enzyme-activated inhibitors are potential drugs. One of them, α -difluoromethylornithine (Box 14-C), is said to bind covalently to only one protein in the body of a rat, namely, ornithine decarboxylase, the target enzyme for the drug.

C. The Specificity of Enzymatic Action

Enzymes are usually impressively specific in their action. The specificity toward substrate is sometimes almost absolute. For many years urea was believed to be the *only* substrate for the enzyme **urease** and succinate the only substrate for **succinate dehydrogenase**. Even after much searching for other substrates, only



one or two closely related compounds could be found that were acted on at all. In other cases enzymes can use a class of compounds as substrates. For example, the **D-amino acid oxidase** of kidney oxidizes a variety of D-amino acids but does not touch L-amino acids.

Almost as impressive as the substrate specificity of enzymes is the specificity for a given type of reaction. Many substrates are capable of undergoing a variety of different chemical reactions, either unimolecular or with water or some other compound present in the cell. The enzyme catalyzes only one of these reactions.

Although side reactions may occur to a small extent, the most impressive thing in comparing an enzyme-catalyzed reaction with an uncatalyzed organic reaction is that the latter often produces large amounts of side reaction products, but the enzymatic reaction does not.

1. Complementarity of Substrate and Enzyme Surfaces

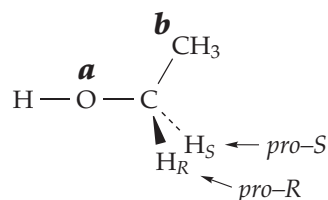
Impressed by the specificity of enzymatic action, biochemists early adopted a “lock-and-key” theory which stated that for a reaction to occur the substrate must fit into an active site precisely. Modern experiments have amply verified the idea. A vast amount of kinetic data on families of substrates and related competitive inhibitors support the idea and numerous X-ray structures of enzymes with bound inhibitors or with very slow substrates have given visual evidence of the reality of the lock-and-key concept. Directed mutation of genes of many enzymes of known three-dimensional structure has provided additional proof.

As anticipated, hydrophobic groups of substrates or inhibitors usually contact hydrophobic regions of the protein and the fit is tight. For example, the active site of chymotrypsin contains a “hydrophobic pocket” designed to hold a large hydrophobic side chain, thus providing the specificity observed with this enzyme (Table 3-2). Likewise, polar groups of substrates contact polar groups of the enzyme. The interactions are complementary, positive charges fitting against negative and with correctly formed hydrogen bonds. Trypsin, whose structure is similar to that of chymotrypsin, exerts its specificity for a positively charged side chain next to the bond cleaved (Table 3-2) by virtue of the presence of a negatively charged carboxylate at the bottom of the hydrophobic pocket. Several C=O and N—H groups of the peptide linkages in the substrate form hydrogen bonds to the edge of a β sheet in the protein, in effect making the substrate an added β strand in the sheet (Fig. 12-10). Aspartate aminotransferase acts on the dicarboxylic amino acids glutamate and aspartate. A pair of arginine side chains bind the two carboxylates of the substrate while the $-\text{NH}_3^+$ of the substrate is attracted to a negatively charged group in the coenzyme pyridoxal phosphate, which is also present in the active site (Eq. 14-39, Fig. 14-10).

2. Stereospecificity and Prochiral Centers

Most enzymes possess an infallible ability to recognize the difference between the right side and the left side of an organic substrate even when the latter has perfect bilateral symmetry. In fact, this ability is limited to **prochiral centers** of molecules and is a natural consequence of their reaction with the chiral

enzyme.^{94–98} Consider carbon atom number 1 of ethanol. The two attached hydrogen atoms are chemically identical and would react identically with a nonchiral reagent. Nevertheless, these atoms are no more equivalent stereochemically than are your right and your left arms. We say that the molecule has a prochiral center at C-1. A prochiral center on a tetrahedrally bonded carbon atom always contains two identical atoms or groups but a total of three different kinds of atoms or groups. If the priority of this H is elevated, e.g., by substitution of ^2H for ^1H the configuration would be *R* (the priority of this ^2H would become **c**).



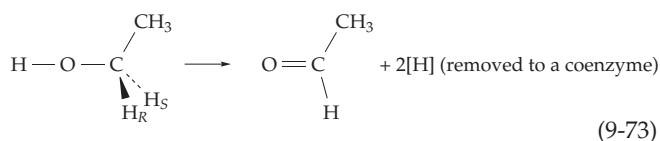
BOX 9-D RECEPTORS, AGONISTS, AND ANTAGONISTS

Inhibition of enzymes provides the basis for many of the effects of antibiotics and other chemotherapeutic substances (e.g., see Box 9-B). However, some drugs act on cell surface **receptors** which have ordinarily not been regarded as enzymes. According to receptor theory, developed around 1937, structurally similar drugs often elicit similar responses because they bind to the same receptor. A receptor may normally bind a hormone, neurotransmitter, or other metabolite whose geometry is partially shared by a drug. Binding of drugs of one class, termed **agonists** in the pharmacological literature, to an appropriate receptor triggers a response in a cell, similar to that of a hormone. On the other hand, compounds of related structure often act as **antagonists**, binding to receptor but failing to elicit a response. Agonist and antagonist often act in a strictly competitive fashion as in competitive inhibition of enzyme action.

We know now that many receptors *are* enzymes, some of which may act quite slowly. The active site may sometimes be far from the receptor binding site and sometimes in a separate subunit. The receptor can be viewed as an allosteric effector which binds at a distant site or as a ligand for a regulatory subunit of the enzyme complex. Alternatively, the active site may be viewed as the site for relaying a signal received from the hormone or other agonist.

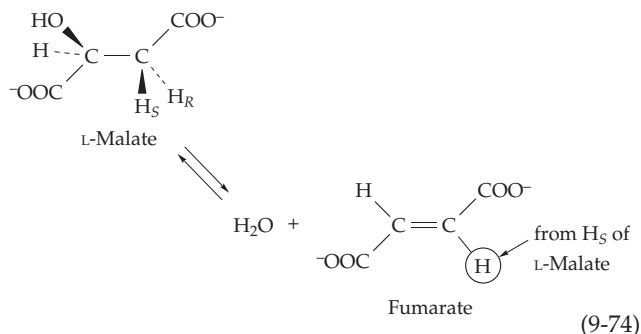
The two hydrogen atoms on C-1 of ethanol are called **enantiotopic** because replacement of one or the other by a fourth different kind of atom or group would produce a pair of enantiomers (page 41). This fact suggests a way of naming the positions occupied by two enantiotopic atoms or groups. We first assign priorities to all of the atoms or groups attached to the central carbon atom according to the *RS* system. Now, we ask whether the configuration will be *R* or *S* when the priority of one of the two identical groups is raised, e.g., by substitution of one of the hydrogen atoms by deuterium. If the configuration becomes *R*, that group occupies a *pro-R* position; if the configuration becomes *S*, it occupies a *pro-S* position. Referring to the preceding diagram, it is easy to see (by viewing down the bond to the group of lowest priority and applying the usual rule for determining configuration) that if the *pro-R* hydrogen (H_R) is replaced by deuterium (^2H), the configuration will be *R*. Conversely, replacement of H_S by deuterium will lead to the *S* configuration.

When ethanol is oxidized by the action of **alcohol dehydrogenase** (Eq. 9-73), only the *pro-R* hydrogen atom is removed. If the reaction is reversed in such a way that deuterium is introduced into ethanol from the reduced coenzyme the optically active *R*-2-deuterioethanol is formed. The ability of an enzyme to

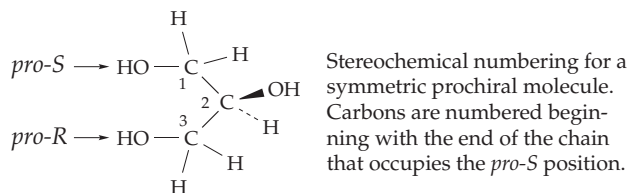


recognize a single hydrogen of a pair of hydrogens on a CH₂ group was at first a surprise to many biochemists.⁹⁹ However, it is a natural result of the complementarity of enzyme and substrate surfaces, just as the fit of a shoe is determined by the complementarity of surfaces of foot and shoe. Only a chiral catalyst can have this ability.

Another example is provided by malic acid, a chiral molecule which also contains a prochiral center (see Eq. 9-74). In this case replacement of the *pro-R* or *pro-S* hydrogen atom by another atom or group would yield a pair of diastereoisomers rather than enantiomers. Therefore, these hydrogen atoms are **diastereotopic**. When L-malic acid is dehydrated by fumarate hydratase (Chapter 13) the hydrogen in the *pro-R* position is removed but that in the *pro-S* position is not touched. This can be demonstrated by allowing the dehydration product, fumarate, to be hydrated to malate in $^2\text{H}_2\text{O}$ (Eq. 9-74). The malate formed contains deuterium in the *pro-R* position. If this malate is now isolated and placed with another portion of enzyme in H₂O, the deuterium is removed cleanly. The fumarate produced contains no deuterium.

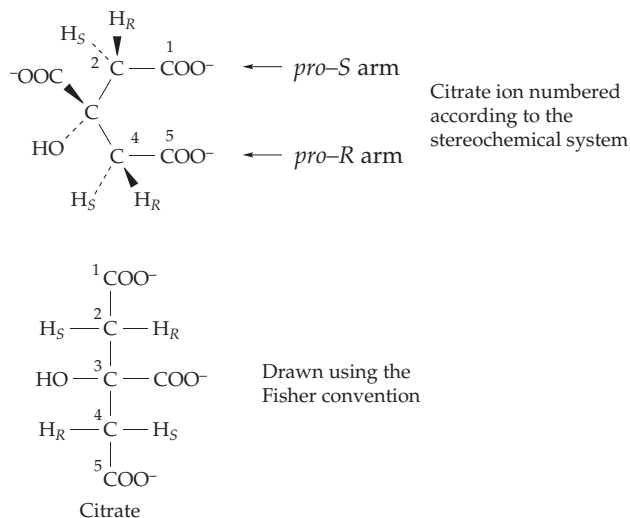


Stereochemical numbering. In some prochiral molecules, such as glycerol, the two ends of the main carbon chain form identical groups. Since the two ends are distinguishable to an enzyme, it is important to decide which should be labeled C-1 and which C-3. Hirschmann proposed a stereochemical numbering system¹⁰⁰ in which the carbons are numbered beginning with the end of the chain that occupies the *pro-S* position.

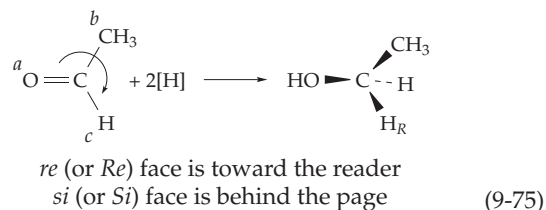


In this numbering system derivatives of the parent prochiral compound are given the prefix *sn*-. Thus, glycerol phosphate, used by cells to construct phospholipids, usually bears a phosphate group on the $-\text{CH}_2\text{OH}$ in the *pro-R* position of glycerol and is therefore *sn*-3-glycerol phosphate.

The **citrate ion**, a very important prochiral metabolic intermediate, has three prochiral centers at C-2, C-3, and C-4, respectively. That at C-3 distinguishes the *pro-R* and *pro-S* arms and determines the stereochemical numbering. Citrate containing ^{14}C in the *sn*-1 position is called *sn*-citrate[1- ^{14}C] and is the form of labeled citrate that is synthesized in living cells from oxaloacetate and [1- ^{14}C]acetyl coenzyme A (see Fig. 10-6). The first step in the further metabolism of citrate is the elimination of the $-\text{OH}$ group from C-3 together with the H_R proton from C-4 through the action of the enzyme **aconitate hydratase** (aconitase). In this case the proton at C-4 (in the *pro-R* arm) is selected rather than that at C-2.



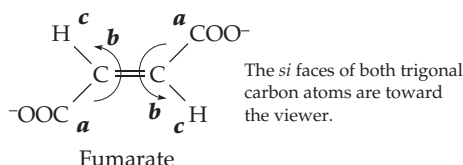
Trigonal prochiral centers. Planar trigonal atoms, such as those of aldehydes, ketones, and alkenes, are also prochiral if they are attached to three different kinds of atoms or groups. The two faces are enantiotopic or diastereotopic, if another chiral center is present. Hanson^{94,94a} proposed that the faces be named as indicated in Eq. 9-75. The trigonal atom is viewed from one side and the three groups surrounding the carbon atom are given priorities, *a*, *b*, and *c*, as in the *RS* system (Chapter 2). If the sequence *a*, *b*, and *c* of priorities is clockwise, the face toward the reader is *re* (*rectus*); if counterclockwise, it is *si* (*sinister*). Priorities can often be assigned on the basis of the atomic numbers of the first atoms in the three groups (as in the example shown)



but, if necessary, replica atoms must be added as described in Chapter 2. Replica atoms attached to an originally trigonal carbon are ignored completely, but those attached to O or N of $\text{C}=\text{O}$ or $\text{C}=\text{N}$ may be required to establish the priorities of the groups around the carbon atom. For $\text{C}=\text{O}$ and $\text{C}=\text{N}$ the faces of the O and N atoms are taken to be the same as that of the attached trigonal carbon atom. In $\text{C}=\text{C}$ the two carbons may have their *re* faces either on the same side or on opposite sides of the group.

Notice that addition of an atom of ^2H to the *re* face of acetaldehyde would give *R*-deuteroethanol (Eq. 9-75; reverse of Eq. 9-73). This is the reaction catalyzed by alcohol dehydrogenase. Addition of ^1H to the *re* face places the entering hydrogen in the *pro-R* position. Addition to the *si* face would place it in the *pro-S*

position. Fumarate has two trigonal carbon atoms. The *si* faces of both are toward the viewer in the following structure (also shown in Eq. 9-74). Referring to Eq. 9-74, we see that an HO^- ion from water adds to the *si* face of one carbon atom of fumarate to give *s*-malate (*L*-malate). At the same time, a proton combines with the *re* face of the adjacent unsaturated carbon atom to enter the *pro-R* position.



3. Induced Fit and Conformational Changes

Results of many X-ray studies indicate that the lock-and-key picture of enzyme action must be modified. If an enzyme is a “lock” and the substrate the “key,” *the entrance of the key into the lock often induces a conformational change in the protein.* Binding of substrates may be imperfect in the initially formed ES complex but may be more nearly perfect a few nanoseconds or microseconds later as the protein readjusts its structure to accommodate the substrate. The substrate has induced a fit. For example, when an amino acid substrate binds to aspartate aminotransferase one whole domain of the enzyme moves inward, packing hydrophobic side chains of the protein against the substrate (Chapter 14). This strengthens the electrostatic interactions between the ion pairs that orient the substrate and align it for reaction. Similar conformational changes have been observed for citrate synthase, glycogen phosphorylase, various kinases (Chapter 12), alcohol dehydrogenase (Chapter 15), and a growing list of other enzymes. Accompanying changes in circular dichroism, ultraviolet spectra, and sedimentation constants are often observed.

For many other enzymes the observed conformational changes are subtle. A single loop of polypeptide chain or even of a side chain moves to cover the bound substrate. With trypsin and other serine proteases a flexible segment of the peptide chain becomes immobilized and forms more hydrogen bonds after substrate binds than before. It is probably quite unusual for a major unfolding and refolding of parts of a protein to take place. The term induced fit usually refers to substrate binding, but as substrates are interconverted within active sites successive changes in geometry and in charge distribution occur. Small conformational changes may be required at several stages of an enzymatic reaction to ensure that complementarity of substrate and enzyme is preserved.¹⁰¹ In line with this idea are the facts that proteins are less tightly packed

at active sites than in other parts of the molecules, and that active sites often lie between domains and usually are formed by several loops of the peptide chain (Chapter 2).¹⁰²

Although conformational changes allow proteins to maintain good complementarity with substrates, it does not follow that substrates are therefore bound very tightly. This is easiest to understand for reversible reactions in which a substrate is the product of the reverse reaction. Not only binding but also the rate of release of product must be rapid. Very tight binding would retard release and cause product inhibition. Furthermore, as we can see from Eq. 9-37, the velocity of catalysis is the product of $[\text{S}][\text{E}] \times k_{\text{cat}} / K_m$, where $[\text{E}]$ is, *free enzyme*. Catalysis can be made more efficient in two ways: by increasing k_{cat} or by decreasing K_m , that is, by tighter binding. However, tighter binding will be advantageous only if K_m is not so low that the enzyme is approaching saturation with substrate. Otherwise, $[\text{E}]$ will fall as K_m is lowered and little increase in rate will be observed.²

In fact, it appears that enzymes have evolved to have high values of k_{cat} and *high* values of K_m , that is, *weak binding* of substrates.² Frequently, in one conformational state of a protein the active site is open, with solvent molecules or substrate surrogates present, while in a second state of nearly equal energy it is closed around the substrate. The functional groups of the enzyme's active site can be bound either to external ions or to ionic groups of substrates and either to water or to hydrogen-bonding groups of substrates. In some cases two ionic groups in the protein may pair with each other in the open conformation and with ionic groups of the substrate in the closed conformation. Thus, the energy changes accompanying the conformational changes can be small but very good complementarity can exist in the ES complex, an important factor in establishing specificity.

4. Specificity and k_{cat}

Enzyme specificity is often observed not only in binding but also in the rate at which ES is converted to products. Thus, it is the values of k_{cat} / K_m that determine specificity. Good examples are provided by chymotrypsin and related serine proteases (Chapter 12),² for which substrates with the shortest chains are often bound as well as those with longer chains but react more slowly. For example, *N*-acetylphenylalanine amide binds to chymotrypsin about as well as does the longer *N*-acetylphenylalanylalanine amide but reacts only 1/47 as fast.² One might anticipate that increasing the length of the substrate would make it bind more tightly because of the greater number of contacts between substrate and enzyme. It has often been suggested that the reason that this does *not* happen is that the

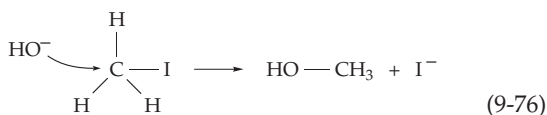
binding of the longer, more specific substrates distorts the enzyme and that *the binding energy is now stored in the enzyme. It is as if the enzyme contained an internal spring which would be compressed when the substrate binds.* This would keep the binding weak but the energy in the spring might then be used to increase the velocity.

5. Proofreading

Although enzymes may be very specific they do make mistakes. This is of particular concern for processes such as protein synthesis in which the correct amino acid is placed at each position in the sequence with an error rate that has been estimated for *E. coli* as only 1 in 10^4 . It would be impossible for an enzyme designed to attach valine to its specific transfer RNA to avoid attaching the smaller alanine if discrimination between the two were based solely on the Gibbs energy differences of binding.¹⁰³ However, it would be easier for an enzyme to exclude *larger* amino acids. This problem may be resolved by use of multistep screening.^{104,105} For example, isoleucyl-tRNA synthetase (Chapter 29) does occasionally attach the smaller valine to the specific tRNA,^{11e} but when it does the enzyme in a “proofreading and editing” step hydrolyzes off the incorrect amino acid. The active site for this hydrolyzing activity, whether at a different place on the enzyme surface or created by a conformational change, may be able to exclude sterically the larger isoleucyl residue while acting on the valyl-tRNA. This editing mechanism for isoleucyl-tRNA synthetase was demonstrated directly in 1998 by X-ray crystallography on complexes of the enzyme with L-isoleucine and L-valine. Both substrates fit into the ATP-requiring synthetic site but neither isoleucine nor isoleucyl-tRNA will fit into the editing site which is located in an adjacent β -barrel domain.^{104,105} Proofreading steps based on differing chemical properties as well as size can also be visualized.^{103,106}

D. Mechanisms of Catalysis

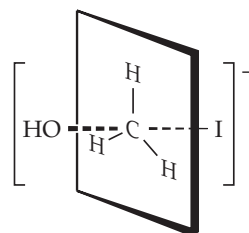
Kinetic studies tell how fast enzymes act but by themselves say nothing about *how* enzymes catalyze reactions. They do not give the **chemical mechanism** of catalysis, the step-by-step process by which a reaction takes place. Most of the individual steps involve the simultaneous breaking of a chemical bond and formation of a new bond. Consider a simple **displacement reaction**, that of a hydroxyl ion reacting with methyl iodide to give the products methanol and iodide ions.



The reaction can be thought of as an “attack” by the OH^- ion on the “back side” of the carbon atom of the methyl groups with a simultaneous displacement of the I^- .

1. The “Transition State”

A reaction such as that of Eq. 9-76 is not instantaneous, and at some time between that at which the reactants exist and that at which products have been formed the C–I bond will be stretched and partially broken and the new C–O bond will be partially formed. The structure at this point is not that of an ordinary compound and is energetically unstable with respect to both the reactant and the products. The intermediate structure of the least stability is known as the **transition state**. Although no one has actually seen a transition state structure, we might represent that for Eq. 9-76 as follows:



Transition state structure

The negative charge is distributed between the attacking HO^- group and the departing iodide. The bonds to the central carbon atom are no longer tetrahedrally arranged but the C–H bonds lie in a single plane and the partial bonds to the OH and iodine atom lie at right angles to that plane.

It is useful in discussing a reaction mechanism to construct a **transition state diagram** in which Gibbs energy G is plotted against **reaction coordinate** (Fig. 9-15A). Energy E or enthalpy H may be plotted in the same way and authors frequently do not state whether G , E , or H is meant. The reaction coordinate is usually not assigned an exact physical meaning but represents the progress from reactants toward products. It is directly related to the extent to which an existing bond has been stretched and broken or a new one formed. The high energy point is the transition state. A somewhat more detailed idea of a transition state is obtained from a contour diagram such as that of Fig. 9-15B. Here, energy is plotted as a function of two distances, e.g., the lengthening C–I bond distance and the shortening C–O distance for Eq. 9-76. The path of minimum energy across the “saddle point” representing the transition state is indicated by the dashed line.

In a reaction coordinate diagram the difference in value of G between reactants and products is the

overall Gibbs energy change ΔG for the reaction, while the difference in G between the transition state and reactants is $\Delta G^\ddagger$, the **Gibbs energy of activation**. The magnitude of $\Delta G^\ddagger$ represents the “energy barrier” to a reaction and largely determines the rate constant.

The diagrams in Fig. 9-15 are too simple because enzymatic reactions usually occur in several intermediate steps. There will be transition states for each step with valleys in between. The valleys correspond to intermediate species, which are sometimes very unstable. The passing from reactants to products in an enzymatic reaction can be likened to wandering through a series of mountain ranges of various heights and finally reaching the other side.

Quantitative transition state theory.^{107–113} In the 1880s Arrhenius observed that the rate of chemical reactions varies with temperature according to Eq. 9-77 in a manner similar to the variation of an equilibrium constant with temperature (integrate Eq. 6-37 and compare). Here, k is a first-order rate constant, the quantity E_a is known as the **Arrhenius activation energy**, and the constant A is referred to as the “pre-exponential factor” or the “frequency factor”.

$$k = A e^{-E_a/RT} \quad (9-77)$$

The Arrhenius equation, together with studies of the effects of salts on reaction rates and observation of quantitative correlations between rates and equilibrium constants, suggested that a rate constant for a reaction might be a product of a constant term which is nearly independent of temperature and a constant $K^\ddagger$ which has the properties of an equilibrium constant for formation of the transition state. Eyring made this quantitative in 1935 with his “absolute rate theory”¹¹² according to which all transition states break down with a rate constant $\kappa k_B T/h$. Eyring reached this conclusion by assuming that *the rate of a chemical reaction is determined by the frequency of stretching of the bond that is being broken in the transition state*. To be more precise, it is the “normal-mode” oscillation of the transition state complex along the reaction coordinate.¹⁰⁹ This frequency ν was deduced by describing the vibrational energy as $h\nu$ (from quantum mechanics) and as $k_B T$ (from classical mechanics) and setting them equal.

$$h\nu = k_B T \quad \text{and} \\ \nu = k_B T/h$$

$$k_B \text{ is the Boltzmann constant} \\ h \text{ is Planck's constant.} \quad (9-78)$$

The right side of Eq. 9-78 is usually multiplied by a **transmission coefficient κ** , which may vary from 1 to 0.1 or even much less. However, for lack of any better value, κ is usually assumed to be 1. From Eq. 9-78, at 25°C $\nu = 6.2 \times 10^{12} \text{ s}^{-1}$. This is the maximum rate for a chemical reaction of molecules in the transition state. This is the rate for a single molecule and must be multiplied by the concentration of the reacting substance X in the transition state. This concentration $[X]^\ddagger$ is determined by the equilibrium constant $K^\ddagger = [X]^\ddagger/[X]$. The velocity of the reaction becomes

$$\frac{-d[X]}{dt} = \nu = \frac{k_B T}{h} \cdot K^\ddagger [X] \quad (9-79)$$

The first-order rate constant is

$$k_1 = \frac{-d[X]/dt}{[X]} = K \frac{k_B T}{h} \cdot K^\ddagger \text{ s}^{-1} \quad (9-80)$$

Since $\Delta G^\ddagger = -RT \ln K^\ddagger$, Eq. 9-80 can be rewritten as Eq. 9-81, in which $\Delta G^\ddagger$ is the Gibbs energy of activation.

$$k_1 \approx \frac{k_B T}{h} \cdot e^{-\Delta G^\ddagger/RT} \text{ s}^{-1} \quad (9-81)$$

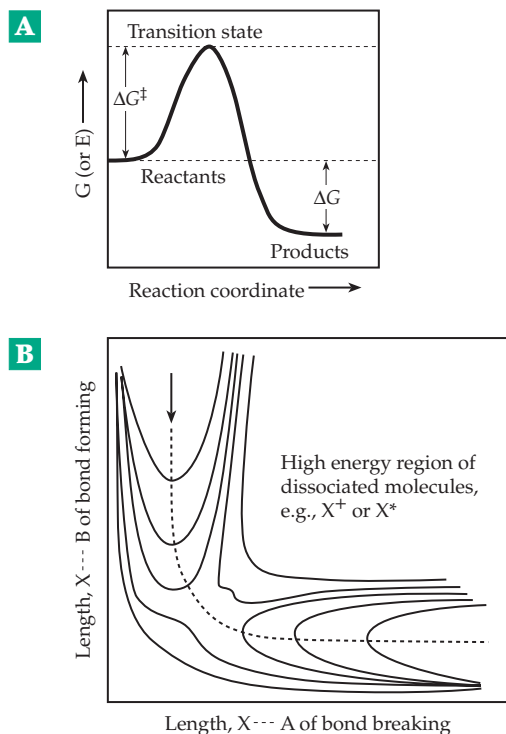


Figure 9-15 (A) Transition state diagram illustrating Gibbs energy vs reaction coordinate for conversion of reactants to products in a chemical reaction. (B) Contour map of Gibbs energy vs interatomic bond distances for reaction $B + X \rightarrow A \rightarrow B + X + A$.

At 25°C, $\Delta G^\ddagger$ in kJ/mol, the following “practical” form of Eq. 9-81 can be written.

$$\begin{aligned} k_1 &= 6.2 \times 10^{12} e^{-\Delta G^\ddagger/2.48} \text{ s}^{-1} \\ \log k_1 &= 12.79 - \Delta G^\ddagger/5.71 \\ \text{and } \Delta G^\ddagger &= 73.0 - 5.71 \log k_1 \end{aligned}$$

Equation 9-81 is approximate and a more correct statistical mechanical treatment is available.^{113,114} See also comment on p. 288 about $\log k$ (and $\log k_1$) being unitless. Employing Eq. 6-14, we may expand Eq. 9-81 as follows:

$$k \approx \left(\frac{k_B T}{h} e^{\Delta S^\ddagger/R} \right) e^{-\Delta H^\ddagger/RT} \text{ s}^{-1} \quad (9-82)$$

From this it appears that $\Delta H^\ddagger = E_a$, the Arrhenius activation energy. A more correct treatment gives $\Delta H^\ddagger = E_a - RT$ for reactions in solution. However, since RT at 25°C is only 2.5 kJ mol⁻¹, the approximation that $\Delta H^\ddagger = E_a$ is often used. The preexponential term, in parentheses in Eq. 9-82, depends principally on $\Delta S^\ddagger$, the entropy change accompanying formation of the transition state. The quantities $\Delta G^\ddagger$, $\Delta H^\ddagger$, and $\Delta S^\ddagger$ are sometimes measured for enzymatic reactions but useful interpretations are difficult.

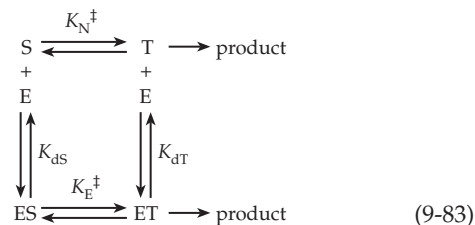
Equations 9-81 and 9-82 also show that a small decrease in transition state energy will give a large increase in rate; stabilization of the transition state by 5.7 kJ/mol (1.4 kcal/mol) will increase the rate 10-fold. If $\Delta G^\ddagger = 400$ kJ/mol, as for a strong covalent single bond, $k_1 = 5 \times 10^{-58} \text{ s}^{-1}$ and the half-life $t_{1/2} = .693/k_1 = 1.3 \times 10^{57} \text{ s}$ (4×10^{49} years, greater than the $\sim 10^{10}$ years estimated age of the universe). If $\Delta G^\ddagger = 100$ kJ/mol, $\log k_1 \approx -4.7$, $k_1 = 1.9 \times 10^{-5} \text{ s}^{-1}$, and $t_{1/2} \approx 10 \text{ h}$. This is about the rate of a typical nonenzymatic “model” reaction at 25°C. If $\Delta G^\ddagger = 50$ kJ/mol, $\log k_1 \approx 4$, as for a *fast* enzyme.

We can conclude that *enzymes make use of relatively small energy differences in catalyzing reactions*. The energies of numerous van der Waals interactions of hydrogen bonds and electrostatic attraction or repulsion of charges are sufficient. Nevertheless, we can see that anything that stabilizes the transition state (decreases $\Delta G^\ddagger$) will increase the rate of reaction. The role of a catalyst is to permit the formation of a transition state of lower energy (higher stability) than that for the uncatalyzed reaction. Stabilization of the transition state of a reaction by an enzyme suggests that the enzyme has a higher affinity for the transition state than it does for substrate or products, an idea that appears to have been expressed first by Haldane⁴¹ and popularized by Pauling.¹¹⁵

I think that enzymes are molecules that are complementary in structure to the activated complexes of the reactions that they catalyze. The attraction of the enzyme

molecule for the activated complex would thus lead to a decrease in its energy, and hence to a decrease in energy of activation of the reaction and to an increase in the rate of the reaction.

Transition state inhibitors. Suppose that a chemical reaction of a compound S takes place with rate constant k_N through transition state T. Let the equilibrium constant for formation of T be $K_N^\ddagger$. Assume that an enzyme E can combine either with S with dissociation constant K_{ds} or with the compound in its transition state structure T with dissociation constant K_{dT} (Eq. 9-83).



If equilibrium is assumed for all four sets of double arrows it is easy to show that Eq. 9-84 holds.

$$K_E^\ddagger / K_N^\ddagger = K_{ds} / K_{dT} \quad (9-84)$$

According to transition state theory, if the transmission coefficient $\kappa = 1$, T and ET will be transformed to products at the same rate. Thus, if the mechanisms of the nonenzymatic and enzymatic reactions are assumed the same, the ratio of maximum velocities for first-order transformation of ES and S will be given by Eq. 9-85. For some enzymes the ratio

$$k_E / k_N = K_E^\ddagger / K_N^\ddagger = K_{ds} / K_{dT} \quad (9-85)$$

k_E / k_N may be 10^8 or more. Thus, if $K_{ds} \approx 10^{-3}$ the constant K_{dT} would be $\approx 10^{-11}$. The enzyme would be expected to bind the transition state structure (T) 10^8 times more tightly than it binds S.

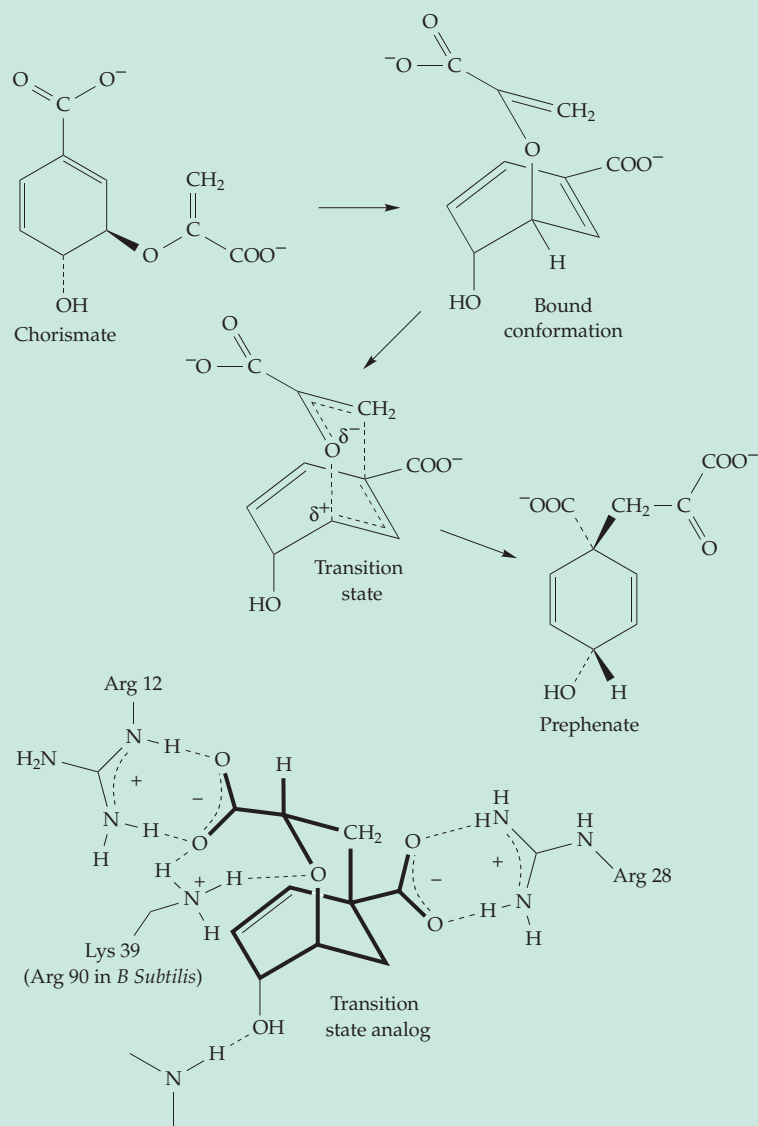
The foregoing reasoning suggests that if structural analogs of T could be found for a particular reaction, they too might be very tightly bound—more so than ordinary substrate analogs.¹¹⁶ Wolfenden¹¹⁷ listed a series of compounds which may be **transition state inhibitors** of this type and many others have been described.^{117a} Nevertheless, very tight, reversible binding is no *proof* that an inhibitor is actually binding to the transition state structure of the enzyme. For example, an inhibitor may bind to a conformational form of the enzyme that lies off to a side rather than directly on the reaction pathway for catalysis.

Describing the transition state. How can we describe the structure of the transition state when we have no direct knowledge of its structure? We can try

BOX 9-E CHORISMATE MUTASE

Bacteria, fungi, and plants convert products of glucose metabolism into the aromatic amino acids phenylalanine, tyrosine, and tryptophan by a complex sequence of reactions that is described in Chapter 25. One of these reactions, the conversion of **chorismate** into **prephenate**, is unique among enzyme-catalyzed reactions. It is a Claisen rearrangement that occurs readily *without catalysis* in an aqueous solution. However, the rate is increased over one million-fold by the enzyme **chorismate mutase**. The enzyme from *Bacillus subtilis* is a trimer of identical, small 127-residue subunits. Many studies, both of the nonenzymatic and enzymatic reactions, have suggested a “pericyclic” mechanism involving a polar chairlike transition state whose presumed structure is shown in the accompanying equation.^{a–f} The transition state analog, whose

structure is also depicted, is a powerful inhibitor. X-ray structural studies show that this inhibitor binds in the conformation shown with a number of interactions with polar side chain groups.^{d,g} Some of these interactions are shown for the *E. coli* enzyme in the figure;^g they are different in enzymes from *B. subtilis*^d and yeast.^h However, in no case are there groups in obvious positions to serve as acid–base catalysts and the maximum velocity $V_{\max}$ is independent of pH.^b NMR studies show that the bound product prephenate displays large shifts in some ¹³C resonances, indicating strong interaction with the polar side chains of the protein.^c Theoretical calculations also suggested that these interactions, especially those with Arg 90 and Glu 78 of the *B. subtilis* enzymes,^{f–i} help to stabilize the partial charge separation in the polar transition state. Studies of mutant enzymes have confirmed that the charged Arg 90 (Lys 39 for *E. coli*) side chain is essential.^k



^a Gray, J. V., Eren, D., and Knowles, J. R. (1990) *Biochemistry* **29**, 8872–8878

^b Turnbull, J., Cleland, W. W., and Morrison, J. F. (1991) *Biochemistry* **30**, 7777–7782

^c Rajagopalan, J. S., Taylor, K. M., and Jaffe, E. K. (1993) *Biochemistry* **32**, 3965–3972

^d Chook, Y. M., Gray, J. V., Ke, H., and Lipscomb, W. N. (1994) *J. Mol. Biol.* **240**, 476–500

^e Gray, J. V., and Knowles, J. R. (1994) *Biochemistry* **33**, 9953–9959

^f Lyne, P. D., Mulholland, A. J., and Richards, W. G. (1995) *J. Am. Chem. Soc.* **117**, 11345–11350

^g Lee, A. Y., Karplus, P. A., Ganem, B., and Clardy, J. (1995) *J. Am. Chem. Soc.* **117**, 3627–3628

^h Xue, Y., Lipscomb, W. N., Graf, R., Schnappauf, G., and Braus, G. (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 10814–10818

ⁱ Wiest, O., and Houk, K. N. (1995) *J. Am. Chem. Soc.* **117**, 11628–11639

^j Kast, P., Hartgerink, J. D., Asif-Ullah, M., and Hilvert, D. (1996) *J. Am. Chem. Soc.* **118**, 3069–3070

^k Lin, S. L., Xu, D., Li, A., Rosen, M., Wolfson, H. J., and Nussinov, R. (1997) *J. Mol. Biol.* **271**, 838–845

^l Ma, J., Zheng, X., Schnappauf, G., Braus, G., Karplus, M., and Lipscomb, W. N. (1998) *Proc. Natl. Acad. Sci. U.S.A.* **95**, 14640–14645

to predict the atomic coordinates of all of the atoms of the reacting substrates and of the protein. The transition state structure of the substrate must be between that of the last ES complex and the first EP complex formed. It should be similar to that of a transition state inhibitor. We can hope to obtain suitable X-ray or NMR structures and deduce an appropriate enzyme structure by molecular modeling. Theoretical calculations using quantum chemical methods are of value in this effort.^{118,119}

Lengths of the breaking and forming bonds in a transition state are often estimated using **kinetic isotope effects** (KIEs) on the velocity of the reaction.^{118–122} This is somewhat indirect. Nevertheless, by measuring these effects for a large number of substrates and several isotopes a picture of the transition state structure that is good enough to use as a model for design of an enzyme inhibitor may be obtained.^{117a,121–124} Other physical measurements such as Raman difference spectroscopy of isotopically labeled inhibitors can also be of value.¹²⁵

Getting to the transition state. Since some bonds must lengthen and some shorten and bond angles must also change, it may not be simple for an enzyme to catalyze necessary steps prior to the rate-limiting breakdown of the transition state. Hydrogen bonds within the protein may need to be broken and reformed with new bonding partners. Nonpolar side chains may need to shift to give a packing arrangement that will allow the movement of nuclei that must occur in forming the transition state structure. The enzyme must not only stabilize the transition state but also hold the substrate and escort it into the transition state configuration.

The attacking hydroxyl ion in Eq. 9-76 carries a negative charge which becomes distributed more or less equally between the hydroxyl and the iodine atom in the transition state. In this state the CH₃ group carries a partial positive charge as well. To provide good complementarity between an enzyme and this transition state, a change in the initial charge distribution within the enzyme will also be required. If a positively charged group initially binds the OH[−] ion it may have to lose part of its charge and a group next to the iodide atom may have to gain positive charge. In the nonenzymatic reaction of Eq. 9-76 the ionic atmosphere provided by positive counterions in solution can continuously readjust to keep the negative charge effectively balanced at every step along the reaction coordinate and through the transition state. Within enzymes this adjustment may occur via redistribution of electrical charges within the polarizable network of internal hydrogen bonds. The enzyme structure must allow this. Because of the complexity of an enzymatic transition state it may be hard to compare it with the transition state of the corresponding nonenzymatic

reaction. Effects of pressure^{125a} and of temperature^{125b} also raise doubts about the simple picture of Eq. 9-83. It also follows that tight binding of substrates in the ground state does not necessarily interfere with transition state stabilization.^{126–129a}

2. Microscopic Reversibility

The important statistical mechanical principle of microscopic reversibility asserts that *the mechanism of any chemical reaction considered in the reverse direction must be exactly the inverse of the mechanism of the forward reaction*. A consequence of this principle is that if the mechanism of a reaction is known, that of the reverse reaction is also known. Furthermore, it follows that *the forward and reverse reactions catalyzed by an enzyme must occur at the same active site on the enzyme and the transition state must be the same in both directions*. The principle of microscopic reversibility is often useful when the likelihood of a given mechanism is being considered. If a mechanism is proposed for a reversible reaction in one direction the principle of microscopic reversibility will give an unambiguous mechanism for the reverse reaction. Sometimes this reverse mechanism will be chemically untenable and, recognizing this, the enzymologist can search for a better one.

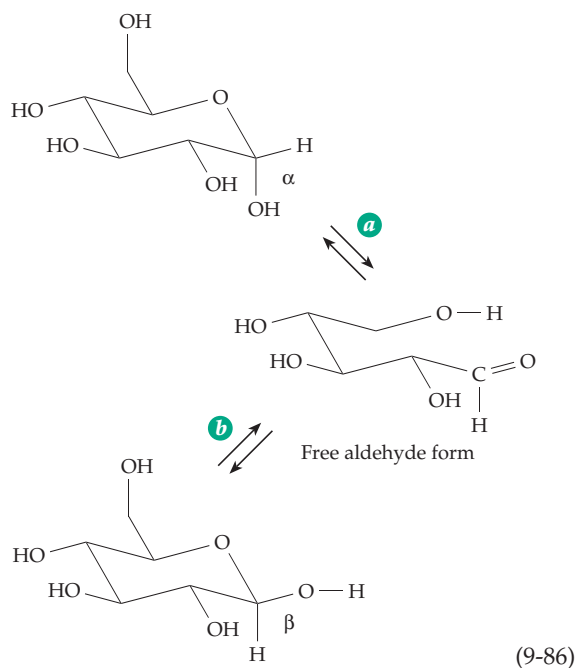
3. Acid and Base Catalysis

Many reactions that are promoted by enzymes can also be catalyzed by acids or bases or by both. An example is **mutarotation**, the reversible interconversion of the α - and β -anomeric forms of sugars (Eqs. 4-1 and 9-86). This reaction is catalyzed by a specific **mutarotase** and also by inorganic acids and bases. The frequently observed bell-shaped curve for the dependence of rate of catalysis on pH (e.g., Eqs. 9-55 to 9-57 and Fig. 9-8) also suggests participation of both protonated and unprotonated acid–base groups present in enzymes.

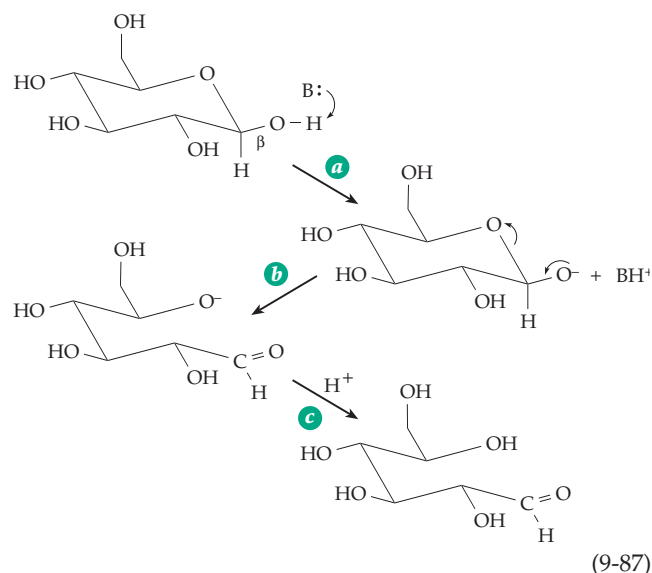
Acidic and basic groups in enzymes. In this discussion the symbols HB or H⁺B will be used for acids and B[−] or B for their conjugate bases. Remember that *strong acids have low pK_a values and that the conjugate bases formed from them are weak bases*. Likewise, *very weak acids have high pK_a values and their conjugate bases are strong*. The following are among the pK_a values that may be important in considering an enzyme mechanism. However, depending upon the environment of the side chain in a protein, these pK_a values may fall substantially outside of the indicated ranges.

-1.74	H ₃ O ⁺ in 55.5 M water
~3.6	-COOH, terminal in peptide
4.3-4.7	-COOH in a glutamic or aspartic acid side chain
6.4-7.0	Imidazole (histidine) and phosphate (-OPO ₃ H ⁻)
7.5-8.0	-NH ₃ ⁺ , terminal in peptide
8.5-10	-SH of cysteine side chains
9.5-11	Phenolic -OH of tyrosine
~10.5	-NH ₃ ⁺ of lysine side chain
~13.6	-CH ₂ OH of serine side chain
15.7	free HO ⁻ in 55.5 M H ₂ O, 25°C

Acid-base catalysis of mutarotation. The mutarotation of glucose proceeds through the free aldehyde form as an intermediate (Eq. 9-86). A hydrogen atom is removed as H⁺ from the oxygen at carbon 1 in step *a* and a proton (probably a different one) is transferred to the oxygen of the ring with cleavage of the O-C bond to the anomeric carbon atom. A similar process in reverse is required for step *b*. Transfers of hydrogen ions between atoms of oxygen, nitrogen, and sulfur atoms are a common feature of biochemical reactions. The bonds between hydrogen and O, N, and S atoms tend to be polarized strongly, leaving a partial positive charge on the hydrogen atoms. Consequently, the groups are weakly acidic and protons can be transferred from them relatively easily. It is reasonable to suppose that acid and base catalysis is related to these proton transfers.



General base and general acid catalysis. Base-catalyzed mutarotation might be formulated as follows: A hydroxyl ion or some other base attacks the proton on the anomeric -OH group of the sugar, removing it to form an anion and the conjugate acid BH⁺ (Eq. 9-87,



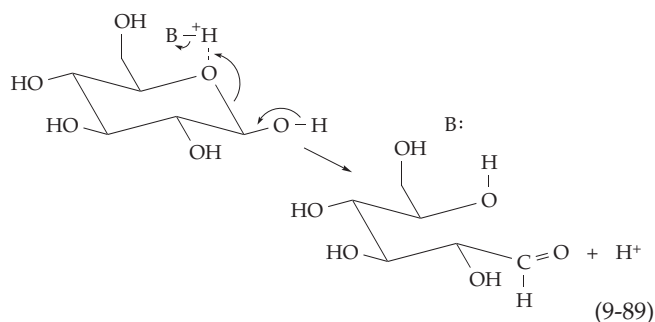
step *a*). The anion is isomerized to a second anion with the ring opened (step *b*). Addition of a proton (transfer of a proton from H₃O⁺) produces the free aldehyde form of the sugar (step *c*).

The catalytic base B: might be HO⁻ or a weaker base such as ammonia or even water. For reactions the rate is proportional only to the concentration of OH⁻ and the presence of other weaker bases has no effect.^{129b} Such catalysis is referred to as **specific hydroxyl ion catalysis**.¹⁹ More commonly, the rate is found to depend both on [OH⁻] and on the concentration of other weaker bases. In such cases the apparent first-order rate constant (*k*_{obs}) for the process is represented by a sum of terms (Eq. 9-88). The term *k*_{H₂O} is the rate in

$$k_{\text{obs}} = k_{\text{H}_2\text{O}} + K_{\text{OH}^-}[\text{OH}^-] + k_{\text{B}}[\text{B}] \quad (9-88)$$

pure water and represents catalysis occurring by the action of water alone as either an acid or a base. The last two terms represent the contributions to the catalysis by OH⁻ and by the other base, respectively. The term *k*_B[B] represents **general base catalysis**.

Catalysis of mutarotation by acids occurs if an acid donates a proton to the oxygen in the sugar ring as

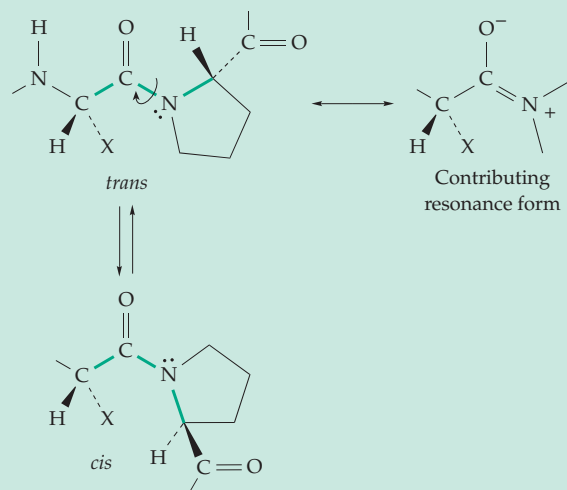
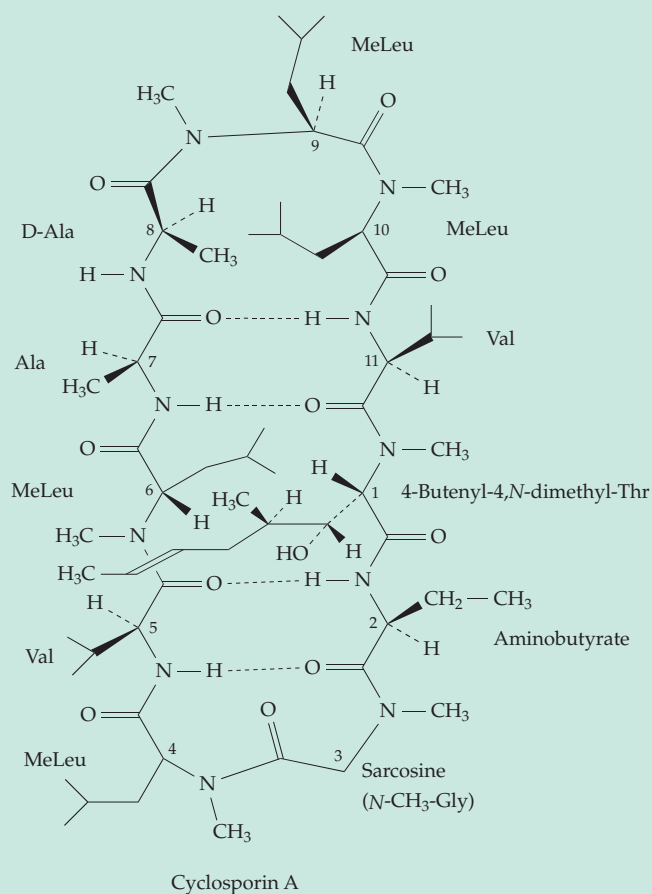


BOX 9-F IMMUNOPHILINS AS ROTAMASES

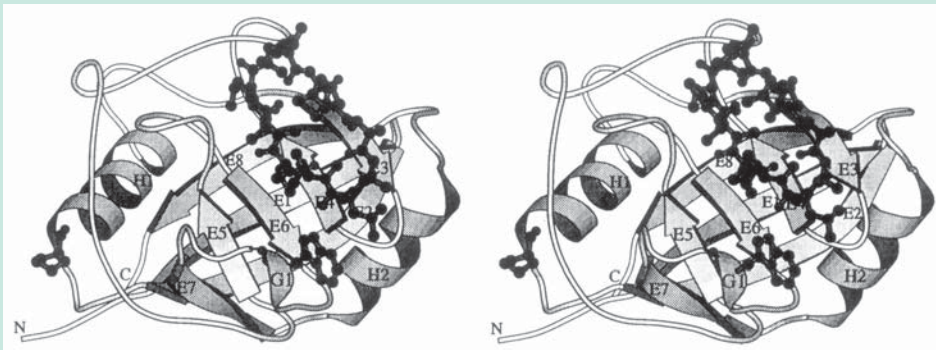
Since its introduction into clinical use in about 1979 the immunosuppressant **cyclosporin** has been responsible for a revolution in human organ transplantation.^{a-c} The exact mechanism of action in suppressing T-lymphocyte-mediated autoimmune responses is still not completely clear, but cyclosporin, a cyclic lipophilic peptide from a fungus, was found to bind to specific proteins that were named **cyclophilins**.^d Human cyclophilin A is a 165-residue protein which associates, in the crystal form, as a decamer with five-fold rotational and dihedral symmetry.^e This protein is also found in almost all

other organisms and has a highly conserved sequence.^f Another immunosuppressant, a synthetic molecule known only as **FK506**, has an action similar to that of cyclosporin but binds to distinctly different proteins, the FK506-binding proteins (FKBPs), of which the 107-residue FKBP-12 is the best known.^{g-i} These proteins also bind another immunosuppressant, **rapamycin**, which, however, has different biological properties than does FK506. Both FK506 and rapamycin are macrocyclic compounds but with structures very different from that of cyclosporin.

It was a surprise to discover that all of the cyclophilins and FK506-binding proteins are **peptidyl prolyl *cis-trans* isomerases** or **rotamases**. They all catalyze the following simple and reversible reaction of a prolyl peptide linkage:



The transition state energy for the reaction is lowered by 33 kJ mol⁻¹ by cyclophilin A^k and by 27 kJ mol⁻¹ by FKBP-12^g with corresponding rate increases of $\sim 6 \times 10^5$ - and 5×10^4 -fold, respectively.



Cyclosporin (black) bound to human cyclophilin. From Pflügl *et al.*^e Courtesy of J. N. Jansonius.

BOX 9-F (continued)

For both enzymes the maximum velocity is independent of pH over a wide region. For cyclophilin the rate is nearly independent of the nature of residue X in the foregoing structure but FKBP prefers a hydrophobic residue.

How do these enzymes work? One possibility would be for the protein to transfer a proton to the nitrogen atom of the proline ring. This would destroy the partial double-bond character of the amide linkage and allow free rotation about a single bond. The same thing could be accomplished if a nucleophilic group from the enzyme formed a covalent adduct with the C=O of the substrate. However, this seems unlikely because there is no suitably placed nucleophile. A third possibility is that the enzyme distorts the substrate and stabilizes the transition state using only noncovalent interactions.^{g,h,l} This would account for the lack of pH dependence and observable solvent isotope effects. However, in human cyclophilin the guanidinium group of arginine 55 hydrogen bonds to the peptide C=O of the substrate's proline residue and could easily shift to place a guanidinium proton against the proline ring nitrogen effectively protonating it and permitting rotation about the peptide linkage.^m

Do cyclosporin and FK506 act as transition state inhibitors? If so, we might learn something about the mechanism from the three-dimensional structures of the inhibited rotamases.^{i,l,n,o} From these structures as well as those with simpler substrate analogs, it is seen that the substrate is desolvated and that many nonbonded (van der Waals) interactions stabilize the binding. Hydrogen bonding is also important.^{p,q} The presence of distinct hydrogen bonds to the peptide NH and C=O groups on the N-terminal side of the substrate X-Pro linkage and less well-defined bonding on the terminal side suggests that the C-terminal part may be rotated while the enzyme holds the N-terminal part.^p It can also be concluded that mechanisms are probably not identical for all rotamases.

Cyclophilins and FKBP are large families of proteins. In *E. coli* there are two cyclophilin genes,^{r,s} three encoding FK506-binding proteins, and one of

a third family of bacterial rotamases (parvulins) that lack sequence similarity to the other two families.^{t,u} Yeast contains at least five cyclophilins^v and even more eukaryotic FKBP are known.^j There is abundant evidence that these proteins play an important role in protein folding *in vivo* (Chapter 29).

^a Kahan, B. D. (1989) *N. Engl. J. Med.* **321**, 1725–1738

^b Schreiber, S. L. (1991) *Science* **251**, 283–287

^c High, K. P., Joiner, K. A., and Handschumacher, R. E. (1994) *J. Biol. Chem.* **269**, 9105–9112

^d Fruman, D. A., Burakoff, S. J., and Bierer, B. E. (1994) *FASEB J.* **8**, 391–400

^e Pflügl, G. M., Kallen, J., Jansonius, J. N., and Walkinshaw, M. D. (1994) *J. Mol. Biol.* **244**, 385–409

^f Lippuner, V., Chou, I. T., Scott, S. V., Ettinger, W. F., Theg, S. M., and Gasser, C. S. (1994) *J. Biol. Chem.* **269**, 7863–7868

^g Fischer, S., Michnick, S., and Karplus, M. (1993) *Biochemistry* **32**, 13830–13837

^h Orozco, M., Tirado-Rives, J., and Jorgensen, W. L. (1993) *Biochemistry* **32**, 12864–12874

ⁱ Van Duyne, G. D., Standaert, R. F., Karplus, P. A., Schreiber, S. L., and Clardy, J. (1993) *J. Mol. Biol.* **229**, 105–124

^j Lam, E., Martin, M. M., Timmerman, A. P., Sabers, C., Fleischer, S., Lukas, T., Abraham, R. T., O'Keefe, S. J., O'Neill, E. A., and Wiederrecht, G. J. (1995) *J. Biol. Chem.* **270**, 26511–26522

^k Eberhardt, E. S., Loh, S. N., Hinck, A. P., and Raines, R. T. (1992) *J. Am. Chem. Soc.* **114**, 5437–5439

^l Kakalis, L. T., and Armitage, I. M. (1994) *Biochemistry* **33**, 1495–1501

^m Zhao, Y., and Ke, H. (1996) *Biochemistry* **35**, 7362–7368

ⁿ Thériault, Y., Logan, T. M., Meadows, R., Yu, L., Olejniczak, E. T., Holzman, T. F., Simmer, R. L., and Fesik, S. W. (1993) *Nature (London)* **361**, 88–91

^o Konno, M., Ito, M., Hayano, T., and Takahashi, N. (1996) *J. Mol. Biol.* **256**, 897–908

^p Kern, D., Kern, G., Scherer, G., Fischer, G., and Drakenberg, T. (1995) *Biochemistry* **34**, 13594–13602

^q Göthel, S. F., Herrler, M., and Marahiel, M. A. (1996) *Biochemistry* **35**, 3636–3640

^r Clubb, R. T., Ferguson, S. B., Walsh, C. T., and Wagner, G. (1994) *Biochemistry* **33**, 2761–2772

^s Edwards, K. J., Ollis, D. L., and Dixon, N. E. (1997) *J. Mol. Biol.* **271**, 258–265

^t Rudd, K. E., Sofia, H. J., Koonin, E. V., Plunkett, G., III, Lazar, S., and Rouviere, P. E. (1995) *Trends Biochem. Sci.* **20**, 12–14

^u Rahfeld, J.-U., Rücknagel, K. P., Stoller, G., Horne, S. M., Schierhorn, A., Young, K. D., and Fischer, G. (1996) *J. Biol. Chem.* **271**, 22130–22138

^v Matouschek, A., Rospert, S., Schmid, K., Glick, B. S., and Schatz, G. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 6319–6323

shown in Eq. 9-89. Again, either **specific acid catalysis** (by H₃O⁺) or **general acid catalysis** is possible.

Enzymes are not able to concentrate protons or hydroxyl ions to the point that *specific* base or acid catalysis would be effective. However, either general acid or general base catalysis can be accomplished by groups present in an enzyme in their normal states of

protonation at the pH of the cell. Thus, if one of the enzymes must be dissociated to its conjugate base and a second must be protonated for reaction to take place (as in Eq. 9-54), it is reasonable to suppose that these two groups participate in acid–base catalysis. Of the acidic and basic groups present in proteins, the imidazole group of histidine would appear to be the

most ideal both for general base catalysis and, because both forms may exist in nearly equal amounts at pH 7, for general acid catalysis. However, carboxyl, thiol, lysyl, tyrosyl, and N-terminal amino groups are all thought to function in various enzymes.

The mutarotase of *E. coli* has a turnover number of 10^4 s^{-1} . The plot of $-\log K_m$ vs pH indicates two pK_a values in the free enzyme at 5.5 and 7.6, while the plot of $\log V_{\max}$ yields a single pK_a of 4.75 for the ES complex.¹³⁰ The latter might represent a carboxyl group on an imidazole group in its conjugate base form. Why doesn't the group having pK_a 7.6 in the free enzyme also show a pK_a in the ES complex? Either the group has no catalytic function so that EH_2S of Eq. 9-54 reacts to form products just as fast as does HES or the pK_a is so strongly shifted by substrate binding that it is not detected in the $\log V_{\max}$ plot.

The Brönsted relationships. The effectiveness of a specific base as a general base catalyst can usually be related to its basicity (pK_a) via the **Brönsted equation** (Eq. 9-90). Here, k_b is defined by Eq. 9-88 and G_b is a

$$\log k_b = \log G_b + \beta (pK_a) \quad (9-90)$$

constant for a particular reaction. A similar equation (9-91) relates the constant k_a for general acid catalysis

$$\log k_{\text{HB}} = \log G_a - \alpha (pK_a) \quad (9-91)$$

to pK_a . These equations are linear Gibbs energy relationships similar to the ones discussed in Box 6-C. For the Brönsted equations to hold, the Gibbs energy of activation for the reaction must be directly related to the basicity or acidity of the catalyst.

The exponents β and α of Eqs. 9-90 and 9-91 measure the *sensitivity* of a reaction toward the basicity or acidity of the catalyst. It is easy to show that as β and α approach 1.0 general base or general acid catalysis is lost and that the rate becomes exactly that of specific hydroxyl ion or specific hydrogen ion catalysis.¹³¹ As β or α approach zero basic or acidic catalysis is undetectable. Thus, general base or general acid catalysis is most significant when β or α is in the neighborhood of 0.5. Under these circumstances it is easy to see how a moderately weak basic group, such as the imidazole group of histidine, can be an unusually effective catalyst at pH 7.

To determine α or β experimentally a plot of $\log k_b$ or $\log k_{\text{HB}}$ vs pK_a (a **Brönsted plot**) is made and the slope is measured. Statistical corrections (Chapter 7) should be applied for dicarboxylic acids and for ammonium ions from which one of three protons may be lost from the nitrogen atom. General base or general acid catalysis implies an important feature of any mechanism for which it is observed, namely, that removal of a proton or addition of a proton is involved in the

rate-determining step of a reaction.

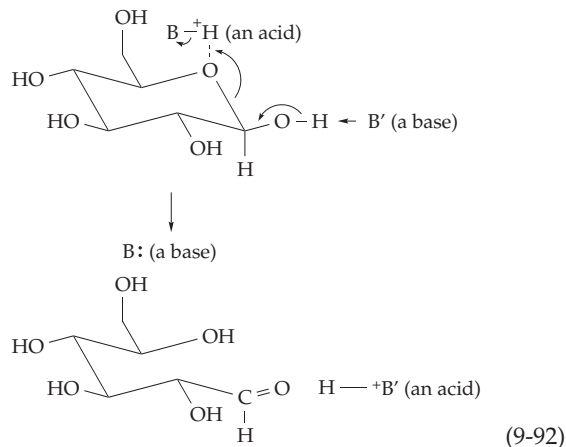
Concerted acid–base or tautomeric catalysis.

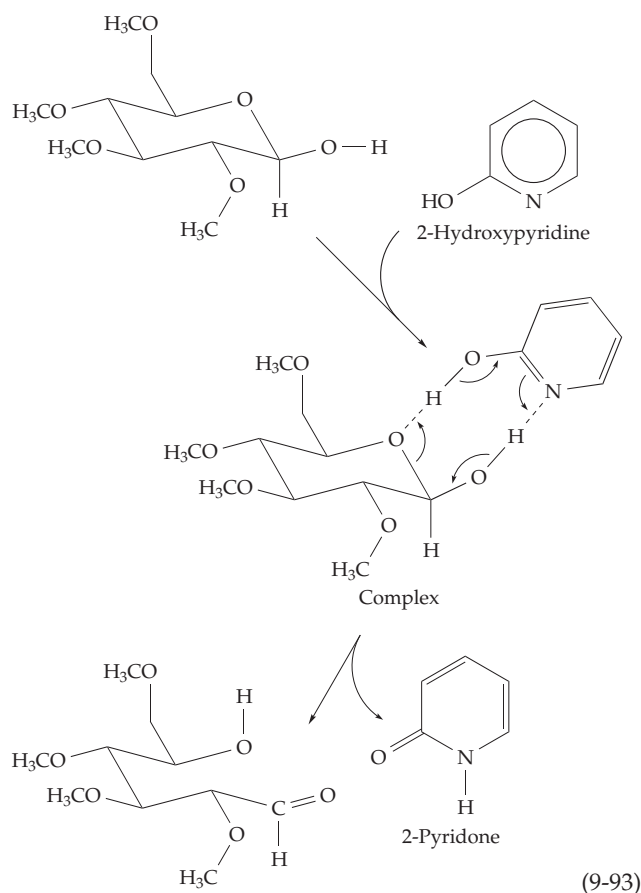
A third possible type of catalysis requires that a base and an acid act *synchronously* to effect the breaking and formation of bonds in a single step. Thus, tetramethylglucose mutarotates very slowly in benzene containing either pyridine (a base) or phenol (an acid). However, when both pyridine and phenol are present, mutarotation is rapid. This suggested to Swain and Brown¹³² a **concerted mechanism** (Eq. 9-92) in which both an acid and a base participate.

During the reaction shown, the acid BH^+ is converted to its conjugate base B and the base B' to its conjugate acid $\text{H}^+\text{B}'$. It might seem that these agents, having been altered by the reaction, are not serving as true catalysts. However, a simple proton exchange will restore the original forms and complete the catalytic cycle. In aqueous solutions, water itself might act as the acid or the base or even both in concerted catalysis.

The original experimental evidence for concerted acid–base catalysis of the mutarotation in benzene is now considered unsound^{133,134} and concerted acid–base catalysis has been difficult to prove for nonenzymatic reactions in aqueous solution. However, measurements of kinetic isotope effects seem to support Swain and Brown's interpretation.¹³⁵ Concerted acid–base catalysis by acetic acid and acetate ions may have been observed for the enolization of acetone¹³⁶ and it may be employed by enzymes.^{136a}

Swain and Brown showed that a more effective catalyst for the mutarotation of sugars than a mixture of an acid and a base can be designed by incorporating the acidic and basic groups into *the same molecule*.^{132,135} Thus, with 0.1 M tetramethylglucose in benzene solution, 0.001 M α -hydroxypyridine is 7000 times as effective a catalyst as a mixture of 0.001 M pyridine + 0.001 M phenol. Swain and Brown suggested the following completely concerted reaction mechanism for the **polyfunctional catalyst α -hydroxypyridine** in which the hydrogen-bonded complex formed (Eq. 9-93) is analogous to an enzyme–substrate complex. The product of the catalyst is 2-pyridone, a tautomer of





2-hydroxypyridine, with which it is in a rapid reversible equilibrium.

Rony called catalysis of the type illustrated in Eq. 9-93 tautomeric catalysis and suggested that its efficiency lies not simply in the close proximity of acidic and basic groups in the same molecule but also in the ability of the catalyst to repeatedly cycle between the two tautomeric states.¹³³ For an enzyme the tautomerization of the free catalyst could sometimes be rate determining (see Section A.4 on isomechanisms).

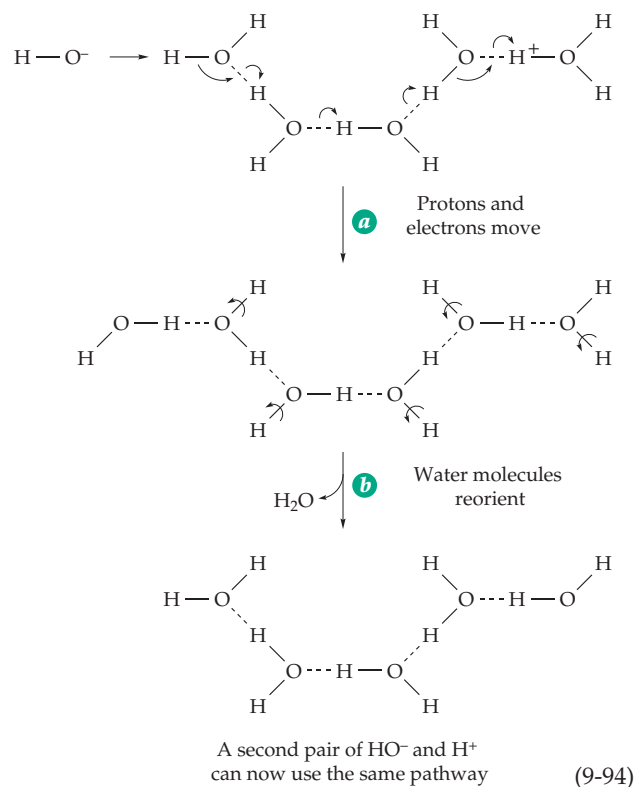
4. Hydrogen Bonding and the Transfer of Protons within Active Sites

An extensive network of hydrogen bonds runs throughout most proteins and may be especially complex within active sites. The network often runs through bound substrate molecules and immobilized water molecules in the active site cavity. This network arises in part because of the frequent occurrence of acidic and basic catalytic groups in active sites and by the fact that many substrates contain polar groups. The structure of an enzyme often seems to be more rigid in a complex with substrates or inhibitors than in the free state. Does this network of linked hydrogen bonds play a role in catalysis? If so, what? Wang

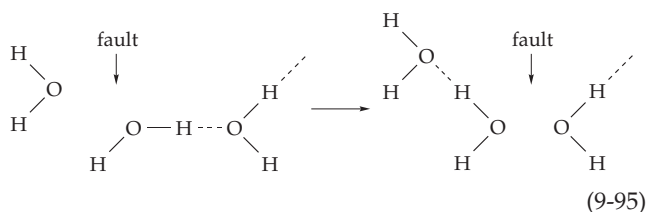
suggested that rapid transfer of protons along rigidly and accurately held hydrogen bonds in the ES complex may be an essential feature of enzymatic catalysis.¹³⁷ This conclusion is supported by many more recent observations. A remaining question is whether this proton transfer can take place in such a way that the transition state barrier is lowered.

Ultrafast proton transfer. The diffusion-controlled limit for second-order rate constants (Section A3) is $\sim 10^{10} \text{ M}^{-1} \text{ s}^{-1}$. In 1956, Eigen, who had developed new methods for studying very fast reactions, discovered that protons and hydroxide ions react much more rapidly when present in a lattice of ice than when in solution.¹³⁸ He observed second-order rate constants of 10^{13} to $10^{14} \text{ M}^{-1} \text{ s}^{-1}$. These represent rates almost as great as those of molecular vibration. For example, the frequency of vibration of the OH bond in water is about 10^{14} s^{-1} . The latter can be deduced directly from the frequency of infrared light absorbed in exciting this vibration: Frequency ν equals wave number (3710 cm^{-1} for $-\text{OH}$ stretching) times c , the velocity of light ($3 \times 10^{10} \text{ cm s}^{-1}$).

This ultrafast transfer of protons can be explained as follows: The OH^- ion and the proton, which is combined with a water molecule to form H_3O^+ , are both hydrogen-bonded to adjacent water molecules. In ice a chain of hydrogen-bonded water molecules links the hydroxide and the hydrogen ions (Eq. 9-94). By synchronous movement of electron pairs from the OH^- ion and from each of the water molecules in the

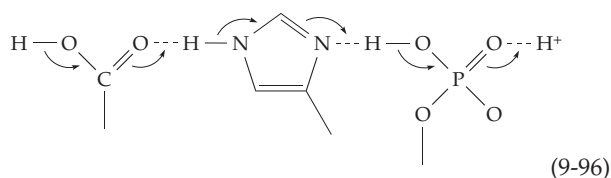


chain (as indicated by the little arrows) the neutralization can take place during the time of one molecular vibration (step *a*, Eq. 9-94). The positions of the oxygen atoms remain unchanged at the end of the reaction but the protons that were engaged in hydrogen bond formation have moved toward the left and are now attached to different oxygen atoms. The central chain of water molecules can be restored to its original state if each molecule rotates around one of its single bonds, swinging a hydrogen from right back to left (Eq. 9-94, step *b*). This must be a slower process which may occur one molecule at a time. The rotation of the leftmost water molecule would leave an empty space between two O-atoms, a "fault" in the H-bonded chain of the ice structure. This fault can be corrected by



rotation of the second molecule (Eq. 9-95); however, this would create a fault at the right of the second water molecule. This would induce rotation of the third molecule, etc., causing the fault to migrate from left to right across the entire chain of water molecules (Eq. 9-94, step *b*) and leaving the chain ready to function again in transferring another proton.

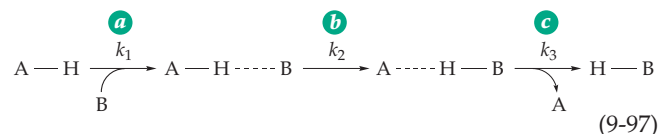
Because of the disorder present in ice, an array of water molecules such as that in Eq. 9-94 wouldn't revert to its exact original form in step *b*. However, active sites of enzymes are highly structured and proton transfers may occur with precision. For example, a synchronous shift of protons in an array of carboxylic acid, imidazolium, and phosphate groups can be envisioned readily (Eq. 9-96). The net effect of the process is to transfer a proton from one end of the chain to the other (as in Eq. 9-94) with facile tauto-



merization reactions providing the pathway. Such a pathway might be constructed by protein side chains to join the two sides of an active center promoting a concerted acid-base catalyzed reaction such as that of Eq. 9-92. Other tautomerization processes are possible within proteins if the existence of less stable minor tautomers of selected amide groups in the peptide backbone is allowed.¹³⁹ Nagle and Morowitz suggested a

process similar to that of Eq. 9-94 but involving side chains of serine residues.¹⁴⁰

Proton transfer rates. Consider the reversible reaction of a proton acceptor B with acid H-A (Eq. 9-97). Eigen pointed out that the reaction will be fastest if the two reactants form a hydrogen-bonded complex (Eq. 9-97, step *a*).¹³⁸ The hydrogen bonding shortens the distance from the proton to B and allows for very rapid transfer of the proton from A to B within the



hydrogen-bonded complex (step *b*). The activation energy is close to zero. The complex dissociates in step *c* to form the products. The reactions are reversible, even though they have been indicated by unidirectional arrows.

The hydrogen-bonded complexes A-H...B and A...H-B can be formed readily if A and B are oxygen or nitrogen bases such as COO^- , OH^- , NH_2 , or imidazole. In such cases, as in ice, the interconversion of the two complexes (step *b* in Eq. 9-97) is very fast. The overall rate of the proton transfer will then be limited by diffusion of B and H-A together or by diffusion of A and H-B apart. It might seem that these two processes should also both be very fast. However, the rates will be determined by the concentration ratio $[\text{A-H...B}]/[\text{A...H-B}]$. If $[\text{A-H...B}]$ equals $[\text{A...H-B}]$ the rate of dissociation of A-H...B will be half what it is if nearly all of the complex is A...H-B. If A-H...B predominates by 1000 to 1, the rate will be slowed much more.

The equilibrium constant for the reaction of Eq. 9-97 will be:

$$\begin{aligned} K_{\text{eq}}^{\text{AB}} &= K_{\text{a}}^{\text{HA}} / K_{\text{a}}^{\text{HB}} = k_{\text{f}} / k_{\text{r}} \\ \text{or} \quad \log K_{\text{eq}} &= \text{p}K_{\text{a}}^{\text{HA}} - \text{p}K_{\text{a}}^{\text{HB}} = \log k_{\text{f}} - \log k_{\text{r}} \end{aligned} \quad (9-98)$$

Here k_{f} and k_{r} refer to the rates in the forward and reverse directions. If the $\text{p}K_{\text{a}}$'s of HA and HB are equal k_{f} and k_{r} will be the same, but if they are very far from equal the reaction will be slowed in one direction. If proton transfer is a step in an enzymatic reaction it may be slowed enough to become rate limiting.

The difference in $\text{p}K_{\text{a}}$ values between the proton donor and the proton acceptor in Eq. 9-97 can be expressed as the Gibbs energy change which at 25°C is equal to $\pm 5.71 \times \Delta \text{p}K_{\text{a}}$. This is often referred to as the **thermodynamic barrier** ΔG° to a reaction and $\Delta G^\ddagger$ can be expressed as the sum of the thermodynamic barrier ΔG° plus an **intrinsic barrier** $\Delta G_{\text{int}}^\ddagger$. For the proton transfer of Eq. 9-97 the intrinsic barrier (for step *b*) is thought to be near zero so that $\Delta G^\ddagger \approx 5.71 \Delta \text{p}K_{\text{a}}$.

From this we can conclude that two pK_a values can be as much as eight units apart and $\Delta G^\ddagger$ will still be less than 50 kJ / mol, low enough to permit rapid enzymatic reactions. However, for transfer of a proton from a C–H bond to a catalytic group, for example, to form an enolate ion for an aldol condensation (Chapter 13), the intrinsic barrier is known to be about 50 kJ / mol.¹⁴¹ In this case, to allow rapid enzymatic reaction either the thermodynamic barrier must be very low, as a result of closely matching pK_a values, or the enzyme must lower the intrinsic barrier. It may do both.

Marcus theory. Discussion of intrinsic barriers is often approached using a quantitative theory proposed by Marcus.^{142–145} It was first applied to electron transfer (Chapter 16) but has been used for a great variety of nonenzymatic and enzymatic reactions. As used by Gerlt and Gassman,¹⁴¹ the Marcus formalism describes the reaction coordinate (Fig. 9-15A) as an inverted parabola whose shape is determined by the overall Gibbs energy change ΔG° and the intrinsic barrier $\Delta G^\ddagger_{\text{int}}$. The value of the Gibbs energy at any point on the curve is designated G and the reaction coordinate x is taken as 0 for the reactants and 1 for the products.

$$G = -4 \Delta G^\ddagger_{\text{int}} (x - 0.5)^2 + \Delta G^\circ (x - 0.5)$$

$$\text{Limits: } \Delta G^\circ / 4 \leq \Delta G^\ddagger_{\text{int}} \leq 4 \Delta G^\circ \quad (9-99)$$

Differentiation of Eq. 9-99 yields the position of the transition state coordinate $x^\ddagger$ as follows:

$$x^\ddagger = 0.5 + \Delta G^\circ / 8 \Delta G^\ddagger_{\text{int}} \quad (9-100)$$

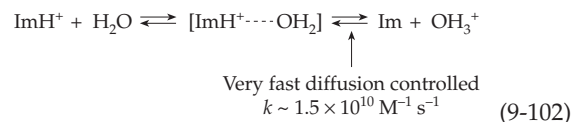
and it follows that

$$\Delta G^\ddagger = \Delta G^\ddagger_{\text{int}} (1 + \Delta G^\circ / 4 \Delta G^\ddagger_{\text{int}})^2 \quad (9-101)$$

Diffusion-controlled dissociation of protons.

The direct proton transfer between C-1 and C-2 during the action of sugar isomerases may seem puzzling. How can a highly mobile proton remain attached to a group in the enzyme for a millisecond or more instead of being transferred out to a solvent molecule? This can mean that the enzyme promotes the transfer of a hydride ion or of a hydrogen atom rather than a proton (see Chapter 13). If so, the observed proton exchange with solvent would be an unimportant side reaction. On the other hand, could the group in the enzyme that removes the proton be out of contact with the aqueous medium and thus able to hold onto the proton more tightly? In recent years, it has been recognized that neither of these explanations may be necessary. An imidazole group is a likely proton-carrying group at many active sites and it is thought that a proton cannot be expected to transfer out from an imidazolium group with a rate constant greater than $\sim 10^3 \text{ s}^{-1}$.

The argument is as follows.¹⁴⁶ The rate of donation of a proton from H_3O^+ to imidazole (reverse of Eq. 9-102) is known to be diffusion controlled with a rate constant of $1.5 \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$.



The equilibrium constant for Eq. 9-102, calculated from the pK_a of 7.0 for imidazole, is 10^{-7} M . Since K_{eq} is also the *ratio of the overall rate constants for the forward and reverse reactions*, we see that for the forward reaction $k_f = 10^{-7} \times 1.5 \times 10^{10} = 1.5 \times 10^3 \text{ s}^{-1}$. This slow rate results from the fact that in the intermediate complex (in brackets in Eq. 9-102) the proton is on the imidazole group most of the time. For a small fraction of the time it is on the coordinated molecule H_2O but reverts to being on the imidazole many times before the imidazole and OH_3^+ separate (see also Eqs. 9-97 and 9-98). Because of this unfavorable equilibrium within the complex, the diffusion-controlled rate of proton transfer from a protonated imidazole to water is far less than for proton transfer in the reverse direction.

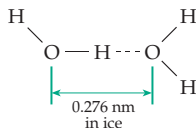
Coupled proton transfers. Enzymatic reactions often require the transfer of two or more protons. They may move individually, one proton at a time, or as in Eq. 9-93 they may move synchronously in a **coupled** or concerted process. Such coupled movement is generally not possible for heavier nuclei.¹⁴⁷ However, studies of solvent isotope effects using a **proton inventory** technique^{111,148,149} have provided evidence favoring coupled proton transfers for a variety of enzymes. Movement of protons along hydrogen-bonded paths, as well as electron transfer, may take place with some participation of **quantum mechanical tunneling**.^{150–152} Coupling to vibrational modes of the hydrogen-bonded protons may provide **vibration-assisted tunneling**.^{153–154d} These reactions are associated with unusually large kinetic isotope effects.

Unusually strong hydrogen bonds. The strength of a hydrogen bond is thought to be directly related to the length, which is ordinarily taken to be the distance between the two surrounding heavier atoms (see Chapter 2, Section B,3). Hydrogen bonds are sometimes classified on the basis of N–H---O distances^{155,156} as:

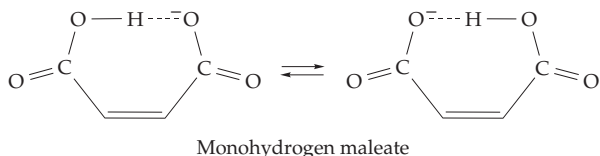
<0.25 nm	very strong
0.25–0.265 nm	strong
>0.28 nm	weak
>0.37 nm	no van der Waals contact but electrostatic interaction still occurs

N–H---O distances may be a little longer than these.

The 0.276-nm hydrogen bonds in ice are regarded as moderately strong. However, if one of the oxygen atoms in an O—H---O hydrogen bond carries a negative charge, as in the maleate monoanion, it will be

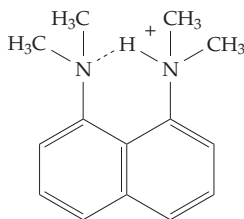


shorter.^{156–158} Although the proton will be closer to one oxygen atom than to the other, it will be able to move between them by passing a very low transition state barrier. For this to occur the microscopic pK_a values for the two groups (when they are protonated) must be similar; for the maleate anion they are identical.

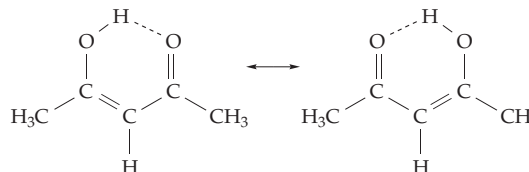


The *strength* of a hydrogen bond can be measured for hydrogen-bonded complexes in the gas phase and range from 10–100 kJ/mol and even higher¹⁵⁵ for such complexes as (FHF)[−]. It is more difficult to establish the strength in the liquid state or within the active site of an enzyme, but shorter hydrogen bonds are usually stronger than longer ones. Hydrogen bond distances in crystals of small molecules can be measured precisely by X-ray or neutron diffraction but greater uncertainty is present in distances within proteins. Lengths and probably strengths of hydrogen bonds in proteins can be measured by NMR methods.^{158a,b} For example, solid-state NMR measurements on crystals of amino acids and other carboxylic acids have provided a plot of ¹H chemical shift vs hydrogen bond length.¹⁵⁹ For very short (0.24- to 0.25-nm) hydrogen bonds the ¹H chemical shift may be as great as 21 ppm. See also Fig. 3-30 and associated discussion. When dissolved in an ¹H₂O–²H₂O mixture strongly hydrogen-bonded protons within a protein become enriched in ¹H. The ¹H/²H ratio of the hydrogen-bonded protons provides another measure of the hydrogen bond strength.^{158c}

A short hydrogen bond is also present in such cations as the following:



The pK_a for dissociation of its proton is 12.3 and the hydrogen-bonded proton is probably located in the *center* of the bond with both amino groups sharing the charge.¹⁶⁰ Enols can also form unusually strong “resonance-assisted” hydrogen bonds:



The structure can be thought of as a resonance-stabilized enolate anion with a proton bound between the two oxygen atoms and equidistant from them.^{156,157}

Do these short “**low-barrier hydrogen bonds**” have a special significance in enzymology? Proposals that they contribute to stabilization of transition states^{161–166} have received some support^{154,167} and aroused controversy.^{158b,168–173} In later chapters we will examine specific enzymes in which low-barrier hydrogen bonds have been observed.

5. Covalent Catalysis

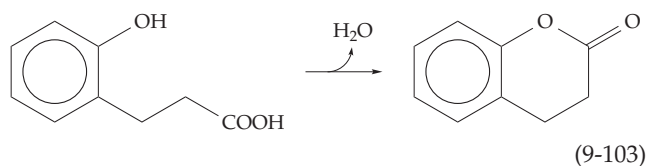
In addition to participating in acid–base catalysis, some amino acid side chains may enter into covalent bond formation with substrate molecules, a phenomenon that is often referred to as covalent catalysis.¹⁷⁴ When basic groups participate this may be called **nucleophilic catalysis**. Covalent catalysis occurs frequently with enzymes catalyzing nucleophilic displacement reactions and examples will be considered in Chapter 12. They include the formation of an acyl-enzyme intermediate by chymotrypsin (Fig. 12-11). Several of the coenzymes discussed in Chapters 14 and 15 also participate in covalent catalysis. These coenzymes combine with substrates to form reactive intermediate compounds whose structures allow them to be converted rapidly to the final products.

6. Proximity and Orientation

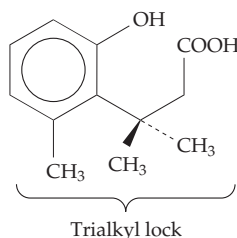
One of the earliest ideas about enzymes was that they simply brought reactants together and bound them side by side for a long enough time that the reactive groups might bump together and finally react. How important is this **proximity factor**? Page and Jencks estimated that rate enhancements by factors of 10³ or more may be expected solely from the loss in the entropy of two reactants when they are bound in close proximity on an enzyme surface.^{107,175,176} In view of the large entropy decrease involved, *the enthalpy of*

binding must be high, and if this explanation is correct the binding of the substrates to the enzyme provides much of the driving force for catalysis. Westheimer described this by stating that enzymes use the substrate-binding force as an **entropy trap**.¹⁰⁸ The losses of translational and rotational entropy, which Page and Jencks estimated as up to -160 to -210 J/deg/mol, overcome the unfavorable entropy of activation that is usual in bimolecular reactions.

How precise must the orientation of substrates be for rapid reaction?^{107,177,177a} Compounds such as the acid shown in Eq. 9-103 form an internal ester (lactone) spontaneously with elimination of water.



However, the following compound reacts at a rate over 10^{11} times that of the acid shown in Eq. 9-102. This is presumably because its conformation is highly restricted and the $-\text{COOH}$ is constrained to frequently collide with the $-\text{OH}$ group.^{178,179} The three methyl

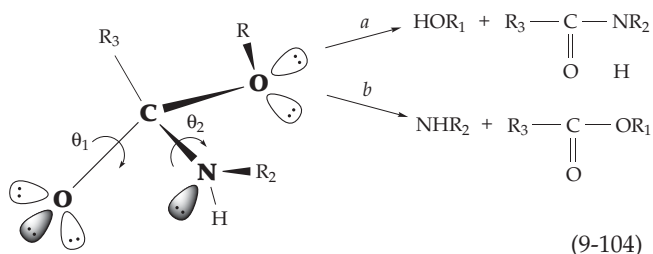


groups interdigitate and form a **trialkyl lock**. Orientation must also play a large role in enzymatic catalysis. As previously emphasized, enzymes often orient substrates precisely by formation of multiple hydrogen bonds. Small distortions by mutation or substitution of an essential metal by a different one can have very great effects. For example, because of an altered coordination pattern isocitrate dehydrogenases with Ca^{2+} in the active site has a maximum velocity of catalysis only 2.5×10^{-3} that with the normal Mg^{2+} -containing enzyme.¹⁸⁰ Bruice and associates concluded that enzymes must bring reacting groups into close proximity with orbitals of the reactants properly aligned in the ground state prior to moving to the transition state. They suggested that enzymes preorganize the enzyme-substrate complex into a **near attack conformation** in which the positions of reacting groups, the arrangements of hydrogen bonds, and the local dielectric constant in the active site, resemble those in the transition state. In this conformation the energy barrier to the transition

state may be very low.^{128-128b} Preorganization of the complex also acts to eliminate the slow components of solvent reorganization required for reaction in aqueous solutions.^{128c,128d}

The necessity for reacting groups of substrates to collide with an orientation that allows productive interaction of electronic orbitals is often called a **stereoelectronic effect**. An example is the addition reaction of Eq. 9-74. The orbital of an unshared pair of electrons on the HO^- ion must be perpendicular to the plane of the double bond. Furthermore, if the proton becomes attached to the adjacent carbon in a synchronous or concerted manner it must enter from the opposite side, as it does in Eq. 9-74.

In many biochemical reactions an alcohol or amine is eliminated from a tetrahedrally bonded intermediate as in Eq. 9-104. Deslongchamps proposed a stereoelectronic theory^{181,182} according to which elimination of either $\text{NH}-\text{R}_2$ or $-\text{O}-\text{R}_1$ from this intermediate will depend upon the values of the torsion angles θ_1 and θ_2 (Eq. 9-104). The theory asserts that for rapid elimination of OR_1 (Eq. 9-104a), unshared electron pairs on both the $-\text{O}^-$ and N atoms must be antiperiplanar to the bond being broken (see Figure 2-2). If rotation around the $\text{C}-\text{O}$ bond (θ_1) occurs an orientation can be found in which an electron pair on each of the two oxygens will be antiperiplanar to the bond to NHR_2 . From this orientation the latter will be eliminated (Eq. 9-104b). The theory has been supported by much experimental evidence involving reaction rates and product distribution among competing reactions of small conformationally restricted organic molecules.¹⁸¹⁻¹⁸⁴ Although more recent experiments¹⁸⁵ suggest that stereoelectronic factors may be of less significance than had been assumed, even a small decrease in transition state energy can be significant in an enzyme-catalyzed reaction. Enzymes may not only orient substrates in accord with stereoelectronic principles but also be able to promote conformational alterations in intermediates that allow them to take advantage of stereoelectronic factors. Examples are considered in Chapter 12.



7. The Microenvironment

A substrate bound at an active site may be in an environment quite different than that in an ordinary

aqueous solution. In fact, the protein often surrounds the substrate to the extent that *the protein is the solvent*. What kind of **microenvironment** does the protein provide and can it assist in the catalytic process? Charged and dipolar groups of the protein provide an electrostatic field that provides part of the binding energy and which may also assist in catalysis, a concept expressed by Quastel as early as 1926.^{183a} The ends of protein helices often seem to point at active centers.^{186,187} In many cases more than one N terminus with its positive electrical potential (see Chapter 2) or more than one C terminus (negative potential) of a helix point to an active site. These helix dipoles may be important in stabilizing transition state structures¹⁸⁸ or in altering pK_a values of functional groups.¹⁸⁶ Fluctuations in charge distribution within a protein and in the hydrogen-bonding pattern of the protein with solvent and substrate may also be important.¹⁸⁹

For some enzymatic reactions a transition state will be favored by a medium of very low dielectric constant and a correctly constructed active site can provide just such a surrounding.^{189a} Hydrophobic groups may be packed around a site where an ion pair or other ionic interaction between enzyme and substrate occurs increasing the strength of that interaction. Conformational changes may enhance such effects. The dehydration of polar groups that must often occur upon binding of substrate may make these groups more reactive.^{187,190,191} It has been suggested that the substantial volume changes ($\Delta V^\ddagger$) that sometimes occur during formation of transition states for enzyme-catalyzed reactions may result largely from changes in hydration of groups on the enzyme surface and that these changes may play an important role in catalysis.

In the past most enzymologists tacitly assumed that the external medium in which enzymes act must be aqueous. However, many enzymes function well in media containing largely hydrocarbons. Enzymes in a dry, powdered form have been suspended directly in organic solvents.^{192,193} Under these conditions, enzymes may contain only tightly bound “structural” water, together with less than one equivalent of a monolayer of water outside the protein. The enzymes often remain active and are able to catalyze reactions with an altered substrate specificity as well as different reactions overall.

8. Strain and Distortion

The fact that enzymes appear to bind their substrates in such a way as to surround and immobilize them means that something other than the kinetic energy of the substrate is needed to provide energy for the ES complex to pass over the transition state barrier. What is the source of this activation energy? As with nonenzymatic reactions, it must come ultimately from

the translational energy of solvent or solute molecules bombarding the complex. Can enzymes act as “energy funnels” that effectively channel kinetic energy from spots on the enzyme surface to the active site?¹⁹⁴ This could either be through strictly mechanical movement or through induction by fluctuations of the electrical field.¹⁹⁵

It is often suggested that enzymes assist in catalysis by distorting bond lengths or angles away from their normal values. If the distorted structure were closer to the transition state geometry than the undistorted structure, catalysis would be assisted (see *lysozyme*, Chapter 12). However, Levitt¹⁹⁶ concluded (see also Fersht²) that forces provided this way by a protein are small and that the protein would become distorted rather than a substrate. On the other hand, if binding of a substrate distorts the protein, could the resultant “stored energy” be used in some way to assist in catalysis? If binding of a substrate bends a “spring” (Section C,4), can the tension in the spring then be used to distort the enzyme to be more exactly complementary to the transition state? This concept has become popular.^{2,113,197}

The amino acid side chains of proteins are usually well packed. However, neither the side chains nor the main chain are rigid and immobile. Some regions of the protein will contain empty spaces – packing defects. Lumry called these **mobile defects** because, as a result of fluctuations in side chain packing, they can move within a protein from one site to another for a considerable distance.¹⁹⁸ Much evidence, including X-ray studies at low temperatures,¹⁹⁹ supports the existence of many **conformational substates** in proteins. Some substates may bind substrates better than others and some may allow conversion to the transition state more readily than do others. Rotation of histidine rings, peptide linkages, –OH groups, or amide side chains^{199a} may be required and has, in fact, been observed for some enzymes. Perhaps in one of the substates of an enzyme–substrate complex an especially favorable vibrational mode²⁰⁰ leads the complex to the transition state. In the transition state the packing of side chains may be especially tight. A mobile defect present in the active site may have moved elsewhere. The binding of substrates to the protein in the transition state will also be tighter because of the conformational alterations that have occurred. The binding energy of the substrates is now being utilized to lower the transition state barrier. The substrate is literally squeezed into the transition state configuration.

Is it possible that the protein domains forming an active site act like **ferroelectric crystals**, which change their dipole moment in response to a change in electric field? The highly polarizable hydrogen-bonded network, the amide linkages, imidazole rings, guanidinium group, etc. of active sites may permit a flip-flop of the dipole moment as in domains of ferroelectric crystals.

In such crystals, e.g., those of the hydrogen-bonded KH_2PO_4 , a 180° change of dipole-moment direction results from very small movements of heavy atoms together with larger movements of the hydrogen-bonded protons.^{200a,b,c} Could a similar flip-flop in a protein domain be coupled to the passage of a substrate over the transition state barrier? I have not seen any discussion of this possibility, but the structure of protein domains would seem to allow it. One could also imagine that with a small change in the hydrogen-binding arrangement of protein groups an active site could become preorganized to favor a flip-flop along a hydrogen-bonded chain in a different direction for a subsequent step in a reaction sequence.

9. Why Oligomeric Enzymes?

As we have seen (Chapter 7), a large fraction of all proteins exist as dimers, trimers, and higher oligomers. Oligomeric proteins raise the osmotic pressure much less than would the same number of monomeric subunits and this may be crucial to a cell. Another advantage of oligomers may be reflected in the fact that active sites of enzymes are often at interfaces between two or more subunits. This may enhance the ability of enzymes to undergo conformational rearrangements that are required during their action, just as hemoglobin changes its oxygen affinity in concert with a change in inter-subunit contacts (Fig. 7-25).

A curious observation is that crystals of the dimeric pig heart malate dehydrogenase bind only one molecule of the substrate NAD^+ per dimer tightly; the second NAD^+ is bound weakly.^{201,202} Similar *anti-cooperativity* has been reported for other crystalline dehydrogenases²⁰³ and various other enzymes. An intriguing idea is that anticooperativity in binding might reflect a cooperative action between subunits during catalysis. Suppose that only conformation A of a dehydrogenase bound reduced substrate and NAD^+ , while conformation B bound NADH and oxidized substrate. If reduced substrate and NAD^+ were present in excess and if oxidized substrate were efficiently removed from the scene by further oxidation, the following cooperative events could occur in the mixed AB dimer. The subunit of conformation A would bind substrates, react, and be converted to conformation B. At the same time, because of the strong AB interaction, the subunit that was originally in conformation B would be converted back to A and would be ready to initiate a new round of catalysis. Since conformation A has a low affinity for NADH , dissociation of the reduced coenzyme, which is often the slow step in dehydrogenase action, would be facilitated.^{204–207} Such a **reciprocating** or **flip-flop** mechanism, suggested first by Harada and Wolfe,²⁰⁴ is attractive because it provides a natural basis for the

existence of the many known dimeric enzymes that do not exhibit evident allosteric properties. Attempts to verify this idea have largely failed. However, recent crystallographic studies of both glyceraldehyde phosphate dehydrogenase^{206,207} and thymidylate synthase²⁰⁸ are consistent with the proposal. There is still a possibility that coordinated reciprocal changes in distribution of electrical charges in the two subunits may also be important.

10. Summary

It appears that enzymes exert their catalytic powers by **first** bringing together substrates and binding them in proper orientations at the active site. **Second**, they often provide acidic and basic groups in the proper orientations to promote proton transfers within the substrate. **Third**, groups within the enzyme (especially nucleophilic groups) may enter into covalent interaction with the substrates to form structures that are more reactive than those originally present in the substrate. **Fourth**, the protein often closes around the substrate to immobilize it and to hold it in an environment, often lacking water, which could impede catalysis. The enzyme is also probably able to make small readjustments of its structure to provide good complementarity to the substrate at every stage and especially to the transition state structure. **Fifth**, the enzyme may be able to induce strain or distortion in the substrate perhaps accompanying a conformational change within the protein. The following question is often asked: "Why are enzymes such large molecules?"²⁰⁹ At least part of the answer is that an enzyme usually interacts, sometimes via special domains, with other proteins.²⁰⁶ Another part of the answer is evident when we consider that the formation of a surface complementary to that of the substrate and possessing reasonable rigidity requires a complex geometry in the peptide backbone. In addition, the enzyme must provide functional groups at the proper places to enter into catalysis. It may require a certain bulk to provide a low dielectric medium. Finally, if essential conformational changes occur during the course of the catalysis, we can only be surprised that nature has succeeded in packing so much machinery into such a small volume.

E. Classification of Enzymes

An official commission of the International Union of Biochemistry (IUB) has classified enzymes in the following six categories:²¹⁰

1. *Oxidoreductases*. Enzymes catalyzing dehydrogenation or other oxidation and reduction reactions.

2. *Transferases*. These catalyze group transfer reactions.
3. *Hydrolases*. Enzymes catalyzing transfer of groups to the HO^- ion of H_2O .
4. *Lyases*. Enzymes promoting addition to double bonds or the reverse.
5. *Isomerases*. Enzymes catalyzing rearrangement reactions.
6. *Ligases* (synthetases). Enzymes that catalyze condensation with simultaneous cleavage of ATP and related reactions.

As an example, chymotrypsin is classified EC 3.4.4.5 according to the IUB system.²¹⁰ In this book a more mechanistically based classification is used. Because some official names are quite long, traditional trivial names for enzymes often have been retained. Remember that to be precise it is always necessary to mention the species from which an enzyme was isolated and, if possible, the strain. Also remember that almost every significant genetic difference is reflected in some change in some protein. It is possible that the enzyme you are working with is slightly different from the same enzyme prepared in a different laboratory.

References

1. Fromm, H. (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York
2. Fersht, A. (1999) *Structure and Mechanism in Protein Science: A Guide to Enzyme Catalysis and Protein Folding*, Freeman, New York
3. Dixon, M., and Webb, E. C., eds. (1979) *Enzymes*, 3rd ed., Academic Press, New York
4. Wharton, C. W., and Eisenthal, R. (1981) *Molecular Enzymology*, Wiley, New York
5. Engel, P. C. (1982) *Enzyme Kinetics, the Steady-State Approach*, Chapman and Hall, London
6. Hammes, G. G. (1982) *Enzyme Catalysis and Regulation*, Academic Press, New York
7. Kull, F. J. (1994) *Principles of Biomolecular Kinetics and Binding*, CRC Press, Boca Raton, Florida
8. Suckling, C. J., ed. (1990) *Enzyme Chemistry*, 2nd ed., Chapman and Hall, New York
9. Price, N. C., and Stevens, L., eds. (1989) *Fundamentals of Enzymology*, 2nd ed., Oxford Univ. Press, Oxford, England
10. Kubo, S. A. (1990) *Enzymes: a Comprehensive Study*, CRC Press, Boca Raton, Florida
11. Schulz, A. R. (1994) *Enzyme Kinetics From Diastase to Multi-enzyme Systems*, Cambridge Univ. Press, New York
12. Segel, I. H. (1975) *Enzyme Kinetics: Behavior and Analysis of Rapid Equilibrium and Steady-State Enzyme Systems*, Wiley, New York
13. Cornish-Bowden, A. (1995) *Fundamentals of Enzyme Kinetics*, 2nd ed., Portland Press, Brookfield, Vermont
- 13a. Purich, D. L., and Allison, R. D. (2000) *Trends Biochem. Sci.* **25**, 455
14. Lowry, O. H., and Passonneau, J. V. (1972) *A Flexible System of Enzymatic Analysis*, Academic Press, New York
15. Passonneau, J., and Lowry, O., eds. (1993) *Enzymatic Analysis*, 1st ed., Humana Press, Totowa, New Jersey
16. Liu, Y.-M., and Sweedler, J. V. (1995) *J. Am. Chem. Soc.* **117**, 8871–8872
17. Liébecq, C., ed. (1992) *Biochemical Nomenclature*, Portland Press, London and Chapel Hill, North Carolina (for the International Union of Biochemistry and Molecular Biology)
18. Caldin, E. F. (1964) *Fast Reactions in Solution*, Wiley, New York
19. Jencks, W. P. (1987) *Catalysis in Chemistry and Enzymology*, Dover, Mineola, New York (p. 393)
20. Glick, N., Landman, A. D., and Roufogalis, B. D. (1979) *Trends Biochem. Sci.* **4**, N82–N83
- 20a. Wilkinson. (1980) *Biochem. J.* **80**, 324–
21. Orsi, B. A., and Tipton, K. E. (1979) *Methods Enzymol.* **63**, 159–183
22. Wharton, C. W., and Szawelski, R. J. (1982) *Biochem. J.* **203**, 351–360
23. Cleland, W. W. (1979) *Methods Enzymol.* **63**, 103–138
24. Frieden, C. (1994) *Methods Enzymol.* **240**, 311–322
25. Schechter, A. N. (1970) *Science* **170**, 273–280
26. Cha, S. (1970) *J. Biol. Chem.* **245**, 4814–4818
27. Martinez, M. B., Flickinger, M. C., and Nelsestuen, G. L. (1996) *Biochemistry* **35**, 1179–1186
28. Bull, H. B. (1971) *An Introduction to Physical Biochemistry*, 2nd ed., Davis Co., Philadelphia, Pennsylvania
29. Eisenstein, B. I. (1987) *Escherichia coli and Salmonella typhimurium*, Am. Soc. Microbiology, Washington, D.C., FC Niedhardt, ed. (pp. 84–90)
30. Smoluchowski, M. (1917) *Z. Phys. Chem* **92**, 129–168
31. Caldin, E. F. (1964) *Fast Reactions in Solution*, Wiley, New York (pp. 10 and 279)
32. Debye, P. (1942) *Trans. Electrochem. Soc.* **82**, 265–272
- 32a. Elcock, A. H., Huber, G. A., and McCammon, J. A. (1997) *Biochemistry* **36**, 16049–16058
- 32b. Selzer, T. and Schreiber, G. (1999) *J. Mol. Biol.* **287**, 409–419
33. Mastro, A. M., Babich, M. A., Taylor, W. D., and Keith, A. D. (1984) *Proc. Natl. Acad. Sci. U.S.A.* **81**, 3414–3418
34. French, D. (1957) *Brewers Digest* **32**, 50–56
35. Cohn, E. J., and Edsall, J. T. (1943) *Proteins, Amino Acids and Peptides as Ions and Dipolar Ions*, Van Nostrand-Reinhold, Princeton, New Jersey (pp. 90–93)
36. Koenig, S. H. (1975) *Biopolymers* **14**, 2421–2423
37. Blacklow, S. C., Raines, R. T., Lim, W. A., Zamore, P. D., and Knowles, J. R. (1988) *Biochemistry* **27**, 1158–1167
38. Alberty, W. J., and Knowles, J. R. (1976) *Biochemistry* **15**, 5627–5631
39. Alberty, W. J., and Knowles, J. R. (1976) *Biochemistry* **15**, 5631–5640
40. Jenkins, W. T. (1982) *Adv. Enzymol.* **53**, 307–344
41. Haldane, J. B. S. (1930) *Enzymes*, Longmans, Green, New York
42. Cleland, W. W. (1970) in *The Enzymes*, 3rd ed., Vol. 2 (Boyer, P. D., ed), Academic Press, New York
43. Dalziel, K. (1957) *Acta Chem. Scand.* **11**, 1706–1723
44. Rebholz, K. L., and Northrop, D. B. (1994) *Arch. Biochem. Biophys.* **312**, 227–233
45. Volkenstein, M. V., and Goldstein, B. N. (1966) *Biokhim.* **31**, 541–547
46. Volkenstein, M. V., and Goldstein, B. N. (1966) *Biokhim. Biophys. Acta.* **115**, 471–477
47. Fromm, H. J. (1970) *Biochem. Biophys. Res. Commun.* **40**, 692–697
48. Seshagiri, N. (1972) *J. Theor. Biol.* **34**, 469–486
49. Huang, C. Y. (1979) *Methods Enzymol.* **63**, 54–84
50. Chou, K., and Forsén, S. (1980) *Biochem. J.* **187**, 829–835
51. Fromm, H. J., Silverstein, E., and Boyer, P. D. (1964) *J. Biol. Chem.* **239**, 3645–3652
52. Fromm, S. J., and Fromm, H. J. (1999) *Biochem. Biophys. Res. Commun.* **265**, 448–452
53. Silverstein, E., and Boyer, P. D. (1964) *J. Biol. Chem.* **239**, 3901–3907
54. Hurst, R. O. (1969) *Can. J. Biochem. Physiol.* **47**, 941–944
- 54a. Cha, S. (1968) *J. Biol. Chem.* **243**, 820–825
55. Srere, P. A. (1967) *Science* **158**, 936–937
56. Hiromi, K., ed. (1979) *Kinetics of Fast Enzyme Reaction-Theory and Practice*, John Wiley, New York
57. Rentzepis, P. M. (1978) *Science* **202**, 174–182
58. Hammes, G. G., and Schimmel, P. R. (1970) in *The Enzymes*, 3rd ed., Vol. 2 (Boyer, P. D., ed), pp. 67–114, Academic Press, New York
59. Ahrens, M.-L., Maass, G., Schuster, P., and Winkler, H. (1970) *J. Am. Chem. Soc.* **92**, 6134–6139
- 59a. Gutfreund, H. (1999) *Trends Biochem. Sci.* **24**, 457–460
- 59b. Beechem, J. M. (1998) *Biophys. J.* **74**, 2141
60. Fink, A. L. (1979) *Trends Biochem. Sci.* **4**, 8–10
61. Balashov, S. P., Imasheva, E. S., Ebrey, T. G., Chen, N., Menick, D. R., and Crouch, R. K. (1997) *Biochemistry* **36**, 8671–8676
62. Fink, A. L., and Gees, M. A. (1979) *Methods Enzymol.* **63**, 336–370
63. Alberty, R. A. (1956) *Adv. Enzymol.* **17**, 1–64
64. Dixon, M., and Webb, E. C., eds. (1979) *Enzymes*, 3rd ed., Academic Press, New York (pp. 138–163)
65. Cleland, W. W. (1977) *Adv. Enzymol.* **45**, 273–387
66. Cleland, W. W. (1982) *Methods Enzymol.* **87**, 390–405
67. Kyte, J. (1995) *Mechanism in Protein Chemistry*, Garland Publ., New York
68. Ono, S., Hiromi, K., and Yashikawa, Y. (1958) *Bull. Chem. Soc. Jap.* **31**, 957–962
69. Brocklehurst, K., and Dixon, H. B. F. (1976) *Biochem. J.* **155**, 61–70
70. Tipton, K. F., and Dixon, H. B. F. (1979) *Methods Enzymol.* **63**, 183–234
71. Dixon, M. (1953) *Biochem. J.* **55**, 170–171
72. Purich, D. L., and Fromm, H. J. (1972) *Biochim. Biophys. Acta.* **268**, 1–3
73. Williams, J. W., and Morrison, J. F. (1979) *Methods Enzymol.* **63**, 437–467
74. Morrison, J. F. (1982) *Trends Biochem. Sci.* **7**, 102–105
75. Frieden, C. (1964) *J. Biol. Chem.* **239**, 3522–3531

References

76. Morrison, J. F., and Walsh, C. T. (1988) *Adv. Enzymol.* **61**, 201–201
77. Gutheil, W. G., and Bachovin, W. W. (1993) *Biochemistry* **32**, 8723–8731
78. Di Cera, E., Hopfner, K.-P., and Dang, Q. D. (1996) *Biophys. J.* **70**, 174–181
79. Botts, J., and Morales, M. (1953) *Trans. Faraday Soc.* **49**, 696–707
80. Monod, J., Wyman, J., and Changeux, J. D. (1965) *J. Mol. Biol.* **12**, 88–118
81. Plapp, B. V., Leidal, K. G., Smith, R. K., and Murch, B. P. (1984) *Arch. Biochem. Biophys.* **230**, 30–38
82. Chadha, V. K., Leidal, K. G., and Plapp, B. V. (1983) *Journal of Medicinal Chemistry* **26**, 916–922
83. Cho, H., Ramaswamy, S., and Plapp, B. V. (1997) *Biochemistry* **36**, 382–389
84. Westley, A. M., and Westley, J. (1996) *J. Biol. Chem.* **271**, 5347–5352
85. Hervé, G., ed. (1989) *Allosteric Enzymes*, CRC Press, Boca Raton, Florida
86. Rubin, M. M., and Changeux, J. D. (1966) *J. Mol. Biol.* **21**, 265–274
87. Grant, G. A., Schuller, D. J., and Banaszak, L. J. (1996) *Protein Sci.* **5**, 34–41
88. Williams, R. O., Young, J. R., and Majiwa, P. A. O. (1979) *Nature (London)* **282**, 847–849
89. Rabin, B. R. (1967) *Biochem. J.* **102**, 22c
90. Newsholme, E. A., and Start, C. (1973) *Regulation in Metabolism*, Wiley, New York
91. Walsh, C. (1979) *Enzymatic Reaction Mechanism*, Freeman, San Francisco, California
- 91a. Plapp, B. V. (1982) *Methods Enzymol.* **87**, 469–499
92. Sandler, M., ed. (1980) *Enzyme Inhibitors as Drugs*, Univ. Park Press, Baltimore, Maryland
93. Bey, P. (1981) *Chem. Ind. (London)*, 139–144
94. Hanson, K. R. (1966) *J. Am. Chem. Soc.* **88**, 2731–2742
- 94a. Prelog, V., and Helmchen, G. (1982) *Angew. Chem. Int. Ed. Engl.* **21**, 567–583
95. Bentley, R. (1969) *Molecular Asymmetry in Biology*, Vol. 1, Academic Press, New York (pp. 49–56)
96. Bentley, R. (1970) *Molecular Asymmetry in Biochemistry*, Vol. 2, Academic Press, New York
97. Alworth, W. L. (1972) *Stereochemistry and its Application in Biochemistry*, Wiley-Interscience, New York
98. Bentley, R. (1978) *Nature (London)* **276**, 673–676
99. Barry, J. M. (1997) *Trends Biochem. Sci.* **22**, 228–230
100. Hirschmann, H. (1960) *J. Biol. Chem.* **235**, 2762–2767
101. Post, C. B., and Ray, W. J. J. (1995) *Biochemistry* **34**, 15881–15885
102. Tsou, C.-L. (1986) *Trends Biochem. Sci.* **11**, 427–429
103. Ferscht, A. R. (1980) *Trends Biochem. Sci.* **5**, 262–265
104. Fersht, A. R. (1998) *Science* **280**, 541
105. Nureki, O., Vassilyev, D. G., Tateno, M., Shimada, A., Nakama, T., Fukai, S., Konno, M., Hendrickson, T. L., Schimmel, P., and Yokoyama, S. (1998) *Science* **280**, 578–582
106. Hopfield, J. J. (1980) *Proc. Natl. Acad. Sci. U.S.A.* **77**, 5248–5252
107. Jencks, W. P. (1975) *Adv. Enzymol.* **43**, 219–410
108. Westheimer, F. H. (1962) *Adv. Enzymol.* **24**, 441–482
109. Kraut, J. (1988) *Science* **242**, 533–539
110. Fersht, A. (1985) *Enzyme Structure and Mechanism*, 2nd ed., Freeman, New York
111. Gandour, R. D., and Schowen, R. L., eds. (1978) *Transition States of Biochemical Processes*, Plenum, New York
112. Laidler, K. J. (1969) *Theories of Chemical Reaction Rates*, McGraw-Hill, New York
113. Kyte, J. (1995) *Mechanism in Protein Chemistry*, Garland Publ., New York (pp. 199–213)
114. Marcus, R. A. (1992) *Science* **256**, 1523–1524
115. Pauling, L. (1948) *Nature (London)* **161**, 707–713
116. Linehard, G. E. (1973) *Science* **180**, 149–154
117. Wolfenden, R. (1976) *Annu. Rev. Biophys. Bioeng.* **5**, 271–306
- 117a. Schramm, V. L. (1998) *Ann. Rev. Biochem.* **67**, 693–720
118. Nielsen, P. A., Glad, S. S., and Jensen, F. (1996) *J. Am. Chem. Soc.* **118**, 10577–10583
119. Zheng, Y.-J., and Bruice, T. C. (1997) *J. Am. Chem. Soc.* **119**, 8137–8145
120. Merkle, D. J., Kline, P. C., Weiss, P., and Schramm, V. L. (1993) *Biochemistry* **32**, 12993–13001
121. Degano, M., Almo, S. C., Sacchetti, J. C., and Schramm, V. L. (1998) *Biochemistry* **37**, 6277–6285
122. Glad, S. S., and Jensen, F. (1997) *J. Am. Chem. Soc.* **119**, 227–232
123. Schramm, V. L., Horenstein, B. A., and Kline, P. C. (1994) *J. Biol. Chem.* **269**, 18259–18262
124. Kline, P. C., and Schramm, V. L. (1994) *J. Biol. Chem.* **269**, 22385–22390
125. Deng, H., Kurz, L. C., Rudolph, F. B., and Callender, R. (1998) *Biochemistry* **37**, 4968–4976
- 125a. Cho, Y.-K., and Northrop, D. B. (1999) *Biochemistry* **38**, 7470–7475
- 125b. Wolfenden, R., Snider, M., Ridgway, C., and Miller, B. (1999) *J. Am. Chem. Soc.* **121**, 7419–7420
126. Menger, F. M. (1992) *Biochemistry* **31**, 5368–5373
127. Goldsmith, J. O., and Kuo, L. C. (1993) *J. Biol. Chem.* **268**, 18481–18484
128. Lightstone, F. C., and Bruice, T. C. (1996) *J. Am. Chem. Soc.* **118**, 2595–2605
- 128a. Bruice, T. C., and Lightstone, F. C. (1999) *Acc. Chem. Res.* **32**, 127–136
- 128b. Bruice, T. C., and Benkovic, S. J. (2000) *Biochemistry* **39**, 6267–6274
- 128c. Cannon, W. R., Singleton, S. F., and Benkovic, S. J. (1996) *Nature Struct. Biol.* **3**, 821–833
- 128d. Cannon, W. R., and Benkovic, S. J. (1998) *J. Biol. Chem.* **273**, 26257–26260
129. Murphy, D. J. (1995) *Biochemistry* **34**, 4507–4510
- 129a. Warshel, A. (1998) *J. Biol. Chem.* **273**, 27035–27038
- 129b. Jencks, W. P. (1987) *Catalysis in Chemistry and Enzymology*, Dover, Mineola, New York (pp. 210–213)
130. Hucho, F., and Wallenfels, K. (1971) *Eur. J. Biochem.* **23**, 489–496
131. Jencks, W. P. (1969) *Catalysis in Chemistry and Enzymology*, McGraw-Hill, New York (pp. 170–199)
132. Swain, C. G., and Brown, J. F., Jr. (1952) *J. Am. Chem. Soc.* **74**, 2534–2537 and 2538–2543
133. Rony, P. R. (1969) *J. Am. Chem. Soc.* **91**, 6090–6096
134. Jencks, W. P. (1987) *Catalysis in Chemistry and Enzymology*, Dover, Mineola, New York (p. 199)
135. Engdahl, K.-Å., Bivehed, H., Ahlberg, P., and Saunders, W. H., Jr. (1983) *J. Am. Chem. Soc.* **105**, 4767–4774
136. Hegarty, A. F., and Jencks, W. P. (1975) *J. Am. Chem. Soc.* **97**, 7188–7189
- 136a. Williams, A. (1999) *Concerted Organic and Bio-Organic Mechanisms*, CRC Press, Boca Raton, Florida
137. Wang, J. H. (1968) *Science* **161**, 328–334
138. Eigen, M. (1964) *Angew. Chem. Int. Ed. Engl.* **3**, 1–19
139. Metzler, D. E. (1979) *Adv. Enzymol.* **50**, 1–40
140. Nagle, J. F., and Morowitz, H. J. (1978) *Proc. Natl. Acad. Sci. U.S.A.* **75**, 298–302
141. Gerlt, J. A., and Gassman, P. G. (1993) *J. Am. Chem. Soc.* **115**, 11552–11568
142. Cohen, A. O., and Marcus, R. A. (1968) *J. Phys. Chem.* **72**, 4249–4256
143. Marcus, R. A. (1969) *J. Am. Chem. Soc.* **91**, 7224–7225
144. Guthrie, J. P. (1996) *J. Am. Chem. Soc.* **118**, 12878–12885
145. Guthrie, J. P. (1996) *J. Am. Chem. Soc.* **118**, 12886–12890
146. Jencks, W. P. (1987) *Catalysis in Chemistry and Enzymology*, Dover, Mineola, New York (pp. 207–213)
147. Dewar, M. J. S. (1984) *J. Am. Chem. Soc.* **106**, 209–219
148. Gandour, R. D., Maggiora, G. M., and Schowen, R. L. (1974) *J. Am. Chem. Soc.* **96**, 6967–6979
149. Quinn, D. M. (1987) *Chem. Rev.* **87**, 955–979
150. Klinman, J. P. (1989) *Trends Biochem. Sci.* **14**, 368–373
151. Rucker, J., Cha, Y., Jonsson, T., Grant, K. L., and Klinman, J. P. (1992) *Biochemistry* **31**, 11489–11499
152. Hwang, J.-K., and Warshel, A. (1996) *J. Am. Chem. Soc.* **118**, 11745–11751
153. Barbara, P. F., Walker, G. C., and Smith, T. P. (1992) *Science* **256**, 975–981
154. Kearley, G. J., Fillaux, F., Baron, M.-H., Bennington, S., and Tomkinson, J. (1994) *Science* **264**, 1285–1289
- 154a. Antoniou, D., and Schwartz, S. D. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 12360–12365
- 154b. Basran, J., Sutcliffe, M. J., and Scrutton, N. S. (1999) *Biochemistry* **38**, 3218–3222
- 154c. Sutcliffe, M. J., and Scrutton, N. S. (2000) *Trends Biochem. Sci.* **25**, 405–408
- 154d. Huang, Y., Rettner, C. T., Auerbach, D. J., and Wodtke, A. M. (2000) *Science* **290**, 111–114
155. Hibbert, F., and Emsley, J. (1990) *Adv. Phys. Org. Chem.* **26**, 255–379
156. Gilli, P., Bertolasi, V., Ferretti, V., and Gilli, G. (1994) *J. Am. Chem. Soc.* **116**, 909–915
157. Perrin, C. L. (1994) *Science* **266**, 1665–1668
158. Garcia-Viloca, M., González-Lafont, A., and Lluch, J. M. (1997) *J. Am. Chem. Soc.* **119**, 1081–1086
- 158a. Bao, D., Huskey, W. P., Kettner, C. A., and Jordan, F. (1999) *J. Am. Chem. Soc.* **121**, 4684–4689
- 158b. Garcia-Viloca, M., Gelabert, R., González-Lafont, A., Moreno, M., and Lluch, J. M. (1998) *J. Am. Chem. Soc.* **120**, 10203–10209
- 158c. Bowers, P. M., and Klevit, R. E. (2000) *J. Am. Chem. Soc.* **122**, 1030–1033
159. McDermott, A., and Ridenour, C. F. (1996) in *Encyclopedia of NMR*, (Emsley, J. W., ed.), pp. 3820–3824, Wiley, Sussex, England
160. Frey, P. A. (1995) *Science* **269**, 104–106
161. Cleland, W. W., Frey, P. A., and Gerlt, J. A. (1998) *J. Biol. Chem.* **273**, 25529–25532
162. Cleland, W. W., and Kreevoy, M. M. (1994) *Science* **264**, 1887–1890
163. Frey, P. A., Whitt, S. A., and Tobin, J. B. (1994) *Science* **264**, 1927–1930
164. Cassidy, C. S., Lin, J., and Frey, P. A. (1997) *Biochemistry* **36**, 4576–4584
165. Halkides, C. J., Wu, Y. Q., and Murray, C. J. (1996) *Biochemistry* **35**, 15941–15948
166. Zheng, Y.-J., and Bruice, T. C. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 4285–4288
167. Richard, J. P. (1998) *Biochemistry* **37**, 4305–4309
168. Scheiner, S., and Kar, T. (1995) *J. Am. Chem. Soc.* **117**, 6970–6975
169. Warshel, A., Papazyan, A., Kollman, P. A., Cleland, W. W., Kreevoy, M. M., and Frey, P. A. (1995) *Science* **269**, 102–106
170. Shan, S.-o., Loh, S., and Herschlag, D. (1996) *Science* **272**, 97–101
171. Ash, E. L., Sudmeier, J. L., De Fabo, E. C., and Bachovchin, W. W. (1997) *Science* **278**, 1128–1132

References

172. Shan, S.-o, and Herschlag, D. (1996) *J. Am. Chem. Soc.* **118**, 5515–5518
173. Warshel, A., and Papazyan, A. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 13665–13670
174. Spector, L. B. (1982) *Covalent Catalysis by Enzymes*, Springer-Verlag, New York
175. Page, M. I., and Jencks, W. P. (1971) *Proc. Natl. Acad. Sci. U.S.A.* **68**, 1678–1683
176. Kirsch, J. F. (1973) *Ann. Rev. Biochem.* **42**, 205–234
177. Koshland, D. E., Jr., Carraway, K. W., Dafforn, G. A., Goss, J. D., and Storm, D. R. (1971) *Cold Spring Harbor Symp. Quant. Biol.* **36**, 13–20
- 177a. Villà, J., Strajbl, M., Glennon, T. M., Sham, Y. Y., Chu, Z. T., and Warshel, A. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 11899–11904
178. Milstien, S., and Cohen, L. A. (1972) *J. Am. Chem. Soc.* **94**, 9158–9165
179. Karle, J. M., and Karle, I. L. (1972) *J. Am. Chem. Soc.* **94**, 9182–9189
180. Mesecar, A. D., Stoddard, B. L., and Koshland, D. E., Jr. (1997) *Science* **277**, 202–206
181. Deslongchamps, P., Lebreux, C., and Taillefer, R. (1973) *Can. J. Chem.* **51**, 1665–1669
182. Deslongchamps, P. (1975) *Tetrahedron* **31**, 2463–2490
183. Perrin, C. L., and Arrhenius, M. L. (1982) *J. Am. Chem. Soc.* **104**, 2839–2842
184. Evans, C. M., Glenn, R., and Kirby, A. J. (1982) *J. Am. Chem. Soc.* **104**, 4706–4707
185. Kuo, L. C., Fukuyama, J. M., and Makinen, M. W. (1983) *J. Mol. Biol.* **163**, 63–105
186. Plogman, J. H., Drenth, G., Kalk, K. H., and Hol, W. G. J. (1979) *J. Mol. Biol.* **127**, 149–162
187. Warshel, A. (1981) *Biochemistry* **20**, 3167–3177
188. Hol, W. G. J., van Duijn, P. T., and Berendsen, H. J. C. (1978) *Nature (London)* **273**, 443–446
189. Welch, G. R., ed. (1986) *The Fluctuating Enzyme*, Wiley, New York
- 189a. Mertz, E. L., and Krishtalik, L. I. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 2081–2086
190. Dewar, M. J. S., and Storch, D. M. (1985) *Proc. Natl. Acad. Sci. U.S.A.* **82**, 2225–2229
191. Dewar, M. J. S., and Dieter, K. M. (1988) *Biochemistry* **27**, 3302–3308
192. Yennawar, H. P., Yennawar, N. H., and Farber, G. K. (1995) *J. Am. Chem. Soc.* **117**, 583–585
193. Schmitke, J. L., Wescott, C. R., and Klibanov, A. M. (1996) *J. Am. Chem. Soc.* **118**, 3360–3365
194. Kell, D. B. (1982) *Trends Biochem. Sci.* **7**, 1
195. Robertson, B., and Astumian, R. D. (1992) *Biochemistry* **31**, 138–141
196. Levitt, M. (1974) in *Peptides, Polypeptides and Proteins* (Blout, E. R., Bouey, F. A., Goodman, M., and Lotan, N., eds), p. 99, Wiley, New York
197. Avis, J. M., and Fersht, A. R. (1993) *Biochemistry* **32**, 5321–5326
198. Lumry, R., and Rosenberg, A. (1974) *Colloques Internationaux du C.N.R.S.* **246**, 53–62
199. Rader, S. D., and Agard, D. A. (1997) *Protein Sci.* **6**, 1375–1386
- 199a. Word, J. M., Lovell, S. C., Richardson, J. S., and Richardson, D. C. (1999) *J. Mol. Biol.* **285**, 1735–1747
200. Bialek, W., and Onuchic, J. N. (1988) *Proc. Natl. Acad. Sci. U.S.A.* **85**, 5908–5912
- 200a. Bystrov, D. S., and Popova, E. A. (1987) *Ferroelectrics* **72**, 147–155
- 200b. Emsley, J., Reza, N. M., Dowes, H. M., Hursthouse, M. B., and Kurodo, R. (1988) *Phosphorus and Sulfur* **35**, 141–149
- 200c. Rakvin, and Dalal. (1992) *Ferroelectrics* **135**, 227–236
201. Banaszak, L. A., and Bradshaw, R. A. (1975) in *The Enzymes*, 3rd ed., Vol. 11 (Boyer, P. D., ed), Academic Press, New York (pp. 369–396)
202. Zimmerle, C. T., and Alter, G. M. (1993) *Biochemistry* **32**, 12743–12748
203. Niefind, K., Hecht, H.-J., and Schomburg, D. (1995) *J. Mol. Biol.* **251**, 256–281
204. Harada, K., and Wolfe, R. G. (1968) *J. Biol. Chem.* **243**, 4131–4137
205. Lazdunski, M., Petitclerc, C., Chappellet, D., and Lazdunski, C. (1971) *Eur. J. Biochem.* **20**, 124–139
206. Skarzynski, T., Moody, P. C. E., and Wonacott, A. J. (1987) *J. Mol. Biol.* **193**, 171–187
207. Yun, M., Park, C.-G., Kim, J.-Y., and Park, H.-W. (2000) *Biochemistry* **39**, 10702–10710
208. Anderson, A. C., O'Neil, R. H., DeLano, W. L., and Stroud, R. M. (1999) *Biochemistry* **38**, 13829–13836
209. Srere, P. A. (1984) *Trends Biochem. Sci.* **9**, 387–390
210. Nomenclature Committee of the International Union of Biochemistry and Molecular Biology. (1992) *Enzyme Nomenclature*, Academic Press, San Diego, California

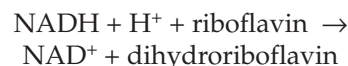
Study Questions

1. Give two reasons why enzymes are important to living organisms.
2. If an enzyme catalyzes the reaction $A \rightarrow B + C$, will it also catalyze the reaction $B + C \rightarrow A$?
3. Can an enzyme use a site to catalyze a reverse reaction different from what it uses to catalyze the reaction in the forward direction? Explain your answer.
4. Define the "steady state" of living cells and contrast with a state of chemical equilibrium.
5. How is the instantaneous initial velocity of an enzymatic reaction measured? What precautions must be taken to ensure that a true instantaneous velocity is being obtained?
6. How is the rate of an enzyme action influenced by changes in:
 - a) temperature
 - b) enzyme concentration
 - c) substrate concentration
 - d) pH
7. In what ways are enzymatic reactions typical of ordinary catalytic reactions in organic or inorganic chemistry, and in what ways are they distinct?
8. In general, for a reaction that can take place with or without catalysis by an enzyme, what is the effect, if any, of the enzyme on
 - a) standard Gibbs energy change of the reaction
 - b) energy of activation of the reaction
 - c) initial velocity of the reaction
 - d) temperature coefficient of the rate constant
9. What are the dimensions of the rate constant for zero-, first-, and second-order reactions? If a first-order reaction is half completed in 2 min, what is its rate constant?
10. A sample of hemin containing radioactive iron, ^{59}Fe , was assayed with a Geiger-Müller counter at intervals of time with the following results:

Time (days)	Radioactive counts per min
0	3981
2	3864
6	3648
10	3437
14	3238
20	2965

Determine the half-life of ^{59}Fe and the value for the decay constant.

11. The kinetics of the aerobic oxidation of enzymatically reduced nicotinamide adenine dinucleotide (NADH) have been investigated at pH 7.38 at 30°C. The reaction rate was followed spectrophotometrically by measuring the decrease in absorbance at 340 nm over a period of 30 min. The reaction may be represented as



Time (days)	A (at 340 nm)
1	0.347
2	0.339
5	0.327
9	0.302
16.5	0.275
23	0.254
27	0.239
30	0.229

Determine the rate constant and the order of the reaction.

12. You have isolated and purified a new enzyme (E) which converts a single substrate (S) into a single product (P). You have determined M_r by gel filtration as $\sim 46,400$. However, in SDS gel electrophoresis, a molecular mass of ~ 23 kDa was indicated for the single protein band observed. A solution of the enzyme was analyzed in the following way. The absorbance at 280 nm was found to be 0.512. A 1.00 ml portion of the same solution was subjected to amino acid analysis and was found to contain 71.3 nmol of tryptophan. N-terminal analysis on the same volume of enzyme revealed 23.8 nmol of N-terminal alanine.
 - a) What is the approximate molecular mass of the enzyme? Discuss this answer. Be sure to use an appropriate number of significant figures in this and other calculations.
 - b) What is the concentration of your enzyme in moles per liter of active sites?
 - c) What is the molar extinction coefficient ϵ at 280 nm where $A = \epsilon cl$; A = absorbance, c = mol / liter, and l = cell width in cm. Assume that all spectrophotometric measurements are made in 1.00 cm cuvettes.

A second preparation of the enzyme had an absorbance at 280 nm of 0.485. This enzyme was diluted very carefully: 1.00 ml into 250 ml and this diluted enzyme was used for the following experiments (I to III).

Study Questions

Experiment I. A 1.00 ml portion of the diluted enzyme was added to 250 ml of buffered substrate at pH 7.0 and was mixed rapidly. The resulting initial substrate concentration $[S]_0$ was 1.000 mM. This reaction mixture was held at 25.0°C and portions were removed periodically at time t for analysis of the product P formed. The results follow. Plot $[P]$ vs. time.

Time t (s)	$[P]$ (mM)	Time t (s)	Remaining $[S]$ (mM)
200	0.104	2800	0.070
400	0.208	3200	0.040
800	0.392	3600	0.022
1200	0.554	4400	0.0060
1600	0.695	6000	0.00048
2000	0.800		
2400	0.881		

The integrated rate equation corresponding to the Michaelis–Menten equation

$v = V_{\max} [S] / (K_m + [S])$ is as follows:

$$V_{\max} \cdot t = [S]_0 - [S] + K_m \ln ([S]_0 / [S])$$

d) Plot $([S]_0 - [S]) / t$ vs. $1/t \times \ln ([S]_0 / [S])$. From this plot evaluate K_m and $V_{\max}$. Make this and other plots to appropriate scale on good quality graph paper.

e) What is k_{cat} ?

Experiment II. In a second experiment, a series of test tubes were set up, each containing a different amount of buffered substrate at pH 7 but each in a volume of exactly 4.00 ml. The same enzyme solution used in part d (absorbance at 280 nm = 0.485) was diluted 2.00 ml in 250 ml as in I, then again 2.00 ml in 200 ml. Portions of 1.00 ml of this twice diluted enzyme were added at $t = 0$ to each of the test tubes of buffered substrate. The reaction was stopped in just 10.0 minutes by adding perchloric acid; a suitable reagent was added to provide for a colorimetric determination of the product. The results were as follows:

$[S]$ (mM)	Amount of product (μmol/tube)	$[S]$ (mM)	Amount of product (μmol/tube)
10.0	2.29	0.600	1.31
5.00	2.18	0.400	1.07
2.50	2.00	0.200	0.686
1.200	1.69	0.100	0.400
0.800	1.48		

- f) Plot $1/v$ vs. $1/[S]$ where v is in units of μmol per tube and $[S]$ in millimoles / liter. Evaluate K_m , $V_{\max}$, and k_{cat} from this plot. Fit your data by eye. It is generally agreed that a linear least squares fit is inappropriate unless suitable weighting factors are used.
- g) Plot the same data as $v/[S]$ vs. v . Again evaluate K_m and $V_{\max}$.

Experiment III. The preceding experiment was repeated but an inhibitor was present in each tube in a concentration equal to 5.00 mM, 10.0 mM, or 25.0 mM. Two different inhibitors were used, A and B. The following results were obtained.

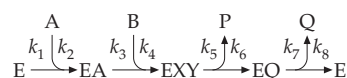
$[S]$ (mM)	μmol of product formed in 10 min/tube				
	$[I] = 5.00 \text{ mM}$		25.0 mM		10.0 mM
	Inhibitor: A	B	Inhibitor: A	B	
5.00	2.00	1.09	1.50	0.727	
2.50	0.71	1.00	1.09	0.667	
1.20	1.31	0.848	0.686	0.565	
0.800	1.07	0.739	0.505	0.492	
0.600	0.900	0.654	0.400	0.436	
0.400	0.686	0.533	0.282	0.356	
0.200	0.400	0.343	0.150	0.229	
0.100	0.218	0.200	0.077	0.133	

h) Plot $1/v$ vs. $1/[S]$ for each of these sets of data.

For each case evaluate $V_{\max}$, apparent K_m , and inhibitor constants $K_I = [I][E]/[EI]$. There may be two K_I values.

- i) For uninhibited enzymes, when $[S] = 0.4 \text{ mM}$ what fraction of the enzyme is ES? Free E?
- j) For enzyme in the presence of 10.00 mM inhibitor and 0.8 mM substrate, what fraction of the total enzyme is ES? EI? free E?
- k) Recall that $v = [ES] k_{\text{cat}} = [E][S] \cdot k_{\text{cat}} / K_m$ in many cases. What will be the relative rates of product formation from your substrate S and another competing substrate S' present at the same concentration which also reacts with your enzyme with $K_m = 0.01 \text{ mM}$ and $k_{\text{cat}} = 200 / \text{s}$ if $[S] = [S']$?

13. Using the method of graphs, write the initial rate equation for the following system with A, B, P, and Q present.



Odd-numbered rate constants are for forward reactions; even-numbered constants are for reverse.

Study Questions

a) Satisfy yourself that

$$\begin{aligned} \frac{-d[A]}{dt} &= k_1[A][E] - k_2[EA] \\ &= \frac{-d[B]}{dt} = k_3[B][EA] - k_4[EXY] \\ &= \frac{d[P]}{dt} = k_5[EXY] - k_6[P][EQ] \\ &= \frac{d[Q]}{dt} = k_7[EQ] - k_8[E][Q] \end{aligned}$$

The determinants (given by the method of graphs) which provide the steady-state concentrations of [E] and of the various enzyme–substrate and enzyme–product complexes are

$$[E] = k_2k_7[k_4 + k_5] + k_3k_5k_7[B] + k_2k_4k_6[P]$$

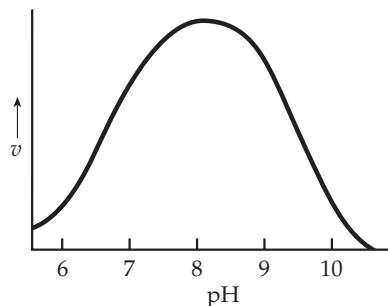
$$[EA] = k_1k_7[k_4 + k_5][A] + k_1k_4k_6[A][P] + k_4k_6k_8[P][Q]$$

$$[EXY] = k_1k_3k_7[A][B] + k_2k_6k_8[P][Q] + k_1k_3k_6[A][B][P] + k_3k_6k_8[B][P][Q]$$

$$[EQ] = k_2k_8[k_4 + k_5][Q] + k_1k_3k_5[A][B] + k_3k_5k_8[B][Q]$$

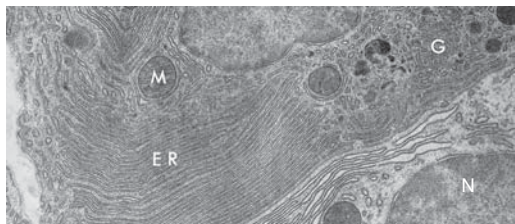
- b) To obtain an expression for v_f the expression for $-d[A]/dt$ may be multiplied by $[E]_t$ and divided by $[E] + [EA] + [EXY] + [EQ] = [E]_t$. Then the above determinants may be substituted.
- c) What is $V_{\max, \text{forward}}$ and $V_{\max, \text{reverse}}$? HINT: $V_{\max, \text{forward}}$ is obtained when $[P] = [Q] = 0$ and $[A] = [B] = \infty$. What are K_{mA} and K_{mB} ? K_{mA} is obtained when $[P] = [Q] = 0$, $[B] = \infty$, and $v = 1/2 V_{\max, \text{forward}}$.
- d) With a knowledge of the kinetic parameters, indicate how the eight rate constants may be obtained if the total concentration of enzyme, $[E]_t$, is known.

15. Interpret, for each of the following cases, the curve showing measured initial velocity at constant substrate concentration (*not* maximum velocity) against pH for an enzyme-catalyzed reaction.



- a) The substrate is neutral and contains no acidic or basic groups. The Michaelis constant is found to be independent of pH over the range studied.
- b) The substrate is neutral and contains no acidic or basic groups, and the maximum velocity is found to be independent of pH.
- c) The substrate is an α -amino acid, and the maximum velocity is found to be independent of pH.

14. Anticooperativity was observed in the plot of velocity vs. substrate concentration for an enzyme. Can this observation be explained by the Monod–Wyman–Changeux model for oligomeric enzymes? By the model of Koshland? Explain.



This electron micrograph of a thin section of pancreatic epithelial cells shows parts of two secretory cells that are synthesizing proenzymes (zymogens). Nuclei (N) are seen at the top center and lower right. Numerous ribosomes (barely seen here) line the many membranes of the endoplasmic reticulum (ER) and are most abundant near lateral and basal (left) surfaces of the cells. A few mitochondria (M) are present. The synthesized proteins move toward the apical surfaces (a small piece is seen at upper right), passing through the Golgi region (G) and accumulating in zymogen granules (Z) before secretion. From Porter, K. R. and Bonneville, M. A., *Fine Structure of Cells and Tissues*, Lea and Febiger, Philadelphia 1973. Courtesy of Mary Bonneville.

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Boxes

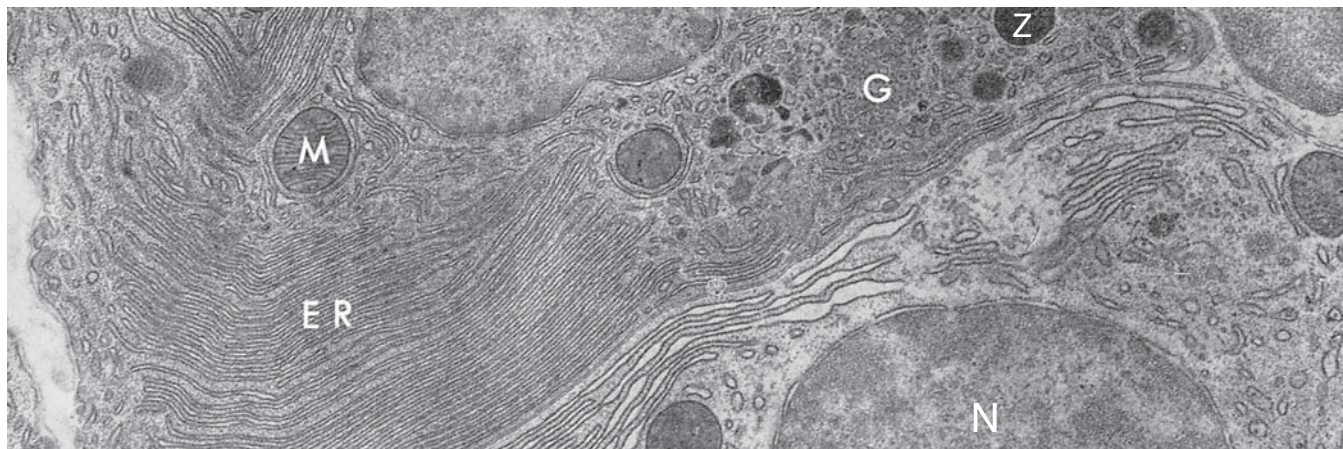
- 513 Box 10-A An Early Labeling Experiment
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An Introduction to Metabolism

10



The first section of this book has dealt with the basic structures and with many of the complexities of the materials formed by living cells. In the next major section we will look at the chemical reactions that build and maintain a cell and that permit it to grow and to be responsive to external stimuli. These reactions are organized into **metabolic sequences** or **pathways** that form a complex, branched, and interconnected network. It would be pointless to try to memorize all of them. However, at this point it will be worthwhile to consider the significance of a few of the major sequences, which describe central pathways of metabolism.

Beginning in the nineteenth century, the investigation of these pathways and of the associated enzymes has provided much of our knowledge about the chemistry of living things. The protein products that are encoded in the structural genes that direct these pathways make up a large fraction of the material in a cell. It will be well worth the reader's effort to examine the chemistry of these metabolic sequences in detail. Regulatory mechanisms that are applicable to them, and to other pathways, are described in Chapter 11 and chemical details of enzyme action are considered in Chapters 12–16. Then, in Chapter 17, the chemical logic of the reaction sequences is considered in more detail.

A. The Starting Materials

All cells of all organisms take in chemical starting materials and give off chemical products. They all have a source of energy and generate heat during their metabolism. They all synthesize complex organic substances and maintain a high degree of organization, that is, a state of relatively low entropy. The materials taken up by cells are often organic compounds which

not only supply material for the synthesis of cellular constituents but also may be degraded to provide energy. A characteristic of all cells is the ability simultaneously to synthesize thousands of complex proteins and other materials and, at the same time, to break down (catabolize) the same types of compounds. Since cells both synthesize and catabolize most cellular components there is a continuous turnover of the very structural components of which they are composed. Metabolism encompasses all of these processes.

The most rapid catabolic reactions are usually those that provide the cell's energy. Organisms vary greatly in the materials used for food. Human beings, as well as many other organisms, break down carbohydrates, lipids, and proteins to obtain energy and starting materials for biosynthesis. In contrast, some organisms use only one or a few simple organic compounds while autotrophs satisfy their needs entirely with inorganic materials. Looking at all of the species of living things we find an extraordinary range of specialization in metabolism as well as in structure. Nevertheless, there are many common features of metabolism. For example, most cells utilize glucose or a close relative as a source of biosynthetic intermediates. Pathways for synthesis of nucleic acids and proteins are similar in all species. Even the control of growth and development depends upon proteins whose structures are often conserved throughout the living world.

1. Digestion

We humans must digest most foods before we can utilize them. The same is true for most bacteria, which need amino acids generated by the breakdown of

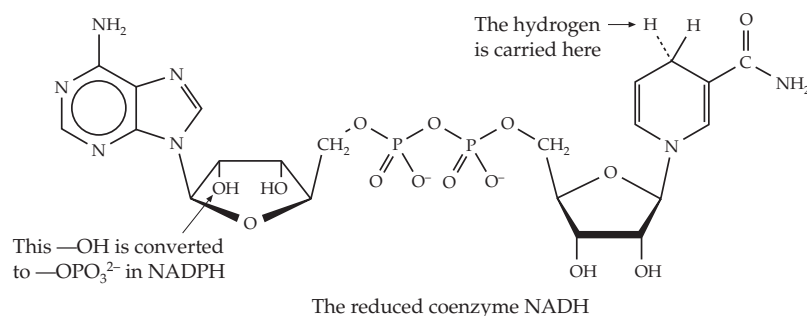


Figure 10-1 An overall view of some metabolic sequences. Several major pathways of catabolism are indicated by heavy lines. The glycolysis pathway, leading to pyruvate and lactate, starts at the top left, while the β oxidation pathway of fatty acids is on the right. Biosynthetic routes are shown in green lines. A few of the points of synthesis and utilization of ATP are indicated by the lighter green lines. Some of the oxidation–reduction reactions that produce or utilize the reduced hydrogen carriers NADH, NADPH, and FADH₂ are also indicated. The citric acid cycle, a major supplier of these molecules, is shown at the bottom right, while photosynthesis, the source of reduced hydrogen carriers in green plants, and the source of nearly all energy for life, is shown at the bottom left.

proteins. Our digestive enzymes, which act on carbohydrates, proteins, lipids, and nucleic acids, are synthesized by cells in the salivary glands, stomach, pancreas, and intestinal lining. The properties of these enzymes are described in Chapter 12. Bacteria secrete digestive enzymes into their surroundings. Although most green plants have no need to digest foods, they too have enzymes closely related to our digestive enzymes. These enzymes break bonds in proteins or carbohydrate polymers to allow expansion of leaves and stems, ripening of fruits, and other physical changes. All cells carry out **processing** or **maturation** of their newly synthesized polymers. This often involves the cutting off of one or more pieces of a protein, polysaccharide, or nucleic acid. The processing enzymes are also relatives of digestive enzymes.

2. Sources of Energy

All cells require a source of chemical energy. This is provided to a large extent by ATP (Chapter 6), whose hydrolysis can be coupled to reactions of synthesis, to transport across membranes, and to other endergonic processes. For this reason, all cells have active pathways for the synthesis of ATP. *Biosynthesis often also involves chemical reduction of intermediates; therefore, cells must have a means of generating suitable reductants. The reagent of choice is often the hydrogen-carrying coenzyme NADPH.* It is a phosphorylated form of NADH, a derivative of the vitamin **nicotinamide** (Box 15-A), whose structure follows. Reduction of metabolic intermediates by NADPH provides a second mechanism by which chemical energy is harnessed for biosynthesis. A third source of energy, which is considered in Chapter 18, is the gradient of hydrogen ions that is set up across cell membranes by oxidative or photochemical processes.



B. Catabolism and the Synthesis of ATP

The catabolic sequences by which cells obtain energy often appear dominant. For animals, fungi, and nonphotosynthetic bacteria these pathways are

used to metabolize large amounts of food and to produce large amounts of ATP. Because they are also related to biosynthetic pathways they have a central importance in virtually all organisms.

1. Priming or Activation of Metabolites

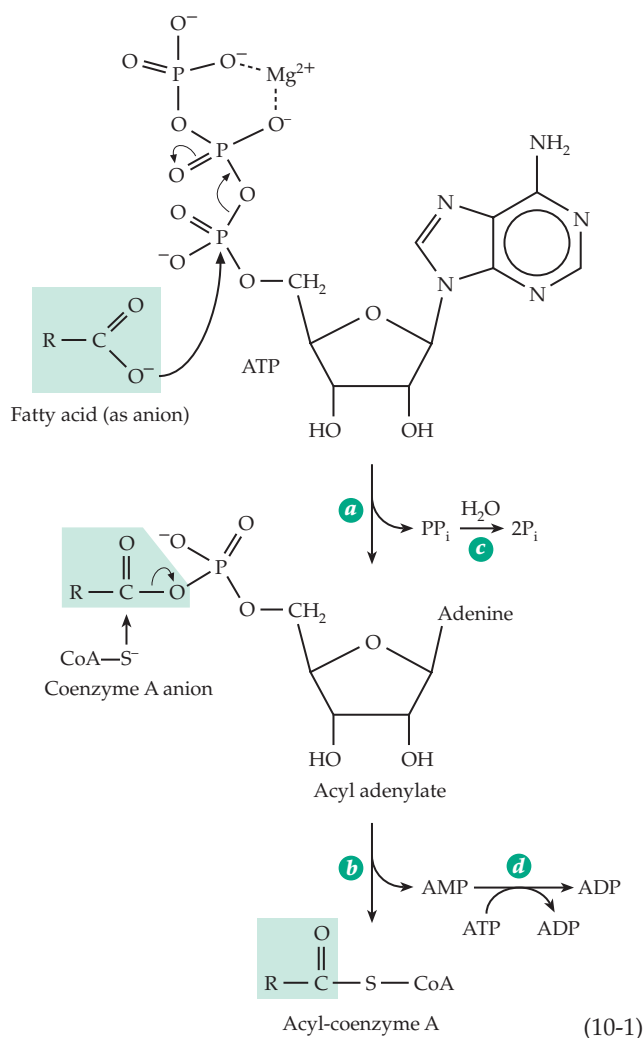
After a polymeric nutrient is digested and the monomeric products are absorbed into a cell, an energy-requiring **priming reaction** is usually required. For example, the hydrolysis of fats (whether in the gut or within cells) produces free fatty acids. Before undergoing further metabolism, these fatty acids are combined with the thiol compound **coenzyme A** to form **fatty acyl-CoA** derivatives. This requires a thermodynamically unfavorable reaction that necessitates the “expenditure” of ATP, that is, its hydrolysis to AMP and inorganic pyrophosphate (PP_i). Likewise, glucose, when taken into cells, is converted into the phosphate ester glucose 6-phosphate (glucose-6-P), again with an associated cleavage of ATP. *The major metabolic pathways often start with one of these two substances: a fatty acyl-CoA derivative or glucose 6-phosphate.* The structures of both are given at the top of Fig. 10-1, which is designed to provide an overall view of metabolism.

The phospho group of glucose-6-P is not very reactive but it provides a “handle” which helps enzymes to recognize and to hold onto this glucose derivative. Likewise, the coenzyme A molecule, whose complete structure is given in Chapter 14, provides a large and complex handle for enzymes. However, from a chemical viewpoint, the formation of the thioester linkage of a fatty acid with the -SH group of coenzyme A in the acyl-CoA is more important. This alters the electronic structure of the fatty acid molecule, “activating” it toward the reactions that it must undergo subsequently.

Thus, priming reactions often provide chemical changes essential to the mechanisms of subsequent reactions.

Activation of a carboxylic acid by formation of an acyl-CoA derivative (Eq. 10-1) is of special importance because of its widespread use by all organisms. The basic reaction occurs by two steps. The first (Eq. 10-1, step *a*) is a displacement reaction on the phosphorus atom of ATP to form an **acyl adenylate**, a mixed anhydride of the carboxylic acid and a substituted phosphoric acid. Such mixed anhydride intermediates, or **acyl phosphates**,

are central to much of cellular energy metabolism because *they preserve the high group-transfer potential of a phospho group of ATP (Chapter 6) while imparting a high group-transfer potential to the acyl group.* This allows the acyl group to be transferred in step *b*



of Eq. 10-1 onto the sulfur atom of coenzyme A to form an acyl-coenzyme A in which the high group transfer potential of the acyl group is conserved (see Table 6-6). Two additional steps are linked to this sequence. In step *c* the inorganic pyrophosphate that is displaced in step *a* is hydrolyzed to two molecules of HPO_4^{2-} (P_i), and in step *d* the AMP formed is phosphorylated by ATP to form ADP. The overall sequence of Eq. 10-1 leads to hydrolysis of two molecules of ATP to $\text{ADP} + \text{P}_i$. In other words, two molecules of ATP are spent in activating a carboxylic acid by conversion to an acyl-coenzyme A.

2. Interconversions of Sugar Phosphates

The strategy employed by most cells in the catabolism of several 6-carbon sugars is to convert them to glucose 6-phosphate and, in the several steps outlined in Fig. 10-2, to cleave this hexose phosphate to two equivalent molecules of **glyceraldehyde 3-phosphate**. This triose phosphate can then be metabolized further. Notice the chemical nature of the reactions involved in

the formation of glyceraldehyde 3-phosphate. The first step (Fig. 10-2, step *a*) is the transfer of the phospho group from ATP to the 6 position of glucose. Step *b* is the reversible opening of the sugar ring and step *c* the isomerization of an aldose to a ketose. Step *d* is a second transfer of a phospho group from ATP, another priming reaction that provides the second of the phosphate handles for the two molecules of triose phosphate that are formed. Step *e* is an aldol cleavage which breaks the C-C bond in the center of the ketose chain, while step *f* is another sugar isomerization that is chemically similar to the one in step *c*.

We see that in the six steps shown in Fig. 10-2 for conversion of glucose to two molecules of glyceraldehyde 3-phosphate, there are only four kinds of reaction: (1) phospho transfer from ATP; (2) opening of a sugar ring; (3) aldose-ketose isomerization; and (4) aldol cleavage. These types of reactions, which are described in more detail in Chapters 12 and 13, occur in many places in metabolism. As is indicated in Fig. 10-2, glucose 6-phosphate can also be generated from glycogen or starch. The first step (Fig. 10-2, step *g*) is phosphorolysis, cleavage by an inorganic phosphate ion. This leads directly to glucose 1-phosphate. The latter is isomerized by a phospho transfer process (step *h*).

3. Glycolysis and Fermentation

A major route of breakdown of carbohydrates is the **Embden–Meyerhof–Parnas** pathway, often referred to simply as **glycolysis**. It is indicated on the left side of Fig. 10-1 and in more detail in Figs. 10-2 and 10-3. The pathway begins with the reactions of Fig. 10-2, with either free glucose or glycogen as starting materials. Its end products may be reduced materials such as **lactic acid** or **ethanol**, which are formed under *anaerobic conditions* (Fig. 10-3). However, under *aerobic conditions* the product is **acetyl-coenzyme A**, whose acetyl group can then be oxidized to carbon dioxide and water in the **citric acid cycle**.

After the sugar chain is cleaved in the glycolysis sequence, the two resultant molecules of glyceraldehyde 3-phosphate are oxidized to 3-phosphoglycerate (Fig. 10-3, steps *a, b*). The oxidant is the hydrogen carrier NAD^+ , the oxidized form of NADH. Cells frequently use NAD^+ to dehydrogenate alcohols to aldehydes or ketones, with one atom of hydrogen and an electron from the alcohol becoming attached to the NAD^+ to give NADH and the other hydrogen being released as H^+ (Eq. 10-2).

The oxidation of glyceraldehyde 3-phosphate is considerably more complex. The oxidation of an aldehyde to a carboxylic acid is a strongly exergonic process and the oxidation of glyceraldehyde 3-phosphate by cells is almost always coupled to the synthesis

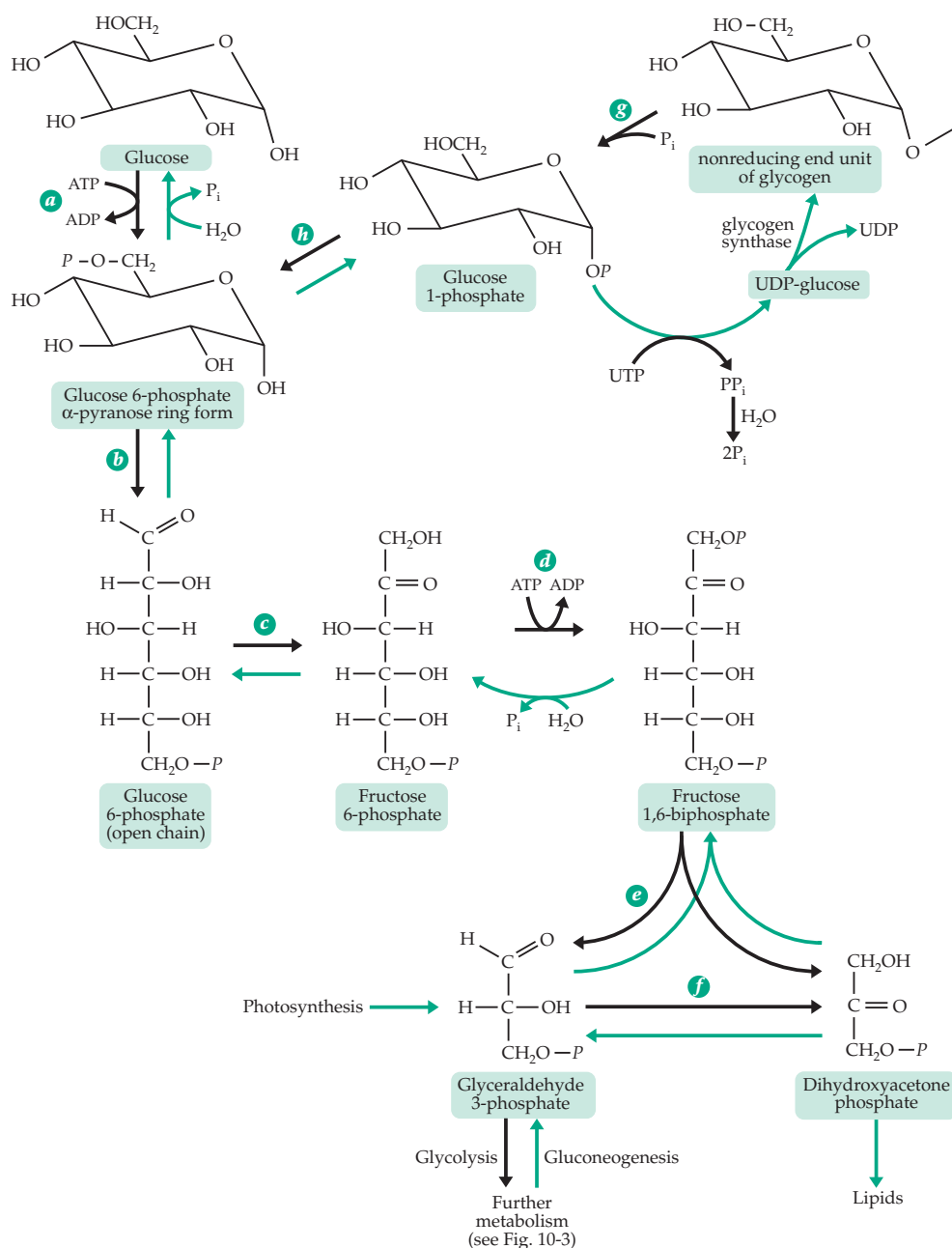


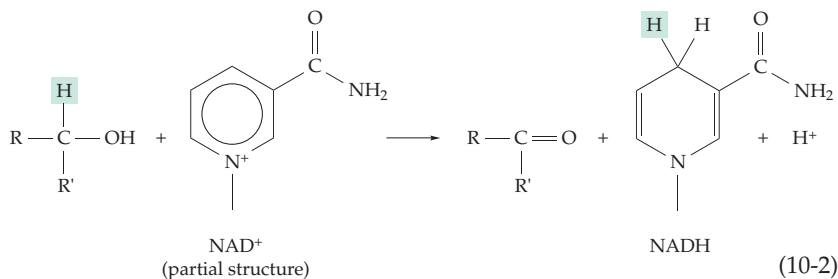
Figure 10-2 The interconversion of glucose, glycogen, and glyceraldehyde-3-phosphate in the pathways of glycolysis, gluconeogenesis, and glycogen synthesis. Pathways of catabolism are indicated with black lines and those of biosynthesis with green lines.

of ATP. The importance of this coupling can be understood by the fact that for many organisms living by anaerobic fermentation reactions, this one oxidation step provides the sole source of ATP. It is the mechanism for coupling this oxidation to synthesis of ATP that accounts for the complexity of the glyceraldehyde 3-phosphate dehydrogenase reaction, whose chemistry is presented in detail in Chapter 15. As is indicated in the simplified version of Fig. 10-3 (step *a*) inorganic phosphate (P_i) is

a reactant and the product released by the enzyme is 1,3-bisphosphoglycerate. This compound, which has been formed by the oxidative process, is an anhydride of phosphoric acid with 3-phosphoglyceric acid. The phospho group of such **acyl phosphates**, like that of ATP, has a high group transfer potential and can be transferred to ADP to generate ATP (Fig. 10-3, step *b*). Since each glucose molecule is cleaved to two molecules of triose phosphate, two molecules of ATP are

generated in this process by the fermentation of one molecule of glucose. This is enough for a bacterium to live on if it ferments enough sugar.

Further metabolism of 3-phosphoglycerate involves



an isomerization by means of a phosphotransferase (mutase reaction, step *c*) to form 2-phosphoglycerate which then loses water in an elimination reaction (step *d*).

Elimination of a hydroxyl group in a β position relative to a carboxyl group together with an α -proton is another very frequently used type of metabolic reaction. In this case the product, **phosphoenolpyruvate** (PEP), is a unique and important metabolite. It is a key intermediate in the biosynthesis of aromatic amino acids and in numerous other biosynthetic sequences.

In the Embden–Meyerhof–Parnas pathway PEP transfers its phospho group to ADP (step *e*) to generate ATP and pyruvate. The latter is shown enclosed in brackets in Fig. 10-3 as the enolic form, which, tautomerizes to the oxo form (step *f*). The fact that the oxo form of pyruvate is much more stable than the enol gives the phospho group of phosphoenolpyruvate its high group transfer potential. The metabolic significance of this step is that the phosphate handles of the glycolytic intermediates, which were initially transferred on from ATP, are now returned to ADP with the regeneration of an equivalent amount of ATP.

The product of this metabolic sequence, **pyruvate**, is a metabolite of central importance. Its fate depends upon the conditions within a cell and upon the type of cell. When oxygen is plentiful pyruvate is usually converted to acetyl-coenzyme A, but under anaerobic conditions it may be reduced by $\text{NADH} + \text{H}^+$ to the alcohol **lactic acid** (Fig. 10-3, step *h*). This reduction exactly balances the previous oxidation step, that is, the oxidation of glyceraldehyde 3-phosphate to 3-phosphoglycerate (steps *a* and *b*). With a balanced sequence of an oxidation reaction, followed by a reduction reaction, glucose can be converted to lactate in the absence of oxygen, a **fermentation** process. The lactic acid fermentation occurs not only in certain bacteria but also in our own muscles under conditions of extremely vigorous exercise. It also occurs continuously in some tissues, e.g., the transparent lens and cornea of the eye.

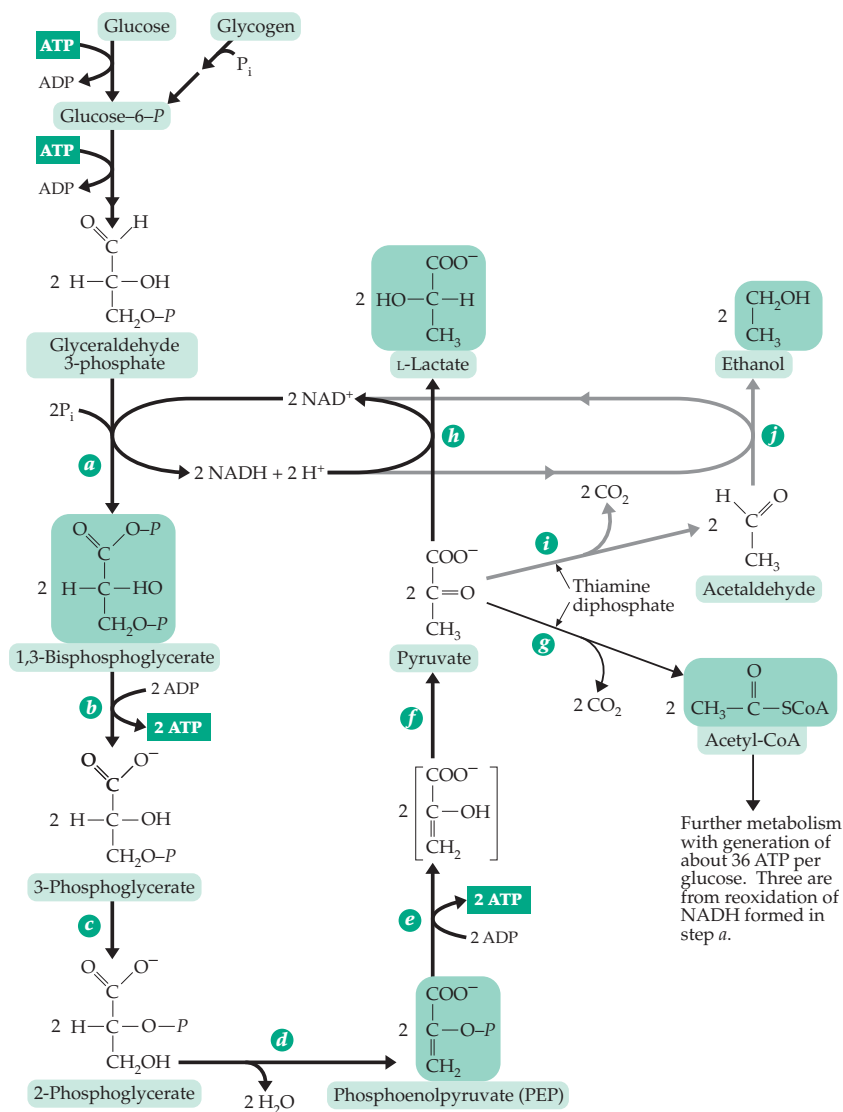
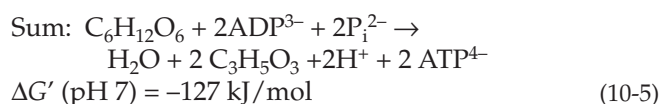
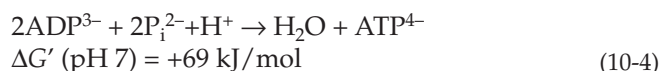
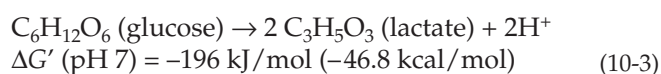


Figure 10-3 Coupling of the reactions of glycolysis with formation of lactic acid and ethanol in fermentations. Steps *a* to *g* describe the Embden–Meyerhof–Parnas pathway. Generation of 2 ATP in step *b* can provide all of the cell's energy.

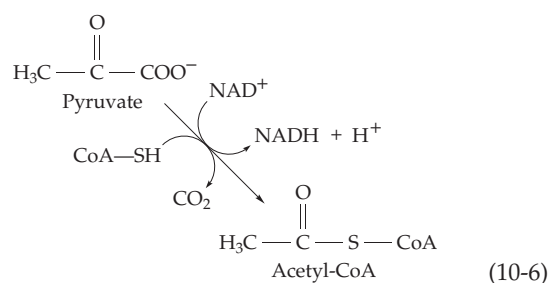
In the well-known fermentation of sugar to ethanol by yeast, the pyruvate generated by glycolysis is first decarboxylated to acetaldehyde (Fig. 10-3, step *i*). This decarboxylation of a 2-oxo acid is chemically difficult and the enzyme catalyzing it makes use of a special reagent known as a **coenzyme**. For this type of reaction, the coenzyme is the diphosphate (pyrophosphate) ester of **thiamin (vitamin B₁)**. Its mode of action is discussed in Chapter 14. It is usually needed when a 2-oxo acid is decarboxylated. The alcoholic fermentation by yeast is completed by reduction of acetaldehyde to ethanol (Fig. 10-3, step *j*), again using the NADH produced in the oxidation of glyceraldehyde 3-phosphate. The conversion of glucose to lactic acid or to ethanol and CO₂ are just two of many fermentation reactions, most of which are carried out by bacteria and which are dealt with further in Chapter 17. An important requirement is that the Gibbs energy change for the overall fermentation reaction be sufficiently negative that ATP synthesis can be coupled to it. Thus, using data from Table 6-4:



A requirement for all fermentations is the existence of a mechanism for coupling ATP synthesis to the fermentation reactions. In the lactic acid and ethanol fermentations this coupling mechanism consists of the formation of the intermediate 1,3-bisphosphoglycerate by the glyceraldehyde 3-phosphate dehydrogenase (Fig. 10-3, step *a*). This intermediate contains parts of both the products ATP and lactate or ethanol.

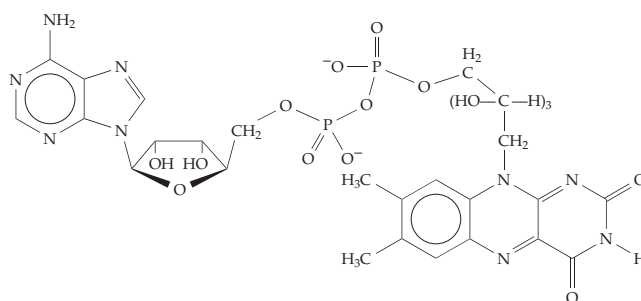
4. Pyruvate Dehydrogenase

In most organisms undergoing aerobic metabolism, pyruvate is oxidized to acetyl-CoA in a complex process involving its decarboxylation (Eq. 10-6). This **oxidative decarboxylation**, like the decarboxylation of pyruvate to acetaldehyde, requires thiamin diphosphate. In addition, an array of other catalysts participate in the process (see Fig. 15-15). Among these are the electron carrier **flavin adenine diphosphate (FAD)**, which is derived from the vitamin **riboflavin**. Like NAD⁺, this



compound can accept two electrons (plus two protons) to form FADH₂. However, it sometimes serves as a one-electron carrier.

Acetyl-CoA is another major metabolic intermediate.



The coenzyme flavin adenine diphosphate (FAD)

It is an acyl-CoA of the type mentioned in Section 1 and can also be formed from acetate, ATP, and coenzyme A. Although the human diet contains some acetic acid, the two major sources of acetyl-CoA in our bodies are the oxidative decarboxylation of pyruvate (Eq. 10-6) and the breakdown of fatty acid chains. Let us consider the latter process before examining the further metabolism of acetyl-CoA.

5. Beta Oxidation

Whether fatty acids are oxidized to obtain energy or are utilized for biosynthesis, they are first converted to their acyl-CoA forms and are then cleaved to the two-carbon units represented by the acetyl groups of acetyl-CoA. The **beta oxidation** sequence, by which this occurs, is represented by the solid vertical arrow on the right side of Fig. 10-1 and is shown in greater detail in Fig. 10-4. We see from the latter figure that there are two dehydrogenation steps. The first (step *b*) removes hydrogen atoms from the α- and β-carbon atoms to produce a *trans* α, β unsaturated fatty acyl-CoA (Enoyl-CoA). The hydrogen acceptor is FAD. It is needed because this reaction requires a more powerful oxidant than does the dehydrogenation of an alcohol, for which NAD⁺ is adequate. Addition of water to the double bond of the unsaturated acyl-CoA (step *c*)

generates an alcohol which is then dehydrogenated (step *d*) by NAD^+ to form a β -oxoacyl-CoA. This is cleaved (step *e*) by a reaction (thiolysis) with another molecule of coenzyme A to form acetyl-CoA and a

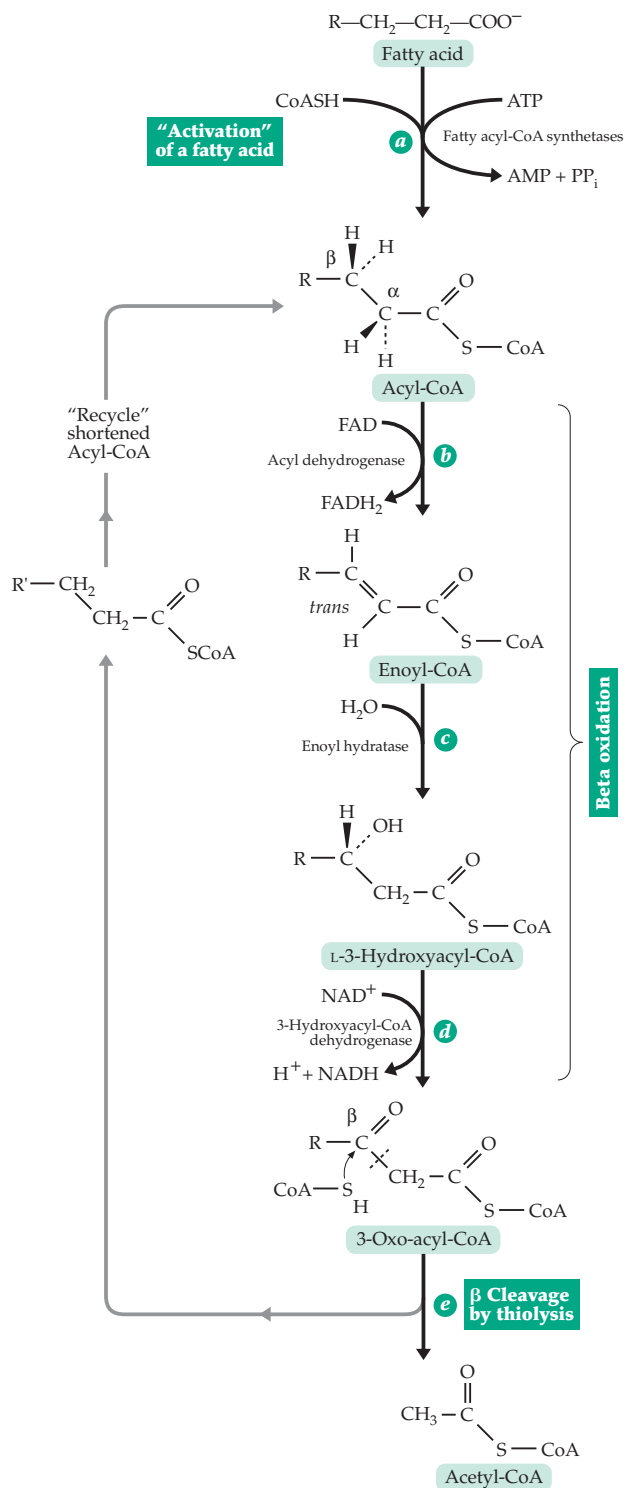


Figure 10-4 Reactions of fatty acid activation and of breakdown by β oxidation.

new acyl-CoA with a shortened chain. The latter is "recycled" by passage through the β oxidation sequence repeatedly until the chain is shortened to a 2- or 3-carbon fragment, acetyl-CoA, or propionyl-CoA. Since most dietary fatty acids contain an even number of carbon atoms, acetyl-CoA is the predominant product.

The beta oxidation of fatty acids, like the dehydrogenation of pyruvate to acetyl-CoA, takes place within the inner **matrix** of the mitochondria in eukaryotic organisms. The reduced hydrogen carriers FADH_2 and NADH transfer their electrons to other carriers located within the **inner membrane** of the mitochondria. In bacteria the corresponding reactions occur with electron carriers present in the plasma membrane. Both FAD and NAD^+ are regenerated in this way and are able to again accept hydrogen from the beta oxidation reactions. This transfer of hydrogen atoms from substrates to hydrogen carriers is typical of biological oxidation processes.

6. The Electron Transport Chain, Oxidative Phosphorylation

Reoxidation of the reduced carriers NADH and FADH_2 actually involves a sequence of electron carriers, the **electron transport chain**, whose function is indicated below the circle near the center of Fig. 10-1. The oxidation of reduced NADH by O_2 (Eq. 10-7) is a highly exergonic process and is accompanied by the



generation of about three molecules of ATP (from ADP and inorganic phosphate). This process, termed **oxidative phosphorylation**, is the principal source of usable energy (in the form of ATP) provided by breakdown of both carbohydrates and fats in the human body.

The mechanism of oxidation of NADH in the electron transport chain appears to occur by transfer of a hydrogen atom together with two electrons (a hydride ion H^-). Oxidation of FADH_2 to FAD might occur by transfer of two hydrogen atoms or by transfer of $\text{H}^- + \text{H}^+$. However, it is useful to talk about all of these compounds as **electron carriers** with the understanding that movement of one or both of the electrons may be accompanied by transfer of H^+ . The electron transport complex is pictured in a very simplified form in Fig. 10-5.

The electrons donated from NADH or other reductants, upon entering this complex, travel from one carrier to the next, with each carrier being a somewhat more powerful oxidant than the previous one. The

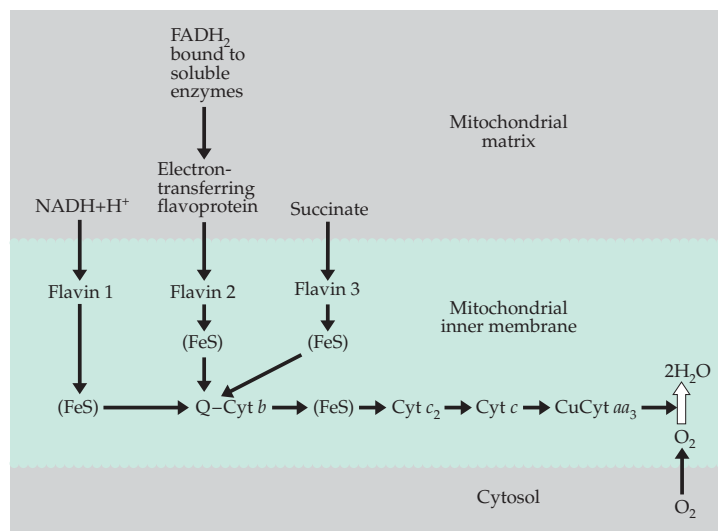


Figure 10-5 An abbreviated version of the electron transport chain of mitochondria. Four electrons reduce O_2 to $2 H_2O$. For details see Figs. 18-5 and 18-6.

final carrier, known as cytochrome aa_3 or **cytochrome oxidase**, reacts with molecular oxygen. Each molecule of O_2 , together with $4 H^+$, is converted to two molecules of water. This stoichiometry requires that two molecules of NADH pass four electrons down the chain for each O_2 reduced.

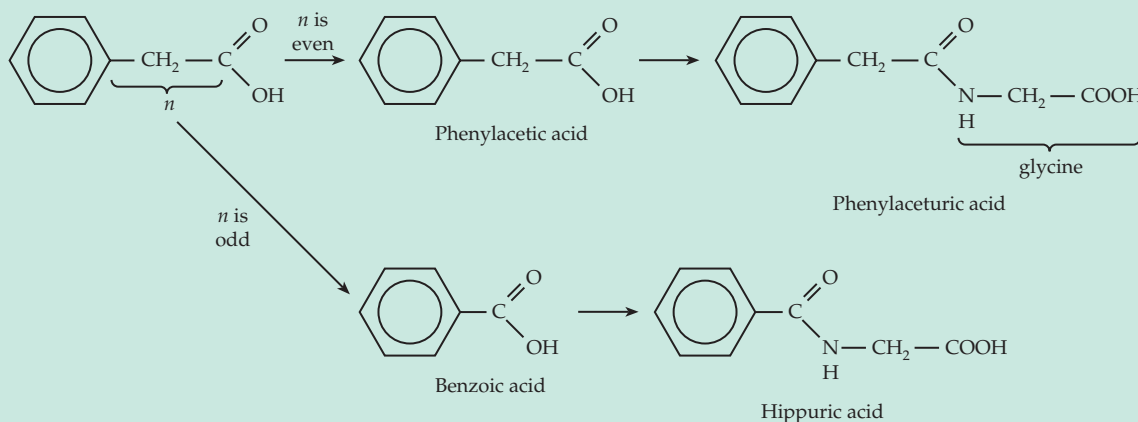
The chemical structures of the components of the mitochondrial electron transport chain are varied but fall into several distinct classes. Most are proteins but these proteins always contain special coenzymes or **prosthetic groups** able to engage in oxidation–reduction reactions. At the substrate end of the chain NADH passes a hydrogen atom together with its bonding electron pair to a riboflavin-containing coenzyme **riboflavin 5'-phosphate** (FMN), which is tightly bound into a protein (designated Flavin 1 in Fig. 10-5). Similar **flavoproteins**, Flavin 2, and Flavin 3, act as oxidants for $FADH_2$ arising during beta

BOX 10-A AN EARLY LABELING EXPERIMENT

In 1904, long before the advent of radioactive tracers, Knoop synthesized fatty acids labeled by chemical attachment of a benzene ring at the end opposite the carboxyl group. He prepared these compounds with both odd and even numbers of carbon atoms in straight chains and fed them to dogs. From the dogs' urine he isolated **hippuric acid** and **phenylaceturic acid**, which are the amides of glycine with benzoic acid and phenylacetic acid, respectively. Knoop showed that the phenylacetic acid was produced from those fatty acids with an

even number of carbon atoms, while the benzoic acid was formed from those with an odd number. From these experimental results, Knoop deduced that *fatty acid degradation occurs two carbon atoms at a time* and proposed his famous **β oxidation theory**.

Later experiments using isotopic labeling with ^{13}C verified Knoop's proposals,^a but study of isolated enzymes was not possible until after the discovery of coenzyme A in 1950. Then, studies of fatty acid oxidation by extracts from isolated mitochondria established the details of the pathway.^b



^a Weinhouse, S. (1995) *FASEB J.* **9**, 820–821

^b Quastel, J. H. (1984) *Trends Biochem. Sci.* **9**, 117–118

oxidation and oxidation of succinate, whose significance to metabolism is discussed in the next section.

A significant difference between NADH and FADH_2 is that the former diffuses freely between the **dehydrogenases** that transfer hydrogen to it and the flavoprotein NADH dehydrogenase (Flavin 1) that reoxidizes it. However, FAD, whether in its oxidized state or as FADH_2 , stays tightly bound to proteins at all times. Only hydrogen atoms or electrons are transferred into or out of these proteins. During beta oxidation the FADH_2 generated remains tied to the fatty acyl-CoA dehydrogenase protein. However, the two hydrogen atoms of this FADH_2 are transferred to another molecule of FAD, which is bound to an **electron-transferring flavoprotein**. This protein carries electrons one at a time, as FADH, by diffusion to the inner surface of the inner mitochondrial membrane. There it transfers the hydrogen atoms that it carries to the flavoprotein designated Flavin 2 in Fig. 10-5. The electron-transferring flavoprotein can also be viewed as a carrier of single electrons, each accompanied by H^+ .

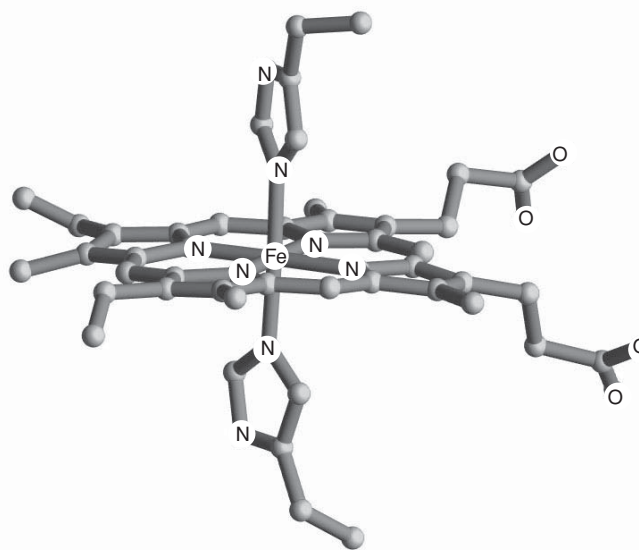
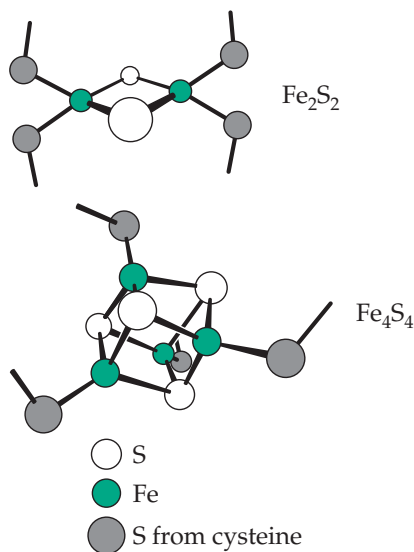
A second group of electron carriers in mitochondrial membranes are the **iron-sulfur [Fe-S] clusters** which are also bound to proteins. Iron-sulfur proteins release Fe^{3+} or Fe^{2+} plus H_2S when acidified. The “inorganic clusters” bound into the proteins have characteristic compositions such as Fe_2S_2 and Fe_4S_4 . The sulfur atoms of the clusters can be regarded as sulfide ions bound to the iron ions. The iron atoms are also attached to other sulfur atoms from cysteine side chains from the proteins. The Fe-S proteins are often tightly associated with other components of the electron transport chain. For example, the flavoproteins Flavin 1, Flavin 2, and Flavin 3 shown in Fig. 10-5 all contain Fe-S clusters as does the Q-cytochrome *b* complex. All of these Fe-S clusters seem to be one-electron carriers.

A third hydrogen carrier of mitochondrial membranes, and the only one that is not unequivocally

associated with a specific protein, is the isoprenoid quinone **ubiquinone** or **coenzyme Q** (Q in Fig. 10-5). Ubiquinone apparently serves as a common carrier, collecting electrons from three or more separate input ends of the chain and directing them along a single pathway to O_2 .

The final group of mitochondrial redox components are one-electron carriers, small proteins (**cytochromes**) that contain iron in the form of the porphyrin complex known as **heme**. These carriers, which are discussed in Chapter 16, exist as several chemically distinct types: *a*, *b*, and *c*. Two or more components of each type are present in mitochondria. The complex cytochrome *aa_3* deserves special comment. Although cytochromes are single-electron carriers, the cytochrome *aa_3* complex must deliver four electrons to a single O_2 molecule. This may explain why the monomeric complex contains two hemes and two **copper** atoms which are also able to undergo redox reactions.^{1,2}

Although the components of the electron transport chain have been studied intensively, there is still some mystery associated with the process by which ATP synthesis is coupled to electron transport (oxidative phosphorylation). A theory originally proposed by Mitchell³ and now generally accepted^{4,5} is that passage of electrons through the chain “pumps” protons from the inside to the outside of the tight inner mitochondrial membrane. As a result, protons accumulate along the outside of this membrane, as do negative counterions along the inside. The membrane becomes charged like a miniature electrical condenser. The synthesis of ATP takes place in the small knoblike **ATP synthase** which is partially embedded in the same membranes. The Gibbs energy for the formation of ATP from ADP and inorganic phosphate is apparently supplied by the flow of protons back to the inside of the membrane through the ATP synthase. Possible



Iron in a heme in a *b* type cytochrome

ways in which this may occur as well as many other aspects of electron transport are dealt with in Chapter 18.

7. The Citric Acid Cycle

The 2-carbon acetyl units removed from fatty acid chains or generated from sugars by glycolysis and the action of pyruvate dehydrogenase must be completely oxidized to carbon dioxide to provide cells with the maximum amount of energy. The oxidation of an acetyl group is a difficult chemical process, and probably for this reason nature has devised an elegant catalytic cycle, the **citric acid cycle**, which is indicated at the lower right in Fig. 10-1 and is shown in detail in Fig. 10-6. The cycle begins when the 4-carbon **oxaloacetate** (also spelled oxalacetate) is condensed with an acetyl group of acetyl-CoA to form the 6-carbon **citrate**. Then, in the remaining reactions of the cycle, two carbon atoms are removed as CO_2 and oxaloacetate is regenerated. Several oxidation steps are involved, each of which feeds additional **reducing equivalents** (i.e., hydrogen atoms removed from substrates) into the pool of hydrogen carriers and allows for more synthesis of ATP via the electron transport chain. The importance of this pathway to an organism can be understood from the fact that when glucose is oxidized completely to CO_2 and water via the citric acid cycle, about 38 molecules of ATP are formed, 19 times as much as by fermentation.

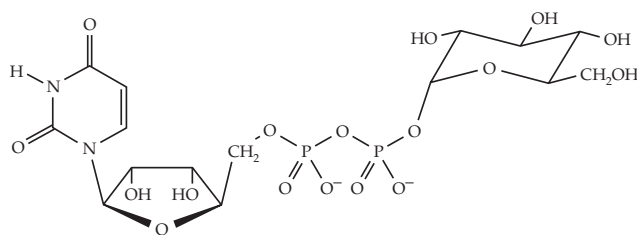
Although oxaloacetate is regenerated and therefore not consumed during the operation of the citric acid cycle, it also enters other metabolic pathways. To replace losses, oxaloacetate can be synthesized from pyruvate and CO_2 in a reaction that uses ATP as an energy source. This is indicated by the heavy gray line leading downward to the right from pyruvate in Fig. 10-1 and at the top center of Fig. 10-6. This reaction depends upon yet another coenzyme, a bound form of the vitamin **biotin**. Pyruvate is formed from breakdown of carbohydrates such as glucose, and the need for oxaloacetate in the citric acid cycle makes the oxidation of fats in the human body dependent on the concurrent metabolism of carbohydrates.

C. Biosynthesis

At the same time that cells break down foodstuffs to obtain energy, they are continuously creating new materials. The green lines in Fig. 10-1 indicate some pathways by which such biosynthesis takes place. Examining the left side of the figure, we see that either pyruvate or 3-phosphoglycerate can be converted back to glucose 6-phosphate and that the latter can be used to synthesize glycogen or other sugars or polysaccharides.

1. Reversing Catabolic Pathways

Breakdown of carbohydrates is thermodynamically spontaneous. Therefore, *cells cannot simply use catabolic pathways operating in reverse without finding ways of coupling the cleavage of ATP to synthesis*. In the formation of glucose from pyruvate in the liver, a process known as **gluconeogenesis**, there are three distinct points at which the enzymes used differ from those used in catabolism: (1) Pyruvate is converted to phosphoenolpyruvate by a mechanism utilizing more than one molecule of ATP, a pathway that is discussed in detail in Chapter 17; (2) as is shown in Fig. 10-2, fructose 1,6-bisphosphate is hydrolyzed to fructose 6-phosphate and inorganic phosphate by a **phosphatase** rather than through reversal of step *d*, which would form ATP; and (3) glucose-6-phosphate is hydrolyzed by a phosphatase rather than following the reverse of step *a* in Fig. 10-2. Furthermore, glycogen is synthesized from glucose 6-phosphate, not by reversal of the phosphorylase reaction (Fig. 10-2, step *g*), but via a new intermediate, **uridine diphosphate glucose** (UDPG), whose formation involves cleavage of UTP. The latter is generated by phospho transfer from ATP. Inorganic **pyrophosphate** (PP_i) is a product of UDPG formation and is removed by hydrolytic cleavage to two molecules of inorganic phosphate by the enzyme **pyrophosphatase**. This reaction helps to make the biosynthesis thermodynamically spontaneous by removing one of the reaction products.



Uridine diphosphate glucose (UDPG)

A similar situation exists in the case of fatty acid synthesis, which proceeds from acetyl-CoA and reverses fatty acid breakdown. However, both carbon dioxide and ATP, a source of energy, are needed in the synthetic pathway. Furthermore, while oxidation of fatty acids requires NAD^+ as one of the oxidants, and generates NADH, the biosynthetic process often requires the related NADPH. These patterns seen in biosynthesis of sugars and fatty acids are typical. *Synthetic reactions resemble the catabolic sequences in reverse, but distinct differences are evident*. These can usually be related to the requirement for energy and often also to control mechanisms.

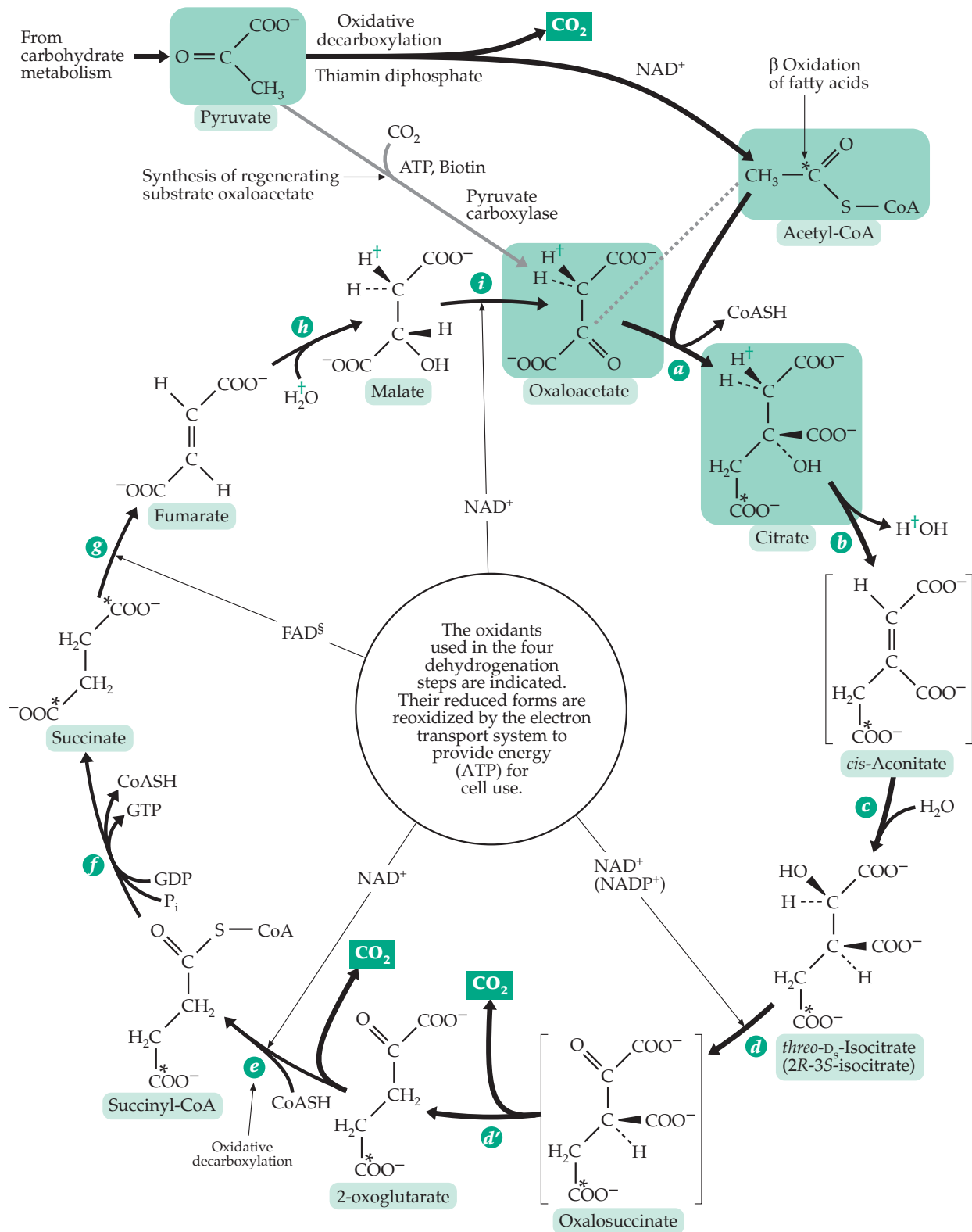


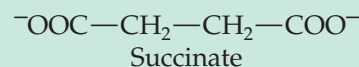
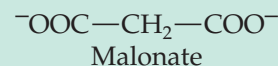
Figure 10-6 Reactions of the citric acid cycle (Krebs' tricarboxylic acid cycle). Asterisks designate positions of isotopic label from entrance of carboxyl-labeled acetate into the cycle. Note that it is *not* the two carbon atoms from acetyl-CoA that are immediately removed as CO_2 but two atoms from oxaloacetate. Only after several turns of the cycle are the carbon atoms of the acetyl-CoA completely converted to CO_2 . Nevertheless, the cycle can properly be regarded as a mechanism of oxidation of acetyl groups to CO_2 . Green daggers (†) designate the position of $^2\text{H}^+$ introduced into malate as $^2\text{H}^+$ from the medium. FAD^{S} designates covalently bound 8-histidyl-FAD (see Chapter 15).

BOX 10-B DISCOVERY OF THE CITRIC ACID CYCLE (KREBS' TRICARBOXYLIC ACID CYCLE)

One of the first persons to study the oxidation of organic compounds by animal tissues was T. Thunberg, who between 1911 and 1920 discovered about 40 organic compounds that could be oxidized by animal tissues. Salts of succinate, fumarate, malate, and citrate were oxidized the fastest. Well aware of Knoop's β oxidation theory, Thunberg proposed a cyclic mechanism for oxidation of acetate. Two molecules of this two-carbon compound were supposed to condense (with reduction) to succinate, which was then oxidized as in the citric acid cycle to oxaloacetate. The latter was decarboxylated to pyruvate, which was oxidatively decarboxylated to acetate to complete the cycle. *One of the reactions essential for this cycle could not be verified experimentally.* It is left to the reader to recognize which one.

In 1935, A. Szent-Györgyi discovered that all of the carboxylic acids that we now recognize as members of the citric acid cycle stimulated respiration of animal tissues that were oxidizing other substrates such as glucose. Drawing on this knowledge, Krebs and Johnson in 1937 proposed the citric acid cycle.^{a,b,c} Krebs provided further confirmation

in 1940 by the observation that malonate, a close structural analog and competitive inhibitor of succinate, in concentrations as low as 0.01 M blocked the respiration of tissues by stopping the oxidation of succinate to fumarate.^c



In muscle, 90% of all respiration was inhibited and succinate was shown to accumulate, powerful proof of the importance of the citric acid cycle in the respiration of animal tissues.

^a Krebs, H. A., and Johnson, W. A. (1937) *Enzymologia* 4, 148–156

^b Krebs, H. A. (1981) *Reminiscences and Reflections* Clarendon Press, Oxford

^c Fruton, J. S. (1999) *Proteins, Enzymes, Genes: the Interplay of Chemistry and Biology* Yale Univ. Press, New Haven, CT

^d Quastel, J. H. (1978) *Trends Biochem. Sci.* 3, 68–69

2. Photosynthesis

The principal means of formation of glucose in nature is through photosynthesis in green plants (Fig. 10-1, lower left). Light energy is captured by chlorophyll and is used to transfer electrons from chlorophyll to other electron carriers, the most important of which is NADP⁺. It is reduced to NADPH which is used to reduce carbon dioxide to sugar phosphates in a complex series of reactions known as the **reductive pentose phosphate pathway**, which is described in Chapter 17. ATP is also required for photosynthetic reduction of CO₂. It is generated by allowing some of the electrons to flow back through an electron transport chain in the membranes of the chloroplasts. This chain closely resembles that from Q to cytochrome *c* in mitochondria (Fig. 10-5), and the generation of ATP in this **photo-synthetic phosphorylation** occurs in a manner analogous to that in the electron transport chain of mitochondria. In green plants the electrons removed from chlorophyll in one light-requiring reaction are replaced by electrons formed during the cleavage of water in a second light-dependent reaction, a reaction that also releases oxygen, O₂, and generates hydrogen ions (H⁺). The first stable product from reduction of CO₂ in photosynthesis is 3-phosphoglycerate. It can be converted to sugars by pathways analogous to those employed by animals in gluconeogenesis. One inter-

esting difference is the use in chloroplasts of NADPH + H⁺ in reduction of 3-phosphoglycerate (reverse of step *a* in Fig. 10-3).

A small number of other biosynthetic pathways, which are used by both photosynthetic and nonphotosynthetic organisms, are indicated in Fig. 10-1. For example, pyruvate is converted readily to the amino acid **L-alanine** and oxaloacetate to **L-aspartic acid**; the latter, in turn, may be utilized in the biosynthesis of pyrimidines. Other amino acids, purines, and additional compounds needed for construction of cells are formed in pathways, most of which branch from some compound shown in Fig. 10-1 or from a point on one of the pathways shown in the figure. In virtually every instance biosynthesis is dependent upon a supply of energy furnished by the cleavage to ATP. In many cases it also requires one of the hydrogen carriers in a reduced form. While Fig. 10-1 outlines in briefest form a minute fraction of the metabolic pathways known, the ones shown are of central importance.

D. Synthesis and Turnover of Macromolecules

Proteins make up the bulk of the catalytic machinery of cells and together with other macromolecules most of the structure. Therefore, the synthesis and degradation of proteins and the control of those processes are of great importance to cells. Although the emphasis

in this section is on proteins, similar considerations apply to nucleic acids, polysaccharides, and other macromolecules.

With the exception of some antibiotics and other short-chain molecules, all polypeptides are formed on ribosomes, which assemble proteins according to the sequences of nucleotides in the messenger RNA (mRNA) molecules. The basic chemistry is simple. The carboxyl groups of the amino acids are converted to reactive acyl adenylates by reaction with ATP, just as in Eq. 10-1. Each “activated” amino acid is carried on a molecule of transfer RNA (tRNA) and is placed in the reactive site of a ribosome when the appropriate codon of the mRNA has moved into the site. The growing peptide chain is then transferred by a displacement reaction onto the amino group of the activated amino acid that is being added to the peptide chain. In this manner, new amino acids are added one at a time to the carboxyl end of the chain, which always remains attached to a tRNA molecule. The process continues until a stop signal in the mRNA ends the process and the completed protein chain is released from the ribosome. Details are given in Chapter 29.

1. Folding and Maturation of Proteins

A newly synthesized peptide chain probably folds quickly. However, the cytosol provides an environment rich in other proteins and other macromolecules that can interact with the new peptide and may catalyze or inhibit folding. Among the most abundant proteins of bacteria or eukaryotes are proteins known as **chaperonins**. They apparently help polypeptide chains to fold correctly, partly by “chaperoning” them through the cytoplasm and across cell membranes, protecting them from becoming entangled with other proteins and macromolecules while they fold.^{6–9} There are several classes of chaperonins. Most are oligomers made up of 10- to 90-kDa subunits. They have a variety of names, which may be somewhat confusing. The first chaperonins were identified only as **heat shock proteins** (Hsp). They are produced in large amounts by bacteria or other cells when the temperature is raised quickly, and are designated Hsp70, Hsp90, etc., where the number is the subunit mass in kDa. Other chaperonins were recognized as products of genes needed for replication of the DNA of bacteriophage λ in *E. coli*. Consequently, one major chaperonin of the Hsp70 class is designated **DnaK**. The Hsp70 protein of mitochondria was named “binding protein” or BiP.¹⁰ Other workers have abbreviated chaperonins as Cpn70, Cpn60, etc.

The abbreviations Hsp70, Cpn70, DnaK, and BiP all refer to a group of similar 70-kDa proteins that are apparently found in all organisms. Their role seems to be to stabilize unfolded proteins prior to final folding

in the cytosol or after translocation into the endoplasmic reticulum (ER) or into mitochondria or other organelles. Each of these proteins consists of two functional domains. A 52-kDa domain at the C terminus binds 7- or 8-residue segments of unfolded peptide chains in an elongated conformation.⁹ At some point in its reaction cycle the N-terminal 40-kDa domain, which binds ATP tightly, causes the ATP to be hydrolyzed to ADP and inorganic phosphate: $\text{ATP} + \text{H}_2\text{O} \rightarrow \text{ADP} + \text{P}_i$. Binding and release of a polypeptide by the *E. coli* DnaK protein is coupled tightly to this exergonic **ATPase** reaction.^{10a,10b} The reaction is dependent upon potassium ions¹¹ and is regulated by two **co-chaperones**, **DnaJ** (Hsp40)^{11a–11c} and **GrpE**.^{10a} Both ATP and extended polypeptides bind weakly to DnaK, and the ATP in the DnaK•polypeptide•ATP complex is hydrolyzed slowly to ADP and inorganic phosphate.^{10a,10b,12} Co-chaperone DnaJ, which shares a largely α -helical structural motif with J-domains in various other proteins, binds to the complex.^{11a, 11b, 11c} It probably induces a conformational change that leads to rapid hydrolysis of ATP. In the resulting complex both ADP and the polypeptide are bound tightly to the DnaK protein and dissociate from it very slowly. The *E. coli* co-chaperone GrpE acts as a **nucleotide exchange factor** that catalyzes rapid loss of ADP from the complex. If ATP binds to this DnaK•polypeptide complex the polypeptide is released.^{10a} This cycle, which can be repeated, accomplishes the function of DnaK in protecting extended polypeptides and releasing them under appropriate conditions. There is also evidence that DnaK participates in refolding of misfolded proteins.^{11d} It cooperates with a ribosome-associated prolyl isomerase in bacteria.^{11e} The three-dimensional structure of the ATPase domain of Hsp70 is strikingly similar to that of the enzyme **hexokinase** (Chapter 12) and to that of the muscle protein **actin**, Fig. 7-10.^{13,14} Archaeal chaperonins lack the Cpn10 ring but have lid-like extensions at the cylinder ends.^{15a}

The Cpn60 class of chaperonins are amazing cage-like structures. Each oligomer is composed of two rings, each made up of seven 60-kDa subunits stacked back-to-back. Cpn60 structures from archaeobacteria are similar but may have 8- or 9-subunit rings.^{15,15a} The best known member of this group is the *E. coli* protein known as **GroEL** whose three-dimensional structure is depicted in Box 7-A.^{16,17} This 14-subunit oligomer of GroEL is a cylinder which is capped by a smaller ring composed of seven 10-kDa subunits of **GroES**, a Cpn10 protein.^{17,18} Like Hsp70, GroEL has ATPase activity and an ATP binding site in its large equatorial domain.¹⁶ Archaeal chaperonins lack the Cpn10 ring but have lid-like extensions at the cylinder ends.^{15a}

Within the GroEL–GroES cage polypeptide chains can fold without becoming entangled with other proteins or being cleaved by protein-hydrolyzing enzymes

of the cytoplasm. An unfolded or partially folded protein may diffuse into the open end of the complex and bind temporarily via noncovalent interactions. The hydrophobic inner surface of the complex may favor the formation of helices within the folding protein. Binding of the polypeptide substrate and capping with GroES causes a major cooperative conformational change with doubling of the internal volume (Box 7-A). The character of the inner wall also changes as a result of exposure of hydrophilic groups.^{18a,18b} Perhaps the expansion also stretches segments of the unfolded polypeptide. Subsequent hydrolysis of ATP may be coupled to release of the bound polypeptide, which may then leave the complex or bind again for another chance to fold correctly.^{17,19} The two ends of the GroEL•GroES complex may function alternately, with each end in turn receiving the GroES cap.¹⁷

Hsp90 is one of the most abundant cytosolic proteins in eukaryotic cells, but it seems to chaperone only a few proteins, among which are steroid-hormone receptors.^{20,20a} Another group of over 30 different chaperone proteins participate in assembly of external bacterial cell-surface structures such as pili (Fig. 7-9). The chaperone PapD consists of two domains, both having an immunoglobulin-like fold.^{20b} PapD binds to the pilin subunits, escorting them to the site of pilus assembly. Folding of integral membrane proteins may be facilitated by a chaperone function of membrane lipids.^{20c}

For most proteins the initial synthesis is followed by a sequence of **processing** or **maturation** reactions. These reactions sometimes involve alteration of amino acid side chains and very often include hydrolytic cleavages by which pieces of the peptide chain are cut off. The initially synthesized polypeptides are often not functional and gain biological activity only after one or more pieces have been removed. For example, digestive enzymes are usually secreted as **zymogens** or **proenzymes** which are activated by hydrolytic cleavages only after secretion into the digestive tract. Many proenzymes are components of the extremely complex cascades of activating reactions involved in **blood clotting** (Chapter 12) and in the defensive **complement system** (Chapter 31).

Most peptide hormones are cut out from larger proteins. For example, human insulin is synthesized as a 110-residue **preproinsulin** which is converted in stages to the active two-chain, 51-residue hormone (Eq. 10-8 and Fig. 10-7).^{20d,20e}

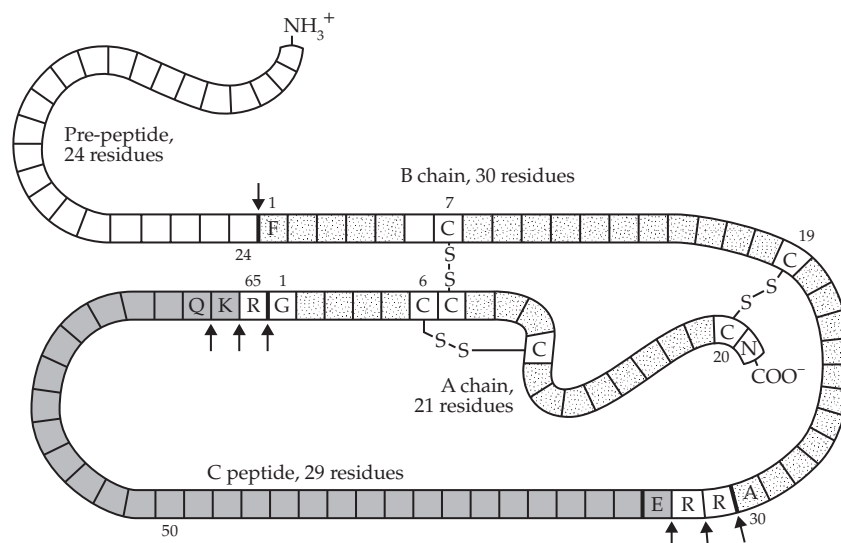
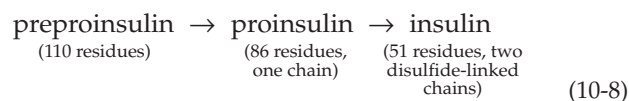


Figure 10-7 Schematic diagram of the structure of human preproinsulin. The 24-residue prepeptide, the 29-residue C-peptide and basic residues 31, 32, 64, and 65 are cut from the peptide upon conversion to insulin as indicated by the small arrows. Some amino acid residues are identified using the one-letter codes. See Fig. 7-17 for details of insulin structure.



The 24-residue prepeptide is cleaved from preproinsulin within a few minutes of synthesis. Then, over a period of about an hour additional cleavages occur to give the final product.²¹ This crystallizes as the zinc hexamer (Fig. 7-18) within the “dense cores” of storage vesicles from which it is released into the bloodstream as needed.²² Other examples are described in Section 3.

2. Transport of Proteins to Their Destinations within a Cell

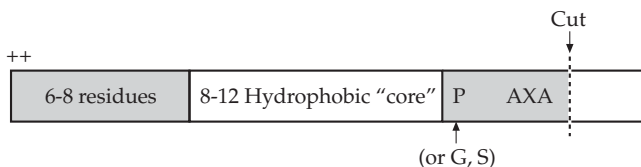
The synthesis of proinsulin on ribosomes in the cytosol and the release of mature insulin from secretion vesicles involve not only the chain cleavages but also transport of the polypeptide across membranes of the ER, Golgi compartments, and secretion vesicles.

Signal sequences and translocation. In 1971, Blobel and Sabatini postulated that for some proteins an N-terminal segment of newly synthesized polypeptide chains contains a signal sequence or leader peptide of 15–30 amino acids which carries information concerning the location of the mature protein in the cell.^{23–28b} The rather nonpolar signal sequences of proteins destined for secretion from cells would interact

with and pass through the membrane of the ER. On the other side of the membrane, a **signal peptidase** would cut off the signal peptide. In many cases, glycosyltransferases would add sugar residues near the N terminus to create a hydrophilic end which would help to pull the rest of the peptide chain through the membrane. A similar situation would hold for secretion of proteins by bacteria.

The signal hypothesis has been proven correct but with considerable added complexity. It now appears that when the N terminus of a new polypeptide chain carrying the proper signal sequence emerges from a ribosome it is intercepted by a **signal recognition particle (SRP)**, a 250-kDa ribonucleoprotein consisting of six polypeptide chains and a small 300-residue 7S RNA chain.^{29–33} Binding to the ribosome, the SRP temporarily blocks further elongation of the peptide chain until it bumps against and binds to a 72-kDa membrane protein called the **SRP receptor** or **docking protein**. Then translation begins again and the protein moves into the ER, where it is cleaved by the signal peptidase and is modified further. Although the bacterial SRP is a single protein called **Ffh** (for 54 homolog),^{33a} the basic machinery for signal recognition and secretion is remarkably conserved from *E. coli* to humans.^{33b} Bacterial proteins that are destined for secretion or for a function in the periplasmic space or in external membranes also contain signal sequences which are cut off by a signal peptidase embedded in the plasma membrane.^{33c}

Signal sequences vary in structure but usually have a net positive charge within the first 5–8 residues at the N terminus. This region is followed by a “hydrophobic core” made up of 8–10 residues with a strong tendency toward α helix formation. This sequence is often followed by one or a few proline, glycine, or serine residues and then a sequence AXA that immediately precedes the cleavage site. Here, A is usually alanine in prokaryotes but may also be glycine, serine, or threonine in eukaryotes. Residue X is any amino acid.^{25,27,28a,28b}



It had often been assumed that a hydrophobic signal sequence, perhaps folded into a hairpin loop, spontaneously inserts itself into an ER membrane to initiate translocation. However, study of the genetics of protein transport suggests otherwise. Over 50 different genetic loci affect the translocation of proteins in yeast.^{31,32,33d} Products of these **secretory genes**,

which are named *SEC 61*, *SEC 62*, *SEC 63*, etc., are proteins with corresponding names, e.g., Sec 61 protein (or Sec 61p). Study of these proteins suggested that the Sec 61, Sec 62, and Sec 63 proteins are all directly involved in transport into the ER.^{31,34,35} Additional proteins are needed for movement of vesicles out from the Golgi and for delivery of secretory vesicles to the plasma membrane.^{33d} The protein Sec 61p has been identified as homologous to the α subunit of a similar trimeric $\alpha\beta\gamma$ Sec 61p complex of mammalian tissues.³² It also appears to correspond to Sec Y, a protein required for secretion of proteins through the plasma membrane of *E. coli* into the periplasmic space. Proteins corresponding to the β and γ subunits of mammalian related proteins have been identified in yeast and *E. coli*³⁶ and in *Arabidopsis*, where it is necessary to bring proteins into the thylakoid membrane of the chloroplasts.³⁷

In *E. coli* the products of genes *SecY*, *SecE*, and *SecG* are integral membrane proteins that, together with additional proteins, form a proteinaceous pore through which proteins pass.^{38–40a} Additional proteins required for translocation of some bacterial proteins are the chaperonin **SecB**, a tetramer of 17.3-kDa subunits,⁴¹ and **SecA**, a soluble ATPase that may participate in docking and serve as an engine for transport through the pore.^{38,40a,42} In eukaryotic cells a membrane-bound Ca^{2+} -dependent chaperonin called **calnexin** assists in bringing the protein into the ER in a properly folded state.^{43,44}

Unlike transport across the membranes of the ER, transport across plasma membranes of bacteria often requires both hydrolysis of ATP and energy provided by the membrane electrical potential.^{33,38,44–48} Secretion into the periplasmic space has been well characterized but less is known about transport of proteins into the external membranes of *E. coli*.⁴⁸ A 16 kDa periplasmic chaperone may be required.^{48a}

Many bacteria have a second complete secretion system.^{48b} This multi-gene type II system is present but usually inactive in *E. coli*. A third (type III) system is present in many pathogenic bacteria, and has evolved for delivery of specialized structural and regulatory proteins into host cells.^{48c,48d}

Ticketing destinations. We have seen that proteins that contain suitable signal sequences are exported from the cytoplasm while other proteins remain. Some proteins are secreted while others take up residence as integral or peripheral membrane proteins or as soluble proteins within an organelle. All of the available evidence indicates that it is the sequence of a protein that determines its destination. Proteins targeted to pass through the inner mitochondrial or chloroplast membrane have 20- to 70-residue presequences that are rich in arginine and lysine and which are removed when the protein reaches its

destination.^{49–53} Proteins meant to go to the intermembrane space or the outer membrane of mitochondria have a sequence containing basic amino acids followed by a long stretch of uncharged residues.⁵⁴ Peroxisomal preproteins may have signal sequences such as SKL at the C terminus^{54–56} and proteins destined to become attached to phosphatidylinositol glycan anchors (Fig. 8-13)⁵⁷ as well as some bacterial surface proteins^{57a} have signal peptides at both N and C termini. Some lysosomal membrane glycoproteins have an LE pair in the N-terminal cytoplasmic tail⁵⁸ and proteins with suitable dileucine and related pairs are often taken up in lysosomes.⁵⁹ Soluble proteins that are resident in the cisternae of the ER often have the C-terminal sequence KDEL or HDEL.^{29,60–64} It may serve both as a **retention signal** and as a **retrieval signal** for return of the protein if it passes on into the Golgi vesicles.^{65,66}

Transport of proteins into mitochondria is dependent upon both cytosolic chaperonins and mitochondrial chaperonins of the Hsp70 and Hsp60 classes.^{67–70} Entrance into mitochondria^{71–73} resembles passage through membranes of the ER. However, entry to the nucleus through the nuclear pore complex requires other localization signals^{73a} as well as specialized proteins.^{74,74a} The sorting of proteins into six different compartments within a chloroplast requires a whole series of recognition signals as well as chaperonins and channel proteins.^{75–79} The various signal sequences that determine a protein's interactions with chaperonins, docking proteins, and proteinaceous pore complexes may be complex and overlapping but evolution has selected sequences that allow cells to live and function. Computer programs for detecting sorting signals are being developed.^{79a} Not only proteins but also other macromolecules are automatically sorted to their correct destination.

Vesicular transport and the Golgi system.

Movement of secreted proteins through the cytosol from the ER to the external surface occurs through the formation and opening up of small vesicles about 70 nm in diameter. It occurs in steps that involve passage from the ER to the various Golgi membranes. Figure 10-8 provides a sketch of the system of ER, Golgi, secretion vesicles, and lysosomes.⁶³ Newly formed secretory proteins flow from the ER into an intermediate compartment where vesicles are formed and are carried to the *cis* Golgi network (CGN). They move step-by-step through the Golgi stack (GS) and into the *trans* Golgi network (TGN). New vesicles are formed to carry proteins between each pair of membranous compartments. In each compartment new glycosylation reactions or other modifications may occur. Finally, vesicles carry the mature proteins to the plasma membrane, lysosomes, or vacuoles.^{79b} A related process is *uptake* of proteins by endocytosis to form, consecutively, early and late endosomes which fuse with lysosomes

(Chapter 8, Section C,6, Fig. 10-8).

At every step in these processes vesicles are formed and are carried to the next destination where the vesicles fuse with the new membrane and discharge their contents.^{80–85a} This remarkable process is complex and highly specific. Rothman proposed that the vesicles to be transported are “docked” on appropriate receptor molecules (called SNARES) on the destination membrane. This is accomplished with the aid of specific soluble marker proteins (called SNAPS) with surfaces complementary to those of the receptors.⁸⁰ Proteins are sorted in this way according to their destination signals. The Golgi system is considered further in Chapter 20.

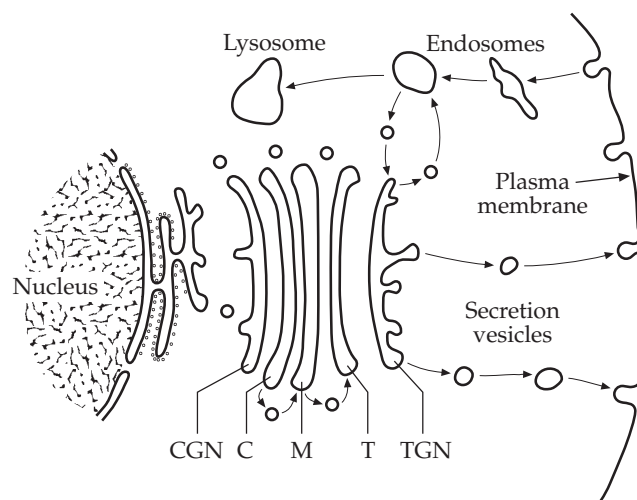
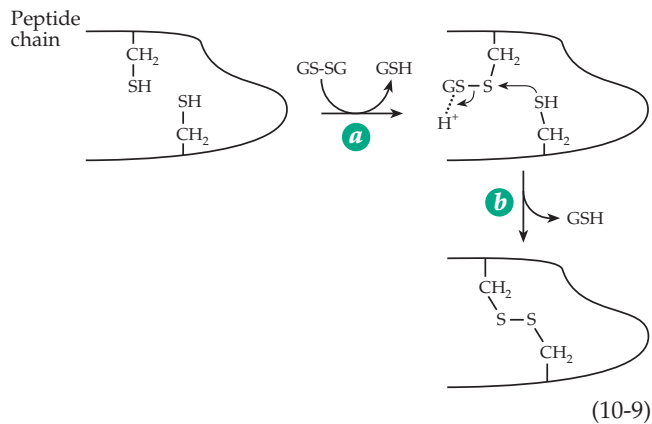


Figure 10-8 Current version of protein synthesis and processing via ER, Golgi, and secretory vesicles. CGN, *cis*-Golgi network; C, T, M are the *cis*, medial, and *trans* compartments of the Golgi stack; TGN, *trans* Golgi network. Arrows indicate some of the movements of transport vesicles.

3. Posttranslational Alterations

Chemical alterations to a protein begins as soon as the peptide chain is formed. Proteins translocated into the periplasmic space of bacteria or into the cisternae of the ER of eukaryotic cells meet several important enzymes. Peptidylprolyl isomerase (discussed in Box 9-F) assists the folding of the peptide chain and protein disulfide isomerases help to form disulfide linkages. The cytosol provides a reducing environment in which many proteins that have cysteine side chains carry free –SH groups. In contrast, the periplasmic space and the ER have more oxidizing environments in which disulfide bridges may form. As shown in Eq. 10-9, the oxidizing agent is often assumed to be the disulfide form of the tripeptide **glutathione** (see Box 11-B),



which is commonly abbreviated GS-SG. It can react with proteins by disulfide exchange reactions to form mixed disulfide groups with release of reduced glutathione, GSH (Eq. 10-9, step *a*). The mixed disulfides can be converted to disulfide bridges by a second exchange reaction (Eq. 10-9, step *b*). However, in the bacterial periplasm the dithiol protein DsbA appears to be the major oxidant (see Box 15-C). It functions together with other proteins, some of which carry electrons from the electron transport chains of the bacteria.^{85a,85b} The rates of these reactions are greatly increased by the protein disulfide isomerases.^{85a-85e,86-91} These enzymes tend to promote formation of disulfide linkages between the correct pairs of SH groups. If there are three or more -SH groups in a chain some incorrect pairing may, and often does, occur. The protein disulfide isomerases break these bonds and allow new ones to form.⁹² The active sites of these isomerases contain pairs of -SH groups which can be oxidized to internal -S-S- bridges by NAD⁺-dependent enzymes. These enzymes and their relatives **thioredoxin** and **glutaredoxin** are discussed further in Box 15-C. Glutathione and oxidation-reduction buffering are considered in Box 11-B.

Proteolytic processing. For proteins exported from the cytosol, a signal peptidase (or leader peptidase) is usually waiting in the bacterial plasma membrane or in membranes of the ER. The *E. coli* leader peptidase is an integral membrane protein with its catalytic domain in the periplasm.^{45,93} For insulin the 24-residue prepiece serves as the signal sequence which is hydrolyzed off in the ER. However, the other cleavages shown in Fig. 10-7 appear to take place in immature secretory granules.^{21,22} For many proteins proteolytic processing begins earlier than this and may occur in more than one location. Eukaryotic processing proteases often resemble the bacterial protease subtilisin^{20e} but cut peptide chains preferentially after dibasic amino acid pairs, e.g., Arg-Arg, Arg-Lys.⁹⁴⁻⁹⁶ There are also other processing enzymes with differing specificities.⁹⁷

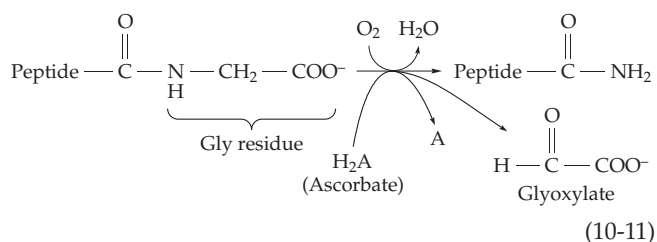
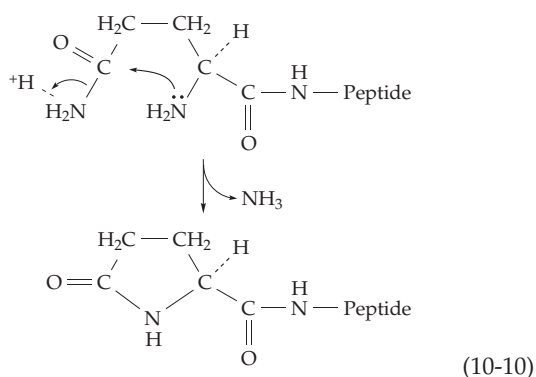
Although most polypeptides leave ribosomes as

single proteins, some are **polypeptides**, which give rise to two or more functional peptides. Polypeptides occur in all cells but are especially prevalent among virally encoded peptides. For example, the polio virus polypeptide is cut into at least ten pieces by proteases, some present in the host cell normally and some encoded by the virus.⁹⁸ Many neurohormones arise from polypeptides that undergo processing as they travel down an axon from the cell body before being secreted into a synapse (Chapter 30). Within our own brains the peptide **prepro-opiomelanocortin** undergoes numerous cleavages to give rise to at least seven different neurohormones (see Fig. 30-2).^{96,99-101}

Altered ends. Proteins often contain “blocked” end groups in place of free -NH₃⁺ or -COO⁻. For many of the cytosolic proteins, which do not carry leader sequences, the N-terminal methionine that is always used to initiate the ribosomal synthesis of proteins in eukaryotes is removed. If the next residue after the N-terminal methionine is small and uncharged, the methionine is usually hydrolyzed off enzymatically after 30–40 residues have been added to the growing peptide chain.¹⁰²⁻¹⁰⁵ An acetyl group is transferred onto the NH₂ termini of about 85% of all cytosolic proteins, whether or not the initiator methionine has been removed.¹⁰⁶ In other cases fatty acyl groups may be transferred to the terminal -NH₂ group or to side chain SH groups. For example, a **myristoyl** (tetradecanoyl) group is frequently combined in amide linkage at the N terminus of cellular and virally encoded proteins. A palmitoyl group is joined in thioester linkage to a cysteine side chain near the N terminus of the *E. coli* periplasmic lipoprotein (Fig. 8-28).¹⁰⁷ Together with the N-terminal myristoyl group it forms a membrane anchor for this protein. Polyprenyl groups are transferred onto cysteine -SH groups at or near the C termini of many eukaryotic proteins to form a different membrane anchor.¹⁰⁸⁻¹¹⁰ Some of the enzymes act on the sequence CAAX, where X = M, S, Q, C but not L or I. After transfer of a farnesyl or geranylgeranyl group onto the cysteine the AAX is removed proteolytically and the new terminal cysteinyl carboxyl group is methylated. Surface proteins of gram-positive bacteria are joined by amide linkage from their C termini to a pentaglycine chain of the peptidoglycan layer (Fig. 8-28).^{57a}

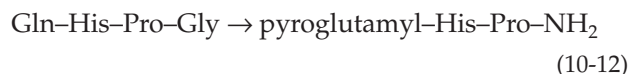
A pyroglutamyl N terminus is found in the thyrotropin-releasing hormone (Fig. 2-4) and in many other peptide hormones and proteins. It presumably arises by attack of the α-NH₂ group of an N-terminal glutamine on the side chain amide group with release of NH₃ (Eq. 10-10).^{111,112}

Another frequent modification at the C terminus of peptide hormones and of other proteins is **amidation**. In this reaction a C-terminal glycine is oxidatively removed as glyoxylate in an O₂⁻, copper- and ascorbate



(vitamin C)-dependent process (Eq. 10-11).^{113–117} See also Chapter 18, Section F.2.

An example of both of these modifications is formation of thyrotropin-releasing hormone shown in Fig. 2-4 from its immediate precursor:



Proteins with long C-terminal hydrophobic signal sequences may become attached to phosphatidylinositol-glycan anchors embedded in the plasma membrane (Fig. 8-13). An example is a human alkaline phosphatase in which the α carboxyl of the terminal aspartate residue forms an amide linkage with the ethanolamine part of the anchor. Attachment may occur by a direct attack of the -NH_2 group of the ethanolamine on a peptide linkage in a transacylation reaction that releases a 29-residue peptide from the C terminus.^{118,119} (See Chapter 29).

Many other covalent modifications of proteins are dealt with in other sections of the book.^{120–122} A few are described in Chapter 2, Eqs. 2-14 to 2-22. Reversible alterations used to regulate enzymes are considered in Chapter 11. Of these, the phosphorylation of -OH groups of serine, threonine, and tyrosine is the most important. A large fraction of all cellular proteins appear to be modified in this way. Protein glycosylation, the transfer of glycosyl groups onto -OH side chain groups of serine and threonine (Chapter 4, and Chapter 20) and nonenzymatic glycation (Eq. 4-8) also affect many proteins, often at turns in the peptide chain. Hydroxylation, glycosylation, and other modifications of collagen are described in Chapter 8. Another common

reversible alteration is formation of sulfate esters of tyrosine -OH groups.^{99,123} Reactions by which cofactors become attached to proteins¹²⁴ are described in Chapters 15–17.

Methylation,¹²⁵ hydroxylation, and other irreversible modifications often affect specific residues in a protein. Oxidative alterations occur during aging of proteins (Chapter 18).¹²⁶ A few proteins even undergo “splicing” that alters the amino acid sequence (Box 29-E).¹²⁷ All of these reactions not only affect the properties of the proteins but also participate in driving the turnover of these macromolecules.

4. Intracellular Degradation of Proteins

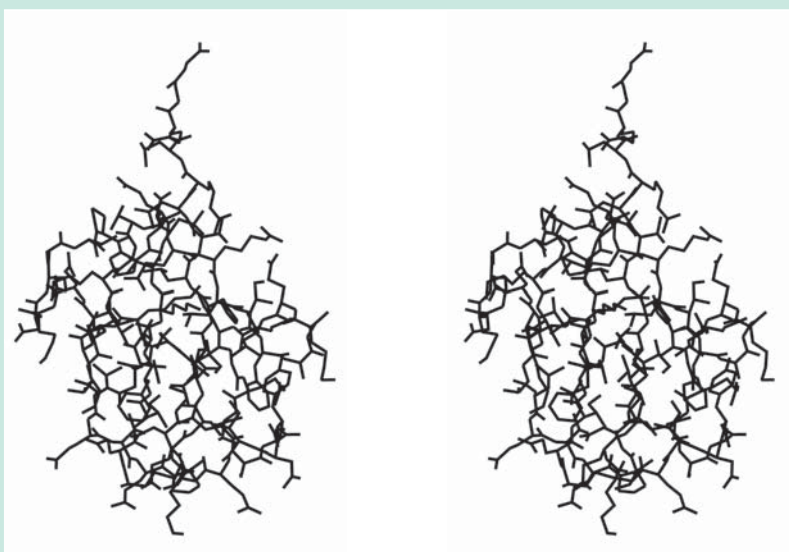
Once a protein has reached its correct location and has acquired its proper function, it usually has a limited lifetime, which may average only a few hours or a few days. The protein is then hydrolytically degraded back to its constituent amino acids.¹²⁸ Defective and damaged proteins are usually degraded much more rapidly than are intact proteins.^{129,130} Under conditions of starvation, proteins are broken down more rapidly than usual to supply the cell with energy. Rapid degradation of proteins is often induced at certain stages of differentiation. For example, spore-forming bacteria contain a protease that becomes activated upon germination of the spore.¹³¹ Within minutes this enzyme digests stored proteins to provide amino acids for the synthesis of new proteins during growth.

Eukaryotic cells degrade proteins within both the cytosol and lysosomes. Lysosomes apparently take up many proteins but have a preference for N-terminal KFERQ¹³² and also for particular types of glycosylation (Chapter 20). Lysosomes act on many long-lived proteins.^{133,133a} Once within the lysosomes, the proteins are broken down into amino acids with a half-life of ~ 8 minutes. During nutritional deprivation, the rate of uptake of proteins by lysosomes increases markedly. The same is true during certain developmental changes, for example, when a tadpole loses its tail.

Many short-lived proteins are degraded within the cytosol in ATP-dependent processes. A major process involves the small protein **ubiquitin** (Box 10-C).¹³⁴ Once “labeled” by formation of an isopeptide linkage to ubiquitin, a peptide is attacked by proteases in the **proteasome** complexes (Box 7-A, Chapter 12). There it is quickly degraded. Other proteases, most of which do not require ATP, are also present in the cytoplasm (Chapter 12). How do these enzymes as well as those within the lysosomes work together to produce a harmonious turnover of the very substance of our tissues? How is it possible that one protein has a long half life of many days while another lasts only an hour or two in the same cell? The answer seems to be that

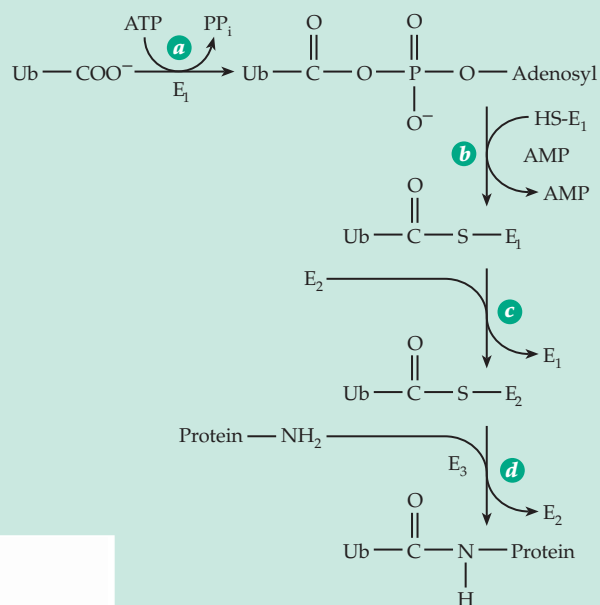
BOX 10-C UBIQUITIN

The small 76-residue protein called ubiquitin^{a-c} is probably present in all eukaryotic cells. Found in the nucleus, cytoplasm, cell surface membranes, and extracellular fluids, ubiquitin is often joined by isopeptide linkages from its C-terminal glycine to the ϵ -amino groups of lysine side chains of other proteins. Ubiquitin has one of the most conserved of all known amino acid sequences. No amino acid substitutions have been found among animal species and only three differences distinguish plant ubiquitins from that of humans. In its three-dimensional structure ubiquitin is compact, tightly hydrogen bonded, and roughly spherical. It contains an α helix, a mixed β sheet, and a distinct hydrophobic core.



Stereoscopic drawing of human ubiquitin. The C terminus is at the top.
From Vijay-Kumar *et al.*^c Courtesy of William J. Cook

Linkage of ubiquitin to other proteins occurs through the action of a ligase system which catalyzes four sequential reactions as shown in the accompanying equation. In step *a* a **ubiquitin activating enzyme** (E_1) forms a C-terminal acyl adenylate by reaction of ubiquitin (Ub) and ATP.^{d,e} In step *b* a sulfhydryl group of the same enzyme then displaces the AMP part to form a thioester linkage to ubiquitin. The chemistry of the reaction is the same as that in Eq. 10-1. The activated ubiquitin is next transferred (step *c*) by transacylation to several **ubiquitin-conjugating enzymes** (E_2), also called ubiquitin-carrier proteins.^{f,i} These in turn (step *d*) transfer the ubiquitin to amino groups of lysine side chains of target proteins (Prot-NH₂). A third protein, **ubiquitin-protein ligase** (E_3), is sometimes required for this



last step. Most E_2 enzymes, acting without E_3 , couple ubiquitin with amines or with small basic proteins whose cellular functions are still unclear. For example, an E_2 which is the product of the yeast gene CDC34 appears to function in the **cell cycle** (Fig. 11-15). Its absence from yeast is lethal^{f,i,j} and it is clear that ubiquitin-mediated hydrolysis of the specialized proteins called **cyclins** is essential to operation of the cycle.^{k,l}

The best understood function of ubiquitin is in nonlysosomal degradation of proteins.^{j,m,n} Protein E_3 appears to select the proteins for degradation, binding them and catalyzing the formation of the isopeptide linkage to substrate. After one ubiquitin molecule has been attached, and while still held by E_3 , a second activated ubiquitin is coupled to Lys 48 of the first ubiquitin. This process may continue until several molecules of ubiquitin are joined by Gly-Lys isopeptide linkages to form a **polyubiquitin** chain. Sometimes more than one lysine of the substrate becomes polyubiquitinated to form branched chains. A free α -NH₃⁺ group on the protein being degraded is essential for rapid conjugation with ubiquitin and certain N-terminal residues such as arginine favor the conjugation and subsequent hydrolytic breakdown. Some proteins become attached to ubiquitin only after arginine is transferred

BOX 10-C (continued)

onto their N-termini from an aminoacyl-tRNA.^o

Polyubiquitin chains serve as recognition markers that induce rapid hydrolysis of the marked proteins in the 2000 kDa 26S proteasome or **multicatalytic** protease.^{P-s} This complex is discussed in Box 7-A and again in Chapter 12. During hydrolytic destruction of the protein the ubiquitin is released for reuse by **ubiquitin carboxyl-terminal hydrolases** or **isopeptidases** which cleave the thioester or isopeptide linkages that tie ubiquitin to proteins.^{t,u}

While proteins may be modified to favor rapid ubiquitination, others may be altered to protect them from ubiquitination. For example, calmodulin produced from a cloned gene in bacteria is a good substrate for ubiquitination but within cells it appears to be protected by the posttranslational conversion of Lys 115 to trimethyllysine.^v

About 10% of the histone H2A present in higher eukaryotes is ubiquitinated at Lys 119.^w In the slime mold *Physarum* the content of ubiquitinated histones H2A and H2B changes rapidly during the various stages of mitosis. Apparently, ubiquitin must be cleaved from the histones to permit packaging of DNA into metaphase chromosomes and must become attached to the histones in some regions of the chromosomes to allow unfolding of the highly packed nucleosomes. A yeast enzyme that attaches ubiquitin to histones is encoded by the gene *RAD6*, which is required for DNA repair, sporulation, and

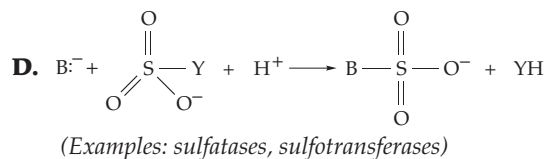
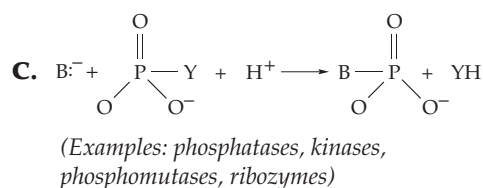
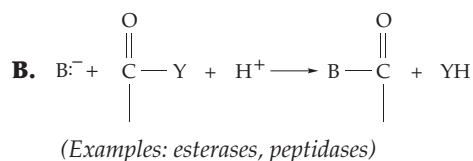
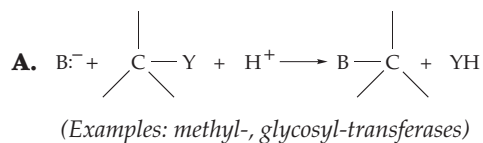
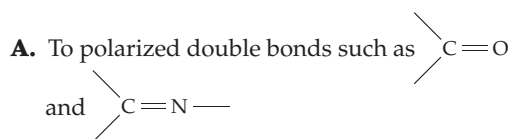
other cellular processes. Terminal differentiation of reticulocytes to form erythrocytes involves loss of specific enzymes as well as of entire mitochondria. These processes also depend upon ubiquitin. Ubiquitin is one of the components of the paired helical filaments present in brains of persons with Alzheimer's disease.^x

In most organisms there are two arrangements for ubiquitin genes. There is a cluster of up to 100 tandemly repeated genes whose transcription gives rise to polyubiquitin, a chain of ubiquitin molecules joined by Gly-Met linkages. These must be cleaved, perhaps by the same ubiquitin C-terminal hydrolase that releases ubiquitin from its conjugates.^y Other ubiquitin genes are fused to genes encoding ribosomal proteins. The resulting polypeptides have the ribosomal peptides fused to the C termini of the ubiquitin sequences and must be proteolytically cleaved to give mature proteins.^z

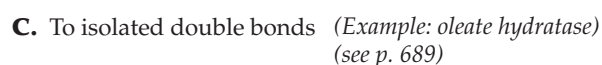
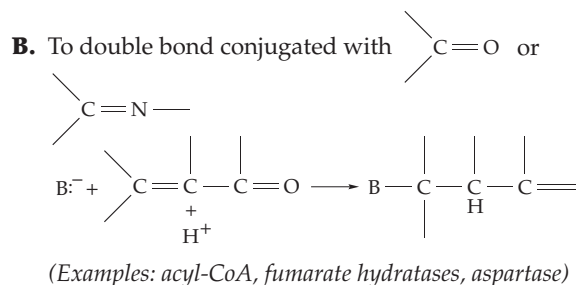
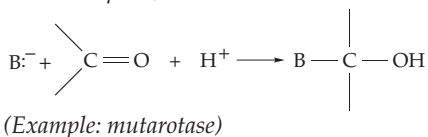
Recently a variety of modifiers of ubiquitin ligases have been discovered^{aa,bb} as have ubiquitin-like domains in other proteins.^{cc} These findings elucidate the complexity of the sorting of proteins and removal of improperly folded and otherwise defective proteins from the secretory pathway and return to the proteasomes in the cytosol.^{dd,ee} They also suggest important roles for ubiquitination in a broad range of metabolic controls.

- ^a Vijay-Kumar, S., Bugg, C. E., Wilkinson, K. D., Vierstra, R. D., Hatfield, P. M., and Cook, W. J. (1987) *J. Biol. Chem.* **262**, 6396–6399
- ^b Vierstra, R. D., Langan, S. M., and Schaller, G. E. (1986) *Biochemistry* **25**, 3105–3108
- ^c Vijay-Kumar, S., Bugg, C. E., and Cook, W. J. (1987) *J. Mol. Biol.* **194**, 531–544
- ^d Hershko, A. (1991) *Trends Biochem. Sci.* **16**, 265–268
- ^e Pickart, C. M., Kasperek, E. M., Beal, R., and Kim, A. (1994) *J. Biol. Chem.* **269**, 7115–7123
- ^f Jentsch, S., Seufert, W., Sommer, T., and Reins, H.-A. (1990) *Trends Biochem. Sci.* **15**, 195–198
- ^g Cook, W. J., Jeffrey, L. C., Sullivan, M. L., and Vierstra, R. D. (1992) *J. Biol. Chem.* **267**, 15116–15121
- ^h Cook, W. J., Jeffrey, L. C., Xu, Y., and Chau, V. (1993) *Biochemistry* **32**, 13809–13817
- ⁱ Blumenfeld, N., Gonen, H., Mayer, A., Smith, C. E., Siegel, N. R., Schwartz, A. L., and Ciechanover, A. (1994) *J. Biol. Chem.* **269**, 9574–9581
- ^j Ciechanover, A., and Schwartz, A. L. (1994) *FASEB J.* **8**, 182–191
- ^k Barinaga, M. (1995) *Science* **269**, 631–632
- ^l Dorée, M., and Galas, S. (1994) *FASEB J.* **8**, 1114–1121
- ^m Pickart, C. M. (2000) *Trends Biochem. Sci.* **25**, 544–548
- ⁿ Johnson, E. S., Ma, P. C. M., Ota, I. M., and Varshavsky, A. (1995) *J. Biol. Chem.* **270**, 17442–17456
- ^o Ciechanover, A., Ferber, S., Ganioth, D., Elias, S., Hershko, A., and Arfin, S. (1988) *J. Biol. Chem.* **263**, 11155–11167
- ^p Goldberg, A. L. (1995) *Science* **268**, 522–523

- ^q Löwe, J., Stock, D., Jap, B., Zwickl, P., Baumeister, W., and Huber, R. (1995) *Science* **268**, 533–539
- ^r Thrower, J. S., Hoffman, L., Rechsteiner, M., and Pickart, C. M. (2000) *EMBO J.* **19**, 94–102
- ^s Peters, J.-M. (1994) *Trends Biochem. Sci.* **19**, 377–382
- ^t Stein, R. L., Chen, Z., and Melandri, F. (1995) *Biochemistry* **34**, 12616–12623
- ^u Wilkinson, K. D., Tashayev, V. L., O'Connor, L. B., Larsen, C. N., Kasperek, E., and Pickart, C. M. (1995) *Biochemistry* **34**, 14535–14546
- ^v Johnston, S. C., Riddle, S. M., Cohen, R. E., and Hill, C. P. (1999) *EMBO J.* **18**, 3877–3887
- ^w Davie, J. R., and Murphy, L. C. (1990) *Biochemistry* **29**, 4752–4757
- ^x Mori, H., Kondo, J., and Ihara, Y. (1987) *Science* **235**, 1641–1644
- ^y Finley, D., and Varshavsky, A. (1985) *Trends Biochem. Sci.* **10**, 343–347
- ^z Baker, R. T., Tobias, J. W., and Varshavsky, A. (1992) *J. Biol. Chem.* **267**, 23364–23375
- ^{aa} Tyers, M., and Willems, A. R. (1999) *Science* **284**, 601–604
- ^{bb} Turner, G. C., Du, F., and Varshavsky, A. (2000) *Nature (London)* **405**, 579–583
- ^{cc} Hochstrasser, M. (2000) *Science* **289**, 563–564
- ^{dd} Ellgaard, L., Molinari, M., and Helenius, A. (1999) *Science* **286**, 1882–1888
- ^{ee} Plemper, R. K., and Wolf, D. H. (1999) *Trends Biochem. Sci.* **24**, 266–270

TABLE 10-1
Types of Biochemical Reactions with Ionic Mechanisms**1. Nucleophilic displacement, often via an addition–elimination sequence****2. Addition**

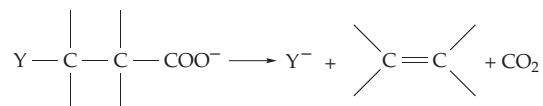
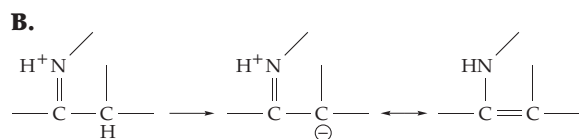
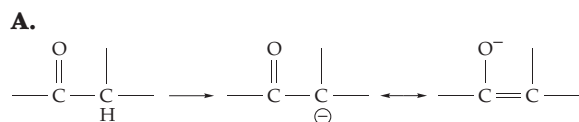
(This reaction most often occurs as a step in an enzymatic process e.g., formation of hemiacetals, hemiketals, hemimercaptals, carbinolamines)

**3. Elimination**

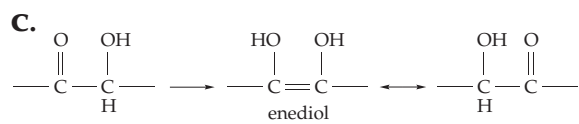
A. and B. Precisely the opposite of *addition*

(Eliminations that are the reverse of type 2A are frequent steps in more complex enzyme mechanisms)

C. Decarboxylative elimination

**4. Formation of stabilized enolate anions and enamines**

Isomerization reactions



(Example: sugar isomerases)

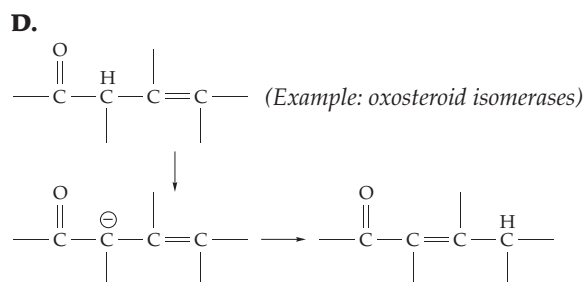
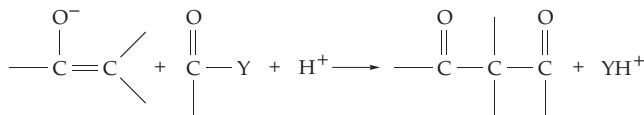
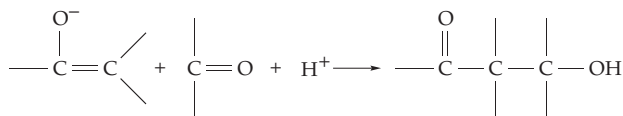
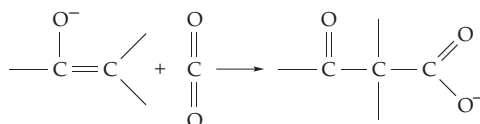


TABLE 10-1

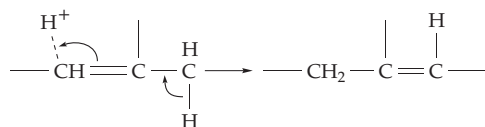
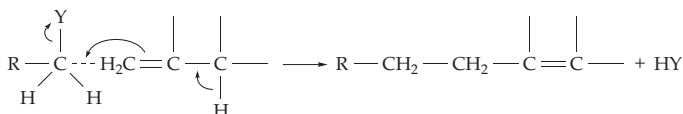
(continued)

5. Stabilized enolate anions as nucleophiles: formation of carbon-carbon bonds (β condensation)**A. Displacement on a carbonyl group**(Example: 3-Oxoacyl-CoA transferase ($\text{Y} = \text{---S---CoA}$))**B. Addition to a carbonyl group: aldol condensation**

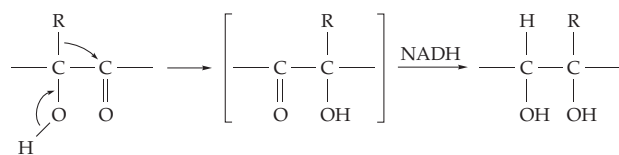
(Examples: aldolases, citrate synthases)

C. Addition to a carbon dioxide (β carboxylation); decarboxylation

(Examples: phosphoenolpyruvate carboxylase, oxaloacetate decarboxylases)

6. Some isomerization and rearrangement reactions**A. Allylic rearrangement (1,3-proton shift)****B. Allylic rearrangement with condensation**

(Example: condensation of dimethylallyl pyrophosphate with isopentenyl pyrophosphate)

C. Rearrangements with alkyl or hydride ion shift

Glyoxalase

(Examples: biosynthesis of leucine and valine, xylose isomerase, glyoxalase)

the turnover rate of a protein is determined in large part by its sequence. Some proteins are tightly folded and have few bends on the outside that have sequences meeting the specificity requirements of intracellular proteolytic enzymes. These have long half-lives. Other proteins may have external loops with sequences susceptible to attack or sequences that favor rapid reaction with the ubiquitin system or uptake into lysosomes. Prematurely terminated proteins and peptide fragments from partial degradation of proteins may tend to be unfolded at the N terminus and to be attacked rapidly.^{128,135} Proteins that have undergone covalent modifications or oxidative damage also seem to be hydrolyzed rapidly.¹³⁶

Regions rich in proline, glutamate, serine, and threonine (PEST regions) may be good substrates for Ca^{2+} -activated cytosolic proteases.¹³⁷ The ubiquitin system appears to act most slowly on a protein when the normal initiation amino acid methionine is present at the N terminus. For example, the half-life of β -galactosidase in yeast is over 20 hours. Replacement of the methionine with S, A, T, V, or G has little effect. However, replacement with other amino acids shortens the half-life as follows: I and E, 30 min.; Y and Q, 10 min.; F, L, D, and K, 3 min.; and R, 2 min.^{138-140a}

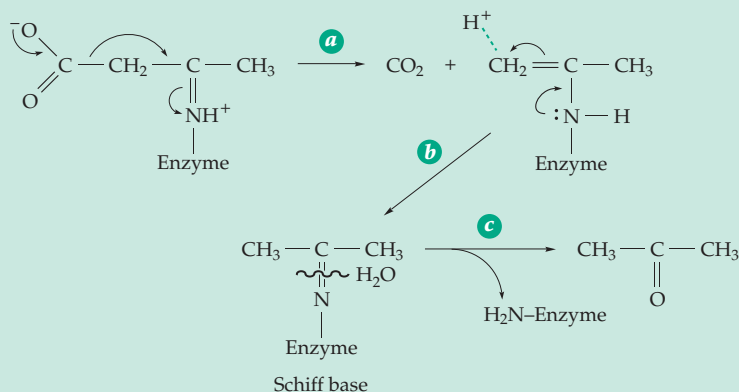
5. Turnover of Nucleic Acids

Because of its special role in carrying genetic information, DNA is relatively stable. An elaborate system of repair enzymes (Chapter 27) act to correct errors and to help DNA to preserve its genetic information. However, in some specialized cells such as those forming immunoglobulin, the DNA too undergoes major rearrangements (Chapter 31). RNA molecules are subject to extensive processing. This includes the conversion of RNA bases to modified forms, chain cleavages during maturation of ribosomal and transfer RNAs, cutting and splicing of gene transcripts to form mRNAs and finally degradation of the mRNA (Chapter 28). Proteins that serve as **RNA chaperones** assist in folding these molecules.¹⁴¹

BOX 10-D DRAWING THOSE LITTLE ARROWS

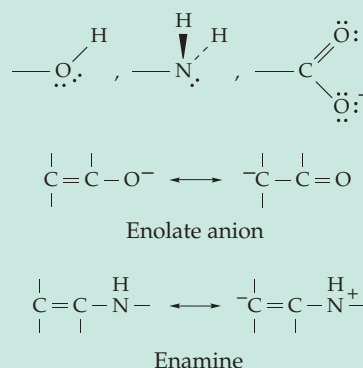
Organic mechanisms are often indicated by arrows that show the flow of electrons in individual steps of a reaction. Many errors are made by students on exams and even in published research papers. The arrows are often drawn backward, are too numerous, or do not clearly indicate electron flow. Here are some tips.

- 1** Always write a mechanism step-by-step. Never combine two steps (e.g., *a* and *b* at right) in which electron flow occurs in opposite directions. Notice that step *c*, the hydrolysis of a Schiff base, is also a two-step process. The reaction is a familiar one that is commonly indicated as shown here. However, this scheme does not show a detailed mechanism for step *c*.

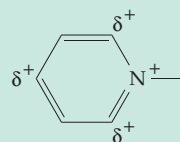
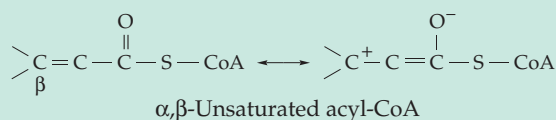
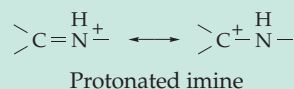
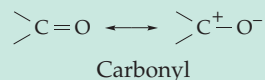


- 2** Identify the **nucleophilic** and **electrophilic** centers before starting to write a mechanism. Oxygen, sulfur, and nitrogen atoms are usually nucleophilic, e.g., those in —OH , —NH_2 , —COO^- , —SH , —OPO_3^{2-} , and enolate anions. The weak nucleophiles —OH and —SH may be converted into strong nucleophiles —O^- —S^- by removal of a proton by a basic group of an enzyme. The nonnucleophilic —NH_3^+ becomes the good nucleophile —NH_2 by loss of a proton. Nucleophilic centers contain unshared electron pairs. Nucleophiles are basic but the basicity, as indicated by proton binding (by the $\text{p}K_a$), is not necessarily proportional to nucleophilic strength (nucleophilicity). Enolate anions and enamines provide nucleophilic centers on carbon atoms, important in formation and cleavage of C—C bonds.

Electrophilic centers include acidic hydrogen atoms, metal ions, the carbon atoms of carbonyl groups, and the β -carbon atoms of α , β unsaturated acids, ketones, or acyl-CoA derivatives. Highly polarized groups such as carbonyl and enamine generate electrophilic centers as indicated by the positive charges. They also affect more distant positions in conjugated systems, e.g., in α , β -unsaturated acyl-CoA derivatives, and in intermediates formed from thiamin diphosphate and pyridoxal phosphate.



Groups with nucleophilic centers indicated by unshared electron pairs and/or negative charge.

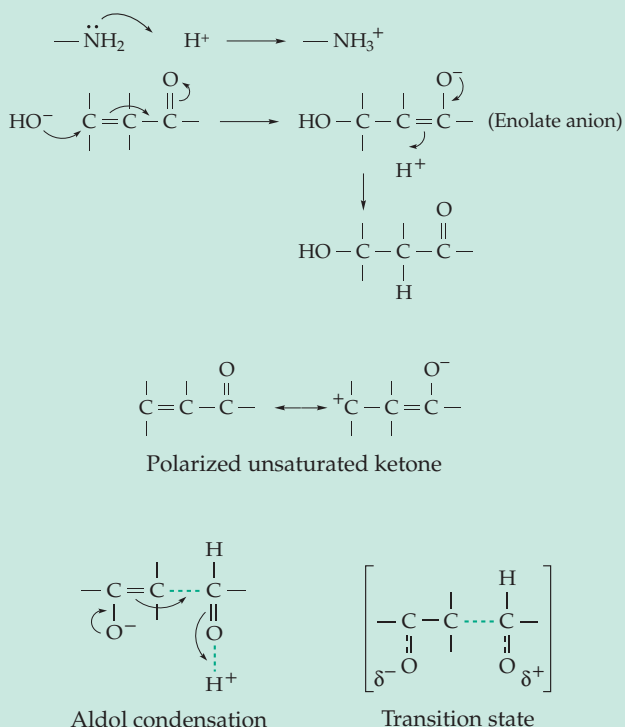


Some electrophilic centers are indicated by + or δ^+

BOX 10-D (continued)

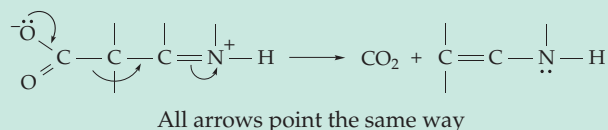
3

When an ionic organic reaction (the kind catalyzed by most enzymes) occurs a nucleophilic center joins with an electrophilic center. We use arrows to show the movement of pairs of electrons. The movement is always *away* from the nucleophile which can be thought of as “attacking” an electrophilic center. Notice the first step in the second example at right. The unsaturated ketone is polarized initially. However, this is not shown as a separate step. Rather, the flow of electrons from the double bond, between the α - and β -carbons into the electron-accepting C=O groups, is coordinated with the attack by the nucleophile. Dotted lines are often used to indicate bonds that will be formed in a reaction step, e.g., in an aldol condensation (right). Dashed or dotted lines are often used to indicate partially formed and partially broken bonds in a transition state, e.g., for the aldol condensation (with prior protonation of the aldehyde). However, *do not put arrows on transition state structures*.



4

In a given reaction step all of the arrows must point the same way. *The arrows point into bonds that are forming or toward atoms that will carry an unshared electron pair in the product. Arrows originate from unshared electron pairs or from bonds that are breaking.*

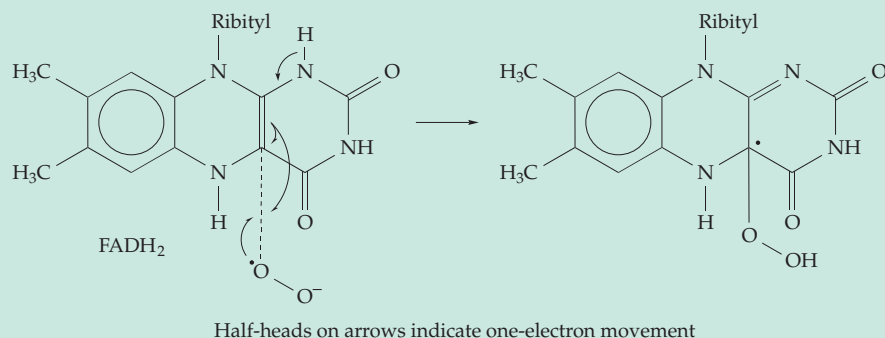


5

Never start an arrow from the same bond that another arrow is forming; i.e., electrons flow out of *alternate* bonds.

6

For reactions of radicals (homolytic reactions) arrows are used to indicate motion of single electrons rather than of electron pairs. It is desirable to use arrows with *half-heads*. For example, reaction of a superoxide radical ($\cdot\text{O}_2^-$) with FADH_2 could occur as follows:



E. Classifying Enzymatic Reactions

The majority of enzymes appear to contain, in their active centers, only the side chains of amino acids. Most of the reactions catalyzed by these enzymes can be classified into a small number of types as is indicated in Table 10-1: (Type 1) **displacement** or **substitution** reactions in which one base or nucleophile replaces another, (Type 2) **addition** reactions in which a reagent adds to a double bond, and (Type 3) **elimination** reactions by which groups are removed from a substrate to create double bonds. Note that the latter are the reverse of addition reactions. Two other groups of reactions depend upon formation of transient **enolic forms**. These include (Type 4) **isomerases** and (Type 5) **lyases**, a large and important group of reactions that form or cleave carbon-carbon bonds. Finally, there is a group (Type 6) of **isomerization** and **rearrangement** reactions that do not appear to fit any of the foregoing categories. Another quite different group of enzyme-catalyzed reactions, which are considered in Chapter 16, function with the participation of free radical intermediates.

Biochemical displacement reactions include all of the hydrolytic reactions by which biopolymers are broken down to monomers as well as most of the reactions by which the monomers are linked together to form polymers. Addition reactions are used to introduce oxygen, nitrogen, and sulfur atoms into biochemical compounds and elimination reactions often

provide the driving force for biosynthetic sequences. Complex enzymatic processes are often combinations of several steps involving displacement, addition, or elimination. The reactions involving formation or cleavage of C-C bonds are essential to biosynthesis and degradation of the various carbon skeletons found in biomolecules, while the isomerization reactions provide connecting links between the other kinds of reactions in the establishment of metabolic pathways.

In Chapters 12 and 13 the individual reactions of metabolism are classified into these types and the enzymes that catalyze them are described in some detail. The chemistry of coenzymes and metalloenzymes are presented systematically in Chapters 14 to 16, and in Chapter 17 the logic of the combining of individual reactions into metabolic sequences is considered. It is not necessary to read Chapters 12-16 in their entirety since much of their content is reference material. In the later chapters on metabolism, cross-references point out the discussions of individual enzymes in Chapters 12-16.

The following are topics that may be especially valuable to the student and which might be read initially: in Chapter 12, lysozyme (Section B,5), chymotrypsin (Section C,1), kinases (Section D,9), multiple displacement, reactions (Section G); in Chapter 13, imines (Section A,2), addition to C=C bonds (Section A, 4,5), beta cleavage and condensation (Section C); in Chapter 14, thiamin diphosphate (Section D), pyridoxal phosphate (Section E); in Chapter 15, NAD (Section A).

References

1. Iwata, S., Ostermeier, C., Ludwig, B., and Michel, H. (1995) *Nature* **376**, 660-669
2. Tsukihara, T., Aoyama, H., Yamashita, E., Tomizaki, T., Yamaguchi, H., Shinzawa-Itoh, K., Nakashima, R., Yaono, R., and Yoshikawa, S. (1995) *Science* **269**, 1069-1074
3. Mitchell, P. (1966) *Biol. Rev. Cambridge Philos. Soc.* **41**, 445-502
4. Mitchell, P. (1979) *Science* **206**, 1148-1159
5. Nicholls, D. G., and Ferguson, S. J. (1992) *Bioenergetics 2*, Academic Press, London
6. Gething, M.-J., and Sambrook, J. (1992) *Science* **355**, 33-45
7. Frydman, J., Nimmesgern, E., Ohtsuka, K., and Hartl, F. U. (1994) *Nature* **370**, 111-117
8. Buchner, J. (1994) *Trends Biochem. Sci.* **19**, 559
9. Hartl, F.-U., Hlodan, R., and Langer, T. (1994) *Trends Biochem. Sci.* **19**, 20-25
10. Flynn, G. C., Pohl, J., Flocco, M. T., and Rothman, J. E. (1991) *Nature* **353**, 726-730
- 10a. Russell, R., Jordan, R., and McMacken, R. (1998) *Biochemistry* **37**, 596-607
- 10b. Harrison, C. J., Hayer-Hartl, M., Di Liberto, M., Hartl, F.-U., and Kuriyan, J. (1997) *Science* **276**, 431-435
11. Wilbanks, S. M., and McKay, D. B. (1995) *J. Biol. Chem.* **270**, 2251-2257
- 11a. Kelley, W. L. (1998) *Trends Biochem. Sci.* **23**, 222-227
- 11b. Martinez-Yamout, M., Legge, G. B., Zhang, O., Wright, P. E., and Dyson, H. J. (2000) *J. Mol. Biol.* **300**, 805-818
- 11c. Westermann, B., and Neupert, W. (1997) *J. Mol. Biol.* **272**, 477-483
- 11d. Mogk, A., Tomoyasu, T., Goloubinoff, P., Rüdiger, S., Röder, D., Langen, H., and Bukau, B. (1999) *EMBO J.* **18**, 6934-6949
- 11e. Deuerling, E., Schulze-Specking, A., Tomoyasu, T., Mogk, A., and Bukau, B. (1999) *Nature (London)* **400**, 693-696
12. McCarty, J. S., Buchberger, A., Reinstein, J., and Bukau, B. (1995) *J. Mol. Biol.* **249**, 126-137
13. Flaherty, K. M., DeLuca-Flaherty, C., and McKay, D. B. (1990) *Nature* **346**, 623-628
14. Flaherty, K. M., Wilbanks, S. M., DeLuca-Flaherty, C., and McKay, D. B. (1994) *J. Biol. Chem.* **269**, 12899-12907
15. Phipps, B. M., Typke, D., Heger, R., Volker, S., Hoffmann, A., Stetter, K. O., and Baumeister, W. (1993) *Nature* **361**, 475-477
- 15a. Schoehn, G., Hayes, M., Cliff, M., Clarke, A. R., and Saibil, H. R. (2000) *J. Mol. Biol.* **301**, 323-332
16. Braig, D., Otwinowski, Z., Hegde, R., Boisvert, D. C., Joachimiak, A., Horwich, A. L., and Sigler, P. B. (1994) *Nature* **371**, 578-586
17. Hayer-Hartl, M. K., Martin, J., and Hartl, F. U. (1995) *Science* **269**, 836-841
18. Chen, S., Roseman, A. M., Hunter, A. S., Wood, S. P., Burston, S. G., Ranson, N. A., Clarke, A. R., and Saibil, H. R. (1994) *Nature* **371**, 261-264
- 18a. Betancourt, M. R., and Thirumalai, D. (1999) *J. Mol. Biol.* **287**, 627-644
- 18b. Wang, Z., Feng, H.-p., Landry, S. J., Maxwell, J., and Gierasch, L. M. (1999) *Biochemistry* **38**, 12537-12546
19. Shtilerman, M., Lorimer, G. H., and Englander, S. W. (1999) *Science* **284**, 822-825
20. Weaver, A. J., Sullivan, W. P., Felts, S. J., Owen, B. A. L., and Toft, D. O. (2000) *J. Biol. Chem.* **275**, 23045-23052
- 20a. Buchner, J. (1999) *Trends Biochem. Sci.* **24**, 136-141
- 20b. Sauer, F. G., Fütterer, K., Pinkner, J. S., Dodson, K. W., Hultgren, S. J., and Waksman, G. (1999) *Science* **285**, 1058-1061
- 20c. Bogdanov, M., and Dowhan, W. (1999) *J. Biol. Chem.* **274**, 36827-36830
- 20d. Orci, L., Vassalli, J.-D., and Perrelet, A. (1988) *Sci. Am.* **259**(Sep), 85-94
- 20e. Zhou, A., Webb, G., Zhu, X., and Steiner, D. F. (1999) *J. Biol. Chem.* **274**, 20745-20748
21. Smeekens, S. P., Montag, A. G., Thomas, G., Albiges-Rizo, C., Carroll, R., Benig, M., Phillips, L. A., Martin, S., Ohagi, S., Gardner, P., Swift, H. H., and Steiner, D. F. (1992) *Proc. Natl. Acad. Sci. U.S.A.* **89**, 8822-8826
22. Huang, X. F., and Arvan, P. (1995) *J. Biol. Chem.* **270**, 20417-20423
23. Blobel, G., and Sabatini, D. D. (1971) in *Biomembranes*, Vol. 2 (Manson, L. A., ed), pp. 193-195, Plenum, New York
24. Blobel, G., and Dobberstein, B. (1975) *J. Cell Biol.* **67**, 852

References

25. Landry, S. J., and Gierasch, L. M. (1991) *Trends Biochem. Sci.* **16**, 159–163
26. Gierasch, L. M. (1989) *Biochemistry* **28**, 923–930
27. Jain, R. G., Rusch, S. L., and Kendall, D. A. (1994) *J. Biol. Chem.* **269**, 16305–16310
28. Andersson, H., and von Heijne, G. (1993) *J. Biol. Chem.* **268**, 21389–21393
- 28a. von Heijne, G. (1998) *Nature (London)* **396**, 111–113
- 28b. Emanuelsson, O., Nielsen, H., Brunak, S., and von Heijne, G. (2000) *J. Mol. Biol.* **300**, 1005–1016
29. Verner, K., and Schatz, G. (1988) *Science* **241**, 1307–1313
30. Rapoport, T. A. (1992) *Science* **258**, 931–935
- 30a. Batey, R. T., Rambo, R. P., Lucast, L., Rha, B., and Doudna, J. A. (2000) *Science* **287**, 1232–1239
31. Sanders, S. L., and Schekman, R. (1992) *J. Biol. Chem.* **267**, 13791–13794
32. Hartmann, E., Sommer, T., Prehn, S., Görlich, D., Jentsch, S., and Rapoport, T. A. (1994) *Nature* **367**, 654–657
- 32a. Clemons, W. M., Jr., Gowda, K., Black, S. D., Zwieb, C., and Ramakrishnan, V. (1999) *J. Mol. Biol.* **292**, 697–705
33. Wickner, W. T. (1994) *Science* **266**, 1197–1198
- 33a. Moser, C., Mol, O., Goody, R. S., and Sinning, I. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 11339–11344
- 33b. Samuelson, J. C., Chen, M., Jiang, F., Möller, I., Wiedmann, M., Kuhn, A., Phillips, G. J., and Dalbey, R. E. (2000) *Nature (London)* **406**, 637–640
- 33c. Paetzel, M., Strynadka, N. C. J., Tschantz, W. R., Casareno, R., Bullinger, P. R., and Dalbey, R. E. (1997) *J. Biol. Chem.* **272**, 9994–10003
- 33d. Guo, W., Grant, A., and Novick, P. (1999) *J. Biol. Chem.* **274**, 23558–23564
34. Noël, P. J., and Cartwright, I. L. (1994) *EMBO J.* **13**, 5253–5261
35. Brodsky, J. L., Goeckeler, J., and Schekman, R. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 9643–9646
36. Dobberstein, B. (1994) *Nature* **367**, 599–600
37. Laidler, V., Chaddock, A. M., Knott, T. G., Walker, D., and Robinson, C. (1995) *J. Biol. Chem.* **270**, 17664–17667
38. Dalbey, R. E., and Robinson, C. (1999) *Trends Biochem. Sci.* **24**, 17–22
39. Douville, K., Price, A., Eichler, J., Economou, A., and Wickner, W. (1995) *J. Biol. Chem.* **270**, 20106–20111
- 39a. Eichler, J., Brunner, J., and Wickner, W. (1997) *EMBO J.* **16**, 2188–2196
- 39b. Matsumoto, G., Yoshihisa, T., and Ito, K. (1997) *EMBO J.* **16**, 6384–6393
- 39c. Scotti, P. A., Urbanus, M. L., Brunner, J., de Gier, J.-W. L., von Heijne, G., van der Does, C., Driessen, A. J. M., Oudega, B., and Lührink, J. (2000) *EMBO J.* **19**, 542–549
40. Meyer, T. H., Ménétret, J.-F., Breitling, R., Miller, K. R., Akey, C. W., and Rapoport, T. A. (1999) *J. Mol. Biol.* **285**, 1789–1800
- 40a. Yahr, T. L., and Wickner, W. T. (2000) *EMBO J.* **19**, 4393–4401
41. Randall, L. L., and Hardy, S. J. S. (1995) *Trends Biochem. Sci.* **20**, 65–69
42. Ulbrandt, N. D., London, E., and Oliver, D. B. (1992) *J. Biol. Chem.* **267**, 15184–15192
43. Bergeron, J. J. M., Brenner, M. B., Thomas, D. Y., and Williams, D. B. (1994) *Trends Biochem. Sci.* **19**, 124–128
44. Jungery, M., Pasvol, G., Newbold, C. I., and Weatherall, D. J. (1983) *Proc. Natl. Acad. Sci. U.S.A.* **80**, 1018–1022
45. Cao, G., Kuhn, A., and Dalbey, R. E. (1995) *EMBO J.* **14**, 866–875
46. Kawasaki, S., Mizushima, S., and Tokuda, H. (1993) *J. Biol. Chem.* **268**, 8193–8198
47. Wickner, W., Driessen, A. J. M., and Hartl, F.-U. (1991) *Ann. Rev. Biochem.* **60**, 101–124
48. Matsuyama, S.-i., Tajima, T., and Tokuda, H. (1995) *EMBO J.* **14**, 3365–3372
- 48a. Schäfer, U., Beck, K., and Müller, M. (1999) *J. Biol. Chem.* **274**, 24567–24574
- 48b. Russel, M. (1998) *J. Mol. Biol.* **279**, 485–499
- 48c. Galán, J. E., and Collmer, A. (1999) *Science* **284**, 1322–1328
- 48d. Tamano, K., Aizawa, S.-I., Katayama, E., Nonaka, T., Imajoh-Ohmi, S., Kuwae, A., Nagai, S., and Sasakawa, C. (2000) *EMBO J.* **19**, 3876–3887
49. Schatz, G. (1993) *Protein Sci.* **2**, 141–146
50. Pfeffer, S. R., and Rothman, J. E. (1987) *Ann. Rev. Biochem.* **56**, 829–852
51. Lodish, H. F. (1988) *J. Biol. Chem.* **263**, 2107–2110
52. Smeekens, S., Bauerle, C., Hageman, J., Keegstra, K., and Weisbeek, P. (1986) *Cell* **46**, 365–375
53. Roise, D., and Schatz, G. (1988) *J. Biol. Chem.* **263**, 4509–4511
54. Hurt, E. C., and van Loon, A. P. G. M. (1986) *Trends Biochem. Sci.* **11**, 204–207
55. Aitchison, J. D., Murray, W. W., and Rachubinski, R. A. (1991) *J. Biol. Chem.* **266**, 23197–23203
56. Wolins, N. E., and Donaldson, R. P. (1994) *J. Biol. Chem.* **269**, 1149–1153
57. Takeda, J., and Kinoshita, T. (1995) *Trends Biochem. Sci.* **20**, 367–371
- 57a. Mazmanian, S. K., Liu, G., Ton-That, H., and Schneewind, O. (1999) *Science* **285**, 760–762
58. Ogata, S., and Fukuda, M. (1994) *J. Biol. Chem.* **269**, 5210–5217
59. Pond, L., Kuhn, L. A., Teyton, L., Schutze, M.-P., Tainer, J. A., Jackson, M. R., and Peterson, P. A. (1995) *J. Biol. Chem.* **270**, 19989–19997
60. Pelham, H. R. B. (1990) *Trends Biochem. Sci.* **15**, 483–486
61. Wilson, D. W., Lewis, M. J., and Pelham, H. R. B. (1993) *J. Biol. Chem.* **268**, 7465–7468
62. Mallabiarrena, A., Jiménez, M. A., Rico, M., and Alarcón, B. (1995) *EMBO J.* **14**, 2257–2268
63. Luzio, J. P., and Banting, G. (1993) *Trends Biochem. Sci.* **18**, 395–398
64. Peter, F., Van, P. N., and Soling, H.-D. (1992) *J. Biol. Chem.* **267**, 10631–10637
65. Beh, C. T., and Rose, M. D. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 9820–9823
66. Hoe, M. H., Slusarewicz, P., Misteli, T., Watson, R., and Warren, G. (1995) *J. Biol. Chem.* **270**, 25057–25063
67. Schneider, H.-C., Berthold, J., Bauer, M. F., Dietmeier, K., Guiard, B., Brunner, M., and Neupert, W. (1994) *Nature* **371**, 768–773
68. Bhattacharyya, T., Karnezis, A. N., Murphy, S. P., Hoang, T., Freeman, B. C., Phillips, B., and Morimoto, R. I. (1995) *J. Biol. Chem.* **270**, 1705–1710
69. Schmitt, M., Neupert, W., and Langer, T. (1995) *EMBO J.* **14**, 3434–3444
70. Stuart, R. A., Cyr, D. M., Craig, E. A., and Neupert, W. (1994) *Trends Biochem. Sci.* **19**, 87–92
71. Lithgow, T., Glick, B. S., and Schatz, G. (1995) *Trends Biochem. Sci.* **20**, 98–101
72. Pfanner, N., Craig, E. A., and Meijer, M. (1994) *Trends Biochem. Sci.* **19**, 368–372
73. Mayer, A., Nargang, F. E., Neupert, W., and Lill, R. (1995) *EMBO J.* **14**, 4204–4211
- 73a. Heard, T. S., and Weiner, H. (1998) *J. Biol. Chem.* **273**, 29389–29393
74. Görlich, D., Vogel, F., Mills, A. D., Hartmann, E., and Laskey, R. A. (1995) *Nature* **377**, 246–248
- 74a. Koehler, C. M., Merchant, S., and Schatz, G. (1999) *Trends Biochem. Sci.* **24**, 428–432
75. Smeekens, S., Weisbeek, P., and Robinson, C. (1990) *Trends Biochem. Sci.* **15**, 73–76
76. Pilon, M., Wienk, H., Sips, W., de Swaaf, M., Talboom, I., van 't Hof, R., de Korte-Kool, G., Demel, R., Weisbeek, P., and de Kruijff, B. (1995) *J. Biol. Chem.* **270**, 3882–3893
77. Schnell, D. J., Kessler, F., and Blobel, G. (1994) *Science* **266**, 1007–1011
78. Voelker, R., and Barkan, A. (1995) *EMBO J.* **14**, 3905–3914
79. Viitanen, P. V., Schmidt, M., Buchner, J., Suzuki, T., Vierling, E., Dickson, R., Lorimer, G. H., Gatenby, A., and Soll, J. (1995) *J. Biol. Chem.* **270**, 18158–18164
- 79a. Nakai, K., and Horton, P. (1999) *Trends Biochem. Sci.* **24**, 34–35
- 79b. Klionsky, D. J. (1998) *J. Biol. Chem.* **273**, 10807–10810
80. Rothman, J. E. (1994) *Nature* **372**, 55–63
81. Pryer, N. K., Wuestehube, L. J., and Schekman, R. (1992) *Ann. Rev. Biochem.* **61**, 471–516
82. Bennett, M. K., and Scheller, R. H. (1993) *Proc. Natl. Acad. Sci. U.S.A.* **90**, 2559–2563
83. Duden, R., Hosobuchi, M., Hamamoto, S., Winey, M., Byers, B., and Schekman, R. (1994) *J. Biol. Chem.* **269**, 24486–24495
84. Edelmann, L., Hanson, P. I., Chapman, E. R., and Jahn, R. (1995) *EMBO J.* **14**, 224–231
- 84a. Dulubova, I., Sugita, S., Hill, S., Hosaka, M., Fernandez, L., Südhof, T. C., and Rizo, J. (1999) *EMBO J.* **18**, 4372–4382
- 84b. Katz, L., Hanson, P. I., Heuser, J. E., and Brennwald, P. (1998) *EMBO J.* **17**, 6200–6209
- 84c. Gerona, R. R. L., Larsen, E. C., Kowalchuk, J. A., and Martin, T. F. J. (2000) *J. Biol. Chem.* **275**, 6328–6336
85. Morgan, A., and Burgoyne, R. D. (1995) *EMBO J.* **14**, 232–239
- 85a. Glockshuber, R. (1999) *Nature (London)* **401**, 30–31
- 85b. Bolhuis, A., Venema, G., Quax, W. J., Bron, S., and van Dijk, J. M. (1999) *J. Biol. Chem.* **274**, 24531–24538
- 85c. Ban, N., Nissen, P., Hansen, J., Moore, P. B., and Steitz, T. A. (2000) *Science* **289**, 905–920
- 85d. Berardi, M. J., and Bushweller, J. H. (1999) *J. Mol. Biol.* **292**, 151–161
- 85e. van den Berg, B., Chung, E. W., Robinson, C. V., Mateo, P. L., and Dobson, C. M. (1999) *EMBO J.* **18**, 4794–4803
86. Freedman, R. B., Hirst, T. R., and Tuite, M. F. (1994) *Trends Biochem. Sci.* **19**, 331–335
87. Martin, J. L., Bardwell, J. C. A., and Kuriyan, J. (1993) *Nature* **365**, 465–468
88. Kishigami, S., Kanaya, E., Kikuchi, M., and Ito, K. (1995) *J. Biol. Chem.* **270**, 17072–17074
89. Jander, G., Martin, N. L., and Beckwith, J. (1994) *EMBO J.* **13**, 5121–5127
90. Kanaya, E., Anaguchi, H., and Kikuchi, M. (1994) *J. Biol. Chem.* **269**, 4273–4278
91. Frech, C., and Schmid, F. X. (1995) *J. Biol. Chem.* **270**, 5367–5374
92. Chivers, P. T., Laboisière, M. C. A., and Raines, R. T. (1996) *EMBO J.* **15**, 2659–2667
93. Tschantz, W. R., Paetzel, M., Cao, G., Suciu, D., Inouye, M., and Dalbey, R. E. (1995) *Biochemistry* **34**, 3935–3941
94. Steiner, D. F., Smeekens, S. P., Ohagi, S., and Chan, S. J. (1992) *J. Biol. Chem.* **267**, 23435–23438
95. De Bie, I., Savaria, D., Roebroek, A. J. M., Day, R., Lazure, C., Van de Ven, W. J. M., and Seidah, N. G. (1995) *J. Biol. Chem.* **270**, 1020–1028
96. Fisher, J. M., and Scheller, R. H. (1988) *J. Biol. Chem.* **263**, 16515–16518
97. Rehfeld, J. F., Hansen, C. P., and Johnsen, A. H. (1995) *EMBO J.* **14**, 389–396
98. Ypma-Wong, M. F., Filman, D. J., Hogle, J. M., and Semler, B. L. (1988) *J. Biol. Chem.* **263**, 17846–17856
99. Bateman, A., Solomon, S., and Bennett, H. P. J. (1990) *J. Biol. Chem.* **265**, 22130–22136
100. Richter, D. (1983) *Trends Biochem. Sci.* **8**, 278–280

References

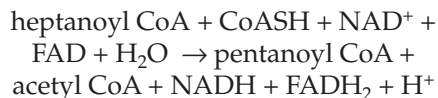
101. Douglass, J., Civelli, O., and Herbert, E. (1984) *Ann. Rev. Biochem.* **53**, 665–715
102. Arfin, S. M., and Bradshaw, R. A. (1988) *Biochemistry* **27**, 7979–7984
103. Tolan, D. R., Amsden, A. B., Putney, S. D., Urdea, M. S., and Penhoet, E. E. (1984) *J. Biol. Chem.* **259**, 1127–1131
104. Rubenstein, P. A., and Martin, D. J. (1983) *J. Biol. Chem.* **258**, 3961–3966
105. Sheff, D. R., and Rubenstein, P. A. (1992) *J. Biol. Chem.* **267**, 20217–20224
106. Kulkarni, M. S., and Sherman, F. (1994) *J. Biol. Chem.* **269**, 13141–13147
107. Gan, K., Gupta, S. D., Sankaran, K., Schmidt, M. B., and Wu, H. C. (1993) *J. Biol. Chem.* **268**, 16544–16550
108. Vogt, A., Sun, J., Qian, Y., Tan-Chiu, E., Hamilton, A. D., and Sebt, S. M. (1995) *Biochemistry* **34**, 12398–12403
109. Parish, C. A., and Rando, R. R. (1994) *Biochemistry* **33**, 9986–9991
110. Pompliano, D. L., Rands, E., Schaber, M. D., Mosser, S. D., Anthony, N. J., and Gibbs, J. B. (1992) *Biochemistry* **31**, 3800–3807
111. Busby, W. H., Jr., Quackenbush, G. E., Humm, J., Youngblood, W. W., and Kizer, J. S. (1987) *J. Biol. Chem.* **262**, 8532–8536
112. Fischer, W. H., and Spiess, J. (1987) *Proc. Natl. Acad. Sci. U.S.A.* **84**, 3628–3632
113. Katopodis, A. G., Ping, D., Smith, C. E., and May, S. W. (1991) *Biochemistry* **30**, 6189–6194
114. Merkler, D. J., Kulathila, R., Consalvo, A. P., Young, S. D., and Ash, D. E. (1992) *Biochemistry* **31**, 7282–7288
115. Ping, D., Katopodis, A. G., and May, S. W. (1992) *J. Am. Chem. Soc.* **114**, 3998–4000
116. Eipper, B. A., Milgram, S. L., Husten, E. J., Yun, H.-Y., and Mains, R. E. (1993) *Protein Sci.* **2**, 489–497
117. Bradbury, A. F., and Smyth, D. G. (1991) *Trends Biochem. Sci.* **16**, 112–115
118. Low, M. G., and Saltiel, A. R. (1988) *Science* **239**, 268–275
119. Lisanti, M. P., and Rodriguez-Boulan, E. (1990) *Trends Biochem. Sci.* **15**, 113–118
120. Graves, D. J., Martin, B. L., and Wang, J. H. (1994) *Co-and post-translational modification of proteins*, Oxford Univ. Press, New York
121. Tuboi, S., Taniguchi, N., and Katunuma, N., eds. (1992) *Post-Translation Modification of Proteins*, CRC Press, Boca Raton, Florida
122. Barrett, G. C., ed. (1985) *Chemistry and Biochemistry of the Amino Acids*, Chapman and Hall, London; New York
123. Niehrs, C., Kraft, M., Lee, R. W. H., and Huttner, W. B. (1990) *J. Biol. Chem.* **265**, 8525–8532
124. Rucker, R. B., and Wold, F. (1988) *FASEB J.* **2**, 2252–2261
125. Klotz, A. V., Leary, J. A., and Glazer, A. N. (1986) *J. Biol. Chem.* **261**, 15891–15894
126. Oliver, C. N., Ahn, B., Moerman, E. J., Goldstein, S., and Stadtman, E. R. (1987) *J. Biol. Chem.* **262**, 5488–5491
127. Cooper, A. A., and Stevens, T. H. (1995) *Trends Biochem. Sci.* **20**, 351–356
128. Rechsteiner, M., Rogers, S., and Rote, K. (1987) *Trends Biochem. Sci.* **12**, 390–394
129. Stadtman, E. R. (1986) *Trends Biochem. Sci.* **11**, 11–12
130. Rivett, A. J. (1985) *J. Biol. Chem.* **260**, 300–305
131. Loshon, C. A., Swerdlow, B. M., and Setlow, P. (1982) *J. Biol. Chem.* **257**, 10838–10845
132. Olden, K., Parent, J. B., and White, S. C. (1982) *Biochim. Biophys. Acta.* **650**, 209–232
133. Chiang, H.-L., and Dice, J. F. (1988) *J. Biol. Chem.* **263**, 6797–6805
- 133a. Dell'Angelica, E. C., Mullins, C., Caplan, S., and Bonifacio, J. S. (2000) *FASEB J.* **14**, 1265–1278
134. Hershko, A., and Ciechanover, A. (1992) *Ann. Rev. Biochem.* **61**, 761–807
135. Dice, J. F. (1987) *FASEB J.* **1**, 349–357
136. Stadtman, E. R. (1990) *Biochemistry* **29**, 6323–6331
137. Rogers, S., Wells, R., and Rechsteiner, M. (1986) *Science* **234**, 364–368
138. Bachmair, A., Finley, D., and Varshavsky, A. (1986) *Science* **234**, 179–186
139. Gonda, D. K., Bachmair, A., Wünnig, I., Tobias, J. W., Lane, W. S., and Varshavsky, A. (1989) *J. Biol. Chem.* **264**, 16700–16712
140. Madura, K., and Varshavsky, A. (1994) *Science* **265**, 1454–1458
- 140a. Davydov, I. V., and Varchavsky, A. (2000) *J. Biol. Chem.* **275**, 22931–22941
141. Herschlag, D. (1995) *J. Biol. Chem.* **270**, 20871–20874

Study Questions

1. Outline in detail, using structural formulas, the enzyme-catalyzed reactions by which cells in the human body convert glyceraldehyde 3-phosphate into pyruvate.
2. Describe, using chemical structural formulas, the reactions involved in the breakdown of glycogen to glucose 1-phosphate and the synthesis of glycogen from glucose 1-phosphate.
3. Describe the reaction steps in gluconeogenesis by which pyruvate is converted into glyceraldehyde 3-phosphate.
4. Compare the reactions of pyruvate that give rise to the following three compounds. List coenzymes or electron-carriers involved in each case and indicate any intermediate compounds.
 - a) Ethanol
 - b) Lactic acid
 - c) Acetyl-Coenzyme A
5. Mammalian sperm cells metabolize D-fructose preferentially as a source of energy. Fructose is formed in cells of the seminal vesicle from D-glucose via reduction to the sugar alcohol sorbitol using NADPH, followed by oxidation of sorbitol to fructose using NAD⁺. The fructose concentration in human semen is about 12 mM, whereas the glucose concentration within cells is usually less than 1 mM. If the ratio [NADPH] / [NADP⁺] is 10⁴ times higher than the ratio [NADH] / [NAD⁺], what is the minimum glucose concentration in cells that could allow formation of 12 mM fructose? The standard Gibbs energies of formation from the elements G^o_f in kJ / mol at 25°C are: D-glucose –917, D-fructose –915.
6. Outline the reactions by which glyceraldehyde 3-phosphate is converted to 3-phosphoglycerate with coupled synthesis of ATP in the glycolysis pathway. Show important mechanistic details.
7. Why can't acetyl CoA be converted to glucose in animals?
8. Describe the parallel reaction sequences between the citric acid cycle and the β oxidation pathway.
9. Contrary to legend, camels do not store water in their humps, which actually consist of a large fat deposit. How can these fat deposits serve as a source of water? Calculate the amount of water (liters) that can be produced by the camel from 500 g of fat. Assume for simplicity that the fat consists entirely of tripalmitin.

Study Questions

10. A little-known microorganism carries out the following net reaction in a series of enzymatic steps:

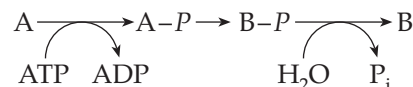


The net result is exactly the same as that of the β oxidation pathway, but for this microorganism, the pathway is demonstrably different in that pentanal, $\text{CH}_3(\text{CH}_2)_3\text{CHO}$, is an intermediate in the sequence. Which enzyme(s) of the β oxidation pathway does this microorganism lack? Propose an enzymatic reaction to account for the formation of this intermediate and a series for its conversion to the acyl CoA. ATP is not required.

11. Most natural fatty acids are even-numbered. What is the product of the *final* thiolase reaction to form an odd-chain fatty acid? Give a brief explanation.
12. It has been calculated that an average man takes in 1.5 kg of solid food and 1.4 kg of water per day. He gives off 3.5 kg of waste and 0.75 kg of sweat. It would thus appear that he should be losing over 1 kg per day through normal activities. How do you account for the fact that his weight remains relatively constant?
13. Suppose that fatty acids, instead of being broken down in two carbon fragments, were metabolized in three carbon units. What product(s) would Dr. Knoop have observed in the urine of his experimental dogs?
14. A renowned pharmacologist has announced the discovery of a new drug that specifically inhibits the fatty acid oxidation pathway within minutes of ingestion. The effects last for several hours only. The drug has no other effects on the subjects. The pharmacologist argues that this will increase athletic performance by shifting oxidative metabolism entirely to the more rapidly mobilized carbohydrate degradation pathway. Assume that the drug does work exactly as he suggests. Explain in a sentence or two how the drug would affect the performance of
a) a sprinter in the 100 meter dash
b) a long-distance runner in a marathon
15. Some bacteria use a "dicarboxylic acid cycle" to oxidize glyoxylate $\text{OHC}-\text{COO}^-$ to CO_2 . The regenerating substrate for this cycle is acetyl-CoA. It is synthesized from glyoxylate by a complex pathway that begins with conversion of two molecules of glyoxylate to tartronic semialdehyde:

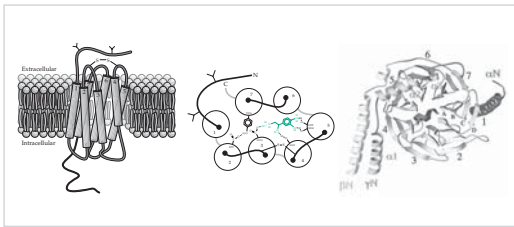
$^-\text{OOC}-\text{CHOH}-\text{CHO}$. The latter is then dehydrogenated to D-glycerate. Write out a detailed scheme for the dicarboxylate cycle. Also indicate how glucose and other cell constituents can be formed from intermediates created in this biosynthetic pathway.

16. Write a balanced fermentation sequence by which glycogen can be converted rapidly to *sn*-glycerol 3-phosphate and pyruvate in insect flight muscle. How many molecules of ATP per glucose unit of glycogen will be formed?
17. Show which parts (if any) of the citric acid cycle are utilized in each of the following reactions and what if any additional enzymes are needed in each case.
a) Oxidation of acetyl-CoA to CO_2
b) Catabolism of glutamate to CO_2
c) Biosynthesis of glutamate from pyruvate
d) Formation of propionate from pyruvate
18. a) If the Gibbs energy change $\Delta G'$ (pH 7) for the reaction $\text{A} \rightarrow \text{B}$ is $+25 \text{ kJ/mol}$ at 25°C , what would the ratio of $[\text{B}]/[\text{A}]$ be at equilibrium?
b) Suppose that the reaction were coupled to the cleavage of ATP as follows:



Suppose further that the group transfer potential ($-\Delta G'$) for the phospho group of A-P at 25°C , pH 7 is 12 kJ/mol and that the equilibrium constant for conversion of A-P to B-P is the same as that for conversion of A to B. Calculate the concentrations of A, B, A-P and B-P at equilibrium if the phosphorylation state ratio R_p is 10^4 M^{-1} .

19. This problem refers to the 13 meter climb described in Chapter 6, Study Question 14.
a) Assuming that muscle accounts for 30% of total body mass, estimate the amounts (mmol/kg, total mmol, and total grams) of each of the following in their coenzyme forms: nicotinamide, pantothenic acid, thiamin. You may be able to obtain rough estimates from the vitamin content of pork.
b) Calculate how many times, on the average, each molecule of nicotinamide would undergo reduction and re-oxidation (i.e., turn over) during the climb. Do the same for pantothenic acid (cycling between acyl-CoA and free CoA forms) and for thiamin through its catalytic cycles.



Much of the control of cellular metabolism is accomplished by hormones and other molecules that bind to receptor proteins embedded in the plasma membrane. Left: Many receptors are seven-helix transmembrane proteins, the best known being the light receptors of the rhodopsin family (Chapter 23). Center: A related adrenergic receptor, viewed here from the extracellular side, has a molecule of the hormone adrenaline (green) bound in the center (Fig. 11-6). Right: These receptors interact with GTP-hydrolyzing “G proteins,” which pass signals, often for short periods of time, to enzymes and other proteins. One structural domain of a G protein (Fig. 11-7C) shown here contains a β propeller domain which binds to the Ras-like domain of the α subunit. A few residues of the latter are visible in the center of this image.

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Tables

543 Table 11-1 A Few Covalent Modification Reactions Utilized to Control Metabolism

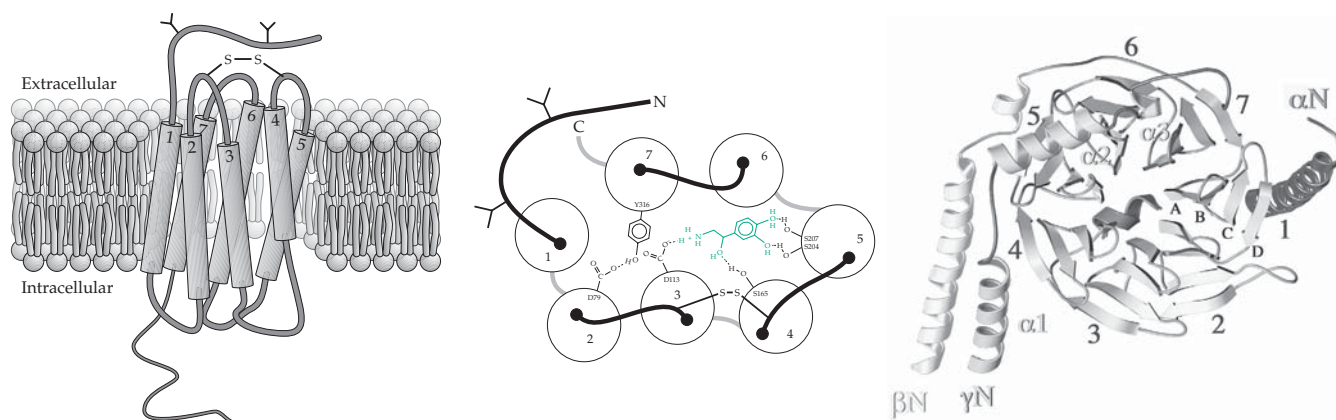
554 Table 11-2 Some Molecules (Second Messengers) That Carry Intracellular Signals

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The Regulation of Enzymatic Activity and Metabolism

11



Living cells must operate with controls that provide a stable environment and a relatively constant supply of materials needed for biosynthesis and for meeting the energy needs of cells. They must also be responsive to changes in their environment and must be able to undergo mitosis and reproduce when appropriate. The necessary control of metabolism and of growth is accomplished largely through mechanisms that regulate the locations, the amounts, and the catalytic activities of enzymes. The purpose of this chapter is to summarize these control mechanisms and to introduce terminology and shorthand notations that will be used throughout this book. Many of the **control elements** considered are summarized in Fig. 11-1.

A. Pacemakers and the Control of Metabolic Flux

Metabolic control can be understood to some extent by focusing attention on those enzymes that catalyze rate-limiting steps in a reaction sequence. Such **pacemaker enzymes**¹⁻⁴ are often involved in reactions that determine the overall respiration rate of a cell, reactions that initiate major metabolic sequences, or reactions that initiate branch pathways in metabolism. Often the first step in a unique biosynthetic pathway for a compound acts as the pacemaker reaction. Such a reaction may be described as the **committed step** of the pathway. It usually proceeds with a large decrease in Gibbs energy and tends to be tightly controlled. Both the rate of synthesis of the enzyme protein and the activity of the enzyme, once it is formed, may be inhibited by **feedback inhibition** which occurs when an end product of a biosynthetic pathway accumulates

and inhibits the enzyme. Enzyme activity may also be turned on or off by the effect of a hormone, by some other external stimulus, or by internal mechanisms that sense the metabolic state of the cell. Enzymes, other than the pacemaker, that catalyze reactions in a pathway may not be regulated and may operate at a steady state close to equilibrium.

If conditions within a cell change, a pacemaker reaction may cease to be rate limiting. A reactant plentiful in one circumstance may, in another, be depleted to the point that the rate of its formation from a preceding reaction determines the overall rate. Thus, as we have seen in Chapter 10, metabolism of glucose in our bodies occurs through the rapidly interconvertible phosphate esters glucose 6-phosphate and fructose 6-phosphate. The pacemaker enzyme in utilization of glucose or of glycogen is often **phosphofructokinase** (Fig. 11-2, step *b*) which catalyzes further metabolism of fructose 6-phosphate. However, if metabolism by this route is sufficiently rapid, the rate of formation of glucose 6-phosphate from glucose catalyzed by **hexokinase** (Fig. 11-2, step *a*) may become rate limiting.

Some catabolic reactions depend upon ADP, but under most conditions its concentration is very low because it is nearly all phosphorylated to ATP. Reactions utilizing ADP may then become the rate-limiting pacemakers in reaction sequences. Depletion of a reactant sometimes has the effect of changing the whole pattern of metabolism. Thus, if oxygen is unavailable to a yeast, the reduced coenzyme NADH accumulates and reduces pyruvate to ethanol plus CO₂ (Fig. 10-3). The result is a shift from oxidative metabolism to fermentation.

Pacemaker enzymes are often identified by the fact that the measured **mass action ratio**, e.g., for the reaction $A + B \rightarrow P + Q$, the ratio $[P][Q]/[A][B]$, is far

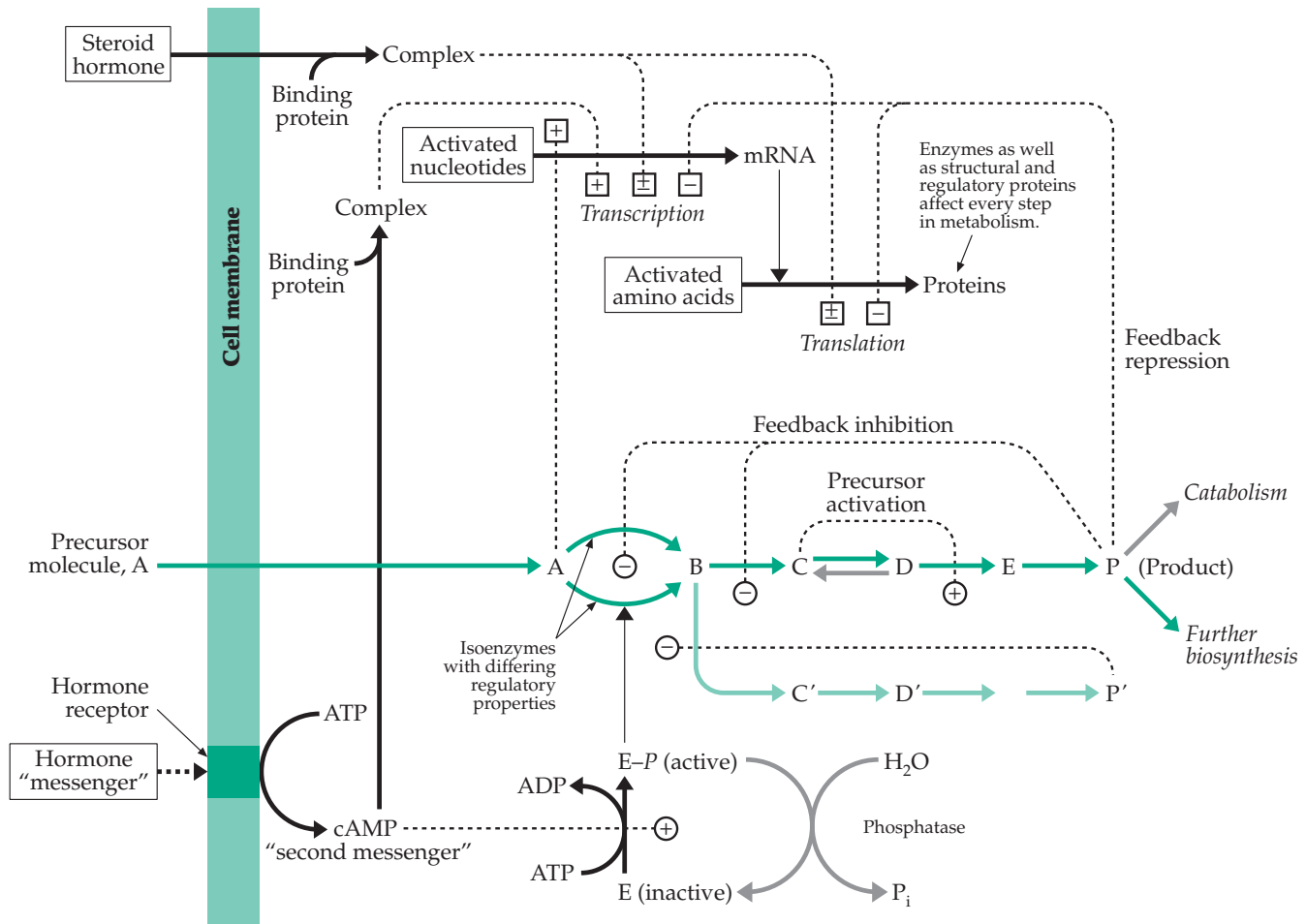


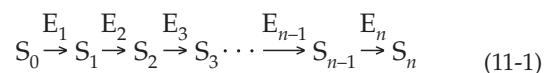
Figure 11-1 Some control elements for metabolic reactions. Throughout the book modulation of the activity of an enzyme by allosteric effectors or of transcription and translation of genes is indicated by dotted lines from the appropriate metabolite. The lines terminate in a minus sign for inhibition or repression and in a plus sign for activation or derepression. Circles indicate direct effects on enzymes, while boxes indicate repression or induction of enzyme synthesis.

from that predicted by the known equilibrium constant for the reaction. Another approach to identifying a pacemaker enzyme is to inhibit it and observe resulting changes in steady-state metabolite concentrations.⁵ If the pacemaker enzyme is inhibited, its substrate and other compounds preceding the step catalyzed by this enzyme will accumulate. At the same time the concentrations of products of the pacemaker reaction will drop as a result of their relatively more rapid removal by enzymes catalyzing subsequent steps. However, conclusions drawn by this approach may sometimes be erroneous.⁶

In spite of its usefulness, the pacemaker concept is oversimplified. It is often impossible to identify a specific pacemaker enzyme. When both catabolism and biosynthesis occur (e.g., as in the scheme in Fig. 11-2) it may be more useful to model the entire system with a computer than to try to identify pacemakers.^{7,8} It is also important to realize that reaction rates may be

determined by the rate of diffusion of a compound through a membrane. Thus, membrane transport processes can serve as pacemakers.

A general approach to analysis of complex metabolic pathways or to cell growth was introduced by Savageau,⁹⁻¹¹ and similar approaches have been followed by others.^{3,12-19} They emphasize the **flux** of material through a pathway under steady-state conditions and recognize that every enzyme in a sequence can have some effect on the overall rate. Consider a chain of enzymes E_1 to E_n acting on substrate S_0 , which is converted via substrates $S_1, S_2 \dots S_{n-1}$ into product S_n :



The flux $F = d[S_n]/dt$ is a constant. If the concentration of E_i were to change by an infinitesimal amount a corresponding change in F might be observed. The

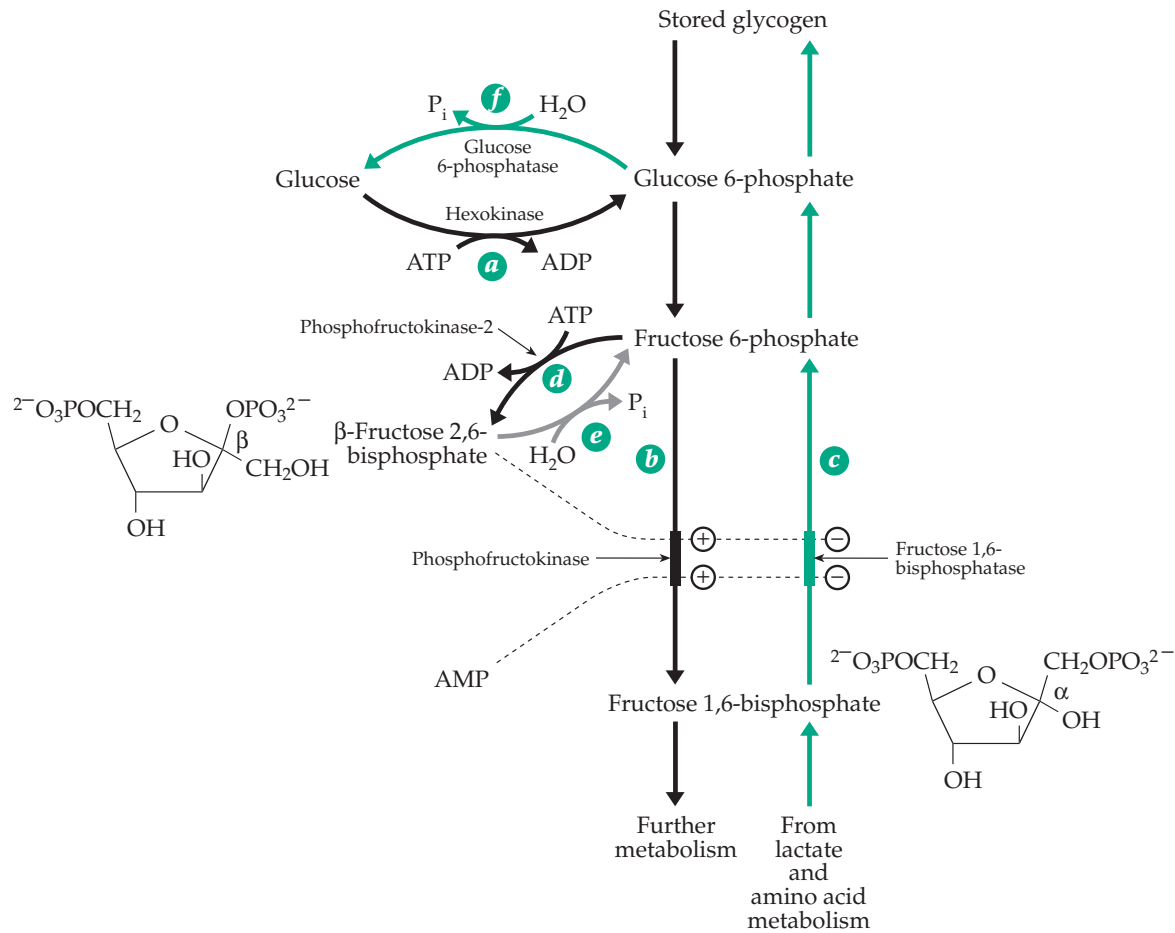


Figure 11-2 Roles of phosphofructose kinase and fructose 1,6-bisphosphatase in the control of the breakdown (→) and storage (←) of glycogen in muscle. The uptake of glucose from blood and its release from tissues is also illustrated. The allosteric effector fructose 2,6-bisphosphate (Fru-2,6- P_2) regulates both phosphofructokinase and fructose 2,6-bisphosphatase. These enzymes are also regulated by AMP if it accumulates. The activity of phosphofructokinase-2 (which synthesizes Fru-2,6- P_2) is controlled by a cyclic AMP-dependent kinase and by dephosphorylation by a phosphatase.

ratio of the change in flux to the change in $[E_i]$ is the sensitivity coefficient Z_i :

$$Z_i = \frac{dF/F}{d[E_i]/[E_i]} = \frac{d \ln F}{d \ln [E_i]} \quad (11-2)$$

The sensitivity coefficient can range from 1 for a pacemaker enzyme to almost zero for a very active enzyme that does not significantly limit the flux. The sum of Z_1 to Z_n is equal to 1:

$$\sum_{i=1}^n Z_i = 1 \quad (11-3)$$

Not only can the concentration $[E_i]$ change but also allosteric effectors can alter the activity. Kacser and Burns defined this in terms of a **controllability coefficient** κ ,

the ratio of the logarithmic change in velocity to the change in the concentration of an effector:

$$\kappa = d \ln v / d \ln [P] \quad (11-4)$$

It follows that the logarithmic change in flux $d \ln F$ can be related to the change in effector concentration $[P]$ in the following way:

$$d \ln F = Z_i \kappa d \ln [P] \quad (11-5)$$

From this equation we see that for the flux to be sensitively dependent upon $[P]$, both Z_i and κ must be reasonably large. This is exactly the case for many of the enzymes that have been identified as pacemakers. This approach can be applied to many aspects of metabolic control including cell growth.²⁰

B. Genetic Control of Enzyme Synthesis

All cellular regulatory mechanisms depend upon the synthesis of proteins, that is, upon the expression of genetic information. Within a given cell many genes are transcribed continuously but others may remain unexpressed. The rates of both transcription and degradation of mRNA are regulated, as are the rates of synthesis of enzymes on the ribosomes in the cytoplasm and the rates of protein turnover. Although a single copy of a gene in each member of a pair of chromosomes is often adequate, there are situations in which extra copies of part of the DNA of a cell are formed. Such **gene amplification**, which is dealt with further in Chapter 27, provides the possibility for very rapid synthesis of an enzyme or other protein. It often happens in highly specialized cells such as those of the silkworm's silk glands which must make enormous amounts of a small number of proteins.²¹

1. One Gene or Many?

Many enzymes exist within a cell as two or more **isoenzymes**, enzymes that catalyze the same chemical reaction and have similar substrate specificities. They are not isomers but are distinctly different proteins which are usually encoded by different genes.^{22,23} An example is provided by aspartate aminotransferase (Fig. 2-6) which occurs in eukaryotes as a pair of cytosolic and mitochondrial isoenzymes with different amino acid sequences and different isoelectric points. Although these isoenzymes share less than 50% sequence identity, their internal structures are nearly identical.²⁴⁻²⁷ The two isoenzymes, which also share structural homology with that of *E. coli*,²⁸ may have evolved separately in the cytosol and mitochondria, respectively, from an ancient common precursor. The differences between them are concentrated on the external surface and may be important to as yet unknown interactions with other protein molecules.

Isoenzymes are designated in various ways. They are often *numbered in the order of decreasing electrophoretic mobility* at pH 7 to 9: Most enzymes are negatively charged in this pH range and the one that migrates most rapidly toward the anode is numbered one. This is the same convention used in the electrophoresis of blood proteins (e.g. see Box 2-A).

Lactate dehydrogenase exists in the cytoplasm of humans and most animals as *five forms* which are easily separable by electrophoresis and are evenly spaced on electropherograms.⁸ This enzyme is a tetramer made of two kinds of subunits. Isoenzyme 1, which has the highest electrophoretic mobility, consists of four identical type B subunits. The slowest moving tetramer (isozyme 5) consists of four type A subunits, while the other three forms, AB₃, A₂B₂, and A₃B, contain

both subunits in different proportions. The two subunits are encoded by separate genes which are expressed to different extents in different tissues. Thus, heart muscle and liver produce mainly subunit B, while skeletal muscle produces principally subunit A. A third subunit type (C) is found only in the testes.²⁹

Why do cells produce isoenzymes? One reason may be that enzymes with differing kinetic parameters are needed.³⁰ Substrate concentrations may vary greatly between different tissues; between different subcellular compartments; and at different developmental stages of an organism. While the need for various isoenzymes of lactate dehydrogenase is not well understood,³¹ it is easier to understand the roles of the multiple forms of hexokinase, the enzyme that catalyzes the reaction of step *a* in Fig. 11-2. The brain enzyme has a high affinity for glucose ($K_m = 0.05$ mM). Thus, it is able to phosphorylate glucose and to make that substrate available to the brain for metabolism, even if the glucose concentration in the tissues falls to low values.^{31a} On the other hand, **glucokinase**, the hexokinase isozyme found in liver, has a much higher K_m of ~10 mM. It functions to remove excess glucose from blood, whose normal glucose content is ~5.5 mM. Glucokinase reaches its maximal activity only when the glucose concentration becomes much higher.³² This happens after a meal when the absorbed glucose passes through the portal circulation directly to the liver.

Another important source of variation in enzymes as well as in other proteins is **alternative splicing** of mRNA.³³ For example, transcription of the mouse α -amylase gene in the salivary gland starts at a different site (promoter) than does transcription in the liver. The two common isoforms of the insulin receptor (Fig. 11-11) arise because a 36-nucleotide (12-amino acid) exon is spliced out of the mRNA for the shorter protein. Isoenzymes of aldolase³⁴ and of many other proteins are formed in a similar manner. **Frame-shifting** during protein synthesis (Chapter 29) and also post-translational alterations may give rise to additional modified forms. They are often synthesized in relatively small amounts but may be essential to the life of the cell. In addition, genetic variants of almost any protein will be found in any population. These often differ in sequence by a single amino acid.

2. Repression, Induction, and Turnover

The synthesis of some enzymes is referred to as **constitutive**, implying that the enzyme is formed no matter what the environmental conditions of the cell. For example, many bacteria synthesize the enzymes required to catabolize glucose under all conditions of growth. Other enzymes, known as **inducible**, are often produced only in small amounts. However, if

cells are grown in specific inducing conditions for these enzymes, they are synthesized in larger quantities. For example, when *E. coli* is cultivated in the presence of lactose, several of the enzymes required for the catabolism of that disaccharide are formed. Synthesis of these enzymes is normally **repressed**. The genes which code for them are kept turned off through the action of protein **repressors** which bind to specific sites on the DNA and block transcription of the genes that they control (Fig. 11-1). Repressors have allosteric properties; in one conformation they bind tightly to DNA but in another they do not bind. For example, the free tryptophan repressor binds to DNA only weakly, but if a high concentration of tryptophan develops within the cell the tryptophan binds to an allosteric site on the repressor protein (Fig. 5-35). This changes the conformation to one that binds tightly to the appropriate control sequence in the DNA. In the case of lactose catabolism the free repressor binds to control sequences in the DNA until the **inducer**, allolactose (Chapters 4 and 28), binds at an allosteric site. This decreases the affinity of the repressor for DNA and the controlled genes are **derepressed**. There are also many protein **transcription factors** that have a positive effect, binding to DNA and promoting transcription of specific genes.

Synthesis of many enzymes is repressed most of the time. The appearance of an enzyme at a particular stage in the life of an organism as well as the differing distributions of isoenzymes within differentiated tissue result from derepression. The control of enzyme synthesis may also be exerted during the splicing of transcripts and at the translational level as well. These control mechanisms are often relatively slow, with response times of hours or even days. However, effects on the synthesis of some hormones, such as insulin (Section G), may be observed within a few minutes.

Genetic factors influence the rate of not only synthesis of proteins but also their breakdown, i.e., the rate of turnover. As we have seen in Chapter 10, some enzymes are synthesized as inactive proenzymes which are later modified to active forms, and active enzymes are destroyed, both by accident and via deliberate hydrolytic pathways. Protein **antienzymes** may not only inhibit enzymes but may promote their breakdown.³⁵ An example is the antienzyme that controls ornithine decarboxylase, a key enzyme in the synthesis of the polyamines that are essential to growth.^{36,37} As with all cell constituents, the synthesis of enzymes and other proteins is balanced by degradation.

3. Differences among Species

Catalytic mechanisms of enzymes have usually been conserved throughout evolution, and certain

residues in an enzyme may be invariant among many species. However, there are usually many differences in the distribution of amino acid residues on the surface of the proteins. Since changes in the surface shape of a protein molecule may alter the sensitivity to a potential allosteric effector, very different regulatory properties are sometimes found between species.

C. Regulation of the Activity of Enzymes

Some regulation of metabolism is provided by the kinetic properties of the enzymes. Thus, the value (mol/l) of the Michaelis constant K_m for an enzyme is usually low if the substrate normally occurs in very low concentration. It is likely to be higher if the enzyme acts on an abundant substrate. A tightly bound product of an enzymatic reaction will be released slowly from an enzyme if the product concentration within the cell is too high. However, while some regulation of metabolism is provided in such simple ways, the rapid changes that result from stimulation by hormones or by nerve impulses depend upon additional specific regulatory mechanisms that are discussed in the following sections.

1. Allosteric Control

Probably the most common and widespread control mechanisms in cells are **allosteric inhibition** and **allosteric activation**. These mechanisms are incorporated into metabolic pathways in many ways, the most frequent being **feedback inhibition**. This occurs when an end product of a metabolic sequence accumulates and turns off one or more enzymes needed for its own formation. It is often *the first enzyme unique to the specific biosynthetic pathway for the product that is inhibited*. When a cell makes two or more isoenzymes, only one of them may be inhibited by a particular product. For example, in Fig. 11-1 product P inhibits just one of the two isoenzymes that catalyzes conversion of A to B; the other is controlled by an enzyme modification reaction. In bacteria such as *E. coli*, three isoenzymes, which are labeled I, II, and III in Fig. 11-3, convert aspartate to β -aspartyl phosphate, the precursor to the end products threonine, isoleucine, methionine, and lysine. Each product inhibits only one of the isoenzymes as shown in the figure.

Feedback can also be positive. Since AMP is a product of the hydrolysis of ATP, its accumulation is a signal to *activate* key enzymes in metabolic pathways that generate ATP. For example, AMP causes allosteric activation of glycogen phosphorylase, which catalyzes the first step in the catabolism of glycogen. As is shown in Fig. 11-5, the allosteric site for AMP or IMP binding is more than 3 nm away from the active site. Only a

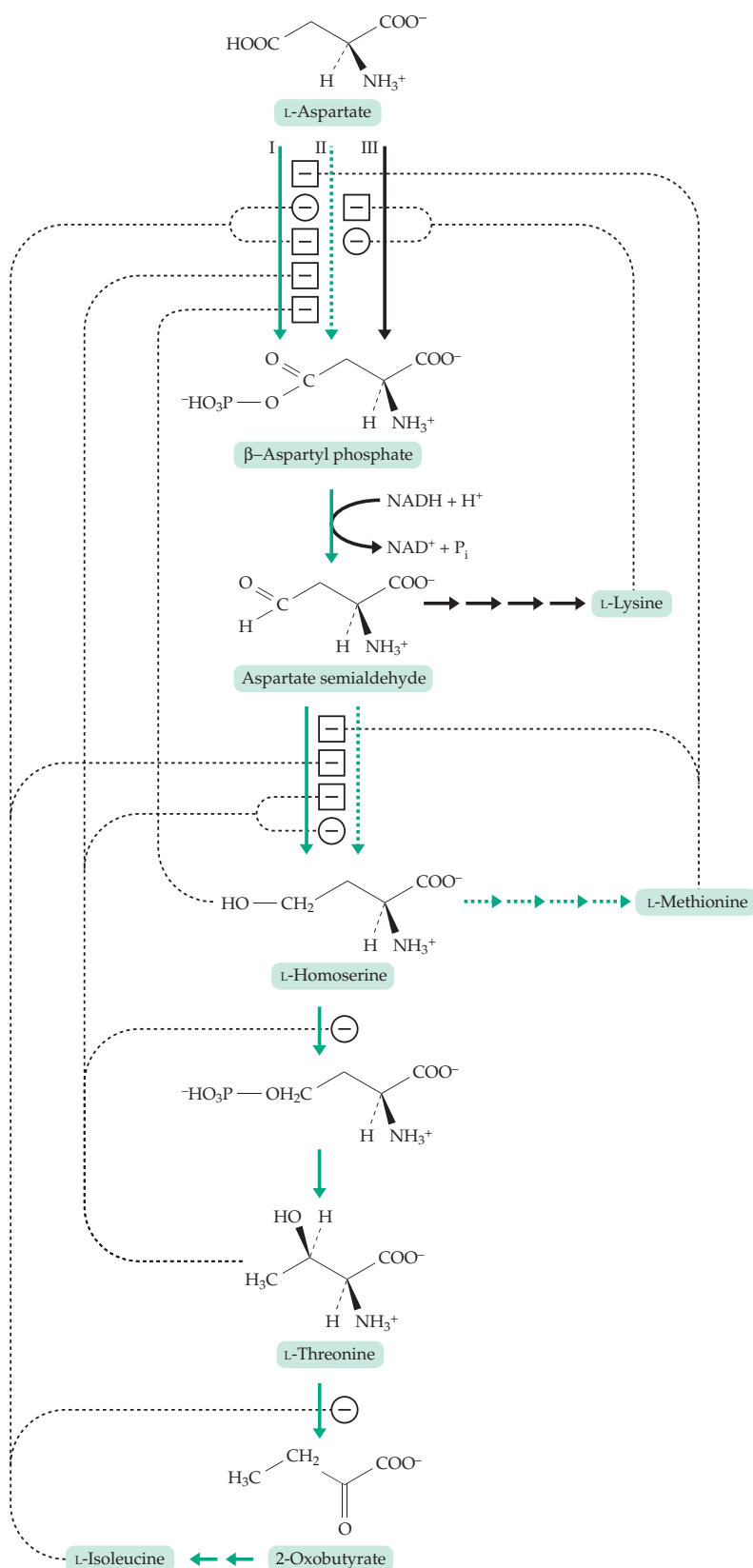


Figure 11-3 Feedback inhibition of enzymes involved in the biosynthesis of threonine, isoleucine, methionine, and lysine in *E. coli*. These amino acids all arise from L-aspartate, which is formed from oxaloacetate generated by the biosynthetic reactions of the citric acid cycle (Fig. 10-6). $\ominus$ Allosteric inhibition. $\boxminus$ Repression of transcription of the enzyme or of its synthesis on ribosomes.

subtle conformational change accompanies binding of IMP.³⁸ However, ³¹P NMR studies indicate that binding of the activator may induce a change in the state of protonation of the phosphate group of the coenzyme pyridoxal phosphate at the active site as explained in Chapter 12.

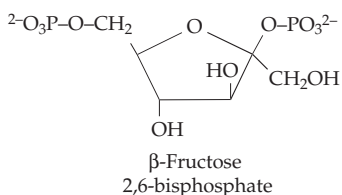
A second pattern of allosteric control may be referred to as **precursor activation** or feed-forward control. A metabolite acting as an allosteric effector turns on an enzyme that either acts directly on that metabolite or acts on a product that lies further ahead in the sequence. For example, in Fig. 11-1, metabolite C activates the enzyme that catalyzes an essentially irreversible reaction of compound D. An actual example is provided by glycogen synthase, whose inactive “dependent” or D form is activated allosterically by the glycogen precursor glucose 6-phosphate.³⁹ See also phosphorylase kinase (Section 2).

Regulatory subunits. Some enzymes consist not only of catalytic subunits, which contain the active sites, but also of **regulatory subunits**. The latter bind to the catalytic subunits and serve as allosteric modifiers. The binding of inhibitors or activators to specific sites on the regulatory subunits induces conformational changes in these subunits, altering their interaction with the catalytic subunits. A well-known example is aspartate carbamoyltransferase (ACTase) from *E. coli* (Fig. 7-20).^{40–42} Its regulatory subunits carry binding sites for cytidine triphosphate (CTP), which acts as a specific allosteric inhibitor of the enzyme. The significance of this fact is that ACTase catalyzes the first reaction specific to the pathway of synthesis of pyrimidine nucleotides (Chapter 25). CTP is an end product of that pathway and exerts feedback inhibition on the enzyme that initiates its synthesis. ATP also binds to the regulatory subunits and causes inhibition of the enzyme.

In the presence of CTP the binding of the substrates is cooperative, as would be anticipated if there is a two-state conformational equilibrium involving the catalytic subunits. This would be similar to the case depicted in Fig. 9-13 except that trimers rather than dimers are involved and the inhibitor is a part of the regulatory subunit and is controlled by binding of CTP.

Binding of ATP to ACTase decreases the cooperativity in substrate binding, again as predicted for a two-state model. However, as suggested by its structure, this enzyme system is more complex, as also indicated by the observed anticooperative binding of the activator CTP. X-ray studies show that binding of ligands causes a movement along subunit interfaces as well as localized conformational changes within the subunits. These are reminiscent of the changes seen upon binding of oxygen to hemoglobin (Fig. 7-25).

Glycolysis and gluconeogenesis. The highly regulated enzymes phosphofructokinase and fructose 1,6-bisphosphatase catalyze steps *b* and *c* of the reactions in Fig. 11-2, reactions that control glucose metabolism in cells. These enzymes have been studied for many years but the important allosteric effector **β -fructose 2,6-bisphosphate** was not discovered until 1980.⁴³⁻⁴⁶



This compound, which is formed from fructose 6-phosphate by a new enzyme, **phosphofructokinase-2** (also called fructose 6-phosphate 2-kinase) in step *d*, Fig. 11-2 activates phosphofructokinase allosterically. At the same time it inhibits fructose 1,6-bisphosphatase, an enzyme required for reversal of glycogen breakdown, that is, for the conversion of various metabolites arising from amino acids into glycogen (Fig. 11-2, step *c*).^{46a} These same two key regulated enzymes are also affected by many other metabolites. For example, ATP in excess inhibits phosphofructokinase, decreasing the overall rate of glucose metabolism and consequently of ATP production. Citrate, which is exported from mitochondria when carbohydrate metabolism is excessive, inhibits the same enzyme. On the other hand, AMP acts together with fructose 2,6-bisphosphate to activate the pathway for glycogen breakdown and to inhibit that for its synthesis (Fig. 11-2). The concentration of the regulator fructose 2,6-bisphosphate is controlled by mechanisms that are discussed in the following section.

2. Covalent Modification by Phosphorylation and Dephosphorylation

Rapid alteration of the activities of enzymes is often accomplished by **reversible covalent modification**.^{39,47} Many different modification reactions are known (Table 11-1) and there are doubtless many more to be discovered. Probably the most widespread and certainly the most studied is **phosphorylation**, the transfer of a phospho group from ATP or other suitable donor to a side chain group on the enzyme. An example is the phosphorylation by ATP of hydroxyl groups of specific serine residues in the two enzymes glycogen phosphorylase and glycogen synthase. These modifications are accomplished through a series or **cascade** of reactions initiated by the binding of hormones to cell surface receptors or by nerve impulses as is shown in Fig. 11-4. In the absence of such a stimulus, glycogen phosphorylase is present in its unphosphorylated or *b* form. Although this form can be activated allosterically by AMP, it is normally nearly inactive. When an appropriate hormone binds to the cell surface a cascade of reactions, as described in Section D, leads to activation of an enzyme called **phosphorylase kinase**.^{47a} This enzyme transfers a phospho group from ATP to the -OH group of the side chain of Ser 14 in each subunit (Fig. 11-4, left center), converting the enzyme into the active glycogen-degrading **phosphorylase a**. This switches the cellular metabolism from that designed to deposit the storage polysaccharide glycogen to one that degrades glycogen to provide the cell with energy. Serine 14 of glycogen phosphorylase is located adjacent to the allosteric AMP binding site (Fig. 11-5) and is surrounded by positively charged arginine and lysine side chains. Phosphorylation induces a rearrangement of hydrogen bonds involving these residues and in some way sends an appropriate signal to the active site. It also increases the affinity for AMP in the allosteric sites.^{48,48a} Phosphorylase kinase is allosterically activated by AMP, a product of its action – a feed-forward activator.

The control of glycogen phosphorylase by the phosphorylation–dephosphorylation cycle was discovered in 1955 by Edmond Fischer and Edwin Krebs⁵⁰ and was at first regarded as peculiar to glycogen breakdown. However, it is now abundantly clear that similar reactions control most aspects of metabolism.⁵¹ Phosphorylation of proteins is involved in control of carbohydrate, lipid, and amino acid metabolism; in control of muscular contraction, regulation of photosynthesis in plants,⁵² transcription of genes,⁵¹ protein syntheses,⁵³ and cell division; and in mediating most effects of hormones.

Protein kinases and cyclic AMP. Phosphorylase kinase is one of hundreds of different protein kinases which differ in specificity toward their substrates, in

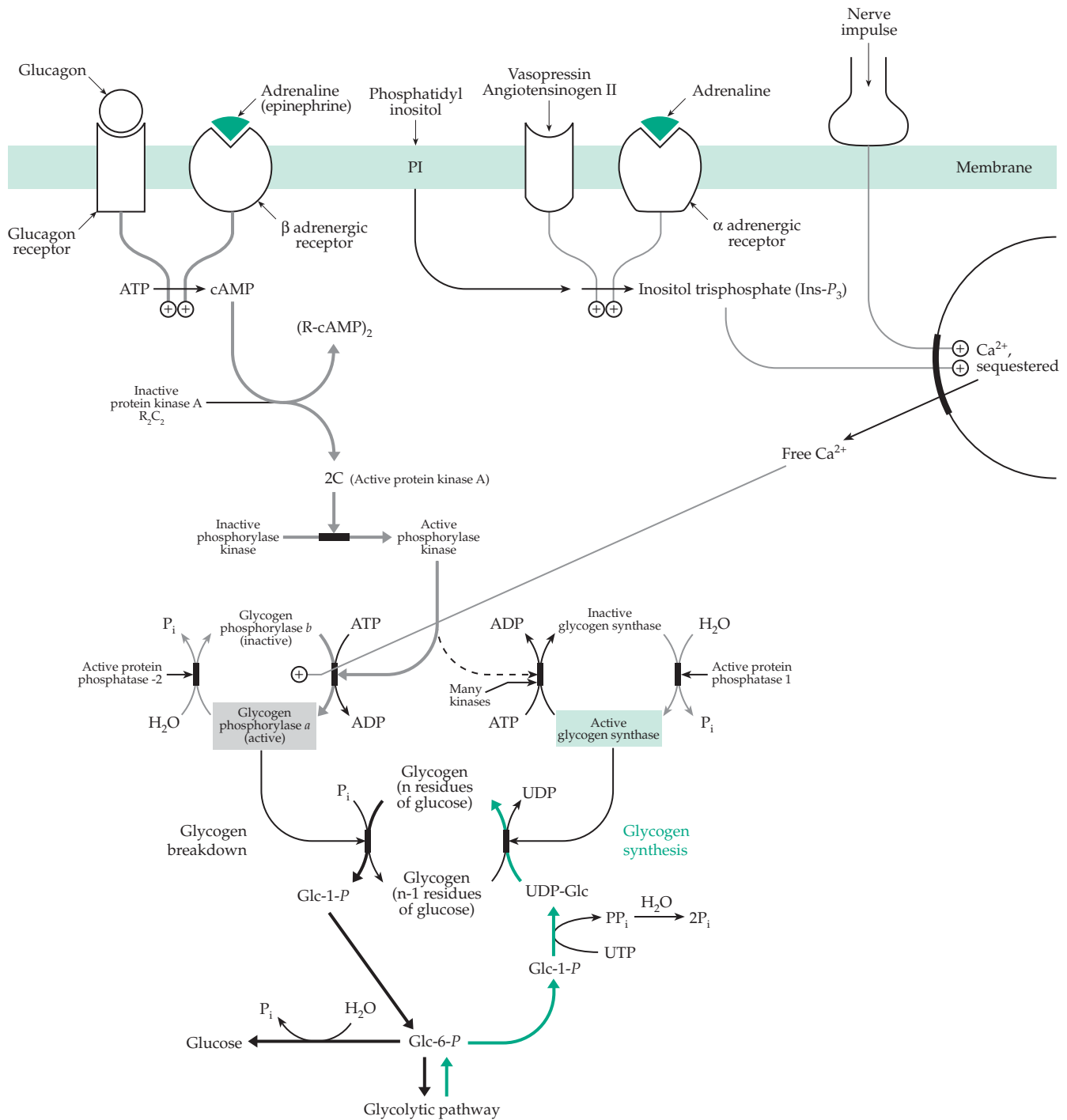


Figure 11-4 Cascades of phosphorylation and dephosphorylation reactions involved in the control of the metabolism of glycogen. Heavy arrows ($\longrightarrow$) show pathways by which glucosyl units of glycogen are converted into free glucose or enter the glycolytic pathway. Green arrows ($\longrightarrow$) trace the corresponding biosynthetic pathways. Gray arrows ($\longrightarrow$) trace the pathway of activation of glycogen phosphorylase by a hormone such as adrenaline (epinephrine) or glucagon and by the action of protein kinases. A few of the related pathways in the network of reactions that affect glycogen metabolism are also shown. This includes protein phosphatases, which remove phospho groups from proteins and allow cells to relax to the state that preceded activation. One of these (protein phosphatase 1) is activated by phosphorylation by an insulin-stimulated protein kinase.⁴⁹ However, the significance is uncertain; control of glycogen synthase is complex.^{49a}

TABLE 11-1
A Few Covalent Modification Reactions Utilized to Control Metabolism

Reaction	Example	Location of discussion
A. Phosphorylation–dephosphorylation		
Phosphorylation of Ser, Thr	Glycogen phosphorylase	This section
Phosphorylation of Tyr	Insulin receptor	Section G
Adenylylation, Uridylylation	Glutamine synthetase	Chapter 25
ADP-ribosylation		This section
B. Methylation of carboxyl groups	Bacterial glutamyl Aspartyl Protein phosphatase 2A Ras	Section D,5 Box 12-A Section C,3 Section D,3
C. Formation of carbamino groups	In hemoglobin In ribulose biphosphate carboxylase	Eq. 7-23 Chapters 13,23
D. Acylation		
Acetylation	Histones	Chapter 27
Palmitoylation	Ras	Section D,3
Prenylation	Ras	Chapter 22
E. Disulfide formation and cleavage		Chapters 10,15 Section C,4

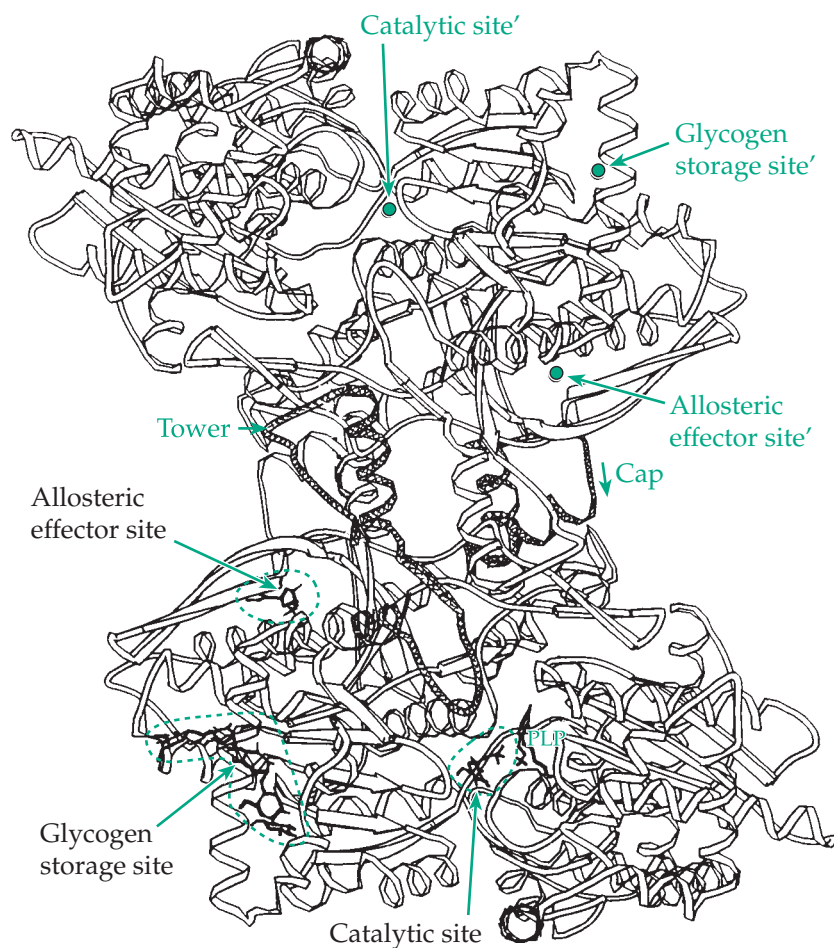
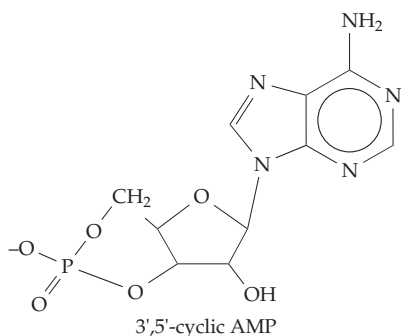


Figure 11-5 Schematic diagram of the glycogen phosphorylase dimer. The view is down the twofold axis of the dimer with the allosteric and Ser-*P* sites toward the viewer. Access to the catalytic site is from the far side of the molecule. The diagram shows the major change in conformation of the amino-terminal residues on phosphorylation. Residues 10–23 of glycogen phosphorylase *b* are shown as a thick solid line. These residues are not well ordered and make intrasubunit contacts. Upon phosphorylation, residues 10–23 change their conformation and are shown dark crosshatched with the position of Ser14-*P* indicated at the intersubunit interface. The fold of residues 24–80 through the α 1 helix, the cap, and the α 2 helix is shown lightly cross-hatched. The AMP allosteric effector site is located between the α 2 helix and the cap region of the other subunit. The glycogen storage site is located on the surface of the subunit and is associated with a long α helix. The catalytic site is at the center of the subunit where the two domains come together. Courtesy of Louise N. Johnson.⁴⁸

the functional groups phosphorylated, and in their allosteric activators.^{39,51,54–56} There may be more than 1000 different protein kinase genes in vertebrate animals, accounting for ~2% of the genome.⁵⁷ Many cytosolic protein kinases transfer a phospho group from ATP to either a serine or threonine side chain at a β bend or other surface feature of the substrate protein. Some sites of phosphorylation in the substrate proteins contain lysine or arginine residues, separated from the serine or threonine by only one residue but many other sequences may also surround phosphorylation sites.^{51,58} In the case of the 750-residue glycogen synthase, seven serine residues in different parts of the chain are phosphorylated by the action of at least five different kinases.^{49,49a,59,60} The various kinases phosphorylate groups at different sites and their effects are roughly additive.

Some of the best known protein kinases (designated PKA or cAPK) are those that depend upon **3', 5'-cyclic AMP (cAMP)** as an allosteric effector. They are oligomeric proteins of composition R_2C_2 , where R is a regulatory subunit and C is a catalytic subunit. Unless



cAMP is present, the regulatory subunits interact with the catalytic subunits, keeping them in an inactive inhibited form. However, when cAMP is present it binds to the regulatory subunit dimer, releasing the two active catalytic units (Eq. 11-6). This reaction is



reversible and as the concentration of cAMP is reduced by hydrolysis (see Eq. 11-8) the regulatory units recombine with the catalytic subunits and again inhibit them.

There are two prominent types of mammalian cAMP-dependent protein kinases.^{51,61} The catalytic subunit is identical for both; the 41-kDa peptide as isolated from beef heart has 350 residues and an N terminus blocked by a myristoyl (tetradecanoyl) group.⁶² One phosphoserine and one phosphothreonine are also present.⁵¹ The ~50-kDa regulatory subunits vary in size and may also be subject to additional regulation by phosphorylation.⁶³ Three-dimensional structures are known for both the catalytic^{62,64,65} and the regulatory⁶⁶ subunits. A **cyclic GMP (cGMP)**-activated protein

kinase is present in some mammalian tissues^{67,68} and is widespread in invertebrates.^{51,67} The cyclic nucleotide-binding domains of these kinases have structures similar to that of the *E. coli* catabolite activator protein (Fig. 28-6)^{51,68} and the catalytic domains are structurally similar to those of many other kinases.

Among the kinases that phosphorylate glycogen synthase is a **casein kinase**, named for the fact that the milk protein casein is also a good substrate. A family of casein kinases are found in the cytoplasm and nuclei of all eukaryotic cells. They phosphorylate serine and threonine side chains but have structures distinct from those of the cAMP-dependent kinases.^{69–72} Casein kinase-2 (CK2) phosphorylates many proteins, including several that function in gene replication, transcription, and cell growth and division.^{72a} **Theileriosis**, a parasitic disease of cattle in Africa, is caused by the tick-borne protist *Theileria parva*. The condition is often fatal as a result of a leukemia-like condition resulting from overexpression of the casein kinase-2 gene.⁷²

Phosphorylase kinase is one of a large group of *specialized* protein kinases, each of which acts on a small number of proteins. It is regulated both by covalent modification and allosterically by **calcium ions**.^{73–78} It contains four different kinds of subunits ranging in size from 17 kDa to about 145 kDa and has the composition $(\alpha\beta\gamma\delta)_4$. Phosphorylation of one serine on each of the 145-kDa α and 120-kDa β regulatory subunits is catalyzed by a cAMP-dependent protein kinase. The δ subunit is the Ca^{2+} -binding protein **calmodulin** (Fig. 6-8) and serves as a regulatory subunit sensitive to Ca^{2+} . The 45-kDa γ subunit contains the catalytic domain as well as a calmodulin-binding domain.⁷⁹ Other Ca^{2+} -dependent protein kinases include the **protein kinase C** family, which is discussed further in Section E, Ca^{2+} /calmodulin-dependent protein kinases,^{80,81} and a plant kinase with a regulatory domain similar to calmodulin.⁸²

Protein tyrosine kinases (PTKs) place phospho groups on the phenolic oxygen atoms of tyrosyl residues of some proteins.^{83–85} The resulting phosphotyrosine accounts for only about 1/3000 of the phospho groups in proteins but has aroused interest for two reasons. First, binding of growth hormones, such as **epidermal growth factor (EGF)**, **platelet-derived growth factor (PDGF)**, and **insulin**, to their receptors stimulates tyrosine-specific protein kinase activity of the receptor proteins (see Figs. 11-11 to 11-13).^{69,83,86} Second, a number of cancer-causing **oncogenes** encode similar kinases (Section H). Tyrosine protein kinases are essential components of the cell division cycle (see Fig. 11-15).

Protein phosphatases. Most regulatory alterations in enzymatic activity are spontaneously reversible. The concentration of the allosteric effector soon drops and the covalent modifications are reversed so that the system relaxes to a state approximating the

original one. Among the enzymes required for this reversal are the protein phosphatases that remove the phospho groups placed on amino acid side chains by kinases.^{54,86–91} For example, four different phosphatases act on the (inactive) phosphorylated glycogen synthase and on the phosphorylated forms of glycogen phosphorylase and phosphorylase kinase. At least one of these (a protein phosphatase 1; see Fig. 11-4) is regulated by phosphorylation that is stimulated by insulin.^{49,49a} The matter is made yet more complex by the presence in tissues of small protein inhibitors which can prevent the action of these phosphatases.^{88,92–94} Another naturally occurring inhibitor is the polyether fatty acid **okadaic acid**, a “shellfish poison,” actually produced by dinoflagellates but which accumulates in sponges and in mussels and other bivalves.⁹⁴ See also Chapter 12, D,4.

Most protein phosphatases are specific toward phosphoserine and phosphothreonine^{89,95} or toward phosphotyrosine residues.^{90,96–98} Some have a dual specificity.⁹⁹ There are Ca^{2+} -activated phosphatases¹⁰⁰ and phosphatases with transmembrane segments connected to receptors on the external cell surface. Some phosphorylation–dephosphorylation regulatory systems utilize bifunctional enzymes consisting of a protein kinase fused to a protein phosphatase. For example, the same *E. coli* enzyme that phosphorylates isocitrate dehydrogenase also hydrolyzes off the phospho group that it has put on.¹⁰¹ Another bifunctional kinase–phosphatase catalyzes both the formation (reaction *d* of Fig. 11-2) and breakdown (reaction *e*) of fructose 2,6-bisphosphate in eukaryotic tissues. This kinase–phosphatase enzyme is, in turn, controlled by phosphorylation by a cAMP-dependent protein kinase. In the liver, this latter phosphorylation strongly inhibits the kinase activity preventing buildup of fructose 2,6-bisphosphate, the allosteric activator of phosphofructokinase-1, which catalyzes reaction *b* of Fig. 11-2. Consequently, when a pulse of cAMP is generated within a hepatocyte, glycolysis is blocked but **glycogenolysis**, the breakdown of glycogen by phosphorylase, is stimulated. The glucose 6-phosphate that is formed is hydrolyzed by glucose-6-phosphatase releasing glucose into the bloodstream. On the other hand, in heart and probably in most other tissues, a different isoenzyme form of the bifunctional enzyme is present. Phosphorylation by cAMP activates the kinase and inhibits the phosphatase. Consequently, in these tissues cAMP induces the activation of glycogenolysis and glycolysis coordinately.^{102–105} In brain a different allosteric activator, **ribose 1, 5-bisphosphate**, may regulate phosphofructokinase-1.¹⁰⁶

Phosphorylation in bacteria. A bacterial enzyme whose activity is controlled by phosphorylation is **isocitrate dehydrogenase**. Transfer of a phospho group to the –OH of Ser 113 completely inactivates the

E. coli enzyme,^{107,108} causing isocitrate to build up in the citric acid cycle (Fig. 10-6) and to be diverted into the glyoxylate pathway, which is depicted in Fig. 17-16. In this instance, it appears likely that the negative charge of the added phospho group causes electrostatic repulsion of the substrate isocitrate. In agreement with this concept, mutation of Ser 113 of this enzyme to Asp (mutant S113 D) also inactivates the enzyme.¹⁰⁹

In addition to kinase–phosphatase cycles, bacteria use at least two other ATP-dependent regulatory mechanisms.¹¹⁰ In the **sensor kinase/response regulator** (or “two-component”) systems^{110–111} a sensor protein, upon being allosterically activated, phosphorylates itself (autophosphorylation) on a specific histidyl residue to form an *N*-phosphohistidine derivative. The phosphohistidine then transfers its phospho group onto a specific aspartyl group in the response regulator causing the regulator to bind to its target protein and to exert its regulatory effect. The best known example involves the control of the motion of bacterial flagella which is discussed briefly in Section D,5 and further in Chapter 19.

The third bacterial regulatory device is the phosphoenolpyruvate:sugar phosphotransferase system (Eq. 8-4). It is involved not only in transport but also in controlling a variety of physiological processes.^{110,112,113}

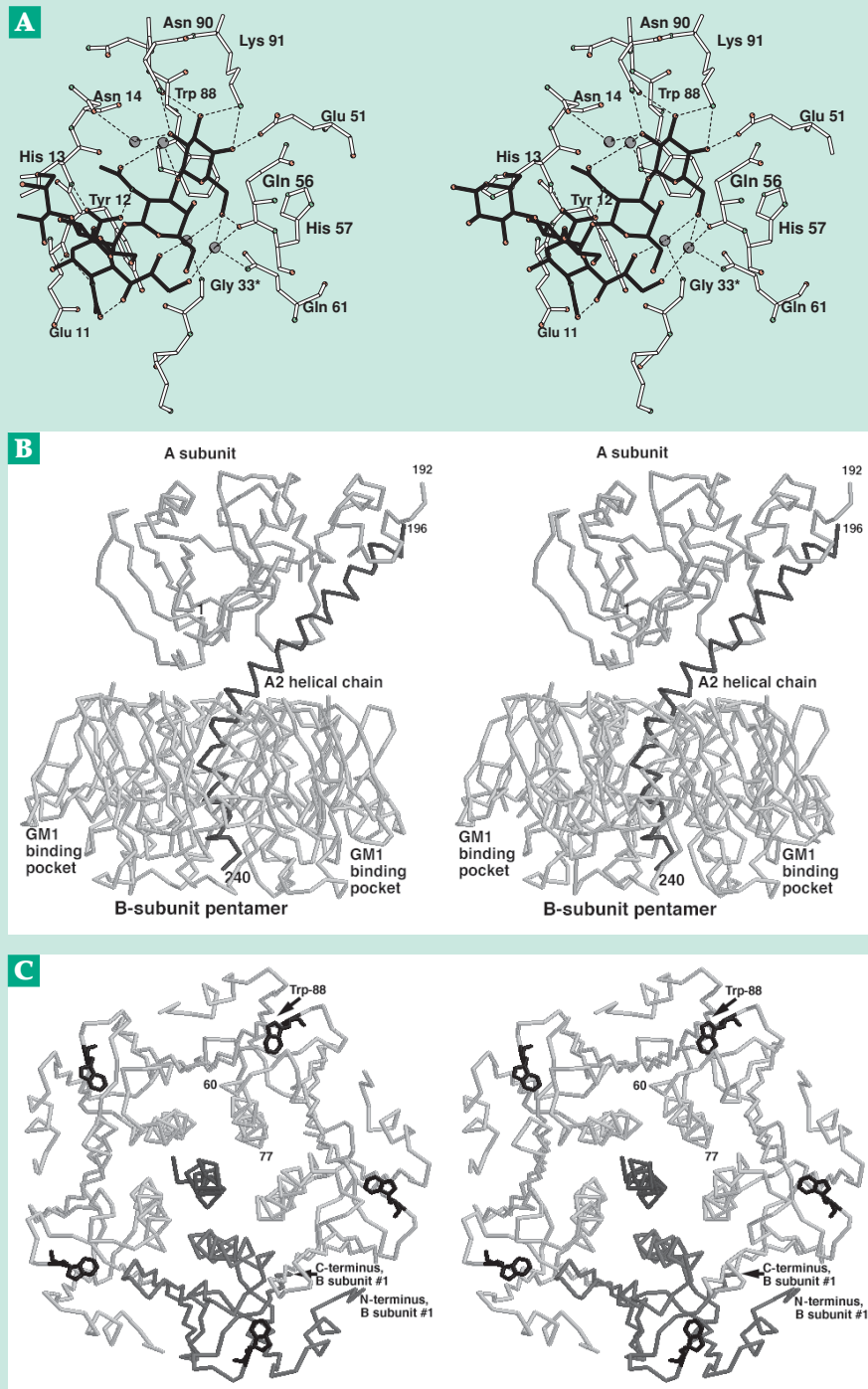
3. Other Modification Reactions Involving Group Transfer

Nucleotidylation, the transfer of an entire nucleotidyl unit, rather than just a phospho group, to a protein is sometimes utilized for regulation. For example, glutamine synthetase of *E. coli* is modified by **adenylylation**, transfer of an adenylyl group from ATP to a tyrosine side chain.^{114–116} Relaxation to the unmodified enzyme is catalyzed by a deadenylylating enzyme. A regulatory protein that undergoes reversible uridylylation¹¹⁷ also functions in this system (see Fig. 24-7). Several mitochondrial and cytoplasmic proteins are modified by attachment of ADP-ribosyl groups to specific guanidino groups of arginine and other side chain groups.^{114,118–123} This **ADP-ribosylation** requires the coenzyme NAD^+ as the ADP-ribosyl donor and is also catalyzed by bacterial toxins such as cholera toxin (Box 11-A) and by diphtheria toxin (Box 29-A). A second kind of ADP-ribosylation occurs within nuclei where the enzyme **poly(ADP-ribose) synthase** catalyzes both an initial ADP-ribosylation of an amino acid side chain group and also addition of more ADP-ribosyl units to form a polymer¹²⁴ (see Chapter 27).

Like inorganic phosphate, inorganic sulfate can be converted into an activated form in which the sulfate resembles the terminal phospho group of ATP (Eq. 17-38). The resulting **activated sulfo group** can be transferred to other compounds including enzymes. Formation of

BOX 11-A (continued)

(A) Stereoscopic view of the G_{M1} pentasaccharide binding site, showing both direct and solvent-mediated hydrogen bonding interactions between sugar and protein residues. The viewpoint is from “underneath” the membrane binding surface of the B pentamer. Starred residues are from an adjacent monomer. The terminal galactose residue of the pentasaccharide (upper right) is most deeply inserted into the binding site and is involved in the greatest number of identifiable binding interactions. The sialic acid residue, near the bottom of the figure, is also involved in several hydrogen-bonding interactions. Hydrophobic interactions include the approach of the sialic acetyl group to the edge of the Tyr 12 phenyl ring and the positioning of the terminal galactose sugar ring parallel to the indole ring of Trp 88. Figures generated using program MolScript (Kraulis, 1991). From Merritt *et al.*^c (B) Side view of the intact AB_5 toxin as an α -carbon trace. The nicked A chain is at the top. The nick can be seen in the upper right corner, as can a disulfide bridge connecting the A_1 and A_2 segments. A single α helix extends into the “pore” in the center of the B pentamer. The side chains of Trp 88 of the B subunits have been added to mark the ganglioside binding sites. These side chains are also seen in (A). (C) View of the AB_5 molecule from the bottom showing the helix from the A_2 fragment surrounded by a tight cage of five long helices from the B subunits. Residues 237–240 of the A_2 fragment have the KDEL sequence and may extend from the pore to contact the membrane. (B) and (C) courtesy of Edwin M. Westbrook. See Zhang *et al.*^c



generating abnormally high concentrations of cAMP. It is the effect of the cAMP on proteins of the intestinal mucosal cell membranes that causes the disastrous excessive secretion of water and salts that are characteristic of cholera.

Most strains of *V. cholerae* are relatively harmless but they may suddenly be transformed into a virulent toxin-producing form by infection with a

bacterial virus similar to M13 (Chapter 5), which carries the toxin gene. Entrance into the *Vibrio* cell occurs with the help of pili which are present in many strains.^{i,j}

Many other bacterial toxins have AB_n structures similar to that of cholera toxin. For example, toxin-producing strains of *E. coli* are also important causes of diarrhea in humans and in domestic animals.^{k-m}

BOX 11-A CHOLERA TOXIN AND OTHER DANGEROUS PROTEINS (continued)

The heat-labile *E. coli* enterotoxin, whose gene is carried on a plasmid, is a close relative of cholera toxin^{n,o} and also catalyzes ADP ribosylation of arginine 201 of the G_{sa} subunit.^m *Bordetella pertussis*, which causes whooping cough, forms a similar toxin that attacks the inhibitory regulatory protein G_i^{p,q} as well as transducin and inactivates them by ADP ribosylation. Diphtheria toxin (Box 29-A), the exotoxin from *Pseudomonas aeruginosa*, and the toxin from *Clostridium botulinum* also catalyze ADP-ribosylation reactions.^{k,r}

Some bacteria produce effects similar to those of cholera toxin in different ways. For example, among a variety of toxic proteins produced by *Bacillus anthracis*, the causative agent of anthrax, is an adenylate cyclase that is able to enter the host's cells.^s Similarly, *B. pertussis*, in addition to

its ADP-ribosylating toxin, produces a calmodulin-stimulated adenylate cyclase which, when taken up by phagocytic cells, disrupts their function in the body's defense system.^{t-w} In addition to the heat-labile enterotoxin, some strains of *E. coli* produce heat-stable toxins, small 18-residue peptides related in structure to the intestinal peptide **guanylin**.^{l,x} These peptides bind to a membrane-bound guanylate cyclase activating fluid secretion into the intestine.

Why do our cells obligingly provide both initial receptors and means of uptake for these dangerous toxic proteins? Some ADP-ribosylation reactions are a natural part of cell function^y and some hormones, for example thyrotropin, seem to stimulate their activity. It may be that the toxins use mechanisms designed to respond to normal hormonal stimulation.

^a Holmgren, J. (1981) *Nature (London)* **292**, 413–417

^b Hirschhorn, N., and Greenough, W. B., III (1991) *Sci. Am.* **264**(May), 50–56

^c Zhang, R.-G., Scott, D. L., Westbrook, M. L., Nance, S., Spangler, B. D., Shipley, G. G., and Westbrook, E. M. (1995) *J. Mol. Biol.* **251**, 563–573

^d Zhang, R.-G., Westbrook, M. L., Westbrook, E. M., Scott, D. L., Otwinowski, Z., Maulik, P. R., Reed, R. A., and Shipley, G. G. (1995) *J. Mol. Biol.* **251**, 550–562

^e Merritt, E. A., Sarfaty, S., van den Akker, F., L'Hoir, C., Martial, J. A., and Hol, W. G. J. (1994) *Protein Sci.* **3**, 166–175

^f Moss, J., Stanley, S. J., Morin, J. E., and Dixon, J. E. (1980) *J. Biol. Chem.* **255**, 11085–11087

^g Janicot, M., Fouque, F., and Desbuquois, B. (1991) *J. Biol. Chem.* **266**, 12858–12865

^h Tsai, S.-C., Adamik, R., Moss, J., and Vaughan, M. (1991) *Biochemistry* **30**, 3697–3703

ⁱ Williams, N. (1996) *Science* **272**, 1869–1870

^j Waldor, M. K., and Mekalanos, J. J. (1996) *Science* **272**, 1910–1914

^k Moss, J., and Vaughan, M., eds. (1990) *ADP-Ribosylating Toxins and G Proteins: Insights into Signal Transduction*, American Society for Microbiology, Washington, DC

^l Sato, T., Ozaki, H., Hata, Y., Kitagawa, Y., Katsube, Y., and Shimonishi, Y. (1994) *Biochemistry* **33**, 8641–8650

^m van den Akker, F., Merritt, E. A., Pizza, M. G., Domenighini,

M., Rappuoli, R., and Hol, W. G. J. (1995) *Biochemistry* **34**, 10996–11004

ⁿ Sixma, T. K., Pronk, S. E., Kalk, K. H., van Zanten, B. A. M., Berghuis, A. M., and Hol, W. G. J. (1992) *Nature (London)* **355**, 561–564

^o Sixma, T. K., Kalk, K. H., van Zanten, B. A. M., Dauter, Z., Kingma, J., Witholt, B., and Hol, W. G. J. (1993) *J. Mol. Biol.* **230**, 890–918

^p Antoine, R., and Loch, C. (1994) *J. Biol. Chem.* **269**, 6450–6457

^q Goodemote, K. A., Mattie, M. E., Berger, A., and Spiegel, S. (1995) *J. Biol. Chem.* **270**, 10270–10277

^r Ohtsuka, T., Nagata, K.-i, Iiri, T., Nozawa, Y., Ueno, K., Ui, M., and Katada, T. (1989) *J. Biol. Chem.* **264**, 15000–15005

^s Arora, N., Klimpel, K. R., Singh, Y., and Leppla, S. H. (1992) *J. Biol. Chem.* **267**, 15542–15548

^t Benz, R., Maier, E., Ladant, D., Ullmann, A., and Sebo, P. (1994) *J. Biol. Chem.* **269**, 27231–27239

^u Heveker, N., Bonnaffé, D., and Ullmann, A. (1994) *J. Biol. Chem.* **269**, 32844–32847

^v Otero, A. S., Yi, X. B., Gray, M. C., Szabo, G., and Hewlett, E. L. (1995) *J. Biol. Chem.* **270**, 9695–9697

^w Hackett, M., Guo, L., Shabanowitz, J., Hunt, D. F., and Hewlett, E. L. (1994) *Science* **266**, 433–435

^x Ozaki, H., Sato, T., Kubota, H., Hata, Y., Katsube, Y., and Shimonishi, Y. (1991) *J. Biol. Chem.* **266**, 5934–5941

^y Moss, J., and Vaughan, M. (1995) *J. Biol. Chem.* **270**, 12327–12330

tyrosine-O-sulfate residues may be a widespread regulatory mechanism.^{125–127} One of the proteins known to contain a tyrosine sulfate residue is the blood protein fibrinogen. Many polysaccharides and oligosaccharides on glycoproteins exist in part as sulfate esters (Fig. 4-11).^{128,129}

Carboxyl groups of certain glutamate side chains in proteins that control bacterial **chemotaxis** are methylated reversibly to form methyl esters.¹³⁰ This **carboxylmethylation** occurs as part of a reaction sequence by which the bacteria sense compounds that can serve as food or that indicate the presence of food

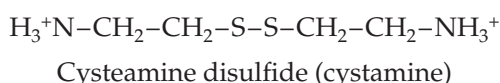
(Section D,5, Fig. 11-8, and Chapter 19). Demethylation occurs by hydrolysis, which is catalyzed by esterases. Carboxylmethylation also occurs in eukaryotic cells but is often substoichiometric and part of a mechanism for repair of isomerized or racemized aspartyl residues in aged proteins (Box 12-A). However, the major eukaryotic protein phosphatase 2A is carboxylmethylated at its C terminus,¹³¹ as are the Ras proteins discussed in Section D,3.

It was pointed out in Chapter 4 that many proteins, especially those secreted from cells or taking up residence within membranes, are **glycosylated**. Specific

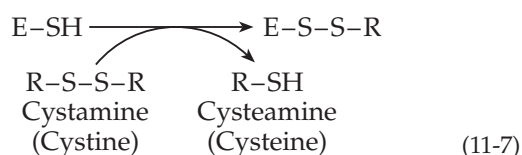
glycosyl groups may be removed or added to a protein. Such alterations may all be regarded as parts of control mechanisms that direct the proteins to their proper locations or determine the length of time that they remain active. The nuclear histones undergo extensive and reversible **acetylation**^{132,133} which is thought to be important to replication of DNA and to transcription (Chapters 27 and 28).

4. Thiol-Disulfide Equilibria

The activity of enzymes is sometimes controlled by the formation or the reductive cleavage of disulfide linkages between cysteine residues within the protein. An example is provided by the effects of light on enzymes of chloroplasts. Light absorbed by the photosynthetic reaction centers generates NADPH, which in turn reduces **thioredoxin**. This is a small protein containing a readily accessible S–S bridge that can be reduced to a pair of SH groups (Box 15-C). Within illuminated chloroplasts, these newly formed SH groups reduce disulfide bridges to activate a series of enzymes including fructose 1,6-bisphosphatase that participate in the photosynthetic incorporation of CO₂ into sugars^{134,135} (see Chapter 23). Thioredoxins also function within bacteria, fungi, and animals, serving as electron carriers for processes, some of which are involved in metabolic control. A number of known enzymes and other proteins, including insulin, contain reducible S–S bridges within peptide loops as in the thioredoxins. Another possible control mechanism for SH-containing enzymes depends upon formation of **mixed disulfides** with small SH-containing metabolites. For example, **cysteamine disulfide**, a minor constituent of cells, cysteine, or some small disulfide, could be released



following hormonal or other stimulation and could react by disulfide exchange (Eq. 11-7) to either inactivate or activate an enzyme.



Almost all cells contain a high concentration (3–9 mM) of the thiol-containing tripeptide **glutathione** (G–SH, Box 11-B). In its disulfide form it participates in forming disulfide bridges in secreted extracellular proteins (Eq. 10-9) via intermediate mixed disulfides. Mixed disulfides with glutathione as well as with other thiols can also be formed within cells by oxidative

reactions. Reduction of these disulfides by reduced glutathione will return the enzymes to their reduced states. The small protein **glutaredoxin**, whose eukaryotic forms are also called **thioltransferase**, resembles thioredoxins. It undergoes reduction by glutathione and, in turn, reduces S–S linkages in a different set of proteins than those reduced by thioredoxin.

Low-molecular-mass thiols such as coenzyme A and protein-bound thiol cofactors such as phosphopantetheine are present in all cells. Their SH groups can also be oxidized to disulfides and it is of interest that in resting bacterial spores these compounds exist largely as disulfides or mixed disulfides. Upon germination of the spores special enzymes reduce the disulfides.¹³⁶ Some proteins involved in control of protein synthesis contain SH groups that add covalently to C-6 atoms of a uracil ring in specific mRNA molecules. Control of their state of reduction may also be important.¹³⁷

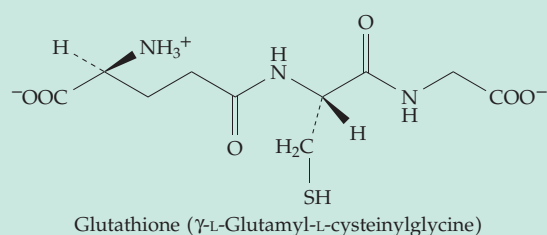
5. Regulatory Effects of H⁺, Ca²⁺, and Other Specific Ions

The pH of the cytoplasm is controlled tightly. Yet transient changes can occur and may affect many aspects of metabolism. For example, rapid glycolysis leads to lactic acid formation with an associated drop in internal pH. Both increases and decreases in pH have been associated with successive stages in embryonic or larval development.¹³⁸ Cytoplasmic pH changes may serve as regulatory signals. A well-understood example is the Bohr effect on oxygenation of hemoglobin (Chapter 7). Another is the protein kinase C-stimulated H⁺/Na⁺ exchange through membranes. Because the Na⁺ concentration is high outside cells and low inside, the exchange leads to an increase in cytosolic pH with many resultant effects on metabolism.^{139,140} Exchange of external Cl[−] for internal HCO₃[−] also affects pH.¹⁴¹

Uptake of Ca²⁺ into cells, or release of this ion from intracellular stores, is a major regulatory mechanism in many if not all cells (see Section E). Mn²⁺ activates **phosphoenolpyruvate carboxykinase** (Eq. 13-46) and may be a regulator of gluconeogenesis.¹⁴² Iron controls the synthesis of ferritin and of transferrin receptors¹³⁷ (Chapter 16). The specific metal ions present in many biological macromolecules are likely to participate in additional regulatory processes.

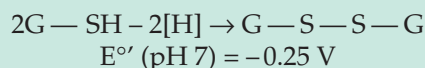
Phosphate and bicarbonate ions are important substrates for many enzymatic processes and as such have regulatory functions. Bicarbonate controls the key enzyme of photosynthesis, **ribulose biphosphate carboxylase**, by carbamate formation (Fig. 13-12). Chloride ions activate amylases and may affect the action of “G proteins” that mediate hormone actions. Other observed effects of ions are too numerous to mention.

BOX 11-B GLUTATHIONE, INTRACELLULAR OXIDATION-REDUCTION BUFFER



In 1888, de Rey-Pailhade discovered the sulfur-containing tripeptide that we now know as glutathione (G-SH).^a By 1929 it had been characterized by F. G. Hopkins and others as an unusual tripeptide present in most, if not all, eukaryotic cells. Within animal cells the concentration is typically 1–5 mM. Lower levels are found in many bacteria.^{b–g}

The most interesting chemical characteristics of glutathione are the γ-glutamyl linkage and the presence of a free SH group. The latter can be oxidized to form a disulfide bridge linking two glutathione molecules.



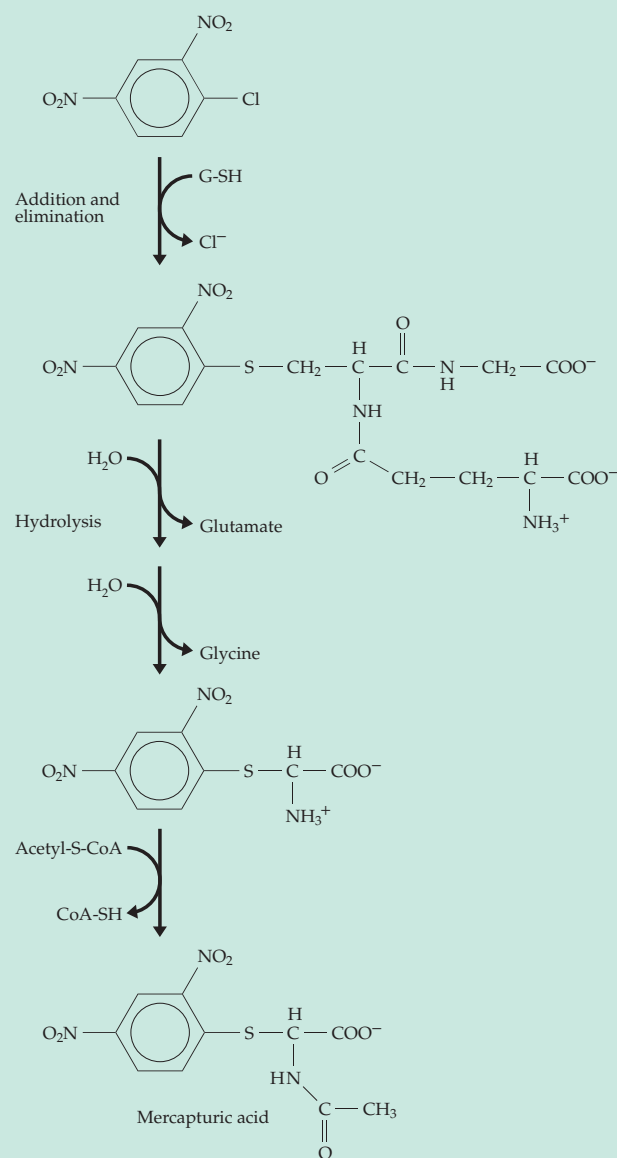
It is this redox reaction that has focused attention on glutathione as an intracellular reducing agent whose primary function may be to keep the SH groups of proteins reduced (see Section C). Glutathione is usually maintained in its reduced form by the flavoprotein glutathione reductase (Fig. 15-12).

Although it is primarily an intracellular compound, glutathione is secreted by epithelial and other cells. It may regulate the redox state of proteins in plasma and other extracellular fluids as well as within cells. In addition, glutathione released from the liver may be an important source of cysteine for other tissues. In the endoplasmic reticulum and the periplasm of bacteria glutathione functions in crosslinking thiol groups in newly formed proteins (Eq. 10-9).

Glutathione also has a series of **protective functions**. It reduces peroxides via the selenium-containing **glutathione peroxidase** (Box 15-H) and is part of a system for trapping and detoxifying harmful free radicals. The importance of this function is suggested by the fact that *Entamoeba histolytica*, an organism that lacks both mitochondria and aerobic respiration, produces no glutathione.^h It may be that the primary function of glutathione in eukaryotes is to protect cells against oxygen toxicity associated with their mitochondria.ⁱ The renewal of free radicals and regeneration of protein -SH groups

may involve the cooperation of glutathione, glutaredoxin (Box 15 - C), and ascorbic acid.^j

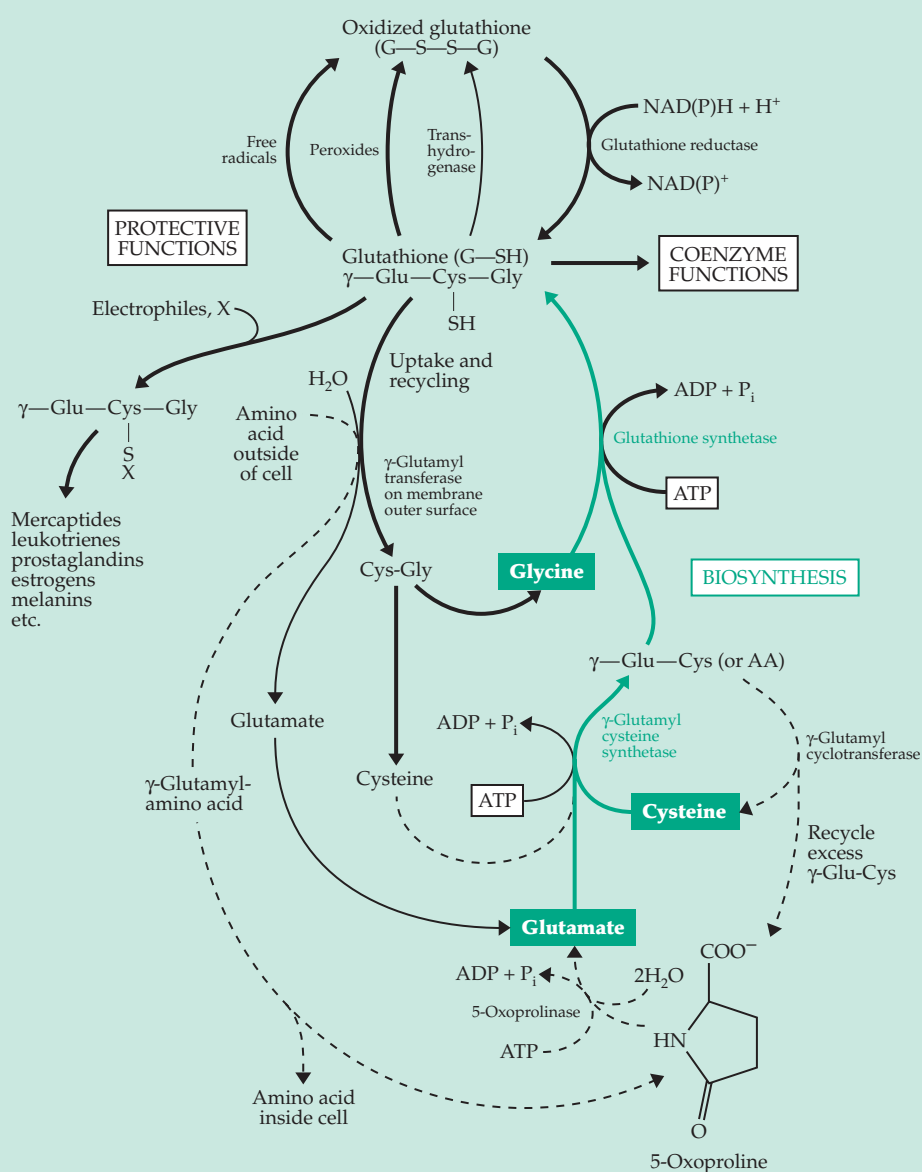
Another protective function is fulfilled by the formation of soluble **mercaptides** and other “conjugates” of glutathione with many foreign substances (**xenobiotics**). These conjugates are made by the action of a large family of **glutathione transferases**^{e,k,l} (see also Chapter 13) which catalyze addition reactions of the thiolate group of glutathione with epoxides, alkylating compounds, and chlorinated aromatic hydrocarbons. The addition step is often followed by elimination, e.g., of chloride ion, as in the following example. Two steps of hydrolysis and an acetylation by acetyl-CoA form a mercapturic acid:



BOX 11-B (continued)

The mercapturic acids and related compounds can then be exported from cells by an ATP-dependent export pump.^m Glutathione is a coenzyme for **glyoxalase** (Eq. 13-33), **maleylacetoacetate isomerase** (Eq. 13-20), and **DDT dehydrochlorinase**. The latter enzyme catalyzes elimination of HCl from molecules of the insecticide and is especially active in DDT-resistant flies.^a Glutathione is said to be the specific factor eliciting the feeding reaction of *Hydra*; that is, the release of glutathione from injured cells causes the little animal to engulf food.

Synthesis of glutathione occurs within cells via the ATP-dependent reactions in the following scheme.^{n,o} Much more γ -glutamylcysteine is synthesized than is converted to glutathione and the excess is degraded by γ -glutamyl cyclotransferase to form the cyclic amide 5-oxoproline. Cleavage of ATP is required to reopen the ring to form glutamate. Although biosynthesis is exclusively intracellular, most glutathione appears to be secreted and degraded by extracellular enzymes. The membrane-bound γ -glutamyl transferase catalyzes hydrolysis of



Scheme illustrating interrelationships of the biosynthesis and protective, coenzymatic, and transport functions of glutathione. See also Meister.^c

BOX 11-B GLUTATHIONE, INTRACELLULAR OXIDATION-REDUCTION BUFFER (continued)

glutathione to cysteinylglycine which is further cleaved by a peptidase. The activity of γ -glutamyltransferase varies among tissues and is especially high in cells of the kidney tubules. The cysteine and glycine released by the peptidase may reenter the cells by a Na^+ -dependent process. Meister proposed that the γ -glutamyltransferase, acting on amino acids, forms γ -glutamyl amino acids (see scheme) which are released *within* cells and are cleaved by γ -glutamyl cyclotransferase.^c The cleavage of glutathione would provide the driving force for amino acid uptake. However, this is probably a minor pathway.^p

Trypanosomes contain little glutathione but a large amount of **trypanothione** [N^1, N^8 -bis (glutathionyl) spermidine].^q This diamide of spermidine (Chapter 24) is in equilibrium with its disulfide, a 24-membered macrocyclic structure, and appears to have functions similar to those of glutathione. Trypanothione reductase, which is unique to trypanosomes, is a potential target for antitrypanosomal drugs,^{q-s} as is trypanothione synthetase.^t Bacteria that do not synthesize glutathione usually accumulate some other thiol, e.g., α -glutamylcysteine or coenzyme A, in high concentrations.^{u,v}

^a Meister, A. (1988) *Trends Biochem. Sci.* **13**, 185–188

^b Bernofsky, C., and Wanda, S.-Y. C. (1982) *J. Biol. Chem.* **257**, 6809–6817

^c Meister, A. (1988) *J. Biol. Chem.* **263**, 17205–17208

^d Inoue, M. (1985) in *Renal Biochemistry* (Kinne, R. K. H., ed), pp. 225–269, Elsevier, Amsterdam

^e Orrenius, S., Ormstad, K., Thor, H., and Jewell, S. A. (1983) *Fed. Proc.* **42**, 3177–3188

^f Viña, J., ed. (1990) *Glutathione: Metabolism and Physiological Functions*, CRC Press, Inc., Boca Raton, Florida

^g Dolphin, D., Poulson, R., and Avramovic, O., eds. (1989) *Glutathione - Chemical, Biochemical and Medical Aspects (Coenzymes and Cofactors)*, Vol. 3, Wiley, New York (Parts A & B)

^h Fahey, R. C., Newton, G. L., Arrick, B., Overdank-Bogart, T., and Aley, S. B. (1984) *Science* **224**, 70–72

ⁱ Garcia de la Asuncion, J., Millan, A., Pla, R., Bruseghini, L., Esteras, A., Pallardo, F. V., Sastre, J., and Viña, J. (1996) *FASEB J.* **10**, 333–338

^j Meister, A. (1994) *J. Biol. Chem.* **269**, 9397–9400

^k Ji, X., Johnson, W. W., Sesay, M. A., Dickert, L., Prasad, S. M., Ammon, H. L., Armstrong, R. N., and Gilliland, G. L. (1994) *Biochemistry* **33**, 1043–1052

^l Hebert, H., Schmidt-Krey, L., and Morgenstern, R. (1995) *EMBO J.* **14**, 3864–3869

^m Ishikawa, T. (1992) *Trends Biochem. Sci.* **17**, 463–468

ⁿ Polekhina, G., Board, P. G., Gali, R. R., Rossjohn, J., and Parker, M. W. (1999) *EMBO J.* **18**, 3204–3213

^o Lu, S. C. (1999) *FASEB J.* **13**, 1169–1183

^p Lee, W., Hawkins, R., Peterson, D., and Viña, J. (1996) *J. Biol. Chem.* **271**, 19129–19133

^q Henderson, G. B., Ulrich, P., Fairlamb, A. H., Rosenberg, I., Pereira, M., Sela, M., and Cerami, A. (1988) *Proc. Natl. Acad. Sci. U.S.A.* **85**, 5374–5378

^r Bollinger, J. M., Jr., Kwon, D. S., Huisman, G. W., Kolter, R., and Walsh, C. T. (1995) *J. Biol. Chem.* **270**, 14031–14041

^s Sullivan, F. X., Sobolov, S. B., Bradley, M., and Walsh, C. T. (1991) *Biochemistry* **30**, 2761–2767

^t Henderson, G. B., Yamaguchi, M., Novoa, L., Fairlamb, A. H., and Cerami, A. (1990) *Biochemistry* **29**, 3924–3929

^u Sundquist, A. R., and Fahey, R. C. (1989) *J. Biol. Chem.* **264**, 719–725

^v Swerdlow, R. D., Green, C. L., Setlow, B., and Setlow, P. (1979) *J. Biol. Chem.* **254**, 6835–6837

6. Compartments and Organized Assemblies

The geometry of cell construction provides another important aspect of metabolic control. In a bacterium, the periplasmic space (Fig. 8-28) provides a compartment that is separate from the cytosol. Some enzymes are localized in this space and do not mix with those within the cell. Other enzymes are fixed within or attached to the membrane. Eukaryotic cells have more compartments: nuclei, mitochondria (containing both matrix and intermembrane spaces), lysosomes, microbodies, plastids, and vacuoles. Within the cytosol the tubules and vesicles of the endoplasmic reticulum (ER) separate off other membrane-bounded compartments. The rate of transport of metabolites through the membranes between compartments is limited and often is controlled tightly.

While many enzymes appear to be dissolved in the cytosol and have no long-term association with other proteins, enzymes that catalyze a series of con-

secutive reactions may form complexes within which substrates are **channeled**.^{143–147} Many enzymes are attached to membranes where they may be held close together as organized assemblies.¹⁴⁶ This appears to be the case for oxidative enzymes of mitochondria (Chapter 18) and for the cytoplasmic fatty acid synthetases (Chapter 21). In bacterial fatty acid synthesis, the product of the first enzyme is covalently attached to a “carrier” and, while so attached, is subjected to the action of a series of other enzymes. In eukaryotes several enzymes form domains of a single fatty acid synthase. Efficient substrate channeling results. Tryptophan synthase (Fig. 25-3) passes indole through a tunnel between subunits.^{146a} Both NH_3 and carbamate ions pass through tunnels between subunits of *E. coli* carbamoyl phosphate synthetase (Eq. 24-22). The product carbamoyl phosphate may then be passed directly to aspartate carbamoyltransferase for synthesis of carbamoylaspartate in the pyrimidine biosynthetic pathway.^{146b} Channeling is sometimes difficult to

prove. Geck and Kirsch have provided a generally useful technique for testing. A large amount of a genetically modified, inactive form of an enzyme is added. Unless channeling occurs, this will decrease the rate of a reaction whose rate is limited by diffusion or by instability of an intermediate.^{146c}

Proteasomes (Box 7-A) have enzymatic sites within a protected box which limits the escape of long peptide fragments. Membrane anchors (Chapter 8), often consisting of acyl or polyprenyl groups, hold many proteins to cell surfaces and strong protein–protein bonds hold many others.^{146d} Enzymes involved in cell signaling, and discussed in the following section, are often anchored close together on membrane surfaces.^{148,149}

D. Hormones and Their Receptors

A major element in the control of the metabolism of a cell is provided by chemical messages sent from *other cells* and sensed by receptors on the cell membrane or in the cytoplasm. Hormones such as insulin, adrenaline, and the sex hormones are released from an organ and travel through the blood, affecting tissues throughout the body. There are many such hormones. A large number of other hormones have more local effects, influencing mostly adjacent cells. When released at nerve endings these substances are called **neurotransmitters**. Chemical messages are probably sent from virtually all cells to their immediate neighbors, affecting both their growth and their behavior. In recent years the very reactive and toxic compounds **nitric oxide (NO)** and **carbon monoxide (CO)** have been identified as important hormones. These compounds can diffuse rapidly and react with many compounds within cells. Their actions are more rapid, more rapidly ended, and probably less specific than those of most hormones.

Hormones bind to **receptors** on their “target” cells. The receptors are often integral membrane proteins on cell surfaces. Binding of the hormone often causes a conformational alteration that “activates” the receptor. In some cases the receptor is an enzyme and the hormone an allosteric activator. In others, the activated receptor interacts with an enzyme in the cytoplasm or on the membrane facing the cytoplasm. This enzyme may generate a **second messenger**, a substance that can diffuse throughout the cell and alter metabolism by exerting allosteric effects on various other enzymes. The best known second messenger is cyclic AMP but there are many others, a few of which are listed in Table 11-2.

Some hormone–receptor complexes enter the cell via endocytosis in coated vesicles. Within the cell both the receptor and the bound hormone, if a peptide, may be degraded by proteases. The initial binding of the hormone may induce the release of a second messenger,

while the degradation of the receptor complex at a later time releases peptides that may be additional second messengers. This is one way in which hormones may elicit a rapid response followed by delayed responses.

While many hormones bind to surface receptors the steroid hormones, which are lipid in nature, pass through the cell membrane and bind to receptor proteins in the nucleus. The resulting hormone–protein complexes induce changes in gene expression through regulation of transcription (Fig. 11-1, top). These receptors are considered in Chapter 22 and hormones are considered further in Chapter 30.

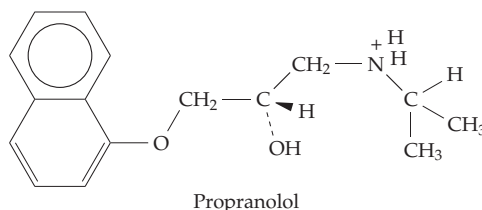
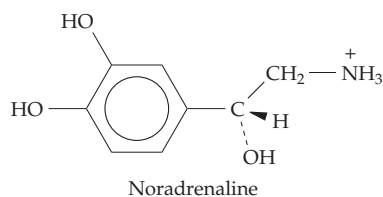
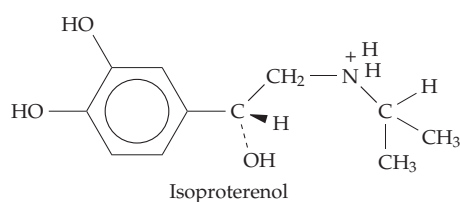
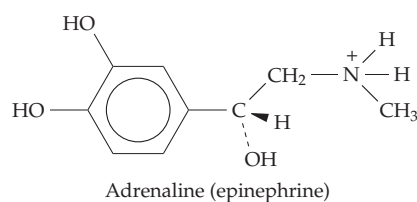
1. Beta Adrenergic Receptors and Related Seven-Helix Proteins

Sites that bind **adrenaline** (epinephrine), **noradrenaline** (norepinephrine), and related **catecholamines** (see Chapter 30) to almost all cell surfaces are classified as either **α adrenergic** or **β adrenergic receptors**. The β receptors, which have been studied the most,¹⁵⁰ occur as two major types. The β_1 receptors have approximately equal affinity for adrenaline and noradrenaline, whereas the β_2 receptors, the most common type, are more nearly specific for adrenaline. Binding of the hormone or other agonist such as **isoproterenol** to any of the β receptors stimulates cAMP formation within the cell.¹⁵³ The receptors belong to a family of integral membrane proteins related in sequence to the **opsins** of the retina and the light-operated proton pump bacteriorhodopsin (Fig. 23-45) and probably have a very similar three-dimensional structure. Based upon the known three-dimensional structures of rhodopsin and bacteriorhodopsin (Chapter 23), sequence comparisons, and much other evidence, all of these receptors are probably folded into groups of seven hydrophobic membrane-spanning helices arranged as closely packed bundles with folded loops protruding into the cytoplasm and into the extracellular space. The N termini of the proteins are thought to be in the extracellular space and the C termini in the cytoplasm as shown in Fig. 11-6.^{153a,b,c} Defects in structure or functioning of human β receptors have been associated with both asthma¹⁵⁴ and heart failure.^{155,155a,b}

Among the proteins phosphorylated in response to formation of cAMP are the β adrenergic receptors themselves. Phosphorylation occurs within a cluster of serine and threonine residues near the C terminus as is indicated in Fig. 11-6A by the action of various kinases including a **receptor kinase**^{156,156a} which may be anchored nearby.¹⁵⁷ The effect of this C-terminal phosphorylation is to decrease the sensitivity of the receptor so that after a few minutes it conveys a diminished response. Thus, cAMP exerts feedback inhibition of its own synthesis. *Desensitization upon continuous occupancy*

TABLE 11-2
Some Molecules (Second Messengers) That Carry Intracellular Signals

Compound	Metabolic state	Response	Location of discussion
Cyclic AMP	Stimulation of β adrenergic receptors	Increased glycogenolysis, glycolysis	Sections C,2 and D,2;
Cyclic GMP	Visual stimulation	Neuronal signal	Chapter 23, Section D
Ca^{2+}	Ca^{2+} channels open	Muscular contraction, others	Box 6-D, Section E Chapter 19, Section B,4
Inositol-1,4,5-trisphosphate and related compounds	Stimulation of α adrenergic receptors, various other stimuli	Opens Ca^{2+} channels in ER	Section E
Diadenosine 5'-tetraphosphate (Ap_4A)	Oxidative or heat stress		Chapters 28, 29
Guanosine 5'-diphosphate, 3-diphosphate (ppGpp)	Nutritional stress		Chapter 29
Mn^{2+}	Low glucose	Increased PECK activity	Chapter 17, Section L
		Increased arginase activity	Chapter 16, Section E
Cyclic ADP-ribose (cADPR) and 2'-P-cADPR			Chapter 15, Section E



by an agonist is a property of many other receptors as well. When a receptor is no longer occupied, phosphatases remove the phospho groups added by the receptor kinase permitting the sensitivity to rise again. This phosphorylation–dephosphorylation cycle appears to be only one of several mechanisms by which cells regulate receptor sensitivity.^{155b,158,158a,158b}

Many other receptors also have a seven-helix structure similar to that of the adrenergic receptors. These include receptors for the following: **glucagon**,

one of the pancreatic hormones regulating glucose metabolism;^{159,160} **vasopressin** (Fig. 2-4);^{161,162} **lutropin**, another pituitary hormone;¹⁶³ other gonadotropins;¹⁶⁴ the **thyrotropin-releasing factor** (TRF; Fig. 2-4);¹⁶⁵ a receptor for **KDEL** peptide sequences. The KDEL receptor functions in the return of soluble proteins containing the KDEL motif from the Golgi to the endoplasmic reticulum.¹⁵²

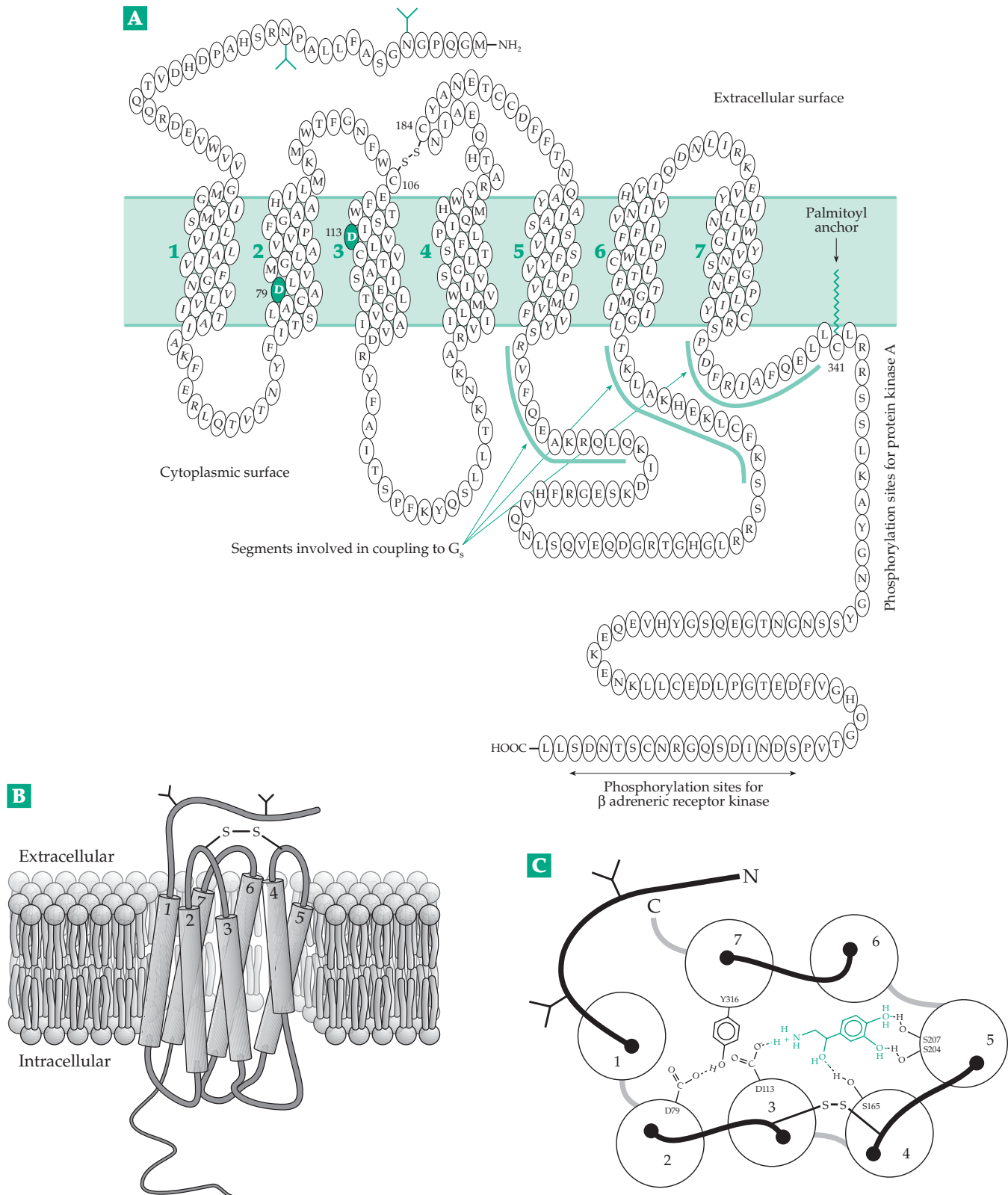


Figure 11-6 (A) Proposed organization of the human β_2 adrenergic receptor. The 413-residue polypeptide chain is arranged according to the model for rhodopsin as seven transmembrane helices. The two N-glycosylation sites, which may carry very large oligosaccharides, are indicated by Y. The palmitoylated cysteine 341 is shown with its alkyl side chain embedded in the membrane. Aspartates 79 and 113, also shown in (C), are shaded. After Strosberg¹⁵¹ and Scheel and Pelham.¹⁵² (B) Arrangement of the seven helices suggested by the rhodopsin structure. (C) Hypothetical view of the receptor from the external membrane surface showing a molecule of noradrenaline bound to hydrophilic residues deep in the cleft between the helices. From Strosberg.¹⁵¹

2. Adenylate Cyclases (Adenylyl Cyclases)

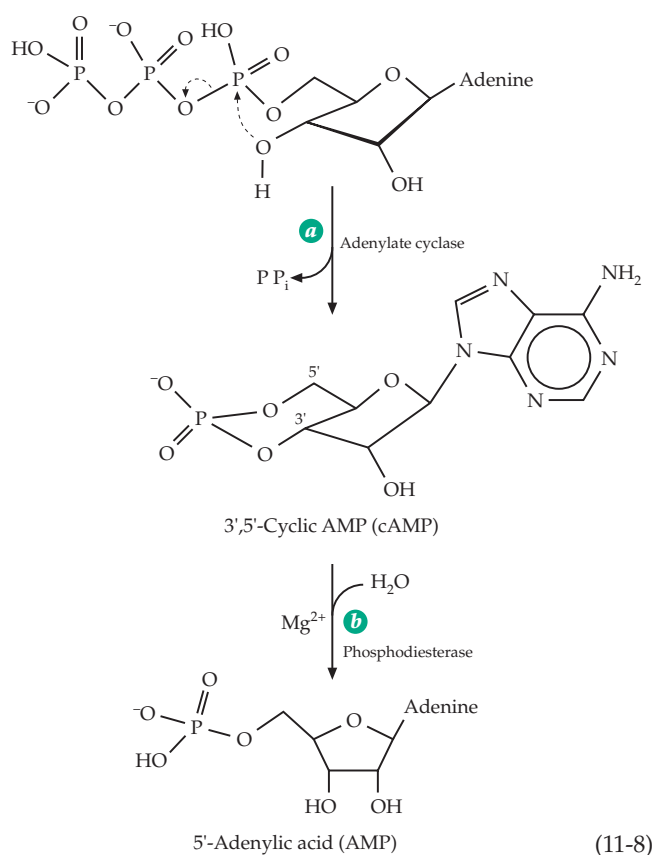
All of the effects of the catecholamines bound to β adrenergic receptors and of **glucagon**, **ACTH**, and many other hormones appear to be mediated by **adenylate cyclase**. This integral membrane protein catalyzes the formation of cAMP from ATP (Eq. 11-8, step *a*). The reaction, whose mechanism is considered in Chapter 12, also produces inorganic pyrophosphate. The released cAMP acts as the second messenger and diffuses rapidly throughout the cell to activate the cAMP-dependent protein kinases and thereby to stimulate phosphorylation of a selected group of proteins (Fig. 11-4). Subsequent relaxation to a low level of cytosolic cAMP is accomplished by hydrolysis of the cAMP by a phosphodiesterase (Eq. 11-8, step *b*).^{166,167} In the absence of phosphodiesterase cAMP is extremely stable kinetically. However, it is thermodynamically unstable with respect to hydrolysis.

The existence of cAMP as a compound mediating the action of adrenaline and glucagon on glycogen phosphorylase was first recognized in 1956 by Sutherland.^{168,169} However, for many years most biochemists regarded cAMP as a curiosity and the regulatory chemistry of phosphorylase as an unusual specialization. That view was altered drastically when cAMP was found to function as a second messenger in the action of over 20 different hormones. Phosphorylation by

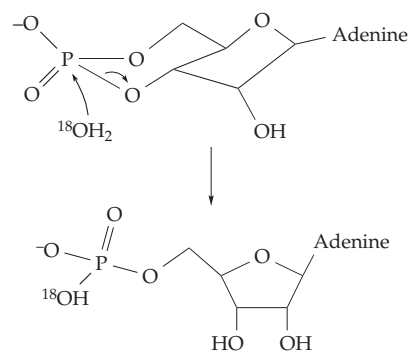
cAMP-dependent protein kinases regulates several enzymes concerned with energy metabolism or with the control of cell division. Phosphorylation reactions can lead to different responses in different specialized cells. Proteins of membranes, microtubules, and ribosomes are phosphorylated, as are nuclear histones. Cyclic AMP mediates the action of some neurotransmitters released at synapses and functions as a signal between cells of some slime molds (Box 11-C). Transcription of genes can be either stimulated or inhibited by cAMP.¹⁷⁰ Even in *E. coli* cAMP is generated and acts as a positive effector for transcription of some genes (Chapter 28). Cyclic AMP may function in signaling stress in plants.¹⁷¹

The isolation and characterization of adenylate cyclases have been difficult because of their location within membranes and because there is so little (usually only 0.001–0.01%) of the membrane protein.¹⁷² The 1060- to 1250-residue, 120 kDa proteins are also easily denatured. However, the sequences of cloned adenylate cyclase genes and the observed patterns of synthesis of the corresponding mRNAs have been revealing. There are at least eight mammalian adenylate cyclase genes with complex regulatory properties.^{173–177} The sequences suggest that most of the isoenzymes are integral membrane proteins, each with 12 transmembrane helices organized as two sets of six with a large ~40-kDa cytoplasmic domain between the sets. This is similar to the organization of the cystic fibrosis transmembrane conductance regulator (Box 26-A) and some other membrane transporters. However, there is no firm evidence that adenylate cyclases contain ion channels. These enzymes are also discussed in Chapter 12, Section D,9.

A quantitative indication of the importance of the cAMP system within cells can be derived from measurement of the kinetics of the incorporation of ¹⁸O from water into the α -phospho groups of AMP, ADP, and ATP. This incorporation will result from hydrolysis of cAMP by the phosphodiesterases that allow relaxation to a low cAMP level (Eq. 11-8). It is thought that in human blood platelets this represents the major pathway of this labeling (Eq. 11-9), which occurs at a rate of about 1.1 μmol of ¹⁸O $\text{kg}^{-1}\text{s}^{-1}$.



(11-8)



(11-9)

Stimulation of the platelets by prostacyclin (prostaglandin I_2 ; see Chapter 22) leads to a 10- to 40- fold increase in cAMP concentration and a 4- to 5-fold increase in the rate of ^{18}O incorporation.¹⁷⁸ A quite different *soluble* adenylate cyclase is present in spermatozoa. It is stimulated directly by bicarbonate ions. It may be a **bicarbonate sensor** in sperm cells and in some other bicarbonate-responsive tissues as well as in cyanobacteria.^{178a,178b}

3. Guanine Nucleotide-Binding Proteins (G Proteins)

The β adrenergic receptors are not coupled directly to adenylate cyclase but interact through an intermediary stimulatory protein **G_s**, which contains three subunits, α , β , and γ .^{179–182} We know that the G_s protein associated with β adrenergic stimulation is only one of a very large number of related **G proteins**, so named because of their property of binding and hydrolyzing GTP. In its unactivated state the α subunit of a $G_{\alpha\beta\gamma}$ heterotrimer carries a molecule of bound GDP. Apparently, the G_s proteins and the hormone receptors,

which are activated by the binding of hormones, diffuse within the membrane until they make contact and form molecules of the $G_s \cdot \text{GDP}$ complex (Eq. 11-10, step *b*). The complex then undergoes a rapid exchange of the bound GDP for GTP, after which the hormone and receptor dissociate. The $G_{\alpha} \cdot \text{GTP}$ complex may also dissociate from the complex, perhaps entering the cytosol as a soluble protein (Eq. 11-10, step *c*). The $G_{\alpha} \cdot \text{GTP}$ complex combines with adenylate cyclase (step *d*) and activates it to generate cAMP. However, the activation is transient. G_{α} also contains GTP-hydrolyzing (GTPase) activity and within a few minutes $G_{\alpha} \cdot \text{GTP}$ is completely converted to $G_{\alpha} \cdot \text{GDP}$ and the adenylate cyclase dissociates (Eq. 11-10, step *e*). The $G_{\alpha} \cdot \text{GDP}$ recombines with the $\beta\gamma$ complex which may serve as a membrane anchor, to complete the regulatory cycle. The overall effect is for hormone binding to cause a rapid release of cAMP in a short burst that may last only about 15 s.

It is not the binding of GTP but a slow subsequent conformational alteration that activates the $G_{\alpha} \cdot \text{GTP}$ adenylate cyclase complex. This was deduced by study of analogs of GTP, such as guanosine 5'-(β , γ -imido) triphosphate (GMP-*P*-(NH)-*P* or GppNp), which are

BOX 11-C THE ATTRACTION OF *Dictyostelium* TO CYCLIC AMP

In higher animals cyclic AMP (cAMP) is an intracellular second messenger, but in the cellular slime mold *Dictyostelium discoideum* it serves to convey signals *between* cells.^{a–c} As was mentioned on p. 20, the organism exists as individual amoebae until the food supply is exhausted. Then the cells begin to signal their lack of food by secreting pulses of cAMP. Because they also secrete a phosphodiesterase, the cAMP is short-lived.^d However, it is present long enough for any other nearby amoeba to sense the gradient of cAMP concentration from one end of the cell to the other. As little as a 2% difference in concentration can induce chemotaxis.^e The amoebae move toward the source of cAMP and emit pulses of the compound. This results in the formation of aggregation centers in which the concentration of cAMP oscillates spontaneously as the cAMP moves outward in waves. The cells move up the concentration gradient until the peak of a wave reaches them. Then they move in a random direction until the next wave reaches them and again orient their motion.

After ~50 movement steps an aggregation center contains ~10⁵ cells which now follow a “development program.” The amoebae adhere to each other to form 1- to 2 mm-long multicellular “slugs” in which all of the cells move forward together. About 30 h after aggregation begins the slugs stop and form stalks with spore-containing fruiting bodies on top. The

lead cells in a slug become the stalk.^{a,c}

Two types of G protein-coupled cAMP receptors in the cell membranes of the amoebae have been identified. One leads to activation of adenylate cyclase and the release of new pulses of cAMP and the other to activation of guanylate cyclase.^e This enzyme causes a rapid 7- to 10-fold increase in intracellular cGMP which plays an important role in controlling chemotaxis.

Some species of cellular slime molds use other chemical attractants (**acrasins**). For example, *D. minutum* secretes an analog of folic acid^f and cells of *Polysphondylium violaceum* are attracted by the ethyl ester of *N*-propionyl- γ -L-glutamyl-L-ornithine- δ -lactam.^g

^a Gerisch, G. (1987) *Ann. Rev. Biochem.* **56**, 853–879

^b Devreotes, P. (1989) *Science* **245**, 1054–1058

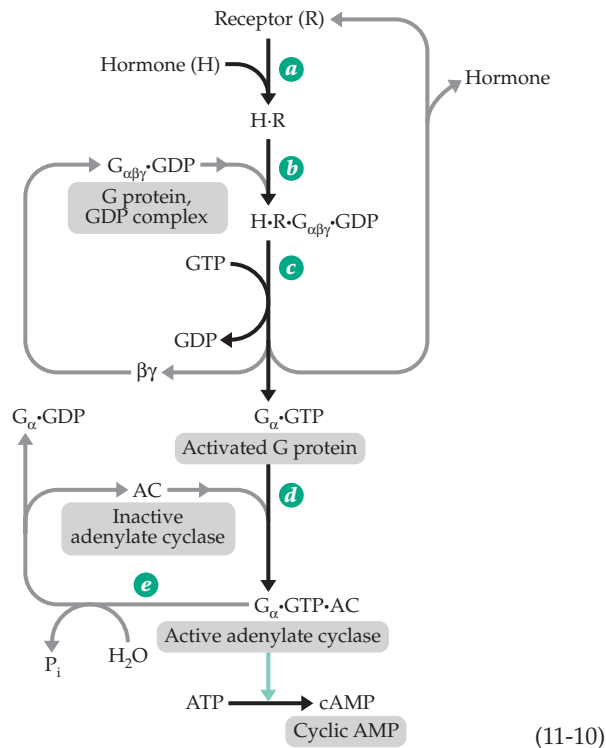
^c Alberts, B., Bray, D., Lewis, J., Raff, M., Roberts, K., and Watson, J. D. (1994) *Molecular Biology of the Cell*, 3rd ed., Garland, New York

^d Levine, H., Aranson, I., Tsimring, L., and Truong, T. V. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 6382–6386

^e Kuwayama, H., and Van Haastert, P. J. M. (1996) *J. Biol. Chem.* **271**, 23718–23724

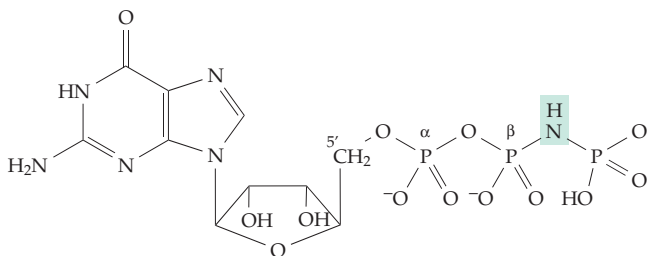
^f Schapp, P., Konijn, T. M., and van Haastert, P. J. M. (1984) *Proc. Natl. Acad. Sci. U.S.A.* **81**, 2122–2126

^g Shimomura, O., Suthers, H. L. B., and Bonner, J. T. (1982) *Proc. Natl. Acad. Sci. U.S.A.* **79**, 7376–7379



(11-10)

not hydrolyzed to GDP. These compounds may require many minutes to activate the complex, but because the GppNp is not hydrolyzed, the complex remains activated and able to catalyze cAMP formation continuously.



Guanosine 5'-(β,γ -imido) triphosphate (GppNp); in place of the NH, O is present in GTP, S is present in GTP γ S

Occupied receptors for adrenaline, glucagon, ACTH, and histamine activate adenylate cyclase via G $_s$ proteins. Other G $_s$ proteins, which contain subunits designated α_{olf} and which exist as a number of subtypes, mediate **olfactory responses**. Subunit α_o is another specialized polypeptide which is located primarily in neural tissues. A variety of additional G proteins have been discovered in organisms ranging from bacteria to mammals.^{179,183–186} All have similar structures with 39- to 45-kDa α subunits, 35- to 36-kDa β subunits and 5- to 8-kDa γ subunits. Whereas the α subunits are unique to each G protein, β and γ subunits may be shared among several G proteins. These proteins appear to function with many kinds of hormone receptors and

in nerve transmission, secretion, and endocytosis.¹⁸⁷ Mammals can form at least 23 different α_s subunits, which are coded for by 16 genes. Some of the forms arise by alternative splicing of mRNA transcribed from a single gene.¹⁸⁰ There are at least 6 β and 12 γ subunits.^{179,180,182,188,189} Other G proteins, found in heart muscle, induce the opening of K $^+$ or Ca $^{2+}$ channels in cell membranes.^{185,190} It has often been assumed that all of the effects of G proteins are mediated by activated α subunits. However, the $\beta\gamma$ complex, which sticks together very tightly, can also have signaling functions.¹⁹¹ For example, an auxiliary protein called **phosducin** binds to the $\beta\gamma$ complex of G $_s$, slowing the reversion to the $\alpha\beta\gamma$ trimer.¹⁹² Phosducin was discovered in rod cells of the retina but has been found to be distributed broadly in other tissues as well.^{192,193} In yeast the $\beta\gamma$ subunit of a G protein linked to the receptor for the yeast mating factor initiates the pheromone response of a cell.^{193a}

There is another large class of receptors whose occupancy by an agonist leads to *inhibition* of adenylate cyclase. These include the α_2 adrenergic receptors, receptors for acetylcholine, adenosine, prostaglandin E $_2$ (Chapter 21), somatostatin, and some receptors for dopamine. Their responses are mediated by inhibitory proteins G $_i$, which closely resemble G $_s$ in their sizes, amino acid sequences, and heterotrimeric structures, but which inhibit adenylate cyclase when activated.¹⁸⁰ A clear distinction between the G $_s$ and G $_i$ proteins is evident in the fact that G $_s$ is irreversibly activated by the action of cholera toxin, while G $_i$ loses its ability to respond to occupied receptors when modified by the action of *Pertussis* toxin (Box 11-A). A specialized heterotrimeric G $_i$ protein known as **transducin** mediates the light-induced activation of a **cyclic GMP phosphodiesterase** in the retina^{194,195} (see Chapter 23). Its α subunit is designated α_t . The related **gustducin** is found in taste buds.¹⁹⁶

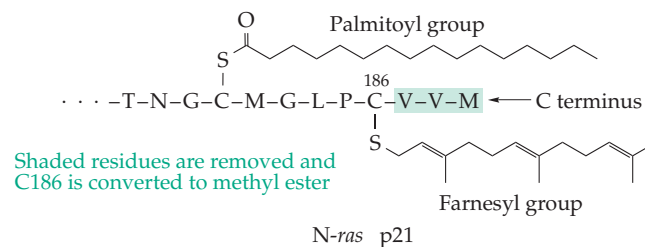
Monomeric G proteins. An entirely different class of G proteins was discovered when it was found that the small 189-residue, 21-kDa protein products of the human oncogenes and proto-oncogenes known as **ras** are monomeric G proteins.^{186,197–200} There are over 80 related proteins of this group of nine families.^{200a} They include the much larger **elongation factor** EF-Tu, which functions in protein synthesis (Chapter 29). Even though there is a large difference in size, the three-dimensional structures of the GTP-binding domains of EF-Tu^{201–204} and of the 21-kDa Ras p21 proteins^{205–209} are very similar (Fig. 11-7). They also resemble that of adenylate kinase (Fig. 12-30).²⁰⁵ The human genome contains four “true” *ras* genes: H-*ras*, N-*ras*, K-*ras*, and K-*rasB*. The occurrence of mutations in these genes in human cancer is discussed in Section H, as are the complex signaling functions of *ras* genes, (see also Fig. 11-13).²¹⁰

Other proteins in the Ras superfamily include some members of the **Rho** family, which function in the cytoskeleton.^{200a,211,212,212a,b,c,213} These include at least 14 mammalian proteins, among them RhoA, RhoB, etc., Rac1, Rac2, etc., and Cdc42. The latter is associated with the formation of fingerlike extensions of the cell membrane containing actin bundles.^{200a,212d,212e} Genes *rac1* and *rac2* encode proteins involved in the initiation of superoxide radical formation by activated phagocytes (Chapter 18).²¹⁴ The **Rab** family of proteins are involved in exocytosis and endocytosis and vesicular transport. There are 24 or more human *rab* genes.^{212,215,216} Yeast cells also contain Ras proteins. One *RAS* gene must be present for yeast to form spores.²¹⁷ **ADP-ribosylation factors** (ARFs), which stimulate the action of cholera toxin A subunit (Box 11-A) are also members of the Ras superfamily.^{120,217a}

Like the trimeric G proteins, the monomeric proteins also serve as “timed switches” that are turned on by GDP–GTP exchange and are turned off by the hydrolysis of the bound GTP. In the absence of other regulatory proteins both the exchange of bound GDP for GTP and the hydrolysis of bound GTP are slow. While the receptor-associated trimeric G proteins are activated by the hormone receptors, the monomeric G proteins are activated toward GDP–GTP exchange by binding to proteins called **guanine nucleotide dissociation stimulators (GDSs)**.^{217b} They speed the release of GDP from the G protein allowing the active GTP complex to be formed. The velocity of hydrolysis of the bound GTP in the activated G protein is greatly increased by **GTPase activating proteins (GAPs)**. These auxiliary proteins can be thought of as signaling molecules that pass their messages to the G protein, controlling the extent of its activation and therefore the strength of a signal that it sends on to the processes that it is controlling.^{200,210,218} One group of GAP proteins are known as **regulators of G-protein signaling (RGS)**.^{218–218b} These proteins, which contain a characteristic 12-residue core, were first recognized as negative regulators of signaling in yeast cells called **GDP-dissociation inhibitors (GDIs)**.²¹⁹ They can also be regarded as a family of GAPs.

Acylation and prenylation. The amino terminus (usually glycine) of the α subunit of any G protein is nearly always converted to an **N-myristoyl group**.^{220–223} This modification occurs in a **cotranslational process** after removal of the initiating methionine (Chapter 29) and can be described as an acyl transfer from coenzyme A.²²⁴ The C termini of the γ subunits of heterotrimeric G proteins, and also of the monomeric proteins of the Ras family, also undergo processing. For example, the C-terminal end of an intact Ras protein contains 18 residues which probably assume a largely α -helical conformation. A cysteine side chain near the terminus and having the sequence CAAX is

converted within cells to a **thioether** with all-*trans*-farnesol or longer polyprenyl alcohols.²²⁵ Another nearby cysteine may form a **thioester** with palmitic acid,^{221,226,227} for example:



The farnesyl (or longer) polyprenyl and palmitoyl groups, together with nearby nonpolar side chains, serve as a membrane anchor. After the prenylation the C-terminal three amino acids (VVM in *N-ras* p21) are removed proteolytically and the new terminal carboxyl group of the farnesylated Cys 186 is converted to a methyl ester.^{212c,225,228} N-terminal myristoylation is usually regarded as irreversible. However, the thioester linkages by which palmitoyl groups are attached to proteins are labile and may be cleaved rapidly by hydrolases. It follows that palmitoylation can be a rapid, regulated modification of proteins,^{229–230a,b} strongly affecting the location and extent of adherence to membrane surfaces.

Three-dimensional structures. The structure of the GTP-binding domain of elongation factor EF-Tu was determined by Jornak in 1985²⁰¹ and that of the complete three-domain structure later.^{202,203} When the structure of the *catalytic domain* of the first Ras protein was determined (Fig. 11-7A) it was clear that it was similar to that of EF-Tu.^{205,207} The same was true for the transducin α_t ,^{194,231,232} for the inhibitory $G_{i\alpha 1}$,^{218b,233,234} and for other G proteins.²³⁵ In every case the differences in structure of the enzyme with GDP or with analogs of GTP were small and limited to a region close to the γ -phospho group of the bound GTP. This group can be seen clearly in Fig. 11-7A adjacent to residue 60 of the protein. See also Fig. 12-36.

The β strands and helices in the GTP-binding domain of the G proteins are connected by a series of loops. The first, second, and fourth loops (residues 10–17, 32–40, and 57–65, respectively) in the Ras structure of Fig. 11-7A form the catalytic site in which the hydrolysis to GDP takes place. The first loop, also called the P loop, is conserved in all GTP-binding proteins. In the Ras proteins it has the following sequence:

10	17	
G-A-G-G-V-G-K-S		Ras
G-X-X-X-X-G-K-S(T)		Consensus

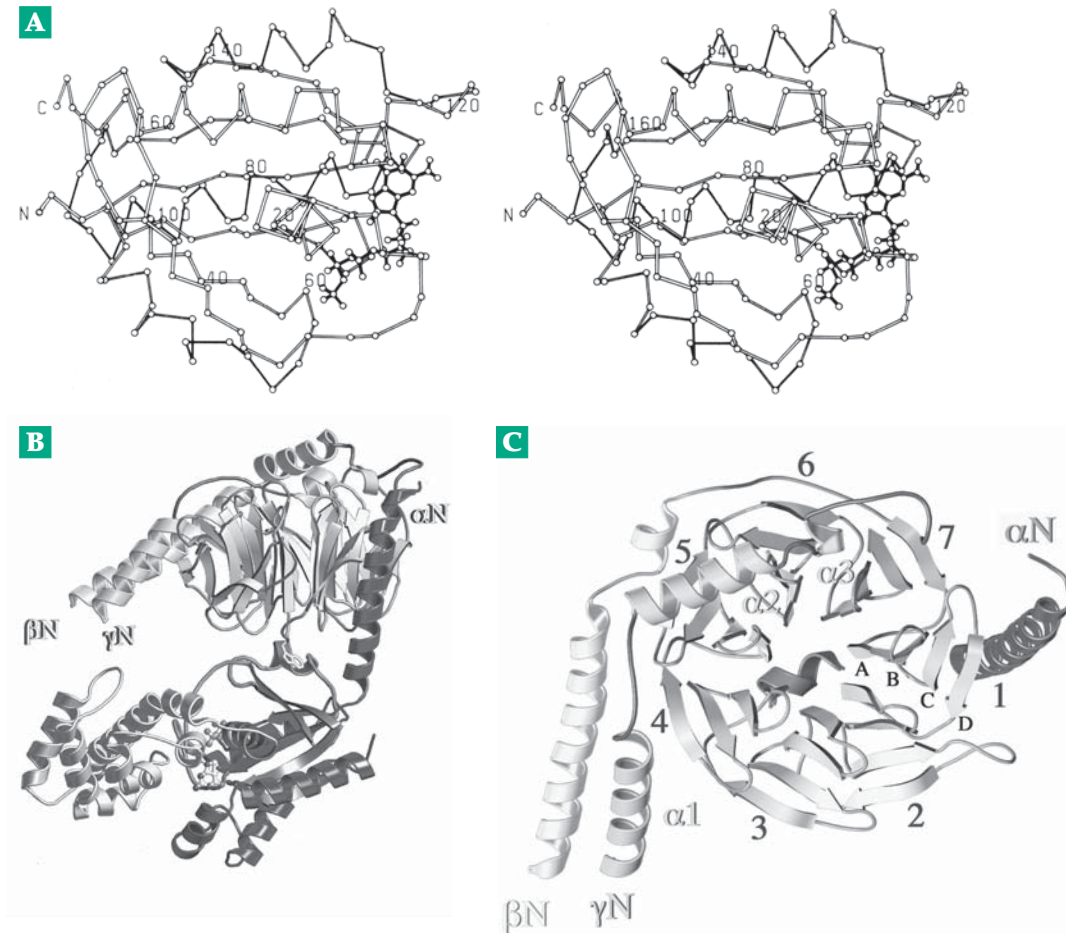


Figure 11-7 (A) Stereoscopic drawing showing α -carbon positions in the 166-residue “catalytic domain” of the human c-H-ras gene product and the bound GTP analog GppNp (solid bond lines). The intact protein contains an additional 23-residue C-terminal extension. From Pai *et al.*²⁴¹ (B), (C) Ribbon drawings showing two nearly orthogonal views of a hetero-trimeric inhibitory G_i brain protein produced using a cloned bovine gene. (B) The amino termini (N) of the three subunits are seen in the left-to-right order: γ , β , α . A side view of the β propeller domain of the β subunit is seen at top center. The Ras-like domain and the additional large helical domain of the α subunit are marked. (C) View from the flared end of the β propeller looking toward the α subunit. The strands of each propeller blade are labeled A, B, C, and D and the seven blades are numbered around the periphery of the propeller. From Wall *et al.*²⁴² (B) and (C) courtesy of Stephen Sprang.

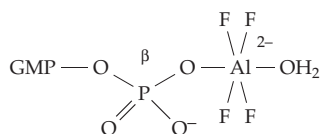
Also shown here is the “consensus” sequence for all GTP-binding proteins. Cancer cells frequently have mutations in Ras at Gly12, which may be substituted by Asp, Lys, Val, or Arg, and at Gly 13 and Gln 61. The latter is in loop 4, a part of the sequence that appears especially mobile but is highly conserved. These mutations all activate the protein, i.e., they decrease the GTPase activity, allowing the G protein to remain in its active conformation longer than normal.

Ras proteins fulfill their functions by interacting closely with two or more proteins in signaling pathways as described in Section H. Other G proteins have additional domains. The 405-residue EF-Tu from *Thermus thermophilus* has three domains: the C-terminal nucleotide-binding domain and two β -barrel domains following it. A major difference in conformation is observed between forms of the protein with bound

GTP or bound GDP, the GDP form being opened up by a hinging motion of $\sim 90^\circ$ between domains I and II.^{202,203} The conformational change is triggered by the GDP–GTP exchange. An enhanced rate of GTP hydrolysis, which results from binding of a transfer RNA and binding to a ribosome, is apparently accomplished by interaction with loop 2 of the catalytic domain (corresponding to residues 32 – 40 in Fig. 11-7A). The function of this protein is considered further in Chapter 29.

Transducin has a 113-residue domain inserted into loop 2 of the catalytic domain. In this case, too, a large hinging movement is associated with the GDP–GTP exchange. The light-activated receptor rhodopsin induces the conformational change.^{194,231} Again, structural changes resulting from the presence or absence of interactions of the protein with the γ -phospho group

of GTP are involved. The GTPase activity of heterotrimeric G proteins can also be activated by aluminum fluoride AlF_4^- .^{232,236} From the X-ray structure of the transducin $\alpha \cdot \text{GDP} \cdot \text{AlF}_4^-$ complex it is seen that hydrated AlF_4^- is covalently bonded in the position of the γ -phospho group and makes hydrogen bonds to active site groups, mimicking a possible transition state for the GTPase reaction²³² (see Chapter 12, Section D).



The β subunit of the $\beta\gamma$ complex of transducin is a **seven-bladed β propeller** (Fig. 11-7C). It is composed of seven GH-WD repeat units: $-\text{[GH} - \text{X}_n - \text{WD}]_{4-8}-$ where GH = Gly-His, WD = Trp-Asp, and X_n is a core repeating sequence, usually 32–42 residues in length. This motif is also found in at least 40 other eukaryotic proteins.^{188,237–240} In the $\beta\gamma$ complex (Fig. 11-7B) the γ subunit assumes an elongated, largely α -helical structure. It is often anchored at its C terminus by a farnesyl or geranylgeranyl chain, while G_α may be myristoylated or palmitoylated.^{195,230,243}

The three-dimensional structure of the GDP complex of the intact transducin heterotrimer¹⁹⁵ also shows a tight interaction between α and β subunits. The major interaction is probably disrupted by replacement of the bound GDP by GTP and the conformational change that occurs around the γ -phospho group. This explains the dissociation of the α subunit from $\beta\gamma$ upon activation. An entirely similar picture has been obtained for the action of the inhibitory G protein, G_{i2} , for which structures of the α subunit and of the $\alpha\beta\gamma$ heterotrimer (Fig. 11-7,B,C) have been determined.^{188,233,234,242} The structures resemble those of transducin, but differ in details.

4. Guanylate Cyclase (Guanylyl Cyclase), Nitric Oxide, and the Sensing of Light

Formation of the less abundant cyclic guanosine monophosphate (cGMP) is catalyzed by guanylate cyclases found in both soluble and particulate fractions of tissue homogenates.^{244–247} However, its significance in metabolism is only now becoming well understood. There are cGMP-dependent protein kinases,²⁴⁷ but, until recently cGMP could not be regarded as a second messenger for any mammalian hormone. Now we know that cGMP has several essential functions. It mediates the effects of the **atrial natriuretic factor**, a peptide hormone causing dilation of blood vessels (Box 23-D),^{245,248–250} and also of a peptide called **guanylin**, which is formed in the intestinal

epithelium.^{245,251} The receptors for both of these peptides are *transmembrane proteins with cytoplasmic domains that have guanylate cyclase activity*. The released cGMP appears to induce relaxation of smooth muscles of the blood vessel walls. Another mediator of smooth muscle relaxation is the **endothelial cell-derived relaxing factor**, which is evidently **nitric oxide, NO**²⁵² (see Chapter 18). A soluble guanylate cyclase appears to be an NO receptor and is activated by binding of the NO to a heme group in one of the two subunits of the cyclase.^{253–255} The cyclase can also be activated by carbon monoxide, CO, in a similar fashion.^{256,257} Carbon monoxide is produced in the body by degradation of heme (Fig. 24-24) and is thought to be a neurohormone.²⁵⁸

Cyclic GMP also plays an important role in vision. Specific phosphodiesterases hydrolyze cGMP to GMP. The cGMP phosphodiesterases of the rod and cone cells of the retina are activated by the G protein transducin, which has been activated by light absorbed by the visual pigments.¹⁹⁴ The resulting decrease in the cGMP concentration is thought to cause the closing of cation channels and thereby to initiate a nerve impulse (see Chapter 23). Cyclic nucleotide forms of the pyrimidine nucleotides are present only in very small amounts in cells.

5. Bacterial Chemoreceptors

Bacteria are attracted to foods with the aid of chemoreceptors that bind certain amino acids such as aspartate. The receptors send signals to the mechanisms that ensure that the bacterium is swimming toward the food^{113,259, 259a} (more details are given in Chapter 19). The aspartate and serine chemoreceptors found in membranes of *E. coli* and *Salmonella typhimurium* are among the most carefully studied of all receptors. Like the seven-helix receptors, they have an extracellular sensory domain to which the signaling molecule, aspartate or serine, binds, and a transmembrane helical bundle. In these receptors the transmembrane part consists of four helical segments coming from the two subunits of the dimeric protein. For the aspartate receptor there is a high-resolution X-ray structure for the cytoplasmic domain^{260,261} including the bound aspartate²⁶² as is shown in Fig. 11-8A,B.^{260,263} The structures of the receptors for aspartate, serine, and many other signaling molecules are evidently very similar. The structure of the cytosolic domain of the serine receptor has been determined and a model for the complete receptor has been proposed.^{259a} The molecular mass of the 188-residue extracellular domain is ~18 kDa, while that of the larger cytoplasmic domains is ~36 kDa. The latter includes a linker region as well as a long four-helical bundle domain which can be divided into a methylation domain and a signaling domain as shown in (Fig. 11-8C).

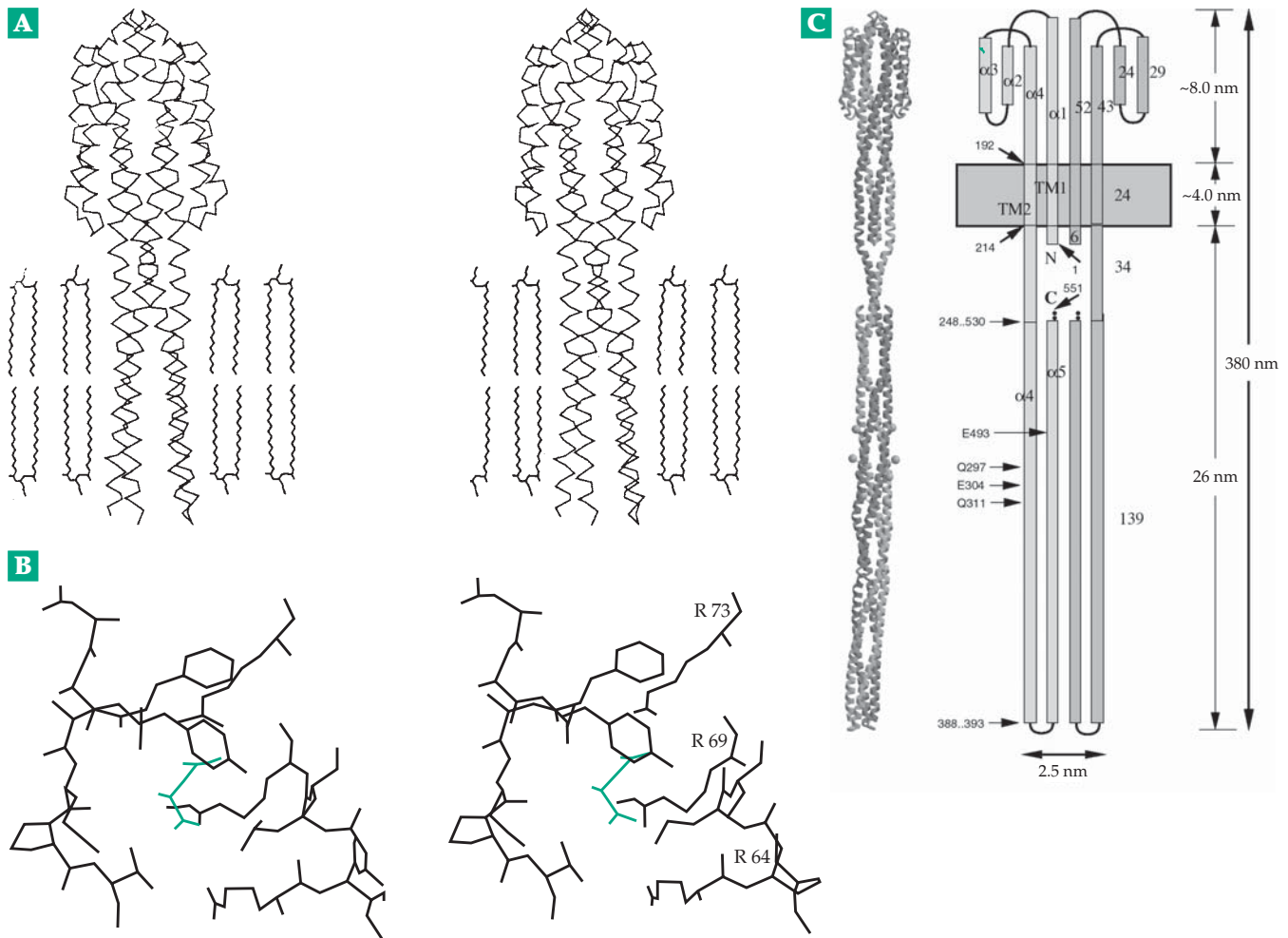


Figure 11-8 (A) Stereoscopic view of the sensory domain of the dimer with modeled transmembrane region. Each monomer contains a four α -helix bundle with two helices continuing through the membrane. From Scott *et al.*²⁶⁰ (B) Stereoscopic image of aspartate in the major binding site of the receptor. The atomic model of aspartate (green) has been fitted into the observed difference electron density map. From Yeh *et al.*²⁶² (C) Model of an intact *E. coli* serine chemoreceptor. Left. Ribbon drawing viewed perpendicular to the molecular twofold axis. Methylation sites are represented by the dark balls by the cytoplasmic domain. The bound serine is drawn as a partially hidden green ball at the upper left in the extracellular domain. Right. Diagram of the receptor. The presumed membrane bilayer is represented by the gray band. Positions of some residues are marked on the left side. On the right, the numbers of residues in various peptide segments are indicated. From Kim *et al.*^{259a} Courtesy of Sung-Hou Kim.

What happens chemically when aspartate binds into a deep pocket in the center of the sensory domain? What kind of signal can be passed from the aspartate to the “signaling domain”? Several possibilities were discussed by Kim,²⁶⁴ who suggested that the binding of aspartate causes the rotation of the sensory domain of one subunit relative to that of the other and that this rotation is transmitted through the membrane to the signaling domain. The signaling domain of the receptor forms a complex with a multimeric **protein histidine kinase** called **CheA** (chemotaxis protein A) together with an auxiliary protein **CheW**. A more likely possibility is that some kind of “piston” action occurs between

the transmembrane helices of the two subunits.^{264a} Binding of aspartate to the receptor, and the associated alterations in the receptor•CheA•CheW complex, causes a strong *decrease* in the catalytic activity of CheA.

In its active form CheA undergoes **autophosphorylation**, that is, the phosphorylation of a histidine imidazole group in one of its subunits by the protein kinase active site of an adjacent subunit. The phospho group is then transferred from phospho-CheA to another protein, **CheY**. Phospho-CheY interacts with the flagellar motor proteins (Chapter 19) periodically causing a reversal of direction of the bacterial flagella. As a result the bacteria tumble and then usually move

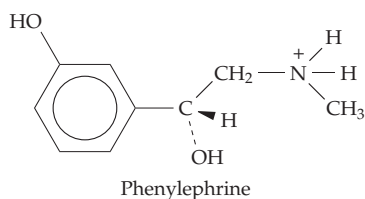
in a new direction. If the attractant molecule aspartate is present on the receptor the signal is weakened and the bacteria continue to swim in the same direction—toward food. Phospho-CheY is spontaneously hydrolyzed to remove the phospho group, but the dephosphorylation is promoted by an additional protein **CheZ**. At least two other auxiliary proteins also function in this control system. **CheR** is a **methyltransferase** that methylates the glutamate side chain carboxylates in the methylation domain of the coiled coil region and **CheB** is a **methyl-esterase** that removes the methyl groups. These two proteins control the methylation level in the coiled coil region of the receptor (Fig. 11-8A). Increased methylation increases the strength of the signal sent from the receptor, perhaps because methylation removes negative carboxylate charges that repel each other and weaken the structure, interfering with signaling.²⁵⁹ See also Fig. 19-3.

E. Calcium, Inositol Polyphosphates, and Diacylglycerols

Calcium ions entering cells from the outside or released from internal stores trigger many biological responses (see Box 6-D). Within cells Ca^{2+} often accumulates in mitochondria, in the ER, or in vesicles called **calciosomes**.²⁶⁵ Release of the stored Ca^{2+} is induced by hormones or by nerve impulses. For example, impulses flow from the nerve endings into the muscle fibers and along the invaginations of the plasma membrane called transverse tubules (Chapter 19). There they induce release of Ca^{2+} from the ER. The released ions activate enzymes²⁶⁶ and induce contraction of the muscle fibers. In many cells, Ca^{2+} causes release of secreted materials, for example, neurotransmitters in the brain.^{267,268}

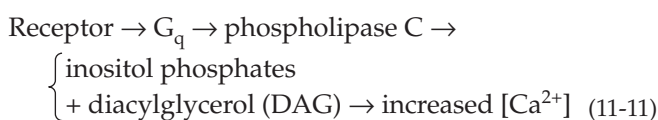
1. Alpha Adrenergic Receptors

The release of stored Ca^{2+} is often triggered by α adrenergic receptors.²⁶⁹ Like β adrenergic receptors, the α receptors are activated by adrenaline. Specific inhibitors distinguish them. For example, the β receptors are inhibited by propranolol, while the α receptors are blocked by phenoxybenzamine. The synthetic agonist **phenylephrine** activates only α receptors and no increases in cAMP or in protein kinase activity are observed.



There are two major α adrenergic receptor subtypes.^{151,269} Activation of the α_2 receptors, which are present in various tissues including blood platelets, causes *inhibition* of adenylate cyclase. This inhibition is evidently mediated by the G_i protein considered in Section D.3. The nucleotide sequences of cloned α_2 receptor genes and other properties suggest close structural similarity to the β receptors.^{270–273} Subtle differences in hydrogen bonding to the serine side chains shown in Fig. 11-6C may distinguish α_2 from β_2 receptors.²⁷³ A characteristic of α_2 receptors is that pertussis toxin (Box 11-A) abolishes the inhibition of adenylate cyclase, which they mediate. In contrast, the action of α_1 adrenergic receptors is not affected by pertussis toxin.

The α_1 receptors are activated not only by catecholamines but also by the hormones **vasopressin** and **angiotensin II**. Binding of these hormones to α_1 receptors induces a complex response that involves rapid hydrolysis of **phosphatidylinositol** derivatives and release of Ca^{2+} into the cytoplasm, and of diacylglycerols into the lipid bilayer of the membrane. The response is mediated by another G protein called G_q .^{274,275} When this G protein is activated it induces the hydrolysis of **phosphatidylinositol 4,5-bisphosphate** (PtdInsP_2), a normal minor component of the lipid bilayer, by **phospholipase C** (phosphoinositidase C).^{265,276–281}



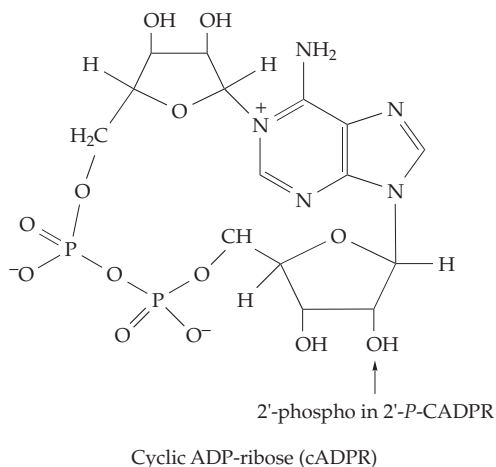
The products, **inositol 1,4,5-trisphosphate** [abbreviated InsP_3 or $\text{Ins}(1,4,5)\text{P}_3$], and **diacylglycerol** (DAG) are both regarded as second messengers for the catecholamines acting on α_1 receptors and for about 20 other hormones, neurotransmitters, and growth factors upon binding to their specific receptors.^{269,282} In addition to vasopressin, the gonadotropin-releasing hormones, histamine, thrombin (upon binding to platelet surfaces), and acetylcholine (upon binding to its “muscarinic” receptors; Chapter 30) all stimulate inositol phosphate release.

2. Phosphatidylinositol and the Release of Calcium Ions

The phosphoinositides constitute ~2–8% of the lipid of eukaryotic cell membranes but are metabolized more rapidly than are other lipids.^{265,278,279,283–285} A simplified picture of this metabolism is presented in Fig. 11-9. Phosphatidylinositol is converted by the consecutive action of two kinases into phosphatidylinositol 4,5-bisphosphate.^{286,287} The InsP_3 released from this precursor molecule by receptor-stimulated phospholipase C is thought to mobilize calcium ions by

opening “gates” of calcium channels in membranes of the ER or of calciosomes.^{282,288–289a} It diffuses across the peripheral cytoplasm to **InsP₃ receptors** which are embedded in the membranes of the ER.^{282,290} One of the several isotypes of InsP₃ receptors is a 2749-residue protein thought to contain a calcium ion channel.²⁹⁰ Similar receptors are also found in inner membranes of the nucleus.^{291,292}

Several uncertainties have complicated our understanding of the role of Ca²⁺ in signaling. What is the source of Ca²⁺? How much of it enters cells from the outside and how much is released from internal stores? Where are the internal stores? What other kinds of ion channels are present and what second messengers regulate them? The sarcoplasmic reticulum of skeletal muscle and also membranes in many other cells contain **ryanodine receptors** as well as InsP₃ receptors.^{282,293} Both of these receptors have similar structures and contain Ca²⁺ channels. However, the ryanodine receptors are activated by **cyclic ADP ribose** (cADPR),^{294,295} which was first discovered as a compound inducing the release of Ca²⁺ in sea urchin eggs.²⁹⁶ The 2-phospho derivative of cADPR may also have a similar function.²⁹⁷



Phospholipase C, which initiates the release of phosphatidylinositol derivatives, also requires Ca²⁺ for activity. It is difficult to determine whether release of Ca²⁺ is a primary or secondary response. There are three isoenzyme types of phospholipase C— β , γ , and δ —and several subforms of each with a variety of regulatory mechanisms.^{298–300a} For example, the γ isoenzymes are activated by binding to the tyrosine kinase domain of receptors such as that for epidermal growth factor (see Fig. 11-13). In contrast, the β forms are often activated by inhibitory G_i proteins and also by G_q, which is specific for inositol phosphate release.

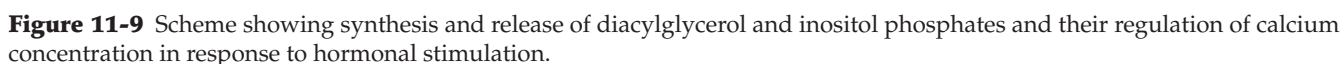
Calcium ions are usually released in distinct pulses or “quanta.” The kinetic characteristics of the system of receptors, diffusing InsP₃, calcium buffers, and calcium pumps in the cell membrane, and the membrane potential may account for this behavior.^{174,301–306} Since

the phosphoinositols are chelators of Ca²⁺, the equilibria involved in this control system are complex.³⁰⁷

In addition to InsP₃, several other inositol derivatives are released by adrenergic stimulation. **Inositol 1,3,4,5-tetrakisphosphate** (InsP₄) is formed from InsP₃ by action of a soluble kinase.^{308,309,309a,b} A controversial suggestion is that InsP₄ may induce opening of Ca²⁺ channels through the outer membrane of the cell^{310–312} and may also function to promote storage of Ca²⁺. It also acts as a transcriptional regulator. Other possible second messengers are inositol 1,2-cyclic-3,4-trisphosphate and related metabolites that arise by the action of phospholipase C on PtdIns 4-P followed by additional actions of kinases and phosphatases.^{278,309,313–315} A different inositol tetrakisphosphate, Ins(3,4,5,6)P₄, may control chloride ion channels.^{316,317} Both inositol 1,3,4,5,6-pentakisphosphate (**InsP₅**) and inositol hexakisphosphate (**InsP₆**) are also found in plants³¹⁸ and animals.³¹⁹ InsP₅ serves as an allosteric effector regulating hemoglobin in avian erythrocytes (Chapter 7). InsP₆, also known as **phytate**, is present in large amounts in cereals and recently has been found to play a role in regulating the export of mRNA from the nucleus.^{319a} Additional phospho groups can be added to InsP₆ to form pyrophosphates^{320,320a} and other complex polyphosphates.³²¹ Both InsP₅ and InsP₆ can also serve as precursors to Ins(1,3,4,5)P₄ and Ins(1,4,5)P₃ by hydrolytic dephosphorylation.³²²

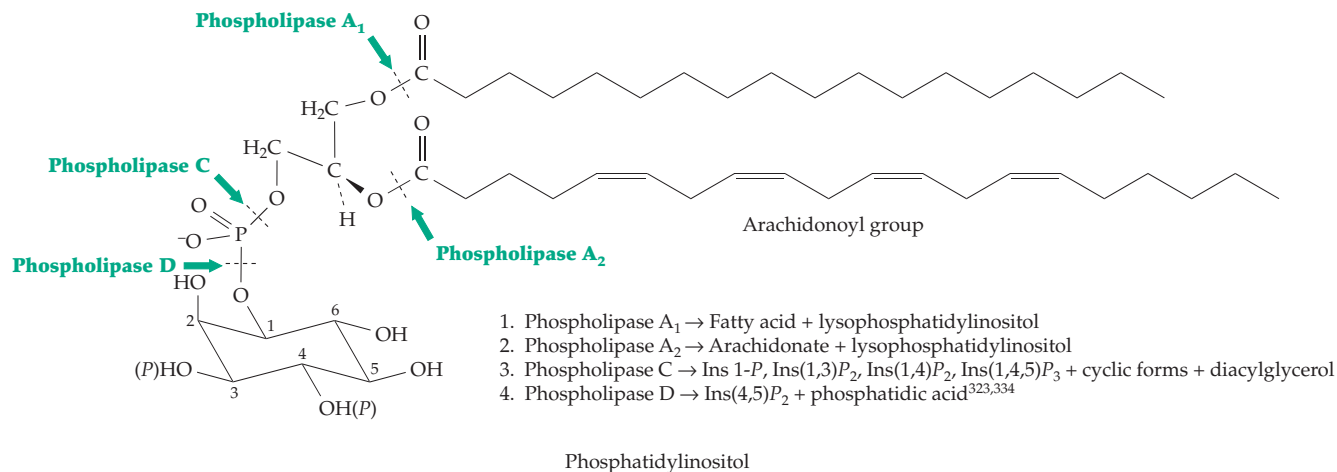
Following stimulation of a cell the induced metabolism of phosphoinositides decays rapidly. The diacylglycerols are converted into phosphatidic acid and resynthesized into phospholipids (Chapter 21). The InsP₄, InsP₃, and other inositol metabolites are hydrolyzed by phosphatases.^{278,309,323–327} Two of these phosphatases are inhibited by Li⁺ as indicated in Fig. 11-9. They may represent one site of action of **lithium ions** in the brain.^{324,327,328} Lithium salts are one of the most important drugs for treatment of **bipolar (manic-depressive) illness**. By blocking the release of free inositol, which can be resynthesized into PtdInsP₂, Li⁺ may prevent neuronal receptors from becoming too active. However, the basis for the therapeutic effect of lithium ions remains uncertain.

The diacylglycerols released by phospholipase C diffuse laterally through the bilayer and, together with the incoming Ca²⁺, activate **protein kinases C**. These kinases also require **phosphatidylserine** for their activity and phosphorylate serine and threonine side chains in a variety of proteins.^{329–330b} They are stimulated by the released unsaturated diacylglycerols. In addition protein kinases C can be activated by **phorbol esters**, which are the best known tumor promoters (Box 11-D). The diacylglycerol requirement favors a function for these protein kinases in membranes. They also appear to cooperate with calmodulin to activate the Ca²⁺-dependent contraction of smooth muscle.³³⁰

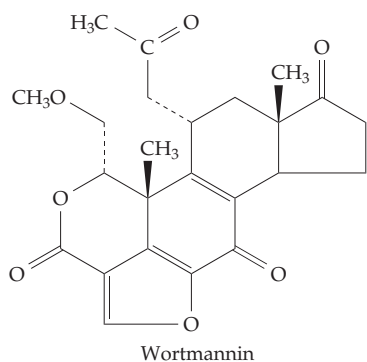


are also indicated in the following structure.^{298,333}

Initially most attention was paid to the water-soluble inositol phosphates that are released from phosphoinositides. However, phosphoinositol derivatives that retain the diacylglycerol part of the molecule have regulatory functions while remaining in membranes. A phosphatidylinositol 3-kinase phosphorylates the 3-OH of inositol in PtdIns, PtdIns 4-*P* and PtdIns (4,5)*P*₂ to give PtdIns 3-*P*, PtdIns(3,4)*P*₂, and PtdIns(3,4,5)*P*₂, respectively.^{335–337d} However, they are ideally suited to function as spatially restricted membrane signals.^{337a} They affect protein kinase C as well as the Ser / Thr protein kinase Akt (discussed in Section F3),^{326b} and have several functions in vesicular membrane transport and in regulation of the cytoskeleton.^{337e,f}



Characteristic of many of the phosphoinositide-regulated proteins is a proline-rich PH domain (Table 7-3) which can transmit regulatory signals to additional proteins in a cascade. Their importance is emphasized by the finding that a 405-residue phosphatase **PTEN** (named after its gene symbol) which catalyses hydrolytic removal of one phospho group from PtdIns(3,4,5) *P*₃ to form PtdIns(3,4) *P*₂, (Fig. 11-9) is a major human tumor suppressor (see Box 11-D).^{338,338a,b} These lipids are also important in insulin action (Section G,3). The steroid-like fungal metabolite wortmannin is a specific inhibitor of the 3-kinase.^{339,339a}



F. Regulatory Cascades

The effect of a regulated change in the activity of an enzyme is often amplified through a cascade mechanism. The first enzyme acts on a second enzyme, the second on a third, etc. The effect is to rapidly create a large amount of the active form of the last enzyme in the series.

We have already considered regulatory cascades initiated respectively by the β and α_2 adrenergic receptors. The effect of these cascades on glycogen phosphorylase is outlined on the left side of Fig. 11-4. One branch of the cascade sequence begins with release of

adrenaline under control of the autonomic nervous system. In muscle, binding of this hormone to the β receptors on the cell membrane releases cAMP which activates a protein kinase. The kinase then phosphorylates phosphorylase kinase. At this point the muscles are prepared for the rapid breakdown of glycogen. However, an additional initiating signal is the release of Ca^{2+} into the cytoplasm in response to impulses to specific muscles via the motor neurons. Calcium ions can also be released in the liver by α adrenergic stimulation. Phosphorylase kinase is activated by the calcium ions, and in their presence it converts inactive phosphorylase *b* to the active phosphorylase *a*. Both protein kinases and phosphorylase kinase also act on glycogen synthase, phosphorylating it and converting it to an inactive form. This turns off the biosynthetic pathway at the same time that glycogenolysis (glycogen breakdown) is turned on. Spontaneous reversion of the enzymes to their resting states occurs through the action of phosphatases that cut off the phospho groups placed on the protein by the kinases. Also essential are phosphodiesterases, which destroy the cAMP, and the calcium ion pump, which reduces the concentration of the activating calcium ion to a low level.

Elaborate cascades initiate the clotting of blood (Chapter 12) and the action of the protective complement system (Chapter 31). Cascades considered later in the book are involved in controlling transcription (Fig. 11-13) and in the regulation of mammalian pyruvate dehydrogenase (Eq. 17-9), 3-hydroxy-3-methylglutaryl-CoA reductase and eicosanoids (Chapter 21), and glutamine synthetase (Chapter 24).

1. Advantages of Regulatory Cascades

Computer simulations as well as studies of experimental models have led to the following conclusions.^{340,341} Even simple cascade mechanisms, such as the one shown in Fig. 11-10, can provide a more flexible response

to allosteric effectors (such as e_1 in the figure) than if the effector acted directly on the enzyme rather than on the protein kinase. The cascade also provides **amplification**. This is especially true if additional cycles are added. A response can result from the binding of only a small number of hormone molecules to a receptor in a cell membrane or from activation of only a few molecules of a protease in the initiation of blood clotting. A striking amplification occurs in visual responses. Under appropriate conditions a single quantum of light falling on a receptor cell in the retina of an eye can initiate a nerve impulse (Chapter 23). The latter requires the flow of a large number of Na^+ ions across the plasma membrane. It would be hard to imagine how absorption of one quantum could initiate a photochemical reaction leading to that much sodium transport without intermediate amplification stages. Another advantage of cascades is that they may provide **ultrasensitive responses**. Not only can a response be sensitive to a higher power than the first of the concentration of a signaling molecule but also the amplification provided by the cascade confers a high sensitivity to the response.³⁴²

Cascade systems also provide for response to more than one allosteric stimulus in a single pathway. Thus, as shown in Fig. 11-4, glycogen catabolism can be initiated in more than one way. Two pathways are known for initiation of both blood clotting and activation of the complement system. Many pathways activate the MAP kinase pathway shown in Fig. 11-13.

2. Substrate Cycles

Although cascade systems offer advantage to cells, there is a distinct energy expenditure associated with the controls.³⁴³ As can be seen in Fig. 11-10, in addition to providing for turning on and turning off the regulated enzyme, the kinase and the phosphatase together catalyze hydrolysis of ATP to ADP and inorganic phosphate. Thus, the regulated enzyme is continually cycling between active and inactive forms. The relative amounts of each are determined by the amounts of the effectors e_1 and e_2 , the concentrations of modifying enzymes, and the kinetic constants. Such cycles have sometimes been called “futile cycles” because they seemingly waste ATP. This is particularly true of cycles that involve major metabolites, sometimes called **substrate cycles**. An example is the conversion of fructose 6- P to fructose 1,6-bisphosphate by phosphofructokinase and hydrolysis of the bisphosphate back to fructose 6- P by a phosphatase (Fig. 11-2). However, *the cycles are not futile* because the hydrolysis of the ATP provides the energy required to maintain the concentrations of modified (phosphorylated) enzyme or enzymes at steady-state levels that are required for efficient catalysis.^{341,344} The energy utilized in this way

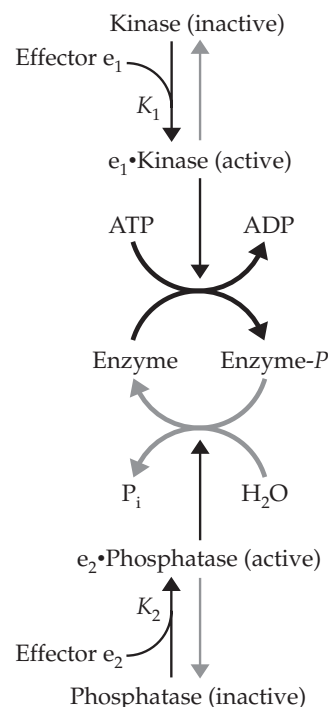


Figure 11-10 A “monocyclic” regulatory cascade involving phosphorylation and dephosphorylation of an enzyme. After Stadtman and Chock.³⁴¹

in regulation of enzymes is often small.³⁴⁵ Substrate cycles are not only an unavoidable consequence of the need to regulate enzymes but also provide for improved sensitivity in metabolic control. Consider Fig. 11-2 again. If the enzymes are set for conversion of fructose 1,6- P_2 to glycogen (backwards direction), the flux of materials flowing in the forward direction (glycolysis) will be efficiently curtailed because any fructose 1,6- P_2 formed will be hydrolyzed rapidly by the active phosphatase. Under these circumstances the flux in the forward direction will be ultrasensitive to the activation of phosphofructokinase and to the inhibition of fructose 1,6-bisphosphatase.^{346,347}

G. Insulin and Related Growth-Regulating Hormones

Since its isolation in 1921, insulin has been the object of an enormous amount of experimentation aimed at clarifying its mode of action. It is produced by the β cells of the pancreatic islets of Langerhans and released into the bloodstream in response to elevated glucose levels. The absence of insulin or of a normal response to insulin results in the condition of **diabetes mellitus**, which is the most prevalent human metabolic disorder (see Box 17-G).³⁴⁸

Insulin or insulin-like material is also produced in ciliated protozoa, vertebrate and invertebrate animals, fungi, green plants,^{349,350} and even in *E. coli*.³⁵¹

1. Metabolic Effects of Insulin

Insulin has many effects on metabolism.^{348,352,353} (Some of these are listed in Table 17-3). They can be summarized by saying that: (1) In most tissues insulin stimulates synthesis of proteins, glycogen, and lipids. (2) It affects the permeability of membranes, promoting the uptake and utilization of glucose and amino acids and of various ions from the blood. (3) It promotes both synthesis of glycogen and the breakdown of glucose by glycolysis. At the same time, it inhibits synthesis of glucose from amino acids by the gluconeogenesis pathway. The effects are not uniformly the same in all tissues. Some can be observed within a few minutes after administration, presumably as a result of regulation of enzymes. Effects on mRNA metabolism, protein synthesis, and cell growth are seen at later times.

The uptake of glucose by brain, liver, kidneys, erythrocytes, and the islets of Langerhans is unaffected by insulin. However, in muscle and adipose tissues insulin stimulates glucose uptake. Part of this effect results from insulin-induced translocation of molecules of the 509-residue glucose transport protein GLUT4 (Chapter 8) from the cytosol into the plasma membrane where it can function.^{354–356a} Insulin apparently also increases the rate of synthesis of the transporters.

Insulin stimulates the phosphorylation of serine side chains of many proteins, including ATP citratelase, acetyl-CoA carboxylase, and ribosomal subunit S6. At the same time it stimulates the dephosphorylation of other proteins, including acetyl-CoA carboxylase, glycogen synthase in skeletal muscle (Fig. 11-4),⁴⁹ pyruvate dehydrogenase, and hormone-sensitive lipase in adipose tissue. Yet another effect of insulin is to alter the amounts of specific messenger RNA molecules. For example, the transcription of the gene for phosphoenolpyruvate carboxykinase (PEPCK; Eq. 13-46) a key enzyme in gluconeogenesis, is inhibited by insulin within seconds after binding. The hexokinase isoenzyme called **glucokinase** phosphorylates glucose to glucose-6-*P* in liver and in pancreatic β cells. Its synthesis in liver is induced by insulin. (However, in the β cells the synthesis of glucokinase is induced by glucose.)

2. Insulin Receptors

All of the effects of insulin appear to result from its binding to insulin receptors, of which $\sim 10^2$ to 10^5 are present in the plasma membranes of most animal cells. First isolated in 1972,³⁵⁷ insulin receptors and

their cloned genes have been studied intensively. The receptors are $\alpha_2\beta_2$ disulfide crosslinked oligomers composed of pairs of identical 120- to 135-kDa α subunits and 95-kDa β subunits. An α and a β subunit are cut from a single precursor chain. The human insulin receptor precursor exists as two isoforms, A and B, which arise as a result of a difference in splicing of the mRNA.³⁵⁸ Twenty-one introns are removed by splicing to form the mRNA for the 1355-residue B precursor.^{348,358a} The mRNA for the A form lacks a 36-nucleotide segment (exon 11) that is discarded during splicing. The α chains come from the N-terminal part of the precursor and the β chains from the C-terminal part. The receptor sequence is numbered as in the longer B form precursor. (However, many authors number the chains of the A form receptor as in its precursor, 12 less than the numbers given here for residues 719 or higher³⁵⁹). The B form receptor has 731-residue α chains while the A form has 719-residue α chains as a result of the missing sequence from exon 11 (see Fig. 11-11A). Four residues (732–735 in the B form) are cut out and discarded, leaving 620-residue β chains for both isoforms. The chains become linked by three or more disulfide crossbridges.³⁶⁰

The two α subunits, which contain the insulin binding sites, are apparently present entirely on the outer surface of the plasma membrane. The β subunits pass through the membrane with their C termini in the cytoplasm.^{361–363} Both α and β subunits are glycoproteins. Study of the amino acid sequences suggests that only one 23-residue segment (residues 930–952 of the B isoform) of hydrophobic amino acid residues in each β subunit is likely to exist as an α helix that spans the membrane. This raises several questions. How can a signal be sent from the cell surface into the cytoplasm through a single pair of α helices? Are there additional parts of the receptor that span the bilayer of the membrane? To make it easier to visualize these questions refer to Fig. 11-11B, which is a more realistic, although fanciful, drawing than that in Fig. 11-11A. Electron microscopy and crystallography are now providing the first direct images of the receptor.^{363a, b, c}

One possible way in which a signal could be sent through the membrane is for binding of insulin to promote aggregation of two or more receptors.³⁶⁴ If insulin induces the receptors to stick together on the outside of the cell, the parts protruding into the cytoplasm would also tend to aggregate. This could induce a response. A second possibility is that a conformational change in the α subunits pulls or pushes on the β subunit allowing the latter to be exposed less or more on the cytoplasmic side. This difference might be sufficient to cause a conformational change in the cytoplasmic domain of the subunit. A third possibility involves activation by a twisting mechanism as proposed for the aspartate receptor (Fig. 11-8).

A large part of the cytoplasmic domain of the insulin

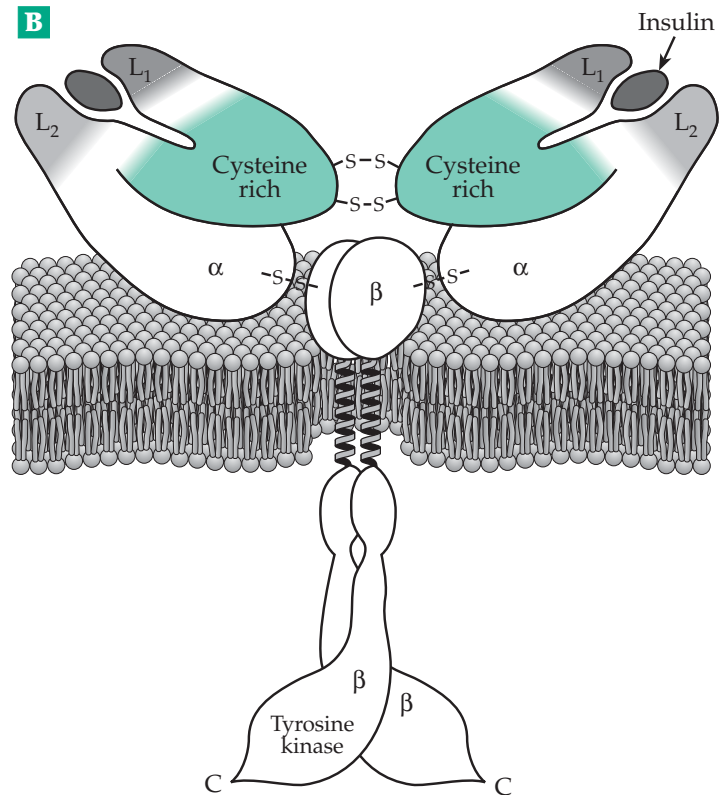
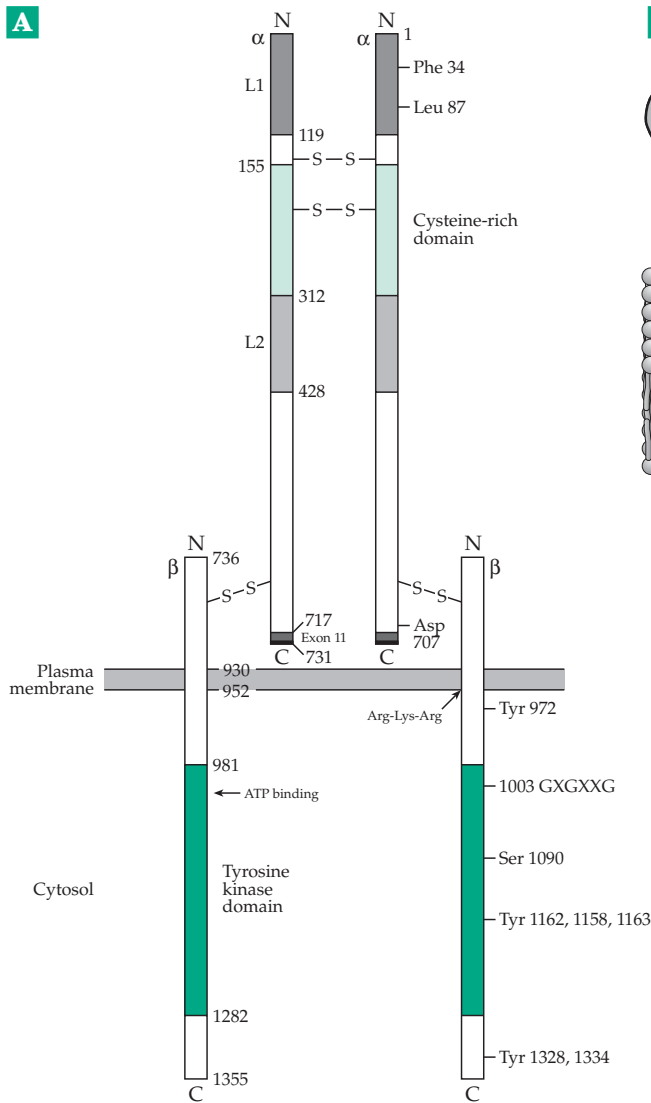


Figure 11-11 Schematic drawings of the insulin receptor. (A) Typical textbook drawing. The bars represent the extended peptide chains with positions of domain boundaries and chain ends mapped (left side). Locations of some residues of special interest are marked on the right side. (B) A fanciful but more realistic picture of the receptor. See Fabry *et al.*³⁶⁷

receptor consists of a tyrosine-specific protein kinase whose three-dimensional structure is depicted in Fig. 11-12.^{365,365a} Not only can it phosphorylate –OH groups on tyrosine side chains of other proteins³⁶² but also it catalyzes ATP-dependent **autophosphorylation** of several residues in the C-terminal region. The most readily phosphorylated residue is Tyr 1158.³⁶⁶ Some diabetic individuals have receptors with impaired tyrosine kinase activity.³⁶²

Using directed mutation of the cloned receptor gene, Lys 1018 in the ATP-binding part of the tyrosine kinase domain was replaced by alanine. This caused a loss both of kinase activity and of biologic response to insulin.³⁶² Thus, both the tyrosine kinase activity and autophosphorylation appear essential. If so, aggregation of two or more receptors may increase the extent of autophosphorylation and initiate a response.

Studies of many mutant proteins indicate that insulin binds to the α chains of the receptor between

the two domains labeled L1 and L2 in Fig. 11-11.³⁶⁸ Among essential residues is Phe 39 (marked).³⁶⁹ Insulin contains three disulfide linkages (Fig. 7-16) and might undergo a thiol–disulfide exchange reaction (Eq. 11-7) with SH groups present in a cytoplasmic cysteine-rich domain of the α subunit of the receptor or with an external thiol compound.^{369a,b} Such an exchange may also be essential for activation of the receptor.³⁷⁰

3. A Second Messenger for Insulin?

The known regulatory effects of insulin (Table 17-3) often involve phosphorylation of serine or threonine side chains on specific proteins. The tyrosine kinase of the activated insulin receptors does not catalyze such phosphorylation. Therefore, it seems likely that one or more second messengers or mediator substances are needed. Much effort has gone into searching for

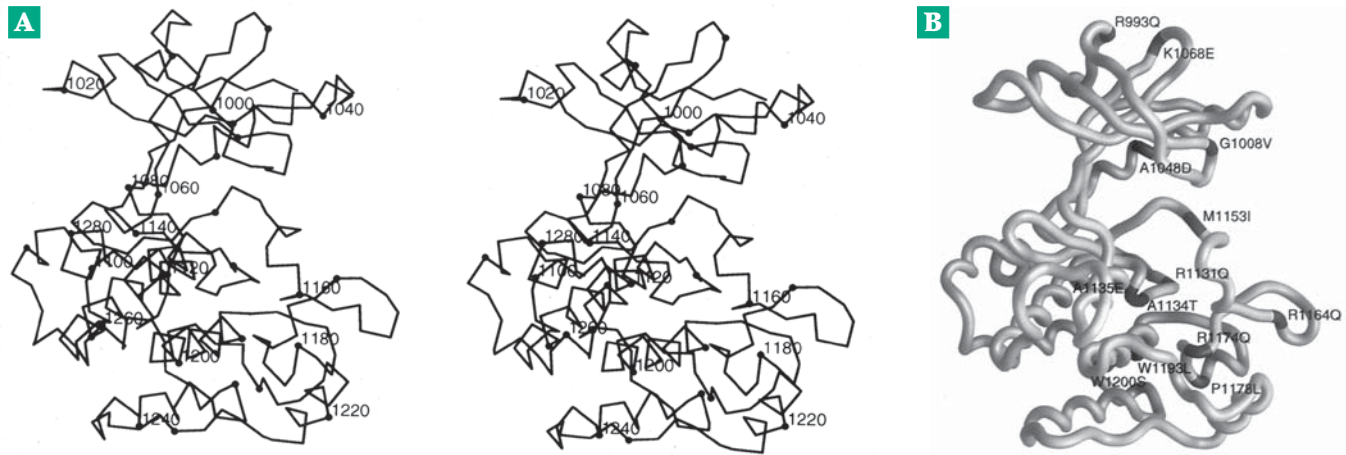


Figure 11-12 (A) Stereoscopic view of an α -carbon trace of the insulin receptor kinase domain. Every tenth residue is marked with a filled circle and every twentieth residue is labeled. (B) Locations of missense mutations in noninsulin-dependent diabetes mellitus patients mapped onto the receptor kinase structure. The mutations are R993Q, G1008V, A1048D, R1164Q, R1174Q, P1178L, W1193L, and W1200S. Here, R993Q is the mutant in which arginine 993 is replaced with glutamine, etc. From Hubbard *et al.*³⁶⁵

substrates for the tyrosine kinase,^{353,362,371,372} whose activity on external protein substrates reaches a maximum after receptor tyrosines 1158, 1162, and 1163 have been autophosphorylated. Phosphorylation of such **insulin receptor substrates (IRSs)** as well as of some smaller “adapter” proteins is thought to initiate a series of complex cascades that serve to pass the insulin signal on to a variety of sites of action within a cell.^{373–376} The large 185-kDa **insulin receptor substrate-1 (IRS-1)**, which is present in most cells, is phosphorylated on several tyrosines, usually within the sequences YXXM or YMXM. Phosphotyrosine within such sequences is known to be a ligand for the recognition domain SH2, which is present in many proteins. Binding of SH2 domains of other proteins to IRS-1 allows the insulin signal to be passed to several different proteins in the branched signaling pathways. Other insulin receptor substrates include **IRS-2**,^{376a,b} protein Gab-1^{376c}, and the smaller protein called **Shc**, which occurs as 46-, 52-, and 66-kDa isoforms and is discussed further on p. 568.^{376,377} Shc is an adapter protein which forms a complex that triggers the activation of the G protein Ras and the MAP kinase cascade shown in Fig. 11-13.^{377,378} This pathway (see Section H,2) is thought to mediate the mitogenic (growth promoting) effects of insulin and also to promote phosphorylation of serine and threonine side chains of many proteins in the cytoplasm, the cytoskeleton, ribosomes, membranes, and the nucleus.^{379,380} One serine kinase specifically phosphorylates the insulin receptor on serine 1078.³⁸¹

In addition to IRS-1, IRS-2, and Shc, there are additional adapter proteins that interact with the

phosphorylated tyrosine kinase domain of the insulin receptor.³⁸² Furthermore, the receptor tyrosine kinase may catalyze *direct* phosphorylation of some proteins, e.g. a cytoplasmic loop of the β_2 adrenergic receptor (Fig. 11-6), without intervention of an adapter.³⁸³

Phosphorylated IRS-1 activates a second signaling pathway by interacting with an 85-kDa SH2-containing protein that is a subunit of phosphatidylinositol 3-kinase.^{384–386} This activates the 110-kDa catalytic subunit of the 3-kinase, which catalyzes formation of phosphatidylinositol 3-phosphate as well as PtdIns (3,4) P_2 and PtdIns (3,4,5) P_3 .^{387,387a} These compounds, which remain within membranes, activate other branches of the signaling cascade, some of which may converge with those of the MAP kinase cascade. However, there appears to be specific activation of a ribosomal Ser/Thr kinase that, among other activities, phosphorylates ribosomal protein S6, a component of the small ribosomal subunit.³⁸⁸ It also phosphorylates some isoforms of protein kinase C and other enzymes. PtdIns 3-kinase may also activate 6-phosphofructo-2-kinase (Fig. 11-2, step *d*).^{384,388}

One of the most important effects of insulin is to increase glucose uptake by cells.^{373,389} The mechanism is thought to depend upon the transporter protein GLUT4, which is stored within the membranes of small cytoplasmic vesicles. Binding of insulin to its receptors induces movement of these vesicles to the plasma membrane where fusion with the plasma membrane makes the GLUT4 molecules available for glucose transport.³⁹⁰ Phosphatidylinositol 3-kinase also plays an important role. The PtdIns(3,4) P_2 and PtdIns(3,4,5) P_3 generated by this enzyme (Fig. 11-9)

remain in the membrane and provide sites to which various proteins, e.g., those containing PH domains (Chapter 7) may bind. Among the proteins “recruited” to the inner membrane surface in this way is a Ser/Thr protein kinase known as PKB/Akt. The Akt abbreviation refers to its relationship to a particular viral oncogene.^{390a–390e} and the phosphatidyl-inositol derivatives generated by the 3-kinase may have a direct effect upon the exocytosis of the GLUT-4 containing vesicles as may Gq-coupled receptors.³⁹¹

Although insulin unquestionably stimulates phosphorylation of many proteins, the major metabolic effect appears to result from *dephosphorylation* by phosphatases of phosphorylated forms of several enzymes.^{379,392} These include **glycogen synthase** and **pyruvate dehydrogenase**, which are both activated by dephosphorylation, and **glycogen phosphorylase**³⁹³ and **hormone-sensitive lipase**, which are deactivated by dephosphorylation. These changes result in increased synthesis and storage of both glycogen and triacylglycerols. Insulin also stimulates a membrane-bound cAMP phosphodiesterase causing a reduction in cAMP concentrations.³⁹⁴

Insulin injected into the human body has a half-life of only 5–10 minutes. Much of the hormone is destroyed by hydrolytic cleavage of the peptide chain by a non-lysosomal protease.^{395–397} The receptors together with the hormone are taken into cells by endocytosis from clathrin-coated pits and both may be degraded in the lysosomes.^{396,398} Some signaling may arise from insulin receptors in endosomes.³⁹⁹ Some of the hormone enters the nucleus, perhaps still bound to the receptors,⁴⁰⁰ and may have a direct effect upon gene transcription. Some receptors are recycled to the cell surface.³⁹⁸ This latter process, together with that of new receptor synthesis, controls the number of receptors on the surface and, in turn, the sensitivity of the cell to insulin. This is only one mechanism used for desensitization or **downregulation** of receptor sensitivity and numbers, something that happens normally for any hormone when its concentration is high.^{401,402}

H. Growth Factors, Oncogenes, and the Cell Cycle

Insulin is just one of a large number of proteins that are secreted by cells and which influence the growth of nearby cells.^{403–405} For example, blood platelets, which aggregate at the site of an injury to a blood vessel, contain granules (α granules) which, when the platelet is “activated” (see Box 8-A) release **platelet-derived growth factor** (PDGF), a 28- to 31-kDa glycosylated peptide which stimulates growth and tissue repair in the injured region.^{406,407} There are many other protein growth factors, a few of which are considered here. Others are discussed in Chapters 30–32.

1. Oncogenes and Proto-oncogenes

Studies of cancer-causing cellular oncogenes and **proto-oncogenes** (**c-onc**) identified in the human genome^{408–411} have contributed greatly to our understanding of the action of growth factors. Proto-oncogenes are segments of DNA that code for proteins that have a normal function but which may become “activated” by a mutation or by chromosomal rearrangement to become cancer-causing oncogenes (Box 11-D). Oncogenes were discovered first in oncogenic (cancer-causing) retroviruses,⁴¹² where they are designated **v-onc**. The **v-onc** genes are usually very similar but not identical to the corresponding **c-onc** genes and are thought to have arisen from them. The **v-onc** genes are often incomplete or have become fused to other genes by genetic recombination. Considerable excitement has attended the discovery that *many solid human tumors contain activated oncogenes*,⁴¹³ several of which are listed in Table 11-3.

In 1983 it was found that the oncogene *sis*, which is carried by the simian sarcoma virus, has a 104-residue sequence nearly identical to that of human PDGF.^{404,414} The PDGF receptor has a large extracellular domain consisting of five immunoglobulin-like domains. Like the insulin receptor, it has tyrosine kinase activity which resides in a C-terminal cytoplasmic domain.^{406,407,415} This suggests that the malignant transformation of cells by the viral *v-sis* gene leads to an excessive production of normal PDGF and, consequently, to excessive growth. The sequences related to *v-sis* are often found in human tumors and are located on chromosome 22.⁴¹⁶ Rearrangements by which part of this chromosome is moved to another location are well known and sometimes lead to conversion of the proto-oncogene to an active cancer-inducing oncogene.

The avian *v-erbB* oncogene and oncogene *neu*^{417–419} both have sequences homologous to that of the gene for the receptor for the 53-residue **epidermal growth factor** EGF.^{420–423a} The corresponding cellular *c-erbB* is the gene for the EGF receptor, a large 170-kDa 1186-residue protein. It also resembles the insulin receptor in having an N-terminal domain outside the cell, a single hydrophobic helix that spans the membrane, and a cytoplasmic domain with tyrosine-specific protein kinase activity.^{403,419,424,425} It differs from the insulin receptor in being a single peptide chain. However, when EGF binds, the receptor dimerizes and the protein kinase is activated.⁴²⁶ It phosphorylates its own Tyr 1173 near the C terminus, as well as tyrosines in the **lipocortins**, 36- to 38-kDa calcium-binding proteins located on the cytosolic face of plasma membranes.⁴²⁷ The activated receptor also stimulates phospholipase C with a resultant increase in concentrations of inositol triphosphate and Ca^{2+} (see Fig. 11-13).⁴²⁸ The active oncogene *v-erbB* encodes a major fragment of the EGF receptor. However, most of the N-terminal domain

TABLE 11-3
A Few Oncogenes That Have Interested Biochemists

Oncogene Symbol	Source	Properties
<i>v-sis</i>	Simian sarcoma virus	Gene product is closely related to the B-chain of platelet-derived growth factor
<i>v-erbB</i>	Avian erythroblastosis virus	Gene product is shortened version of the EGF receptor, a tyrosine kinase
<i>neu</i>	(Her-2) Rat neuroblastomas A similar gene is found in human breast cancers and adenocarcinomas	Homologous to EGF receptor; has tyrosine kinase activity; may control phosphatidylinositol 3-kinase
<i>v-src</i>	Rous avian sarcoma virus	Gene product is another tyrosine-specific protein kinase
<i>abl</i>	Chronic myelogenous leukemia	Similar to <i>src</i> ; characteristic chromosomal translocations yield cancer
<i>ras</i> , <i>has</i> , <i>v-H-ras</i> <i>v-K-ras</i> <i>N-ras</i>	Human bladder, colon, lung carcinomas	Homology with G _s and G _i regulatory proteins; several human proto-oncogenes exist; a single mutation in <i>H-ras</i> may lead to cancer
<i>bas</i> , <i>kis</i>	Murine sarcoma viruses	
<i>v-myc</i>	Burkitt's lymphoma, mouse plasmacytomas, avian retrovirus MC29	Nuclear location for gene product; which forms complex with protein Jun
<i>mos</i>	Burkitt's lymphoma, mouse plasmacytomas	
<i>v-fos</i>	Osteosarcoma virus (mouse)	Nuclear location for gene product, which forms complex with protein Jun
<i>v-jun</i>		Protein product is subunit of transcription factor AP-1
<i>bcl1</i>		Cyclin D1

including the EGF binding site is missing and the C-terminal end has been shortened. The tyrosine kinase domain is intact but Tyr 1173 is missing.

The rat *neu* oncogene (also called *erbB-2* and *HER-2*), which apparently is derived from the gene for another growth factor receptor, differs from normal *neu* by a single nucleotide. This change causes valine to be substituted for glutamic acid at position 664 in the membrane-spanning domain of the 185-kDa protein.⁴¹⁸ This evidently gives an overactive and perhaps uncontrolled tyrosine-specific protein kinase. The *c-erbA* proto-oncogene appears to be the nuclear receptor for the thyroid hormone **triiodothyronine** (Chapter 25).

The Rous sarcoma virus oncogene *v-src* and a family of related oncogenes are derived from protein tyrosine kinases that are attached with the aid of a

myristoyl anchor to the inner, cytoplasmic surfaces of membranes.^{429–431} They may be activated by interaction with an occupied surface receptor. It has been difficult to understand the functioning of the normal Src protein. However, inactivation of *c-src* in mice caused the serious bone disease **osteopetrosis** in which the osteoclasts fail to function properly in resorbing the bone matrix, thereby allowing excessive accumulation of calcium phosphate.^{432,433} A *c-src* deficiency also decreases formation of the bone adhesion protein **osteopontin**, an RGD protein.⁴³⁴ There is actually a family of src proteins, some of which have important functions in lymphocytes.^{435–438} The gene for one of these is mutated in the β cell disorder **agammaglobulinemia**.⁴³⁷

Another oncogene derived from a tyrosine kinase

BOX 11-D CANCER

Although cancer occurs in about 200 clinically distinct types, most cancers can be classified into four categories. In **leukemias**, which account for 3% of the ~700,000 cases of cancer diagnosed per year in the United States, an abnormal number of leukocytes are produced by the bone marrow. **Lymphomas**, such as Hodgkin's disease and Burkitt's lymphoma, arise from lymphocytes. They account for ~5% of human cancers. In these diseases malignant cells are produced in the spleen and lymph nodes and sometimes aggregate in lymphoid tissues. **Sarcomas**, solid tumors of bone or other connective tissue, contribute ~2% to the total of human cancers, while **carcinomas**, cancers of epithelial tissue, account for 85%. Carcinomas may develop from either the external or the internal epithelia, including the glands, lungs, and nerves.^{a-d} More than one-third of all cancers in the United States are nonmelanoma carcinomas of the skin;^e for these the mortality rate is low. Lung, colorectal, breast, and prostate tumors account for 55% of cancer deaths.^c Carcinomas predominate in humans, but lymphomas, leukemias, and sarcomas are much more prevalent in laboratory animals and fowl.

An important characteristic of cancer cells is their uncontrolled proliferation. They don't respond to the normal signals from adjacent cells that indicate that cell division should stop. Cancer cells also differ dramatically from those present in warts and other benign tumors and in psoriasis. These conditions also result in excessive proliferation of cells and partial derangement of normal regulatory processes.

A second characteristic of cancer cells is that they usually appear less differentiated than the tissues from which they arise and are more like embryonic cells. Many cancers produce **ectopic proteins**, proteins inappropriate to the tissue involved and often identical to proteins synthesized by embryonic or fetal cells. A well-known example is **α -fetoprotein**, a 72-kDa glycoprotein normally present in serum in almost undetectable amounts but present in large amounts when some types of cancer are present.^f A third property of cancers is the tendency toward **metastasis**, the detachment of cells from the cancer and their development in distant parts of the body.^{g-i}

Cancer cells don't grow any faster than normal cells but they continue to divide when normal cells would not. For this reason, a cancer can grow rapidly and its demands for nutrients can literally starve the host. Cancer tends to weaken the immune system, making the host more susceptible to infections. In addition, cancers often interfere directly with the functioning of various organs and may cause death in this way.

What initiates a cancer? We know that cancers can arise from only one or a very small number of cells and that cancer can be induced by carcinogenic chemical compounds, by certain viruses, and by radiation. Use

of tobacco appears to be responsible for about 30% of all human tumors.^{c,j} Diet also affects the likelihood of developing cancer. For example, diets containing less animal fat and more fruits and vegetables are associated with lower levels of colon cancer.^{c,k} Many carcinogenic compounds are naturally present in foods. Genetic factors help to determine susceptibility to cancer^{b,l} and characteristic chromosomal aberrations are usually associated with cancer.^{m-o} The incidence of cancer increases markedly with age.

A common feature of all the agents that induce cancer is the production of mutations and cancer probably always involves some alteration in the cell's DNA. The long lag between exposure to carcinogenic materials and development of cancers, often 20 years or more, suggested that more than one mutation or chromosomal rearrangement is required for production of a cancer. Recent evidence confirms that several mutations are required.^{b,p} Relevant to this conclusion is the fact that carcinogenic compounds can be applied to the skin in amounts sufficient to cause a number of mutations in the epithelial cells but insufficient to actually induce cancer. Then, even many years later, irritant compounds known as **cancer promoters** can be applied and cancer will develop promptly. The promoters apparently induce cell proliferation which leads to more errors in DNA replication, converting an initially mutated cell to a cancerous cell. The most studied promoters are the phorbol esters, which are known to activate the protein kinase C isoenzymes (Section E,2). Any factors that increase rates of cell division such as some hormones, excess calories, or chronic inflammation cause increased cancer.^c Cell divisions are accompanied by errors in replication of DNA and sometimes by translocation or deletion of parts of chromosomes. Chronic infection by bacteria, viruses, or other organisms may cause cancer as a result of continuing inflammation. Other cancers arise from integration of viral DNA into the host's DNA.

Virally induced cancers are often epidemic among poultry and rodents. When infected with cancer-causing viruses from these animals, cells in culture often become **transformed**. Whereas normal cells tend to respond to **contact inhibition** and grow as a monolayer, transformed cells continue to divide after the monolayer is complete. In laboratory studies, transformation of cells is often taken as the equivalent to an early step in cancer production in an animal. Studies of transformation led to the identification and characterization of several **viral oncogenes** which cause the transformation. These are designated by abbreviations such as *v-src* (the oncogene of Rous sarcoma virus, which induces cancer in chickens) and *v-sis* (the oncogene of simian sarcoma virus, which causes cancers in monkeys). Some other oncogenes are described in the main text. Naturally occurring gene

BOX 11-D CANCER (continued)

sequences closely homologous to those of the viral oncogenes have been found in many solid human tumors. Study of oncogenes has shown that they are related to and derived from **proto-oncogenes**, normal cellular genes that are involved in control of growth and differentiation.

Oncogenes are often “amplified” in tumor cells so that their copy number is greater than that of the corresponding genes in normal cells. For example an oncogene related to the viral oncogene *neu* is amplified in many human breast and ovarian cancers^q and oncogene *src* in many colon cancers.^r Mutated *ras* genes have been found in over one-third of human colorectal cancers.^s Amplified oncogenes *ras*, *myc*, and *myb* have been observed in other cancers.^t Oncogenes are often overexpressed or are responsible for over-expression of other genes. One idea that developed from these observations is that cancer cells may secrete new or mutated growth factors that stimulate their own receptors (**autocrine** stimulation) in a way that promotes uncontrolled growth.^u

Cancer develops in stages and in many cases defective proto-oncogenes appear before truly malignant cells appear. The latter must result from additional mutations that often involve *loss* of parts of chromosomes. A major breakthrough in our understanding of cancer and how it is induced by loss of genes has come from studies of some rare cancers that are inherited in a Medelian fashion. One of these is **retinoblastoma**, an intraocular tumor which affects 1 child in 20,000. Homozygotes always develop the disease between the ages of 1 and 5. The hereditary defect has been traced to the absence of a functional retinoblastoma gene *RB1*, which is found in band q14 of chromosome 13.^v Additional mutational events are required to induce cancer. The *RB1* gene was the first **tumor-suppressor** gene (anti-oncogene) identified. These suppressor genes encode proteins that inhibit growth.^{w-z} The retinoblastoma gene encodes a 105-kDa DNA-binding phosphoprotein (*Rb-P*).^{aa-bb} The Rb protein is phosphorylated and dephosphorylated in a cyclic fashion that is synchronized with the cell replication cycle (Fig. 11-15). It forms a complex with a transcription factor E2F that functions in transcription of the adenovirus genes and is also involved in control of the cell replication cycle.^{cc} Deletion of the Rb gene from mice leads to death of embryos homozygous for the mutation.^{dd}

Study of other rare hereditary cancers has led to the location of 20 or more additional probable tumor-suppressor genes. One of these, **p53**, is inactive in over 50% of all human cancers and over 90% of squamous cell carcinomas of the skin.^{ee} In small-cell lung cancers and osteosarcoma *both RB* and *p53* are inactive.^z Protein p53 is a stronger tumor suppressor than protein Rb. Results of a variety of experiments have suggested that p53, a DNA-binding protein of known structure,^{ff} plays a key role in checking DNA for damage at the G₁

to S-phase checkpoint in the cell cycle. If the DNA has too many defects the cycle is stopped in the G₁ stage and the cell may be killed by the process known as **apoptosis**.^{gg-ij} Protein p53 has been called the “guardian of the genome.” The mechanisms by which it functions are complex and poorly understood. It may act with the assistance of Rb and many other proteins. Mutations in DNA and their repair are discussed in Chapter 27 and the cell cycle is discussed in this chapter and further in Chapters 26 and 32.

Many other cancer susceptibility genes also encode suppressors. Mutation in genes *BRCA1*^{kk} and *BRCA2*^{ll} are responsible for early onset ovarian and breast cancer. Gene *DPC4* may be a suppressor of pancreatic cancer.^{mmm} The gene *ptc* (patched), first studied as a developmental gene in *Drosophila*, may encode a suppressor of basal cell carcinoma, the commonest form of human skin cancer.ⁿⁿ Gene *p16* (also called *CDKN2*) may be a major suppressor that is mutated in many cancers including the dangerous skin melanoma.^{oo} Mutations in the *NF* gene, which may be a cytoskeletal protein, are associated with **neurofibromatosis**,^{pp} a relatively common hereditary disease causing tumors of the nervous system. The tumors are usually not malignant but are numerous and disfiguring. Several cancer susceptibility genes are associated with faulty mismatch repair of DNA. Among these is the *APC* gene, whose malfunction is associated with human **familial adenomatous polyposis** which causes thousands of benign tumors in the lining of the large intestine and often colorectal cancer.^{qq,rr,ss} In the much more common nonpolyposis colon cancer a complex pattern of instability in several genes is associated with DNA repair,^{tt} a topic dealt with further in Chapter 27. The *ATM* gene defective in **ataxia telangiectasia** (Chapter 27) may encode a phosphatidylinositol kinase that is in some way involved in repair of DNA.^{ss,uu,vv} A transcription factor gene *nm23* may be a suppressor gene for metastasis^{ww} and the cell-cell adhesion molecule **E-cadherin** may suppress tumor invasion in some kinds of breast cancer.^{xx} The **VHL** (van Hippel-Lindau cancer syndrome) suppressor protein is defective in the majority of kidney cancers. It normally binds to the **elongin complex**, a DNA-binding complex that functions in control of transcription.^{zz}

In addition to treatment by surgery there are numerous chemical approaches to combating cancer.^{yy} They usually exploit the tendency of cancers to grow continuously. For example, a toxic analog of a metabolite needed for growth, such as methotrexate (Chapter 16) and 5-fluorouridine (Box 28-C) may be taken up more rapidly by tumor cells than by normal cells. A variety of DNA-binding compounds are useful in chemotherapy (Box 5-B). Alkylating agents such as the nitrogen mustards (Eq. 5-20) and certain antibiotic compounds are also used widely, as are intercalating compounds

BOX 11-D (continued)

such as adriamycin (14-hydroxydaunomycin; Fig. 5-22) and cisplatin (Box 5-B). A disadvantage to most present-day chemotherapy is that normal proliferation of cells, especially of glandular tissues, intestinal epithelium, hair, etc., is severely damaged. A possibility for circumventing this problem is to put normal growth "on hold" temporarily while cancer growth is being inhibited. Some inhibitors of **topoisomerase I** (see Chapter 27) have low toxicity and could be useful. Another therapeutic approach is based on the fact that one function of the immune system is to destroy cancerous or pre-cancerous cells. The immune system tends to weaken with age, which may be one reason that the incidence of cancers rises rapidly in older age. Are there ways of stimulating the immune system into increased activity against cancer cells? Another approach is to find ways of increasing the activity of tumor-suppressor genes.

Can drugs be developed to *prevent* cancer? Such "chemoprevention"^{yz} may be appropriate for persons carrying genes that make them highly susceptible to cancer. The antiestrogenic drug tamoxifen (Chapter 22) is currently being tested on women with a high risk for breast cancer. Use of oral contraceptives appears to have cut the risk of endometrial cancer substantially.^{yy} Newer approaches to contraception may help prevent breast cancer and chemoprevention may also be possible for prostate cancer.^{yy}

^a Cairns, J. (1978) *Cancer: Science and Society*, Freeman, San Francisco, California

^b Cavenee, W. K., and White, R. L. (1995) *Sci. Am.* **272**(Mar), 72–79

^c Ames, B. N., Gold, L. S., and Willett, W. C. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 5258–5265

^d Ruddon, R. W. (1987) *Cancer Biology*, 2nd ed., Oxford Univ. Press, London

^e Preston, D. S., and Stern, R. S. (1992) *N. Engl. J. Med.* **327**, 1649–1662

^f Zhang, D., Hoyt, P. R., and Papaconstantinou, J. (1990) *J. Biol. Chem.* **265**, 3382–3391

^g Packard, B. (1986) *Trends Biochem. Sci.* **11**, 490–491

^h Feldman, M., and Eisenbach, L. (1988) *Sci. Am.* **259**(Nov), 60–85

ⁱ Marx, J. (1993) *Science* **259**, 626–629

^j zur Hausen, H. (1991) *Science* **254**, 1167–1173

^k Willett, W. (1989) *Nature (London)* **338**, 389–394

^l Dragani, T. A., Canzian, F., and Pierotti, M. A. (1996) *FASEB J.* **10**, 865–870

^m Solomon, E., Borrow, J., and Goddard, A. D. (1991) *Science* **254**, 1153–1160

ⁿ Nowell, P. C. (1994) *FASEB J.* **8**, 408–413

^o Pennisi, E. (1996) *Science* **272**, 649

^p Marx, J. (1989) *Science* **246**, 1386–1388

^q Slamon, D. J., Godolphin, W., Jones, L. A., Holt, J. A., Wong, S. G., Keith, D. E., Levin, W. J., Stuart, S. G., Udove, J., Ullrich, A., and Press, M. F. (1989) *Science* **244**, 707–712

^r Cartwright, C. A., Meisler, A. I., and Eckhart, W. (1990) *Proc. Natl. Acad. Sci. U.S.A.* **87**, 558–562

^s Bos, J. L., Fearon, E. R., Hamilton, S. R., Verlaan-de Vries, M., van Boom, J. H., van der Eb, A. J., and Vogelstein, B. (1987) *Nature (London)* **327**, 293–297

^t Yokota, J., Tsunetsugu-Yokota, Y., Battifora, H., Le Fevre, C., and Cline, M. J. (1986) *Science* **231**, 261–265

^u Sporn, M. B., and Roberts, A. B. (1985) *Nature (London)* **313**, 745–747

^v Weinberg, R. A. (1990) *Trends Biochem. Sci.* **15**, 199–202

^w Stanbridge, E. J. (1990) *Science* **247**, 12–13

^x Weinberg, R. A. (1991) *Science* **254**, 1138–1146

^y Knudson, A. G. (1993) *Proc. Natl. Acad. Sci. U.S.A.* **90**, 10914–10921

^z Yokota, J., and Sugimura, T. (1993) *FASEB J.* **7**, 920–925

^{aa} Wiman, K. G. (1993) *FASEB J.* **7**, 841–845

^{bb} Cobrinik, D., Dowdy, S. F., Hinds, P. W., Mitnacht, S., and Weinberg, R. A. (1992) *Trends Biochem. Sci.* **17**, 312–315

^{cc} Nevins, J. R. (1992) *Science* **258**, 424–429

^{dd} Jacks, T., Fazeli, A., Schmitt, E. M., Bronson, R. T., Goodell, M. A., and Weinberg, R. A. (1992) *Nature (London)* **359**, 295–300

^{ee} Ziegler, A., Jonason, A. S., Leffell, D. J., Simon, J. A., Sharma, H. W., Kimmelman, J., Remington, L., Jacks, T., and Brash, D. E. (1994) *Nature (London)* **372**, 773–776

^{ff} Cho, Y., Gorina, S., Jeffrey, P. D., and Pavletich, N. P. (1994) *Science* **265**, 346–355

^{gg} Marx, J. (1993) *Science* **262**, 1644–1645

^{hh} Enoch, T., and Norbury, C. (1995) *Trends Biochem. Sci.* **20**, 426–430

ⁱⁱ Kaufmann, W. K., and Paules, R. S. (1996) *FASEB J.* **10**, 238–247

^{jj} Hartwell, L. H., and Kastan, M. B. (1994) *Science* **266**, 1821–1828

^{kk} Futreal, P. A., and 26 other authors (1994) *Science* **266**, 120–122

^{ll} Wooster, R., and 30 other authors (1994) *Science* **265**, 2088–2090

^{mm} Hahn, S. A., Schutte, M., Hoque, A. T. M. S., Moskaluk, C. A., da Costa, L. T., Rozenblum, E., Weinstein, C. L., Fischer, A., Yeo, C. J., Hruban, R. H., and Kern, S. (1996) *Science* **271**, 350–353

ⁿⁿ Pennisi, E. (1996) *Science* **272**, 1583–1584

^{oo} Marx, J. (1994) *Science* **265**, 1364–1365

^{pp} Rouleau, G. A., and 20 other authors (1993) *Nature (London)* **363**, 515–521

^{qq} Peltomäki, P., Aaltonen, L. A., Sistonen, P., Pylkkänen, L., Mecklin, J.-P., Järvinen, H., Green, J. S., Jass, J. R., Weber, J. L., Leach, F. S., Petersen, G. M., Hamilton, S. R., de la Chapelle, A., and Vogelstein, B. (1993) *Science* **260**, 810–819

^{rr} Huang, J., Papadopoulos, N., McKinley, A. J., Farrington, S. M., Curtis, L. J., Wyllie, A. H., Zheng, S., Willson, J. K. V., Markowitz, S. D., Morin, P., Kinzler, K. W., Vogelstein, B., and Dunlop, M. G. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 9049–9054

^{ss} Kolodner, R. D. (1995) *Trends Biochem. Sci.* **20**, 397–401

^{tt} Papadopoulos, N., and 19 other authors (1994) *Science* **18**, 1625–1629

^{uu} Keith, C. T., and Schreiber, S. L. (1995) *Science* **270**, 50–51

^{vv} Sanchez, Y., Desany, B. A., Jones, W. J., Liu, Q., Wang, B., and Elledge, S. J. (1996) *Science* **271**, 357–360

^{ww} Marx, J. (1993) *Science* **261**, 428–429

^{xx} Berx, G., Cleton-Jansen, A.-M., Nollet, F., de Leeuw, W. J. F., van de Vijver, M. J., Cornelisse, C., and van Roy, F. (1995) *EMBO J.* **14**, 6107–6115

^{yy} Hoffman, E. J. (1999) *Cancer and the Search for Selective Biochemical Inhibitors*, CRC Press, Boca Raton, Florida

^{zz} Henderson, B. E., Ross, R. K., and Pike, M. C. (1993) *Science* **259**, 633–638

^{yz} Young, R. C. (2000) *Nature (London)* **408**, 141

^{zz} Stebbins, C. E., Kaelin, W. G., Jr., and Pavletich, N. P. (1999) *Science* **284**, 455–461

gene is the Abelson murine leukemia virus *v-abl*.^{438a} In the mouse the *c-abl* gene is split into at least ten exons. However, *v-abl* contains all of these as a correctly spliced sequence suggesting that *v-abl* was derived from a *c-abl* messenger RNA. Part of one exon of the cellular gene is missing in *v-abl* and there is at least one base substitution mutation as well. Human *c-abl* is located on the long arm q of chromosome 9. This arm has long been known to be translocated to chromosome 22 (the “Philadelphia translocation”) in patients with chronic myelogenous leukemia. There the *c-abl* gene is fused with another gene.⁴³⁹ The human *c-sis* gene is also located in the region of chromosome 22 that is translocated to chromosome 9 in the same patients.

A group of human pituitary tumors have been shown to contain an oncogene that is apparently a mutated gene for the stimulatory protein G_s .⁴⁴⁰ This is the protein that activates adenylate cyclase in response to hormonal activation. The oncogenic mutations inhibit the GTPase activity that normally turns off this activation. These tumors secrete growth hormone which binds to receptors on the tumor cells activating the defective G_s proteins and causing excessive synthesis of cAMP. This in turn promotes growth of the tumor.

The *ras* oncogenes. Activated *ras*-oncogenes have been found in at least 25% of all human tumors. The proto-oncogenes, which are designated *c-H-ras*, *c-K-ras*, and *c-N-ras*, are found on the short arms of chromosomes 11, 12, and 1, respectively. A single base substitution ($G \rightarrow T$) at position 35 of any of the genes, resulting in a Ras protein containing valine instead of glycine at position 12 of the 21 kDa protein product (usually designated p21) produces an active oncogene. Substitutions at position 13 or at positions 59 and 61, which are adjacent to Gly 12 in the three-dimensional structure, can also activate the oncogenes.⁴⁴¹ From the drawing in Fig. 11-7A the locations of glycines 12 and 13 and of residues 59/61 are seen to be close to the β phospho group of bound GTP. Proteins encoded by activated *ras* oncogenes are less active in catalyzing GTP hydrolysis than are the corresponding normal proteins. In addition, the GTPase-activating protein (GAP) that binds to normal *ras* proteins and stimulates their GTPase activity does not affect the mutant oncogenic proteins.

A single-base alteration is capable of activating a *ras* gene with respect to cell transformation. However, initiation of a malignant tumor requires the additional presence of at least a second activated oncogene such as *myc*^{442,443} or *fos*⁴⁴⁴ or previous transformation of a fibroblast into a nonmalignant but “immortalized” form by treatment with carcinogens. The *c-myc* gene, which is found in active form in plasmacytomas (tumors of B lymphocytes) of mice as well as in the human Burkitt’s lymphoma, is normally located on human chromosome 8. In most Burkitt’s lymphomas a trans-

location has brought the *c-myc* gene into the locus of the immunoglobulin heavy chains on chromosome 14. There its transcription may be subject to different controls than in its original location.^{445,446} During the translocation process the *c-myc* gene is often broken within the first intron. Thus, the activated gene lacks the first exon and is placed after a new controlling sequence that may drastically alter its transcription rate. Viral *myc* genes are found in at least one human virus, **cytomegalovirus**, which has been associated with carcinomas.

Transcription factors. The proto-oncogenes *c-myc*^{447–451a}, *c-myb*,^{452–454} *c-fos*, *c-jun* and *c-ets*⁴⁵⁵ all encode nuclear proteins involved in regulation of transcription. The 39 kDa protein Jun, which is encoded by *c-jun*, is a major component of the **transcriptional activator** called **AP-1**.^{456–459} It binds to palindromic **enhancer** sites (Chapter 28) in DNA promoters to increase the transcription rate for a group of genes. Jun is actually a multigene family whose encoded proteins bind to DNA as complexes formed with the 62 kDa phosphoprotein Fos, the product of the *c-fos* gene.⁴⁶⁰ The heterodimeric Fos/Jun complex is held together, at least in part, by interactions between leucine side chains lying along a pair of parallel α -helices in a “leucine zipper” (Fig. 5-36; Fig. 2-21).^{461,462}

Regulation of the synthesis of Jun is complex, but growth factors such as Neu, EGF,⁴⁶³ and PDGF stimulate transcription of *c-jun* in cultured cells. Messenger RNA for synthesis of Fos appears within a few minutes of stimulation of PDGF receptors.^{464,465} This is one of the earliest known nuclear reactions to a mitogenic stimulus and suggested that a Ras p21 protein is involved in stimulating transcription of *c-fos* in the signaling pathway from PDGF.⁴⁶⁶ Synthesis of Fos is also induced by a variety of other stimuli.⁴⁵⁶ Upon translocation into the nucleus Fos combines with pre-existing Jun to form AP-1, which binds to sites on DNA and induces the transcription of a large number of proteins (Chapter 28). Deletion of the *c-fos* gene in mice leads to defects in developing bone, teeth, and blood cells,⁴⁶⁷ while excessive synthesis of Fos has been associated with the human bone disease **fibrous dysplasia**.⁴⁶⁸

2. The MAP Kinase Cascade

Insulin, platelet-derived growth factor (PDGF), epidermal growth factor (EGF), and many other proteins have **mitogenic activity**, that is, they induce cells to transcribe genes, to grow, and to divide. How is this accomplished? Study of such oncogenes as *src* and *ras* suggested that the proteins that they encode also participate in the process, as do transcription factors, including those encoded by the proto-oncogenes

Table 11-4
A Few Abbreviations Used in Discussions of Cell Signaling

AP-1	A major transcriptional activator protein	PKC, PKC α , β , γ	Protein kinase C, α , β , γ subforms
β ARK	Beta adrenergic receptor kinase	PKR	dsRNA-activated protein kinase
EGF	Epidermal growth factor	PLA	Phospholipase A
EGFR	Epidermal growth factor receptor	PLC	Phospholipase C
ERK-1, ERK-2	Extracellular signal-regulated protein kinases (proline-directed protein kinases)	PP2A	Protein phosphatase, type 2A
Fos	Protein encoded by proto-oncogene <i>fos</i> (Table 11-3)	PTK	Protein tyrosine kinase
Grb2	Adapter protein containing one SH2 and two SH3 domains	PTP (PTPase, PTP 1B, etc.)	Protein tyrosine phosphatases
IGF-1	Insulin-like growth factor 1	pY	Phosphotyrosine residue
InsRTK	Insulin receptor tyrosine kinase	Raf-1	A cytoplasmic serine/threonine protein kinase; also called MAP kinase kinase kinase (MAP3K)
InsP ₃ or IP ₃	Inositol (1,3,5)-trisphosphate	Ras	A monomeric G protein encoded by the proto-oncogene <i>ras</i>
Jun	Protein encoded by proto-oncogene <i>jun</i> , an AP-1 gene	RSK	Ribosomal protein S6 kinase
MAPK	Mitogen-activated protein kinase (also designated ERK)	RTK	Receptor tyrosine kinase
MAPKK	A kinase acting on MAPK	RTK-P	Phosphorylated receptor tyrosine kinase
MEK-1, MEK-2	Mitogen-activated ERK-activating kinases; dual function Ser/Thr and Tyr protein kinases; also called MAP kinase kinases	SH2, SH3	Src homology domains 2 and 3
<i>c-myc</i>	A cellular proto-oncogene	Sos	"son of sevenless," a GDP – GTP exchange factor named for a similarity to the protein encoded by the <i>Drosophila</i> sevenless gene
PDGF	Platelet-derived growth factor	Src	Protein tyrosine kinase encoded by oncogene <i>src</i> (Table 11-3); contains recognition domains SH2 and SH3
PtdIns (or PI) 3-kinase	Phosphatidylinositol 3-kinase	V2R	Type 2 vasopressin receptor
PKA (cAPK)	cyclic AMP-dependent protein kinase	Shc	A proline-rich adapter protein containing SH2 and PH domains

myc, *fos*, and *jun*. The Src and Ras proteins are anchored on the inner surfaces of cytoplasmic membranes. Although the exact functional relationships of all of the components, one to another, has not been completely established, a general picture, usually described as the **mitogen-activated protein kinase** (MAP kinase) **cascade**, has emerged.^{69,380,469-470b} This is sketched in simplified form in Fig. 11-13. The many amplification steps provide for ultrasensitive responses.³⁴²

The binding of a hormone or growth factor (a ligand) to a dimeric receptor activates the protein kinase domain of the receptor which phosphorylates a number of tyrosine hydroxyl groups of the receptor itself. This autophosphorylation is followed by a variety of events, which include phosphorylation of tyrosine side chains of various other proteins.⁴²⁶ An-

other major event is the *binding of a variety of different protein molecules containing recognition domains to the phosphotyrosyl groups of the activated receptors*.⁴⁷¹ The major recognition motif is the SH2 domain. See Figs. 7-30 and 11-14.⁴⁷²⁻⁴⁷⁵ Proteins containing SH2 domains can bind to the phosphotyrosyl groups of the activated receptors and while bound become phosphorylated by the receptor tyrosine kinase action and/or be activated allosterically.

Other proteins interact with the receptors indirectly through adapter molecules which have no catalytic activity. Two well-known adapter proteins are **Grb2**⁴⁷⁶ and **Shc**.^{477,478} The 25-kDa protein Grb2 consists entirely of recognition domains, one SH2 and two SH3 domains (Fig. 11-14). The larger Shc, which is found in all mammalian tissues, contains a 200-residue phosphotyrosyl-binding PH domain (Chapter 7) at the N terminus, a

collagen-like domain that binds to Grb2 and an SH2 domain at the C terminus.^{376,377,477}

Adapter Grb2 binds to a phosphotyrosine side chain of an activated receptor, such as that for EGF, and simultaneously binds to the GDP-GTP exchange protein called **Sos** (Table 11-4). This signals Sos to activate the membrane-bound Ras protein by converting it into the GTP form. The second adapter Shc may also participate in formation of the receptor kinase - Grb2-Sos complex,⁴⁷⁹ perhaps permitting formation of a more robust complex that may receive signals from more than one kind of receptor. Functions of other members of the Grb adapter family are being discovered.^{479a}

Activated Ras binds to and activates the cytoplasmic serine/threonine protein kinase called **Raf-1**.^{380,480,481} This kinase becomes transiently activated within 2–3 min of the binding of a mitogen to a receptor. Raf-1 initiates a cascade of other protein kinases by acting on the dual-function Ser/Thr and tyrosine protein kinases called **MEK-1** and **MEK-2**. The phosphorylated, active MEK proteins phosphorylate the mitogen-activated protein kinases **MAPK** which act on a variety of other proteins. Two of the best known MAPK proteins are designated **ERK-1** and **ERK-2**. These are *proline-directed* kinases which phosphorylate serines and threonines that are neighbors to prolines, e.g. in the sequence PLS/TP.³⁸⁰ The activated ERKs are able to phosphorylate a large number of different proteins including nuclear proteins that control the transcription of such protein transcription factors as AP-1 and Myc. A protein known as the **serum response factor** binds to nucleotide sequences CC(A/T)₆GC in the DNA to locate initiation sites for transcription. Other proteins that have been phosphorylated by the MAP kinase cascade then induce transcription.⁴⁸² The induction of *c-fos* mRNA is one of the earliest identified responses to growth factors.^{375,456,483} Protein Jun, whose synthesis is induced independently by almost all growth factors, is usually present in excess.

The Fos/Jun complex is transcription factor AP-1, which induces transcription of many genes needed for cell growth. However, transcription of specific genes often depends upon additional nucleotide sequences. For example, the sequence CGGAAA is present in an **insulin response element** found in the promoter sequences of genes encoding such proteins as phosphoenolpyruvate carboxykinase, glyceraldehyde phosphate dehydrogenase, and prolactin—proteins whose synthesis is induced by insulin.⁴⁸⁴

The MAPK cascade also has direct effects upon protein synthesis, i.e., on the translation of mRNA messages. For example, insulin stimulates phosphorylation of proteins that regulate a translation initiation factor, a protein called eIF-4E (see Chapter 29). Phosphorylation of inhibitory proteins allows them to dissociate from the initiation factor so that protein synthesis can proceed.^{485,486}

The scheme in Fig. 11-13 is complex, but in reality it is *much* more complex than is shown. Each protein kinase (receptor kinase, Raf-1, MEK, and MAPK) will phosphorylate not only the proteins indicated in this scheme but also any others that meet the specificity requirements of the kinases. Thus, there will be branches diverging from the pathways shown.^{486a,b,c} There are isoenzymes that provide further divergence and interaction.⁴⁸⁷ Not only do pathways diverge but also others *converge*. Thus, binding of many different ligands to their receptors activates the same MAPK cascade. For example, seven-helix G protein-coupled receptors release their $\beta\gamma$ subunits which may also activate Ras as indicated on the right edge of Fig. 11-13A. At the same time the α subunits of the heterotrimer G proteins can affect not only adenylate cyclase but also phospholipases C which can, in some cases, also activate the MAP kinase pathway.^{488,489} Sphingosine 1-phosphate may be released from membrane spingolipids and activate the same cascade.⁴⁹⁰ However, hormones do not all affect cells in the same way. In view of all the converging pathways, how is this possible? Part of the answer lies in the proximity or spatial separation of components of the pathway. The kinases exist in complexes with other signaling proteins and may phosphorylate them, sending a signal back, as well as forward via other protein substrates. There are also unknown kinetic considerations. Hormones, neurotransmitters, and calcium ions are often released in pulses. The signaling system must integrate effects of all the stimuli that arise from different parts of the cell, at different times, and from differing receptors. At the same time all of the phosphorylated proteins are acted upon by phosphatases that may either activate or deactivate the proteins and by proteases that process newly formed peptides and modify or destroy mature proteins. The various modifying enzymes act on cytosolic proteins, proteins of membranes, of the cytoskeleton, and of the nucleus. The regulatory processes that we discuss in such minute detail involve the very substance of living cytoplasm which is ever-changing and responding to its surroundings. The flow of energy, provided by synthesis of ATP and by the use of ATP by kinases, phosphatases, and the protein synthetic machinery, goes along with the flow of information and drives the signaling network. Evolution has shaped this system to allow it to respond appropriately for every species.

3. The Cell Cycle and Control of Growth

When a cell divides it is of utmost importance that the DNA be replicated reliably. This requires that the dividing cell be large enough and contain enough biosynthetic precursor materials to complete the elaborate process of DNA synthesis and of mitosis. The **cell**

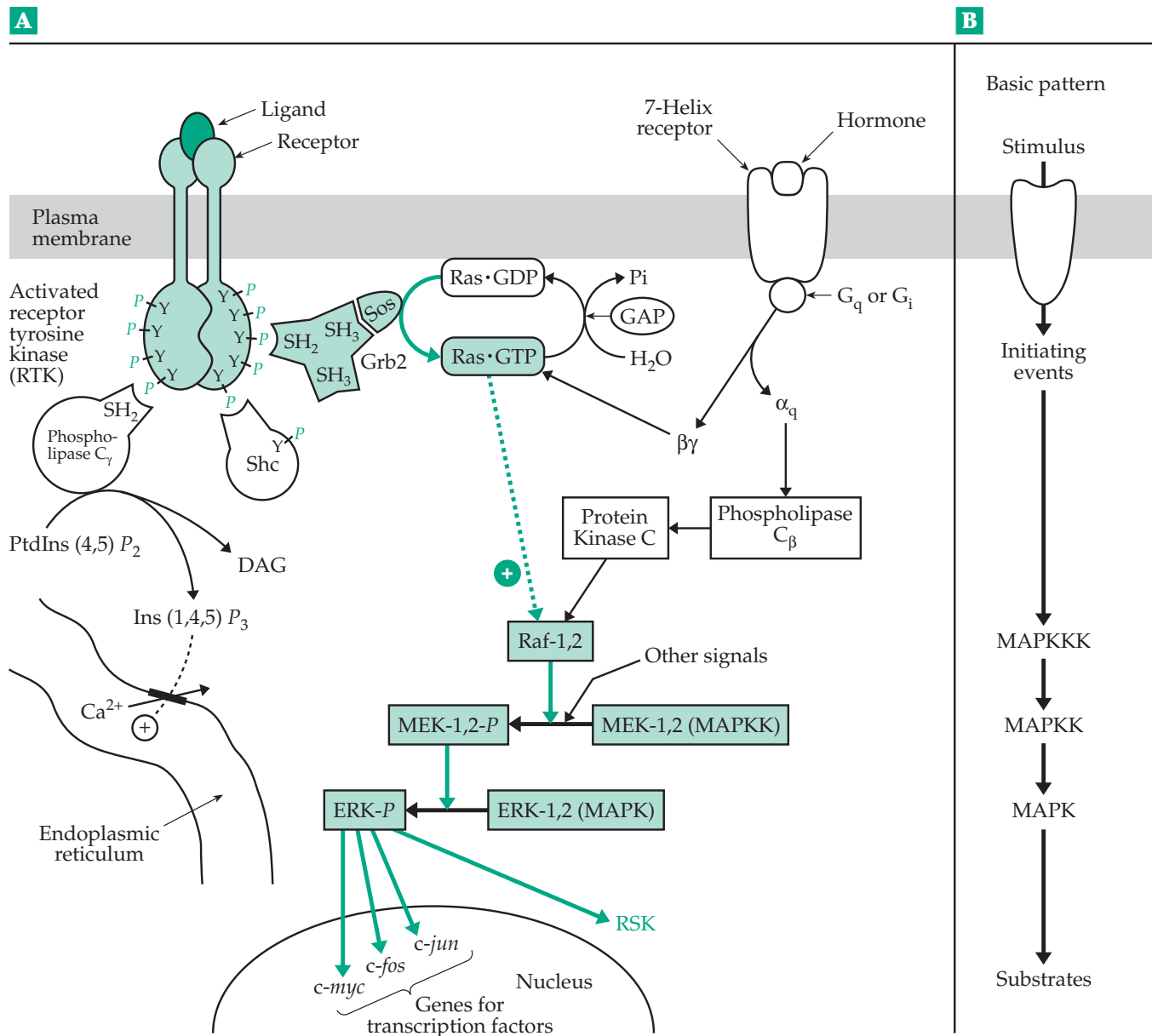


Figure 11-13 (A) A simplified version of the mitogen-activated kinase (MAPK) signaling cascade. At left is shown a hormone receptor, e.g., that for the epidermal growth factor (EGF). The receptor tyrosine kinase undergoes autophosphorylation on numerous tyrosines. The resulting phosphotyrosyl (Y-P) groups bind to SH2 domains of adapters such as Grb2 and Shc. Two pathways from the activated receptor are shown. At the left is activation of phospholipase C_γ and formation, at a membrane-bound site, of inositol trisphosphate and diacylglycerol (DAG). The main pathway, in the center, activates Ras with the aid of the G protein Sos. Activated Ras, in turn, activates Raf and successive components of the MAPK cascade. At the right a seven-helix receptor activates both phospholipase C_β and Ras via interaction with a βγ subunit. (B) A generalized scheme for the MAP kinase pathway. See Seger and Krebs.³⁸⁰

replication cycle (or simply cell cycle) is commonly shown as a circle in which the time from one cell division to the next, in a rapidly growing organism or tissue, is represented by the circumference. The time required for DNA synthesis is the **S-phase** and the time required for mitosis the mitotic or **M-phase** (Fig. 11-15). After metaphase there is a **gap** in time denoted **G₁**. A second

gap **G₂** separates the synthetic S-phase and the M-phase. The total time required for one cycle varies with conditions. It may be as short as 8–60 min in an early embryo but is usually two hours or more. A slowly growing cell may pause before the G₁ phase in a nongrowing **G₀ phase**.^{491,492}

What controls the cell replication cycle? Signals

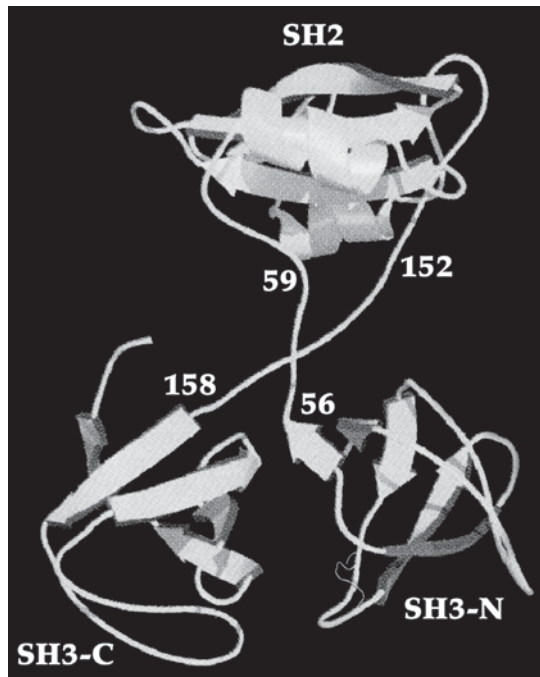


Figure 11-14 Ribbon drawing of the three-dimensional structure of adapter protein Grb2. The two SH3 domains at the N and C termini are labeled, as is the central SH2 domain. Produced with programs MolScript and Raster3D. From Maignan *et al.*⁴⁷⁶ Courtesy of Arnaud Ducruix.

from growth factors can stimulate a cell to leave G_0 and enter G_1 . However, to go further the cell depends upon a large number of proteins of which two types are prominent. The **cyclins** are labile 45- to 60-kDa proteins which are degraded by the ubiquitin system as part of the cycle. The cyclins associate with a series of protein Ser/Thr kinases known as **cyclin-dependent kinases (CDKs)**. It is the cyclin-CDK complexes that signal the start of the next step in the cycle.⁴⁹³⁻⁴⁹⁵

The number of different cyclins and CDK enzymes needed varies with the organism, often being greater for more complex species. As a rule, there are at least two types of cyclins. The **G_1 or start cyclins** initiate the passage through the start (G_1 checkpoint) into the S-phase, while **mitotic cyclins** initiate the passage from the S-phase into the M-phase. Both vertebrates and *Drosophila* utilize at least four different types of cyclin (A, B, D, and E) and also four or more CDKs (CDK1, CDK2, CDK4, and CDK6).⁴⁹² A somewhat different set of these proteins are found in yeast.

Control of cell growth is directly related to the cell cycle. Mitogenic signals from growth factors act to initiate progression through G_1 , apparently by stimulating transcription of D-type cyclins.⁴⁹⁶ They may affect other steps as well. Oncogenic signals can arise from such oncogenes as *ras* and *abl*.^{497,498} As pointed out in Box 11-D, the tumor-suppressor protein Rb becomes phosphorylated in synchrony with the cell

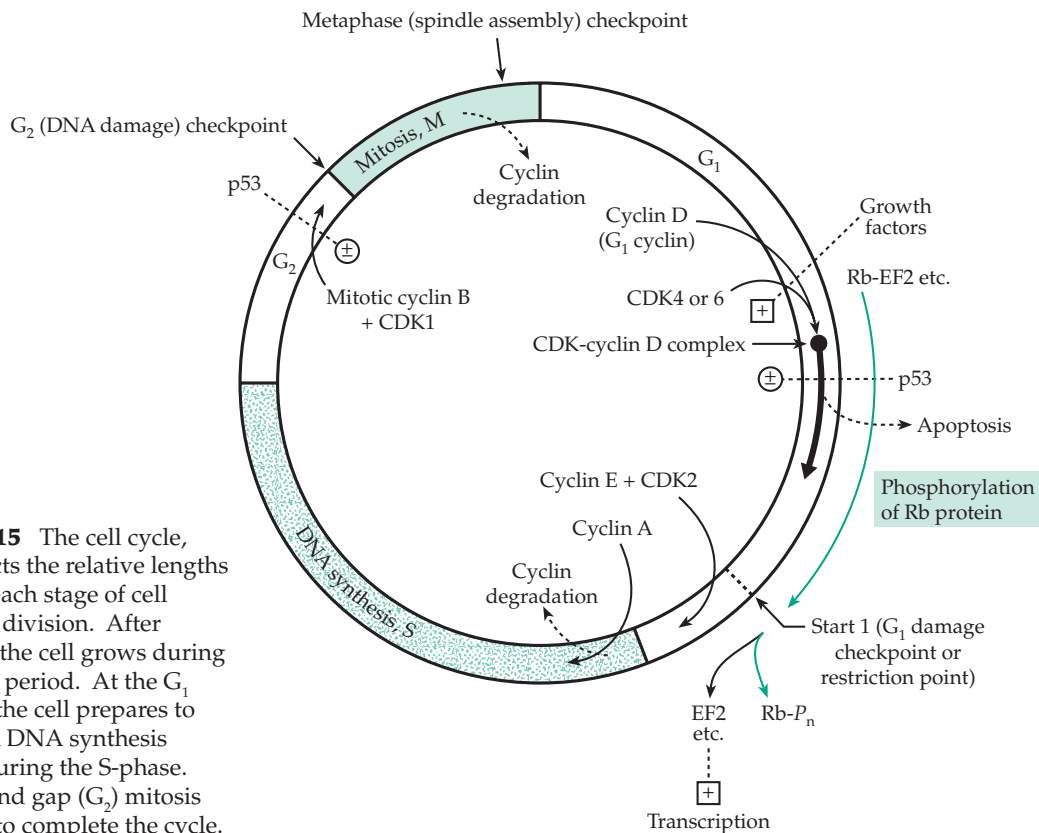


Figure 11-15 The cell cycle, which depicts the relative lengths of time for each stage of cell growth and division. After mitosis (M) the cell grows during the G_1 (gap) period. At the G_1 checkpoint the cell prepares to divide, with DNA synthesis occurring during the S-phase. After a second gap (G_2) mitosis takes place to complete the cycle.

cycle. This seems to be an essential process for passage through the G₁ "checkpoint." The checkpoints in the cycle should be viewed not as points but as essential interlocking processes that must be accomplished before transcription of genes essential for the next step in the cycle can take place.⁴⁹⁹ A possibility is that the cyclin-dependent kinase in the cyclin D•CDK4 (or CDK6) complex phosphorylates the Rb protein which in its unphosphorylated state forms a complex with transcriptional regulators of the E2F family. Phosphorylation of Rb allows this complex to dissociate and permits E2F to induce transcription of genes encoding essential proteins for the next step in the cycle.^{499a,b,c} Each CDK may also phosphorylate its associated cyclin, a modification that may be essential to the

proteolysis of the cyclin and progression to the next stage of the cycle.

The powerful cancer suppressor, p53, which is also described in Box 11-D, in some way senses DNA damage. It prevents passage through the G₁ checkpoint and also through the G₂ checkpoint if the DNA has not been adequately repaired.^{496,497} Protein p53, whose three-dimensional structure is known,^{500,501} binds to DNA and also induces transcription of genes that cause arrest of the cell cycle. It may also induce cell death (apoptosis), a process that may also require the protein product of protooncogene *c-myc*.⁵⁰²⁻⁵⁰⁵ A variety of protein kinases and phosphatases act on p53 and influence its activity.⁵⁰⁶

References

- Krebs, H. A. (1957) *Endeavour* **16**, 125–132
- Newsholme, E. A., and Start, C. (1973) *Regulation in Metabolism*, Wiley, New York
- Heinrich, R., Rapoport, S. M., and Rapoport, T. A. (1977) *Prog. Biophys. and Mol. Biol.* **32**, 1–82
- Martin, B. R. (1987) *Metabolic Regulation*, Blackwell Scientific Publ., Oxford
- Chance, B., and Williams, G. R. (1955) *J. Biol. Chem.* **217**, 477–488
- Heinrich, R., and Rapoport, T. A. (1974) *Eur. J. Biochem.* **42**, 89–95; 97–105
- Mendes, P. (1997) *Trends Biochem. Sci.* **22**, 361–363
- Wright, B. E. (1986) *Trends Biochem. Sci.* **11**, 164–165
- Savageau, M. A. (1987) *Trends Biochem. Sci.* **12**, 219–220
- Voit, E. O., and Savageau, M. A. (1987) *Biochemistry* **26**, 6869–6880
- Hlavacek, W. S., and Savageau, M. A. (1996) *J. Mol. Biol.* **255**, 121–139
- Voit, E. O. (1987) *Trends Biochem. Sci.* **12**, 221
- Kacser, H., and Burns, J. A. (1979) *Biochem. Soc. Trans.* **7**, 1149–1166
- Crabtree, B., and Newsholme, E. A. (1987) *Trends Biochem. Sci.* **12**, 4–12
- Kholodenko, B. N., and Westerhoff, H. V. (1995) *Trends Biochem. Sci.* **20**, 52–54
- Fell, D. (1997) *Understanding the Control of Metabolism*, Portland Press, London
- Quant, P. A. (1993) *Trends Biochem. Sci.* **18**, 26–30
- Salter, M., Knowles, R. G., and Pogson, C. I. (1994) in *Essays in Biochemistry*, Vol. 28 (Tipton, K. F., ed), Portland Press, London
- Schulze, E.-D. (1994) *Flux Control in Biological Systems*, Academic Press, San Diego, California
- Kacser, H., and Burns, J. A. (1973) *Symp. Soc. Exp. Biol.* **XXVII**, 65–104
- Stark, G. R., and Wahl, G. M. (1984) *Ann. Rev. Biochem.* **53**, 447–491
- Wilkinson, J. H. (1970) *Isoenzymes*, 2nd ed., Lippincott, Philadelphia, Pennsylvania
- Markert, C. L., ed. (1975) *Isozymes*, Vol. 4, Academic Press, New York
- Christen, P., and Metzler, D. E., eds. (1985) *Transaminases*, Wiley, New York
- Salerno, C., Giartosio, A., and Fasella, P. (1986) in *Vitamin B₆ Pyridoxal Phosphate*, Vol. 1b (Dolphin, D., Poulson, R., and Avramovic, O., eds), Wiley, New York
- McPhalen, C. A., Vincent, M. G., and Jansonius, J. N. (1992) *J. Mol. Biol.* **225**, 495–517
- Mehta, P. K., Hale, T. I., and Christen, P. (1989) *Eur. J. Biochem.* **186**, 249–253
- Kamitori, S., Okamoto, A., Hirotsu, K., Higuchi, T., Kuramitsu, S., Kagamiyama, H., Matsuura, Y., and Katsube, Y. (1990) *J. Biochem.* **108**, 175–184
- Hogrefe, H. H., Griffith, J. P., Rossman, M. G., and Goldberg, E. (1987) *J. Biol. Chem.* **262**, 13155–13162
- Purich, D. L., and Fromm, H. J. (1972) *Curr. Top. Cell. Regul.* **6**, 131–167
- Huijing, F. (1979) *Trends Biochem. Sci.* **4**, N132–N134
- Aleshin, A. E., Kirby, C., Liu, X., Bourenkov, G. P., Bartunik, H. D., Fromm, H. J., and Honzatko, R. B. (2000) *J. Mol. Biol.* **296**, 1001–1015
- Purich, D. L., Fromm, H. J., and Rudolph, F. R. (1973) *Adv. Enzymol.* **39**, 249–326
- Breitbar, R. E., Andreadis, A., and Nadal-Ginard, B. (1987) *Ann. Rev. Biochem.* **56**, 467–495
- Joh, K., Arai, Y., Mukai, T., and Hori, K. (1986) *J. Mol. Biol.* **190**, 401–410
- Haldane, J. B. S. (1930) *Enzymes*, Longmans, Green, New York
- Hayashi, S.-i, Murakami, Y., and Matsufuji, S. (1996) *Trends Biochem. Sci.* **21**, 27–30
- Li, X., Stebbins, B., Hoffman, L., Pratt, G., Rechsteiner, M., and Coffino, P. (1996) *J. Biol. Chem.* **271**, 4441–4446
- Sprang, S. R., Acharya, K. R., Goldsmith, E. J., Stuart, D. I., Varvill, K., Fletterick, R. J., Madsen, N. B., and Johnson, L. N. (1988) *Nature (London)* **336**, 215–221
- Cohen, P. (1983) *Control of Enzyme Activity*, 2nd ed., Chapman and Hall, London
- Kantrowitz, E. R., and Lipscomb, W. N. (1990) *Trends Biochem. Sci.* **15**, 53–59
- Zhou, B.-B., and Schachman, H. K. (1993) *Protein Sci.* **2**, 103–112
- Xi, X. G., De Staercke, C., Van Vliet, F., Trinolles, F., Jacobs, A., Stas, P. P., Ladjimi, M. M., Simon, V., Cunin, R., and Hervé, G. (1994) *J. Mol. Biol.* **242**, 139–149
- Uyeda, K., Furuya, E., and Sherry, A. D. (1981) *J. Biol. Chem.* **256**, 8679–8684
- Hers, H., Hue, L., and Schaftingen, E. (1982) *Trends Biochem. Sci.* **7**, 329–331
- Lively, M. O., El-Maghrabi, M. R., Pilkis, J., D'Angelo, G., Colosia, A. D., Ciavola, J.-A., Fraser, B. A., and Pilkis, S. J. (1988) *J. Biol. Chem.* **263**, 839–849
- Kitamura, K., and Uyeda, K. (1988) *J. Biol. Chem.* **263**, 9027–9033
- Choe, J.-Y., Fromm, H. J., and Honzatko, R. B. (2000) *Biochemistry* **39**, 8565–8574
- Graves, D. J., Martin, B. L., and Wang, J. H. (1994) *Co- and Post-Translational Modification of Proteins*, Oxford Univ. Press, New York
- Skamniaki, V. T., Owen, D. J., Noble, M. E. M., Lowe, E. D., Lowe, G., Oikonomakos, N. G., and Johnson, L. N. (1999) *Biochemistry* **38**, 14718–14730
- Johnson, L. N. (1992) *FASEB J.* **6**, 2274–2282
- Cheng, A., Fitzgerald, T. J., Bhatnagar, D., Roskoski, R., Jr., and Carlson, G. M. (1988) *J. Biol. Chem.* **263**, 5534–5542
- Dent, P., Lavoie, A., Nakielnny, S., Caudwell, F. B., Watt, P., and Cohen, P. (1990) *Nature (London)* **348**, 302–308
- Brady, M. J., Bourbonnais, F. J., and Saltiel, A. R. (1998) *J. Biol. Chem.* **273**, 14063–14066
- Krebs, E. G., and Fischer, E. H. (1956) *Biochim. Biophys. Acta.* **20**, 150–157
- Edelman, A. M., Blumenthal, D. K., and Krebs, E. G. (1987) *Ann. Rev. Biochem.* **56**, 567–613
- Allen, J. F. (1992) *Trends Biochem. Sci.* **17**, 12–17
- Hershey, J. W. B. (1989) *J. Biol. Chem.* **264**, 20823–20826
- Kennelly, P. J., and Krebs, E. G. (1991) *J. Biol. Chem.* **266**, 15555–15558
- Hanks, S. K., and Hunter, T. (1995) *FASEB J.* **9**, 576–596
- Woodgett, J. R., ed. (1994) *Protein Kinases*, IRL Press, Oxford
- Hunter, T. (1994) *Sem. Cell Biol.* **5**, 367–376
- Kemp, B. E., Parker, M. W., Hu, S., Tiganis, T., and House, C. (1994) *Trends Biochem. Sci.* **19**, 440–444
- Imazu, M., Stricklund, W. G., Chrisman, T. D., and Exton, J. H. (1984) *J. Biol. Chem.* **259**, 1813–1821
- He, X., Saint-Jannet, J.-P., Woodgett, J. R., Varmus, H. E., and David, I. B. (1995) *Nature (London)* **374**, 617–622
- Hartl, F. T., and Roskoski, R., Jr. (1983) *J. Biol. Chem.* **258**, 3950–3955
- Zheng, J., Knighton, D. R., Xuong, N.-H., Taylor, S. S., Sowadski, J. M., and Eyck, L. F. T. (1993) *Protein Sci.* **2**, 1559–1573
- Beebe, S. J., Reimann, E. M., and Schlender, K. K. (1984) *J. Biol. Chem.* **259**, 1415–1422
- Taylor, S. S., Knighton, D. R., Zheng, J., Sowadski, J. M., Gibbs, C. S., and Zoller, M. J. (1993) *Trends Biochem. Sci.* **18**, 84–89
- Narayana, N., Cox, S., Shaltiel, S., Taylor, S. S., and Xuong, N.-h. (1997) *Biochemistry* **36**, 4438–4448

References

66. Su, Y., Dostmann, W. R. G., Herberg, F. W., Durick, K., Xuong, N.-h., Eyck, L. T., Taylor, S. S., and Varughese, K. I. (1995) *Science* **269**, 807–813
67. Lincoln, T. M., Thompson, M., and Cornwell, T. L. (1988) *J. Biol. Chem.* **263**, 17632–17637
68. Weber, I. T., Shabb, J. B., and Corbin, J. D. (1989) *Biochemistry* **28**, 6122–6127
69. Fry, M. J., Panayotou, G., Booker, G. W., and Waterfield, M. D. (1993) *Protein Sci.* **2**, 1785–1797
70. Xu, R.-M., Carmel, G., Sweet, R. M., Kuret, J., and Cheng, X. (1995) *EMBO J.* **14**, 1015–1023
71. Allende, J. E., and Allende, C. C. (1995) *FASEB J.* **9**, 313–323
72. Seldin, D. C., and Leder, P. (1995) *Science* **267**, 894–897
- 72a. Niefind, K., Guerra, B., Pinna, L. A., Issinger, O.-G., and Schomburg, D. (1998) *EMBO J.* **17**, 2451–2462
73. Chan, K.-F., and Graves, D. J. (1984) in *Calcium and Cell Function*, Vol. 5 (Cheung, W. Y., ed), pp. 1–32, Academic Press, New York
74. Heilmeyer, L. M. G., Jr. (1991) *Biochim. Biophys. Acta.* **1094**, 168–174
75. Norcum, M. T., Wilkinson, D. A., Carlson, M. C., Hainfeld, J. F., and Carlson, G. M. (1994) *J. Mol. Biol.* **241**, 94–102
76. Kee, S. M., and Graves, D. J. (1987) *J. Biol. Chem.* **262**, 9448–9453
77. Paudel, H. K., and Carlson, G. M. (1987) *J. Biol. Chem.* **262**, 11912–11915
78. Cheng, A., Fitzgerald, T. J., Bhatnagar, D., Roskoski, R., Jr., and Carlson, G. M. (1988) *J. Biol. Chem.* **263**, 5534–5542
79. Owen, D. J., Papageorgiou, A. C., Garman, E. F., Noble, M. E. M., and Johnson, L. N. (1995) *J. Mol. Biol.* **246**, 374–381
80. Ohmstede, C.-A., Bland, M. M., Merrill, B. M., and Sahyoun, N. (1991) *Proc. Natl. Acad. Sci. U.S.A.* **88**, 5784–5788
81. Hanson, P. I., and Schulman, H. (1992) *Ann. Rev. Biochem.* **61**, 559–601
82. Harper, J. F., Sussman, M. R., Schaller, G. E., Putnam-Evans, C., Charbonneau, H., and Harmon, A. C. (1991) *Science* **252**, 951–954
83. Ullrich, A., Bell, J. R., Chen, E. Y., Herrera, R., Petruzzelli, L. M., Dull, T. J., Gray, A., Coussens, L., Liao, Y.-C., Tsubokawa, M., Mason, A., Seeburg, P. H., Grunfield, C., Roden, O. M., and Ramachandran, J. (1985) *Nature (London)* **313**, 756–761
84. Wang, J. Y. J. (1994) *Trends Biochem. Sci.* **19**, 373–376
85. Clark, E. A., Shattil, S. J., and Brugge, J. S. (1994) *Trends Biochem. Sci.* **19**, 464–469
86. Sun, H., and Tonks, N. K. (1994) *Trends Biochem. Sci.* **19**, 480–485
87. Ingebritsen, T. S., and Cohen, P. (1983) *Science* **221**, 331–338
88. Cohen, P. (1989) *Ann. Rev. Biochem.* **58**, 453–508
89. Goldberg, J., Huang, H., Kwon, Y., Greengard, P., Nairn, A. C., and Kuriyan, J. (1995) *Nature (London)* **376**, 745–753
90. Fischer, E. H., Charbonneau, H., and Tonks, N. K. (1991) *Science* **253**, 401–406
91. Yuvanityama, J., Denu, J. M., Dixon, J. E., and Saper, M. A. (1996) *Science* **272**, 1328–1331
92. MacKintosh, C., and MacKintosh, R. W. (1994) *Trends Biochem. Sci.* **19**, 444–448
93. Hidaka, H., and Kobayashi, R. (1994) in *Essays in Biochemistry*, Vol. 28 (Tipton, K. F., ed), pp. 73–98, Portland Press, London
94. Cohen, P., Holmes, C. F. B., and Tsukitani, Y. (1990) *Trends Biochem. Sci.* **15**, 98–102
95. Price, N. E., and Mumby, M. C. (2000) *Biochemistry* **39**, 11312–11318
96. Zhang, M., Van Etten, R. L., and Stauffacher, C. V. (1994) *Biochemistry* **33**, 11097–11105
97. Barford, D., Flint, A. J., and Tonks, N. K. (1994) *Science* **263**, 1397–1403
98. Stone, R. L., and Dixon, J. E. (1994) *J. Biol. Chem.* **269**, 31323–31326
99. Chen, W., Wilborn, M., and Rudolph, J. (2000) *Biochemistry* **39**, 10781–10789
100. Kissinger, C. R., Parge, H. E., Knighton, D. R., Lewis, C. T., Pelletier, L. A., Tempczyk, A., Kalish, V. J., Tucker, K. D., Showalter, R. E., Moomaw, E. W., Gastinel, L. N., Habuka, N., Chen, X., Maldonado, F., Barker, J. E., Bacquet, R., and Villafranca, J. E. (1995) *Nature (London)* **378**, 641–644
101. LaPorte, D. C., and Chung, T. (1985) *J. Biol. Chem.* **260**, 15291–15297
102. Pilgis, S. J., Claus, T. H., Kurland, I. J., and Lange, A. J. (1995) *Ann. Rev. Biochem.* **64**, 799–835
103. Istvan, E. S., Hasemann, C. A., Kurumbail, R. G., Uyeda, K., and Deisenhofer, J. (1995) *Protein Sci.* **4**, 2439–2441
104. Vertommen, D., Bertrand, L., Sontag, B., Di Pietro, A., Louckx, M. P., Vidal, H., Hue, L., and Rider, M. H. (1996) *J. Biol. Chem.* **271**, 17875–17880
105. Abe, Y., Minami, Y., Li, Y., Nguyen, C., and Uyeda, K. (1995) *Biochemistry* **34**, 2553–2559
106. Ogushi, S., Lawson, J. W. R., Dobson, G. P., Veech, R. L., and Uyeda, K. (1990) *J. Biol. Chem.* **265**, 10943–10949
107. Nimmo, H. G. (1984) *Trends Biochem. Sci.* **9**, 475–478
108. Dean, A. M., and Koshland, D. E., Jr. (1993) *Biochemistry* **32**, 9302–9309
109. Dean, A. M., and Koshland, D. E., Jr. (1990) *Science* **249**, 1044–1046
110. Saier, M. H., Jr., Wu, L.-F., and Reizer, J. (1990) *Trends Biochem. Sci.* **15**, 391–395
- 110a. Song, H. K., Lee, J. Y., Lee, M. G., Moon, J., Min, K., Yang, J. K., and Suh, S. W. (1999) *J. Mol. Biol.* **293**, 753–761
111. Swanson, R. V., Alex, L. A., and Simon, M. I. (1994) *Trends Biochem. Sci.* **19**, 485–490
112. Saier, M. H., Jr. (1989) *Microbiol. Rev.* **53**, 109–120
113. Stock, J. B., Stock, A. M., and Mottonen, J. M. (1990) *Nature (London)* **344**, 395–400
114. Boon, P., Chock, P. B., Rhee, S. G., and Stadtman, E. R. (1980) *Ann. Rev. Biochem.* **49**, 813–843
115. Mura, U., Chock, P. B., and Stadtman, E. R. (1981) *J. Biol. Chem.* **256**, 13022–13029
116. Almassy, R. J., Janson, C. A., Hamlin, R., Xuong, N.-H., and Eisenberg, D. (1986) *Nature (London)* **323**, 304–309
117. Garcia, E., and Rhee, S. G. (1983) *J. Biol. Chem.* **258**, 2246–2253
118. Moss, J., and Vaughan, M. (1995) *J. Biol. Chem.* **270**, 12327–12330
119. Amor, J. C., Harrison, D. H., Kahn, R. A., and Ringe, D. (1994) *Nature (London)* **372**, 704–708
120. Boman, A. L., and Kahn, R. A. (1995) *Trends Biochem. Sci.* **20**, 147–150
121. Lee, F.-J. S., Stevens, L. A., Hall, L. M., Murtagh, J. J., Jr., Kao, Y. L., Moss, J., and Vaughan, M. (1994) *J. Biol. Chem.* **269**, 21555–21560
122. Moss, J., Stanley, S. J., Nightingale, M. S., Murtagh, J. J., Jr., Monaco, L., Mishima, K., Chen, H.-C., Williamson, K.-C., and Tsai, S.-C. (1992) *J. Biol. Chem.* **267**, 10481–10488
123. Guse, A. H., da Silva, C. P., Weber, K., Ashamu, G. A., Potter, B. V. L., and Mayr, G. W. (1996) *J. Biol. Chem.* **271**, 23946–23953
124. Gaal, J. C., Smith, K. R., and Pearson, C. K. (1987) *Trends Biochem. Sci.* **12**, 129–130
125. Huttner, W. B. (1987) *Trends Biochem. Sci.* **12**, 361–363
126. Sakakibara, Y., Takami, Y., Zwieb, C., Nakayama, T., Suiko, M., Nakajima, H., and Liu, M.-C. (1995) *J. Biol. Chem.* **270**, 30470–30478
127. Rosenquist, G. L., and Nicholas, H. B., Jr. (1993) *Protein Sci.* **2**, 215–222
128. Sundaram, K. S., and Lev, M. (1992) *J. Biol. Chem.* **267**, 24041–24044
129. Hooper, L. V., Manzella, S. M., and Baenziger, J. U. (1996) *FASEB J.* **10**, 1137–1146
130. Terwilliger, T. C., and Koshland, D. E., Jr. (1984) *J. Biol. Chem.* **259**, 7719–7725
131. Favre, B., Zolnierowicz, S., Turowski, P., and Hemmings, B. A. (1994) *J. Biol. Chem.* **269**, 16311–16317
132. Edmondson, D. G., and Roth, S. Y. (1996) *FASEB J.* **10**, 1173–1182
133. Kuo, M.-H., Brownell, J. E., Sobel, R. E., Ranalli, T. A., Cook, R. G., Edmondson, D. G., Roth, S. Y., and Allis, C. D. (1996) *Nature (London)* **383**, 269–272
134. Brandes, H. K., Larimer, F. W., and Hartman, F. C. (1996) *J. Biol. Chem.* **271**, 3333–3335
135. Brandes, H. K., Hartman, F. C., Lu, T.-Y. S., and Larimer, F. W. (1996) *J. Biol. Chem.* **271**, 6490–6496
136. Swerdlow, R. D., Green, C. L., Setlow, B., and Setlow, D. (1979) *J. Biol. Chem.* **254**, 6835–6837
137. Hentze, M. W., Rouault, T. A., Harford, J. B., and Klausner, R. D. (1989) *Science* **244**, 357–359
138. Wadsworth, W. G., and Riddle, D. L. (1988) *Proc. Natl. Acad. Sci. U.S.A.* **85**, 8435–8438
139. Boron, W. F. (1984) *Nature (London)* **312**, 312
140. Siffert, W., and Akkerman, J. W. N. (1988) *Trends Biochem. Sci.* **13**, 148–151
141. Reinertsen, K. V., Tonnessen, T. I., Jacobsen, J., Sandvig, K., and Olsnes, S. (1988) *J. Biol. Chem.* **263**, 11117–11125
142. Schramm, V. L. (1982) *Trends Biochem. Sci.* **7**, 369–371
143. Srere, P. A. (1987) *Ann. Rev. Biochem.* **56**, 89–124
144. Srivastava, D. K., and Bernhard, S. A. (1987) *Ann. Rev. Biophys. Chem.* **16**, 175–204
145. Ovádi, J. (1988) *Trends Biochem. Sci.* **13**, 486–490
146. Kaprelyants, A. S. (1988) *Trends Biochem. Sci.* **13**, 43–46
- 146a. Miles, E. W., Rhee, S., and Davies, D. R. (1999) *J. Biol. Chem.* **274**, 12193–12196
- 146b. Serre, V., Guy, H., Penverne, B., Lux, M., Rotgeri, A., Evans, D., and Hervé, G. (1999) *J. Biol. Chem.* **274**, 23794–23801
- 146c. Geck, M. K., and Kirsch, J. F. (1999) *Biochemistry* **38**, 8032–8037
- 146d. Dell'Acqua, M. L., and Scott, J. D. (1997) *J. Biol. Chem.* **272**, 12881–12884
147. Ovádi, J., and Srere, P. A. (1992) *Trends Biochem. Sci.* **17**, 445–447
148. Inagaki, N., Ito, M., Nakano, T., and Inagaki, M. (1994) *Trends Biochem. Sci.* **19**, 448–452
149. Faux, M. C., and Scott, J. D. (1996) *Trends Biochem. Sci.* **21**, 312–315
150. Fraser, C. M. (1989) *J. Biol. Chem.* **264**, 9266–9270
151. Strosberg, A. D. (1993) *Protein Sci.* **2**, 1198–1209
152. Scheel, A. A., and Pelham, H. R. B. (1996) *Biochemistry* **35**, 10203–10209
153. Levitzki, A. (1988) *Science* **241**, 800–806
- 153a. Stacey, M., Lin, H.-H., Gordon, S., and McKnight, A. J. (2000) *Trends Biochem. Sci.* **25**, 284–289
- 153b. Leurs, R., Smit, M. J., Alewijnse, A. E., and Timmerman, H. (1998) *Trends Biochem. Sci.* **23**, 418–422
- 153c. Bockaert, J., and Pin, J. P. (1999) *EMBO J.* **18**, 1723–1729
154. Insel, P. A., and Wasserman, S. I. (1990) *FASEB J.* **4**, 2732–2736
155. Turki, J., Lorenz, J. N., Green, S. A., Donnelly, E. T., Jacinto, M., and Liggett, S. B. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 10483–10488

References

- 155a. Rohrer, D. K., Chruscinski, A., Schauble, E. H., Bernstein, D., and Kobilka, B. K. (1999) *J. Biol. Chem.* **274**, 16701–16708
- 155b. Limbird, L. E., and Vaughan, D. E. (1999) *Proc. Natl. Acad. Sci. U.S.A.* **96**, 7125–7127
156. Premont, R. T., Inglese, J., and Lefkowitz, R. J. (1995) *FASEB J.* **9**, 175–182
- 156a. Pitcher, J. A., Freedman, N. J., and Lefkowitz, R. J. (1998) *Ann. Rev. Biochem.* **67**, 653–692
157. Mochly-Rosen, D. (1995) *Science* **268**, 247–251
158. Benovic, J. L., DeBlasi, A., Stone, W. C., Caron, M. G., and Lefkowitz, R. J. (1989) *Science* **246**, 235–240
- 158a. Danner, S., Frank, M., and Lohse, M. J. (1998) *J. Biol. Chem.* **273**, 3223–3229
- 158b. Gabilondo, A. M., Hegler, J., Krasel, C., Boivin-Jahns, V., Hein, L., and Lohse, M. J. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 12285–12290
159. Rodbell, M., Birnbaumer, L., Pohl, S. L., and Krans, H. M. J. (1971) *J. Biol. Chem.* **246**, 1877–1882
160. Unson, C. G., Cypess, A. M., Wu, C.-R., Goldsmith, P. K., Merrifield, R. B., and Sakmar, T. P. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 310–315
161. Rosenthal, W., Antaramian, A., Gilbert, S., and Birnbaumer, M. (1993) *J. Biol. Chem.* **268**, 13030–13033
162. Sharif, M., and Hanley, M. R. (1992) *Nature (London)* **357**, 279–280
163. McFarland, K. C., Sprengel, R., Phillips, H. S., Köhler, M., Rosenblit, N., Nikolics, K., Segaloff, D. L., and Seeburg, P. H. (1989) *Science* **245**, 494–499
164. Kudo, M., Osuga, Y., Kobilka, B. K., and Hsueh, A. J. W. (1996) *J. Biol. Chem.* **271**, 22470–22478
165. Perlman, J. H., Laakkonen, L., Osman, R., and Gershengorn, M. C. (1994) *J. Biol. Chem.* **269**, 23883–23886
166. Conti, M., Iona, S., Cuomo, M., Swinnen, J. V., Odeh, J., and Svoboda, M. E. (1995) *Biochemistry* **34**, 7979–7987
167. Smith, K. J., Scotland, G., Beattie, J., Trayer, I. P., and Houslay, M. D. (1996) *J. Biol. Chem.* **271**, 16703–16711
168. Sutherland, E. W. (1972) *Science* **177**, 401–408
169. Pastan, I. (1972) *Sci. Am.* **227**(Aug), 97–105
170. Lalli, E., and Sassone-Corsi, P. (1994) *J. Biol. Chem.* **269**, 17359–17362
171. Bolwell, G. P. (1995) *Trends Biochem. Sci.* **20**, 492–495
172. Taussig, R., and Gilman, A. G. (1995) *J. Biol. Chem.* **270**, 1–4
173. Dessauer, C. W., Tesmer, J. J. G., Sprang, S. R., and Gilman, A. G. (1998) *J. Biol. Chem.* **273**, 25831–25839
174. Cooper, D. M. F., Mons, N., and Karpen, J. W. (1995) *Nature (London)* **374**, 421–424
175. Désaubry, L., Shoshani, I., and Johnson, R. A. (1996) *J. Biol. Chem.* **271**, 14028–14034
176. Hurley, J. H. (1999) *J. Biol. Chem.* **274**, 7599–7602
177. Hellevuo, K., Yoshimura, M., Mons, N., Hoffman, P. L., Cooper, D. M. F., and Tabakoff, B. (1995) *J. Biol. Chem.* **270**, 11581–11589
178. Walseth, T. F., Gander, J. E., Eide, S. J., Krick, T. P., and Goldberg, N. D. (1983) *J. Biol. Chem.* **258**, 1544–1558
- 178a. Chen, Y., Cann, M. J., Litvin, T. N., Lourgenko, V., Sinclair, M. L., Levin, L. R., and Buck, J. (2000) *Science* **289**, 625–628
- 178b. Kaupp, U. B., and Weyand, I. (2000) *Science* **289**, 559–560
179. Hamm, H. E. (1998) *J. Biol. Chem.* **273**, 669–672
180. Neer, E. J. (1994) *Protein Sci.* **3**, 3–14
181. Collins, S., Caron, M. G., and Lefkowitz, R. J. (1992) *Trends Biochem. Sci.* **17**, 37–39
182. Hepler, J. R., and Gilman, A. G. (1992) *Trends Biochem. Sci.* **17**, 383–387
183. Levitzki, A. (1988) *Trends Biochem. Sci.* **13**, 298–301
184. Rawls, R. L. (1987) *Chem. Eng. News* **65**, December 21, 26–39
185. Wang, N., Yan, K., and Rasenick, M. M. (1990) *J. Biol. Chem.* **265**, 1239–1242
186. Wang, Q., Mullah, B. K., and Robishaw, J. D. (1999) *J. Biol. Chem.* **274**, 17365–17371
187. Mayorga, L. S., Diaz, R., and Stahl, P. D. (1989) *Science* **244**, 1475–1477
188. Coleman, D. E., and Sprang, S. R. (1996) *Trends Biochem. Sci.* **21**, 41–44
189. Ray, K., Kunsch, C., Bonner, L. M., and Robishaw, J. D. (1995) *J. Biol. Chem.* **270**, 21765–21771
190. Brown, A. M. (1991) *FASEB J.* **5**, 2175–2179
191. Chen, J., DeVivo, M., Dingus, J., Harry, A., Li, J., Sui, J., Carty, D. J., Blank, J. L., Exton, J. H., Stoffel, R. H., Inglese, J., Lefkowitz, R. J., Logothetis, D. E., Hildebrandt, J. D., and Iyengar, R. (1995) *Science* **268**, 1166–1169
192. Schulz, K., Danner, S., Bauer, P., Schröder, S., and Lohse, M. J. (1996) *J. Biol. Chem.* **271**, 22546–22551
193. Yoshida, T., Willardson, B. M., Wilkins, J. F., Jensen, G. J., Thornton, B. D., and Bitensky, M. W. (1994) *J. Biol. Chem.* **269**, 24050–24057
- 193a. Whiteway, M., Hougau, L., Dignard, D., Thomas, D. Y., Bell, L., Saari, G. C., Grant, F. J., O'Hara, P., and MacKay, V. L. (1989) *Cell* **56**, 467–477
194. Noel, J. P., Hamm, H. E., and Sigler, P. B. (1993) *Nature (London)* **366**, 654–663
195. Lambright, D. G., Sondek, J., Bohm, A., Skiba, N. P., Hamm, H. E., and Sigler, P. B. (1996) *Nature (London)* **379**, 311–319
196. McLaughlin, S. K., McKinnon, P. J., and Margolskee, R. F. (1992) *Nature (London)* **357**, 563–569
197. Hall, A. (1990) *Science* **249**, 635–640
198. Valencia, A., Chardin, P., Wittinghofer, A., and Sander, C. (1991) *Biochemistry* **30**, 4637–4648
199. Macara, I. G., Lounsbury, K. M., Richards, S. A., McKiernan, C., and Bar-Sagi, D. (1996) *FASEB J.* **10**, 625–630
200. Boguski, M. S., and McCormick, F. (1993) *Nature (London)* **366**, 643–654
- 200a. Scita, G., Tenca, P., Frittoli, E., Tocchetti, A., Innocenti, M., Giardina, G., and Di Fiore, P. P. (2000) *EMBO J.* **19**, 2393–2398
201. Jurnak, F. (1985) *Science* **230**, 32–36
202. Sprinzl, M. (1994) *Trends Biochem. Sci.* **19**, 245–250
203. Berchtold, H., Reshetnikova, L., Reiser, C. O. A., Schirmer, N. K., Sprinzl, M., and Hilgenfeld, R. (1993) *Nature (London)* **365**, 126–132
204. Kawashima, T., Berthet-Colominas, C., Wulff, M., Cusack, S., and Leberman, R. (1996) *Nature (London)* **379**, 511–518
205. Jurnak, F. (1988) *Trends Biochem. Sci.* **13**, 195–198
206. Downward, J. (1990) *Trends Biochem. Sci.* **15**, 469–472
207. Wittinghofer, A., and Pai, E. F. (1991) *Trends Biochem. Sci.* **16**, 382–387
208. Schlichting, I., Almo, S. C., Rapp, G., Wilson, K., Petratos, K., Lentfer, A., Wittinghofer, A., Kabsch, W., Pai, E. F., Petsko, G. A., and Goody, R. S. (1990) *Nature (London)* **345**, 309–315
209. Kraulis, P. J., Domaille, P. J., Campbell-Burk, S. L., Van Aken, T., and Laue, E. D. (1994) *Biochemistry* **33**, 3515–3531
210. Bokoch, G. M., and Der, C. J. (1993) *FASEB J.* **7**, 750–759
211. Takai, Y., Sasaki, T., Tanaka, K., and Nakanishi, H. (1995) *Trends Biochem. Sci.* **20**, 227–231
212. Fischer von Mollard, G., Stahl, B., Li, C., Südhof, T. C., and Jahn, R. (1994) *Trends Biochem. Sci.* **19**, 164–168
- 212a. Burridge, K. (1999) *Science* **283**, 2028–2029
- 212b. Lin, R., Cerione, R. A., and Manor, D. (1999) *J. Biol. Chem.* **274**, 23633–23641
- 212c. Desrosiers, R. R., Gauthier, F., Lanthier, J., and Béliveau, R. (2000) *J. Biol. Chem.* **275**, 14949–14957
- 212d. Schwartz, M. A., and Shattil, S. J. (2000) *Trends Biochem. Sci.* **25**, 388–391
- 212e. Zhang, B., Zhang, Y., Wang, Z.-x., and Zheng, Y. (2000) *J. Biol. Chem.* **275**, 25299–25307
213. Symons, M. (1996) *Trends Biochem. Sci.* **21**, 178–181
214. Ando, S., Kaibuchi, K., Sasaki, T., Hiraoka, K., Nishiyama, T., Mizuno, T., Asada, M., Nunoi, H., Matsuda, I., Matsuura, Y., Polakis, P., McCormick, F., and Takai, Y. (1992) *J. Biol. Chem.* **267**, 25709–25713
215. Nuoffer, C., and Balch, W. E. (1994) *Ann. Rev. Biochem.* **63**, 949–990
216. Rybin, V., Ullrich, O., Rubino, M., Alexandrov, K., Simon, I., Seabra, M. C., Goody, R., and Zerial, M. (1996) *Nature (London)* **383**, 266–269
217. Garrett, M. D., Self, A. J., van Oers, C., and Hall, A. (1989) *J. Biol. Chem.* **264**, 10–13
- 217a. Moss, J., and Vaughan, M. (1998) *J. Biol. Chem.* **273**, 21431–21434
- 217b. Hutchinson, J. P., and Eccleston, J. F. (2000) *Biochemistry* **39**, 11348–11359
218. Scheffzek, K., Ahmadian, M. R., and Wittinghofer, A. (1998) *Trends Biochem. Sci.* **23**, 257–262
- 218a. Berman, D. M., and Gilman, A. G. (1998) *J. Biol. Chem.* **273**, 1269–1272
- 218b. Coleman, D. E., and Sprang, S. R. (1999) *J. Biol. Chem.* **274**, 16669–16672
219. Schalk, I., Zeng, K., Wu, S.-K., Stura, E. A., Matteson, J., Huang, M., Tandon, A., Wilson, I. A., and Balch, W. E. (1996) *Nature (London)* **381**, 42–48
220. Buss, J. E., Mumby, S. M., Casey, P. J., Gilman, A. G., and Sefton, B. M. (1987) *Proc. Natl. Acad. Sci. U.S.A.* **84**, 7493–7497
221. Wedegaertner, P. B., Wilson, P. T., and Bourne, H. R. (1995) *J. Biol. Chem.* **270**, 503–506
222. Song, J., Hirschman, J., Gunn, K., and Dohlman, H. G. (1996) *J. Biol. Chem.* **271**, 20273–20283
223. Hepler, J. R., Biddlecome, G. H., Kleuss, C., Camp, L. A., Hofmann, S. L., Ross, E. M., and Gilman, A. G. (1996) *J. Biol. Chem.* **271**, 496–504
224. Pesceckis, S. M., and Resh, M. D. (1994) *J. Biol. Chem.* **269**, 30888–30892
225. Giner, J.-L., and Rando, R. R. (1994) *Biochemistry* **33**, 15116–15123
226. Liu, L., Dudler, T., and Gelb, M. H. (1996) *J. Biol. Chem.* **271**, 23269–23276
227. Casey, P. J. (1995) *Science* **268**, 221–225
228. Maltese, W. A., Sheridan, K. M., Repko, E. M., and Erdman, R. A. (1990) *J. Biol. Chem.* **265**, 2148–2155
229. Milligan, G., Parenti, M., and Magee, A. I. (1995) *Trends Biochem. Sci.* **20**, 181–186
230. Duncan, J. A., and Gilman, A. G. (1996) *J. Biol. Chem.* **271**, 23594–23600
- 230a. Iiri, T., Backlund, P. S., Jr., Jones, T. L. Z., Wedegaertner, P. B., and Bourne, H. R. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 14592–14597
- 230b. Lee, T. W., Seifert, R., Guan, X., and Kobilka, B. K. (1999) *Biochemistry* **38**, 13801–13809
231. Lambright, D. G., Noel, J. P., Hamm, H. E., and Sigler, P. B. (1994) *Nature (London)* **369**, 621–628
232. Sondek, J., Lambright, D. G., Noel, J. P., Hamm, H. E., and Sigler, P. B. (1994) *Nature (London)* **372**, 276–279

References

233. Coleman, D. E., Berghuis, A. M., Lee, E., Linder, M. E., Gilman, A. G., and Sprang, S. R. (1994) *Science* **265**, 1405–1412
234. Mixon, M. B., Lee, E., Coleman, D. E., Berghuis, A. M., Gilman, A. G., and Sprang, S. R. (1995) *Science* **270**, 954–960
235. Kjeldgaard, M., Nyborg, J., and Clark, B. F. C. (1996) *FASEB J.* **10**, 1347–1368
236. Higashijima, T., Graziano, M. P., Suga, H., Kainosho, M., and Gilman, A. G. (1991) *J. Biol. Chem.* **266**, 3396–3401
237. Sondek, J., Bohm, A., Lambright, D. G., Hamm, H. E., and Sigler, P. B. (1996) *Nature (London)* **379**, 369–374
238. Clapham, D. E. (1996) *Nature (London)* **379**, 297–299
239. Neer, E. J., Schmidt, C. J., Nambudripad, R., and Smith, T. F. (1994) *Nature (London)* **371**, 297–300
240. Faber, X. X. (1994) *Structure* **3**, 551–559
241. Pai, E. F., Kabsch, W., Kregel, U., Holmes, K. C., John, J., and Wittinghofer, A. (1989) *Nature (London)* **341**, 209–214
242. Wall, M. A., Posner, B. A., and Sprang, S. R. (1998) *Structure* **6**, 1169–1183
243. Wilson, P. T., and Bourne, H. R. (1995) *J. Biol. Chem.* **270**, 9667–9675
244. Zhao, Y., Brandish, P. E., DiValentin, M., Schelvis, J. P. M., Babcock, G. T., and Marletta, M. A. (2000) *Biochemistry* **39**, 10848–10854
245. Garbers, D. L., and Lowe, D. G. (1994) *J. Biol. Chem.* **269**, 30741–30744
246. Subbaraya, I., Ruiz, C. C., Helekar, B. S., Zhao, X., Gorczyca, W. A., Pettenati, M. J., Rao, P. N., Palczewski, K., and Baehr, W. (1994) *J. Biol. Chem.* **269**, 31080–31089
247. Lincoln, T. M., and Cornwell, T. L. (1993) *FASEB J.* **7**, 328–338
248. Houslay, M. D. (1985) *Trends Biochem. Sci.* **10**, 465–466
249. Garbers, D. L. (1989) *J. Biol. Chem.* **264**, 9103–9106
250. Lowe, D. G., Chang, M.-S., Hellmiss, R., Chen, E., Singh, S., Garbers, D. L., and Goeddel, D. V. (1989) *EMBO J.* **8**, 1377–1384
251. Hamra, F. K., Forte, L. R., Eber, S. L., Pidhorodeckyj, N. V., Krause, W. J., Freeman, R. H., Chin, D. T., Tompkins, J. A., Fok, K. F., Smith, C. E., Duffin, K. L., Siegel, N. R., and Currie, M. G. (1993) *Proc. Natl. Acad. Sci. U.S.A.* **90**, 10464–10468
252. Stamler, J. S., Singel, D. J., and Loscalzo, J. (1992) *Science* **258**, 1898–1902
253. Yuen, P. S. T., Doolittle, L. K., and Garbers, D. L. (1994) *J. Biol. Chem.* **269**, 791–793
254. Stone, J. R., and Marletta, M. A. (1996) *Biochemistry* **35**, 1093–1099
255. Mayer, B., Schrammel, A., Klatt, P., Koesling, D., and Schmidt, K. (1995) *J. Biol. Chem.* **270**, 17355–17360
256. Stone, J. R., and Marletta, M. A. (1995) *Biochemistry* **34**, 16397–16403
257. Deinum, G., Stone, J. R., Babcock, G. T., and Marletta, M. A. (1996) *Biochemistry* **35**, 1540–1547
258. Verma, A., Hirsch, D. J., Glatt, C. E., Ronnett, G. V., and Snyder, S. H. (1993) *Science* **259**, 381–384
259. Stock, J. B., Surette, M. G., McCleary, W. R., and Stock, A. M. (1992) *J. Biol. Chem.* **267**, 19753–19756
- 259a. Kim, K. K., Yokota, H., and Kim, S.-H. (1999) *Nature (London)* **400**, 787–792
260. Scott, W. G., Milligan, D. L., Milburn, M. V., Privé, G. G., Yeh, J., Koshland, D. E., Jr., and Kim, S.-H. (1993) *J. Mol. Biol.* **232**, 555–573
261. Danielson, M. A., Biemann, H.-P., Koshland, D. E., Jr., and Falke, J. J. (1994) *Biochemistry* **33**, 6100–6109
262. Yeh, J. I., Biemann, H.-P., Privé, G. G., Pandit, J., Koshland, D. E., Jr., and Kim, S.-H. (1996) *J. Mol. Biol.* **262**, 186–201
263. Danielson, M. A., Bass, R. B., and Falke, J. J. (1997) *J. Biol. Chem.* **272**, 32878–32888
264. Kim, S.-H. (1994) *Protein Sci.* **3**, 159–165
- 264a. Ottemann, K. M., Xiao, W., Shin, Y.-K., and Koshland, D. E., Jr. (1999) *Science* **285**, 1751–1754
265. Berridge, M. J., and Irvine, R. F. (1989) *Nature (London)* **341**, 197–205
266. Schulman, H., and Lou, L. L. (1989) *Trends Biochem. Sci.* **14**, 62–66
267. Carafoli, E. (1987) *Ann. Rev. Biochem.* **56**, 395–433
268. Rasmussen, H. (1989) *Sci. Am.* **261**(Oct), 66–73
269. Kobilka, B. K., Matsui, H., Kobilka, T. S., Yang-Feng, T. L., Francke, U., Caron, M. G., Lefkowitz, R. J., and Regan, J. W. (1987) *Science* **238**, 650–656
270. Cotecchia, S., Kobilka, B. K., Daniel, K. W., Nolan, R. D., Lapetina, E. Y., Caron, M. G., Lefkowitz, R. J., and Regan, J. W. (1990) *J. Biol. Chem.* **265**, 63–69
271. Ceresa, B. P., and Limbird, L. E. (1994) *J. Biol. Chem.* **269**, 29557–29564
272. Scheer, A., Fanelli, F., Costa, T., De Benedetti, P. G., and Cotecchia, S. (1996) *EMBO J.* **15**, 3566–3578
273. Hwa, J., and Perez, D. M. (1996) *J. Biol. Chem.* **271**, 6322–6327
274. Boyer, J. L., Waldo, G. L., Evans, T., Northup, J. K., Downes, C. P., and Harden, T. K. (1989) *J. Biol. Chem.* **264**, 13917–13922
275. Wu, D., Jiang, H., and Simon, M. I. (1995) *J. Biol. Chem.* **270**, 9828–9832
276. Nishizuka, Y. (1995) *FASEB J.* **9**, 484–496
277. Majerus, P. W. (1992) *Ann. Rev. Biochem.* **61**, 225–250
278. Majerus, P. W., Connolly, T. M., Bansal, V. S., Inhorn, R. C., Ross, T. S., and Lips, D. L. (1988) *J. Biol. Chem.* **263**, 3051–3054
279. Rhee, S. G., Suh, P.-G., Ryu, S.-H., and Lee, S. Y. (1989) *Science* **244**, 546–550
280. Hough, E., Hansen, L. K., Birknes, B., Jynge, K., Hansen, S., Hordvik, A., Little, C., Dodson, E., and Derewenda, Z. (1989) *Nature (London)* **338**, 357–360
281. Essen, L.-O., Perisic, O., Cheung, R., Katan, M., and Williams, R. L. (1996) *Nature (London)* **380**, 595–602
282. Berridge, M. J. (1993) *Nature (London)* **361**, 315–325
283. Parthasarathy, R., and Eisenberg, F. J. (1986) *Biochem. J.* **235**, 313–322
284. Majerus, P. W., Connolly, T. M., Deckmyn, H., Ross, T. S., Bross, T. E., Ishii, H., Bansal, V. S., and Wilson, D. B. (1986) *Science* **234**, 1519–1526
285. Houslay, M. D. (1987) *Trends Biochem. Sci.* **12**, 1–2
286. Nickels, J. T., Jr., Buxeda, R. J., and Carman, G. M. (1994) *J. Biol. Chem.* **269**, 11018–11024
287. Boronkov, I. V., and Anderson, R. A. (1995) *J. Biol. Chem.* **270**, 2881–2884
288. Hughes, A. R., Takemura, H., and Putney, J. W., Jr. (1988) *J. Biol. Chem.* **263**, 10314–10319
289. Furuichi, T., Yoshikawa, S., Miyawaki, A., Wada, K., Maeda, N., and Mikoshiba, K. (1989) *Nature (London)* **342**, 32–38
- 289a. Zimmermann, B., and Walz, B. (1999) *EMBO J.* **18**, 3222–3231
290. Yoshikawa, F., Morita, M., Monkawa, T., Michikawa, T., Furuichi, T., and Mikoshiba, K. (1996) *J. Biol. Chem.* **271**, 18277–18284
291. Humbert, J.-P., Matter, N., Artault, J.-C., Köppler, P., and Malviya, A. N. (1996) *J. Biol. Chem.* **271**, 478–485
292. Hennager, D. J., Welsh, M. J., and DeLisle, S. (1995) *J. Biol. Chem.* **270**, 4959–4962
293. Taylor, C. W., and Marshall, I. C. B. (1992) *Trends Biochem. Sci.* **17**, 403–407
294. Gu, Q.-M., and Sih, C. J. (1994) *J. Am. Chem. Soc.* **116**, 7481–7486
295. Berridge, M. J. (1993) *Nature (London)* **365**, 388–389
296. Lee, H. C., Walseth, T. F., Bratt, G. T., Hayes, R. N., and Clapper, D. L. (1989) *J. Biol. Chem.* **264**, 1608–1615
297. Vu, C. Q., Lu, P.-J., Chen, C.-S., and Jacobson, M. K. (1996) *J. Biol. Chem.* **271**, 4747–4754
298. Dennis, E. A., Rhee, S. G., Billah, M. M., and Hannun, Y. A. (1991) *FASEB J.* **5**, 2068–2077
299. Sternweis, P. C., and Smrcka, A. V. (1992) *Trends Biochem. Sci.* **17**, 502–506
300. Rhee, S. G., and Choi, K. D. (1992) *J. Biol. Chem.* **267**, 12393–12396
- 300a. Kim, M. J., Chang, J.-S., Park, S. K., Hwang, J.-I., Ryu, S. H., and Suh, P.-G. (2000) *Biochemistry* **39**, 8674–8682
301. Jafri, M. S., and Keizer, J. (1995) *Biophys. J.* **69**, 2139–2153
302. Combettes, L., Cheek, T. R., and Taylor, C. W. (1996) *EMBO J.* **15**, 2086–2093
303. van de Put, F. H. M. M., De Pont, J. J. H. H. M., and Willems, P. H. G. M. (1994) *J. Biol. Chem.* **269**, 12438–12443
304. Kukuljan, M., Rojas, E., Catt, K. J., and Stojilkovic, S. S. (1994) *J. Biol. Chem.* **269**, 4860–4865
305. Berridge, M. J., and Galione, A. (1988) *FASEB J.* **2**, 3074–3082
306. Meyer, T., Wensel, T., and Stryer, L. (1990) *Biochemistry* **29**, 32–37
307. Luttrell, B. M. (1993) *J. Biol. Chem.* **268**, 1521–1524
308. Wilson, M. P., and Majerus, P. W. (1996) *J. Biol. Chem.* **271**, 11904–11910
309. Shears, S. B. (1989) *J. Biol. Chem.* **264**, 19879–19886
- 309a. Zhu, D.-M., Tekle, E., Huang, C. Y., and Chock, P. B. (2000) *J. Biol. Chem.* **275**, 6063–6066
- 309b. Odom, A. R., Stahlberg, A., Wente, S. R., and York, J. D. (2000) *Science* **287**, 2026–2029
310. Ryu, S. H., Lee, S. Y., Lee, K.-Y., and Rhee, S. G. (1987) *FASEB J.* **1**, 388–393
311. Fukuda, M., and Mikoshiba, K. (1996) *J. Biol. Chem.* **271**, 18838–18842
312. Bird, G. St. J., and Putney, J. W., Jr. (1996) *J. Biol. Chem.* **271**, 6766–6770
313. Dixon, J. F., and Hokin, L. E. (1987) *J. Biol. Chem.* **262**, 13892–13895
314. Heinz, D. W., Ryan, M., Bullock, T. L., and Griffith, O. H. (1995) *EMBO J.* **14**, 3855–3863
315. Ali, N., Craxton, A., and Shears, S. B. (1993) *J. Biol. Chem.* **268**, 6161–6167
316. Xie, W., Kaetzel, M. A., Bruzik, K. S., Dedman, J. R., Shears, S. B., and Nelson, D. J. (1996) *J. Biol. Chem.* **271**, 14092–14097
317. Ismailov, I. I., Fuller, C. M., Berdiev, B. K., Shlyonsky, V. G., Benos, D. J., and Barrett, K. E. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 10505–10509
318. Dasgupta, S., Dasgupta, D., Sen, M., Biswas, S., and Biswas, B. B. (1996) *Biochemistry* **35**, 4994–5001
319. Voglmaier, S. M., Bembenek, M. E., Kaplin, A. I., Dormán, G., Olszewski, J. D., Prestwich, G. D., and Snyder, S. H. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 4305–4310
- 319a. Chi, T. H., and Crabtree, G. R. (2000) *Science* **287**, 1937–1939
320. Menniti, F. S., Oliver, K. G., Putney, J. W., Jr., and Shears, S. B. (1993) *Trends Biochem. Sci.* **18**, 53–56
- 320a. Yang, X., Safrany, S. T., and Shears, S. B. (1999) *J. Biol. Chem.* **274**, 35434–35440

References

321. Stephens, L., Radenberg, T., Thiel, U., Vogel, G., Khoo, K.-H., Dell, A., Jackson, T. R., Hawkins, P. T., and Mayr, G. W. (1993) *J. Biol. Chem.* **268**, 4009–4015
322. Van Dijken, P., de Haas, J.-R., Craxton, A., Erneux, C., Shears, S. B., and Van Haastert, P. J. M. (1995) *J. Biol. Chem.* **270**, 29724–29731
323. Bansal, V. S., Caldwell, K. K., and Majerus, P. W. (1990) *J. Biol. Chem.* **265**, 1806–1811
324. York, J. D., Ponder, J. W., Chen, Z.-w., Mathews, F. S., and Majerus, P. W. (1994) *Biochemistry* **33**, 13164–13171
325. Norris, F. A., Auethavekiat, V., and Majerus, P. W. (1995) *J. Biol. Chem.* **270**, 16128–16133
326. Jefferson, A. B., and Majerus, P. W. (1995) *J. Biol. Chem.* **270**, 9370–9377
- 326a. Norris, F. A., Wilson, M. P., Wallis, T. S., Galyov, E. E., and Majerus, P. W. (1998) *Proc. Natl. Acad. Sci. U.S.A.* **95**, 14057–14059
- 326b. Munday, A. D., Norris, F. A., Caldwell, K. K., Brown, S., Majerus, P. W., and Mitchell, C. A. (1999) *Proc. Natl. Acad. Sci. U.S.A.* **96**, 3640–3645
327. Bone, R., Springer, J. P., and Atack, J. R. (1992) *Proc. Natl. Acad. Sci. U.S.A.* **89**, 10031–10035
328. Ganzhorn, A. J., Lepage, P., Pelton, P. D., Strasser, F., Vincendon, P., and Rondeau, J.-M. (1996) *Biochemistry* **35**, 10957–10966
329. Burgoyne, R. D. (1989) *Trends Biochem. Sci.* **14**, 87–88
330. Huang, K.-P., Huang, F. L., Nakabayashi, H., and Yoshida, Y. (1988) *J. Biol. Chem.* **263**, 14839–14845
- 330a. Ron, D., and Kazanietz, M. G. (1999) *FASEB J.* **13**, 1658–1676
- 330b. Ward, N. E., Stewart, J. R., Ionnides, C. G., and O'Brian, C. A. (2000) *Biochemistry* **39**, 10319–10329
331. Pessin, M. S., and Raben, D. M. (1989) *J. Biol. Chem.* **264**, 8729–8738
332. Exton, J. H. (1990) *J. Biol. Chem.* **265**, 1–4
- 332a. Perisic, O., Paterson, H. F., Mosedale, G., Lara-González, S., and Williams, R. L. (1999) *J. Biol. Chem.* **274**, 14979–14987
333. Roberts, M. F. (1996) *FASEB J.* **10**, 1159–1172
334. Liscovitch, M., Chalifa, V., Pertile, P., Chen, C.-S., and Cantley, L. C. (1994) *J. Biol. Chem.* **269**, 21403–21406
335. Stein, R. L., Melandri, F., and Dick, L. (1935) *Biochemistry* **35**, 3899–3908
336. Ward, S. G., Mills, S. J., Liu, C., Westwick, J., and Potter, B. V. L. (1995) *J. Biol. Chem.* **270**, 12075–12084
337. Woscholski, R., Kodaki, T., Palmer, R. H., Waterfield, M. D., and Parker, P. J. (1995) *Biochemistry* **34**, 11489–11493
- 337a. Odorizzi, G., Babst, M., and Emr, S. D. (2000) *Trends Biochem. Sci.* **25**, 229–235
- 337b. Walker, E. H., Perisic, O., Ried, C., Stephens, L., and Williams, R. L. (1999) *Nature (London)* **402**, 313–320
- 337c. Ching, T.-T., Wang, D.-S., Hsu, A.-L., Lu, P.-J., and Chen, C.-S. (1999) *J. Biol. Chem.* **274**, 8611–8617
- 337d. Bertsch, U., Deschermeier, C., Fanick, W., Girkontaite, I., Hillemeier, K., Johnen, H., Weglöhner, W., Emmrich, F., and Mayr, G. W. (2000) *J. Biol. Chem.* **275**, 1557–1564
- 337e. Jackson, T. R., Kearns, B. G., and Theibert, A. B. (2000) *Trends Biochem. Sci.* **25**, 489–495
- 337f. Gillooly, D. J., Morrow, I. C., Lindsay, M., Gould, R., Bryant, N. J., Gaullier, J.-M., Parton, R. G., and Stenmark, H. (2000) *EMBO J.* **19**, 4577–4588
338. Pennisi, E. (1997) *Science* **275**, 1876–1878
- 338a. Stambolic, V., Suzuki, A., de la Pompa, J. L., Brothers, G. M., Mirtsos, C., Sasaki, T., Ruland, J., Penninger, J. M., Siderovski, D. P., and Mak, T. W. (1998) *Cell* **95**, 29–39
- 338b. Toker, A., and Cantley, L. C. (1997) *Nature (London)* **387**, 673–675
339. Ui, M., Okada, T., Hazeki, K., and Hazeki, O. (1995) *Trends Biochem. Sci.* **20**, 303–307
- 339a. Takasuga, S., Katada, T., Ui, M., and Hazeki, O. (1999) *J. Biol. Chem.* **274**, 19545–19550
340. Chock, P. B., Rhee, S. G., and Stadtman, E. R. (1980) *Ann. Rev. Biochem.* **49**, 813–843
341. Stadtman, E. R., and Chock, P. B. (1977) *Proc. Natl. Acad. Sci. U.S.A.* **74**, 2761–2765
342. Huang, C.-Y. F., and Ferrell, J. E., Jr. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 10078–10083
343. Goldbeter, A., and Koshland, D. E., Jr. (1987) *J. Biol. Chem.* **262**, 4460–4471
344. Schacter-Noiman, E., Chock, P. B., and Stadtman, E. R. (1983) *Philos. Trans. R. Soc. London B* **302**, 157–166
345. Schacter, E., Chock, P. B., and Stadtman, E. R. (1984) *J. Biol. Chem.* **259**, 12260–12264
346. Newsholme, E. A., Challiss, R. A. J., and Crabtree, B. (1984) *Trends Biochem. Sci.* **9**, 277–280
347. Koshland, D. E., Jr. (1987) *Trends Biochem. Sci.* **12**, 225–228
348. Taylor, S. I. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 1 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 843–896, McGraw-Hill, New York
349. Müller, G., Rouveyre, N., Crecelius, A., and Bandlow, W. (1998) *Biochemistry* **37**, 8683–8695
350. Smit, A. B., Vreugdenhil, E., Ebberink, R. H. M., Geraerts, W. P. M., Klootwijk, J., and Joosse, J. (1988) *Nature (London)* **331**, 535–538
351. LeRoith, D., Shiloach, J., Roth, J., and Lesniak, M. A. (1983) *J. Biol. Chem.* **256**, 6533–6536
352. Rosen, O. M. (1987) *Science* **237**, 1452–1458
353. White, M. F., and Kahn, C. R. (1994) *J. Biol. Chem.* **269**, 1–4
354. Barnard, R. J., and Youngren, J. F. (1992) *FASEB J.* **6**, 3238–3244
355. Bell, G. I., Burant, C. F., Takeda, J., and Gould, G. W. (1993) *J. Biol. Chem.* **268**, 19161–19164
356. Mueckler, M. (1994) *Eur. J. Biochem.* **219**, 713–725
- 356a. Li, J., Houseknecht, K. L., Stenbit, A. E., Katz, E. B., and Charron, M. J. (2000) *FASEB J.* **14**, 1117–1125
357. Cuatrecasas, P. (1972) *Proc. Natl. Acad. Sci. U.S.A.* **69**, 1277–1281
358. Pashmforoush, M., Yoshimasa, Y., and Steiner, D. F. (1994) *J. Biol. Chem.* **269**, 32639–32648
359. Ullrich, A., Bell, J. R., Chen, E. Y., Herrera, R., Petruzzelli, L. M., Dull, T. J., Gray, A., Coussens, L., Liao, Y.-C., Tsubokawa, M., Mason, A., Seeburg, P. H., Grunfeld, C., Rosen, O. M., and Ramachandran, J. (1985) *Nature (London)* **313**, 756–761
360. Xu, Q.-Y., Paxton, R. J., and Fujita-Yamaguchi, Y. (1990) *J. Biol. Chem.* **265**, 18673–18681
361. Herrera, R., Petruzzelli, L., Thomas, N., Bramson, H. N., Kaiser, E. T., and Rosen, O. M. (1985) *Proc. Natl. Acad. Sci. U.S.A.* **82**, 7899–7903
362. McClain, D. A., Maegawa, H., Lee, J., Dull, T. J., Ullrich, A., and Olefsky, J. M. (1987) *J. Biol. Chem.* **262**, 14663–14671
363. Perlman, R., Bottaro, D. P., White, M. F., and Kahn, C. R. (1989) *J. Biol. Chem.* **264**, 8946–8950
- 363a. Rouard, M., Bass, J., Grigorescu, J., Garrett, T. P. J., Ward, C. W., Lipkind, G., Jaffiole, C., Steiner, D. F., and Bell, G. I. (1999) *J. Biol. Chem.* **274**, 18487–18491
- 363b. Luo, R. Z.-T., Beniac, D. R., Fernandes, A., Yip, C. C., and Ottensmeyer, F. P. (1999) *Science* **285**, 1077–1080
- 363c. Woldin, C. N., Hing, F. S., Lee, J., Pilch, P. F., and Shipley, G. G. (1999) *J. Biol. Chem.* **274**, 34981–34992
364. Lemmon, M. A., and Schlessinger, J. (1994) *Trends Biochem. Sci.* **19**, 459–563
365. Hubbard, S. R., Wei, L., Ellis, L., and Hendrickson, W. A. (1994) *Nature (London)* **372**, 746–754
- 365a. Hubbard, S. R. (1997) *EMBO J.* **16**, 5572–5581
366. White, M. F., Shoelson, S. E., Keutmann, H., and Kahn, C. R. (1988) *J. Biol. Chem.* **263**, 2969–2980
367. Fabry, M., Schaefer, E., Ellis, L., Kojro, E., Fahrenholz, F., and Brandenburg, D. (1992) *J. Biol. Chem.* **267**, 8950–8956
368. Williams, P. F., Mynarcik, D. C., Yu, G. Q., and Whittaker, J. (1995) *J. Biol. Chem.* **270**, 3012–3016
369. Kjeldsen, T., Wiberg, F. C., and Andersen, A. S. (1994) *J. Biol. Chem.* **269**, 32942–32946
- 369a. Sparrow, L. G., McKern, N. M., Gorman, J. J., Strike, P. M., Robinson, C. P., Bentley, J. D., and Ward, C. W. (1997) *J. Biol. Chem.* **272**, 29460–29467
- 369b. Garant, M. J., Kole, S., Maksimova, E. M., and Bernier, M. (1999) *Biochemistry* **38**, 5896–5904
370. Clark, S., and Harrison, L. C. (1983) *J. Biol. Chem.* **258**, 11434–11437
371. Saltiel, A. R. (1994) *FASEB J.* **8**, 1034–1040
372. Carter, W. G., Asamoah, K. A., and Sale, G. J. (1995) *Biochemistry* **34**, 9488–9499
373. Myers, M. G., Jr., Sun, X. J., and White, M. F. (1994) *Trends Biochem. Sci.* **19**, 289–293
374. Yenush, L., Makati, K. J., Smith-Hall, J., Ishibashi, O., Myers, M. G. J., and White, M. F. (1996) *J. Biol. Chem.* **271**, 24300–24306
375. Kowalski-Chauvel, A., Pradayrol, L., Vaysse, N., and Seva, C. (1996) *J. Biol. Chem.* **271**, 26356–26361
376. Paz, K., Voliovitch, H., Hadari, Y. R., Roberts, C. T., Jr., LeRoith, D., and Zick, Y. (1996) *J. Biol. Chem.* **271**, 6998–7003
- 376a. Rother, K. I., Imai, Y., Caruso, M., Beguinot, F., Formisano, P., and Accili, D. (1998) *J. Biol. Chem.* **273**, 17491–17497
- 376b. Clark, S. F., Molero, J.-C., and James, D. E. (2000) *J. Biol. Chem.* **275**, 3819–3826
- 376c. Lehr, S., Kotzka, J., Herkner, A., Sikkman, A., Meyer, H. E., Krone, W., and Müller-Wieland, D. (2000) *Biochemistry* **39**, 10898–10907
377. Ricketts, W. A., Rose, D. W., Shoelson, S., and Olefsky, J. M. (1996) *J. Biol. Chem.* **271**, 26165–26169
378. Okada, S., and Pessin, J. E. (1996) *J. Biol. Chem.* **271**, 25533–25538
379. Czech, M. P., Klarlund, J. K., Yagaloff, K. A., Bradford, A. P., and Lewis, R. E. (1988) *J. Biol. Chem.* **263**, 11017–11020
380. Seger, R., and Krebs, E. G. (1995) *FASEB J.* **9**, 726–735
381. Carter, W. G., Sullivan, A. C., Asamoah, K. A., and Sale, G. J. (1996) *Biochemistry* **35**, 14340–14351
382. Kasus-Jacobi, A., Perdureau, D., Auzan, C., Clauser, E., Van Obberghen, E., Mauvais-Jarvis, F., Girard, J., and Burnol, A.-F. (1998) *J. Biol. Chem.* **273**, 26026–26035
383. Baltensperger, K., Karoor, V., Paul, H., Ruoho, A., Czech, M. P., and Malbon, C. C. (1996) *J. Biol. Chem.* **271**, 1061–1064
384. Hara, K., Yonezawa, K., Sakaue, H., Ando, A., Kotani, K., Kitamura, T., Kitamura, Y., Ueda, H., Stephens, L., Jackson, T. R., Hawkins, P. T., Dhand, R., Clark, A. E., Holman, G. D., Waterfield, M. D., and Kasuga, M. (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 7415–7419
385. Lam, K., Carpenter, C. L., Ruderman, N. B., Friel, J. C., and Kelly, K. L. (1994) *J. Biol. Chem.* **269**, 20648–20652
386. Sutherland, C., O'Brien, R. M., and Granner, D. K. (1995) *J. Biol. Chem.* **270**, 15501–15506
387. Domin, J., Dhand, R., and Waterfield, M. D. (1996) *J. Biol. Chem.* **271**, 21614–21621
- 387a. Kosaki, A., Yamada, K., Suga, J., Otaka, A., and Kuzuya, H. (1998) *J. Biol. Chem.* **273**, 940–944

References

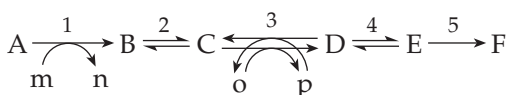
388. Lefebvre, V., Méchin, M.-C., Louckx, M. P., Rider, M. H., and Hue, L. (1996) *J. Biol. Chem.* **271**, 22289–22292
389. Quon, M. J., Butte, A. J., Zarnowski, M. J., Sesti, G., Cushman, S. W., and Taylor, S. I. (1994) *J. Biol. Chem.* **269**, 27920–27924
390. Kandror, K. V., and Pilch, P. F. (1996) *J. Biol. Chem.* **271**, 21703–21708
- 390a. Marte, B. M., and Downward, J. (1997) *Trends Biochem. Sci.* **22**, 355–358
- 390b. Hemmings, B. A. (1997) *Science* **275**, 628–630
- 390c. Chan, T. O., Rittenhouse, S. E., and Tschlis, P. N. (1999) *Ann. Rev. Biochem.* **68**, 965–1014
- 390d. Aguirre, V., Uchida, T., Yenush, L., Davis, R., and White, M. F. (2000) *J. Biol. Chem.* **275**, 9047–9054
- 390e. Nakae, J., Park, B.-C., and Accili, D. (1999) *J. Biol. Chem.* **274**, 15982–15985
391. Kishi, K., Hayashi, H., Wang, L., Kamohara, S., Tamaoka, K., Shimizu, T., Ushikubi, F., Narumiya, S., and Ebina, Y. (1996) *J. Biol. Chem.* **271**, 26561–26568
392. Yamauchi, K., Ribon, V., Saliel, A. R., and Pessin, J. E. (1995) *J. Biol. Chem.* **270**, 17716–17722
393. Zhang, J., Hiken, J., Davis, A. E., and Lawrence, J. C., Jr. (1989) *J. Biol. Chem.* **264**, 17513–17523
394. Robinson, F. W., Smith, C. J., Flanagan, J. E., Shibata, H., and Kono, T. (1989) *J. Biol. Chem.* **264**, 16458–16464
395. Najjar, S. M., Choice, C. V., Soni, P., Whitman, C. M., and Poy, M. N. (1998) *J. Biol. Chem.* **273**, 12923–12928
396. Haft, C. R., Klausner, R. D., and Taylor, S. I. (1994) *J. Biol. Chem.* **269**, 26286–26294
397. Safavi, A., Miller, B. C., Cottam, L., and Hersh, L. B. (1996) *Biochemistry* **35**, 14318–14325
398. Kublaoui, B., Lee, J., and Pilch, P. F. (1995) *J. Biol. Chem.* **270**, 59–65
399. Bevan, A. P., Burgess, J. W., Drake, P. G., Shaver, A., Bergeron, J. J. M., and Posner, B. I. (1995) *J. Biol. Chem.* **270**, 10784–10791
400. Podlecki, D. A., Smith, R. M., Kao, M., Tsai, P., Huecksteadt, T., Brandenburg, D., Lasher, R. S., Jarrett, L., and Olefsky, J. M. (1987) *J. Biol. Chem.* **262**, 3362–3368
401. Cherniack, A. D., Klarlund, J. K., Conway, B. R., and Czech, M. P. (1995) *J. Biol. Chem.* **270**, 1485–1488
402. Unoue, G., Cheatham, B., and Kahn, C. R. (1996) *J. Biol. Chem.* **271**, 28206–28211
403. Yarden, Y., and Ullrich, A. (1988) *Biochemistry* **27**, 3113–3119
404. James, R. (1984) *Ann. Rev. Biochem.* **53**, 259–292
405. Nilsen-Hamilton, M., ed. (1994) *Growth Factors and Signal Transduction in Development*, Wiley-Liss, New York
406. Claesson-Welsh, L. (1994) *J. Biol. Chem.* **269**, 32023–32026
407. Williams, L. T. (1989) *Science* **243**, 1564–1570
408. Hesketh, R. (1995) *The Oncogene Facts Book*, Academic Press, San Diego, California
409. Hunter, T. (1984) *Sci. Am.* **251**(Aug), 70–79
410. Marx, J. (1994) *Science* **266**, 1942–1944
411. Cooper, G. M., ed. (1990) *Oncogenes*, Jones & Bartlett, Boston, Massachusetts
412. Bishop, J. M. (1996) *FASEB J.* **10**, 362–364
413. Weinberg, R. A. (1983) *Sci. Am.* **249**(Nov), 126–142
414. Hickman, C. P. (1973) *Biology of the Invertebrates*, Mosby, St. Louis, Missouri
415. Yarden, Y., Escobedo, J. A., Kuang, W. J., Yang-Feng, T. L., Daniel, T. O., Tremble, P. M., Chen, E. Y., Ando, M. E., Harkins, R. N., Francke, V., Fried, V. A., Ullrich, A., and Williams, L. T. (1986) *Nature (London)* **323**, 226–232
416. Robbins, K. C., Antoniades, H. N., Devare, S. G., Hunkapillar, M. W., and Aaronson, S. A. (1983) *Nature (London)* **305**, 605–608
417. Yarden, Y., and Peles, E. (1991) *Biochemistry* **30**, 3543–3550
418. Peles, E., Lamprecht, R., Ben-Levy, R., Tzahar, E., and Yarden, Y. (1992) *J. Biol. Chem.* **267**, 12266–12274
419. Carpenter, G., and Cohen, S. (1990) *J. Biol. Chem.* **265**, 7709–7712
420. Suen, T.-C., and Goss, P. E. (2000) *J. Biol. Chem.* **275**, 6600–6607
421. Stover, D. R., Becker, M., Liebetanz, J., and Lydon, N. B. (1995) *J. Biol. Chem.* **270**, 15591–15597
422. Kauffmann-Zeh, A., Thomas, G. M. H., Ball, A., Prosser, S., Cunningham, E., Cockcroft, S., and Hsuan, J. J. (1995) *Science* **268**, 1188–1190
- 422a. Jones, J. T., Ballinger, M. D., Pisacane, P. I., Lofgren, J. A., Fitzpatrick, V. D., Fairbrother, W. J., Wells, J. A., and Sliwkowski, M. X. (1998) *J. Biol. Chem.* **273**, 11667–11674
423. Montelione, G. T., Wüthrich, K., and Scheraga, H. A. (1988) *Biochemistry* **27**, 2235–2243
424. Schlessinger, J. (1988) *Biochemistry* **27**, 3119–3123
425. Montelione, G. T., Wüthrich, K., and Scheraga, H. A. (1988) *Biochemistry* **27**, 2235–2243
426. Chantry, A. (1995) *J. Biol. Chem.* **270**, 3068–3073
427. Karasik, A., Pepinsky, R. B., Shoelson, S. E., and Kahn, C. R. (1988) *J. Biol. Chem.* **263**, 11862–11867
428. Kauffmann-Zeh, A., Klinger, R., Endemann, G., Waterfield, M. D., Wetzker, R., and Hsuan, J. J. (1994) *J. Biol. Chem.* **269**, 31243–31251
429. Kaplan, K. B., Bibbins, K. B., Swedlow, J. R., Arnaud, M., Morgan, D. O., and Varmus, H. E. (1994) *EMBO J.* **13**, 4745–4756
- 429a. Williams, J. C., Weijland, A., Gonfloni, S., Thompson, A., Courtneidge, S. A., Superti-Furga, G., and Wierenga, R. K. (1997) *J. Mol. Biol.* **274**, 757–775
430. Okada, M., and Nakagawa, H. (1989) *J. Biol. Chem.* **264**, 20886–20893
431. Buss, J. E., and Sefton, B. M. (1985) *Journal of Virology* **53**, 7–12
432. Soriano, P., Montgomery, C., Geske, R., and Bradley, A. (1991) *Cell* **64**, 693–702
433. Lowe, C., Yoneda, T., Boyce, B. F., Chen, H., Mundy, G. R., and Soriano, P. (1993) *Proc. Natl. Acad. Sci. U.S.A.* **90**, 4485–4489
434. Tezuka, K.-i., Denhardt, D. T., Rodan, G. A., and Harada, S.-i. (1996) *J. Biol. Chem.* **271**, 22713–22717
435. Bolen, J. B., and Veillette, A. (1989) *Trends Biochem. Sci.* **14**, 404–407
436. Mustelin, T., and Burn, P. (1993) *Trends Biochem. Sci.* **18**, 215–220
437. Vetrie, D., Vorechovsky, I., Sideras, P., Holland, J., Davies, A., Flinter, F., Hammarström, L., Kinnon, C., Levinsky, R., Bobrow, M., Smith, C. I. E., and Bentley, D. R. (1993) *Nature (London)* **361**, 226–233
438. Hatakeyama, M., Kono, T., Kobayashi, N., Kawahara, A., Levin, S. D., Perlmutter, R. M., and Taniguchi, T. (1991) *Science* **252**, 1523–1528
- 438a. Zou, X., and Calame, K. (1999) *J. Biol. Chem.* **274**, 18141–18144
439. Daley, G. Q., Van Etten, R. A., and Baltimore, D. (1990) *Science* **247**, 824–830
440. Landis, C. A., Masters, S. B., Spada, A., Pace, A. M., Bourne, H. R., and Vallar, L. (1989) *Nature (London)* **340**, 692–696
441. Barbacid, M. (1987) *Ann. Rev. Biochem.* **56**, 779–827
442. Kato, G. J., and Dang, C. V. (1992) *FASEB J.* **6**, 3065–3072
443. Marcu, K. B., Bossone, S. A., and Patel, A. J. (1992) *Ann. Rev. Biochem.* **61**, 809–860
444. Greenhalgh, D. A., Welty, D. J., Player, A., and Yuspa, S. H. (1990) *Proc. Natl. Acad. Sci. U.S.A.* **87**, 643–647
445. Rabbitts, T. H., Forster, A., Hamlyn, P., and Baer, R. (1984) *Nature (London)* **309**, 592–597
446. Shima, E. A., Le Beau, M. M., McKeithan, T. W., Minowada, J., Showe, L. C., Mak, T. W., Minden, M. D., Rowley, J. D., and Diaz, M. O. (1986) *Proc. Natl. Acad. Sci. U.S.A.* **83**, 3439–3443
447. Cramer, C. J., and Truhlar, D. G. (1993) *J. Am. Chem. Soc.* **115**, 5745–5753
448. Perrin, C. L., Armstrong, K. B., and Fabian, M. A. (1994) *J. Am. Chem. Soc.* **116**, 715–722
449. Gabb, H. A., and Harvey, S. C. (1993) *J. Am. Chem. Soc.* **115**, 4218–4227
450. Plavec, J., Tong, W., and Chattopadhyaya, J. (1993) *J. Am. Chem. Soc.* **115**, 9734–9746
451. Ellervik, U., and Magnusson, G. (1994) *J. Am. Chem. Soc.* **116**, 2340–2347
- 451a. Hermeking, H., Rago, C., Schuhmacher, M., Li, Q., Barrett, J. F., Obaya, A. J., O'Connell, B. C., Mateyak, M. K., Tam, W., Kohlhuber, F., Dang, C. V., Sedivy, J. M., Eick, D., Vogelstein, B., and Kinzler, K. W. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 2229–2234
452. Eden, A., Simchen, G., and Benvenisty, N. (1996) *J. Biol. Chem.* **271**, 20242–20245
453. Postel, E. H., Berberich, S. J., Flint, S. J., and Ferrone, C. A. (1993) *Science* **261**, 478–480
454. Ying, G.-G., Proost, P., van Damme, J., Brusch, M., Introna, M., and Golay, J. (2000) *J. Biol. Chem.* **275**, 4152–4158
455. Macleod, K., Leprince, D., and Stehelin, D. (1992) *Trends Biochem. Sci.* **17**, 251–256
456. Karin, M. (1995) *J. Biol. Chem.* **270**, 16483–16486
457. Turner, R., and Tjian, R. (1989) *Science* **243**, 1689–1694
458. Vogt, P. K., and Bos, T. J. (1989) *Trends Biochem. Sci.* **14**, 172–175
459. Neuberger, M., Schuermann, M., Hunter, J. B., and Müller, R. (1989) *Nature (London)* **338**, 589–590
460. Treisman, R. (1995) *EMBO J.* **14**, 4905–4913
461. Abel, T., and Maniatis, T. (1989) *Nature (London)* **341**, 24–25
462. Glover, J. N. M., and Harrison, S. C. (1995) *Nature (London)* **373**, 257–261
463. Quantin, B., and Breathnach, R. (1988) *Nature (London)* **334**, 538–539
464. Greenberg, M. E., and Ziff, E. B. (1984) *Nature (London)* **311**, 433–438
465. Müller, R., Bravo, R., Burckhardt, J., and Curran, T. (1984) *Nature (London)* **312**, 716–720
466. Fukumoto, Y., Kaibuchi, K., Oku, N., Hori, Y., and Takai, Y. (1990) *J. Biol. Chem.* **265**, 774–780
467. Wang, Z.-Q., Ovitt, C., Grigoriadis, A. E., Möhle-Steinlein, U., Rütger, U., and Wagner, E. F. (1992) *Nature (London)* **360**, 741–745
468. Candelieri, G. A., Glorieux, F. H., Prud'homme, J., and St.-Arnaud, R. (1995) *N. Engl. J. Med.* **332**, 1546–1551
469. Egan, S. E., and Weinberg, R. A. (1993) *Nature (London)* **365**, 781–783
470. Davis, R. J. (1994) *Trends Biochem. Sci.* **19**, 470–473
- 470a. Gutkind, J. S. (1998) *J. Biol. Chem.* **273**, 1839–1842
- 470b. Whitmarsh, A. J., and Davis, R. J. (1998) *Trends Biochem. Sci.* **23**, 481–485
471. Pawson, T. (1995) *Nature (London)* **373**, 573–580
472. Daly, R. J., Sanderson, G. M., Janes, P. W., and Sutherland, R. L. (1996) *J. Biol. Chem.* **271**, 12502–12510
473. Feller, S. M., Ren, R., Hanafusa, H., and Baltimore, D. (1994) *Trends Biochem. Sci.* **19**, 453–458
474. Pei, D., Lorenz, U., Klingmüller, U., Neel, B. G., and Walsh, C. T. (1994) *Biochemistry* **33**, 15483–15493

References

475. Ladbury, J. E., Hensmann, M., Panayotou, G., and Campbell, I. D. (1996) *Biochemistry* **35**, 11062–11069
476. Maignan, S., Guilloteau, J.-P., Fromage, N., Arnoux, B., Becquart, J., and Ducruix, A. (1995) *Science* **268**, 291–293
477. Mikol, V., Baumann, G., Zurini, M. G. M., and Hommel, U. (1995) *J. Mol. Biol.* **254**, 86–95
478. Zhou, M.-M., Ravichandran, K. S., Olejniczak, E. T., Petros, A. M., Meadows, R. P., Sattler, M., Harlan, J. E., Wade, W. S., Burakoff, S. J., and Fesik, S. W. (1995) *Nature (London)* **378**, 584–592
479. Bonfini, L., Migliaccio, E., Pelicci, G., Lanfrancone, L., and Pelicci, P. G. (1996) *Trends Biochem. Sci.* **21**, 257–261
- 479a. Nantel, A., Mohammad-Ali, K., Sherk, J., Posner, B. I., and Thomas, D. Y. (1998) *J. Biol. Chem.* **273**, 10475–10484
480. Burgering, B. M. T., and Bos, J. L. (1995) *Trends Biochem. Sci.* **20**, 18–22
481. Pumiglia, K. M., LeVine, H., Haske, T., Habib, T., Jove, R., and Decker, S. J. (1995) *J. Biol. Chem.* **270**, 14251–14254
482. Thompson, M. J., Roe, M. W., Malik, R. K., and Blackshear, P. J. (1994) *J. Biol. Chem.* **269**, 21127–21135
483. Jhun, B. H., Haruta, T., Meinkoth, J. L., Leitner, J. W., Draznin, B., Saltiel, A. R., Pang, L., Sasaoka, T., and Olefsky, J. M. (1995) *Biochemistry* **34**, 7996–8004
484. Jacob, K. K., Ouyang, L., and Stanley, F. M. (1995) *J. Biol. Chem.* **270**, 27773–27779
485. Pause, A., Belsham, G. J., Gingras, A.-C., Donzé, O., Lin, T.-A., Lawrence, J. C., Jr., and Sonenberg, N. (1994) *Nature (London)* **371**, 762–767
486. Proud, C. G. (1994) *Nature (London)* **371**, 747–748
- 486a. Wartmann, M., Hofer, P., Turowski, P., Saltiel, A. R., and Hynes, N. E. (1997) *J. Biol. Chem.* **272**, 3915–3923
- 486b. Weng, G., Bhalla, U. S., and Lyengar, R. (1999) *Science* **284**, 92–96
- 486c. Petosa, C., Masters, S. C., Bankston, L. A., Pohl, J., Wang, B., Fu, H., and Liddington, R. C. (1998) *J. Biol. Chem.* **273**, 16305–16310
487. Cobb, M. H., and Goldsmith, E. J. (1995) *J. Biol. Chem.* **270**, 14843–14846
488. Inglese, J., Koch, W. J., Touhara, K., and Lefkowitz, R. J. (1995) *Trends Biochem. Sci.* **20**, 151–156
489. van Biesen, T., Hawes, B. E., Raymond, J. R., Luttrell, L. M., Koch, W. J., and Lefkowitz, R. J. (1996) *J. Biol. Chem.* **271**, 1266–1269
490. Wu, J., Spiegel, S., and Sturgill, T. W. (1995) *J. Biol. Chem.* **270**, 11484–11488
491. Nurse, P. (2000) *Science* **289**, 1711–1716
492. Edgar, B. A., and Lehner, C. F. (1996) *Science* **274**, 1646–1652
493. Kirschner, M. (1992) *Trends Biochem. Sci.* **17**, 281–285
494. Horne, M. C., Goolsby, G. L., Donaldson, K. L., Tran, D., Neubauer, M., and Wahl, A. F. (1996) *J. Biol. Chem.* **271**, 6050–6061
495. Tyson, J. J., Novak, B., Odell, G. M., Chen, K., and Thron, C. D. (1996) *Trends Biochem. Sci.* **21**, 89–96
496. Sherr, C. J. (1996) *Science* **274**, 1672–1677
497. Hartwell, L. H., and Kastan, M. B. (1994) *Science* **266**, 1821–1828
498. Hennig, M., Jansonius, J. N., Terwisscha van Scheltinga, A. C., Dijkstra, B. W., and Schlesier, B. (1995) *J. Mol. Biol.* **254**, 237–246
499. Elledge, S. J. (1996) *Science* **274**, 1664–1672
- 499a. Chin, L., Pomerantz, J., and DePinho, R. A. (1998) *Trends Biochem. Sci.* **23**, 291–296
- 499b. Nead, M. A., Baglia, L. A., Antinore, M. J., Ludlow, J. W., and McCance, D. J. (1998) *EMBO J.* **17**, 2342–2352
- 499c. Bakiri, L., Lallemand, D., Bossy-Wetzel, E., and Yaniv, M. (2000) *EMBO J.* **19**, 2056–2068
500. Kussie, P. H., Gorina, S., Marechal, V., Elenbaas, B., Moreau, J., Levine, A. J., and Pavletich, N. P. (1996) *Science* **274**, 948–953
501. Gorina, S., and Pavletich, N. P. (1996) *Science* **274**, 1001–1005
502. Galaktionov, K., Chen, X., and Beach, D. (1996) *Nature (London)* **382**, 511–517
503. Agarwal, M. L., Taylor, W. R., Chernov, M. V., Chernova, O. B., and Stark, G. R. (1998) *J. Biol. Chem.* **273**, 1–4
504. Carr, A. M. (2000) *Science* **287**, 1765–1766
505. Hirao, A., Kong, Y.-Y., Matsuoka, S., Wakeham, A., Ruland, J., Yoshida, H., Liu, D., Elledge, S. J., and Mak, T. W. (2000) *Science* **287**, 1824–1827
506. Gatti, A., Li, H.-H., Traugh, J. A., and Liu, X. (2000) *Biochemistry* **39**, 9837–9842

Study Questions

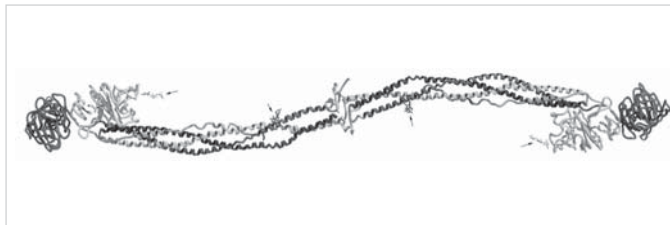
1. Illustrated is a generalized metabolic pathway in which capital letters indicate major metabolites in the pathway, small letters indicate cofactors and numbers indicate enzymes catalyzing the reactions.



- List and describe four different ways in which the pathway might be regulated, referring to the specific enzymes, reactants, and cofactors indicated in the diagram. NOTE: Do not just refer to four different reactions as possible sites of regulation, but give four different general methods for regulation.
2. Describe the role of β -fructose-2,6-bisphosphate in the control of the further breakdown and resynthesis of glucose 1-phosphate.
3. It has been proposed that the substrate cycle involving phosphofructokinase and fructose biphosphatase is used by bumblebees to warm their flight muscles to 30°C before flight begins.

Clark *et al.* (1973) *Biochem. J.* **134**, 589–597 found maximal rates of catalytic activity for both enzymes to be about 44 $\mu\text{mol} / \text{min} / \text{g}$ of fresh tissue. In flying bees glycolysis occurred at a rate of about 20 $\mu\text{mol} / \text{min} / \text{g}$ of tissue with no substrate cycling. In resting bees at 27°C no cycling was detected, but at 5°C substrate cycling occurred at the rate of 10.4 $\mu\text{mol} / \text{min} / \text{g}$ while glycolysis had slowed to 5.8 $\mu\text{mol} / \text{min} / \text{g}$. If the cycling provides heat to warm the insect, estimate how long it would take to reach 30°C if a cold (5°C) bee could carry out cycling at the maximum rate of 40 $\mu\text{mol} / \text{min} / \text{g}$ and if no heat were lost to the surroundings.

4. a) Compare the reaction cycle of a small GTPase (G protein) that is regulated by the actions of a GTPase-activating protein (GAP) and a guanine-nucleotide exchange factor (GEF) with that of a G protein linked to a receptor.
- b) Some G proteins have been described as “timed switches” and others as “triggered switches.” In what ways might these two groups differ? See Kjeldgaard *et al.* (1996) *FASEB J.* **10**, 1347–1368.



View of a modified bovine fibrinogen molecule. The 45-nm-long disulfide-linked dimer is composed of three nonidentical polypeptide chains. The N termini of the six chains from the two halves come together in the center in a small globular “disulfide knot.” The C termini form globular domains at the ends. The 340-kDa molecule has been treated with a lysine-specific protease which has removed portions of two chains to give the ~285-kDa molecule whose crystal structure is shown. Arrows point to attached oligosaccharides. From Brown *et al.* (2000) *Proc. Natl. Acad. Sci. U. S. A.* 97, 85–90. Courtesy of Carolyn Cohen.

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Transferring Groups by Displacement Reactions

12



The majority of enzymes that are apt to be mentioned in any discussion of metabolism catalyze nucleophilic displacement reactions (Type 1, Table 10-1). These include most of the reactions by which the energy of ATP cleavage is harnessed and by which polymers are assembled from monomers. They include reactions by which pieces, large or small, are transferred onto or off of polymers as well as the reactions by which polymers are cleaved into pieces.^{1-3d}

In these reactions a **nucleophilic group** (a base), designated B^- in Table 10-1, approaches an **electrophilic center**, often an electron-deficient carbon or phosphorus atom. It forms a bond with this atom, at the same time displacing some other atom, usually O, N, S, or C. The displaced atom leaves with its bonding electron pair and with whatever other chemical group is attached, the entire unit being called the **leaving group**. This is designated YH in Table 10-1. Simultaneous or subsequent donation of a proton from an acidic group of the enzyme, or from water, to the O, N, or S atom of the leaving group is usually required to complete the reaction. The entering base B, which may or may not carry a negative charge, must often be generated by enzyme-catalyzed removal of a proton from the conjugate acid BH. If BH is water, the enzyme is a **hydrolase**; otherwise, it is called a **transferase**.

For purposes of classifying the reactions of metabolism, in this book the nucleophilic displacements are grouped into four subtypes (Table 10-1). These are displacements on (A) a saturated carbon atom, often from a methyl group or a glycosyl group; (B) a carbonyl group of an ester, thioester, or amide; (C) a phospho group; or (D) a sulfur atom. In addition, many enzymes employ in sequence a displacement on a carbon atom followed by a second displacement on a phosphorus atom (or vice versa).

A. Factors Affecting Rates of a Displacement Reaction

In Chapter 9, the displacement of an iodide ion from methyl iodide by a hydroxide ion (Eq. 9-76) was considered. Can we similarly displace a methyl group from ethane, $\text{CH}_3\text{--CH}_3$, to break the C–C bond and form CH_3OH ? The answer is no. Ethane is perfectly stable in sodium hydroxide and is not cleaved by a simple displacement process within our bodies. Likewise, long hydrocarbon chains such as those in the fatty acids cannot be broken by a corresponding process during metabolism of fatty acids. Not every structure will allow a nucleophilic displacement reaction to occur and not every anion or neutral base can act to displace another group.

At least four factors affect the likelihood of a displacement reaction:^{1-3d}

(1) *The position of the equilibrium in the overall reaction.* An example is provided by the hydrolases that catalyze cleavage of amide, ester, and phosphodiester linkages using water as the entering nucleophile. Because enzymes usually act in an environment of high water content, the equilibrium almost always favors hydrolysis rather than the reverse reactions of synthesis. However, in a nonaqueous solvent the same enzyme will catalyze synthetic reactions.

(2) *The reactivity of the entering nucleophile.* Nucleophilic reactivity or **nucleophilicity** is partly determined by basicity. Compounds that are strong bases tend to react more rapidly in nonenzymatic displacement reactions than do weaker bases; the hydroxyl ion, HO^- , is a better nucleophile than --COO^- . However, enzymes usually act optimally near a neutral pH. Under these conditions the --COO^- group may be

more reactive than strongly basic groups such as -NH_2 or HO^- because at pH 7 the latter groups will be almost completely protonated and the active nucleophile will be present in low concentrations.

A second factor affecting nucleophilic reactivity is **polarizability**, which is the ease with which the electronic distribution around an atom or within a chemical group can be distorted. A large atomic radius and the presence of double bonds in a group both tend to increase polarizability. In most cases, the more highly polarizable a group, the more rapidly it will react in a nucleophilic displacement, apparently because a polarizable group is able to form a partial bond at a greater distance than can a nonpolarizable group. Thus, I^- is more reactive than Br^- , which is more reactive than Cl^- . Polarizable bases such as imidazole are often much more reactive than nonpolarizable ones such as -NH_2 . Sulfur compounds also tend to have a high nucleophilic reactivity. In displacement reactions on carbonyl groups (reaction type 1B), the less polarizable “hard” nucleophiles are more reactive than polarizable ones such as I^- . Attempts have been made to relate quantitatively nucleophilic reactivity to basicity plus polarizability.¹

Certain chemical groups, e.g., those in which an atom with unpaired electrons is directly bonded to the nucleophilic center undergoing reaction, are more reactive than others of similar basicity. This **α effect** has been invoked to explain the high reactivity of the poisons hydroxylamine (NH_2OH) and cyanide ion⁴ and other puzzling results.¹

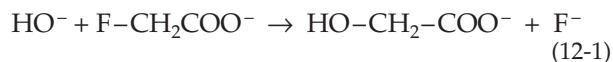
(3) *The chemical nature of the leaving group that is displaced.* The chemistry of the leaving group affects both the rate and the equilibrium position in nucleophilic displacements. The leaving group must accommodate a pair of electrons and often must bear a negative charge. A methyl group displaced from ethane or methane as CH_3^- would be an extremely poor leaving group; the $\text{p}K_a$ of methane as an acid⁵ has been estimated as 47. Iodide ion is a good leaving group, but F^- is over 10^4 times poorer.⁵ In an aqueous medium, phosphate is a much better leaving group than OH^- , and pyrophosphate and tripolyphosphate are even better.

(4) *Special structural features present in the substrate.* Enzymes are usually constructed so that they recognize unique features in substrates. As a consequence, they have many ways of lowering the energy of the transition state and increasing the apparent nucleophilic reactivity.

B. Nucleophilic Displacements on Singly Bonded Carbon Atoms

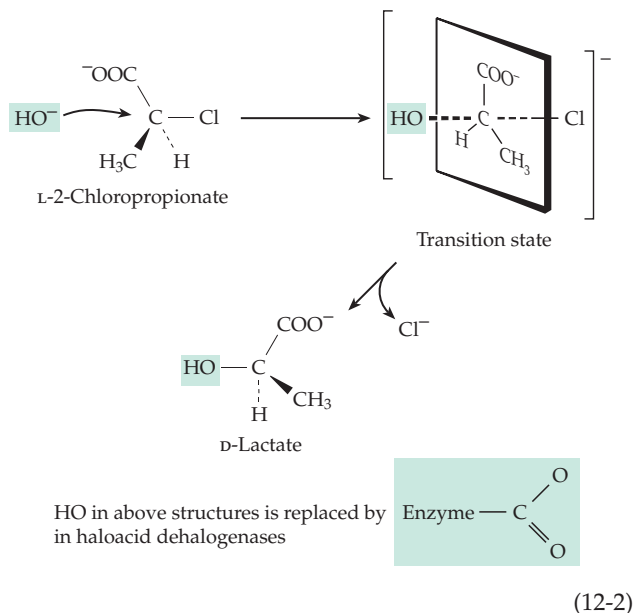
The enzymatic counterpart of Eq. 9-76, the displacement of I^- from a saturated carbon atom by HO^- ,

is catalyzed by the **haloacid dehalogenases** of soil pseudomonads (Eq. 12-1).⁶⁻⁸ Even the poor leaving group F^- can be displaced by OH^- in the active site of these enzymes.

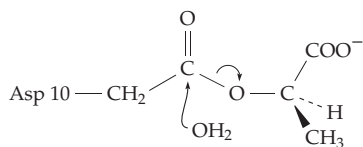


1. Inversion as a Criterion of Mechanism

An interesting result was obtained when a haloacid dehalogenase was tested with a substrate containing a chiral center.⁶ Reaction of L-2-chloropropionate with hydroxyl ion gave only D-lactate, a compound with a chirality opposite to that of the reactant. A plausible explanation is that the hydroxyl ion attacks the central carbon from behind the chlorine atom (Eq. 12-2). The resulting five-bonded transition state (center) loses a chloride ion to form the product D-lactate in which inversion of configuration has occurred. *Inversion always accompanies single displacement reactions in which bond breaking and bond formation occur synchronously, as in Eq. 12-2.* This is true for both enzymatic and nonenzymatic reactions. However, the occurrence of inversion does not rule out more complex mechanisms. Indeed, in this case the displacing group is evidently not HO^- but a carboxylate group from the enzyme.



In one of the haloacid dehalogenases, a 232-residue protein for which the three-dimensional structure is known,^{8,9} Asp 10 is in a position to carry out the initial attack which would give an enzyme-bound intermediate with an ester linkage:



Hydrolytic cleavage, as indicated in the diagram, yields the product D-lactate.⁷ Thus, we have a direct displacement with inversion followed by an additional hydrolytic step. This is an example of **covalent catalysis**, the enzyme providing a well-oriented reactive group instead of generating a hydroxyl group from a bound water molecule.

Related **haloalkane dehalogenases** as well as **epoxide hydrolases** and a large superfamily of other enzymes utilize similar mechanisms.^{10,11,11a} In the active site of a haloalkane dehalogenase from *Xanthobacter* (Fig. 12-1) the carboxylate of Asp 124 acts as the attacking nucleophile that displaces Cl^- from the substrate dichloroethane, which is held in a small cavity with the aid of two tryptophan indole rings.¹²⁻¹⁴ However, the histidine-aspartate pair and the bound water molecule shown in Fig. 12-1 are essential for the subsequent hydrolysis of the intermediate ester. (see also Fig. 12-11).¹⁵ The substrate shown in Fig. 12-1 is 1,2-dichloroethane, a widespread environmental pollutant that is not known to occur naturally. An interesting question considered by Pries *et al.*¹⁶ is how this dehalogenase has evolved in the years since 1922 when industrial production of dichloroethane began.

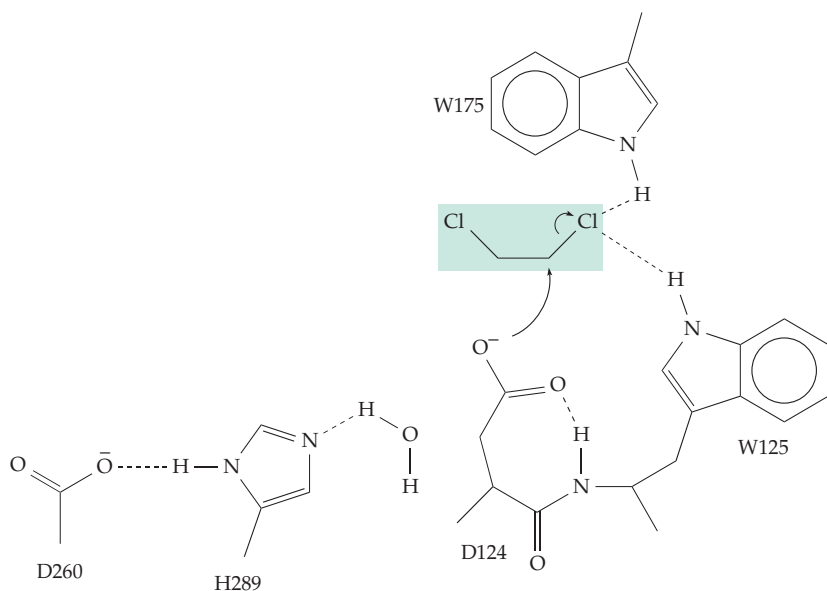


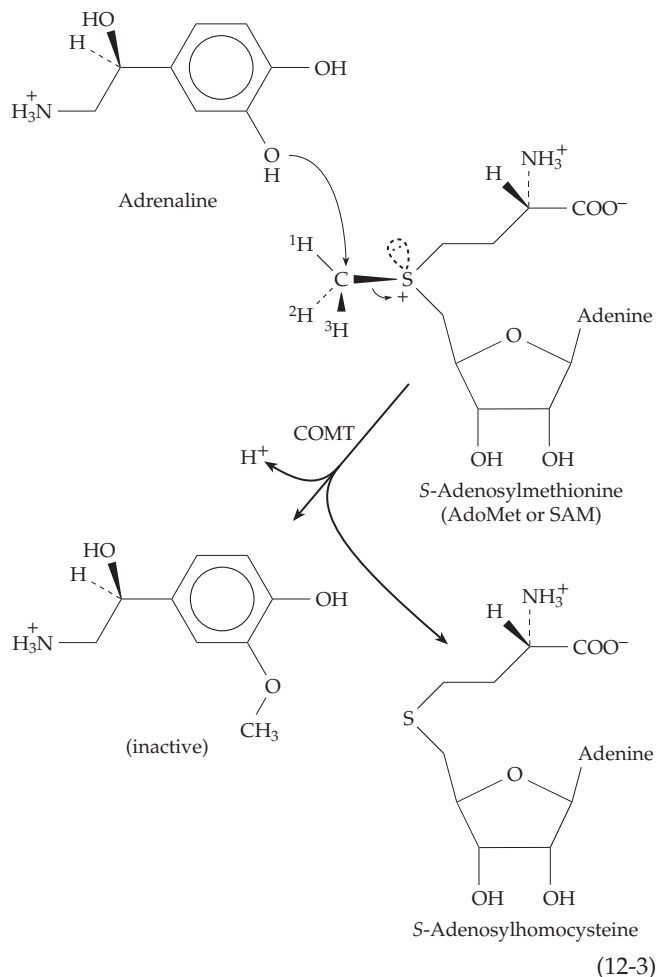
Figure 12-1 The active site structure of haloalkane dehalogenase from *Xanthobacter autotrophicus* with a molecule of bound dichloroethane. See Pries *et al.*¹³ The arrows illustrate the initial nucleophilic displacement. The D260 – H289 pair is essential for the subsequent hydrolysis of the intermediate ester formed in the initial step.

2. Transmethylation

Nucleophilic attack on a methyl group or other alkyl group occurs most readily if the carbon is attached to an atom bearing a positive charge, for example, the sulfur atom of a sulfonium ion such as that present in **S-adenosylmethionine** (abbreviated **AdoMet** or SAM). The ensuing **transmethylation** reaction results in the transfer of the methyl group from the sulfur to the attacking nucleophile (Eq. 12-3). Transmethylation is an important metabolic process by which various oxygen, nitrogen, and other atoms at precise positions in proteins, nucleic acids, phospholipids, and other small molecules are methylated.¹⁷ The methyl group donor is usually S-adenosylmethionine. This compound has two chiral centers, one at the α -carbon of the amino acid and one at the sulfur atom, with an unshared electron pair serving as the fourth group around the S atom. In the naturally occurring AdoMet both centers have the S configuration.¹⁸ The reaction of Eq. 12-3, which is catalyzed by **catechol O-methyltransferase** (COMT), inactivates the neurotransmitters adrenaline, dopamine, and related compounds by methylation. When the substrate contains a chiral methyl group ($-\text{C}^1\text{H}^2\text{H}^3\text{H}$; see also Chapter 13), the inversion of the methyl group expected for a simple $\text{S}_{\text{N}}2$ -like reaction is observed.¹⁹

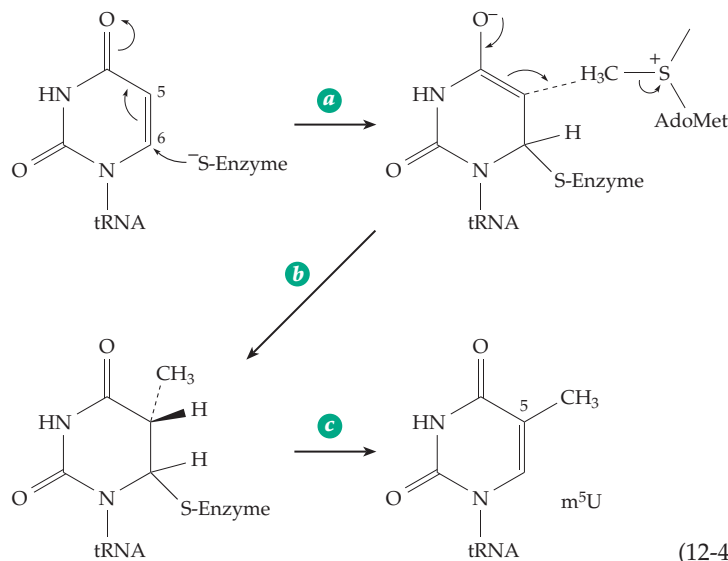
Structural features similar to those of COMT²⁰ are found in glycine N-methyltransferase²¹ and guanidinoacetate methyltransferase²² from liver, and transferases that place methyl groups on N-6 of adenine^{23-25a} and N-4 of cytosine or on C-5 of cytosine²⁶ in nucleic acids.²⁷ A stereoscopic view of the active site of glycine N-methyltransferase is shown in Fig. 12-2. An acetate ion present in the active site in the crystal has been converted into glycine by computer-assisted modeling. The amino group of the bound glycine is adjacent to the methyl group of AdoMet. A nearby glutamic acid side chain (E15) may have removed a proton from the glycine dipolar ion to create the free $-\text{NH}_2$ group required for the reaction. In COMT a magnesium Mg^{2+} ion binds to the two aromatic hydroxyl groups of the substrate and helps to hold it correctly in the active site.^{28-29a}

In the case of the more difficult C-methylation of uracil 54 in transfer RNA, an $-\text{SH}$ group from Cys 324 of the methyltransferase adds to the C-6 position of U54 of the substrate, which may be any of the *E. coli* tRNAs (Eq. 12-4, step a).³⁰ In the



adduct carbon atom C-5 acquires substantial nucleophilic character which permits the transfer of the methyl group in step *b*. The adduct breaks up (step *c*) and the product is released.

Methylation of nucleic acids is considered further in Chaps 27 and 28. Methylation of carboxyl groups



(12-4)

of certain proteins regulates motion of bacterial flagella (Chapter 19) and other aspects of metabolism³¹ while methylation of isomerized aspartyl residues is part of a protein repair process (Box 12-A). Methyl groups are usually transferred from *S*-adenosylmethionine, but sometimes from a folic acid derivative (Chapter 15) or from a cobalt atom of a corrin ring (Chapter 16).

3. Kinetic Isotope Effects

An *S*-¹²C bond breaks a little faster than an *S*-¹³C bond in a nucleophilic displacement reaction. This **primary kinetic isotope effect** (KIE)^{3a,3b,31a-d} is usually discussed first for breakage of a C–H bond. In a linear transition state, in which the C–H bond is being stretched, then cleaved, the difference between the transition state barrier for a C–¹H bond and a C–²H bond is thought to arise principally from a difference in the energies of the C–H stretching vibration.^{3a} This vibrational energy, in the ground state of a molecule (the zero-point energy) is equal to $1/2 h\nu_0$ where ν_0 may be observed in the infrared absorption spectrum (see Fig. 23-2). For a C–²H bond, with a stretching vibration at a wave number of $\sim 2900\text{ cm}^{-1}$, the zero point energy is about $+17.4\text{ kJ mol}^{-1}$. For a C–²H bond, with a stretching wave number of $\sim 2200\text{ cm}^{-1}$, it is about $+17.4\text{ kJ mol}^{-1}$. This difference is a result of the difference in mass of ¹H ($1.67 \times 10^{-24}\text{ g}$) and ²H ($2.34 \times 10^{-24}\text{ g}$). The isotope effect arises because the vibration occurs along the axis of the bond being broken, and the vibrational energy is converted into translational motion along the reaction coordinate, in effect lowering the transition state barrier by $\Delta G^{\ddagger}$. The difference in this effect between ¹H and ²H ($\Delta\Delta G^{\ddagger}$) gives the ratio of expected rate constants as follows:

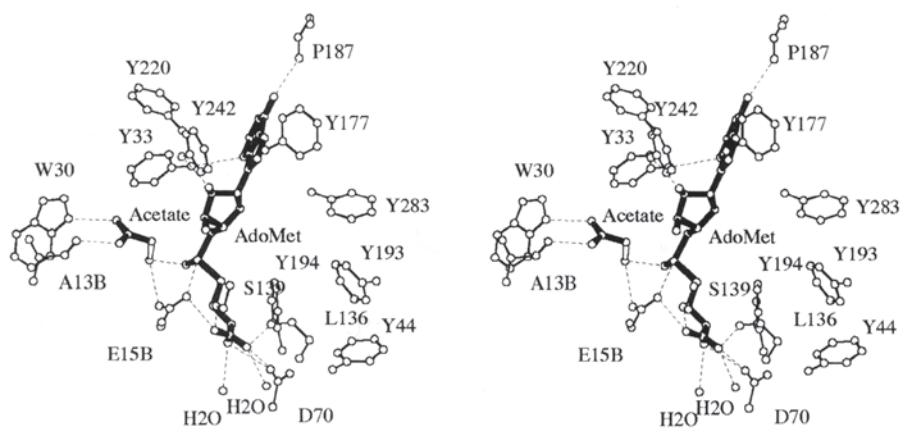
$$\frac{k^{\text{C-H}}}{k^{\text{C-D}}} = e^{-\Delta\Delta G^{\ddagger}/RT} = 7$$

For heavier nuclei, such as C, N, O the kinetic isotope effect is much smaller.

When ¹³C was introduced into the methyl group of *S*-adenosylmethionine and its rate of reaction with catechol *O*-methyltransferase was compared with that of the normal ¹²C-containing substrate, the expected effect on V_{max} , expressed as a first-order rate constant, was seen: $k_{12}/k_{13} = 1.09 \pm 0.05$. This effect is small but it can be measured reliably and establishes that the methyl transfer step rather than substrate binding, product release, or a conformational change in the protein is rate limiting.³²

Substitution of ¹H by ²H in the CH₃ group has a larger effect. This **secondary kinetic isotope effect** (or α -deuterium

Fig. 12-2 S-Adenosylmethionine (AdoMet, solid bonds) bound in the active site of glycine N-methyltransferase together with an acetate ion bound in the glycine site. Glycine was built by attaching an amino group (open bond) to the acetate (solid bonds). Possible polar interactions (O–O and O–N < 0.31 nm and O–S < 0.4 nm) are indicated by dotted lines. Tyrosine residues located at the inner surface of the active site are also shown. From Fu *et al.*²¹



effect) arises because of small differences in the vibrational energies of the methyl group resulting from the differences in mass of ^1H and ^2H . It often leads to a more rapid reaction for the ^1H -containing substrates. In a model nonenzymatic displacement reaction of a similar type³² the ratio of rate constants was $k_{1\text{H}}/k_{2\text{H}} = 1.17 \pm 0.02$. However, for COMT an inverse α -deuterium effect was seen: $k_{(\text{C}^1\text{H}_3)}/k_{(\text{C}^2\text{H}_3)} = 0.83 \pm 0.05$. Theoretical calculations suggested that such an effect might be observed if the enzyme compresses an $\text{S}_{\text{N}}2$ -like transition state, shortening the bonds from the central carbon atom to both the oxygen atom of the entering nucleophile and the sulfur atom of the leaving group.^{33,34} However, more recent calculations suggest that it is difficult to draw such conclusions from secondary isotope effects.^{35,36} Calculations by Zheng, Kahn, and Bruice predict that in the gas phase the two substrates, upon collision, will react with a very low energy barrier.^{29,29a} For this enzyme, as for the haloalkane dehalogenase (Fig. 12-1), it has been concluded that the enzyme excludes water from the active site and binds the two substrates very close together and in a correct orientation for reaction. The computations predicted secondary isotope effects for the enzymatic reaction similar to those measured by Hegazi *et al.*³² suggesting that the transition states for the enzymatic reactions closely resemble those for the gas phase.

4. Glycosyltransferases

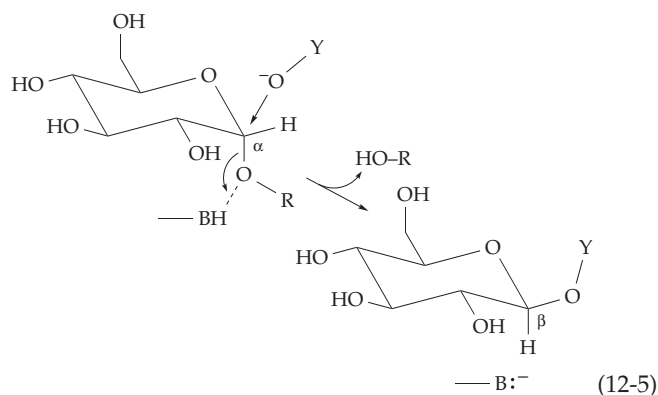
The polarization of a single C–O bond in an ether is quantitatively much less than that of the C–S⁺ bond of S-adenosylmethionine, and simple ethers are not readily cleaved by displacement reactions. However, glycosides, which contain a carbon atom attached to *two* oxygen atoms, do undergo displacement reactions which lead to **hydrolysis**, **phosphorolysis**, or **glycosyl exchange**. The corresponding enzymes are glycosylhydrolases, phosphorylases, and glycosyltransferases. Two characteristics are commonly used to classify

glycosylhydrolases that act on polysaccharides:

Endoglycanases cut at random locations within the chains of sugar units, while **exoglycanases** cut only at one end or another, usually at the nonreducing end. **Inverting enzymes** invert the configuration at the anomeric carbon atom which they attack, while **retaining enzymes** do not. Although simple glycosides undergo acid-catalyzed hydrolysis readily, uncatalyzed hydrolysis is extremely slow, the estimated first order rate constant³⁷ at 25°C being about $2 \times 10^{-15} \text{ s}^{-1}$. Polysaccharides may be the most stable of all biopolymers.

Glycosyltransferases are numerous. For example, the amino acid sequences for about 500 glucosidases, which hydrolyze linkages between glucose residues, have been determined and this one group of enzymes has been classified into over 60 families^{38,39} and eight larger groups.⁴⁰ Many three-dimensional structures have been reported and numerous studies of the reaction mechanism for both enzymatic and nonenzymatic hydrolysis of glycosides have been conducted. Most of the experimental results have been carefully verified. Nevertheless, they serve to illustrate how difficult it is to understand how enzymes catalyze this simple type of reaction.⁴¹

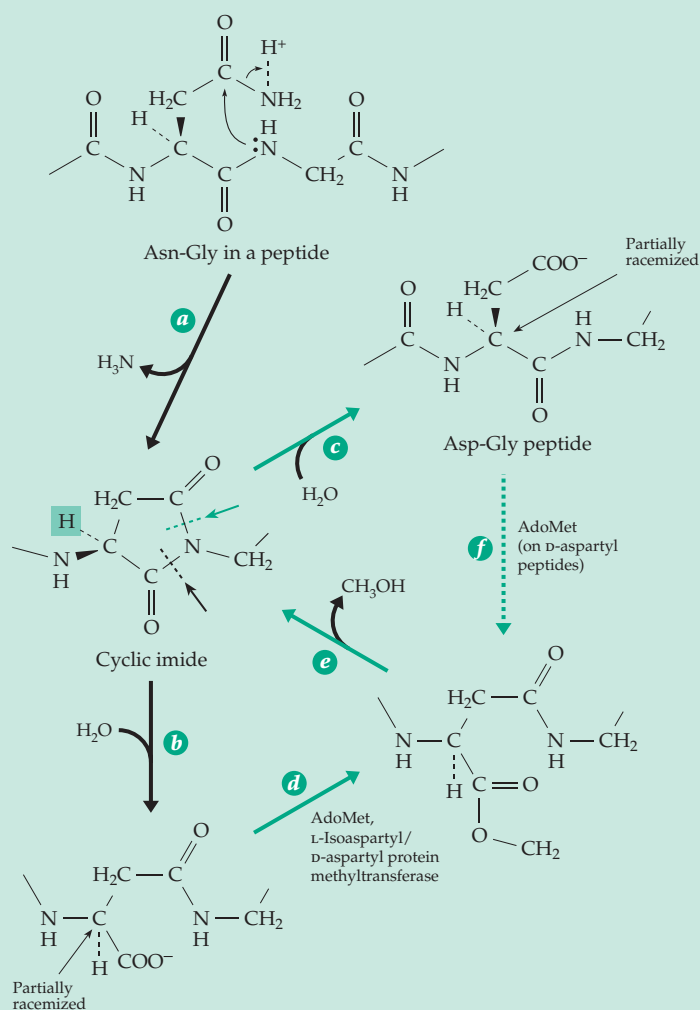
Inversion or retention? Equation 12-5 pictures the reaction of a glycoside (such as a glucose unit at



BOX 12-A CARBOXYMETHYLATION OF PROTEINS

Methylation and demethylation of the carboxyl groups of the side chains of specific glutamyl residues play a role in bacterial chemotaxis (Chapter 19, Section A). However, similar functions in eukaryotes have not been found. There is a specific methyltransferase that methylates C-terminal carboxyl groups on prenylated and sometimes palmitoylated peptides.^a Another type of carboxylmethyltransferase acts on only a small percentage of many proteins and forms labile methyl esters. These methyltransferases have a dual specificity, acting on **L-isospartyl** residues and usually also on **D-aspartyl** residues. Both of these amino acids can arise from deamidation of asparagine, especially in Asn-Gly sequences (Eq. 2-24 and steps *a* and *b* of following scheme).^{b–g} A similar isomerization of aspartyl residues occurs more slowly. Asn-Gly sequences are present in many proteins and provide weak linkages whose isomerization is an inevitable aspect of aging.^{e,h} Furthermore, the α proton of the cyclic imide (green box) is more readily dissociated than in a standard peptide, leading to racemization (step *c*). The isoaspartyl and D-aspartyl residues provide kinks in the peptide chain which may interfere with normal function and turnover.

The L-isoaspartyl / D-aspartyl methyltransferase (step *d*), which is especially abundant in the brain,^{i,j} provides for partial repair of these defects.^{k–o} The methyl esters of isoaspartyl residues readily undergo demethylation with a return to a cyclic imide (step *e*). The cyclic imide is opened hydrolytically (step *c*) in part to an isoaspartyl residue, but in part to a normal aspartyl form. The combined action of the carboxylmethyltransferase and the demethylation reaction tends to repair the isomerized linkages. Methylation of D-aspartyl residues returns them to the cyclic imide (steps *f*, *e*), allowing them to also return to the normal L configuration. Nevertheless, the protein will have a different net charge than it did originally and must be considered a new modified form.



^a Dai, Q., Choy, E., Chiu, V., Romano, J., Slivka, S. R., Steitz, S. A., Michaelis, S., and Philips, M. R. (1998) *J. Biol. Chem.* **273**, 15030–15034

^b Geiger, T., and Clarke, S. (1987) *J. Biol. Chem.* **262**, 785–794

^c Tyler-Cross, R., and Schirch, V. (1991) *J. Biol. Chem.* **266**, 22549–22556

^d Brennan, T. V., and Clarke, S. (1993) *Protein Sci.* **2**, 331–338

^e Paranandi, M. V., Guzzetta, A. W., Hancock, W. S., and Aswad, D. W. (1994) *J. Biol. Chem.* **269**, 243–253

^f Donato, A. D., Ciardiello, M. A., de Nigris, M., Piccoli, R., Mazzarella, L., and D'Alessio, G. (1993) *J. Biol. Chem.* **268**, 4745–4751

^g Tomizawa, H., Yamada, H., Ueda, T., and Imoto, T. (1994) *Biochemistry* **33**, 8770–8774

^h Man, E. H., Sandhouse, M. E., Burg, J., and Fisher, G. H. (1983) *Science* **220**, 1407–1408

ⁱ Orpiszewski, J., and Aswad, D. W. (1996) *J. Biol. Chem.* **271**, 22965–22968

^j Najbauer, J., Orpiszewski, J., and Aswad, D. W. (1996) *Biochemistry* **35**, 5183–5190

^k Johnson, B. A., Murray, E. D. J., Clarke, S., Glass, D. B., and Aswad, D. W. (1987) *J. Biol. Chem.* **262**, 5622–5629

^l Brennan, T. V., Anderson, J. W., Jia, Z., Waygood, E. B., and Clarke, S. (1994) *J. Biol. Chem.* **269**, 24586–24595

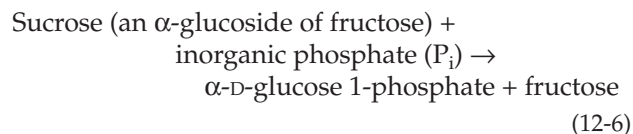
^m Mudgett, M. B., and Clarke, S. (1993) *Biochemistry* **32**, 11100–11111

ⁿ Mudgett, M. B., and Clarke, S. (1994) *J. Biol. Chem.* **269**, 25605–25612

^o Aswad, D. W., ed. (1995) *Deamidation and Isoaspartate Formation in Peptides and Proteins*, CRC Press, Boca Raton, Florida

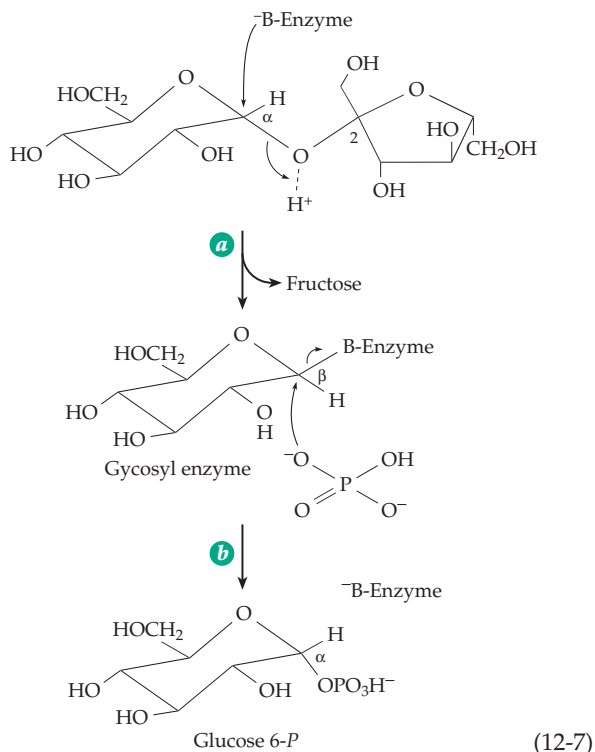
the end of a starch chain) with a nucleophile $Y-O^-$ as the displacing group. An enzyme-bound acidic group $-BH$ is shown assisting the process. Inversion of configuration with formation of a product of the β configuration at the anomeric carbon atom would be predicted and is observed for many of these enzymes. However, many others do *not* cause inversion.^{41,42}

An example is the reaction catalyzed by **sucrose phosphorylase** from *Pseudomonas saccharophila*:



Two possible explanations for the lack of inversion during this reaction are that the enzyme acts by a **double-displacement** reaction or through a stabilized **carbocationic intermediate**. Let us consider these possibilities in turn.

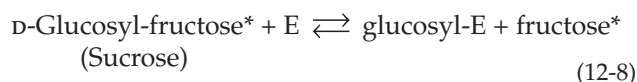
Double-displacement mechanisms. In a double-displacement mechanism sucrose phosphorylase would catalyze two consecutive single displacements, each with inversion. A nucleophilic group of the enzyme would react in Eq. 12-7, step *a*. In step *b*, a phosphate would react to regenerate the enzyme with its free nucleophilic group $-B^-$.



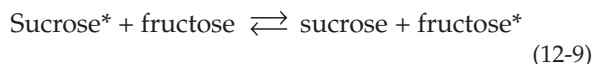
Four kinds of experiments were used to identify this double-displacement mechanism.

(1) *Kinetics.* In a double-displacement mechanism the enzyme shuttles between free enzyme and the intermediate carrying the substrate fragment (here, the glycosyl enzyme). With sucrose phosphorylase the maximum velocity varies with the concentrations of sucrose and HPO_4^{2-} in the characteristic fashion expected for this "ping-pong" mechanism (Eq. 9-47).⁴³

(2) *Exchange reactions.* In a double-displacement mechanism sucrose containing ^{14}C in the fructose portion of the molecule should react with free enzyme *E* to form glycosyl enzyme and free radioactive fructose (Eq. 12-8). The ^{14}C -containing groups are designated here by the asterisks.



If a very low molar concentration of enzyme is present, and a large excess of nonradioactive fructose is added, the enzyme will catalyze no net reaction but will change back and forth repeatedly between the free enzyme and glucosyl enzyme. Each time, in the reverse reaction, it will make use primarily of unlabeled fructose. The net effect will be catalysis of a sucrose–fructose exchange:



This exchange reaction, as well as other predicted exchanges, has been observed.⁴⁴ Although the exchange criterion of the mechanism is often applied to enzymatic processes, the observation of exchange reactions does not *prove* the existence of a covalently bound intermediate. Furthermore, enzymes using double-displacement mechanisms may not always catalyze the expected exchanges.

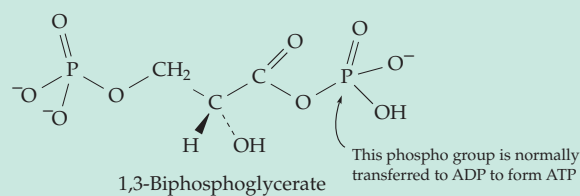
(3) *Arsenolysis.* Sucrose phosphorylase also catalyzes the cleavage of sucrose by arsenate and promotes a rapid cleavage (**arsenolysis**) of glucose 1-phosphate to free glucose. This reaction is evidently a result of a displacement by arsenate on a glycosyl enzyme intermediate. The resulting unstable glucose 1-arsenate (see Box 12-B) is hydrolyzed rapidly. Arsenolysis is a general way of trapping reactive enzyme-bound intermediates that normally react with phosphate groups. Arsenate is one of many substrate analogs that can be used to siphon off reactive enzyme-bound intermediates into nonproductive side paths.

(4) *Identification of glycosyl-enzyme intermediates.* Studies with pure enzymes often make it possible to confirm directly the existence of enzyme-bound intermediates. The intermediates detected are frequently **glycosyl esters** of glutamate or aspartate side chain

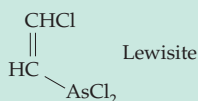
BOX 12-B ARSENIC

Arsenate, AsO_4^{3-} , is chemically similar to phosphate in size and geometry and in its ability to enter into biochemical reactions. However, arsenate esters are far less stable than phosphate esters. If formed in the active site of an enzyme, they are quickly hydrolyzed upon dissociation from the enzyme. This fact accounts for some of the toxicity of arsenic compounds.^a

Arsenate will replace phosphate in all phosphorylytic reactions, e.g., in the cleavage of glycogen by glycogen phosphorylase, of sucrose by sucrose phosphorylase, and in the action of purine nucleoside phosphorylase.^b Glucose 1-arsenate or ribose 1-arsenate is presumably a transient intermediate which is hydrolyzed to glucose. The overall process is called **arsenolysis**. Another reaction in which arsenate can replace phosphate is the oxidation of glyceraldehyde 3-phosphate in the presence of P_i to form 1,3-bisphosphoglycerate:



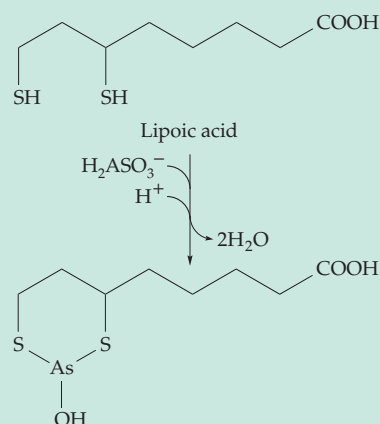
The subsequent transfer of the 1-phospho group to ADP is an important energy-yielding step in metabolism (Chapter 12). When arsenate substitutes for phosphate the acyl arsenate (1-arseno-3-phosphoglycerate) is hydrolyzed to 3-phosphoglycerate. As a consequence, in the presence of arsenate oxidation of 3-phosphoglyceraldehyde continues but ATP synthesis ceases. Arsenate is said to **uncouple** phosphorylation from oxidation. Arsenate can also partially replace phosphate in stimulating the respiration of mitochondria and is an uncoupler of oxidative phosphorylation (Chapter 18). Enzymes that normally act on a phosphorylated substrate will usually catalyze a slow reaction of the corresponding unphosphorylated substrate in the presence of arsenate. Apparently, the arsenate ester of the substrate forms transiently on the enzyme surface, permitting the reaction to occur.



Arsenite is noted for its tendency to react rapidly with thiol groups,^c especially with pairs of adjacent (vicinal) or closely spaced $-\text{SH}$ groups^d as in lipoic acid. By blocking oxidative enzymes requiring

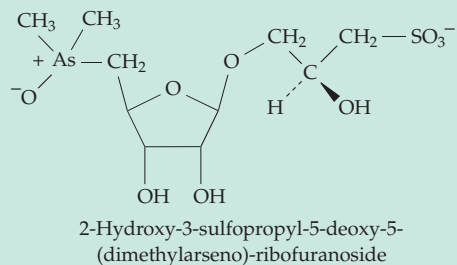
lipoic acid (Chapter 15), arsenite causes the accumulation of pyruvate and of other 2-oxo acids. Similar chemistry underlies the action of the mustard gas Lewisite, which also attacks lipoic acid.^{de}

Many, perhaps all, organisms have enzyme systems for protection against arsenic compounds.



In *E. coli* arsenate is reduced to arsenite by a glutaredoxin- and NADH-dependent system.^{e-g} The arsenite as well as antimonite and tellurite are pumped out by an ATP-dependent transporter. The genes for reductase, periplasmic-binding protein, and transporter components are encoded in a conjugative plasmid.^{h,i} A quite similar system functions in yeast.^j

Marine algae as well as some higher aquatic plants detoxify and excrete arsenate by conversion to various water-soluble organic forms such as trimethylarsonium lactic acid (Chapter 8) and the following ribofuranoside.^a



Arsenic-containing phospholipids are also formed (Chapter 8).^k

Arsenic is present at high levels in some soils and contamination of drinking water with arsenic is a major problem in some areas of the world. In West Bengal, India, millions of people drink contaminated well water. As a result hundreds of thousands have developed debilitating nodular keratoses on their feet.^{l,m} The problem is made worse by the increasing

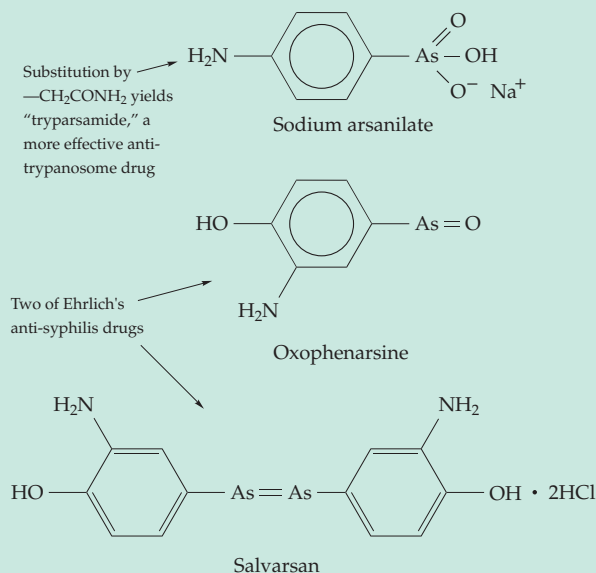
BOX 12-B (continued)

use of fresh water for irrigation and the difficulty of removing arsenic contamination.

Although it is most known for its toxicity, arsenic may be an essential nutrient. Data from feeding of chicks, goats, rats, and miniature pigs suggest a probable human need for arsenic of $\sim 12 \mu\text{g} / \text{day}$.ⁿ

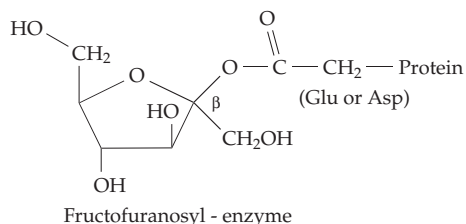
Compounds of arsenic have been used in medicine for over 2000 years, but only in the past century

have specific arsenicals been created as drugs. In 1905, it was discovered that sodium arsinite is toxic to trypanosomes. The development by P. Ehrlich of **arsenicals** for the treatment of syphilis (in 1909) first focused attention on the possibility of effective chemotherapy against bacterial infections.



- ^a Knowles, F. C., and Benson, A. A. (1983) *Trends Biochem. Sci.* **8**, 178–180
- ^b Kline, P. C., and Schramm, V. L. (1995) *Biochemistry* **34**, 1153–1162
- ^c Lam, W.-C., Tsao, D. H. H., Maki, A. H., Maegley, K. A., and Reich, N. O. (1992) *Biochemistry* **31**, 10438–10442
- ^d Li, J., and Pickart, C. M. (1995) *Biochemistry* **34**, 15829–15837
- ^{de} Ord, M. G., and Stocken, L. A. (2000) *Trends Biochem. Sci.* **25**, 253–256
- ^e Silver, S., Nucifora, G., Chu, L., and Misra, T. K. (1989) *Trends Biochem. Sci.* **14**, 76–80
- ^f Ji, G., Garber, E. A. E., Armes, L. G., Chen, C.-M., Fuchs, J. A., and Silver, S. (1994) *Biochemistry* **33**, 7294–7299
- ^g Rosen, B. P., Weigel, U., Karkaria, C., and Gangola, P. (1988) *J. Biol. Chem.* **263**, 3067–3070
- ^h Gladysheva, T. B., Oden, K. L., and Rosen, B. P. (1994) *Biochemistry* **33**, 7288–7293
- ⁱ Chen, Y., and Rosen, B. P. (1997) *J. Biol. Chem.* **272**, 14257–14262
- ^j Wysocki, R., Bobrowicz, P., and Ulaszewski, S. (1997) *J. Biol. Chem.* **272**, 30061–30066
- ^k Cooney, R. V., Mumma, R. O., and Benson, A. A. (1978) *Proc. Natl. Acad. Sci. U.S.A.* **75**, 4262–4264
- ^l Bagla, P., and Kaiser, J. (1996) *Science* **274**, 174–175
- ^m Saha, D. P., and Subramanian, K. S. (1996) *Science* **274**, 1287–1288
- ⁿ Nielsen, F. H. (1991) *FASEB J.* **5**, 2661–2667

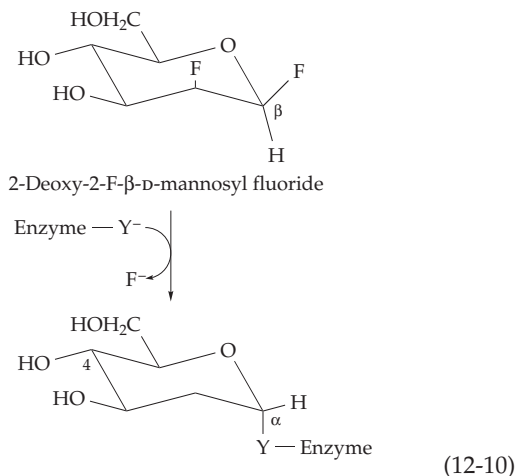
carboxyl groups. Such an intermediate, fructofuranosyl-enzyme, was identified tentatively for sucrose phosphorylase and also for a related levan sucrase.⁴⁵ Recently, identification of glycosyl-enzyme intermediates has been accomplished for many other glycosyltransferases. Among these are human pancreatic and salivary α -amylases,^{42,46} α -glucosidases, and some cellulases and xylanases.



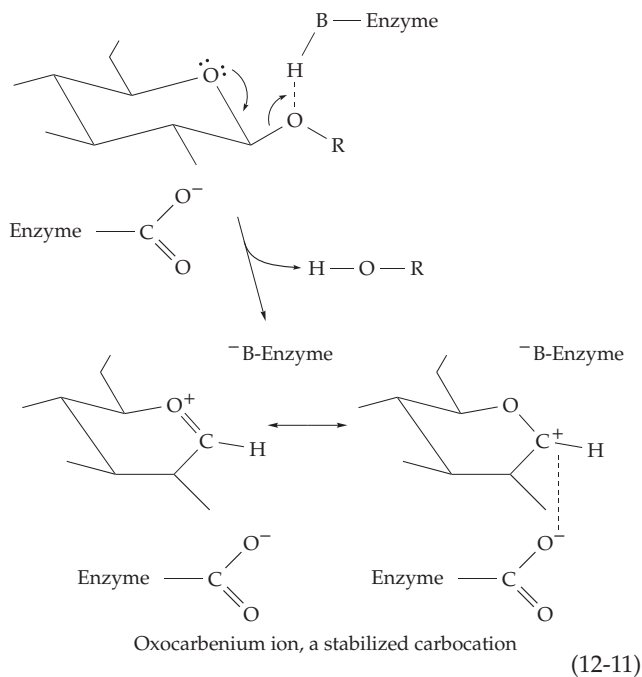
For these hydrolytic enzymes the glycosyl-enzyme would be attacked by a hydroxyl ion derived from H_2O , whose deprotonation would presumably be assisted by the conjugate base ($-\text{B}$: in Eq. 12-5) of the catalytic acidic group.

A convenient way to form and identify covalently linked glucosyl-enzyme intermediates, developed by Withers and coworkers, employs enzyme-activated inhibitors such as 2-deoxy-2-fluoroglycosyl fluorides (Eq. 12-10).^{42,47,48} The 2-F substituent greatly decreases both the rate of formation of a glycosyl-enzyme and its rate of breakdown by hydrolysis or transglycosylation. This may be in part because these compounds lack the 2-OH group which helps to stabilize, by hydrogen bonding, the complexes of normal substrates. A second factor is the high electronegativity of $-\text{F}$ versus $-\text{OH}$, which leads to significant loss of stability for a transition state in which the anomeric carbon atom carries a significant amount of positive charge (see next section). Having a good leaving group, such as fluorine or 2,4-dinitrophenyl, the compounds react rapidly to give stable glucosyl-enzymes which can be characterized. In the example shown in Eq. 12-10, the mannosyl-enzyme was identified by the chemical shifts and line-widths of the ^{19}F resonances of the intermediates and the anomeric configuration was established. More recently, mass spectrometry has been used.⁴⁹ For example, a maltotriosyl-enzyme intermediate in the

action of glycogen-debranching enzyme was identified by separation of an active site peptide by HPLC followed by mass spectrometry (Chapter 3).⁵⁰



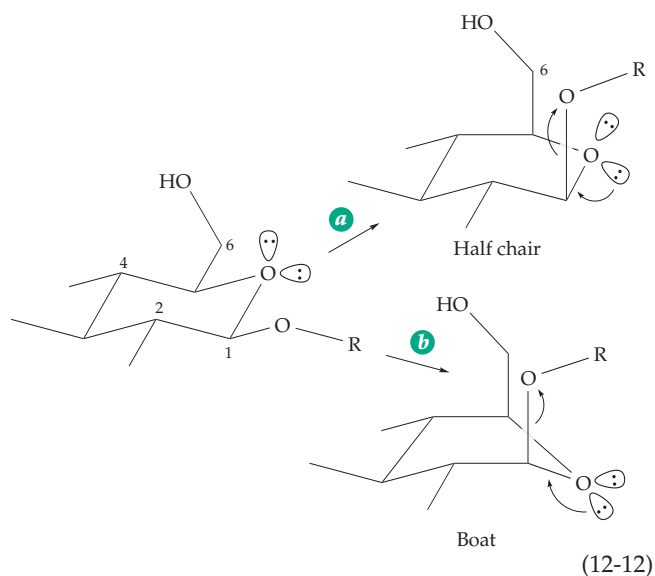
Carbocationic intermediates. In a second mechanism of nucleophilic displacement the leaving group departs (often in a protonated form) before the entering nucleophile reacts.



A carbocation is formed as shown in Eq. 12-11, which represents just one-half of the overall displacement reaction. In the common terminology of physical organic chemistry this is an S_N1 reaction rather than an S_N2 reaction of the kind shown in Eqs. 12-3 and 12-5. This terminology is not quite appropriate for enzymes because the breakdown of ES complexes to product is usually a zero-order process and the numbers 1 and 2

in the symbols S_N1 and S_N2 refer to the order or molecularity of the reaction. It is better to speak of S_N1 -like or S_N2 -like reactions.

The carbocation in Eq. 12-11 is depicted as a resonance hybrid of two forms. One of these is an **oxocarbenium** (or oxonium) ion which contains a double bond between carbon and oxygen.^{50a} This double-bonded structure can be visualized as arising from the original structure by an internal displacement or elimination by the unshared electrons on oxygen, as indicated by the small arrows. In the oxocarbenium ion, the ring atoms C-2, C-1, and C-5 and the oxygen atom are almost coplanar and the ring conformation is described as **half-chair** or **sofa**. As pointed out in Chapter 9, the theory of stereoelectronic control predicts that elimination of the group $-OR$ of Eq. 12-11 should occur most easily if a lone pair of electrons on the ring oxygen atom are antiperiplanar to the $O-R$ bond. This is impossible for a β -glycoside with a ring in the chair conformation shown in Eq. 12-11. This fact suggested that enzymes cleaving β -glycosides may preferentially bind substrate with the appropriate sugar ring in a less stable half-chair or flexible boat conformation prior to bond cleavage (Eq. 12-12).^{51,52} This would allow an unshared pair of electrons on oxygen antiperiplanar to the $C-1$ to $O-R$ bond to participate in the elimination reaction as is indicated in the following diagrams.



On the other hand, Bennet and Sinnott provided evidence that an antiperiplanar lone electron pair is *not* needed in acid-catalyzed cleavage of glycosides via a carbocationic intermediate.^{53,54} Theories of stereoelectronic control must be applied to enzymes with caution!

5. Lysozymes and Chitinases

Polysaccharide chains in the peptidoglycan layer (Fig. 8-29) of the cell walls of bacteria are attacked and cleaved by lysozymes,⁵⁵ enzymes that occur in tears and other body secretions and in large amounts in egg white. Some bacteria and fungi, and even viruses, contain lysozymes.⁵⁶ Their function is usually to protect against bacteria, but lysozyme of phage T4 is a component of the baseplate of the virus tail (Box 7-C). Its role is to cut a hole in the bacterial cell wall to permit injection of the virus' own DNA. Egg white lysozyme, the first enzyme for which a complete three-dimensional structure was determined by X-ray diffraction,⁵⁵ is a 129-residue protein. The active site is in a cleft between a large domain with a nonpolar core and a smaller β -sheet domain that contains many hydrogen-bonded polar side chains (Figs. 12-3, 12-4). Human lysozyme has a similar structure and properties.^{57–59} The T4 lysozyme has an additional C-terminal domain whose function may be to bind the crosslinking peptide of the *E. coli* peptidoglycan. Goose lysozyme is similar in part to both hen lysozyme and T4 lysozyme. All three enzymes, as well as that of our own tears, may have evolved from a common ancestral protein.⁶⁰ On the other hand, *Streptomyces erythraeus* has developed its own lysozyme with a completely different structure.⁶¹ An extensive series of T4 lysozyme mutants have been studied in efforts to understand protein folding and stability.^{61–63}

Catalytic side chain groups. Six *N*-acetylglucosamine (GlcNAc) or *N*-acetylmuramic acid (MurNAc) rings of the polysaccharide substrate are able to fit precisely into six subsites (designated A to F) in a groove between the two structural domains of egg white lysozyme (Fig. 12-4). The bond between the fourth and fifth rings (subsites D and E) is then cleaved. At the active site, the side chain carboxyl group of Glu 35 is positioned correctly to serve as the proton donor BH of Eq. 12-11, while the carboxyl of Asp 52 lies on the opposite side of the groove. Both Glu 35 and Asp 52 have relatively high pK_a values for carboxyl (microscopic pK_a 's are ~ 5.3 and 4.6 , respectively, in the fully protonated active site when the ionic strength is ~ 0.2)⁶⁴ as a result of the hydrophobic environment and hydrogen bonding to other groups. If the pH is raised from an initially low value, a proton from Asp 52 usually dissociates first and the electrostatic field of the resulting anion keeps Glu 35 largely protonated until the pH

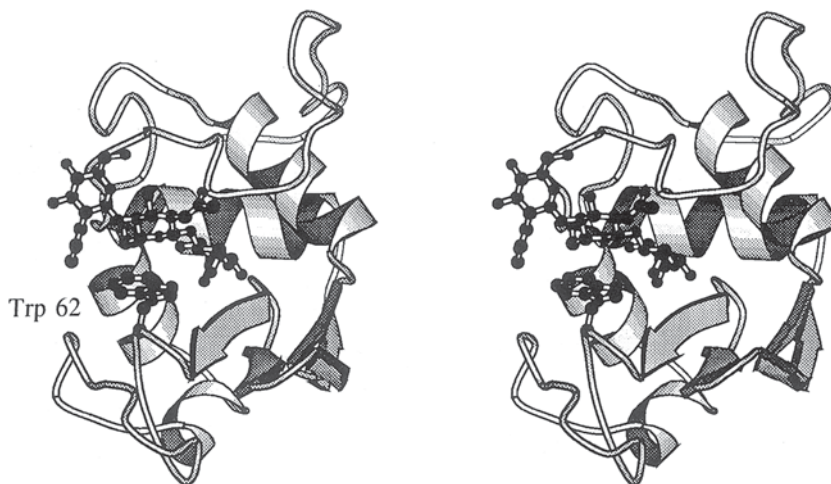


Figure 12-3 Stereoscopic MolScript view of hen egg white lysozyme complexed with a trisaccharide of *N*-acetylglucosamine (GlcNAc)₃ in binding subsites A, B, and C. The side chain of Trp 62 and the trisaccharide above it are shown in ball-and-stick form. From Maenaka *et al.*⁶⁵ Courtesy of Izumi Kumagai.

approaches 6. Positively charged basic groups nearby affect the pK_a values; hence, the behavior of the enzyme is sensitive to the ionic strength of the medium.⁶³ The Asp 52 anion lies only ~ 0.3 nm from the center of positive charge expected for an oxocarbenium ion intermediate and presumably stabilizes it. Replacement of Glu 35 by Gln destroys all catalytic activity and replacement of Asp 52 by Ala or Asn decreases activity to ~ 4 – 5% of the original.^{66–68} Less than 1% activity remained for the D52S mutant.⁶⁹

Nevertheless, Asp 52 is not absolutely essential for lysozyme activity. Goose lysozyme lacks this catalytic aspartate.⁷⁰ Matsumura and Kirsch suggested that carboxyl groups of glycine residues covalently attached to *N*-acetylmuramic acid rings in the natural substrates may participate in catalysis.⁶⁶ Many mutant forms of lysozyme have been studied. Of special interest is the D52E mutant. In this enzyme the carboxylate of the longer glutamate side chain reacts with the oxocarbenium ion intermediate to form a covalent adduct and apparently alter the basic mechanism.⁷¹ Replacement of asparagine 46, which can hydrogen bond to Asp 52, also decreases k_{cat} greatly, suggesting a role in catalysis.⁷²

The lysozyme-catalyzed reaction is completed by stereospecific addition of a hydroxyl ion to the oxocarbenium ion with the original β configuration being retained in the product. Such stereospecificity for reactions of enzyme-bound carbocations is not surprising because the enzyme probably assists in generation, on the appropriate side of the sugar ring, of the attacking hydroxyl ion.

Kinetic isotope effect for lysozyme. A secondary kinetic isotope effect is expected because a molecule with ^1H in the number 1 position can be converted to the corresponding oxocarbenium ion somewhat more easily than the molecule with ^2H in the same position (Eq. 12-13). For example, in the nonenzymatic acid-catalyzed hydrolysis of a methyl- α -glucoside, a reaction also believed to proceed through a carbocation intermediate,^{41,75} the ratio $k_{1\text{H}}/k_{2\text{H}}$ is 1.14 for the α anomer and 1.09 for the β anomer.⁵³ In the base-catalyzed hydrolysis of the same compound, which is believed to occur by a double-displacement reaction involving participation of the neighboring OH group on C-2, the ratio $k_{1\text{H}}/k_{2\text{H}}$ is 1.03. The corresponding ratio measured

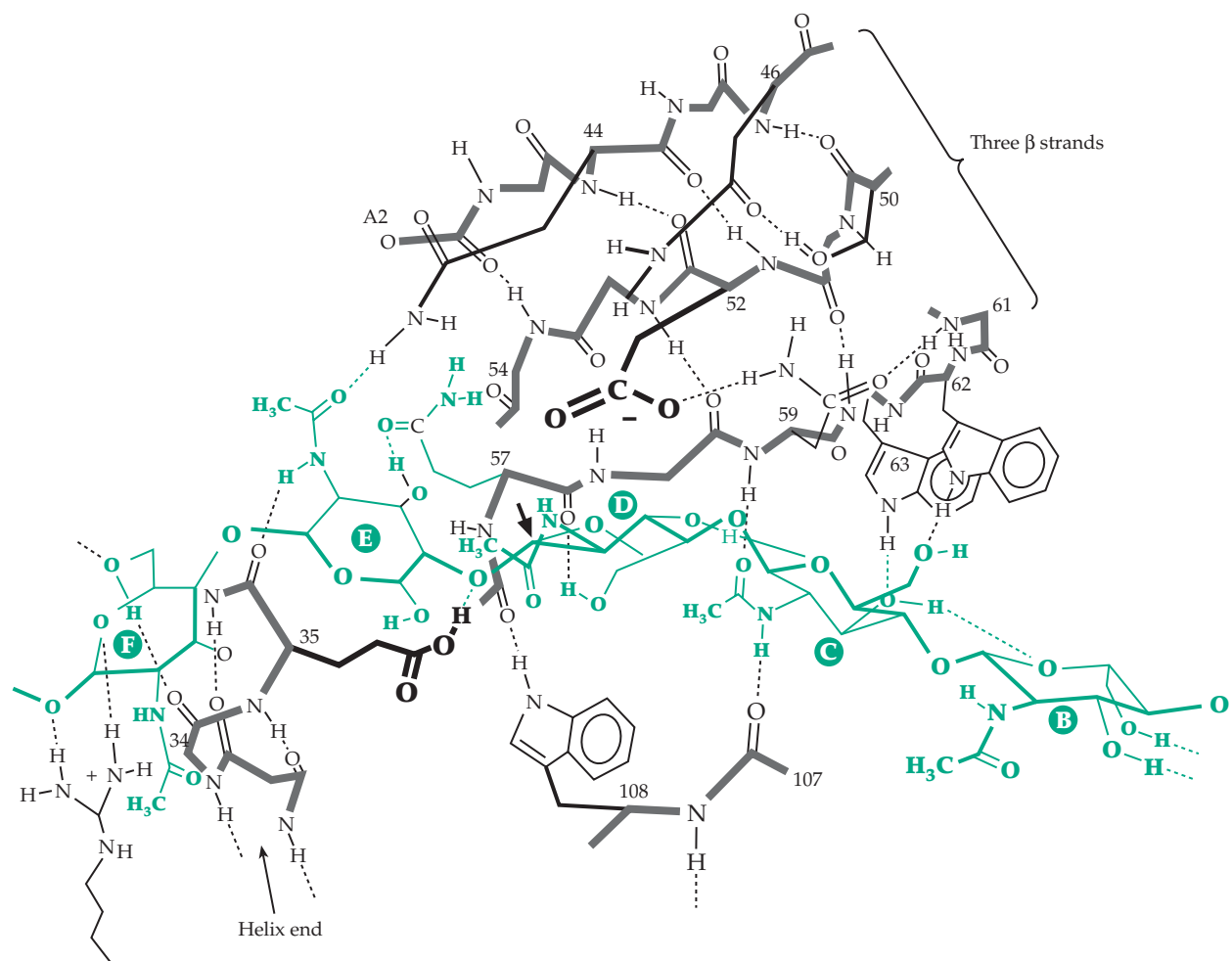
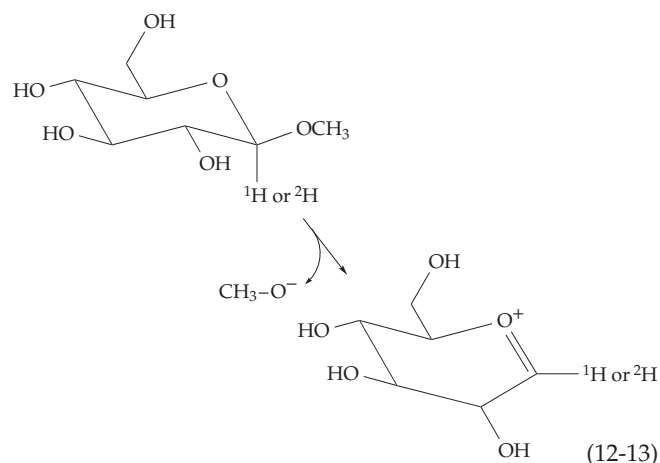


Figure 12-4 Schematic drawing of the active site of egg white lysozyme with substrate in place and about to be cleaved. Three strands of the small β -sheet domain, which contains an extensive hydrogen bond network, are seen at the top. Notice that this view into the active site is different from that in Fig. 12-3. A 180° rotation of either figure will make the views more similar. The three-strand β sheet can be seen (in stereo) at the lower right in Fig. 12-3 and the helix end carrying Glu 35 in the right center foreground. A segment of a chitin oligosaccharide (green), whose reducing end is to the left, is bound into subsites B–F. Cleavage occurs between rings D and E as indicated by the heavy arrow which points to the anomeric carbon atom of ring D. The side chain of Glu 35 is shown protonating the bridge oxygen. Ring D has been distorted into a twist conformation to facilitate cleavage. A larger domain, which contains one long α helix, is at the bottom of the drawing. The active site lies in a cleft between the domains. Notice the chain of hydrogen bonds between the carbonyl of residue 107 in the large domain and the peptide NH of residue 59 in the small domain. Also notice the aromatic side chains which are usually present in carbohydrase active sites. Drawing is based in part on those of Irving Geis⁷³ and Levitt⁷⁴ and on a sketch by author from a three-dimensional model.

for the action of lysozyme is 1.11, much closer to that of the oxocarbenium ion mechanism than to that of the double-displacement mechanism. Similar observations have been made with amylases.⁷⁶ Kinetic isotope effects have also been measured for ^{12}C vs ^{13}C in the anomeric position, ^{16}O vs ^{18}O in the leaving group ($-\text{OCH}_3$) of the methyl glucosides, and for other locations⁵³ as well as for hydrolysis of glucosyl fluorides.^{77–79} The results are generally supportive of the oxocarbenium ion mechanism for lysozyme. However, as mentioned in Section B.3, the interpretation of secondary isotope effects is difficult. Such effects cannot reliably identify a carbohydrase mechanism.^{80,81}

For acid-catalyzed hydrolysis of methyl glucosides⁵³ the kinetic isotope effect observed for the oxygen of the leaving group was $k_{^{16}\text{O}}/k_{^{18}\text{O}} = 1.024\text{--}1.026$. Observation of similar effects for enzymes supports the participation of an acidic group of the protein (Glu 35 of lysozyme) in catalysis but does not eliminate the possibility of concerted involvement of a nucleophilic group, e.g., Asp 52 in lysozyme.^{81,82}

Does lysozyme distort its substrate? An early study of models indicated that for six sugar rings of a substrate to bind tightly into the active site of lysozyme, the ring in subsite D, which contains the carbon atom on which the displacement occurs, had to be distorted from its normal chair conformation into the half-chair conformation.⁸³ This is illustrated in Fig. 12-4. It was suggested that by binding the substrate chain at six different sites the enzyme provides leverage to distort the ring in subsite D into a conformation similar to that of the transition state. This idea was criticized on the basis that an enzyme would be too flexible to act in this manner.⁷⁴ Furthermore, the non-hydrolyzed trisaccharide MurNAc–GlcNAc–MurNAc was shown to fit into subsites B, C, and D of the active site groove without distortion.⁸⁴ However, it is bound weakly.

Can electrical forces acting within the active site help to distort the substrate and assist in formation of the carbocation intermediate? Levitt and Warshel suggested this possibility and proposed that the necessary electrostatic force arises from the arrangement of dipoles within the peptide backbone of the protein and in the amino acid side chains.⁷⁴ As can be seen from Fig. 12-4, the enzyme forms many hydrogen bonds with the substrates. Of special interest is a chain of hydrogen bonds that passes from the backbone carbonyl of Ala 107 through the 2-acetamido group of the substrate in subsite C and into the edge of the β sheet at the backbone NH of Asn 59. This interaction provides specificity toward GlcNAc-containing substrates. Perhaps oscillation of charge within such polarizable chains of hydrogen bonds can also help a substrate to move toward its transition state structure.^{85,86}

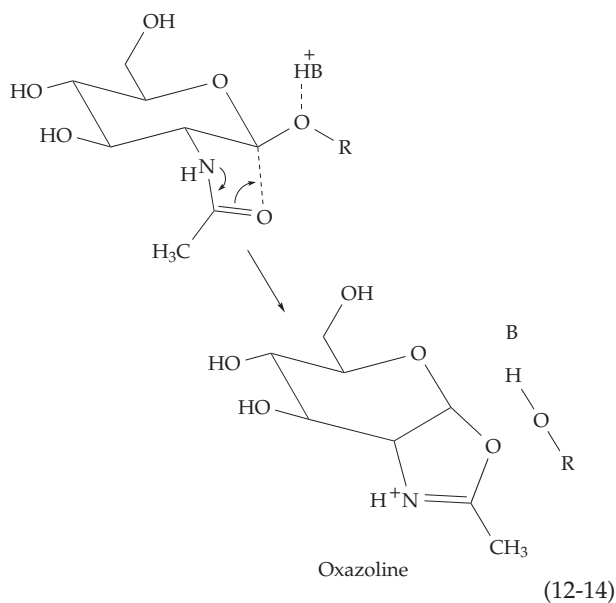
From simulations of lysozyme action by the methods of molecular dynamics, Post and Karplus observed

that Glu 35 tended not to hydrogen bond to the exocyclic bridge oxygen, even though that would be a logical step in the protonation (by $-\text{BH}$) shown in Eq. 12-11. They suggested that Glu 35 may protonate the ring oxygen.⁸⁷ The ring could then open to form an exocyclic oxonium ion, which could hydrolyze and cyclize to the final product. The initial elimination could receive stereoelectronic assistance from the anti-periplanar lone pair of electrons on the bridge oxygen. This proposal has been criticized.^{81,88} For example, it is hard to explain the rapid cleavage of glucosyl fluorides by this exocyclic oxonium mechanism.

As attractive as the arguments for a oxocarbenium mechanism in lysozyme are, is it still possible to explain the experimental observations by a double displacement in which Asp 52 serves as the nucleophile to form a glycosyl enzyme?^{38,82,82a} Sucrose phosphorylase and other glucosyl transferases, like lysozyme, apparently have a carboxylate ion at the active site. In some enzymes, the carboxylate ion forms a covalent glucosyl enzyme, whereas in lysozyme it apparently only stabilizes a carbocation. Is there really a difference in mechanism? Do glucosyl enzymes form only with certain substrates or upon denaturation of the enzymes? Nature hides her secrets well. The difficulty in pinning down the fine mechanistic details of enzymatic action makes it essential to be skeptical, to examine data critically, and to try to imagine all possible alternatives—even when things seem to be proven beautifully.

Help from a neighboring group. In the 1960s it was suggested that the acetamido group of *N*-acetylglucosamine residues might participate as a nucleophile, either stabilizing an oxocarbenium ion or forming an oxazoline intermediate.^{89–91} The proposal received little support, as applied to lysozyme, until sequences and structures of many larger carbohydrases were determined. Among these are **chitinases**, enzymes that act on the same substrates as the small lysozymes. One group of plant chitinases have a structure similar to that of egg white lysozyme. The 243-residue enzyme from barley seeds apparently has Glu 67 as a proton donor and Glu 89 as a possible stabilizing nucleophile.⁹² Another group of chitinases from both plants and bacteria have active sites at ends of $(\alpha\beta)_6$ barrels.^{93–95} They all have a proton-donating glutamate in a conserved position but *no aspartate* that could serve as a nucleophile. Study of complexes with substrates and inhibitors with these enzymes has provided direct evidence of ring distortion and of the probable role of the acetamido group as indicated in Eq. 12-14.^{94,95} The distortion is beyond that in a sofa conformation and allows for maximum stereoelectronic assistance as well as participation of the acetamido group.

In these relatively large enzymes the substrate is deeply buried and cannot reach a conformation



approaching that of the transition state without having a strained ring conformation. These enzymes may use binding forces exerted on many parts of the substrate to stabilize the transition state structure.

6. Cellulases and Other β -Glycosidases

Cellulose, the most abundant of all biopolymers, is extremely stable but is attacked by a host of bacterial and fungal β -glycanases.⁹⁶ Animals do not ordinarily produce cellulases but some termites do.⁹⁷ Cellulase structures are varied, being represented by 10 of 57 different glycosylhydrolase families.⁹⁸ Most, like lysozyme, retain the β configuration in their products but some invert.^{98–100}

Among the cellulases are **endoglycanases**, which cleave chains at random positions, and **exoglycanases** (also called **cellobiohydrolases**) that cleave cellobiose units from ends of chains. Some act on the nonreducing ends and some on reducing ends; a mixture of enzymes is most effective.^{99,101,102} In some bacteria a whole series of different cellulases, together with a large (197-kDa) organizing protein, form a **cellulosome**, a complex with high catalytic activity for crystalline cellulose.^{103–105} Cellulases usually have tightly packed catalytic domains which may vary in size from ~200 residues⁹⁸ to over 400.¹⁰⁶ In most cases the catalytic domain is connected by a flexible linker to one or more small, globular **cellulose-binding domains**. These vary in size from 36 to 200 residues and often have a β -barrel fold.^{104,107–108a} Their function is to hold the enzymes to the cellulose surfaces. They may also facilitate disruption of the tightly hydrogen bonded cellulose structure (Fig. 4-5). As with other carbohydrases, carboxyl groups of amino acid side chains provide the major catalytic groups (Table 12-1). An

extensive hydrogen-bond network which often includes imidazole groups may influence activity.^{109,110}

A striking feature of a 411-residue endoglucanase from *Fusarium* was revealed by the binding of a non-hydrolyzable thiooligosaccharide substrate analog. The pyranose ring at the cleavage site was distorted in an identical manner to that mentioned in the preceding section for a chitinase thought to use the substrate's acetamido group as a nucleophile (Eq. 12-14).¹⁰⁶ The distortion observed is beyond that required for a sofa conformation and allows for the maximum stereoelectronic assistance (Eq. 12-12b). An oxygen atom of the E197 carboxylate, the catalytic nucleophile, occupies a position in the complex that is coincident with that of the C2 acetamido oxygen in the catalytic site of the chitinase discussed in the previous section.

The *inverting* β -glucanases also have two catalytic acid base groups but they are ~0.9 – 1.0 nm apart rather than ~0.6 nm for retaining enzymes. This allows space for a water molecule whose ^-OH is the nucleophilic reactant (^-OY in Eq. 12-5) and in which a carboxylate group assists in dissociating the water molecule.⁹⁸ (This mechanism is illustrated for glucoamylase in Fig. 12-7).

Structural features met in some cellulases include an α,α barrel¹¹¹ similar to that of glucoamylase (Fig. 2-29) and, in a cellobiohydrolase,¹⁰¹ a 5-nm-long tunnel into which the cellulose chains must enter. Ten well-defined subsites for glycosyl units are present in the tunnel.¹⁰¹ A feature associated with this tunnel is **processive** action, movement of the enzyme along the chain without dissociation,¹⁰⁵ a phenomenon observed long ago for amylases (see Section 9) and often observed for enzymes acting on nucleic acids.

Another group of **β -glucanases**, found in plants and their seeds, hydrolyze β -1,3-linked glucans¹¹² and, in some cases, also mixed 1,3- and 1,4-linked polysaccharides. A characteristic enzyme is found in barley (Table 12-1).^{113,114} Similar enzymes are produced by some bacteria¹¹³ and also by molluscs.¹¹⁵

Xylanases act on the β -1,4-linked xylan, the most abundant of the **hemicelluloses** that constitutes over 30% of the dry weight of terrestrial plants.¹¹⁶ They resemble cellulases and cooperate with cellulases and **xylosidases**¹¹⁷ in digestion of plant cell walls.^{110,116,118–121} **Galactanase** digests the β -1,4-linked component of pectins.¹²²

The final step in degradation of cellulose is hydrolysis of cellobiose to glucose. This is accomplished by **β -glucosidases**, enzymes that also hydrolyze lactose, phosphorylated disaccharides, and cyanogenic glycosides.^{123–126} Lactose is also cleaved by **β -galactosidases**.¹²⁷ The large 1023-residue β -galactosidase from *E. coli*¹²⁸ is famous in the history of molecular biology as a component of the *lac* operon.^{129–132} Its properties are also employed to assist in the cloning of genes (Chapter 26).

TABLE 12-1

Acidic and Basic Catalytic Groups in a Few Glycosyltransferases^a

Enzyme	Number of residues	Inverting?	Glycosyl-enzyme identified?	Nucleophile (–COO [–])	Electrophile (proton donor) * Assisting group
Lysozyme					
Human, hen ^b	130	No		E35	D53(52)
Bacteriophage T4 ^c		No		E11	D20
Chitinase					
Rubber plant ^d	273	No			E127
Cellulases					
E1 endocellulase, <i>Acidothermus</i> ^e	521				
catalytic domain	358	No		E282	E162, D252*
Endoglucanase I <i>Fusarium</i> ^f	411	No		E197	E202
Endoglucanase Cen A <i>Cellulomonas</i> ^g	>351	Yes		D392	D78
1,3-β-D-Glucanase					
Barley ^h		No		E231	E288
Xylanase					
<i>B. circulans</i> ⁱ		No	Yes	E78	E172
β-Glucosidase					
<i>Agrobacterium</i> ^j	458	No	Yes	E358	E170, Y298*
β-Galactosidase					
<i>E. coli</i> ^k	1023	No	Yes	E537	E461, Y503*
α-Amylase					
Human and pig ^l	496	No		D197	E233, D300*
Barley ^m		No		D179	E204, D289*
<i>Aspergillus</i> ⁿ		No		D206	E230
Cyclodextrin glucosyltransferase					
<i>B. circulans</i> ^o				D229	E257, D328
α-Glucosidase					
<i>Saccharomyces</i> ^p	No	Yes		D214	E233, D300
Glycogen debranching enzyme, rabbit ^s	No	Yes		D549	
Glucoamylase					
<i>Aspergillus</i> ^q	616	Yes		H ₂ O, E400	E179
β-Amylase					
Soybean ^r		Yes		E186	E380
Glucocerebrosidase					
Human ^t				E340	

^a For classification of glycosyl hydrolases into families, see Henrissat, B., Callebaut, I., Fabrega, S., Lehn, P., Mornon, J.-P., and Davies, G. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 7090–7094

^b Matsumura, I., and Kirsch, J. F. (1996) *Biochemistry* **35**, 1881–1889; Harata, K., Muraki, M., Hayashi, Y., and Jigami, Y. (1992) *Protein Sci.* **1**, 1447–1453

^c Brzozowski, A. M., and Davies, G. J. (1997) *Biochemistry* **36**, 10837–10845; Hardy, L. W., and Poteete, A. R. (1991) *Biochemistry* **30**, 9457–9463; Kuroki, R., Weaver, L. H., and Matthews, B. W. (1993) *Science* **262**, 2030–2033

^d Tews, I., Terwisscha van Scheltinga, A. C., Perrakis, A., Wilson, K. S., and Dijkstra, B. W. (1997) *J. Am. Chem. Soc.* **119**, 7954–7959

^e Sakon, J., Adney, W. S., Himmel, M. E., Thomas, S. R., and Karplus, P. A. (1996) *Biochemistry* **35**, 10648–10660; Ghidoni, R., Sonnino, S., Tettamanti, G., Baumann, N., Reuter, G., and Schauer, R. (1980) *J. Biol. Chem.* **255**, 6990–6995

^f Sulzenbacher, G., Schüle, M., and Davies, G. J. (1997) *Biochemistry* **36**, 5902–5911; Sulzenbacher, G., Driguez, H., Henrissat, B., Schüle, M., and Davies, G. J. (1996) *Biochemistry* **35**, 15280–15287; Mackenzie, L. F., Davies, G. J., Schüle, M., and Withers, S. G. (1997) *Biochemistry* **36**, 5893–5901

^g Damude, H. G., Withers, S. G., Kilburn, D. G., Miller, R. C., Jr., and Warren, R. A. J. (1995) *Biochemistry* **34**, 2220–2224

^h Chen, L., Garrett, T. P. J., Fincher, G. B., and Hoj, P. B. (1995) *J. Biol. Chem.* **270**, 8093–8101

ⁱ Wakarchuk, W. W., Campbell, R. L., Sung, W. L., Davoodi, J., and Yaguchi, M. (1994) *Protein Sci.* **3**, 467–475; Lawson, S. L., Wakarchuk, W. W., and Withers, S. G. (1997) *Biochemistry* **36**, 2257–2265; Sidhu, G., Withers, S. G., Nguyen, N. T., McIntosh, L. P., Ziser, L., and Brayer, G. D. (1999) *Biochemistry* **38**, 5346–5354

^j Wang, Q., Trimbur, D., Graham, R., Warren, R. A. J., and Withers, S. G. (1995) *Biochemistry* **34**, 14554–14562

^k Gebler, J. C., Aebersold, R., and Withers, S. G. (1992) *J. Biol. Chem.* **267**, 11126–11130; Jacobson, R. H., Zhang, X.-J., DuBose, R. F., and Matthews, B. W. (1994) *Nature (London)* **369**, 761–766; Richard, J. P., Huber, R. E., Lin, S., Heo, C., and Amyes, T. L. (1996) *Biochemistry* **35**, 12377–12386

^l Brayer, G. D., Luo, Y., and Withers, S. G. (1995) *Protein Sci.* **4**, 1730–1742; Qian, M., Haser, R., Buisson, G., Duée, E., and Payan, F. (1994) *Biochemistry* **33**, 6284–6294

^m Kadziola, A., Sogaard, M., Svensson, B., and Haser, R. (1998) *J. Mol. Biol.* **278**, 205–217

ⁿ Brzozowski, A. M., and Davies, G. J. (1997) *Biochemistry* **36**, 10837–10845; Matsuura, Y., Kusunoki, M., Harada, W., and Kakudo, M. (1984) *J. Biochem.* **95**, 697–702

^o Knegetel, R. M. A., Strokopytov, B., Penninga, D., Faber, O. G., Rozeboom, H. J., Kalk, K. H., Dijkhuizen, L., and Dijkstra, B. W. (1995) *J. Biol. Chem.* **270**, 29256–29264

^p McCarter, J. D., and Withers, S. G. (1996) *J. Biol. Chem.* **271**, 6889–6894

^q Christensen, U., Olsen, K., Stoffer, B. B., and Svensson, B. (1996) *Biochemistry* **35**, 15009–15018

^r Mikami, B., Degano, M., Hehre, E. J., and Sacchettini, J. C. (1994) *Biochemistry* **33**, 7779–7787; Adachi, M., Mikami, B., Katsube, T., and Utsumi, S. (1998) *J. Biol. Chem.* **273**, 19859–19865

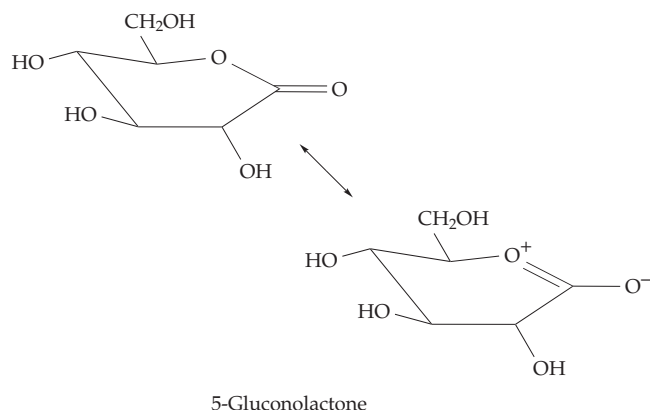
^s Braun, C., Lindhorst, T., Madsen, N. B., and Withers, S. G. (1996) *Biochemistry* **35**, 5458–5463

^t Withers, S. G., and Aebersold, R. (1995) *Protein Sci.* **4**, 361–372

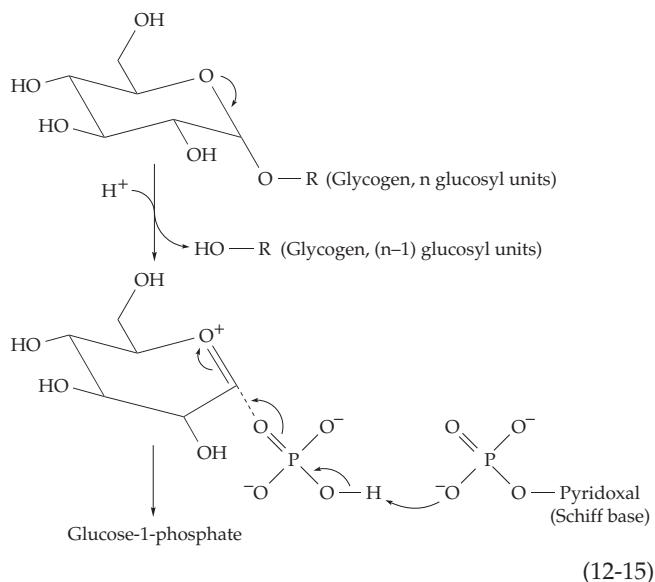
7. Glycogen Phosphorylase

A large number of different glycosyltransferases act on the α 1,4 linkages of glycogen, starch, and related polysaccharides. Among these, one of the most studied is glycogen phosphorylase. It is not a hydrolase, but it catalyzes cleavage of α 1,4 linkages at the nonreducing ends of glycogen molecules by displacement with inorganic phosphate to give α -D-glucose-1-phosphate. It is a very large enzyme (841 residues in rabbit muscle) whose structure is shown in Fig. 11-5. Its complex regulatory mechanisms were discussed briefly in Chapter 11.

Neither partial exchange reactions nor inversion of configuration occur when glycogen phosphorylase acts on its substrates. The enzyme apparently does nothing until both substrates are present. These are glycogen + inorganic phosphate or, for the reverse reaction, glycogen (shifted over by one sugar binding subsite) + glucose-1-phosphate. The active site is in a deep groove in the enzyme. Evidently, the protein must close and fold around the substrates before it becomes active.¹³³ An oxocarbenium ion mechanism has also been proposed for this enzyme, partly on the basis of strong inhibition of phosphorylase by **5-gluconolactone**,^{134,135} a compound having a half-chair conformation and perhaps acting as a transition state inhibitor (Chapter 9). This gluconolactone also inhibits many other carbohydrases.^{123,136}



An unexpected discovery was that glycogen phosphorylase contains a molecule of the coenzyme pyridoxal 5'-phosphate (PLP) bound into the center of the protein behind the active site^{134,135,137-139} with its phosphate group adjacent to the binding site of the phosphate of glucose-1-phosphate. It probably serves as a general base catalyst, e.g., assisting the attack of a phosphate ion on the oxocarbenium ion formed by cleavage of the glycogen chain (Eq. 12-15).^{137,140-142} A key observation by Graves and associates was that pyridoxal alone does not activate apo-phosphorylase but that pyridoxal



plus phosphite, phosphate, or fluorophosphate does provide up to 20% of full activity.¹⁴³ X-ray crystallography confirmed that these activating anions are bound into the active site at the approximate position of the phosphate group of PLP.^{139,143a} Glycogen phosphorylase is being studied by X-ray diffraction techniques that allow observation of structural changes in as short a time as a few milliseconds (see Chapter 3, Section H).¹⁴⁴

The regulation of glycogen phosphorylase, like that of many other allosteric proteins, depends upon the existence of two distinct conformational states, whose structures have been established by crystallography.¹³⁷ It is not immediately evident how they can affect the active site. In the R-state the enzyme has a low affinity for both substrates and activators such as AMP. In the T-state the affinities are much higher. For example, that of inorganic phosphate is raised by a factor of fifteen.¹³⁷ As we have already seen (Figs. 11-4, 11-5) the relatively inactive phosphorylase *b* is converted to the active phosphorylase *a* by phosphorylation of the side chain of serine 14, a structural change which favors the R-state. In the inactive T-state of phosphorylase *b*, an 18-residue N-terminal segment of the polypeptide is not seen by X-ray diffraction, presumably because it does not assume a fixed conformation but projects into a solvent channel in the crystal and moves freely within it. However, in the active R-state of phosphorylase *a*, in which Ser 14 has been phosphorylated, the N-terminal segment is rigid. The phospho group on Ser 14 binds to Arg 69 and other arginine, lysine, and histidine side chains from both subunits.^{137,145} The phosphorylation of Ser 14 occurs 1.5 nm from the active sites. The conformational changes induced by phosphorylation of Ser 14 cause a rotation of about 10° between subunits, somewhat reminiscent of the changes accompanying oxygenation of hemoglobin (Fig. 7-25).

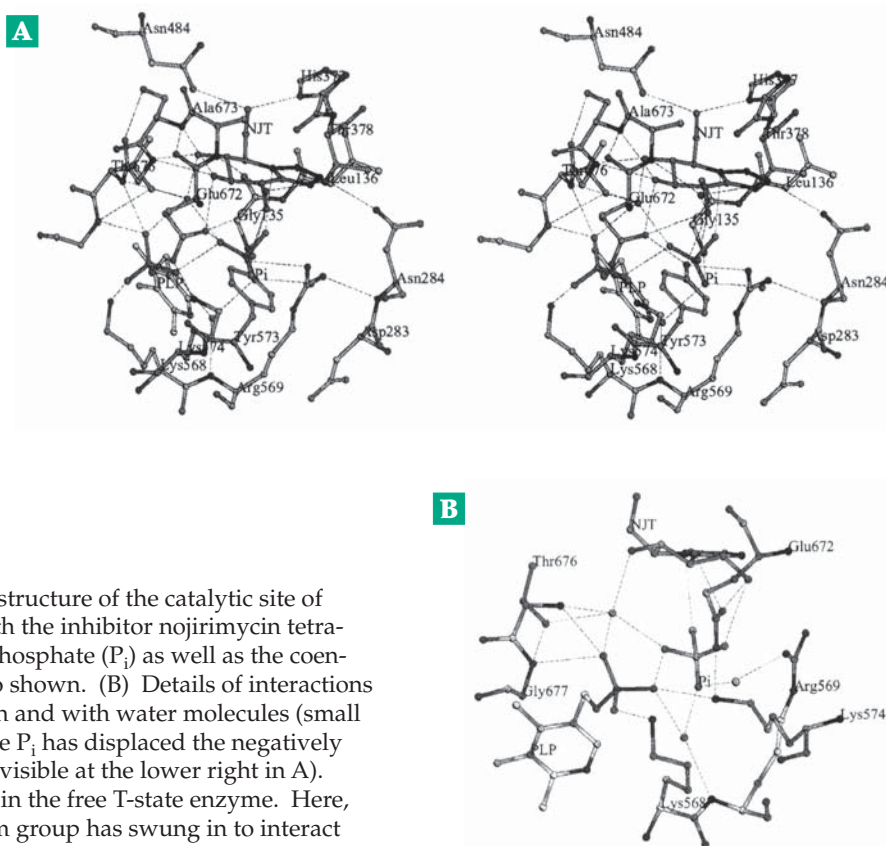
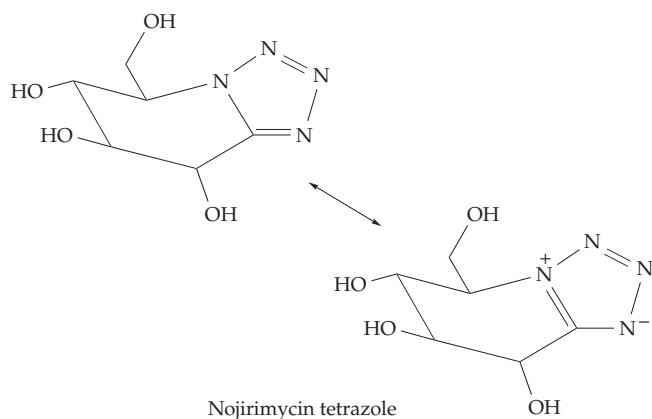


Figure 12-5 (A) Stereoscopic view of the structure of the catalytic site of phosphorylase *b* in the inhibited T-state with the inhibitor nojirimycin tetrazole bound into the active site. Inorganic phosphate (P_i) as well as the coenzyme pyridoxal 5'-phosphate (PLP) are also shown. (B) Details of interactions of the inhibitor, P_i , and PLP with the protein and with water molecules (small circles). This is a weak-binding state but the P_i has displaced the negatively charged side chain carboxylate of Asp 283 (visible at the lower right in A). This carboxylate blocks access to the P_i site in the free T-state enzyme. Here, the positively charged Arg 569 guanidinium group has swung in to interact with the P_i . From Mitchell *et al.*¹³³ Courtesy of Louise N. Johnson.

A related structural change accompanies binding of caffeine, adenosine, or AMP at high concentrations. These substances inhibit the enzyme by binding at a site next to the catalytic site (Fig. 11-5), stabilizing the T-state and causing a loop of protein to move into the catalytic site and to block it.¹³⁷ The catalytic site lies deeply buried in the protein between the two large structural domains (Fig. 11-5). The substrate is held by a network of hydrogen bonds, some of which are shown in Fig. 12-5, in which the active site contains an inhibitory substrate analog **nojirimycin tetrazole**, which is viewed toward the edges of the two rings.



The enzyme is in the weak-binding T-state but inorganic phosphate (P_i) is bound below it and next to the phosphate group of PLP, as required by the mechanism of Eq. 12-15.

The difference in binding affinities for the T- and R-states lies in a flexible loop of residues 280–288, which in the T-state blocks access to the substrate-binding cleft. The universally conserved Asp 238 behaves as a substrate mimic, occupying the P_i site.^{137,146,147} In the R-state this residue moves, allowing P_i to enter and bind (Fig. 12-5).

Most of the structure of mammalian phosphorylases is conserved in species as diverse as *E. coli* and the potato.¹⁴⁸ However, *E. coli* maltodextrin phosphorylase^{143a,149–151} and potato phosphorylases have less sophisticated regulatory mechanisms than do the animal enzymes. Another feature of glycogen phosphorylase is a “glycogen storage site” about 2.5 nm from the active site (Fig. 11-5).¹⁵² This provides a means for the enzyme to hold onto the giant glycogen molecule while “nibbling off” the outside ends of nearby branches.

8. Starch-Hydrolyzing Enzymes

Among the hydrolases are the widely distributed **α -amylases**, *endo*-glycosidases which hydrolyze

starch chains by random attack at points far from chain ends to form short polysaccharide chains known as **dextrins** as well as simpler sugars.¹⁵³ The catalyzed reactions proceed with retention of the original α configuration. Alpha-amylases are found in fungi, plants, and animals. One powerful enzyme of this class is present in the saliva of most humans and other isoenzymes are formed by the pancreas.¹⁵⁴ They are encoded by a family of genes, a fact that accounts for the existence of some healthy individuals completely lacking salivary amylase.¹⁵⁵

Structures are known for human¹⁵⁴ and porcine^{156–159} α -amylases as well as for corresponding enzymes from barley,¹⁶⁰ mealworms,¹⁶¹ fungi, and bacteria.^{162–164} The α -amylase from *Aspergillus oryzae* (Taka-amylase), widely used in laboratory work, was the first for which a structure was determined.^{165,166} The α -amylases fold into three domains (Fig. 12-6), with the active site in the center of an $(\alpha/\beta)_8$ barrel. All of the α -amylases contain one or more bound Ca^{2+} ions. Some, including the human α -amylases, also require a **chloride ion**. The Cl^- is held by a pair of arginine guanidinium groups¹⁵⁴ and interacts with adjacent carboxyl groups, inducing pK_a shifts and allosteric activation.¹⁶³

The sequences of α -amylases vary widely but a conserved cluster of one glutamate and two aspartates is usually present (Table 12-1).

Studies of the **action patterns**, i.e., the distribution of products formed by α -amylases when acting on a variety of α 1,4-linked oligosaccharides, suggested that the substrate binding region of the porcine pancreatic enzyme has five subsites, each binding one glucose residue.¹⁵³ The α -glucan chain is cleaved between the residues bound at the second and third subsites (numbered from the reducing end of the substrate)

by a lysozyme-like mechanism. Endolytic enzymes, which cleave biopolymer chains internally, are usually thought to carry out random attack. However, after the initial catalytic reaction one of the polysaccharide products of porcine pancreatic amylase action often does not leave the enzyme. The polysaccharide simply “slides over” until it fully occupies all of the subsites of the substrate binding site and a second “attack” occurs. An average of seven catalytic events occur each time it forms a complex with amylase.¹⁶⁷ Is there a mechanism by which the enzyme deliberately promotes the “sliding” of the substrate in this **multiple attack** or **processive** mechanism^{168,169} or does the dissociated product simply diffuse a short distance while enclosed in a solvent cage? The latter explanation may be adequate for some enzymes. In contrast to pancreatic α -amylase a bacterial “maltogenic” α -amylase produces principally maltose as a product.^{169a,b}

Digestion of starch and glycogen by α -amylases produces a mixture of glucose, oligosaccharides, and dextrins, which in the human body are further degraded by **α -glucosidases** of the brush border membrane of the small intestine.¹⁵⁴ A lysosomal form of the enzyme is missing in Pompe’s disease (Box 20-D).^{170,171} The α -glucosidases are also members of the α -amylase family.^{46,172} For digestion of the branched amylopectin and glycogen a **debranching enzyme** and **α -1,6-glucosidase** activity are required. In mammals these activities are found in a single polypeptide chain with separate but adjacent active sites.^{50,173,174} The debranching enzyme catalyzes transfer of oligosaccharide chains from α 1,6-linked branch positions to new locations at ends of chains with α 1,4 linkages. A bacterial oligo-1,6-glucosidase has a catalytic site formed from

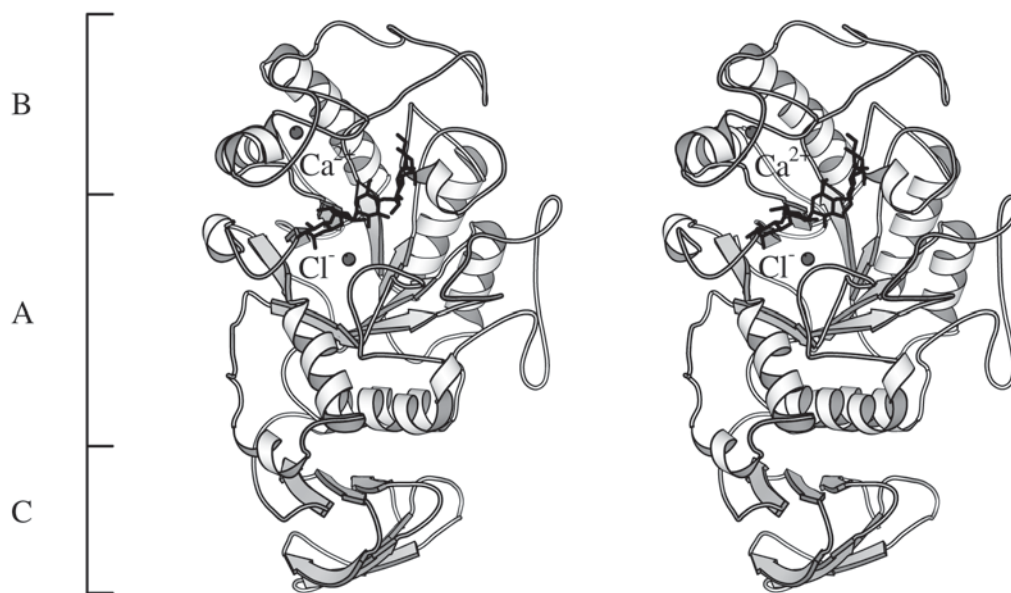


Figure 12-6 Drawing showing the overall polypeptide chain fold and relative positioning of the three structural domains of human pancreatic α -amylase. Also drawn are the locations of the calcium and chloride binding sites. Overlaid is the placement of a modified form of the inhibitor acarbose (p. 607) that binds in the active site cleft. MolScript drawing courtesy of G. Sidhu and G. Brayer.

D199, E255, and D329 similar to that of α -amylases.¹⁷⁵

Another member of the α -amylase family has principally a glycosyltransferase activity. **Cyclodextrin glucanotransferase** forms cyclodextrins (Box 4-A) by a transferase reaction in which a 6- to 8- member oligo-saccharyl group is transferred from a straight amylase chain onto a protein side chain (Glu 257) and then joins the ends of the oligosaccharide to form the cyclo-dextrin rings (an overall double-displacement process).^{176–177b} However, large circular dextrans are produced initially and are then converted into the smaller cyclodextrins.¹⁷⁸

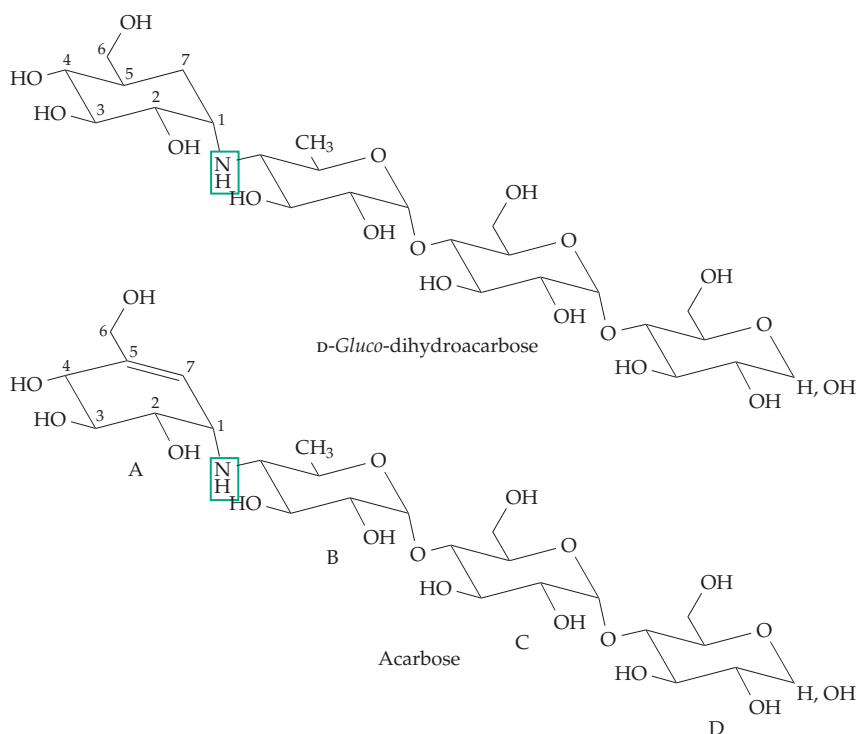
A starch-digesting enzyme of great industrial importance is **glucoamylase**, whose $(\alpha, \alpha)_6$ -barrel structure is shown in Fig. 2-29. That figure also shows the tetrasaccharide inhibitor **acarbose** in the active site. The ring at the nonreducing end is deeply embedded in the protein and, as shown in Fig. 12-7 for the related *D*-gluco-dihydroacarbose, is held by many hydrogen bonds. This slow enzyme ($k_{\text{cat}} \sim 50 \text{ s}^{-1}$ at 45°C)¹⁷⁹ cuts off a single glucose unit, then releases

inverting enzyme.

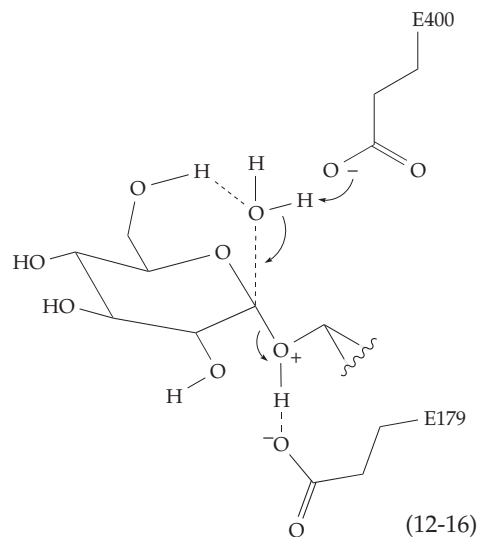
Notice in Fig. 12-7 the abundance of polar groups, many of which are charged. Four carboxyl groups, from D55, E179, E180, and E400, are present in the active site region and participate in the hydrogen bonded network. Active sites of other carbohydrases differ from that of glucoamylase. However, the presence of many hydrogen-bonded polar groups, including two or more carboxyl groups, is characteristic. As illustrated in Figs. 9-8 and 9-9, the pH dependence of the maximum velocity v_{max} is often determined by these carboxyl groups. At least one of them must be protonated for maximum catalytic activity and if two are protonated activity may fall again. Under these circumstances a single proton bound to a molecule of enzyme may spend a fraction of its time on each of several different carboxylate, imidazole, or other groups.¹⁸¹ In the case of glucoamylase it was concluded from the X-ray structure that the catalytic acid group, E179, is probably *unprotonated* most of the time.¹⁸¹ However, it can still bind to starch. Then, after a fraction of a second a

proton may be transferred onto the Glu 179 carboxylate and from there to the bridge oxygen of the substrate, inducing reaction according to Eq. 12-16. It may even be necessary to have enzyme protonated initially on a group other than E179 to allow small conformational changes to occur prior to formation of the final activated complex. Such essential conformational changes have often been invoked for glucoamylase.¹⁸²

Just as most cellulases have special cellulose-binding domains, glucoamylase has a compact C-terminal starch-binding domain (residues 509–616) similar to the



itself from the starch before releasing the glucose and rebinding to the starch. The catalytic acid has been identified as the carboxyl group of Glu 179. In Fig. 12-7 it is seen, presumably as a carboxylate group, tightly hydrogen bonded to what is doubtless the bridge -NH_2^+ between rings A and B of the inhibitor *D*-gluco-dihydroacarbose. Thus, the complex mimics that of a true substrate protonated on the bridge oxygen, a possible first step in normal catalysis. In accord with this mechanism (Eq. 12-16), glucoamylase is an



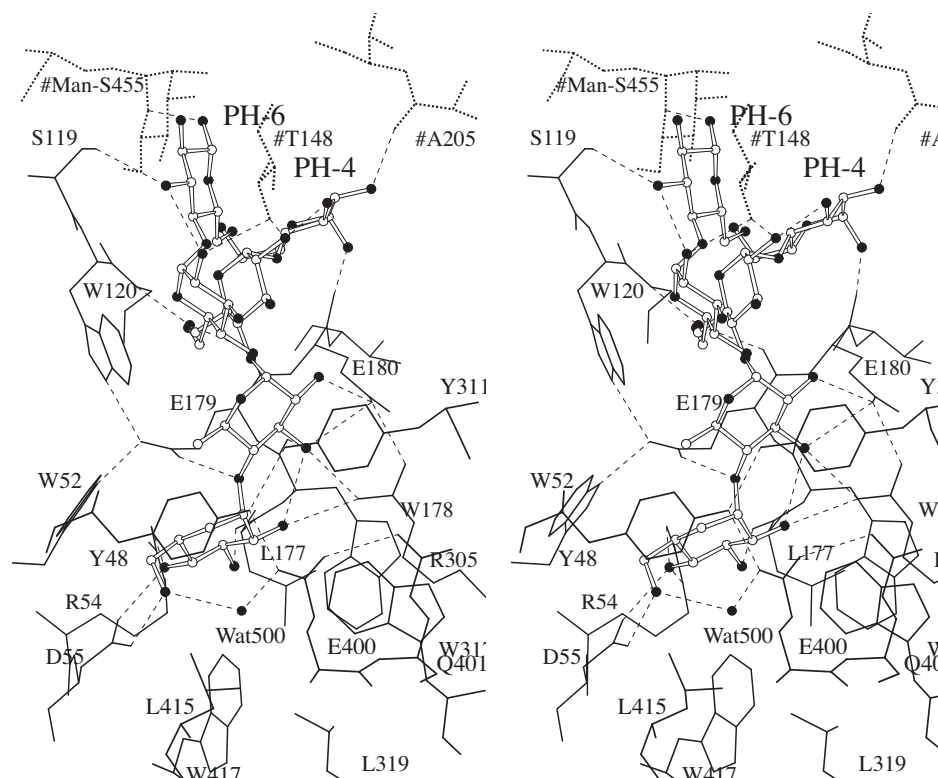


Figure 12-7 Stereoscopic view of the inhibitor D-glucodihydroacarbose in its complex with glucoamylase of *Aspergillus*. Residues from symmetry-related molecules of the enzyme are shown as dotted lines. Ring A at the nonreducing end of an amylose chain is thought to bind in a similar way (with ring A at the bottom in this figure). Cleavage is between the A and B rings E178 acting as a proton donor to the bridge oxygen (or NH for the inhibitor). The attacking nucleophile is thought to be a water molecule, which is labeled Wat 500 and is held by the assisting carboxylate of E400.¹⁸⁰ Courtesy of Alexander Aleshin.

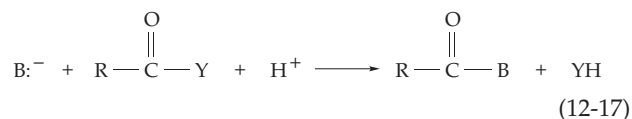
corresponding domain of cellulases.¹⁸³ It is connected by a glycosylated linker to the 470-residue catalytic subunit. Cyclodextrin glucanotransferase also has a starch-binding domain.¹⁸⁴

Beta-amylases, characteristic plant and bacterial enzymes^{184a}, have an *exo* action, cutting off chain ends two sugar units at a time as *maltose*. The original α linkage is inverted, the product being β -maltose. The $(\alpha/\beta)_8$ -barrel structure is unlike that of glucoamylase, but the spacing of active site carboxyl groups suggests that a water molecule is held and activated as in glucoamylase.^{185,186} The multiple subsite structure of the active site may permit the substrate amylase to slip forward after a maltose product molecule leaves the site. This may account for the observed processive action.¹⁸⁶

Many specialized glycosyl transferases synthesize glycogen, starch, cellulose, and other polysaccharides and add glycosyl groups to glycoproteins and glycolipids.^{187,188} Often the glycosyl group is transferred from a carrier such as uridine diphosphate (UDP). An example is glycogen synthase (Fig. 11-4), which transfers glucosyl groups of UDP-glucose to the 4'-OH groups at the nonreducing ends of the bushlike glycogen molecules. Other similar synthetic reactions are considered in Chapter 20.

C. Displacement Reactions on Carbonyl Groups

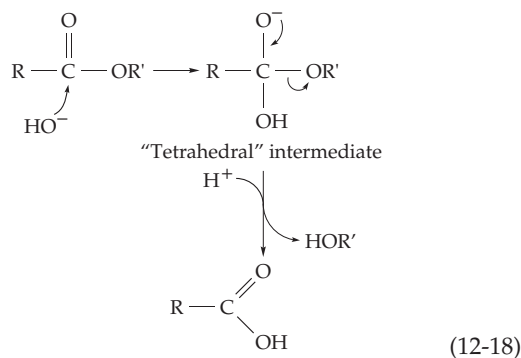
A second major class of nucleophilic displacement reactions (Type 1B in Table 10-1) involve replacement of a group Y attached to a carbonyl carbon:



Group Y may be -OR (esters), -SR (thioesters) or -NHR (amides or peptides). If B⁻ is a hydroxyl ion formed from H₂O, the reaction is **hydrolysis**. If B⁻ is the anion of an alcohol, thiol, or amine, the reaction is **transacylation**. Transacylation is an essential process in biosynthesis of proteins and lipids, but it is the digestive enzymes, which catalyze hydrolysis, that have been studied most intensively.

Uncatalyzed hydrolysis of a peptide linkage is very slow with $t_{1/2}$ at neutral pH and 25°C of ~300–600 years.¹⁸⁹ Both acids and bases catalyze hydrolysis, but enzymes are needed for rapid digestion. The carbonyl group C=O is highly polarized, with the resonance form C⁺-O⁻ contributing substantially to its structure. An attack by a base will take place readily on the electrophilic carbon atom. While the reactivity of the carbonyl group in esters and amides is relatively low

because of the resonance stabilization of these groups, the carbon atom still maintains an electrophilic character and may combine with basic groups. Thus, in the base-catalyzed hydrolysis of esters a hydroxyl ion adds to the carbonyl group to form a transient single-bonded “tetrahedral” intermediate (Eq. 12-18). Similar intermediates are believed to form during the action of many enzymes. However, for purposes of classification, we can regard them as simple displacement reactions on a carbon atom with the understanding that there are probably transient single-bonded intermediates.



A large number of hydrolytic enzymes, the **proteases**, and **peptidases** act on peptide linkages of proteins.¹⁹⁰ At present the traditional name protease, which implies proteolysis, is most often used. However, the IUB encourages use of the name **proteinase**. Although this seems less specific in meaning, its use will probably increase. Some proteases trim newly formed peptide chains, others convert proteins from precursor forms into biologically active molecules, and others digest proteins. In addition to endopeptidases, such as trypsin and chymotrypsin, which cleave at positions *within* a long peptide chain, there are many enzymes that cleave amino acids from the *ends* of chains. These are usually called peptidases and are designated **aminopeptidases** if they cleave from the N terminus and **carboxypeptidases** if they cleave from the C terminus. Most of these enzymes can be classified into **serine** proteases, **cysteine** proteases, **aspartic** proteases, **metallo** proteases, or **N-terminal nucleophile** hydrolases, depending upon the chemical nature of the active site. These groups are further divided into families or “clans.”¹⁹⁰

1. The Serine Proteases

The digestive enzymes **trypsin**, **chymotrypsin**, **elastase**, and **proteinase E** are related serine proteases. All three are synthesized in the pancreas which secretes 5–10 g per day of proteins, mostly the inactive proenzymes (zymogens) of digestive enzymes.^{191,192}

These proenzymes are synthesized and “packaged” as **zymogen granules** which travel to the surfaces of the secretory cells. The contents of the granules are secreted into the extracellular medium and are discharged via the pancreatic duct into the small intestine. At their sites of action, the zymogens are converted into active enzymes by the cutting out of one or more pieces from the precursor. This occurs in a cascade-type process triggered by **enteropeptidase** (historically enterokinase), another serine protease which is secreted by the intestinal lining.^{192a} Human enteropeptidase consists of a 235-residue catalytic subunit bonded through a disulfide bridge to a larger 784-residue membrane-anchoring subunit.^{193–195} It attacks specifically **trypsinogen**, converting it to active trypsin.^{196,197} Trypsin in turn activates the other zymogens, as is indicated in Fig. 12-8. Trypsin can also activate its own zymogen, trypsinogen, in an autocatalytic process.

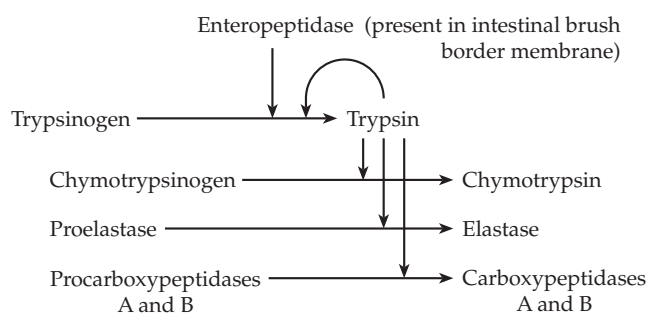
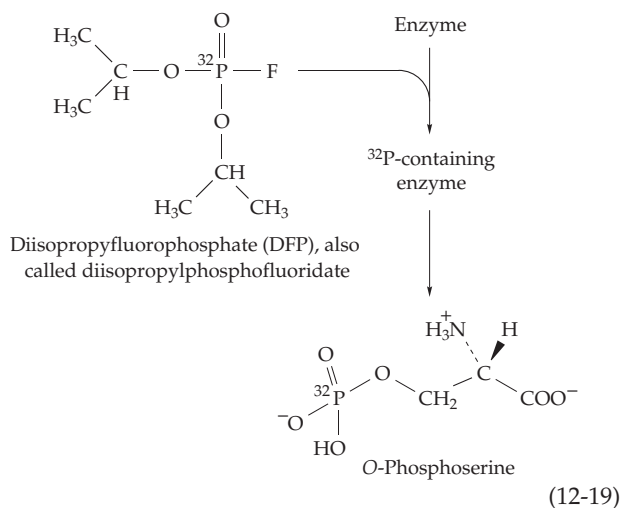


Figure 12-8 Cascade of reactions that activate pancreatic proteases. Enteropeptidase, or trypsin, cleaves the proenzyme (zymogen) at specific sites.

Chymotrypsinogen consists of a single 245-residue chain. The amino acid residues in chymotrypsin, trypsin, and elastase are usually all numbered according to their position in this zymogen. Inactive proenzymes are formed as precursors to enzymes of many different classes and are activated in a variety of ways. A part of the polypeptide chain of the proenzymes is often folded over the active site, interacting in a nonsubstrate-like fashion and blocking the site.^{197a}

Serine as a nucleophile. An early clue to the mechanism of action of chymotrypsin came from investigation of the related **acetylcholinesterase**. This key enzyme of the nervous system is inactivated irreversibly by powerful phosphorus-containing poisons that had been developed as insecticides and as war gases (nerve gases, Box 12-C). Around 1949, the nerve gas diisopropylfluorophosphate (DFP) was shown also to inactivate chymotrypsin. When radioactive ³²P-containing DFP was allowed to react the ³²P became

covalently attached to the enzyme. When the labeled enzyme was denatured and subjected to acid hydrolysis the phosphorus stuck tightly; the radioactive fragment was identified as *O*-phosphoserine. It was evident that this product could be formed by an attack of the hydroxyl group of the serine side chain on the phosphorus with displacement of the fluoride ion. This is a nucleophilic displacement on phosphorus, occurring on an enzyme that normally catalyzes displacement on $C=O$. The DFP molecule acts as a pseudosubstrate which reacts with the enzyme in a manner analogous to that of a true substrate but which does not complete the reaction sequence normally.



From study of peptides formed by partial hydrolysis of the ^{32}P -labeled chymotrypsin, the sequence of amino acids surrounding the reactive serine was established and serine 195 was identified as the residue whose side chain hydroxyl group became phosphorylated. The same sequence Gly-Asp-Ser-Gly was soon discovered around reactive serine residues in trypsin, thrombin, elastase, and in the trypsin-like **cocoonase** used by silkmoths to escape from their cocoons.¹⁹⁸ We know now that these are only a few of the enzymes in a very large family of serine proteases, most of which have related active site sequences.^{199,200} Among these are **thrombin** and other enzymes of the blood-clotting cascade (Fig. 12-17), proteases of lysosomes, and secreted proteases.

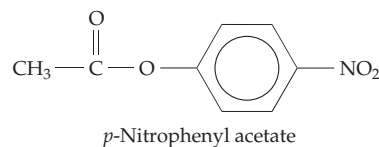
Numerous serine proteases, including trypsin-like enzymes called **tryptases**^{201–204a} and chymotrypsin-like chymases,^{205–207} are found within tissues in which they are stored in granules of mast cells,²⁰⁸ neutrophils, lymphocytes, and cytotoxic T cells.²⁰⁵ Secretory granules of mast cells present in skin and other tissues contain high concentrations of tryptase and chymase precursors^{202,206} which may be released as part of an inflammatory response. Tryptase may be involved in asthma and other allergic responses.²⁰¹ **Cathepsin G**²⁰⁹

(proteinase II), **neutrophil elastase**, and **proteinase III**²¹⁰ are found in granules of neutrophils and monocytes as well as in mast cells.²⁰⁹ Cytoplasmic granules of cytotoxic T cells contain at least seven proteases called **granzymes** that can be released to attack target cells.^{211–213a}

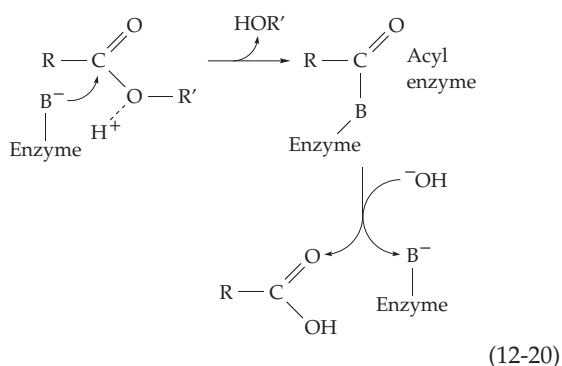
Many secreted proteins, as well as smaller peptide hormones, are acted upon in the endoplasmic reticulum by tryptases and other serine proteases. They often cut between pairs of basic residues such as KK, KR, or RR.^{214–216} A subtilisin-like protease cleaves adjacent to methionine.²¹⁷ Other classes of proteases (e.g., zinc-dependent carboxypeptidases) also participate in this processing. **Serine carboxypeptidases** are involved in processing human prohormones.²¹⁸ Among the serine carboxypeptidases of known structure is one from wheat²¹⁹ and **carboxypeptidase Y**, a vacuolar enzyme from yeast.²²⁰ Like the pancreatic metallocarboxypeptidases discussed in Section 4, these enzymes remove one amino acid at a time, a property that has made carboxypeptidases valuable reagents for determination of amino acid sequences. Carboxypeptidases may also be used for modification of proteins by removal of one or a few amino acids from the ends.

The variety of bacterial serine proteases known include the 198-residue **α -lytic protease** of *Myxobacter*,²²¹ a family of at least 80 **subtilisins** which are produced by various species of *Bacillus*^{222–225a} as well as by many other organisms,²²⁶ and a trypsinlike enzyme from *Streptomyces griseus*.^{227,228} **Tripeptidyl peptidases**, subtilisin-like enzymes, cut tripeptides from the N-termini of proteins.^{228a,b,c} One participates in lysosomal protein degradation and the other, an oligomer of 138 kDa subunits, cuts precursor proteins to form neuropeptides and other hormones (Chapter 30).

Acyl-enzyme intermediates. Serine proteases are probably the most studied of any group of enzymes.²²⁹ Early work was focused on the digestive enzymes. The pseudosubstrate, *p*-nitrophenyl acetate, reacts with chymotrypsin at pH 4 (far below the optimum pH for hydrolysis) with rapid release of *p*-nitrophenol and formation of acetyl derivative of the enzyme.



This acetyl enzyme hydrolyzes very slowly at pH 4 but rapidly at higher pH. These experiments suggested a double displacement mechanism:

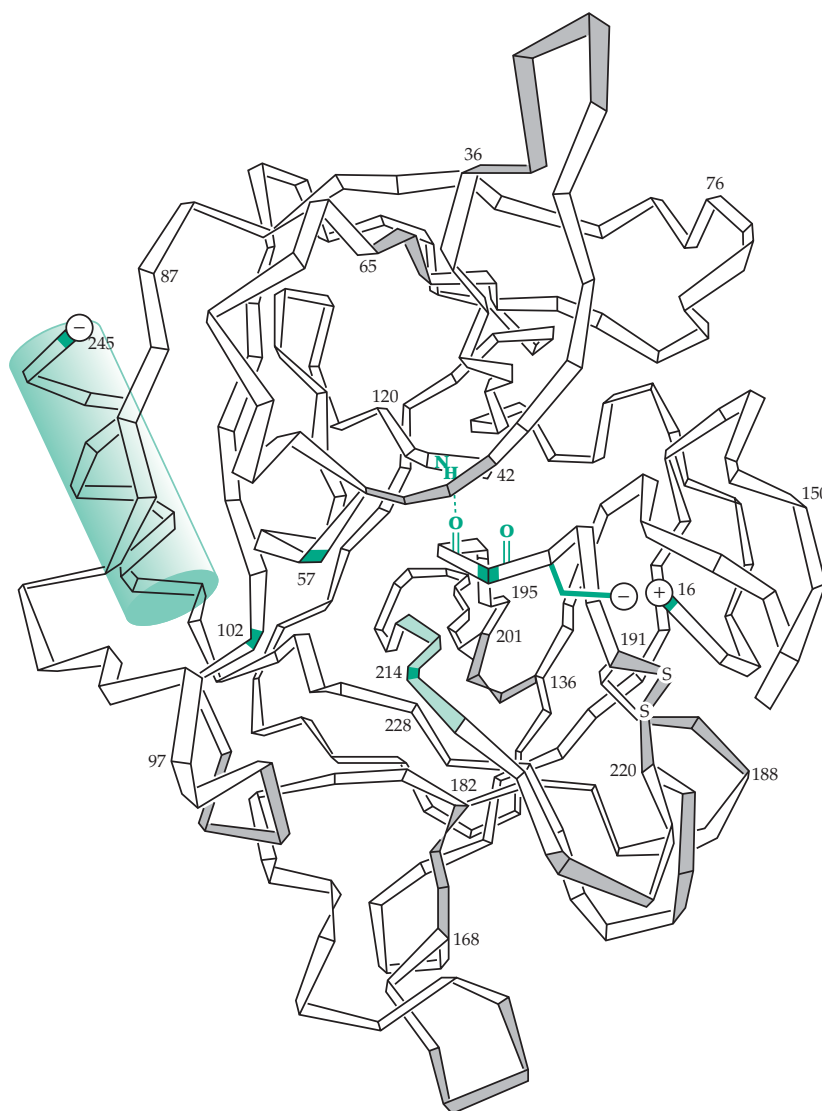


Although the experiments with DFP suggested that the $-O^-$ group from Ser 195 might be the base B^- in this equation, there was reluctance to accept this deduction because of the very weak acidity of the $-CH_2OH$ group. Furthermore, the pH dependence of catalysis suggested an imidazole group of a histidine side chain as the attacking nucleophile B. Indeed, imidazoles catalyze the nonenzymatic hydrolysis

of *p*-nitrophenyl acetate with formation of unstable *N*-acetyl imidazoles as intermediates. Thus, while the stable end products of reactions with pseudosubstrates were unquestionably derivatives of serine, the possibility remained that these were side products and that histidine was involved in transient, rapidly forming, and reacting intermediates. It remained for the results of the then newly developed science of X-ray crystallography to clarify this question.²³⁰

Three-dimensional structures. The structures of chymotrypsin,^{199,230,231} trypsin,^{232,233} elastase,^{234,235} thrombin,²³⁶ kallikrein,^{237,238} and many other enzymes are similar, with the basic fold shown in Fig. 12-9.^{199,228} Both Ser 195 and His 57 (or corresponding residues) are present in the active site (Figs. 12-9, 12-10). From the observed positions of competitive inhibitors occupying the active site, the modes of binding depicted in Fig. 12-10A for the chymotrypsin family and in Fig. 12-10B for the subtilisin family have been deduced. Bear in mind that the X-ray diffraction results do not

Figure 12-9 Alpha-carbon diagram of the three-dimensional structure of pancreatic elastase. A principal structural feature is a pair of β cylinders. One of these, at the top of the figure, is viewed end-on, while the other, at the bottom of the figure, is viewed from a side. The prominent interface between them is seen in the center. The α -carbon positions of the catalytic triad serine 195, histidine 57, and aspartate 102 are marked. As shown in Fig. 12-10 the catalytic triad is located across the interface between β cylinders, which may allow for easier conformational alterations during the action of the enzyme. The four disulfide bridges and the long C-terminal helix are also emphasized. The cross-hatched loop regions are residues present in elastase but not present in chymotrypsin. The green shaded strand (residues 214–216) is a segment that joins residues P_1 and P_2 of the substrate to the edge of the lower β cylinder in an extended β structure. The ion pair formed by aspartate 194 and isoleucine 16 during zymogen activation is also shown, as are two peptide carbonyl groups that protrude into the interface area, one forming a hydrogen bond across the interface. Modified from a drawing of Sawyer *et al.*²³⁴ which was in turn based on the original chymotrypsin drawing of Annette Snazle.²³⁰



A

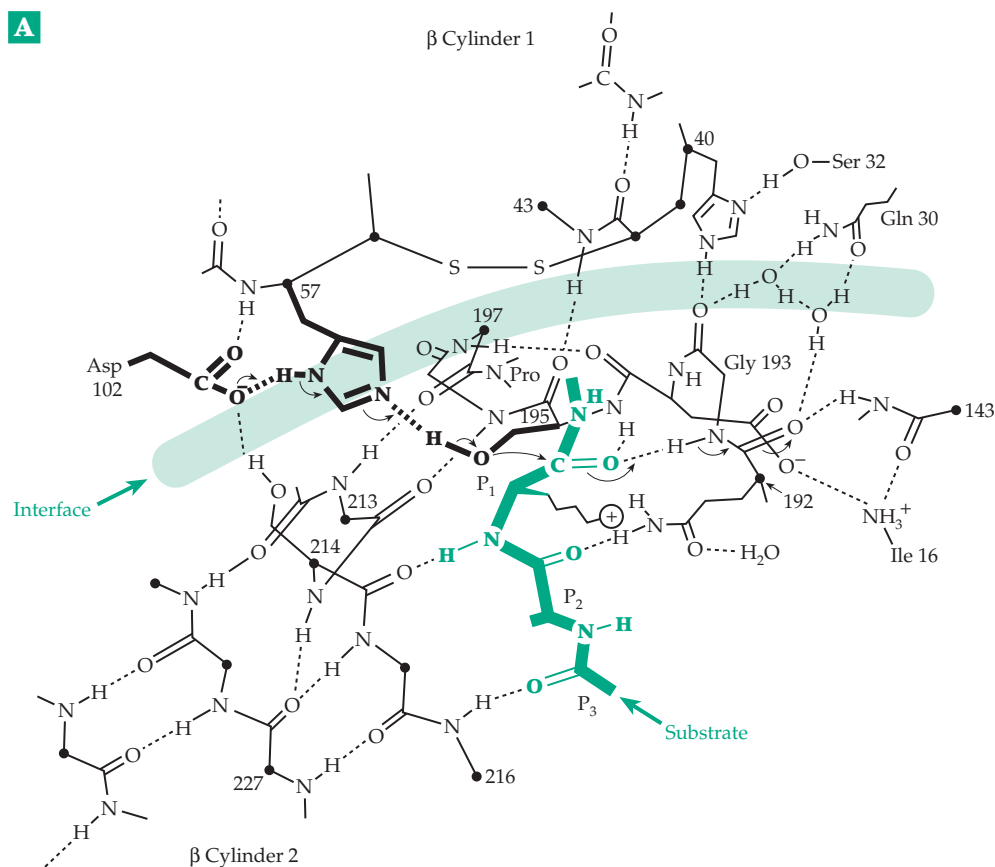
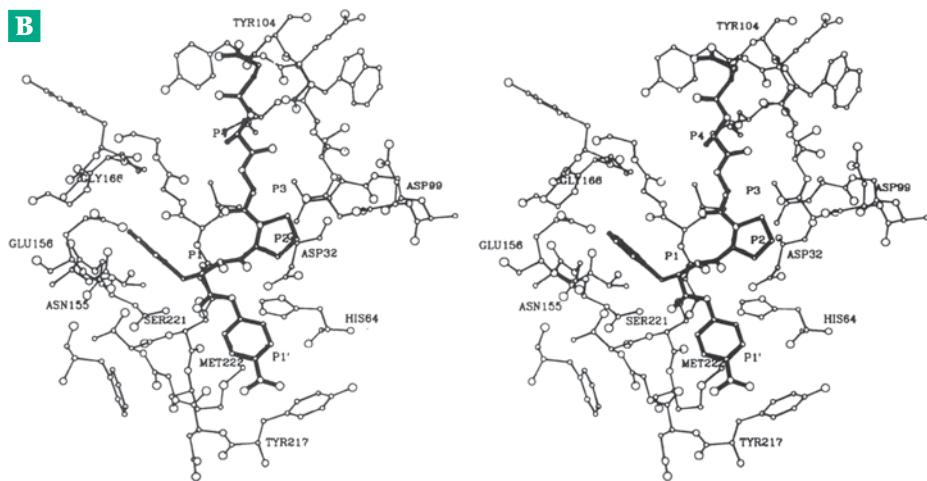


Figure 12-10 (A) Part of the hydrogen-bonding network of trypsin and other serine proteases with a bound trypsin substrate (green). Residues P_1 to P_3 , as defined in Fig. 12-14, have been marked. The view is similar to that in Fig. 12-9 but some background lines have been omitted. Note the two competing hydrogen-bonded chains passing through C=O of residue 227 in β cylinder 2. One of these chains passes through the backbone of the substrate and the other through residues 214 and 195 across the interface between domains into β cylinder 2. Asp 102, His 57, and Ser 195 of the catalytic triad are emphasized. Arrows indicate probable movement of electrons from the negative charge of Asp 102 into the “oxanion hole.” After Metzler.⁸⁵ Based on papers of Sawyer *et al.*²³⁴ and Huber and Bode.²⁴⁰ (B) Stereoscopic view showing the model substrate *N*-succinyl-L-Ala-L-Ala-L-Pro-L-Phe-*p*-nitroanilide bound into the active site of subtilisin BPN'. Residues of the catalytic triad (Ser 221, His 64, Asp 32) and some others are labeled. Subsites P_1' , P_1 , P_2 , P_3 , and P_4 (see Fig. 12-14) are also labeled. Notice that site P_3 , which is near the top in this drawing, is at the bottom in (A). Based on X-ray data of R. Bott and M. Ultsch at 0.2 nm resolution. From Wells and Estell.²²³

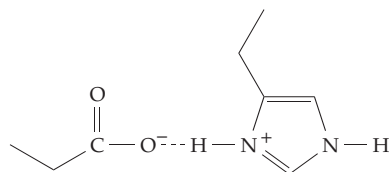
B



show where the hydrogen atoms are and that these have been added in Fig. 12-10. The imidazole group of His 57 is located next to the side chain –OH group of Ser 195 and is able to form an N---H–O hydrogen bond to it. The obvious conclusion is that His 57 acts as a general base catalyst that assists in removing the proton from the –OH of Ser 195, making that hydroxyl group more nucleophilic than it would be otherwise. This may happen after the –OH group has started to add to the substrate carbonyl.

The second nitrogen atom of His 57 is hydrogen bonded to the carboxylate group of Asp 102, which is in turn hydrogen bonded to two other groups. Aspartate 102 has one of the few carboxylate side chains that is buried inside the protein. To Blow,^{230,231,239} the structure suggested a **charge-relay system** by which negative charges might move synchronously from Asp 102 to the imidazole which could then deprotonate the hydroxyl group of Ser 195, allowing the serine oxygen to add to the substrate carbonyl to form the tetrahedral

or **oxyanion** intermediate, which is depicted in step *b* of Fig. 12-11. The small arrows in Fig. 12-10 also indicate the movement of charge. Blow suggested that in the extreme case a *proton* might be transferred also from the His 57 imidazole to the Asp 102 carboxylate. However, a variety of experiments, including studies by ^{15}N NMR^{241,242} and by neutron diffraction,²⁴³ suggest that the imidazole does not transfer its proton to the carboxylate of Asp 102,^{244,245} unless it does so transiently, e.g., in a transition state complex. Instead, the carboxylate and the imidazolium ions formed by protonation of His 57 probably exist as a tight ion pair:



Support for this concept is provided by ^1H NMR studies which have identified a downfield resonance of the hydrogen-bonded proton in this pair at ~ 18 ppm in chymotrypsinogen and chymotrypsin at low pH and at ~ 14.9 – 15.5 ppm at high pH values.^{246,247} Similar resonances are seen in the α -lytic protease,²⁴⁸ in subtilisin,²⁴⁹ in adducts of serine proteases with boronic acids^{250,251} or peptidyl trifluoromethyl ketones,²⁵² in alkylated derivative of the active site histidine,²⁵³ and in molecular complexes that mimic the Asp-His pair in the active sites of serine proteases.²⁵⁴

The catalytic cycle. Figure 12-11 depicts the generally accepted sequence of reactions for a serine protease. If we consider both the formation and the subsequent hydrolysis of the acyl-enzyme intermediate with appropriate oxyanion intermediates, there are at least seven distinct steps. As indicated in this figure, His 57 not only accepts a proton from the hydroxyl

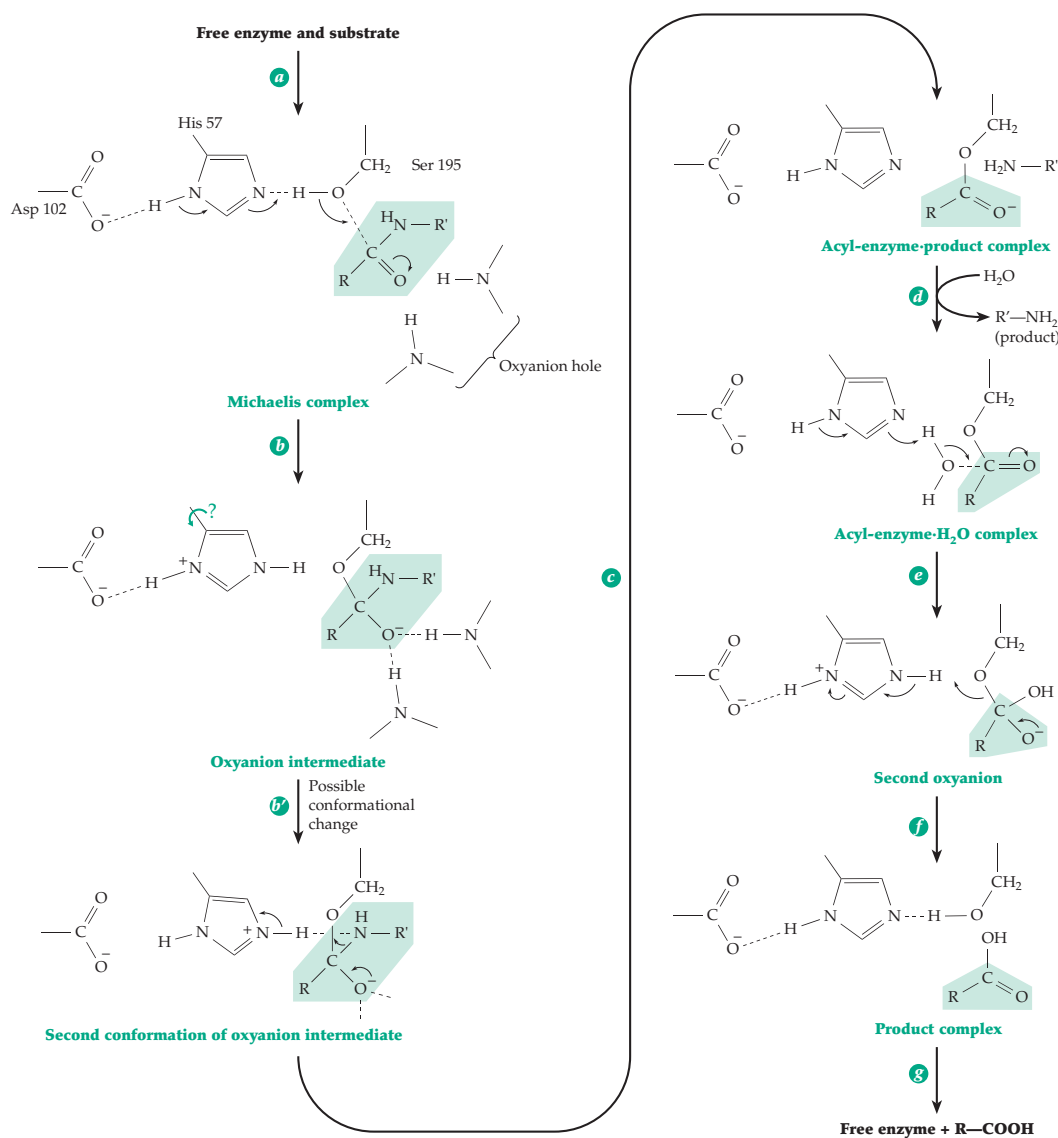


Figure 12-11 Sequence of chemical reactions involved in the action of a serine protease. The oxyanion hole structure has been omitted in the right-hand column. The imidazole ring may rotate, as indicated by the green arrow, to provide two different conformations.

group of Ser 195 (step *b*) but probably also functions in protonation of the --NH--R' leaving group (step *c*). An unprotonated --NH--R' would be such a poor leaving group that the oxyanion intermediate would not go on to acyl-enzyme. However, donation of a proton to that leaving group from the protonated His 57 (general acid catalysis) permits elimination of $\text{H}_2\text{N--R'}$ (step *c*). The product of this step is the acyl-enzyme intermediate which must be hydrolyzed to complete the catalytic cycle. This is accomplished through steps *d--f* of Fig. 12-11. The product R'--NH_2 is replaced by H_2O , which, in steps paralleling steps *b* and *c*, is converted to an HO^- ion that serves as the attacking nucleophile to form (in step *e*) a second oxyanion intermediate which is cleaved to the second product R--COOH . The water molecule that enters in step *d*, and which participates in hydrolysis of the acyl enzyme, has apparently been observed directly by time-resolved Laue crystallography at low temperature.²⁵⁵

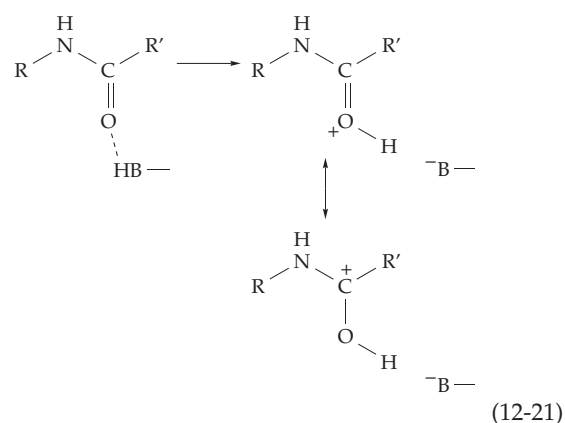
The catalytic triad. The significance of the charge-relay effect may not be fully understood but the importance of the Ser•His•Asp cluster, which has become known as the catalytic triad, cannot be doubted. It has evolved independently in several subfamilies of bacterial and plant proteases, a well-known example of “convergent evolution.” This triad is also found throughout a broad range of many different kinds of enzymes and other proteins. In pancreatic enzymes the catalytic triad consists of Ser 195•His 57•Asp 102 but in the bacterial subtilisin it consists of Ser 221•His 64•Asp 32 and in a wheat serine carboxypeptidase it is Ser 146•His 397•Asp 338.²¹⁹ In all three cases the folding patterns of the polypeptides are entirely different but the geometry of the triad is the same. Another folding pattern is seen in a protease encoded by cytomegalovirus, which contains an active site Ser 132•His 63 pair.^{256,257}

Investigation of a host of mutant proteins also demonstrates the importance of the catalytic triad. For example, if either the histidine or the serine of the triad of subtilisin was replaced by alanine the catalytic activity decreased by a factor of 2×10^6 and replacement of the aspartate of the triad by alanine decreased activity by a factor of 3×10^4 .^{229,258} When Asp 102 of trypsin is replaced by asparagine the catalytic activity falls by four orders of magnitude.²⁵⁹ This may be in part because the histidine in this mutant is hydrogen bonded to Asn 102 as the tautomer with a proton on N^ϵ , the nitrogen that should serve as the catalytic base in step *b* (Fig. 12-11).²⁶⁰ A mutant in which Ser 214 (see Fig. 12-10) was replaced with alanine is fully active but charged residues in this position interfere with catalysis.²⁶¹

Does the “low-barrier hydrogen bond” in the catalytic triad play any special role in catalysis? Blow’s suggestion of a charge relay from Asp 102 to

Ser 195 of chymotrypsin is probably correct. Some theoretical calculations have indicated the possibility of synchronous movement of the two protons in the system during step *a* of the sequence shown in Fig. 12-11, with the proton in the strong hydrogen bond to Asp 102 moving away from His 57 and toward the midpoint distance of the hydrogen bond. Could the presumed high energy of the short hydrogen bond be harnessed to lower the transition state energy? A realistic possibility is that the hydrogen bond increases the polarizability of the catalytic triad, facilitating movement of the substrate to the transition state. Cassidy *et al.* suggested that the strong hydrogen bond may be formed by compression of the triad resulting from binding of substrate in the S_1 and S_2 subsites.²⁵² They suggested that this would raise the pK_a of His 57 from ~ 7 in the free enzyme to 10–12, high enough to enable it to remove the proton from the Ser 195 --OH group and low enough to allow the protonated form to be the proton donor to the leaving group (step *c*, Fig. 12-11).

The “oxyanion hole.” A third mechanism by which an enzyme can assist in a displacement reaction on a carbonyl group is through protonation of the carbonyl oxygen atom by an acidic group of the enzyme (Eq. 12-21). This will greatly increase the positive charge on the carbon atom making attack by a nucleophile easier and will also stabilize the tetrahedral



intermediate. Although the carbonyl oxygen is very weakly basic, it can interact with a suitably oriented acidic group of the enzyme. In many serine proteases this acidic function is apparently fulfilled by NH groups of two amide linkages. In chymotrypsin these are the backbone NH groups of Ser 195 and Gly 193 (Figs. 12-10, 12-12). No actual transfer of a proton to the carbonyl oxygen of the substrate is expected. However, the NH groups are positive ends of amide dipoles and can interact electrostatically with the negative charge that develops on the oxyanion. The fit of substrate into the **oxyanion hole** between the two

NH groups is apparently good only for the tetrahedrally bonded oxyanion intermediate,²⁶² a structure thought to be close to that of the transition state. The importance of the oxyanion hole for catalysis has been supported by theoretical calculations²⁴⁵ and by the study of mutant enzymes.²⁶³ In subtilisin, in which a side chain of asparagine forms part of the oxyanion hole, replacement of Asn with the isosteric Leu causes k_{cat} to fall by a factor of about 200 while K_m is unaffected.²⁶⁴ It is also of interest that thiono ester substrates, in which the C=O of an oxygen ester has been replaced by C=S, bind with normal affinity to chymotrypsin but are not hydrolyzed at significant rates.²⁶⁵

The formation of the oxyanion intermediate during serine protease action is also supported by the existence of tetrahedral forms of enzymes inhibited by substrate-like aldehydes. The –OH group of Ser 195 can add to the carbonyl group to form hemiacetals. For example, a ¹³C-enriched aldehyde whose carbonyl carbon had a chemical shift of 204 ppm gave a 94 ppm resonance as it formed the tetrahedral hemiacetal with one of the inhibitory aldehydes, *N*-acetyl-L-Leu-L-Leu-L-arginal

(**leupeptin**; Box 12-C). A natural product from a species of *Streptomyces*, this aldehyde inhibits trypsin and several other enzymes strongly.²⁶⁶ Adducts of wheat serine carboxypeptidases with aldehyde inhibitors have also been observed.²⁶⁷ While the carbonyl group of the substrate amide linkage that is to be cleaved apparently can't form strong hydrogen bonds to the NH groups of the oxyanion hole, that of the acyl-enzyme intermediate can, as judged by resonance Raman spectroscopy.^{268,269} The strength of the hydrogen bonding, as judged by the stretching frequency of the C=O group, is correlated with the reactivity of the acyl group.²⁶⁹

Chymotrypsinogen and related proenzymes have extremely low catalytic activity even though a major part of the substrate binding site as well as the catalytic triad system are already in place. However, the oxyanion hole is created during activation of the proenzyme by a subtle conformational change^{197,262,271} that involves the chain segment containing Gly 193 (Fig. 12-12). This is further evidence of the importance of this part of the active site structure.

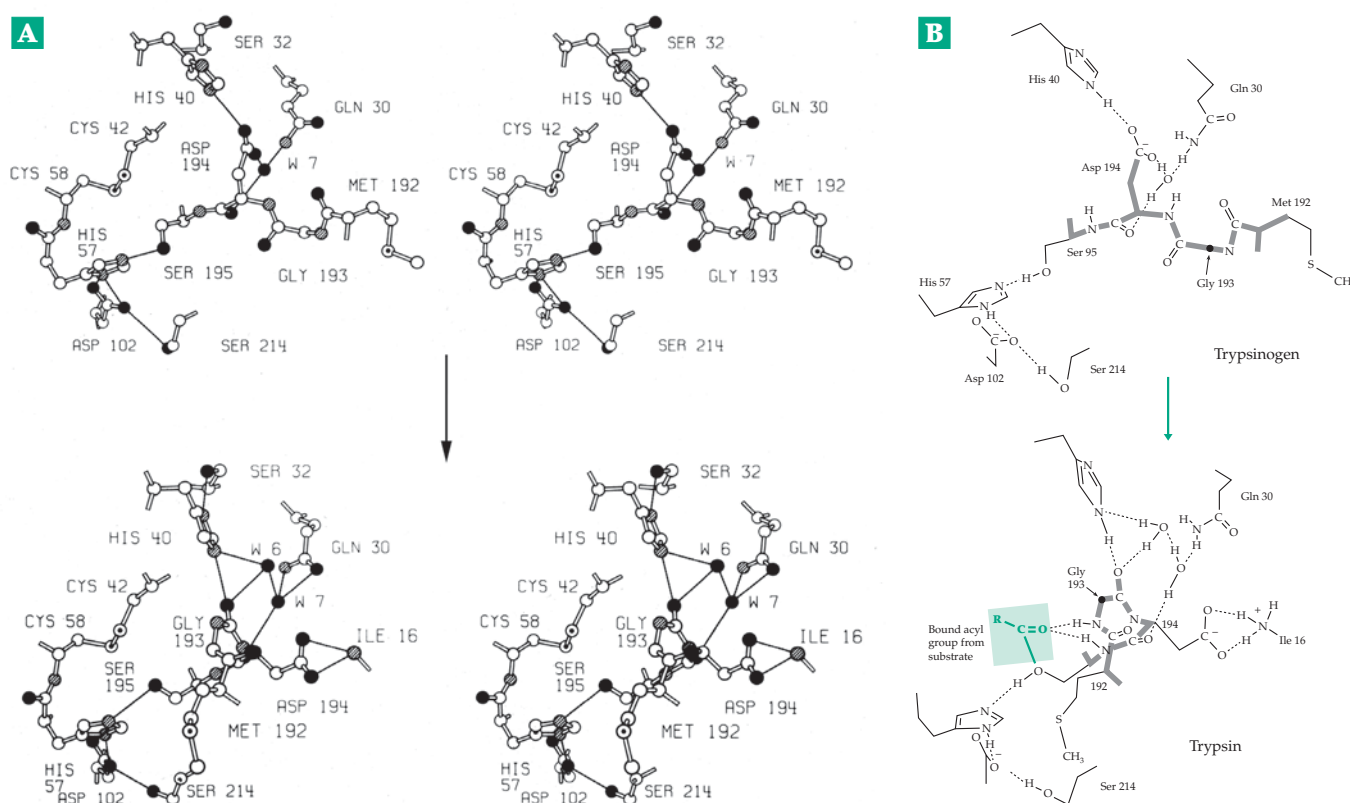


Figure 12-12 Formation of the oxyanion hole following cleavage of trypsinogen between Lys 15 and Ile 16. (A) Stereoscopic view. (B) Schematic representation. The newly created terminal --NH_3^+ of Ile 16 forms a hydrogen-bonded ion pair with the carboxylate of Asp 194. This breaks the hydrogen bond between Asp 194 and His 40 in trypsinogen, inducing the peptide segment 192–194 to shift from an extended conformation to a helical form in which the NH groups of Gly 193 and Ser 195 form the oxyanion hole. Notice that the positions and interactions of Asp 102, His 57, and Ser 195, the catalytic triad, are very little changed. From Birktoft *et al.*²⁷⁰

Stereoelectronic considerations. The amide group that is cleaved by a protease is a resonance hybrid of structures A and B of Fig. 12-13. The unshared pair of electrons on the nitrogen atom of structure A and the third unshared pair on the oxygen atoms of structure B (shaded orbitals) have been drawn in this figure in such a way that they are anti-periplanar to the entering serine oxygen. This is required in the transition state according to stereoelectronic theory. The newly created (shaded) electron pair on oxygen is one of those that hydrogen bonds to an NH group of the oxyanion hole. The tetrahedral intermediate has another unshared pair, which is not hydrogen bonded, and is antiperiplanar to the HN-R leaving group. Thus, it appears to be set up for easy elimination. Nevertheless, there is some doubt about the need for adherence to the stereoelectronic “rule” that two antiperiplanar lone pairs are necessary for elimination.

There is another complication. The leaving group $-\text{NH}-\text{R}'$ cannot be eliminated from the oxyanion until it is protonated, presumably by the imidazolium group of His 57. However, the proton on His 57 will be adjacent to the proton that is already on this nitrogen rather than to the unshared pair of electrons on the same nitrogen atom. A conformational change in which the tilt of the catalytic imidazole ring is altered (step b' , Fig. 12-11) or in which the ring rotates, may have to precede the protonation of the leaving group (step c , Fig. 12-11).^{272–274b} This change may be assisted by the presence of positive charge on the protonated imidazole, but it still does not solve the problem.

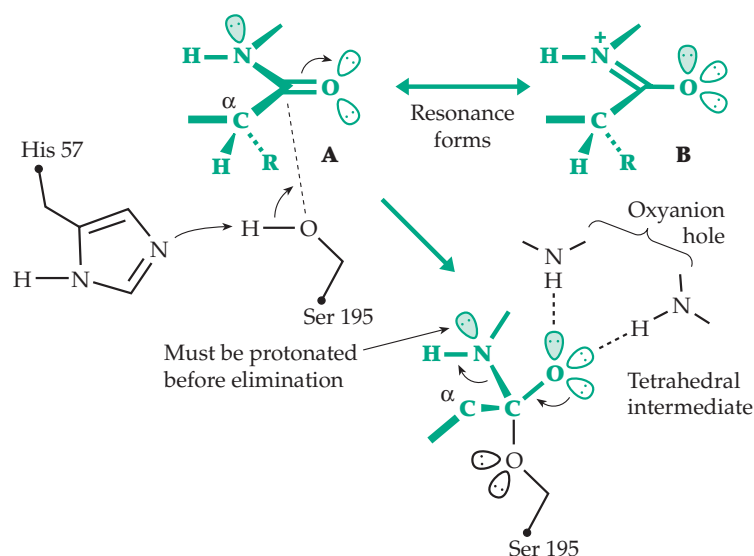


Figure 12-13 The stereochemistry of formation of a tetrahedral intermediate by a serine protease. The most probable orientation of groups as deduced by model building is shown. The shaded orbitals in A and B are antiperiplanar to the entering oxygen of Ser 195. See Polgár and Halász.²⁷²

Bizzozero and Butler suggested that a rapid inversion of this chiral center on the $-\text{NH}-\text{R}'$ group may be required prior to protonation and elimination.²⁷³ This would explain their observation that *N*-alkylated peptide linkages of otherwise fast substrates are not cleaved by chymotrypsin. A high-resolution structure determination demonstrated that a very slow elastase substrate that contains *N*-methylleucine at the cleavage site forms a normal ES complex.²⁷⁵ However, inversion would be hindered by the *N*-methyl group. An alternative in inversion would be a torsional rearrangement of the substrate during binding.²⁷⁶ The structure of an acyl-enzyme intermediate with elastase has been determined by crystallographic cryoenzymology. The intermediate was allowed to accumulate at -26°C , after which the temperature was lowered to -50°C and the structure determined. The structure shows the carbonyl group in the oxyanion hole as anticipated.²⁷⁷

pH dependence. A plot of k_{cat}/K_m for chymotrypsin is bell shaped with a maximum around pH 7.8 and pK_a values of 6.8 and 8.8. These represent pK_a s of the free enzyme (Eq. 9-57). That of 6.8 has been shown to represent His 57. As the pH is lowered from the optimum the affinity for substrate falls off as His 57 becomes protonated. The high pK_a of 8.8 is thought to belong to the N-terminal amino group of Ile 16, a group that is generated during the conversion of the proenzyme to active enzyme.²⁷⁸ The Ile 16 amino group forms an ion pair with Asp 194 (Fig. 12-12B) which is next to the serine at the active center. This salt linkage helps to hold the enzyme in the required conformation for reaction and its deprotonation at high pH causes a decrease in substrate affinity.²⁷⁹

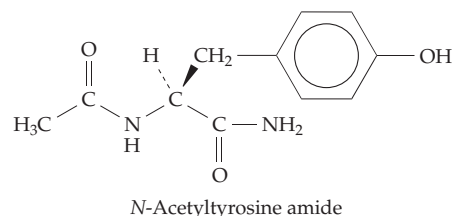
The value of k_{cat} is also pH dependent and falls off at low pH around a pK_a value of from 6–7 depending upon the substrate. However, no higher pK_a affects k_{cat} in the experimentally accessible range. These results provide the basis for believing that an unprotonated His 57 is needed in the ES complex for catalysis to occur. Similar conclusions have been reached for other serine proteases.

Substrate specificity. Like most other enzymes, proteases display distinct preferences for certain substrates. These are often discussed using the nomenclature of Fig. 12-14. The substrate residue contributing the carbonyl of the amide group to be cleaved is designated P_1 and residues toward the N terminus as P_2 , P_3 , etc., as is shown in Fig. 12-14. Residues toward the C terminus from the peptide linkage to be cleaved are designated P_1' , P_2' , etc. Chymotrypsin acts

most rapidly if the P_1 residue is one of the aromatic amino acids.²⁸⁰ Thus, the S_1 part of the substrate binding site must bind preferentially to large, flat aromatic rings. The crystal structure of chymotrypsin showed this site to be composed of nonpolar side chain groups. On the other hand, in trypsin the specificity portion of the S_1 site is a deep “specificity pocket” containing a fixed negative charge provided by the carboxylate side chain group of Asp 189. This explains why trypsin acts only upon peptide linkages containing the positively charged arginine or lysine residues in the P_1 position. In elastase the specificity pocket is partly filled by nonpolar side chains and the enzyme can accommodate only small P_1 side chains such as the methyl group of alanine. Replacement of Asp 189 of trypsin by lysine led to the predicted loss of specificity for basic side chains. However, the mutant enzyme did not become specific for negatively charged side chains.²⁸¹ This is presumably because the lysine $-NH_3^+$ group was not located in the same position as the Asp 189 carboxylate as a result of a different packing of side chains between native and mutant enzymes. Residues P_1' and P_2' also have major effects on substrate binding by serine proteases.^{282–284} Now many mutant forms of subtilisin^{225,285} and other serine proteases are being made and are yielding a more sophisticated understanding of the basis of the specificity of these enzymes. An important factor that has emerged is flexibility of surface loops in allowing an enzyme to adjust its structure to give a better fit to some substrates.^{286,287} The specificity of the serine proteases is also being exploited in the design of specific inhibitors (Box 12-D).

Many serine proteases react with *p*-nitrophenylacetate to give acetyl enzymes. However, its rate of

hydrolysis to give acetate is orders of magnitude slower than that of acyl-enzymes derived from small substrates such as the chymotrypsin substrate *N*-acetyltyrosine amide.



In addition to the specificity-determining P_1 aromatic side chain, the amide groups of this substrate can form specific hydrogen bonds to the protein (Fig. 12-10). These hydrogen bonds presumably help the enzyme to recognize the compound, which is bound with K_m of ~ 0.03 M and is hydrolyzed (with liberation of NH_3)^{288,289} with $k_{cat} \sim 0.17$ s⁻¹.

What happens when the length of the substrate is extended in the direction of the N terminus? The tyrosyl residue in the foregoing compound may be designated P_1 . For extended substrates which contain P_2 , P_3 , and additional residues (this includes most natural substrates) the K_m values decrease very little from that of short substrates despite the larger number of “subsites” to which the substrate is bound. However, the maximum velocity is often much greater for the extended substrates than for short ones. Thus, for *N*-acetyltyrosyl-glycine amide K_m is 0.017 M, only a little less than for *N*-acetyltyrosine amide, but k_{cat} is 7.5 s⁻¹, 440 times greater than for the shorter substrate.^{229,288,289} Other examples have been tabulated by Fersht.²⁷⁹

These observations suggest that the binding energy that would be expected to increase the tightness of binding is, instead, causing an increase in V_{max} ^{279,290} that is, it is reducing the Gibbs energy of activation. How can this be? Imagine that as the extended substrate binds, for example, into the subsite S_2 (which binds residue P_2), it must compress a spring in the enzyme. Could not the compressed spring now provide a source of energy for assisting in peptide bond cleavage? If this is the case, we must ask “what are the springs?” Are there amide linkages of the peptide backbone that are distorted when the substrate hydrogen bonds into the site? How is the distortion transmitted to the active site and how does it stabilize the transition

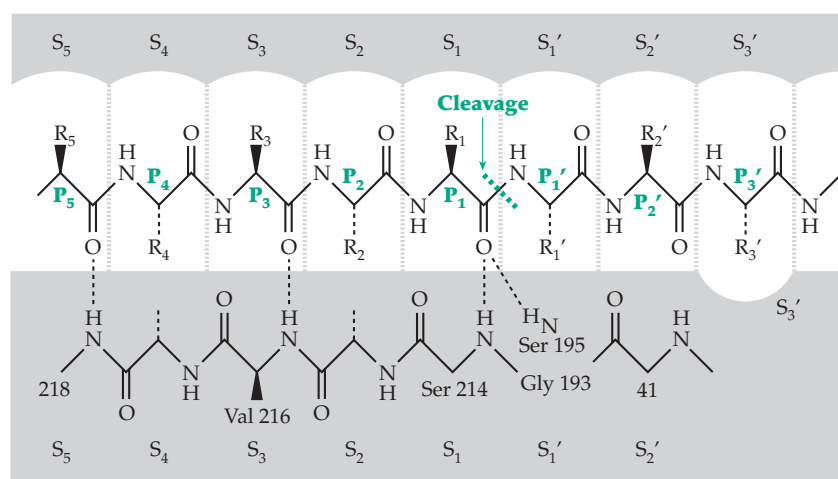


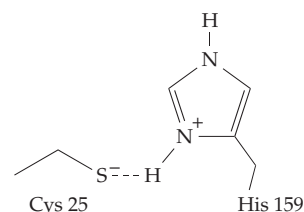
Figure 12-14 Standard nomenclature used to define the residues P_1 , P_2 . . . toward the N terminus and P_1' , P_2' . . . toward the C terminus of a peptide substrate for a protease. The corresponding subsites of the protease are designated S_1 , S_2 . . . S_1' , S_2' .

state? We are far from understanding the answers to these questions. However, it is of interest that one chain of H-bonds that passes through the amide between P_1 and P_2 also passes through a carbonyl group of a β bulge (Fig. 12-10). A second chain of H-bonds that connects to the same β bulge passes through the backbone amide of the active site Ser 195, across the interface between the two domains, and through the backbone of the second β cylinder.⁸⁵ Perhaps when substrate binds, the strengthening of the first H-bond chain weakens the second through competition at the β bulge, allowing some subtle rearrangement in the active site structure. We are talking here about the P_1 – P_2 amide. What happens at P_2 – P_3 , and further along the chain in both directions? It is important to understand these effects; the phenomenon of increased reaction rate for longer, more specific substrates is observed with many proteases and other enzymes as well.

Another difference between small substrates and longer, more specific substrates has been found in studies of the effects of changing the solvent from pure H_2O to mixtures containing an increasing mole fraction of 2H_2O . This is called a **proton inventory**.²⁹¹ For the serine proteases the rates are decreased as the 2H content of the solvent increases, a fact that suggests that some step involving a proton transfer, for example, the deprotonation of Ser 195, is rate limiting. For simple substrates, the effect is directly proportional to the mole fraction of 2H . However, for extended substrates, a quadratic dependence on the mole fraction of 2H is observed. This suggests that a process involving synchronous transfer of two protons, as in the postulated charge relay system, may be more important in extended substrates than in simple ones and may account for the more rapid action on these substrates.^{290,292} However, other interpretations of the data are possible, leaving this conclusion uncertain.^{229,291}

2. The Cysteine Proteases (Thiol Proteases)

Papain from the papaya is one of a family of enzymes that includes **bromelain** of the pineapple, **ficin** of the fig, and **actinidin** of the kiwifruit.²⁹³ Additional cysteine proteases²⁹⁴ from the latex of the papaya tree *Carica papaya* are known as **caricain**, **chymopapain**,²⁹⁴ and **glycyl endopeptidase**.²⁹⁵ All are members of a large superfamily which includes at least 12 mammalian enzymes and many others from both eukaryotic and prokaryotic organisms.²⁹⁶ All share with papain a characteristic structure which was determined by Drenth and coworkers in 1968.²⁹⁷ The participating nucleophile in the active sites of these enzymes is an SH group, that of Cys 25 in papain. An adjacent imidazole from His 159 removes the proton from the SH group to form a thiolate-imidazolium pair.^{298–300}



The close proximity of the imidazolium group greatly lowers the microscopic pK_a of the Cys 25 thiol group and the proximity of the resulting hydrogen-bonded S^- group greatly raises the microscopic pK_a of the imidazolium group.

Studies of the pH dependence of V_{max}/k_{cat} (Eq. 9-57) reveal a bell-shaped dependence on pH with pK_a values^{301,302} of ~ 4 and ~ 9 . However, the ion pair is formed at a pH below four with apparent pK_a values of 2.5, 2.9, and 3.3 for ficin, caricain, and papain, respectively.³⁰² These low values can be assigned principally to Cys 25 with only very small contributions from His 159 (see Eq. 6-75). A third pK_a , perhaps of a nearby carboxylate from Glu 50, affects the rate. For caricain the nearby Asp 158 (Fig. 12-15) has been implicated.^{303,304}

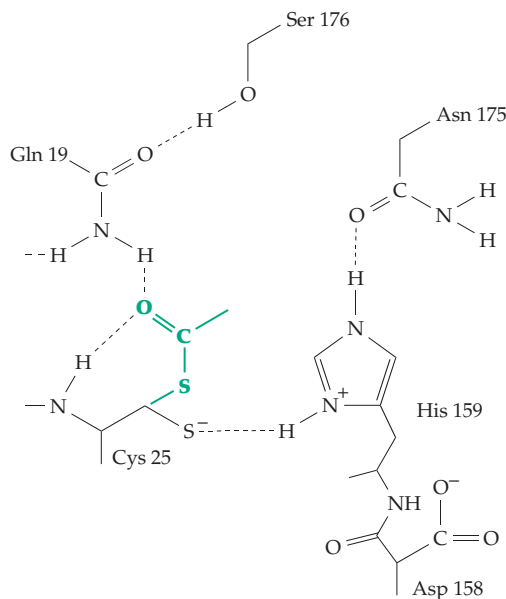


Figure 12-15 Schematic drawing of the active site of a cysteine protease of the papain family with a partial structure of an acyl-enzyme intermediate in green. The thiolate-imidazolium pair of Cys 25•His 159 lies deep in the substrate-binding cleft and bridges an interface between two major structural domains, just as the Ser•His pair does in serine proteases (Fig. 12-10). This may facilitate small conformational changes during the catalytic cycle. Asn 175 provides a polarizable acceptor for positive charge, helping to stabilize the preformed ion pair, and allows easy transfer of an imidazolium proton to the product of substrate cleavage. The peptide NH of Cys 25 and the side chain of Gln 19 form an oxyanion hole.

As shown in Fig. 12-15, the side chain of Asn 175 provides papain with a third member of a catalytic triad analogous to that of serine proteases.³⁰⁵ Glutamine 19, together with the peptide backbone NH of Cys 25, provides an oxyanion hole.^{306–308} Many studies, including structure determinations on bound aldehyde and other inhibitors, and observation by ¹³C NMR³⁰⁹ indicate that thiol proteases act by addition of the thiolate anion to the peptide carbonyl of the P₁ residue, just as in step *b* of Fig. 12-11 (see also Fig. 12-12) for serine proteases.^{308,310} However, alternative sequences of proton transfer have been suggested.³¹¹ A possible role for a strong hydrogen bond has also been proposed.³¹⁰ For a detailed discussion see Brocklehurst *et al.*³¹²

Most of the lysosomal proteases called **cathepsins** are small 20- to 40-kDa glycoproteins found in all animal tissues.³¹³ Most are cysteine proteases which function best and are most stable in the low pH reducing environment of lysosomes. They resemble papain in size, amino acid sequence, and active site structures. Papain is nonspecific but most cathepsins have definite substrate preferences. Cathepsin B is the most abundant. There are smaller amounts of related cathepsins H (an aminopeptidase)³¹⁴ and L³¹⁵ and still less of cathepsins C, K, and others. Cathepsin B is both an endopeptidase and an exopeptidase.³¹⁶ It acts on peptides with arginine at either P₁ or P₂ but also accepts bulky hydrophobic residues in P₁ and prefers tyrosine at P₃.³¹⁷ Cathepsin S is less stable at higher pH than other cathepsins and has a more limited tissue distribution, being especially active in the immune system.^{318,319}

Cathepsin K is especially abundant in the bone resorbing osteoclasts (Chapter 8). It is essential to normal bone structure and its absence is associated with the rare hereditary disease pycnodysostosis (pycno) which causes short stature, fragile bones, and skull and skeletal deformities.³²⁰ It may also play a role in the very common bone condition osteoporosis. Cathepsin C is also called **dipeptidyl peptidase**. It removes N-terminal dipeptides from many intracellular proteins activating many enzymes, including some other cathepsins.^{321,322} A **prohormone thiol protease** cleaves peptide chains between pairs of basic residues, e.g., in the brain peptide precursor proenkephalin (Chapter 30), and also on the N-terminal side of arginine residues.³²³ **Pyroglutamate aminopeptidase** removes pyroglutamyl (5-oxoprolyl) groups from amino termini of some peptides and proteins (see Fig. 2-4).³²⁴ Another cysteine protease cleaves **isopeptide** linkages such as those formed by transglutaminase or those involving ubiquitin (Box 10-C).³²⁵ Another cysteine protease present in animal tissues was recognized by its ability to hydrolyze the anticancer drug bleomycin. This **bleomycin hydrolase** is a hexamer with a central channel lined with papainlike active sites as in the proteasome structure (Box 7-A).³²⁶ The enzyme also binds to DNA.³²⁷ Unfortunately, cancer tissues often

contain high levels of the enzyme, whereas it is low in skin and lung tissues, which are damaged by bleomycin.

The Ca²⁺-dependent neutral proteases called **calpains** are found within the cells of higher animals. The 705-residue multidomain peptide chain of a chicken calpain contains a papain-like domain as well as a calmodulin-like domain.³²⁸ It presumably arose from fusion of the genes of these proteins. At least six calpains with similar properties are known.³²⁹ Some have a preference for myofibrillar proteins or neurofilaments.³³⁰ They presumably function in normal turnover of these proteins and may play a role in numerous calcium-activated cellular processes.^{331–332a}

A group of **cysteinyl aspartate-specific proteases (caspases)** play an essential role in programmed cell death (apoptosis).^{333–335} Recall that nematodes are cell-constant organisms. For maturation of *Caenorhabditis elegans* 131 programmed cell deaths must occur at specific stages of development. An essential gene for this process was identified and named **CED-3**. Deletion of this gene completely blocked the death of these cells. *CED-3* encodes a cysteine protease that is highly homologous to the mammalian **interleukin-1 β -converting enzyme (ICE or caspase 1)** which cleaves the 31-kDa **pro-interleukin-1 β** to form the active 175 kDa species of this cytokine (Chapter 30). At least ten caspases are known and many observations have confirmed their role in apoptosis (see also Chapter 32). In caspases 1 and 2 the side chains of Cys 285 and His 237 form the catalytic dyad and peptide NH groups of Cys 285 and Gly 238 form the oxyanion hole.³³³

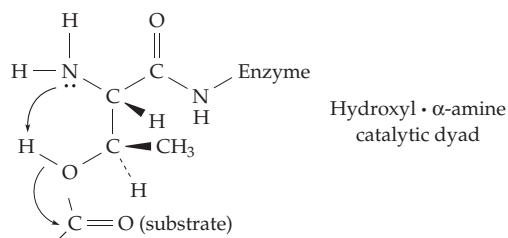
Parasites often use proteases in attacks on their hosts. The cysteine protease **cruzain** is secreted by the trypanosomes that cause Chagas' disease and is essential to their survival within the human body.^{336,337} Cruzain is consequently an attractive target for development of drugs for treating this major disease.

Among cysteine proteases of bacteria is a papain-like enzyme from *Clostridium histolyticum* with a specificity similar to that of trypsin.³³⁸ The anaerobic *Porphyromonas gingivalis*, which is implicated in periodontal disease, produces both arginine- and lysine-specific cysteine proteases designated **gingipains**.^{339,339a} Some virally encoded cysteine proteases, including one from the polio virus, have trypsin-like sequences with the serine of the catalytic triad replaced by cysteine.^{340,341} A human adenovirus protease also has a Cys•His•Glu triad but a totally different protein fold.³⁴²

Zymogens of cysteine proteases usually have a long terminal extension which is removed, sometimes by autoactivation. Propapain has a 107-residue extension.³⁴³ The 322-residue cathepsin B carries an unusually short 62-residue extension in its proenzyme form.^{315,343,344} In every case the N-terminal extension folds into a domain, one of whose functions is to block the active site cleft.

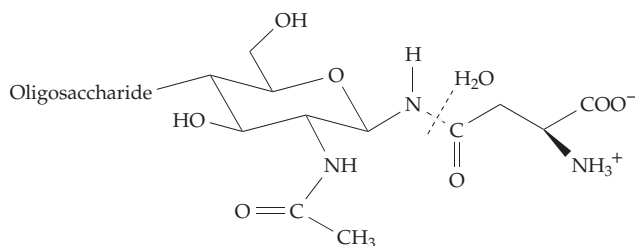
3. N-terminal Nucleophile Hydrolases and Related Enzymes

The most recently discovered group of proteases are the N-terminal threonine hydrolases of the **multi-catalytic protease complex** (MPC) of proteasomes. The enzymes are arranged in a regular array inside proteasomal compartments as shown in Box 7-A. The active site is a catalytic dyad formed from the amino group at the N terminus of the β subunits.^{345–346a}



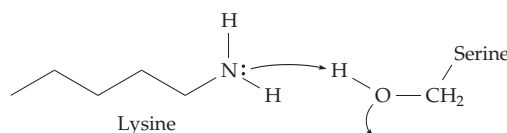
Proteasomes of *Thermoplasma* contain a single type of β subunit but eukaryotic proteasomes contain subunits with at least three distinct substrate preferences.^{347–349c} They all appear to use the same hydrolytic mechanism but in their substrate specificities they are chymotrypsin-like, peptidylglutamyl-peptide hydrolyzing, branched chain amino acid preferring, and small neutral amino acid preferring based on the P_1 amino acid residue. In the spleen some of the β subunits of the proteasomes appear to have been replaced by proteins encoded by the major histocompatibility complex of the immune system (Chapter 31).³⁴⁷ This may alter the properties of the proteasome to favor their function in antigen processing. Proteasomes are also ATP- and ubiquitin-dependent, as discussed in Section 6.

The enzyme glucosylasparaginase (aspartylglucosaminidase) is one of a group of other enzymes that use N-terminal thronyl groups as catalytic dyads.^{346,350–353} It removes N-linked glycosyl groups from asparagine side chains of proteolytically degraded proteins and, as indicated on the accompanying structural formula, releases free aspartate and a 1-amino-*N*-acetylglucosamine-containing oligosaccharide. The amino group is then released as NH_3 from the product by acid catalysis in the lysosome.³⁵⁰



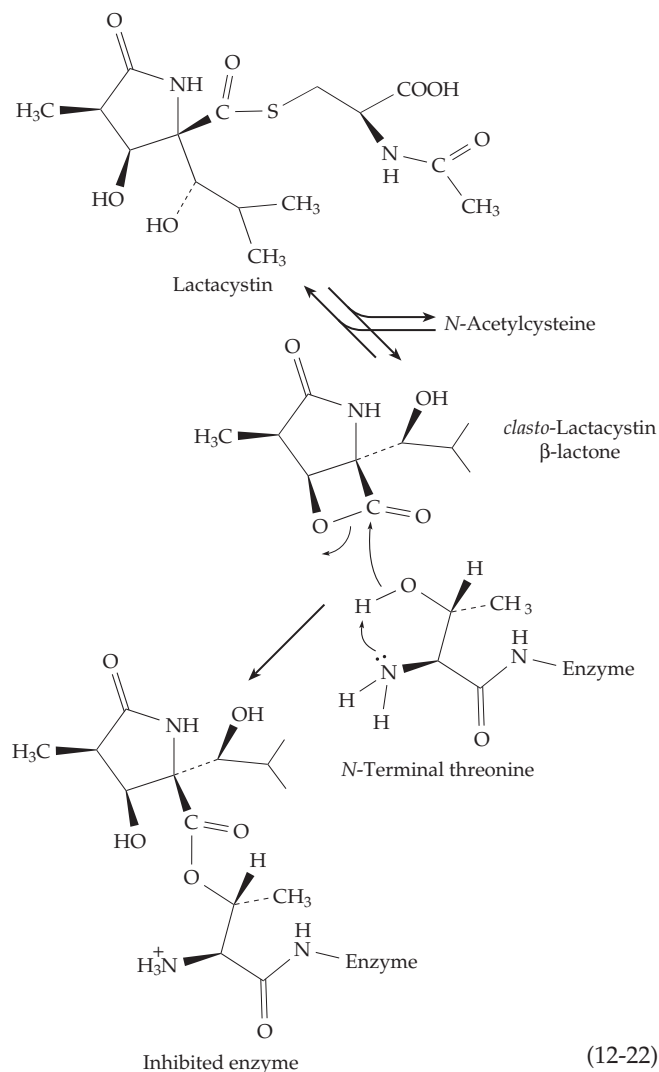
Glutamine PRPP amidotransferase (Fig. 25-15) and a penicillin acylase have similar active sites and overall

structures.³⁵⁴ The $-\text{NH}_2$ group is basic enough in the environment of the protein to remove the proton from the threonine $-\text{OH}$ group, activating it as indicated below. Several serine proteases use an $-\text{NH}_2$ group of a lysine to form a serine•lysine dyad.³⁴⁶

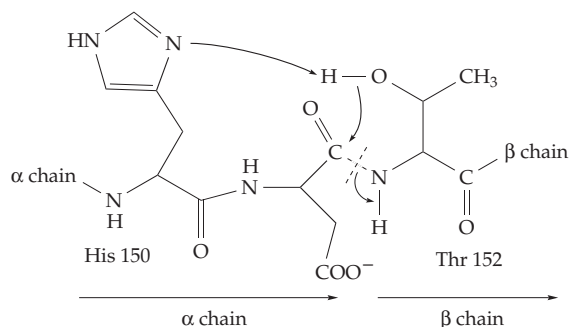


Among these are the well-known *E. coli* **leader peptidase**^{355,356} and other signal peptidases.³⁵⁷ These are integral membrane proteins that cleave N-terminal signal sequences from proteins incorporated into plasma membranes. Another enzyme of this class is the **lexA** repressor and protease discussed in Chapter 28.

A specific inhibitor of the major proteasomal activities is **lactacystin**, a compound formed by *Streptomyces*. Lactacystin is converted reversibly, by loss of *N*-acetyl-cysteine, into a β -lactone known as *clasto*-lactacystin. The N-terminal amino group attacks the reactive four-membered ring of the lactone (Eq. 12-22).^{358,359}



The active sites of the N-terminal nucleophile hydrolases are generated *autocatalytically*.^{360–362} A single peptide chain is cleaved to form α and β chains as in the following diagram. An activating nucleophile such as histidine removes a proton from an adjacent threonine –OH and the resulting alkoxide ion attacks the adjacent peptide linkage, presumably via a tetrahedral intermediate, to form an ester linkage. Compare this sequence with reactions in Box 12-A and Eq. 14-41. Hydrolysis generates the N-terminal threonine as indicated:

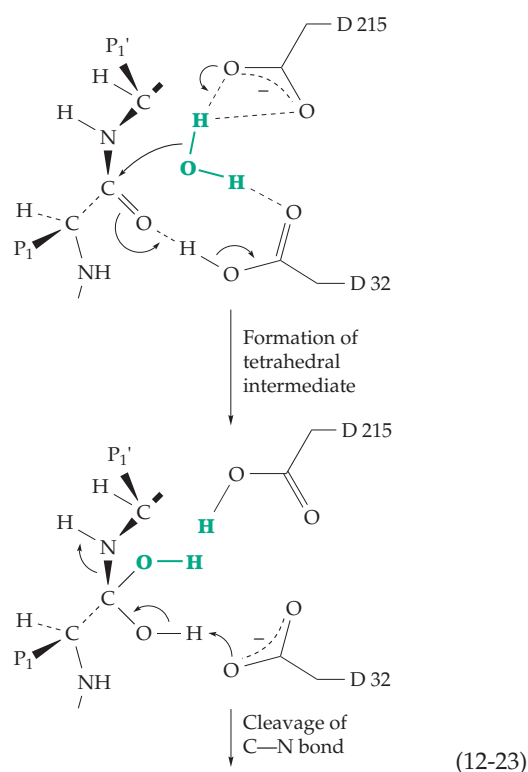


4. The Aspartic Proteases

A fourth large group of protein-hydrolyzing enzymes consists of **pepsin** of the stomach³⁶³ and related enzymes.³⁶⁴ Each of these ~320-residue proteins is folded into two domains which associate with a pseudotwofold axis of symmetry that passes through the active site.³⁶⁵ A second human gastric aspartic proteinase is **gastricsin**.³⁶⁵ The related **chymosin** (rennin),³⁶⁶ which is obtained from the fourth stomach of the calf, causes a rapid clotting of milk and is widely used in manufacture of cheese. Pepsin has a broad specificity but cleaves preferentially between pairs of hydrophobic residues, converting proteins into soluble fragments. It is unusual in being able to cleave X-Pro peptide bonds.³⁶⁷ Chymosin has a more restricted specificity, cutting the κ -casein of milk between a Phe–Met bond. This decreases the stability of the milk micelles, inducing clotting.³⁶⁶ The serum protein **renin** (distinct from rennin), the lysosomal **cathepsins D** and **E**,^{368–370} an **aspartyl aminopeptidase**,³⁷¹ and various fungal proteases are also closely related.^{372–375} Renin,^{376,377} which is synthesized largely in the kidneys, is involved in blood pressure regulation (Box 22-D). More distantly related aspartic proteases are encoded by retroviruses.

The pepsin family is most active in the low pH range 1–5. All of the enzymes contain two especially reactive aspartate carboxyl groups.³⁷⁸ One of them (Asp 215 in pepsin) reacts with site-directed diazonium compounds and the other (Asp 32) with site-directed epoxides.³⁷⁹ It is attractive to think that one of these carboxyl groups might be the nucleophile in a double displacement mechanism. The second carboxyl could then be the proton donor to the cleaving group.

The acyl-enzyme would be an acid anhydride. However, X-ray studies on pepsin^{363,380} and on related fungal enzymes such as **penicillopepsin**³⁸¹ and others^{373,375} suggested a different possibility: A water molecule, hydrogen bonded to one of the active site Asp carboxylates or bridging between them, becomes the nucleophile. A proton, held by the carboxylate pair, protonates the substrate carbonyl to facilitate nucleophilic attack.³⁸¹ This is illustrated in Eq. 12-23 but without detail. Several intermediate sequences of reaction steps are possible.³⁸² At the beginning of the sequence one of the two symmetrically placed carboxylates is protonated while the other is not. Also notice that although the active site is symmetric, the substrate is bound asymmetrically as determined by its hydrogen bonding into an extended binding site.



A characteristic feature of catalysis by the aspartic proteases is a tendency, with certain substrates, to catalyze transpeptidation reactions of the following type.



Here Phe* is an isotopically labeled residue. Although such reactions suggested the possibility of some kind of activated amino group on the tyrosine that is cut off in the initial cleavage of the unlabeled substrate, it is more likely that the released tyrosine stays in the active site,³⁸³ while the acetyl-Phe fragment exchanges with acetyl*-Phe.

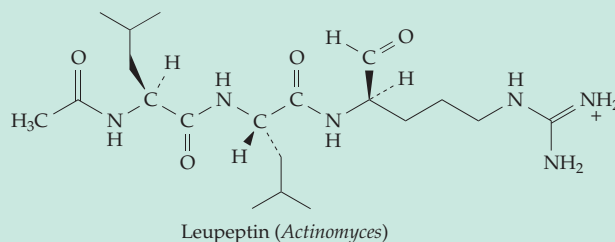
BOX 12-C SYNTHETIC PROTEASE INHIBITORS

One of the goals of synthetic medicinal chemistry is to design potent inhibitors of clinically important proteases. Elastase inhibitors may be useful for treatment of emphysema, pancreatitis, and arthritis,^{a,b} while inhibitors of the angiotensinogen-converting enzyme or of renin (Box 22-D) can help control blood pressure. Inhibition of thrombin, factor Xa, or other blood clotting factors (Fig. 12-17) may prevent blood clots and inhibition of the cytosolic tryptase may provide a new treatment for asthma. Inhibition of the cysteine protease cathepsin K may help combat osteoporosis and inhibition of cysteine proteases of corona viruses may fight the common cold. Cysteine proteases of schistosomes are also targets for protease inhibitors.^c

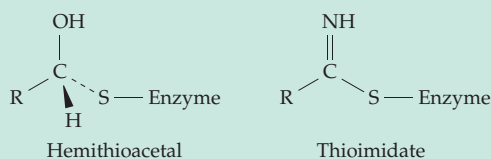
Many chemical approaches are used in designing inhibitors. Often, a naturally occurring inhibitor provides a starting point. The availability of high-resolution structures of the target enzymes and of various enzyme-inhibitor complexes assists in the rational design of tight-binding inhibitors. Use of combinatorial chemistry (Chapter 3)^d is another source of potential inhibitors. To be of practical use in medicine many criteria of stability, solubility, and low toxicity must be met. While most inhibitors are disappointing as drugs, their use in laboratory experimentation has clarified a great deal of biochemistry.

A straightforward approach is to hunt for short polypeptides that meet the specificity requirement of an enzyme but which, because of peculiarities of the sequence, are acted upon very slowly. Such a peptide may contain unusual or chemically modified amino acids. For example, the peptide **Thr-Pro-nVal-NMeLeu-Tyr-Thr** (nVal = norvaline; NMeLeu = *N*-methylleucine) is a very slow elastase substrate whose binding can be studied by X-ray diffraction and NMR spectroscopy.^e Thiol proteases are inhibited by **succinyl-Gln-Val-Val-Ala-Ala-*p*-nitroanilide**, which includes a sequence common to a number of naturally occurring peptide inhibitors called **cystatins**.^f They are found in various animal tissues where they inhibit cysteine proteases.

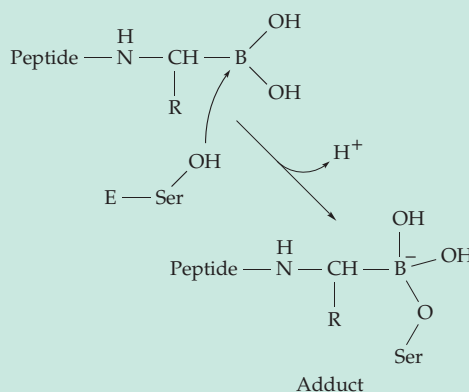
A group of inhibitors such as **leupeptin** have C-terminal aldehyde groups. Small oligopeptides with this structure and with appropriate specificity-determining side chains form tetrahedral hemiacetals, which may mimic transition state structures, at the active sites of the target enzymes.^{g,h}



Leupeptin is a slow, tight-binding inhibitor of trypsin. Some peptide aldehydes are potent, reversible inhibitors of cysteine proteases forming hemithioacetals with the active site cysteine.ⁱ Similarly, peptide nitriles form thioimide adducts.^h



The peptide **boronic acids** form adducts with the active site serine of serine proteases.^{j-1}

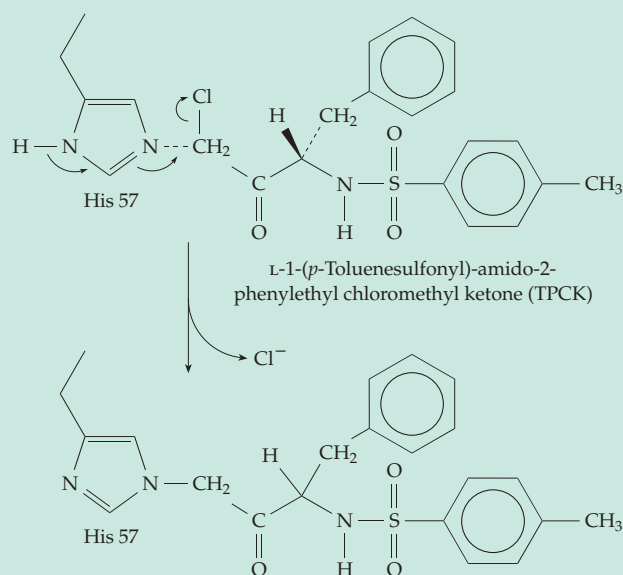


However, both X-ray crystallography^m and ¹¹B NMRⁿ have shown that imidazole of the catalytic triad may also add to the boronic acid and that a tetrahedral adduct with both the serine oxygen and histidine nitrogen covalently bonded to boron can also be formed.^m Thus, in reversibly inhibited enzymes a mixture of different chemical species may exist. Inhibitors can be designed to bind more tightly by providing additional bonding opportunities. For example, a suitably placed cyano group on a phenylalanine ring in the P₁ position of thrombin

BOX 12-C (continued)

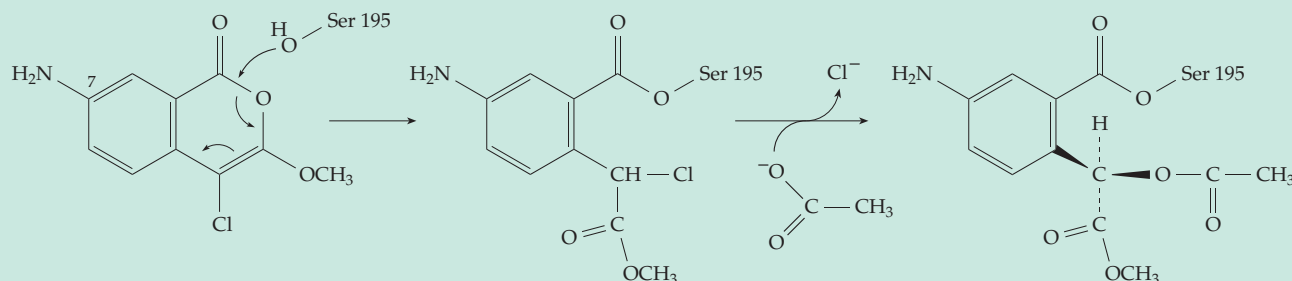
or other serine proteases can form a hydrogen bond to the peptide NH of Gly 219 (see Fig. 12-9).^o

Numerous synthetic active-site directed or enzyme-activated irreversible inhibitors have been designed.^p For example, the following chloroketone inhibits chymotrypsin but does not act on trypsin. The corresponding structure with a lysine side chain (TLCK) inhibits only trypsin. These **affinity labeling compounds** initially bind noncovalently at the active site.



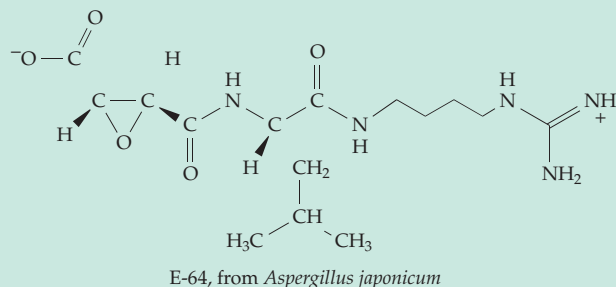
However, α -chloroketones are powerful alkylating agents and the bound inhibitor attacks His 57 of the catalytic triad system. The reaction is probably more complex than is indicated in the foregoing equation and may involve an epoxy ether intermediate.^q Many other peptide chloromethyl ketone inhibitors have been devised.^{qr}

Iscoumarins inactivate many serine proteases. For example, 7-amino-4-chloro-3-methoxyisocoumarin acylates serine 195 of elastases as follows.^s

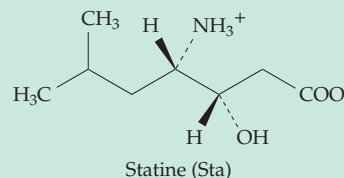


Many other enzyme-activated inhibitors are being developed.^{c,d,t}

Epoxy groups, such as that of E-64, a compound isolated from the culture medium of a species of *Aspergillus*, react irreversibly with the active site thiolate group of cysteine proteases.^{i,u,v} Related epoxides may become useful medications against abnormal cathepsin levels.



All of the aspartic proteases are inhibited by **pepstatin**, a peptide produced by some species of *Actinomyces* and which contains two residues of the unusual amino acid **statine** (**sta**).^w Pepstatin has the sequence Isovaleryl-L-Val-L-Val-Sta-L-Ala-Sta.



The statine residue mimics the noncovalently bonded tetrahedral intermediate, permitting formation of a very tight complex. Pepstatin is a poor inhibitor of human renin but its existence has inspired the synthesis of numerous related compounds, some of which are effective renin inhibitors.^{x,y} Some of these inhibitors use the human angiotensinogen sequence with a secondary alcohol group mimicking the tetrahedral intermediate.^{x,z}

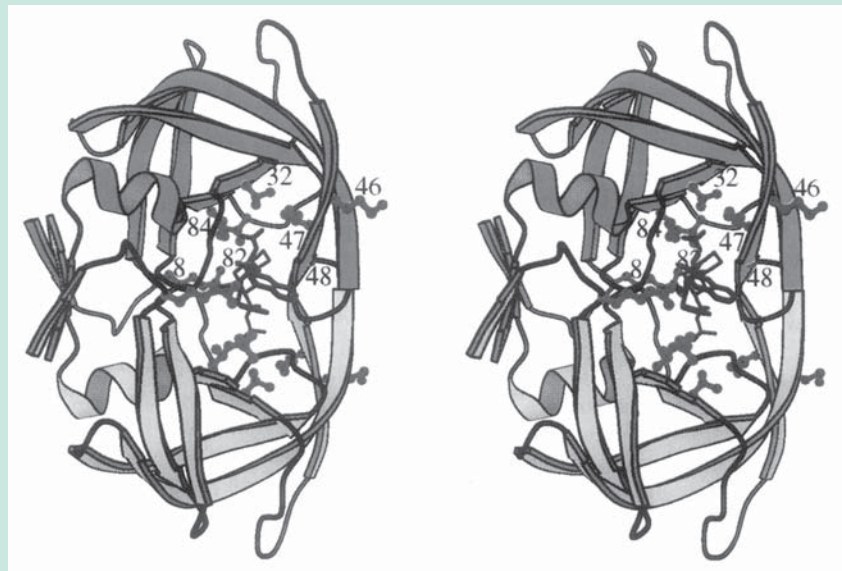
BOX 12-C SYNTHETIC PROTEASE INHIBITORS (continued)

These aspartic protease inhibitors are also “lead compounds” in the development of inhibitors of HIV protease.^{aa–cc} As in statine-based inhibitors, the site of occupancy by the catalytic H₂O (green in Eq. 12-23) is occupied in the inhibitor by something that mimics a tetrahedral intermediate with –CHOH–, –PO₂H–, etc.^{bb} Tremendous efforts are being expended in designing these inhibitors and considerable

success has been achieved. However, the rapid development of mutant strains of the virus with drug-resistant proteases presents a major challenge.^{cc,dd}

Mercaptans of suitable structure bind tightly to Zn²⁺ in the active sites of metalloproteases. For example, **captopril**^{ee} is a tight-binding competitive inhibitor of the angiotensinogen-converting enzyme that is effective in lowering blood pressure and the first of many related inhibitors.^{ff}

As mentioned in the text, a variety of inhibitors that mimic the geometry of a tetrahedral intermediate or transition state are also potent inhibitors of metalloproteases.



Stereoscopic ribbon structure of the HIV-1 protease with the synthetic inhibitor Sequinivir^{cc} bound in the active site. One of the two identical subunits (top) is shaded darker than the second (bottom). When mutated, the amino acid side chains shown in ball-and-stick form with residue numbers shown for the top subunit led to drug-resistant viruses. Courtesy of Alex Wlodawer, National Cancer Institute.^{cc}

- ^a Powers, J. C., Oleksyszyn, J., Narasimhan, S. L., Kam, C.-M., Radhakrishnan, R., and Meyer, E. F., Jr. (1990) *Biochemistry* **29**, 3108–3118
- ^b Mattos, C., Giammona, D. A., Petsko, G. A., and Ringe, D. (1995) *Biochemistry* **34**, 3193–3203
- ^c Seife, C. (1997) *Science* **277**, 1602–1603
- ^d Peisach, E., Casebier, D., Gallion, S. L., Furth, P., Petsko, G. A., Hogan, J. C., Jr., and Ringe, D. (1995) *Science* **269**, 66–69
- ^e Meyer, E. F., Jr., Clore, G. M., Gronenborn, A. M., and Hansen, H. A. S. (1988) *Biochemistry* **27**, 725–730
- ^f Yamamoto, A., Tomoo, K., Doi, M., Ohishi, H., Inoue, M., Ishida, T., Yamamoto, D., Tsuboi, S., Okamoto, H., and Okada, Y. (1992) *Biochemistry* **31**, 11305–11309
- ^g Ortiz, C., Tellier, C., Williams, H., Stolowich, N. J., and Scott, A. I. (1991) *Biochemistry* **30**, 10026–10034
- ^h Dufour, E., Storer, A. C., and Ménard, R. (1995) *Biochemistry* **34**, 9136–9143
- ⁱ Mehdi, S. (1991) *Trends Biochem. Sci.* **16**, 150–153
- ^j Bone, R., Shenvi, A. B., Kettner, C. A., and Agard, D. A. (1987) *Biochemistry* **26**, 7609–7614
- ^k Takahashi, L. H., Radhakrishnan, R., Rosenfield, R. E., Jr., and Meyer, E. F., Jr. (1989) *Biochemistry* **28**, 7610–7617
- ^l Nienaber, V. L., Mersinger, L. J., and Kettner, C. A. (1996) *Biochemistry* **35**, 9690–9699
- ^m Stoll, V. S., Eger, B. T., Hynes, R. C., Martichonok, V., Jones, J. B., and Pai, E. F. (1998) *Biochemistry* **37**, 451–462
- ⁿ Zhong, S., Jordan, F., Kettner, C., and Polgar, L. (1991) *J. Am. Chem. Soc.* **113**, 9429–9435
- ^o Lee, S.-L., Alexander, R. S., Smallwood, A., Triebel, R., Mersinger, L., Weber, P. C., and Kettner, C. (1997) *Biochemistry* **36**, 13180–13186
- ^p Bode, W., Meyer, E., Jr., and Powers, J. C. (1989) *Biochemistry* **28**, 1951–1963
- ^q Kreutter, K., Steinmetz, A. C. U., Liang, T.-C., Prorok, M., Abeles, R. H., and Ringe, D. (1994) *Biochemistry* **33**, 13792–13800

- ^r Wolf, W. M., Bajorath, J., Müller, A., Raghunathan, S., Singh, T. P., Hinrichs, W., and Saenger, W. (1991) *J. Biol. Chem.* **266**, 17695–17699
- ^s Meyer, E. F., Jr., Presta, L. G., and Radhakrishnan, R. (1985) *J. Am. Chem. Soc.* **107**, 4091–4094
- ^t Groutas, W. C., Kuang, R., Venkataraman, R., Epp, J. B., Ruan, S., and Prakash, O. (1997) *Biochemistry* **36**, 4739–4750
- ^u Yamamoto, D., Matsumoto, K., Ohishi, H., Ishida, T., Inoue, M., Kitamura, K., and Mizuno, H. (1991) *J. Biol. Chem.* **266**, 14771–14777
- ^v Varughese, K. I., Su, Y., Cromwell, D., Hasnain, S., and Xuong, N.-h. (1992) *Biochemistry* **31**, 5172–5176
- ^w Gómez, J., and Freire, E. (1995) *J. Mol. Biol.* **252**, 337–350
- ^x Cooper, J., Quail, W., Frazao, C., Foundling, S. I., Blundell, T. L., Humblet, C., Lunney, E. A., Lowther, W. T., and Dunn, B. M. (1992) *Biochemistry* **31**, 8142–8150
- ^y Tong, L., Pav, S., Lamarre, D., Pilote, L., LaPlante, S., Anderson, P. C., and Jung, G. (1995) *J. Mol. Biol.* **250**, 211–222
- ^z Cooper, J. B., Foundling, S. I., Blundell, T. L., Boger, J., Jupp, R. A., and Kay, J. (1989) *Biochemistry* **28**, 8596–8603
- ^{aa} Hui, K. Y., Manetta, J. V., Gygi, T., Bowdon, B. J., Keith, K. A., Shannon, W. M., and Lai, M.-H. T. (1991) *FASEB J.* **5**, 2606–2610
- ^{bb} Abdel-Meguid, S. S., Zhao, B., Murthy, K. H. M., Winborne, E., Choi, J.-K., Desjarlais, R. L., Minnich, M. D., Culp, J. S., Debouck, C., Tomaszek, T. A., Meek, T. D., and Dreyer, G. B. (1993) *Biochemistry* **32**, 7972–7980
- ^{cc} Ridky, T., and Leis, J. (1995) *J. Biol. Chem.* **270**, 29621–29623
- ^{dd} Chen, Z., Li, Y., Schock, H. B., Hall, D., Chen, E., and Kuo, L. C. (1995) *J. Biol. Chem.* **270**, 21433–21436
- ^{ee} Vidt, D. G., Bravo, E. L., and Fouad, F. M. (1982) *N. Engl. J. Med.* **306**, 214–219
- ^{ff} Gros, C., Noël, N., Souque, A., Schwartz, J.-C., Danvy, D., Plaquevent, J.-C., Duhamel, L., Duhamel, P., Lecomte, J.-M., and Bralet, J. (1991) *Proc. Natl. Acad. Sci. U.S.A.* **88**, 4210–4214

Pepsin is secreted as the inactive pepsinogen, which is activated by H^+ ions at a pH below 5. Determination of its crystal structure revealed that in the proenzyme the N-terminal 44-residue peptide segment lies across the active site, blocking it.³⁸⁴ At low pH the salt bridges that stabilize the proenzyme are disrupted and the active site is opened up to substrates.

While the cellular aspartate proteases are over 300 residues in length, the retroviral proteases are less than one-half this size.^{385–388} That of the human HIV-I protease contains only 99 residues. These enzymes are cut from a polypeptide (encoded by the viral *gag* and *pol* genes (Fig. 28-26). The *pol* gene encodes four other essential enzymes as well, and these are cut apart at eight different sites by action of the protease.³⁸⁸ Despite its small size, it displays sequence homologies with the larger cellular aspartate proteases and has a related three-dimensional structure. Each chain has only one active site aspartate; the functional enzymes are dimers and the catalytic mechanism appears to be similar to that of pepsin.^{388–392} A great deal of effort is being devoted to designing synthetic HIV protease inhibitors, which are used in the treatment of AIDS (Box 12-C).

5. Metalloproteases

The pancreatic **carboxypeptidases** are characterized by the presence of one firmly bound **zinc ion** in each molecule. The Zn^{2+} can be removed and can be replaced by other metal ions such as Co^{2+} and Ni^{2+} , in some cases with reconstitution of catalytic activity. The human pancreas synthesizes and secretes proenzymes for two forms of carboxypeptidase A, with a preference for C-terminal hydrophobic residues, as well as carboxypeptidases B, which prefer C-terminal basic residues. Additional A and B forms³⁹³ as well as more specialized nondigestive carboxypeptidases are also known. In eukaryotic cells the latter participate in processing of proteins. Following removal of N-terminal signal sequences, processing often continues with removal of basic residues from the C termini by **carboxypeptidases N, H** (also called E or enkephalin convertase),³⁹⁴ and **M**, all of which are metalloenzymes.^{395,396} Carboxypeptidase N removes C-terminal arginines from many biologically important peptides. It also circulates in the plasma and protects the body by inactivating such potent inflammatory peptides as the kinins and anaphylotoxins.³⁹⁷ Carboxypeptidase H is located in secretory granules, while carboxypeptidase M is membrane associated.³⁹⁶ Dipeptidyl carboxypeptidase (**angiotensin-converting enzyme**) removes the C-terminal Pro-Phe dipeptide from angiotensinogen to generate the potent pressor agent angiotensin I (Box 22-D) and cleaves dipeptides from many other substrates as well.³⁹⁸ A **D-alanyl-D-alanyl carboxypeptidase** cleaves D-alanine from the ends of

cell wall peptides (Chapter 20).³⁹⁹

Another well-known zinc-containing enzyme is **thermolysin**, a nonspecific *endopeptidase* widely used in laboratories. Produced by a thermophilic bacterium, it is unusually resistant to heat. It contains four bound calcium ions in addition to the active site zinc.^{400–402} The active site structure resembles that of pancreatic carboxypeptidase A and the two enzymes appear to act by similar mechanisms.^{401,403,404} The mammalian zinc endopeptidase **neprilysin**, an integral membrane protein involved in inactivation of enkephalins and other signaling peptides, also resembles thermolysin.⁴⁰⁵ A related neutral endopeptidase is the product of a gene called *PEX* (phosphate-regulating gene with homologies to endopeptidases on the X chromosome). The absence of the *PEX* gene product causes **X-linked hypophosphatemic rickets** which leads to excessive loss of phosphate from the body with defective mineralization of bone.⁴⁰⁶ The **mitochondrial processing peptidase**, which removes signal sequences from the N termini of mitochondrial proteins, also contains Zn^{2+} at its active site.^{407,408} However, it is an $\alpha\beta$ heterodimer and a member of an additional family of enzymes, one of which includes human insulin-degrading enzyme.

In both carboxypeptidase A and thermolysin the active site Zn^{2+} is chelated by two imidazole groups and a glutamate side chain (Fig. 12-16). In carboxypeptidase A, Arg 145, Tyr 248, and perhaps Arg 127 form hydrogen bonds to the substrate. A water molecule is also bound to the Zn^{2+} ion. The presence of the positively charged side chain of Arg 145 and of a hydrophobic pocket accounts for the preference of the enzyme for C-terminal amino acids with bulky, nonpolar side chains. The Zn^{2+} in thermolysin is also bound to two imidazole groups and that in D-alanyl-D-alanyl carboxypeptidase to three.

The presence of a zinc ion in the metalloproteases immediately suggested a role in catalysis. Unlike protons, which have a weak affinity for the oxygen of an amide carbonyl group, a metal ion can form a strong complex. If held in position by other ligands from the protein, a properly placed zinc ion might be expected to greatly enhance the electrophilic nature of the carbon atom of the $C=O$ group. However, it has been difficult to establish the exact mechanism of action.⁴¹⁰ Carboxypeptidase A cleaves both peptides and ester substrates. For peptides, K_m is the same for various metals while k_{cat} changes, but the converse is true for ester substrates.⁴¹¹ From its position Glu 270 (Glu 143 in thermolysin) seems to be the logical nucleophile to attack the substrate to form an acyl-enzyme intermediate, an anhydride. Using the following specific *ester* substrate, Makinen *et al.* showed that at very low temperatures of -40° to $-60^\circ C$, in solvents such as 50:50 ethylene glycol:water, an acyl-enzyme intermediate can be detected spectroscopically. It could even be separated from free enzyme⁴¹² by gel

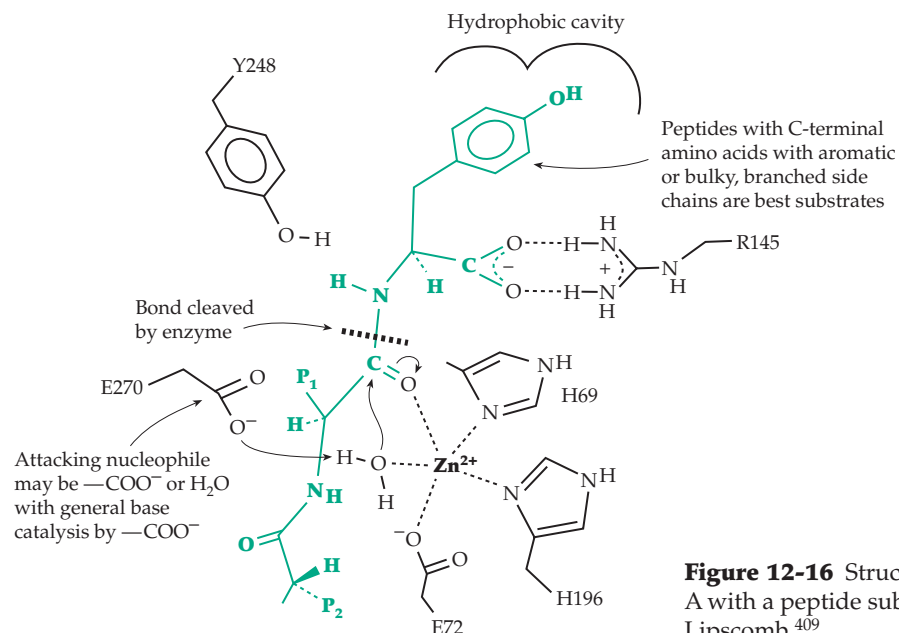
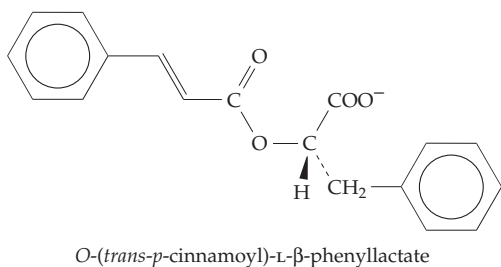
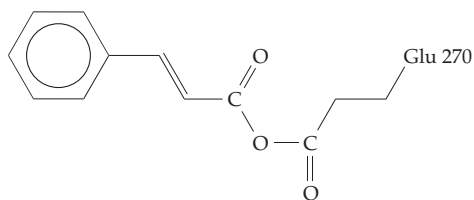


Figure 12-16 Structure of the active site of carboxypeptidase A with a peptide substrate present. See Christianson and Lipscomb.⁴⁰⁹

filtration at -60°C and its cyanoborohydride reduction product was characterized.⁴¹³



The intermediate appears to be the acid anhydride formed by Glu-270.^{414,415} The conformation of the intermediate was deduced by ENDOR spectroscopy and its formation and reaction interpreted according to stereoelectronic principles.⁴¹⁶



If the anhydride mechanism is correct, the water molecule bound to the Zn^{2+} probably provides an HO^- ion necessary for the hydrolysis of the intermediate anhydride.⁴¹⁷

Although these results seem convincing there are objections.⁴¹⁰ The mechanism deduced for hydrolysis of an ester may be different than that for a peptide.⁴¹⁸

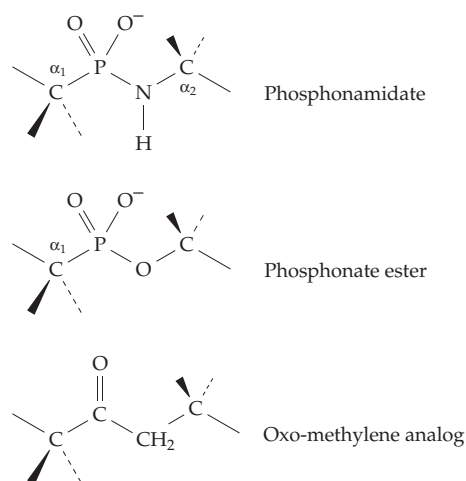
Furthermore, many observations favor an alternative mechanism. A hydroxide ion derived from a water molecule bound to the zinc ion may be the initial attacking nucleophile.^{404,419–422} Both X-ray crystallographic studies and EPR investigations^{403,414} show that the zinc in carboxypeptidases can coordinate at least five surrounding atoms. As shown in Fig. 12-16, the Zn^{2+} could hold the attacking water molecule and, at the same time, provide the positive charge for stabilizing the resulting tetrahedral intermediate. Glu 270 (or in thermolysin His 231) acts as a base to deprotonate the bound H_2O . Mock and Stanford argue that the H_2O is probably displaced when the peptide carbonyl binds and that the H_2O is then deprotonated and adds to the polarized carbonyl.⁴⁰⁴

The pH dependence of the action of carboxypeptidase A is determined by pK_a values of ~ 6 and ~ 9.5 for the free enzyme⁴²² and of ~ 6.4 and ~ 9 for k_{cat} .⁴²⁰ For thermolysin the values for k_{cat} are ~ 5 and ~ 8 .⁴⁰⁴ Assignment of pK_a values has been controversial. They may all be composites of two or more microscopic constants but probably, at least for carboxypeptidase, the low pK_a is largely that of Glu 270 while the high one represents largely the dissociation of a proton from the zinc-bound H_2O .

Earlier studies of carboxypeptidase had indicated that Tyr 248 moves its position dramatically upon substrate binding, and it was suggested that its phenolic $-\text{OH}$ group protonates the leaving group in the acylation step. However, a mutant in which Tyr 248 was replaced by phenylalanine still functions well.⁴²³

Phosphoramidates,^{424,425} phosphonate esters, and oxo-methylene substrate analogs, which presumably mimic the geometry of the tetrahedral intermediate or transition state of the intermediate or transition state

of the catalytic cycle, are effective inhibitors of zinc proteases. X-ray studies of complexes of such inhibitors with both thermolysin and carboxypeptidase A support the suggestion that the Zn^{2+} binds the carbonyl oxygen of the amide bond that is to be cleaved in a

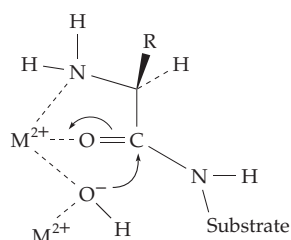


substrate and that a glutamate side chain could activate a Zn^{2+} -bound water to form an attacking nucleophile. A similar conclusion was reached from the structure of a thermolysin-product complex.⁴²⁶

Another large group of zinc proteases are the **matrix metalloproteases** which act on proteins of the extracellular matrix, such as collagen, proteoglycans, and fibronectin.^{427–429c} These enzymes have been classified as **collagenases**, **gelatinases** (which act on denatured collagen), **stromelysins** (which are activated in response to inflammatory stimuli),^{430–432} and a membrane type. The group also includes the digestive enzyme **astacin**, from the crayfish,⁴³³ and metalloproteases from snake venoms.^{429,434} The matrix metalloproteases are essential to the remodeling of the extracellular matrix that occurs during wound healing, tissue growth, differentiation, and cell death. An example of tissue remodeling is the development of dental enamel (Box 8-G). The proteinaceous matrix formed initially must be digested away, perhaps by the metalloprotease **enamelysin**, and replaced by the dense mineral of enamel.⁴³⁵ Excessive secretion of collagenase by fibroblasts is observed in **rheumatoid arthritis** and other inflammatory conditions.

At least 64 different matrix metalloproteins are known.⁴²⁷ Each enzyme consists of three domains. An 80- to 90-residue N-terminal propeptide domain contains a cysteine whose $-\text{S}^-$ group binds to the active site zinc, screening it from potential substrates. The central catalytic domain is followed by a hinge region and a C-terminal domain that resembles the serum iron binding and transporting **hemopexin**.^{427,436} The mechanism of action is probably similar to that of thermolysin.⁴³⁰

Many aminopeptidases are metalloenzymes.⁴³⁷ Most studied is the cytosolic **leucine aminopeptidase** which acts rapidly on N-terminal leucine and removes other amino acids more slowly. Each of the subunits of the hexameric enzyme contains *two* divalent metal ions, one of which must be Zn^{2+} or Co^{2+} .^{438,439} A methionine aminopeptidase from *E. coli* contains two Co^{2+} ions^{440,441} and a proline-specific aminopeptidase from the same bacterium two Mn^{2+} .⁴⁴² In all of these enzymes the metal ions are present as dimetal pairs similar to those observed in phosphatases and discussed in Section D.4 and to the Fe-Fe pairs of hemerythrin and other diiron proteins (Fig. 16-20). A hydroxide ion that bridges the metal ions may serve as the nucleophile in the aminopeptidases.⁴³⁸ A bound bicarbonate ion may assist.^{438a}



A metalloenzyme **peptide deformylase** removes the formyl groups from the N termini of bacterial proteins. Although the active site is similar to that of thermolysin,⁴⁴³ the Zn^{2+} form of peptide deformylase is unstable. Both Ni^{2+} and Fe^{2+} form active, stable enzymes.^{444,445}

6. ATP-Dependent Proteases

Much of metabolism is driven by the Gibbs energy of hydrolysis of ATP so it shouldn't be surprising that ATP is sometimes rather directly involved in hydrolytic degradation of polypeptide chains. Much of protein processing occurs in the endoplasmic reticulum,⁴⁴⁶ which also assists in sorting unneeded and defective proteins for degradation in the ubiquitin-proteasome system. Proteasomal degradation occurs in the cytosol, in the nucleus, and along the ER. However, the predominant location is the *centrosome*.^{446a}

Polyubiquitination of proteins requires ATP (Box 10-C) and additional ATP is utilized in the proteasomes (Box 7-A) during the selection of polyubiquitinated proteins for hydrolysis.^{447,448} With 28 subunits the 26S proteasome is complex and not fully understood.⁴⁴⁸ The ATP-hydrolyzing subunits appear to all be in the cap regions. Is the ATP used to open and close the entry pores? To induce conformational changes in all subunits as part of a catalytic cycle? Or to unfold folded proteins to help them enter the proteasomes?⁴⁴⁹

Some answers may be obtained from smaller bacterial, mitochondrial, and chloroplast ATP-dependent proteases. Cells of *E. coli* contain at least nine proteases, which have been named after the musical scale as Do, Re, Mi, Fa, So, La, Ti, Di, and Ci.^{450,451} Most are serine proteases but two, Ci and Pi, are metalloproteins. Protease **La** (**Lon protease**, encoded by gene *lon*) has attracted particular attention because the hydrolysis of two molecules of ATP occurs synchronously with cleavage of a peptide linkage in the protein chain.⁴⁵² This enzyme, as well as protease **Ti** (more often called **Clp**, for caseinolytic protease),^{451,453,454} is ATP-dependent.⁴⁵¹

The Lon protease of *E. coli* is a large 88-kDa serine protease with the catalytic domain, containing active site Ser 679, in the C-terminal half. The N-terminal portion contains two ATP-binding motifs and a linker region.^{455,456} A homologous protein known either as Lon or as PIM1 is present in mitochondria.^{457,458} The *E. coli* proteases Clp (Ti) include ClpAP and ClpXP, which are heterodimers, each containing the catalytic subunit ClpP and either ATPase ClpA or ClpX. The active enzyme may be designated **ClpAP**.^{459–461} A homolog of ClpP has been found in human mitochondria.^{461a} The ATPase Clp forms seven-subunit rings resembling the rings of proteasomes (Box 7-A), while the catalytic subunit ClpA forms six-membered rings.^{451,454} Despite the mismatch in symmetry and the fact that they are serine proteases, these enzymes appear similar to proteasomes. However, each catalytic subunit has an ATPase neighbor. Why is it needed? The related **ClpB** is both an ATPase and a chaperonin. It is essential for survival of *E. coli* at high temperatures.^{461b}

Another ATP-dependent protease identified among heat shock proteins of *E. coli* is known as **Hs1V–Hs1U** or (ClpQ–ClpY). It has a threonine protease active site and is even more closely reminiscent of eukaryotic proteasomes.^{462–463a} Also active in *E. coli* is another ring-like protease, a membrane-bound zinc endopeptidase **FtsH** (or HflB).^{463b} Similar eukaryotic proteases also exist.⁴⁶⁰

7. The Many Functions of Proteases

Many of the enzymes considered in the preceding sections function within the *digestive* tract. Others function in the *processing* of newly formed peptide chains, while others act in the *intracellular degradation* of proteins. Yet others are secreted from cells and function in the external surroundings. Both processing and intracellular degradation can be viewed as parts of *biosynthetic loops* (Chapter 17) that synthesize mature proteins, and then degrade them when they have served their function or have become damaged. The pathways involved are varied and complex, a natural result of thousands of enzymes and other compounds

acting on accessible and chemically appropriate parts of the proteins. Evolutionary selection has evidently led to the particular set of pathways that we observe for any organisms.

Because they must often cleave large **polypeptides**, many viruses encode processing proteases.^{464–466b} For example, the entire RNA genome of the poliovirus encodes a large polypeptide which is cut by two virally encoded chymotrypsin-like cysteine proteases within Tyr-Gly and Gln-Gly sequences;^{341,465} Asn-Ser sequences are also cut, apparently autocatalytically. As we have already seen, retroviruses encode their own aspartic protease. Most cellular and secreted proteins of bacteria or eukaryotes also undergo processing. This ranges from removal of *N*-formyl groups to cutting off signal peptides, addition or formation of prosthetic groups, internal cleavages, and modifications at the C termini. A variety of peptidases are required.

Degradation of proteins, which converts them back to amino acids as well as other products, takes place in part in the cytosol via the ubiquitin-protease system. Proteolysis also occurs in ER and external cell spaces through the action of membrane-bound and secreted proteases. Loosely folded proteins, which can arise in various ways, are subject to rapid degradation. For example, synthesis of a polypeptide chain on a ribosome may be accidentally disrupted with formation of a protein with a shorted chain. Mature proteins may become damaged. For example, certain histidine residues are readily attacked by oxidizing reagents. Oxidative damage may mark proteins for rapid degradation.⁴⁵⁰ In *E. coli*, proteases So and Re attack oxidized glutamine synthetase much more rapidly than the intact native enzyme.^{450,467} In mammalian cells the proteasomes degrade oxidized as well as poorly folded proteins.⁴⁶⁸ However, proteasomal digestion is not complete and peptide fragments may be secreted. Peptide fragments of proteins are also formed in the ER and may be secreted. Thus, a variety of small peptide hormones and other biologically active peptides are both generated and inactivated in or around cells.^{469–471} Some receptors are activated by proteases.^{471a}

Circulating proenzymes of the blood clotting factors, of the complement system (Chapter 31), represent a specialized group of secreted **signaling proteins** that are able to initiate important defensive cascades. Proteases also act more directly in defense systems of the body. For example, serine proteases cause lysis of the target cells of cytotoxic T lymphocytes⁴⁷² (Chapter 31) and activated neutrophils⁴⁷³ (Chapter 18). At the same time, pathogenic bacteria often secrete proteases that assist in attack on their hosts⁴⁷⁴ and schistosomes secrete an elastase that helps them penetrate skin and invade their hosts.⁴⁷⁵

8. Protease Inhibitors of Animals and Plants

Premature conversion of proenzymes such as trypsinogen into active proteases in the pancreas would be disastrous. To prevent this, the pancreas also produces inhibitors. The complex cascades that control blood clotting would also be unstable were it not for the presence in blood of numerous inhibitory proteins. Indeed, inhibitors of proteases are found everywhere in animals, plants, and microorganisms.^{476–478} They are usually proteins but small antibiotic protease inhibitors are also produced by microorganisms. Protein inhibitors are usually specific for a given type of enzyme: The **serpins** inhibit serine proteases, and the **cystatins** inhibit cysteine proteases.^{479–483} Other inhibitors block the action of metalloproteases^{484–486} and aspartic proteases. Inhibitors help not only host organisms but also parasites. For example, *Ascaris suum*, a very large nematode that is thought to infect 1/4 of the human population of the earth and nearly all of the pigs, secretes a pepsin inhibitor.⁴⁸⁷ However, the large 720-kDa serum **α -macroglobulins** (Box 12-D) inhibit all classes of proteases.

About 20 families of protein inhibitors of proteases have been described.⁴⁸⁸ The egg white **ovomucoids** comprise one family. Turkey ovomucoid is a three-domain protein whose 56-residue third domain is a potent inhibitor of most serine proteases.^{488,489} The 58-residue **pancreatic trypsin inhibitor**⁴⁹⁰ is a member of another family of small proteins. A 36-residue insect (locust) protease inhibitor is even smaller.⁴⁹¹

Inhibitors that block the action of trypsin and other proteases are found in many plants. The inhibitor activity is usually highest in seeds and tubers, but synthesis of inhibitors can be induced in other parts of plants e.g. tomato and potato by wounding.⁴⁹² See also Chapter 31. Legumes contain small 60- to 76-residue inhibitors (Bowman–Birk inhibitors) each containing seven disulfide linkages, and they are relatively stable toward cooking and toward acid denaturation in the stomach.^{493,494} Although they interfere with protein digestion they seem to have an anticarcinogen effect.⁴⁹³ Soybeans also contain a larger 181-residue trypsin inhibitor (Kunitz type).⁴⁹⁵

Several inhibitor–protease complexes have been crystallized and details of their interactions are known. For example, the pancreatic trypsin inhibitor binds at the active site of trypsin with $K_f > 10^{13} \text{ M}^{-1}$ at neutral pH.⁴⁹⁶ The two molecules fit snugly together,^{490,497} the inhibitor being bound as if it were a peptide substrate with one edge of the inhibitor molecule forming an antiparallel β structure with a peptide chain in the enzyme. Lysine 15, which forms part of this β structure, enters the specific P_1 binding site for a basic amino acid in a substrate. Thus, the protease inhibitor is a modified substrate which may actually undergo attack at the active site. However, the fit between the two

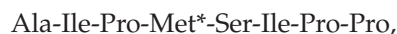
molecules is so tight that it is hard for a water molecule to enter and complete the catalytic act. The complex reacts very slowly, keeping the enzyme inactive. There is not enough inhibitor to interfere with the large amounts of trypsin formed from trypsinogen in the small intestine so that trypsin can function there.

Most protease inhibitors act by mechanisms similar to that of the pancreatic trypsin inhibitor. They are very slow substrates with a **reactive loop** that carries suitable P_1 , P_2 , and P_1' residues that meet the specificity requirements of the enzyme. Additional noncovalent interactions prevent dissociation and make the energy barrier for hydrolysis so high that the reaction is extremely slow.^{488,494,495}

The serpins are larger ~400-residue proteins.^{498–500} They also form complexes in which hydrolysis and release of the serpin occurs very slowly. However, structural analysis before reaction and after release showed that a major rearrangement occurs in the serpin structure. The P_1 through P_{15} residues of the cleaved reactive loop are inserted into the center of the main β sheet of the serpin, leaving the P_1 and P_1' residues ~7 nm apart.^{499,500} It seems likely that the rearrangement begins during formation of the tight inhibited complex, which cannot be dissociated by boiling in a sodium dodecyl sulfate (SDS) solution and which may be an acyl-enzyme.^{501,502} A serpin molecule can act only once.

Blood contains several serpins. They are abundant, accounting for about 10% of the total protein of human plasma.^{478,500} The most abundant is the **α_1 -protease inhibitor** or α_1 -antitrypsin, a 394-residue glycoprotein component of the α -globulin fraction of blood serum.^{500a} There is no trypsin in tissues, but this inhibitor blocks the action of other serine proteases, including cathepsin G and **leukocyte elastase**.⁵⁰³ Hereditary absence of α_1 -protease inhibitor often leads to severe **pulmonary emphysema** at an early age. Elastase released by neutrophils at sites of inflammation degrades many components of connective tissue including elastin, collagen, and proteoglycans. Without the presence of protease inhibitor too much damage is done to surrounding tissue. A lack of this inhibitor is one of the commonest genetic defects among persons of European ancestry, affecting 1 in 750 persons born.^{504–506} The most serious known mutation is a replacement of guanine by adenine at a specific point in the DNA and a resultant replacement of Glu 342 with lysine in the protease inhibitor. This in some manner adversely affected the processing and secretion of the protein.

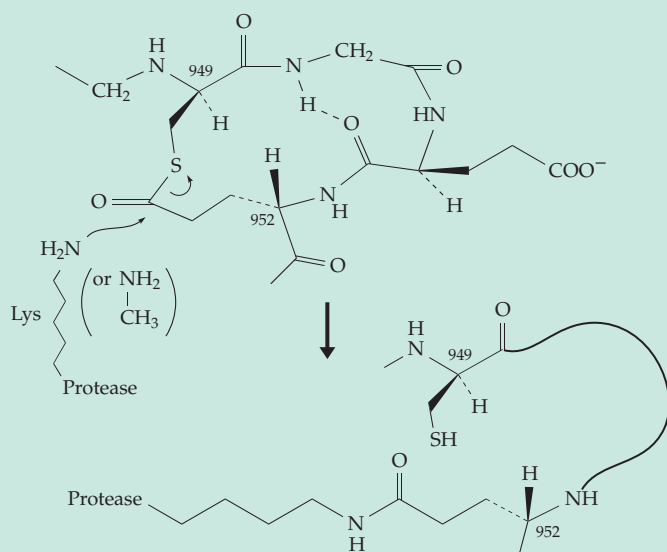
At its reactive site the α_1 -protease inhibitor has the sequence



the Met-Ser pair marked by the asterisk fitting into the P_1 - P_1' sites (Fig. 12-14). The methionine (Met 358) in

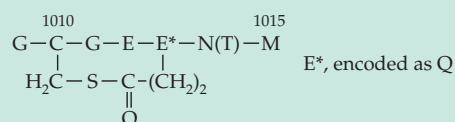
BOX 12-D MOLECULAR MOUSETRAPS

The large 720-kD α_2 -macroglobulin of human serum, as well as related proteins of vertebrate and invertebrate circulatory systems and of egg whites of birds and reptiles, is a trap for proteases.^{a-f} Human macroglobulin is a homotetramer consisting of two pairs of identical 180-kDa subunits, each pair being held together in an antiparallel configuration by two disulfide bridges. Each subunit contains a “bait region” with cleavage sites appropriate for nearly all known endoproteases^f and also a thioester linkage as explained later. Electron microscopic reconstructions of the native protein and its complexes with proteases show that a major structural transformation occurs.^{e,f} The macroglobulin traps two protease molecules of the size of trypsin, or one larger one such as plasmin, in an internal cavity. The internal thioesters, which are formed between Cys 949 and Gln 952 (with loss of NH_3) in each subunit, become reactive^g and form covalent bonds with ϵ -amino groups of various lysine side chains of the trapped proteases.

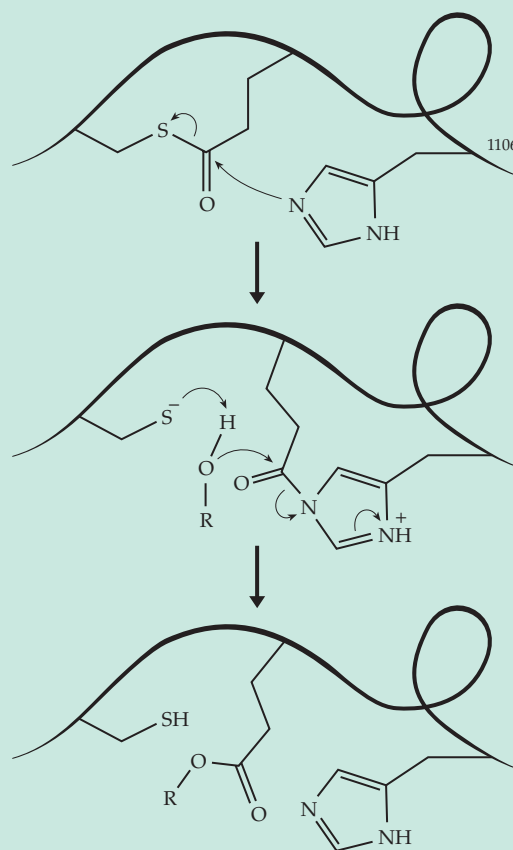


The serum proteins C3 and C4, members of the complement system (Chapter 31), are converted into their active forms, C3b and C4b, by proteolytic removal of short N-terminal peptide fragments. Both C3b and C4b bind tightly to cell surfaces, a feature that helps to direct the complement system's attack to the surfaces of invading organisms. This tight binding also involves covalent attachment of macromolecules by reaction with a preformed thioester just as with α_2 -macroglobulin.^{h-j} In fact, the thioester linkage was first discovered in the complement proteins. Both C3 and C4 contain the thioester

within the following sequence, which is the same as that in α_2 -macroglobulin:



Here the side chains have been added for the thioester-forming cysteine and glutamate and the sequence numbers are for C3. The thioester-forming glutamate is labeled E* because it is not encoded as glutamate but as glutamine, suggesting a mechanism of thioester formation. Protein C4 exists as two subforms, C4A and C4B. Both C3 and C4A react predominately with lysine amino groups, but C4B reacts with $-\text{OH}$ groups of cell surface polysaccharides.^h It has a histidine at position 1106. There is good evidence that it is adjacent to the $\text{C}=\text{O}$ group of the thioester and reacts to form an acyl-imidazole which is more reactive with hydroxyl groups than is a thioester:



Activation of C3 and C4 apparently allows the preformed thioester, which is buried in the interior

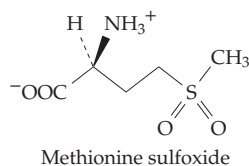
BOX 12-D (continued)

of these proteins, to become exposed on the external surface where it can react. The high group transfer potential of the thioester ensures that the reaction will go to completion.

- ^a Sottrup-Jensen, L. (1989) *J. Biol. Chem.* **264**, 11539–11542
^b Fothergill, J. (1982) *Nature (London)* **298**, 705–706
^c Jacobsen, L., and Sottrup-Jensen, L. (1993) *Biochemistry* **32**, 120–126
^d Andersen, G. R., Koch, T. J., Dolmer, K., Sottrup-Jensen, L., and Nyborg, J. (1995) *J. Biol. Chem.* **270**, 25133–25141

- ^e Boisset, N., Taveau, J.-C., Pochon, F., and Lamy, J. (1996) *J. Biol. Chem.* **271**, 25762–25769
^f Qazi, U., Gettins, P. G. W., Strickland, D. K., and Stoops, J. K. (1999) *J. Biol. Chem.* **274**, 8137–8142
^g Gettins, P. G. W. (1995) *Biochemistry* **34**, 12233–12240
^h Law, S. K. A., and Dodds, A. W. (1997) *Protein Sci.* **6**, 263–274
ⁱ Khan, S. A., Sekulski, J. M., and Erickson, B. W. (1986) *Biochemistry* **25**, 5165–5171
^j Dodds, A. W., Ren, X.-D., Willis, A. C., and Law, S. K. A. (1996) *Nature (London)* **379**, 177–179

this sequence, as well as another nearby methionine residue, is very susceptible to oxidation to sulfoxides:



The oxidation, which may be caused by such agents as myeloperoxidase (Chapter 16) released from leukocytes,⁵⁰⁷ inactivates the inhibitor. This may be physiologically important in permitting the proteases to be *uninhibited* in the immediate vicinity of the leukocyte. Cigarette smoke also inactivates α_1 -protease inhibitor by oxidation of the same methionine residues and the lungs of smokers contain the oxidized inhibitor.⁵⁰⁸ However, the major cause of emphysema among smokers appears to be an increase in released neutrophil elastase.⁵⁰⁹ One approach to the treatment of emphysema involves weekly intravenous injection of α_1 -antitrypsin.⁵¹⁰ This treatment may be improved by use of genetically engineered oxidation-resistant variants of the antitrypsin such as Met 385→Val.^{504,511} Efforts are also being made to introduce an α_1 -antitrypsin gene into lung epithelial cells.^{510,512}

Blood plasma also contains at least nine other protease inhibitors. One of these, the thrombin inhibitor **antithrombin III** (Section 9), contains the sequence Arg-Ser-Leu at the P_1 , P_1' , and P_2' sites. A tragic case of a person born with a Met 385 → Arg mutation in α_1 -antitrypsin has been reported.⁵¹³ This converted the antitrypsin to an antithrombin causing a bleeding disorder that was eventually fatal.

9. Coagulation of Blood

The clotting of blood following injury and the subsequent dissolving of the clot are familiar phenomena

that involve several cascades of proteolytic enzymes together with a number of accessory **cofactors**.^{514–516} The first step in clotting results from “activation” of blood platelets which aggregate to form a platelet plug that slows bleeding.^{514,517} The clot, which is formed by the insoluble **fibrin**, grows on the platelet surfaces and strengthens the plug. The initial rapid formation of a clot occurs via the **tissue factor pathway** (or extrinsic pathway; right side of Fig. 12-17) which is triggered by the exposure in injured tissues of **tissue factor** (TF), a transmembrane glycoprotein^{518–522} and a member of the cytokine receptor superfamily.⁵¹⁸ Human TF is a 263-residue protein with a single membrane-spanning region and a small 20-residue C-terminal cytoplasmic domain. The 219-residue extracellular domain consists largely of two IgG-like domains (Fig. 12-18).^{519,523} This protein stimulates the conversion of fibrinogen^{524–526} into the insoluble fibrin through the action of three proteases—factor VIIa, factor Xa, and thrombin. These enzymes are generated from proenzymes VII, X, and prothrombin, respectively in a cascade mechanism.

Factor VII binds tightly to TF,^{527,527a,b} which also binds Ca^{2+} and phospholipid of the cell membranes. Within this complex a plasma protease, such as thrombin or factor VIIa or Xa, cleaves a single Arg-Ile bond in VII to form active VIIa.^{528–530} The TF•VIIa complex is a very active protease which cleaves a specific peptide bond in factor X to form Xa^{531–533} which continues the cascade. Notice that there are autocatalytic features: VII can be converted to the active VIIa by the action of Xa and the accessory factor Va is generated from the precursor, factor V, in part through the action of thrombin.^{514,534–535} Factors Xa and Va together with Ca^{2+} and phospholipid form the active **prothrombinase** complex which attacks prothrombin to form the active enzyme thrombin.^{536,537} The roles of factor Va and Ca^{2+} appear to be to hold the prothrombin and the activated protease Xa together on the phospholipid surface.⁵³⁸ This localizes the clotting. Factor Xa is unusually specific, cleaving only after arginine in the sequence. Its activation of prothrombin results from cleavage of two

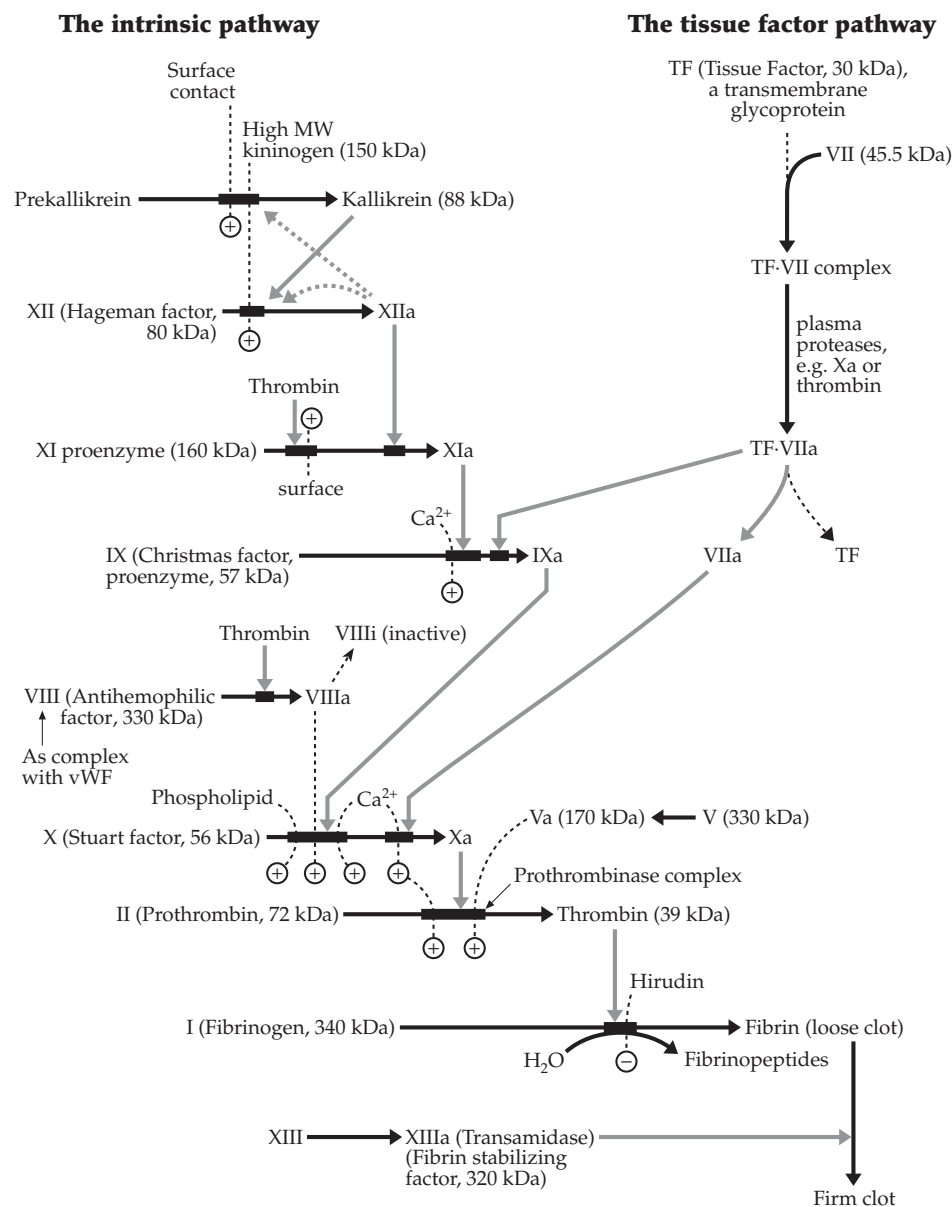


Figure 12-17 Major components of the human blood clotting cascades. The site of action of the leech anti-coagulant protein hirudin is also indicated.

peptide bonds, which releases the 39-kDa thrombin from the much larger, immobilized 72-kDa proenzyme.^{236,531}

Thrombin, like most other clotting factors, is also a serine protease. However, the clotting factors are multi-domain proteins that are more elaborate and more specific⁵³⁹ than the digestive enzymes. Prothrombin, as well as factors VII, IX, and X and the anticoagulant protein C, contain at their N-terminal ends several residues of γ -carboxyglutamate (Gla), an amino acid generated in a posttranslational modification that depends upon vitamin K (Chapter 15). In human prothrombin there are ten of these in the following N-terminal sequence, where E* represents Gla. Since many of these enzymes are also dependent upon activation by calcium ions (see Fig. 12-17), it is thought that the function of Gla is to assist in the binding of

10 20 30
ANTFLE•E•VRKGNLE•RE•CVE•E•TCSYE•E•AAFE•ALE•SS

Ca²⁺ which helps to tie these proteins to the phospholipids of platelet surfaces. In factors VII, IX, X, and protein C this Ca²⁺-binding domain is followed by two epidermal growth factor (EGF)-like domains, each containing one residue of *erythro*- β -hydroxyaspartate or hydroxyasparagine formed by hydroxylation of an aspartate or asparagine residue in the first EGF-like domain.^{540,540a,b} The C-terminal catalytic domain of each enzyme contains the protease active site.

Fibrinogen is an elongated molecule with an $(\alpha\beta\gamma)_2$ structure.^{524,541,541a} Thrombin cleaves specific Arg-Gly bonds in the α and β chains releasing short (14- to 16-residue) “fibrinopeptides” from the N termini of the peptide chains. This leaves Gly-Pro-Arg “knobs” at

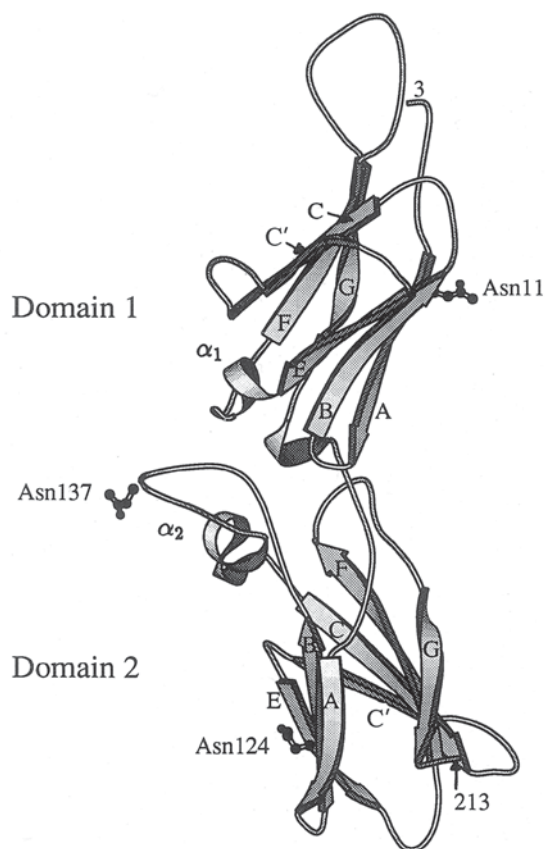


Figure 12-18 A ribbon drawing of the three-dimensional structure of the extracellular region (residues 3 to 213) of human tissue factor. Strands of domain 1 and 2 are labeled A to G. Two helices are labeled α_1 and α_2 and the three asparagines that provide the glycosylation sites are also marked. MolScript drawing from Harlos *et al.*⁵²⁰

the N termini of the α -chains and these fit into complementary “holes” in the γ chains to form noncovalently linked aggregates.^{542–544} The clot that forms is unstable, but it is soon crosslinked by the action of the **transamidase** (transglutaminase or **factor XIIIa**; Eq. 2-23).^{545,546} The fibrin monomers, von Willebrand factor (discussed in following paragraphs), fibronectin, collagen, and other proteins all become crosslinked.

The slower **intrinsic mechanism** first described in 1964^{547,548} consists of a cascade involving six proteases (Fig. 12-17, left side). Again, autocatalytic cycles are present in the activation by XIIIa of both prekallikrein⁵⁴⁹ and XII and in activation by thrombin of factors XI and VIII. This intrinsic pathway is initiated by the serine protease proenzymes prekallikrein and factor XII together with the accessory protein high-molecular-mass-H- kininogen.⁵⁴⁹ Activation occurs when blood contacts surfaces such as glass or kaolin (a clay).^{547,548,550} Factor XI can also be activated by thrombin. Hereditary absence of factor XI leads to bleeding problems, especially

after surgery, but absence of XII does not. This fact suggests that direct activation of XI by thrombin is important and that the kallikrein–factor XII pathway is usually less important.

Factor IX (Christmas factor) is next in the intrinsic mechanism cascade. It can be activated either by XIa or by VIIa of the tissue factor pathway. The absence of a functional factor IX leads to the inherited X-linked bleeding disorder **hemophilia B** which affects 1 in 30,000 males. The condition can be mild or very serious^{551,552} and may be caused by a variety of mutations or by incorrect splicing of the messenger RNA for the 416-residue factor IX. The level of factor IX in blood increases with age, almost doubling by old age.^{552a}

Factor IXa causes a rapid activation of factor X only if Ca^{2+} , phospholipid,^{553,554} and the accessory factor VIIIa⁵⁵⁵ are present. The IXa•VIIIa complex acts on X about 2×10^5 times faster than does IXa alone. This complex cleaves the same bonds in X as does the VIIa•Va complex formed in the tissue factor pathway.⁵¹⁴ The 2332-residue factor VIII and factor V have similar structures that include three repeats of a domain homologous to the blue copper-containing plasma protein **ceruloplasmin** (Chapter 16).^{556–559} Tyrosine 1680 of VIII apparently must be converted to a sulfate ester for full activity.⁵⁶⁰

The absence of factor VIII in about 1/10,000 males born, leads to the severe X-linked bleeding disorder **hemophilia A**. Human factor VIII is encoded by a 186-kb gene containing 26 coding exons. Severe cases of hemophilia are usually a result of point mutations that produce stop codons in this gene and therefore a shortened protein. Milder cases may result from an amino acid substitution.⁵⁶¹ Factor VIII circulates in the plasma as a complex with the **von Willebrand factor**, (vWF), a large multimeric protein derived by proteolytic cleavage, glycosylation, and sulfation of a 2813-residue precursor.^{562–566} The mature 2050-residue vWF stabilizes factor VIII in the blood. Another important function of vWF is to bind platelets to damaged endothelial surfaces.^{567,568} Like fibrinogen, vWF contains RGD sequences specific for binding to adhesion receptors.^{569,570} It also binds to collagen. A serious inherited bleeding disorder caused by deficiency of vWF was first discovered among inhabitants of islands in the Gulf of Bothnia, Finland, in 1926 by von Willebrand.^{563,571} If mild forms of the disease are included, vWD deficiency is the commonest bleeding problem. However, abnormalities have been identified in almost every one of the proteins in the coagulation cascades.⁵¹⁴

Until recently, these bleeding diseases were treated by regular injections of the appropriate factors isolated from human plasma. This was both costly and carried a high risk of infection by HIV, hepatitis, or other contaminating viruses. Now, cloned genes are used for commercial production of the factors.^{572,573} Experiments in animals, designed to lead to eventual gene

replacement therapy, have been somewhat successful.^{573,574} It is of interest to compare the relative concentrations of several of the clotting factors in plasma and to compare these with their positions in the cascade of the tissue factor (extrinsic) pathway.⁵¹⁵

	mg / L	Mass kDa	μM
VII	0.5	45.5	0.01
V	10.0	330	0.03
Prothrombin	150	72	2.1
Fibrinogen	3,000	340	8.8

Why do we have the intrinsic pathway when the tissue factor pathway provides rapid clot formation? The answer seems to be that the tissue factor pathway is needed immediately after injury but that it is turned off quickly by the **anticoagulation systems** of the body. As a result the protease **plasmin** begins to dissolve (lyse) the clot within a few hours. The intrinsic pathway is apparently needed to maintain the clot for a longer period.⁵¹⁴

What prevents the clotting mechanism, with its autocatalytic cycles, from running out of control? In part the answer lies in the localization of the enzymatic activation to tissue surfaces near a wound. The flow of blood rapidly dilutes components that escape into the general circulation and liver cells take up and destroy the active proteases. A variety of circulating antiproteases including the **tissue factor pathway inhibitor**⁵⁷⁵ act on these escaped enzymes. Two anticoagulation systems that are localized on vessel walls⁵⁷⁶ also come into action very quickly. The circulating polysaccharide heparin (Chapter 4) forms a complex with the serpin **antithrombin**. Antithrombin traps thrombin as an inactive complex or compound^{577,578a} and heparin greatly accelerates the inactivation.⁵⁷⁸ Kallikrein and factors IXa, Xa, XIa, and XIIa are also inhibited.

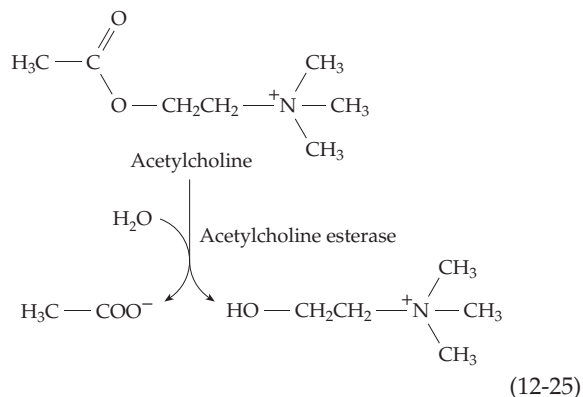
Thrombin is an allosteric protein which exists as a mixture, in nearly equal amounts, of a fast-acting form that cleaves fibrinogen and is stabilized by Na^+ , and a slow-acting form that initiates the second anticoagulant cascade.⁵⁷⁹ The slow acting form, bound to **thrombomodulin**,^{527a,540b,580,580a,b,581} an endothelial cell surface protein, attacks the proenzyme of another serum protease called **protein C**.^{582,582a} Activated protein C (APC) inactivates the accessory clotting factors Va and VIIIa. The accessory factor **protein S**⁵⁸³ is also needed for rapid inactivation. A blood clot is temporary and its dissolution begins as soon as it is formed, largely through the action of plasmin, a protease derived from the circulating 791-residue **plasminogen**^{584–588} through the action of yet other proteases. Plasminogen often becomes crosslinked to cell surface proteins by transglutaminase.⁵⁸⁶ Plasminogen activators include **urokinase**,⁵⁸⁹ a

protease present in kidney tissue and urine, and **tissue plasminogen activator** (tPA),^{590,591} a 527-residue protease which is now produced by recombinant DNA technology. It is sometimes used to dissolve blood clots in emergency situations, such as myocardial infarction and pulmonary embolism,⁵⁹² but can also cause serious bleeding problems. Plasmin is inhibited by the plasma α_2 **antiplasmin** and is inactivated by action of clotting factor XIIa. Likewise, tPA is inhibited by several protease inhibitors present in tissues and in plasma. One is a fast-acting serpin called plasminogen activator inhibitor.⁵⁹³ Another anticoagulant compound of medical interest is **hirudin**, a 65-residue peptide from the leech. It binds very tightly to thrombin ($K_d = 1 \text{ pM}$) preventing its action.^{594–596} Insects also produce antithrombin.^{597,598} Ticks^{599,600} and some insects⁵⁹⁸ inject proteins that inhibit Xa selectively.⁵⁹⁹ Anticoagulants are of great medical importance and much effort is being devoted to the design of better inhibitors of thrombin,^{601–603} factor Xa,⁶⁰⁴ and other components of the blood coagulation cascade.

Inherited deficiencies of the anticoagulant pathways with associated problems of thrombosis are known. These include problems with protein C,^{576,605} plasminogen,⁶⁰⁶ and antithrombin.^{607,608}

10. Esterases and Lipases

A group of esterases hydrolyze simple oxygen esters. Some of these are designed to hydrolyze a particular ester or small group of esters, while others have a more nonspecific action. **Acetylcholinesterase**^{609–611a} is specific for acetylcholine (Eq. 12-25), a neurotransmitter that is released at many nerve synapses and neuromuscular junctions (Chapter 30). The acetylcholine, which is very toxic in excess, must be destroyed rapidly to prepare the synapse for transmission of another impulse:



The more widely distributed **butyrylcholinesterase**⁶¹² is less specific but prefers butyrylcholine. Acetylcholinesterase is a very efficient catalyst:^{613–615} $k_{\text{cat}} = 1.6 \times 10^4 \text{ s}^{-1}$

and $k_{\text{cat}}/K_m = 2 \times 10^8 \text{ M}^{-1}\text{s}^{-1}$ at 25°C . It exists as a series of molecular forms containing varying numbers of 68-kDa catalytic subunits, 100-kDa structural subunits and subunits with triple-helical ~ 120 -kDa collagen-like "tails."^{609,616} The subunits are joined by disulfide bridges to give aggregates that range from simple dimers of catalytic subunits to tailed forms containing 8–12 catalytic subunits and non-collagenous structural subunits as well. The tailed forms are secreted from cells and may be designed to take up residence in the basal lamina of synapses, whereas the dimers are apparently attached to phosphatidylinositol anchors in the membranes (Fig. 8-13).⁶¹⁷ Human liver **carboxylesterases** are relatively nonspecific enzymes that hydrolyze ester groups of various drugs and toxins including cocaine and heroin. Products are often excreted in the urine.⁶¹⁸ **Thioesterases** function in biosynthesis of fatty acids, polyketides (Chapter 21), and many other substances.⁶¹⁹

Lipases^{620,621} hydrolyze triacylglycerols. The pancreatic digestive lipase^{622,623} acts faster on emulsified fats than on glycerol esters in true solution but requires the cooperation of a small 10-kDa **colipase**.⁶²⁴

Cholesterol esterase hydrolyzes not only cholesterol esters but also esters of fat-soluble vitamins, phospholipids, and triacylglycerols.^{625,626} Other lipases include gastric⁶²⁷ and hepatic lipases and a **lysosomal acid lipase**⁶²⁸ which also attacks neutral lipids. Plasma **lipoprotein lipase**^{629,630} digests fats in the chylomicrons and from the very low-density lipoproteins of blood.

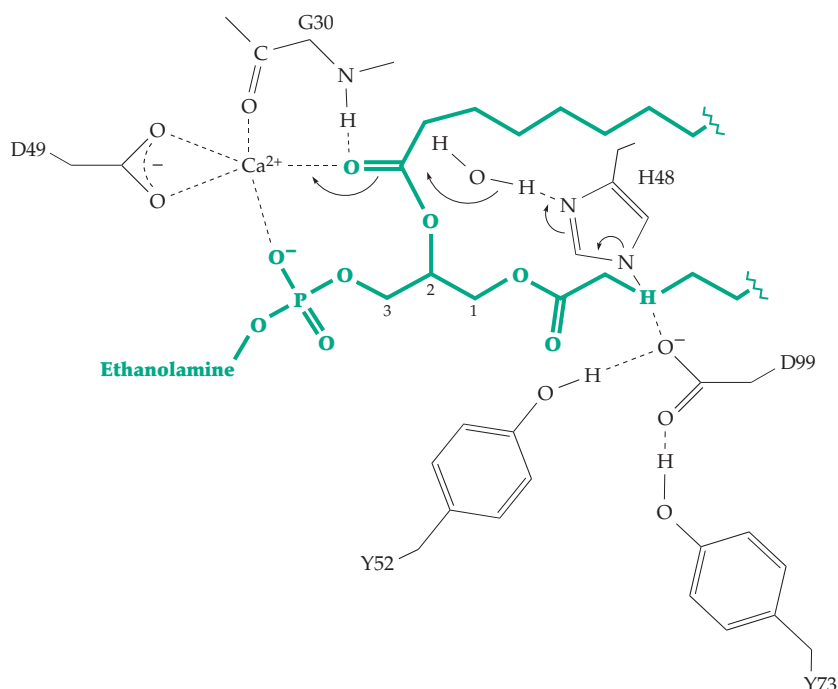
Hormone-sensitive lipase hydrolyzes stored triacylglycerols in the cytosol in response to catecholamines, ACTH, and other hormones.⁶³¹ The **phospholipases** attack phospholipids, while **cutinase**,⁶³²

produced by some fungi, cleaves the ester linkages in the cutin (Chapter 21) of plant surfaces. Fungal lipases are important industrial commodities. Numerous structural and mechanistic studies have been made with them.^{620,629,633–635} The gene for a lipase from *Candida rugosa* has been synthesized using codons that maximize its expression in *Saccharomyces cerevisiae* and which allow for further genetic engineering of the lipase.⁶³⁶

All of these esterases appear to act by mechanisms closely related to those of proteases. Acetylcholinesterase contains an active site serine that reacts with organophosphorus compounds (Box 12-E) and is part of an Asp-His-Ser catalytic triad which lies in a deep "gorge" as well as an oxyanion hole.⁶³⁷ A surprise is the absence of an essential carboxylate group that might bind the positively charged trimethylammonium

group of acetylcholine. Instead, the lining of the gorge is rich in aromatic side chains which may interact with the methyl groups of the substrate and by their polarizability stabilize the charge.^{611,638} Most lipases, including cutinase, also have an Asp-His-Ser or Glu-His-Ser triad as well as some form of oxyanion hole.^{620,632,639} Like the serine proteases, the lipases have bell-shaped pH optima.

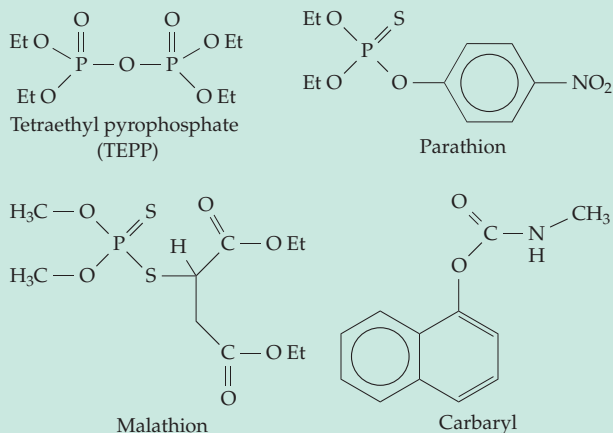
Phospholipase A₂ cleaves the ester linkage at the 2 position in phospholipids.^{640–642} One isoenzyme form is secreted by the pancreas as a proenzyme whose N-terminal seven residues are removed by trypsin to give an active 125-residue enzyme. Phospholipase A₂ of a similar type is abundant in venoms of reptiles and bees. The venom and pancreatic enzymes have closely similar three-dimensional structures.^{643–645} Although the folding pattern is different from that of chymotrypsin, imidazole (His 48) and carboxylate (Asp 99) groups are present in the active site in an orientation resembling that of the catalytic triad of serine protease. The enzyme requires calcium ions, one of which binds at an appropriate point for complexation with the substrate carbonyl as it is converted to an oxyanion intermediate. The backbone NH of Gly 30 may also serve as an oxyanion ligand. However, in most phospholipases there is no active site serine. Instead, a water molecule is positioned to serve as the attacking nucleophile in formation of the oxyanion as is indicated in the following scheme, which shows a truncated phosphatidylethanolamine as the substrate.⁶⁴³ Phospholipase A₂ is up to 1000 or more times as active on phospholipids in micelles as on dissolved substrates.^{621,646,647} Apparently, the



BOX 12-E INSECTICIDES

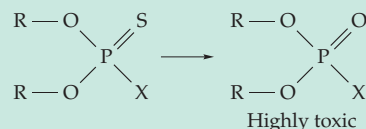
Over 200 organic insecticides, designed to kill insects without excessive danger to humans and animals, are presently in use.^{a-e} Many of these compounds act by inhibiting cell respiration; others uncouple ATP synthesis from electron transport. The chlorinated hydrocarbons such as DDT act on nerves in a manner that is still not fully understood. One of the largest classes of insecticides acts on the enzyme acetylcholinesterase of nerve synapses. Like chymotrypsin, it contains an active site serine residue that reacts with organophosphorus compounds. The extreme toxicity of esters of pyrophosphate and of dialkylphosphonofluoridates was recognized in the 1930s and led to their development in Germany and in England as insecticides and as nerve gases. Among the most notorious is diisopropylphosphonofluoridate (diisopropylfluorophosphate; DFP), for which the LD₅₀ (dose lethal to 50% of the animals tested) is only 0.5 mg kg⁻¹ intravenously. This exceedingly dangerous compound can cause rapid death by absorption through the skin.

The following structures are a few of the organophosphorus compounds and other acetylcholinesterase inhibitors that are selectively toxic to insects.



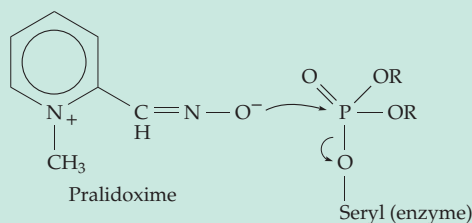
The characteristic high group transfer potential of a phospho group in pyrophosphate linkage, which makes ATP so useful in cells, also permits tetraethyl pyrophosphate to phosphorylate active sites of acetylcholinesterases. While TEPP is very toxic, it is rapidly hydrolyzed; all harmful residues are gone within a few hours after use.

Two insecticides that have been used widely are **parathion** and **malathion**. They are less toxic than DFP or TEPP and do not become effective insecticides until they undergo bioactivation during which conversion from a P = S to a P = O compound occurs:



The desulfuration reaction involves microsomal oxidases of the liver, the sulfur being oxidized ultimately to sulfate.^f The reactivity of parathion with cholinesterases depends upon the high group transfer potential imparted by the presence of the excellent leaving group, the *p*-nitrophenolate anion. If the P–O linkage to this group is hydrolyzed before the desulfuration takes place, the phosphorus compound is rendered harmless. Thus, the design of an effective insecticide involves finding a compound which insects activate rapidly but which is quickly degraded by higher animals. Other factors, such as rate of penetration of the insect cuticle and rate of excretion from the organism, are also important.

The phosphorylated esterases formed by the action of organophosphorus inhibitors are very stable, but some antidotes can reverse the inhibition. The oxime of 2-formyl-1-methylpyridinium ion (pralidoxime) is very effective.^g Its positive charge permits it to bind to the site normally occupied by the quaternary nitrogen of acetylcholine and to displace the dialkylphospho group:



Carbaryl, a widely used methyl carbamate, is a pseudosubstrate of acetylcholinesterase that reacts 10⁵ to 10⁶ times more slowly than do normal substrates. The carbamoylated enzyme formed is not as stable as the phosphorylated enzymes and the inhibition is reversible.

A basic problem is that most insecticides are designed to attack the central nervous system of the insect, the system that depends heavily upon acetylcholine. However, in human beings the readily accessible peripheral nervous system also depends upon acetylcholine, e.g., in neuronuscular junctions. The danger of poisoning is great. Another approach is to attack the glutamate-dependent peripheral system in insects, e.g., with inhibitors of glutamate decarboxylase. Glutamate functions as a neurotransmitter in the human body but only in the well-

BOX 12-E (continued)

protected central nervous system.

Another important problem is the development of insects resistant to insecticides. This often arises as a result of increased levels of carboxylesterases which hydrolyze both organophosphates and carbaryl.^{h,i} A mutation that changed a single active site glycine to aspartate in a carboxylesterase of a blowfly changed the esterase to an organophosphorus hydrolase which protected the fly against insecticides.^j

^a Büchel, K. H., ed. (1983) *Chemistry of Pesticides*, Wiley, New York

^b Hassall, K. A. (1990) *The Biochemistry and Uses of Pesticides*, 2nd ed., VCH Publ., Weinheim

^c Kamrin, M. A., ed. (1997) *Pesticide Profiles*, CRC Press, Boca Raton, Florida

^d Casida, J. E. (1973) *Ann. Rev. Biochem.* **42**, 259–278

^e Wilkinson, C. F., ed. (1976) *Insect Biochemistry and Physiology*, Plenum, New York

^f Nakatsugawa, T., Tolman, N. M., and Dahm, P. A. (1969) *Biochem. Pharmacol.* **18**, 1103–1114

^g Wilson, I. B., and Ginsburg, S. (1955) *Biochim. Biophys. Acta.* **18**, 168–170

^h Raymond, M., Callaghan, A., Fort, P., and Pasteur, N. (1991) *Nature (London)* **350**, 151–153

ⁱ Karunaratne, S. H. P. P., Hemingway, J., Jayawardena, K. G. I., Dassanayaka, V., and Vaughan, A. (1995) *J. Biol. Chem.* **270**, 31124–31128

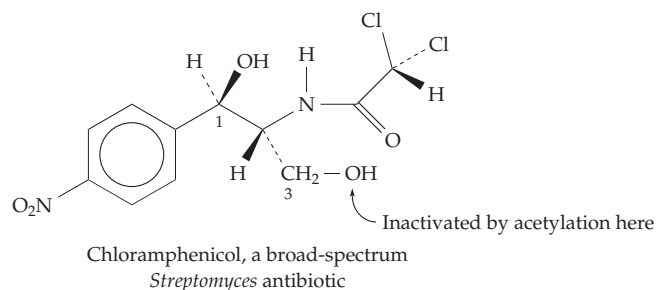
^j Newcomb, R. D., Campbell, P. M., Ollis, D. L., Cheah, E., Russell, R. J., and Oakeshott, J. G. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 7464–7468

substrate-binding cavity of the protein is designed to accommodate phospholipid molecules in the preferred conformation found in the micelles.⁶⁴³ Most of the other lipases have lids which close over the active sites and impede access of substrates. Binding to a phospholipid surface apparently induces a conformational change that opens the lid. This allows substrate to enter from the lipid surface.^{620,621,646} Cutinases, which do not display interfacial activation, do not have lids. Phospholipase A₂ activity is stimulated by an applied electrical field, a result that suggests that its activity *in vivo* may be regulated in part by the membrane potential.⁶⁴⁸

11. Other Acyltransferases

Acyl groups are frequently transferred from amides or esters to various acceptors in biosynthetic reactions. Among the many known acyltransferases are the ribosomal **peptidyl transferases** (Chapter 29), a transacylase involved in bacterial peptidoglycan synthesis (Chapter 20), transglutaminase (Eq. 2-23),^{649,650} γ -glutamylcyclotransferase (Box 11-B), and transacylation reactions involving acyl-CoA derivatives. Examples of the latter are *N*-acetylation⁶⁵¹ or myristoylation (Chapter 8) of proteins, the formation of acetylcholine from choline⁶⁵² and of acetylcarnitine from carnitine (Eq. 17-4), and acetylation of the antibiotic chloramphenicol. The high group transfer potential of thioesters ensures that these reactions proceed to completion. **Chloramphenicol acetyltransferase (CAT)**,⁶⁵³ the enzyme that catalyzes acetylation and inactivation of the antibiotic by bacteria, is much used in studies of gene expression (Chapter 28). A catalytic histidine removes the proton from the 3-OH group and a serine hydroxyl provides an oxyanion hole to accommodate the anticipated tetrahedral intermediate.^{652,654} The

steroidal antibiotic fusidic acid (Chapter 22, Section G) is a competitive inhibitor.⁶⁵⁵ A related transferase is aspartate carbamoyltransferase (Fig. 7-20; Chapter 24).



Penicillin and related antibiotics are inactivated by β -lactamases (Box 20-G), some of which resemble serine proteases in forming acyl enzymes with active site serine side chains.^{656,657} Others are zinc metalloenzymes.^{658,659} **Amidohydrolases** such as **asparaginase** and **glutaminase**,^{660,661} deacetylases,⁶⁶² and many other hydrolases can also be described as acyltransferases.

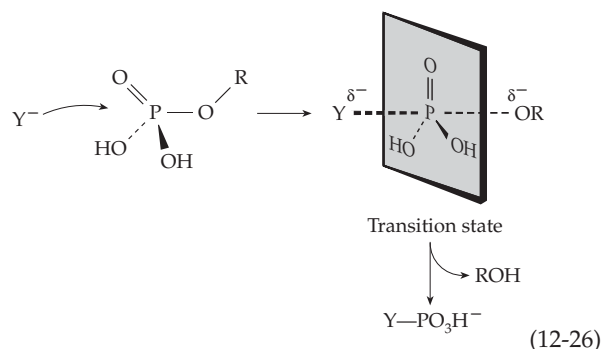
D. Displacement on a Phosphorus Atom

Nucleophilic displacements on phosphorus (Table 10-1, reaction type 1C) are involved in virtually every aspect of cellular energetics and in many aspects of biosynthesis. One large group of such enzymes are **phosphotransferases**, which transfer **phospho** (also called **phosphono** or, traditionally in biochemistry, **phosphoryl**) groups from one nucleophilic center to another. When transfer is to water the enzymes are called **phosphatases**, and when from one group in a molecule to another in the same molecule, **mutases**.

Enzymes that transfer a phospho group from ATP to water are **ATPases** and those that transfer the phospho group from ATP to some other nucleophile are **kinases**. Substituted phospho groups can also be transferred. Thus, **nucleases**, members of a larger class of **phosphodiesterases**, hydrolyze nucleic acids by transfer of nucleotidyl groups to a hydroxyl group of water. **Polynucleotide polymerases** transfer nucleotidyl groups to growing polynucleotide chains. An intramolecular nucleotidyl transferase is adenylate cyclase. **Topoisomerases** carry out a sequence of phosphotransferase reactions.

1. Questions about Mechanisms

Consider the following general equation for transfer to nucleophile Y^- of a phospho group attached in an ester or an anhydride linkage to form ROH, which could be an alcohol, carboxylic acid, or a phosphoric acid such as ATP.



This equation could also represent a half-reaction in a double-displacement process. As with displacements on saturated carbon atoms, two basic mechanisms can be imagined. The first is S_N2 -like or **associative**.^{663–667} The transition state might be represented where the bonds from the phosphorous atom to Y and to O are approximately equally formed. In an **in-line** displacement, where Y, P, and $-OR$ are colinear, this mechanism leads to inversion if a chiral phospho group is used. There is another possibility for the associative mechanism. Whereas a carbon atom can form only four stable covalent bonds, phosphorus is able to form five. While nucleophilic attack on carbon leads to a *transient* five-bonded transition state (Eq. 12-2), attack on phosphorus could produce a relatively long-lived pentacovalent intermediate (Eq. 12-27). Notice that two transition states are involved. Remember that our conventional way of drawing phosphate esters with a double bond from phosphorus to one of the oxygens is misleading. All of the P–O bonds share some of the double-bond character and the phosphate group has many characteristics of a completely single-bonded structure:

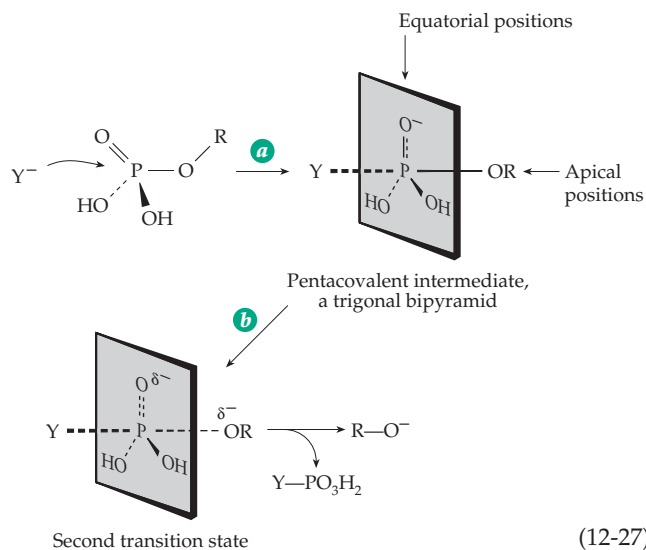


Likewise, in the transition state structures of Eq. 12-27, step *b*, the P–O bonds, except for that to $-OR$, are equivalent and all have partial double-bond character.

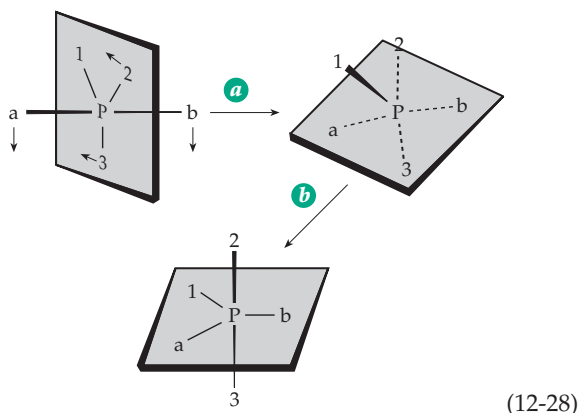
Geometric complexities. The geometry of the pentacovalent intermediate in Eq. 12-27 is that of a **trigonal bipyramid**. In this structure the bond angles in the **equatorial** plane are 120° , whereas all of the angles between any of those bonds and the two that attach to the groups in the **apical** positions are 90° . This disparity arises naturally from the fact that it is impossible to place five points on the surface of a sphere all equidistant one from the other. The attack of Y^- from the side opposite O–R (an in-line attack) leads to a trigonal bipyramid in which $-O-R$ and $-Y$ occupy the two apical positions. However, if Y^- attacks a face opposite one of the other oxygens (**adjacent** attack), $-O-R$ will take an equatorial position.

The chemical reactivities of groups in the apical and equatorial positions of pentacovalent intermediates are different.⁶⁶⁴ In particular, elimination of a nucleophilic group to form a tetrahedral phosphate is easier from an apical position than from an equatorial position. For the in-line displacement of Eq. 12-27 elimination of RO^- should be easy. However an adjacent attack would leave $-OR$ in an equatorial position. Before it could be eliminated, the intermediate would probably have to undergo a **permutational rearrangement** by which $-OR$ would be transferred from an equatorial to an apical position.

One type of permutational rearrangement, known as **pseudorotation**, can be visualized as in Eq. 12-28.^{668,669} The axial groups *a* and *b* move back while equatorial groups 2 and 3 move forward, still in the same equato-

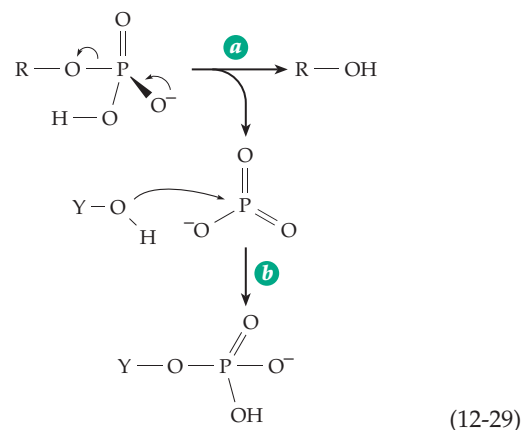


rial plane. Equatorial group 1 does not move. This decreases the 120° bond angles between the equatorial groups and increases the bond angles between group 1 and the axial groups until all four bond angles to group 1 are equivalent. The resulting **square pyramid** is a high-energy transition state structure in the pseudorotation process and can either revert to the original structure or, by continued motion of the groups in the same directions as before, to the structures shown at



the bottom of Eq. 12-28. In the final structure, groups 2 and 3 are axial and the original axial groups *a* and *b* are equatorial. Pseudorotation is slow enough that it could be rate limiting in enzymatic reactions. Stereoelectronic effects could also affect these rates.^{670,671} However, much evidence indicates that enzymes almost always avoid these complexities by using in-line mechanisms.

Metaphosphate ions. An alternative to an associative mechanism is an S_N1 -like or **dissociative** mechanism which can occur by elimination of a **metaphosphate ion** (Eq. 12-29, step *a*). A nucleophilic reagent can then add to the eliminated metaphosphate in step *b*. The formation of metaphosphate is analogous to formation of carbocation during the action of lysozyme (Eq. 12-11). This dissociative mechanism could lead either to racemization or to inversion of a chiral phosphate. Metaphosphate ions have been shown to exist. They are generated in certain nonenzymatic elimination reactions in aprotic solvents^{672–674} and they are reactive electrophiles able to react as in Eq. 12-29. However, there is some doubt that they can exist free in aqueous media.⁶⁷⁵ Jencks and associates concluded from studies of linear Gibbs energy relationships that in the transition state for nonenzymatic phospho transfer reactions there is a large amount of bond breaking and a small amount of new bond formation in the transition state⁶⁷⁶ but no free metaphosphate. There is a large dependence of the rate on the pK_a of the leaving group, but there is still a small dependence on the pK_a of the entering nucleophile. Thus, we have *a*



dissociative mechanism but without free metaphosphate. The concept is supported by studies of kinetic isotope effects.^{677,678}

Coping with negative charges. We visualize phospho group transfers as involving attack by a nucleophile bearing at least a partial negative charge. However, phospho groups also carry one or two negative charges and that in ATP even more. Therefore, it does not seem surprising that many phosphotransferases are metalloenzymes, sometimes containing bound Zn^{2+} which neutralizes some of the negative charge. Furthermore, enzymes usually accept ATP as a substrate only when it is accompanied by a divalent metal ion,⁶⁷⁹ usually Mg^{2+} . Another way in which enzymes deal with the negative charges on phospho groups is to have arginine side chains in appropriate positions to interact by forming strong ion paired-hydrogen bonds. It is often assumed that it is essential to neutralize the charge on the phospho group to avoid electrostatic repulsion from a partial or complete negative charge on the attacking nucleophilic center. However, with a dissociative mechanism and a metaphosphate-like intermediate the transition state may be reached without charge neutralization.^{679,680} Interactions of Mg^{2+} with phosphorus-containing substrates, like those of fixed positive charges in the protein may also be essential for binding the substrate correctly to an enzyme.

2. Magnetic Resonance Studies

There have been many investigations of phosphotransferases by NMR and EPR methods.^{681,682} One approach is to use paramagnetic ions such as Mn^{2+} , Cu^{2+} , or Cr^{3+} to induce nuclear relaxation in substrate and coenzyme molecules at active sites of enzymes. Flavin radicals and specifically introduced nitroxide spin labels can serve as well. Paramagnetic ions greatly increase the rate of magnetic relaxation of nearby nuclei (Chapter 3, Section I). Thus, small amounts of

Mn^{2+} in a sample lead to broadening of lines in ordinary ^1H , ^{13}C , and ^{31}P NMR spectra.

Useful information about enzymes can sometimes be obtained by observing effects of paramagnetic ions on the NMR signal of protons in the solvent water. The relaxation time of solvent protons is usually greater than 1 s. However, in the ion $\text{Mn}(\text{H}_2\text{O})_6^{2+}$ the protons of the coordinated water molecules relax much more rapidly, both T_1 and T_2 being $\sim 10^{-1}$ s. Since the coordinated water molecules usually exchange rapidly with the bulk solvent, a small number of manganese ions can cause a significant increase in relaxation rate for all of the water protons. Broadening of the proton band is observed and differences in T_1 and T_2 can be measured by appropriate methods. Paramagnetic relaxation effects usually increase as the inverse sixth power of the internuclear distance. Knowing the Mn^{2+} –H distance to be 0.287 ± 0.005 nm for hydrated Mn^{2+} , it has been possible to relate the effects on T_1 and T_2 to the number of H_2O molecules coordinated at any one time to a protein-bound metal ion and to their rate of exchange with the bulk solvent.

Relaxation effects on ^1H , ^{13}C , and ^{31}P , while more difficult to observe, can provide geometric information about active sites. The theory is complex, but under some conditions the paramagnetically induced relaxation can be described adequately by Eq. 12-30.

$$r = C [T_{1M} f(\tau_c)]^{1/6} \quad (12-30)$$

Here r is the internuclear distance, C is a combination of physical constants, and T_{1M} is the longitudinal relaxation time. The complex function $f(\tau_c)$ depends upon the correlation time τ_c , the resonance frequency of the nucleus being observed, and the frequency of precession of the electron spins at the paramagnetic centers. The value of τ_c can be estimated (Chapter 3) and, in turn, the distance r according to Eq. 12-30.

Such studies on creatine kinase (Eq. 12-31) utilized both a bound Mn^{2+} ion and a nitroxide spin label to estimate distances of various protons from the nitroxide.⁶⁸³ Together with EPR measurements (Box 8-C), which gave the Mn^{2+} –nitroxide distance, a model of the $\text{ATP} \cdot \text{Mn}^{2+}$ complex in the active site was constructed. Additional EPR experiments on Mn^{2+} complexes with ATP and ADP containing ^{17}O in the α , β , or γ phospho groups showed that in the enzyme•ATP•creatine complex the metal ion is bound to all three phospho groups of ATP. It remained coordinated with the two phospho groups of ADP and also that of the phospho-creatine product in the enzyme•ADP•creatine- P complex as well as in the transition state, which is pictured occurring via a metaphosphate ion.⁶⁸⁴

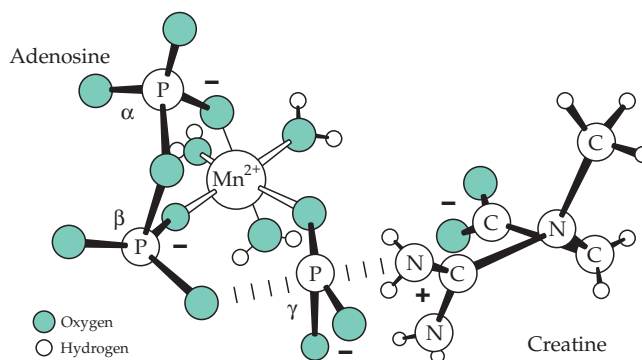
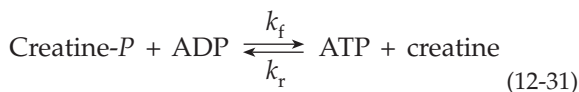
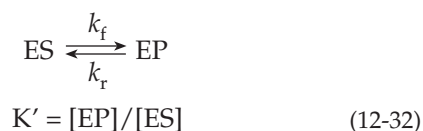
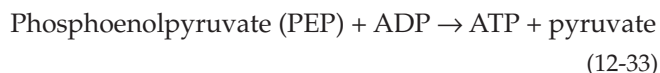


Figure 12-19 Proposed transition state structure formed from Mn^{2+} , ATP, and creatine bound in the active site of muscle creatine kinase. Based on EPR spectroscopy with regiospecifically ^{17}O -labeled substrates. The electrical charges have been added in one possible constellation. However, hydrogen atoms bound to phospho groups are not shown. After Leyh *et al.*⁶⁸⁴

Phosphorus-31 NMR has been used to measure **internal equilibrium constants** within enzyme-substrate (ES) complexes.^{663,685–687} By having both substrate and product concentrations high enough to saturate the enzyme, all of the enzyme exists as ES and enzyme-product (EP) complexes in equilibrium with each other. For a phosphotransferase at least one substrate and one product contain phosphorus. Although the NMR resonances are broadened by binding to the large, slowly tumbling protein, their areas can be measured satisfactorily and can be used to calculate an equilibrium constant such as that for Eq. 12-32:



An example is illustrated in Fig. 12-20. In this experiment⁶⁸⁵ the relative areas of the ^{31}P signals of ADP (one for free ADP and one, slightly more intense, for MgADP) and of the signal for phosphoenolpyruvate (PEP) were measured in the absence of enzyme and in the presence of a catalytic amount of pyruvate kinase (Fig. 12-20A). The results verified that the equilibrium constant for the overall reaction (Eq. 12-33) is very high (3300).



However, with an excess of enzyme (Fig. 12-20B) the internal constant was estimated as

$$K' = \frac{[E \cdot \text{MgATP} \cdot \text{pyruvate}]}{[E \cdot \text{MgADP} \cdot \text{PEP}]} = 0.5 - 1.0 \quad (12-34)$$

This is a difficult measurement and reinvestigation by another group⁶⁸⁶ indicated that the amount of the PEP-containing complex had been overestimated and that $K' \approx 10$.

If the rates of the forward and backward reactions in Eq. 12-34 are of the same order as the spin-lattice relaxation times (T_1) of the ^{31}P in the bound substrate and product, the rate constants k_f and k_r can be evaluated by **saturation-transfer NMR**.⁶⁸⁸

This is done by irradiating one resonance, e.g., that of the γ -P of ATP, and observing whether this causes a loss of intensity of the resonance for the product which is receiving its phospho group from ATP. This technique was used to observe the creatine kinase reaction (Eq. 12-31) in living muscle in both relaxed and contracting states. For resting muscle the observed forward flux was $1.7 \times 10^{-3} \text{ M s}^{-1}$ and the backward flux $1.2 \times 10^{-3} \text{ M s}^{-1}$. Thus, this reaction, which supplies ATP for contraction of the muscle from stored phosphocreatine, appears to be operating at or near equilibrium. This had been assumed but had previously been difficult to prove. Two-dimensional NMR techniques can now be used for this kind of measurement.⁶⁸⁹

When ^{31}P is bonded to ^{18}O the chemical shift of the ^{31}P is altered by 0.0206 ppm from that when the phosphorus is bonded to ^{16}O . This allows ^{18}O labels introduced into phospho groups to serve as tracers which can be followed continuously during reactions.⁶⁸³ The technique is useful in studies of stereochemistry (see Section 2) and for examination of **positional isotope exchange**.⁶⁹⁰

This latter technique is often used with ATP containing ^{18}O in the β, γ -bridge position. If an enzyme transfers the terminal (γ) phospho group to an acceptor via a phosphoenzyme but without loss of the ADP, we may expect positional isomerization. The ^{18}O will move between the β, γ -bridge position and

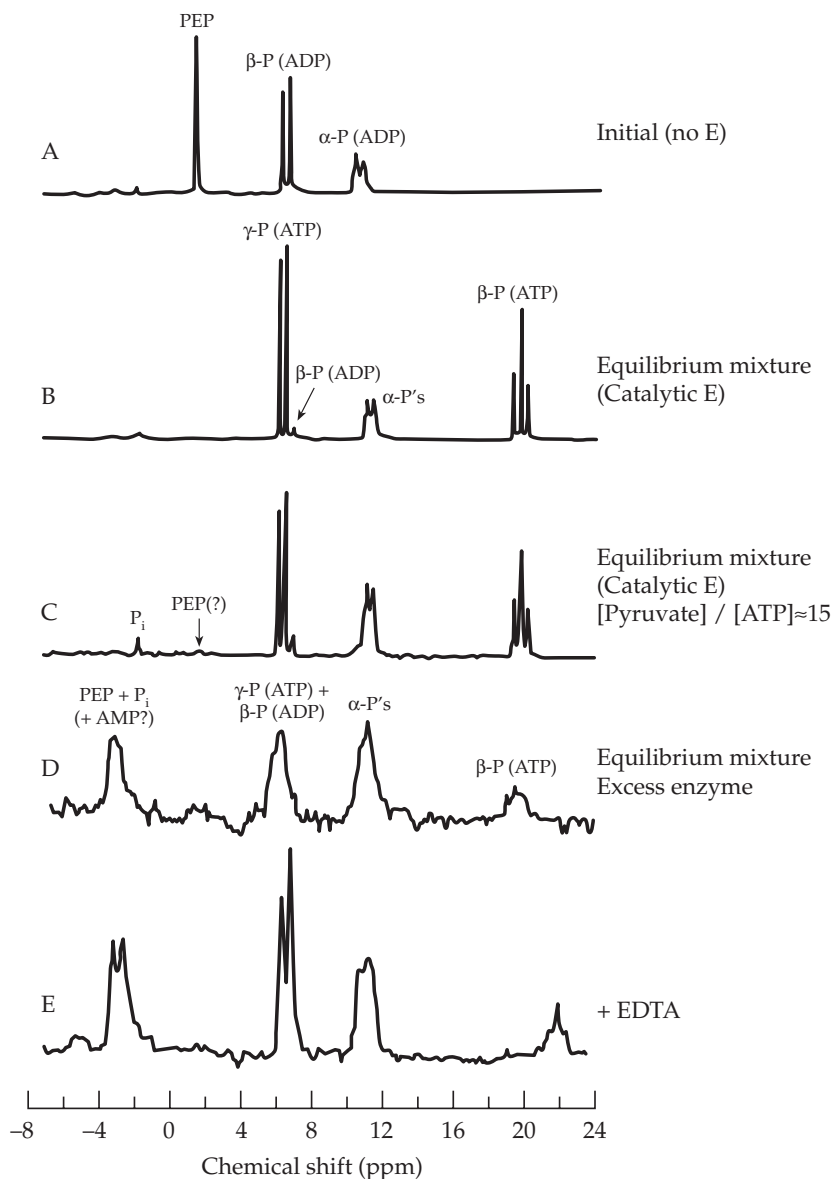


Figure 12-20 Equilibria in pyruvate kinase reaction as studied by ^{31}P NMR at 40.3 MHz, pH 8.0, 15°C. (A–C) Equilibria with low enzyme in levels ~15% $^2\text{H}_2\text{O}$. (A) ^{31}P NMR spectrum of 1.5 ml of reaction mixture; PEP, 13.3 mM; ADP, 14.1 mM; MgCl_2 , 20 mM; potassium Hepes buffer, 100 mM; KCl, 50 mM without enzyme. (B) Equilibrium mixture after the addition of ~1 mg of pyruvate kinase to the reaction mixture. (C) Equilibrium after the addition of potassium pyruvate (final concentration of 200 mM) to the sample of the spectrum in (B). (D,E) Equilibrium with enzyme concentrations in excess of the substrates. Sample volumes ~1.1 ml with 10% $^2\text{H}_2\text{O}$. (D) Equilibrium mixture set up with enzyme (2.8 mM active sites); 2.8 mM PEP; 2.4 mM ADP; 5.7 mM MgCl_2 ; 100 mM potassium Hepes; 100 mM KCl. (E) Spectrum after the addition of 50 μl of 400 mM EDTA (pH readjusted to 8.0) to the sample of spectrum D. The EDTA removes metal ions, stopping the catalytic reactions and sharpening the resonances. From Nageswara Rao *et al.*⁶⁸⁵

a nonbonding position as the phospho group is repeatedly transferred back and forth between ATP and the acceptor and as the phospho group rotates.^{682,690}

Equation 12-35 shows one part of this isomerization. The negative oxygens have been omitted here to avoid implying a known state of protonation or a localization of charge.

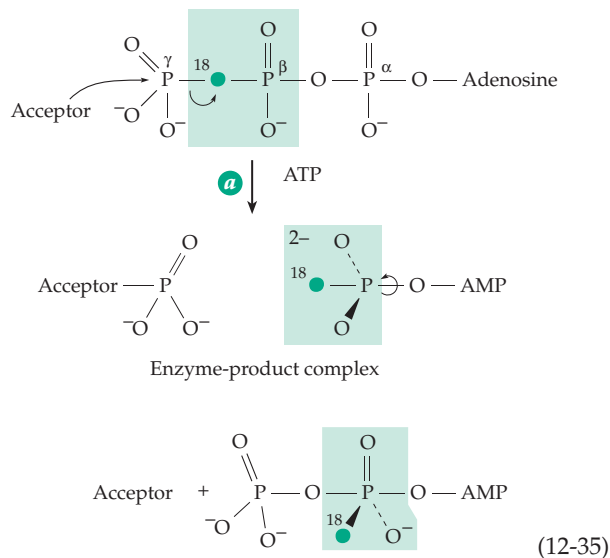
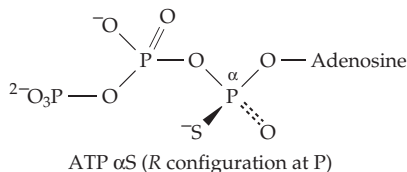


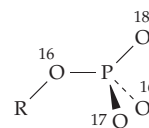
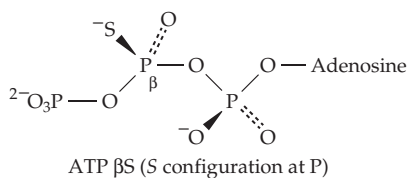
Figure 12-21 illustrates the use of the technique in investigating the possible participation of metaphosphate in a nonenzymatic reaction.

3. Stereochemistry

Evidence for an in-line S_N2 -like mechanism of most enzymatic phospho group transfer reactions comes largely from study of chiral phospho groups.^{663,692–695} A chiral phosphate can be introduced at either the α or β phosphorus of ATP by substitution of one of the oxygen atoms by sulfur. A chiral phospho group in the β position can be formed by substituting one oxygen by S and a second by ^{18}O .



Notice that the negative charge is largely localized on sulfur in these phosphorothioate compounds.⁶⁹⁶ More general is the use of ^{17}O and ^{18}O to form a chiral phospho group:



An ester chiral at the phosphorus atom; *R* configuration

Considerable ingenuity was required in both the synthesis of these chiral compounds^{695,697} and the stereochemical analysis of the products formed from them by enzymes.^{698–700} In one experiment the phospho group was transferred from chiral phenyl phosphate to a diol acceptor using *E. coli* alkaline phosphatase as a catalyst (Eq. 12-36). In this reaction transfer of the phospho group occurred without inversion. The chirality of the product was determined as follows. It was cyclized by a nonenzymatic in-line displacement to give equimolar ratios of three isomeric cyclic diesters. These were methylated with diazomethane to a mixture of three pairs of diastereoisomers triesters. These diastereoisomers were separated and the chirality was determined by a sophisticated mass spectrometric analysis.⁶⁹² A simpler analysis employs ^{31}P NMR spectroscopy and is illustrated in Fig. 12-22. Since alkaline phosphatase is relatively nonspecific, most phosphate esters produced by the action of phosphotransferases can have their phospho groups transferred without inversion to 1,2-propanediol and the chirality can be determined by this method.

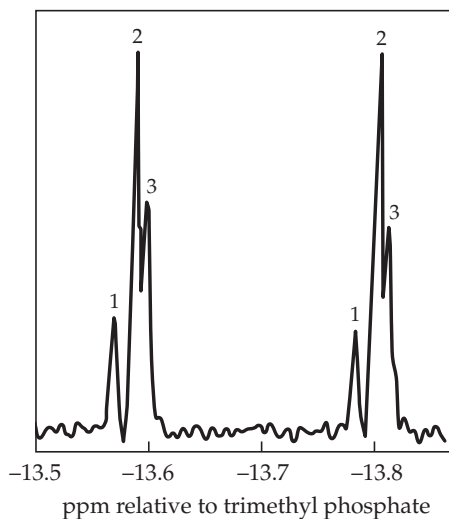
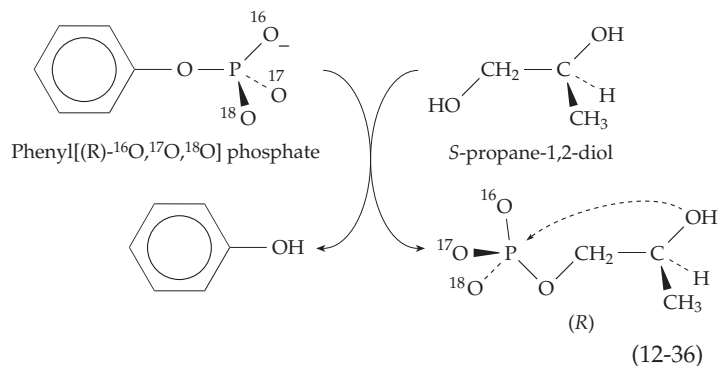
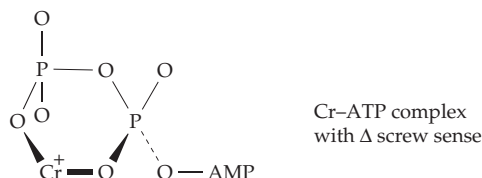
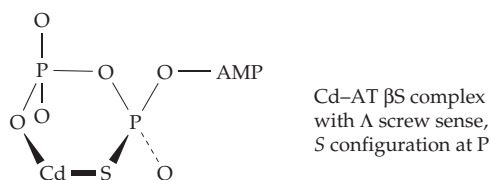
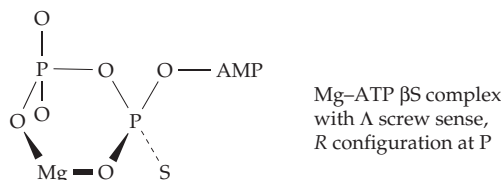


Figure 12-21 The ^{31}P NMR spectrum at 101.2 MHz of P_α of isotopically labeled ADP. This was recovered from an experiment in which ADP containing 87 atom % of ^{18}O in all four oxygens around P_β was allowed to undergo partial (~20%) nonenzymatic hydrolysis to AMP and P_i . Peaks 1 represent the species containing no ^{18}O bonded to P_α . Peaks 2 represent all species with ^{18}O in the $\text{P}_\alpha\text{--O--P}_\beta$ bridge, and peaks 3 represent species with ^{18}O in nonbridging positions at P_α . These have undergone positional isotope exchange. From Lowe and Tuck.⁶⁹¹



Although inversion was not observed with the *E. coli* alkaline phosphatase, it has been observed for ribonucleases and many other hydrolytic enzymes and for most kinases transferring phospho groups from ATP. The difference lies in the existence of a phospho-enzyme intermediate in the action of alkaline phosphatase (see Eq. 12-38). Each of the two phosphotransferase steps in the phosphatase action apparently occurs with inversion. The simplest interpretation of all the experimental results is that *phosphotransferases usually act by in-line S_N2-like mechanisms which may involve metaphosphate-ion-like transition states that are constrained to react with an incoming nucleophile to give inversion*. An adjacent attack with pseudorotation would probably retain the original configuration and is therefore excluded.

The substrate for many phosphotransferases is MgATP. Which of the possible isomers of this chelate complex is utilized by these enzymes? Since Mg²⁺ associates and dissociates rapidly from the complexes there are several possibilities: a tridentate complex with oxygens from α , β , and γ phospho groups coordinated with the metal ion, an α,β -bidentate, a β,γ -bidentate, or a monodentate complex. Most evidence suggests that β,γ -bidentate complexes of the following types are the true substrates.

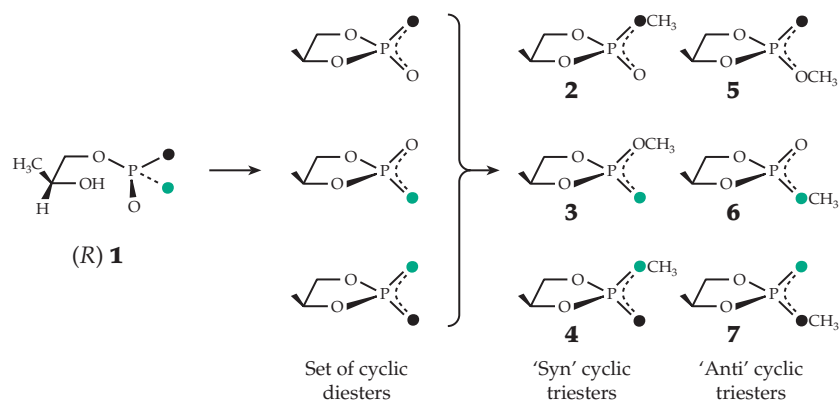


The first structure drawn is for a Mg-ATP S complex with a chiral β -phosphothioate group. The Mg²⁺ is expected to bond to oxygen. However, in the second complex, in which Cd²⁺ has been substituted for Mg²⁺, it is expected that the Cd²⁺ will bond to sulfur. Therefore, the stereochemically equivalent structure will be obtained only when the compound has the S configuration at the phosphorus. In the third of the foregoing complexes, Mg²⁺ has been replaced by Cr³⁺ to give an **exchange-inert** complex of ATP in which the Cr³⁺ will remain attached firmly to

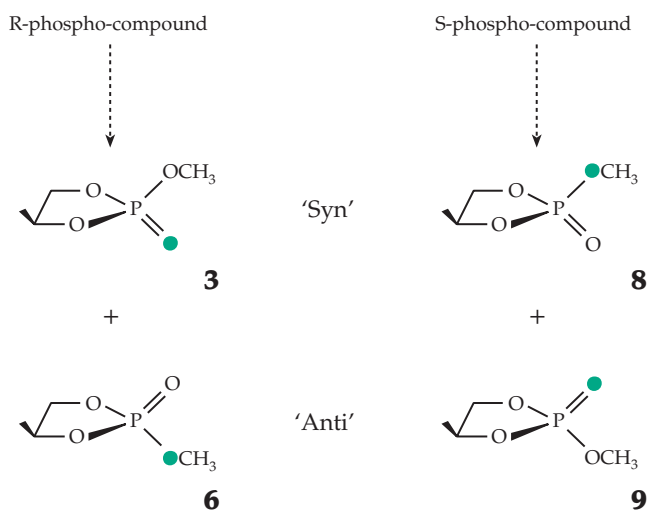
the same two oxygen atoms under most experimental conditions. Notice also that in this complex, the AMP portion occupies a different position than in the first two complexes. It is called the Δ **screw-sense isomer**,⁷⁰² while the upper two complexes are Λ screw-sense isomers. We may reasonably expect that an enzyme, which may provide additional ligands to the metal ions, will prefer one or the other of these screw-sense isomers. For α,β -bidentate ATP complexes both α and β phosphorous atoms become chiral centers and even more isomers are possible.

The use of exchange inert Cr³⁺ and Co³⁺ complexes of ATP has been developed by Cleland and associates.⁷⁰²⁻⁷⁰⁵ The β,γ -bidentate chromium complexes were separated into the Λ and Δ isomers.⁷⁰⁴ Each was separated further into a pair of "ring-puckering isomers". These metal complexes are all competitive inhibitors of MgATP and both ring puckering isomers of the Δ screw sense are very slow substrates of various kinases. The Λ isomers serve as very slow substrates for pyruvate kinase, adenylate kinase, and fructose-6-phosphate kinase.⁷⁰³ The Δ exo isomer is the strongest inhibitor of creatine kinase, suggesting this same conformation for the MgATP substrate. The diastereoisomers of ATP α S and ATP β S have also been tested with kinases in the presence of either Mg²⁺ or Cd²⁺. With creatine kinase^{695,706} the isomers with the R configuration at phosphorus are preferred in the presence of Mg²⁺ but those with the S configuration are preferred in the presence of Cd²⁺. This might be related to the previously mentioned preferences of the metals for O vs. S ligands, and would suggest that this enzyme prefers the Λ isomer (see foregoing structures). However, EPR studies with the corresponding Mn²⁺ complexes suggest that the Δ isomer is preferred by the enzyme.⁶⁹³ There are 12 isomers of monoammine Cr(III)ATP. Their use has provided information about the location of water molecules in metal complexes of kinases.⁷⁰⁷

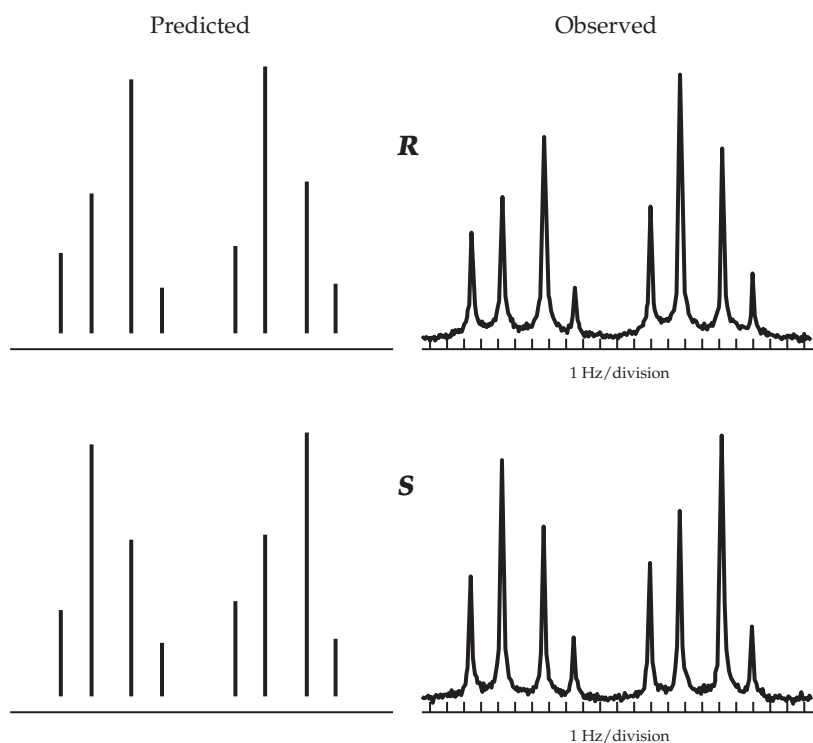
In the following sections we will consider several individual phosphotransferases.



The product 1-phospho-(S)-propane-1,2-diol **1** (here shown as *R* at phosphorus) is converted by in-line ring closure to an equimolar mixture of three cyclic diesters. These are methylated to give six cyclic triesters. Of these, only **3** and **6** give sharp ^{31}P resonance because the ^{17}O in the others broadens the lines.



Two cyclic triesters **3** and **6** from an *R*-phospho compound and two others **8** and **9** from an *S*-phospho compound will give sharp ^{31}P NMR resonances.

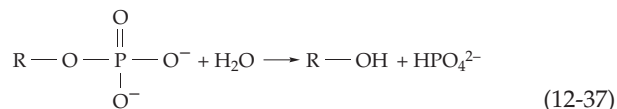


Predicted and observed ^{31}P NMR spectra of the mixtures of syn and anti cyclic triesters derived from labeled samples of 1 phospho-(S)-propane-1,2-diols that are *R* and *S* at phosphorus.

Figure 12-22 Method for determining chirality of phospho groups containing ^{16}O , ^{17}O , and ^{18}O . From Buchwald and Knowles.⁷⁰¹

4. Phosphatases

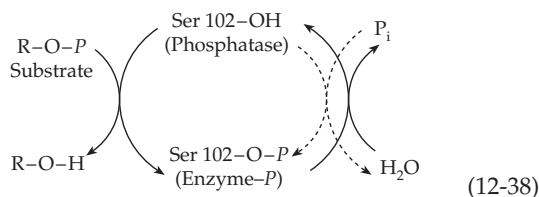
The phosphatases catalyze hydrolysis of phosphate esters to produce inorganic phosphate:⁶⁶⁷



The **acid phosphatases** and **alkaline phosphatases** are nonspecific and cleave many different phosphate esters, whereas **glucose-6-phosphatase**, **fructose-1,6-bisphosphatase**, and many others are specific for single substrates. The nonspecific phosphatases may provide inorganic phosphate ions in places where they are needed, e.g., in mineralizing bone. All phosphatases help to drive metabolic cycles (Chapter 17).

The alkaline phosphatases are found in bacteria, fungi, and higher animals but not in higher plants. In *E. coli* alkaline phosphatase is concentrated in the periplasmic space. In animals it is found in the brush border of kidney cells, in cells of the intestinal mucosa, and in the osteocytes and osteoblasts of bone. It is almost absent from red blood cells, muscle, and other tissues which are not involved extensively in transport of nutrients.

The alkaline phosphatase of *E. coli* is a dimer of 449-residue subunits which requires Zn^{2+} , is allosterically activated by Mg^{2+} , and has a pH optimum above 8.^{667,708–711} At a pH of ~ 4 , incubation of the enzyme with inorganic phosphate leads to formation of a phosphoenzyme. Using ^{32}P -labeled phosphate, it was established that the phosphate becomes attached in ester linkages to serine 102. The same active site sequence Asp-Ser-Ala is found in mammalian alkaline phosphatases. These results, as well as the stereochemical arguments given in Section 2, suggest a double-displacement mechanism of Eq. 12-38:



The active site contains two Zn^{2+} ions and one Mg^{2+} ion which are held by imidazole and carboxylate groups. The inorganic phosphate in an enzyme-product complex is bound to both zinc ions (Fig. 12-23). The Ser 102 side chain is above one Zn. In the enzyme-*P* intermediate it would be linked to the phospho group as an ester which would then be hydrolyzed, reversibly, by a water molecule bound to Zn.^{712–713a} This water presumably dissociates to Zn^+-OH and its bound hydroxyl ion carries out the displacement. This reaction may be preceded by a proton transfer to an oxygen atom of the phospho group.⁷¹⁴

Acid phosphatases, which have pH optima of ~ 5 and are inhibited by fluoride ion, occur in bacteria, fungi, plants, and animals. In bone, the acid phosphatase content is high in the osteoclasts which function in the resorption of calcium from bone. The highest content of acid phosphatase in humans is in the prostate and an elevated serum level has long been used as a diagnostic indicator of prostatic cancer.⁷¹⁵ Acid phosphatase is a periplasmic enzyme in *E. coli*.⁷¹⁶ Phosphoenzymes have been trapped from both plant and animal acid phosphatases.⁷¹⁷ For example, a brief incubation with ^{32}P -labeled *p*-nitrophenyl phosphate followed by rapid denaturation in an alkaline medium gave a covalently labeled protein from which ^{32}P -containing N δ -phosphohistidine was isolated. In agreement with kinetic evidence that a phosphoenzyme is a true intermediate, there is no inversion of chiral phospho groups by liver or prostate acid phosphatases.⁷¹⁸

Some acid phosphatases from animals and plants are violet in color and contain iron (Chapter 16) and an Mn^{3+} -containing acid phosphatase has been isolated from sweet potatoes.⁷²⁰ These enzymes have dimetal centers, often containing one Zn^{2+} and one Fe^{3+} with bridging carboxylate and hydroxide ions between the metals. Imidazole, tyrosinate, and carboxylate side chains hold the metals as in Fig. 16-20. A water molecule bound to the Fe^{3+} is thought to dissociate with a low pK_a of 4.8 to give an $\text{Fe}^{3+} \cdot ^-\text{OH}$ complex. The hydroxyl ion can then attack the phospho groups, one

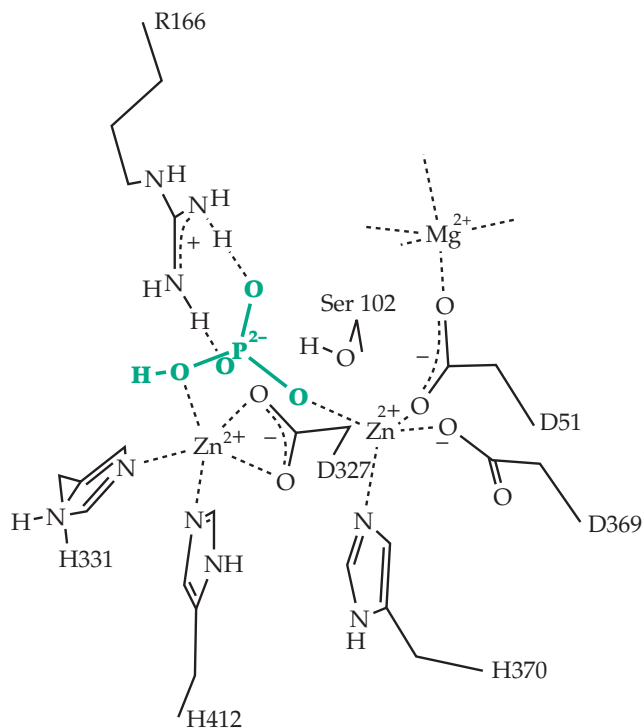


Figure 12-23 Schematic drawing of the product inorganic phosphate bound in the active site of *E. coli* alkaline phosphatase. See Ma and Kantrowitz.⁷¹⁹

of whose oxygen atoms is coordinated with the Zn^{2+} .

The mechanism resembles that proposed for a **phosphotriesterase** (Fig. 12-24). The triesterase catalyzes detoxification of organophosphorus toxins such as parathion (Box 12-E) and seems to have evolved rapidly from a homologous protein of unknown function.⁷²¹ The phosphotriesterase contains two Zn^{2+} ions in a dimetal center. An unusual structural feature is a carbamate group, formed from Lys 169 and CO_2 , which provides a bridging ligand for the metal pair.⁷²¹⁻⁷²⁵ A carbamylated lysine also functions in ribulose biphosphate carboxylase (Fig. 13-11).

Phosphatases specific for such substrates as glucose-6-phosphate, fructose-1,6-bisphosphate, and phosphoglycolate help to drive metabolic cycles (Chapter 17). The 335-residue **fructose-1,6-bisphosphatase** associates to form a tetramer with D_2 symmetry.⁷²⁶⁻⁷³⁰ The allosteric enzyme exists in two conformational states (see Chapter 11). Activity is dependent upon Mg^{2+} or other suitable divalent cation, e.g., Mn^{2+} or Zn^{2+} , and is further enhanced by K^+ or NH_3^+ . While the dimetal sites depicted in Figs. 12-23 and 12-24 are quite rigid and undergo little change upon formation of complexes with substrates or products, the active site of fructose-1,6-bisphosphatase is more flexible. There are three metal-binding sites but they contain no histidine side chains and have been seen clearly only in a product complex.^{727,728} Perhaps because of the need for

flexibility involved in allosteric changes, the active site is not fully formed until the substrate binds.

Fructose-2,6-bisphosphatase forms one domain of a bifunctional kinase-phosphatase (Chapter 11). It has two imidazole rings, as well as side chains from a glutamate and two arginine residues at the catalytic and substrate-binding site.^{728a}

The 357-residue mammalian glucose-6-phosphatase plays an important role in metabolism (Chapter 17). Defects in the enzyme cause a glycogen storage disease (Box 20-D) and severe disruption of metabolism.⁷³¹ However, the molecular basis of its action is not well-known. Furthermore, the active site of the enzyme is located in the lumen of the endoplasmic reticulum⁷³² and glucose-6-phosphate must pass in through the plasma membrane. An additional glucose-6-phosphate transporter subunit may be required to allow the substrate to leave the cytoplasm.⁷³

Pyrophosphatases, which are present in all cells, and catalyze hydrolysis of inorganic pyrophosphate (PP_i) to orthophosphate (P_i) (see Chapter 6, Section D), also drive metabolic sequences. The very active pyrophosphatase of *E. coli* has a turnover number of over $2 \times 10^4 \text{ s}^{-1}$ at 37°C . The 1000 molecules per cell are sufficient to immediately hydrolyze any pyrophosphate produced by bacterial metabolism.⁷³³ The much studied soluble pyrophosphatases of *E. coli*,^{734,735} yeast,⁷³⁶ and other organisms^{736a,b} are metalloenzymes that are most active with Mg^{2+} . Two Mg^{2+} ions are held, mostly by carboxylate side chains, while a third apparently enters the active site as magnesium pyrophosphate, perhaps MgP_2O_7^- . As with other metallohydrolases, a metal-bound hydroxyl ion may serve as the attacking nucleophile.

At least three distinct families of **protein phosphatases** remove phosphate groups from serine, threonine, and tyrosine side chains in proteins.⁷³⁷ Their role in control of numerous biochemical processes has been discussed in Chapter 11, Section C.2. The catalytic domains or subunits of the protein phosphatases act together with regulatory domains or separate regulatory subunits to control thousands of reactions. For example, protein phosphatase 1 (PP1) together with a glycogen-targeting subunit dephosphorylates inactive glycogen kinase (see Fig. 11-4). Belonging to the same family is **calcineurin** (PP2B), a phosphatase activated by Ca^{2+} through binding to calmodulin (Box 6-D). There are two families of Ser/Thr phosphatases. Their polypeptide folding patterns differ, but the active sites have similar dimetal centers resembling those in Figs. 12-24 and 16-20 with $\text{Mn}^{2+} + \text{Fe}^{2+}$, $\text{Zn}^{2+} + \text{Fe}^{3+}$, and probably other pairs of metals.⁷³⁷⁻⁷⁴⁰ The family containing PP1 has weak sequence homology with the purple acid phosphatases.⁷⁴⁰ Another common feature of these enzymes is a conserved His-Asp dyad (His 125 and Asp 95 in PP1) which is thought to be a proton donor, protonating the leaving group ($-\text{O}^-$) in a manner

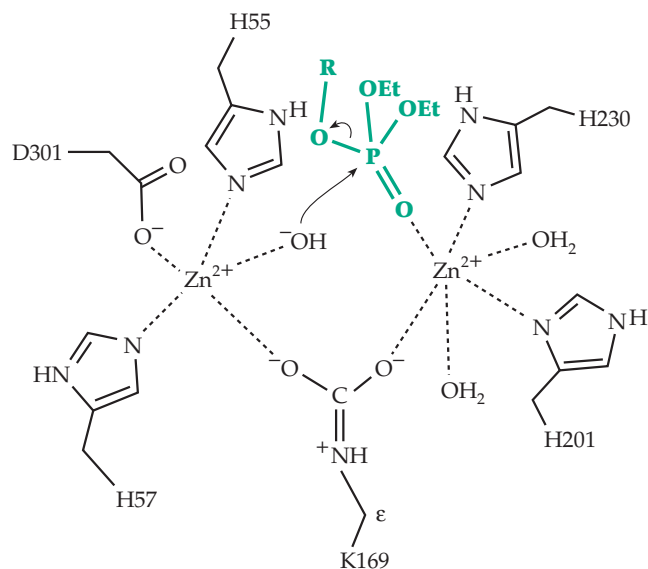


Figure 12-24 Hypothetical event in the action of a phosphotriesterase. A carbamylated lysine (lower center), as well as a water molecule, bridge the two Zn^{2+} ions, which are held by imidazole and aspartate carboxylate groups. The bound H_2O can be deprotonated to give the HO^- complex shown. The substrate may displace the HO^- ion from the right-hand zinc and thereby move close to the bound HO^- which attacks as indicated. Based on Cd^{2+} -containing structure and discussion by Benning *et al.*⁷²²

reminiscent of the serine proteases. The reaction is evidently initiated by attack of an OH^- ion held by the dimetal center. This would resemble the mechanism pictured in Fig. 12-24 except that the phospho group would carry two negative charges.

The protein tyrosine phosphatases also exist as several families with numerous functions in control of transcription, growth, differentiation, and metabolism.^{741–743} These enzymes function by a double-displacement mechanism, as in Eq. 12-38, but with a cysteine side chain rather than serine. The cysteine is present in the conserved sequence (H/V)CX₅R(S/T). The arginine binds the phospho group and helps to stabilize the transition state, which probably is metaphosphate-like.⁷⁴²

5. Ribonucleases (RNases)

Many hydrolases act on phosphodiester linkages, which abound in nature.^{725,744} Some are digestive enzymes but others serve more specific metabolic functions. **Ribonuclease A (RNase A)**, the pancreatic digestive enzyme responsible for breakdown of RNA, was one of the first enzymes for which a structure was deduced. By 1963 Moore and Stein and their associates, who had earlier developed ion exchange methods for separating amino acids and peptides (Fig. 3-6), had determined the sequence of the 124-residue bovine enzyme.⁷⁴⁵ They observed that Lys 41 was unusually reactive with dinitrofluorobenzene and that photo-oxidation of His 12 and His 119, which are almost at opposite ends of the peptide chain, inactivates the enzyme. They concluded that both histidines are at the active site, a conclusion that was later substantiated by X-ray crystallography.^{746–748} A segment 12 nucleotides in length can fit into the cleft in the enzyme that contains the active site. The negatively charged phosphates of the RNA backbone form 8–9 electrostatic bonds to lysine and arginine side chains of the enzyme.^{749,750} However, the only close interactions of the nucleic acid bases with the enzyme occur at the site of cleavage as shown in Fig. 12-25. The four residues His 12, Lys 41, Thr 45, and His 119 are strictly conserved in the RNase A superfamily.⁷⁴⁹ The carboxylate of Asp 121 apparently helps to orient the proper tautomer of His 119 for catalysis.⁷⁵¹

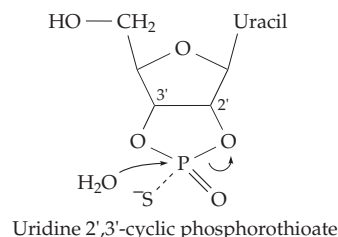
Ribonuclease A was the first enzyme to be synthesized in the laboratory. Fully active ribonuclease has been synthesized,⁷⁵² as have new modified enzymes. For example a 63-residue peptide made up of five segments of the native RNase sequence retained measurable catalytic activity.⁷⁵³ Using total synthesis, unnatural amino acids, such as 4-fluorohistidine, have been incorporated at specific positions in RNase.⁷⁵²

Cleavage of a phosphodiester linkage in the substrate chain occurs in two steps. In the first or *trans-*

esterification step, the hydroxyl group on the 2' position of the ribose ring is thought to be deprotonated by attack of either the imidazole of His 12, as shown in Fig. 12-25, or by the adjacent amino group of Lys 41. (In the latter case His 12 would have to remove a proton from the $-\text{NH}_3^+$ group of Lys 41 before it could attack.) In either case the positive charge of Lys 41 would help to neutralize the negative charge on the phosphate. The deprotonation of the 2'-OH may occur synchronously with its attack on the adjacent 3' phospho group. An in-line displacement of the oxygen attached to the 5' carbon of the next nucleotide unit is thought to be assisted by His 119. Its protonated imidazolium group may transfer a proton to a phosphate oxygen atom prior to or synchronously with formation of the new P–O bond in step *a* (Fig. 12-25).⁷⁵⁴ The intermediate formed in step *a* is a cyclic 2', 3'-diphosphate which then undergoes hydrolysis by attack of a water molecule in step *b* to give the free nucleoside 3'-phosphate. The overall reaction is a two-step double-displacement, analogous to that with chymotrypsin, except that a neighboring group in the substrate rather than an amino acid side chain is the nucleophilic catalyst. The pH dependence of the enzyme is in agreement with this mechanism because there are two pK_a values of ~ 5.4 and ~ 6.4 which regulate the catalytic activity. Microscopic pK_a values of His 12 and His 119 have been measured by NMR spectroscopy as ~ 6.1 and ~ 6.3 and are shifted somewhat by binding of nucleotides.

A bacterial peptidase splits a 20-residue fragment containing His 12 from the N-terminal end of RNase A. This “S-peptide” can be recombined with the rest of the molecule, which is inactive, to give a functional enzyme called ribonuclease S. In a similar way, residues 119–124 of RNase A can be removed by digestion with carboxypeptidase to give an inactive protein which lacks His 119. In this case, a synthetic peptide with the sequence of residues 111–124 of RNase A forms a complex with the shortened enzyme restoring full activity.⁷⁵⁵

Stereochemical studies support in-line mechanisms for both the transesterification and hydrolysis steps of ribonuclease catalysis. For example, chiral uridine 2',3'-cyclic phosphorothioates are hydrolyzed with inversion of configuration, with the diastereoisomer shown yielding a 2'-monophosphothioate of the *R* configuration at phosphorus.



Transesterification step is also in-line⁷⁵⁷ as it is shown in Fig. 12-25. Study of kinetic isotope effects in H₂O–D₂O mixtures suggested that two protons may move synchronously as the enzyme–substrate complex passes through the transition state.⁷⁵⁸ Although RNase A is one of the most studied of all enzymes, there are still uncertainties about the mechanism. Is a proton removed first by His 12 (Fig. 12-25), as has long been assumed, or is a proton transferred first from His 119 to the oxygen of the phospho group?^{759,760} Is the reaction concerted, as suggested by kinetic isotope effects,⁷⁶¹ or is there a pentacovalent intermediate?

The specificity of RNase A for a pyrimidine on the 3' side of the phosphodiester bond that is cleaved is evidently ensured by the pair of hydrogen bonds from O-2' of the pyrimidine to the backbone NH of Thr 45 and a second from the N-4' proton to the side chain OH of the same threonine (Fig. 12-25). Other nucleases, such as ribonuclease T₂,⁷⁶² with different specificities also make use of hydrogen bonding of the base at the 3' side of the cleavage point with backbone amide groupings.

Various bacterial ribonucleases as well as the fungal ribonucleases T₁, U₁, and U₂ (see also Fig. 5-43) have amino acid sequences related to that of RNase A^{763,764,764a} but with distinctly different three-dimensional structures. The active sites contain Glu, His, and Arg side chains. For RNase T₁, Glu 58 and His 92 appear to provide acid–base catalysis with assistance from Tyr 38, Arg 77, and His 40.^{763,765} A glutamate carboxylate also appears to be the catalytic base in the related RNase, called **barnase**, from *Bacillus amyloliquefaciens*.⁷⁶⁶

In addition to extracellular digestive enzymes, the RNase family contains many intracellular enzymes that are involved in turnover of RNA.^{767,768} RNase H digests away RNA primers during DNA synthesis (Chapter 27). RNase H activity is also present in a domain of viral reverse transcriptases and is absolutely essential for the replication of HIV and other retroviruses.⁷⁶⁹ The structures of the reverse transcriptase RNase H domain and of the *E. coli* enzyme are similar.^{769–771} Unlike RNase A, the RNases H are metallo-enzymes which apparently contain two Mg²⁺ ions held by carboxylate groups and utilize a metal bound HO[–] ion as in previously discussed phosphatases. Secreted RNases sometimes have specific functions. For example, the 123-residue **angiogenin** is homologous to pancreatic RNases but acts to induce formation of new blood vessels (angiogenesis).^{772–774} This is essential to growth of solid cancers as well as for normal growth. The enzyme is a very poor catalyst but its RNase activity appears essential for its biological function. Mutation of any of the catalytic residues His 13, Lys 40, or His 114 abolishes all angiogenic activity. A neurotoxin secreted by eosinophils⁷⁷⁵ is one of a group of selectively toxic RNases.^{776,777} Intracellular RNases are often found as complexes of specific inhibitor proteins.⁷⁷⁸

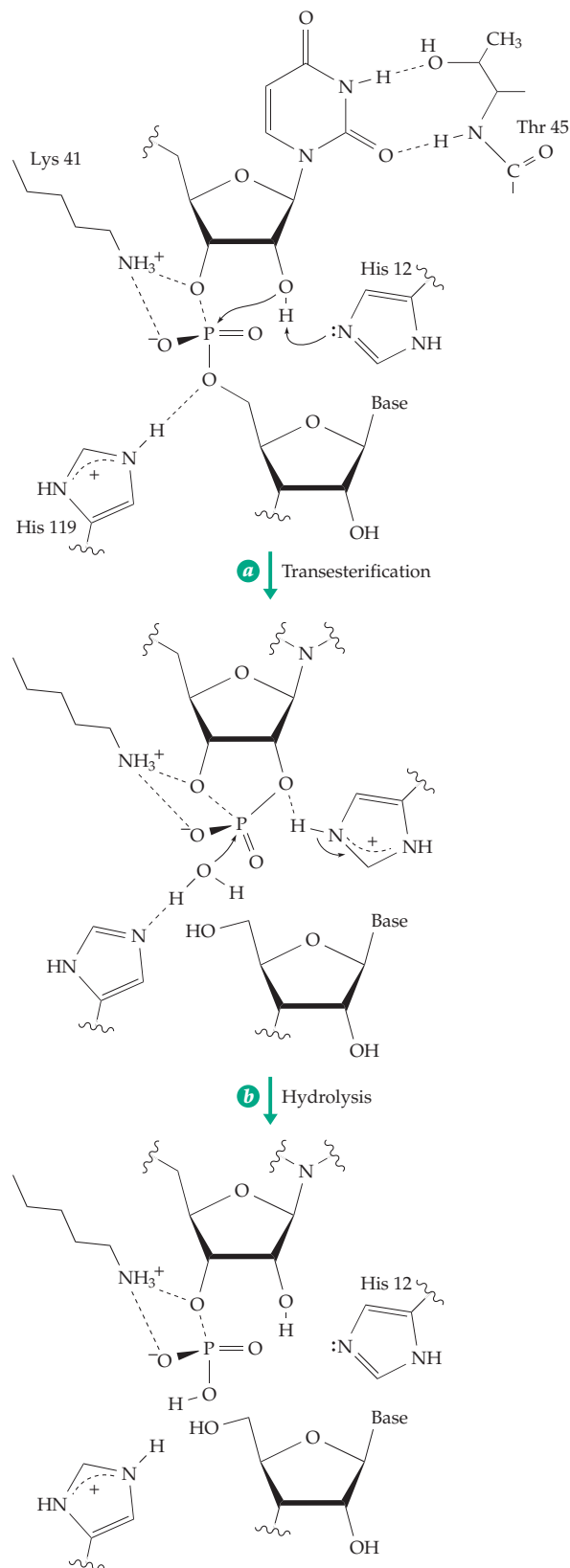


Figure 12-25 Proposed two-step in-line reaction mechanism for ribonuclease A. The hydrogen bonding that provides recognition of the pyrimidine base at the 3' end created by the cleavage is also shown. See Wladkowski *et al.*⁷⁵⁶

6. Ribonuclease P, Ribozymes, and Peptidyl Transferase

A very different ribonuclease participates in the biosynthesis of all of the transfer RNAs of *E. coli*.

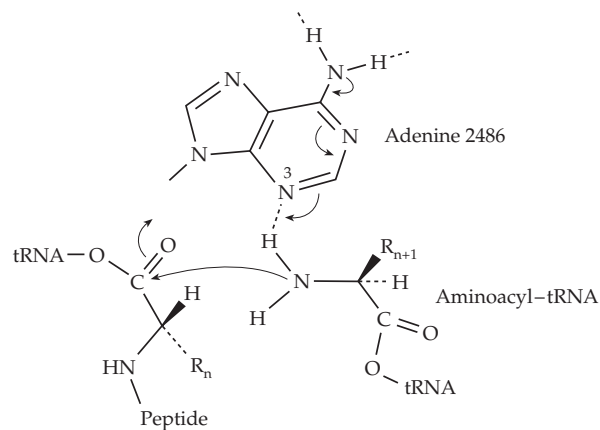
Ribonuclease P cuts a 5' leader sequence from precursor RNAs to form the final 5' termini of the tRNAs. Sidney Altman and coworkers in 1980 showed that the enzyme consists of a 13.7-kDa protein together with a specific 377-nucleotide RNA component (designated M1 RNA) that is about five times more massive than the protein.⁷⁷⁹ Amazingly, the M1 RNA alone is able to catalyze the ribonuclease reaction with the proper substrate specificity.^{780–782a} The protein apparently accelerates the reaction only about twofold for some substrates but much more for certain natural substrates. The catalytic center is in the RNA, which functions well only in a high salt concentration. A major role of the small protein subunit may be to provide counterions to screen the negative charges on the RNA and permit rapid binding of substrate and release of products.⁷⁸³ Eukaryotes, as well as other prokaryotes, have enzymes similar to the *E. coli* RNase P. However, the eukaryotic enzymes require the protein part as well as the RNA for activity.⁷⁸⁴

Thomas Cech and associates independently discovered another class of catalytic RNA molecules. These are **self-splicing RNAs** that cut out intervening sequences from themselves to generate ribosomal RNA precursors (see Chapter 28).^{785–787} They act only once and are therefore not enzymes. However, the introns that are cut out during self-splicing are **ribozymes** which, like the RNA from ribonuclease P, can act catalytically and have properties similar to those of protein enzymes. They exhibit the kinetic properties of enzymes and are denatured by heat. The RNAs are folded into compact structures resembling those of globular proteins. Like tRNA, they contain loops and hydrogen-bonded stems. Phylogenetic comparisons (Chapter 29) of the M1 RNA of ribonuclease P isolated from various species have allowed prediction of precise secondary structures.^{783,788,789} A simplified M1 RNA consisting of 263 nucleotides from conserved regions of the molecule is catalytically effective.⁷⁸³ Tetrahymena ribozyme also has a complex structure with a 247-nucleotide catalytic core formed by two structural domains (Fig. 12-26).^{790,791} The crystal structure of a third ribozyme, one found in the RNA of the human pathogen hepatitis delta virus (HDV), has also been determined.⁷⁹² It is a smaller 72-nucleotide self-cleaved molecule with a very different structure from that in Fig. 12-26. It makes use of a double pseudoknot (see Fig. 5-29) to bind the RNA into a compact, tightly hydrogen-bonded structure with a deep active site cleft. It is the fastest known naturally occurring self-cleaving RNA and is able to react at a rate of more than 1 s^{-1} at its optimum temperature of 65°C.

Smaller self-cleaving RNAs have been found among plant viruses and viroids. Many of them have a common catalytic core which can be converted into 30- to 40-nucleotide ribozymes. Only 17 nucleotides and three hydrogen-bonded helical stems are required to form the self-cleaving “hammerhead” domain, which has a structural similarity to the catalytic core of the *Tetrahymena* ribozyme. The **hammerhead ribozymes** (Fig. 12-27) represent one form of small ribozyme.^{793–797} Another is the **hairpin ribozyme** shown in Fig. 12-28,^{798,801} which also shows the even smaller lead-dependent “**leadzyme**,” a ribozyme that doesn't occur in nature.

In an intact viral self-cleaving RNA the entire catalytic center is formed from a single strand. Stems II and III of the hammerhead ribozyme (Fig. 12-27C) are closed by large loops. In the ribozyme shown, loop III has been cut off and stem II has been closed by a tight loop to form a compact catalytic RNA that will cut a substrate having a suitable nucleotide sequence for binding to the ribozyme. Only 12 bases in this ribozyme are highly conserved. By varying the sequences in the ribozyme half of stems I and III, catalysts that cleave after any sequence GUX, where X=A, C, or U, can be designed. Such catalysts are useful in the laboratory and potentially also in medicine.

What groups of a ribozyme bind to substrates and what groups participate in catalysis? Like peptides, RNAs have amide groups that can hydrogen bond to substrates. Adenine and cytosine can supply protonated amino groups which could participate in acid–base catalysis. This is evidently the case in the **peptidyl transferase** centers of ribosomes. The RNA in these centers catalyzes a transesterification in which an aminoacyl group is transferred from an aminoacyl-tRNA onto the growing polypeptide chain attached to a second tRNA molecule. The reaction is evidently catalyzed by a universally conserved adenine ring located at position 2486 in the *Haloarcula marismortii* 23S RNA (position 2451 in *E. coli*). There are no protein groups within 1.8 nm of the location of peptide bond synthesis.^{798a} The active site adenine appears to be much more basic than normal. A high pK_a of 7.6 controls the peptidyl transferase, and also controls the methylation of the active site adenine by dimethylsulfate.^{798b} The site of protonation is thought to be largely N3 of adenine 2486, which is probably the basic center involved in catalysis. The peptidyltransferase reaction may be initiated as follows.^{798a}



A similar catalytic mechanism is probably used by the small 85-nucleotide hepatitis delta virus ribozyme whose catalytic base is thought to be N3 of cytosine 75, which is associated with a pK_a of ~ 6.1 .^{798c} Both this HDV ribozyme and the ribosomal RNA resemble serine proteases with histidine as the catalytic base. However, the self-splicing RNA of *Tetrahymena* initiates a nucleophilic attack with the 3'-OH group of a guanosine molecule that is bound to a site in the P7 region (Fig. 12-16A) and which acts as a cofactor (see Fig. 28-18). Ribonuclease P and all group I and II self-cleaving introns also use an external nucleophile such as a guanosine -OH and form 3'-OH and O-phosphate or

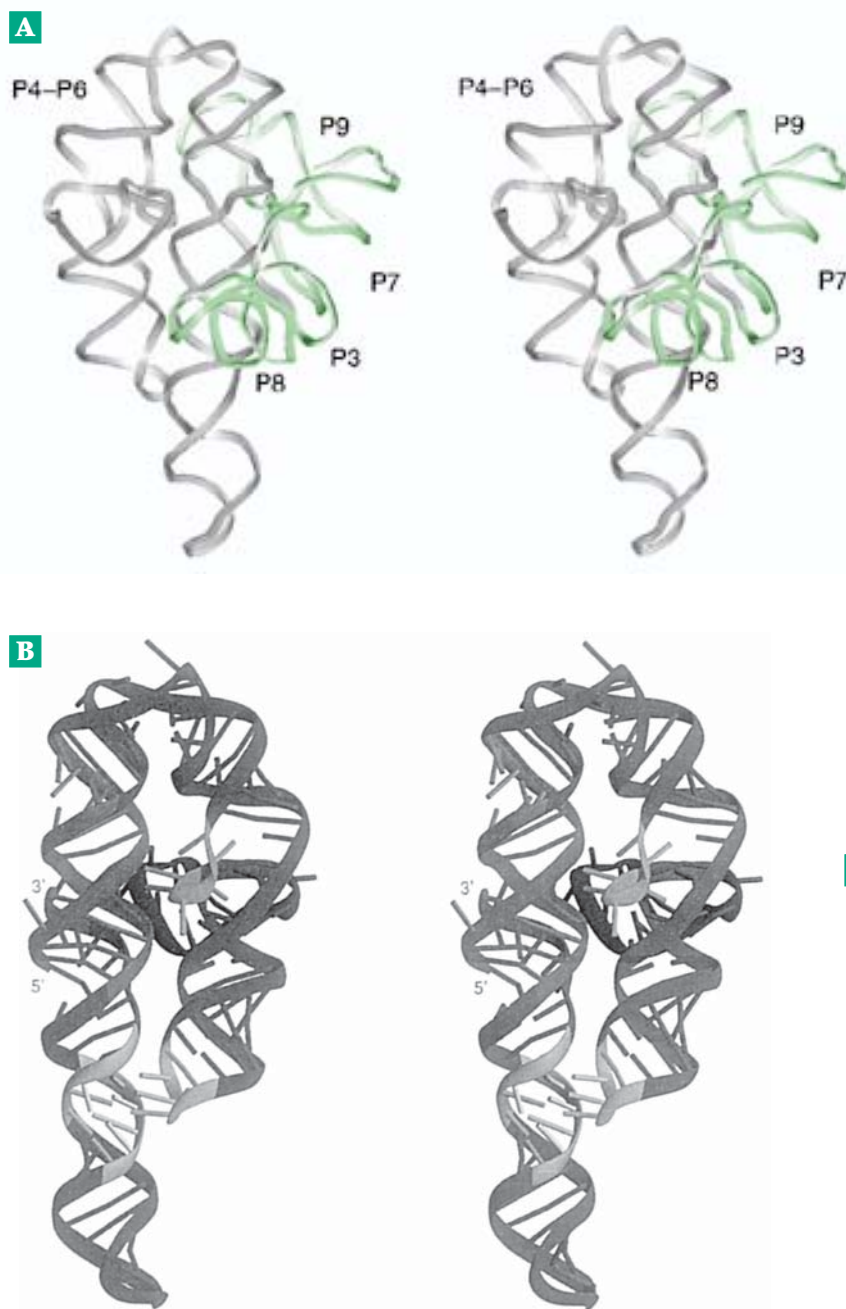
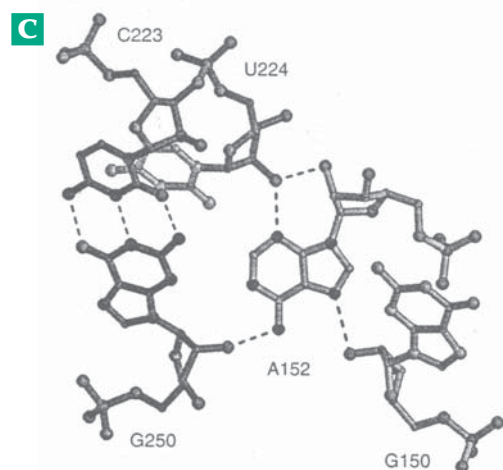
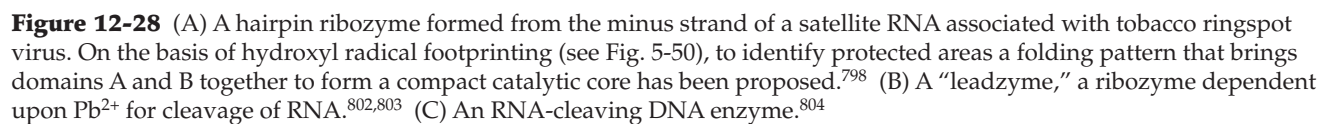
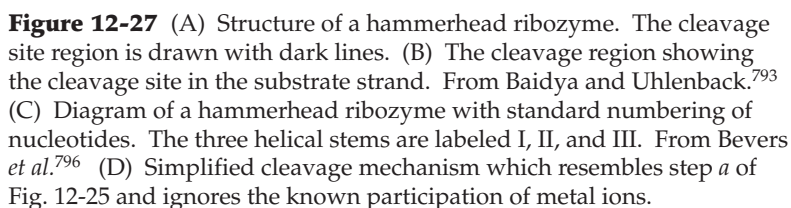


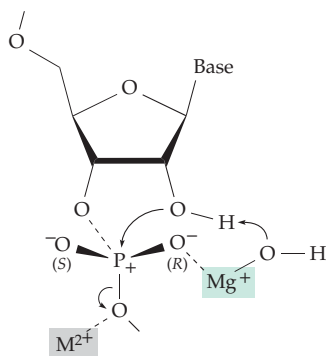
Figure 12-26 (A,B) Stereoscopic views of the *Tetrahymena* ribozyme. (A) Ribbon tracing of the phosphodiester backbone. The successive conserved structural elements, from the 5' and 3' ends are designated P1 to P9. The P4 – P6 region forms one major domain, while the P3 and P5 – P9 regions form a second domain which folds around the first. From Golden *et al.*⁷⁹⁰ (B) Structure of the P4 – P6 domain viewed from the back side as pictured in (A). From Cate *et al.*⁷⁹¹ The active site lies between this domain and the larger one which folds over it leaving the active site in a deep crevice.⁷⁹⁰ The active site is centered above the “tetraloop” GAAA (residues G150 to A153), which is shown with lighter shading (in B) near the upper center of the P4 – P6 domain. (C). Details of one of three layers of hydrogen-bonded interactions between purine and pyrimidine bases involved in interaction of the tetraloop with the adjacent helix in the P5 region. Cleavage occurs in the P1 domain, which folds into the active site and is not shown in these drawings. Courtesy of Thomas Cech.





phosphodiester ends at the cleavage points. Hammerhead ribozymes utilize the 2'-OH of the ribose at the cleavage site to form a 2',3'-cyclic phosphate ester as in step *a* of the ribonuclease A reaction (Fig. 12-25) and as indicated in Fig. 12-27D. Ribozymes act by in-line mechanisms, causing inversion of the configuration at the phosphorus.^{799,800}

Most ribozymes, as well as RNase P,^{805,806} require one or two metal ions for activity.⁸⁰⁷ Magnesium ions predominate and many Mg^{2+} ions are bound at distinct sites in crystalline ribozymes. Hammerhead and hairpin ribozymes work reasonably well with monovalent ions. One proposed mechanism is for an HO^- ion bound to the Mg^{2+} to remove H^+ from the 2'-OH of the ribose ring as follows:



When the *pro-R* oxygen of the phospho group (labeled *R* in the accompanying structure) was replaced by sulfur the rate of cleavage decreased 10^3 -fold. However, the rate was restored fully when Cd^{2+} was added, presumably because of the high affinity of cadmium ions for sulfur ligands.⁸⁰⁸ This supports the possibility of a dual role for Mg^{2+} in activating a water molecule to provide HO^- and in stabilizing negative charges on the phospho group by interaction with the *pro-R* oxygen in the transition state. Several investigators have suggested that *two* metal ions may be needed. One possibility is for a mechanism similar to that proposed for alkaline phosphatase (Fig. 12-25) and other phosphotransferases.^{809,810} The *Tetrahymena* ribozyme functions best if both Mg^{2+} and Mn^{2+} are present.⁸¹¹ A second metal may act as a Lewis acid facilitating loss of the 5'-OH as indicated by M^{2+} (in gray) in the preceding diagram. Metal ions may also participate in conformational changes as well as have structural functions in ribozymes.^{812,813} Studies of the kinetics of action of hammerhead ribozymes have suggested that the intrinsic ribozyme-substrate binding energy is utilized for catalysis.⁸¹⁴ This may be possible because the ribozyme is only partially folded in the ground state but it folds into a tighter conformation in the transition state.⁸¹⁵

Methods have been devised for generating enormous numbers of RNA molecules with random sequences and for selecting those with unusual catalytic

activities.^{816–819} Among the new catalysts produced in this way are very small ribozymes that cleave RNA specifically in the presence of Pb^{2+} (Fig. 12-28B).^{802,803} The leadzyme is more active with neodymium (Nd^{3+}) + Pb^{2+} than with lead alone, suggesting a two-metal mechanism.⁸²⁰ Other artificial ribozymes include RNA ligases,⁸¹⁷ acyltransferases,⁸²¹ and DNA hydrolases.^{822,823} Is it possible to find a DNA enzyme? Without the 2'-OH of ribose to form hydrogen bonds it seemed doubtful, but an RNA-cleaving DNA enzyme has been selected from a population of $\sim 10^{14}$ different small DNA molecules. The DNA enzyme (Fig. 12-28C) will cleave RNA, whose sequence fulfills the base pairing requirements of two 8-deoxynucleotide recognition domains. Cleavage occurs between an unpaired purine and a paired pyrimidine using a metal-dependent mechanism that gives a 2',3'-cyclic phosphate as in ribonuclease A cleavage.⁸⁰⁴

7. Deoxyribonucleases (DNases)

A multitude of nucleases cleave DNA, single- or double-stranded. They range from the pancreatic digestive enzyme DNase I through specialized nucleases that function during DNA repair and the hundreds of restriction endonucleases that have become so valuable in modern laboratory work. Some nucleases leave a 3'-phosphate ester at a cut end in a DNA chain, while others leave a 5'-phosphate end.⁸²⁴ Many nucleases are dealt with in later chapters. Only a few will be mentioned here.

One of the most studied enzymes of this group is the 149-residue micrococcal (staphylococcal) nuclease from *Micrococcus* which cleaves either RNA or single- or double-stranded DNA. The relatively nonspecific enzyme cuts nearly randomly at the 5' side of the phosphodiester linkages, leaving 3'-phosphate groups. It enhances the uncatalyzed hydrolysis rate at least 10^{16} -fold.^{825–827} The crystal structure showed that the majority of the acidic and basic side chains of the protein interact with each other through clusters of hydrogen bonds. At the active site the side chains of both Arg 35 and Arg 87 form pairs of hydrogen bonds to the 5'-phosphate group of the specific inhibitor deoxythymidine 3',5'-diphosphate (Fig. 12-29). While Arg 87 appears to be in a position to protonate the leaving group $-\text{O}^-$, ^{13}C NMR experiments showed that all of the arginine side chains had pK_a values above 11.6. However, Tyr 85 has a pK_a of 9.5, which appears to control k_{cat}/K_m .⁸²⁸ A Y85F mutant lacks this pK_a . The X-ray structure also suggests that Glu 43 may be the attacking nucleophile and that it may deprotonate a water molecule bound to the Glu 43 carboxylate. The resulting HO^- probably carries out a direct in-line attack as shown in Fig. 12-29. Mutants such as E43D, E43Q, and E43S have greatly decreased activity,⁸²⁹ in

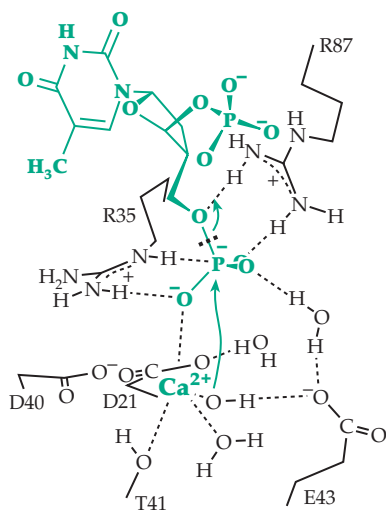


Figure 12-29 Drawing showing the hydrogen-bonding interactions between the guanidinium ions of arginines 35 and 87 of the micrococcal (staphylococcal) nuclease with the 5'-phosphate of the inhibitor thymidine 3',5'-diphosphate in the complex of E + I + Ca^{2+} . A possible mechanism is illustrated. A hydroxyl ion bound to Ca^{2+} carries out an in-line attack on the phosphorus. See Libson *et al.*⁸²⁶

agreement with this mechanism. The nearby Ca^{2+} is essential. For this nonspecific nuclease there is no hydrogen bonding of a purine or pyrimidine base of the substrate to the enzyme.

The digestive enzyme, pancreatic DNase I, makes single-stranded cuts in double-stranded (ds) DNA. An exposed strand of peptide chain from the enzyme binds into the minor groove of B-type DNA.⁸³⁰ Because this groove becomes too narrow in long (A+T) rich sequences, they are cleaved slowly. Certain hypersensitive sites are cleaved very rapidly, perhaps because the DNA at these regions is bent or is able to bend to give a very good fit to the enzyme active site. A histidine which is hydrogen bonded to a nearby carboxylate of a glutamate side chain appears to be a catalytic base that acts upon a water molecule as in phospholipase A (Section D,10), displacing the 3' oxygen of the phosphodiester linkage. An imidazolium group from a second histidine is hydrogen bonded to an aspartate carboxylate and a tyrosine -OH to form a catalytic triad that can protonate the 3' -O⁻ as it is displaced.⁸³¹ Two Mg^{2+} ions are also required. Both are held by different carboxylate side chains and may also interact with oxygen atoms of the phospho group to neutralize charge and stabilize the transition state.

In contrast to DNase I, the **restriction endonucleases**, which are discussed in Chapter 5, Section H,2 and in Chapter 26, have precise substrate sequence specificities. Three of the best known restriction endonucleases are called *EcoRI*,^{832–834} an enzyme which binds

to and cuts both strands of the palindromic sequence 5'-GAATTC; *EcoRV*,⁸³⁵ which cuts both strands in the center of the sequence 5'-GATATC; and *BamHI*, which binds to the sequence 5'-GGATCC and cleaves after the 5' G on each strand.⁸³⁶ A high-resolution structure is also known for *Cfr10I*, which recognizes the less strict sequence 5'-PuCCGGPy and cleaves both strands after the 5' Pu.⁸³⁷ All of these enzymes require Mg^{2+} and have active sites containing carboxylate groups. Two-metal mechanisms have been suggested.

Restriction endonuclease *EcoRI* is able to cut a chain in dsDNA which has a chiral phosphorothioate group at the specific cleavage site.⁸³⁸ The reaction occurs with inversion of configuration at phosphorus, suggesting direct in-line attack by a hydroxyl ion generated from H_2O .

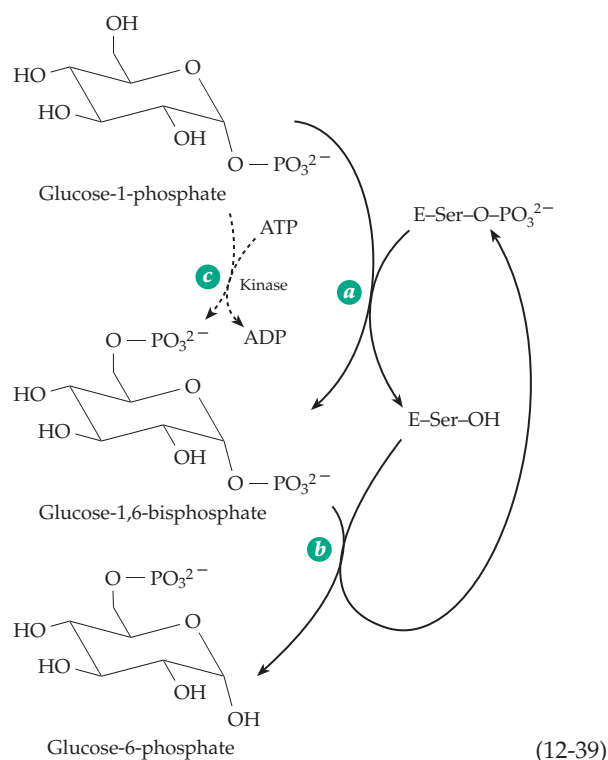
Attempts are being made to design semisynthetic restriction endonucleases specific for single-stranded DNA or RNA. For example, an oligonucleotide with a sequence complementary to a sequence adjacent the linkage that is to be cut can be covalently linked to a relatively nonspecific nuclease. Such an enzyme derived from micrococcal nuclease cuts a single-stranded chain of either DNA or RNA adjacent to the double-stranded region of the ES complex.⁸³⁹

8. Mutases

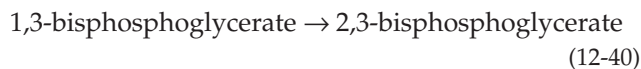
Phosphotransferases that shift phospho groups from one position within a substrate to another are often called **mutases**. For example, **phosphoglucumutase** catalyzes the interconversion of glucose 1-phosphate and glucose 6-phosphate, an important reaction that bridges glycogen metabolism and glycolysis (Fig. 11-2). This 561-residue protein operates through formation of an intermediate **phosphoenzyme**.^{840–842} The phospho group becomes attached to the OH of Ser 116 and can be transferred either to the 6 or the 1 position of a glucose phosphate (step *a* and reverse of step *b* in Eq. 12-39).

The two-step reaction accomplishes the reversible isomerization of glucose 1-phosphate and glucose 6-phosphate via **glucose 1,6-bisphosphate**. Evidently, the glucose bisphosphate, without leaving the enzyme is reoriented to allow transfer of the phospho group to either the 1- or 6-position.⁸⁴¹ The phospho enzyme is relatively unstable and can undergo hydrolysis to free enzyme and P_i . To prevent loss of active enzyme in this way, a separate reaction (catalyzed by a kinase; Eq. 12-39, step *c*)⁸⁴³ generates glucose 1,6-bisphosphate, which rephosphorylates any free enzyme formed by hydrolysis of the phospho enzyme. The glucose 1,6-bisphosphate can be regarded as a cofactor or **cosubstrate** for the reaction.

Phosphoglycerate mutase, which interconverts 2-phosphoglycerate and 3-phosphoglycerate in glycolysis (Fig. 10-3, step *c*), functions by a similar mecha-



nism.^{843–845} However, the enzyme-bound phospho group is carried on an imidazole group. The essential cosubstrate required by some phosphoglycerate mutases is 2,3-bisphosphoglycerate.⁸⁴⁵ It is formed from the glycolytic intermediate 1,3-bisphosphoglycerate (Fig. 10-3) by action of another mutase, **bisphosphoglycerate mutase**.^{843,846}



This is also the pathway for synthesis of 2,3-bisphosphoglycerate in red blood cells where it serves as an important allosteric regulator (Chapter 7). Two human phosphoglycerate mutase isoenzymes are known. One is found in muscle and the other in brain and other tissues.⁸⁴⁴ A hereditary lack of the muscle type enzyme is one of the known types of glycogen storage diseases (Box 20-D).

Human **phosphomannomutase**, which catalyzes the interconversion of mannose 1- and 6- phosphates, appears to carry the phospho group on an aspartate side chain in the sequence **DXDX** (T/V), which is conserved in a family of phosphomutases and phosphatases.⁸⁴⁷ The first aspartate in the sequence is phosphorylated during the enzymatic reaction.

9. Molecular Properties of Kinases

Kinases transfer phospho groups from polyphosphates such as ATP to oxygen, nitrogen, or sulfur

atoms of a second substrate.⁸⁴⁸ Examples include **hexokinase**, the enzyme responsible for synthesis of glucose 6-phosphate from free glucose and ATP (Fig. 11-2, step *a*); **phosphofructokinase**, which forms fructose 1,6 bisphosphate in the glycolysis pathway (Fig. 11-2, step *b*); and **phosphoglycerate kinase**, and **pyruvate kinase**, both of which form ATP from ADP in the glycolysis pathway (Fig. 10-3, steps *b*, *c*, and *f*). There are many others. Kinases vary greatly in size and in three-dimensional structure. For example, a small **adenylate kinase**, which phosphorylates AMP to ADP (Eq. 6-65), is a 22-kDa monomer of 194 residues. Pyruvate kinase is a tetramer of 60-kDa subunits and muscle phosphofructokinases are tetramers of 75- to 85-kDa subunits. The three-dimensional structures also vary. While all kinases consist of two domains built around central β sheets (Fig. 12-30), there are several different folding patterns.⁸⁴⁹ The two-domain structures all have deep clefts which contain the active sites. Both adenylate kinase (Fig. 12-30) and hexokinase crystallize in two or more forms with differing conformations.⁸⁵⁰ This and other evidence suggests that as a kinase binds and recognizes its two correct substrates, the active site cleft closes by a hinging action that brings together the reacting molecules in the correct orientation.^{851–853} In the crystal structure shown in Fig. 12-30 both ADP and AMP are bound in a nonproductive complex. If the ADP were replaced with ATP (or the AMP with a second ADP) to form a productive complex the two reacting phospho groups would be ~ 0.8 nm apart. A reaction could not occur without further closing of the active site cleft.⁸⁵⁴ Evidence for domain closure has been obtained for many other kinases. For example, substrate complexes of phosphoglycerate kinase have been crystallized in both “open” forms and “closed” forms in which a 30° hinge-bending movement has brought the ligands together for an in-line phospho group transfer.⁸⁵⁵

For many enzymes an ATP binding site has been revealed by study of nonhydrolyzable analogs of ATP such as “AMP-PNP” whose structure is shown in Fig. 12-31.⁸⁵⁶ AMP-PNP has been used in thousands of investigations of ATP-dependent processes.⁸⁵⁷ For example, the structure of a phosphoglycerate $\cdot \text{Mg}^{2+} \cdot \text{AMP-PNP}$ complex in the active site of phosphoglycerate kinase has been determined.⁸⁵⁸ Modeling of a transition state complex indicates that all three negatively charged oxygens of the ATP portion of structure are stabilized by hydrogen bonding.⁸⁵² Related analogs such as AMP-PCH₂P (Fig. 12-3) have been used in similar ways.⁸⁵⁹ Another analog $\text{Mg}^{2+} \cdot \text{Ap}_5\text{A}$ (Fig. 12-31) is a bisubstrate inhibitor which binds to adenylate kinases, fixing the enzymes in a closed conformation that is thought to resemble the transition state.^{859,860}

The Mg^{2+} complex of ATP is regarded as the true substrate for kinases. The metal usually also binds both to the phospho groups of ATP and to groups on

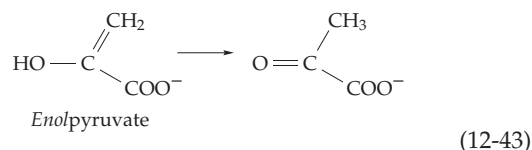
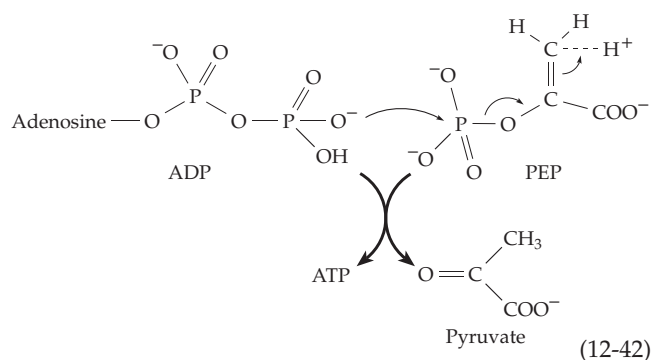
Hexokinase, the enzyme that phosphorylates glucose to glucose 6-phosphate, exists as four isoenzymes in mammals. Hexokinases I, II, and III are large ~100-kDa monomers with similar amino acid sequences, a single active site, and complex allosteric regulatory properties. The three-dimensional structure is known for hexokinase I, which is the pacemaker of glycolysis in the brain.^{874,875} Hexokinase IV (glucokinase) is a 50-kDa protein that is found in liver and in the β cells of the pancreas. It has a low K_m and is not inhibited by glucose 6-phosphate, properties that allow for rapid uptake of glucose after a meal.⁸⁷⁶ Its properties are similar to those of a major isoenzyme of yeast,⁸⁷⁷ a dimer of identical 50-kDa subunits whose structure is also known.⁸⁷⁸ The glucose-binding residues are conserved in yeast hexokinase, in glucokinase, and in brain hexokinase. The sequence of the latter suggests that it may have arisen by a doubling of a shorter hexokinase gene.⁸⁷⁹

Fructose 6-phosphate kinase (phosphofructokinase) has attracted much attention because of its regulatory properties (Chapter 11).^{880–882} Prokaryotic forms are somewhat simpler.⁸⁸³ The related fructose 6-phosphate 2-kinase is a component of a bifunctional kinase-phosphatase (Fig. 11-2, steps *d* and *e*) and has a structure similar to that of adenylate kinase.⁸⁸⁴

Phosphoglycerate kinase is encoded by the mammalian X chromosome. Several mutant forms are associated with hemolytic anemia and mental disorders.^{885,886} Mutation of Glu 190, which is in the hinge region far from the active site, to Gln or Asn markedly reduces enzymatic activity.⁸⁸⁷

Creatine kinase transfers a phospho group to a nitrogen atom of the guanidinium group of creatine (Eq. 12-31 and Fig. 12-19). Several isoenzymes participate in its function of buffering the ATP level in tissues such as muscle fibers, neurons, photoreceptors, and spermatozoa which experience high and fluctuating energy needs.⁸⁸⁸ A form from the mitochondrial intermembrane space of chicken heart is an octomer of 380-residue subunits.⁸⁸⁹ The structurally and mechanistically similar **arginine kinase** has an analogous function in many invertebrates, e.g., in the horseshoe crab, which provided enzyme for a structure determination.⁸⁹⁰

The ~500-residue subunits of **pyruvate kinase** consist of four domains,⁸⁹¹ the largest of which contains an 8-stranded barrel similar to that present in triose phosphate isomerase (Fig. 2-28). Although these two enzymes catalyze different types of reactions, a common feature is an enolic intermediate. One could imagine that pyruvate kinase protonates its substrate phosphoenolpyruvate (PEP) synchronously with the phospho group transfer (Eq. 12-42). However, the enzyme catalyzes the rapid conversion of the enolic form of pyruvate to the oxo form (Eq. 12-43) adding the proton stereospecifically to the *si* face. This and other evidence favors the enol as a true intermediate



and a product of the phosphotransfer step.⁸⁹¹ Pyruvate kinase requires not only two equivalents of a divalent cation such as Mg^{2+} or Mn^{2+} but also a monovalent cation, usually K^+ . However, Li^+ , Na^+ , NH_4^+ , Rb^+ , Tl^+ , and others can substitute. The monovalent cation induces an essential conformational change. Using ^{205}Tl NMR it was found that the thallium ion binds about 0.6 nm from Mn^{2+} that is also present in the active site.⁸⁹² All three metals interact directly with the γ phospho group of ATP.⁸⁹¹ Pyruvate kinase is a regulated allosteric enzyme present in four isoenzymic forms in mammals.^{893–894a}

Protein kinases, which were discussed in Chapter 11, phosphorylate selected –OH groups of serine, threonine, and tyrosine side chains in proteins. Examination of the sequences in the complete genome of yeast (*Saccharomyces cerevisiae*) indicates the presence of at least 113 protein kinase genes, which account for ~2% of the total DNA.⁸⁹⁵ Higher eukaryotes have more. While structures of these enzymes vary widely, they share a common two-domain catalytic core structure.^{896,897} The best known, and one of the simplest of them, is the catalytic subunit of cyclic AMP-dependent protein kinase.^{897,898} The substrate $MgATP$ binds into the active site cleft with the γ -phospho group protruding to meet the appropriate site of a bound protein substrate (Fig. 12-32). Most protein kinases are regulated by an “activation loop” that must be correctly placed before the ES complex can be formed. As discussed in Chapter 11, regulation of the cAMP-dependent kinase depends upon inhibition by a regulatory subunit. In the cAMP-dependent kinases and many other protein kinases the activation loop (not shown in Fig. 12-32), which helps to form the substrate site, contains a phosphothreonine residue which is essential for activity. It is a stable feature of the cAMP-activated kinase, incorporated at position 197, but for some tyrosine kinases it is generated by autophosphorylation.⁸⁹⁹

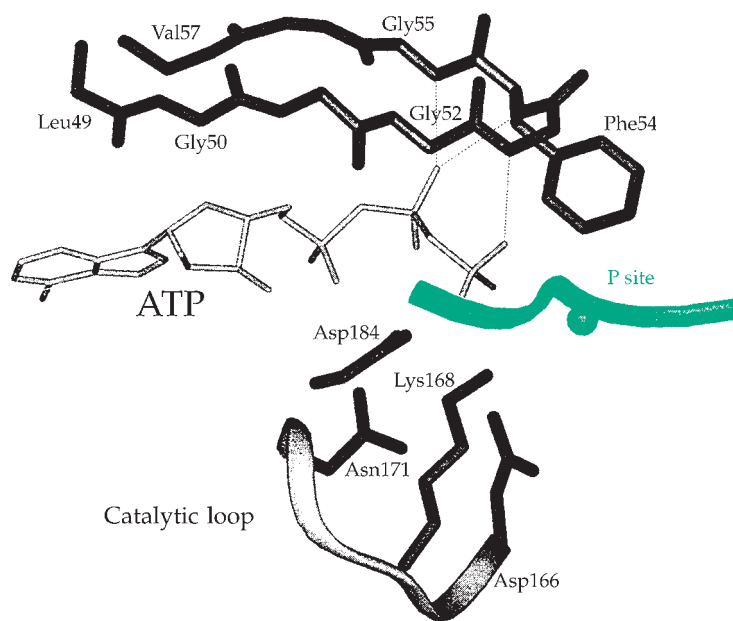


Figure 12-32 The active site of cyclic AMP-dependent protein kinase with bound ATP and a segment of an inhibitory peptide (green) blocking the substrate site. The small ball is an alanine side chain; it would be Ser or Thr in a substrate. The ATP is clamped by the glycine-rich loop at the top of the figure. Hydrogen bonds from peptide NH groups bind the β and γ phospho groups of ATP. A magnesium ion (not shown) also binds to the β and γ phosphate oxygens and to the invariant Asp 184. The Asp 166 carboxylate is probably the catalytic base for deprotonation of the substrate -OH . From Grant *et al.*⁸⁹⁶ Courtesy of Susan S. Taylor.

For the enzyme to be activated the phosphothreonine must form a hydrogen-bonded ion pair with Lys 189 and be hydrogen bonded to His 87, tying together critical regions of the catalytic domain. The regulatory subunits, unless occupied by cAMP (Chapter 11) are competitive inhibitors of the substrates.⁹⁰⁰ Cyclic GMP-activated kinases also have distinct functions.⁹⁰¹ In **tryosine kinases**^{902–904} C-terminal src-homology domains (Fig. 7-30 and Chapter 11) fold over and interact in an inhibitory fashion until an appropriate activating signal is received (see Fig. 11-13).

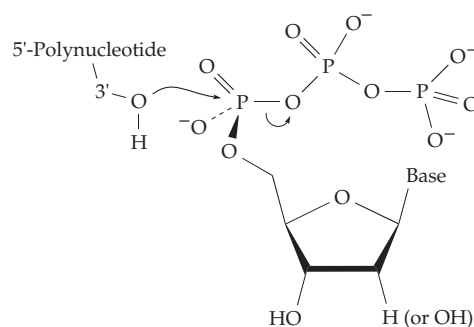
A relative of the kinases is **adenylate cyclase**, whose role in forming the allosteric effector 3',5'-cyclic AMP (cAMP) was considered in Chapter 11. This enzyme catalyzes a displacement on P_{α} of ATP by the 3'-hydroxyl group of its ribose ring (see Eq. 11-8, step *a*). The structure of the active site is known.⁹⁰⁵ Studies with $\text{ATP}\alpha\text{S}$ suggest an in-line mechanism resembling that of ribonuclease (step *a*, Eq. 12-25). However, it is Mg^{2+} dependent, does not utilize the two-histidine mechanism of ribonuclease A, and involves an aspartate carboxylate as catalytic base.⁹⁰⁶ All isoforms of adenylate cyclase are activated by the α subunits of some G proteins (Chapter 11). The structures⁹⁰⁷ of $G_{s\alpha}$ and of its complex with adenylate kinase⁹⁰⁵ have been determined. The $G_{s\alpha}$ activator appears to serve as an allosteric effector.

Guanylate cyclases, which form cyclic GMP, occur in particulate and soluble forms.⁹⁰⁸ The latter have been of great interest because they are activated by nitric oxide (NO). The soluble guanylate cyclases are $\alpha\beta$ heterodimers. The C-terminal regions of both α and β subunits are homologous to the catalytic domain of adenylate cyclase. The N-terminal domain of the α subunits contains heme whose Fe atom is coordinated

by a histidine imidazole.^{908,908a} This iron atom is apparently the receptor for NO, a major gaseous hormone, which is discussed in Chapter 18.

10. Nucleotidyl Transferases

An important group of enzymes transfer substituted phospho groups, most often nucleotidyl groups. The nucleases, ATPases, and GTPases, which have already been discussed, belong to this group as do the nucleic acid synthesizing enzymes, the **DNA** and **RNA polymerases**,^{909–911} **reverse transcriptase**,⁹¹² and **topoisomerases**. As with other phosphotransferases, the nucleotidyl transfers occur with inversion^{913,913a} and crystallographic investigations also support in-line mechanisms as illustrated in the following scheme. Two metal ions assist.



The inorganic pyrophosphate formed is hydrolyzed to inorganic phosphate by pyrophosphatase. Specific information about the polymerases and topoisomerases is given in Chapters 27 and 28.

E. The Adenylate Kinase Fold, the P Loop, and ATPases and GTPases

A magnesium–ATP-binding fragment consisting of ~40 residues at the N terminus of adenylate kinase contains sequences homologous to those in the GTP-

binding “G proteins” such as the protooncogenes *ras* (Fig. 11-7A) and also to sequences in myosin and in mitochondrial ATP synthase.⁹¹⁴ This includes the glycine-rich “P loop” which extends from Gly 15 to Gly 22 in the porcine cytosolic enzyme and contains a highly conserved lysine [Lys 21 of porcine adenylate

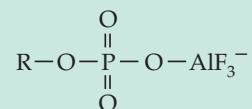
BOX 12-F THE TOXICITY OF ALUMINUM

Aluminum, in the form of oxides and silicate minerals, is the most abundant element in the earth's crust. Yet it appears to be actively excluded from most living organisms. It seems surprising that a naturally occurring Al^{3+} -dependent enzyme hasn't been found, but there is no evidence that Al is an essential element. Until recently, it was usually regarded as harmless. The aluminum salts known as alums are used in baking powders and have been added to pickles and other foods. Aluminum sulfate is often used as a coagulant to clarify turbid drinking water. Because of the insolubility of $\text{Al}(\text{OH})_3$ and of aluminum phosphate the concentration of Al^{3+} is very low at neutral pH. However, small amounts of AlOH^{2+} , $\text{Al}(\text{OH})_2^+$, and $\text{Al}(\text{OH})_4^-$ are present in water. Fluoride complexes such as AlF_2^{2+} may also be present. Soluble complexes, such as that with citrate, may permit some Al^{3+} to be absorbed by the body.^{a,b}

The toxicity of aluminum has been recognized most clearly by the development of bone disease caused by deposition of Al in bones of patients on hemodialysis^{a-c} and in infants on intravenous therapy.^{d,e} Excessive Al in the water used for dialysis may also cause brain damage. Dietary aluminum may be one cause of Alzheimer's disease,^{f-h} but this is controversial as is a possible role of aluminum in vaccines in causing inflammation in muscle.^{i,j} Solubilization of soil aluminum by acid rain has been blamed for the decline of forests in Europe and North America,^j for the death of fish in acid waters,^k and for very large reductions in yield for many crops.^{l,m} An aluminum-resistant strain of buckwheat makes and secretes from its roots large amounts of oxalate which binds and detoxifies the Al^{3+} ions.^m

Al^{3+} has a radius somewhat less than that of Fe^{3+} (Table 6-10) and it may sometimes occupy empty Fe^{3+} binding sites. Thus, the transferrinⁿ in blood carries some Al^{3+} , although citrate is probably a more important carrier.^o Al^{3+} binds preferentially to oxygen ligands and can compete with Mg^{2+} . However, the slower rate of ligand exchange reactions with Al^{3+} may interfere with the proper functioning of the metal. Brain hexokinase is strongly inhibited^p by Al^{3+} and the binding of Al^{3+} to tubulin decreases

the rates of GTP hydrolysis and of Ca^{2+} -induced depolymerization of microtubules.^q Aluminofluoride ions, such as AlF_4^- , react with phosphates to form ions such as,



which may be potent competitive inhibitors of enzymes acting on ATP, GTP, or other phosphate-containing substrate.^{r,s} However, Fe^{3+} can be replaced by Al^{3+} in a purple acid phosphatase (Chapter 16) with retention of good catalytic activity.^t

^a Martin, R. B. (1986) *Clinical Chemistry* **32**, 1797–1806

^b Macdonald, T. L., and Martin, R. B. (1988) *Trends Biochem. Sci.* **13**, 15–19

^c Address, D. L., Kopp, J. B., Maloney, N. A., Coburn, J. W., and Sherrard, D. J. (1987) *N. Engl. J. Med.* **316**, 292–296

^d Sedman, A. B., Klein, G. L., Merritt, R. J., Miller, N. L., Weber, K. O., Gill, W. L., Anand, H., and Alfrey, A. C. (1985) *N. Engl. J. Med.* **312**, 1337–1342

^e Bishop, N. J., Morley, R., Chir, B., Day, J. P., and Lucas, A. (1997) *N. Engl. J. Med.* **336**, 1557–1561

^f Good, P. F., and Perl, D. P. (1993) *Nature (London)* **362**, 418

^g Shen, Z. M., Perczel, A., Hollósi, M., Nagypál, I., and Fasman, G. D. (1994) *Biochemistry* **33**, 9627–9636

^h Walker, P. R., LeBlanc, J., and Sikorska, M. (1989) *Biochemistry* **28**, 3911–3915

ⁱ Landsberg, J. P., McDonald, B., and Watt, F. (1992) *Nature (London)* **360**, 65–68

^{ij} Malakoff, D. (2000) *Science* **288**, 1323–1324

^j Godbold, D. L., Fritz, E., and Hüttermann, A. (1988) *Proc. Natl. Acad. Sci. U.S.A.* **85**, 3888–3892

^k Birchall, J. D., Exley, C., Chappell, J. S., and Phillips, M. J. (1989) *Nature (London)* **338**, 146–148

^l Barinaga, M. (1997) *Science* **276**, 1497

^m Ma, J. F., Zheng, S. J., Matsumoto, H., and Hiradate, S. (1997) *Nature (London)* **390**, 569–570

ⁿ Roskams, A. J., and Connor, J. R. (1990) *Proc. Natl. Acad. Sci. U.S.A.* **87**, 9024–9027

^o Martin, R. B., Savory, J., Brown, S., Bertholf, R. L., and Wills, M. R. (1987) *Clinical Chemistry* **33**, 405–407

^p Viola, R. E., Morrison, J. F., and Cleland, W. W. (1987) *Biochemistry* **19**, 3131–3137

^q Macdonald, T. L., Humphreys, W. G., and Martin, R. B. (1987) *Science* **236**, 183–186

^r Troullier, A., Girardet, J.-L., and Dupont, Y. (1992) *J. Biol. Chem.* **267**, 22821–22829

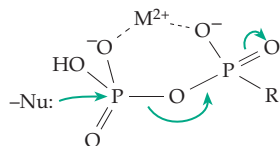
^s Chabre, M. (1990) *Trends Biochem. Sci.* **15**, 6–10

^t Merx, M., and Averill, B. A. (1999) *J. Am. Chem. Soc.* **121**, 6683–6689

kinase and Lys 17 of the archaeal enzyme (Fig. 12-30)] and of the *ras* oncogene product).⁹¹⁵ The lysine side chain appears to interact with the β and / or γ phospho group of ATP or GTP. The peptide chain of the P loop wraps around the β phospho group as the chain turns from the first β strand into the first helix. This can be seen in Fig. 12-30, in which the P loop surrounds the β phospho group of ADP. Two peptide NH groups also bind to the β phospho group on a side opposite to that occupied by Mg^{2+} . A similar loop in the dinucleotide-binding domain of a dehydrogenase can be seen hydrogen bonded to the P_α and P_β phospho groups of NAD^+ in Fig. 2-13. Consensus sequences for three groups of glycine-rich loops are:⁸⁹⁶

Dinucleotide-binding	G X G X X G
P loop	G X X X X G K (S/T)
Protein kinase	G X G X X G X V

The protein kinase loop is seen at the top of Fig. 12-32 and extends from Gly 50 to Val 57. These conserved loops help to hold the ATP in place and to orient it correctly. Do they have any other significance? The answer is not clear. These glycine-rich loops fold across the β - γ diphosphate linkage that is broken when ATP or GTP is hydrolyzed. Cleavage of both of these molecules is associated with movement. Kinases close around ATP, and parts of G proteins (Fig. 11-7) move when these regulatory devices function.^{915,916} Cleavage of ATP causes movement in the ATPase heads of muscle myosin, providing the force for muscular contraction (Chapter 19). Somewhat the opposite occurs in mitochondrial ATP synthase (Chapter 18) when movement in the synthase heads snaps ADP and inorganic phosphate together to form ATP. Common to all of these processes is a movement of charge within the ATP (or its cleavage products) from the attack nucleophile into the neighboring phospho group, as is indicated by the arrows in the following diagram.



Not shown in this diagram is an accompanying flow of positive charge which may include movement of a metal ion, addition and loss of protons, and which may induce conformational changes.⁹¹⁴ The latter are essential not only to muscle contraction and ATP synthesis but also to many other processes that depend upon the Gibbs energy of cleavage of ATP and related compounds. This includes the pumping of ions against concentration gradients (Chapter 8), the action of **topoisomerases** (Chapter 27) which function to alter the supercoiling in DNA, and the functioning of

the phosphotransferase system by which sugars and other compounds are brought into bacterial cells (Chapter 8).

F. Displacements on Sulfur Atoms

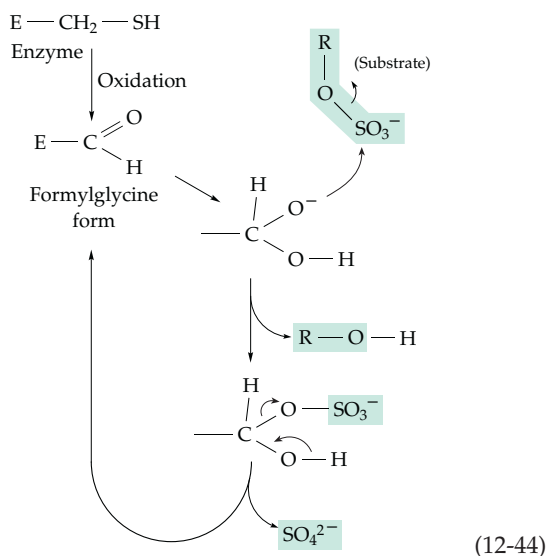
Nucleophilic displacement reactions occur on sulfur atoms in various oxidation states. A common reaction is **thiol-disulfide** exchange (Eqs. 10-9, 11-7), a reaction in which a nucleophilic thiolate anion attacks one of the sulfur atoms of a disulfide. Proteins such as **thioredoxin** of *E. coli* and **thioltransferases** (Box 15-C), which contain internal disulfide bridges, can be reduced by disulfide exchange with thiols such as glutathione (Box 11-B). The reduced proteins may then undergo similar exchange reactions that cleave disulfide linkages in other molecules. An example is glutathione reductase (Fig. 15-12). Thioltransferases may also serve as **protein disulfide isomerases** (Chapter 10, Section D,3). Nucleophilic displacements on sulfur or on selenium atoms are steps in a variety of enzymatic reactions. Among these are glutathione peroxidase (Eq. 15-59) and thiosulfate: cyanide sulfo-transferase (Eq. 24-45).

While esters of sulfuric acid do not play as central a role in metabolism as do phosphate esters, they occur widely. Both oxygen esters ($R-O-SO_3^-$, often referred to as **O-sulfates**) and derivatives of sulfamic acid ($R-NH-SO_3^-$, **N-sulfates**) are found, the latter occurring in mucopolysaccharides such as heparin. Sulfate esters of mucopolysaccharides and of steroids are ubiquitous and sulfation is the most abundant known modification of tyrosine side chains. Choline sulfate and ascorbic acid 2-sulfate are also found in cells. Sulfate esters of phenols and many other organic sulfates are present in urine.

Sulfotransferases^{917-920a} transfer sulfo groups to O and N atoms of suitable acceptors (reaction type 1D, Table 10-1). Usually, transfer is from the "active sulfate," **3'-phosphoadenosine 5'-phosphosulfate (PAPS)**,⁹²¹ whose formation is depicted in Eq. 17-38. **Sulfatases** catalyze hydrolysis of sulfate esters. The importance of such enzymes is demonstrated by the genetic **mucopolysaccharidoses**. In four of these disease-specific sulfatases that act on iduronate sulfate, heparan N-sulfate, galactose-6-sulfate, or N-acetylglucosamine-4-sulfate are absent. Some of these, such as heparan N-sulfatase deficiency, lead to severe mental retardation, some cause serious skeletal abnormalities, while others are mild in their effects.⁹²²

Sulfatases are unusual in having a residue of **formylglycine** at the active site. This is generated oxidatively from cysteine in human enzymes^{923,923a} and from serine in some bacterial sulfatases.^{924,924a} Absence of this modification results in a multiple sulfatase deficiency disease. A probable mechanism of sulfatase

based on a crystal structure determination is given by Eq. 12-44.⁹²³



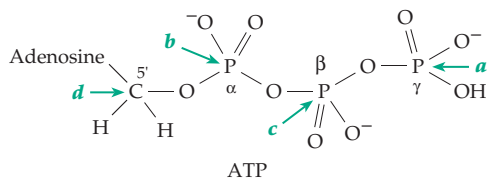
G. Multiple Displacement Reactions and the Coupling of ATP Cleavage to Endergonic Processes

A combination of successive displacement reactions of two types is required in many enzymatic reactions, including most of those by which the cleavage of ATP is coupled to biosynthesis. To harness the group transfer potential of ATP to drive an endergonic metabolic process there must be a mechanism of **coupling**. Otherwise, hydrolysis of ATP within a cell would simply generate heat. *An essential part of the coupling mechanism usually consists of a nucleophilic displacement on phosphorus followed by displacement on carbon.* Likewise, the synthesis of ATP and related compounds often begins with a displacement on carbon followed by one on phosphorus.

1. Transfer of Phospho, Pyrophospho, and Adenylyl Groups from ATP

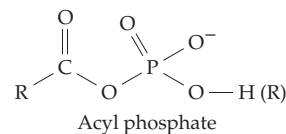
The first step in coupling ATP cleavage to any process is transfer of part of the ATP molecule to a nucleophile Y, usually by displacement on one of the three phosphorus atoms. The nucleophilic attack may be (a) on the terminal phosphorus (P_γ) with displacement of ADP or (b) on the internal phosphorus (P_α) with displacement of inorganic pyrophosphate. In the first case, $Y-PO_3H^-$ is formed; in the latter, **Y-adenylyl** (sometimes shortened to Y-adenyl) is formed. More rarely, displacement occurs (c) on the central phosphorus (P_β) with transfer of a pyrophospho group to the

nucleophile. Still less frequent (d) is a displacement on C-5' as shown in Eq. 17-37. If the nucleophile Y in any of these displacement reactions is H_2O , the resulting hydrolysis tends to go to completion, i.e., the phospho, adenylyl, and pyrophospho groups of ATP all have high group transfer potentials (Table 6-6). If Y is an $-OH$ group in an ordinary alcohol the transfer reaction also tends to go to completion because the group transfer potential of a simple phosphate ester is relatively low. Consequently, phosphorylation by ATP is often used as a means of introducing an essentially irreversible step in a metabolic pathway.



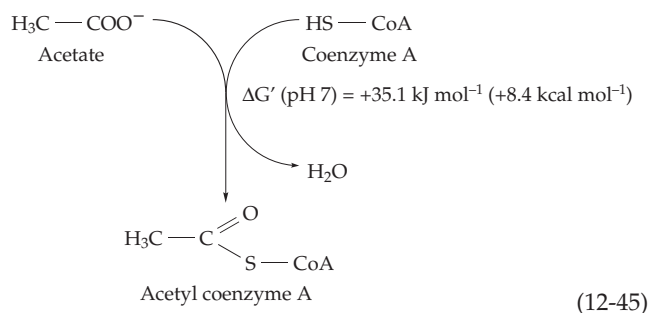
2. Acyl Phosphates

Transfer of a phospho or adenylyl group from ATP to the oxygen atom of a carboxylate group yields an **acyl phosphate**, a type of metabolic intermediate of special significance. Acyl phosphates are mixed anhydrides of carboxylic and phosphoric acids in which *both the acyl group and the phospho group have high group*

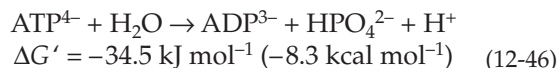


transfer potentials. As a consequence, acyl phosphates can serve as metabolic intermediates through which the group transfer potential of ATP is transferred into other molecules and is harnessed to do chemical work.

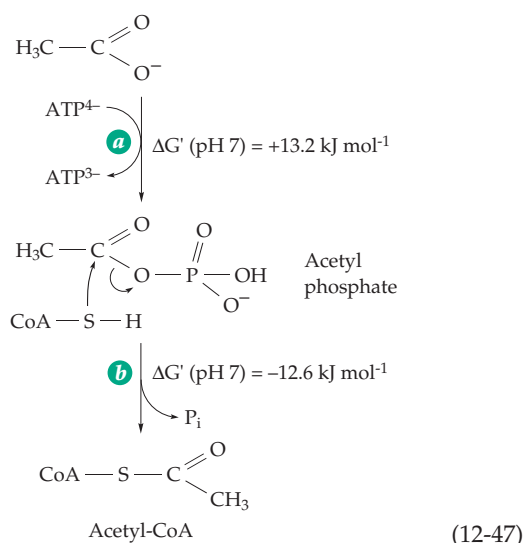
A typical example is the synthesis of acetyl coenzyme A (Eq. 12-45). See Fig. 14-1 for the complete structure of $-SH$ group-containing coenzyme A.



Because the acetyl group in the product also has a high group transfer potential, $\Delta G'$ for this reaction is highly positive and the formation of acetyl-CoA will not occur spontaneously. However, the sum of $\Delta G'$ for the reaction in Eq. 12-45 plus that for the hydrolysis of ATP (Eq. 12-46) is nearly zero ($+0.6 \text{ kJ mol}^{-1}$).



Coupling of the two reactions is accomplished by first letting an oxygen atom of the nucleophilic carboxylate group attack P_γ of ATP to form acetyl phosphate (Eq. 12-47, step *a*). In the second step (step *b*) the sulfur atom of the $-\text{SH}$ group of coenzyme A (often abbreviated CoA-SH) attacks the carbon atom of the acetyl phosphate with displacement of the good leaving group P_i . While $\Delta G'$ for Eq. 12-47, step *a*, is moderately positive (meaning that a relatively low concentration of acetyl phosphate will accumulate unless the $[\text{ATP}] / [\text{ADP}]$ ratio is high), the equilibrium in Eq. 12-47, step *b*, favors the products.

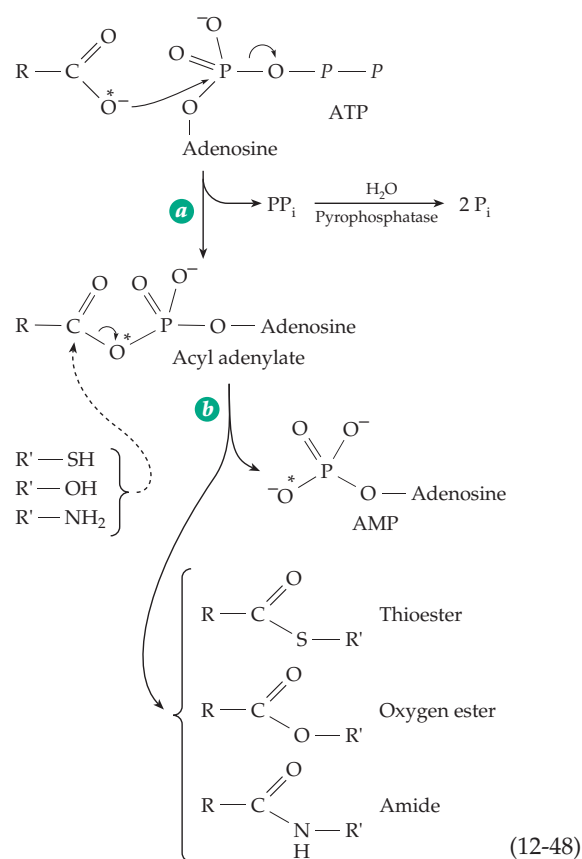


The two reactions of Eq. 12-47 are catalyzed by **acetate kinase**⁹²⁵ and an **S-acetyltransferase**, respectively. The sequence represents an essential first stage in bacterial utilization of acetate for growth. It is also used in a few bacteria in reverse as a way of *generating* ATP in fermentation reactions. On the other hand, most eukaryotic cells make acetyl-CoA from acetate by coupling the synthesis to cleavage of ATP to AMP and P_i . A single enzyme **acetyl-CoA synthetase** (acetate thiokinase) catalyzes both steps in the reaction (Eq. 10-1). The sequence parallels that of Eq. 12-47, but the initial displacement is on P_α of ATP to form **acetyl adenylate**. This intermediate remains tightly bound

to the enzyme until the second step in the sequence takes place. When ^{18}O is present in the acetate (designated by the asterisks in Eq. 12-48) it appears in the phospho group of AMP as expected for the indicated mechanism.

3. General Mechanism of Formation of Thioesters, Esters, and Amides

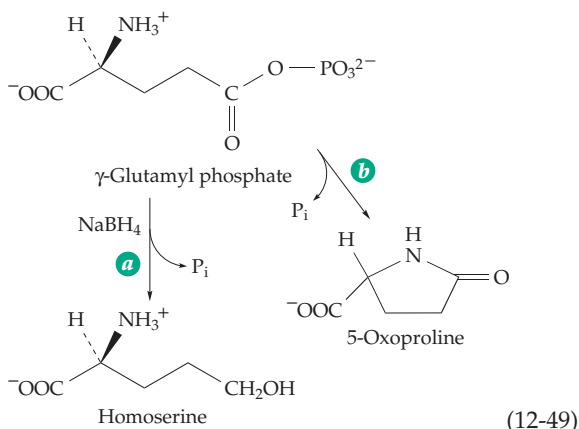
The sequences of Eqs. 12-47 and 12-48 are general ones used by cells for linking carboxylic acids to $-\text{OH}$, $-\text{SH}$, and $-\text{NH}_2$ groups to form oxygen esters, thioesters, or amides, respectively. ATP can be cleaved at either P_α or P_γ . If cleavage is at P_α (Eq. 12-48) the hydrolysis of inorganic pyrophosphate (PP_i) to P_i provides an additional coupling of ATP cleavage to biosynthesis as is discussed in Chapter 17, Section H.



An enzyme catalyzing a reaction similar to that of acetyl-CoA synthetase is **succinyl-CoA synthetase** (succinate thiokinase).^{926,927} The enzyme from *E. coli* has been studied most. The first step is formation of a phosphoenzyme by transfer of the γ phospho group from ATP to N^ϵ of histidine 246 in the α subunit of the 140-kDa $\alpha_2\beta_2$ tetramer.⁹²⁶ The phospho group is then transferred to succinate to form succinyl phosphate, which reacts with coenzyme A, as in step *b*, Eq. 12-48,

to form succinyl-CoA. Crystallographic studies suggest that for this step the succinyl phosphate and the coenzyme A may be bound to opposite α subunits in the tetramer.⁹²⁶ However, in some bacteria and in eukaryotes the enzyme appears to operate as an $\alpha\beta$ heterodimer.⁹²⁷

Glutamine synthetase,^{928,929} a large enzyme containing 12 identical 468-residue subunits with 622 symmetry (as in Fig. 7-12), has a major regulatory function in nitrogen metabolism, which is discussed in Chapter 24. Apparently, the intermediate acyl phosphate (γ -glutamyl phosphate) has a transient existence and all three reactants—glutamate, NH_4^+ , and ATP—must be bound to the enzyme concurrently before the active site becomes functional. Early evidence for the acyl phosphate^{930,931} included reduction by sodium borohydride to an alcohol (Eq. 12-49a) and isolation of the internal amide of glutamic acid **5-oxoproline** (Eq. 12-49b).



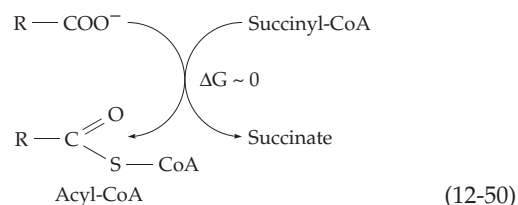
Ammonium ions appear to be bound at a specific site and to be deprotonated by the Asp 50 carboxylate.⁹²⁹

The **aminoacyl-tRNA synthetases** join amino acids to their appropriate tRNA molecules for protein synthesis. They have the very important task of selecting both a specific amino acid and a specific tRNA and joining them. The enzymes differ in size and other properties. However, they all appear to function by a common basic chemistry that makes use of cleavage of ATP at P_α (Eq. 12-48) via an intermediate aminoacyl adenylate and that is outlined also in Eq. 17-36. These enzymes are discussed in Chapter 29.

4. Coenzyme A Transferases

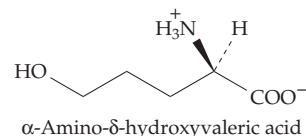
The following problem in energy transfer arises occasionally: A thioester, such as succinyl-CoA, is available to a cell and the energy available in its unstable linkage is needed for synthesis of a different thioester. It would be possible for a cell to first form ATP or GTP,

using a synthetase reaction in reverse; then the ATP or GTP formed could be used to make the new linkage by the action of another acyl-CoA synthetase. However, special enzymes, the **CoA transferases**, function more directly (Eq. 12-50). The mechanism is not obvious.



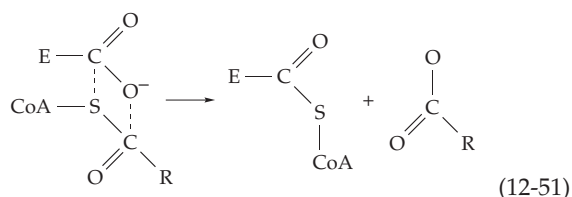
How can the CoA be transferred from one acyl group to another while still retaining the high group transfer potential of the acyl group?

The following experiments shed some light. Kinetic studies of **succinyl-CoA-acetoacetate CoA transferase** indicate a ping-pong mechanism. The enzyme alternates between two distinct forms, one of which has been shown to contain bound CoA.⁹³²⁻⁹³⁴ The E-CoA intermediate formed from enzyme plus acetoacetyl-CoA was reduced with ^3H -containing sodium borohydride and the protein was completely hydrolyzed with HCl. Tritium-containing α -amino- δ -hydroxyvaleric acid was isolated. Since thioesters (as well as oxygen esters) are cleaved in a two-step process



to alcohols by reduction with borohydride, it was concluded that the intermediate E-CoA is a thioester of the Glu 344 side chain. In exchange reactions ^{18}O from labeled succinate entered both the E-CoA intermediate and the carboxyl group of acetoacetate.

A mechanism involving formation of a transient anhydride is similar to reactions discussed in preceding sections.^{717,720} The student should be able to write out the step-by-step detail. Does this mechanism explain the ^{18}O exchange data? A second possibility is a **4-center reaction** (Eq. 12-51). However, mechanisms of this type have not been demonstrated for enzymatic reactions.



References

1. Jencks, W. P. (1987) *Catalysis in Chemistry and Enzymology*, Dover, Mineola, New York (pp. 78–110)
2. Lowe, J. N., and Ingraham, L. L. (1974) *An Introduction to Biochemical Reaction Mechanisms*, Prentice-Hall, Englewood Cliffs, New Jersey
3. Bruice, T. C., and Benkovic, S. J. (1966) *Bioorganic Mechanisms*, Benjamin, New York (2 vols.)
- 3a. Kyte, J. (1995) *Mechanism in Protein Chemistry*, Garland Publ., New York
- 3b. Fersht, A. (1999) *Structure and Mechanism in Protein Science: A Guide to Enzyme Catalysis and Protein Folding*, Freeman, New York
- 3c. Silverman, R. B. (1999) *The Organic Chemistry of Enzyme-Catalyzed Reactions*, Academic Press, San Diego, California
- 3d. Sinnott, M., ed. (1998) *Comprehensive Biological Catalysis. A Mechanistic Reference*, Vol. I, Academic Press, San Diego, California
4. Dixon, J. E., and Bruice, T. C. (1971) *J. Am. Chem. Soc.* **93**, 6592–6597
5. Kosower, E. M. (1968) *An Introduction to Physical Organic Chemistry*, Wiley, New York (pp. 28 and 81)
6. Goldman, P., Milne, G. W. A., and Keister, D. B. (1968) *J. Biol. Chem.* **243**, 428–434
7. Liu, J.-Q., Kurihara, T., Miyagi, M., Esaki, N., and Soda, K. (1995) *J. Biol. Chem.* **270**, 18309–18312
8. Hisano, T., Hata, Y., Fujii, T., Liu, J.-Q., Kurihara, T., Esaki, N., and Soda, K. (1996) *J. Biol. Chem.* **271**, 20322–20330
9. Li, Y.-F., Hata, Y., Fujii, T., Hisano, T., Nishihara, M., Kurihara, T., and Esaki, N. (1998) *J. Biol. Chem.* **273**, 15035–15044
10. Grimmelikhuijzen, C. J. P., and Schaller, H. C. (1979) *Trends Biochem. Sci.* **4**, 265–267
11. Koonin, E. V., and Tatusov, R. L. (1994) *J. Mol. Biol.* **244**, 125–132
- 11a. Nardini, M., Ridder, I. S., Rozeboom, H. J., Kalk, K. H., Rink, R., Janssen, D. B., and Dijkstra, B. W. (1999) *J. Biol. Chem.* **274**, 14579–14586
12. Verschueren, K. H. G., Seljée, F., Rozeboom, H. J., Kalk, K. H., and Dijkstra, B. W. (1993) *Nature (London)* **363**, 693–698
- 12a. Ridder, I. S., Rozeboom, H. J., Kalk, K. H., and Dijkstra, B. W. (1999) *J. Biol. Chem.* **274**, 30672–30678
13. Pries, F., Kingma, J., Krooshof, G. H., Jeronimus-Stratingh, C. M., Bruins, A. P., and Janssen, D. B. (1995) *J. Biol. Chem.* **270**, 10405–10411
- 13a. Schindler, J. F., Naranjo, P. A., Honaberger, D. A., Chang, C.-H., Brainard, J. R., Vanderberg, L. A., and Unkefer, C. J. (1999) *Biochemistry* **38**, 5772–5778
14. Lightstone, F. C., Zheng, Y.-J., Maulitz, A. H., and Bruice, T. C. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 8417–8420
15. Lightstone, F. C., Zheng, Y.-J., and Bruice, T. C. (1998) *J. Am. Chem. Soc.* **120**, 5611–5621
16. Pries, F., van den Wijngaard, A. J., Bos, R., Pentenga, M., and Janssen, D. B. (1994) *J. Biol. Chem.* **269**, 17490–17494
17. Chiang, P. K., Gordon, R. K., Tal, J., Zeng, G. C., Doctor, B. P., Pardhasaradhi, K., and McCann, P. P. (1996) *FASEB J.* **10**, 471–480
18. Cornforth, J. W., Reichard, S. A., Talalay, P., Carrell, H. L., and Glusker, J. P. (1977) *J. Am. Chem. Soc.* **99**, 7292–7301
19. Woodard, R. W., Tsai, M.-D., Floss, H. G., Crooks, P. A., and Coward, J. K. (1980) *J. Biol. Chem.* **255**, 9124–9127
20. Vidgren, J., Svensson, L. A., and Liljas, A. (1994) *Nature (London)* **368**, 354–357
21. Fu, Z., Hu, Y., Konishi, K., Takata, Y., Ogawa, H., Gomi, T., Fujioka, M., and Takusagawa, F. (1996) *Biochemistry* **35**, 11985–11993
22. Takata, Y., Konishi, K., Gomi, T., and Fujioka, M. (1994) *J. Biol. Chem.* **269**, 5537–5542
23. Schluckebier, G., O'Gara, M., Saenger, W., and Cheng, X. (1995) *J. Mol. Biol.* **247**, 16–20
24. Szczelkun, M. D., Jones, H., and Connolly, B. A. (1995) *Biochemistry* **34**, 10734–10743
25. O'Gara, M., Roberts, R. J., and Cheng, X. (1996) *J. Mol. Biol.* **263**, 597–606
- 25a. Reddy, Y. V. R., and Rao, D. N. (2000) *J. Mol. Biol.* **298**, 597–610
26. Patel, D. J. (1994) *Nature (London)* **367**, 688–690
27. Jeltsch, A., Christ, F., Fatemi, M., and Roth, M. (1999) *J. Biol. Chem.* **274**, 19538–19544
28. Lotta, T., Vidgren, J., Tilgmann, C., Ulmanen, I., Melén, K., Julkunen, I., and Taskinen, J. (1995) *Biochemistry* **34**, 4202–4210
29. Zheng, Y.-J., and Bruice, T. C. (1997) *J. Am. Chem. Soc.* **119**, 8137–8145
- 29a. Kahn, K., and Bruice, T. C. (2000) *J. Am. Chem. Soc.* **122**, 46–51
30. Kealey, J. T., and Santi, D. V. (1995) *Biochemistry* **34**, 2441–2446
31. Hrycyna, C. A., Yang, M. C., and Clarke, S. (1994) *Biochemistry* **33**, 9806–9812
- 31a. Cleland, W. W., O'Leary, M. H., and Northrop, D. B., eds. (1977) *Isotope Effects on Enzyme-Catalyzed Reactions*, Univ. Park Press, Baltimore, Maryland
- 31b. Klinman, J. P. (1978) in *Transition States of Biochemical Processes* (Gandour, R. D., and Schowen, R. L., eds), pp. 165–200, Plenum, New York
- 31c. Gandour, R. D., and Schowen, R. L., eds. (1978) *Transition States of Biochemical Processes*, Plenum, New York
- 31d. Schramm, V. L., Horenstein, B. A., and Kline, P. C. (1994) *J. Biol. Chem.* **269**, 18259–18262
32. Hegazi, M. F., Borchardt, R. T., and Schowen, R. L. (1979) *J. Am. Chem. Soc.* **101**, 4359–4365
33. Mihel, I., Knipe, J. O., Coward, J. K., and Schowen, R. L. (1979) *J. Am. Chem. Soc.* **101**, 4349–4351
34. Rodgers, J., Femec, D. A., and Schowen, R. L. (1982) *J. Am. Chem. Soc.* **104**, 3263–3268
35. Boyd, R. J., Kim, C.-K., Shi, Z., Weinberg, N., and Wolfe, S. (1993) *J. Am. Chem. Soc.* **115**, 10147–10152
36. Glad, S. S., and Jensen, F. (1997) *J. Am. Chem. Soc.* **119**, 227–232
37. Wolfenden, R., Lu, X., and Young, G. (1998) *J. Am. Chem. Soc.* **120**, 6814–6815
38. Henrissat, B., and Bairoch, A. (1993) *Biochem. J.* **293**, 781–788
39. Henrissat, B., and Bairoch, A. (1996) *Biochem. J.* **316**, 695–696
40. Henrissat, B., Callebaut, I., Fabrega, S., Lehn, P., Morion, J.-P., and Davies, G. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 7090–7094
41. Sinnott, M. L. (1990) *Chem. Rev.* **90**, 1171–1202
42. Withers, S. G., and Aebersold, R. (1995) *Protein Sci.* **4**, 361–372
43. Silverstein, R., Voet, J., Reed, D., and Abeles, R. H. (1967) *J. Biol. Chem.* **242**, 1338–1346
44. Doudoroff, M., Barker, H. A., and Hassid, W. Z. (1947) *J. Biol. Chem.* **168**, 725–732 and 733–746
45. LeBrun, E., and van Rapenbusch, R. (1980) *J. Biol. Chem.* **255**, 12034–12036
46. McCarter, J. D., and Withers, S. G. (1996) *J. Biol. Chem.* **271**, 6889–6894
47. Withers, S. G., Rupitz, K., and Street, I. P. (1988) *J. Biol. Chem.* **263**, 7929–7932
48. Withers, S. G., and Street, I. P. (1988) *J. Am. Chem. Soc.* **110**, 8551–8553
49. Miao, S., Ziser, L., Aebersold, R., and Withers, S. G. (1994) *Biochemistry* **33**, 7027–7032
50. Braun, C., Lindhorst, T., Madsen, N. B., and Withers, S. G. (1996) *Biochemistry* **35**, 5458–5463
- 50a. Namchuk, M. N., McCarter, J. D., Becalski, A., Andrews, T., and Withers, S. G. (2000) *J. Am. Chem. Soc.* **122**, 1270–1277
51. Gorenstein, D. G., Findlay, J. B., Luxon, B. A., and Kar, D. (1977) *J. Am. Chem. Soc.* **99**, 3473–3479
52. Kirby, A. J. (1983) *The Anomeric Effect and Related Stereoelectronic Effects at Oxygen*, Springer, Berlin
53. Bennet, A. J., and Sinnott, M. L. (1986) *J. Am. Chem. Soc.* **108**, 7287–7294
54. Sinnott, M. L. (1986) in *Mechanisms of Enzymatic Reactions: Stereochemistry A. Steenbock Symposium* (Frey, P. A., ed), pp. 293–305, Elsevier, New York
55. Phillips, D. C. (1966) *Sci. Am.* **215** (Nov), 78–90
56. Sanz, J. M., García, P., and García, J. L. (1992) *Biochemistry* **31**, 8495–8499
57. Artymicek, P. J., and Blake, C. C. F. (1981) *J. Mol. Biol.* **152**, 737–762
58. Harata, K., Muraki, M., and Jigami, Y. (1993) *J. Mol. Biol.* **233**, 524–535
59. Harata, K., Muraki, M., Hayashi, Y., and Jigami, Y. (1992) *Protein Sci.* **1**, 1447–1453
60. Grütter, M. G., Weaver, L. H., and Matthews, B. W. (1983) *Nature (London)* **303**, 828–831
61. Zhang, X.-J., Baase, W. A., and Matthews, B. W. (1991) *Biochemistry* **30**, 2012–2017
62. Dao-pin, S., Sauer, U., Nicholson, H., and Matthews, B. W. (1991) *Biochemistry* **30**, 7142–7153
63. Matsumura, M., Becktel, W. J., Levitt, M., and Matthews, B. W. (1989) *Proc. Natl. Acad. Sci. U.S.A.* **86**, 6562–6566
64. Parsons, S. M., and Raftery, M. A. (1972) *Biochemistry* **11**, 1623–1628
65. Maenaka, K., Matsushima, M., Song, H., Sunada, F., Watanabe, K., and Kumagai, I. (1995) *J. Mol. Biol.* **247**, 281–293
66. Matsumura, I., and Kirsch, J. F. (1996) *Biochemistry* **35**, 1881–1889
67. Kuroki, R., Yamada, H., Moriyama, T., and Imoto, T. (1986) *J. Biol. Chem.* **261**, 13571–13574
68. Malcolm, B. A., Rosenberg, S., Corey, M. J., Allen, J. S., de Baetselier, A., and Kirsch, J. F. (1989) *Proc. Natl. Acad. Sci. U.S.A.* **86**, 133–137
69. Hadfield, A. T., Harvey, D. J., Archer, D. B., MacKenzie, da, Jeenes, D. J., Radford, S. E., Lowe, G., Dobson, C. M., and Johnson, L. N. (1994) *J. Mol. Biol.* **243**, 856–872
70. Weaver, L. H., Grütter, M. G., and Matthews, B. W. (1995) *J. Mol. Biol.* **245**, 54–68
71. Kuroki, R., Ito, Y., Kato, Y., and Imoto, T. (1997) *J. Biol. Chem.* **272**, 19976–19981
72. Matsumura, I., and Kirsch, J. F. (1996) *Biochemistry* **35**, 1890–1896
73. Dickerson, R. E., and Geis, I. (1969) *The Structure and Action of Proteins*, Harper and Row, New York (p. 71)
74. Levitt, M. (1974) in *Peptides, Polypeptides and Proteins* (Blout, E. R., Bouey, F. A., Goodman, M., and Lotan, N., eds), pp. 99–113, Wiley, New York
75. Huang, X., Surry, C., Hiebert, T., and Bennet, A. J. (1995) *J. Am. Chem. Soc.* **117**, 10614–10621
76. Matsui, H., Blanchard, J. S., Brewer, C. F., and Hehre, E. J. (1989) *J. Biol. Chem.* **264**, 8714–8716
77. Banait, N. S., and Jencks, W. P. (1991) *J. Am. Chem. Soc.* **113**, 7951–7958
78. Zhang, Y., Bommuwamy, J., and Sinnott, M. L. (1994) *J. Am. Chem. Soc.* **116**, 7557–7563
79. Tanaka, Y., Tao, W., Blanchard, J. S., and Hehre, E. J. (1994) *J. Biol. Chem.* **269**, 32306–32312
80. Knier, B. L., and Jencks, W. P. (1980) *J. Am. Chem. Soc.* **102**, 6789–6798
81. Huang, X., Tanaka, K. S. E., and Bennet, A. J. (1997) *J. Am. Chem. Soc.* **119**, 11147–11154

References

82. Hardy, L. W., and Poteete, A. R. (1991) *Biochemistry* **30**, 9457–9463
- 82a. Davies, G. J., Mackenzie, L., Varrot, A., Dauter, M., Brzozowski, A. M., Schülein, M., and Withers, S. G. (1998) *Biochemistry* **37**, 11707–11713
- 82b. Vocadlo, D. J., Mayer, C., He, S., and Withers, S. G. (2000) *Biochemistry* **39**, 117–126
83. Banerjee, S. K., and Rupley, J. A. (1975) *J. Biol. Chem.* **250**, 8267–8274
84. Kelly, J. A., Sielecki, A. R., Sykes, B. D., James, M. N. G., and Phillips, D. C. (1979) *Nature (London)* **282**, 875–878
85. Metzler, D. E. (1979) *Adv. Enzymol.* **50**, 1–40
86. Dao-pin, S., Liao, D.-I., and Remington, S. J. (1989) *Proc. Natl. Acad. Sci. U.S.A.* **86**, 5361–5365
87. Post, C. B., and Karplus, M. (1986) *J. Am. Chem. Soc.* **108**, 1317–1319
88. Fife, T. H., Jaffe, S. H., and Natarajan, R. (1991) *J. Am. Chem. Soc.* **113**, 7646–7653
89. Piskiewicz, D., and Bruice, T. C. (1968) *J. Am. Chem. Soc.* **90**, 2156–2163
90. Lowe, G., and Sheppard, G. (1968) *J. Chem. Soc. Chem. Commun.*, 529–530
91. Capon, B. (1969) *Chem. Rev.* **69**, 407–498
92. Hart, P. J., Pfluger, H. D., Monzingo, A. F., Hollis, T., and Robertus, J. D. (1995) *J. Mol. Biol.* **248**, 402–413
93. Terwisscha van Scheltinga, A. C., Armand, S., Kalk, K. H., Isogai, A., Henrissat, B., and Dijkstra, B. W. (1995) *Biochemistry* **34**, 15619–15623
94. Tews, I., Terwisscha van Scheltinga, A. C., Perrakis, A., Wilson, K. S., and Dijkstra, B. W. (1997) *J. Am. Chem. Soc.* **119**, 7954–7959
95. Brameld, K. A., and Goddard, W. A., III (1998) *J. Am. Chem. Soc.* **120**, 3571–3580
96. Klyosov, A. A. (1990) *Biochemistry* **29**, 10577–10585
97. Watanabe, H., Noda, H., Tokuda, G., and Lo, N. (1998) *Nature (London)* **394**, 330–331
98. Davies, G. J., Tolley, S. P., Henrissat, B., Hjort, C., and Schülein, M. (1995) *Biochemistry* **34**, 16210–16220
99. Damude, H. G., Withers, S. G., Kilburn, D. G., Miller, R. C., Jr., and Warren, R. A. J. (1995) *Biochemistry* **34**, 2220–2224
100. Barr, B. K., Wolfgang, D. E., Piens, K., Claeysens, M., and Wilson, D. B. (1998) *Biochemistry* **37**, 9220–9229
101. Divine, C., Ståhlberg, J., Teeri, T. T., and Jones, T. A. (1998) *J. Mol. Biol.* **275**, 309–325
102. Barr, B. K., Hsieh, Y.-L., Ganem, B., and Wilson, D. B. (1996) *Biochemistry* **35**, 586–592
103. Choi, S. K., and Jungdahl, L. G. (1996) *Biochemistry* **35**, 4906–4910
104. Tormo, J., Lamed, R., Chirino, A. J., Morag, E., Bayer, E. A., Shoham, Y., and Steitz, T. A. (1996) *EMBO J.* **15**, 5739–5751
105. Sulzenbacher, G., Schülein, M., and Davies, G. J. (1997) *Biochemistry* **36**, 5902–5911
106. Sulzenbacher, G., Driguez, H., Henrissat, B., Schülein, M., and Davies, G. J. (1996) *Biochemistry* **35**, 15280–15287
107. Johnson, P. E., Joshi, M. D., Tomme, P., Kilburn, D. G., and McIntosh, L. P. (1996) *Biochemistry* **35**, 14381–14394
108. Fukuda, M., Spooner, E., Oates, J. E., Dell, A., and Klock, J. C. (1984) *J. Biol. Chem.* **259**, 10925–10935
- 108a. Ponyi, T., Szabó, L., Nagy, T., Orosz, L., Simpson, P. J., Williamson, M. P., and Gilbert, H. J. (2000) *Biochemistry* **39**, 985–991
109. Domínguez, R., Souchon, H., Lascombe, M.-B., and Alzari, P. M. (1996) *J. Mol. Biol.* **257**, 1042–1051
110. Roberge, M., Shareck, F., Morosoli, R., Kluepfel, D., and Dupont, C. (1997) *Biochemistry* **36**, 7769–7775
111. Juy, M., Amit, A. G., Alzari, P. M., Poljak, R. J., Claeysens, M., Béguin, P., and Aubert, J.-P. (1992) *Nature (London)* **357**, 89–91
112. Chen, L., Fincher, G. B., and Hoj, P. B. (1993) *J. Biol. Chem.* **268**, 13318–13326
113. Chen, L., Garrett, T. P. J., Fincher, G. B., and Hoj, P. B. (1995) *J. Biol. Chem.* **270**, 8093–8101
114. Müller, J. J., Thomsen, K. K., and Heinemann, U. (1998) *J. Biol. Chem.* **273**, 3438–3446
115. Lindley, M. G., Shallenberger, R. S., and Herbert, S. M. (1976) *Food Chemistry* **1**, 149–159
116. Krengel, U., and Dijkstra, B. W. (1996) *J. Mol. Biol.* **263**, 70–78
117. Kasumi, T., Tsumuraya, Y., Brewer, C. F., Kersters-Hilderson, H., Claeysens, M., and Hehre, E. J. (1987) *Biochemistry* **26**, 3010–3016
118. Lawson, S. L., Wakarchuk, W. W., and Withers, S. G. (1996) *Biochemistry* **35**, 10110–10118
119. Derewenda, U., Swenson, L., Green, R., Wei, Y., Morosoli, R., Shareck, F., Kluepfel, D., and Derewenda, Z. S. (1994) *J. Biol. Chem.* **269**, 20811–20814
120. McIntosh, L. P., Hand, G., Johnson, P. E., Joshi, M. D., Körner, M., Plesniak, L. A., Ziser, L., Wakarchuk, W. W., and Withers, S. G. (1996) *Biochemistry* **35**, 9958–9966
121. He, X. M., and Carter, D. C. (1992) *Nature (London)* **358**, 209–215
122. Braithwaite, K. L., Barna, T., Spurway, T. D., Charnock, S. J., Black, G. W., Hughes, N., Lakey, J. H., Virden, R., Hazlewood, G. P., Henrissat, B., and Gilbert, H. J. (1997) *Biochemistry* **36**, 15489–15500
123. Sanz-Aparicio, J., Hermoso, J. A., Martínez-Ripoll, M., Lequerica, J. L., and Polaina, J. (1998) *J. Mol. Biol.* **275**, 491–502
124. Wang, Q., Trimbur, D., Graham, R., Warren, R. A. J., and Withers, S. G. (1995) *Biochemistry* **34**, 14554–14562
125. Namchuk, M. N., and Withers, S. G. (1995) *Biochemistry* **34**, 16194–16202
126. Hrmova, M., MacGregor, E. A., Biely, P., Stewart, R. J., and Fincher, G. B. (1998) *J. Biol. Chem.* **273**, 11134–11143
127. Febbraio, F., Barone, R., D'Auria, S., Rossi, M., Nucci, R., Piccialli, G., De Napoli, L., Orrù, S., and Pucci, P. (1997) *Biochemistry* **36**, 3068–3075
128. Jacobson, R. H., Zhang, X.-J., DuBose, R. F., and Matthews, B. W. (1994) *Nature (London)* **369**, 761–766
129. Gebler, J. C., Aebersold, R., and Withers, S. G. (1992) *J. Biol. Chem.* **267**, 11126–11130
130. Richard, J. P., Westerfeld, J. G., Lin, S., and Beard, J. (1995) *Biochemistry* **34**, 11713–11724
131. Richard, J. P., Huber, R. E., Lin, S., Heo, C., and Amyes, T. L. (1996) *Biochemistry* **35**, 12377–12386
132. Roth, N. J., Rob, B., and Huber, R. E. (1998) *Biochemistry* **37**, 10099–10107
133. Mitchell, E. P., Withers, S. G., Ermer, P., Vasella, A. T., Garman, E. F., Oikonomakos, N. G., and Johnson, L. N. (1996) *Biochemistry* **35**, 7341–7355
134. Tu, J.-I., Jacobson, G. R., and Graves, D. J. (1971) *Biochemistry* **10**, 1229–1236
135. Gold, A. M., Legrand, E., and Sanchez, G. R. (1971) *J. Biol. Chem.* **246**, 5700–5706
136. Levvy, G. A., and Snaith, S. M. (1972) *Adv. Enzymol.* **36**, 151–181
137. Barford, D., and Johnson, L. N. (1989) *Nature (London)* **340**, 609–616
138. Oikonomakos, N. G., Johnson, L. N., Acharya, K. R., Stuart, D. I., Barford, D., Hajdu, J., Varvill, K. M., Melpidou, A. E., Papageorgiou, T., Graves, D. J., and Palm, D. (1987) *Biochemistry* **26**, 8381–8389
139. Oikonomakos, N. G., Zographos, S. E., Tsitsanou, K. E., Johnson, L. N., and Acharya, K. R. (1996) *Protein Sci.* **5**, 2416–2428
140. Helmreich, E. J. M., and Klein, H. W. (1980) *Angew. Chem. Int. Ed. Engl.* **19**, 441–455
141. Klein, H. W., Im, M. J., Palm, D., and Helmreich, E. J. M. (1984) *Biochemistry* **23**, 5853–5861
142. Street, I. P., Rupitz, K., and Withers, S. G. (1989) *Biochemistry* **28**, 1581–1587
143. Parrish, R. F., Uhing, R. J., and Graves, D. J. (1977) *Biochemistry* **16**, 4824–4831
- 143a. Watson, K. A., McCleverty, C., Geremia, S., Cottaz, S., Driguez, H., and Johnson, L. N. (1999) *EMBO J.* **18**, 4619–4632
144. Duke, E. M. H., Wakatsuki, S., Hadfield, A., and Johnson, L. N. (1994) *Protein Sci.* **3**, 1178–1196
145. Buchbinder, J. L., Luong, C. B. H., Browner, M. F., and Fletterick, R. J. (1997) *Biochemistry* **36**, 8039–8044
146. Buchbinder, J. L., and Fletterick, R. J. (1996) *J. Biol. Chem.* **271**, 22305–22309
147. Gregoriou, M., Noble, M. E. M., Watson, K. A., Garman, E. F., Krulle, T. M., De La Fuente, C., Fleet, G. W. J., Oikonomakos, N. G., and Johnson, L. N. (1998) *Protein Sci.* **7**, 915–927
148. Nakano, K., and Fukui, T. (1986) *J. Biol. Chem.* **261**, 8230–8236
149. Watson, K. A., Schinzel, R., Palm, D., and Johnson, L. N. (1997) *EMBO J.* **16**, 1–14
150. Becker, S., Palm, D., and Schinzel, R. (1994) *J. Biol. Chem.* **269**, 2485–2490
151. Lin, K., Hwang, P. K., and Fletterick, R. J. (1995) *J. Biol. Chem.* **270**, 26833–26839
152. Goldsmith, E. J., Fletterick, R. J., and Withers, S. G. (1987) *J. Biol. Chem.* **262**, 1449–1455
153. Robyt, J. F., and French, D. (1970) *J. Biol. Chem.* **245**, 3917–3927
154. Brayer, G. D., Luo, Y., and Withers, S. G. (1995) *Protein Sci.* **4**, 1730–1742
155. MacDonald, R. J., Crerar, M. M., Swain, W. F., Pictet, R. L., Thomas, G., and Rutter, W. J. (1980) *Nature (London)* **287**, 117–122
156. Qian, M., Haser, R., Buisson, G., Dué, E., and Payan, F. (1994) *Biochemistry* **33**, 6284–6294
157. Qian, M., Haser, R., and Payan, F. (1995) *Protein Sci.* **4**, 747–755
158. Larson, S. B., Greenwood, A., Cascio, D., Day, J., and McPherson, A. (1994) *J. Mol. Biol.* **235**, 1560–1584
159. Machius, M., Vértessy, L., Huber, R., and Wiegand, G. (1996) *J. Mol. Biol.* **260**, 409–421
160. Kadziola, A., Sogaard, M., Svensson, B., and Haser, R. (1998) *J. Mol. Biol.* **278**, 205–217
161. Strobl, S., Maskos, K., Betz, M., Wiegand, G., Huber, R., Gomis-Rüth, F. X., and Glockshuber, R. (1998) *J. Mol. Biol.* **278**, 617–628
162. Fujimoto, Z., Takase, K., Doui, N., Momma, M., Matsumoto, T., and Mizuno, H. (1998) *J. Mol. Biol.* **277**, 393–407
163. Feller, G., le Bussy, O., Houssier, C., and Gerday, C. (1996) *J. Biol. Chem.* **271**, 23836–23841
164. Aghajari, N., Feller, G., Gerday, C., and Haser, R. (1998) *Protein Sci.* **7**, 564–572
165. Matsuura, Y., Kusunoki, M., Harada, W., and Kakudo, M. (1984) *J. Biochem.* **95**, 697–702
166. Brzozowski, A. M., and Davies, G. J. (1997) *Biochemistry* **36**, 10837–10845
167. Robyt, J. F., and French, D. (1970) *Arch. Biochem. Biophys.* **138**, 662–670
168. Robyt, J. F., and French, D. (1967) *Arch. Biochem. Biophys.* **122**, 8–16
169. delCardayré, S. B., and Raines, R. T. (1994) *Biochemistry* **33**, 6031–6037
- 169a. Dauter, Z., Dauter, M., Brzozowski, A. M., Christensen, S., Borchert, T. V., Beier, L., Wilson, K. S., and Davies, G. J. (1999) *Biochemistry* **38**, 8385–8392
- 169b. Park, K. H., Kim, M. J., Lee, H. S., Han, N. S., Kim, D., and Robyt, J. F. (1998) *Carbohydr. Res.* **313**, 235–246

References

170. Hermans, M. M. P., Kroos, M. A., van Beeuman, J., Oostru, B. A., and Reuser, A. J. J. (1991) *J. Biol. Chem.* **266**, 13507–13512
171. Raben, N., Nagaraju, K., Lee, E., Kessler, P., Byrne, B., Lee, L., LaMarca, M., King, C., Ward, J., Sauer, B., and Plotz, P. (1998) *J. Biol. Chem.* **273**, 19086–19092
172. Howard, S., and Withers, S. G. (1998) *Biochemistry* **37**, 3858–3864
173. Jespersen, H. M., MacGregor, E. A., Henrissat, B., Sierks, M. R., and Svensson, B. (1993) *Journal of Protein Chemistry* **12**, 791–805
174. Liu, W., Madsen, N. B., Fan, B., Zucker, K. A., Glew, R. H., and Fry, D. E. (1995) *Biochemistry* **34**, 7056–7061
175. Watanabe, K., Hata, Y., Kizaki, H., Katsube, Y., and Suzuki, Y. (1997) *J. Mol. Biol.* **269**, 142–153
176. Knegtel, R. M. A., Strokopytov, B., Penninga, D., Faber, O. G., Rozeboom, H. J., Kalk, K. H., Dijkhuizen, L., and Dijkstra, B. W. (1995) *J. Biol. Chem.* **270**, 29256–29264
177. Schmidt, A. K., Cottaz, S., Driguez, H., and Schulz, G. E. (1998) *Biochemistry* **37**, 5909–5915
- 177a. Uitdehaag, J. C. M., Mosi, R., Kalk, K. H., van der Veen, B. A., Dijkhuizen, L., Withers, S. G., and Dijkstra, B. W. (1999) *Nature Struct. Biol.* **6**, 432–436
- 177b. van der Veen, B. A., Uitdehaag, J. C. M., Penninga, D., van Alebeek, G.-J. W. M., Smith, L. M., Dijkstra, B. W., and Dijkhuizen, L. (2000) *J. Mol. Biol.* **296**, 1027–1038
178. Terada, Y., Yanase, M., Takata, H., Takaha, T., and Okada, S. (1997) *J. Biol. Chem.* **272**, 15729–15733
179. Fierobe, H.-P., Clarke, A. J., Tull, D., and Svensson, B. (1998) *Biochemistry* **37**, 3753–3759
180. Aleshin, A. E., Stoffer, B., Firsov, L. M., Svensson, B., and Honzatko, R. B. (1996) *Biochemistry* **35**, 8319–8328
181. Firsov, L. M., Neustroev, K. N., Aleshin, A. E., Metzler, C. M., Metzler, D. E., Scott, R. D., Stoffer, B., Christensen, T., and Svensson, B. (1994) *Eur. J. Biochem.* **223**, 293–302
182. Natarajan, S. K., and Sierks, M. R. (1997) *Biochemistry* **36**, 14946–14955
183. Williamson, M. P., Le Gal-Coëffet, M.-F., Sorimachi, K., Furniss, C. S. M., Archer, D. B., and Williamson, G. (1997) *Biochemistry* **36**, 7535–7539
184. Penninga, D., van der Veen, B. A., Knegtel, R. M. A., van Hijum, S. A. F. T., Rozeboom, H. J., Kalk, K. H., Dijkstra, B. W., and Dijkhuizen, L. (1996) *J. Biol. Chem.* **271**, 32777–32784
- 184a. Mikami, B., Adachi, M., Kage, T., Sarikaya, E., Nanmori, T., Shinke, R., and Utsumi, S. (1999) *Biochemistry* **38**, 7050–7061
185. Mikami, B., Degano, M., Hehre, E. J., and Sacchettini, J. C. (1994) *Biochemistry* **33**, 7779–7787
186. Adachi, M., Mikami, B., Katsube, T., and Utsumi, S. (1998) *J. Biol. Chem.* **273**, 19859–19865
187. Amado, M., Almeida, R., Carneiro, F., Levery, S. B., Holmes, E. H., Nomoto, M., Hollingsworth, M. A., Hassan, H., Schwientek, T., Nielsen, P. A., Bennett, E. P., and Clausen, H. (1998) *J. Biol. Chem.* **273**, 12770–12778
188. Baenziger, J. U. (1994) *FASEB J.* **8**, 1019–1025
189. Radzicka, A., and Wolfenden, R. (1996) *J. Am. Chem. Soc.* **118**, 6105–6109
190. Barrett, A. J., Rawlings, N. D., and Woessner, J. F., eds. (1998) *Handbook of Proteolytic Enzymes*, Academic Press, San Diego, California
191. Fushiki, T., and Iwai, K. (1989) *FASEB J.* **3**, 121–126
192. Pignol, D., Granon, S., Chapus, C., and Carlos, J. (1995) *J. Mol. Biol.* **252**, 20–24
- 192a. Lu, D., Fütterer, K., Korolev, S., Zheng, X., Tan, K., Waksman, G., and Sadler, J. E. (1999) *J. Mol. Biol.* **292**, 361–373
193. Light, A., and Janska, H. (1989) *Trends Biochem. Sci.* **14**, 110–112
194. Matsushima, M., Ichinose, M., Yahagi, N., Kakei, N., Tsukada, S., Miki, K., Kurokawa, K., Tashiro, K., Shiokawa, K., Shinomiya, K., Umeyama, H., Inoue, H., Takahashi, T., and Takahashi, K. (1994) *J. Biol. Chem.* **269**, 19976–19982
195. Kitamoto, Y., Veile, R. A., Donis-Keller, H., and Sadler, J. E. (1995) *Biochemistry* **34**, 4562–4568
196. Khan, A. R., and James, M. N. G. (1998) *Protein Sci.* **7**, 815–836
197. Hedstrom, L., Lin, T.-Y., and Fast, W. (1996) *Biochemistry* **35**, 4515–4523
- 197a. Khan, A. R., Khazanovich-Bernstein, N., Bergman, E. M., and James, M. N. G. (1999) *Proc. Natl. Acad. Sci. U.S.A.* **96**, 10968–10975
198. Kramer, K. J., Felsted, R. L., and Law, J. H. (1973) *J. Biol. Chem.* **248**, 3021–3028
199. Lesk, A. M., and Fordham, W. D. (1996) *J. Mol. Biol.* **258**, 501–537
200. Cygler, M., Schrag, J. D., Sussman, J. L., Harel, M., Silman, I., Gentry, M. K., and Doctor, B. P. (1993) *Protein Sci.* **2**, 366–382
201. Pereira, P. J. B., Bergner, A., Macedo, R., S., Huber, R., Matschner, G., Fritz, H., Sommerhoff, C. P., and Bode, W. (1998) *Nature (London)* **392**, 306–311
- 201a. Sommerhoff, C. P., Bode, W., Pereira, P. J. B., Stubbs, M. T., Stürzebecher, J., Piechottka, G. P., Matschner, G., and Bergner, A. (1999) *Proc. Natl. Acad. Sci. U.S.A.* **96**, 10984–10991
202. Schechter, N. M., Eng, G. Y., Selwood, T., and McCaslin, D. R. (1995) *Biochemistry* **34**, 10628–10638
203. Johnson, D. A., and Barton, G. J. (1992) *Protein Sci.* **1**, 370–377
204. Kido, H., Yokogoshi, Y., Sakai, K., Tashiro, M., Kishino, Y., Fukutomi, A., and Katunuma, N. (1992) *J. Biol. Chem.* **267**, 13573–13579
- 204a. Wong, G. W., Tang, Y., Feyfant, E., Sali, A., Li, L., Li, Y., Huang, C., Friend, D. S., Krilis, S. A., and Stevens, R. L. (1999) *J. Biol. Chem.* **274**, 30784–30793
205. Remington, S. J., Woodbury, R. G., Reynolds, R. A., Matthews, B. W., and Neurath, H. (1988) *Biochemistry* **27**, 8097–8105
206. McGrath, M. E., Mirzadegan, T., and Schmidt, B. F. (1997) *Biochemistry* **36**, 14318–14324
207. Sali, A., Matsumoto, R., McNeil, H. P., Karplus, M., and Stevens, R. L. (1993) *J. Biol. Chem.* **268**, 9023–9034
208. Springman, E. B., Dikov, M. M., and Serafin, W. E. (1995) *J. Biol. Chem.* **270**, 1300–1307
209. Hof, P., Mayr, I., Huber, R., Korzus, E., Potempa, J., Travis, J., Powers, J. C., and Bode, W. (1996) *EMBO J.* **15**, 5481–5491
210. Rao, N. V., Wehner, N. G., Marshall, B. C., Gray, W. R., Gray, B. H., and Hoidal, J. R. (1991) *J. Biol. Chem.* **266**, 9540–9548
211. Hershberger, R. J., Gershenfeld, H. K., Weissman, I. L., and Su, L. (1992) *J. Biol. Chem.* **267**, 25488–25493
212. Beresford, P. J., Kam, C.-M., Powers, J. C., and Lieberman, J. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 9285–9290
213. Caputo, A., Garner, R. S., Winkler, U., Hudig, D., and Bleackley, R. C. (1993) *J. Biol. Chem.* **268**, 17672–17675
- 213a. Harris, J. L., Peterson, E. P., Hudig, D., Thornberry, N. A., and Craik, C. S. (1998) *J. Biol. Chem.* **273**, 27364–27373
214. Fuller, R. S., Brake, A. J., and Thorner, J. (1989) *Science* **246**, 482–486
215. Cromlish, J. A., Seidah, N. G., Marcinkiewicz, M., Hamelin, J., Johnson, D. A., and Chrétien, M. (1987) *J. Biol. Chem.* **262**, 1363–1373
216. Schaner, P., Todd, R. B., Seidah, N. G., and Nillni, E. A. (1997) *J. Biol. Chem.* **272**, 19958–19968
217. Rose, C., Vargas, F., Facchinetti, P., Bourgeat, P., Bambal, R. B., Bishop, P. B., Chan, S. M. T., Moore, A. N. J., Ganellin, C. R., and Schwartz, J.-C. (1996) *Nature (London)* **380**, 403–409
218. Shilton, B. H., Thomas, D. Y., and Cygler, M. (1997) *Biochemistry* **36**, 9002–9012
219. Bullock, T. L., Branchaud, B., and Remington, S. J. (1994) *Biochemistry* **33**, 11127–11134
220. Mortensen, U. H., Remington, S. J., and Breddam, K. (1994) *Biochemistry* **33**, 508–517
221. Brayer, G. D., Delbaere, L. T. J., and James, M. N. G. (1979) *J. Mol. Biol.* **131**, 743–775
222. Plou, F. J., Kowlessur, D., Malthouse, J. P. G., Mellor, G. W., Hartshorn, M. J., Pinitglang, S., Patel, H., Topham, C. M., Thomas, E. W., Verma, C., and Brocklehurst, K. (1996) *J. Mol. Biol.* **257**, 1088–1111
223. Wells, J. A., and Estell, D. A. (1988) *Trends Biochem. Sci.* **13**, 291–297
224. Kossiakoff, A. A., Ultsch, M., White, S., and Eigenbrot, C. (1991) *Biochemistry* **30**, 1211–1221
225. Sorensen, S. B., Bech, L. M., Meldal, M., and Breddam, K. (1993) *Biochemistry* **32**, 8994–8999
- 225a. Smith, C. A., Toogood, H. S., Baker, H. M., Daniel, R. M., and Baker, E. N. (1999) *J. Mol. Biol.* **294**, 1027–1040
226. Koszelak, S., Ng, J. D., Day, J., Ko, T. P., Greenwood, A., and McPherson, A. (1997) *Biochemistry* **36**, 6597–6604
227. Delbaere, L. T. J., and Brayer, G. D. (1980) *J. Mol. Biol.* **139**, 45–51
228. Blanchard, H., and James, M. N. G. (1994) *J. Mol. Biol.* **241**, 574–587
- 228a. Tomkinson, B. (1999) *Trends Biochem. Sci.* **24**, 355–359
- 228b. Rose, C., Vargas, F., Facchinetti, P., Bourgeat, P., Bambal, R. B., Bishop, P. B., Chan, S. M. T., Moore, A. N. J., Ganellin, C. R., and Schwartz, J.-C. (1996) *Nature (London)* **380**, 403–409
- 228c. Renn, S. C. P., Tomkinson, B., and Taghert, P. H. (1998) *J. Biol. Chem.* **273**, 19173–19182
229. Wharton, C. W. (1998) in *Comprehensive Biological Catalysis. A Mechanistic Reference*, Vol. I (Sinnott, M., ed), pp. 345–379, Academic Press, San Diego, California
230. Blow, D. M. (1997) *Trends Biochem. Sci.* **22**, 405–408
231. Tsukada, H., and Blow, D. M. (1985) *J. Mol. Biol.* **184**, 703–711
232. Bode, W., and Schwager, P. (1975) *J. Mol. Biol.* **98**, 693–717
233. Gaboriaud, C., Serre, L., Guy-Crotte, O., Forest, E., and Fontecilla-Camps, J.-C. (1996) *J. Mol. Biol.* **259**, 995–1010
234. Sawyer, L., Shotten, D. M., Campbell, J. W., Wendell, P. L., Muirhead, H., and Watson, H. C. (1978) *J. Mol. Biol.* **118**, 137–208
235. Bode, W., Meyer, E., Jr., and Powers, J. C. (1989) *Biochemistry* **28**, 1951–1963
236. Stubbs, M. T., and Bode, W. (1995) *Trends Biochem. Sci.* **20**, 23–28
237. Katz, B. A., Liu, B., Barnes, M., and Springman, E. B. (1998) *Protein Sci.* **7**, 875–885
238. Timm, D. E. (1997) *Protein Sci.* **6**, 1418–1425
239. Blow, D. M. (1971) in *The Enzymes*, 3rd ed., Vol. 3 (Boyer, P. D., ed), pp. 185–212, Academic Press, New York
240. Huber, R., and Bode, W. (1977) in *NMR in Biology* (Dwek, R. A., Campbell, I. D., Richards, R. E., and Williams, R. J. P., eds), pp. 1–31, Academic Press, New York
241. Bachovchin, W. W., and Roberts, J. D. (1978) *J. Am. Chem. Soc.* **100**, 8041–8047

References

242. Smith, S. O., Farr-Jones, S., Griffin, R. G., and Bachovchin, W. W. (1989) *Science* **244**, 961–964
243. Kossiakoff, A. A., and Spencer, S. A. (1981) *Biochemistry* **20**, 6462–6474
244. Polgár, L. (1989) *Mechanisms of Protease Action*, CRC Press, Boca Raton, Florida
245. Warshel, A., Naray-Szabo, G., Sussman, F., and Hwang, J.-K. (1989) *Biochemistry* **28**, 3630–3637
246. Robillard, G., and Schulman, R. G. (1972) *J. Mol. Biol.* **71**, 507–511
247. Markley, J. L., and Westler, W. M. (1996) *Biochemistry* **35**, 11092–11097
248. Ash, E. L., Sudmeier, J. L., De Fabo, E. C., and Bachovchin, W. W. (1997) *Science* **278**, 1128–1132
249. Halkides, C. J., Wu, Y. Q., and Murray, C. J. (1996) *Biochemistry* **35**, 15941–15948
250. Zhong, S., Haghighi, K., Kettner, C., and Jordan, F. (1995) *J. Am. Chem. Soc.* **117**, 7048–7055
251. Bachovchin, W. W., Wong, W. Y. L., Farr-Jones, S., Shenvi, A. B., and Kettner, C. A. (1988) *Biochemistry* **27**, 7689–7697
252. Cassidy, C. S., Lin, J., and Frey, P. A. (1997) *Biochemistry* **36**, 4576–4584
253. Tsilikounas, E., Rao, T., Gutheil, W. G., and Bachovchin, W. W. (1996) *Biochemistry* **35**, 2437–2444
254. Tobin, J. B., Whitt, S. A., Cassidy, C. S., and Frey, P. A. (1995) *Biochemistry* **34**, 6919–6924
255. Singer, P. T., Smalås, A., Carty, R. P., Mangel, W. F., and Sweet, R. M. (1993) *Science* **259**, 669–673
256. Shieh, H.-S., Kurumbail, R. G., Stevens, A. M., Stegeman, R. A., Sturman, E. J., Pak, J. Y., Wittwer, A. J., Palmier, M. O., Wiegand, R. C., Holwerda, B. C., and Stallings, W. C. (1996) *Nature (London)* **383**, 279–282
257. Tong, L., Quin, C., Massariol, M.-J., Bonneau, P. R., Cordingley, M. G., and Lagacé, L. (1996) *Nature (London)* **383**, 272–275
258. Carter, P., and Wells, J. A. (1988) *Nature (London)* **332**, 564–568
259. Corey, D. R., and Craik, C. S. (1992) *J. Am. Chem. Soc.* **114**, 1784–1790
260. Sprang, S., Standing, T., Fletterick, R. J., Stroud, R. M., Finer-Moore, J., Xuong, N.-H., Hamlin, R., Rutter, W. J., and Craik, C. S. (1987) *Science* **237**, 905–909
261. McGrath, M. E., Vásquez, J. R., Craik, C. S., Yang, A. S., Honig, B., and Fletterick, R. J. (1992) *Biochemistry* **31**, 3059–3064
262. Robertus, J. D., Kraut, J., Alden, R. A., and Birktoft, J. J. (1972) *Biochemistry* **11**, 4293–4303
263. Whiting, A. K., and Peticolas, W. L. (1994) *Biochemistry* **33**, 552–561
264. Bryan, P., Pantoliano, M. W., Quill, S. G., Hsiao, H.-Y., and Poulos, T. (1986) *Proc. Natl. Acad. Sci. U.S.A.* **83**, 3743–3745
265. Asbóth, B., and Polgár, L. (1983) *Biochemistry* **22**, 117–122
266. Ortiz, C., Tellier, C., Williams, H., Stolowich, N. J., and Scott, A. I. (1991) *Biochemistry* **30**, 10026–10034
267. Bullock, T. L., Breddam, K., and Remington, S. J. (1996) *J. Mol. Biol.* **255**, 714–725
268. Tonge, P. J., and Carey, P. R. (1989) *Biochemistry* **28**, 6701–6709
269. Tonge, P. J., and Carey, P. R. (1992) *Biochemistry* **31**, 9122–9125
270. Birktoft, J. J., Kraut, J., and Freer, S. T. (1976) *Biochemistry* **15**, 4481–4485
271. Brünger, A. T., Huber, R., and Karplus, M. (1987) *Biochemistry* **26**, 5153–5162
272. Polgár, L., and Halász, R. (1982) *Biochem. J.* **207**, 1–10
273. Bizzozero, S. A., and Dutler, H. (1981) *Bioorg. Chem.* **10**, 46–62
274. Cruickshank, W. H., and Kaplan, H. (1974) *J. Mol. Biol.* **83**, 267–274
- 274a. Ash, E. L., Sudmeier, J. L., Day, R. M., Vincent, M., Torchilin, E. V., Haddad, K. C., Bradshaw, E. M., Sanford, D. G., and Bachovchin, W. W. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 10371–10376
- 274b. Kidd, R. D., Sears, P., Huang, D.-H., Witte, K., Wong, C.-H., and Farber, G. K. (1999) *Protein Sci.* **8**, 410–417
275. Meyer, E. F., Jr., Clore, G. M., Gronenborn, A. M., and Hansen, H. A. S. (1988) *Biochemistry* **27**, 725–730
276. Wells, G. B., Mustafi, D., and Makinen, M. W. (1994) *J. Biol. Chem.* **269**, 4577–4586
277. Ding, X., Rasmussen, B. F., Petsko, G. A., and Ringe, D. (1994) *Biochemistry* **33**, 9285–9293
278. Oppenheimer, H. L., Labouesse, B., and Hess, G. P. (1966) *J. Biol. Chem.* **241**, 2720–2730
279. Fersht, A. (1985) *Enzyme Structure and Mechanism*, 2nd ed., Freeman, San Francisco, California
280. Perona, J. J., Hedstrom, L., Rutter, W. J., and Fletterick, R. J. (1995) *Biochemistry* **34**, 1489–1499
281. Graf, L., Craik, C. S., Patthy, A., Rocznik, S., Fletterick, R. J., and Rutter, W. J. (1987) *Biochemistry* **26**, 2616–2623
282. Schellenberger, V., Turck, C. W., and Rutter, W. J. (1994) *Biochemistry* **33**, 4251–4257
283. Kurth, T., Ullmann, D., Jakubke, H.-D., and Hedstrom, L. (1997) *Biochemistry* **36**, 10098–10104
284. Le Bonniec, B. F., Myles, T., Johnson, T., Knight, C. G., Tapparelli, C., and Stone, S. R. (1996) *Biochemistry* **35**, 7114–7122
285. Wangikar, P. P., Rich, J. O., Clark, D. S., and Dordick, J. S. (1995) *Biochemistry* **34**, 12302–12310
286. Perona, J. J., and Craik, C. S. (1995) *Protein Sci.* **4**, 337–360
287. Davis, J. H., and Agard, D. A. (1998) *Biochemistry* **37**, 7696–7707
288. Baumann, W. K., Bizzozero, S. A., and Dutler, H. (1970) *FEBS Letters* **8**, 257–260
289. Baumann, W. K., Bizzozero, S. A., and Dutler, H. (1973) *Eur. J. Biochem.* **39**, 381–387
290. Stein, R. L., Elrod, J. P., and Schowen, R. L. (1983) *J. Am. Chem. Soc.* **105**, 2446–2452
291. Chang, T. K., Chiang, Y., Guo, H.-X., Kresge, A. J., Mathew, L., Powell, M. F., and Wells, J. A. (1996) *J. Am. Chem. Soc.* **118**, 8802–8807
292. Stein, R. L., and Stimpler, A. M. (1987) *J. Am. Chem. Soc.* **109**, 4387–4390
293. Baker, E. N. (1980) *J. Mol. Biol.* **141**, 441–484
294. Maes, D., Bouckaert, J., Poortmans, F., Wyns, L., and Looze, Y. (1996) *Biochemistry* **35**, 16292–16298
295. O'Hara, B. P., Hemmings, A. M., Buttle, D. J., and Pearl, L. H. (1995) *Biochemistry* **34**, 13190–13195
296. Berti, P. J., and Storer, A. C. (1995) *J. Mol. Biol.* **246**, 273–283
297. Drenth, J., Jansonius, J. N., Koekoek, R., and Wolthers, B. G. (1971) *Adv. Prot. Chem.* **25**, 79–115
298. Lewis, S. D., Johnson, F. A., and Shafer, J. A. (1981) *Biochemistry* **20**, 48–51
299. Roberts, D. D., Lewis, S. D., Ballou, D. P., Olson, S. T., and Shafer, J. A. (1986) *Biochemistry* **25**, 5595–5601
300. Johnson, F. A., Lewis, S. D., and Shafer, J. A. (1981) *Biochemistry* **20**, 44–48
301. Keillor, J. W., and Brown, R. S. (1992) *J. Am. Chem. Soc.* **114**, 7983–7989
302. Pinitglang, S., Watts, A. B., Patel, M., Reid, J. D., Noble, M. A., Gul, S., Bokth, A., Naem, A., Patel, H., Thomas, E. W., Sreedharan, S. K., Verma, C., and Brocklehurst, K. (1997) *Biochemistry* **36**, 9968–9982
303. Ménard, R., Khouri, H. E., Plouffe, C., Laflamme, P., Dupras, R., Vernet, T., Tessier, D. C., Thomas, D. Y., and Storer, A. C. (1991) *Biochemistry* **30**, 5531–5538
304. Katerelos, N. A., and Goodenough, P. W. (1996) *Biochemistry* **35**, 14763–14772
305. Vernet, T., Tessier, D. C., Chatellier, J., Plouffe, C., Lee, T. S., Thomas, D. Y., Storer, A. C., and Ménard, R. (1995) *J. Biol. Chem.* **270**, 16645–16652
306. Ménard, R., Carrière, J., Laflamme, P., Plouffe, C., Khouri, H. E., Vernet, T., Tessier, D. C., Thomas, D. Y., and Storer, A. C. (1991) *Biochemistry* **30**, 8924–8928
307. Ménard, R., Laflamme, P., Plouffe, C., Vernet, T., Tessier, D. C., Thomas, D. Y., and Storer, A. C. (1995) *Biochemistry* **34**, 464–471
308. Doran, J. D., Tonge, P. J., Mort, J. S., and Carey, P. R. (1996) *Biochemistry* **35**, 12487–12494
309. Gamcsik, M. P., Malthouse, J. P. G., Primrose, W. U., Mackenzie, N. E., Boyd, A. S. F., Russell, R. A., and Scott, A. I. (1983) *J. Am. Chem. Soc.* **105**, 6324–6325
310. Zheng, Y.-J., and Bruce, T. C. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 4285–4288
311. Arad, D., Langridge, R., and Kollman, P. A. (1990) *J. Am. Chem. Soc.* **112**, 491–502
312. Brocklehurst, K., Watts, A. B., Patel, M., Verma, C., and Thomas, E. W. (1998) in *Comprehensive Biological Catalysis. A Mechanistic Reference*, Vol. I (Sinnott, M., ed), Academic Press, San Diego, California
313. Bond, J. S., and Butler, P. E. (1987) *Ann. Rev. Biochem.* **56**, 333–364
314. Takahashi, T., Dehdarani, A. H., and Tang, J. (1988) *J. Biol. Chem.* **263**, 10952–10957
315. Coulombe, R., Grochulski, P., Sivaraman, J., Ménard, R., Mort, J. S., and Cygler, M. (1996) *EMBO J.* **15**, 5492–5503
316. Jia, Z., Hasnain, S., Hiram, T., Lee, X., Mort, J. S., To, R., and Huber, C. P. (1995) *J. Biol. Chem.* **270**, 5527–5533
317. Taralp, A., Kaplan, H., Sytwu, I.-I., Vlattas, I., Bohacek, R., Knap, A. K., Hiram, T., Huber, C. P., and Hasnain, S. (1995) *J. Biol. Chem.* **270**, 18036–18043
318. McGrath, M. E., Palmer, J. T., Brömme, D., and Somoza, J. R. (1998) *Protein Sci.* **7**, 1294–1302
319. Shi, G.-P., Webb, A. C., Foster, K. E., Knoll, J. H. M., Lemere, C. A., Munger, J. S., and Chapman, H. A. (1994) *J. Biol. Chem.* **269**, 11530–11536
320. Gelb, B. D., Shi, G.-P., Chapman, H. A., and Desnick, R. J. (1996) *Science* **273**, 1236–1238
321. Ishidoh, K., Munro, D., Sato, N., and Kominami, E. (1991) *J. Biol. Chem.* **266**, 16312–16317
322. Wolters, P. J., Raymond, W. W., Blount, J. L., and Caughey, G. H. (1998) *J. Biol. Chem.* **273**, 15514–15520
323. Schiller, M. R., Mende-Mueller, L., Moran, K., Meng, M., Miller, K. W., and Hook, V. Y. H. (1995) *Biochemistry* **34**, 7988–7995
324. Friedman, T. C., Kline, T. B., and Wilk, S. (1985) *Biochemistry* **24**, 3907–3913
325. Matsui, S.-I., Sandberg, A. A., Negoro, S., Seon, B. K., and Goldstein, G. (1982) *Proc. Natl. Acad. Sci. U.S.A.* **79**, 1535–1539
326. Joshua-Tor, L., Xu, H. E., Johnston, S. A., and Rees, D. C. (1995) *Science* **269**, 945–950
327. Richards, F. M. (1991) *Sci. Am.* **264** (Jan), 54–63
328. Ohno, S., Emori, Y., Imaizumi, S., Kawasaki, H., Kisaragi, M., and Suzuki, K. (1984) *Nature (London)* **312**, 566–570
329. Saido, T. C., Sorimachi, H., and Suzuki, K. (1994) *FASEB J.* **8**, 814–822
330. Chian, H.-L., and Dice, J. F. (1988) *J. Biol. Chem.* **263**, 6797–6805
331. Tompa, P., Baki, A., Schád, É., and Friedrich, P. (1996) *J. Biol. Chem.* **271**, 33161–33164

References

332. Stabach, P. R., Cianci, C. D., Glantz, S. B., Zhang, Z., and Morrow, J. S. (1997) *Biochemistry* **36**, 57–65
- 332a. Hosfield, C. M., Elce, J. S., Davies, P. L., and Jia, Z. (1999) *EMBO J.* **18**, 6880–6889
333. Nicholson, D. W., and Thornberry, N. A. (1997) *Trends Biochem. Sci.* **22**, 299–306
334. Villa, P., Kaufmann, S. H., and Earnshaw, W. C. (1997) *Trends Biochem. Sci.* **22**, 388–393
335. Munday, N. A., Vaillancourt, J. P., Ali, A., Casano, F. J., Miller, D. K., Molineaux, S. M., Yamin, T.-T., Yu, V. L., and Nicholson, D. W. (1995) *J. Biol. Chem.* **270**, 15870–15876
336. McGrath, M. E., Eakin, A. E., Engel, J. C., McKerrow, J. H., Craik, C. S., and Fletterick, R. J. (1995) *J. Mol. Biol.* **247**, 251–259
337. Gillmor, S. A., Craik, C. S., and Fletterick, R. J. (1997) *Protein Sci.* **6**, 1603–1611
338. Porter, W. H., Cunningham, L. W., and Mitchell, W. M. (1971) *J. Biol. Chem.* **246**, 7675–7682
339. Pavloff, N., Pemberton, P. A., Potempa, J., Chen, W.-C. A., Pike, R. N., Prochazka, V., Kiefer, M. C., Travis, J., and Barr, P. J. (1997) *J. Biol. Chem.* **272**, 1595–1600
- 339a. Nelson, D., Potempa, J., Kordula, T., and Travis, J. (1999) *J. Biol. Chem.* **274**, 12245–12251
340. Malcolm, B. A. (1995) *Protein Sci.* **4**, 1439–1445
341. Mosimann, S. C., Cherney, M. M., Sia, S., Plotch, S., and James, M. N. G. (1997) *J. Mol. Biol.* **273**, 1032–1047
342. Ding, J., McGrath, W. J., Sweet, R. M., and Mangel, W. F. (1996) *EMBO J.* **15**, 1778–1783
343. Fox, T., de Miguel, E., Mort, J. S., and Storer, A. C. (1992) *Biochemistry* **31**, 12571–12576
344. Podobnik, M., Kuhelj, R., Turk, V., and Turk, D. (1997) *J. Mol. Biol.* **271**, 774–788
345. Seemüller, E., Lupas, A., Stock, D., Löwe, J., Huber, R., and Baumeister, W. (1995) *Science* **268**, 579–582
346. Paetzel, M., and Dalbey, R. E. (1997) *Trends Biochem. Sci.* **22**, 28–31
- 346a. Kisselev, A. F., Songyang, Z., and Goldberg, A. L. (2000) *J. Biol. Chem.* **275**, 14831–14837
347. Eleuteri, A. M., Kohanski, R. A., Cardozo, C., and Orlowski, M. (1997) *J. Biol. Chem.* **272**, 11824–11831
348. Kopp, F., Kristensen, P., Hendil, K. B., Johnsen, A., Sobek, A., and Dahlmann, B. (1995) *J. Mol. Biol.* **248**, 264–272
349. Cardozo, C., Michaud, C., and Orlowski, M. (1999) *Biochemistry* **38**, 9768–9777
- 349a. Groll, M., Heinemeyer, W., Jäger, S., Ullrich, T., Bochtler, M., Wolf, D. H., and Huber, R. (1999) *Proc. Natl. Acad. Sci. U.S.A.* **96**, 10976–10983
- 349b. Witt, E., Zantopf, D., Schmidt, M., Kraft, R., Kloetzel, P.-M., and Krüger, E. (2000) *J. Mol. Biol.* **301**, 1–9
- 349c. Orlowski, M., and Wilk, S. (2000) *Arch. Biochem. Biophys.* **383**, 1–16
350. Tikkanen, R., Riikonen, A., Oinonen, C., Rouvinen, J., and Pelttonen, L. (1996) *EMBO J.* **15**, 2954–2960
351. Xuan, J., Tarentino, A. L., Grimwood, B. G., Plummer, T. H., Jr., Cui, T., Guan, C., and Van Roey, P. (1998) *Protein Sci.* **7**, 774–781
352. Liu, Y., Guan, C., and Aronson, N. N., Jr. (1998) *J. Biol. Chem.* **273**, 9688–9694
353. Guan, C., Liu, Y., Shao, Y., Cui, T., Liao, W., Ewel, A., and Whitaker, R. (1998) *J. Biol. Chem.* **273**, 9695–9702
354. Brannigan, J. A., Dodson, G., Duggleby, H. J., Moody, P. C. E., Smith, J. L., Tomchick, D. R., and Murzin, A. G. (1995) *Nature (London)* **378**, 416–419
355. Dalbey, R. E., and Wickner, W. (1988) *J. Biol. Chem.* **263**, 404–408
356. Tschantz, W. R., Sung, M., Delgado-Partin, V. M., and Dalbey, R. E. (1993) *J. Biol. Chem.* **268**, 27349–27354
357. Dalbey, R. E., Lively, M. O., Bron, S., and Van Dijk, J. M. (1997) *Protein Sci.* **6**, 1129–1138
358. Fenteany, G., and Schreiber, S. L. (1998) *J. Biol. Chem.* **273**, 8545–8548
359. Bogoy, M., McMaster, J. S., Gaczynska, M., Tortorella, D., Goldberg, A. L., and Ploegh, H. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 6629–6634
360. Schmidtke, G., Kraft, R., Kostka, S., Henklein, P., Frömmel, C., Löwe, J., Huber, R., Kloetzel, P. M., and Schmidt, M. (1996) *EMBO J.* **15**, 6887–6898
361. Heinemeyer, W., Fischer, M., Krimmer, T., Stachon, U., and Wolf, D. H. (1997) *J. Biol. Chem.* **272**, 25200–25209
362. Ditzel, L., Huber, R., Mann, K., Heinemeyer, W., Wolf, D. H., and Groll, M. (1998) *J. Mol. Biol.* **279**, 1187–1191
363. Andreeva, N. S., Zdanov, A. S., Gustchina, A. E., and Fedorov, A. A. (1984) *J. Biol. Chem.* **259**, 11353–11365
364. Fusek, M., and Vetvicka, V., eds. (1995) *Aspartic Proteinases: Physiology and Pathology*, CRC Press, Boca Raton, Florida
365. Moore, S. A., Sielecki, A. R., Chernaia, M. M., Tarasova, N. I., and James, M. N. G. (1995) *J. Mol. Biol.* **247**, 466–485
366. Strop, P., Sedlacek, J., Stys, J., Kaderabkova, Z., Blaha, I., Pavlickova, L., Pohl, J., Fabry, M., Kostka, V., Newman, M., Frazao, C., Shearer, A., Tickle, I. J., and Blundell, T. L. (1990) *Biochemistry* **29**, 9863–9871
367. Vance, J. E., LeBlanc, D. A., and London, R. E. (1997) *Biochemistry* **36**, 13232–13240
368. Majer, P., Collins, J. R., Gulnik, S. V., and Erickson, J. W. (1997) *Protein Sci.* **6**, 1458–1466
369. Krieger, T. J., and Hook, V. Y. H. (1992) *Biochemistry* **31**, 4223–4231
370. Fineschi, B., and Miller, J. (1997) *Trends Biochem. Sci.* **22**, 377–382
371. Wilk, S., Wilk, E., and Magnusson, R. P. (1998) *J. Biol. Chem.* **273**, 15961–15970
372. Tong, L., Pav, S., Lamarre, D., Pilote, L., LaPlante, S., Anderson, P. C., and Jung, G. (1995) *J. Mol. Biol.* **250**, 211–222
373. Yang, J., Teplyakov, A., and Quail, J. W. (1997) *J. Mol. Biol.* **268**, 449–459
374. Gómez, J., and Freire, E. (1995) *J. Mol. Biol.* **252**, 337–350
375. Symersky, J., Monod, M., and Foundling, S. I. (1997) *Biochemistry* **36**, 12700–12710
376. Green, D. W., Aykent, S., Gierse, J. K., and Zupce, M. E. (1990) *Biochemistry* **29**, 3126–3133
377. Baldwin, E. T., Bhat, T. N., Gulnik, S., Hosur, M. V., Sowder, R. C., Cachau, R. E., Collins, J., Silva, A. M., and Erickson, J. W. (1993) *Proc. Natl. Acad. Sci. U.S.A.* **90**, 6796–6800
378. Goldblum, A. (1988) *Biochemistry* **27**, 1653–1658
379. Subramanian, E. (1978) *Trends Biochem. Sci.* **3**, 1–3
380. Andreeva, N. S., Zdanov, A. S., Gustchina, A. E., and Fedorev, A. A. (1984) *J. Biol. Chem.* **259**, 11353–11365
381. James, M. N. G., Sielecki, A. R., Hayakawa, K., and Gelb, M. H. (1992) *Biochemistry* **31**, 3872–3886
382. Meek, T. D. (1998) in *Comprehensive Biological Catalysis. A Mechanistic Reference*, Vol. I (Sinnott, M., ed), pp. 327–344, Academic Press, San Diego, California
383. Blum, M., Cunningham, A., Pang, H., and Hofmann, T. (1991) *J. Biol. Chem.* **266**, 9501–9507
384. James, M. N. G., and Sielecki, A. R. (1986) *Nature (London)* **319**, 33–38
385. Blundell, T. L., Lapatto, R., Wilderspin, A. F., Hemmings, A. M., Hobart, P. M., Danley, D. E., and Whittle, P. J. (1990) *Trends Biochem. Sci.* **15**, 425–430
386. Katz, R. A., and Skalka, A. M. (1994) *Ann. Rev. Biochem.* **63**, 133–173
387. Wlodawer, A., Miller, M., Jaskólski, M., Sathyanarayana, B. K., Baldwin, E., Weber, I. T., Selk, L. M., Clawson, L., Schneider, J., and Kent, S. B. H. (1989) *Science* **245**, 616–620
388. Rose, R. B., Craik, C. S., Douglas, N. L., and Stroud, R. M. (1996) *Biochemistry* **35**, 12933–12944
389. Chatfield, D. C., and Brooks, B. R. (1995) *J. Am. Chem. Soc.* **117**, 5561–5572
390. Rodriguez, E. J., Angeles, T. S., and Meek, T. D. (1993) *Biochemistry* **32**, 12380–12385
391. Liu, H., Müller-Plathe, F., and van Gunsteren, W. F. (1996) *J. Mol. Biol.* **261**, 454–469
392. Silva, A. M., Cachau, R. E., Sham, H. L., and Erickson, J. W. (1996) *J. Mol. Biol.* **255**, 321–346
393. Reverter, D., Ventura, S., Villegas, V., Vendrell, J., and Avilés, F. X. (1998) *J. Biol. Chem.* **273**, 3535–3541
394. Varlamov, O., Leiter, E. H., and Fricker, L. (1996) *J. Biol. Chem.* **271**, 13981–13986
395. Rodríguez, C., Brayton, K. A., Brownstein, M., and Dixon, J. E. (1989) *J. Biol. Chem.* **264**, 5988–5995
396. Tan, F., Chan, S. J., Steiner, D. F., Schilling, J. W., and Skidgel, R. A. (1989) *J. Biol. Chem.* **264**, 13165–13170
397. Tan, F., Weerasinghe, D. K., Skidgel, R. A., Tamei, H., Kaul, R. K., Roninson, I. B., Schilling, J. W., and Erdős, E. G. (1990) *J. Biol. Chem.* **265**, 13–19
398. Ehlers, M. R. W., and Riordan, J. F. (1989) *Biochemistry* **28**, 5311–5318
399. Dideberg, O., Charlier, P., Dive, G., Joris, B., Frere, J. M., and Ghuyssen, J. M. (1982) *Nature (London)* **299**, 469–470
400. Weaver, L. H., Kester, W. R., and Matthews, B. W. (1977) *J. Mol. Biol.* **114**, 119–132
401. Hausrath, A. C., and Matthews, B. W. (1994) *J. Biol. Chem.* **269**, 18839–18842
402. Holland, D. R., Hausrath, A. C., Juers, D., and Matthews, B. W. (1995) *Protein Sci.* **4**, 1955–1965
403. Holden, H. M., Tronrud, D. E., Monzingo, A. F., Weaver, L. H., and Matthews, B. W. (1987) *Biochemistry* **26**, 8542–8553
404. Mock, W. L., and Stanford, D. J. (1996) *Biochemistry* **35**, 7369–7377
405. Marie-Claire, C., Ruffet, E., Antonczak, S., Beaumont, A., O'Donohue, M., Roques, B. P., and Fournié-Zaluski, M.-C. (1997) *Biochemistry* **36**, 13938–13945
406. Lipman, M. L., Panda, D., Bennett, H. P. J., Henderson, J. E., Shane, E., Shen, Y., Goltzman, D., and Karaplis, A. C. (1998) *J. Biol. Chem.* **273**, 13729–13737
407. Ogishima, T., Niidome, T., Shimokata, K., Kitada, S., and Ito, A. (1995) *J. Biol. Chem.* **270**, 30322–30326
408. Luciano, P., Tokatlidis, K., Chabre, L., Germanique, J.-C., and Géli, V. (1998) *J. Mol. Biol.* **280**, 193–199
409. Christianson, D. W., and Lipscomb, W. N. (1986) *Proc. Natl. Acad. Sci. U.S.A.* **83**, 7568–7572
410. Mock, W. L. (1998) in *Comprehensive Biological Catalysis. A Mechanistic Reference*, Vol. I (Sinnott, M., ed), pp. 425–453, Academic Press, San Diego, California
411. Campbell, P., and Nashed, N. T. (1982) *J. Am. Chem. Soc.* **104**, 5221–5226
412. Makinen, M. W., Fukuyama, J. M., and Kuo, L. C. (1982) *J. Am. Chem. Soc.* **104**, 2667–2669
413. Sander, M. E., and Witzel, H. (1985) *Biochem. Biophys. Res. Commun.* **132**, 681–687
414. Kuo, L. C., Fukuyama, J. M., and Makinen, M. W. (1983) *J. Mol. Biol.* **163**, 63–105

References

415. Suh, J., Hong, S.-B., and Chung, S. (1986) *J. Biol. Chem.* **261**, 7112–7114
416. Mustafi, D., and Makinen, M. W. (1994) *J. Biol. Chem.* **269**, 4587–4595
417. Kuo, L. C., and Makinen, M. W. (1985) *J. Am. Chem. Soc.* **107**, 5255–5261
418. Cleland, W. W. (1977) *Adv. Enzymol.* **45**, 273–286
419. Kim, H., and Lipscomb, W. N. (1990) *Biochemistry* **29**, 5546–5555
420. Osumi, A., Rahmo, A., King, S. W., Przystas, T. J., and Fife, T. H. (1994) *Biochemistry* **33**, 14750–14757
421. Mock, W. L., and Zhang, J. Z. (1991) *J. Biol. Chem.* **266**, 6393–6400
422. Zhang, K., and Auld, D. S. (1995) *Biochemistry* **34**, 16306–16312
423. Hilver, D., Gardell, S. J., Rutter, W. J., and Kaiser, E. T. (1986) *J. Am. Chem. Soc.* **108**, 5298–5304
424. Kam, C.-M., Nishino, N., and Powers, J. C. (1979) *Biochemistry* **18**, 3032–3038
425. Bartlett, P. A., and Marlowe, C. K. (1983) *Biochemistry* **22**, 4618–4624
426. Holden, H. M., and Matthews, B. W. (1988) *J. Biol. Chem.* **263**, 3256–3260
427. Massova, I., Kotra, L. P., Fridman, R., and Mobashery, S. (1998) *FASEB J.* **12**, 1075–1095
428. Bond, J. S., and Beynon, R. J. (1995) *Protein Sci.* **4**, 1247–1261
429. Stöcker, W., Grams, F., Baumann, U., Reinemer, P., Gomis-Rüth, F.-X., McKay, D. B., and Bode, W. (1995) *Protein Sci.* **4**, 823–840
- 429a. Morgunova, E., Tuuttila, A., Bergmann, U., Isupov, M., Lindqvist, Y., Schneider, G., and Tryggvason, K. (1999) *Science* **284**, 1667–1670
- 429b. Tortorella, M. D., and 27 other authors. (1999) *Science* **284**, 1664–1666
- 429c. Nagase, H., and Weissner, J. F., Jr. (1999) *J. Biol. Chem.* **274**, 21491–21494
430. Becker, J. W., Marcy, A. I., Rokosz, L. L., Axel, M. G., Burbaum, J. J., Fitzgerald, P. M. D., Cameron, P. M., Esser, C. K., Hagmann, W. K., Hermes, J. D., and Springer, J. P. (1995) *Protein Sci.* **4**, 1966–1976
431. Wetmore, D. R., and Hardman, K. D. (1996) *Biochemistry* **35**, 6549–6558
432. Arumugam, S., Hemme, C. L., Yoshida, N., Suzuki, K., Nagase, H., Berjanskii, M., Wu, B., and Van Doren, S. R. (1998) *Biochemistry* **37**, 9650–9657
433. Gomis-Rüth, F. X., Stöcker, W., Huber, R., Zwilling, R., and Bode, W. (1993) *J. Mol. Biol.* **229**, 945–968
434. Gomis-Rüth, F. X., Kress, L. F., Kellermann, J., Mayr, I., Lee, X., Huber, R., and Bode, W. (1994) *J. Mol. Biol.* **239**, 513–544
435. Llano, E., Pendás, A. M., Knäuper, V., Sorsa, T., Salo, T., Salido, E., Murphy, G., Simmer, J. P., Bartlett, J. D., and López-Otín, C. (1997) *Biochemistry* **36**, 15101–15108
436. Gomis-Rüth, F. X., Gohlke, U., Betz, M., Knäuper, V., Murphy, G., López-Otín, C., and Bode, W. (1996) *J. Mol. Biol.* **264**, 556–566
437. Taylor, A. (1993) *FASEB J.* **7**, 290–298
438. Sträter, N., and Lipscomb, W. N. (1995) *Biochemistry* **34**, 14792–14800
- 438a. Sträter, N., Sun, L., Kantrowitz, E. R., and Lipscomb, W. N. (1999) *Proc. Natl. Acad. Sci. U.S.A.* **96**, 11151–11155
439. Chen, G., Edwards, T., D'souza, V. M., and Holz, R. C. (1997) *Biochemistry* **36**, 4278–4286
440. Lowther, W. T., Orville, A. M., Madden, D. T., Lim, S., Rich, D. H., and Matthews, B. W. (1999) *Biochemistry* **38**, 7678–7688
441. Bradshaw, R. A., Bricey, W. W., and Walker, K. W. (1998) *Trends Biochem. Sci.* **23**, 263–267
442. Wilce, M. C. J., Bond, C. S., Dixon, N. E., Freeman, H. C., Guss, J. M., Lilley, P. E., and Wilce, J. A. (1998) *Proc. Natl. Acad. Sci. U.S.A.* **95**, 3472–3477
443. Chan, M. K., Gong, W., Rajagopalan, P. T. R., Hao, B., Tsai, C. M., and Pei, D. (1997) *Biochemistry* **36**, 13904–13909
444. Dardel, F., Ragusa, S., Lazennec, C., Blanquet, S., and Meinel, T. (1998) *J. Mol. Biol.* **280**, 501–513
445. Becker, A., Schlichting, I., Kabsch, W., Schultz, S., and Wagner, A. F. V. (1998) *J. Biol. Chem.* **273**, 11413–11416
446. Andrews, D. W., and Johnson, A. E. (1996) *Trends Biochem. Sci.* **21**, 365–369
- 446a. Fabunmi, R. P., Wigley, W. C., Thomas, P. J., and DeMartino, G. N. (2000) *J. Biol. Chem.* **275**, 409–413
447. Sommer, T., and Wolf, D. H. (1997) *FASEB J.* **11**, 1227–1233
448. Ciechanover, A., and Schwartz, A. L. (1998) *Proc. Natl. Acad. Sci. U.S.A.* **95**, 2727–2730
449. Rubin, D. M., Glickman, M. H., Larsen, C. N., Dhruvakumar, S., and Finley, D. (1998) *EMBO J.* **17**, 4909–4919
450. Lee, Y. S., Park, S. C., Goldberg, A. L., and Chung, C. H. (1988) *J. Biol. Chem.* **263**, 6643–6646
451. Shin, D. H., Lee, C. S., Chung, C. H., and Suh, S. W. (1996) *J. Mol. Biol.* **262**, 71–76
452. Chin, D. T., Goff, S. A., Webster, T., Smith, T., and Goldberg, A. L. (1988) *J. Biol. Chem.* **263**, 11718–11728
453. Flanagan, J. M., Wall, J. S., Capel, M. S., Schneider, D. K., and Shanklin, J. (1995) *Biochemistry* **34**, 10910–10917
454. Kessel, M., Maurizi, M. R., Kim, B., Kocsis, E., Trus, B. L., Singh, S. K., and Steven, A. C. (1995) *J. Mol. Biol.* **250**, 587–594
455. Suzuki, C. K., Rep, M., Maarten van Dijk, J., Suda, K., Grivell, L. A., and Schatz, G. (1997) *Trends Biochem. Sci.* **22**, 118–123
456. Roudiak, S. G., and Shrader, T. E. (1998) *Biochemistry* **37**, 11255–11263
457. Wang, N., Maurizi, M. R., Emmert-Buck, L., and Gottesman, M. M. (1994) *J. Biol. Chem.* **269**, 29308–29313
458. Savel'ev, A. S., Novikova, L. A., Kovaleva, I. E., Luzikov, V. N., Neupert, W., and Langer, T. (1998) *J. Biol. Chem.* **273**, 20596–20602
459. Singh, S. K., Guo, F., and Maurizi, M. R. (1999) *Biochemistry* **38**, 14906–14915
460. Kihara, A., Akiyama, Y., and Ito, K. (1998) *J. Mol. Biol.* **279**, 175–188
461. Lupas, A., Flanagan, J. M., Tamura, T., and Baumeister, W. (1997) *Trends Biochem. Sci.* **22**, 399–404
- 461a. de Sagarra, M. R., Mayo, I., Marco, S., Rodríguez-Vilarino, S., Oliva, J., Carrascosa, J. L., and Castano, J. G. (1999) *J. Mol. Biol.* **292**, 819–825
- 461b. Kim, K. I., Cheong, G.-W., Park, S.-C., Ha, J.-S., Woo, K. M., Choi, S. J., and Chung, C. H. (2000) *J. Mol. Biol.* **303**, 655–666
462. Rohrwild, M., Coux, O., Huang, H.-C., Moerschell, R. P., Yoo, S. J., Seol, J. H., Chung, C. H., and Goldberg, A. L. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 5808–5813
463. Missiakas, D., Schwager, F., Betton, J.-M., Georgopoulos, C., and Raina, S. (1996) *EMBO J.* **15**, 6899–6909
- 463a. Bochtler, M., Hartmann, C., Song, H. K., Bourenkov, G. P., Bartunik, H. D., and Huber, R. (2000) *Nature (London)* **403**, 800–805
- 463b. Shotland, Y., Telf, D., Koby, S., Kobilier, O., and Oppenheim, A. B. (2000) *J. Mol. Biol.* **299**, 953–964
464. Kräusslich, H.-G., and Wimmer, E. (1988) *Ann. Rev. Biochem.* **57**, 701–754
465. Ypma-Wong, M. F., Filman, D. J., Hogle, J. M., and Semler, B. L. (1988) *J. Biol. Chem.* **263**, 17846–17856
466. López-Otín, C., Simón-Mateo, C., Martínez, L., and Vinuela, E. (1989) *J. Biol. Chem.* **264**, 9107–9110
- 466a. Wu, Z., Yao, N., Le, H. V., and Weber, P. C. (1998) *Trends Biochem. Sci.* **23**, 92–93
- 466b. Barbato, G., Cicero, D. O., Cordier, F., Narjes, F., Gerlach, B., Sambucini, S., Grzesiek, S., Matassa, V. G., De Francesco, R., and Bazzo, R. (2000) *EMBO J.* **19**, 1195–1206
467. Roseman, J. E., and Levine, R. L. (1987) *J. Biol. Chem.* **262**, 2101–2110
468. Grune, T., Reinheckel, T., and Davies, K. J. A. (1997) *FASEB J.* **11**, 526–534
469. Erdős, E. G., and Skidgel, R. A. (1989) *FASEB J.* **3**, 145–151
470. Hui, K.-S. (1988) *J. Biol. Chem.* **263**, 6613–6618
471. Zisfein, J. B., Graham, R. M., Dreskin, S. V., Willey, G. M., Fischman, A. J., and Honey, C. J. (1987) *Biochemistry* **26**, 8690–8697
- 471a. Loew, D., Perrault, C., Morales, M., Moog, S., Ravanat, C., Schuhler, S., Arcone, R., Pietropaolo, C., Cazenave, J.-P., van Dorsselaer, A., and Lanza, F. (2000) *Biochemistry* **39**, 10812–10822
472. Lobe, C. G., Finlay, B. B., Paranchych, W., Paetkau, V. H., and Bleackley, R. C. (1986) *Science* **232**, 858–861
473. Melloni, E., Pontremoki, S., Salamino, F., Sparatore, B., Michetti, M., Sacco, O., and Horecker, B. L. (1986) *J. Biol. Chem.* **261**, 11437–11439
474. Pohlner, J., Halter, R., Beyreuther, K., and Meyer, T. F. (1987) *Nature (London)* **325**, 458–462
475. Newport, G. R., McKerrow, J. H., Hedstrom, R., Pettit, M., McGarrigle, L., Barr, D. J., and Agabian, N. (1988) *J. Biol. Chem.* **263**, 13179–13184
476. Folk, J. E. (1980) *Ann. Rev. Biochem.* **49**, 517–531
477. McGrath, M. E., Gillmor, S. A., and Fletterick, R. J. (1995) *Protein Sci.* **4**, 141–148
478. Salier, J.-P. (1990) *Trends Biochem. Sci.* **15**, 435–439
479. Laskowski, M., Jr., and Kato, I. (1980) *Ann. Rev. Biochem.* **49**, 593–626
480. Barrett, A. J. (1987) *Trends Biochem. Sci.* **12**, 193–196
481. Freije, J. P., Balbin, M., Abrahamson, M., Velasco, G., Dalboge, H., Grubb, A., and López-Otín, C. (1993) *J. Biol. Chem.* **268**, 15737–15744
482. Martin, J. R., Craven, C. J., Jerala, R., Kroon-Zitko, L., Zverovnik, E., Turk, V., and Waltho, J. P. (1995) *J. Mol. Biol.* **246**, 331–343
483. Brown, W. M., and Dziegielewska, K. M. (1997) *Protein Sci.* **6**, 5–12
484. Williamson, R. A., Carr, M. D., Frenkiel, T. A., Feeney, J., and Freedman, R. B. (1997) *Biochemistry* **36**, 13882–13889
485. Apte, S. S., Olsen, B. R., and Murphy, G. (1995) *J. Biol. Chem.* **270**, 14313–14318
486. Baumann, U., Bauer, M., Létóffé, S., Deleplaire, P., and Wandersman, C. (1995) *J. Mol. Biol.* **248**, 653–661
487. Martzen, M. R., McMullen, B. A., Smith, N. E., Fujikawa, K., and Peanasky, R. J. (1990) *Biochemistry* **29**, 7366–7372
488. Huang, K., Lu, W., Anderson, S., Laskowski, M., Jr., and James, M. N. G. (1995) *Protein Sci.* **4**, 1985–1997
489. Lu, W., Qasim, M. A., and Kent, S. B. H. (1996) *J. Am. Chem. Soc.* **118**, 8518–8523
490. Huber, R., Kukla, D., Bode, W., Schwager, P., Bartels, K., Dessenhofer, J., and Steigemann, W. (1974) *J. Mol. Biol.* **89**, 73–101

References

491. Mer, G., Hietter, H., Kellenberger, C., Renatus, M., Luu, B., and Lefèvre, J.-F. (1996) *J. Mol. Biol.* **258**, 158–171
492. Conconi, A., and Ryan, C. A. (1993) *J. Biol. Chem.* **268**, 430–435
493. Werner, M. H., and Wemmer, D. E. (1991) *Biochemistry* **30**, 3356–3364
494. McBride, J. D., Brauer, A. B. E., Nievo, M., and Leatherbarrow, R. J. (1998) *J. Mol. Biol.* **282**, 447–457
495. Song, H. K., and Suh, S. W. (1998) *J. Mol. Biol.* **275**, 347–363
496. Brandt, P., and Woodward, C. (1987) *Biochemistry* **26**, 3156–3162
497. Huber, R., and Bode, W. (1978) *Acc. Chem. Res.* **11**, 114–122
498. Potempa, J., Korzus, E., and Travis, J. (1994) *J. Biol. Chem.* **269**, 15957–15960
499. Whistock, J., Skinner, R., and Lesk, A. M. (1998) *Trends Biochem. Sci.* **23**, 63–67
500. Lukacs, C. M., Rubin, H., and Christianson, D. W. (1998) *Biochemistry* **37**, 3297–3304
- 500a. Whistock, J. C., Skinner, R., Carrell, R. W., and Lesk, A. M. (2000) *J. Mol. Biol.* **296**, 685–699
501. Lawrence, D. A., Ginsburg, D., Day, D. E., Berkenpas, M. B., Verhamme, I. M., Kvassman, J.-O., and Shore, J. D. (1995) *J. Biol. Chem.* **270**, 25309–25312
502. Kaslik, G., Kardos, J., Szabó, E., Szilágyi, L., Závodszy, P., Westler, W. M., Markley, J. L., and Gráf, L. (1997) *Biochemistry* **36**, 5455–5464
503. Bode, W., Wei, A.-Z., Huber, R., Meyer, E., Travis, J., and Neumann, S. (1986) *EMBO J.* **5**, 2453–2458
504. Boswell, D. R., and Carrell, R. (1986) *Trends Biochem. Sci.* **11**, 102–103
505. Curiel, D. T., Holmes, M. D., Okayama, H., Brantly, M. L., Vogelmeier, C., Travis, W. D., Stier, L. E., Perks, W. H., and Crystal, R. G. (1989) *J. Biol. Chem.* **264**, 13938–13945
506. Lomas, D. A., Finch, J. T., Seyama, K., Nukiwa, T., and Carrell, R. W. (1993) *J. Biol. Chem.* **268**, 15333–15335
507. Carrell, R. W., Jeppson, J.-O., Laurell, C.-B., Brennan, S. O., Owen, M. C., Vaughan, L., and Boswell, D. R. (1982) *Nature (London)* **298**, 329–334
508. Carp, H., Miller, F., Hoidal, J. R., and Janoff, A. (1982) *Proc. Natl. Acad. Sci. U.S.A.* **79**, 2041–2045
509. Radinsky, L. (1983) *Science* **221**, 1187–1191
510. Culliton, B. J. (1989) *Science* **246**, 750–751
511. Matheson, N. R., Gibson, H. E., Hallewell, R. A., Barr, P. J., and Travis, J. (1986) *J. Biol. Chem.* **261**, 10404–10409
512. Rosenfeld, M. A., Siegfried, W., Yoshimura, K., Yoneyama, K., Fukayama, M., Stier, L. E., Pääkkö, P. K., Gilardi, P., Stratford-Perricaudet, L. D., Perricaudet, M., Jallat, S., Pavirani, A., Lecocq, J.-P., and Crystal, R. G. (1991) *Science* **252**, 431–434
513. Owen, M. C., Brennan, S. O., Lewis, J. H., and Carrell, R. W. (1983) *N. Engl. J. Med.* **309**, 694–698
514. Davie, E. W., Fujikawa, K., and Kisiel, W. (1991) *Biochemistry* **30**, 10364–10370
515. Smith, E. L., Hill, R. L., Lehman, I. R., Lefkowitz, R. J., Handler, P., and White, A. (1983) in *Principles of Biochemistry, Mammalian Biochemistry*, 7th ed., pp. 17–37, McGraw-Hill, New York
516. Mann, K. G., Jenny, R. J., and Krishnaswamy, S. (1988) *Ann. Rev. Biochem.* **57**, 915–956
517. Zucker, M. B. (1980) *Sci. Am.* **242** (Jun), 86–103
518. Gibbs, C. S., McCurdy, S. N., Leung, L. L. K., and Paborsky, L. R. (1994) *Biochemistry* **33**, 14003–14010
519. Muller, Y. A., Ultsch, M. H., Kelley, R. F., and de Vos, A. M. (1994) *Biochemistry* **33**, 10864–10870
520. Harlos, K., Martin, D. M. A., O'Brien, D. P., Jones, E. Y., Stuart, D. I., Polikarpov, I., Miller, A., Tuddenham, E. G. D., and Boys, C. W. G. (1994) *Nature (London)* **370**, 662–666
521. Martin, D. M. A., Boys, C. W. G., and Ruf, W. (1995) *FASEB J.* **9**, 852–859
522. Ruf, W., Kelly, C. R., Schullek, J. R., Martin, D. M. A., Polikarpov, I., Boys, C. W. G., Tuddenham, E. G. D., and Edgington, T. S. (1995) *Biochemistry* **34**, 6310–6315
523. Muller, Y. A., Ultsch, M. H., and de Vos, A. M. (1996) *J. Mol. Biol.* **256**, 144–159
524. Doolittle, R. F. (1981) *Sci. Am.* **245** (Dec), 126–135
525. Fu, Y., Weissbach, L., Plant, P. W., Oddoux, C., Cao, Y., Liang, T. J., Roy, S. N., Redman, C. M., and Grieninger, G. (1992) *Biochemistry* **31**, 11968–11972
526. Hunziker, E. B., Straub, P. W., and Haeblerli, A. (1990) *J. Biol. Chem.* **265**, 7455–7463
527. Banner, D. W., D'Arcy, A., Chène, C., Winkler, F. K., Guha, A., Konigsberg, W. H., Nemerson, Y., and Kirchhofer, D. (1996) *Nature (London)* **380**, 41–46
- 527a. Banner, D. W. (2000) *Nature (London)* **404**, 449–450
- 527b. Chang, Y.-J., Hamaguchi, N., Chang, S.-C., Ruf, W., Shen, M.-C., and Lin, S.-W. (1999) *Biochemistry* **38**, 10940–10948
528. Muranyi, A., Finn, B. E., Gippert, G. P., Forsén, S., Stenflo, J., and Drakenberg, T. (1998) *Biochemistry* **37**, 10605–10615
529. Andrews, B. S. (1991) *Trends Biochem. Sci.* **16**, 31–36
530. Neuenschwander, P. F., and Morrissey, J. H. (1992) *J. Biol. Chem.* **267**, 14477–14482
531. Altieri, D. C. (1995) *FASEB J.* **9**, 860–865
532. Brandstetter, H., Kühne, A., Bode, W., Huber, R., von der Saal, W., Wirthensohn, K., and Engh, R. A. (1996) *J. Biol. Chem.* **271**, 29988–29992
533. Sabharwal, A. K., Padmanabhan, K., Tulinsky, A., Mathur, A., Gorka, J., and Bajaj, S. P. (1997) *J. Biol. Chem.* **272**, 22037–22045
534. Dharmawardana, K. R., and Bock, P. E. (1998) *Biochemistry* **37**, 13143–13152
- 534a. Macedo-Ribeiro, S., Bode, W., Huber, R., Quinn-Allen, M. A., Kim, S. W., Ortel, T. L., Bourenkov, G. P., Bartunik, H. D., Stubbs, M. T., Kane, W. H., and Fuentes-Prior, P. (1999) *Nature (London)* **402**, 434–439
535. Xue, J., Kalafatis, M., Silveira, J. R., Kung, C., and Mann, K. G. (1994) *Biochemistry* **33**, 13109–13116
536. Comfurios, P., Smeets, E. F., Willems, G. M., Bevers, E. M., and Zwaal, R. F. A. (1994) *Biochemistry* **33**, 10319–10324
537. Betz, A., and Krishnaswamy, S. (1998) *J. Biol. Chem.* **273**, 10709–10718
538. Bottenus, R. E., Ichinose, A., and Davie, E. W. (1990) *Biochemistry* **29**, 11195–11209
539. Slon-Usakiewicz, J. J., Purisima, E., Tsuda, Y., Sulea, T., Pedyczak, A., Féthière, J., Cygler, M., and Konishi, Y. (1997) *Biochemistry* **36**, 13494–13502
540. Hughes, P. E., Morgan, G., Rooney, E. K., Brownlee, G. G., and Handford, P. (1993) *J. Biol. Chem.* **268**, 17727–17733
- 540a. Mathur, A., and Bajaj, S. P. (1999) *J. Biol. Chem.* **274**, 18477–18486
- 540b. Tolkatheev, D., Ng, A., Zhu, B., and Ni, F. (2000) *Biochemistry* **39**, 10365–10372
541. Xu, W.-f., Chung, D. W., and Davie, E. W. (1996) *J. Biol. Chem.* **271**, 27948–27953
- 541a. Brown, J. H., Volkman, N., Jun, G., Hensch-Edman, A. H., and Cohen, C. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 85–90
542. Doolittle, R. F., Everse, S. J., and Spraggon, G. (1996) *FASEB J.* **10**, 1464–1470
543. Everse, S. J., Spraggon, G., Veerapandian, L., Riley, M., and Doolittle, R. F. (1998) *Biochemistry* **37**, 8637–8642
544. Spraggon, G., Everse, S. J., and Doolittle, R. F. (1997) *Nature (London)* **389**, 455–462
545. Murthy, S. N. P., Wilson, J. H., Lukas, T. J., Veklich, Y., Weisel, J. W., and Lorand, L. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 44–48
546. Pedersen, L. C., Yee, V. C., Bishop, P. D., Le Trong, I., Teller, D. C., and Stenkamp, R. E. (1994) *Protein Sci.* **3**, 1131–1135
547. Bernardo, M. M., Day, D. E., Olson, S. T., and Shore, J. D. (1993) *J. Biol. Chem.* **268**, 12468–12476
548. Beaubien, G., Rosinski-Chupin, I., Mattei, M. G., Mbikay, M., Chrétien, M., and Seidah, N. G. (1991) *Biochemistry* **30**, 1628–1635
549. Herwald, H., Renné, T., Meijers, J. C. M., Chung, D. W., Page, J. D., Colman, R. W., and Müller-Esterl, W. (1996) *J. Biol. Chem.* **271**, 13061–13067
550. Baglia, F. A., Jameson, B. A., and Walsh, P. N. (1993) *J. Biol. Chem.* **268**, 3838–3844
551. Hamaguchi, N., Charifson, P. S., Pedersen, L. G., Brayer, G. D., Smith, K. J., and Stafford, D. W. (1991) *J. Biol. Chem.* **266**, 15213–15220
552. Kurachi, S., Hitomi, Y., Furukawa, M., and Kurachi, K. (1995) *J. Biol. Chem.* **270**, 5276–5281
- 552a. Kurachi, S., Deyashiki, Y., Takeshita, J., and Kurachi, K. (1999) *Science* **285**, 739–743
553. Freedman, S. J., Furie, B. C., Furie, B., and Baleja, J. D. (1995) *Biochemistry* **34**, 12126–12137
554. Freedman, S. J., Blostein, M. D., Baleja, J. D., Jacobs, M., Furie, B. C., and Furie, B. (1996) *J. Biol. Chem.* **271**, 16227–16236
555. Lenting, P. J., Christophe, O. D., ter Maat, H., Rees, D. J. G., and Mertens, K. (1996) *J. Biol. Chem.* **271**, 25332–25337
556. Tagliavacca, L., Moon, N., Dunham, W. R., and Kaufman, R. J. (1997) *J. Biol. Chem.* **272**, 27428–27434
557. Pittman, D. D., Wang, J. H., and Kaufman, R. J. (1992) *Biochemistry* **31**, 3315–3325
558. Gilbert, G. E., and Baleja, J. D. (1995) *Biochemistry* **34**, 3022–3031
559. Gilbert, G. E., and Drinkwater, D. (1993) *Biochemistry* **32**, 9577–9585
560. Leyte, A., van Schijndel, H. B., Niehrs, C., Huttner, W. B., Verbeet, M. P., Mertens, K., and van Mourik, J. A. (1991) *J. Biol. Chem.* **266**, 740–746
561. Gitschier, J., Wood, W. I., Shuman, M. A., and Lawn, R. M. (1986) *Science* **232**, 1415–1416
562. Titani, K., and Walsh, K. A. (1988) *Trends Biochem. Sci.* **13**, 94–97
563. Sadler, J. E. (1998) *Ann. Rev. Biochem.* **67**, 395–424
564. Ruggeri, Z. M., and Ware, J. (1993) *FASEB J.* **7**, 308–316
565. Huizinga, E. G., van der Plas, R. M., Kroon, J., Sixma, J. J., and Gros, P. (1997) *Structure* **5**, 1147–1156
566. Emsley, J., Cruz, M., Handin, R., and Liddington, R. (1998) *J. Biol. Chem.* **273**, 10396–10401
567. Sugimoto, M., Dent, J., McClintock, R., Ware, J., and Ruggeri, Z. M. (1993) *J. Biol. Chem.* **268**, 12185–12192
568. George, J. N., Nurden, A. T., and Phillips, D. R. (1984) *N. Engl. J. Med.* **311**, 1084–1098
569. Cruz, M. A., Yuan, H., Lee, J. R., Wise, R. J., and Handin, R. I. (1995) *J. Biol. Chem.* **270**, 10822–10827
570. Beacham, D. A., Wise, R. J., Turci, S. M., and Handin, R. I. (1992) *J. Biol. Chem.* **267**, 3409–3415
571. Zhang, Z. P., Blombäck, M., Nyman, D., and Anvret, M. (1993) *Proc. Natl. Acad. Sci. U.S.A.* **90**, 7937–7940

References

572. Kaufman, R. J. (1989) *Nature (London)* **342**, 207–208
573. Dwarki, V. J., Belloni, P., Nijjar, T., Smith, J., Couto, L., Rabier, M., Clift, S., Berns, A., and Cohen, L. K. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 1023–1027
574. Kay, M. A., Rothenberg, S., Landen, C. N., Bellinger, D. A., Leland, F., Toman, C., Finegold, M., Thompson, A. R., Read, M. S., Brinkhous, K. M., and Woo, S. L. C. (1993) *Science* **262**, 117–119
575. Jesty, J., Wun, T.-C., and Lorenz, A. (1994) *Biochemistry* **33**, 12686–12694
576. Hajjar, K. A. (1994) *N. Engl. J. Med.* **331**, 1585–1587
577. Björk, I., Nordling, K., and Olson, S. T. (1993) *Biochemistry* **32**, 6501–6505
578. Streusand, V. J., Björk, I., Gettins, P. G. W., Petitou, M., and Olson, S. T. (1995) *J. Biol. Chem.* **270**, 9043–9051
- 578a. Whistock, J. C., Pike, R. N., Jin, L., Skinner, R., Pei, X. Y., Carrell, R. W., and Lesk, A. M. (2000) *J. Mol. Biol.* **301**, 1287–1305
579. Di Cera, E., Guinto, E. R., Vindigni, A., Dang, Q. D., Ayala, Y. M., Wuyi, M., and Tulinsky, A. (1995) *J. Biol. Chem.* **270**, 22089–22092
580. Weisel, J. W., Nagaswami, C., Young, T. A., and Light, D. R. (1996) *J. Biol. Chem.* **271**, 31485–31490
- 580a. Fuentes-Prior, P., Iwanaga, Y., Huber, R., Pagila, R., Rumennik, G., Seto, M., Morser, J., Light, D. R., and Bode, W. (2000) *Nature (London)* **404**, 518–525
- 580b. Baerga-Ortiz, A., Rezaie, A. R., and Komives, E. A. (2000) *J. Mol. Biol.* **296**, 651–658
581. Esmon, C. T. (1995) *FASEB J.* **9**, 946–955
582. Colpitts, T. L., Prorok, M., and Castellino, F. J. (1995) *Biochemistry* **34**, 2424–2430
- 582a. Shen, L., Dahlbäck, B., and Villoutreix, B. O. (2000) *Biochemistry* **39**, 2853–2860
583. Öhlin, A., Landes, G., Bourden, P., Oppenheimer, C., Wydro, R., and Stenflo, J. (1988) *J. Biol. Chem.* **263**, 19240–19248
584. Plow, E. F., Herren, T., Redlitz, A., Miles, L. A., and Hoover-Plow, J. L. (1995) *FASEB J.* **9**, 939–945
585. Menhart, N., Hoover, G. J., McCance, S. G., and Castellino, F. J. (1995) *Biochemistry* **34**, 1482–1488
586. Bendixen, E., Harpel, P. C., and Sottrup-Jensen, L. (1995) *J. Biol. Chem.* **270**, 17929–17933
587. Pirie-Shepherd, S. R., Jett, E. A., Andon, N. L., and Pizzo, S. V. (1995) *J. Biol. Chem.* **270**, 5877–5881
588. Marti, D. N., Hu, C.-K., An, S. S. A., von Haller, P., Schaller, J., and Llinás, M. (1997) *Biochemistry* **36**, 11591–11604
589. Bogusky, M. J., Dobson, C. M., and Smith, R. A. G. (1989) *Biochemistry* **28**, 6728–6735
590. Byeon, I.-J. L., Kelley, R. F., Mulkerrin, M. G., An, S. S. A., and Llinás, M. (1995) *Biochemistry* **34**, 2739–2750
591. Parry, M. A. A., Zhang, X. C., and Bode, W. (2000) *Trends Biochem. Sci.* **25**, 53–59
592. Oates, J. A., Wood, A. J. J., Loscalzo, J., and Braunwald, E. (1988) *N. Engl. J. Med.* **319**, 925–931
593. Mottonen, J., Strand, A., Symersky, J., Sweet, R. M., Danley, D. E., Geoghegan, K. F., Gerard, R. D., and Goldsmith, E. J. (1992) *Nature (London)* **355**, 270–273
594. Szyperski, T., Antuch, W., Schick, M., Betz, A., Stone, S. R., and Wüthrich, K. (1994) *Biochemistry* **33**, 9303–9310
595. Betz, A., Hofsteenge, J., and Stone, S. R. (1992) *Biochemistry* **31**, 4557–4562
596. Vitali, J., Martin, P. D., Malkowski, M. G., Robertson, W. D., Lazar, J. B., Winant, R. C., Johnson, P. H., and Edwards, B. F. P. (1992) *J. Biol. Chem.* **267**, 17670–17678
597. van de Locht, A., Lamba, D., Bauer, M., Huber, R., Friedrich, T., Kröger, B., Höffken, W., and Bode, W. (1995) *EMBO J.* **14**, 5149–5157
598. Stark, K. R., and James, A. A. (1998) *J. Biol. Chem.* **273**, 20802–20809
599. Lim-Wilby, M. S. L., Hallenga, K., De Maeyer, M., Lasters, I., Vlasuk, G. P., and Brunck, T. K. (1995) *Protein Sci.* **4**, 178–186
600. van de Locht, A., Stubbs, M. T., Bode, W., Friedrich, T., Bollschweiler, C., Höffken, W., and Huber, R. (1996) *EMBO J.* **15**, 6011–6017
601. Tabernero, L., Chang, C. Y. Y., Ohringer, S. L., Lau, W. F., Iwanowicz, E. J., Han, W.-C., Wang, T. C., Seiler, S. M., Roberts, D. G. M., and Sack, J. S. (1995) *J. Mol. Biol.* **246**, 14–20
602. Cheng, Y., Slon-Usakiewicz, J. J., Wang, J., Purisima, E. O., and Konishi, Y. (1996) *Biochemistry* **35**, 13021–13029
603. Krishnan, R., Tulinsky, A., Vlasuk, G. P., Pearson, D., Vallar, P., Bergum, P., Brunck, T. K., and Ripka, W. C. (1996) *Protein Sci.* **5**, 422–433
604. Kamata, K., Kawamoto, H., Honma, T., Iwama, T., and Kim, S.-H. (1998) *Proc. Natl. Acad. Sci. U.S.A.* **95**, 6630–6635
605. Lu, D., Bovill, E. G., and Long, G. L. (1994) *J. Biol. Chem.* **269**, 29032–29038
606. Ichinose, A. (1992) *Biochemistry* **31**, 3113–3118
607. Olds, R. J., Lane, D. A., Chowdhury, V., De Stefano, V., Leone, G., and Thein, S. L. (1993) *Biochemistry* **32**, 4216–4224
608. Watton, J., Longstaff, C., Lane, D. A., and Barrowcliffe, T. W. (1993) *Biochemistry* **32**, 7286–7293
609. Taylor, P. (1991) *J. Biol. Chem.* **266**, 4025–4028
610. Sussman, J. L., Harel, M., Frolow, F., Oefner, C., Goldman, A., Tokor, L., and Silman, I. (1991) *Science* **253**, 872–879
611. Maelicke, A. (1991) *Trends Biochem. Sci.* **16**, 355–356
- 611a. Koellner, G., Kryger, G., Millard, C. B., Silman, I., Sussman, J. L., and Steiner, T. (2000) *J. Mol. Biol.* **296**, 713–735
612. Vellom, D. C., Radic, Z., Li, Y., Pickering, N. A., Camp, S., and Taylor, P. (1993) *Biochemistry* **32**, 12–17
613. Kovach, I. M., Huber, J. H.-A., and Schowen, R. L. (1988) *J. Am. Chem. Soc.* **110**, 590–593
614. Radic, Z., Kirchhoff, P. D., Quinn, D. M., McCammon, J. A., and Taylor, P. (1997) *J. Biol. Chem.* **272**, 23265–23277
615. Wlodek, S. T., Antosiewicz, J., and Briggs, J. M. (1997) *J. Am. Chem. Soc.* **119**, 8159–8165
616. Haas, R., Marshall, T. L., and Rosenberry, T. L. (1988) *Biochemistry* **27**, 6453–6457
617. Krejci, E., Thomine, S., Boschetti, N., Legay, C., Sketelj, J., and Massoulié, J. (1997) *J. Biol. Chem.* **272**, 22840–22847
618. Pindel, E. V., Kedishvili, N. Y., Abraham, T. L., Brzezinski, M. R., Zhang, J., Dean, R. A., and Bosron, W. F. (1997) *J. Biol. Chem.* **272**, 14769–14775
619. Li, J., Sztittner, R., Derewenda, Z. S., and Meighen, E. A. (1996) *Biochemistry* **35**, 9967–9973
620. Derewenda, Z. S., and Sharp, A. M. (1993) *Trends Biochem. Sci.* **18**, 20–25
621. Quinn, D. M., and Feaster, S. R. (1998) in *Comprehensive Biological Catalysis. A Mechanistic Reference*, Vol. I (Sinnott, M., ed), pp. 455–482, Academic Press, San Diego, California
622. Bourne, Y., Martinez, C., Kerfelec, B., Lombardo, D., Chapus, C., and Cambillau, C. (1994) *J. Mol. Biol.* **238**, 709–732
623. Egloff, M.-P., Marguet, F., Buono, G., Verger, R., Cambillau, C., and van Tilbeurgh, H. (1995) *Biochemistry* **34**, 2751–2762
624. Dahim, M., and Brockman, H. (1998) *Biochemistry* **37**, 8369–8377
625. Jentoft, N. (1990) *Trends Biochem. Sci.* **15**, 291–294
626. Chen, J. C.-H., Miercke, L. J. W., Krucinski, J., Starr, J. R., Saenz, G., Wang, X., Spilburg, C. A., Lange, L. G., Ellsworth, J. L., and Stroud, R. M. (1998) *Biochemistry* **37**, 5107–5117
627. Gargouri, Y., Moreau, H., Pieroni, G., and Verger, R. (1988) *J. Biol. Chem.* **263**, 2159–2162
628. Warner, T. G., Dambach, L. M., Shin, J. H., and O'Brien, J. S. (1981) *J. Biol. Chem.* **256**, 2952–2957
629. Emmerich, J., Beg, O. U., Peterson, J., Previato, L., Brunzell, J. D., Brewer, J., HB, and Santamarina-Fojo, S. (1992) *J. Biol. Chem.* **267**, 4161–4165
630. Kobayashi, J., Applebaum-Bowden, D., Dugi, K. A., Brown, D. R., Kashyap, V. S., Parrott, C., Duarte, C., Maeda, N., and Santamarina-Fojo, S. (1996) *J. Biol. Chem.* **271**, 26296–26301
631. Shen, W.-J., Patel, S., Natu, V., and Kraemer, F. B. (1998) *Biochemistry* **37**, 8973–8979
632. Nicolas, A., Egmond, M., Verrips, C. T., de Vlieg, J., Longhi, S., Cambillau, C., and Martinez, C. (1996) *Biochemistry* **35**, 398–410
633. Grochulski, P., Bouthillier, F., Kazlauskas, R. J., Serre, A. N., Schrag, J. D., Ziomek, E., and Cygler, M. (1994) *Biochemistry* **33**, 3494–3500
634. Lang, D., Hofmann, B., Haalck, L., Hecht, H.-J., Spener, F., Schmid, R. D., and Schomburg, D. (1996) *J. Mol. Biol.* **259**, 704–717
635. Uppenberg, J., Öhrner, N., Norin, M., Hult, K., Kleywegt, G. J., Patkar, S., Waagen, V., Anthonsen, T., and Jones, T. A. (1995) *Biochemistry* **34**, 16838–16851
636. Brocca, S., Schmidt-Dannert, C., Lotti, M., Alberghina, L., and Schmid, R. D. (1998) *Protein Sci.* **7**, 1415–1422
637. Ördentlich, A., Barak, D., Kronman, C., Ariel, N., Segall, Y., Velan, B., and Shafferman, A. (1998) *J. Biol. Chem.* **273**, 19509–19517
638. Axelsen, P. H., Harel, M., Silman, I., and Sussman, J. L. (1994) *Protein Sci.* **3**, 188–197
639. Brady, L., Brzozowski, A. M., Derewenda, Z. S., Dodson, E., Dodson, G., Tolley, S., Turkenburg, J. P., Christiansen, L., Høj Jensen, B., Nørskov, L., Thim, L., and Menge, U. (1990) *Nature (London)* **343**, 767–770
640. Dennis, E. A. (1997) *Trends Biochem. Sci.* **22**, 1–2
641. Balsinde, J., and Dennis, E. A. (1997) *J. Biol. Chem.* **272**, 16069–16072
642. Leslie, C. C. (1997) *J. Biol. Chem.* **272**, 16709–16712
643. Scott, D. L., White, S. P., Otwinowski, Z., Yuan, W., Gelb, M. H., and Sigler, P. B. (1990) *Science* **250**, 1541–1546
644. Brunie, S., Bolin, J., Gewirth, D., and Sigler, P. B. (1985) *J. Biol. Chem.* **260**, 9742–9749
645. Scott, D. L., Achari, A., Vidal, J. C., and Sigler, P. B. (1992) *J. Biol. Chem.* **267**, 22645–22657
646. van den Berg, B., Tessari, M., de Haas, G. H., Verheij, H. M., Boelens, R., and Kaptein, R. (1995) *EMBO J.* **14**, 4123–4131
647. Gelb, M. H., Jain, M. K., Hanel, A. M., and Berg, O. G. (1995) *Ann. Rev. Biochem.* **64**, 653–688
648. Thuren, T., Tulkki, A.-P., Virtanen, J. A., and Kinnunen, P. K. J. (1987) *Biochemistry* **26**, 4907–4910
649. Huber, M., Yee, V. C., Burri, N., Vikersfors, E., Lavrijsen, A. P. M., Paller, A. S., and Hohl, D. (1997) *J. Biol. Chem.* **272**, 21018–21026
650. Yee, V. C., Pedersen, L. C., Le Trong, I., Bishop, P. D., Stenkamp, R. E., and Teller, D. C. (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 7296–7300

References

651. Kulkarni, M. S., and Sherman, F. (1994) *J. Biol. Chem.* **269**, 13141–13147
652. Wu, D., and Hersh, L. B. (1995) *J. Biol. Chem.* **270**, 29111–29116
653. Schwartz, B., and Drueckhammer, D. G. (1996) *J. Am. Chem. Soc.* **118**, 9826–9830
654. Ellis, J., Bagshaw, C. R., and Shaw, W. V. (1995) *Biochemistry* **34**, 16852–16859
655. Murray, I. A., Cann, P. A., Day, P. J., Derrick, J. P., Sutcliffe, M. J., Shaw, W. V., and Leslie, A. G. W. (1995) *J. Mol. Biol.* **254**, 993–1005
656. Guillaume, G., Vanhove, M., Lamotte-Brasseur, J., Ledent, P., Jamin, M., Joris, B., and Frère, J.-M. (1997) *J. Biol. Chem.* **272**, 5438–5444
657. Maveyraud, L., Pratt, R. F., and Samama, J.-P. (1998) *Biochemistry* **37**, 2622–2628
658. Crowder, M. W., Wang, Z., Franklin, S. L., Zovinka, E. P., and Benkovic, S. J. (1996) *Biochemistry* **35**, 12126–12132
659. Orellano, E. G., Girardini, J. E., Cricco, J. A., Ceccarelli, E. A., and Vila, A. J. (1998) *Biochemistry* **37**, 10173–10180
660. Lubkowski, J., Wlodawer, A., Ammon, H. L., Copeland, T. D., and Swain, A. L. (1994) *Biochemistry* **33**, 10257–10265
661. Stewart, A. E., Arfin, S. M., and Bradshaw, R. A. (1995) *J. Biol. Chem.* **270**, 25–28
662. Nakamura, N., Inoue, N., Watanabe, R., Takahashi, M., Takeda, J., Stevens, V. L., and Kinoshita, T. (1997) *J. Biol. Chem.* **272**, 15834–15840
663. Knowles, J. R. (1980) *Ann. Rev. Biochem.* **49**, 877–919
664. Westheimer, F. H. (1968) *Acc. Chem. Res.* **1**, 70–78
665. Westheimer, F. H. (1987) *Science* **235**, 1173–1178
666. Hengge, A. C. (1998) in *Comprehensive Biological Catalysis. A Mechanistic Reference*, Vol. 1 (Sinnott, M., ed), pp. 517–542, Academic Press, San Diego, California
667. Vincent, J. B., Crowder, M. W., and Averill, B. A. (1992) *Trends Biochem. Sci.* **17**, 105–110
668. Mislow, K. (1970) *Acc. Chem. Res.* **3**, 321–331
669. Bunton, C. A. (1970) *Acc. Chem. Res.* **3**, 257–265
670. Gorenstein, D. G., Luxon, B. A., and Findlay, J. B. (1979) *J. Am. Chem. Soc.* **101**, 5869–5875
671. Taira, K., Mock, W. L., and Gorenstein, D. G. (1984) *J. Am. Chem. Soc.* **106**, 7831–7835
672. Calvo, K. C., and Westheimer, F. H. (1984) *J. Am. Chem. Soc.* **106**, 4205–4210
673. Friedman, J. M., Freeman, S., and Knowles, J. R. (1988) *J. Am. Chem. Soc.* **110**, 1268–1275
674. Wu, Y.-D., and Houk, K. N. (1993) *J. Am. Chem. Soc.* **115**, 11997–12002
675. Buchwald, S. L., Friedman, J. M., and Knowles, J. R. (1984) *J. Am. Chem. Soc.* **106**, 4911–4916
676. Herschlag, D., and Jencks, W. P. (1989) *J. Am. Chem. Soc.* **111**, 7579–7586
677. Hengge, A. C., Edens, W. A., and Elsing, H. (1994) *J. Am. Chem. Soc.* **116**, 5045–5049
678. Jankowski, S., Quin, L. D., Paneth, P., and O'Leary, M. H. (1994) *J. Am. Chem. Soc.* **116**, 11675–11677
679. Herschlag, D., and Jencks, W. P. (1990) *Biochemistry* **29**, 5172–5179
680. Admiraal, S. J., and Herschlag, D. (1999) *J. Am. Chem. Soc.* **121**, 5837–5845
681. Cohn, M., and Reed, G. H. (1982) *Ann. Rev. Biochem.* **51**, 365–394
682. Villafranca, J. J., and Raushel, F. M. (1980) *Annu Rev Biophys Bioeng.* **9**, 363–392
683. McLaughlin, A. C., Leigh, J. S., Jr., and Cohn, M. (1976) *J. Biol. Chem.* **251**, 2777–2787
684. Leyh, T. S., Goodhart, P. J., Nguyen, A. C., Kenyon, G. L., and Reed, G. H. (1985) *Biochemistry* **24**, 308–316
685. Nageswara Rao, B. D., Kayne, F. J., and Cohn, M. (1979) *J. Biol. Chem.* **254**, 2689–2696
686. Stackhouse, J., Nambiar, K. P., Burbaum, J. J., Stauffer, D. M., and Benner, S. A. (1985) *J. Am. Chem. Soc.* **107**, 2757–2763
687. Rao, B. D. N., and Cohn, M. (1981) *J. Biol. Chem.* **256**, 1716–1721
688. Gadian, D. G., Radda, G. K., Brown, T. R., Chance, E. M., Dawson, M. J., and Wilke, D. R. (1981) *Biochem. J.* **194**, 215–228
689. Mendz, G. L., Robinson, G., and Kuchel, P. W. (1986) *J. Am. Chem. Soc.* **108**, 169–173
690. Midelfort, C. F., and Rose, I. A. (1976) *J. Biol. Chem.* **251**, 5881–5887
691. Lowe, G., and Tuck, S. P. (1986) *J. Am. Chem. Soc.* **108**, 1300–1301
692. Abbott, S. J., Jones, S. R., Weinman, S. A., Bockhoff, F. M., McLafferty, F. W., and Knowles, J. R. (1979) *J. Am. Chem. Soc.* **101**, 4323–4332
693. Pluura, D. H., Schomburg, D., Richard, J. P., Frey, P. A., and Knowles, J. R. (1980) *Biochemistry* **19**, 325–329
694. Sammons, R. D., and Frey, P. A. (1982) *J. Biol. Chem.* **257**, 1138–1141
695. Eckstein, F. (1980) *Trends Biochem. Sci.* **5**, 157–159
696. Baraniak, J., and Frey, P. A. (1988) *J. Am. Chem. Soc.* **110**, 4059–4060
697. Richard, J. P., and Frey, P. A. (1982) *J. Am. Chem. Soc.* **104**, 3476–3481
698. Hassett, A., Blättler, W., and Knowles, J. R. (1982) *Biochemistry* **21**, 6335–6340
699. Mehdi, S., and Gerlt, J. A. (1984) *Biochemistry* **23**, 4844–4852
700. Sammons, R. D., Frey, P. A., Bruzik, K., and Tsai, M.-D. (1983) *J. Am. Chem. Soc.* **105**, 5455–5461
701. Buchwald, S. L., and Knowles, J. R. (1980) *J. Am. Chem. Soc.* **102**, 6602–6604
702. Corneliuss, R. D., and Cleland, W. W. (1978) *Biochemistry* **17**, 3279–3286
703. Dunaway-Mariano, D., and Cleland, W. W. (1980) *Biochemistry* **19**, 1506–1515
704. Speckhard, D. C., Pecoraro, V. L., Knight, W. B., and Cleland, W. W. (1986) *J. Am. Chem. Soc.* **108**, 4167–4171
705. Lin, I., and Dunaway-Mariano, D. (1988) *J. Am. Chem. Soc.* **110**, 950–956
706. Burgers, P. M. J., and Eckstein, F. (1980) *J. Biol. Chem.* **255**, 8229–8233
707. Lester, L. M., Rusch, L. A., Robinson, G. J., and Speckhard, D. C. (1998) *Biochemistry* **37**, 5349–5355
708. Matlin, A. R., Kendall, D. A., Carano, K. S., Banzon, J. A., Klecka, S. B., and Solomon, N. M. (1992) *Biochemistry* **31**, 8196–8200
709. Simopoulos, T. T., and Jencks, W. P. (1994) *Biochemistry* **33**, 10375–10380
710. Han, R., and Coleman, J. E. (1995) *Biochemistry* **34**, 4238–4245
711. Craig, D. B., Arriaga, E. A., Wong, J. C. Y., Lu, H., and Dovichi, N. J. (1996) *J. Am. Chem. Soc.* **118**, 5245–5253
712. Kimura, E., Kodama, Y., Koike, T., and Shiro, M. (1995) *J. Am. Chem. Soc.* **117**, 8304–8311
713. Stec, B., Hehir, M. J., Brennan, C., Nolte, M., and Kantrowitz, E. R. (1998) *J. Mol. Biol.* **277**, 647–662
- 713a. Stec, B., Holtz, K. M., and Kantrowitz, E. R. (2000) *J. Mol. Biol.* **299**, 1303–1311
714. Florián, J., and Warshel, A. (1997) *J. Am. Chem. Soc.* **119**, 5473–5474
715. Herschman, H. R. (1980) *Trends Biochem. Sci.* **5**, 82–84
716. Ostanin, K., and Van Etten, R. L. (1993) *J. Biol. Chem.* **268**, 20778–20784
717. VanEtten, R. L., Waymack, P. P., and Rehkop, D. M. (1974) *J. Am. Chem. Soc.* **96**, 6783–6785
718. Buchwald, S. L., Saini, M. S., Knowles, J. R., and Van Etten, R. L. (1984) *J. Biol. Chem.* **259**, 2208–2213
719. Ma, L., and Kantrowitz, E. R. (1996) *Biochemistry* **35**, 2394–2402
720. Sugiyama, Y., Kawabe, H., Tanaka, H., Fujimoto, S., and Ohara, A. (1981) *J. Biol. Chem.* **256**, 10664–10670
721. Hong, S.-B., Kuo, J. M., Mullins, L. S., and Raushel, F. M. (1995) *J. Am. Chem. Soc.* **117**, 7580–7581
722. Benning, M. M., Kuo, J. M., Raushel, F. M., and Holden, H. M. (1995) *Biochemistry* **34**, 7973–7978
723. Vanhooke, J. L., Benning, M. M., Raushel, F. M., and Holden, H. M. (1996) *Biochemistry* **35**, 6020–6025
- 723a. Benning, M. M., Hong, S.-B., Raushel, F. M., and Holden, H. M. (2000) *J. Biol. Chem.* **275**, 30556–30560
724. Kuo, J. M., Chae, M. Y., and Raushel, F. M. (1997) *Biochemistry* **36**, 1982–1988
- 724a. Shim, H., and Raushel, F. M. (2000) *Biochemistry* **39**, 7357–7364
725. Williams, N. H. (1998) in *Comprehensive Biological Catalysis. A Mechanistic Reference*, Vol. 1 (Sinnott, M., ed), pp. 543–561, Academic Press, San Diego, California
726. Villeret, V., Huang, S., Zhang, Y., and Lipscomb, W. N. (1995) *Biochemistry* **34**, 4307–4315
727. Kurbanov, F. T., Choe, J.-y., Honzatko, R. B., and Fromm, H. J. (1998) *J. Biol. Chem.* **273**, 17511–17516
728. Choe, J.-Y., Fromm, H. J., and Honzatko, R. B. (2000) *Biochemistry* **39**, 8565–8574
- 728a. Lee, Y.-H., Ogata, C., Pflugrath, J. W., Levitt, D. G., Sarma, R., Banaszak, L. J., and Pilakis, S. J. (1996) *Biochemistry* **35**, 6010–6019
729. Lu, G., Giroux, E. L., and Kantrowitz, E. R. (1997) *J. Biol. Chem.* **272**, 5076–5081
730. Villeret, V., Huang, S., Zhang, Y., Xue, Y., and Lipscomb, W. N. (1995) *Biochemistry* **34**, 4299–4306
731. Lei, K.-J., Pan, C.-J., Liu, J.-L., Shelly, L. L., and Chou, J. Y. (1995) *J. Biol. Chem.* **270**, 11882–11886
732. Clottes, E., and Burchell, A. (1998) *J. Biol. Chem.* **273**, 19391–19397
733. Josse, J., and Wong, S. C. K. (1971) in *The Enzymes*, 3rd ed., Vol. 4 (Boyer, P. D., ed), pp. 499–541, Academic Press, New York
734. Harutyunyan, E. H., Oganessyan, V. Y., Oganessyan, N. N., Awaeva, S. M., Nazarova, T. I., Vorobyeva, N. N., Kurilova, S. A., Huber, R., and Mather, T. (1997) *Biochemistry* **36**, 7754–7760
735. Salminen, A., Efimova, I. S., Parfenyev, A. N., Magretova, N. N., Mikalahti, K., Goldman, A., Baykov, A. A., and Lahti, R. (1999) *J. Biol. Chem.* **274**, 33898–33904
736. Pohjanjoki, P., Lahti, R., Goldman, A., and Cooperman, B. S. (1998) *Biochemistry* **37**, 1754–1761
- 736a. Lappänen, V.-M., Nummelin, H., Hansen, T., Lahti, R., Schäfer, G., and Goldman, A. (1999) *Protein Sci.* **8**, 1218–1231
- 736b. Baykov, A. A., Fabrichniy, I. P., Pohjanjoki, P., Zyryanov, A. B., and Lahti, R. (2000) *Biochemistry* **39**, 11939–11947
737. Barford, D. (1996) *Trends Biochem. Sci.* **21**, 407–412
738. Goldberg, J., Huang, H., Kwon, Y., Greengard, P., Nairn, A. C., and Kuriyan, J. (1995) *Nature (London)* **376**, 745–753
739. Egloff, M.-P., Cohen, P. T. W., Reinemer, P., and Barford, D. (1995) *J. Mol. Biol.* **254**, 942–959
740. Zhang, J., Zhang, Z., Brew, K., and Lee, E. Y. C. (1996) *Biochemistry* **35**, 6276–6282

References

741. Zhang, M., Zhou, M., Van Etten, R. L., and Stauffacher, C. V. (1997) *Biochemistry* **36**, 15–23
742. Hengge, A. C., Sowa, G. A., Wu, L., and Zhang, Z.-Y. (1995) *Biochemistry* **34**, 13982–13987
743. Yuvaniyama, J., Denu, J. M., Dixon, J. E., and Saper, M. A. (1996) *Science* **272**, 1328–1331
744. D'Alessio, G., and Riordan, J. F., eds. (1997) *Ribonuclease Structure and Functions*, Academic Press, San Diego, California
745. Moore, S., and Stein, W. H. (1973) *Science* **180**, 458–464
746. Wlodawer, A., Svensson, L. A., Sjölin, L., and Gilliland, G. L. (1988) *Biochemistry* **27**, 2705–2717
747. Santoro, J., González, C., Bruix, M., Neira, J. L., Nieto, J. L., Herranz, J., and Rico, M. (1993) *J. Mol. Biol.* **229**, 722–734
748. Tilton, R. F., Jr., Dewan, J. C., and Petsko, G. A. (1992) *Biochemistry* **31**, 2469–2481
749. Fedorov, A. A., Joseph-McCarthy, D., Fedorov, E., Sirakova, D., Graf, I., and Almo, S. C. (1996) *Biochemistry* **35**, 15962–15979
750. Boix, E., Nogués, M. V., Schein, C. H., Benner, S. A., and Cuchillo, C. M. (1994) *J. Biol. Chem.* **269**, 2529–2534
751. Schultz, L. W., Quirk, D. J., and Raines, R. T. (1998) *Biochemistry* **37**, 8886–8898
752. Jackson, D. Y., Burnier, J., Quan, C., Stanley, M., Tom, J., and Wells, J. A. (1994) *Science* **266**, 243–247
753. Gutte, B. (1977) *J. Biol. Chem.* **252**, 663–670
754. Breslow, R., Dong, S. D., Webb, Y., and Xu, R. (1996) *J. Am. Chem. Soc.* **118**, 6588–6600
755. Martin, P. D., Coscha, M. S., and Edwards, B. F. P. (1987) *J. Biol. Chem.* **262**, 15930–15938
756. Wladkowski, B. D., Svensson, L. A., Sjölin, L., Ladner, J. E., and Gilliland, G. L. (1998) *J. Am. Chem. Soc.* **120**, 5488–5498
757. Eckstein, F. (1979) *Acc. Chem. Res.* **12**, 204–210
758. Matta, M. S., and Vo, D. T. (1986) *J. Am. Chem. Soc.* **108**, 5316–5318
759. Herschlag, D. (1994) *J. Am. Chem. Soc.* **116**, 11631–11635
760. Breslow, R., and Chapman, W. H., Jr. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 10018–10021
761. Sowa, G. A., Hengge, A. C., and Cleland, W. W. (1997) *J. Am. Chem. Soc.* **119**, 2319–2320
762. Kawata, Y., Sakiyama, F., Hayashi, F., and Kyogoku, Y. (1990) *Eur. J. Biochem.* **187**, 255–262
763. Pletincx, J., Steyaert, J., Zegers, I., Choe, H.-W., Heinemann, U., and Wyls, L. (1994) *Biochemistry* **33**, 1654–1662
764. Kurihara, H., Nonaka, T., Mitsui, Y., Ohgi, K., Irie, M., and Nakamura, K. T. (1996) *J. Mol. Biol.* **255**, 310–320
- 764a. Noguchi, S., Satow, Y., Uchida, T., Sasaki, C., and Matsuzaki, T. (1995) *Biochemistry* **34**, 15583–15591
765. Cordes, F., Starikov, E. B., and Saenger, W. (1995) *J. Am. Chem. Soc.* **117**, 10365–10372
766. Buckle, A. M., and Fersht, A. R. (1994) *Biochemistry* **33**, 1644–1653
767. Deutscher, M. P. (1988) *Trends Biochem. Sci.* **13**, 136–139
768. Benner, S. A., and Alleman, R. K. (1989) *Trends Biochem. Sci.* **14**, 396–397
769. Davies, J. F., II, Hostomska, Z., Hostomsky, Z., Jordan, S. R., and Matthews, D. A. (1991) *Science* **252**, 88–95
770. Katayanagi, K., Miyagawa, M., Matsushima, M., Ishikawa, M., Kanaya, S., Ikehara, M., Matsuzaki, T., and Morikawa, K. (1990) *Nature (London)* **347**, 306–309
771. Yang, W., Hendrickson, W. A., Crouch, R. J., and Satow, Y. (1990) *Science* **249**, 1398–1405
772. Acharya, K. R., Shapiro, R., Riordan, J. F., and Vallee, B. L. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 2949–2953
773. Lequin, O., Albaret, C., Bontems, F., Spik, G., and Lallemand, J.-Y. (1996) *Biochemistry* **35**, 8870–8880
774. Shapiro, R. (1998) *Biochemistry* **37**, 6847–6856
775. Rosenberg, H. F., Tenen, D. G., and Ackerman, S. J. (1989) *Proc. Natl. Acad. Sci. U.S.A.* **86**, 4460–4464
776. Leland, P. A., Schultz, L. W., Kim, B.-M., and Raines, R. T. (1998) *Proc. Natl. Acad. Sci. U.S.A.* **95**, 10407–10412
777. D'Alessio, G., Di Donato, A., Parente, A., and Piccoli, R. (1991) *Trends Biochem. Sci.* **16**, 104–106
778. Hofsteenge, J., Kieffer, B., Matthies, R., Hemmings, B. A., and Stone, S. R. (1988) *Biochemistry* **27**, 8537–8544
779. Kole, R., and Altman, S. (1981) *Biochemistry* **20**, 1902–1906
780. Altman, S., Baer, M., Guerrier-Takada, C., and Vioque, A. (1986) *Trends Biochem. Sci.* **11**, 515–518
781. Altman, S., Kirsebom, L., and Talbot, S. (1993) *FASEB J.* **7**, 7–14
782. Yuan, Y., and Altman, S. (1995) *EMBO J.* **14**, 159–168
- 782a. Massire, C., Jaeger, L., and Westhof, E. (1998) *J. Mol. Biol.* **279**, 773–793
783. Waugh, D. S., Green, C. J., and Pace, N. R. (1989) *Science* **244**, 1569–1571
784. True, H. L., and Celandier, D. W. (1998) *J. Biol. Chem.* **273**, 7193–7196
785. Cech, T. R. (1987) *Science* **236**, 1532–1539
786. Zaug, A. J., and Cech, T. R. (1986) *Science* **231**, 470–475
787. Cech, T. R. (1986) *Sci. Am.* **255** (Nov), 64–75
788. Brown, J. W., Haas, E. S., Gilbert, D. G., and Pace, N. R. (1994) *Nucleic Acids Res.* **22**, 3660–3662
789. Pan, T. (1995) *Biochemistry* **34**, 902–909
790. Golden, B. L., Gooding, A. R., Podell, E. R., and Cech, T. R. (1998) *Science* **282**, 259–264
791. Cate, J. H., Gooding, A. R., Podell, E., Zhou, K., Golden, B. L., Kundrot, C. E., Cech, T. R., and Doudna, J. A. (1996) *Science* **273**, 1678–1685
792. Ferré-D'Amaré, A. R., Zhou, K., and Doudna, J. A. (1998) *Nature (London)* **395**, 567–574
793. Baidya, N., and Uhlenbeck, O. C. (1997) *Biochemistry* **36**, 1108–1114
794. Pley, H. W., Flaherty, K. M., and McKay, D. B. (1994) *Nature (London)* **372**, 68–74
795. Scott, W. G., Murray, J. B., Arnold, J. R. P., Stoddard, B. L., and Klug, A. (1996) *Science* **274**, 2065–2069
796. Bevers, S., Xiang, G., and McLaughlin, L. W. (1996) *Biochemistry* **35**, 6483–6490
- 796a. Bevers, S., Ha, S. B., and McLaughlin, L. W. (1999) *Biochemistry* **38**, 7710–7718
- 796b. Kore, A. R., and Eckstein, F. (1999) *Biochemistry* **38**, 10915–10918
- 796c. Lyne, P. D., and Karplus, M. (2000) *J. Am. Chem. Soc.* **122**, 166–167
- 796d. Torres, R. A., and Bruice, T. C. (2000) *J. Am. Chem. Soc.* **122**, 781–791
797. Simorre, J.-P., Legault, P., Baidya, N., Uhlenbeck, O. C., Maloney, L., Wincott, F., Usman, N., Beigelman, L., and Pardi, A. (1998) *Biochemistry* **37**, 4034–4044
798. Sargueil, B., Pecchia, D. B., and Burke, J. M. (1995) *Biochemistry* **34**, 7739–7748
- 798a. Nissen, P., Hansen, J., Ban, N., Moore, P. B., and Steitz, T. A. (2000) *Science* **289**, 920–930
- 798b. Muth, G. W., Ortoleva-Donnelly, L., and Strobel, S. A. (2000) *Science* **289**, 947–950
- 798c. Nakano, S.-i., Chadalavada, D. M., and Bevilacqua, P. C. (2000) *Science* **287**, 1493–1497
799. Scott, W. G., and Klug, A. (1996) *Trends Biochem. Sci.* **21**, 220–224
800. Grasby, J. A. (1998) in *Comprehensive Biological Catalysis. A Mechanistic Reference*, Vol. I (Sinnott, M., ed), pp. 563–571, Academic Press, San Diego, California
801. Hampel, K. J., Walter, N. G., and Burke, J. M. (1998) *Biochemistry* **37**, 14672–14682
802. Pan, T., and Uhlenbeck, O. C. (1992) *Nature (London)* **358**, 560–563
803. Chartrand, P., Usman, N., and Cedergren, R. (1997) *Biochemistry* **36**, 3145–3150
804. Santoro, S. W., and Joyce, G. F. (1998) *Biochemistry* **37**, 13330–13342
805. Beebe, J. A., Kurz, J. C., and Fierke, C. A. (1996) *Biochemistry* **35**, 10493–10505
806. Torres, R. A., and Bruice, T. C. (1998) *Proc. Natl. Acad. Sci. U.S.A.* **95**, 11077–11082
807. Pyle, A. M. (1993) *Science* **261**, 709–714
808. Peracchi, A., Beigelman, L., Scott, E. C., Uhlenbeck, O. C., and Herschlag, D. (1997) *J. Biol. Chem.* **272**, 26822–26826
809. Pontius, B. W., Lott, W. B., and von Hippel, P. H. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 2290–2294
810. Bruice, T. C., Tsubouchi, A., Dempcy, R. O., and Olson, L. P. (1996) *J. Am. Chem. Soc.* **118**, 9867–9875
811. Weinstein, L. B., Jones, B. C. N. M., Cosstick, R., and Cech, T. R. (1997) *Nature (London)* **388**, 805–808
812. Christian, E. L., and Yarus, M. (1993) *Biochemistry* **32**, 4475–4480
813. Orita, M., Vinayak, R., Andrus, A., Warashina, M., Chiba, A., Kaniwa, H., Nishikawa, F., Nishikawa, S., and Taira, K. (1996) *J. Biol. Chem.* **271**, 9447–9454
814. Hertel, K. J., Peracchi, A., Uhlenbeck, O. C., and Herschlag, D. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 8497–8502
815. Peracchi, A., Karpeisky, A., Maloney, L., Beigelman, L., and Herschlag, D. (1998) *Biochemistry* **37**, 14765–14775
816. Joyce, G. F. (1992) *Sci. Am.* **267** (Dec), 90–97
817. Ekland, E. H., Szostak, J. W., and Bartel, D. P. (1995) *Science* **269**, 364–370
818. Wilson, C., and Szostak, J. W. (1995) *Nature (London)* **374**, 777–782
819. Wright, M. C., and Joyce, G. F. (1997) *Science* **276**, 614–617
820. Ohmichi, T., and Sugimoto, N. (1997) *Biochemistry* **36**, 3514–3521
821. Suga, H., Cowan, J. A., and Szostak, J. W. (1998) *Biochemistry* **37**, 10118–10125
822. Tsang, J., and Joyce, G. F. (1996) *J. Mol. Biol.* **262**, 31–42
823. Geyer, C. R., and Sen, D. (1998) *J. Mol. Biol.* **275**, 483–489
824. Ceska, T. A., and Sayers, J. R. (1998) *Trends Biochem. Sci.* **23**, 331–336
825. Weber, D. J., Libson, A. M., Gittis, A. G., Lebowitz, M. S., and Mildvan, A. S. (1994) *Biochemistry* **33**, 8017–8028
826. Libson, A. M., Gittis, A. G., and Lattman, E. E. (1994) *Biochemistry* **33**, 8007–8016
827. Hale, S. P., Poole, L. B., and Gerlt, J. A. (1993) *Biochemistry* **32**, 7479–7487
828. Grissom, C. B., and Markley, J. L. (1989) *Biochemistry* **28**, 2116–2124
829. Serpersu, E. H., Hibler, D. W., Gerlt, J. A., and Mildvan, A. S. (1989) *Biochemistry* **28**, 1539–1548
830. Suck, D., Lahm, A., and Oefner, C. (1988) *Nature (London)* **332**, 464–468
831. Jones, S. J., Worrall, A. F., and Connolly, B. A. (1996) *J. Mol. Biol.* **264**, 1154–1163
832. Frederick, C. A., Grable, J., Melia, M., Samudzi, C., Jen-Jacobson, L., Wang, B.-C., Greene, P., Boyer, H. W., and Rosenberg, J. M. (1984) *Nature (London)* **309**, 327–331

References

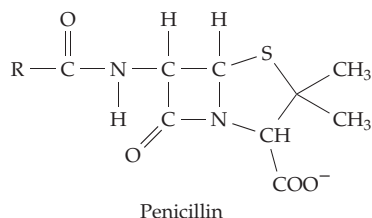
833. McClarin, J. A., Frederick, C. A., Wang, B.-C., Greene, P., Boyer, H. W., Grable, J., and Rosenberg, J. M. (1986) *Science* **234**, 1526–1541
834. McCarthy, A. D., and Hardie, D. G. (1984) *Trends Biochem. Sci.* **9**, 60–63
835. Kostrewa, D., and Winkler, F. K. (1995) *Biochemistry* **34**, 683–696
836. Newman, M., Strzelecka, T., Dorner, L. F., Schildkraut, I., and Aggarwal, A. K. (1995) *Science* **269**, 656–663
837. Bozic, D., Grazulis, S., Siksnys, V., and Huber, R. (1996) *J. Mol. Biol.* **255**, 176–186
838. Connolly, B. A., Echstein, F., and Pingoud, A. (1984) *J. Biol. Chem.* **259**, 10760–10763
839. Corey, D. R., Pei, D., and Schultz, P. G. (1989) *Biochemistry* **28**, 8277–8286
840. Ray, W. J., Jr., Hermodson, M. A., Puvathingal, J. M., and Mahoney, W. C. (1983) *J. Biol. Chem.* **258**, 9166–9174
841. Percival, M. D., and Withers, S. G. (1992) *Biochemistry* **31**, 505–512
842. Dai, J.-B., Liu, Y., Ray, W. J., Jr., and Konno, M. (1992) *J. Biol. Chem.* **267**, 6322–6337
843. Rose, Z. B. (1986) *Trends Biochem. Sci.* **11**, 253–255
844. Shanske, S., Sakoda, S., Hermodson, M. A., DiMauro, S., and Schon, E. A. (1987) *J. Biol. Chem.* **262**, 14612–14617
845. Rigden, D. J., Alexeev, D., Phillips, S. E. V., and Fothergill-Gilmore, L. A. (1998) *J. Mol. Biol.* **276**, 449–459
846. Ravel, P., Craescu, C. T., Arous, N., Rosa, J., and Garel, M. C. (1997) *J. Biol. Chem.* **272**, 14045–14050
847. Collet, J.-F., Stroobant, V., Pirard, M., Delpierre, G., and Van Schaftingen, E. (1998) *J. Biol. Chem.* **273**, 14107–14112
848. Kalckar, H. M. (1985) *Trends Biochem. Sci.* **10**, 291–293
849. Reed, G. H., and Leyh, T. S. (1980) *Biochemistry* **19**, 5472–5480
850. Schulz, G. E., and Schirmer, R. H. (1979) *Principles of Protein Structure*, Springer-Verlag, New York (pp. 222–226)
851. Gerstein, M., Schulz, G., and Chothia, C. (1993) *J. Mol. Biol.* **229**, 494–501
852. Bernstein, B. E., and Hol, W. G. J. (1998) *Biochemistry* **37**, 4429–4436
853. Matte, A., Tari, L. W., and Delbaere, L. T. J. (1998) *Structure* **6**, 413–419
854. Vornrhein, C., Bönisch, H., Schäfer, G., and Schulz, G. E. (1998) *J. Mol. Biol.* **282**, 167–179
855. Bernstein, B. E., Michels, P. A. M., and Hol, W. G. J. (1997) *Nature (London)* **385**, 275–278
856. Yount, R. G., Babcock, D., Ballantyne, W., and Ojala, D. (1971) *Biochemistry* **10**, 2484–2489
857. Yount, R. G. (1975) *Adv. Enzymol.* **43**, 1–56
858. McPhillips, T. M., Hsu, B. T., Sherman, M. A., Mas, M. T., and Rees, D. C. (1996) *Biochemistry* **35**, 4118–4127
859. Schlauderer, G. J., Proba, K., and Schulz, G. E. (1996) *J. Mol. Biol.* **256**, 223–227
860. Byeon, I.-J. L., Shi, Z., and Tsai, M.-D. (1995) *Biochemistry* **34**, 3172–3182
861. Ray, B. D., Chau, M. H., Fife, W. K., Jarori, G. K., and Nageswara Rao, B. D. (1996) *Biochemistry* **35**, 7239–7246
862. Schlauderer, G. J., and Schulz, G. E. (1996) *Protein Sci.* **5**, 434–441
863. Schricker, R., Magdolen, V., Strobel, G., Bogengruber, E., Breitenbach, M., and Bandlow, W. (1995) *J. Biol. Chem.* **270**, 31103–31110
864. Teplyakov, A., Sebastiao, P., Obmolova, G., Perrakis, A., Brush, G. S., Bessman, M. J., and Wilson, K. S. (1996) *EMBO J.* **15**, 3487–3497
865. Zhang, Y., Li, Y., Wu, Y., and Yan, H. (1997) *J. Biol. Chem.* **272**, 29343–29350
866. Müller-Dieckmann, H.-J., and Schulz, G. E. (1995) *J. Mol. Biol.* **246**, 522–530
867. Schlichting, I., and Reinstein, J. (1997) *Biochemistry* **36**, 9290–9296
868. Runquist, J. A., Harrison, D. H. T., and Mizioroko, H. M. (1998) *Biochemistry* **37**, 1221–1226
869. Moréra, S., Chiadmi, M., LeBras, G., Lascu, I., and Janin, J. (1995) *Biochemistry* **34**, 11062–11070
870. Webb, P. A., Perisic, O., Mendola, C. E., Backer, J. M., and Williams, R. L. (1995) *J. Mol. Biol.* **251**, 574–587
871. Abdulaev, N. G., Karaschuk, G. N., Ladner, J. E., Kakuev, D. L., Yakhyayev, A. V., Tordova, M., Gaidarov, I. O., Popov, V. I., Fujiwara, J. H., Chinchilla, D., Eisenstein, E., Gilliland, G. L., and Ridge, K. D. (1998) *Biochemistry* **37**, 13958–13967
872. Mesnildrey, S., Agou, F., Karlsson, A., Bonne, D. D., and Véron, M. (1998) *J. Biol. Chem.* **273**, 4436–4442
873. Turano, A., Furey, W., Pletcher, J., Sax, M., Pike, D., and Kluger, R. (1982) *J. Am. Chem. Soc.* **104**, 3089–3095
874. Aleshin, A. E., Zeng, C., Bourenkov, G. P., Bartunik, H. D., Fromm, H. J., and Honzatko, R. B. (1998) *Structure* **6**, 39–50
875. Aleshin, A. E., Zeng, C., Bartunik, H. D., Fromm, H. J., and Honzatko, R. B. (1998) *J. Mol. Biol.* **282**, 345–357
876. Lundblad, V. (1998) *Proc. Natl. Acad. Sci. U.S.A.* **95**, 8415–8416
877. Behlke, J., Heidrich, K., Naumann, M., Müller, E.-C., Otto, A., Reuter, R., and Kriegl, T. (1998) *Biochemistry* **37**, 11989–11995
878. Anderson, C. M., Stenkamp, R. E., and Steitz, T. A. (1978) *J. Mol. Biol.* **123**, 15–23
879. Schwab, D. A., and Wilson, J. E. (1989) *Proc. Natl. Acad. Sci. U.S.A.* **86**, 2563–2567
880. Schirmer, T., and Evans, P. R. (1990) *Nature (London)* **343**, 140–145
881. Auzat, I., Le Bras, G., and Garel, J.-R. (1995) *J. Mol. Biol.* **246**, 248–253
882. Blake, C. C. F., and Evans, P. R. (1974) *J. Mol. Biol.* **484**, 585–601
883. Berger, S. A., and Evans, P. R. (1992) *Biochemistry* **31**, 9237–9242
884. Mizuguchi, H., Cook, P. F., Hasemann, C. A., and Uyedda, K. (1997) *Biochemistry* **36**, 8775–8784
885. Fujii, H., Krietsch, W. K. G., and Yoshida, A. (1980) *J. Biol. Chem.* **255**, 6421–6423
886. Tanaka, K. R., and Paglia, D. E. (1995) in *The Metabolic and Molecular Bases of Inherited Disease* (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 3485–3511, McGraw-Hill, New York
887. Mas, M. T., Bailey, J. M., and Resplandor, Z. E. (1988) *Biochemistry* **27**, 1168–1172
888. Forstner, M., Müller, A., Stolz, M., and Wallimann, T. (1997) *Protein Sci.* **6**, 331–339
889. Fritz-Wolf, K., Schnyder, T., Wallimann, T., and Kabsch, W. (1996) *Nature (London)* **381**, 341–345
890. Zhou, G., Somasundaram, T., Blanc, E., Parthasarathy, G., Ellington, W. R., and Chapman, M. S. (1998) *Proc. Natl. Acad. Sci. U.S.A.* **95**, 8449–8454
891. Larsen, T. M., Benning, M. M., Rayment, I., and Reed, G. H. (1998) *Biochemistry* **37**, 6247–6255
892. Loria, J. P., and Nowak, T. (1998) *Biochemistry* **37**, 6967–6974
893. Lovell, S. C., Mullick, A. H., and Muirhead, H. (1998) *J. Mol. Biol.* **276**, 839–851
894. Cheng, X., Friesen, R. H. E., and Lee, J. C. (1996) *J. Biol. Chem.* **271**, 6313–6321
- 894a. Valentini, G., Chiarelli, L., Fortin, R., Speranza, M. L., Galizzi, A., and Mattevi, A. (2000) *J. Biol. Chem.* **275**, 18145–18152
895. Hunter, T., and Plowman, G. D. (1997) *Trends Biochem. Sci.* **22**, 18–22
896. Grant, B. D., Hemmer, W., Tsigelny, I., Adams, J. A., and Taylor, S. S. (1998) *Biochemistry* **37**, 7708–7715
897. Cheng, X., Shaltiel, S., and Taylor, S. S. (1998) *Biochemistry* **37**, 14005–14013
898. Engh, R. A., Girod, A., Kinzel, V., Huber, R., and Bossemeyer, D. (1996) *J. Biol. Chem.* **271**, 26157–26164
899. Herberg, F. W., Zimmermann, B., McGlone, M., and Taylor, S. S. (1997) *Protein Sci.* **6**, 569–579
900. Su, Y., Dostmann, W. R. G., Herberg, F. W., Durick, K., Xuong, N.-h., Eyck, L. T., Taylor, S. S., and Varughese, K. I. (1995) *Science* **269**, 807–813
901. Lohmann, S. M., Vaandrager, A. B., Smolenski, A., Walter, U., and DeJonge, H. R. (1997) *Trends Biochem. Sci.* **22**, 307–312
902. Xu, W., Harrison, S. C., and Eck, M. J. (1997) *Nature (London)* **385**, 595–602
903. Sicheri, F., Moarefi, I., and Kuriyan, J. (1997) *Nature (London)* **385**, 602–609
904. Adams, J. A. (1996) *Biochemistry* **35**, 10949–10956
905. Tesmer, J. J. G., Sunahara, R. K., Gilman, A. G., and Sprang, S. R. (1997) *Science* **278**, 1907–1916
906. Tesmer, J. J. G., Sunahara, R. K., Johnson, R. A., Gosselin, G., Gilman, A. G., and Sprang, S. R. (1999) *Science* **285**, 756–760
907. Sunahara, R. K., Tesmer, J. J. G., Gilman, A. G., and Sprang, S. R. (1997) *Science* **278**, 1943–1947
908. Zhao, Y., Schelvis, J. P. M., Babcock, G. T., and Marletta, M. A. (1998) *Biochemistry* **37**, 4502–4509
- 908a. Zhao, Y., Brandish, P. E., DiValentin, M., Schelvis, J. P. M., Babcock, G. T., and Marletta, M. A. (2000) *Biochemistry* **39**, 10848–10854
909. Joyce, C. M., and Steitz, T. A. (1994) *Ann. Rev. Biochem.* **63**, 777–822
910. Kim, Y., Eom, S. H., Wang, J., Lee, D.-S., Suh, S. W., and Steitz, T. A. (1995) *Nature (London)* **376**, 612–616
911. Doublé, S., Tabor, S., Long, A. M., Richardson, C. C., and Ellenberger, T. (1998) *Nature (London)* **391**, 251–258
912. Patel, P. H., Jacobo-Molina, A., Ding, J., Tantillo, C., Clark, A. D., Jr., Raag, R., Nanni, R. G., Hughes, S. H., and Arnold, E. (1995) *Biochemistry* **34**, 5351–5363
913. Bartlett, P. A., and Eckstein, F. (1988) *J. Biol. Chem.* **257**, 8879–8884
- 913a. Stivers, J. T., Nawrot, B., Jagadeesh, G. J., Stec, W. J., and Shuman, S. (2000) *Biochemistry* **39**, 5561–5572
914. Smith, C. A., and Rayment, I. (1996) *Biophys. J.* **70**, 1590–1602
915. Wittinghofer, A., and Pai, E. F. (1991) *Trends Biochem. Sci.* **16**, 382–387
916. Coleman, D. E., and Sprang, S. R. (1998) *Biochemistry* **37**, 14376–14385
917. Falany, C. N. (1997) *FASEB J.* **11**, 206–216
918. Varin, L., Marsolais, F., Richard, M., and Rouleau, M. (1997) *FASEB J.* **11**, 517–525
919. Kakuta, Y., Petrotchenko, E. V., Pedersen, L. C., and Negishi, M. (1998) *J. Biol. Chem.* **273**, 27325–27330
920. Zhang, H., Varmalova, O., Vargas, F. M., Falany, C. N., and Leyh, T. S. (1998) *J. Biol. Chem.* **273**, 10888–10892
- 920a. Hiraoka, N., Nakagawa, H., Ong, E., Akama, T. O., Fukuda, M. N., and Fukuda, M. (2000) *J. Biol. Chem.* **275**, 20188–20196
921. Klaassen, C. D., and Boles, J. W. (1997) *FASEB J.* **11**, 404–418

References

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922. Neufeld, E. F., and Muenzer, J. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 2 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 2465–2494, McGraw-Hill, New York
923. Lukatela, G., Krauss, N., Theis, K., Selmer, T., Gieselmann, V., von Figura, K., and Saenger, W. (1998) *Biochemistry* **37**, 3654–3664
- 923a. Dierks, T., Lecca, M. R., Schlotterhose, P., Schmidt, B., and von Figura, K. (1999) *EMBO J.* **18**, 2084–2091
924. Waldow, A., Schmidt, B., Dierks, T., von Bülow, R., and von Figura, K. (1999) *J. Biol. Chem.* **274**, 12284–12288
- 924a. Szameit, C., Miech, C., Balleininger, M., Schmidt, B., von Figura, K., and Dierks, T. (1999) *J. Biol. Chem.* **274**, 15375–15381
925. Fox, D. K., and Roseman, S. (1986) *J. Biol. Chem.* **261**, 13487–13497
926. Joyce, M. A., Fraser, M. E., James, M. N. G., Bridger, W. A., and Wolodko, W. T. (2000) *Biochemistry* **39**, 17–25
927. Johnson, J. D., Muhonen, W. W., and Lambeth, D. O. (1998) *J. Biol. Chem.* **273**, 27573–27579
928. Liaw, S.-H., and Eisenberg, D. (1994) *Biochemistry* **33**, 675–681
929. Alibhai, M., and Villafranca, J. J. (1994) *Biochemistry* **33**, 682–686
930. Meister, A. (1968) *Adv. Enzymol.* **31**, 183–218
931. Meister, A. (1974) in *The Enzymes*, 3rd ed., Vol. 10 (Boyer, P. D., ed), pp. 669–754, Academic Press, New York
932. Whitty, A., Fierke, C. A., and Jencks, W. P. (1995) *Biochemistry* **34**, 11678–11689
933. Rochet, J.-C., and Bridger, W. A. (1994) *Protein Sci.* **3**, 975–981
934. Selmer, T., and Buckel, W. (1999) *J. Biol. Chem.* **274**, 20772–20778

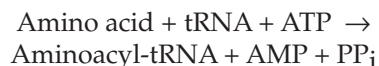
Study Questions

- Outline the reactions by which glyceraldehyde 3-phosphate is converted to 3-phosphoglycerate with coupled synthesis of ATP in the glycolysis pathway. Show important mechanistic details.
- Papain is a protein-hydrolyzing (proteolytic) enzyme with an $-SH$ group and an imidazole group at the active site. Write a reasonable structure for a "tetrahedral intermediate" that would be expected to arise during formation of an acyl enzyme intermediate.
- Adenylate kinase catalyzes the interconversion of ATP, AMP, and ADP.
 - Draw a reasonable structure for a penta-covalent intermediate derived from ATP and AMP.
 - Draw a reasonable structure for the transition state leading from ATP + AMP to two molecules of ADP in an S_N2 -like reaction.
- Penicillin inhibits a D-alanyl-D-alanine transpeptidase that catalyzes the reaction

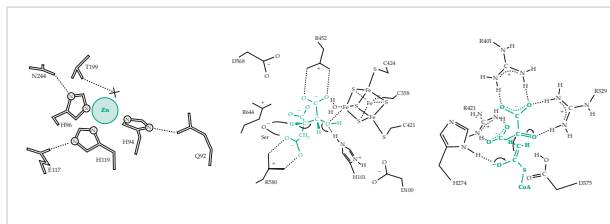


where R and R' are different parts of a bacterial peptidoglycan. Write a step-by-step mechanism for this reaction and indicate how penicillin may inhibit the enzyme by combining with it irreversibly. See Strydom *et al.*, *Nature* **359**, 700–705, 1992, for related reaction of penicillin with a penicillinase.

- Write out step-by-step chemical mechanisms for the following enzymatic reaction. Use small arrows to indicate directions of electron flow. Remember to have all electrons move in the same direction in any single structure.



- Trypsin in which Asp 102 has been replaced by Asn has 10^4 times less catalytic activity than natural trypsin at neutral pH. From the crystal structure of the mutant enzyme it appears that the imidazole group of His 57 is held by the Asn side chain in the wrong tautomeric form for catalysis. Explain. Compare this incorrect tautomeric form with that in the initial structure shown in Fig. 12-11.
- A recent discovery in biochemistry is that RNA can act as an enzyme in chemical reactions, usually reactions involving RNA hydrolysis. Discuss the features of RNA structure that might favor evolution of enzymes composed entirely of a single polyribonucleotide chain, and describe a proposed mechanism for RNA-catalyzed hydrolysis of RNA molecules.



One of the simplest biochemical addition reactions is the hydration of carbon dioxide to form carbonic acid, which is released from the zinc-containing **carbonic anhydrase** (left, Fig. 13-1) as HCO_3^- . **Aconitase** (center, Fig. 13-4) is shown here removing a water molecule from isocitrate, an intermediate compound in the citric acid cycle. The H_2O that is removed will become bonded to an iron atom of the Fe_4S_4 cluster at the active site as indicated by the black H_2O . An enolate anion derived from acetyl-CoA adds to the carbonyl group of oxaloacetate to form citrate in the active site of **citrate synthase** (right, Fig. 13-9) to initiate the citric acid cycle.

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712 D. Some Isomerization and Rearrangement Reactions

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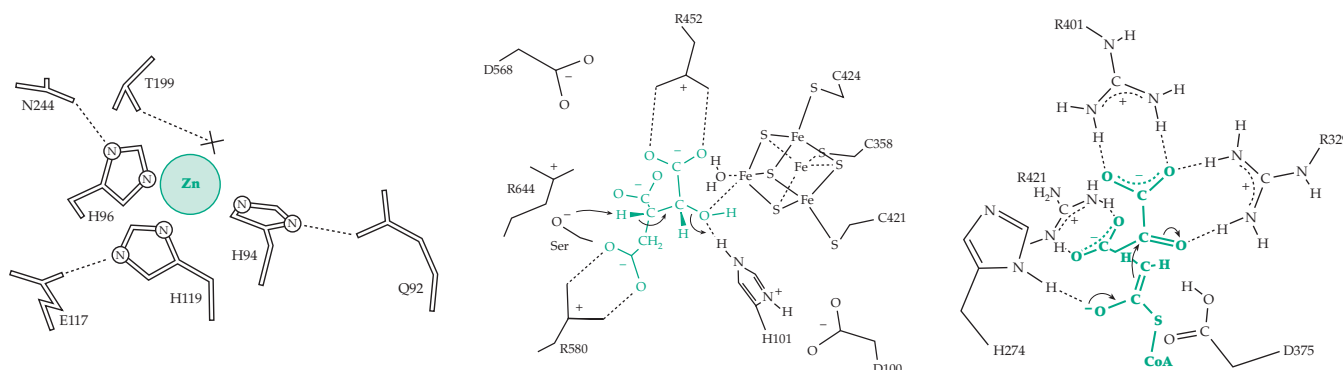
- 680 Box 13-A Zinc
- 687 Box 13-B EPSP Synthase and the Herbicide Glyphosate

Tables

- 701 Table 13-1 Products Arising from Reactions of Acetyl-CoA with a Second Substrate with Catalysis by a Polycarboxylate Synthase

Enzymatic Addition, Elimination, Condensation, and Isomerization: Roles for Enolate and Carbocation Intermediates

13



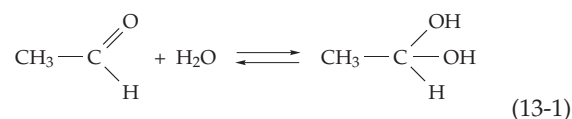
In Chapter 12 we considered reactions by which living cells are able to transfer groups from one molecule to another using nucleophilic displacements. We also showed how transfer reactions can be utilized by ligases to join two molecules together with the Gibbs energy of cleavage of ATP or of a related molecule driving the reaction. In this chapter we will examine addition reactions, which provide a simple way of joining two molecules by means of C–O, C–N, C–S, or C–C bonds. Among these are the aldol and Claisen-type condensations by which C–C bonds are formed. We will also consider elimination reactions and decarboxylations, which are the reverse of addition and condensation reactions, as well as mechanistically related isomerizations. Many reactions of these types occur in the major pathways of metabolism.

A. Addition of R–OH, R–NH₂, and R–SH to Polarized Double Bonds

Next to nucleophilic displacement, the commonest mechanistic processes in enzymatic catalysis are addition to double bonds and elimination to form double bonds. These often involve addition of a nucleophile together with a proton to a highly polarized double bond such as C=O or C=N–. In other reactions, which are discussed in Section C.2, the nucleophile attacks one end of a C=C bond that is polarized by conjugation with C=O or C=N.

Alcohols, amines, and thiols add readily to the electrophilic carbon of the carbonyl group to form **hemiacetals**, **carbinolamines**, **hemiketals**, and **hemimercaptals**. An example is the formation of ring structures of sugars (Eq. 4-1). Water can also add to carbonyl groups and most aliphatic carbonyl compounds

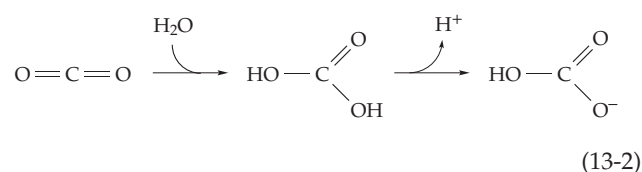
exist in water as an equilibrium mixture with a covalent hydrate (Eq. 13-1). For example, acetaldehyde in aqueous solution consists of a mixture of about 50% free aldehyde and 50% hydrate in rapidly reversible equilibrium^{1,2} and formaldehyde is over 99.9% hydrated.³



Addition reactions often occur as parts of more complex reactions. For example, a thiol group of glyceraldehyde-3-phosphate dehydrogenase reacts with the aldehyde substrate to form a hemimercaptal, which is subsequently oxidized to a thioester (see Fig. 15-6).

1. Carbonic Anhydrase

Another simple addition reaction is the hydration of CO₂ to form the bicarbonate ion. Without catalysis the reaction may require several seconds,^{4,5} the apparent first-order rate constant being $\sim 0.03 \text{ s}^{-1}$ at 25°C. Cells must often hasten the process. The specific catalyst carbonic anhydrase is widespread in its distribution



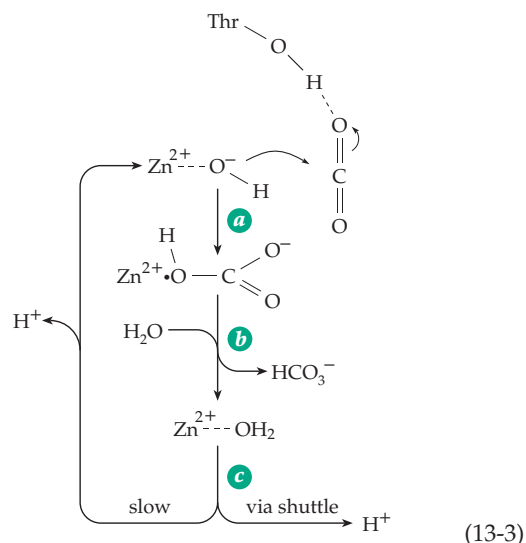
and is especially active in tissues (e.g., red blood cells, and lungs) that are involved in respiration. One liter of mammalian blood contains 1–2 g of this enzyme, a monomeric 30-kDa protein containing ~260 amino acids and one tightly bound ion of Zn^{2+} . Erythrocytes contain two isoenzymes (I and II) of carbonic anhydrase and the human body contains at least eight distinct isoenzymes (I – VIII).^{5–8} They are found wherever there is a high demand for CO_2 or bicarbonate. Isoenzyme I, II, III, and VII are cytosolic. Carbonic anhydrase I is specific to erythrocytes, while isoenzyme II is present in most cells. Hereditary lack of carbonic anhydrase II is associated with **osteopetrosis** (marble bone disease), a condition involving failure of bone resorption and the calcification of other tissues.⁹ The generation of acidity according to Eq. 13-2 is presumably required for dissolution of bone by the osteoclasts.

Isoenzymes III and VII have a more specialized distribution. Carbonic anhydrase III is abundant in adipocytes which use bicarbonate in fatty acid synthesis.⁷ Isoenzyme V is present in the mitochondrial matrix and is also abundant in both adipocytes and liver.^{7,8} Isoenzyme IV is a larger membrane-associated form, while VI is secreted into the saliva.¹⁰ Carbonic anhydrase has also been identified in *E. coli*,¹¹ in a methanobacterium,¹² and in green plants.^{13,13a} A 60-kDa carbonic anhydrase called **nacrein** is found in the organic matrix of the nacreous layer of the pearl oyster, the layer that forms aragonite (orthorhombic calcium carbonate) in the shell and in pearls.¹⁴

X-ray studies of carbonic anhydrases I and II, from human blood, revealed that both have an ellipsoidal shape of dimensions $\sim 4.1 \times 4.1 \times 4.7$ nm.^{15,16} The zinc atom in each molecule lies in a deep pocket ~ 1.2 nm from the surface and is surrounded by three histidine side chains and one H_2O or OH^- ion, the four ligands forming a distorted tetrahedron (Fig. 13-1). The coordinating imidazole group from His 119 is hydrogen bonded to a carboxylate group of Glu 117, a feature reminiscent of the charge-relay system of serine protease. This carboxylate group is also bound into a more extended hydrogen-bonded network, part of which is indicated in Fig. 13-1. The other imidazole groups also form hydrogen bonds to protein groups and the zinc-bound H_2O is involved in an extensive hydrogen-bonded network with several other bound water molecules and protein side chains.^{17a} Most of these structural features are conserved in the other mammalian isoenzymes.⁵ However, from X-ray absorption spectroscopy (EXAFS) it appears that spinach carbonic anhydrase contains one or more sulfur ligands to the zinc,¹³ while the enzymes from the archaeon *Methanosarcina thermophila* have a left-handed β helix structure (see Figs. 2-17 and 13-3).¹² Nevertheless, this enzyme has the same three-histidine Zn–OH structure found in the mammalian enzymes.

Carbonic anhydrase II is among the most rapid

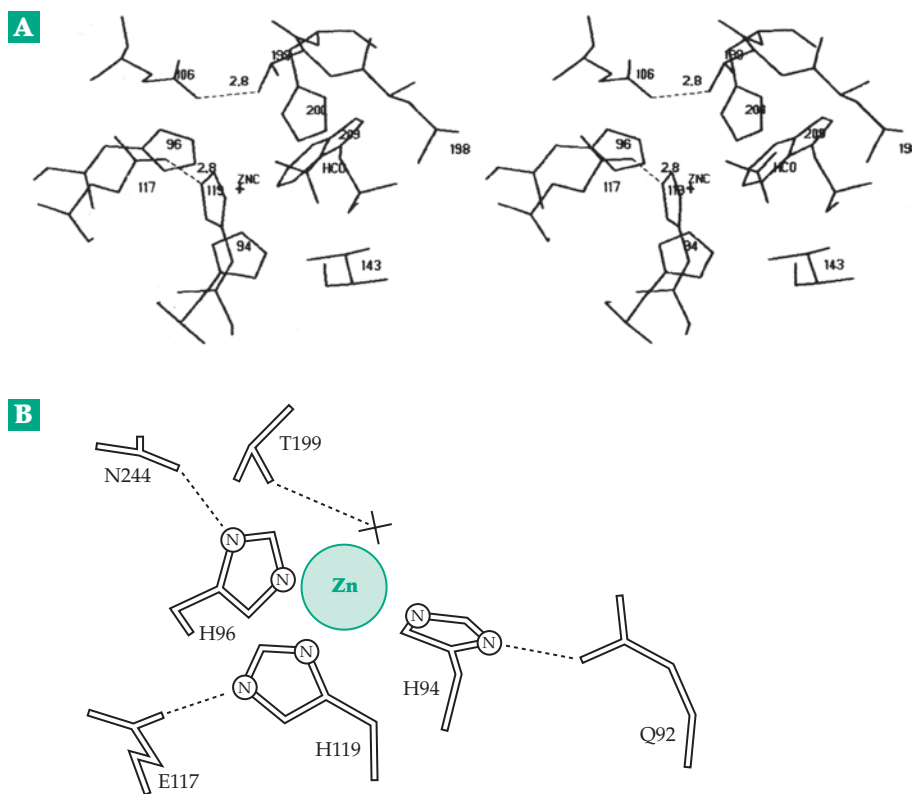
enzymes known, with the turnover number at 25°C for hydration of CO_2 being $\sim 10^6 \text{ s}^{-1}$. The same enzyme catalyzes hydration of acetaldehyde (Eq. 13-1) but at a 1000-fold slower rate. A pK_a of ~ 7 controls the activity. This appears to represent the loss of H^+ from the Zn^{2+} – OH_2 complex¹⁸ to give Zn^+ – OH . The latter is in effect a stabilized hydroxide ion existing at a pH at which OH^- is normally not present in quantity. It is this hydroxide ion that adds to the CO_2 or to the aldehyde substrate (Eq. 13-3, step a).^{18–20} In step b a water molecule replaces the departing bicarbonate.¹⁷ A variety of data



indicate that proton transfers mediated by the enzyme are essential parts of the carbonic anhydrase mechanism.^{18–19b,21–23} One proposal is that the nearby imidazole group of His 64 (not shown in Fig. 13-1) deprotonates the bound H_2O via a hydrogen-bonded network of bound water molecules (Eq. 13-3, step c). The side chain of Thr 199 may function in a cyclic proton transfer in step c.¹⁹ The proton generated in step c is released to the solvent in a process that is catalyzed by buffer anions or by amines such as histamine. The latter binds at the edge of the active site and forms an additional hydrogen-bonded pathway to the zinc-bound H_2O .²⁴ A different proton shuttle pathway has been proposed for the slower carbonic anhydrase III.²⁵

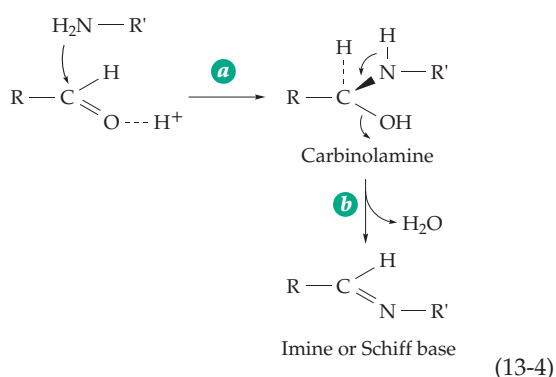
Related to the reaction catalyzed by carbonic anhydrase is the addition of an amino group to CO_2 (Eq. 2-21) to form a carbamino group ($-\text{NH}-\text{COO}^-$). This reaction is essential to the functioning of hemoglobin, which must carry large amounts of CO_2 , as carbamino groups, from tissues to the lungs, (Eq. 7-47) and to some enzymes such as ribulose biphosphate carboxylase (Figs. 13-10, 13-11).

Figure 13-1 (A) Stereoscopic view showing the binding pocket for HCO₃⁻ in the active site of human carbonic acid anhydrase I. Also shown are the hydrogen-bonded pairs E117–H119 and E106–T199. ZNC, zinc ion; HCO, bicarbonate. From Kumar and Kannan.¹⁶ (B) View of the active site of carbonic anhydrase II. The orientation is a little different than in (A). The location of H₂O or ⁻OH bound to the zinc ion is marked X. Hydrogen bonds from the three histidines that form coordinate bonds to the zinc, Q92, E117, the backbone carbonyl of N244, and the hydrogen bond from the zinc-bound hydroxyl to the T199 side chain are shown as dashed lines. From Kiefer *et al.*¹⁷



2. Imines (Schiff Bases)

As we have seen already, many enzymatic reactions depend upon formation of **imines**, which are commonly called Schiff bases. The two-step formation of Schiff bases consists of addition of an amino group to a carbonyl group to form a carbinolamine followed by elimination of water (Eq. 13-4).²⁶ One group of **aldolases** (Section D) have, at their active centers,



lysine side chains which form Schiff bases with the ketone substrates prior to the principal reaction of breaking or forming a C–C bond. Similarly, the initial reaction of the aldehyde coenzyme **pyridoxal phosphate** with amino acid substrates is the formation of Schiff

bases (Fig. 14-4). Indeed, *the groups C=O and H₂N– are inherently complementary* and their interaction through imine formation is extremely common.

Schiff bases often form within a fraction of a second, but one or both steps of Eq. 13-4 may require catalysis to achieve enzymatic velocities.²⁷ The reaction is usually completely reversible and formation constants are often low enough that a carbonyl compound present in small amounts will not react extensively with an amine unless the two are brought together on an enzyme surface. If the amino group is hydrogen bonded to the enzyme, the proton may remain on the Schiff base nitrogen, enhancing the electron-accepting properties of the C=N group. The pK_a values of Schiff bases with aliphatic aldehydes are usually 1–2 units lower than those of the corresponding primary amines. However, the local environment may sometimes cause the pK_a to be increased. Most Schiff bases are reduced readily by sodium borohydride or sodium cyanoborohydride to form secondary amines in which the original aldehydes are bound covalently to the original amino groups (Eq. 3-34). This provides a method for locating sites of Schiff base formation in enzymes. An isotopically substituted aldehyde or amine can be employed, or an isotopic label can be introduced, by using ²H- or ³H-containing sodium borohydride in the reduction.

3. Stereochemistry of Addition to Trigonal Carbon Atoms

Adducts formed by enzymatic addition of nucleophiles to carbonyl groups are usually chiral. For example,

addition of an amino group to the “front” side of the carbonyl carbon in Eq. 13-5 (the *si* face as defined in Chapter 9) creates a carbinolamine of the *S* configuration. Addition of the amino group from behind the plane of the paper (the *re* face) would lead to a carbinolamine of the *R* configuration.

BOX 13-A ZINC

The average human ingests 10–15 mg of zinc a day.^a Although it is poorly absorbed, the concentrations of zinc in tissues are relatively high and the metal plays an essential role in a multitude of enzymes. The total zinc content of a 70-kg person is 1.4–2.3 g. A typical tissue concentration of Zn^{2+} is 0.3–0.5 mM; an unusually high content of ~15 mM is found in the prostate gland.

Zinc ion is much more tightly bound to most organic ligands than is Mg^{2+} (Table 6-9). It has a filled *3d* shell and tends to form four ionic bonds with a tetrahedral geometry, often with nitrogen- or sulfur-containing ligands.^b Unlike Mg^{2+} , which interacts rapidly and reversibly with enzymes, Zn^{2+} tends to be tightly bound within over 300 **metallo-enzymes**.^{c–e} A common feature is the surrounding of the Zn^{2+} at the active center by three imidazole groups, the fourth coordination position being free for interaction with substrate. The second nitrogen of the imidazole ring in many instances is hydrogen bonded to a main chain carbonyl group of the peptide, a feature that is also shared by histidines in other metalloproteins.

The most important chemical function of Zn^{2+} in enzymes is probably that of a Lewis acid providing a concentrated center of positive charge at a nucleophilic site on the substrate.^f This role for Zn^{2+} is discussed for carboxypeptidases (Fig. 12-16) and thermolysin,^g alkaline phosphatase (Fig. 12-23),^h RNA polymerases, DNA polymerases, carbonic anhydrase (Fig. 13-1),ⁱ class II aldolases (Fig. 13-7), some alcohol dehydrogenases (Fig. 15-5), and superoxide dismutases (Fig. 16-22). Zinc ions in enzymes can often be replaced by Mn^{2+} , Co^{2+} , and other ions with substantial retention of catalytic activity.^{f,j}

In addition to its function in catalysis, zinc often plays an important structural role, e.g., in the **zinc finger** transcriptional regulators (Fig. 5-38).^k Zinc ions bind to insulin and stabilize its hexameric structure (Fig. 7-18).^l Six Zn^{2+} ions are present in the hexagonal tail plate of the T-even bacteriophage (Box 7-C) and appear to be essential for invasion of bacteria.^m In carnivores, the **tapetum**, the reflecting layer behind the retina of the eye of many animals, contains crystals of the Zn^{2+} –cysteine complex.

Since zinc ions have no color their presence has often been overlooked. Zinc ions will doubtless be found in many more places within cells. Zinc is usually the major component of the bound metals in the **metallothioneins** (Box 6-E). These small 6.6-kDa proteins which contain ~33% cysteine and bind as many as six ions of Cd^{2+} , Hg^{2+} , Cu^{2+} , or Zn^{2+} per molecule are present in all animal tissues as well as in plants and some bacteria.^j

From a nutritional viewpoint, Cu^{2+} competes with zinc ion, as does the very toxic Cd^{2+} . The latter accumulates in the cortex of the kidney. Dietary cadmium in concentrations less than those found in human kidneys shortens the lives of rats and mice. However, some marine diatoms contain a cadmium-dependent carbonic anhydrase.ⁿ Although zinc deficiency was once regarded as unlikely in humans, it is now recognized as occurring under a variety of circumstances^{o,p} and is well-known in domestic animals.^q Consumption of excessive amounts of protein as well as alcoholism, malabsorption, sickle cell anemia, and chronic kidney disease can all be accompanied by zinc deficiency.

^a O'Dell, B. L., and Campbell, B. J. (1971) *Comprehensive Biochemistry* **21**, 179–216

^b Bock, C. W., Katz, A. K., and Glusker, J. P. (1995) *J. Am. Chem. Soc.* **117**, 3754–3765

^c Berg, J. M., and Shi, Y. (1996) *Science* **271**, 1081–1085

^d Vallee, B. L., and Auld, D. S. (1993) *Biochemistry* **32**, 6493–6500

^e Coleman, J. E. (1992) *Ann. Rev. Biochem.* **61**, 897–946

^f Mildvan, A. S. (1974) *Ann. Rev. Biochem.* **43**, 357–399

^g Holland, D. R., Hausrath, A. C., Juers, D., and Matthews, B. W. (1995) *Protein Sci.* **4**, 1955–1965

^h Kimura, E., Kodama, Y., Koike, T., and Shiro, M. (1995) *J. Am. Chem. Soc.* **117**, 8304–8311

ⁱ Lesburg, C. A., and Christianson, D. W. (1995) *J. Am. Chem. Soc.* **117**, 6838–6844

^j Kagi, J. H. R., Himmelhoch, S. R., Whanger, P. D., Bethune, J. L., and Vallee, B. L. (1974) *J. Biol. Chem.* **249**, 3537–3542

^k Berg, J. M. (1990) *J. Biol. Chem.* **265**, 6513–6516

^l Hill, C. P., Dauter, Z., Dodson, E. J., Dodson, G. G., and Dunn, M. F. (1991) *Biochemistry* **30**, 917–924

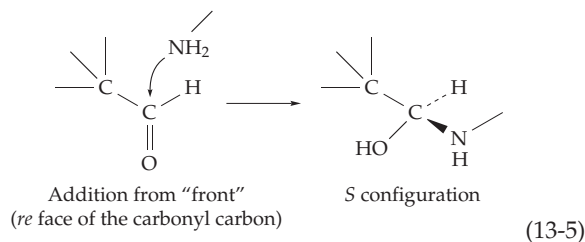
^m Kozloff, L. M., and Lute, M. (1977) *J. Biol. Chem.* **252**, 7715–7724

ⁿ Lane, T. W., and Morel, F. M. M. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 4627–4631

^o Prasad, A. S. (1984) *Fed. Proc.* **43**, 2829–2834

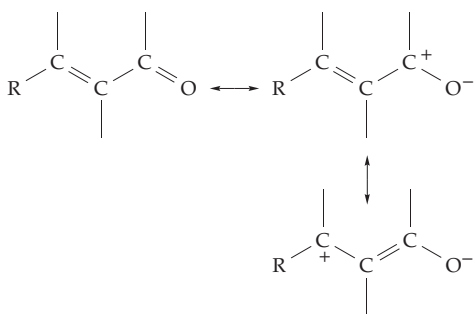
^p Day, H. G. (1991) *FASEB J.* **5**, 2315–2316

^q Luecke, R. W. (1984) *Fed. Proc.* **43**, 2823–2828



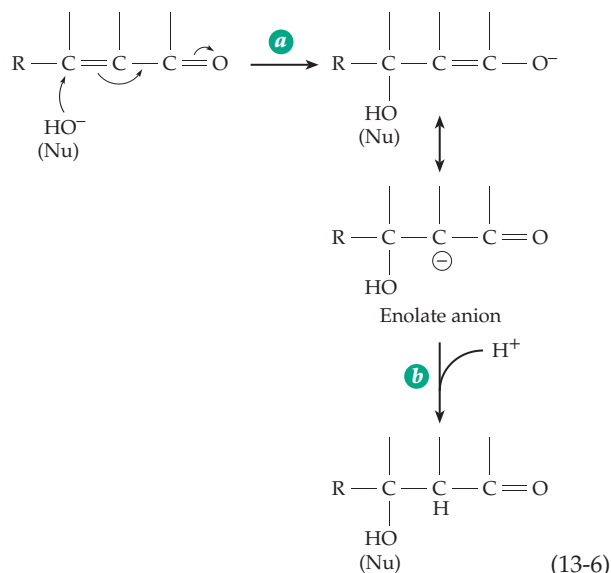
4. Addition to Carbon-Carbon Double Bonds, Often Reversible Reactions

Most of the reactions that we will consider in this chapter involve addition of a proton to a carbon atom or removal of a proton attached to a carbon atom. A frequent metabolic reaction is addition of water to a carbon-carbon double bond that is conjugated with a carbonyl group. This transmits the polarization of the carbonyl group to a position located two carbon atoms further along the chain.



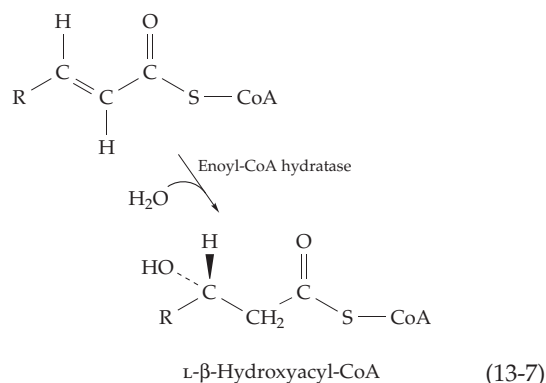
Because of this effect bases can add to a carbon-carbon double bond in a position β to the carbonyl group as in Eq. 13-6, step *a*. The product of addition of HO⁻ is the anion of an **enol** in which the negative charge is distributed by resonance between the oxygen of the carbonyl group and the carbon adjacent to the carbonyl. A stable product results if a proton adds to the latter position (Eq. 13-6, step *b*). Many nucleophilic groups (Nu in Eq. 13-6) other than HO⁻, may add, the reaction being known as a **Michael addition**. If a neutral nucleophile Nu adds, the product may lose a proton, or transfer it to the adjacent carbon in Eq. 13-6, step *b*, to give a neutral end product. The reverse of an addition reaction of this type is also known as an **elimination reaction**.

Enzymatic reactions involving addition to a C=C bond adjacent to a carbonyl group (or in which elimination occurs α,β to a carbonyl) are numerous. Except for some enzymes acting by free radical mechanisms, the nucleophilic group always adds at the β position suggesting that the mechanism portrayed by Eq. 13-7 is probable. It is noteworthy that *frequently in a metabolic sequence a carbonyl group is deliberately introduced to*



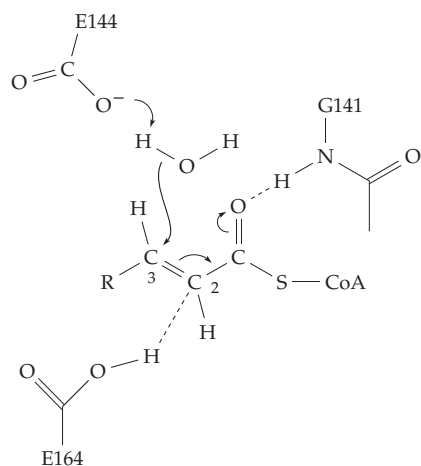
facilitate elimination or addition at adjacent carbon atoms. The carbonyl may be formed by oxidation of a hydroxyl group or it may be provided by a thioester formed with coenzyme A or with an **acyl carrier protein** (Chapter 14).

Enoyl-CoA hydratase. A specific example of the reaction in Eq. 13-6 is the addition of water to *trans*- α,β -unsaturated CoA derivatives (Eq. 13-7). It is catalyzed by enoyl-CoA hydratase (crotonase) from mitochondria and is a step in the β oxidation of fatty acids (Fig. 10-4).



The enzyme is a hexamer, actually a dimer of trimers made up of 291-residue polypeptide chains.²⁸ Acetoacetyl-CoA is a competitive inhibitor which binds into the active site and locates it. From the X-ray structure of the enzyme-inhibitor complex it can be deduced that the carboxylate group of E144 abstracts a proton from a water molecule to provide the hydroxyl ion that binds to the β position (Eq. 13-6, step *a*) and that the E164 carboxyl group donates a proton to the intermediate enolate anion in step *b*.²⁸ The hydroxyl group

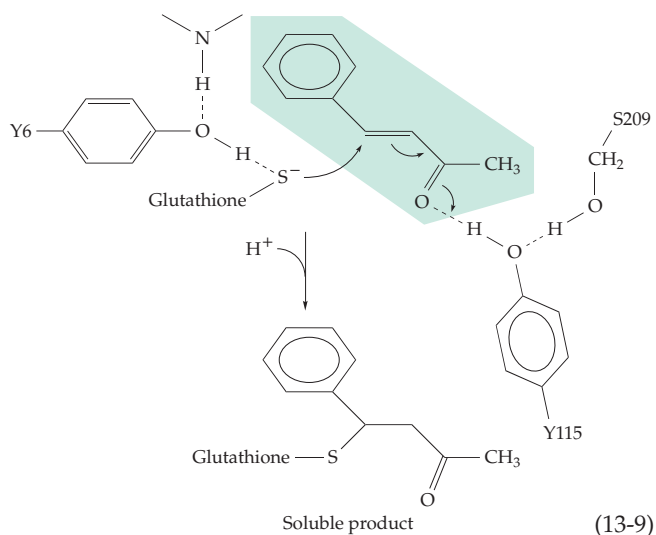
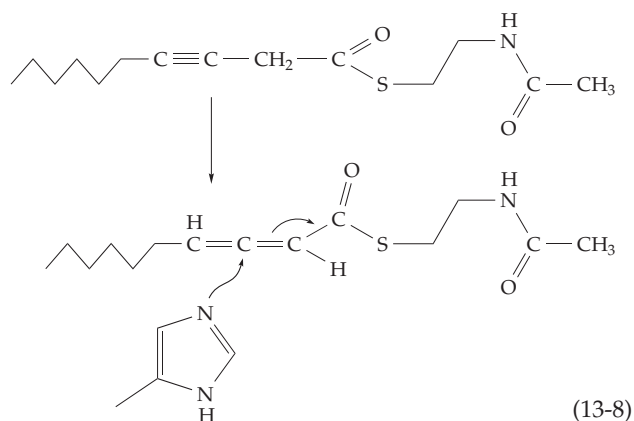
and the proton enter from the same side of the double bond in a *syn* addition. The X-ray structure shows that the E144 and E164 side chains are on the same side of the double bond, accounting for this stereochemistry.^{28,28a}



Cinnamoyl-CoA thiol esters (R = phenyl in the foregoing structure) contain a good light-absorbing group (chromophore). Binding to the protein induces distinct shifts in the ultraviolet absorption spectrum and also in ¹³C NMR and Raman spectra. These shifts suggest that binding induces an enhanced positive charge at C-3 and formation of a strong hydrogen bond to the carbonyl group.²⁹ The X-ray studies indicate that this bond is to the G141 peptide nitrogen as shown. These results favor an enolate anion intermediate as in Eq. 13-6. However, kinetic isotope effects³⁰ as well as studies of proton exchange have suggested a concerted mechanism with water adding at the same time that a proton binds at C-2.^{30–31a}

A closely related *E. coli* protein is a 79-kDa multifunctional enzyme that catalyzes four different reactions of fatty acid oxidation (Chapter 17). The amino-terminal region contains the enoyl hydratase activity.³² A quite different enzyme catalyzes dehydration of thioesters of β -hydroxyacids such as 3-hydroxydecanoyl-acyl carrier protein (see Eq. 21-2) to both form and isomerize enoyl-ACP derivatives during synthesis of unsaturated fatty acids by *E. coli*. Again, a glutamate side chain is the catalytic base but an imidazole group of histidine has also been implicated.³³ This enzyme is inhibited irreversibly by the *N*-acetylcysteamine thioester of 3-decynoic acids (Eq. 13-8). This was one of the first enzyme-activated inhibitors to be studied.³⁴

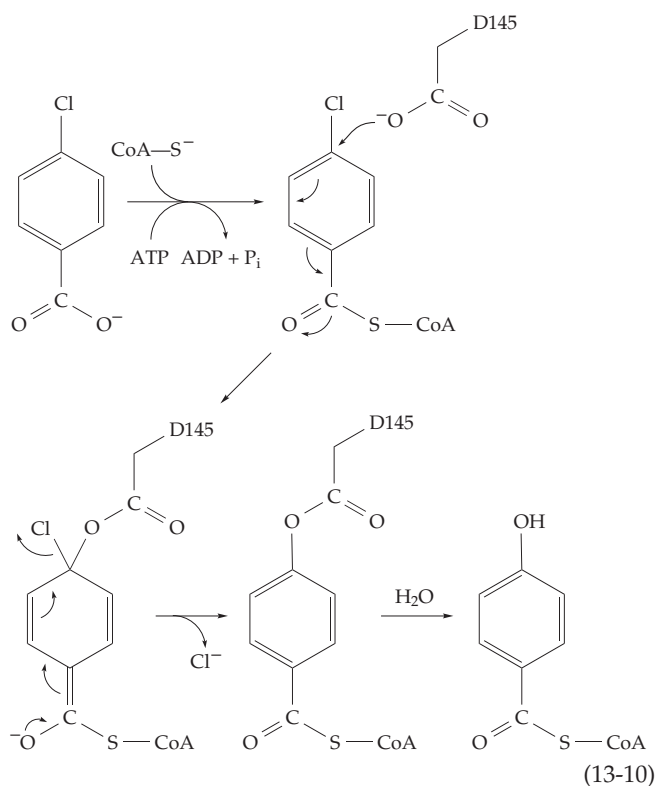
Glutathione S-transferases. Addition of glutathione (Box 11-B) to a large variety of different substrates containing electrophilic centers, such as that at the β position in an α,β -unsaturated ketone (Eq. 13-9), is catalyzed by the ubiquitous group of enzymes called glutathione S-transferases. There are six classes of



eukaryotic glutathione S-transferases—one membrane-associated microsomal³⁵ and five cytosolic.^{36–40} They play an important role in detoxifying many xenobiotics and other relatively hydrophobic compounds by converting them to more soluble compounds that can be degraded and excreted easily (see Box 11-B). A pair of tyrosines appear to participate in catalysis, as is indicated in Eq. 13-9.^{38,41} Glutathione S-transferases have attracted attention because of their role in detoxifying anticancer drugs. Cancer cells can become resistant to drugs as a result of excessive synthesis of these detoxifying enzymes. At the same time glutathione transferases protect human patients from drugs. In plants these enzymes may provide protection from insecticides and herbicides.^{41a} Glutathione transferases of pathogenic organisms such as schistosomes are appropriate targets for new drugs as well as for vaccines.⁴² Nonenzymatic Michael addition of glutathione to such compounds as **4-hydroxynonenal**, a product of peroxidation of the polyunsaturated arachidonate (see Eq. 21-15) are also biochemically important.

Chlorobenzoyl-CoA dehalogenase. The enzymatic release of chloride ion from 4-chloroxybenzoyl-CoA

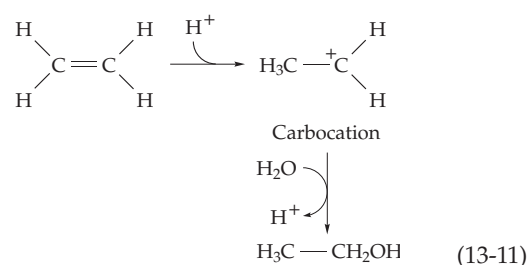
would be hard to explain by a simple nucleophilic displacement by HO⁻. However, *addition* of HO⁻, followed by elimination of Cl⁻, would be chemically reasonable. Nevertheless, single-turnover studies revealed a more complex mechanism involving formation of a covalent adduct with the enzyme.^{43,44} A carboxylate group adds and Cl⁻ is eliminated to form of an oxygen ester, which is then hydrolyzed to the final product (Eq. 13-10). Studies of mutant enzymes⁴⁵ together with X-ray crystallography⁴⁶ support this mechanism. As with enoyl-CoA hydratase, binding of substrate to chlorobenzoyl-CoA dehalogenase causes alterations in ultraviolet, NMR, and Raman spectra that can be interpreted as indicating enhanced polarization of the benzoyl group.^{47,48} For example, changes in the C=O stretching frequency suggest that the bond is elongated, presumably as a result of hydrogen bonding to the N-H of G114, which lies at the N terminus of an α helix and experiences the additional polarizing effect of the helix dipole (Fig. 2-20).^{48,49} This effect can be compared with that mentioned previously for enoyl-CoA hydratase.



In fact, close sequence and structural homologies show that chlorobenzoyl-CoA dehalogenase, enoyl-CoA hydratase, and a variety of other hydratases, isomerases, synthases, lyases, and hydrolases belong to a large family of related proteins.^{49a,49b,49c}

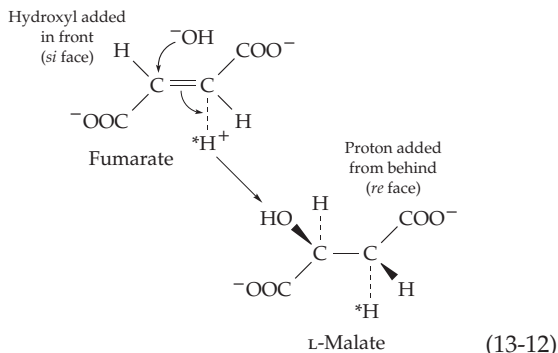
5. Addition to Double Bonds Adjacent to Carboxylate Groups

Biochemical reactions often involve addition to C=C bonds that are not conjugated with a true carbonyl group but with the poorer electron acceptor -COO⁻. While held on an enzyme a carboxylate group may be protonated, making it a better electron acceptor. Nevertheless, there has been some doubt as to whether the carbanion mechanism of Eq. 13-6 holds for these enzymes. Some experimental data suggested a quite different mechanism, one that has been established for the nonenzymatic hydration of alkenes. An example is the hydration of ethylene by hot water with dilute sulfuric acid as a catalyst (Eq. 13-11), an industrial method of preparation of ethanol. The electrons of the double bond form the point of attack by a proton, and the resulting carbocation readily abstracts a hydroxyl



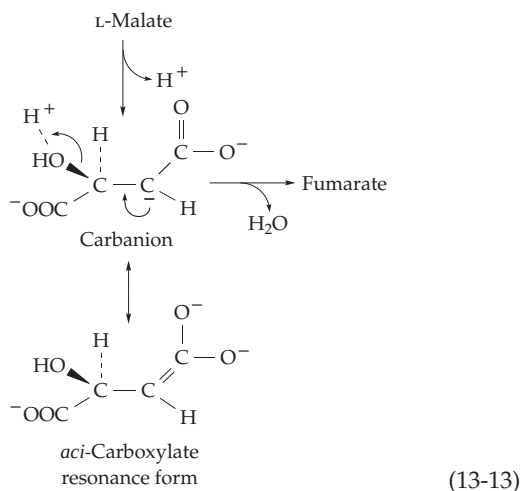
ion from water. Direct addition of OH⁻ to ethylene to form a carbanion is not favored, because there is no adjacent carbonyl group to stabilize the negative charge.

Fumarate hydratase. The most studied enzyme of this group is probably the porcine mitochondrial fumarate hydratase (fumarase; see also Chapter 9), a tetramer of 48.5-kDa subunits⁵⁰ with a turnover number of $\sim 2 \times 10^3 \text{ s}^{-1}$. It accelerates the hydration reaction more than 10^{15} -fold.⁵¹ A similar enzyme, the 467-residue **fumarase C** whose three-dimensional structure is known,^{52,53} is found in cells of *E. coli* when grown aerobically. The product of the fumarate hydratase reaction is L-malate (S-malate). The stereospecificity is extremely high. If the reaction is carried out in ²H₂O an atom of ²H is incorporated into the *pro-R* position, i.e., the proton is added strictly from the *re* face of the trigonal carbon (Eq. 13-12). To obtain L-malate the hydroxyl must have been added from the opposite side of the double bond. Such *anti* (*trans*) addition is much more common in both nonenzymatic and enzymatic reactions than is addition of both H and OH (or -Y) from the same side (*syn*, *cis*, or adjacent addition).⁵⁴ For concerted addition it is a natural result of stereoelectronic control. Almost *all* enzymatic addition and elimination reactions involving free carboxylic acids are *anti* with the proton entering from the *re* face,



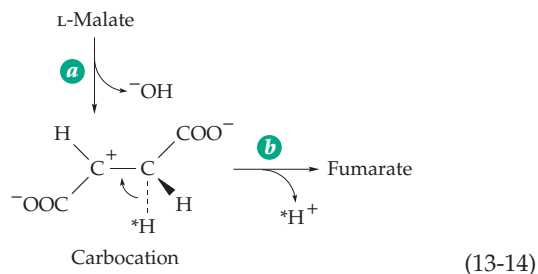
suggesting that there has been conservation throughout evolution of a single basically similar mechanism.^{54,55}

The pH dependence of the action of fumarate hydratase indicates participation of both an acidic and a basic group with pK_a values of 5.8 and 7.1.⁵⁶ See Chapter 9 for additional information. However, either anion or carbocation mechanisms might be possible. That the cleavage of the C–H bond is not rate limiting is suggested by the observation that malate containing ^2H in the *pro-R* position is dehydrated at the same rate as ordinary malate. If the anion mechanism (Eq. 13-13) is correct, the ^2H from the *pro-R* position of specifically labeled malate might be removed rapidly, while the loss of OH^- could be slower. If so, the ^2H would be “washed out” of L-malate faster than could happen by conversion to fumarate followed by rehydration to malate. In fact, the opposite was observed.



The hydroxyl group was lost rapidly and the ^2H more slowly. This result suggested the carbocation ion mechanism of Eq. 13-14.

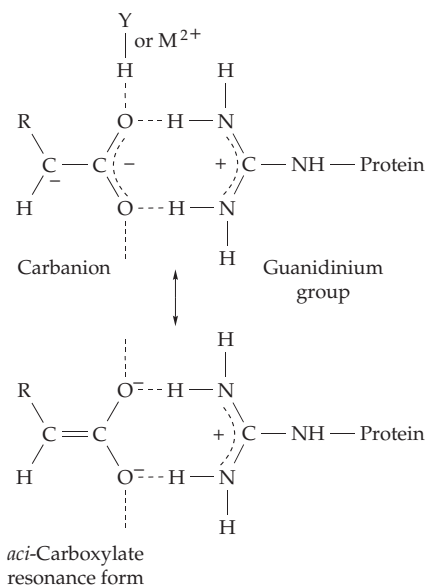
While no *primary* isotope rate effect was observed, when the hydrogen at C-2 or the *pro-S* hydrogen at C-3 of malate was replaced by ^2H or ^3H distinct secondary isotope effects were seen. Thus, $k(^1\text{H})/k(^2\text{H}) = 1.09$ for both the *pro-S* and the C-3 hydrogen atoms.⁵⁷ These findings appeared to support the carbocation



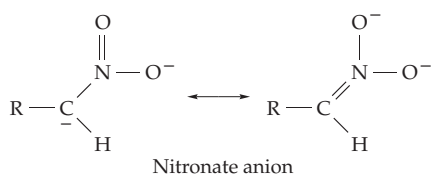
mechanism for the reasons considered in Chapter 12 and suggested that step *b* of Eq. 13-14 is rate limiting. The relative values of V_{max} for hydration of fumarate, fluorofumarate, and difluorofumarate (104, 410, and 86 $\text{mol ml}^{-1} \text{min}^{-1} \text{mg}^{-1}$) also seemed to support the carbocation mechanism.⁵⁸

However, it has become clear that protons removed from a substrate to a basic group in a protein need not exchange rapidly with solvent (see Eq. 9-102). In fact, the proton removed by fumarate hydratase from malate is held by the enzyme for relatively long periods of time. Its rate of exchange between malate and solvent is slower than the exchange of a bound fumarate ion on the enzyme surface with another substrate molecule from the medium.⁵⁹ Thus, the overall rate is determined by the speed of dissociation of products from the enzyme and we cannot yet decide whether removal of a proton precedes or follows loss of OH^- .

Two new lines of evidence suggest that proton abstraction comes first. A careful study of both ^{18}O and ^2H isotope effects⁶⁰ supports the carbanion intermediate, as does the strong inhibition by anions of 3-nitropropionate and 3-nitro-2-hydroxypropionate.⁶¹ To provide a good electron sink the carboxylate group adjacent to the proton that is removed by fumarate hydratase must either be actually protonated in the enzyme–substrate (ES) complex or paired with and hydrogen bonded to a positively charged group.



The anion formed by removal of the 3-H is analogous to the enolate anion of Eq. 13-6 and has a strong structural similarity to the readily formed anions of organic nitro compounds. The nitronate anions may, perhaps, be regarded as transition state inhibitors.

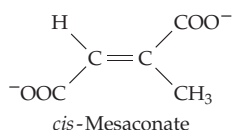


There is still a third possible mechanism for the fumarate hydratase reaction. The proton and hydroxyl groups may be added *simultaneously* in a concerted reaction.⁶² However, observed kinetic isotope effects are not consistent with this mechanism.⁶⁰ In 1997 the structure of fumarase C of *E. coli* was reported.^{52,53} Each active site of the tetrameric enzyme is formed using side chains from three different subunits. The H188 imidazole is hydrogen bonded to an active site water molecule and is backed up by the E331 carboxylate which forms a familiar catalytic pair. However, these results have not clarified the exact mode of substrate binding nor the details of the catalytic mechanism. Structural studies of fumarate hydratase from yeast^{53a} and the pig^{53b} are also in progress.

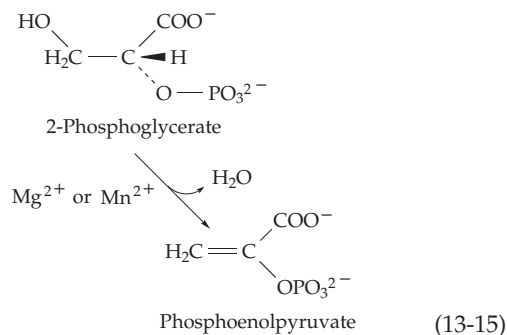
Some other fumarate-forming reactions.

Other enzymes that catalyze elimination reactions that produce fumarate are **aspartate ammonia-lyase** (aspartase),⁶³ **argininosuccinate lyase** (Fig. 24-10, reaction g),^{64,65} and **adenylosuccinate lyase** (Fig. 25-15). In every case it is NH₃ or an amine, rather than an OH group, that is eliminated. However, the mechanisms probably resemble that of fumarate hydratase. Sequence analysis indicated that all of these enzymes belong to a single **fumarate-aspartase family**.^{64,65} The three-dimensional structure of aspartate ammonia-lyase resembles that of fumarate hydratase, but the catalytic site lacks the essential H188 of fumarate hydratase. However, the pK_a values deduced from the pH dependence of V_{max} are similar to those for fumarase.⁶⁴

3-Methylaspartate lyase catalyzes the same kind of reaction to produce ammonia plus *cis*-mesaconate.⁶³ Its sequence is not related to that of fumarase and it may contain a dehydroalanine residue (Chapter 14).⁶⁶



Enolase. A key reaction in the metabolism of sugars is the dehydration of 2-phosphoglycerate to form **phosphoenolpyruvate** (PEP), the phospho derivative of the enolic form of pyruvic acid:



A carbanionic intermediate has often been suggested for this enzyme.^{67,68} However, despite measurements of kinetic isotope effects and many other experiments it has been difficult to establish a detailed mechanism.⁶⁸

Enolase has a complex metal ion requirement,^{68a,69} usually met by Mg²⁺ and Mn²⁺. From NMR studies of the relaxation of water protons, it was concluded that a Mn²⁺ ion coordinates two rapidly exchangeable water molecules in the free enzyme. When substrate binds, one of these water molecules may be immobilized and may participate in an addition reaction that forms phosphoenolpyruvate (reverse of reaction 13-15). A tightly bound “conformational” metal ion is located in the known three-dimensional structure in such a

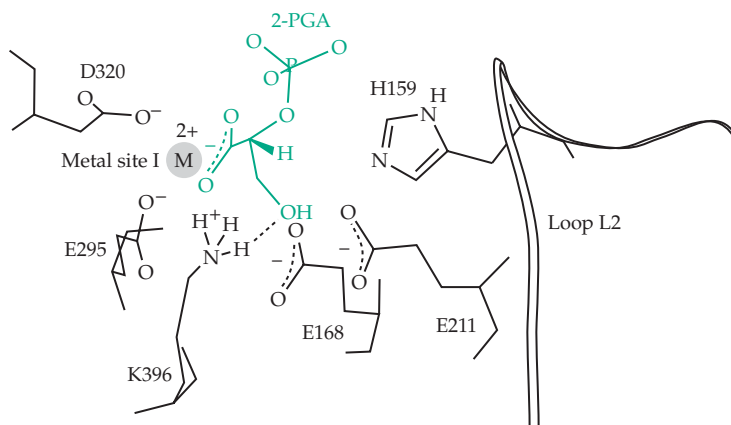


Figure 13-2 View of the active site of yeast enolase containing a bound molecule of 2-phospho-D-glycerate. The catalytic magnesium ion is at the left but the “conformational” metal is not visible here. The imidazole group of His 159 serves as the catalytic base and the -NH₃⁺ of Lys 396 or Lys 345^{73b} as the catalytic acid. From Vinarov and Nowak.⁶⁹

way that it might function in this manner. A more loosely bound “catalytic” metal ion is also essential.^{69–72}

Figure 13-2 shows a view of the active site of yeast enolase occupied by a molecule of bound 2-phosphoglycerate. Histidine 159 is probably the catalytic base that removes the α -proton to form an *aci*-anion which is stabilized by interaction with the catalytic Mg^{2+} ion. A protonated lysine 396 amino group may be the catalytic acid.^{68a,69} The active site is surrounded by a complex hydrogen-bonded network.⁷³ As is discussed in Section B, a large number of other enzymes belong to an **enolase superfamily** of enzymes. Among them is **glucarate dehydratase**, which initiates a pathway for catabolism of D-glucarate and galactarate in *E. coli*.^{73a}

Pectate lyase and related enzymes. A group of polysaccharide lyases cleave the chains of polymers of uronic acids with 1,4 linkages such as pectins, hyaluronan, heparin,^{74,75} and dermatan sulfate (Fig. 4-11). These bacterial enzymes also employ an elimination mechanism.⁷⁶ The geometry of the β -linked galacturonic acid units of pectin is favorable for *anti* elimination of the 5-H and the O-glycosyl group in the 4 position (Eq. 13-16). However, the corresponding **hyaluronate lyase** (hyaluronidase) acting on glucuronic acid residues causes a *syn* elimination. These results suggest the formation of anionic intermediates which can eliminate a substituent from either the equatorial or axial position of the sugar ring. Hyaluronate lyase from the pathogenic *Streptococcus pneumoniae* apparently utilizes an imidazole group as the catalytic base and a tyrosine side chain as the proton donor in the reaction.^{76a,76b}

Pectate lyase C from the plant pathogen *Erwinia*, which causes soft-rot in many different plants, has a parallel β barrel structure (Fig. 13-3)⁷⁷ which is similar to that of the tailspike protein shown in Fig. 2-17 and represents what may be a very large structural family of proteins.⁷⁸ The location of the active site is not

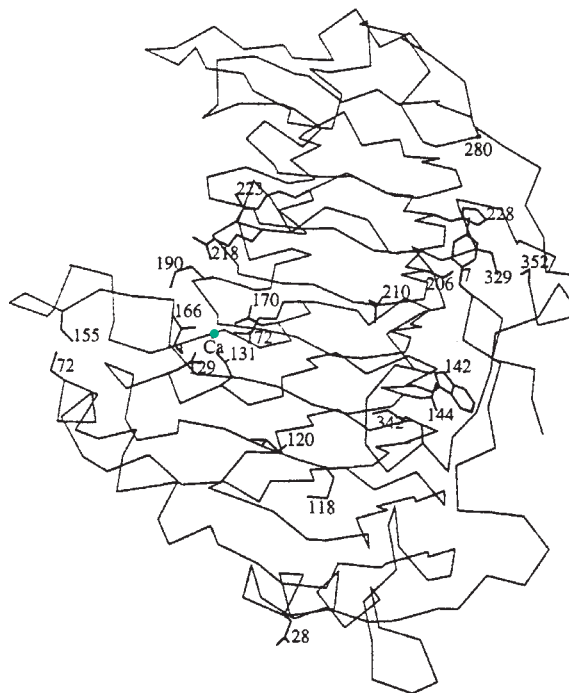


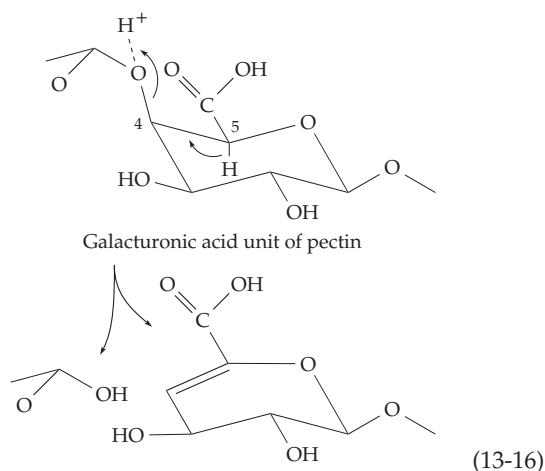
Figure 13-3 The three-dimensional structure of pectate lyase C showing locations of amino acids substituted by oligonucleotide-directed mutation of the cloned gene. The green dot labeled Ca is the Ca^{2+} -binding site.⁷⁹ Courtesy of Frances Jurnak.

obvious from the structure but on the basis of the effects of a large number of mutations, it is thought to be near the Ca^{2+} -binding site labeled in Fig. 13-3. Replacement of any of three Asp or Glu and three Lys residues in this region leads to loss of or reduction in catalytic activity.⁷⁹

6. Aconitase and Related Iron – Sulfur Proteins

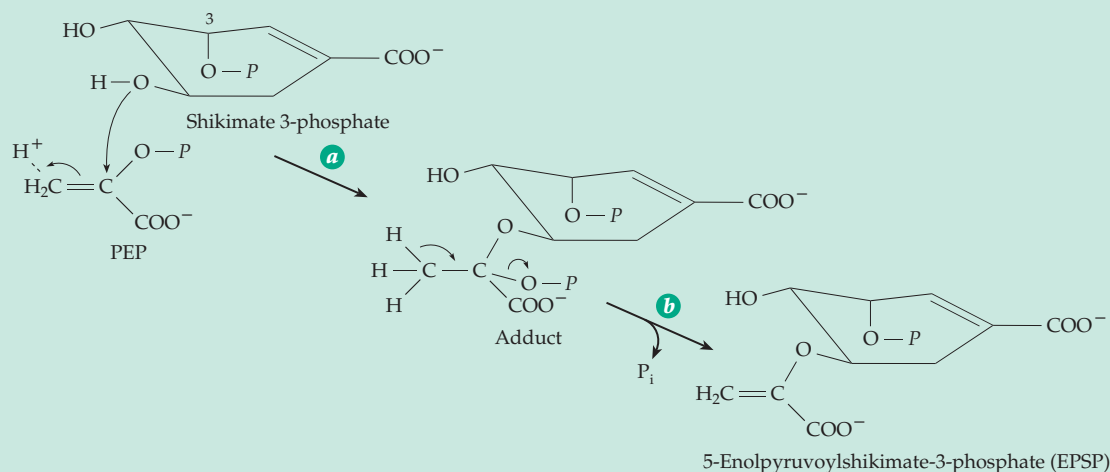
Two consecutive reactions of the citric acid cycle (Fig. 10-6), the dehydration of citrate to form *cis*-aconitate and the rehydration in a different way to form isocitrate (Eq. 13-17), are catalyzed by aconitase (aconitate hydratase). Both reactions are completely stereospecific. In the first (Eq. 13-17, step *a*), the *pro-R* proton from C-4 (stereochemical numbering) of citrate is removed and in step *c* isocitrate is formed. Proton addition is to the *re* face in both cases.

As with fumarate hydratase, the enzyme holds the abstracted proton (for up to 7×10^{-5} s) long enough so that a *cis*-aconitate molecule sometimes diffuses from the enzyme and (if excess *cis*-aconitate is present) is replaced by another. The result is that the new *cis*-aconitate molecule sometimes receives the proton (intermolecular proton transfer). The proton removed



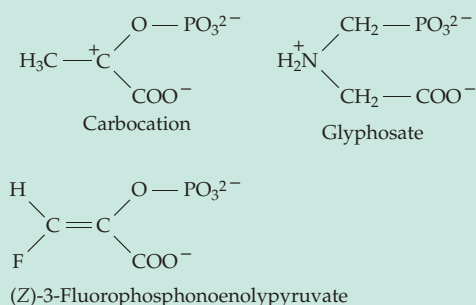
BOX 13-B EPSP SYNTHASE AND THE HERBICIDE GLYPHOSATE

The reversible reaction of phosphoenolpyruvate (PEP) with shikimate 3-phosphate is a step in the synthesis of the aromatic amino acids (see Fig. 25-1). The chemical mechanism indicated



product.^{f-h} Glyphosate was for many years viewed as a transition-state analog but more recently has been shown to be a tight-binding noncompetitive inhibitor.ⁱ

was proposed by Leaven and Sprinson in 1964.^a Step *a* is unusual^b because it involves protonation on the methylene carbon of PEP and addition of a nucleophile at C-2, the opposite of the addition in the enolase reaction (Eq. 13-15, reverse). It is likely that formation of a cationic intermediate precedes that of the adduct shown. However, the structure of the adduct has been confirmed by isolation from the active site and by synthesis^c and it has been observed in the active site by NMR spectroscopy.^d The three-dimensional structure of EPSP synthase is known.^{e,m} It is inhibited by the commercial herbicide **glyphosate** [N-(phosphonomethyl) glycine] whose structure is somewhat similar to that of the proposed carbocation that arises from PEP. The inhibitor (Z)-3-fluorophosphoenolpyruvate is a pseudosubstrate that reacts in step *a* to give a stable adduct unable to go through step *b* to form a



A related mechanism is utilized in the biosynthesis of UDP-muramic acid (Eq. 20-6).^j There is an enolpyruvoyl adduct analogous to that of EPSP synthase; a proposed enolpyruvoyl-enzyme adduct with Cys 115 is not on the major path.^{k,l} However, this enzyme is not inhibited by glyphosate.ⁱ

- ^a Levin, J. G., and Sprinson, D. B. (1964) *J. Biol. Chem.* **239**, 1142–1150
- ^b Barlow, P. N., Appleyard, R. J., Wilson, B. J. O., and Evans, J. N. S. (1989) *Biochemistry* **28**, 7985–7991
- ^c Anderson, K. S., Sikorski, J. A., Benesi, A. J., and Johnson, K. A. (1988) *J. Am. Chem. Soc.* **110**, 6577–6579
- ^d Appleyard, R. J., Shuttleworth, W. A., and Evans, J. N. S. (1994) *Biochemistry* **33**, 6812–6821
- ^e Stallings, W. C., Abdel-Meguid, S. S., Lim, L. W., Shieh, H.-S., Dayringer, H. E., Leimgruber, N. K., Stegeman, R. A., Anderson, K. S., Sikorski, J. A., Padgett, S. R., and Kishore, G. M. (1991) *Proc. Natl. Acad. Sci. U.S.A.* **88**, 5046–5050
- ^f Alber, D. G., Lauhon, C. T., Nyfeler, R., Fässler, A., and Bartlett, P. A. (1992) *J. Am. Chem. Soc.* **114**, 3535–3546
- ^g Walker, M. C., Jones, C. R., Somerville, R. L., and Sikorski, J. A. (1992) *J. Am. Chem. Soc.* **114**, 7601–7603
- ^h Ream, J. E., Yuen, H. K., Frazier, R. B., and Sikorski, J. A. (1992) *Biochemistry* **31**, 5528–5534
- ⁱ Sammons, R. D., Gruys, K. J., Anderson, K. S., Johnson, K. A., and Sikorski, J. A. (1995) *Biochemistry* **34**, 6433–6440
- ^j Samland, A. K., Amrhein, N., and Macheroux, P. (1999) *Biochemistry* **38**, 13162–13169
- ^k Skarzynski, T., Kim, D. H., Lees, W. J., Walsh, C. T., and Duncan, K. (1998) *Biochemistry* **37**, 2572–2577
- ^l Jia, Y., Lu, Z., Huang, K., Herzberg, O., and Dunaway-Mariano, D. (1999) *Biochemistry* **38**, 14165–14173
- ^m Lewis, J., Johnson, K. A., and Anderson, K. S. (1999) *Biochemistry* **38**, 7372–7379

from citrate is often returned to the molecule in Eq. 13-17, step *b*, but the position of reentry is different from that of removal. Apparently, after the initial proton removal, the *cis*-aconitate that is formed “flips over” so that it can be rehydrated (Eq. 13-17, step *c*) with participation of the same groups involved in dehydration but with formation of the new product.^{80,80a}

Aconitase contains iron in the form of an Fe_4S_4 iron-sulfur cluster (Fig. 13-4).^{81–83} However, the enzyme is usually isolated in a form that does not show its maximum activity until it has been incubated with ferrous iron (Fe^{2+}). The inactive form of the enzyme is thought to contain an Fe_3S_4 cluster (Chapter 16) which is converted back to the Fe_4S_4 cluster by the incubation

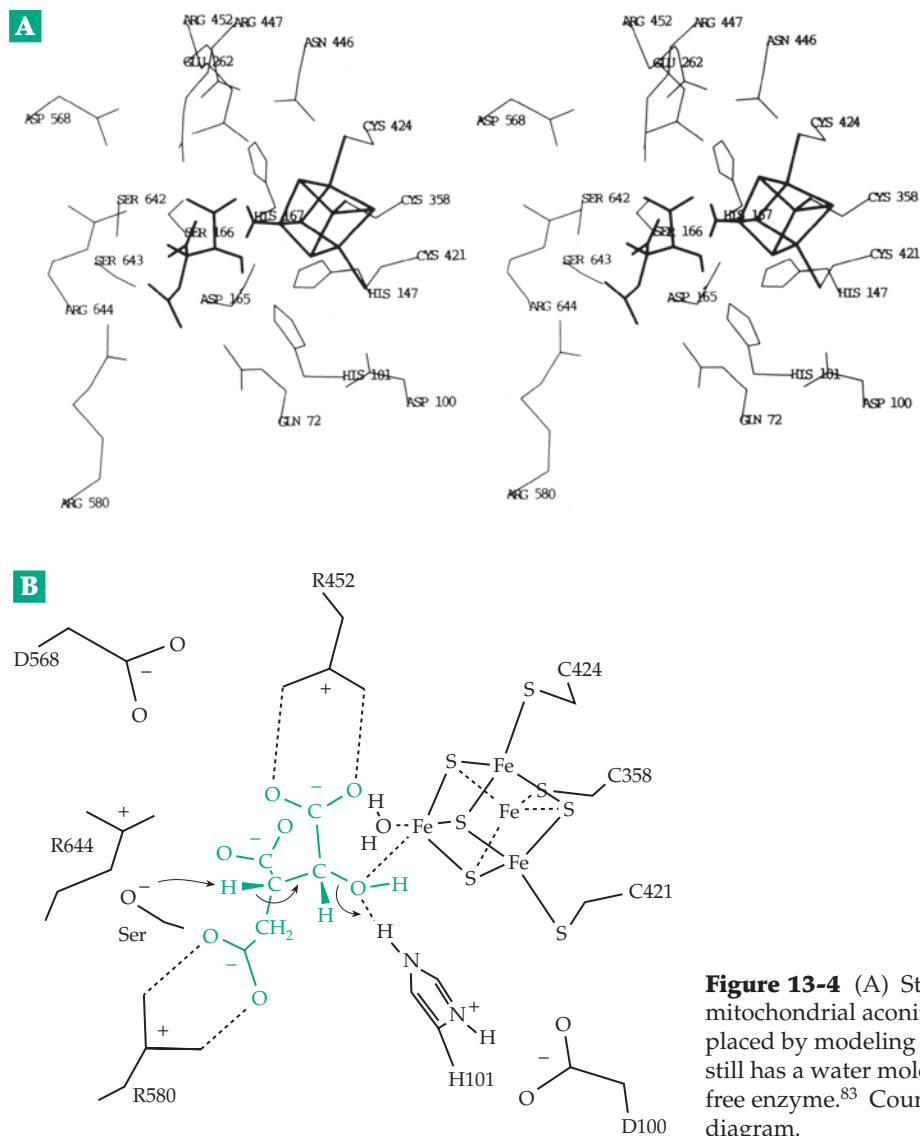
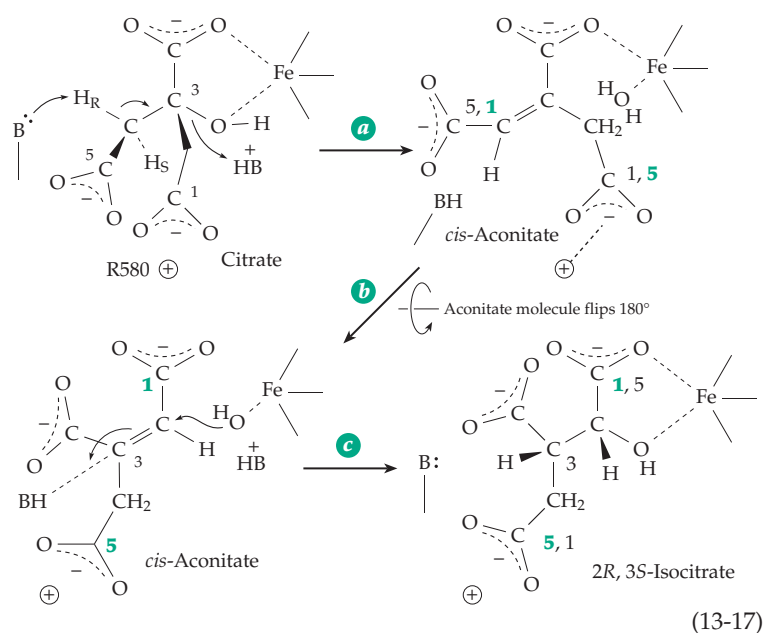


Figure 13-4 (A) Stereoscopic view of the active site of mitochondrial aconitase with a molecule of L-isocitrate placed by modeling next to the Fe_4S_4 cluster. This cluster still has a water molecule bound to one iron atom as in the free enzyme.⁸³ Courtesy of C. D. Stout. (B) Interpretive diagram.

with Fe²⁺.^{84–87} From Mössbauer spectroscopy it was deduced that the iron atom that is lost upon inactivation is the binding site for the –OH group of citrate or isocitrate and of an adjacent carboxylate.⁸⁷ Since iron can engage in oxidation–reduction processes, aconitase could act by a mechanism different from others discussed here. However, an Fe–OH group could act much as does the Zn–OH of carbonic anhydrase (Eq. 13-4). Any redox chemistry may be involved in control of the enzyme.

X-ray studies have confirmed the proximity of the Fe₄S₄ cluster to bound substrate analogs.^{83,88} The isocitrate shown in Fig. 13-4 was fitted into the active site by modeling. It shows the water molecule bound to the Fe₄S₄ cluster as observed for the free enzyme.⁸³ Notice that this cluster is held by three cysteine side chains (C358, C421, and C424) rather than the usual four (Chapter 16). The fourth iron atom is free to bind to water. When the enzyme acts on isocitrate this water must be displaced by the one generated from the substrate. The Fe probably acts as a polarizing electrophile that assists in the elimination but it is apparently the histidine–aspartate pair (H101–D100) that serves as the catalytic acid in generating the H₂O. What is the catalytic base? The only candidate seen in the structure is the Ser 642 –OH, dissociated to –O[–]. Can this be correct? Mutational analysis supports the essential role of this side chain.^{80a} The peptide NH and guanidinium groups of R644 appear to provide an “oxyanion hole” (Chapter 12) that stabilizes the negative charge.⁸⁶ Mutations also support the role of the His–Asp pair.^{88,89} Another His–Asp pair (H167–D165) is located directly behind the bound substrate and Fe–OH₂ in Fig. 13-4A and is apparently also essential.⁸⁸

Aconitase exists as both mitochondrial and cytosolic isoenzyme forms of similar structure. However, the cytosolic isoenzyme has a second function. In its apoenzyme form, which lacks the iron–sulfur cluster, it acts as the much-studied **iron regulatory factor**, or iron-responsive element binding protein (IRE-BP). This protein binds to a specific stem-loop structure in the messenger RNA for proteins involved in iron transport and storage (Chapter 28).^{86,90}

Other enzymes in the aconitase family include **isopropylmalate isomerase** and **homoaconitase** enzymes functioning in the chain elongation pathways to leucine and lysine, both of which are pictured in Fig. 17-18.⁹⁰ There are also iron–sulfur dehydratases, some of which may function by a mechanism similar to that of aconitase. Among these are the two fumarate hydratases, **fumarases A and B**, which are formed in place of fumarase C by cells of *E. coli* growing anaerobically.^{91,92} Also related may be bacterial L-serine and L-threonine dehydratases. These function without the coenzyme pyridoxal phosphate (Chapter 14) but contain iron–sulfur centers.^{93–95} A **lactyl-CoA**

dehydratase and some related iron-sulfur enzymes (pp. 861–862) may act via a mechanism related to that of vitamin B₁₂.

7. Addition to or Formation of Isolated Double Bonds

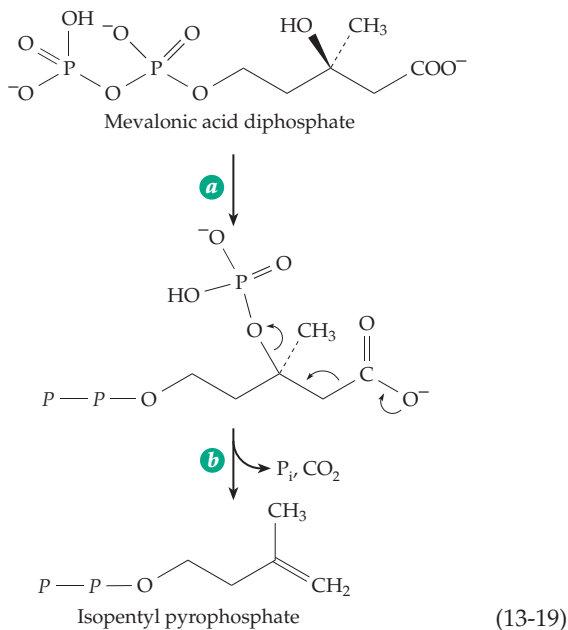
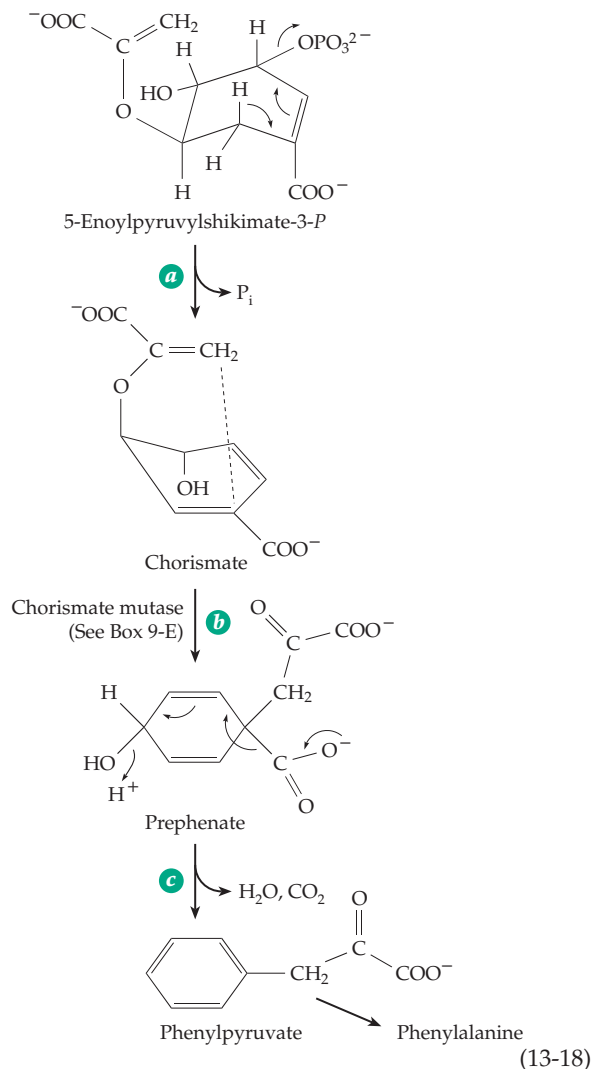
Only a few examples are known in which an enzyme induces addition to a double bond that is *not conjugated* with a carbonyl or carboxyl group. Pseudomonads have been observed to catalyze stereospecific hydration of oleic acid to D-10-hydroxystearate.⁹⁶ The addition is *anti* and the proton enters from the *re* face.

8. Conjugative and Decarboxylative Elimination Reactions

Elimination can occur if the electrophilic and nucleophilic groups to be removed are located not on adjacent carbon atoms but rather are separated from each other by a pair of atoms joined by a double bond. Such a conjugative elimination of phosphate is the last step in the biosynthesis of **chorismate** (Eq. 13-18, step *a*). Chorismate is converted to **prephenate** (see Box 9-E) which undergoes a conjugative and decarboxylative elimination (Eq. 13-18, step *c*) with loss of both water and CO₂ to form **phenylpyruvate**, the immediate biosynthetic precursor of phenylalanine. These reactions provide a good example of how *elimination reactions can be used to generate aromatic groups*. In fact, this is the usual method of synthesis of aromatic rings in nature.

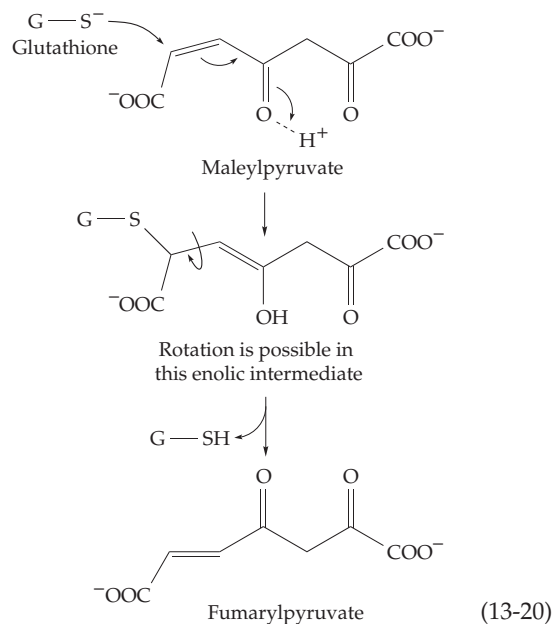
Notice the stereochemistry of Eq. 13-18, step *a*. Orbital interaction rules predict that if the elimination is a concerted process it should be syn. The observed anti elimination suggests a more complex mechanism involving participation of a nucleophilic group of the enzyme.⁹⁷

Elimination usually involves loss of a proton together with a nucleophilic group such as –OH, –NH₃⁺, phosphate, or pyrophosphate. However, as in Eq. 13-18, step *c*, electrophilic groups such as –COO[–] can replace the proton. Another example is the conversion of **mevalonic acid-5-pyrophosphate** to **isopentenyl pyrophosphate** (Eq. 13-19): This is a key reaction in the biosynthesis of isoprenoid compounds such as cholesterol and vitamin A (Chapter 22). The phosphate ester formed in step *a* is a probable intermediate and the reaction probably involves a carbocationic intermediate generated by the loss of phosphate prior to the decarboxylation.



9. Isomerization Assisted by Addition

An interesting use is made of addition to a double bond by glutathione-dependent **cis-trans isomerases**.⁷⁶ One of them converts **maleate** to fumarate with a turnover number of 300 s^{-1} . Similar enzymes, which participate in bacterial breakdown of aromatic compounds (Fig. 25-7), isomerize **maleylacetoacetate** and **maleylpyruvate** to the corresponding fumaryl derivatives (Eq. 13-20). The $-SH$ group of bound glutathione is thought to add to the double bond. Rotation can then occur in the enolic intermediate. Thiocyanate ion catalyzes the isomerization of maleic acid nonenzymatically, presumably by a similar mechanism.

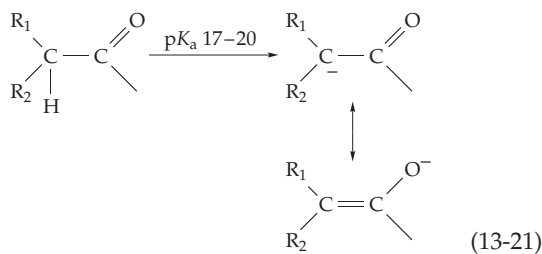


10. Reversibility of Addition and Elimination Reactions

Many addition and elimination reactions, e.g., the hydration of aldehydes and ketones, and reactions catalyzed by lyases such as fumarate hydratase are strictly reversible. However, biosynthetic sequences are often nearly irreversible because of the elimination of inorganic phosphate or pyrophosphate ions. Both of these ions occur in low concentrations within cells so that the reverse reaction does not tend to take place. In decarboxylative eliminations, carbon dioxide is produced and reversal becomes unlikely because of the high stability of CO_2 . Further irreversibility is introduced when the major product is an aromatic ring, as in the formation of phenylpyruvate.

B. Enolic Intermediates in Enzymatic Reactions

Enzymologists have freely proposed enolate anions, enols, and enamines as intermediates for many years. Such intermediates have been demonstrated for some nonenzymatic acid- or base-catalyzed reactions, but how can enzymes form enolates at pH 7 without the use of strong acids or bases? The microscopic pK_a value of an α -hydrogen in a ketone or aldehyde is about 17–20.^{72,98,99}



It will be similar for an acyl-CoA. However, for a carboxylic acid, protonated on oxygen, the pK_a will be much higher (~ 22 – 25) and for carboxylate anions even higher (~ 29 – 32).¹⁰⁰

1. Mandelate Racemase and Related Enzymes

The degradation of mandelic acid by the bacterium *Pseudomonas putida* (Chapter 25) is initiated by mandelate racemase, another $(\alpha/\beta)_8$ -barrel protein.¹⁰¹ X-ray structures of bound inhibitors together with modeling suggest that the side chain of Lys 264 is the catalytic base that abstracts the α -H from *S*-mandelate (Fig. 13-5) and that the catalytic pair of His 297 and Asp 270 acts as proton donor, or, in the reverse direction, as catalytic

base for deprotonation of *R*-mandelate.^{102–104} The enzyme is structurally a member of the enolase superfamily¹⁰⁵ and requires Mg^{2+} for activity. The pH dependence of the reaction velocity k_{cat} reveals two pK_a values in the ES complex; ~ 6.4 and ~ 10.0 . These are the same for *S*- and *R*- isomers. How can the pK_a values be assigned? Do they each belong in part to His 297 and in part to Lys 166? Do other adjacent groups also share? In the K166R mutant the lower pK_a is raised to 8.0. The D270N mutant has lost all but 0.01% of its catalytic activity.¹⁰⁴

The carboxylate group of the mandelate interacts with side chains of E317, K264, and a Mg^{2+} ion (Fig. 13-5). These may serve both to protonate the carboxylate and also to help stabilize an *aci* anion formed upon dissociation of the α -hydrogen (Eq. 13-22). Both pK_a values for mandelic acid and its enolic form and the equilibrium constants for enolization in water are known¹⁰⁷ and are given beside the arrows in Eq. 13-22. It is difficult to imagine how a base with a pK_a near neutrality could remove an α -proton with a pK_a of 22.0, which is 14 pK units away from the pK_a of the catalytic base. This corresponds to a thermodynamic barrier (Eq. 9-97) of $14 \times 5.7 = 80$ kJ/mol, making the reaction impossibly slow. In addition to this thermodynamic barrier the *rates* of dissociation of carbon acids are known to be slow, presumably because of the lack of hydrogen bond formation between the C–H proton and the catalytic base (Eq. 9-97). This *intrinsic* barrier (Chapter 9) for simple ketones has been estimated as 45 kJ/mol for a total barrier of ~ 125 kJ. However, the observed $\Delta G^\ddagger$ as estimated from Eq. 9-81 is only ~ 57 kJ/mol. The enzyme must catalyze the reaction by lowering the very high thermodynamic barrier and perhaps also by lowering the intrinsic barrier.^{72,108}

Protonation of the carboxylate greatly decreases the microscopic pK_a for loss of the α -hydrogen as a proton to form the enolic *aci* acid (Eq. 13-22). Double protonation, although depending upon a pK_a of ~ 8 ,

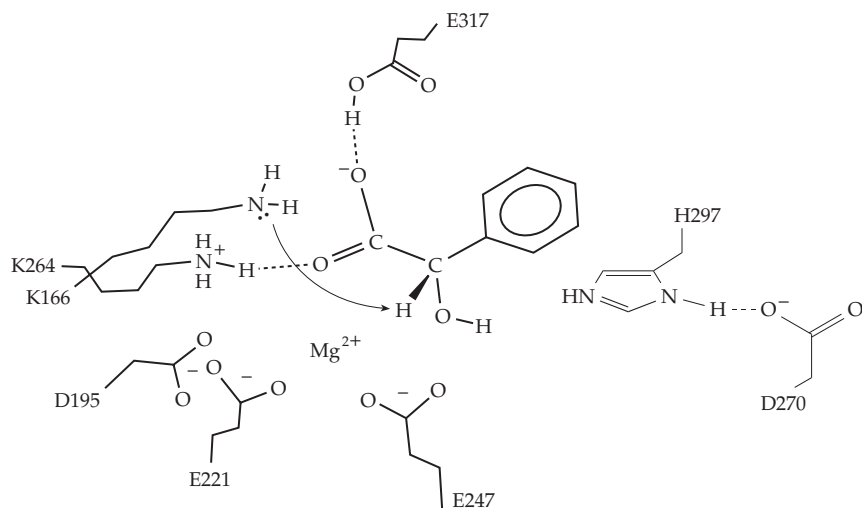
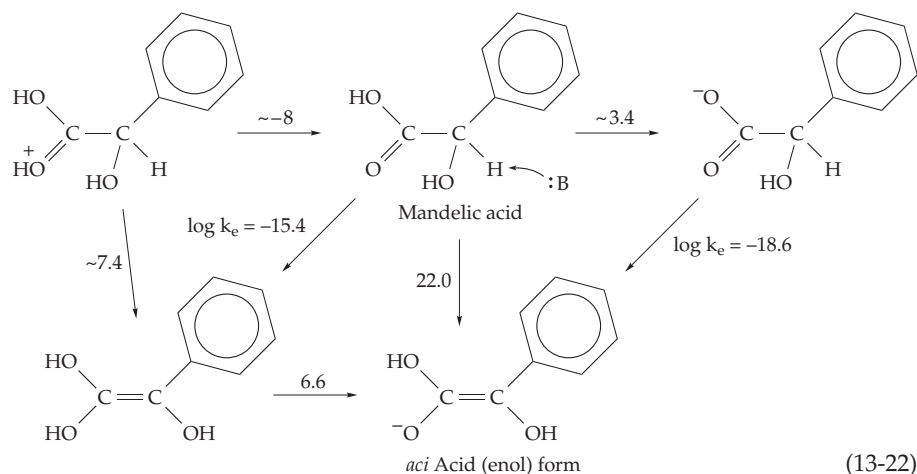


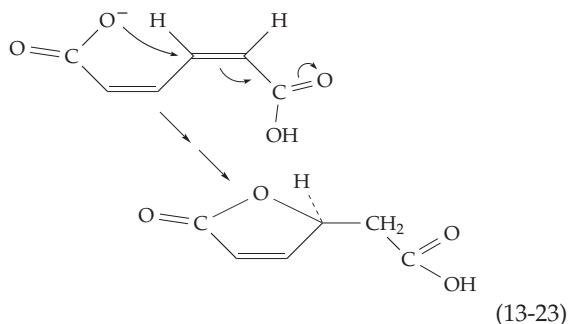
Figure 13-5 An *S*-mandelate ion in the active site of mandelate racemase. Only some of the polar groups surrounding the active site are shown. The enzyme has two catalytic acid–base groups. Lysine 166 is thought to deprotonate *S*-mandelate to form the *aci* anion, while His 297 deprotonates *R*-mandelate to form the same anion.¹⁰⁶



would reduce the pK_a of the α -hydrogen to ~ 7.4 , making it very easy to enolize¹⁰⁹ and solving the thermodynamic problem. However, what forces could keep the substrate molecule in this unlikely state of protonation? Gerlt and Gassman proposed formation of a strong, short hydrogen bond to a carboxylate oxygen.¹¹⁰ Formation of such a bond would lead to an increased positive charge on the α carbon, lowering the pK_a and therefore the thermodynamic barrier. It could also lower the intrinsic barrier,¹⁰⁸ for example, by permitting, to some extent, the formation of a C—H---N hydrogen bond in the transition state.¹¹¹ Polarization by the Mg^{2+} ion may also be involved. The very strong electrostatic forces within the active site may be sufficient to explain the formation of enolate anions or enols as intermediates in many different enzymes.^{108,112}

A number of other racemases and epimerases may function by similar mechanisms. While some amino acid racemases depend upon pyridoxal phosphate (Chapter 14), several others function without this coenzyme. These include racemases for aspartate,¹¹³ glutamate,^{114–115a} proline, phenylalanine,¹¹⁶ and diamino-pimelate epimerase.¹¹⁷ Some spiders are able to interconvert D and L forms of amino acid residues in intact polypeptide chains.^{118,119}

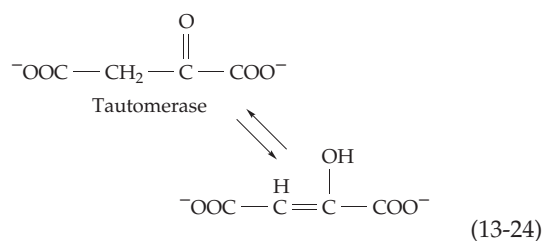
Another enzyme of the mandelate pathway of degradation of aromatic rings (Fig. 25-8) is the **cis,cis-muconate lactonizing enzyme** which catalyzes the reaction of Eq. 13-23. It has a three-dimensional struc-



ture almost identical to that of mandelate racemase but has incorporated an additional feature that allows formation, by addition, of the intermediate *aci*-acid.^{101,105}

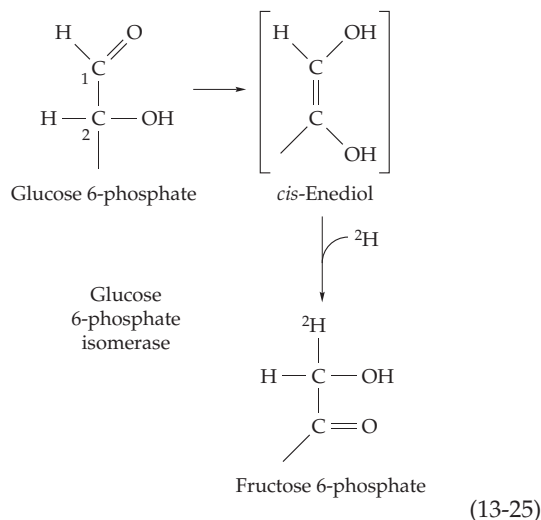
2. Isomerases

The isomerases that catalyze the simplest reactions are **tautomerases** that promote the oxo-enol (keto-enol) transformation. The widely distributed oxaloacetate tautomerase (Eq. 13-24) is especially active in animal tissues.^{97,120} Oxaloacetate exists to a substantial extent in the enolic form: at 38°, $\sim 6\%$ enol, 13% oxo, and $\sim 81\%$ covalent hydrate.^{120,121} A mammalian **phenylpyruvate tautomerase** has also been investigated.¹²²



The oxidation of one functional group of a molecule by an adjacent group in the same molecule is a feature of many metabolic sequences. In most cases an enolic intermediate, formed either from a ketone as in Eq. 13-25 or by a dehydration reaction (see Eq. 13-32), is postulated.

Aldose-ketose interconversions. A metabolically important group of enzymes catalyze the interconversion of aldose sugars with the corresponding 2-ketoses (Table 10-1, reaction 4C). Several sugar phosphates undergo rapid isomerization. **Glucose 6-phosphate isomerase** (Eq. 13-25) appears to function in all cells with a high efficiency.^{123–125a} The 132 kDa dimeric protein from muscle converts glucose 6-phosphate to



fructose 6-phosphate with a turnover number of $\sim 10^3 \text{ s}^{-1}$. Hereditary defects in this enzyme cause a variety of problems that range from mild to very severe.¹²⁶

Mannose 6-phosphate isomerase also forms fructose 6-phosphate, while **ribose 5-phosphate isomerase**¹²⁷ interconverts the 5-phosphates of D-ribose and D-xylulose in the pentose phosphate pathways (Chapter 17). Other enzymes, most often metalloenzymes, catalyze the isomerization of free sugars.

These enzymes vary widely in secondary and tertiary structure.^{127a} Mannose-6-phosphate isomerase is a 45 kDa Zn^{2+} -containing monomer. The larger 65 kDa **L-fucose isomerase**, which also acts on D-arabinose, is a hexameric Mn^{2+} -dependent enzyme.^{127a} **L-Arabinose isomerase** of *E. coli*, which interconverts arabinose and L-ribulose, is a hexamer of 60-kDa subunits¹²⁸ while the **D-xylose isomerase** of *Streptomyces* is a tetramer of 43-kDa subunits.¹²⁹ The nonenzymatic counterpart of the isomerization catalyzed by the enzyme is the base-catalyzed **Lobry deBruyn–Alberda van Ekenstein transformation** (Eq. 13-25).¹³⁰

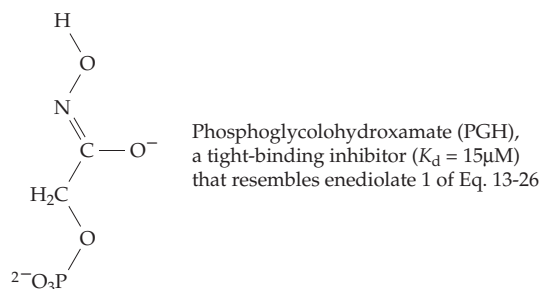
In 1895, Emil Fischer proposed an enediol intermediate for this isomerization. As would be expected, the enzyme-catalyzed isomerization of glucose-6-phosphate in $^2\text{H}_2\text{O}$ is accompanied by incorporation of deuterium into the product fructose 6-phosphate at C-1. In the reverse reaction ^2H -containing fructose 6-phosphate was found to react at only 45% of the rate of the ^1H -containing compound. Thus, the primary deuterium isotope effect expected for a rate-limiting cleavage of the C–H bond was observed (see Chapter 12, Section B,3).

When fructose 6-phosphate containing both ^2H and ^{14}C in the 1 position was isomerized in the presence of a large amount of nonlabeled fructose 6-phosphate, the product glucose 6-phosphate contained not only ^{14}C but also ^2H , and the distribution indicated that the ^2H had been transferred from the C-1 position into the C-2 position.¹²³ It was concluded that in over half the

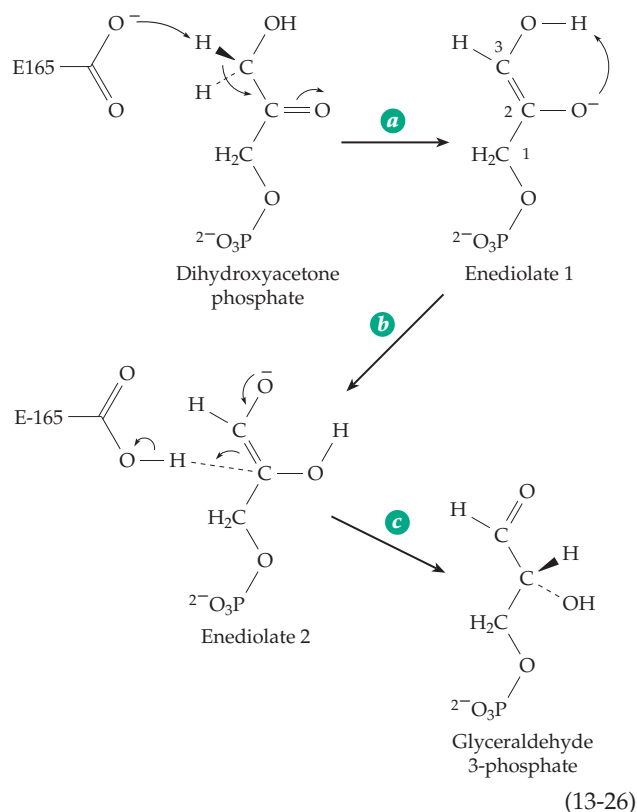
turnovers of the enzyme, ^2H removed from C-1 is put back on the same molecule at C-2. This intramolecular transfer of a proton suggests a syn transfer, the proton being removed and put back on the same side of the molecule. The carrier of the proton may be a histidine side chain.¹³¹ This result, together with the known configuration of glucose at C-2, indicates that the intermediate is the *cis*-enediol and that addition of a proton at either C-1 or C-2 of the enediol is to the *re* face. However, mannose-6-phosphate isomerase catalyzes addition to the *si* face.

Catalysis of ring opening by isomerases. Glucose-6-phosphate isomerase catalyzes a second reaction, namely, the opening of the ring of the α -anomer of glucose 6-phosphate (one-half of the mutarotation reaction; Eq. 10-88). Noltmann suggested a concerted acid–base catalysis by two side chain groups as is indicated in Eq. 9-90. An NMR study showed that the isomerization of the α anomer occurs at least ten times faster than that of the β . Thus, β -glucose 6-phosphate is first converted to α -glucose 6-phosphate before it can be isomerized to fructose 6-phosphate.¹³²

Triose phosphate isomerase. This dimeric 53-kDa enzyme interconverts the 3-phosphate esters of glyceraldehyde and dihydroxyacetone and is the fastest enzyme participating in glycolysis. Its molecular activity at 25° is $\sim 2800 \text{ s}^{-1}$ in the direction shown in Eq. 13-26 and $\sim 250 \text{ s}^{-1}$ in the reverse direction (the predominant direction in metabolism)¹³³ and is thought to operate at the diffusion-controlled limit (Chapter 9).¹³³ Each of the identical subunits consists of a striking $(\alpha/\beta)_8$ barrel (Fig. 13-6) with an active site at the carboxyl ends of the β strands.^{134–137} Structures of the enzyme containing bound inhibitors such as phosphoglycolohydroxamate have also been determined (Fig. 13-6).¹³⁸ Triose phosphate isomerase is also one of the most investigated of all enzymes. Not only are its catalytic properties unusual but also there are known defects in the human enzyme¹³⁷ and it is also a potential target for antitrypanosomal drugs.¹³⁹



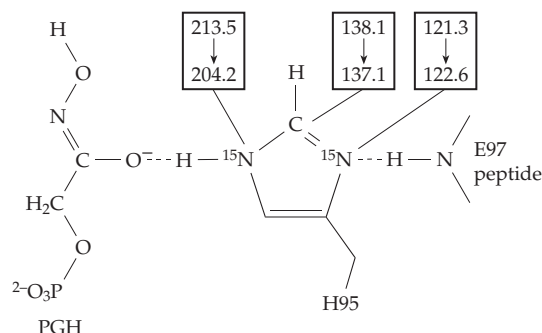
Although its high catalytic activity might favor intramolecular transfer of the proton removed by triose phosphate isomerase, little such transfer has



been observed.¹⁴⁰ This suggested that a relatively weak base such as a carboxylate group might serve as the proton acceptor at the C-3 position of dihydroxyacetone phosphate. Covalent labeling experiments and later X-ray studies have implicated Glu 165, whose carboxylate group is thought to remove the *pro-R* hydrogen atom from the hydroxymethyl group of dihydroxyacetone phosphate as indicated in Eq. 13-26. When this residue is replaced by Asp, most activity is lost.¹⁴¹ Kinetic studies suggest that this carboxyl group has a pK_a of ~ 3.9 in the free enzyme. However, k_{cat} is independent of pH up to pH 10.¹⁴⁰ The pK_a of the phosphate also affects the binding (K_m), with only the phosphate dianion being a substrate. Study of the infrared spec-

trum of dihydroxyacetone bound to the enzyme revealed a shift of the carbonyl bands at 1733 cm^{-1} by about 20 cm^{-1} to $\sim 1713\text{ cm}^{-1}$. This might indicate a polarization and stretching of the carbonyl group by a positively charged histidine or lysine side chain of the protein.¹⁴² The His 95 side chain is appropriately placed (Fig. 13-6) to function in this way¹⁴³ and its replacement by glutamine decreases catalytic activity by a factor of 400.¹⁴⁴ The $-\text{NH}_3^+$ group of lysine 12, another essential residue, is apparently needed for substrate binding.^{136,145} After product is released, an isomerization within the enzyme is usually required to prepare it for acceptance of a new substrate. This may involve movement of protons between side chain groups or a conformational change or both. In the case of triose phosphate isomerase, a very rapid isomerization requiring about $\sim 1\text{ ms}$ has been detected.¹⁴⁶

To investigate further the function of His 95, Lodi and Knowles recorded the ^{13}C and ^{15}N NMR spectra of the three histidine rings¹⁴⁸ both in unligated enzyme and in the phosphoglycolohydroxamate (PGH) complex. The results were a surprise. The His 95 resonances

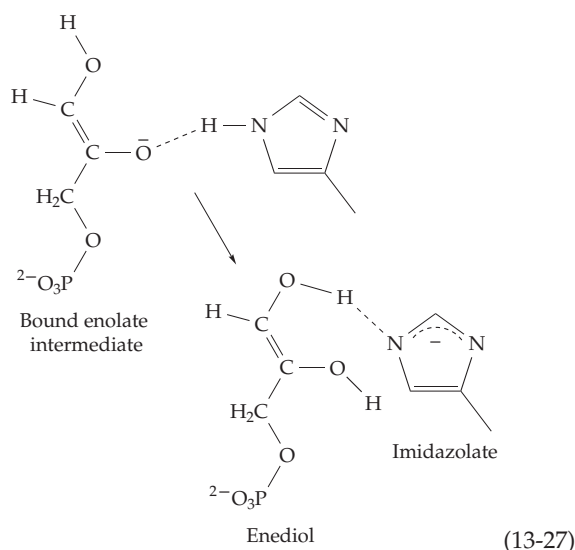


did not change at all when the pH was changed from below 5 to 9.9, and the chemical shift values (shown above) indicate clearly that the ring is **unprotonated** in both free enzyme and in the complex. The key chemical shift values are shown on the diagram to the



Figure 13-6 Stereoscopic view into the active site of triose phosphate isomerase showing side chains of some charged residues; PGH, a molecule of bound phosphoglycolohydroxamate, an analog of the substrate enolate.¹³⁸ The peptide backbone, as an alpha-carbon plot, is shown in light lines.¹⁴⁷ The $(\alpha/\beta)_8$ -barrel structure is often called a TIM barrel because of its discovery in this enzyme. Courtesy of M. Karplus.

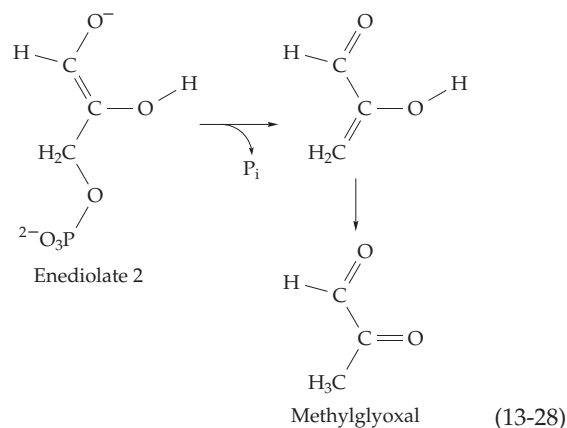
left; the arrows indicate the changes upon formation of the PGH complex. These results suggest that H95 is never protonated and that if it acts as the catalytic acid it does so by dissociating to an imidazolate ion (Eq. 13-27).¹⁴⁸



The imidazole would donate a proton to the enolate ion and then remove a proton from the resulting enediol. Although the proposed chemistry is unusual, it is argued that the high pK_a of the neutral imidazole acting as a proton donor would be matched with the high pK_a of the enol, permitting rapid reaction.¹⁴⁸ Nevertheless, two other possibilities exist. Theoretical calculations support the idea that the proton transfer between the two oxygens in the enediolate may occur *without catalysis by a proton donor*¹⁴⁹ as indicated in step *b* of Eq. 13-26. Another possibility is that a transient protonation of H95 by the adjacent E97 carboxyl group occurs and allows the histidine to participate in the proton transfer. Study of kinetic isotope effects has indicated coupled motion of protons and proton tunneling.¹⁴³

Another detail should be mentioned. The active site of triose phosphate isomerase is formed by a series of loops connecting the α helices and β strands of the barrel. One of those loops, consisting of residues 167–176, folds over the active site after the substrate is bound to form a hinged lid that helps to hold the substrate in the correct orientation for reaction.^{150–152} When the lid, which can be seen in Fig. 13-6, closes, the peptide NH of G171 forms a hydrogen bond to a phosphate oxygen atom of the substrate. This is only one of many known enzymes with deeply buried active sites that close in some similar fashion before a rapid reaction occurs.

Although enzymes tend to be extremely specific they are not always completely able to avoid side reactions. Triose phosphate isomerase releases small amounts of **methylglyoxal** (Eq. 13-28), presumably as

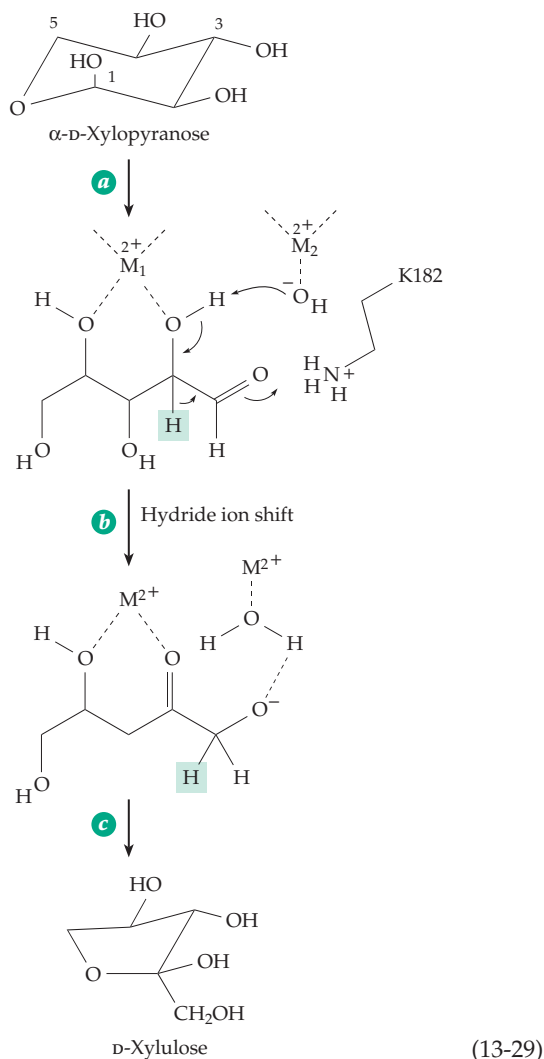


a result of elimination of phosphate from enediolate 2 of Eq. 13-26. Deletion of four residues from the hinged lid produced an enzyme in which this side reaction was increased 5.5-fold.¹⁴⁶

Xylose isomerase and the hydride shift

mechanism. This bacterial enzyme, which isomerizes D-xylose to D-xylulose, has an $(\alpha/\beta)_8$ barrel similar to that of triose phosphate isomerase. It is of industrial importance because it also catalyzes isomerization of D-glucose to the sweeter sugar D-fructose, a reaction used in preparation of high-fructose syrup. For this purpose the isomerase is often immobilized on an insoluble matrix such as diethylaminoethyl cellulose.¹⁵³ This enzyme, as well as other isomerases that act on free sugars, requires a metal ion such as Mg^{2+} , Co^{2+} , or Mn^{2+} . The three-dimensional structures, solved in several different laboratories,^{129,154–158} show that there are two metal ions (ordinarily Mg^{2+}) about 0.5 nm apart and held by an array of glutamate and aspartate side chains. One glutamate carboxylate forms short ionic bonds to both metal ions and an essential histidine is also coordinated with metal ion 2. The three-dimensional structure is superficially similar to that of triose phosphate isomerase but there are major differences in properties. Xylose isomerase requires metal ions, acts on unphosphorylated non-ionic substrates, and does not catalyze detectable exchange of protons with the solvent. Furthermore, *X-ray structures do not show the presence of any catalytic base that could initiate formation of an intermediate enolate ion*. These facts suggested an alternative isomerization mechanism, one involving a hydride ion shift and well-known from studies of the Cannizzaro reaction.¹⁵⁹ Because the non-ionic substrates and inhibitors bind weakly it has been difficult to obtain a clear picture of events in the active site. The substrate is the α -anomer of D-xylopyranose¹⁵⁵ and the enzyme catalyzes the opening of the sugar ring (step *a*, Eq. 13-29).¹⁵⁷ The details of this process are not clear but acid-base catalysis as in Eq. 9-87 is probable.

There is a consensus^{156–158} that the open form of

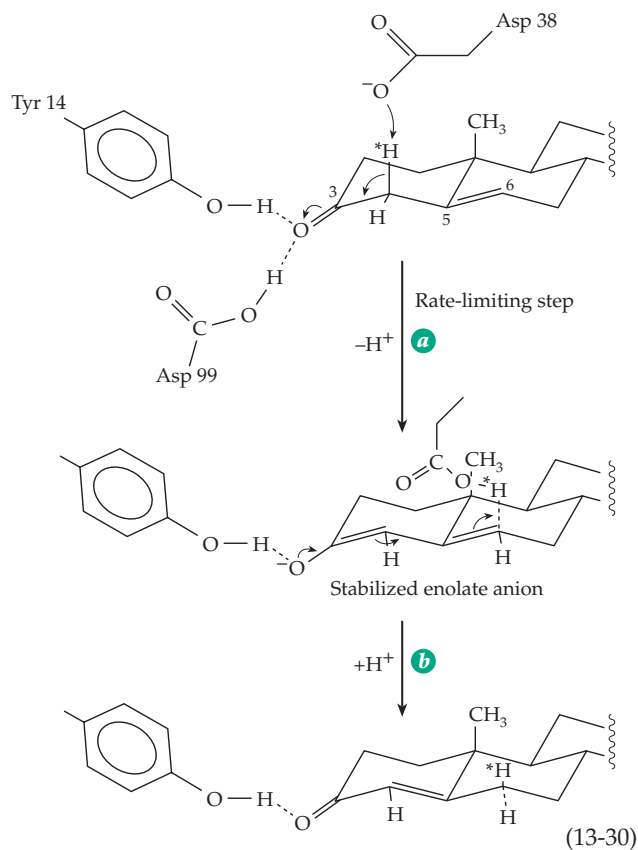


the sugar is bound to one of the two metal ions as shown in Eq. 13-29 and that a bound hydroxide ion on the second metal provides the catalytic base to remove a proton from the -OH at position 2. The isomerization occurs with the shaded hydrogen atom being shifted as H^- to the carbon atom of the carbonyl group. During the reaction metal ion 2 moves apart from metal ion 1 by $\sim 0.09 \text{ nm}$,^{129,157,158} another fact that has made analysis of X-ray data difficult. Protonation of the alkoxide ion at C-2 is followed by ring closure (step *c*, Eq. 13-29).

3-Oxosteroid isomerases. Cholesterol serves in the animal body as a precursor of all of the steroid hormones, including the 3-oxosteroids progesterone and testosterone (Chapter 22). The 3-hydroxyl group of cholesterol is first oxidized to an oxo group. This is followed by an essentially irreversible migration of the double bond in the 5,6-position into conjugation with the carbonyl group (Eq. 13-30) catalyzed by a 3-oxo- Δ^5 -steroid isomerase. The small 125-residue enzyme from the bacterium *Pseudomonas testosteroni* has been studied by a great variety of methods.^{160–164} The subunits

associate as dimers or higher oligomers. The enzyme has a high content of nonpolar amino acid residues and is soluble in high concentrations of ethanol. This property is compatible with the location of the eukaryotic enzyme in the endoplasmic reticulum. The substrate binding site is a hydrophobic cavity.

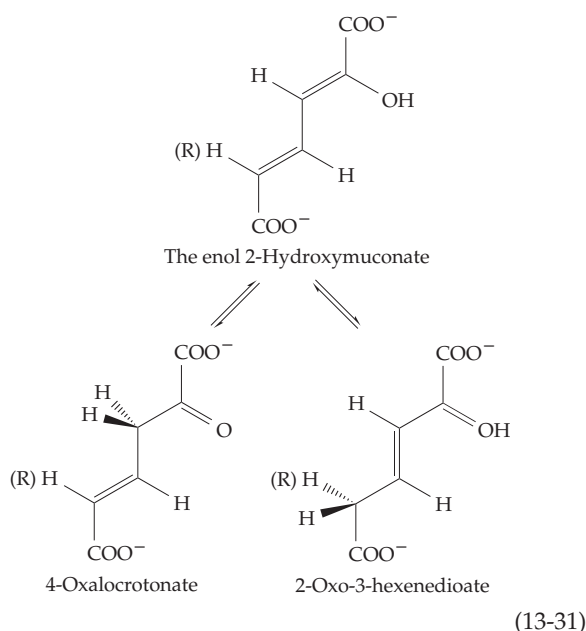
Oxosteroid isomerase has a remarkably high molecular activity ($\sim 0.75 \times 10^5 \text{ s}^{-1}$). Substrates containing ^2H in the 4 position react only one-fourth as rapidly as normal substrates. The large isotope effect suggested that cleavage of the C-H bond to form an enzyme-stabilized enolate anion is rate limiting (Eq. 13-30, step *a*).¹⁶⁰ The proton in the axial position at C-4 is removed preferentially but without complete stereospecificity.¹⁶⁵ The abstracted proton must be carried by a group on the enzyme and returned to the 6 position of the substrate, again in an axial orientation (Eq. 13-30, step *b*), a suprafacial transfer. Studies of the binding of spin-labeled substrate analogs, molecular modeling, and directed mutation indicate that Asp 38 is this proton carrier.^{164,166} Little exchange of the proton with solvent is observed, presumably because of the extreme rapidity of the isomerase action. However, competitive inhibitors such as nortestosterone, whose double-bond arrangement is that of the product in Eq. 13-30, undergo exchange of one of the hydrogens at C-4 with the medium. The ultraviolet absorption band of the inhibitor at 248 nm is shifted to 258 nm upon combination with the enzyme, presumably as a result of formation of the



enolate anion. Tyrosine 14 has been identified as the catalytic acid, able to stabilize the enolate anion as depicted in Eq. 13-30.^{161,162} A high-resolution NMR structure¹⁶⁷ and X-ray crystal structures^{167a,b} have revealed an adjacent Asp 99 side chain that may assist.^{167c}

Enzymatic isomerization of *cis*-aconitate to *trans*-aconitate apparently also involves proton abstraction,¹⁶⁵ with resonance in the anion extending into the carboxylic acid group. Its mechanism may be directly related to that of the oxosteroid isomerase. However, there are other 1,3-proton shifts in which neither a carbonyl nor a carboxyl group is present in the substrate (Eqs. 13-55, 13-56).

4-Oxalocrotonate tautomerase. This bacterial enzyme, which functions in the degradation of toluene (Chapter 25), is actually an isomerase. It catalyzes rapid interconversion of an unconjugated unsaturated α -oxoacid such as 4-oxalocrotonate with an intermediate enol (which may leave the enzyme) and the isomeric conjugated oxoacid (Eq. 13-31).¹⁶⁸⁻¹⁷⁰ A related 5-carboxymethyl-2-hydroxymuconate isomerase



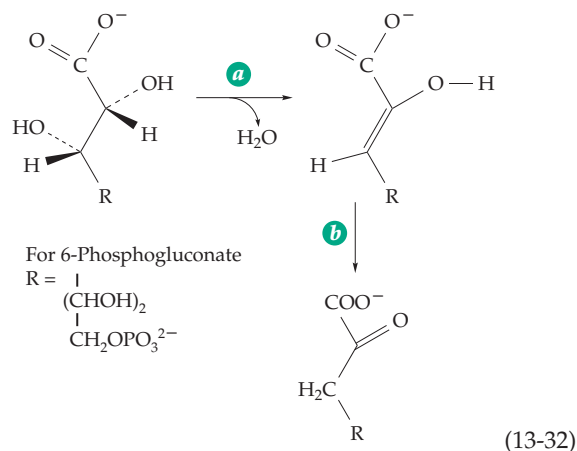
catalyzes the same reaction when $R = -CH_2COO^-$ in Eq. 13-31. It is an unusually small enzyme consisting of only 62 amino acid residues. The pH dependence reveals pK_a values of 6.2 and 9.0 in the free enzyme and 7.7 and 8.5 in the ES complex.¹⁷¹

The pK_a of 6.2 has been associated with the amino-terminal proline 1.¹⁷² 4-Oxalocrotonate tautomerase is one of a small group of enzymes that have been synthesized nonenzymatically with both L amino acids and as a mirror image constructed with D amino acids.¹⁷³

3. Internal Oxidation–Reduction by Dehydration of Dihydroxyacids

When a carboxylic acid contains hydroxyl groups in both the α and β positions, dehydration leads to formation of an enol that can tautomerize to 2-oxo-3-deoxy derivatives of the original acid (Eq. 13-32). Thus, phosphogluconate dehydratase yields 2-oxo-3-deoxyphosphogluconate as the product. When the reaction is carried out in 2H_2O the 2H is incorporated with a random configuration at C-3, indicating that the enzyme catalyzes only the dehydration and that the tautomerization of the enol to the ketone is nonenzymatic.

The reaction of Eq. 13-32 initiates a unique pathway of sugar breakdown, the Entner–Doudoroff pathway (Eq. 17-18), in certain organisms. The 6-phosphogluconate is formed by oxidation of the aldehyde group of glucose 6-phosphate. This pattern of oxidation of a sugar to an aldonic acid followed by dehydration according to Eq. 13-32 occurs frequently in metabolism. A related reaction is the dehydration of 2-phosphoglycerate by enolase (Eq. 13-15). In this case the product is phosphoenolpyruvate, a stabilized form of the enolic intermediate of Eq. 13-32 (when $R = H$).



4. Formation and Metabolism of Methylglyoxal (Pyruvaldehyde)

The rather toxic methylglyoxal is formed in many organisms and within human tissues.¹⁷⁴ It arises in part as a side reaction of triose phosphate isomerase (Eq. 13-28) and also from oxidation of acetone (Eq. 17-7) or aminoacetone, a metabolite of threonine (Chapter 24).¹⁷⁵ In addition, yeast and some bacteria, including *E. coli*, have a **methylglyoxal synthase** that converts dihydroxyacetone to methylglyoxal, apparently using a mechanism similar to that of triose phosphate isomerase. It presumably forms enediolate 2 of Eq. 13-26, which eliminates inorganic phosphate to yield methyl-

glyoxal as in Eq. 13-28.^{176,176a} Methylglyoxal is converted to D-lactate by the two-enzyme **glyoxalase** system (Eq. 13-33).¹⁷⁷ The combined action of methylglyoxal synthase and the glyoxalases provides a bypass to the usual glycolysis pathway (Fig. 10-3). Although it does not provide energy to the cell, it releases inorganic phosphate from sugar phosphates that may accumulate under conditions of low phosphate because the free P_i concentration is too low to support the glyceraldehyde 3-phosphate dehydrogenase reaction (step *a* of Fig. 10-3).

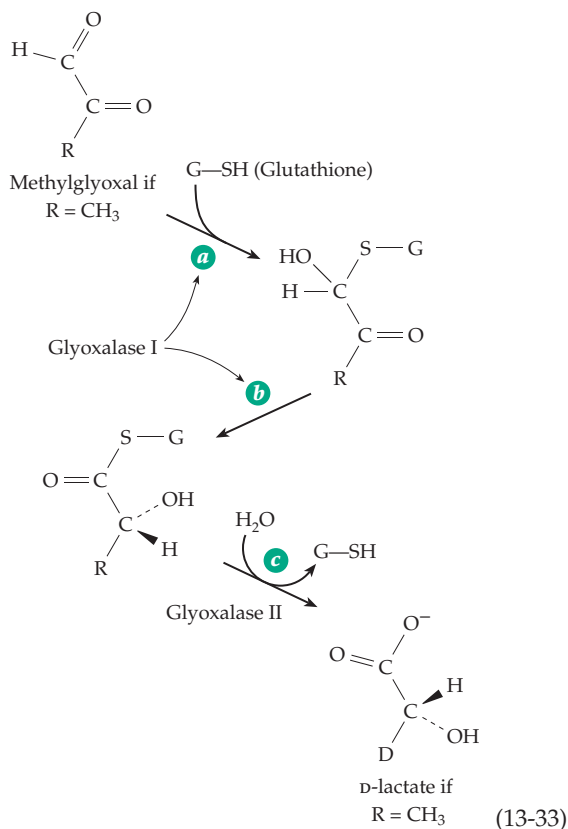
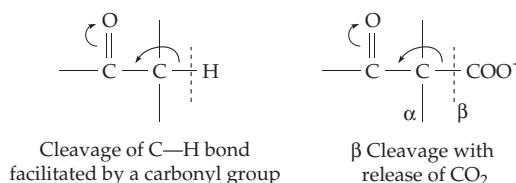
Glyoxalase acts not only on methylglyoxal but also on other α -oxo-aldehydes. It is thought to be an important enzyme system that protects cells against these potentially dangerous metabolites. Glyoxalase consists of a pair of enzymes, **glyoxalase I** and **glyoxalase II**, which catalyze the reactions of Eq. 13-33. Each subunit of the 183-residue human glyoxalase I contains one tightly bound Zn^{2+} ion.¹⁷⁸ However, the *E. coli* enzyme^{178a} is inactive with Zn^{2+} and maximally active with Ni^{2+} . The enzyme requires glutathione as a co-factor, and in step *a* of the reaction (Eq. 13-33) the glutathione adds to form a thiohemiacetal¹⁷⁹ which is then isomerized in step *b*. During this step retention of the abstracted proton is so complete that it was earlier thought to function by an intramolecular hydride ion shift as in xylose isomerase. More recent evidence favors an enolate anion intermediate. However, the three-dimensional structure is not related to that of other isomerases and the exact mechanism remains

uncertain. Glyoxylase II, which catalyzes step *c* of Eq. 13-33, is an esterase.^{175,180}

C. Beta Cleavage and Condensation

In Chapter 12 and in the preceding sections of this chapter we examined displacement and addition reactions involving nucleophilic centers on O, N, or S. Bonds from carbon to these atoms can usually be broken easily by acidic or basic catalysis. The breaking and making of C–C bonds does not occur as readily and the “carbon skeletons” of organic molecules often stick together tenaciously. Yet living cells must both form and destroy the many complex, branched carbon compounds found within them.

A major mechanistic problem in cleavage or formation of carbon–carbon bonds is the creation of a nucleophilic center on a carbon atom. The problem is most often solved by using the *activating influence of a carbonyl group to generate a resonance-stabilized enolate anion*. Just as the presence of a carbonyl group facilitates cleavage of an adjacent C–H bond, so it can also assist the cleavage of a C–C bond. The best known reactions of this type are the **aldol cleavage** and the

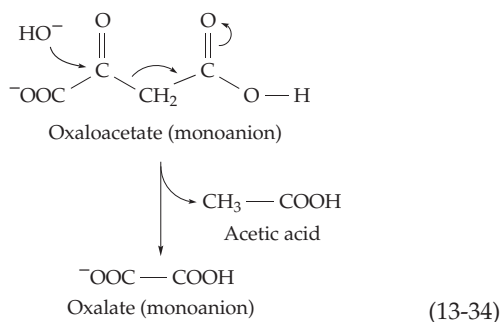


decarboxylation of β -oxo acids. The latter has been referred to as β decarboxylation and its reverse as β carboxylation. In this book these terms have been extended to include other reactions by which bonds between the α and β carbon atoms of a carbonyl compound are broken or formed, and these will be referred to as **β cleavage and β condensation**.

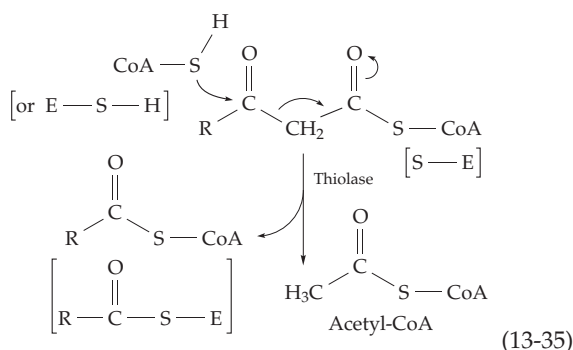
The β *condensation* reactions consist of displacement or addition reactions in which an enzyme-bound enolate anion acts as the nucleophile. We can group these condensation reactions into three categories as indicated by reaction types 5A, 5B, and 5C of Table 10-1.

1. Displacement on a Carbonyl Group

A β -oxo acid, with proper catalysis, is susceptible to hydrolysis by attack of water on the carbonyl group. An example is the reaction catalyzed by **oxaloacetate acetylhydrolase** which has been isolated from *Aspergillus niger* (Eq. 13-34). A related cleavage is catalyzed by ribulose biphosphate carboxylase (see Eq. 13-48).



The **thiolases**¹⁸¹ are lyases that cleave β -oxoacyl derivatives of CoA by displacement with a thiol group of another CoA molecule (Eq. 13-35). This is the chain cleavage step in the β oxidation sequence by which fatty acid chains are degraded (Fig. 10-4). Biosynthetic thiolases catalyze the condensation of two molecules of acetyl-CoA to form acetoacetyl-CoA, (see Eq. 17-5), a precursor to cholesterol and related compounds and to poly- β -hydroxybutyrate (Box 21-D). Because acetyl-CoA is a thioester, the reaction is usually described as a **Claisen condensation**. A related reaction, involving decarboxylation of malonyl-CoA, is a step in fatty acid synthesis. Since the thiolases are inhibited by $-\text{SH}$ reagents it has been suggested that a thiol group in the enzyme reacts initially with the β -carbonyl group as in Eq. 13-35 to give an enzyme-bound *S*-acyl intermediate. The acyl group is then transferred to CoA in a second step.



A very similar reaction is catalyzed by 3-hydroxy-3-methylglutaryl-CoA lyase (HMG-CoA lyase), which functions in the formation of acetoacetate in the human body (Eq. 17-5, step *c*) and also in the catabolism of leucine (Fig. 24-18)^{182,183} and in the synthesis of **3-hydroxy-3-methylglutaryl-CoA**, the precursor of cholesterol (Eq. 17-5, step *b*)^{183a}

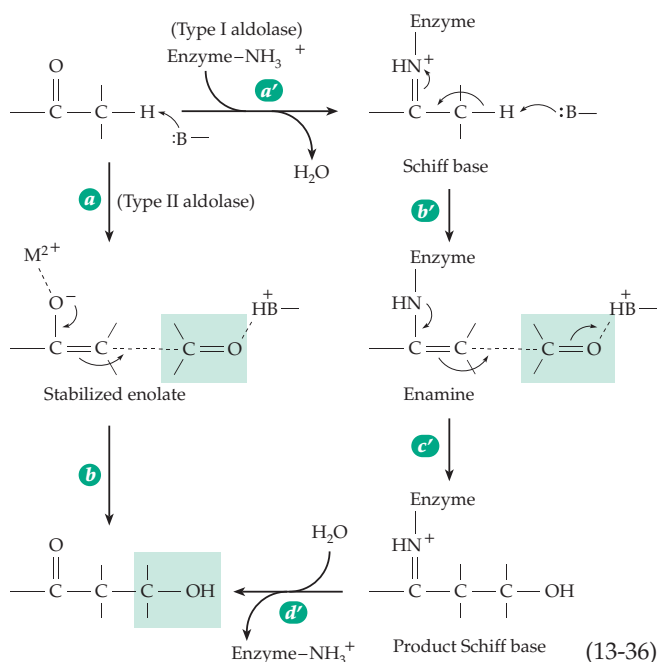
2. Addition of an Enolate Anion to a Carbonyl Group or an Imine; Aldolases

The **aldol condensation** (Eq. 13-36), which is also illustrated in Box 10-D, is one of the more common

reactions by which C–C bonds are formed^{183b} and, in the reverse reaction, cleaved in metabolism. Aldolases are classified into two major types. The type II aldolases are metal-ion dependent, the metal ion stabilizing the intermediate enolate ion (Eq. 13-36, steps *a* and *b*). Type I aldolases, which include the most studied mammalian enzymes, have a more complex mechanism involving intermediate Schiff base forms (Eq. 13-36, steps *a'*, *b'*, *c'*, *d'*).¹⁸⁴ The best known members of this group are the **fructose biphosphate aldolases** (often referred to simply as aldolases), which cleave fructose-1,6-*P*₂ during glycolysis (Fig. 10-2, step *e*).

These enzymes have been found in all plant and animal tissues examined and are absent only from a few specialized bacteria. Three closely related isoenzymes are found in vertebrates.^{185,186} The much studied rabbit muscle aldolase A is a 158-kDa protein tetramer of identical peptide chains.^{186,187} Aldolase B, which is lacking in hereditary fructose intolerance, predominates in liver and isoenzyme C in brain.¹⁸⁵

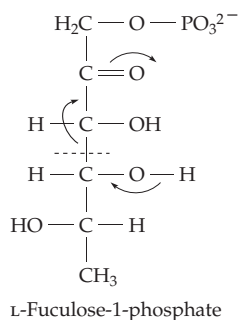
Treatment with sodium borohydride of the enzyme-substrate complex of aldolase A and dihydroxyacetone phosphate leads to formation of a covalent linkage between the protein and substrate. This and other evidence suggested a Schiff base intermediate (Eq. 13-36). When ¹⁴C-containing substrate was used, the borohydride reduction (Eq. 3-34) labeled a lysine side chain in the active site. The radioactive label was followed through the sequence determination and was found on Lys 229 in the chain of 363 amino acids.^{186,188–188b} The enzyme is another (α/β)₈-barrel protein and the side chain of Lys 229 projects into the interior of the barrel which opens at the C-terminal ends of the strands. The conjugate base form of another lysine, Lys 146, may represent the basic group B in Eq. 13-36,



step *b*. Another possibility is that the adjacent phosphate group of the substrate acts as the acid–base catalyst for this step.¹⁸⁹ Aldolase A has been altered by mutations into a monomeric form that retains high catalytic activity,¹⁹⁰ something that has not often been accomplished for oligomeric enzymes.

The **type II aldolases** are not inactivated by sodium borohydride in the presence of substrate. A probable function of the essential metal ion is to polarize the carbonyl group as indicated in Eq. 13-36. In both yeast and *E. coli* aldolase the Zn^{2+} is held by 3 imidazole groups.^{191,191a} An arginine side chain is a conserved residue involved in substrate binding in several class II aldolases.¹⁹² Some blue-green algae contain both types of aldolase, as do the flagellates *Euglena* and *Chlamydomonas*.

The catabolism of L-fucose by *E. coli* requires cleavage of L-fucose-1-phosphate to form dihydroxyacetone phosphate and D-lactaldehyde by a class II aldolase.¹⁹³



The mechanism of chain cleavage proposed on the basis of the structure and modeling¹⁹³ is illustrated in Fig. 13-7.

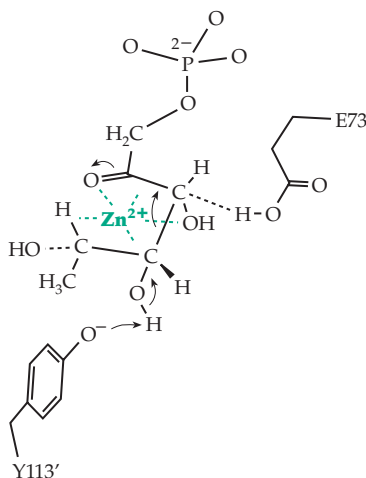
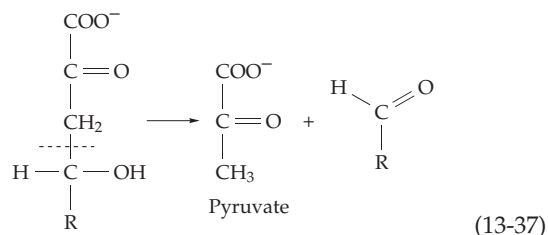


Figure 13-7 Interaction of the bound zinc ion of L-fucose-1-phosphate aldolase and catalytic side chains with the substrate in the active site of the enzyme as revealed by X-ray crystallography and modeling. See Dreyer and Schulz.¹⁹³

Special aldolases of both classes cleave and form C–C bonds throughout metabolism. Several of them act on 2-oxo-3-deoxy substrates forming pyruvate as one product (Eq. 13-37). The aldehyde product varies. In the Entner–Doudoroff pathway of carbohydrate metabolism (Chapter 21) 3-deoxy-2-oxo-6-phosphogluconate (KDPG) is cleaved to pyruvate and 3-phosphoglyceraldehyde.¹⁹⁴ The same products arise from the corresponding phosphogalactonate derivative.¹⁹⁵ The subunits of the trimeric KDPG aldolase have an $(\alpha/\beta)_8$ -barrel structure similar to that of eukaryotic fructose 1,6-bisphosphate aldolase.¹⁹⁴ The 8-carbon sugar acid “KDO” of bacterial cell walls (Fig. 4-26) is cleaved by another aldolase. The catabolism of hydroxyproline leads to 4-hydroxy-2-oxoglutarate, which is cleaved to pyruvate and glyoxylate.¹⁹⁶ In the catabolism of deoxynucleotides, another aldolase converts 2-deoxyribose 5-phosphate to acetaldehyde and glyceraldehyde 3-phosphate.¹⁹⁷



Closely related to aldolases is **transaldolase**, an important enzyme in the pentose phosphate pathways of sugar metabolism and in photosynthesis. The mechanism of the transaldolase reaction (Eq. 17-15) is similar to that used by fructose-1,6-bisphosphate aldolase with a lysine side chain forming a Schiff base and catalytic aspartate and glutamate side chains.¹⁹⁸

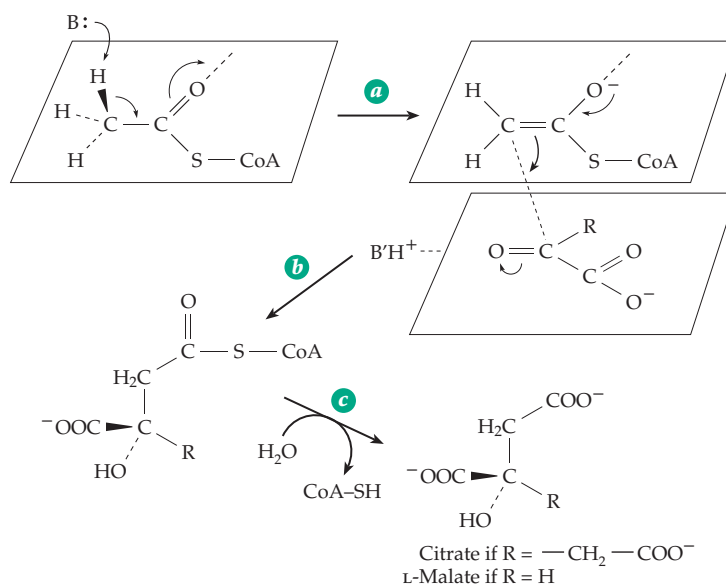
Polycarboxylic acid synthases. Several enzymes, including **citrate synthase**, the key enzyme which catalyzes the first step of the citric acid cycle, promote condensations of acetyl-CoA with ketones (Eq. 13-38). An α -oxo acid is most often the second substrate, and a thioester intermediate (Eq. 13-38) undergoes hydrolysis to release coenzyme A.¹⁹⁹ Because the substrate acetyl-CoA is a thioester, the reaction is often described as a Claisen condensation. The same enzyme that catalyzes the condensation of acetyl-CoA with a ketone also catalyzes the second step, the hydrolysis of the CoA thioester. These polycarboxylic acid synthases are important in biosynthesis. They carry out the initial steps in a *general chain elongation process* (Fig. 17-18). While one function of the thioester group in acetyl-CoA is to activate the methyl hydrogens toward the aldol condensation, the subsequent hydrolysis of the thioester linkage provides for overall irreversibility and “drives” the synthetic reaction.

TABLE 13-1
Products Arising from Reactions of Acetyl-CoA with a Second Substrate with Catalysis by a Polycarboxylate Synthase

Ketone substrate	Product	Further metabolites
$\begin{array}{c} \text{H} \\ \diagup \\ \text{O}=\text{C} \\ \diagdown \\ \text{COO}^- \end{array}$ Glyoxylate	$\begin{array}{c} \text{OOC}^- \quad \text{CH}_2\text{COO}^- \\ \diagup \quad \diagdown \\ \text{C} \\ \diagdown \quad \diagup \\ \text{HO} \quad \text{H} \end{array}$ L-Malate	Carbohydrates, etc., via glyoxylate pathway
$\begin{array}{c} \text{CH}_2\text{COO}^- \\ \diagup \\ \text{O}=\text{C} \\ \diagdown \\ \text{COO}^- \end{array}$ Oxaloacetate	$\begin{array}{c} \text{OOC}^- \quad \text{CH}_2\text{COO}^- \\ \diagup \quad \diagdown \\ \text{C} \\ \diagdown \quad \diagup \\ \text{HO} \quad \text{CH}_2\text{COO}^- \end{array}$ Citrate	2-Carbon unit from acetyl-CoA occupies <i>pro-S</i> position
$\begin{array}{c} \text{CH}_2\text{CH}_2\text{COO}^- \\ \diagup \\ \text{O}=\text{C} \\ \diagdown \\ \text{COO}^- \end{array}$ 2-Oxoglutarate	$\begin{array}{c} \text{OOC}^- \quad \text{CH}_2\text{COO}^- \\ \diagup \quad \diagdown \\ \text{C} \\ \diagdown \quad \diagup \\ \text{HO} \quad \text{CH}_2\text{CH}_2\text{COO}^- \end{array}$ Homocitrate	Lysine via α -aminoacid
$\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{O}=\text{C} \\ \diagdown \\ \text{COO}^- \end{array}$ 2-Oxoisovalerate	$\begin{array}{c} \text{OOC}^- \quad \text{CH}_2\text{COO}^- \\ \diagup \quad \diagdown \\ \text{C} \\ \diagdown \quad \diagup \\ \text{HO} \quad \text{CH}(\text{CH}_3)_2 \end{array}$ α -Isopropylmalate	Leucine
$\begin{array}{c} \text{CH}_2\text{C}(=\text{O})\text{S-CoA} \\ \diagup \\ \text{O}=\text{C} \\ \diagdown \\ \text{CH}_3 \end{array}$ Acetoacetyl-CoA	$\begin{array}{c} \text{HO} \quad \text{CH}_2\text{COO}^- \\ \diagup \quad \diagdown \\ \text{C} \\ \diagdown \quad \diagup \\ \text{H}_3\text{C} \quad \text{CH}_2\text{C}(=\text{O})\text{S-CoA} \end{array}$ S-3-Hydroxy-3-methylglutaryl-CoA (HMG-CoA)	Free acetoacetate Isoprenoid compounds

The stereochemistry of the reaction is also illustrated in Eq. 13-38. These enzymes may be classified by designating the face of the carbonyl group to which the enolate anion adds. The *si* face is up in Eq. 13-38. The common citrate synthase of animal tissues²⁰⁰ and that of *E. coli*²⁰¹ condense with the *si* face and are designated (*si*)-citrate synthases. A few anaerobic bacteria use citrate (*re*)-synthase having the opposite stereochemistry.²⁰² Many citrate synthases are ~100-kDa dimers^{203,204} but some are hexamers^{203,205} and are allosterically inhibited by NADH. The second substrates and products of several related reactions of acetyl-CoA are summarized in Table 13-1.

The *si*-citrate synthase of pigs is a dimer of 437-residue chains, each of which is organized into a large rigid domain and a smaller



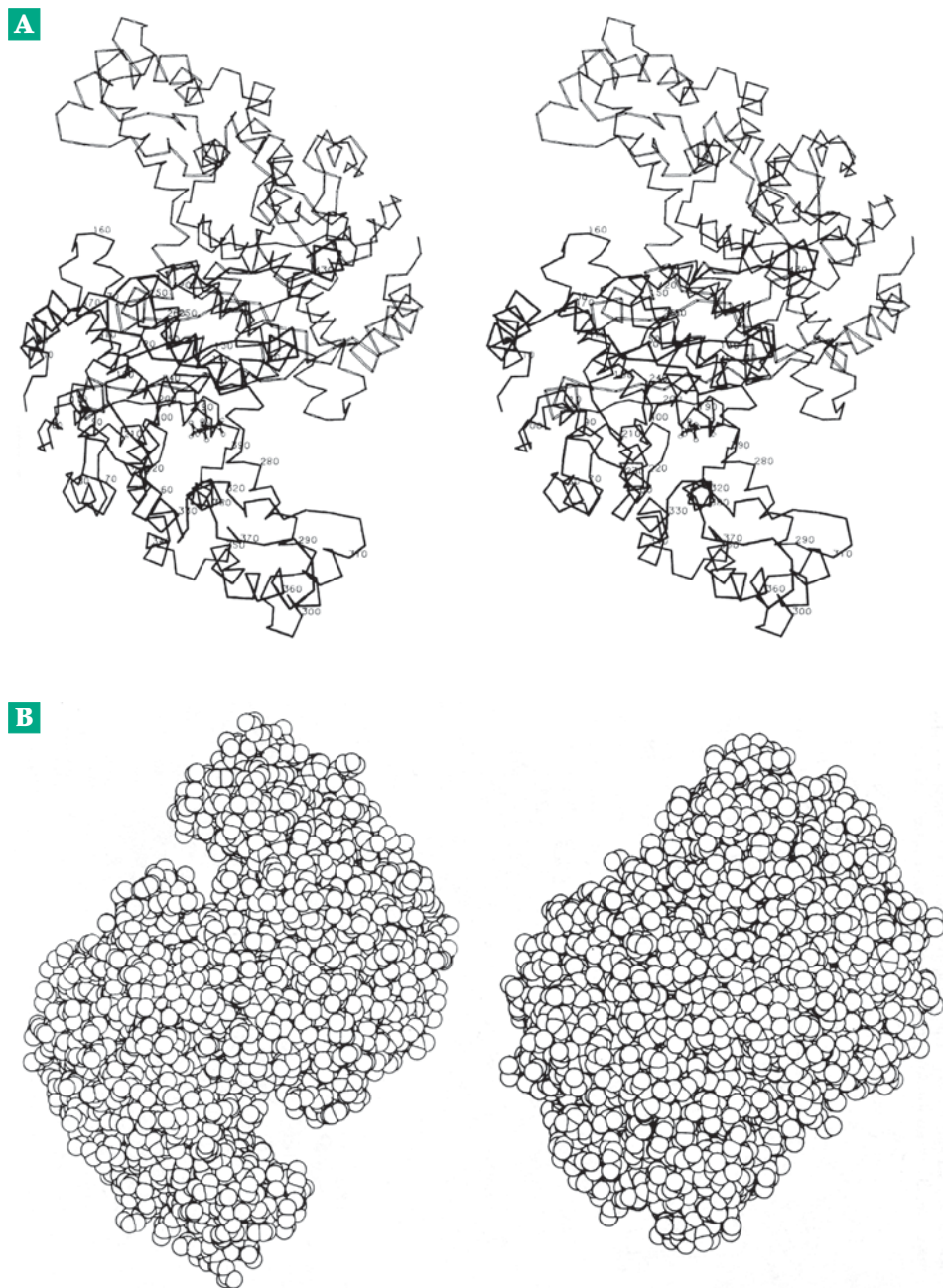


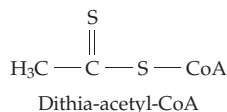
Figure 13-8 Three-dimensional structure of citrate synthase. (A) Stereoscopic alpha carbon trace of the dimeric pig enzyme in its open conformation.²⁰⁹ A molecule of citrate is shown in the lower subunit. The view is down the two-fold axis. Courtesy of Robert Huber. (B) Space-filling representation of the open (left) and closed (right) forms of the same enzyme.²⁰⁰ Courtesy of Stephen J. Remington.

flexible domain. The active site lies between the domains.²⁰⁶ When oxaloacetate binds into the cleft the smaller domain undergoes a complex motion that closes the enzyme tightly around the substrate (Fig. 13-8).^{200,207,208} The bound oxaloacetate is shown in Fig. 13-9. It is surrounded by a large number of polar side chains, including several from histidine and arginine residues. Of these, Arg 401 and Arg 421 of the second subunit bind the substrates' two carboxylate groups. In this tight complex the imidazole of His 320 is in the correct position to protonate the carbonyl oxygen of the oxaloacetate. The domain movement has also brought the groups that bind the acetyl-CoA into their proper position creating, by "induced fit,"

the acetyl-CoA binding site. This conformational change also accounts for the observation of an ordered kinetic mechanism with oxaloacetate binding before acetyl-CoA. The imidazole of His 274 is correctly oriented to abstract a proton from the methyl group of acetyl-CoA to generate the intermediate enolate anion. When oxaloacetate alone is bound into the active site the carbonyl stretching frequency, observed by infrared spectroscopy, is shifted from 1718 cm^{-1} for free oxaloacetate to 1697 cm^{-1} . This decrease of 21 cm^{-1} suggests a strong polarization of the C=O bond by its interaction with His 320 in the ground state.²¹¹ This interaction is seen also in the ^{13}C NMR spectrum, the carbonyl resonance being shifted downfield by 6.8 ppm upon

binding to the protein.²¹² The enolate anion mechanism is also supported by kinetic isotope effects.²¹³

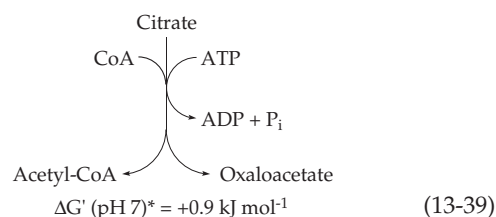
The methyl protons of dithia-acetyl-CoA are much more reactive than those of acetyl-CoA ($pK_a \sim 12.5$ vs ~ 20). When citrate acts on this acetyl-CoA analog together with oxaloacetate, the expected enolate anion



forms rapidly as is indicated by the appearance of a 306 nm absorption band. However, it condenses very slowly with the oxaloacetate.²¹⁴ The binding equilibria, kinetics, and X-ray structures of complexes for a variety of other analogs of acetyl-CoA have also been studied.^{210,215-216a} It appears that Asp 375 and His 274 work together to generate the enolate anion (Fig. 13-9). NMR measurements indicate formation of an unusually short hydrogen bond, but its significance is uncertain.^{216a,216b} Malate synthase (Table 13-1) operates with a very similar mechanism but has an entirely different amino acid sequence and protein fold.^{216c}

Other Claisen condensations are involved in synthesis of fatty acids and polyketides²¹⁷ (Chapter 21) and in formation of 3-hydroxy-3-methylglutaryl-CoA, the precursor to the polyprenyl family of compounds (Chapter 22). In these cases the acetyl group of acetyl-CoA is transferred by a simple displacement mechanism onto an -SH group at the active site of the synthase to form an acetyl-enzyme.^{218,219} The acetyl-enzyme is the actual reactant in step *b* of Eq. 17-5 where this reaction, as well as that of HMG-CoA lyase, is illustrated.

Citrate cleaving enzymes. In eukaryotic organisms the synthesis of citrate takes place within the mitochondria, but under some circumstances citrate is exported into the cytoplasm. There it is cleaved by **ATP-citrate lyase**. To ensure that the reaction goes to completion, cleavage is coupled to the hydrolysis of ATP to ADP and inorganic phosphate (Eq. 13-39). The value of G' given here is extremely dependent upon the concentration of Mg^{2+} as a result of strong chelation of Mg^{2+} by citrate.²²⁰ The reaction sequence is complex but can be understood in terms of an initial ATP-dependent synthesis of



citryl-CoA using a mechanism similar to that in Eq. 12-47. There is evidence for both phosphoenzyme and citryl enzyme intermediates (Eq. 13-40).²²¹ Native ATP-citrate lyase is a tetramer of 110-kDa subunits. It usually contains some phosphoserine and phosphothreonine residues but they apparently have little effect on activity.²²² Phosphorylation is catalyzed by cAMP-dependent and by insulin-dependent protein kinases.^{223,224} A related reaction is the ATP-dependent cleavage of malate to acetyl-CoA and glyoxylate. It requires two enzymes, malyl-CoA being an intermediate.²²⁵

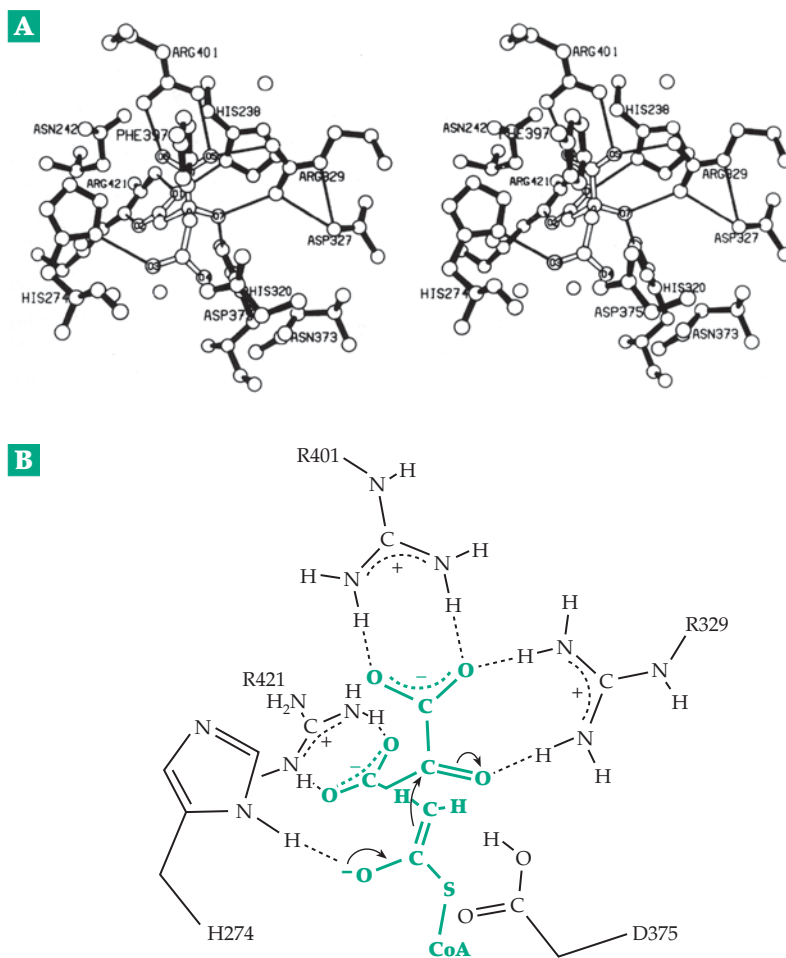
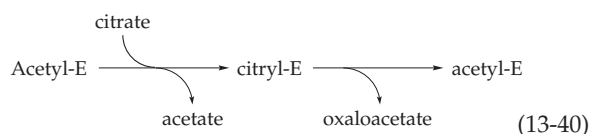
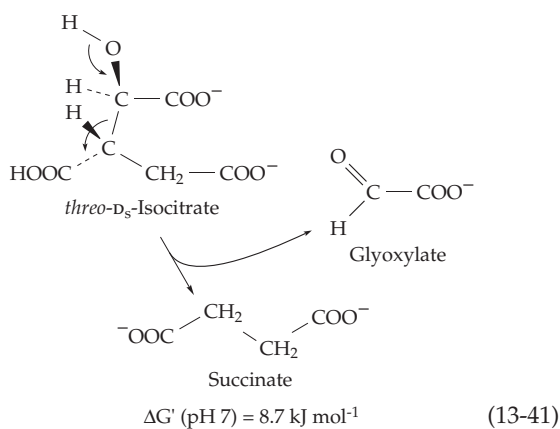


Figure 13-9 Active site of pig citrate synthase. (A) Stereoscopic view with a molecule of citrate in the active site.²⁰⁰ Courtesy of Stephen J. Remington. (B) Interpretive view of the enolate anion of acetyl-CoA and oxaloacetate bound in the active site. Based on work by Kurz *et al.*²¹⁰

A substrate-induced citrate lyase found in bacteria such as *E. coli* and *Klebsiella* promotes the anaerobic dissimilation of citrate splitting it to oxaloacetate and acetate.²²⁶ The large ~585-kDa protein from *Klebsiella*²²⁵ has the composition $(\alpha\beta\gamma)_6$, where the α , β , and γ subunits have masses of ~55-, 30-, and 10-kDa, respectively. The subunit carries an unusual covalently bound derivative of coenzyme A²²⁷ (see also Chapter 14). The 10-kDa γ subunit serves as an **acyl-carrier protein**, the -SH of its prosthetic group being acetylated by a separate ATP-dependent ligase. The resulting acetyl-enzyme undergoes an acyl exchange (analogous to a CoA-transferase reaction; Eq. 12-50) to form a citryl enzyme before the aldol cleavage takes place (Eq. 13-40).



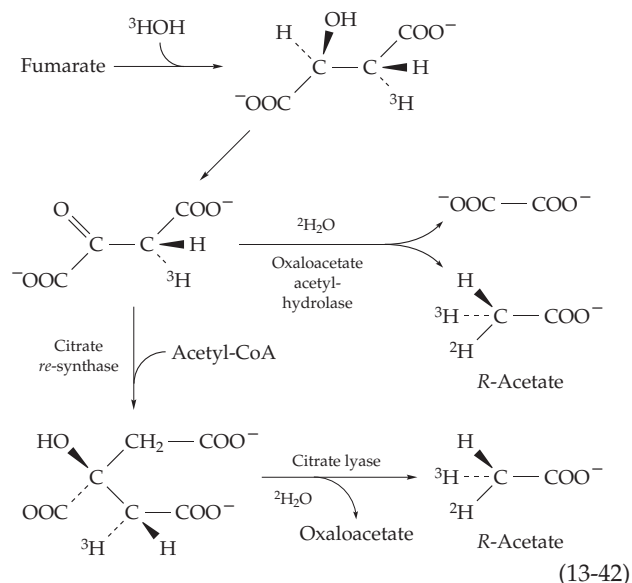
The first unique enzyme of the important glyoxylate pathway (Chapter 17), **isocitrate lyase**, cleaves isocitrate to succinate and glyoxylate (Eq. 13-41).²²⁸ The carboxylate group that acts as electron acceptor would presumably be protonated by the enzyme.



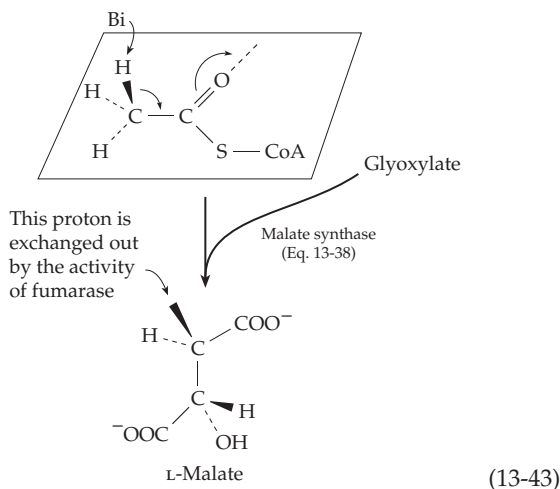
3. Chiral Acetates and Their Use in Stereochemical Studies

Consider the series of enzyme-catalyzed reactions shown in Eq. 13-42. Fumarate is hydrated in ³H-containing water to malate which is oxidized to oxaloacetate. Hydrolysis of the latter with oxaloacetate acetylhydrolase (Eq. 13-34) in ²H₂O gives oxalate and chiral (*R*) acetate. The identical product can be obtained by condensing oxaloacetate with acetyl-CoA using citrate (*re*)-synthetase. The resulting citrate is cleaved in ²H₂O using a citrate lyase having the *si* specificity.^{229,230} Acetate of the opposite chirality can be formed enzymatically beginning with [2,3-³H]fumarate hydrated

by fumarate hydratase in ordinary water. Chiral acetates have also been prepared nonenzymatically,²²⁹ and their configuration has been established unequivocally.



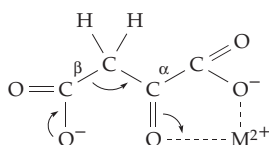
During the action of both oxaloacetate acetylhydrolase and citrate lyase (Eq. 13-42) inversion of configuration occurs about the carbon atom that carries the negative charge in the departing enolate anion. That inversion also occurs during catalysis by citrate (*re*)-synthase and other related enzymes has been demonstrated through the use of chiral acetates.^{229,230} The findings with malate synthase¹⁹⁹ are illustrated in Eq. 13-38. Presumably, a basic group B of the enzyme removes a proton to form the planar enolate anion. The second substrate glyoxylate approaches from the other side of the molecule and condenses as is shown. Since any one of the three protons in either *R* or *S* chiral acetyl-CoA might have been abstracted by base B, several possible combinations of isotopes are possible in the L-malate formed. One of the results of the experiment using chiral (*R*) acetyl-CoA is illustrated in Eq. 13-43. The reader can easily tabulate the results of removal of the ²H or ³H. However, notice that if the base -B: removes ²H (D) or ³H (T) the reaction will be much slower because of the kinetic isotope effects which are expected to be $^3k/^2k \approx 7$ and $^3k/^1k \approx 16$. A second important fact is that the *pro-R* hydrogen at C-3 in malate is specifically exchanged out into water by the action of fumarate hydratase. From the distribution of tritium in the malate and fumarate formed using the two chiral acetates, the inversion by malate synthase was established. See Kyte²³¹ for a detailed discussion.



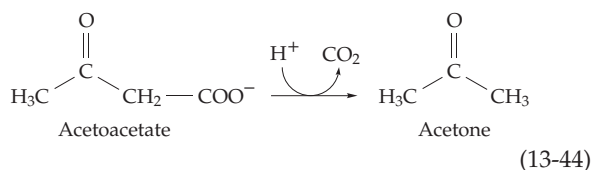
4. Addition of an Enolate Ion to Carbon Dioxide and Decarboxylation

The addition of an enolate anion to CO_2 to form a β -oxoacid represents one of the commonest means of incorporation of CO_2 into organic compounds. The reverse reaction of **decarboxylation** is a major mechanism of biochemical formation of CO_2 . The equilibrium constants usually favor decarboxylation but the cleavage of ATP can be coupled to drive carboxylation when it is needed, e.g., in photosynthesis.

Decarboxylation of β -oxoacids. Beta-oxoacids such as oxaloacetic acid and acetoacetic acid are unstable, their decarboxylation being catalyzed by amines, metal ions, and other substances. Catalysis by amines depends upon Schiff base formation,²³² while metal ions form chelates in which the metal assists in electron withdrawal to form an enolate anion.^{233–235}

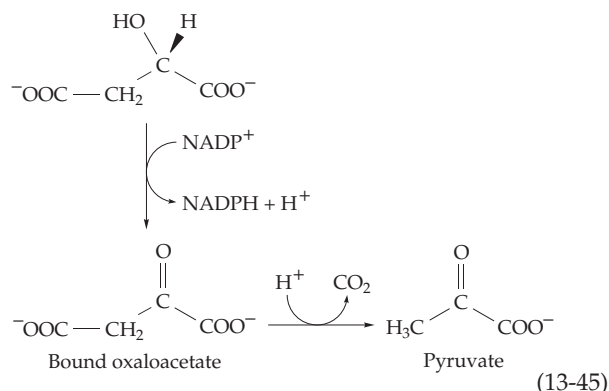


Can we apply any of this information from non-enzymatic catalysis to decarboxylating enzymes? Some decarboxylases do form Schiff bases with their substrates, and some are dependent on metal ions.²³⁵ The acetone-forming fermentation of *Clostridium acetobutylicum* requires large amounts of acetoacetate decarboxylase (Eq. 13-44).



The enzyme is inactivated by borohydride in the presence of substrate, and acid hydrolysis of the inactivated enzyme yielded ϵ -N-isopropyllysine. Decarboxylation occurs from a Schiff base by a mechanism analogous to that of the aldol cleavage shown in Eq. 13-36.²³⁶ Mechanistically related is 4-oxalocrotonate decarboxylase.^{236a}

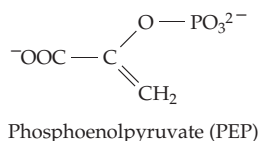
Linked oxidation and decarboxylation. Metabolic pathways often make use of oxidation of a β -hydroxy acid to a β -oxoacid followed by decarboxylation in the active site of the same enzyme. An example is conversion of L-malate to pyruvate (Eq. 13-45). The Mg^{2+} or Mn^{2+} -dependent decarboxylating malic dehydrogenase that catalyzes the reaction is usually called **the malic enzyme**. It is found in most organisms.^{237–240} While a concerted decarboxylation and dehydrogenation may sometimes occur,^{241–242} the enzymes of this group appear usually to operate with bound oxoacid intermediates as in Eq. 13-45.



Other reactions of this type are the oxidation of isocitrate to 2-oxoglutarate in the citric acid cycle (Fig. 17-4, steps *d* and *e*),²⁴³ oxidation of 6-phosphogluconate to ribulose 5-phosphate (Eq. 17-12),²⁴⁴ and corresponding reactions of isopropylmalate dehydrogenase^{245,246} and tartrate dehydrogenase.^{247,248} Crystallographic studies of isocitrate dehydrogenase using both photolabile “caged” isocitrate²⁴⁷ and slow mutant forms²⁴³ with polychromatic Laue crystallography (Chapter 3) have demonstrated the rapid formation of the anticipated intermediate **oxalosuccinate**.

Phosphoenolpyruvate, a key metabolic intermediate. A compound of central importance in metabolism is the phosphate ester of the enol form of pyruvate, commonly known simply as phosphoenolpyruvate (PEP).²⁴⁹ It is formed in the glycolysis pathway by dehydration of 2-phosphoglycerate (Eq. 13-15) or by decarboxylation of oxaloacetate. Serving as a preformed enol from which a reactive enolate anion can be released for condensation reactions,^{250,251} PEP

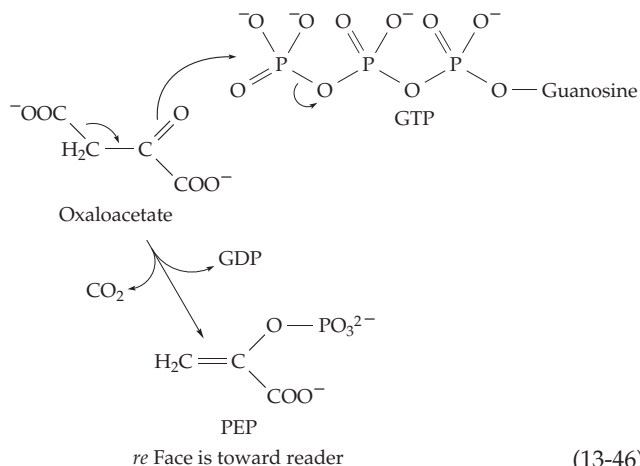
is utilized in metabolism in many ways.



In animals and in many bacteria, PEP is formed by decarboxylation of oxaloacetate. In this reaction, which is catalyzed by **PEP carboxykinase** (PEPCK), a molecule of GTP, ATP, or inosine triphosphate captures and phosphorylates the enolate anion generated by the decarboxylation (Eq. 13-46).²⁵² The stereochemistry is such that CO₂ departs from the *si* face of the forming enol.²⁵³ The phospho group is transferred from GTP with inversion at the phosphorus atom.²⁵⁴ The enzyme requires a divalent metal ion, preferably Mn²⁺. In fact, kinetic studies of the GTP-dependent avian mitochondrial enzyme indicate two metal-binding sites, one on the polyphosphate group of the bound GTP and one on carboxylate side chains of the protein.^{252,255} The three-dimensional structure of the ATP-dependent *E. coli* enzyme reveals a nucleotide binding site similar to the ATP site of adenylate kinase (Fig. 12-30).²⁵⁶ A definite binding site for CO₂ is also present in the enzyme.²⁵⁷

PEPCK is also activated by low concentrations of Fe²⁺ and this activation depends upon a protein that has been identified as glutathione peroxidase (Eq. 15-58). By destroying H₂O₂ the latter may allow the Fe²⁺ to prevent oxidation of an SH group on PEP carboxykinase.²⁵⁸ Synthesis of PEPCK is stimulated by glucagon, evidently through a direct action of cAMP on transcription of the structural gene.²⁵⁹ Transcription is also stimulated by glucocorticoids and thyroid hormone and is inhibited by insulin.

In some organisms, such as the parasitic *Ascaris suum*, PEPCK functions principally as a means of synthesis of oxaloacetate by reaction of PEP with CO₂

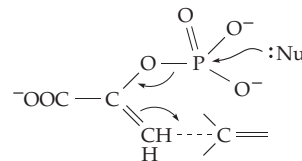


and GDP. These organisms lack pyruvate kinase, which allows for buildup of the high PEP concentration needed to drive this reaction.²⁶⁰ In a similar way PEP can be converted to oxaloacetate by **PEP carboxy-transphosphorylase**, an enzyme found only in prokaryotic bacteria and in *Entamoeba*. The reaction (Eq. 13-47) is accompanied by synthesis of inorganic pyrophosphate which may be cleaved to “pull” the reaction in the indicated direction.



Oxaloacetate is also decarboxylated without phosphorylation of the enolate anion formed but with release of free pyruvate. Both pyruvate kinase and PEPCK can act as oxaloacetate decarboxylases.²⁶¹

In the important reactions discussed in the following sections enolate ions are intermediates in carbon-carbon bond formation. Other examples are given in Eqs. 20-7 and 20-8, and Fig. 25-1, in which C–C bonds are formed by action of PEP as a carbon nucleophile,



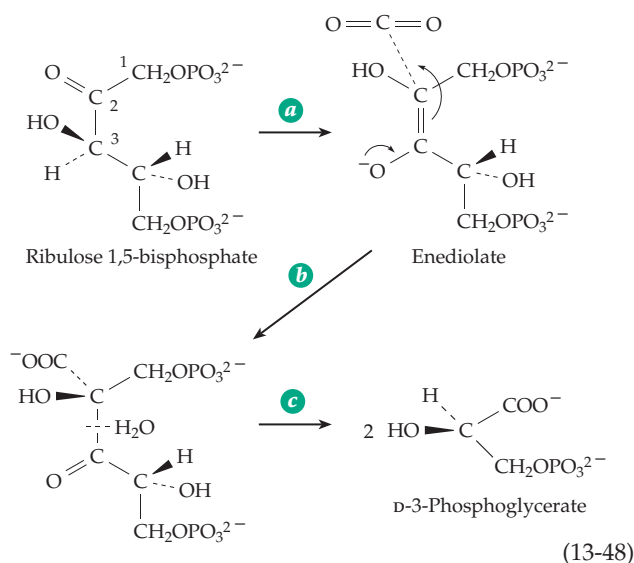
e.g., in aldol-like condensations.^{242a} Yet another unusual type of reaction, involving formation of an enol ether linkage is illustrated in Box 13-B and in Eq. 25-3.

Ribulose biphosphate carboxylase. The major route of incorporation of CO₂ into organic compounds is via photosynthesis. When ¹⁴C-labeled CO₂ enters chloroplasts of green plants, the first organic ¹⁴C-containing compound detected is 3-phosphoglycerate. Two molecules of this compound are formed through the action of **ribulose-1,5-bisphosphate carboxylase** (abbreviated **rubisco**), an enzyme present in chloroplasts and making up 16% of the protein of spinach leaves. This enzyme, whose structure is illustrated in Fig. 13-10, is thought to be the most abundant protein on earth. Because O₂ competes with CO₂ as a substrate, the enzyme also catalyzes an “oxygenase” reaction. It is therefore often called **ribulose-1,5-bisphosphate carboxylase/oxygenase**.²⁶² The rubisco from most plants is a 500- to 560-kDa L₈S₈ oligomer as shown in Fig. 13-10. The large subunit is encoded in the chloroplast DNA. However, a family of nuclear genes encode the small subunits, which are synthesized as larger precursors, with N-terminal extensions being removed to give the mature subunits.^{263,264} Two types of small subunits occupy different positions in the

quasi-symmetric spinach rubisco.²⁶⁴ In *Euglena* the small subunit is synthesized as a polypeptide precursor containing eight copies of the subunit.²⁶⁵ Rubisco from the hydrogen-oxidizing bacterium *Alcaligenes eutrophus* has a similar quaternary structure,²⁶⁶ but the enzyme from *Rhodospirillum rubrum* is a simple dimer.²⁶⁷ In dinoflagellates the rubisco gene is present in nuclear DNA rather than in the chloroplasts.²⁶⁸

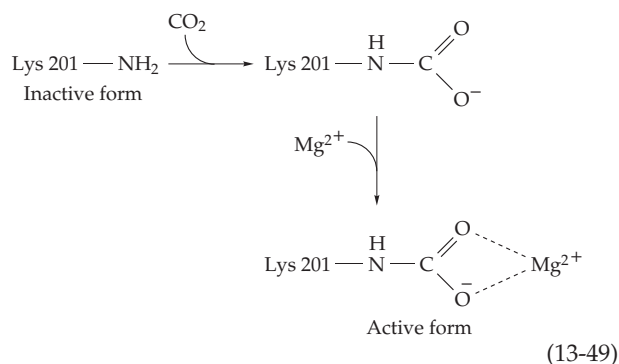
The carboxylation reaction catalyzed by rubisco differs from others that we have considered in that the carboxylated product is split by the same enzyme (Eq. 13-48). The mechanism shown in Eq. 13-48 was suggested by Bassham *et al.*²⁷¹ Ribulose biphosphate, for which the enzyme is absolutely specific, is first converted to its enolic form, a 2,3-enediol. Loss of a proton from the 3-OH group forms the enediolate anion needed for the carboxylation in step *b*. The product of this step is a β -oxoacid which undergoes enzyme-catalyzed hydrolytic cleavage (Eq. 13-48, step *c*; see also Eq. 13-34). Support for this mechanism came from the observation that 2-carboxyarabinitol biphosphate (Fig. 13-11B) is a potent inhibitor, possibly a transition state analog.²⁷² This inhibitor, bound into the active site of the enzyme from spinach, is seen in both Figs. 13-10B and 13-11A. An expanded version of Eq. 13-48 is given in Fig. 13-12, which is based upon modeling together with a variety of X-ray structures including that shown in Fig. 13-11.

An essential Mg^{2+} ion is held by carboxylates of D203 and E204 and by modified K201. It also coordinates three molecules of H_2O in the free enzyme.^{272a} Catalytic roles for the H294 and H327 imidazole groups are still being elucidated.^{272b} Lysine 175 protonates C2 of the aci anion generated by C–C bond cleavage (step *e*, Fig. 13-12).^{272c,276} Like many enzymes, rubisco exists in two major conformational states: open and closed.^{272d}



Chemical studies also support the indicated mechanism. For example, the β -oxoacid intermediate formed in step *b* of Eq. 13-48 or Fig. 13-12 has been identified as a product released from the enzyme by acid denaturation during steady-state turnover.^{273,274} Isotopic exchange with 3H in the solvent²⁷⁵ and measurement of ^{13}C isotope effects²⁷⁷ have provided additional verification of the mechanism. The catalytic activity of the enzyme is determined by ionizable groups with pK_a values of 7.1 and 8.3 in the ES complex.²⁷⁸

The apparent value of K_m for total CO_2 ($CO_2 + HCO_3^-$) is high, 11–30 mM, but for the true substrate CO_2 it is only 0.45 mM. In intact chloroplasts the affinity for substrate is distinctly higher, with the K_m for total CO_2 dropping to ~ 0.6 mM. The difference appears to result largely from a regulatory reaction of CO_2 in which the side chain amino group of lysine 201 of the large subunit forms a carbamate (Eq. 13-49). Although carbamylation is spontaneous, it is enhanced by an ATP-dependent process catalyzed by **rubisco activase**.²⁷⁹ The carbamylation converts the side chain of Lys 201 into a negatively charged group that binds to an essential divalent metal, usually Mg^{2+} , in the active center²⁸² as is shown in Fig. 13-11, A and B. The nature of the reaction is uncertain. One possibility is that the activase is a chaperonin.²⁸⁰ It appears to assist the enzyme in removing inhibitory sugars that arise by side reactions in the active site.²⁸¹ Rubisco is also regulated by the level of a natural inhibitor which has been identified as 2-carboxyarabinitol 1-phosphate. This is the same as the inhibitor shown in Fig. 13-11 but with one less phosphate group and consequent weaker binding.^{273a}



In most plants photosynthesis is also strongly inhibited by O_2 . This observation led to the discovery that O_2 competes directly for CO_2 at the active site of rubisco in a process called **photorespiration**. Chloroplasts inhibited by oxygen produce **glycolate** in large amounts^{282a} as a result of the reaction of the intermediate enediolate ion formed in step *b* of Eq. 13-48 with O_2

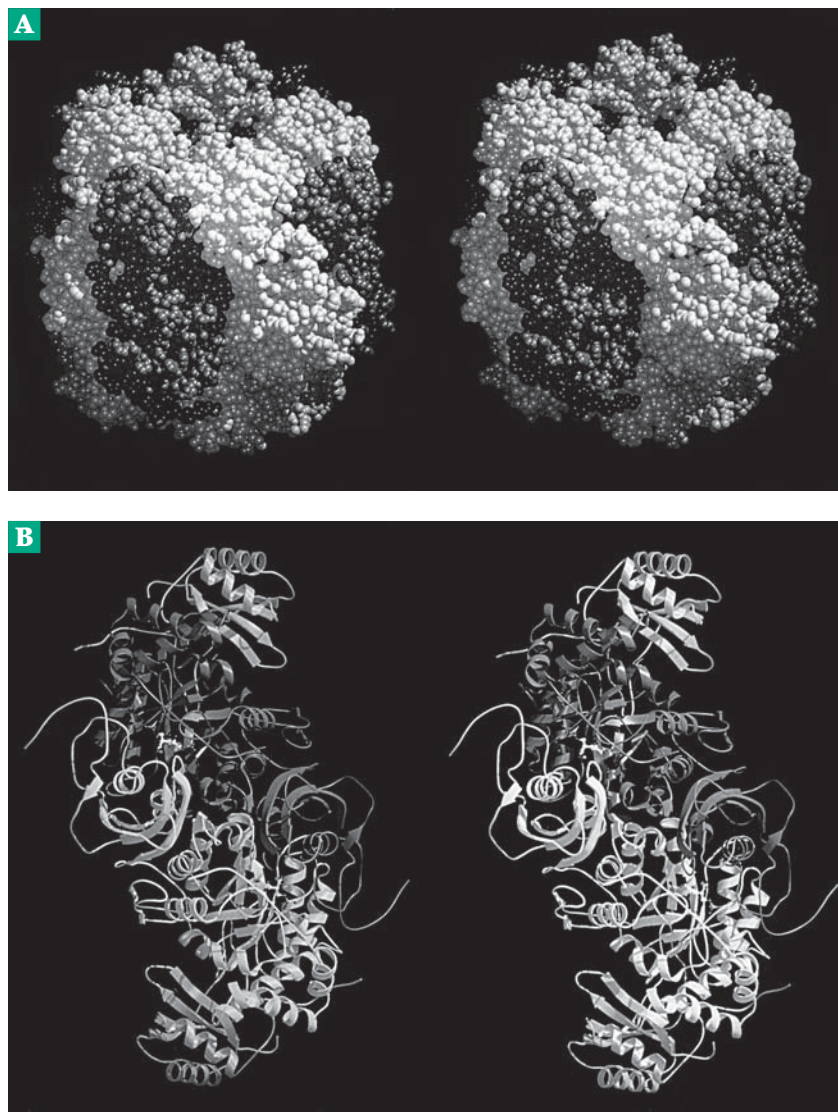


Figure 13-10 Stereoscopic view of ribulose biphosphate carboxylase (rubisco) from spinach. (A) The symmetric L_8S_8 molecule contains eight 475-residue large subunits (in two shades of gray) and eight 123-residue small subunits (lighter gray in upper half of image). (B) One L_2S_2 substructure containing two active sites shared between adjacent large subunits with the bound inhibitor 2-carboxy-D-arabinitol 1,5-bisphosphate.²⁶⁹ In the upper LS unit the S subunit (top) is light and the L subunit is dark. Courtesy of Inger Andersson. Similar structures have been determined for enzymes from tobacco^{269a} and from the cyanobacterium *Synechoccus*.²⁷⁰

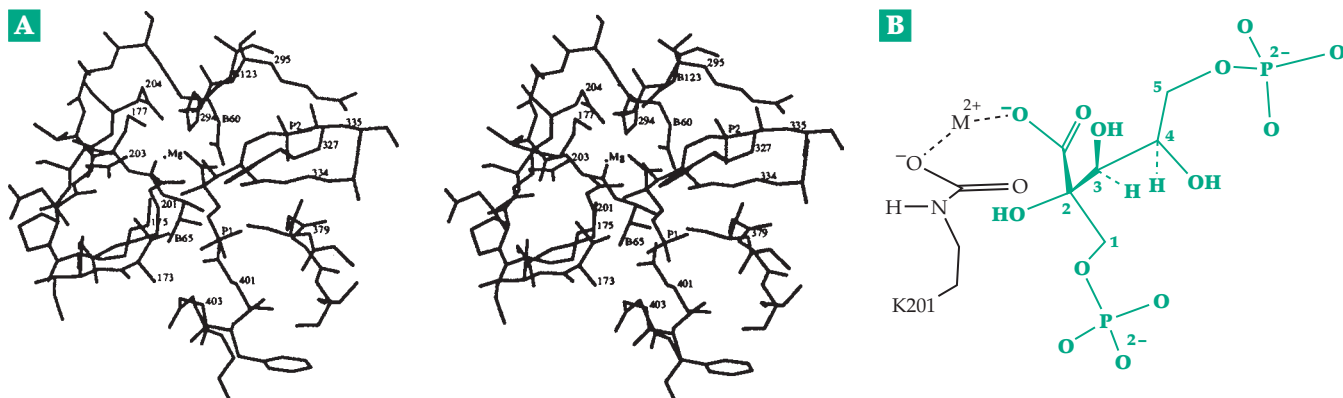


Figure 13-11 (A) Overview of the active site of spinach rubisco showing bound 2-carboxy-D-arabinitol 1,5-bisphosphate and Mg^{2+} and residues within hydrogen-bonding distance of these ligands. The hydroxyl groups at C2 and C3 of the inhibitor are in *cis* conformation.²⁶⁹ Courtesy of Inger Andersson. (B) Structure of the inhibitor 2-carboxy-D-arabinitol 1,5-bisphosphate. A part of the carbamylated lysine 201 and the essential metal ion are also shown.

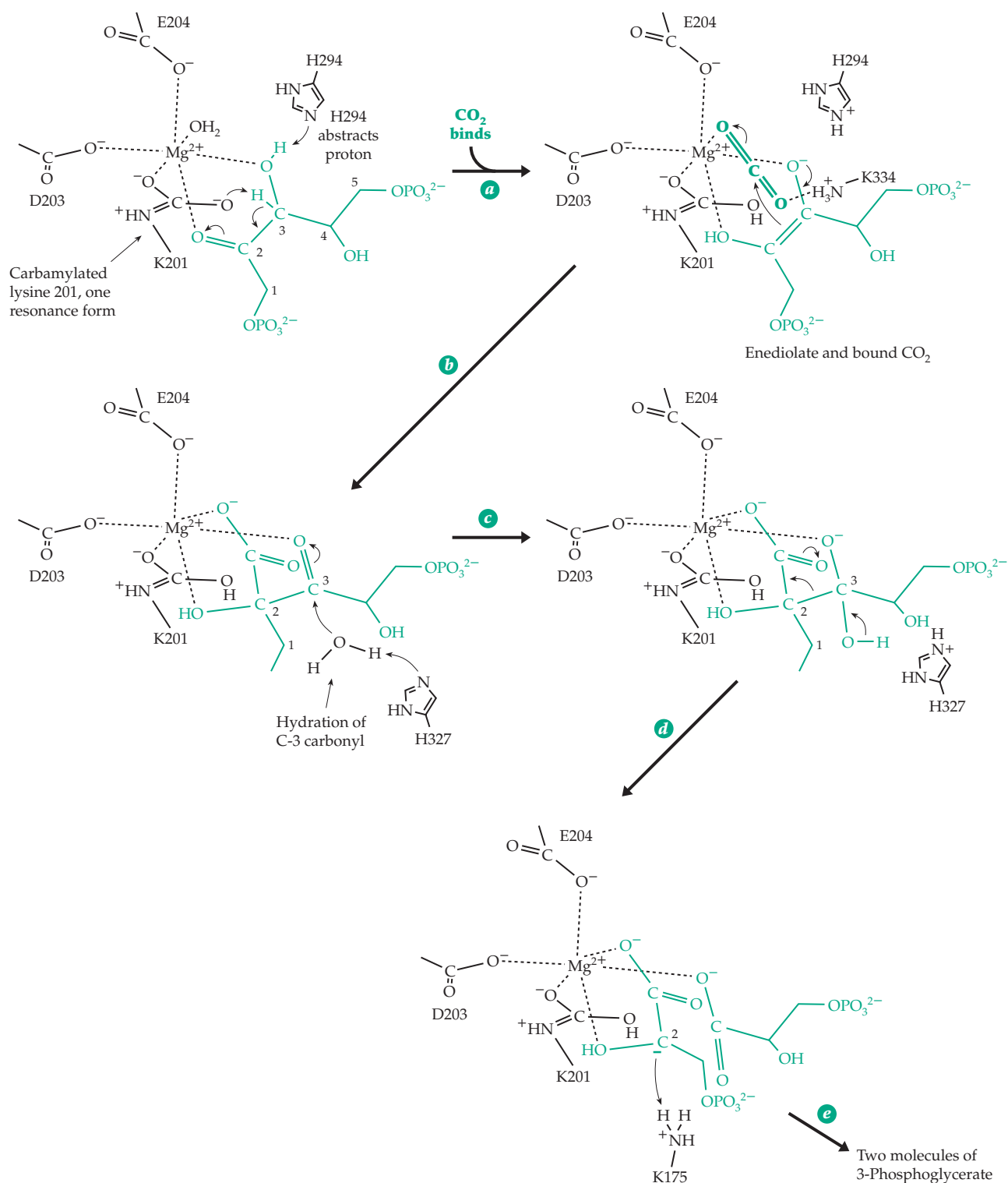
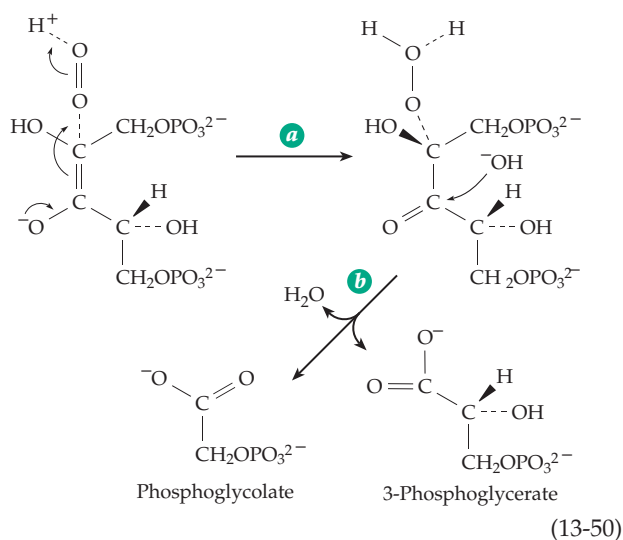


Figure 13-12 Proposed mechanism of action of ribulose biphosphate carboxylase (rubisco). This is an abbreviated version of the mechanism as presented by Taylor and Andersson.²⁷⁶ The binding of ribulose 1,5-bisphosphate occurs after carbamylation of lysine 201 and binding of a magnesium ion. Formation of an enediolate intermediate in step *a* is probably catalyzed by the carbamate group as indicated. Removal of a proton from the 3-OH, perhaps by H294, and addition of a proton to form an -OH group at C2 are also necessary. The CO_2 may bind to the Mg^{2+} and be polarized by interaction with other side chains prior to reaction. Carboxylation occurs by addition of the enediolate to CO_2 in step *b*. The hydration of the resulting 3-oxo group (step *c*) is necessary for cleavage of the C-C bond in step *d*. The participation of the newly formed carboxylate as an *aci* anion coordinated to Mg^{2+} is presumably involved. Protonation, with stereochemical inversion, is thought to involve K175, as shown. The two product molecules dissociate in step *e*.

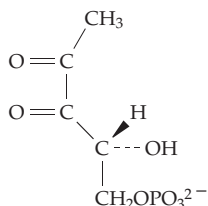
(Eq. 13-50, step *a*). The peroxide formed in this way breaks up under the hydrolytic action of the enzyme to form phosphoglycolate and 3-phosphoglycerate (Eq. 13-50, step *b*).



Molecular oxygen usually reacts rapidly with only those organic substrates, such as dihydroflavins, that are able to form stable free radicals. However, the enolate anion of Eq. 13-50 may be able to donate a single electron to O_2 to form a superoxide-organic radical pair prior to formation of the peroxide (see also Eq. 15-30). Similar oxygenase side reactions have been observed for a variety of other enzymes that utilize carbanion mechanisms.²⁸³ The reaction of rubisco with O_2 is of both theoretical and practical interest, the latter because of its significance in lowering the yield in photosynthesis (Chapter 23).

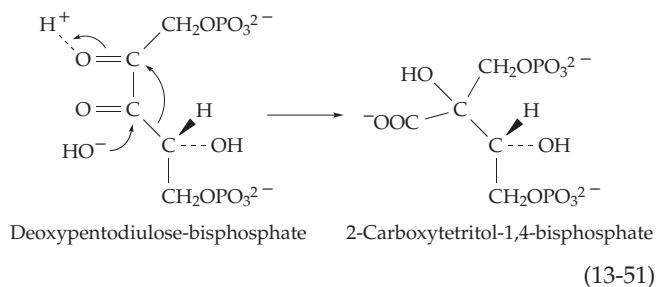
The simpler dimeric rubisco from *Rhodospirillum* is very inefficient in carboxylation and catalyzes much more oxygenation than do rubiscos of higher plants.^{283a} Mutant enzymes that have impaired carboxylase and enhanced oxygenase are also known.^{284,284a}

The small subunits of rubisco may help suppress undesirable side reactions.²⁸⁵ For example, the following deoxypentodiulose phosphate can be formed by β elimination from the second intermediate of Eq. 13-48.

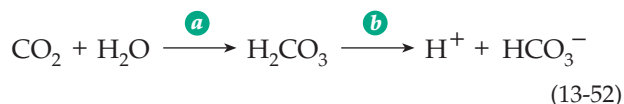


Some side products of the normal oxygenase reaction arise by elimination of a peroxide ion from the first

intermediate of Eq. 13-50. The resulting dicarbonyl bisphosphate can rearrange to give a carboxytetritol bisphosphate (Eq. 13-51).²⁸⁴



Carbon dioxide or bicarbonate ion? An important question in the consideration of carboxylation and decarboxylation reactions is whether the reactant or the product is CO_2 or HCO_3^- . An approach to answering the question was suggested by Krebs and Roughton,²⁸⁶



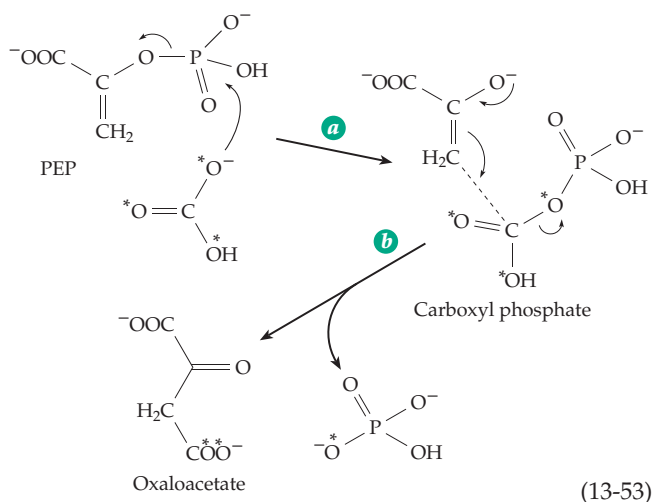
who pointed out that the attainment of equilibrium between free CO_2 and HCO_3^- (Eq. 13-52) may require several seconds. If an enzyme produces CO_2 as a product and progress of the reaction is followed manometrically, the pressure will rise higher than the equilibrium value as the CO_2 is evolved. Later when substrate is exhausted and the CO_2 equilibrates with bicarbonate, the pressure will fall again. The addition of carbonic anhydrase, which catalyzes Eq. 13-52, step *a*, abolishes the "overshoot." If bicarbonate is the primary product of a decarboxylation, there is a lag in the appearance of free CO_2 .

A second approach is to use $[^{18}O]$ bicarbonate and to follow the incorporation of ^{18}O into a carboxylated substrate. If CO_2 is the primary substrate only two labeled oxygen atoms enter the compound, whereas if HCO_3^- is the reactant three are incorporated.²⁸⁷ A third technique is measurement of the rate of incorporation of CO_2 or bicarbonate in the carboxylated product. Over a short interval of time, e.g., 1 min, different kinetics will be observed for the incorporation of CO_2 and of bicarbonate.²⁸⁸ Using these methods, it was established that the product formed in Eq. 13-46 and the reactant in Eq. 13-47 is CO_2 . However, the carboxylation enzymes considered in the next section use bicarbonate as the substrate.

5. Incorporation of Bicarbonate into Carboxyl Groups

An important enzyme with a biosynthetic function in many bacteria and in all higher plants is **PEP carboxylase**,²⁸⁹ which catalyzes the reaction of Eq. 13-53. This enzyme, in effect, accomplishes the reverse of Eq. 13-46 by converting the three-carbon PEP, by reaction with bicarbonate, into the four-carbon oxaloacetate. The latter is needed for “priming” of the citric acid cycle and for biosynthesis of such amino acids as aspartate and glutamate. That the enzyme functions in this way is indicated by the fact that mutants of *Salmonella* defective in the enzyme do not grow unless oxaloacetate or some other intermediate in the citric acid cycle is added to the medium. The enzyme from *S. typhimurium* is a 400-kDa tetramer with complex regulatory properties. The corresponding enzyme from spinach has 12 subunits and 12 bound Mn^{2+} ions. The enzyme also has a special function in the C_4 plants,²⁹⁰ in which it is a component of a carbon dioxide concentrating system (Chapter 23).

When $[\text{^{18}O}]$ bicarbonate is a substrate, two labeled oxygen atoms enter the oxaloacetate, while the third appears in P_i . A concerted, cyclic mechanism could explain these results. However, study of kinetic isotope effects,²⁹¹ use of a substrate with a chiral thiophospho group,²⁹² and additional ^{18}O exchange studies²⁹³ have ruled out this possibility. A transient carboxyl phosphate (Eq. 13-53) is evidently an intermediate.^{294,295} The incorporation of the ^{18}O from bicarbonate into phosphate is indicated by the asterisks. The carboxyl group enters on the *si* face of PEP. However, there is another possibility.^{295,296} The carboxyl phosphate, while in the active site adjacent to the enolate anion, may eliminate phosphate, the enolate ion adding to the resulting CO_2 to form the final product. According to this mechanism the group transfer potential of the phospho group in PEP is

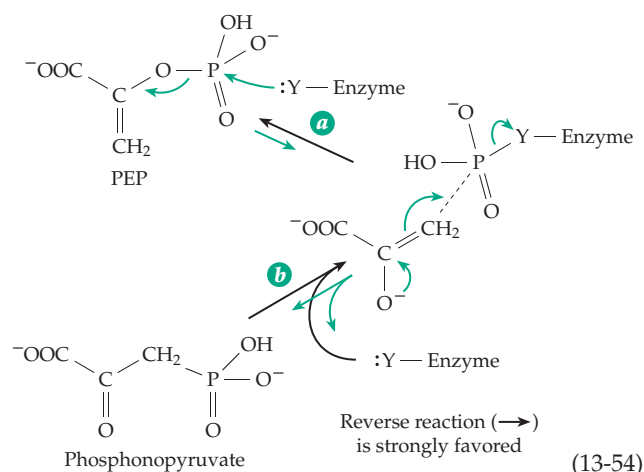


utilized to concentrate the CO_2 and to localize it next to the enolate anion generated by the same process.

PEP carboxylase is lacking from animal tissues and fungi. In these creatures PEP is converted to pyruvate, which is then carboxylated to oxaloacetate with coupled cleavage of ATP by the action of **pyruvate carboxylase** (Eq. 14-3), an enzyme that not only utilizes bicarbonate ion but also contains **biotin**. However, there are mechanistic similarities between its action and that of PEP carboxylase.

PEP mutase and the synthesis of phosphonates.

The lipids of some organisms, such as *Tetrahymena*, contain aminoethylphosphonate, a compound with a C-P bond (Chapter 8). There are also many other naturally occurring phosphono compounds and huge quantities of synthetic phosphonates, present in detergents, herbicides, and insecticides, are metabolized by bacteria.²⁹⁷ Here we will consider only one step in the biosynthesis of phosphonates, the conversion of PEP into phosphonopyruvate (Eq. 13-54), a reaction catalyzed by **PEP mutase**. The phospho group is moved

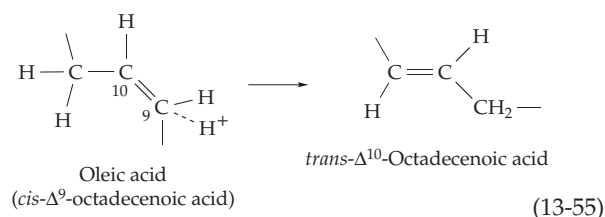


from the oxygen atom of PEP to the methylene carbon atom. When attached to carbon the phospho group is designated **phosphono**. The reaction occurs with retention of the configuration around phosphorus and a phosphoenzyme intermediate seems to be involved^{298,299} as shown in Eq. 13-54. The equilibrium in this reaction strongly favors PEP. One further product formed from phosphonopyruvate is phosphonoethylamine, a component of phosphonolipids.³⁰⁰ It is easy to imagine a synthetic route involving transamination followed by decarboxylation. Another mutase, similar to PEP mutase, shifts the carboxyphospho group $^-\text{OOC}-\text{PO}_2^-$ exactly as in Eq. 13-54 as part of the biosynthetic pathway of a natural herbicide formed by *Streptomyces hygroscopicus*.^{301,302} Soil bacteria contain specialized enzymes that catalyze the hydrolytic cleavage of P-C bonds.^{302a}

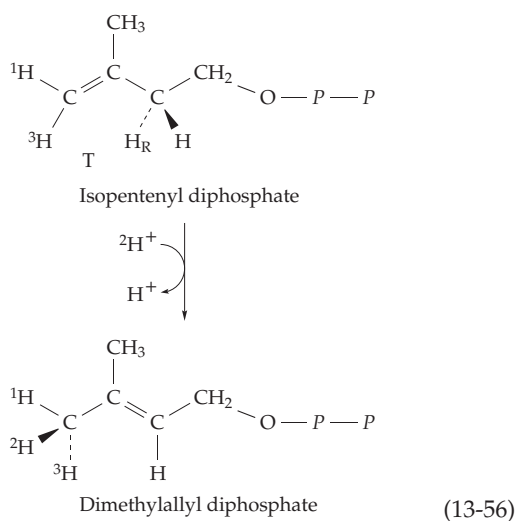
D. Some Isomerization and Rearrangement Reactions

A few metabolic reactions do not fit into any of the categories discussed so far and apparently do not depend upon a coenzyme. Some of these involve transfer of alkyl groups or of hydrogen atoms from one carbon to another. The hydrogen atoms move by direct transfer without exchange with the medium. All of the reactions could involve carbocations but there is often more than one mechanistic possibility.

A simple 1,3-proton shift is shown in Table 10-1 as reaction type 6A. An example is the isomerization of oleic acid to *trans*- Δ^{10} -octadecenoic acid (Eq. 13-55) catalyzed by a soluble enzyme from a pseudomonad.³⁰³



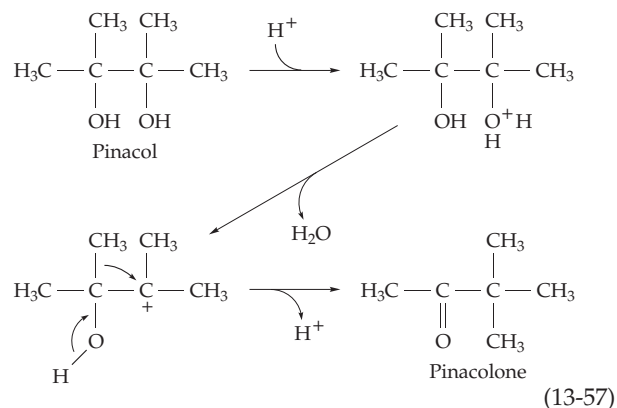
A second example is isomerization of **isopentenyl diphosphate** to **dimethylallyl diphosphate** (Eq. 13-56).^{304–307} The stereochemistry has been investigated using the ^3H -labeled compound shown in Eq. 13-56. The *pro-R* proton is lost from C-2 and a proton is added to the *re* face at C-4. When the reaction was carried out in $^2\text{H}_2\text{O}$ a chiral methyl group was produced as shown.³⁰⁴ A concerted proton addition and abstraction is also possible, the observed *trans* stereochemistry being expected for such a mechanism. However, the



fact that the enzyme is strongly inhibited by the cation of 2-(diethylamino) ethyl pyrophosphate, an analog of a probable carbocation intermediate. Cysteine and glutamate side chains are essential.³⁰⁷

Reaction type 6B of Table 10-1 is allylic rearrangement with simultaneous condensation with another molecule. The reaction, which is catalyzed by **prenyl-transferases**,^{307a} occurs during the polymerization of polyprenyl compounds (Fig. 22-1, Eqs. 22-2, 22-3). Experimental evidence favors a carbocation mechanism for all of these reactions.^{308,309} See Chapter 22.

Reaction type 6C (Table 10-1) occurs during the biosynthesis of leucine and valine (Fig. 24-17). The rearrangement is often compared with the nonenzymatic acid-catalyzed pinacol–pinacolone rearrangement in which a similar shift of an alkyl group takes place (Eq. 13-57). The enzyme-catalyzed rearrangement



presumably gives the structure drawn in brackets in Table 10-1. The same enzyme always catalyzes reduction with NADH to the diol, the Mg^{2+} -dependent enzyme being called **acetohydroxy acid isomeroreductase**. Rearrangement has never been observed without the accompanying reduction.^{310–313} More complex rearrangements that occur during biosynthesis of sterols are described in Chapter 22.

References

1. Jencks, W. P. (1987) *Catalysis in Chemistry and Enzymology*, Dover, Mineola, N. Y. (p. 465)
2. Sorensen, P. E., and Jencks, W. P. (1987) *J. Am. Chem. Soc.* **109**, 4675–4690
3. Kallen, R. G., and Jencks, W. P. (1966) *J. Biol. Chem.* **241**, 5845–5850, 5851–5863
4. Edsall, J. T., and Wyman, J. (1958) *Biophysical Chemistry*, Vol. I, Academic Press, New York (p. 550ff)
5. Fernley, R. T. (1988) *Trends Biochem. Sci.* **13**, 356–359
6. Sly, W. S., and Hu, P. Y. (1995) *Ann. Rev. Biochem.* **64**, 375–401
7. Hazen, S. A., Waheed, A., Sly, W. S., LaNoue, K. F., and Lynch, C. J. (1996) *FASEB J.* **10**, 481–490
8. Heck, R. W., Boriack-Sjodin, P. A., Qian, M., Tu, C., Christianson, D. W., Laipis, P. J., and Silverman, D. N. (1996) *Biochemistry* **35**, 11605–11611
9. Maren, T. H. (1985) *N. Engl. J. Med.* **313**, 179–181
10. Stams, T., Nair, S. K., Okuyama, T., Waheed, A., Sly, W. S., and Christianson, D. W. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 13589–13594
11. Guilloton, M. B., Korte, J. J., Lamblin, A. F., Fuchs, J. A., and Anderson, P. M. (1992) *J. Biol. Chem.* **267**, 3731–3734
12. Kisker, C., Schindelin, H., Alber, B. E., Ferry, J. G., and Rees, D. C. (1996) *EMBO J.* **15**, 2323–2330
13. Rowlett, R. S., Chance, M. R., Wirt, M. D., Sidelinger, D. E., Royal, J. R., Woodroffe, M., Wang, Y.-F. A., Saha, R. P., and Lam, M. G. (1994) *Biochemistry* **33**, 13967–13976
- 13a. Kimber, M. S., and Pai, E. F. (2000) *EMBO J.* **19**, 1407–418
14. Miyamoto, H., Miyashita, T., Okushima, M., Nakano, S., Morita, T., and Matsushiro, A. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 9657–9660
15. Håkansson, K., Carlsson, M., Svensson, L. A., and Liljas, A. (1992) *J. Mol. Biol.* **227**, 1192–1204
16. Kumar, V., and Kannan, K. K. (1994) *J. Mol. Biol.* **241**, 226–232
17. Kiefer, L. L., Paterno, S. A., and Fierke, C. A. (1995) *J. Am. Chem. Soc.* **117**, 6831–6837
- 17a. Lesburg, C. A., Huang, C.-c., Christianson, D. W., and Fierke, C. A. (1997) *Biochemistry* **36**, 15780–15791
18. Liang, J.-Y., and Lipscomb, W. N. (1987) *Biochemistry* **26**, 5293–5301
19. Pocker, Y., and Janjic, N. (1989) *J. Am. Chem. Soc.* **111**, 731–733
- 19a. Earnhardt, J. N., Qian, M., Tu, C., Lakkis, M. M., Bergenhem, N. C. H., Laipis, P. J., Tashian, R. E., and Silverman, D. N. (1998) *Biochemistry* **37**, 10837–10845
- 19b. Denisov, V. P., Jonsson, B.-H., and Halle, B. (1999) *J. Am. Chem. Soc.* **121**, 2327–2328
20. Paneth, P., and O'Leary, M. H. (1985) *J. Am. Chem. Soc.* **107**, 7381–7384
21. Scolnick, L. R., and Christianson, D. W. (1996) *Biochemistry* **35**, 16429–16434
22. Toba, S., Colombo, G., and Merz, K. M., Jr. (1999) *J. Am. Chem. Soc.* **121**, 2290–2302
23. Merz, K. M., Jr., and Banci, L. (1997) *J. Am. Chem. Soc.* **119**, 863–871
24. Briganti, F., Mangani, S., Orioli, P., Scozzafava, A., Vernagione, G., and Supuran, C. T. (1997) *Biochemistry* **36**, 10384–10392
25. Ren, X., Tu, C., Laipis, P. J., and Silverman, D. N. (1995) *Biochemistry* **34**, 8492–8498
26. Malatesta, V., and Cocivera, M. (1978) *Journal of Organic Chemistry* **43**, 1737–1742
27. Jencks, W. P. (1987) *Catalysis in Chemistry and Enzymology*, Dover, Mineola, New York
28. Engel, C. K., Mathieu, M., Zeelen, J. P., Hiltunen, J. K., and Wierenga, R. K. (1996) *EMBO J.* **15**, 5135–5145
- 28a. Engel, C. K., Kiema, T. R., Hiltunen, J. K., and Wierenga, R. K. (1998) *J. Mol. Biol.* **275**, 847–859
29. D'Ordine, R. L., Tonge, P. J., Carey, P. R., and Anderson, V. E. (1994) *Biochemistry* **33**, 12635–12643
30. Bahnson, B. J., and Anderson, V. E. (1991) *Biochemistry* **30**, 5894–5906
31. D'Ordine, R. L., Bahnson, B. J., Tonge, P. J., and Anderson, V. E. (1994) *Biochemistry* **33**, 14733–14742
- 31a. Hofstein, H. A., Feng, Y., Anderson, V. E., and Tonge, P. J. (1999) *Biochemistry* **38**, 9508–9516
32. Yang, S.-Y., He, X.-Y., and Schulz, H. (1995) *Biochemistry* **34**, 6441–6447
33. Annand, R. R., Kozlowski, J. F., Davisson, V. J., and Schwab, J. (1993) *J. Am. Chem. Soc.* **115**, 1088–1094
34. Brock, D. J. H., Kass, L. R., and Bloch, K. (1967) *J. Biol. Chem.* **242**, 4432–4440
35. Hebert, H., Schmidt-Krey, I., Morgenstern, R., Murata, K., Hirai, T., Mitsuoka, K., and Fujiyoshi, Y. (1997) *J. Mol. Biol.* **271**, 751–758
36. Hu, L., Borleske, B. L., and Colman, R. F. (1997) *Protein Sci.* **6**, 43–52
37. Rushmore, T. H., and Pickett, C. B. (1993) *J. Biol. Chem.* **268**, 11475–11478
38. Ji, X., Johnson, W. W., Sesay, M. A., Dickert, L., Prasad, S. M., Ammon, H. L., Armstrong, R. N., and Gilliland, G. L. (1994) *Biochemistry* **33**, 1043–1052
39. Oakley, A. J., Rossjohn, J., Lo Bello, M., Caccuri, A. M., Federici, G., and Parker, M. W. (1997) *Biochemistry* **36**, 576–585
40. Reinemer, P., Prade, L., Hof, P., Neufeind, T., Huber, R., Zettl, R., Palme, K., Schell, J., Koelln, I., Bartunik, H. D., and Bieseler, B. (1996) *J. Mol. Biol.* **255**, 289–309
41. Xiao, G., Liu, S., Ji, X., Johnson, W. W., Chen, J., Parsons, J. F., Stevens, W. J., Gilliland, G. L., and Armstrong, R. N. (1996) *Biochemistry* **35**, 4753–4765
- 41a. Neufeind, T., Huber, R., Reinemer, P., Knäblein, J., Prade, L., Mann, K., and Bieseler, B. (1997) *J. Mol. Biol.* **274**, 577–587
42. McTigue, M. A., Williams, D. R., and Tainer, J. A. (1995) *J. Mol. Biol.* **246**, 21–27
43. Yang, G., Liang, P.-H., and Dunaway-Mariano, D. (1994) *Biochemistry* **33**, 8327–8331
44. Crooks, G. P., Xu, L., Barkley, R. M., and Copley, S. D. (1995) *J. Am. Chem. Soc.* **117**, 10791–10798
45. Yang, G., Liu, R.-Q., Taylor, K. L., Xiang, H., Price, J., and Dunaway-Mariano, D. (1996) *Biochemistry* **36**, 10879–10885
46. Benning, M. M., Taylor, K. L., Liu, R.-Q., Yang, G., Xiang, H., Wesenberg, G., Dunaway-Mariano, D., and Holden, H. M. (1996) *Biochemistry* **35**, 8103–8109
47. Taylor, K. L., Liu, R.-Q., Liang, P.-H., Price, J., Dunaway-Mariano, D., Tonge, P. J., Clarkson, J., and Carey, P. R. (1995) *Biochemistry* **34**, 13881–13888
48. Clarkson, J., Tonge, P. J., Taylor, K. L., Dunaway-Mariano, D., and Carey, P. R. (1997) *Biochemistry* **36**, 10192–10199
49. Taylor, K. L., Xiang, H., Liu, R.-Q., Yang, G., and Dunaway-Mariano, D. (1997) *Biochemistry* **36**, 1349–1361
- 49a. Benning, M. M., Haller, T., Gerlt, J. A., and Holden, H. M. (2000) *Biochemistry* **39**, 4630–4639
- 49b. Xiang, H., Luo, L., Taylor, K. L., and Dunaway-Mariano, D. (1999) *Biochemistry* **38**, 7638–7652
- 49c. Babbitt, P. C., and Gerlt, J. A. (1997) *J. Biol. Chem.* **272**, 30591–30594
50. Sacchettini, J. C., Meininger, T., Rodrick, S., and Banaszak, L. J. (1986) *J. Biol. Chem.* **261**, 15183–15185
51. Bearne, S. L., and Wolfenden, R. (1995) *J. Am. Chem. Soc.* **117**, 9588–9589
52. Weaver, T., and Banaszak, L. (1996) *Biochemistry* **35**, 13955–13965
53. Weaver, T., Lees, M., and Banaszak, L. (1997) *Protein Sci.* **6**, 834–842
- 53a. Weaver, T., Lees, M., Zaitsev, V., Zaitseva, I., Duke, E., Lindley, P., McSweeney, S., Svensson, A., Keruchenko, J., Keruchenko, I., Gladilin, K., and Banaszak, L. (1998) *J. Mol. Biol.* **280**, 431–442
- 53b. Beeckmans, S., and Van Driessche, E. (1998) *J. Biol. Chem.* **273**, 31661–31669
54. Mohrig, J. R., Moerke, K. A., Cloutier, D. L., Lane, B. D., Person, E. C., and Onasch, T. B. (1995) *Science* **269**, 527–529
55. Rose, I. A. (1972) *CRC Critical Review of Biochemistry* **1**, 33–57
56. Brant, D. A., Barnett, L. B., and Alberty, R. A. (1963) *J. Am. Chem. Soc.* **85**, 2204–2209
57. Schmidt, D. E., Jr., Nigh, W. G., Tanzer, C., and Richards, J. H. (1969) *J. Am. Chem. Soc.* **91**, 5849–5854
58. Nigh, W. G., and Richards, J. H. (1969) *J. Am. Chem. Soc.* **91**, 5847–5848
59. Hansen, J. N., Dinovo, E. C., and Boyer, P. D. (1969) *J. Biol. Chem.* **244**, 6270–6279
60. Blanchard, J. S., and Cleland, W. W. (1980) *Biochemistry* **19**, 4506–4513
61. Porter, D. J. T., and Bright, H. J. (1980) *J. Biol. Chem.* **255**, 4772–4780
62. Jones, V. T., Lowe, G., and Potter, B. V. L. (1980) *Eur. J. Biochem.* **108**, 433–437
63. Botting, N. P., and Gani, D. (1992) *Biochemistry* **31**, 1509–1520
64. Shi, W., Dunbar, J., Jayasekera, M. M. K., Viola, R. E., and Farber, G. K. (1997) *Biochemistry* **36**, 9136–9144
65. Jayasekera, M. M. K., Shi, W., Farber, G. K., and Viola, R. E. (1997) *Biochemistry* **36**, 9145–9150
66. Goda, S. K., Minton, N. P., Botting, N. P., and Gani, D. (1992) *Biochemistry* **31**, 10747–10756
67. Weiss, P. M., Boerner, R. J., and Cleland, W. W. (1987) *J. Am. Chem. Soc.* **109**, 7201–7202
68. Anderson, S. R., Anderson, V. E., and Knowles, J. R. (1994) *Biochemistry* **33**, 10545–10555
- 68a. Vinarov, D. A., and Nowak, T. (1998) *Biochemistry* **37**, 15238–15246
69. Vinarov, D. A., and Nowak, T. (1999) *Biochemistry* **38**, 12138–12149
70. Zhang, E., Hatada, M., Brewer, J. M., and Lebioda, L. (1994) *Biochemistry* **33**, 6295–6300
71. Wedekind, J. E., Reed, G. H., and Rayment, I. (1995) *Biochemistry* **34**, 4325–4330
72. Duquerroy, S., Camus, C., and Janin, J. (1995) *Biochemistry* **34**, 12513–12523
73. Larsen, T. M., Wedekind, J. E., Rayment, I., and Reed, G. H. (1996) *Biochemistry* **35**, 4349–4358
- 73a. Hubbard, B. K., Koch, M., Palmer, D. R. J., Babbitt, P. C., and Gerlt, J. A. (1998) *Biochemistry* **37**, 14369–14375
- 73b. Thompson, T. B., Garrett, J. B., Taylor, E. A., Meganathan, R., Gerlt, J. A., and Rayment, I. (2000) *Biochemistry* **39**, 10662–10676
74. Linhardt, R. J., Turnbull, J. E., Wang, H. M., Loganathan, D., and Gallagher, J. T. (1990) *Biochemistry* **29**, 2611–2617
75. Desai, U. R., Wang, H.-M., and Linhardt, R. J. (1993) *Biochemistry* **32**, 8140–8145
76. Kiss, J. (1974) *Adv. Carbohydrate Chem. Biochem.* **29**, 229–303
- 76a. Li, S., Kelly, S. J., Lamani, E., Ferraroni, M., and Jedrzejewski, M. J. (2000) *EMBO J.* **19**, 1228–1240

References

- 76b. Ponnuraj, K., and Jedrzejewski, M. J. (2000) *J. Mol. Biol.* **299**, 885–895
77. Yoder, M. D., Keen, N. T., and Jurnak, F. (1993) *Science* **260**, 1503–1507
78. Cohen, F. E. (1993) *Science* **260**, 1444–1445
79. Kita, N., Boyd, C. M., Garrett, M. R., Jurnak, F., and Keen, N. T. (1996) *J. Biol. Chem.* **271**, 26529–26535
80. Kuo, D. J., and Rose, I. A. (1987) *Biochemistry* **26**, 7589–7596
- 80a. Lloyd, S. J., Lauble, H., Prasad, G. S., and Stout, C. D. (1999) *Protein Sci.* **8**, 2655–2662
81. Werst, M. M., Kennedy, M. C., Houseman, A. L. P., Beinert, H., and Hoffman, B. M. (1990) *Biochemistry* **29**, 10533–10540
82. Kilpatrick, L. K., Kennedy, M. C., Beinert, H., Czernuszewicz, R. S., Qiu, D., and Spiro, T. G. (1994) *J. Am. Chem. Soc.* **116**, 4053–4061
83. Lauble, H., Kennedy, M. C., Beinert, H., and Stout, C. D. (1992) *Biochemistry* **31**, 2735–2748
84. Kennedy, M. C., Emptage, M. H., and Beinert, H. (1984) *J. Biol. Chem.* **259**, 3145–3151
85. Kennedy, M. C., and Beinert, H. (1988) *J. Biol. Chem.* **263**, 8194–8198
86. Beinert, H., and Kennedy, M. C. (1993) *FASEB J.* **7**, 1442–1448
87. Kent, T. A., Emptage, M. H., Merkle, H., Kennedy, M. C., Beinert, H., and Münck, E. (1985) *J. Biol. Chem.* **260**, 6871–6881
88. Lauble, H., Kennedy, M. C., Beinert, H., and Stout, C. D. (1994) *J. Mol. Biol.* **237**, 437–451
89. Zheng, L., Kennedy, M. C., Beinert, H., and Zalkin, H. (1992) *J. Biol. Chem.* **267**, 7895–7903
90. Gruer, M. J., Artymiuk, P. J., and Guest, J. R. (1997) *Trends Biochem. Sci.* **22**, 3–6
91. Flint, D. H. (1993) *Biochemistry* **32**, 799–805
92. Flint, D. H., and McKay, R. G. (1994) *J. Am. Chem. Soc.* **116**, 5534–5539
93. Hofmeister, A. E. M., Grabowski, R., Linder, D., and Buckel, W. (1993) *Eur. J. Biochem.* **215**, 341–349
94. Grabowski, R., Hofmeister, A. E. M., and Buckel, W. (1993) *Trends Biochem. Sci.* **18**, 297–300
95. Hofmeister, A. E. M., Berger, S., and Buckel, W. (1992) *Eur. J. Biochem.* **205**, 743–749
96. Schroepfer, G. J., Jr. (1966) *J. Biol. Chem.* **241**, 5441–5447
97. Hill, R. K., and Newkome, G. R. (1969) *J. Am. Chem. Soc.* **91**, 5893–5894
98. Chiang, Y., and Kresge, A. J. (1991) *Science* **253**, 395–400
99. Vellom, D. C., Radic, Z., Li, Y., Pickering, N. A., Camp, S., and Taylor, P. (1993) *Biochemistry* **32**, 12–17
100. Gerlt, J. A., and Gassman, P. G. (1993) *Biochemistry* **32**, 11943–11952
101. Petsko, G. A., Kenyon, G. L., Gerlt, J. A., Ringe, D., and Kozarich, J. W. (1993) *Trends Biochem. Sci.* **18**, 372–376
102. Neidhart, D. J., Howell, P. L., Petsko, G. A., Powers, V. M., Li, R., Kenyon, G. L., and Gerlt, J. A. (1991) *Biochemistry* **30**, 9264–9273
103. St. Maurice, M., and Bearne, S. L. (2000) *Biochemistry* **39**, 13324–13335
104. Schafer, S. L., Barrett, W. C., Kallarakal, A. T., Mitra, B., Kozarich, J. W., Gerlt, J. A., Clifton, J. G., Petsko, G. A., and Kenyon, G. L. (1996) *Biochemistry* **35**, 5662–5669
105. Babbitt, P. C., Hasson, M. S., Wedekind, J. E., Palmer, D. R. J., Barrett, W. C., Reed, G. H., Rayment, I., Ringe, D., Kenyon, G. L., and Gerlt, J. A. (1996) *Biochemistry* **35**, 16489–16501
106. Babbitt, P. C., Mrachko, G. T., Hasson, M. S., Huisman, G. W., Kolter, R., Ringe, D., Petsko, G. A., Kenyon, G. L., and Gerlt, J. A. (1995) *Science* **267**, 1159–1161
107. Chiang, Y., Kresge, A. J., Pruszyński, P., Schepp, N. P., and Wirz, J. (1990) *Angew. Chem. Int. Ed. Engl.* **29**, 792–794
108. Guthrie, J. P., and Kluger, R. (1993) *J. Am. Chem. Soc.* **115**, 11569–11572
109. Gerlt, J. A., Kozarich, J. W., Kenyon, G. L., and Gassman, P. G. (1991) *J. Am. Chem. Soc.* **113**, 9667–9669
110. Gerlt, J. A., and Gassman, P. G. (1993) *J. Am. Chem. Soc.* **115**, 11552–11558
111. Guthrie, J. P. (1997) *J. Am. Chem. Soc.* **119**, 1151–1152
112. Tobin, J. B., and Frey, P. A. (1996) *J. Am. Chem. Soc.* **118**, 12253–12260
113. Yamauchi, T., Choi, S.-Y., Okada, H., Yohda, M., Kumagai, H., Esaki, N., and Soda, K. (1992) *J. Biol. Chem.* **267**, 18361–18364
114. Glavas, S., and Tanner, M. E. (1999) *Biochemistry* **38**, 4106–4113
115. Ho, H.-T., Falk, P. J., Ervin, K. M., Krishnan, B. S., Discotto, L. F., Dougherty, T. J., and Pucci, M. J. (1995) *Biochemistry* **34**, 2464–2470
- 115a. Hwang, K. Y., Cho, C.-S., Kim, S. S., Sung, H.-C., Yu, Y. G., and Cho, Y. (1999) *Nature Struct. Biol.* **6**, 422–426
116. Stein, T., Kluge, B., Vater, J., Franke, P., Otto, A., and Wittmann-Liebold, B. (1995) *Biochemistry* **34**, 4633–4642
117. Wiseman, J. S., and Nichols, J. S. (1984) *J. Biol. Chem.* **259**, 8907–8914
118. Shikata, Y., Watanabe, T., Teramoto, T., Inoue, A., Kawakami, Y., Nishizawa, Y., Katayama, K., and Kuwada, M. (1995) *J. Biol. Chem.* **270**, 16719–16723
119. Heck, S. D., Faraci, W. S., Kelbaugh, P. R., Saccomano, N. A., Thadeio, P. F., and Volkman, R. A. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 4036–4039
120. Johnson, J. D., Creighton, D. J., and Lambert, M. R. (1986) *J. Biol. Chem.* **261**, 4535–4541
121. Cooper, A. J. L., Gines, J. Z., and Meister, A. (1983) *Chem. Rev.* **83**, 321–358
122. Pirrung, M. C., Chen, J., Rowley, E. G., and McPhail, A. T. (1993) *J. Am. Chem. Soc.* **115**, 7103–7110
123. Noltmann, E. A. (1972) in *The Enzymes*, 3rd ed., Vol. 6 (Boyer, P. D., ed), pp. 271–354, Academic Press, New York
124. McGee, D. M., Hathaway, G. H., Palmieri, R. H., and Noltmann, E. A. (1980) *J. Mol. Biol.* **142**, 29–42
125. Mushegian, A. R., and Koonin, E. V. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 10268–10273
- 125a. Jeffery, C. J., Bahnson, B. J., Chien, W., Ringe, D., and Petsko, G. A. (2000) *Biochemistry* **39**, 955–964
126. Tanaka, K. R., and Paglia, D. E. (1995) in *The Metabolic and Molecular Bases of Inherited Disease* (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 3485–3511, McGraw-Hill, New York
127. Woodruff, W. W., III, and Wolfenden, R. (1979) *J. Biol. Chem.* **254**, 5866–5867
- 127a. Seemann, J. E., and Schulz, G. E. (1997) *J. Mol. Biol.* **273**, 256–268
128. Wallace, L. J., Eiserling, F. A., and Wilcox, G. (1978) *J. Biol. Chem.* **253**, 3717–3720
129. Bogumil, R., Kappl, R., Hüttermann, J., and Witzel, H. (1997) *Biochemistry* **36**, 2345–2352
130. Speck, J. C., Jr. (1958) *Adv. Carbohydrate Chem.* **13**, 63–103
131. Gibson, D. R., Gracy, R. W., and Hartman, F. C. (1980) *J. Biol. Chem.* **255**, 9369–9374
132. Balaban, R. S., and Ferretti, J. A. (1983) *Proc. Natl. Acad. Sci. U.S.A.* **80**, 1241–1245
133. Blacklow, S. C., Raines, R. T., Lim, W. A., Zamore, P. D., and Knowles, J. R. (1988) *Biochemistry* **27**, 1158–1167
134. Muirhead, H. (1983) *Trends Biochem. Sci.* **8**, 326–330
135. Lolis, E., Alber, T., Davenport, R. C., Rose, D., Hartman, F. C., and Petsko, G. A. (1990) *Biochemistry* **29**, 6609–6618
136. Joseph-McCarthy, D., Lolis, E., Komives, E. A., and Petsko, G. A. (1994) *Biochemistry* **33**, 2815–2823
137. Mande, S. C., Mainfroid, V., Kalk, K. H., Goraj, K., Martial, J. A., and Hol, W. G. J. (1994) *Protein Sci.* **3**, 810–821
138. Davenport, R. C., Bash, P. A., Seaton, B. A., Karplus, M., Petsko, G. A., and Ringe, D. (1991) *Biochemistry* **30**, 5821–5826
139. Verlinde, C. L. M. J., Witmans, C. J., Pijning, T., Kalk, K. H., Hol, W. G. J., Callens, M., and Oppendoor, F. R. (1992) *Protein Sci.* **1**, 1578–1584
140. Rose, I. A., Fung, W.-J., and Warms, J. V. B. (1990) *Biochemistry* **29**, 4312–4317
- 140a. Harris, T. K., Cole, R. N., Comer, F. I., and Mildvan, A. S. (1998) *Biochemistry* **37**, 16828–16838
141. Joseph-McCarthy, D., Rost, L. E., Komives, E. A., and Petsko, G. A. (1994) *Biochemistry* **33**, 2824–2829
142. Belasco, J. G., and Knowles, J. R. (1980) *Biochemistry* **19**, 472–477
143. Alston, W. C., II, Kanska, M., and Murray, C. J. (1996) *Biochemistry* **35**, 12873–12881
144. Nickbarg, E. B., Davenport, R. C., Petsko, G. A., and Knowles, J. R. (1988) *Biochemistry* **27**, 5948–5960
145. Lodi, P. J., Chang, L. C., Knowles, J. R., and Komives, E. A. (1994) *Biochemistry* **33**, 2809–2814
146. Raines, R. T., and Knowles, J. R. (1987) *Biochemistry* **26**, 7014–7020
147. Bash, P. A., Field, M. J., Davenport, R. C., Petsko, G. A., Ringe, D., and Karplus, M. (1991) *Biochemistry* **30**, 5826–5832
148. Lodi, P. J., and Knowles, J. R. (1991) *Biochemistry* **30**, 6948–6956
149. Alagona, G., Ghio, C., and Kollman, P. A. (1995) *J. Am. Chem. Soc.* **117**, 9855–9862
150. Pompliano, D. L., Peyman, A., and Knowles, J. R. (1990) *Biochemistry* **29**, 3186–3194
151. Yüksel, K. Ü., Sun, A.-Q., Gracy, R. W., and Schnackerz, K. D. (1994) *J. Biol. Chem.* **269**, 5005–5008
152. Williams, J. C., and McDermott, A. E. (1995) *Biochemistry* **34**, 8309–8319
153. Carrell, H. L., Rubin, B. H., Hurley, T. J., and Glusker, J. P. (1984) *J. Biol. Chem.* **259**, 3230–3236
154. Farber, G. K., Glasfeld, A., Tiraby, G., Ringe, D., and Petsko, G. A. (1989) *Biochemistry* **28**, 7289–7297
155. Collyer, C. A., Goldberg, J. D., Viehmann, H., Blow, D. M., Ramsden, N. G., Fleet, G. W. J., Montgomery, F. J., and Grice, P. (1992) *Biochemistry* **31**, 12211–12218
156. Jenkins, J., Janin, J., Rey, F., Chiadmi, M., van Tilbeurgh, H., Lasters, I., De Maeyer, M., Van Belle, D., Wodak, S. J., Lauwereys, M., Stanssens, P., Mrabet, N. T., Snauwaert, J., Matthyssens, G., and Lambey, A.-M. (1992) *Biochemistry* **31**, 5449–5458
157. Whitaker, R. D., Cho, Y., Cha, J., Carrell, H. L., Glusker, J. P., Karplus, P. A., and Batt, C. A. (1995) *J. Biol. Chem.* **270**, 22895–22906
158. Allen, K. N., Lavie, A., Petsko, G. A., and Ringe, D. (1995) *Biochemistry* **34**, 3742–3749
159. Hall, S. S., Dowsky, A. M., and Jordan, F. (1978) *J. Am. Chem. Soc.* **100**, 5934
160. Xue, L., Talalay, P., and Mildvan, A. S. (1991) *Biochemistry* **30**, 10858–10865
161. Brooks, B., and Benisek, W. F. (1994) *Biochemistry* **33**, 2682–2687
162. Austin, J. C., Zhao, Q., Jordan, T., Talalay, P., Mildvan, A. S., and Spiro, T. G. (1995) *Biochemistry* **34**, 4441–4447

References

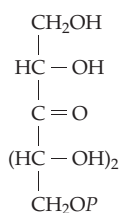
163. Zhao, Q., Li, Y.-K., Mildvan, A. S., and Talalay, P. (1995) *Biochemistry* **34**, 6562–6572
164. Hawkinson, D. C., Pollack, R. M., and Ambulos, N. P., Jr. (1994) *Biochemistry* **33**, 12172–12183
165. Viger, A., Coustal, S., and Marquet, A. (1983) *J. Am. Chem. Soc.* **103**, 451–458
166. Holman, C. M., and Benisek, W. F. (1994) *Biochemistry* **33**, 2672–2681
167. Wu, Z. R., Ebrahimian, S., Zawrotny, M. E., Thornburg, L. D., Perez-Alvarado, G. C., Brothers, P., Pollack, R. M., and Summers, M. F. (1997) *Science* **276**, 415–418
- 167a. Kim, S. W., Cha, S.-S., Cho, H.-S., Kim, J.-S., Ha, N.-C., Cho, M.-J., Joo, S., Kim, K. K., Choi, K. Y., and Oh, B.-H. (1997) *Biochemistry* **36**, 14030–14036
- 167b. Choi, G., Ha, N.-C., Kim, S. W., Kim, D.-H., Park, S., Oh, B.-H., and Choi, K. Y. (2000) *Biochemistry* **39**, 903–909
- 167c. Thornburg, L. D., Hénot, F., Bash, D. P., Hawkinson, D. C., Bartel, S. D., and Pollack, R. M. (1998) *Biochemistry* **37**, 10499–10506
168. Chen, L. H., Kenyon, G. L., Curtin, F., Harayama, S., Bembenek, M. E., Hajipour, G., and Whitman, C. P. (1992) *J. Biol. Chem.* **267**, 17716–17721
169. Subramanya, H. S., Roper, D. I., Dauter, Z., Dodson, E. J., Davies, G. J., Wilson, K. S., and Wigley, D. B. (1996) *Biochemistry* **35**, 792–802
170. Taylor, A. B., Czerwinski, R. M., Johnson, W. H., Jr., Whitman, C. P., and Hackert, M. L. (1998) *Biochemistry* **37**, 14692–14700
171. Stivers, J. T., Abeygunawardana, C., Mildvan, A. S., Hajipour, G., and Whitman, C. P. (1996) *Biochemistry* **35**, 814–823
172. Czerwinski, R. M., Harris, T. K., Johnson, W. H., Jr., Legler, P. M., Stivers, J. T., Mildvan, A. S., and Whitman, C. P. (1999) *Biochemistry* **38**, 12358–12366
173. Fitzgerald, M. C., Chernushevich, I., Standing, K. G., Kent, S. B. H., and Whitman, C. P. (1995) *J. Am. Chem. Soc.* **117**, 11075–11080
174. Lo, T. W. C., Westwood, M. E., McLellan, A. C., Selwood, T., and Thornalley, P. J. (1994) *J. Biol. Chem.* **269**, 32299–32305
175. Ridderström, M., Saccucci, F., Hellman, U., Bergman, T., Principato, G., and Mannervik, B. (1996) *J. Biol. Chem.* **271**, 319–323
176. Richard, J. P. (1991) *Biochemistry* **30**, 4581–4585
- 176a. Saadat, D., and Harrison, D. H. T. (2000) *Biochemistry* **39**, 2950–2960
177. Lan, Y., Lu, T., Lovett, P. S., and Creighton, D. J. (1995) *J. Biol. Chem.* **270**, 12957–12960
178. Cameron, A. D., Ridderström, M., Olin, B., Kavarana, M. J., Creighton, D. J., and Mannervik, B. (1999) *Biochemistry* **38**, 13480–13490
- 178a. He, M. M., Clugston, S. L., Honek, J. F., and Matthews, B. W. (2000) *Biochemistry* **39**, 8719–8727
179. Rae, C., O'Donoghue, S. I., Bubb, W. A., and Kuchel, P. W. (1994) *Biochemistry* **33**, 3548–3559
180. Bito, A., Haider, M., Hadler, I., and Breitenbach, M. (1997) *J. Biol. Chem.* **272**, 21509–21519
181. Mathieu, M., Modis, Y., Zeelen, J. P., Engel, C. K., Abagyan, R. A., Ahlberg, A., Rasmussen, B., Lamzin, V. S., Kunau, W. H., and Wierenga, R. K. (1997) *J. Mol. Biol.* **273**, 714–728
- 181a. Modis, Y., and Wierenga, R. K. (2000) *J. Mol. Biol.* **297**, 1171–1182
182. Roberts, J. R., Narasimhan, C., and Miziorko, H. M. (1995) *J. Biol. Chem.* **270**, 17311–17316
183. Narasimhan, C., Roberts, J. R., and Miziorko, H. M. (1995) *Biochemistry* **34**, 9930–9935
- 183a. Vinarov, D. A., and Miziorko, H. M. (2000) *Biochemistry* **39**, 3360–3368
- 183b. Richard, J. P., and Nagorski, R. W. (1999) *J. Am. Chem. Soc.* **121**, 4763–4770
184. Marsh, J. J., and Lebherz, H. G. (1992) *Trends Biochem. Sci.* **17**, 110–113
185. Cox, T. M. (1994) *FASEB J.* **8**, 62–71
186. Morris, A. J., and Tolan, D. R. (1994) *Biochemistry* **33**, 12291–12297
187. Sygusch, J., Beaudry, D., and Allaire, M. (1987) *Proc. Natl. Acad. Sci. U.S.A.* **84**, 7846–7850
188. Lai, C. Y., and Oshima, T. (1971) *Arch. Biochem. Biophys.* **144**, 363–
- 188a. Choi, K. H., Mazurkie, A. S., Morris, A. J., Utheza, D., Tolan, D. R., and Allen, K. N. (1999) *Biochemistry* **38**, 12655–12664
- 188b. Dalby, A., Dauter, Z., and Littlechild, J. A. (1999) *Protein Sci.* **8**, 291–297
189. Periana, R. A., Motiu-DeGroot, R., Chiang, Y., and Hupe, D. J. (1980) *J. Am. Chem. Soc.* **102**, 3923–3927
190. Beernink, P. T., and Tolan, D. R. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 5374–5379
191. Kadonaga, J. T., and Knowles, J. R. (1983) *Biochemistry* **22**, 130–136
- 191a. Hall, D. R., Leonard, G. A., Reed, C. D., Watt, C. I., Berry, A., and Hunter, W. N. (1999) *J. Mol. Biol.* **287**, 383–394
192. Qamar, S., Marsh, K., and Berry, A. (1996) *Protein Sci.* **5**, 154–161
193. Dreyer, M. K., and Schulz, G. E. (1996) *J. Mol. Biol.* **259**, 458–466
194. Lebioda, L., Hatada, M. H., Tulinsky, A., and Mavridis, I. M. (1982) *J. Mol. Biol.* **162**, 445–458
195. Meloche, H. P. (1981) *Trends Biochem. Sci.* **6**, 38–41
196. Dekker, E. E., and Kitson, R. P. (1992) *J. Biol. Chem.* **267**, 10507–10514
197. Wong, C.-H., Garcia-Junceda, E., Chen, L., Blanco, O., Gijzen, H. J. M., and Steensma, D. H. (1995) *J. Am. Chem. Soc.* **117**, 3333–3339
198. Jia, J., Schörken, U., Lindqvist, Y., Sprenger, G. A., and Schneider, G. (1997) *Protein Sci.* **6**, 119–124
199. Higgins, M. J. P., Kornblatt, J. A., and Rudney, H. (1972) in *The Enzymes*, 3rd ed., Vol. 7 (Boyer, P. D., ed), pp. 407–434, Academic Press, New York
200. Wiegand, G., and Remington, S. J. (1986) *Ann. Rev. Biophys. Biophys. Chem.* **15**, 97–117
201. Anderson, D. H., and Duckworth, H. W. (1988) *J. Biol. Chem.* **263**, 2163–2169
202. Wiegand, G., Remington, S., Deisenhofer, J., and Huber, R. (1984) *J. Mol. Biol.* **174**, 205–219
203. Rault-Leonardon, M., Atkinson, M. A. L., Slaughter, C. A., Moomaw, C. R., and Srere, P. A. (1995) *Biochemistry* **34**, 257–263
204. Russell, R. J. M., Ferguson, J. M. C., Hough, D. W., Danson, M. J., and Taylor, G. L. (1997) *Biochemistry* **36**, 9983–9994
205. Pereira, F. D., Donald, L. J., Hosfield, D. J., and Duckworth, H. W. (1994) *J. Biol. Chem.* **269**, 412–417
206. Karpusas, M., Holland, D., and Remington, S. J. (1991) *Biochemistry* **30**, 6024–6031
207. Chothia, C., and Lesk, A. M. (1985) *Trends Biochem. Sci.* **10**, 116–118
208. Evans, C. T., Kurz, L. C., Remington, S. J., and Srere, P. A. (1996) *Biochemistry* **35**, 10661–10672
209. Remington, S., Wiegand, G., and Huber, R. (1982) *J. Mol. Biol.* **158**, 111–152
210. Kurz, L. C., Roble, J. H., Nakra, T., Drysdale, G. R., Buzan, J. M., Schwartz, B., and Drueckhammer, D. G. (1997) *Biochemistry* **36**, 3981–3990
211. Kurz, L. C., and Drysdale, G. R. (1987) *Biochemistry* **26**, 2623–2627
212. Kurz, L. C., Ackerman, J. J. H., and Drysdale, G. R. (1985) *Biochemistry* **24**, 452–457
213. Clark, J. D., O'Keefe, S. J., and Knowles, J. R. (1988) *Biochemistry* **27**, 5961–5971
214. Wlassics, I. D., and Anderson, V. E. (1989) *Biochemistry* **28**, 1627–1633
215. Usher, K. C., Remington, S. J., Martin, D. P., and Drueckhammer, D. G. (1994) *Biochemistry* **33**, 7753–7759
- 215a. Kurz, L. C., Nakra, T., Stein, R., Plungkhen, W., Riley, M., Hsu, F., and Drysdale, G. R. (1998) *Biochemistry* **37**, 9724–9737
216. Schwartz, B., Drueckhammer, D. G., Usher, K. C., and Remington, S. J. (1995) *Biochemistry* **34**, 15459–15466
- 216a. Gu, Z., Drueckhammer, D. G., Kurz, L., Liu, K., Martin, D. P., and McDermott, A. (1999) *Biochemistry* **38**, 8022–8031
- 216b. Mulholland, A. J., Lyne, P. D., and Karplus, M. (2000) *J. Am. Chem. Soc.* **122**, 534–535
- 216c. Howard, B. R., Endrizzi, J. A., and Remington, S. J. (2000) *Biochemistry* **39**, 3156–3168
217. Smith, S. (1994) *FASEB J.* **8**, 1248–1259
218. Misra, I., Narasimhan, C., and Miziorko, H. M. (1993) *J. Biol. Chem.* **268**, 12129–12135
219. Misra, I., and Miziorko, H. M. (1996) *Biochemistry* **35**, 9610–9616
220. Guynn, R. W., and Veech, R. L. (1979) *J. Biol. Chem.* **254**, 1691–1698
221. Linn, T. C., and Srere, P. A. (1979) *J. Biol. Chem.* **254**, 1691–1698
222. Ranganathan, N. S., Linn, T. C., and Srere, P. A. (1982) *J. Biol. Chem.* **257**, 698–702
223. Elshourbagy, N. A., Near, J. C., Kmetz, P. J., Sathe, G. M., Southan, C., Strickler, J. E., Gross, M., Young, J. F., Wells, T. N. C., and Groot, P. H. E. (1990) *J. Biol. Chem.* **265**, 1430–1435
224. Pentyala, S. N., and Benjamin, W. B. (1995) *Biochemistry* **34**, 10961–10969
225. Hersch, L. B. (1973) *J. Biol. Chem.* **248**, 7295–7303
226. Nilekani, S., and SivaRaman, C. (1983) *Biochemistry* **22**, 4657–4663
227. Oppenheimer, N. J., Singh, M., Sweeley, C. C., Sung, S.-J., and Srere, P. A. (1979) *J. Biol. Chem.* **254**, 1000–1002
228. Ko, Y. H., Vanni, P., Munske, G. R., and McFadden, B. A. (1991) *Biochemistry* **30**, 7451–7456
229. Lenz, H., Buckel, W., Wunderwald, P., Biedermann, G., Buschmeier, V., Eggerer, H., Cornforth, J. W., Redmond, J. W., and Mallaby, R. (1971) *Eur. J. Biochem.* **24**, 207–215
230. Retej, J., Luthy, J., and Arigoni, D. (1970) *Nature (London)* **226**, 519–521
231. Kyte, J. (1995) *Mechanism in Protein Chemistry*, Garland Publ., New York (pp. 293–313)
232. Leussing, D. L., and Raghavan, N. V. (1980) *J. Am. Chem. Soc.* **102**, 5635–5643
233. Steinberger, R., and Westheimer, F. H. (1951) *J. Am. Chem. Soc.* **73**, 429–435
234. Kubala, G., and Martell, A. E. (1982) *J. Am. Chem. Soc.* **104**, 6602–6609
235. Waldrop, G. L., Braxton, B. F., Urbauer, J. L., Cleland, W. W., and Kiick, D. M. (1994) *Biochemistry* **33**, 5262–5267
236. Highbarger, L. A., Gerlt, J. A., and Kenyon, G. L. (1996) *Biochemistry* **35**, 41–46
- 236a. Stanley, T. M., Johnson, W. H., Jr., Burks, E. A., Whitman, C. P., Hwang, C.-C., and Cook, P. F. (2000) *Biochemistry* **39**, 718–726
237. Loeber, G., Infante, A. A., Maurer-Fogy, I., Krystek, E., and Dworkin, M. B. (1991) *J. Biol. Chem.* **266**, 3016–3021
238. Chou, W.-Y., Liu, M.-Y., Huang, S.-M., and Chang, G.-G. (1996) *Biochemistry* **35**, 9873–9879
239. Winning, B. M., Bourguignon, J., and Leaver, C. J. (1994) *J. Biol. Chem.* **269**, 4780–4786
240. Wei, C.-H., Chou, W.-Y., and Chang, G.-G. (1995) *Biochemistry* **34**, 7949–7954
241. Karsten, W. E., and Cook, P. F. (1994) *Biochemistry* **33**, 2096–2103

References

- 241a. Liu, D., Karsten, W. E., and Cook, P. F. (2000) *Biochemistry* **39**, 11955–11960
242. Edens, W. A., Urbauer, J. L., and Cleland, W. W. (1997) *Biochemistry* **36**, 1141–1147
- 242a. Gruys, K. J., and Sikorski, J. A. (1998) in *Comprehensive Biological Catalysis. A Mechanistic Reference*, Vol. I (Sinnott, M., ed), pp. 273–291, Academic Press, San Diego, California
243. Bolduc, J. M., Dyer, D. H., Scott, W. G., Singer, P., Sweet, R. M., Koshland, D. E., Jr., and Stoddard, B. L. (1995) *Science* **268**, 1312–1318
244. Berdis, A. J., and Cook, P. F. (1993) *Biochemistry* **32**, 2041–2046
245. Dean, A. M., and Dvorak, L. (1995) *Protein Sci.* **4**, 2156–2167
246. Wallon, G., Kryger, G., Lovett, S. T., Oshima, T., Ringe, D., and Petsko, G. A. (1997) *J. Mol. Biol.* **266**, 1016–1031
247. Brubaker, M. J., Dyer, D. H., Stoddard, B., and Koshland, D. E., Jr. (1996) *Biochemistry* **35**, 2854–2864
248. Tipton, P. A. (1993) *Biochemistry* **32**, 2822–2827
249. Kalckar, H. M. (1985) *Trends Biochem. Sci.* **10**, 132–133
250. Peliska, J. A., and O'Leary, M. H. (1991) *J. Am. Chem. Soc.* **113**, 1841–1842
251. Seeholzer, S. H., Jaworowski, A., and Rose, I. A. (1991) *Biochemistry* **30**, 727–732
252. Matte, A., Tari, L. W., Goldie, H., and Delbaere, L. T. J. (1997) *J. Biol. Chem.* **272**, 8105–8108
253. Hwang, S. H., and Nowak, T. (1986) *Biochemistry* **25**, 5590–5595
254. Konopka, J. M., Lardy, H. A., and Frey, P. A. (1986) *Biochemistry* **25**, 5571–5575
255. Hlavaty, J. J., and Nowak, T. (1997) *Biochemistry* **36**, 3389–3403
256. Matte, A., Goldie, H., Sweet, R. M., and Delbaere, L. T. J. (1996) *J. Mol. Biol.* **256**, 126–143
257. Arnelles, D. R., and O'Leary, M. H. (1992) *Biochemistry* **31**, 4363–4368
258. Puneekar, N. S., and Lardy, H. A. (1987) *J. Biol. Chem.* **262**, 6714–6719
259. Roseler, W. J., Vandenbark, G. R., and Hanson, R. H. (1989) *J. Biol. Chem.* **264**, 9657–9664
260. Rohrer, S. D., Saz, H. J., and Nowak, T. (1986) *J. Biol. Chem.* **261**, 13049–13055
261. Hebdo, C. A., and Nowak, T. (1982) *J. Biol. Chem.* **257**, 5503–5514
262. Hartman, F. C., and Harpel, M. R. (1994) *Ann. Rev. Biochem.* **63**, 197–234
263. Wasmann, C. C., Reiss, B., and Bohnert, H. J. (1988) *J. Biol. Chem.* **263**, 617–619
264. Shibata, N., Inoue, T., Fukuhara, K., Nagara, Y., Kitagawa, R., Harada, S., Kasai, N., Uemura, K., Kato, K., Yokota, A., and Kai, Y. (1996) *J. Biol. Chem.* **271**, 26449–26452
265. Tessier, L. H., Paulus, F., Keller, M., Vial, C., and Imbault, P. (1995) *J. Mol. Biol.* **245**, 22–33
266. Holzenburg, A., Mayer, F., Harauz, G., van Heel, M., Tokuoka, R., Ishida, T., Harata, K., Pal, G. P., and Saenger, W. (1987) *Nature (London)* **325**, 730–732
267. Erijman, L., Lorimer, G. H., and Weber, G. (1993) *Biochemistry* **32**, 5187–5195
268. Morse, D., Salois, P., Markovic, P., and Hastings, J. W. (1995) *Science* **268**, 1622–1624
269. Andersson, I. (1996) *J. Mol. Biol.* **259**, 160–174
- 269a. Schreuder, H. A., Knight, S., Curmi, P. M. G., Andersson, I., Cascio, D., Sweet, R. M., Brändén, C.-L., and Eisenberg, D. (1993) *Protein Sci.* **2**, 1136–1146
270. Newman, J., and Gutteridge, S. (1993) *J. Biol. Chem.* **268**, 25876–25886
271. Bassham, J. A., Benson, A. A., Kay, L. D., Harris, A. Z., Wilson, A. T., and Calvin, M. (1954) *J. Am. Chem. Soc.* **76**, 1760–1770
272. Siegel, M. I., and Lane, M. D. (1973) *J. Biol. Chem.* **248**, 5486–5498
- 272a. Taylor, T. C., and Andersson, I. (1996) *Nature Struct. Biol.* **3**, 95–101
- 272b. Harpel, M. R., Larimer, F. W., and Hartman, F. C. (1998) *Protein Sci.* **7**, 730–738
- 272c. Harpel, M. R., and Hartman, F. C. (1996) *Biochemistry* **35**, 13865–13870
- 272d. Duff, A. P., Andrews, T. J., and Curmi, R. M. G. (2000) *J. Mol. Biol.* **298**, 903–916
273. Pierce, J., Andrews, T. J., and Lorimer, G. H. (1986) *J. Biol. Chem.* **261**, 10248–10256
- 273a. Berry, J. A., Lorimer, G. H., Pierce, J., Seeman, J. R., Meeks, J., and Freas, S. (1987) *Proc. Natl. Acad. Sci. U.S.A.* **84**, 734–738
274. Jaworowski, A., Hartman, F. C., and Rose, I. A. (1984) *J. Biol. Chem.* **259**, 6783–6789
275. Saver, B. G., and Knowles, J. R. (1982) *Biochemistry* **21**, 5398–5403
276. Taylor, T. C., and Andersson, I. (1997) *J. Mol. Biol.* **265**, 432–444
277. Roeske, C. A., and O'Leary, M. H. (1984) *Biochemistry* **23**, 6275–6284
278. Van Dyke, D. E., and Schloss, J. V. (1986) *Biochemistry* **25**, 5145–5156
279. van de Loo, F. J., and Salvucci, M. E. (1996) *Biochemistry* **35**, 8143–8148
280. de Jiménez, E. S., Medrano, L., and Martínez-Barajas, E. (1995) *Biochemistry* **34**, 2826–2831
281. Larson, E. M., O'Brien, C. M., Zhu, G., Spreitzer, R. J., and Portis, A. R., Jr. (1997) *J. Biol. Chem.* **272**, 17033–17037
282. Mueller, D. D., Schmidt, A., Pappan, K. L., McKay, R. A., and Schaefer, J. (1995) *Biochemistry* **34**, 5597–5603
- 282a. Ogren, W. L. (1984) *Ann. Rev. Plant Physiol.* **35**, 415–442
283. Tse, J. M.-T., and Schloss, J. V. (1993) *Biochemistry* **32**, 10398–10403
- 283a. Sugawara, H., Yamamoto, H., Shibata, N., Inoue, T., Okada, S., Miyake, C., Yokota, A., and Kai, Y. (1999) *J. Biol. Chem.* **274**, 15655–15661
284. Harpel, M. R., Serpersu, E. H., Lamerdin, J. A., Huang, Z.-H., Gage, D. A., and Hartman, F. C. (1995) *Biochemistry* **34**, 11296–11306
- 284a. Gutteridge, S., Rhoades, D. F., and Herrmann, C. (1993) *J. Biol. Chem.* **268**, 7818–7824
285. Morell, M. K., Wilkin, J.-M., Kane, H. J., and Andrews, T. J. (1997) *J. Biol. Chem.* **272**, 5445–5451
286. Krebs, H. A., and Roughton, F. J. W. (1948) *Biochem. J.* **43**, 550–555
287. Kaziro, Y., Hass, L. F., Boyer, P. D., and Ochoa, S. (1962) *J. Biol. Chem.* **237**, 1460–1468
288. Cooper, T. G., Tchen, T. T., Wood, H. G., and Benedict, C. R. (1968) *J. Biol. Chem.* **243**, 3857–3863
289. González, D. H., and Andreo, C. S. (1988) *Biochemistry* **27**, 177–183
290. Harpster, M. H., and Taylor, W. C. (1986) *J. Biol. Chem.* **261**, 6132–6136
291. O'Leary, M. H., Rife, J. E., and Slater, J. D. (1981) *Biochemistry* **20**, 7308–7314
292. Hansen, D. E., and Knowles, J. R. (1982) *J. Biol. Chem.* **257**, 14795–14798
293. Fujita, N., Izui, K., Nishino, I., and Katsuki, H. (1984) *Biochemistry* **23**, 1774–1779
294. González, D. H., and Andreo, C. S. (1989) *Trends Biochem. Sci.* **14**, 24–27
295. Knowles, J. R. (1989) *Ann. Rev. Biochem.* **58**, 195–221
296. Janc, J. W., Urbauer, J. L., O'Leary, M. H., and Cleland, W. W. (1992) *Biochemistry* **31**, 6432–6440
297. Lee, S.-L., Hepburn, T. W., Swartz, W. H., Ammon, H. L., Mariano, P. S., and Dunaway-Matiano, D. (1992) *J. Am. Chem. Soc.* **114**, 7346–7354
298. McQueney, M. S., Lee, S.-I., Bowman, E., Mariano, P. S., and Dunaway-Mariano, D. (1989) *J. Am. Chem. Soc.* **111**, 6885–6887
299. Seidel, H. M., and Knowles, J. R. (1994) *Biochemistry* **33**, 5641–5646
300. Bowman, E., McQueney, M., Barry, R. J., and Dunaway-Mariano, D. (1988) *J. Am. Chem. Soc.* **110**, 5575–5576
301. Hidaka, T., and Seto, H. (1989) *J. Am. Chem. Soc.* **111**, 8012–8013
302. Freeman, S., Pollack, S. J., and Knowles, J. R. (1992) *J. Am. Chem. Soc.* **114**, 377–378
- 302a. Morais, M. C., Zhang, W., Baker, A. S., Zhang, G., Dunaway-Mariano, D., and Allen, K. N. (2000) *Biochemistry* **39**, 10385–10396
303. Mortimer, C. E., and Niehaus, W. G., Jr. (1974) *J. Biol. Chem.* **249**, 2833–2842
304. Clifford, K., Cornforth, J. W., Mallaby, R., and Phillips, G. T. (1971) *J. Chem. Soc. Chem. Commun.*, 1599–1600
305. Reardan, J. E., and Abeles, R. H. (1986) *Biochemistry* **25**, 5609–5616
306. Poulter, C. D., Muehlbacker, M., and Davis, D. R. (1989) *J. Am. Chem. Soc.* **111**, 3740–3742
307. Street, I. P., Coffman, H. R., Baker, J. A., and Poulter, C. D. (1994) *Biochemistry* **33**, 4212–4217
- 307a. Oh, S. K., Han, K. H., Ryu, S. B., and Kang, H. (2000) *J. Biol. Chem.* **275**, 18482–18488
308. Cane, D. E., Abell, C., Harrison, P. H., Hubbard, B. R., Kane, C. T., Lattman, R., Oliver, J. S., and Weiner, S. W. (1991) *Philos. Trans R Soc Lond B Biol Sci* **332**, 123–129
309. Cane, D. E., Pawlak, J. L., and Horak, R. M. (1990) *Biochemistry* **29**, 5476–5490
310. Aulabaugh, A., and Schloss, J. V. (1990) *Biochemistry* **29**, 2824–2830
311. Biou, V., Dumas, R., Cohen-Addad, C., Douce, R., Job, D., and Pebay-Peyroula, E. (1997) *EMBO J.* **16**, 3405–3415
312. Halgand, F., Dumas, R., Biou, V., Andrieu, J.-P., Thomazeau, K., Gagnon, J., Douce, R., and Forest, E. (1999) *Biochemistry* **38**, 6025–6034
313. Proust-De Martin, F., Dumas, R., and Field, M. J. (2000) *J. Am. Chem. Soc.* **122**, 7688–7697

Study Questions

1. Discuss the role of carbonyl groups in facilitating reactions of metabolism.
2. Write a step-by-step sequence showing the chemical mechanisms involved in the action of a type I aldolase that catalyzes cleavage of fructose 1,6-bisphosphate. The enzyme is inactivated by sodium borohydride in the presence of the substrate. Explain this inactivation.
3. Some methylotrophic bacteria dehydrogenate methanol to formaldehyde. The latter undergoes an aldol condensation to form a hexulose-6-phosphate:



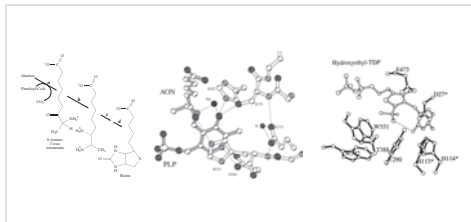
Write a reasonable cycle of biochemical reactions (utilizing this compound) by which three molecules of methanol can yield one molecule of glyceraldehyde 3-phosphate, a compound that can be either catabolized for energy or utilized for biosynthesis.

4. Malate is formed in the glyoxylate pathway by reaction of glyoxylate with Acetyl-CoA. Indicate the chemical mechanism of the reactions involved and structure of an intermediate species.
5. Acetyl-CoA is condensed with acetoacetyl-CoA to give an intermediate which is eventually converted to mevalonate.
 - a) Show the sequence of reactions leading from acetyl-CoA to mevalonate with the structures of important intermediates.
 - b) Mevalonate is an important intermediate on the pathway leading to what *class* of compounds?
6. a) Show the structures of the reactants for the 3-hydroxy-3-methylglutaryl-CoA synthase reaction.
 b) Free coenzyme A is liberated in the above reaction. From which of the two reactants did it come? Explain the metabolic significance of the liberation of free CoA.
7. Phosphoenolpyruvate carboxykinase catalyzes the following reaction:

$$\text{Oxaloacetate} + \text{GTP} \rightarrow \text{CO}_2 + \text{phosphoenolpyruvate} + \text{GDP}$$
 Illustrate the probable mechanism of the reaction.
8. Write a complete step-by-step mechanism for the action of ATP-citrate lyase which catalyzes the following reaction (Eq. 13-39):

$$\text{ATP} + \text{citrate} + \text{CoA-SH} \rightarrow \text{Acetyl-CoA} + \text{oxaloacetate} + \text{ADP} + \text{P}_i$$
9. Malonyl-CoA synthetase forms its product from free malonate and MgATP. Both phospho-enzyme and malonyl-enzyme intermediates have been detected. Suggest a sequential mechanism of action. See Kim, Y. S., and Lee, J. K. (1986) *J. Biol. Chem.* **261**, 16295–16297.
10. The carboxylation of acetone with HCO_3^- to form acetoacetate (the reverse of Eq. 13-44) is not a thermodynamically spontaneous process ($\Delta G^\circ \sim 17 \text{ kJ mol}^{-1}$). An anaerobic strain of *Xanthobacter* couples this reaction to the cleavage of ATP to AMP + 2 P_i . Suggest possible mechanisms. See Sluis, M. K., and Ensign, S. A. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 8456–8461.
11. The reaction of Eq. 10-11 begins with a hydroxylation, which can be viewed as insertion of an oxygen atom into a C–H bond. Draw the probable intermediate formed in this reaction and indicate how it can be converted to the final products.
12. Cyanase from *E. coli* decomposes cyanate (NCO^-) to ammonia and CO_2 . Isotopic labeling studies show that the second oxygen atom in the product CO_2 comes not from H_2O but from a second substrate, bicarbonate ion. The correct equation for the enzymatic process is:

$$\text{NCO}^- + \text{H}^+ + \text{HCO}_3^- \rightarrow \text{NH}_2 - \text{COO}^- + \text{CO}_2$$
 Propose a sequence of steps for this enzymatic reaction, which has been investigated by Anderson and coworkers. Johnson, W. V., and Anderson, P. M. (1987) *J. Biol. Chem.* **262**, 9021–9025 and Anderson, P. M., Korte, J. J., Holcomb, T. A., Cho, Y.-g, Son, C.-m, and Sung, Y.-c. (1994) *J. Biol. Chem.* **269**, 15036–15045.



The vitamin **biotin** is formed in nature (left) by condensation of L-alanine with pimeloyl-CoA to form 8-amino-7-oxononanoate (AON). This compound is seen at the upper left of the center structure joined as a Schiff base with the coenzyme **pyridoxal phosphate** (PLP). This is a product complex of the enzyme AON synthase (see Webster *et al.*, *Biochemistry* **39**, 516-528, 2000). Courtesy of D. Alexeev, R. L. Baxter, and L. Sawyer. Biotin synthesis requires three other enzymes (steps *b*, *c*, *d*). Step *b* is catalyzed by a PLP-dependent transaminase. At the left is **thiamin diphosphate**, in the form of its 2-(1-hydroxyethyl) derivative, an intermediate in the enzyme pyruvate decarboxylase (Dobritzsch *et al.*, *J. Biol. Chem.* **273**, 20196-20204, 1998). Courtesy of Guoguang Lu. Thiamin diphosphate functions in all living organisms to cleave C–C bonds adjacent to C=O groups.

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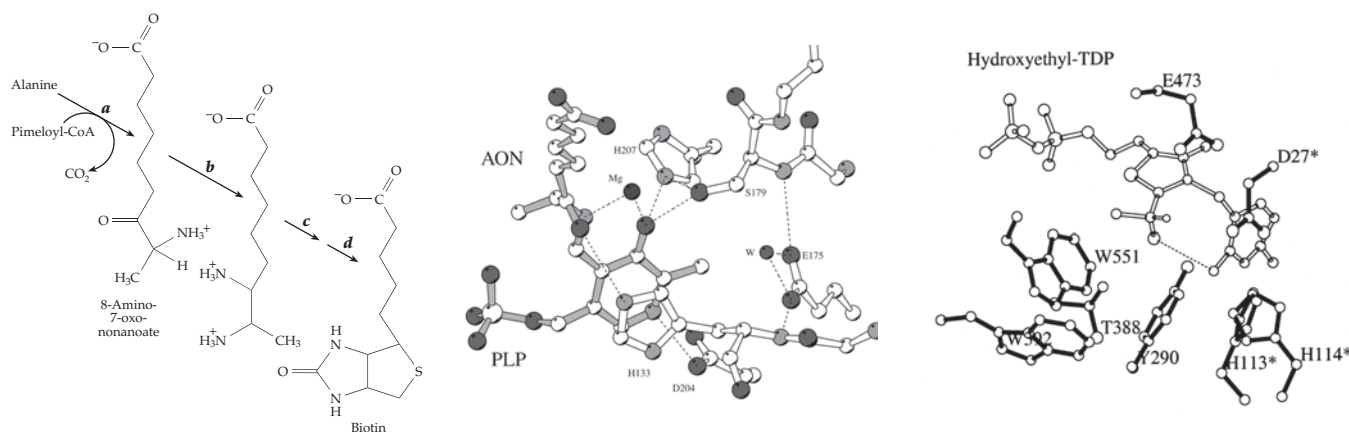
721	Box 14-A	Discovery of the Vitamins
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Coenzymes: Nature's Special Reagents

14



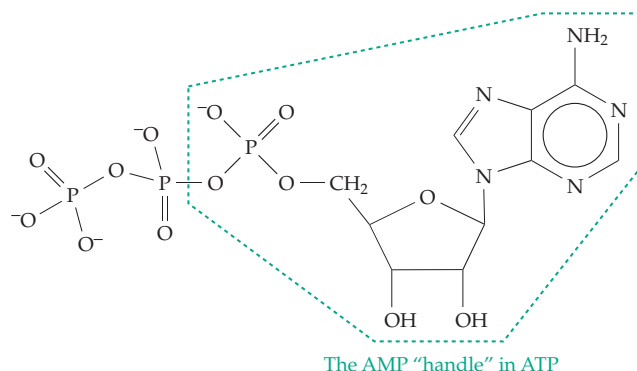
Most of the reactions discussed in Chapters 12 and 13 are catalyzed by enzymes that contain only those functional groups found in the side chains of the constituent amino acids. **Coenzymes** are nonprotein molecules that function as essential parts of enzymes. Coenzymes often serve as “special reagents” needed for reactions that would be difficult or impossible using only simple acid–base catalysis. In many instances, they also serve as **carriers**, alternating catalysts that accept and donate chemical groups, hydrogen atoms, or electrons. Coenzymes will be considered here in three groups:

1. Compounds of high group transfer potential such as ATP and GTP that function in energy coupling within cells. Because it is cleaved and then dissociates from the enzyme to which it is bound, ATP may be regarded as a substrate rather than a coenzyme. However, as a phosphorylated form of ADP it may also be viewed as a carrier of high-energy phospho groups.
2. Compounds, often derivatives of **vitamins** that, while in the active site of the enzyme, alter the structure of a substrate in a way that permits it to react more readily. Coenzyme A, pyridoxal phosphate, thiamin diphosphate, and vitamin B₁₂ coenzymes fall into this group.
3. Oxidative coenzymes with structures of precisely determined oxidation–reduction potential. Examples are NAD⁺, NADP⁺, FAD, and lipoic acid. They serve as carriers of hydrogen atoms or of

electrons. Some of these coenzymes, such as NAD⁺ and NADP⁺, can usually dissociate rapidly and reversibly from the enzymes with which they function. Others, including FAD, are much more tightly bound and rarely if ever dissociate from the protein catalyst. Heme groups are covalently linked to proteins such as cytochrome *c* and cannot be dissociated without destroying the enzyme. Very tightly bound coenzyme groups are often called **prosthetic groups**, but there is no sharp line that divides prosthetic groups from the loosely bound coenzymes. For example, NAD⁺ is bound weakly to some proteins but tightly to others. Oxidative coenzymes are discussed in Chapter 15.

A. ATP and the Nucleotide “Handles”

The role of ATP in “driving” biosynthetic reactions has been considered in Chapter 12, where attention was focused on the polyphosphate group which undergoes cleavage. What about the adenosine end? Here is a shapely structure borrowed from the nucleic acids. What is it doing as a carrier of phospho groups? At least part of the answer seems to be that the adenosine monophosphate (AMP) portion of the molecule is a “handle” which can be “grasped” by catalytic proteins. For some enzymes, such as acetyl-CoA synthetase (Eq. 12-45), the handle is important because the intermediate acyl adenylate must remain bound to the protein. Without the large adenosine group, there would be little for the protein to hold onto.



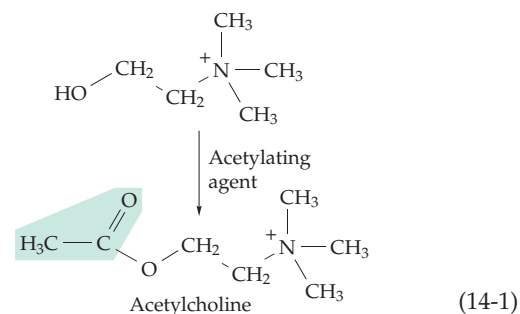
AMP is only one of several handles to which nature attaches phospho groups to form di- and triphosphate derivatives. Like AMP, the other handles are nucleotides, the monomer units of nucleic acids. Thus, one enzyme requiring a polyphosphate as an energy source selects ATP, another CTP, or GTP. The nucleotide handles not only carry polyphosphate groups but also are present in other coenzymes, such as CoA, NAD^+ , NADP^+ , and FAD. In addition, they serve as carriers for small organic molecules. For example, **uridine diphosphate glucose** (UDP-Glc) Chapter 10, is a carrier of active glucosyl groups important in sugar metabolism and **cytidine diphosphate choline** is an intermediate in synthesis of phospholipids.

Recalling that acetyl adenylate (acetyl-AMP) is an intermediate in synthesis of acetyl-CoA, and comparing the biosynthesis of sugars, phospholipids, and acetyl-CoA, we see that in each case the enzyme involved requires a different nucleotide handle. The handle may provide a means of recognition which can help an enzyme to pick the right bit of raw material out of the sea of molecules surrounding it. Figure 5-6 shows the shapes of the four purine and pyrimidine bases forming the most common nucleotide handles. The distinctive differences both in shape and in hydrogen bond patterns are obvious. In binding to proteins, the hydrogen bond-forming groups in the purine and pyrimidine bases sometimes interact with precisely positioned groups in the protein. However, in some enzymes the "handle" is not precisely bound. This seems to be the case for adenine, which makes surprisingly few hydrogen bonds to proteins.¹ Hydroxyl groups of the ribose or deoxyribose ring also often form hydrogen bonds and the negatively charged oxygen atoms of the 5'-phosphate may interact with positively charged protein side chains.

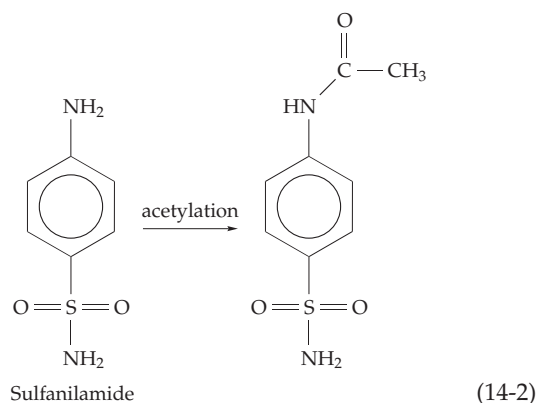
B. Coenzyme A and Phosphopantetheine

The existence of a special coenzyme required in biological acetylation was recognized by Fritz Lipmann in 1945.^{2-3a} The joining of acetyl groups to other

molecules is a commonplace reaction within living cells, one example being the formation of the neurotransmitter **acetylcholine**. In the laboratory acetylation is carried out with reactive compounds such as acetic anhydride or acetyl chloride.



Lipmann wondered what nature used in their place. His approach in seeking the biological "active acetate" is one that has been used successfully in solving many biochemical problems. He first set up a test system to examine the ability of extracts prepared from fresh liver tissue to catalyze the acetylation of sulfanilamide (Eq. 14-2). A specific color test was available for quantitative determination of very small amounts of the



product. The rate of acetylation of sulfanilamide under standard conditions was taken as a measure of the activity of the biochemical acetylation system. Lipmann soon discovered that the reaction required ATP and that the ATP was cleaved to ADP concurrently with the formation of acetyl-sulfanilamide. He also found that dialysis or ultrafiltration rendered the liver extract almost inactive in acetylation. Apparently some essential material passed out through the semipermeable dialysis membrane. When the dialysate or ultrafiltrate was concentrated and added back, acetylation activity was restored. The unknown material was not destroyed by boiling, and Lipmann postulated that it was a new coenzyme which he called **coenzyme A** (CoA). Now the test system was used to estimate the amount of the

BOX 14-A DISCOVERY OF THE VITAMINS

Several mysterious and often fatal diseases which resulted from vitamin deficiencies were prevalent until the past century. Sailors on long sea voyages were often the victims. In the Orient, the disease **beriberi** was rampant and millions died of its strange paralysis called “polyneuritis.” In 1840, George Budd predicted that the disease was caused by the lack of some organic compound that would be discovered in “a not too distant future.”^a In 1893, C. Eijkman, a Dutch physician working in Indonesia, observed paralysis in chicks fed white rice consumed by the local populace. He found that the paralysis could be relieved promptly by feeding an extract of rice polishings. This was one of the pieces of evidence that led the Polish biochemist Casimir Funk to formulate the “vitamine theory” in about 1912. Funk suggested that the diseases **beriberi**, **pellagra**, **rickets**, and **scurvy** resulted from lack in the diet of four different vital nutrients. He imagined them all to be amines, hence the name **vitamine**.

In the same year in England, F. G. Hopkins announced that he had fed rats on purified diets and discovered that amazingly small amounts of **accessory growth factors**, which could be obtained from milk, were necessary for normal growth.^b By 1915, E. V. McCollum and M. Davis at the University of Wisconsin had recognized that rat growth depended on not one but at least two accessory factors. The first, soluble in fatty solvents, they called **A**, and the other, soluble in water, they designated **B**. Factor B cured beriberi in chicks. Later, when it was shown that vitamine A was not an amine, the “e” was dropped and **vitamin** became a general term.

Progress in isolation of the vitamins was slow, principally because of a lack of interest. According to R. R. Williams, when he started his work on isolation of the antiberiberi factor in 1910 most people were convinced that his efforts were doomed to failure, so ingrained was Pasteur’s idea that diseases were caused only by bacteria. In 1926, Jansen isolated a small amount of thiamin, but it was not until 1933 that Williams, working almost without financial support, succeeded in preparing a large amount of the crystalline compound from rice polishings. Characterization and synthesis followed rapidly.^{c,d}

It was soon apparent that the new vitamin alone would not satisfy the dietary need of rats for the B factor. A second thermostable factor (**B₂**) was required in addition to thiamin (**B₁**), which was labile and easily destroyed by heating. When it became clear that factor **B₂** contained more than one component, it was called **vitamin B complex**. There was some confusion until relatively specific animal tests for each one of the members had been devised.

Riboflavin was found to be most responsible for the stimulation of rat growth, while **vitamin B₆** was needed to prevent a facial dermatitis or “rat pellagra.” **Pantothenic acid** was especially effective in curing a chick dermatitis, while **nicotinamide** was required to cure human pellagra. **Biotin** was required for growth of yeast.

The antiscorvy (antiscorbutic) activity was called **vitamin C**, and when its structure became known it was called **ascorbic acid**. The fat-soluble factor preventing rickets was designated **vitamin D**. By 1922, it was recognized that another fat-soluble factor, **vitamin E**, is essential for full-term pregnancy in the rat. In the early 1930s **vitamin K** and the **essential fatty acids** were added to the list of fat-soluble vitamins. Study of the human blood disorders “tropical macrocytic anemia” and “pernicious anemia” led to recognition of two more water-soluble vitamins, **folic acid** and **vitamin B₁₂**. The latter is required in minute amounts and was not isolated until 1948. Have all the vitamins been discovered? Rats can be reared on an almost completely synthetic diet. However, there is the possibility that for good health humans require some as yet undiscovered compounds in our diet. Furthermore, it is quite likely that we receive some essential nutrients that we cannot synthesize from bacteria in our intestinal tracts. An example may be the pyrroloquinoline quinone (PQQ).^e

Why do we need vitamins? Early clues came in 1935 when nicotinamide was found in NAD⁺ by H. von Euler and associates and in NADP⁺ by Warburg and Christian. Two years later, K. Lohman and P. Schuster isolated pure **coccarboxylase**, a dialyzable material required for decarboxylation of pyruvate by an enzyme from yeast. It was shown to be thiamin diphosphate (Fig. 15-3). Most of the water-soluble vitamins are converted into coenzymes or are covalently bound into active sites of enzymes. Some lipid-soluble vitamins have similar functions but others, such as vitamin D and some metabolites of vitamin A, act more like hormones, binding to receptors that control gene expression or other aspects of metabolism.

^a Hughes, R. E. (1973) *Medical History* **17**, 127–134

^b Harris, L. D. (1937) *Vitamins in Theory and Practice*, Cambridge Univ. Press, London and New York

^c Karlson, P. (1984) *Trends Biochem. Sci.* **9**, 536–537

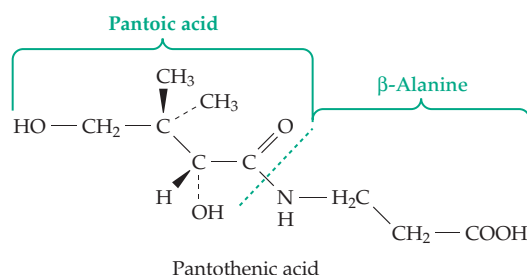
^d Williams, R. R., and Spies, T. D. (1938) *Vitamin B₁ and Its Use in Medicine*, Macmillan, New York

^e Killgore, J., Smidt, C., Duich, L., Romero-Chapman, N., Tinker, D., Reiser, K., Melko, M., Hyde, D., and Rucker, R. B. (1989) *Science* **245**, 850–852

coenzyme in a given volume of dialysate or in any other sample. When small amounts of CoA were supplied to the test system, only partial restoration of the acetylation activity was observed and the amount of restoration was proportional to the amount of CoA. With test system in hand to monitor various fractionation methods, Lipmann soon isolated the new coenzyme in pure form from yeast and liver.

Coenzyme A (Fig. 14-1) is a surprisingly complex molecule. The handle is AMP with an extra phospho group on its 3'-hydroxyl. The phosphate of the 5'-carbon is linked in anhydride (pyrophosphate) linkage to another phosphoric acid, which is in turn esterified with **pantoic acid**. Pantoic acid is linked to **β -alanine** and the latter to **β -mercaptoethylamine** through amide linkages, the reactive SH group being attached to a long (1.9-nm) semi-flexible chain. Coenzyme A can be cleaved by hydrolysis to **pantetheine** (Fig. 14-1), **pantetheine 4'-phosphate**, and **pantothenic acid**. These three compounds are all **growth factors**.

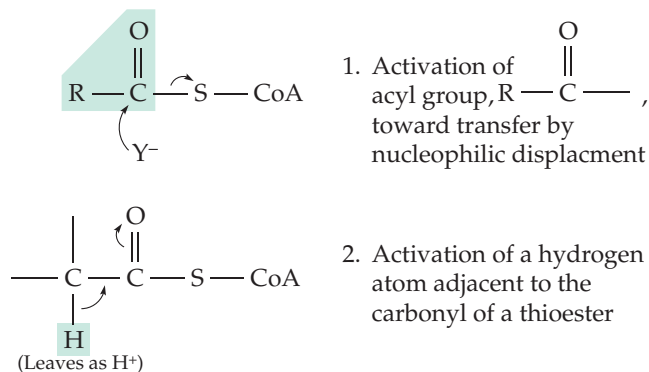
Pantothenic acid is a vitamin, which is essential to human life. Its name is derived from a Greek root that reflects its universal occurrence in living things. The bacterium *Lactobacillus bulgaricus*, which converts milk



to yogurt, needs the more complex pantetheine for growth. It finds a ready supply of pantetheine in milk and has lost its ability to synthesize this compound. However, it can convert pantetheine to CoA. Pantetheine 4'-phosphate is required for growth of *Acetobacter suboxydans*. While it is not a dietary essential for most organisms, it is found in a covalently bound form in several enzymes.

While CoA was discovered as the "acetylation coenzyme," it has a far more general function. It is required, in the form of acetyl-CoA, to catalyze the synthesis of citrate in the citric acid cycle. It is essential to the β oxidation of fatty acids and carries propionyl and other acyl groups in a great variety of other metabolic reactions. About 4% of all known enzymes require CoA or one of its esters as a substrate.⁴

Coenzyme A has two distinctly different biochemical functions, which have already been considered briefly in Chapters 12 and 13 and can be summarized as follows:



These functions depend, to a considerable extent, on the fact that the properties of the carbonyl group of a thioester are closely similar to those of an isolated carbonyl group in a ketone.⁵

Synthesis of fatty acids in bacteria requires a small **acyl carrier protein (ACP)** whose functions are similar to those of CoA. However, it contains pante-

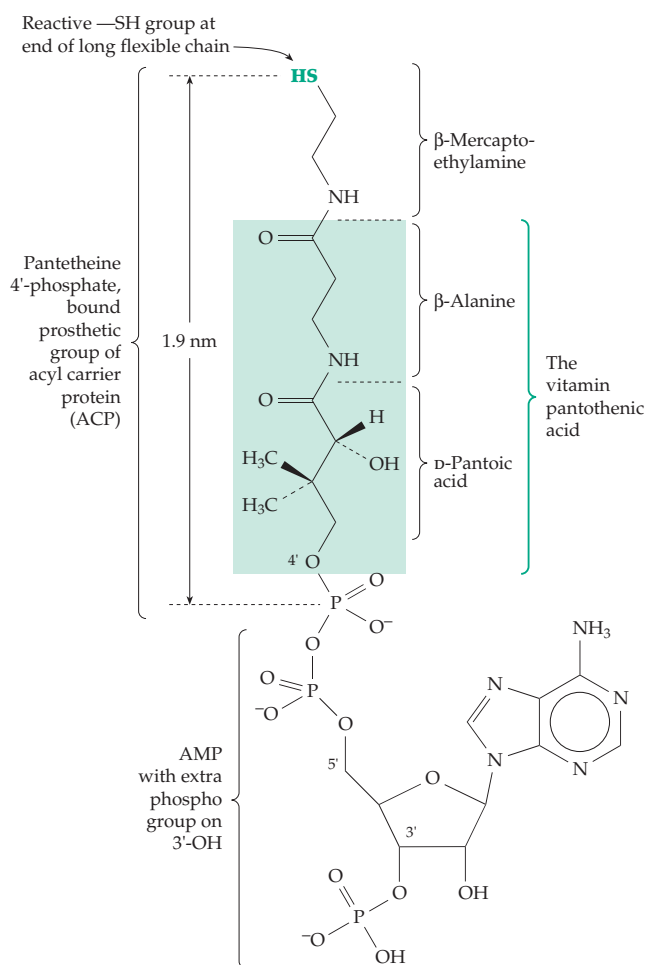
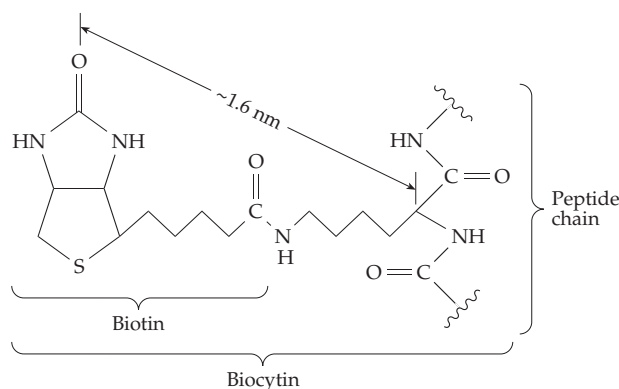


Figure 14-1 Coenzyme A, an acyl-activating coenzyme containing the vitamin pantothenic acid.

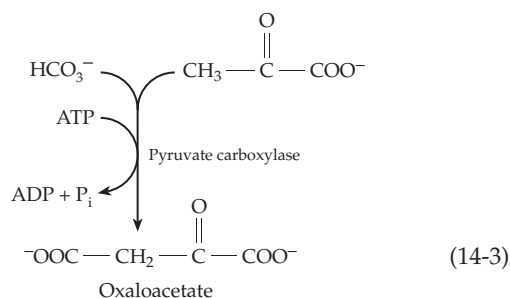
synthase that catalyzes step *a* (banner) and a transaminase that catalyzes step *b*, contain the coenzyme pyridoxal phosphate (PLP).^{20b} The final step is insertion of a sulfur atom from an iron-sulfur center.

1. Biotin-Containing Enzymes

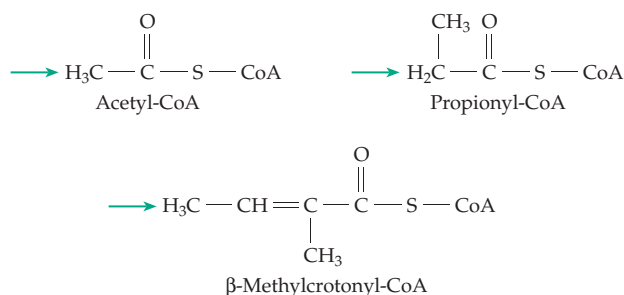
Within cells the biotin is covalently bonded to proteins, its double ring being attached to a 1.6-nm flexible arm. The first clue to this fact was obtained from isolation of a biotin-containing material **biocytin**, ϵ -N-biotinyl-L-lysine, from autolysates (self-digests) of rapidly growing yeast.²¹ It was subsequently shown that the lysine residue of the biocytin was originally present in proteins at the active sites of biotin-containing enzymes usually within the sequence AMKM²² or VMKM.²³ Other conserved features also mark the attachment site.²⁴



Biotin acts as a **carboxyl group carrier** in a series of carboxylation reactions, a function originally suggested by the fact that aspartate partially replaces biotin in promoting the growth of the yeast *Torula cremonis*. Aspartate was known to arise by transamination from oxaloacetate, which in turn could be formed by carboxylation of pyruvate. Subsequent studies showed that biotin was needed for an enzymatic ATP-dependent reaction of pyruvate with bicarbonate ion to form oxaloacetate (Eq. 14-3). This is a β carboxylation coupled to the hydrolysis of ATP.



In addition to pyruvate carboxylase, other biotin-requiring enzymes act on **acetyl-CoA**, **propionyl-CoA**, and **β -methylcrotonyl-CoA**, using HCO_3^- to add carboxyl groups at the sites indicated by the arrows in the accompanying structures. Because of the presence of the C=C double bond conjugated with the carbonyl group, the carboxylation of β -methylcrotonyl-CoA is electronically analogous to β carboxylation.



Human cells, as well as those of other higher eukaryotes, carry genes for all four of these enzymes.²⁵⁻²⁹ Acetyl-CoA carboxylase is a cytosolic enzyme needed for synthesis of fatty acids but the other three enzymes enter mitochondria when they function. The biotin-dependent carboxylases, which are listed in Table 14-1, have a variety of molecular sizes and subunits but show much evolutionary conservation in their sequences and chemical mechanisms.^{22,25} In higher eukaryotes all of the catalytic apparatus of the enzymes is present in single large 190- to 200-kDa subunits. The 251-kDa subunit of yeast acetyl-CoA carboxylase consists of 2337 amino acid residues.³⁰ That of rats contains 2345³¹ and that of the alga *Cyclotella* 2089.³² Human cytosolic acetyl-Co carboxylase has 2347 residues while a mitochondria-associated form has an extra 136 residues, most of them in a hydrophobic N-terminal extension.^{32a,b} Plants have two forms of the enzyme, one cytosolic and one located in plastids. In wheat, they have 2260 and 2311 residues, respectively.^{32c,32d} Animal and fungal pyruvate carboxylases are also large $\sim$ 500-kDa tetramers.^{32e} The yeast enzyme consists of 1178-residue monomers.³³ In contrast, the 560-kDa human propionyl-CoA carboxylase is an $\alpha_4\beta_4$ tetramer.^{34,35}

In bacteria and in at least some plant chloroplasts,^{36,37} acetyl-CoA carboxylase consists of three different kinds of subunit and four different peptide chains. The much studied *E. coli* enzyme is composed of a 156-residue **biotin carboxyl carrier protein**,³⁸ a 449-residue **biotin carboxylase**, whose three-dimensional structure is known,^{39,39a} and a **carboxyltransferase** subunit consisting of 304 (α)- and 319 (β)- residue chains. These all associate as a dimer of the three subunits (eight peptide chains).⁴⁰⁻⁴²

Biotin becomes attached to the proper ϵ -amino groups at the active centers of biotin enzymes by the action of **biotin holoenzyme synthetase** (biotinyl

protein ligase), which utilizes ATP to form an intermediate biotinyl-AMP.^{43–47a} Hereditary deficiency of this enzyme has been observed in a few children and has been treated by administration of extra biotin.⁴⁸ The *E. coli* biotin holoenzyme synthetase, whose three-dimensional structure is known, has a dual function. It is also a repressor of transcription of the biotin biosynthetic operon.⁴⁵ Intracellular degradation of biotin-containing proteins yields biotin-containing oligopeptides as well as biocytin. These are acted on by **biotinidase** to release free biotin.^{49,50} The action of this enzyme in recycling biotin may be a controlling factor in the rate of formation of new biotin-dependent enzymes.

2. The Mechanism of Biotin Action

It may seem surprising that a coenzyme is needed for these carboxylation reactions. However, unless the cleavage of ATP were coupled to the reactions, the equilibria would lie far in the direction of decarboxylation. For example, the measured apparent equilibrium constant K' for conversion of propionyl-CoA to *S* methylmalonyl-CoA at pH 8.1 and 28°C⁵¹ is given by Eq. 14-4.

$$K' = \frac{[\text{ADP}][\text{P}_i][\text{methylmalonyl-CoA}]}{[\text{ATP}][\text{HCO}_3^-][\text{propionyl-CoA}]} = 5.7$$

$$\Delta G' = -4.36 \text{ kJ mol}^{-1} \quad (14-4)$$

TABLE 14-1
Enzymes Containing Bound Biotin

1. Catalyzing beta carboxylation using HCO_3^- with coupled cleavage of ATP to $\text{ADP} + \text{P}_i$

Acetyl-CoA carboxylase
Propionyl-CoA carboxylase
Pyruvate carboxylase
 β -Methylcrotonyl-CoA carboxylase
(δ carboxylation)

2. Carboxyl group transfer without cleavage of ATP

Carboxyltransferase of *Propionobacterium*

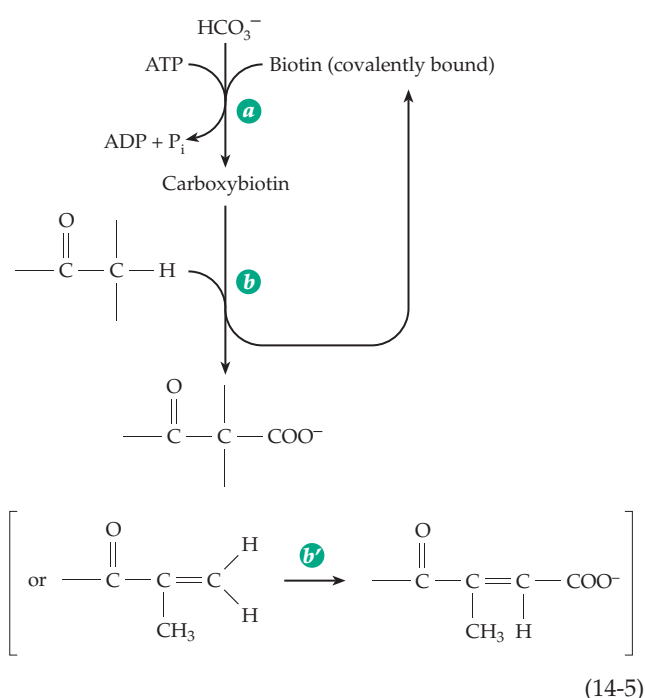
3. Biotin-dependent Na^+ pumps

Oxaloacetate decarboxylase
Methylmalonyl-CoA decarboxylase
Glutaconyl-CoA decarboxylase

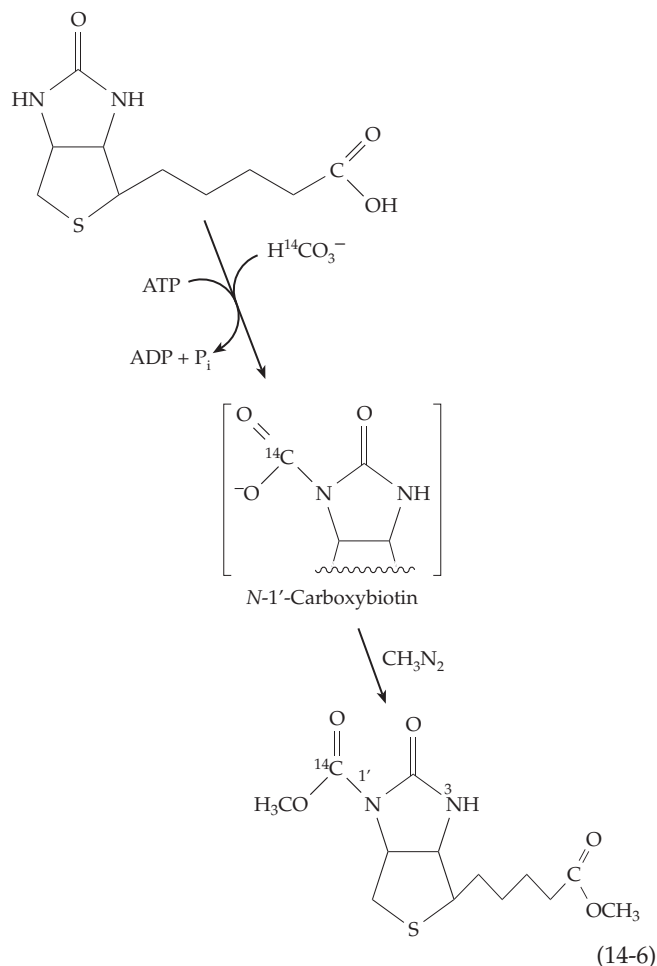
4. Other

Malonate decarboxylase
Urea carboxylase

The function of biotin is to mediate the coupling of ATP cleavage to the carboxylation, making the overall reaction exergonic. This is accomplished by a two-stage process in which a **carboxybiotin** intermediate is formed (Eq. 14-5). There is *one known biotin-containing enzyme that does not utilize ATP*. Propionic acid bacteria contain a **carboxyltransferase** which transfers a carboxyl group reversibly from methylmalonyl-CoA to pyruvate to form oxaloacetate and propionyl-CoA (see Fig. 17-10). This huge enzyme consists of a central hexameric core of large 12S subunits to which six 5S dimeric subunits and twelve 123-residue biotinylated peptides are attached.^{25,52} No ATP is needed because free HCO_3^- is not a substrate. However, biotin serves as the carboxyl group carrier in this enzyme too.



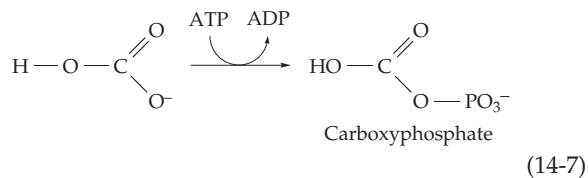
Carboxybiotin. The structure of biotin suggested that bicarbonate might be incorporated reversibly into its position 2'. However, this proved not to be true and it remained for F. Lynen and associates to obtain a clue from a "model reaction." They showed that purified β -methylcrotonyl-CoA carboxylase promoted the carboxylation of *free* biotin with bicarbonate ($\text{H}^{14}\text{CO}_3^-$) and ATP. While the carboxylated biotin was labile, treatment with diazomethane (Eq. 14-6) gave a stable dimethyl ester of **N-1'-carboxybiotin**.^{53,54} The covalently bound biotin at active sites of enzymes was also successfully labeled with $^{14}\text{CO}_2$. Treatment of the labeled enzymes with diazomethane followed by hydrolysis with trypsin and pepsin gave authentic N-1'-carboxybiocytin. It was now clear that the cleavage of ATP is required to couple the CO_2 from HCO_3^- to the biotin to form carboxybiotin. The enzyme must



then transfer the carboxyl group from carboxybiotin to the substrate that is to be carboxylated. Enzymatic transfer of a carboxyl group from chemically synthesized carboxybiotin onto specific substrates confirmed the proposed mechanism.⁵⁵

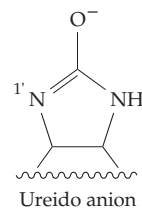
The biotin carboxyl carrier subunit of *E. coli* acetyl-CoA carboxylase contains the covalently bound biotin.^{55a,b} The larger biotin carboxylase subunit catalyzes the ATP-dependent attachment of CO₂ to the biotin and the carboxyltransferase subunit catalyzes the final transcarboxylation step (Eq. 14-5, step *b*) by which acetyl-CoA is converted into malonyl-CoA. The biotin, which is attached to the carrier protein, is presumably able to move by means of its flexible arm from a site on the carboxylase to a site on the transcarboxylase.

Carboxyphosphate. During the initial carboxylation step ¹⁸O from labeled bicarbonate enters the P_i that is split from ATP. This suggested transient formation of **carboxyphosphate** by nucleophilic attack of HCO₃⁻ on ATP (Eq. 14-7). The carboxyl group of this reactive mixed anhydride⁵⁶ could then be transferred to biotin. This mechanism is supported by the fact that biotin carboxylase catalyzes the transfer of a

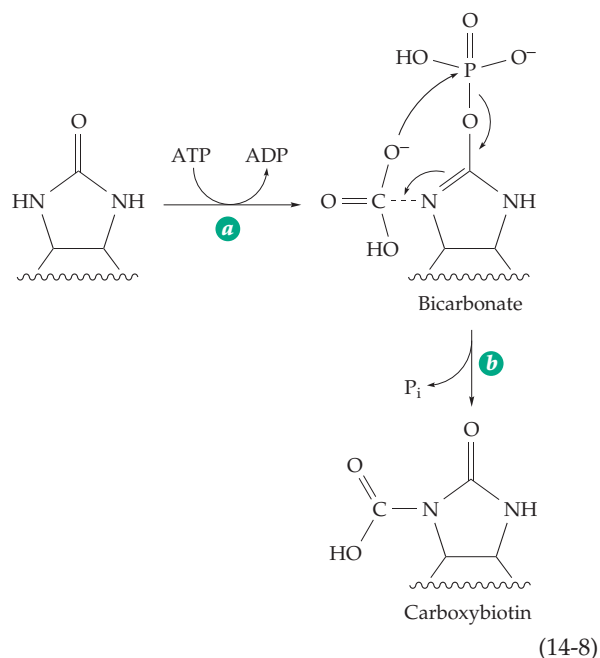


phospho group to ADP from carbamoyl phosphate, an analog of carboxyphosphate in a reaction that is analogous to the reverse of that in Eq. 14-7 and also by a slow bicarbonate-dependent ATPase activity that does not depend upon biotin.^{57,58,58a}

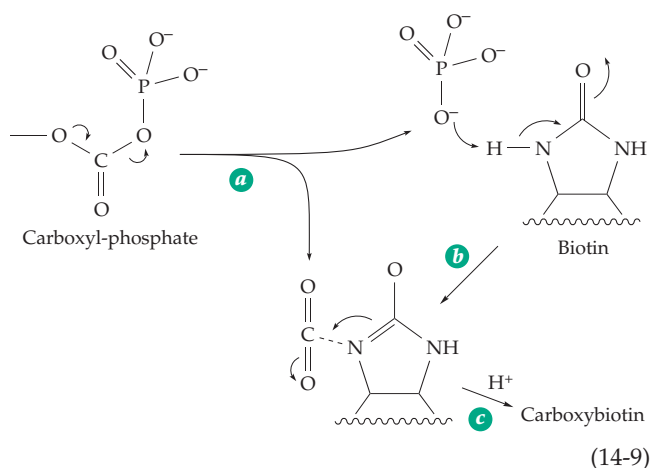
The simplest mechanism for transfer of the carboxyl group of carboxyphosphate to biotin would appear to be nucleophilic displacement of the phosphate leaving group by N1' of biotin. The enzyme could presumably first catalyze removal of the N1' hydrogen to form a ureido anion.⁵⁹ Another reasonable possibility would be for the terminal phospho group of ATP to be transferred to biotin to form an O-phosphate^{60,61}



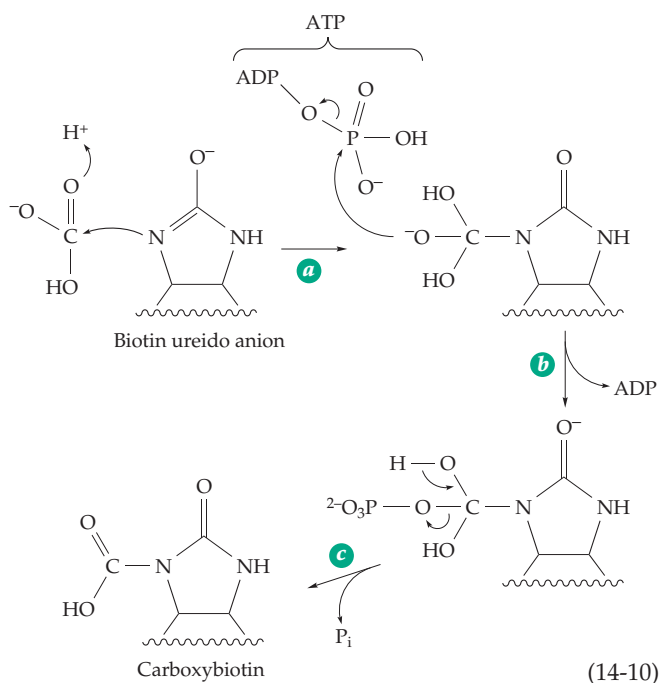
(Eq. 14-8, step *a*) which could react with bicarbonate as in Eq. 14-8, step *b*. Cleavage of the enol phosphate by attack of HCO₃⁻ would simultaneously create a nucleophilic center at N1' and carboxyphosphate ready to react with N1'. However, this could not easily explain the ATPase activity in the absence of biotin.



Either of the foregoing mechanisms requires that the ureido anion of biotin attack the rather unreactive carbon atom of carboxyphosphate. Another alternative, which is analogous to that suggested for PEP carboxylase (Eq. 13-53) is for carboxyphosphate to eliminate inorganic phosphate to give the more electrophilic CO_2 (Eq. 14-9, step *a*). The very basic inorganic phosphate trianion PO_4^{3-} that is eliminated could remove the proton from N1' of biotin to create the biotin ureido anion (step *b*), which could then add to CO_2 (step *c*).²²

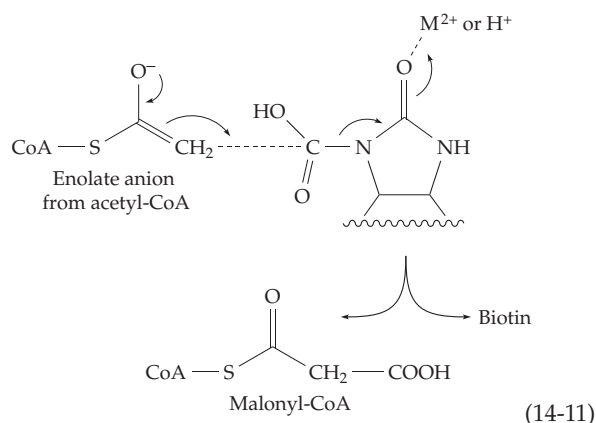


Have we checked all of the possibilities for the mechanism of biotin carboxylation? Kruger and associates^{62,63} suggested that biotin, as a ureido anion, might add to bicarbonate to form a highly unstable intermediate which, however, could be phosphorylated by ATP (Eq. 14-10, steps *a* and *b*). This intermediate could undergo elimination of inorganic phosphate

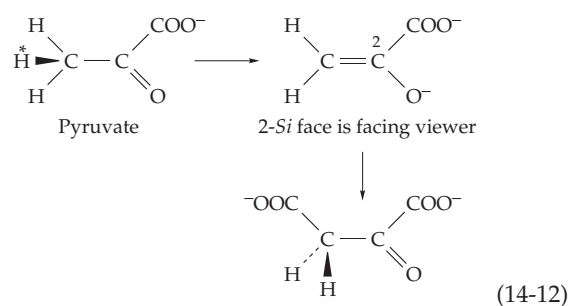


(step *c*), driving the reaction to completion. The observed transfer of isotope from ^{18}O -containing bicarbonate into ADP would be observed.

The β carboxylation step. Once formed, the carboxybiotin "head group" could swing to the carboxyltransferase site where transfer of the carboxyl group into the final product takes place. This might occur either by nucleophilic attack of an enolate anion on the carbonyl carbon (Eq. 14-11) or on CO_2 generated by reversal of the reactions of Eq. 14-9, steps *b* and *c*.



When pyruvate with a chiral methyl group is carboxylated by pyruvate carboxylase the configuration at C-3 is retained. The carboxyl enters from the 2-*si* side, the same side from which the proton (marked H^*) was removed to form the enolate anion (Eq. 14-12). Comparable stereochemistry has been established for other biotin-dependent enzymes.^{64,65}

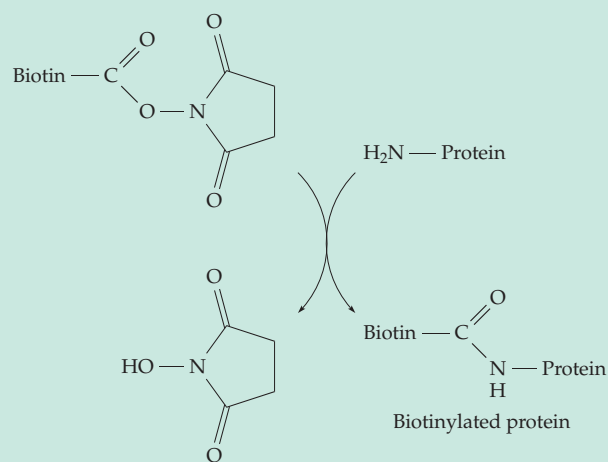


These enzymes do not catalyze any proton exchange at C-3 of pyruvate or at C-2 of an acyl-CoA unless the biotin is first carboxylated. This suggested that removal of the proton to the biotin oxygen and carboxylation might be synchronous. However, ^{13}C and ^2H kinetic isotope effects and studies of ^3H exchange⁶⁶ support the existence of a discrete enolate anion intermediate as shown in Eq. 14-11.^{65,67} This mechanism is also consistent with the observation that propionyl-CoA

BOX 14-B THE BIOTIN-BINDING PROTEINS AVIDIN AND STREPTAVIDIN

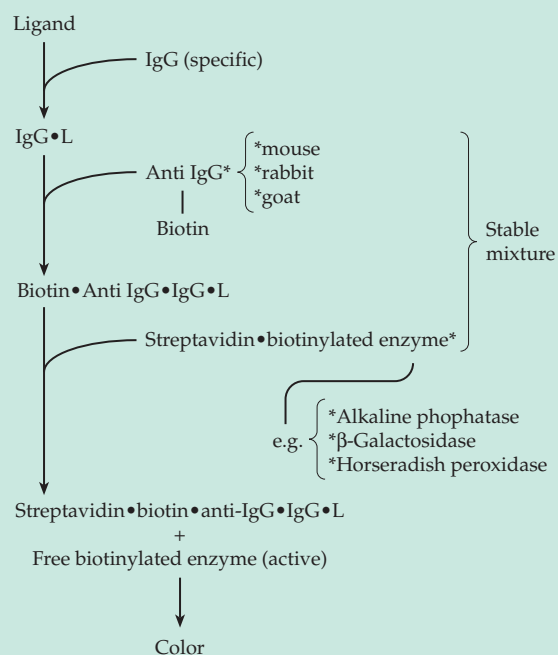
A biochemical curiosity is the presence in egg white of the glycoprotein **avidin**.^{a,b} Each 68-kDa subunit of this tetrameric protein binds one molecule of biotin tenaciously with $K_f \sim 10^{15} \text{ M}^{-1}$. Nature's purpose in placing this unusual protein in egg white is uncertain. Perhaps it is a storage form of biotin, but it is more likely an antibiotic that depletes the environment of biotin. A closely similar protein **streptavidin** is secreted into the culture medium by *Streptomyces avidinii*.^c Its sequence is homologous to that of avidin. It has a similar binding constant for biotin and the two proteins have similar three-dimensional structures.^{a,d-j} Biotin binds at one end of a β barrel formed from antiparallel strands and is held by multiple hydrogen bonds and a conformational alteration that allows a peptide loop to close over the bound vitamin.

Historically, avidin was important to the discovery of biotin. The bonding between avidin and biotin is so tight that inclusion of raw egg white in the diet of animals is sufficient to cause a severe biotin deficiency. Avidin has also been an important tool to enzymologists interested in biotin-containing enzymes. Avidin invariably inhibits these enzymes and inhibition by avidin is diagnostic of a biotin-containing protein. Recently, avidin and streptavidin have found widespread application in affinity chromatography, in immunoassays, and in the staining of cells and tissues.^{d,k-p} These uses are all based on the ability to attach biotin covalently to side chain groups of proteins, polysaccharides, and other substances. The carboxyl group of the biotin "arm," which lies at the surface of the complex with avidin or streptavidin, can be converted to any of a series of reactive derivatives. For example, *p*-nitrophenyl or *N*-hydroxysuccinimide esters of biotin can be used to attach biotin to amino groups of proteins to form biotinylated proteins.



Other reactive derivatives can be used to attach biotin to phenolic, thiol, or carbonyl groups.

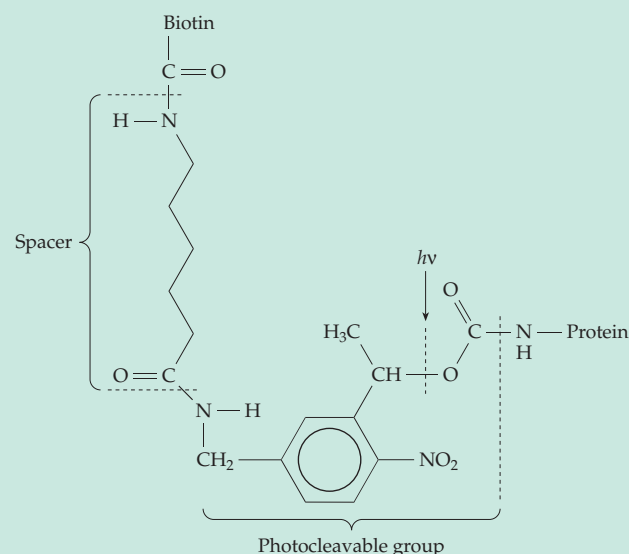
The affinity of avidin or streptavidin for the resulting biotinylated materials is still very high. This fact has permitted the application of this "biotin-avidin technology" to numerous aspects of research and diagnostic medicine. For example, a specific antibody (IgG) can be utilized for immunoassay of a hormone or other ligand. A second antibody, specific for the IgG-ligand complex, can be produced in a biotinylated form. A streptavidin complex of a biotinylated enzyme such as alkaline phosphatase, β -galactosidase, or horseradish peroxidase, for which a sensitive colorimetric assay is available, is allowed to react with the biotin•anti-IgG•IgG•ligand complex. The streptavidin now releases enzyme in proportion to the amount of ligand originally present. Since the biotinylated anti IgG and streptavidin•biotinylated enzyme can be stored as a stable mixture, the assay is simple and fast.



Avidin technology can also be applied to the isolation of proteins and other materials from cells. Because the irreversibility of the binding of biotin may be a problem, photocleavable biotin derivatives have been developed.^q In the following structure, the biotin derivative has been joined to a protein (as in the first equation in this box) and is ready for separation, perhaps on a column containing immobilized avidin or streptavidin. After separation the biotin together with the linker and photocleavable group are cut off by a short irradiation with ultra-

BOX 14-B (continued)

violet light, leaving the protein in a free form.



- ^a Livnah, O., Bayer, E. A., Wilshek, M., and Sussman, J. L. (1993) *Proc. Natl. Acad. Sci. U.S.A.* **90**, 5076–5080
^b Pugliese, L., Coda, A., Malcovati, M., and Bolognesi, M. (1993) *J. Mol. Biol.* **231**, 698–710
^c Meslar, H. W., Camper, S. A., and White, H. B., III. (1978) *J. Biol. Chem.* **253**, 6979–6982

- ^d Wilchek, M., and Bayer, E. A. (1989) *Trends Biochem. Sci.* **14**, 408–412
^e Puneekar, N. S., and Lardy, H. A. (1987) *J. Biol. Chem.* **262**, 6714–6719
^f Henderson, W. A., Pähler, A., Smith, J. L., Satow, Y., Merritt, E. A., and Phizackerley, R. P. (1989) *Proc. Natl. Acad. Sci. U.S.A.* **86**, 2190–2194
^g Weber, P. C., Pantoliano, M. W., and Thompson, L. D. (1992) *Biochemistry* **31**, 9350–9354
^h Schmidt, T. G. M., Koepke, J., Frank, R., and Skerra, A. (1996) *J. Mol. Biol.* **255**, 753–766
ⁱ Weber, P. C., Ohlendorf, D. H., Wendoloski, J. J., and Salemme, F. R. (1989) *Science* **243**, 85–88
^j Sano, T., and Cantor, C. R. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 3180–3184
^k Bayer, E. A., and Wilchek, M. (1980) *Meth. Biochem. Anal.* **26**, 1–46
^l Childs, G. V., Naor, Z., Hazum, E., Tibolt, R., Westlund, K. N., and Hancock, M. B. (1983) *J. Histochem. Cytochem.* **31**, 1422–1425
^m Wilchek, M., and Bayer, E. A., eds. (1990) *Methods in Enzymology*, Vol. 184, Academic Press, San Diego, California
ⁿ Savage, M. D., Mattson, G., Desai, S., Nielander, G. W., Morgensen, S., and Conklin, E. J. (1992) *Avidin-Biotin Chemistry: A Handbook*, Pierce, Rockford, Illinois
^o Donnellson, J. E., and Wu, R. (1972) *J. Biol. Chem.* **247**, 4661–4668
^p Laundon, C. H., and Griffith, J. D. (1987) *Biochemistry* **26**, 3759–3762
^q Olejnik, J., Sonar, S., Krzyżmańska-Olejnik, E., and Rothschild, K. J. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 7590–7594

carboxylase and transcarboxylase both catalyze elimination of HF from β -fluoropropionyl-CoA to form the unsaturated acrylyl-CoA.⁶⁷ The elimination presumably occurs via an enolate anion intermediate as in Eq. 13-28.

A bound divalent metal ion, usually Mn^{2+} , is required in the transcarboxylation step. A possible function is to assist in enolization of the carboxyl acceptor. However, measurement of the effect of the bound Mn^{2+} on ¹³C relaxation times in the substrate for pyruvate carboxylase indicated a distance of ~0.7 nm between the carbonyl carbon and the Mn^{2+} , too great for direct coordination of the metal to the carbonyl oxygen.⁶⁸ Another possibility is that the metal binds to the carbonyl of biotin as indicated in Eq. 14-11. Pyruvate carboxylase utilizes two divalent metal ions and at least one monovalent cation.^{68a}

What is the role of the sulfur atom in biotin? Perhaps it interacts with CO₂, helping to hold it in a correct orientation for reaction. Perhaps it helps to keep the ureido ring of biotin planar, or perhaps it has no special function.⁶⁹

3. Control Mechanisms

Most pyruvate carboxylases of animal and of yeast are allosterically activated by acetyl-CoA, but those of bacteria are usually not. The enzyme from chicken liver has almost no activity in the absence of acetyl-CoA, which appears to increase greatly the rate of formation of carboxyphosphate and to slow the side reaction by which carboxyphosphate is hydrolyzed to bicarbonate and phosphate.⁷⁰ The acetyl-CoA carboxylases of rat or chicken liver aggregate in the presence of citrate to form ~8000-kDa rods. Citrate is an allosteric activator for this enzyme but it acts only on a phosphorylated form and the primary control mechanism.^{71,72} This enzyme in plants is a target for a group of herbicides that are selectively toxic to grasses.⁷³

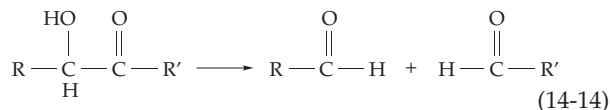
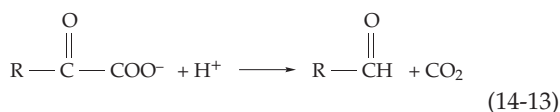
4. Pumping Ions with the Help of Biotin

Biotin-dependent **decarboxylases** act as sodium ion pumps in *Klebsiella*⁷⁴ and in various anaerobes.^{22,75} For example, oxaloacetate is converted to pyruvate and bound carboxybiotin.^{74,74a} The latter is decarboxylated

to CO_2 at the same time that two Na^+ ions are transported from the inside to the outside of the cell. The function of this pump, like that of the Na^+ , K^+ -ATPase (Fig. 8-25) is to provide an electrochemical gradient that drives the transport of other ions and molecules through the membrane. Similar ion pumps are operated by decarboxylation of methylmalonyl-CoA⁷⁶ and glutaconyl-CoA.⁷⁷ Yeast (*Saccharomyces cerevisiae*) cannot make biotin and requires an unusually large amount of the vitamin when urea, allantoin, allantoic acid, and certain other compounds are supplied as the sole source of nitrogen for growth. The reason is that in this organism urea must first be carboxylated by the biotin-containing **urea carboxylase**⁷⁸ (see Eq. 24-25) before it can be hydrolyzed to NH_3 and CO_2 .

D. Thiamin Diphosphate

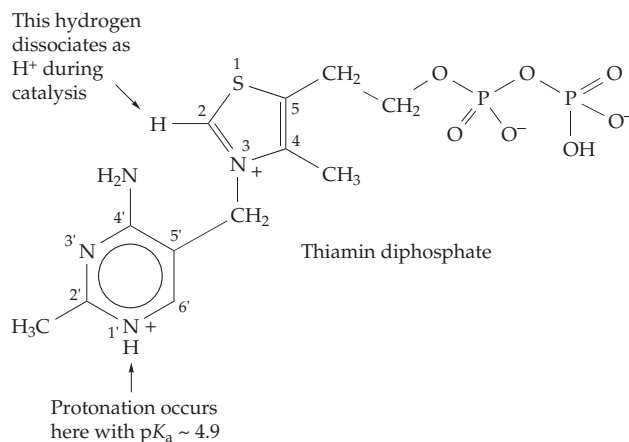
In Chapter 13 we considered the breaking of a bond between two carbon atoms, one of which is also bonded to a carbonyl group. These β cleavages are catalyzed by simple acidic and basic groups of the protein side chains. On the other hand, the decarboxylation of 2-oxo acids (Eq. 14-13) and the cleavage and formation of α -hydroxyketones (Eq. 14-14) depend upon thiamin diphosphate (TDP).^{79–85} These reactions represent a second important method of making and breaking carbon-carbon bonds which we will designate **α condensation and α cleavage**. The common feature of all thiamin-catalyzed reactions is that the bond broken (or formed) is *immediately adjacent to the carbonyl group*, not one carbon removed, as in β cleavage reactions. No simple acid-base catalyzed mechanisms can be written; hence the need for a coenzyme.



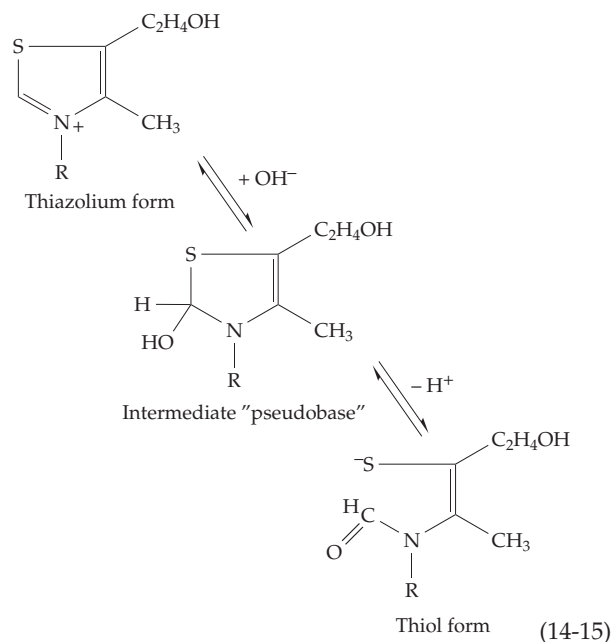
1. Chemical Properties of Thiamin

The weakly basic portion of thiamin or of its coenzyme forms is protonated at low pH, largely on N-1 of the pyrimidine ring.^{86–88} The pK_a value is ~ 4.9 . In basic solution, thiamin reacts in two steps with an opening of the thiazole ring (Eq. 14-15) to give the anion of a thiol form which may be crystallized as the sodium salt.^{79,84} This reaction, like the competing reaction described in Eq. 7-19, and which leads to a yellow

unstable form of the thiamin anion, is an example of a cooperative two-proton dissociation with linked structural changes. A very low concentration of the

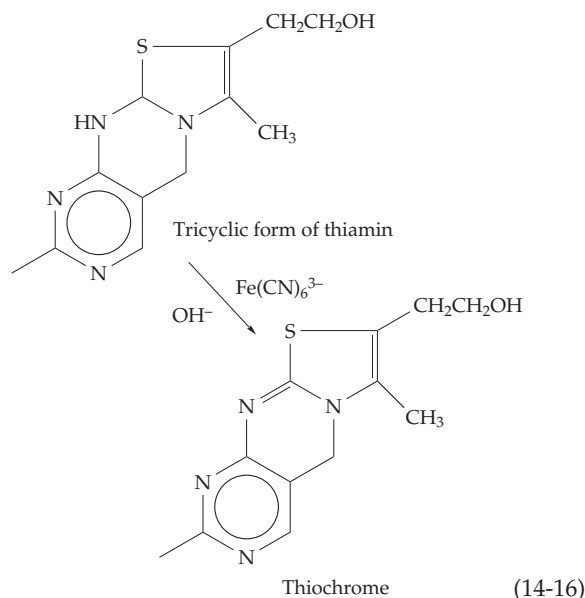


intermediate “pseudobase” is present during the titration. This property, which is unusual among small molecules, was instrumental in leading Williams *et al.* to the correct structure for the vitamin.⁸⁹ A still unanswered question is, What biological significance is associated with these reactions? Perhaps the thiol form depicted in Eq. 14-15 or the “yellow form” (Eq. 7-19) becomes attached to active sites of some proteins through disulfide linkages.

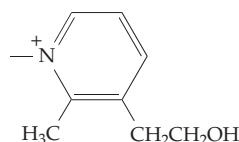


Thiamin is unstable at high pH ^{90,91} and is destroyed by the cooking of foods under mildly basic conditions. The thiol form undergoes hydrolysis and oxidation by air to a disulfide. The tricyclic form (Eq. 7-19) is oxidized to **thiochrome**, a fluorescent compound

whose formation from thiamin by treatment with alkaline hexacyanoferrate (III) (ferricyanide; Eq. 14-16) is the basis of a much used fluorimetric assay.

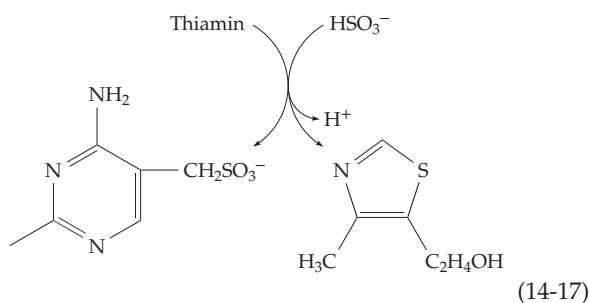


Treatment of thiamin with boiling 5N HCl deaminates it to the hydroxy analogue **oxythiamin**, a potent antagonist. **Pyriethiamin**, another competitor containing



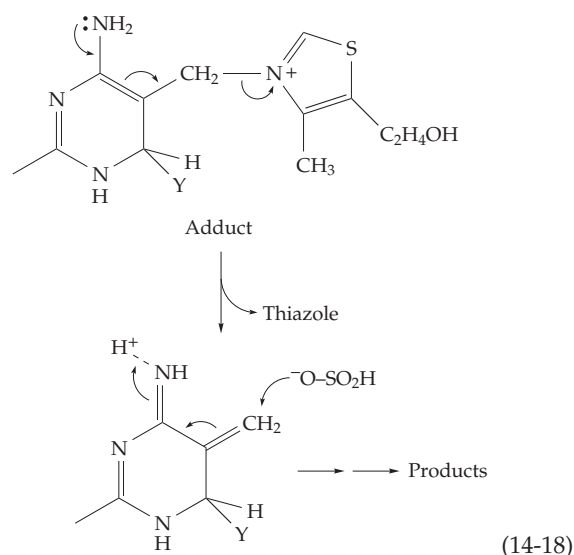
in place of the thiazolium ring, is very toxic especially to the nervous system.

In a solution of sodium sulfite at pH 5, thiamin is cleaved by what appears to be a nucleophilic displacement reaction on the methylene group to give the free thiazole and a sulfonic acid.



In fact the mechanism of the reaction is more complex and is evidently initiated by addition of a nucleophile Y, such as OH^- or bisulfite, followed by elimination of the thiazole (Eq. 14-18).

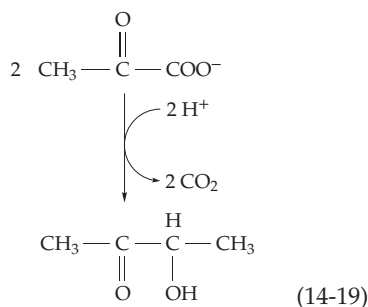
A similar cleavage is catalyzed by thiamin-degrading enzymes known as thiaminases which are found in a number of bacteria, marine organisms, and plants. In a bacterial thiaminase, group Y of Eq. 14-18 is a cysteine -SH .^{92,92a}



Thiamin is synthesized in bacteria, fungi, and plants from 1-deoxyxylulose 5-phosphate (Eq. 25-21), which is also an intermediate in the nonmevalonate pathway of polyprenyl synthesis. However, thiamin diphosphate is a coenzyme for synthesis of this intermediate (p. 736), suggesting that an alternative pathway must also exist. Each of the two rings of thiamin is formed separately as the esters 4-amino-5-hydroxymethylpyrimidine diphosphate and 4-methyl-5-(β -hydroxyethyl) thiazole monophosphate. These precursors are joined with displacement of pyrophosphate to form thiamin monophosphate.^{92b} In eukaryotes this is hydrolyzed to thiamin, then converted to thiamin diphosphate by transfer of a diphospho group from ATP.^{92b,c} In bacteria thiamin monophosphate is converted to the diphosphate by ATP and thiamin monophosphate kinase.^{92b}

2. Catalytic Mechanisms

The first real clue to the mechanism of thiamin-dependent cleavage came in about 1950 when Mizuhara showed that at pH 8.4 thiamin catalyzes the nonenzymatic conversion of pyruvate into acetoin (Eq. 14-19).⁹³ Following Mizuhara's lead, Breslow investigated the same reaction using the then new NMR method.⁹⁴ He made the surprising discovery that the hydrogen atom in the 2 position of the thiazolium ring, between the sulfur and the nitrogen atoms, exchanged easily with deuterium of $^2\text{H}_2\text{O}$. The pK_a of this proton has been estimated as ~ 18 , low enough to permit rapid

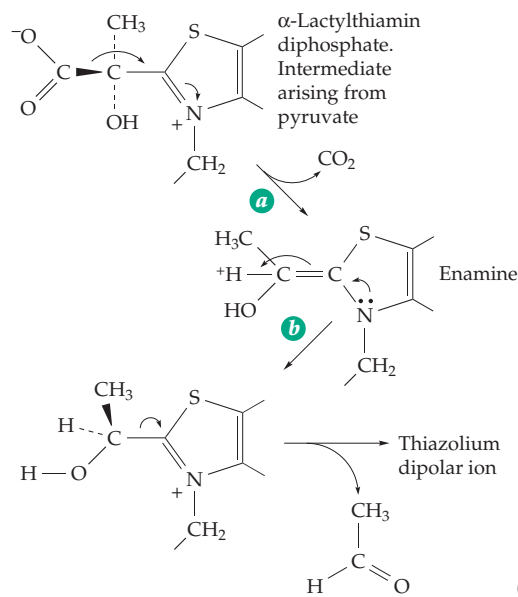
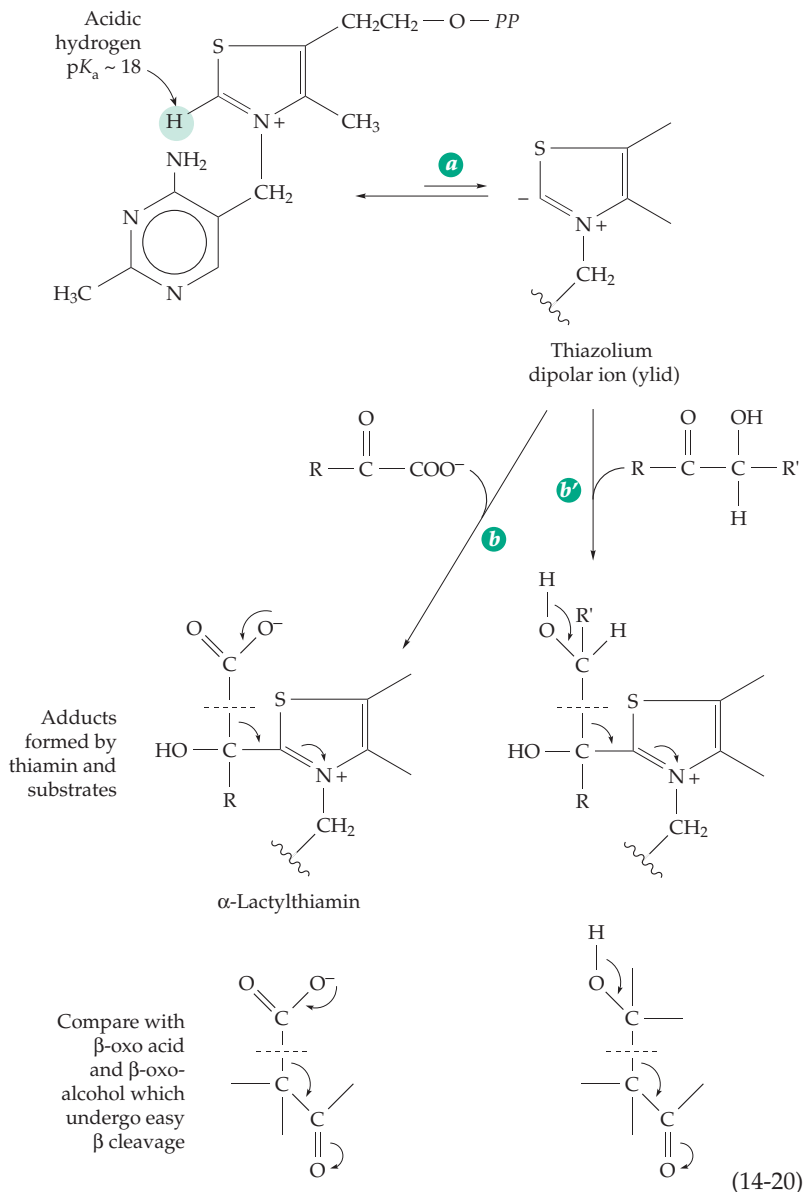


dissociation and replacement with ^2H .^{84,95,96} The resulting **thiazolium dipolar ion** (or **ylid**) formed by this dissociation (Eq. 14-20, step *a*) is stabilized by the electrostatic interaction of the adjacent positive and negative charges. Breslow suggested that this dipolar ion is the key intermediate in reactions of thiamin-dependent enzymes.^{94,97} The anionic center of the dipolar ion can react with a substrate such as an 2-oxo acid or 2-oxo alcohol by addition to the carbonyl group (Eq. 14-20, step *b* or 14-20, step *b'*). The resulting adducts are able to undergo cleavage readily, as indicated by the arrows showing the electron flow toward the $=\text{N}^+$ group.

Below the structures of the adducts in Eq. 14-20 are those of a 2-oxo acid and a β -ketol with arrows indicating the electron flow in decarboxylation and in the aldol cleavage. The similarities to the thiamin-dependent cleavage reaction are especially striking if one remembers that in some aldolases and decarboxylases the substrate carbonyl group is first converted to an N-protonated Schiff base before the bond cleavage.

We see that *the essence of the action of thiamin diphosphate as a coenzyme is to convert the substrate into a form in which electron flow can occur from the bond to be broken into the structure of the coenzyme*. Because of this alteration in structure, a bond breaking reaction that would not otherwise have been possible occurs readily. To complete the catalytic cycle, the electron flow has to be reversed again. The thiamin-bound cleavage product (an enamine) from either of the adducts in Eq. 14-20 can be reconverted to the thiazolium dipolar ion and an aldehyde as shown in step *b* of Eq. 14-21 for decarboxylation of pyruvate to acetaldehyde.

The adducts α -lactylthiamin and α -lactylthiamin diphosphate have both been synthesized.^{84,98-100} As long as α -lactylthiamin is kept as a dry solid or at low pH, it is stable. However, it decarboxylates readily in neutral solution (Eq. 14-21). Decarboxylation is much

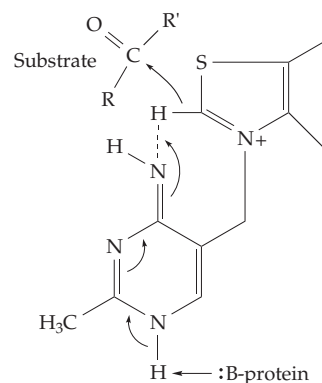


(14-21)

more rapid in methanol, a fact that was predicted by Lienhard and associates.¹⁰¹ They suggested that decarboxylation is easier in a solvent of low polarity because the transition state has a lower polarity than does lactylthiamin. An enzyme could assist the reaction by providing a relatively nonpolar environment.

The crystal structures of thiamin-dependent enzymes (see next section) as well as modeling^{102,103} suggest that lactylthiamin pyrophosphate has the conformation shown in Eq. 14-21. If so, it would be formed by the addition of the ylid to the carbonyl of pyruvate in accord with stereoelectronic principles, and the carboxylate group would also be in the correct orientation for elimination to form the enamine in Eq. 14-21, step *b*.^{82–83a} A transient 380- to 440-nm absorption band arising during the action of pyruvate decarboxylase has been attributed to the enamine.

What is the role of the pyrimidine portion of the coenzyme in these reactions? The pyrimidine ring has a large inductive effect on the basicity of the thiazolium nitrogen and may increase the rate of dissociation of the C-2 proton somewhat.¹⁰⁴ More significant is the fact that the -NH_2 group is properly placed to function as a basic catalyst in the generation of the thiazolium dipolar ion. However, the amino group of thiamin is not very basic ($\text{pK}_a \sim 4.9$) and the site of protonation at low pH is largely N-1 of the pyrimidine ring. In the protonated form the -NH_2 group is even less basic because of electron withdrawal into the ring. Studies of thiamin analogs suggested another possibility. Schellenberger¹⁰⁵ found pyruvate decarboxylase inactive when TDP was substituted by analogs with modified aminopyrimidine rings, e.g., with methylated or dimethylated amino groups or with N1 of the ring replaced by carbon (an aminopyridyl analog). More recently the experiment has been repeated with additional enzymes¹⁰⁶ and X-ray studies have shown that the analogs bind into the active site of transketolase in a normal way.¹⁰⁷ Of the compounds studied *only an aminopyridyl analog of TDP having a nitrogen atom at 1' (but CH at 3') had substantial catalytic activity*. Jordan and Mariam showed that N-1'-methylthiamin is a superior catalyst in non-enzymatic catalysis.¹⁰⁸ These results are consistent with the speculative scheme illustrated in the following drawing from the first edition of this book.¹⁰⁹ The -NH_2 group of the N1-protonated aminopyrimidine has lost a proton to form a normally *minor tautomer* in which the resulting imino group would be quite basic. Assisted by a basic group from the protein, it could abstract the proton from the thiazolium ring to form the ylid.



Crystallographic studies show that the catalytic base (:B-protein) is the carboxylate group of a conserved glutamate side chain (E59' in Fig. 14-2). Kern *et al.* used NMR spectroscopy of thiamin diphosphate present in native and mutant pyruvate decarboxylase and transketolase to monitor the exchange rates of the C2-H proton of the thiazolium ring.^{109a} The results confirmed the importance of the conserved glutamate side chain for dissociation of the C2-H proton. Participation of other catalytic groups from the enzyme may also be important. However, these groups are not conserved in the whole family of enzymes. For example, glutamine 122, which is within hydrogen-bonding distance of both the substrate and thiamin amino group, is replaced by histidine in transketolase.¹¹⁰ A variety of kinetic studies involving mutants,^{111,111a} alternative substrates,^{112,113} and isotope effects in substrates^{114–116} and solvent^{117,118} have not yet resolved the details of the proton transfers that occur within the active site.

3. Structures of Thiamin-Dependent Enzymes

By 1998, X-ray structures had been determined for four thiamin diphosphate-dependent enzymes: (1) a bacterial pyruvate oxidase,^{119,120} (2) yeast and bacterial pyruvate decarboxylases,^{121–122c} (3) transketolase,^{110,123,124} and (4) benzoylformate decarboxylase.^{124a} The reactions catalyzed by these enzymes are all quite different, as are the sequences of the proteins. However, the thiamin diphosphate is bound in a similar way in all of them. A conserved pattern of hydrogen bonds holds the diphosphate group to the protein and also provides ligands to a metal ion. This is normally Mg^{2+} , which is held in nearly perfect octahedral coordination by two phosphate oxygen atoms, a conserved aspartate carboxylate, a conserved asparagine amide, and a water molecule. The thiamin rings are in a less polar region. The amino group of the pyrimidine is adjacent to the 2-CH of the thiazole and N1' of the pyrimidine is apparently protonated and hydrogen bonded to the carboxylate group of a conserved glutamate side

chain. This is shown in Fig. 14-2. Substitution of the corresponding glutamate 51 of yeast pyruvate decarboxylase by glutamine or alanine greatly reduced or eliminated catalytic activity.¹²⁵

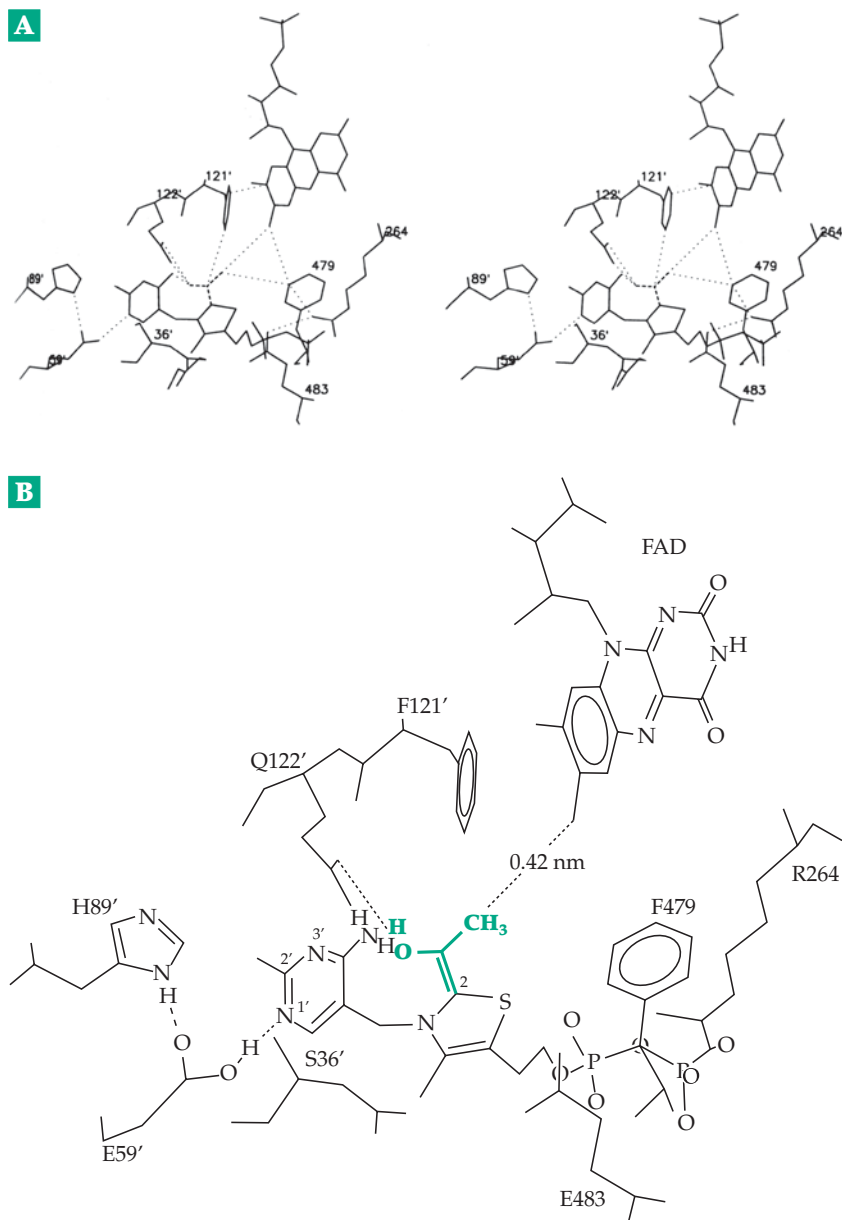
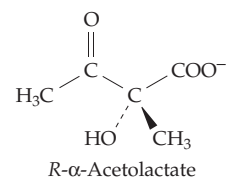


Figure 14-2 (A) Stereoscopic view of the active site of pyruvate oxidase from the bacterium *Lactobacillus plantarium* showing the thiamin diphosphate as well as the flavin part of the bound FAD. The planar structure of the part of the intermediate enamine that arises from pyruvate is shown by dotted lines. Only some residues that may be important for catalysis are displayed: G35', S36', E59', H89', F121', Q122', R264, F479, and E483. Courtesy of Georg E. Schulz.¹¹⁹ (B) Simplified view with some atoms labeled and some side chains omitted. The atoms of the hypothetical enamine that are formed from pyruvate, by decarboxylation, are shown in green.

4. The Variety of Enzymatic Reactions Involving Thiamin

Most known thiamin diphosphate-dependent reactions (Table 14-2) can be derived from the five half-reactions, *a* through *e*, shown in Fig. 14-3. Each half-reaction is an α cleavage which leads to a thiamin-bound enamine (center, Fig. 14-3). The decarboxylation of an α -oxo acid to an aldehyde is represented by step *b* followed by *a* in reverse. The most studied enzyme catalyzing a reaction of this type is yeast **pyruvate decarboxylase**, an enzyme essential to alcoholic fermentation (Fig. 10-3). There are two ~250-kDa isoenzyme forms, one an α_4 tetramer and one with an $(\alpha\beta)_2$ quaternary structure. The isolation of α -hydroxyethylthiamin diphosphate from reaction mixtures of this enzyme with pyruvate⁵² provided important verification of the mechanisms of Eqs. 14-14, 14-15. Other decarboxylases produce aldehydes in specialized metabolic pathways: indolepyruvate decarboxylase¹²⁶ in the biosynthesis of the plant hormone **indole-3-acetate** and benzoylformate decarboxylase in the mandelate pathway of bacterial metabolism (Chapter 25).^{124a,127}

Formation of α -ketols from α -oxo acids also starts with step *b* of Fig. 14-3 but is followed by condensation with another carbonyl compound in step *c*, in reverse. An example is decarboxylation of pyruvate and condensation of the resulting active acetaldehyde with a second pyruvate molecule to give *R*- α -acetolactate, a reaction catalyzed by **acetoxy acid synthase** (acetolactate synthase).¹²⁸ Acetolactate is the precursor to valine and leucine. A similar ketol condensation, which is catalyzed by the same synthase, is



required in the biosynthesis of isoleucine (Fig. 24-17). Since this

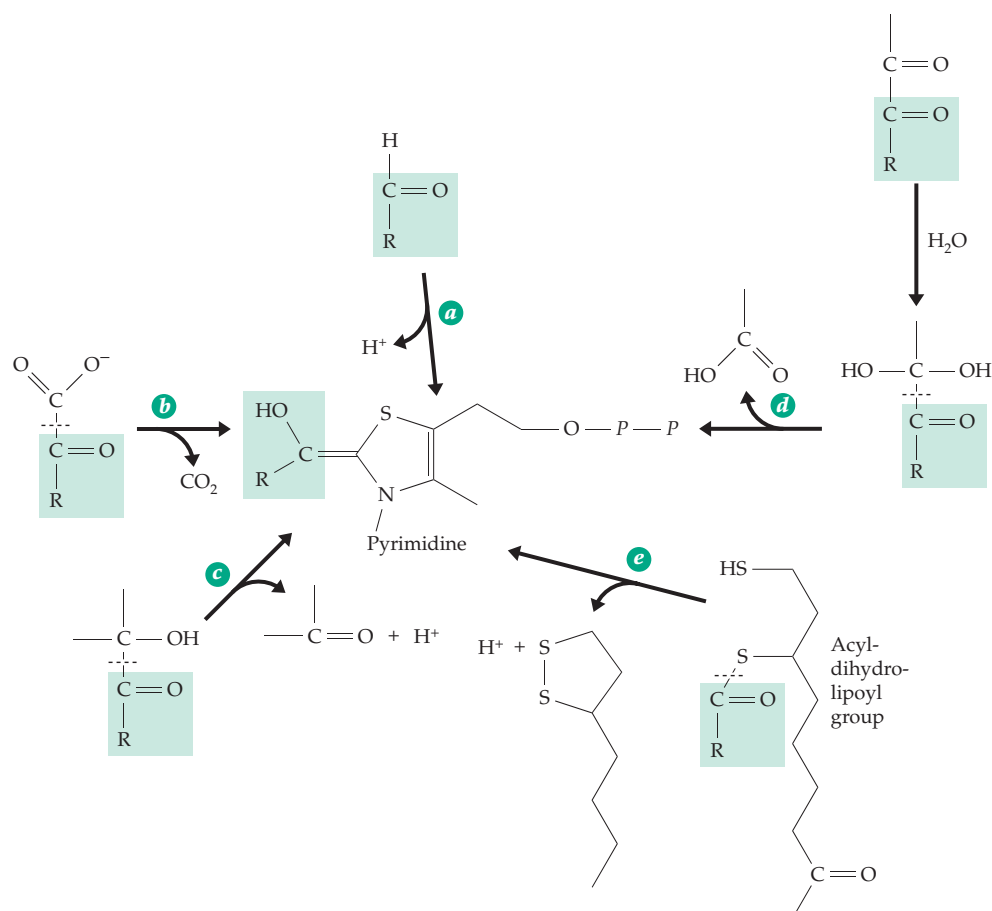


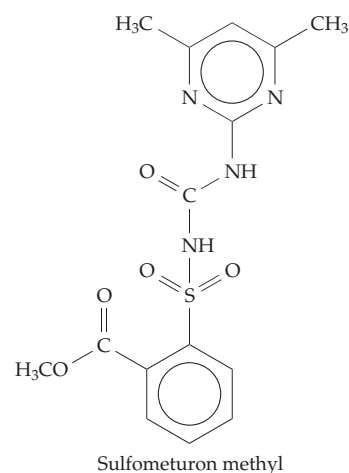
Figure 14-3 Half-reactions making up the thiamin-dependent α cleavage and α condensation reactions.

TABLE 14-2
Enzymes Dependent upon Thiamin Diphosphate as a Coenzyme

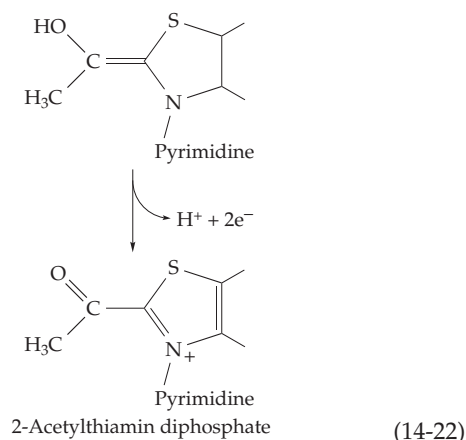
1. Nonoxidative
 - Pyruvate decarboxylase*
 - Indolepyruvate decarboxylase
 - Benzoylformate decarboxylase*
 - Glyoxylate carboligase
 - Acetohydroxy acid synthase (acetolactate synthase)
 - 1-Deoxy-D-xylulose 5-phosphate synthase
 - Transketolase*
 - Phosphoketolase
2. Oxidative decarboxylase
 - Pyruvate oxidase (FAD)*
 - Pyruvate dehydrogenase (Lipoyl, FAD, NAD⁺) multienzyme complex
 - Pyruvate:ferredoxin oxidoreductase
 - Indolepyruvate:ferredoxin oxidoreductase

* Three-dimensional structures for these enzymes had been determined by 1998.

synthase is not present in mammals it is a popular target for herbicides.^{129,130} It is inhibited by many of the most widely used herbicides including sulfometuron methyl, whose structure is shown here.



Acetolactate is a β -oxo acid and is readily decarboxylated to acetoin, a reaction of importance in bacterial fermentations (Eq. 17-26). Acetoin, of both R and S



configurations, is also formed by pyruvate decarboxylases acting on acetaldehyde.^{103,131} The ketol condensation of two molecules of glyoxylate with decarboxylation to form tartronic semialdehyde (see Fig. 17-6) is an important reaction in bacterial metabolism. It is catalyzed by **glyoxylate carboligase**,¹³² another thiamin diphosphate-dependent enzyme. Formation of 1-deoxy-D-xylulose 5-phosphate, an intermediate in the nonmevalonate pathway of isoprenoid synthesis, is formed in a thiamin diphosphate-catalyzed condensation of pyruvate with glyceraldehyde 3-phosphate (Fig. 22-2)^{132a} However, there is an unresolved problem. As previously mentioned, the same intermediate is thought to be a precursor to thiamin diphosphate (Eq. 25-21). This suggests the presence of an alternative pathway.

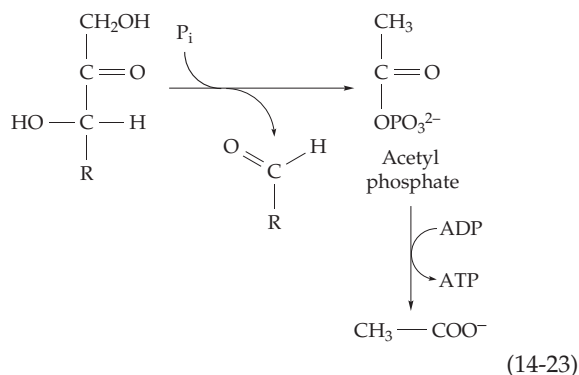
Ketols can also be formed enzymatically by cleavage of an aldehyde (step *a*, Fig. 14-3) followed by condensation with a second aldehyde (step *c*, in reverse). An enzyme utilizing these steps is **transketolase** (Eq. 17-15),^{132b} which is essential in the pentose phosphate pathways of metabolism and in photosynthesis. α -Diketones can be cleaved (step *d*) to a carboxylic acid plus active aldehyde, which can react either via *a* or *c* in reverse. These and other combinations of steps are often observed as side reactions of such enzymes as pyruvate decarboxylase. A related thiamin-dependent reaction is that of pyruvate and acetyl-CoA to give the α -diketone, **diacetyl**, $\text{CH}_3\text{COCOCH}_3$.¹³³ The reaction can be viewed as a displacement of the CoA anion from acetyl-CoA by attack of thiamin-bound active acetaldehyde derived from pyruvate (reverse of step *d*, Fig. 14-3 with release of CoA).

5. Oxidative Decarboxylation and 2-Acetylthiamin Diphosphate

The oxidative decarboxylation of pyruvate to form acetyl-CoA or acetyl phosphate plays a central role in the metabolism of our bodies and of most other

organisms. This reaction is usually formulated as the reverse of step *e* of Fig. 14-3, which shows the cleavage of an **acyl-dihydrolipoyl** derivative. However, there is a possibility that the lipoyl group functions not as shown in Fig. 14-3 but as an oxidant that converts the TDP enamine to 2-acetylthiamin diphosphate (Eq. 14-22)^{134,135} and only after that as an acyl group carrier. A related reaction that is known to proceed through acetyl-TDP is the previously mentioned bacterial pyruvate oxidase. As seen in Fig. 14-2, this enzyme has its own oxidant, FAD, which is ready to accept the two electrons of Eq. 14-22 to produce bound acetyl-TDP. The electrons may be able to jump directly to the FAD, with thiamin and flavin radicals being formed at an intermediate stage.^{135a} The electron transfers as well as other aspects of oxidative decarboxylation are discussed in Chapter 15, Section C.

A reaction that is related to that of transketolase but is likely to function via acetyl-TDP is **phosphoketolase**, whose action is required in the energy metabolism of some bacteria (Eq. 14-23). A product of phosphoketolase is acetyl phosphate, whose cleavage can be coupled to synthesis of ATP. Phosphoketolase presumably catalyzes an α cleavage to the thiamin-containing enamine shown in Fig. 14-3. A possible mechanism of formation of acetyl phosphate is elimination of H_2O from this enamine, tautomerization to 2-acetylthiamin, and reaction of the latter with inorganic phosphate.



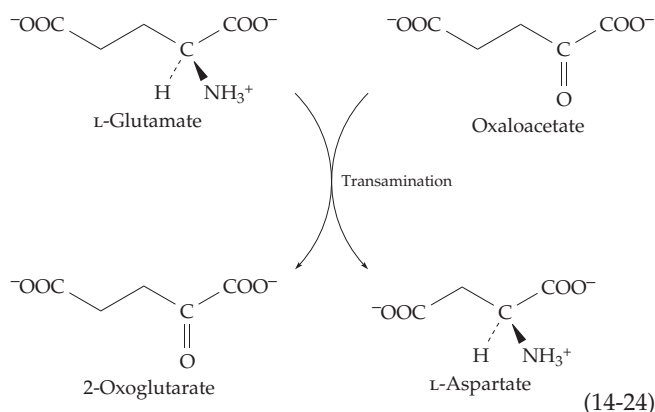
6. Thiamin Coenzymes in Nerve Action

The striking paralysis caused by thiamin deficiency together with studies of thiamin analogs as metabolites suggested a special action for this vitamin in nerves.¹³⁶ The thiamin analog **pyrithiamin** (p. 731) both induces paralytic symptoms and displaces thiamin from nerve preparations. The nerve poison **tetrodotoxin** (Chapter 30) blocks nerve conduction by inhibiting inward diffusion of sodium, but it also promotes release of thiamin from nerve membranes. Evidence for a metabolic significance of thiamin triphosphate comes from identification of soluble and membrane-associated thiamin triphosphatases¹³⁷ as well as a kinase that

forms protein-bound thiamin triphosphate in the brain.¹³⁸ Mono-, tri-, and tetraphosphates also occur naturally in smaller amounts. One might speculate about a possible role for the rapid interconversion of cationic and yellow anionic forms of thiamin via the tricyclic form (Eq. 7-19) in some aspect of nerve conduction.

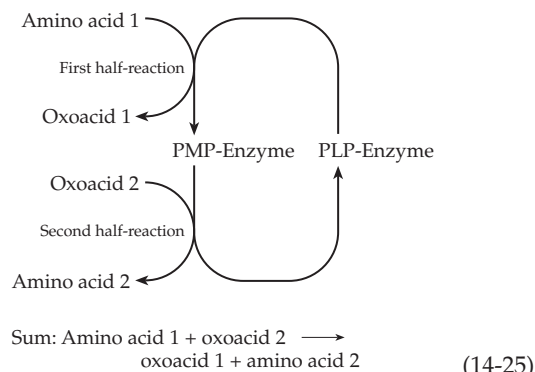
E. Pyridoxal Phosphate

The phosphate ester of the aldehyde form of vitamin B₆, **pyridoxal phosphate** (pyridoxal-*P* or PLP), is required by many enzymes catalyzing reactions of amino acids and amines. The reactions are numerous, and pyridoxal phosphate is surely one of nature's most versatile catalysts. The story begins with biochemical **transamination**, a process of central importance in nitrogen metabolism. In 1937, Alexander Braunstein and Maria Kritzmann, in Moscow, described the transamination reaction by which amino groups can be transferred from one carbon skeleton to another.^{139,140} For example, the amino group of glutamate can be transferred to the carbon skeleton of oxaloacetate to form aspartate and 2-oxoglutarate (Eq. 14-24).



This transamination reaction is a widespread process of importance in many aspects of the nitrogen metabolism of organisms. A large series of **transaminases** (**aminotransferases**), for which glutamate is most often one of the reactants, have been shown to catalyze the reactions of other oxoacids and amino acids.¹⁴¹⁻¹⁴³

In 1944, Esmond Snell reported the nonenzymatic conversion of pyridoxal into pyridoxamine (Box 14-C) by heating with glutamate. He recognized that this was also transamination and proposed that pyridoxal might be a part of a coenzyme needed for aminotransferases and that these enzymes might act via two half-reactions that interconverted pyridoxal and pyridoxamine (Eq. 14-25). The hypothesis was soon verified and the coenzyme was identified as pyridoxal 5'-phosphate or pyridoxamine 5'-phosphate (Fig. 14-5).^{144,145}

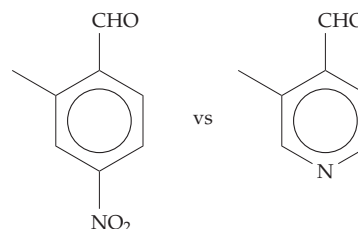


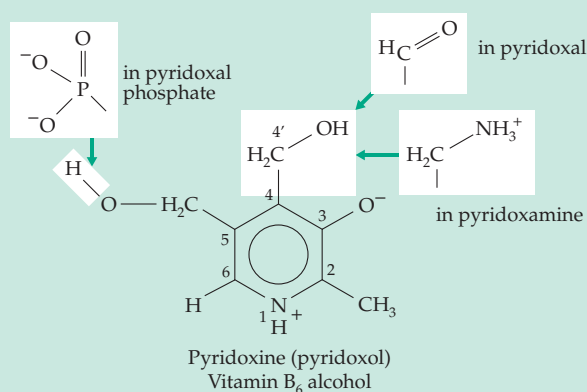
At about the same time, Gunsalus and coworkers noticed that the activity of **tyrosine decarboxylase** produced by lactic acid bacteria was unusually low when the medium was deficient in pyridoxine. Addition of pyridoxal plus ATP increased the decarboxylase activity of cell extracts.¹⁴⁶ PLP was synthesized and was found to be the essential coenzyme for this and a variety of other enzymes.¹⁴⁷

1. Nonenzymatic Models

Pyridoxal or PLP, in the complete absence of enzymes, not only undergoes slow transamination with amino acids but also catalyzes many other reactions of amino acids that are identical to those catalyzed by PLP-dependent enzymes. Thus, *the coenzyme itself can be regarded as the active site of the enzymes* and can be studied in nonenzymatic reactions. The latter can be thought of as *models* for corresponding enzymatic reactions. From such studies Snell and associates drew the following conclusions.¹⁴⁸

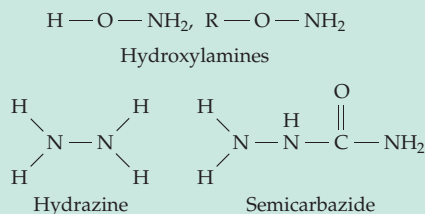
- The aldehyde group of PLP reacts readily and reversibly with amino acids to form Schiff bases (Fig. 14-4) which react further to give products.
- For an aldehyde to be a catalyst, a strong electron-attracting group, e.g., the ring nitrogen of pyridine (as in PLP), must be *ortho* or *para* to the -CHO group. A nitro group, also strongly electron attracting, can replace the pyridine nitrogen in model reactions.



BOX 14-C THE VITAMIN B₆ FAMILY: PYRIDOXINE, PYRIDOXAL, AND PYRIDOXAMINE

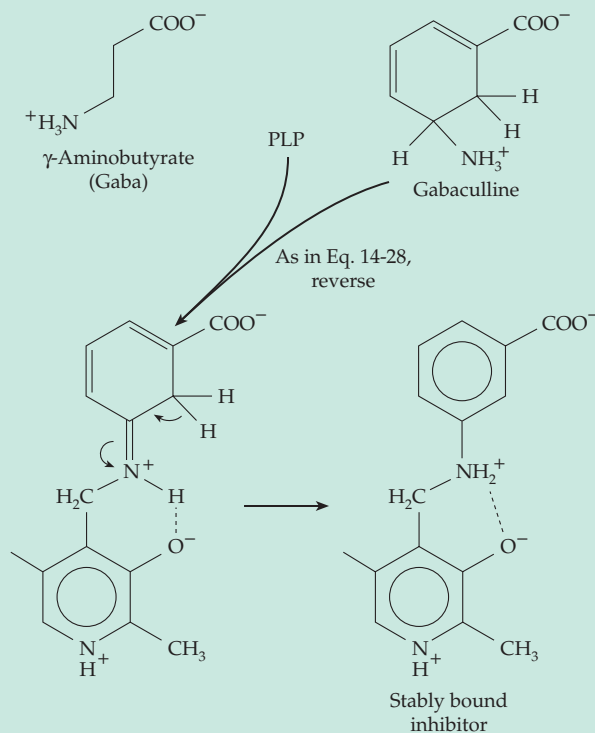
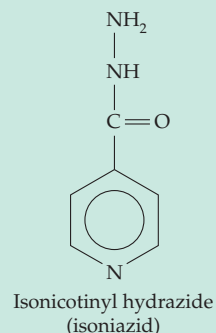
Pyridoxine, the usual commercial form of vitamin B₆, was isolated and synthesized in 1938. However, studies of bacterial nutrition soon indicated the existence of other naturally occurring forms of the new vitamin which were more active than pyridoxine in promoting growth of certain lactic acid bacteria. The amine **pyridoxamine** and the aldehyde **pyridoxal** were identified by Snell, who found that pyridoxal could be formed from pyridoxine by mild oxidation and that pyridoxamine could be formed from pyridoxal by heating in a solution with glutamic acid via a transamination reaction. These simple experiments also suggested the correct structures of the new forms of vitamin B₆. Animal tissues contain largely pyridoxal, pyridoxamine, and their phosphate esters. The lability of the aldehyde explains the ease of destruction of the vitamin by excessive heat or by light. On the other hand, plant tissues contain mostly pyridoxine, which is more stable. Kinases use ATP to form the phosphate esters, which are interconvertible within cells.^{a-e} Pyridoxine 5'-phosphate can be oxidized to PLP^{d,e,f} and the latter may undergo transamination to PMP. The acid-base chemistry and tautomerism of pyridoxine were discussed in Chapter 6, Section E,2.

Many poisonous substances as well as useful drugs react with PLP-requiring enzymes. Thus, much of the toxic effect of the "carbonyl reagents" hydroxylamine, hydrazine, and semicarbazide stems from their formation of stable derivatives analogous to Schiff bases with PLP.



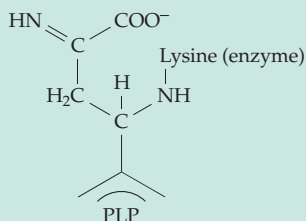
Isonicotinyl hydrazide (INH), one of the most effective drugs against tuberculosis, is inhibitory to pyridoxal kinase, the enzyme that converts pyridoxal to PLP.^c Apparently, the drug reacts with pyridoxal to form a hydrazone which blocks the enzyme. Pyridoxal kinase is not the primary target of INH in mycobacteria. However, patients on long-term isonicotinyl hydrazide therapy sometimes suffer symptoms of vitamin B₆ deficiency.^g

PLP-dependent enzymes are inhibited by a great variety of enzyme-activated inhibitors that react by several distinctly different chemical mechanisms.^h Here are a few. The naturally occurring **gabaculline** mimics γ -aminobutyrate (Gaba) and inhibits γ -aminobutyrate aminotransferase as well as other PLP-dependent enzymes. The inhibitor follows the normal catalytic pathway as far as the ketimine. There, a proton is lost from the inhibitor permitting formation of a stable benzene ring and leaving the inhibitor stuck in the active site:



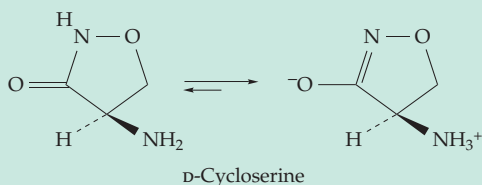
BOX 14-C (continued)

Beta-chloroalanine and serine *O*-sulfate can undergo β elimination (as in Eq. 14-29) in active sites of glutamate decarboxylase or aspartate aminotransferase. The enzymes then form free aminoacrylate, a reactive molecule that can undergo an aldol-type condensation with the external aldimine to give the following product.ⁱ

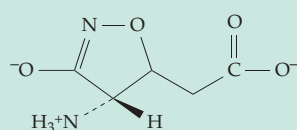


Nucleophilic groups from enzymes can add to double bonds, e.g., in an aminoacrylate Schiff base, or to multiple bonds present in the inhibitor. An example is γ -vinyl γ -aminobutyrate (4-amino-5-hexenoic acid), another inhibitor of brain γ -aminobutyrate aminotransferase which is a useful anti-convulsant drug.

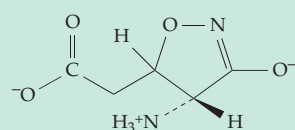
Another enzyme-activated inhibitor is the streptomycin antibiotic **D-cycloserine** (oxamycin), an antitubercular drug that resembles D-alanine in structure. A potent inhibitor of alanine racemase, it also inhibits the non-PLP, ATP-dependent, **D-alanyl-D-alanine synthetase** which is needed in the biosynthesis of the peptidoglycan of bacterial cell walls.



L-Cycloserine inhibits many PLP enzymes and is toxic to humans. This observation led Khomutov *et al.* to synthesize the following more specific “cycloglutamates,” structural analogs of glutamic acid with fixed conformations.^{j,k} Nature apparently anticipated the synthetic chemist, because it has been reported that the mushroom *Tricholoma muscar-*



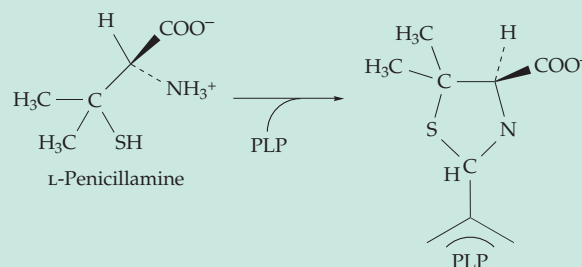
A cycloglutamate that inhibits aspartate aminotransferase



An isomeric cycloglutamate tricholomic acid found in certain mushrooms

ium contains one of the same compounds. It is said to impart two interesting properties to the mushroom: a delicious flavor and a lethal action on flies that alight on the mushroom's surface!^l

The substituted cysteine derivative L-penicillamine causes convulsions and low glutamate decarboxylase levels in the brain, presumably because the Schiff base formed with PLP can then undergo cyclization, the SH group adding to the C=N to form a stable thiazolidine ring.



Toxopyrimidine, the alcohol derived from the pyrimidine portion of the thiamin molecule, is a structural analog of pyridoxal. When fed to rats or mice it induces running fits which can be stopped by administration of vitamin B₆. Phosphorylation of toxopyrimidine by pyridoxal kinase may produce an antagonistic analog of PLP. In a similar fashion, 4-deoxypyridoxine, which was tested as a possible anticancer drug, caused convulsions and other symptoms of vitamin B₆ deficiency in humans. A host of synthetic PLP derivatives have been made, some of which are effective in blocking PLP enzymes.^h

^a Lepkovsky, S. (1979) *Fed. Proc.* **38**, 2699–2700

^b McCormick, D. B., Gregory, M. E., and Snell, E. E. (1961) *J. Biol. Chem.* **236**, 2076–2084

^c Snell, E. E., and Haskell, B. E. (1970) *Comprehensive Biochemistry* **21**, 47–71

^d McCormick, D. B. and Chen, H. (1999) *J. Nutr.* **129**, 325–327

^e Hanna, M. L., Turner, A. J., and Kirkness, E. F. (1997) *J. Biol. Chem.* **272**, 10756–10760

^f Ngo, E. O., LePage, G. R., Thanassi, J. W., Meisler, N., and Nutter, L. M. (1998) *Biochemistry* **37**, 7741–7748

^g Lui, A., and Lumeng, L. (1986) in *Vitamin B₆, Pyridoxal Phosphate: Chemical, Biochemical and Medical Aspects*, Vol. 1B (Dolphin, D., Poulson, R., and Avramovic, O., eds), pp. 601–674, Wiley, New York

^h Walsh, C. T. (1986) in *Vitamin B₆, Pyridoxal Phosphate: Chemical, Biochemical and Medical Aspects*, Vol. 1B (Dolphin, D., Poulson, R., and Avramovic, O., eds), pp. 43–70, Wiley, New York

ⁱ Likos, J. J., Ueno, H., Fedhaus, R. W., and Metzler, D. E. (1982) *Biochemistry* **21**, 4377–4386

^j Khomutov, R. M., Koveleva, G. K., Severin, E. S., and Vdovina, L. V. (1967) *Biokhim.* **32**, 900–907

^k Sastchenko, L. P., Severin, E. S., Metzler, D. E., and Khomutov, R. M. (1971) *Biochemistry* **10**, 4888–4894

^l Iwasaki, H., Kamiya, T., Oka, O., and Veyanagi, J. (1969) *Chem. Pharm. Bull.* **17**, 866–872

- c. The presence of an –OH group adjacent to the –CHO group greatly enhances the catalytic activity. Since certain metal ions, such as Cu^{2+} and Al^{3+} , increase the rates in model systems and are known to chelate with Schiff bases of the type formed with PLP, it was concluded that either a metal ion or a proton formed a chelate ring and helped to hold the Schiff base in a planar conformation (Fig. 14-6). *However, such a function for metal ions has not been found in PLP-dependent enzymes.*
- d. In model systems the 5-hydroxymethyl and 2-methyl groups are not needed for catalysis. However, in enzymes the 5- CH_2OH group is essential for attachment of the phosphate handle. The 2- CH_3 group is usually not necessary for coenzymatic activity.

Many investigations of nonenzymatic reactions of PLP and related compounds have been and are still being conducted^{149,150}

2. A General Mechanism of Action of PLP

Based upon consideration of the various known PLP-dependent enzymes of amino acid metabolism, Braunstein and Shemyakin in 1952 proposed a general mechanism of PLP action^{151,152} which, in most details, was the same as the one proposed independently by Snell and associates on the basis of the nonenzymatic reactions.¹⁴⁸ The general mechanism, which has been verified by studies of many enzymes, can be stated as follows: *Pyridoxal phosphate reacts to convert the amino group of a substrate into a Schiff base that is electronically the equivalent of an adjacent carbonyl* (Fig. 14-4). However, a Schiff base of an amino acid with a simple aldehyde (for example, acetaldehyde) has the opposite polarity from that of $\text{C}=\text{O}$ (see the following structures). Such an imine could not substitute for a carbonyl group in activating an α -hydrogen nor in facilitating $\text{C}-\text{C}$ bond cleavage in the amino acid. It is necessary to have the strongly electron-attracting pyridine group conjugated with the $\text{C}=\text{N}$ group in such a way that electrons can flow from the substrate into the coenzyme.

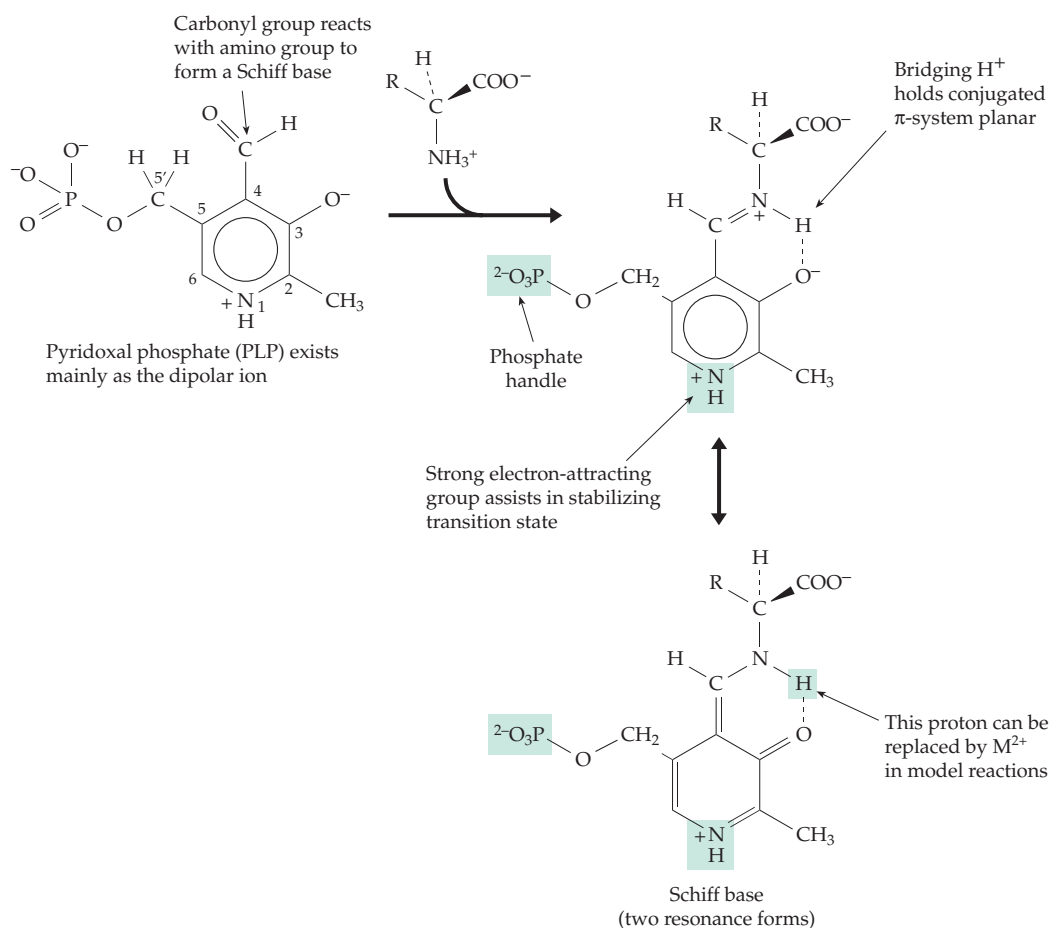
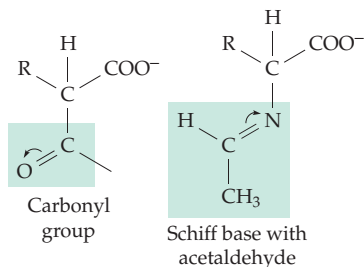
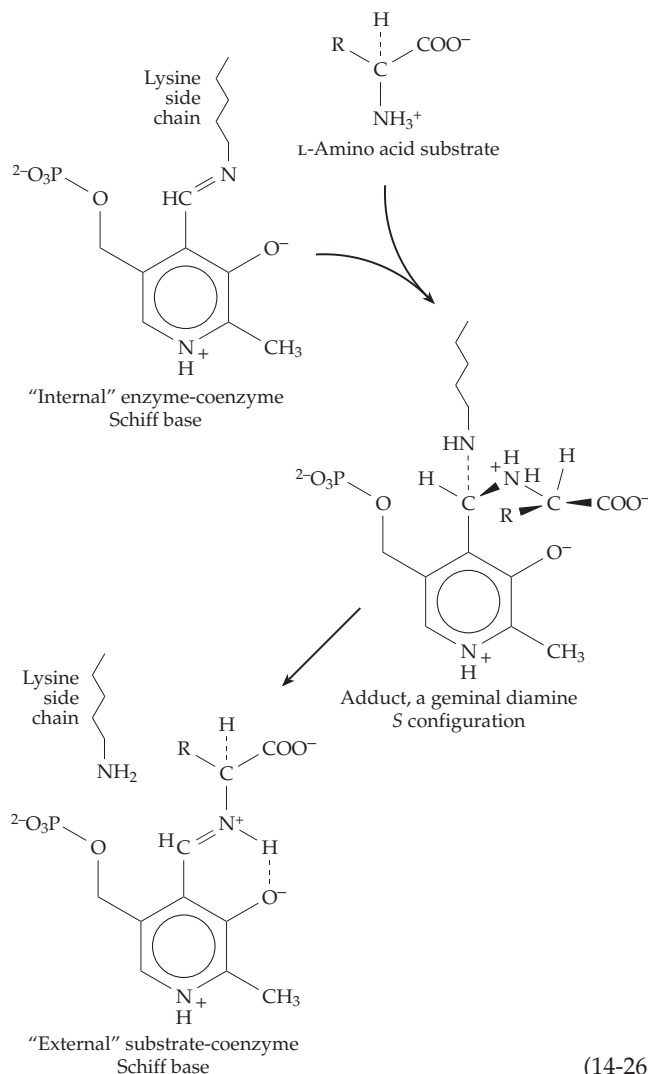


Figure 14-4 Pyridoxal 5'-phosphate (PLP), a special coenzyme for reactions of amino acids.



Before discussing the reactions of Schiff bases of PLP we should consider one fact that was not known in 1952. PLP is bound into an enzyme's active site as a Schiff base with a specific lysine side chain before a substrate binds. This is often called the **internal aldimine**. When the substrate binds it reacts with the internal Schiff base by a two-step process called **transimination** (Eq. 14-26) to form the substrate Schiff base, which is also called the **external aldimine**.



3. The Variety of PLP-Dependent Reactions

In Fig. 14-5 the reactions of PLP-amino acid Schiff bases are compared with those of β -oxo-acids. Beta-hydroxy- α -oxo acids and Schiff bases of PLP with β -hydroxy- α -amino acids can react in similar ways. The reactions fall naturally into three groups (*a, b, c*) depending upon whether the bond cleaved is from the α -carbon of the substrate to the hydrogen atom, to the carboxyl group, or to the side chain. A fourth group of reactions of PLP-dependent enzymes (*d*) also involve removal of the α -hydrogen but are mechanistically more complex. Some of the many reactions catalyzed by these enzymes are listed in Table 14-3.

Loss of the α -hydrogen (Group a). Dissociation of the α -hydrogen from the Schiff base leads to a **quinonoid-carbanionic intermediate** whose structure is depicted in Fig. 14-5. The name reflects the characteristics of the two resonance forms drawn. Like an enolate anion, this intermediate can react in several ways (1-4).

(1) **Racemization.** A proton can be added back to the original alpha position but without stereospecificity. A racemase which does this is important to bacteria. They must synthesize D-alanine and D-glutamic acid from the corresponding L-isomers for use in formation of their peptidoglycan envelopes.^{153-154a} The combined actions of alanine racemase plus D-alanine aminotransferase, which produces D-glutamate as a product, provide bacteria with both D amino acids. A fungal alanine racemase is necessary for synthesis of the immunosuppressant cyclosporin (Box 9-F).^{155,155a} High concentrations of free D-alanine are found in certain regions of the brain and also in various glands.¹⁵⁶

The carboxyl group of an amino acid can also activate the α -hydrogen. This may be the basis for an aspartate racemase and other racemases that are *not* dependent upon PLP.¹⁵⁶⁻¹⁵⁸ See also Chapter 13, Section B,4.

(2) **Cyclization.** A second kind of reaction is represented by the conversion of S-adenosylmethionine to **aminocyclopropanecarboxylic acid**, a precursor to the plant hormone **ethylene** (see Chapter 24).¹⁵⁹ The quinonoid intermediate cyclizes with elimination of methylthioadenosine to give a Schiff base of the product (Eq. 14-27).^{160-161a} The cyclization step appears to be a simple S_N2 -like reaction.¹⁶²

(3) **Transamination.** A proton can add to the carbon attached to the 4 position of the PLP ring (Fig. 14-5) to form a second Schiff base, often referred to as a **ketimine** (Eq. 14-28). The latter can readily undergo hydrolysis to **pyridoxamine phosphate** (PMP) and an α -oxo acid. This sequence represents one of the

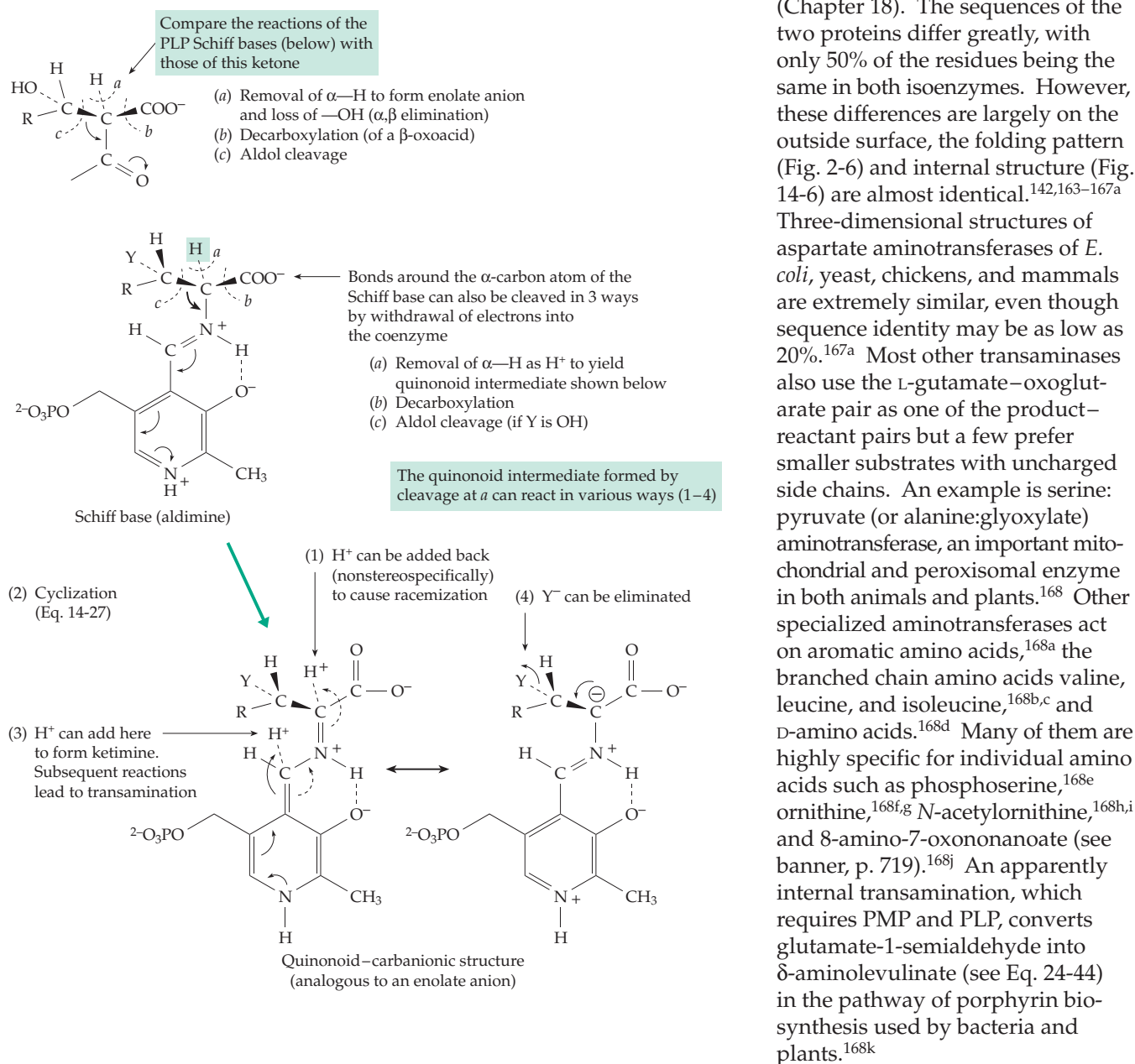


Figure 14-5 Some reactions of Schiff bases of pyridoxal phosphate. (a) Formation of the quinonoid intermediate, (b) elimination of a β substituent, and (c) transamination. The quinonoid-carbanionic intermediate can react in four ways (1–4) if enzyme specificity and substrate structure allow.

two half-reactions (Eqs. 14-24 and 14-25) required for enzymatic transamination.

Transaminases participate in metabolism of most of the amino acids, over 60 different enzymes have been identified.^{142,163} Best studied are the **aspartate aminotransferases**, a pair of cytosolic and mitochondrial isoenzymes which can be isolated readily from animal hearts. Their presence in heart muscle and brain in high concentration is thought to be a result of their functioning in the malate–aspartate shuttle

(Chapter 18). The sequences of the two proteins differ greatly, with only 50% of the residues being the same in both isoenzymes. However, these differences are largely on the outside surface, the folding pattern (Fig. 2-6) and internal structure (Fig. 14-6) are almost identical.^{142,163–167a} Three-dimensional structures of aspartate aminotransferases of *E. coli*, yeast, chickens, and mammals are extremely similar, even though sequence identity may be as low as 20%.^{167a} Most other transaminases also use the L-glutamate–oxoglutarate pair as one of the product–reactant pairs but a few prefer smaller substrates with uncharged side chains. An example is serine: pyruvate (or alanine:glyoxylate) aminotransferase, an important mitochondrial and peroxisomal enzyme in both animals and plants.¹⁶⁸ Other specialized aminotransferases act on aromatic amino acids,^{168a} the branched chain amino acids valine, leucine, and isoleucine,^{168b,c} and D-amino acids.^{168d} Many of them are highly specific for individual amino acids such as phosphoserine,^{168e} ornithine,^{168f,g} N-acetylornithine,^{168h,i} and 8-amino-7-oxononanoate (see banner, p. 719).^{168j} An apparently internal transamination, which requires PMP and PLP, converts glutamate-1-semialdehyde into δ -aminolevulinate (see Eq. 24-44) in the pathway of porphyrin biosynthesis used by bacteria and plants.^{168k}

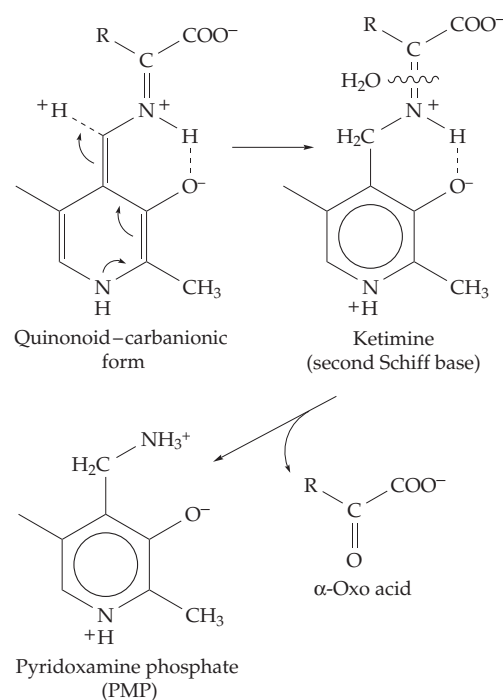
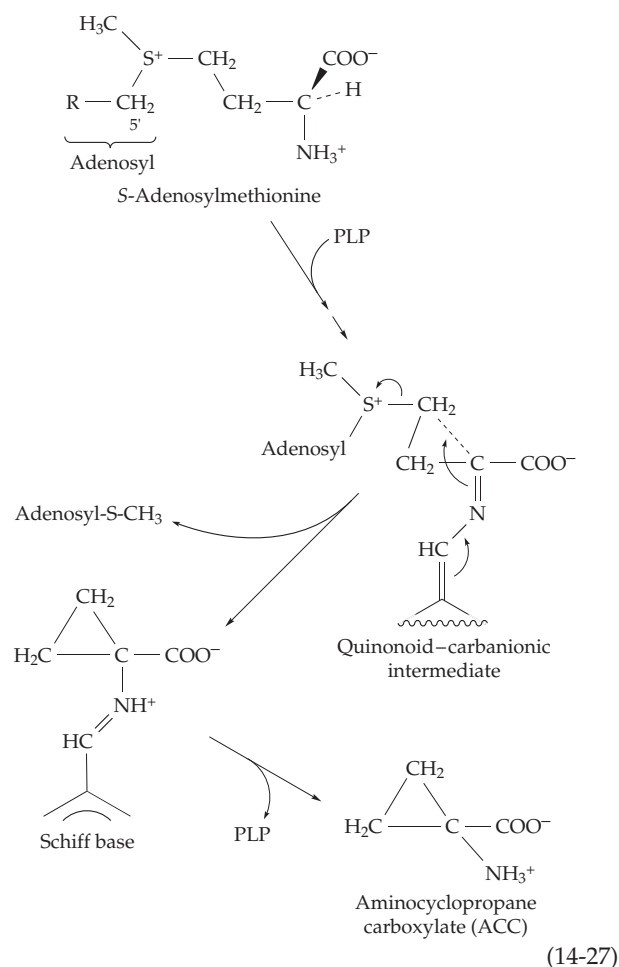
(4) Elimination and β replacement. When a good leaving group is present in the β position of the amino acid it can be eliminated (Fig. 14-5, Eq. 14-29).¹⁷⁰ A large number of enzymes catalyze such

reactions. Among them are **serine** and **threonine dehydratases**, which eliminate OH^- as H_2O ;^{171–173a} **tryptophan indole-lyase** (tryptophanase) of bacteria, which eliminates indole;^{174–176c} **tyrosine phenol-lyase** (elimination of phenol);^{177–178a} and **alliinase** of garlic (elimination of 1-propenylsulfenic acid).^{179,180} Cystathionine, a precursor to methionine, eliminates L-homocysteine through the action of **cystathionine β lyase** (cystathionase).^{181,182} Ammonia is eliminated from the β position of 2,3-diaminopropionate by a bacterial lyase.¹⁸³

TABLE 14-3
Some Enzymes That Require Pyridoxal Phosphate as a Coenzyme

- (a) Removing alpha hydrogen as H⁺
- (1) Racemization
 Alanine racemase*
 - (2) Cyclization
 Aminocyclopropane carboxylate synthase
 - (3) Amino group transfer
 Aspartate aminotransferase*
 Alanine aminotransferase
 D-Amino acid aminotransferase*
 Branched chain aminotransferase
 Gamma-aminobutyrate aminotransferase
 ω-Amino acid:pyruvate aminotransferase*
 Tyrosine aminotransferase
 Serine:pyruvate aminotransferase
 - (4) Beta elimination or replacement
 D- and L- Serine dehydratases (deaminases)
 Tryptophan indole-lyase (tryptophanase)*
 Tyrosine phenol-lyase*
 Alliinase
 Cystathionine β-lyase (cystathionase)*
 O-Acetylserine sulphydrylase (cysteine synthase)
 Cystathionine β-synthase
 Tryptophan synthase*
- (b) Removal of alpha carboxylate as CO₂
- Diaminopimelate decarboxylase
 Glycine decarboxylase (requires lipoyl group)
 Glutamate decarboxylase
 Histidine decarboxylase
 Dopa decarboxylase
 Ornithine decarboxylase*
 Tyrosine decarboxylase
 Dialkylglycine decarboxylase (a decarboxylating transaminase)*
- (c) Removal or replacement of side chain (or -H) by aldol cleavage
- Serine hydroxymethyltransferase
 Threonine aldolase
 δ-Aminolevulinate synthase
 Serine palmitoyltransferase
 2-Amino-3-oxobutyrate-CoA ligase
- (d) Reactions of ketimine intermediates
- Aspartate γ-decarboxylase
 Selenocysteine lyase
 Nif S protein of nitrogenase
 Gamma elimination and replacement
 Cystathionine γ-synthase
 Cystathionine γ-lyase
 Threonine synthase
- (e) Other enzymes
- Lysine 2,3-aminomutase
 Glycogen phosphorylase*
 Pyridoxamine phosphate (PMP) in synthesis of 3,6-dideoxy hexoses

* The three-dimensional structures of these and other PLP-dependent enzymes were determined by 2000.



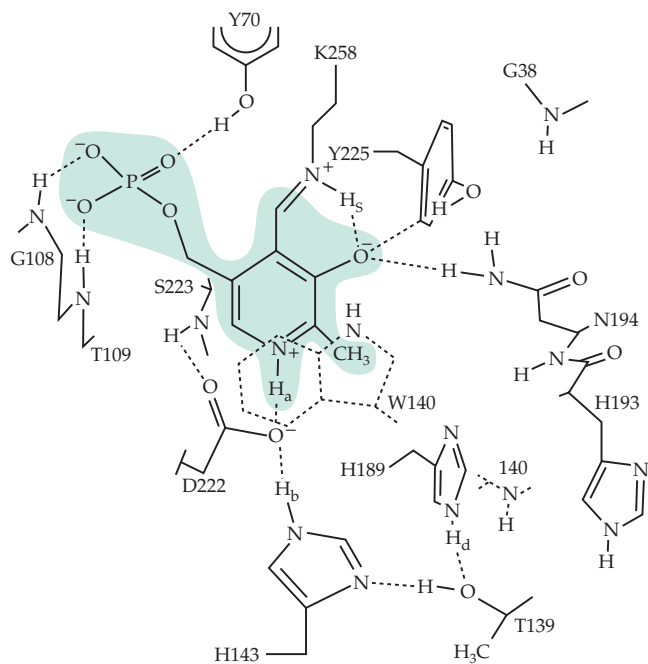
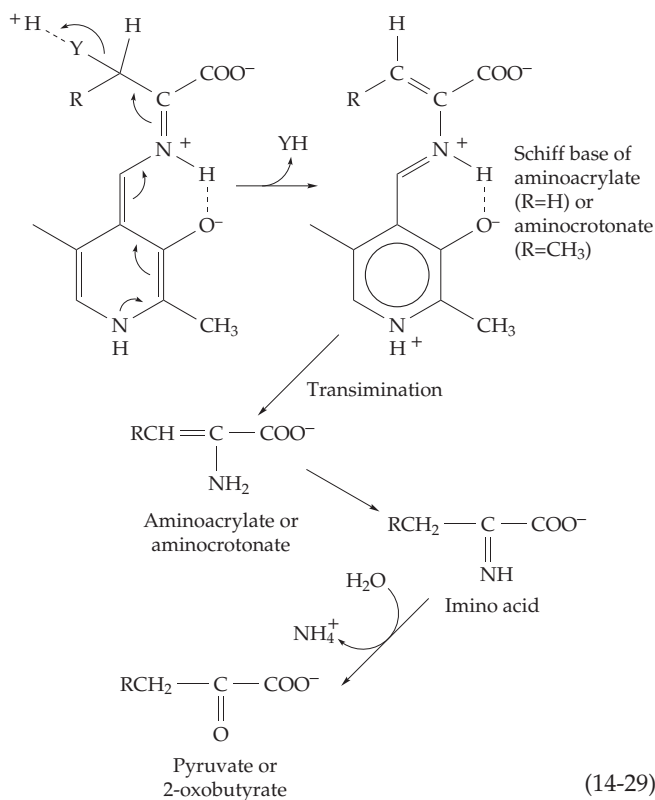
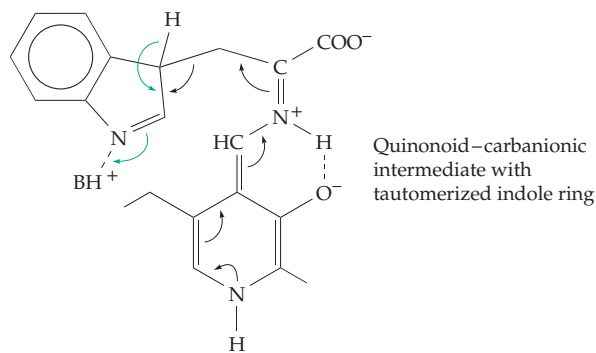


Figure 14-6 Drawing showing pyridoxal phosphate (shaded) and some surrounding protein structure in the active site of cytosolic aspartate aminotransferase. This is the low pH form of the enzyme with an *N*-protonated Schiff base linkage of lysine 258 to the PLP. The tryptophan 140 ring lies in front of the coenzyme. Several protons, labeled H_a , H_b , and H_d , are represented in ^1H NMR spectra by distinct resonances whose chemical shifts are sensitive to changes in the active site.¹⁶⁹



Beta replacement is catalyzed by such enzymes of amino acid biosynthesis as **tryptophan synthase** (Chapter 25),¹⁸⁴ **O-acetylserine sulfhydrylase** (cysteine synthase),^{185–186a} and **cystathionine β -synthase** (Chapter 24).^{187–188c} In both elimination and β replacement an unsaturated Schiff base, usually of aminoacrylate or aminocrotonate, is a probable intermediate (Eq. 14-29). Conversion to the final products is usually assumed to be via hydrolysis to free aminoacrylate, tautomerization to an imino acid, and hydrolysis of the latter, e.g., to pyruvate and ammonium ion (Eq. 14-29). However, the observed stereospecific addition of a



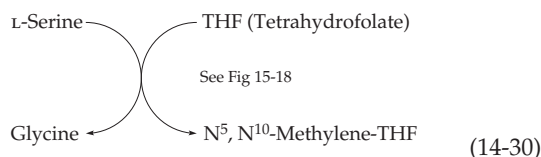
proton at the β -C atom of 2-oxobutyrate¹⁸⁹ suggests that these steps may occur with the participation of groups from the enzyme. Before indole can be eliminated by tryptophan indolelyase the indole ring must presumably be tautomerized to the following form of the quinonoid intermediate. The same species may be created by tryptophan synthase upon addition of indole to the enzyme-bound aminoacrylate. The green arrows on the structure indicate the tautomerization that would occur to convert the indole ring to the structure found in tryptophan. The three-dimensional structure of tryptophan synthase is shown in Fig. 25-3. It is a complex of two enzymes with a remarkable tunnel through which the intermediate indole can pass.^{184,190} An unusual PLP-dependent β replacement is used to synthesize a transfer RNA ester of **seleno-cysteine** prior to its insertion into special locations in a few proteins (Chapter 15, Section G, and Chapter 29).

Decarboxylation (Group b). The bond to the carboxyl group of an amino acid substrate is broken in reactions catalyzed by **amino acid decarboxylases**.^{191,192} These also presumably lead to a transient quinonoid-carbanionic intermediate. Addition of a proton at the original site of decarboxylation followed by breakup of the Schiff base completes the sequence. Decarboxylation of amino acids is nearly irreversible and frequently appears as a final step in synthesis of amino compounds. For example, in the brain glutamic acid is decarboxylated to **γ -aminobutyric acid** (Gaba),^{193–196b} while 3,4-dihydroxyphenylalanine (dopa) and 5-hydroxy-

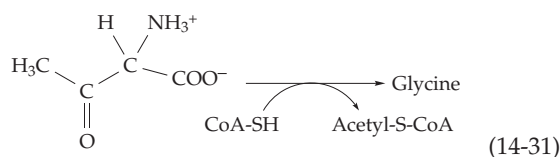
tryptophan are acted upon by an **aromatic amino acid decarboxylase** to form, respectively, the neurotransmitters **dopamine** and **serotonin**.^{197–199b} Histidine is decarboxylated to **histamine**.^{200–202} However, not all histidine decarboxylases use PLP as a coenzyme (Section F).

Arginine is converted by a PLP-dependent decarboxylase to agmatine (Fig. 24-12) which is hydrolyzed to **1,4-diaminopropane**.¹⁹¹ This important cell constituent is also formed by hydrolysis of arginine to **ornithine** (Fig 24-10) and decarboxylation of the latter.^{203–206c} **Lysine** is formed in bacteria by decarboxylation of *meso*-diamino-pimelic acid (Fig. 24-14). **Glycine** is decarboxylated oxidatively in mitochondria in a sequence requiring lipoic acid and tetrahydrofolate as well as PLP (Fig. 15-20).^{207–209b} A **methionine** decarboxylase has been isolated in pure form from a fern.²¹⁰ The bacterial **dialkylglycine** decarboxylase is both a decarboxylase and an aminotransferase which uses pyruvate as its second substrate forming a ketone and L-alanine as products (See Eq. 14-37)^{210a, 210b}

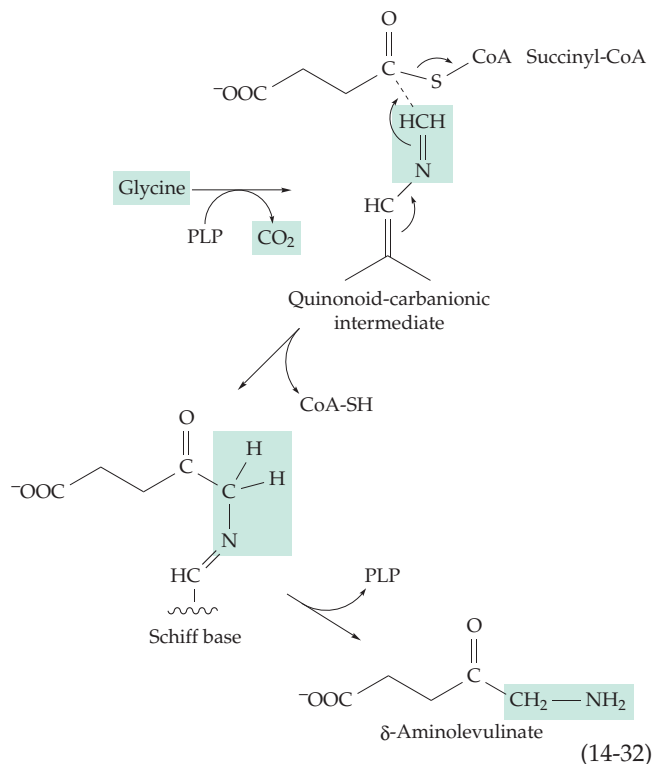
Side chain cleavage (Group c). In a third type of reaction the side chain of the Schiff base of Fig. 14-5 undergoes aldol cleavage. Conversely, a side chain can be added by β condensation. The best known enzyme of this group is **serine hydroxymethyltransferase**, which converts serine to glycine and formaldehyde.^{211–213b} The latter is not released in a free form but is transferred by the same enzyme specifically to **tetrahydrofolic acid** (Eq. 14-30), with which it forms a cyclic adduct.



Threonine is cleaved to acetaldehyde by the same enzyme. A related reaction is indicated in Fig. 24-27 (top). In a more important pathway of degradation of threonine the hydroxyl group of its side chain is dehydrogenated to form 2-amino-3-oxobutyrates which is cleaved by a PLP-dependent enzyme to glycine and acetyl-CoA (Eq. 14-31).^{214,215}



Conversely, ester condensation reactions join acyl groups from CoA derivatives to Schiff bases derived from glycine or serine. Succinyl-CoA is the acyl donor



in Eq. 14-32 for the second known pathway for biosynthesis of **δ -aminolevulinic acid**, an intermediate in porphyrin synthesis (Chapter 24).^{216–218b} The enzyme does not catalyze decarboxylation of glycine in the absence of succinyl-CoA, and the decarboxylation probably follows the condensation as indicated in Eq. 14-32.²¹⁹ In a similar reaction in the biosynthesis of **sphingosine** serine is condensed with palmitoyl-CoA and decarboxylated to form an aminoketone intermediate (Fig. 21-6).^{219a} 8-Amino-7-oxonanonoate synthase forms a precursor of biotin (see banner, p. 719).²²⁰

Ketimine intermediate as electron acceptor (Group d). The fourth group of PLP-dependent reactions are thought to depend upon formation of the ketimine intermediate of Eq. 14-28. In this form the original α -hydrogen of the amino acid has been removed and the $C=NH^+$ bond of the ketimine is polarized in a direction that favors electron withdrawal from the amino acid into the imine group. This permits another series of enzymatic reactions analogous to those of the β -oxo acid shown at the top of Fig. 14-5. Both elimination and C–C bond cleavage α, β to the $C=N$ group of the ketimine can occur.

Enzymes of this group catalyze elimination of γ substituents from amino acids as illustrated in Fig. 14-7. Eliminated groups may be replaced by other substituents, either in the α or the β positions. The ketimine formed initially by such an enzyme (step *a*) undergoes elimination of the γ substituent (β with respect to the $C=N$ group) along with a proton from the β position of the original amino acid to form an

unsaturated intermediate which can react in one of three ways, depending upon the enzyme. Addition of HY' leads to γ replacement (step *c*), while addition of a proton at the α position leads, via reaction step *d*, to an α,β -unsaturated Schiff base. The latter can react by addition of HY' (β replacement, step *e*) or it can break down to an α -oxo acid and ammonium ion (step *f*), just as in the β elimination reactions of Eq. 14-29. An important γ replacement reaction is conversion of *O*-acetyl-, *O*-succinyl-, or *O*-phosphohomoserine to cystathionine (Eq. 14-33). This **cystathionine γ -synthase** reaction^{220a} lies on the pathway of biosynthesis of methionine by bacteria, fungi, and

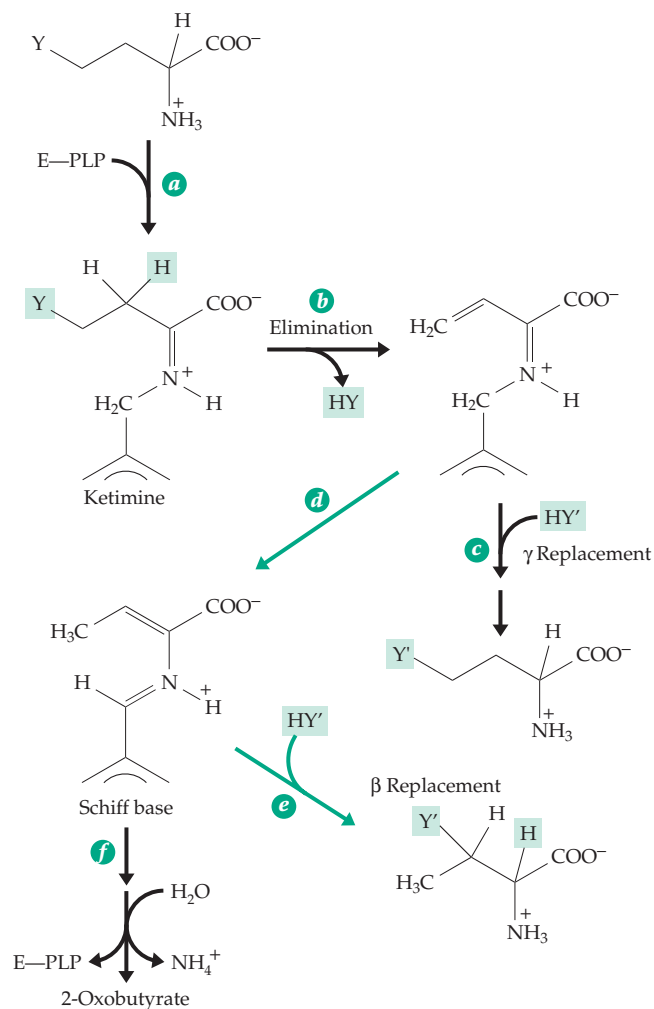
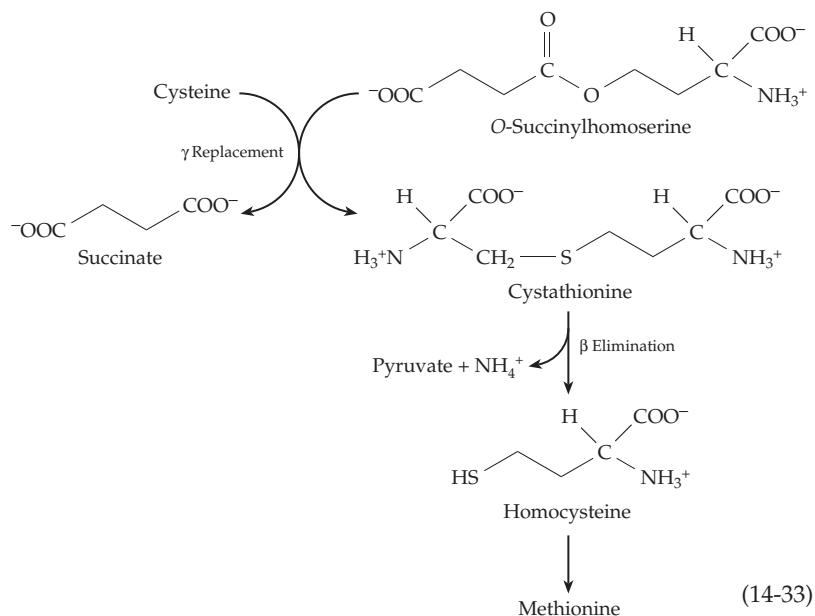


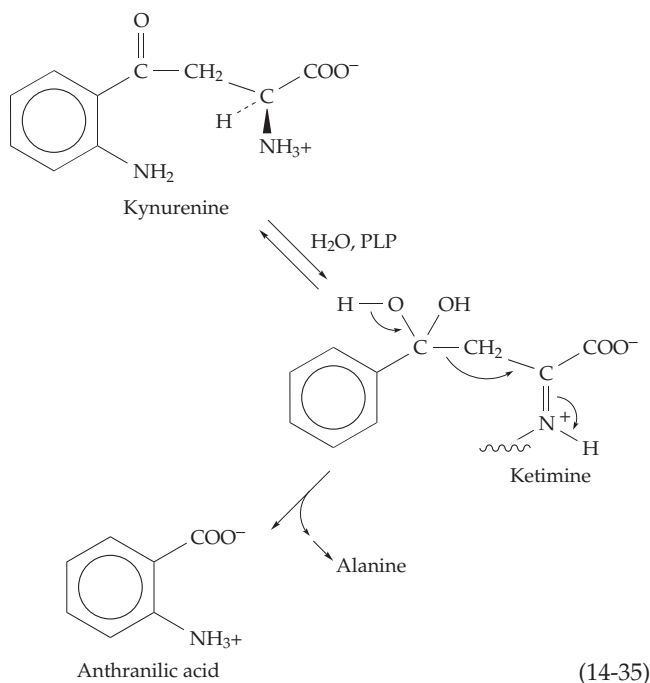
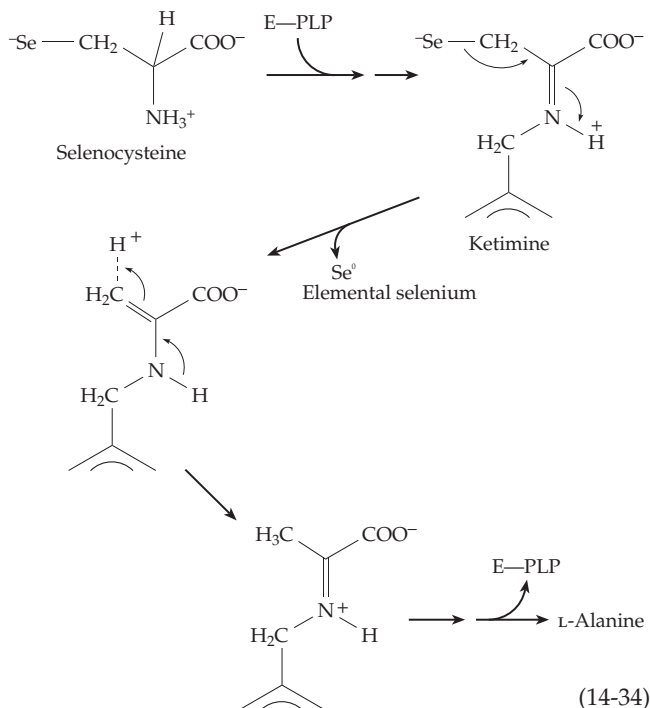
Figure 14-7 Some PLP-dependent reactions involving elimination of a γ substituent. Replacement by another γ substituent or by a substituent in the β position is possible, as is deamination to an α -oxo acid.

higher plants. Subsequent reactions include β elimination from cystathionine of **homocysteine**^{220b} which is then converted to methionine (Eq. 14-33). Threonine is formed from ***O*-phosphohomoserine** via γ elimination followed by β replacement with HO^- , a reaction catalyzed by **threonine synthase** (Fig. 24-13).^{220c}

The loss of a β -carboxyl group as CO_2 can also occur through a ketimine or quinonoid intermediate. For example, the bacterial **aspartate β -decarboxylase**²²¹ converts aspartate to alanine and CO_2 . **Selenocysteine** is utilized to create the active sites of several enzymes (Chapter 15). Excess selenocysteine is degraded by the PLP-dependent **selenocysteine lyase**,^{222–223b} which evidently eliminates elemental selenium from a ketimine or quinonoid state of an intermediate Schiff base (Eq. 14-34). A similar reaction may occur in the biosynthesis of iron–sulfur clusters. The **Nif S** protein is essential for formation of Fe_4S_4 clusters in the nitrogen-fixing enzyme nitrogenase. This enzyme is in some way involved in transferring the sulfur atom of cysteine into an iron–sulfur cluster.²²⁴ Alanine is the other product suggesting transfer of S^0 into the cluster using the sequence of Eq. 14-34.

Another related reaction that goes through a ketimine is the conversion of the amino acid **kynurenine** to alanine and anthranilic acid.²²⁵ It presumably depends upon hydration of the carbonyl group prior to β cleavage (Eq. 14-35). An analogous thiolytic cleavage utilizes CoA to convert 2-amino-4-ketopentanoate to acetyl-CoA and alanine.²²⁶

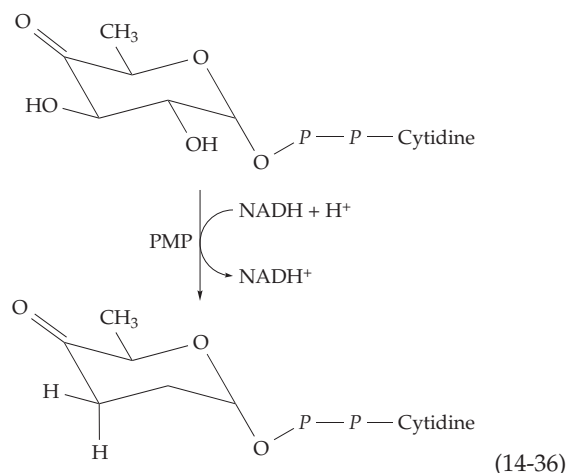
Glycogen phosphorylase. While PLP is ideally designed to catalyze reactions of amino compounds it was surprising to find it as an essential cofactor for glycogen phosphorylase (Fig. 11-5). The PLP is linked as a Schiff base in the same way as in other PLP-depen-



dent enzymes, but there is no obvious function for the coenzyme ring. As suggested in Chapter 12, the phosphate group probably acts as an acid-base catalyst. It has been estimated that 50% of the vitamin B₆ in our body is present as PLP in muscle phosphorylase.²²⁷ Studies of vitamin B₆-deficient rats suggest that PLP in phosphorylase serves as a reserve supply, much of which can be taken for other purposes during times of deficiency.

4. Pyridoxamine Phosphate as a Coenzyme

If PLP is a cofactor designed to react with amino groups of substrates, might not pyridoxamine phosphate (PMP) act as a coenzyme for reactions of carbonyl compounds? An example of this kind of function has been found^{228–230} in the formation of 3,6-dideoxyhexoses needed for bacterial cell surface antigens (Fig. 4-15). Glucose (as cytidine diphosphate glucose; CDP-glucose) is first converted to 4-oxo-6-deoxy-CDP-glucose. The conversion of the latter to 3,6-dideoxy-CDP-glucose (Eq. 14-36) requires PMP as well as NADH or NADPH.



The student may find it of interest to propose a mechanism for this reaction, taking into account the expected direct transfer of a hydrogen from NADH as described in Chapter 15, before consulting published papers. Part of the reaction cycle appears to involve a free radical derived from the PMP. This is discussed further in Chapter 20 together with free radical-forming PLP enzymes

5. Stereochemistry of PLP-Requiring Enzymes

According to stereoelectronic principles, the bond in the substrate amino acid that is to be broken by a PLP-dependent enzyme should lie in a plane perpendicular to the plane of the cofactor-imine π system (Fig. 14-8). This would minimize the energy of the transition state by allowing maximum σ - π overlap between the breaking bond and the ring-imine π system. It also would provide the geometry closest to that of the planar quinonoid intermediate to be formed, thus minimizing molecular motion in the approach to the transition state. Figure 14-8 shows three orientations of an amino acid in which the α -hydrogen, the carboxyl group, and the side chain, respectively, are positioned for cleavage. For each orientation shown, another geometry suitable for cleavage of the same bond is

In each case the bond to be broken lies perpendicular to π system of Schiff base

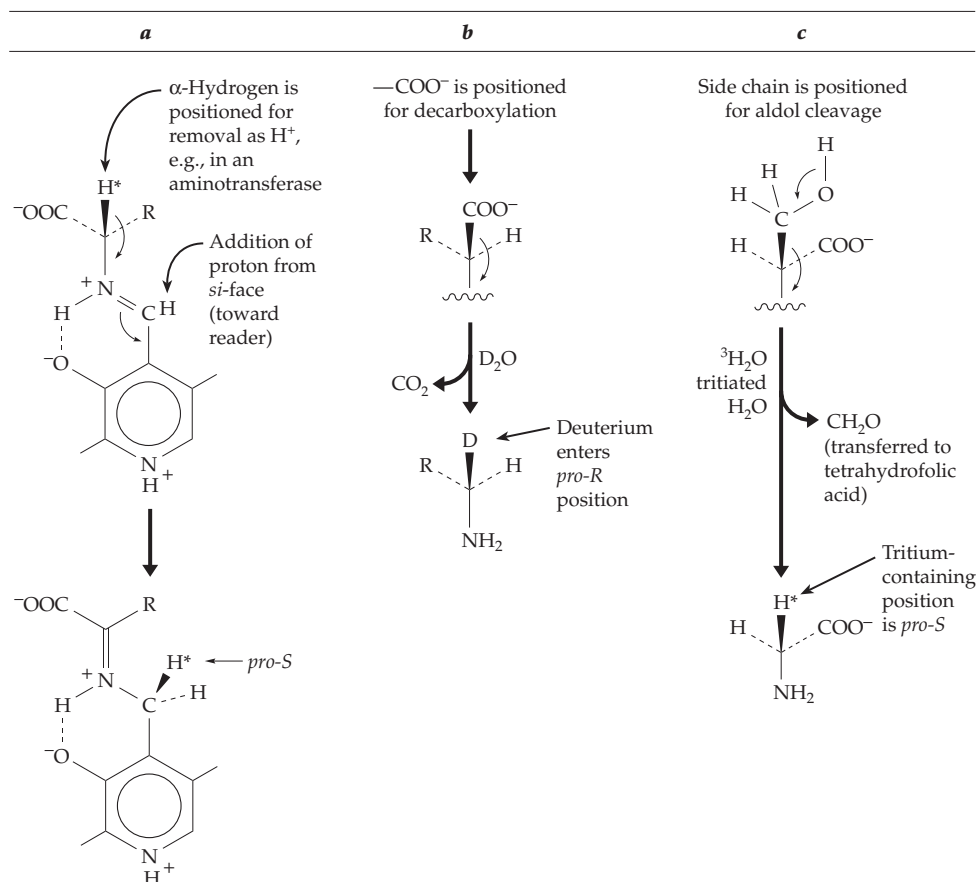
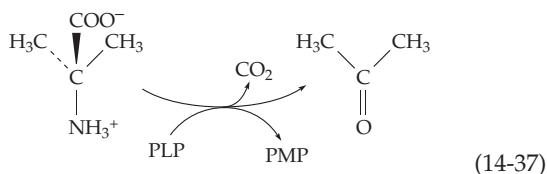


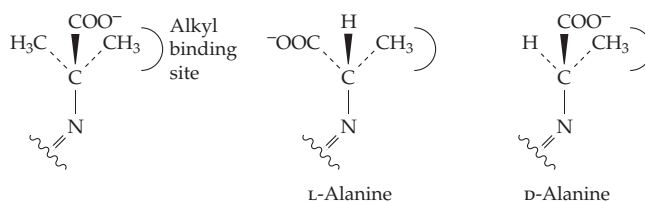
Figure 14-8 Some stereochemical aspects of catalysis of PLP-requiring enzymes.

obtained by rotating the amino acid through 180° .

Dunathan suggested that this stereoelectronic requirement explains certain side reactions observed with PLP-requiring enzymes. The idea also received support from experiments with a bacterial α -dialkylglycinedecarboxylase.^{231,232} The enzyme ordinarily catalyzes, as one half-reaction, the combination decarboxylation–transamination reaction shown in Eq. 14-37. It also acts on both D- and L-alanine, decarboxylating the former but catalyzing only removal of the α -H



from L-alanine. The results can be rationalized by assuming that the enzyme possesses a definite site for one alkyl group but that the position of the second alkyl group can be occupied by $-H$ or $-COO^-$ and that the group labilized lies perpendicular to the π system:



Glycine is unreactive, suggesting that occupation of the alkyl binding site is required for catalysis.

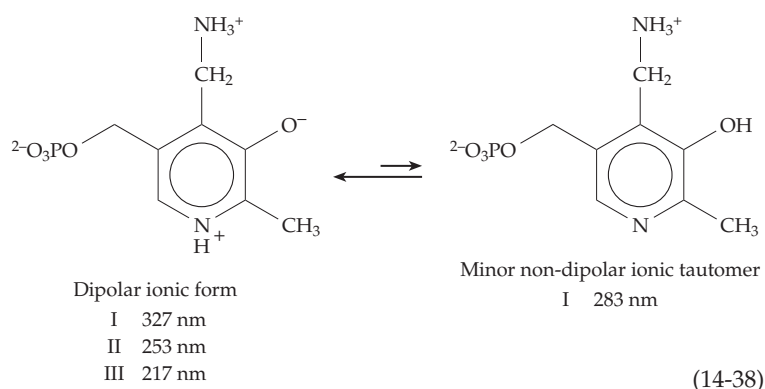
According to Dunathan's postulate, there are only two possible orientations of the amino acid substrate in an aminotransferase. One is shown in Fig. 14-8. In the other, the amino acid is rotated 180° so that the α -hydrogen protrudes *behind* the plane of the paper. Dunathan studied **pyridoxamine:pyruvate aminotransferase**, an enzyme closely related to PLP-requiring aminotransferases and which catalyzes the transamination of pyridoxal with L-alanine to form pyridoxamine and pyruvate. The same reaction is catalyzed by the apoenzyme of aspartate aminotransferase. In both cases, when the alanine contained 2H in the α position the 2H was transferred stereospecifically into

the *pro-S* position at C-4' of the pyridoxamine (indicated by asterisks in Fig. 14-8).²³³ The results suggested that a group from the protein abstracts a proton from the α position and transfers it on the same side of the π system (**suprafacial transfer**), adding it to the *si* face of the C=N group as shown in Fig. 14-8. Later, the same stereospecific proton transfer was demonstrated for the PLP present in the holoenzyme.²³⁴ Not surprisingly, the D-amino acid aminotransferase adds the proton to the *re* face of the C=N group.²³⁵

When a decarboxylase acts on an amino acid in $^2\text{H}_2\text{O}$, an atom of ^2H is incorporated in the *pro-R* position, the position originally occupied by the carboxyl group (Fig. 14-8). Cleavage of serine by serine hydroxymethyltransferase in ^3H -containing water leads to incorporation of ^3H in the *pro-S* position. Stereospecific introduction of ^2H or ^3H has been observed in the β position of 2-oxobutyrate formed in β or γ elimination reactions. Conversion of serine to tryptophan by tryptophan synthetase occurs without inversion at C-3.²³⁶ These and many other observations on PLP-dependent enzymes^{189,237,238} can be generalized by saying that enzymatic reactions of PLP Schiff bases usually take place on only one face of the relatively planar structure. This is the *si* face at C-4' of the coenzyme (see Fig. 14-8). This result is expected if a single acid-base group serves as proton acceptor in one step and as proton donor in a later step. This leads naturally to the observed retention of configuration in steps involving replacement and the suprafacial transfer of protons from one position on that face to another.

6. Seeing Changes in the Optical Properties of the Coenzyme

The absorption of light in the ultraviolet and visible regions is a striking characteristic of many coenzymes. It can be measured accurately and displayed as an absorption spectrum and may also give rise to circular dichroism and to fluorescence (Chapter 23). The optical properties of the vitamin B₆ coenzymes are sensitive to changes both in environment and in the state of protonation of groups in the molecule. For example, PMP in the neutral dipolar ionic form, which exists at pH 7, has three strong light absorption bands centered at 327, 253, and 217 nm.²³⁹ The other ionic forms of PMP and other derivatives of vitamin B₆ also each have three absorption bands spaced at roughly similar intervals, but with varying positions and intensities. The minor tautomer of PMP containing an uncharged ring (Eq. 14-38) has its low-energy (long-wavelength) band at 283 nm. When both the ring nitrogen and phenolic oxygen are protonated, the band shifts again



to 294 nm and if both groups are deprotonated the resulting anion absorbs at 312 nm. Thus, observation of the absorption spectrum of the coenzyme bound to an enzyme surface can tell us whether particular groups are protonated or unprotonated. The peak of bound PMP at 330 nm in aspartate aminotransferase (Fig. 14-9) is indicative of the dipolar ionic form (Eq. 14-38). However, the 5-nm shift from the position of free PMP suggests a distinct change in environment.

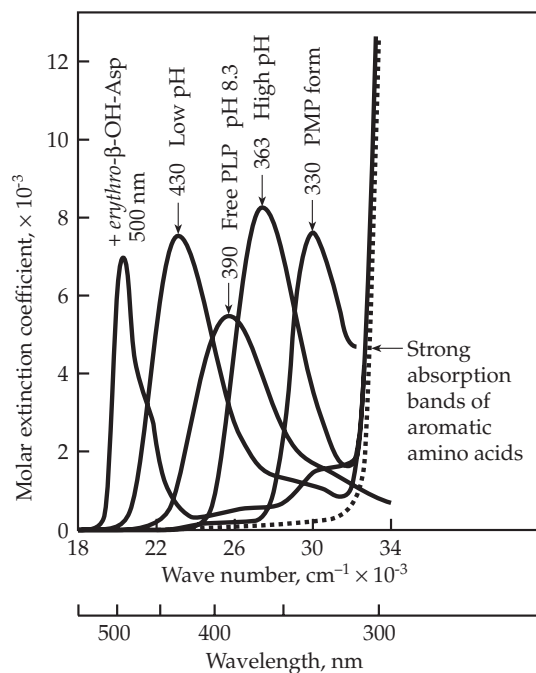
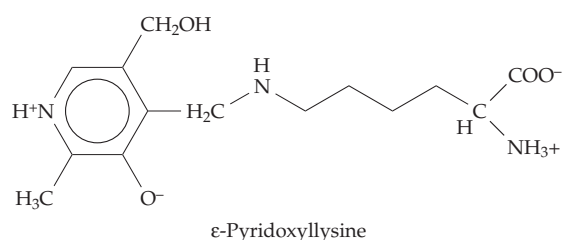


Figure 14-9 Absorption spectra of various forms of aspartate aminotransferase compared with that of free pyridoxal phosphate. The low pH form of the enzyme observed at pH < 5 is converted to the high pH form with $pK_a \sim 6.3$. Addition of *erythro*-3-hydroxyaspartate produces a quinonoid form whose spectrum here is shown only 1/3 its true height. The spectrum of free PLP at pH 8.3 is also shown. The spectrum of the apoenzyme (---) contains a small amount of residual absorption of uncertain origin in the 300- to 400-nm region.

Pyridoxal phosphate exists in an equilibrium between the aldehyde and its covalent hydrate (as in Eq. 13-1). The aldehyde has a yellow color and absorbs at 390 nm (Fig. 14-9), while the hydrate absorbs at nearly the same position as does PMP. The absorption bands of Schiff bases of PLP are shifted even further to longer wavelengths, with N-protonated forms absorbing at 415–430 nm. Forms with an unprotonated C=N group absorb at shorter wavelengths.^{149,240}

Imine groups in free enzymes. When, in 1957, W. T. Jenkins examined one of the first highly purified aspartate aminotransferase preparations he noted a surprising fact: The bound coenzyme, at pH 5, absorbed not at 390 nm, as does PLP, but at 430 nm, like a Schiff base (Fig. 14-9). When the pH was raised, the absorption band shifted to 363 nm. The result suggests dissociation of a proton (with a pK_a of ~6.3) from the hydrogen-bonded position in a Schiff base of the type shown in Fig. 14-6. It was quickly demonstrated for this enzyme and for many other PLP-dependent enzymes that reduction with sodium borohydride caused the spectrum to revert to one similar to that of PMP and fixed the coenzyme to the protein. After complete HCl digestion of such borohydride-reduced proteins a fluorescent amino acid containing the reduced pyridoxyl group was obtained and in every case was identified as ϵ -pyridoxyl-lysine. Thus, PLP-containing enzymes in the absence of substrates usually exist as Schiff bases with lysine side chains of the proteins. Even the PLP in glycogen phosphorylase is joined in this way. However, its absorption maximum at 330 nm shows



that in phosphorylase it is present as the nondipolar ionic tautomer with a 3-OH group on the ring as in Eq. 14-38.

Absorption bands at 500 nm. With many PLP enzymes certain substrates and inhibitors cause the appearance of intense and unusually narrow bands at ~500 nm. Such a band is observed with aspartate aminotransferases acting on *erythro*-3-hydroxyaspartate (Fig. 14-9). This substrate undergoes transamination very slowly, and the 500-nm absorbing form which accumulates is probably an intermediate in the normal reaction sequence. A similar spectrum is produced by tryptophan indole-lyase acting on the competitive inhibitor L-alanine. Under the same conditions the

enzyme promotes a rapid exchange of the α -hydrogen of the alanine with ^2H of $^2\text{H}_2\text{O}$. Serine hydroxymethyl-transferase gives a 495- to 500-nm band with both D-alanine and the normal product glycine.²⁴¹ Similar spectra have been produced in nonenzymatic model reactions²⁴² and probably represent the postulated quinonoid–carbanionic intermediates.

7. Atomic Structures

The three-dimensional structures of aspartate aminotransferases from *E. coli* to humans are very similar.^{163–167a} The folding pattern (Fig. 2-6) and active site structure (Figs. 14-6 and 14-10) are completely conserved. The major domain of the protein contains a central β sheet surrounded by helices with coenzyme attached to a lysine at the C terminus of one of the β strands. The protein is a dimer with the two major domains held together by both polar and nonpolar interactions. The two active sites are located at the interface between the subunits and residues from both subunits participate in forming the active site (Fig. 14-10). The internal Schiff base is formed with Lys 258. The protonated ring nitrogen of the dipolar ionic PLP forms an ion pair with the carboxylate of Asp 222 which protrudes from a central seven-stranded β sheet. The phenolic $-\text{O}^-$ forms a hydrogen bond with the $-\text{OH}$ of Tyr 225. The interactions of Asp 222 and Tyr 225 fix the ring as the dipolar ionic tautomer.

The phosphate group of the coenzyme, which ^{31}P NMR shows to be predominantly dianionic,²⁴³ forms an ion pair with the side chain of Arg 266 and hydrogen bonds to a backbone N–H at the N terminus of a long helix where it can interact with the positive end of the helix dipole. In addition, the phosphate forms hydrogen bonds to four OH groups of Ser, Thr, and Tyr side chains. In front of the coenzyme ring are two guanidinium groups from the side chains of Arg 386 and of Arg 292* (from the second subunit). These have been shown by X-ray crystallography to bind the two carboxylate groups of a substrate such as glutamate, 2-oxoglutarate, aspartate, oxaloacetate, or cysteine sulfinate; of quasi-substrates such as 2-methylaspartate and *erythro*- β -hydroxyaspartate; or of dicarboxylic inhibitors (Fig. 14-10).

Comparison of amino acid sequences suggests that many other PLP-dependent enzymes have folding patterns similar to those of aspartate aminotransferase but that there are four or more additional different folding patterns.^{205,243a,b} Among the enzymes resembling aspartate aminotransferase are ω -amino acid: pyruvate aminotransferase,²⁴⁴ 2,3-dialkylglycine decarboxylase,²³² tyrosine phenol-lyase,¹⁷⁷ a bacterial ornithine decarboxylase,^{182,204} and cystathionine β -lyase.¹⁸² The tryptophan synthase β subunit has a second folding pattern,¹⁸⁴ while alanine racemase¹⁵³ and eukaryotic

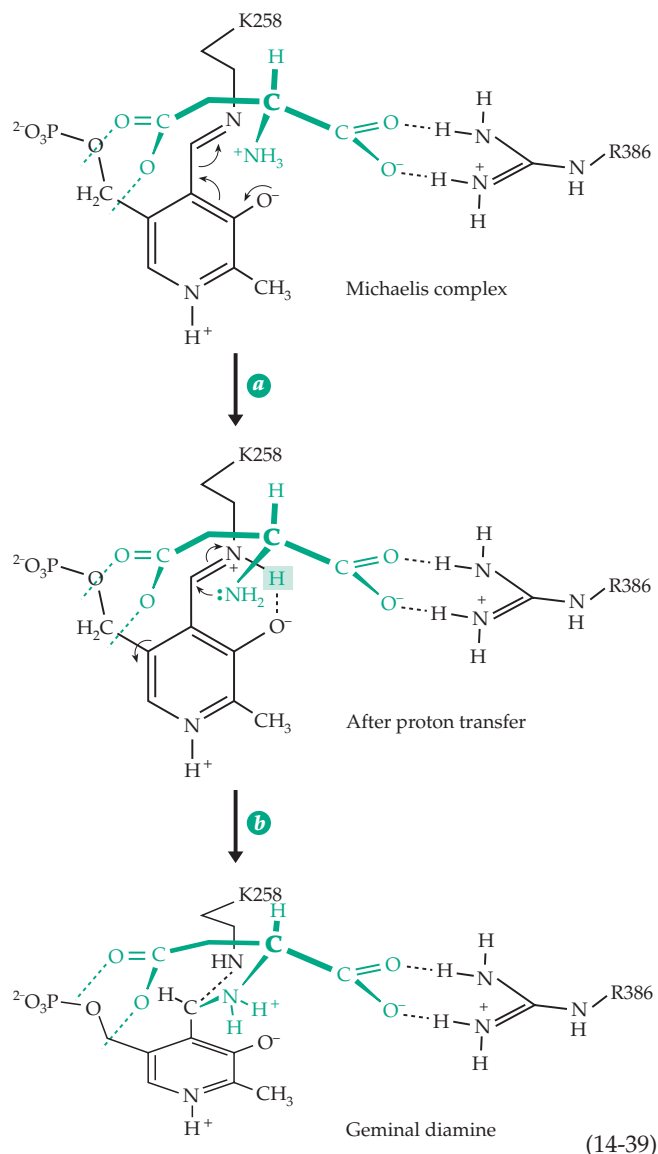
ornithine decarboxylases²⁰⁵ have $(\alpha\beta)_8$ -barrel structures resembling that in Fig. 2-28. A fourth structural pattern is that of D-amino acid aminotransferase.²³⁵ It is anticipated that the branched-chain aminotransferase²⁴⁵ will have a similar structure. Glycogen phosphorylase (Fig 11-5) has a fifth folding pattern.

8. Constructing a Detailed Picture of the Action of a PLP Enzyme

Consider the number of different steps that must occur in about one-thousandths of a second during the action of an aminotransferase. First, the substrate binds to form the “Michaelis complex.” Then the transimination (Eq. 14-26) takes place in two steps and is followed by the removal of the α -hydrogen to form the quinonoid intermediate. An additional four steps are needed to form the ketimine, to hydrolyze it, and to release the oxoacid product to give the PMP form of the enzyme. The reaction sequences in some of the other enzymes are even more complex. How can one enzyme do all this?

The first step in the sequence is the binding of the substrate to form the “Michaelis complex.” The positive charges on Arg 386 and Arg 292 doubtless attract the carboxylate groups of the substrate and aid in guiding it toward a correct fit. In a similar manner the $-O^-$ of the coenzyme, which is distributed by resonance into the $-C=N$ of the Schiff base linkage, attracts the $-NH_3^+$ of the substrate. When a substrate or inhibitor binds to the two guanidinium groups a small structural domain of the enzyme moves and closes around the substrate which now has very little contact with the external solvent. In the initial “Michaelis complex” the $-NH_3^+$ group of the substrate lies directly in front of the C-4' carbon of the coenzyme (Fig. 14-10), where it can initiate the transimination reaction of Eq. 14-26. However, before this can happen a proton must be removed to convert the $-NH_3^+$ to $-NH_2$.

Long before the three-dimensional structures were known, Ivanov and Karpeisky²⁴⁶ suggested that in the free enzyme the positively charged group (Arg 386) that binds the α -carboxylate of the substrate interacts electrostatically with the $-O^-$ of the coenzyme. This is one of the factors that keeps the pK_a of the $-CH=N^+H-$ that is conjugated with this $-O^-$ at a low value of ~ 6.3 (at 0.1 M anion concentration). However, in the Michaelis complex this interaction of the $+$ charge of Arg 386 with the imine group must be weakened because of the pairing of the α - COO^- of the substrate with the $+$ charge. This will increase the basicity of the imine nitrogen and will also cause a decrease in the pK_a of the substrate $-NH_3^+$, making it easier for a proton to jump from the $-NH_3^+$ to the imine group. This proton transfer is shown in Eq. 14-39, step *a*. Thus, the nucleophilic $-NH_2$ group is generated by a



process that at the same time increases the electrophilic properties of the carbon atom of the imine group. This favors the immediate addition of $-NH_2$ to $-C=N^+H-$ to give the adduct a **geminal diamine** (Eq. 14-39, step *b*), which is shown in three dimensions in Fig. 14-10B.

Notice that each step in the overall sequence changes the electronic or steric characteristics of the complex in a way that facilitates the next step.²⁴⁶ This is an important principle that is applicable throughout enzymology: *For an enzyme to be an efficient catalyst each step must lead to a change that sets the stage for the next.* These consecutive steps often require proton transfers, and each such transfer will influence the subsequent step in the sequence. Some steps also require alterations in the conformation of substrate, coenzyme, and enzyme. One of these is the transimination sequence (Eqs. 14-26, 14-39). On the basis of the observed loss of circular dichroism in the external aldimine, Ivanov and Karpeisky suggested that a

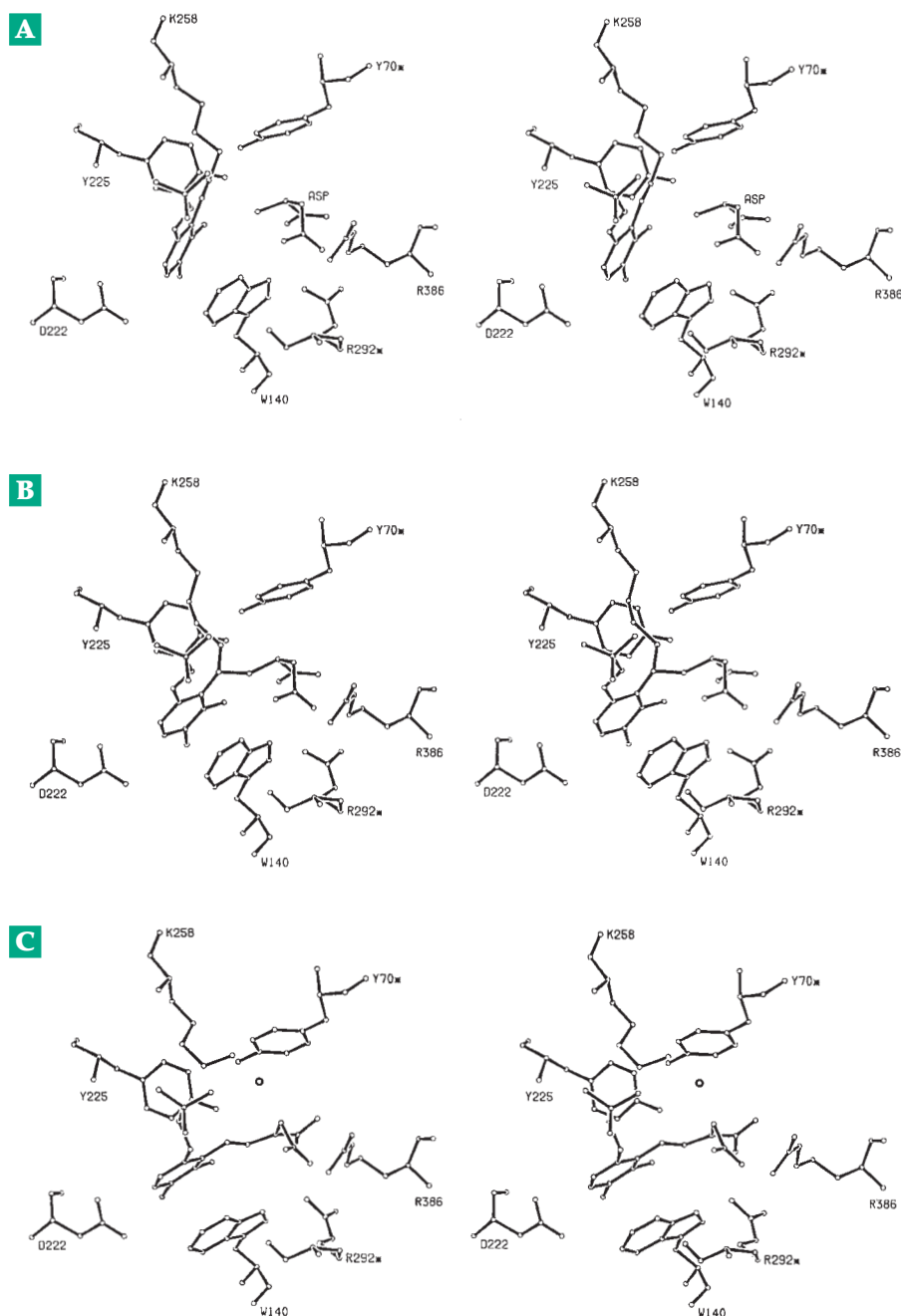


Figure 14-10 Models of catalytic intermediates for aspartate aminotransferase in a half-transamination reaction from aspartate to oxaloacetate. For clarity, only a selection of the active site groups are shown. (A) Michaelis complex of PLP enzyme with aspartate. (B) Geminal diamine. (C) Ketimine intermediate. The circle indicates a bound water molecule. See Jansonius and Vincent in Jurnak and McPherson.¹⁶³ Courtesy of J.N. Jansonius.

rotation of the coenzyme occurs as the -NH_2 of the substrate adds to the $\text{C}=\text{N}$ bond during transimination (Eq. 14-39, step *b*). This accomplishes the essential shortening by ~ 0.15 nm of the distance between C-4' and the N atom from a van der Waals contact distance to a covalent bond distance while the carboxylate

groups of the substrate remain bound in their initial positions.

That the coenzyme really does change its orientation was suggested by a dramatic change in the absorption spectrum of a crystal recorded with plane polarized light (linear dichroism) when 2-methylaspartate was soaked into a crystal (Fig. 23-9).²⁴⁷ X-ray crystallography confirmed rotation of the ring by $\sim 30^\circ$.^{163,242,247} To complete the transimination sequence, which is shown only partially by Eq. 14-39, another proton transfer is needed to move the positive charge on the substrate $\text{-N}^+\text{H}_2\text{-}$ to that of lysine 258, whose amino group is then eliminated. This requires additional tilting of the ring. The crystal structure of the external aldimine with α -methylaspartate has been determined²⁴⁸ as have those of ketimines with glutamate and aspartate,^{249,250} a carbinolamine,²⁵¹ and quinonoid complexes of related enzymes.^{212a}

The ϵ -amino group, which is eliminated in Eq. 14-26, is basic and functions in the next several steps of catalysis, including the abstraction of the α -hydrogen and in its transfer to the 4'-carbon (Eq. 14-28). This amino group can be seen in Fig. 14-10C, where it is positioned beside a water molecule that is needed to hydrolyze the ketimine. From this figure it can be seen that the group is able to move from site to site on one side of the planar external aldimine, quinonoid-carbanionic, and nearly planar ketimine forms. If the Lys 258 amino group is the catalytic base for these reactions it must be present at pH 7. What is its pK_a ? And why doesn't this pK_a show up in the plot of V_{max} vs pH as in Fig 9-9? In fact, the maximum

velocity of aspartate aminotransferase and many other PLP-dependent enzymes is independent of pH over a broad range. The answer is probably that the pK_a 's of both the -NH_2 groups of Lys 258 and the amino acid in the Michaelis complex are high when there is no positive charge on the adjacent Schiff base -C=N- but

very low when the Schiff base is protonated ($-\text{C}=\text{NH}^+$). This is a result of the large electrostatic effect of a closely adjacent charge in a medium of low dielectric constant (see Chapter 7, Section A). When the Schiff base is unprotonated, as in the first structure of Eq. 14-39, the adjacent amino group has a *high* pK_a and is mostly protonated. However, the proton can jump as in step *a* of that equation to give a protonated Schiff base. Now the unprotonated $-\text{NH}_2$ group has a very *low* pK_a , but it is still strongly nucleophilic and can readily add to the Schiff base double bond in step *b* of Eq. 14-39. In a similar fashion the amino group of Lys 258 will alternately be unprotonated and then protonated, its microscopic pK_a alternating between low and high, as it catalyzes the steps of abstracting the α -hydrogen and forming and hydrolyzing the ketimine. Only two microscopic pK_a values, one very high and one very low appear in the $V_{\max}$ vs pH profile. This alternation of microscopic pK_a values may be a common characteristic of enzymes that bind ionized substrates and can make good use of the strong electrostatic effects that arise in the active sites to facilitate essential proton transfers.

Aspartate aminotransferases are distinguished from most other PLP-dependent enzymes including transaminases by the relatively low pK_a of ~ 6.3 for the *free* enzyme. The unprotonated Schiff base in the free enzyme can then react with the protonated amino group of the substrate. How can other PLP enzymes react with amino acids when they have protonated Schiff bases even at relatively high pH? A logical answer is that some basic group with a low pK_a is close to the Schiff base and acts to deprotonate the substrate $-\text{NH}_3^+$ so that transamination can occur. Clausen *et al.* suggested that in cystathionine β -lyase this is probably tyrosine (Y111), which is adjacent to the Schiff base $-\text{C}=\text{NH}^+$ of the external aldimine and is thought to be ionized in the active enzyme.¹⁸² NMR evidence suggests that protonation of an adjacent catalytic base occurs upon substrate binding in D-serine dehydratase as well.²⁵² Many other variations in active site environments are seen among PLP-dependent enzymes. As with aspartate aminotransferases, there is often an essential carboxylate group that holds a proton onto the pyridine ring of the coenzyme. However, in alanine racemase a guanidinium group from arginine is hydrogen bonded to the coenzyme ring.¹⁵³

Below the active site of aspartate aminotransferase, as shown in Fig. 14-6, is a cluster of three buried histidine side chains in close contact with each other. The imidazole of H143 is hydrogen bonded to the D222 carboxylate, the same carboxylate that forms an ion pair with the coenzyme. This system looks somewhat like the catalytic triad of the serine proteases in reverse. As with the serine proteases, the proton-labeled H_b in Fig. 14-6 can be “seen” by NMR spectroscopy (Fig. 3-30). So can the proton H_a on the PLP ring. These protons

act as built-in sensors able to detect small changes in the electronic environment. For example, when the Schiff base proton dissociates around the pK_a of ~ 6.2 the NMR resonance of H_a shifts upfield from 17.2 to 15.2 ppm as a result of donation of electrons into the ring from the $-\text{O}^-$ of the coenzyme.¹⁶⁹ This shift illustrates the reality of the strong electrostatic forces that operate across heterocyclic aromatic rings within active sites of proteins. In alanine racemase a different histidine cluster is present beneath the active site and constitutes part of the “solvent” in which the catalyzed reaction takes place. At least in the case of aspartate aminotransferase, none of the histidines are absolutely essential for activity but the hydrogen-bonded network, which can be altered in mutant forms, may be important.

The detailed description of a reaction sequence given here has to be altered for each specific enzyme. A vast amount of work, only a little of which is cited here, has been done on PLP enzymes.^{253–254a} These studies involve calorimetry,²⁵⁵ kinetics,^{210,256–259} crystallography, optical spectroscopy,^{188,260,261} NMR,^{243,262–264} and genetic engineering and chemical modification.^{178,185,186,265}

F. Pyruvoyl Groups and Other Unusual Electrophilic Centers

A few enzymes that might be expected to have PLP at their active sites have instead a prosthetic group consisting of pyruvic acid bound by an amide linkage, a **pyruvoyl group** (Table 14-4). These and several apparently related enzymes are the subject of this section.

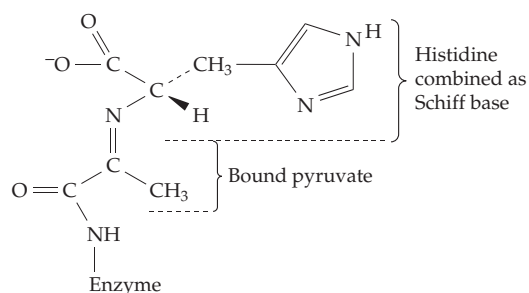
TABLE 14-4
Some Pyruvoyl Enzymes

Decarboxylases	Product
Histidine (bacterial)	
S-Adenosylmethionine	
Aspartate α - decarboxylase	β -Alanine
Phosphatidylserine	Phosphatidyl-ethanolamine
4' - Phosphopantothencylcysteine	4' - Phosphopantetheine
Reductases (clostridial)	
Proline	
Glycine	

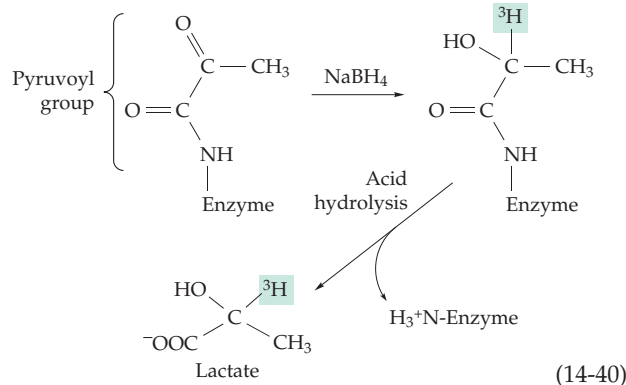
Adapted from van Peolje and Snell.²⁶⁷

1. Decarboxylases

Mammalian **histidine decarboxylase** contains PLP but the enzyme from many bacteria contains a pyruvoyl group, as do a few other decarboxylases both of bacterial and eukaryotic origin.^{266,267} These enzymes are inhibited by carbonyl reagents and by borohydride. When ^3H -containing borohydride was used to reduce the histidine decarboxylase of *Lactobacillus*, ^3H was incorporated and was recovered in lactic acid following hydrolysis. This suggested the presence of a pyruvoyl group attached by an amide linkage and undergoing the chemical reactions that are shown in Eq. 14-40. Reduction in the presence of the substrate histidine resulted in covalent binding of the histidine to the bound pyruvate. Thus, as with the PLP-containing decarboxylases, a Schiff base is formed with the substrate. Decarboxylation is presumably accomplished by using the electron-attracting properties of the carbonyl group of the amide:



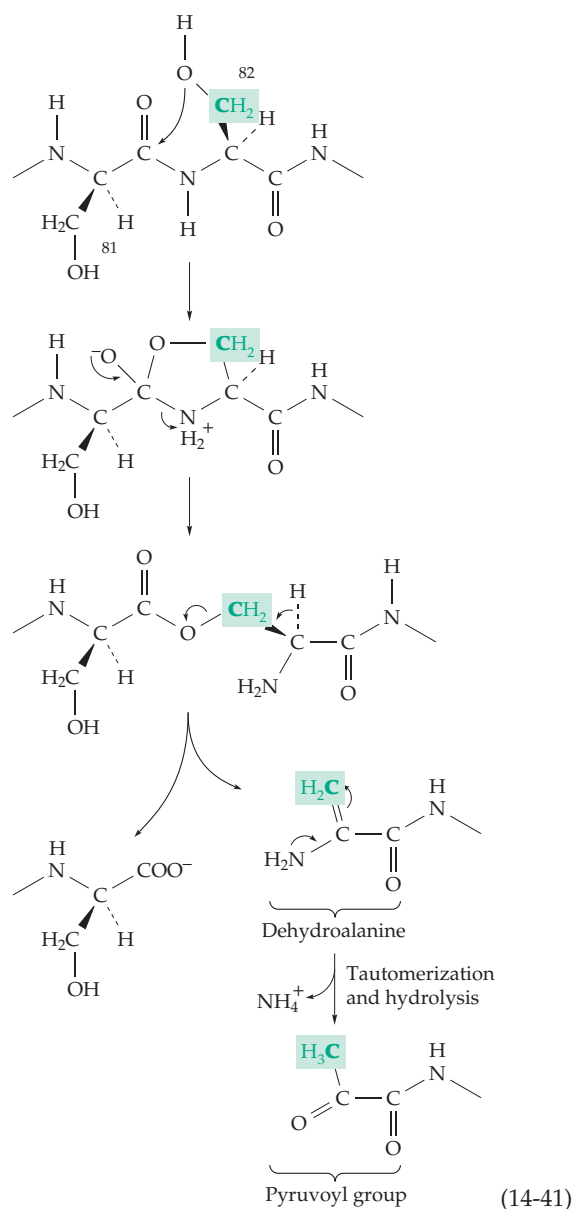
When ^{14}C -labeled serine was fed to organisms producing histidine decarboxylase, ^{14}C was incorporated into the bound pyruvoyl group (Fig. 14-11). Thus, serine is a precursor of the bound pyruvate. The enzyme is manufactured in the cell as a longer 307-residue proenzyme which associates as hexamers (designated π_6). The active enzyme was found to be formed by cleavage of the π chains between Ser 81 and Ser 82 to form 226-residue α chains and 81-residue β chains which associate as $(\alpha\beta)_6$.^{270,271} The α chains



carry the N-terminal pyruvoyl group, which was formed from Ser 82. The activation occurs spontaneously by incubations of the proenzyme for 24–48 h at pH ~7 in the presence of divalent metal ions.

Substituted serine residues under mildly alkaline conditions readily undergo α,β elimination to form **dehydroalanine** residues. When prohistidine decarboxylase containing ^{18}O in its serine side chains was activated ^{18}O was found in the carboxylate group of Ser 81 of the β chains. It was shown that it had been transferred from the side chain of Ser 82. This suggested the formation of an intermediate oxygen ester of Ser 81 during formation of the pyruvoyl group (Eq. 14-41).²⁷²

S-Adenosylmethionine decarboxylase is the first enzyme in the biosynthetic pathway to spermidine (Chapter 24). Whether isolated from bacteria, yeast, animals, or other eukaryotes, this enzyme always contains a bound pyruvoyl group.^{273–274b} Both the



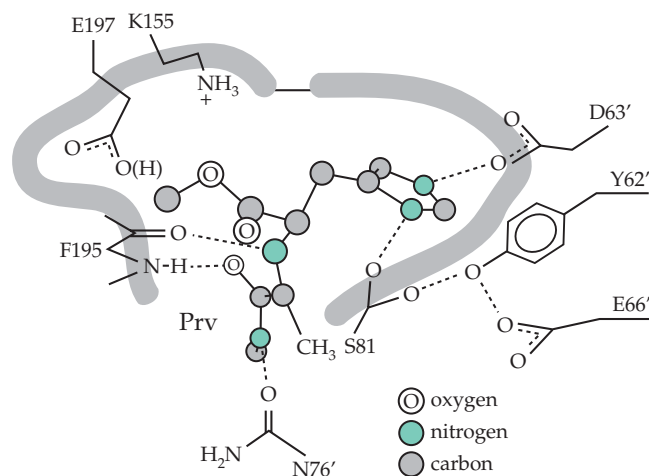
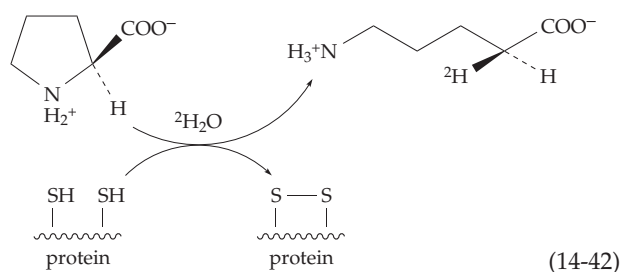


Figure 14-11 Schematic diagram of the active site of the pyruvoyl enzyme histidine decarboxylase showing key polar interactions between the pyruvoyl group and groups of the inhibitor *O*-methylhistidine and surrounding enzyme groups. Aspartate 63 appears to form an ion pair with the imidazolium group of the substrate.²⁶⁸ Hydrogen bonds are indicated by dotted lines. See Gallagher *et al.*²⁶⁹

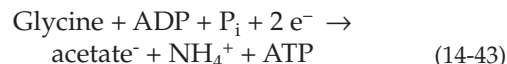
mammalian enzyme and that from *E. coli* are $(\alpha\beta)_4$ tetramers formed in a manner similar to that of bacterial histidine decarboxylase. Other pyruvoyl decarboxylases are **phosphatidylserine decarboxylase**,^{275,276} an intrinsic membrane protein used to form phosphatidylethanolamine, **aspartate α -decarboxylase**,²⁷⁷ which forms β alanine needed for biosynthesis of coenzyme A, and **4'-phosphopantethenoylcysteine decarboxylase**,²⁷⁸ the second of two decarboxylases required in the synthesis of coenzyme A. Because of the lack of a primary amino group, its mechanism must be somewhat different from that of other enzymes in this group.²⁶⁷

2. Proline and Glycine Reductases

An enzyme required in the anaerobic breakdown of proline by clostridia utilizes a dithiol-containing protein to reductively open the ring (Eq. 14-42).^{279,280}



This enzyme also contains an N-terminal pyruvoyl residue as does one subunit of a selenium-containing glycine reductase which utilizes a dithiol to convert glycine into acetate with coupled formation of ATP:

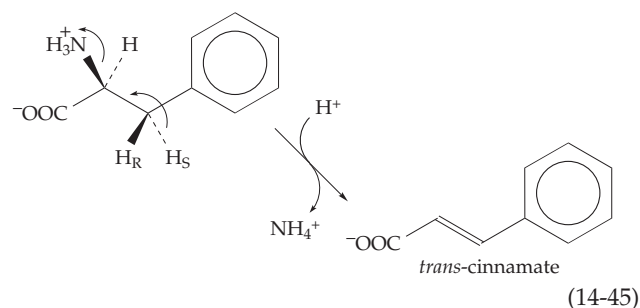
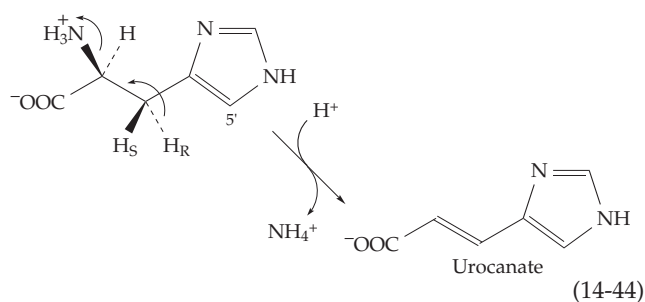


In both cases it has been proposed that the pyruvoyl group forms a positively charged Schiff base with the substrate. The bond-breaking mechanisms are not obvious but some ideas have been proposed.^{281-283a} The ATP may arise from reaction with an intermediate acetyl phosphate.²⁸² See Eq. 15-61.

3. Dehydroalanine and Histidine and Phenylalanine Ammonia-Lyases

Catabolism of histidine in most organisms proceeds via an initial elimination of NH_3 to form **urocanic acid** (Eq. 14-44). The absence of the enzyme **L-histidine ammonia-lyase** (histidase) causes the genetic disease histidinemia.^{284,285} A similar reaction is catalyzed by the important plant enzyme **L-phenylalanine ammonia-lyase**. It eliminates $-\text{NH}_3^+$ along with the *pro-S* hydrogen in the β position of phenylalanine to form *trans*-cinnamate (Eq. 14-45). Tyrosine is converted to *p*-coumarate by the same enzyme. Cinnamate and coumarate are formed in higher plants and are converted into a vast array of derivatives (Box 21-E, Fig. 25-8).

The reactions of Eqs. 14-44 and 14-45 are unusual because the nucleophilic substituent eliminated is on the α -carbon atom rather than the β . There is nothing in the structures of the substrates that would permit an



easy elimination of the α -amino group. Thus, it should not be surprising that both enzymes contain a special active center. When ^2H is introduced into the β position of phenylalanine, no isotope effect on the rate is observed. Rather, the rate-limiting step appears to be the release of ammonia from the coenzyme group. It appears that the enzyme must in some way make the amino group a much better leaving group than it would be otherwise.

Both enzymes are inhibited by sodium borohydride and also by nitromethane. After reduction with NaB^3H_4 and hydrolysis, ^3H -containing alanine was isolated. This suggested that they contain **dehydroalanine**, which could arise by dehydration of a specific serine residue.^{286,287} For phenylalanine ammonia-lyase from *Pseudomonas putida* this active site residue has been identified as S143. Replacement by cysteine in the S143C mutant also gave active enzyme while S143A

BOX 14-D DIETARY REQUIREMENTS FOR B VITAMINS

It is difficult to determine the amounts of vitamins needed for good health and they may differ considerably from one individual to another. The quantities listed below are probably adequate for most young adults but must be increased during pregnancy and lactation and after very strenuous exercise.

Pantothenic acid: 10–15 mg/day. Deficiency causes apathy, depression, impaired adrenal function, and muscular weakness. ω -Methylpantothenic acid is a specific antagonist. The calcium salt, calcium pantothenate, is the usual commercial form.

Biotin: 0.15–0.3 mg/day. The discovery that biotin deficiency in young chickens can lead to sudden death resulted in a recommendation to supplement infant formulations with biotin.^a

Desthiobiotin, in which the sulfur has been removed and replaced by two hydrogen atoms, can replace biotin in some organisms and appears to lie on one pathway of biosynthesis.^{b,c} **Oxybiotin**, in which the sulfur has been replaced by oxygen, is active for many organisms and partially active for others. No evidence for conversion to biotin itself has been reported, and oxybiotin may function satisfactorily in at least some enzymes.

Thiamin: 0.23 mg or more per 1000 kcal of food consumed and a minimum total of 0.8 mg/day. Replacement of the methyl group on the pyrimidine ring by ethyl, propyl, or isopropyl gives compounds with some vitamin activity, but replacement by hydrogen cuts activity to 5% of the original. The butyl analog is a competitive inhibitor.

Vitamin B₆: 1.5–2 mg/day; 0.4 mg/day for infants. Vitamin B₆ is widely distributed in foods, and symptoms of severe deficiency are seldom observed. However, a number of cases of convulsions have been attributed to partial destruction of vitamin B₆ in infant liquid milk formulas. Convulsions occurred when the vitamin B₆ content was reduced to about one-half that normally present in human milk.

Several cases of children with an abnormally high vitamin B₆ requirement (2–10 mg/day) have

been reported, and rare metabolic diseases are known^{d-f} in which specific enzymes, such as cystathionine synthase, have a reduced affinity for PLP. Patients with these diseases also benefit from a higher than normal intake of the vitamin. Excessive excretion of the vitamin may also occur, an example being provided by a strain of laboratory mice that require twice the normal amount of vitamin B₆ and which die in convulsions after a brief period of vitamin B₆ depletion.^d Dietary supplementation with large amounts of vitamin B₆ for treatment of medical conditions such as carpal tunnel syndrome has been controversial.^g Amounts of pyridoxine over 50 mg per day may damage peripheral nerves, probably because of the chemical reactivity of pyridoxal and PLP.

Nicotinamide: About 7.5 mg/day. Tryptophan can substitute to some extent. See also Box 15-A.

Riboflavin: About 1.5 mg/day. See also Box 15-B.

Folic acid: About 0.2–0.4 mg/day. See also Box 15-D.

Vitamin B₁₂: About 2 μg /day. See also Box 16-B.

Vitamin C (ascorbic acid): 50–200 mg/day. See also Box 18-D.

^a Parry, R. J., and Kunitni, M. G. (1976) *J. Am. Chem. Soc.* **98**, 4024–4025

^b Gibson, K. J., Lorimer, G. H., Rendina, A. R., Taylor, W. S., Cohen, G., Gatenby, A. A., Payne, W. G., Roe, D. C., Lockett, B. A., Nudelman, A., Marcovici, D., Nachum, A., Wexler, B. A., Marsilii, E. L., Turner, S., IM, Howe, L. D., Kalbach, C. E., and Chi, H. (1995) *Biochemistry* **34**, 10976–10984

^c Huang, W., Jia, J., Gibson, K. J., Taylor, W. S., Rendina, A. R., Schneider, G., and Lindqvist, Y. (1995) *Biochemistry* **34**, 10985–10995

^d Mudd, S. H., Levy, H. L., and Skovby, F. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 1 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 1279–1327, McGraw-Hill, New York

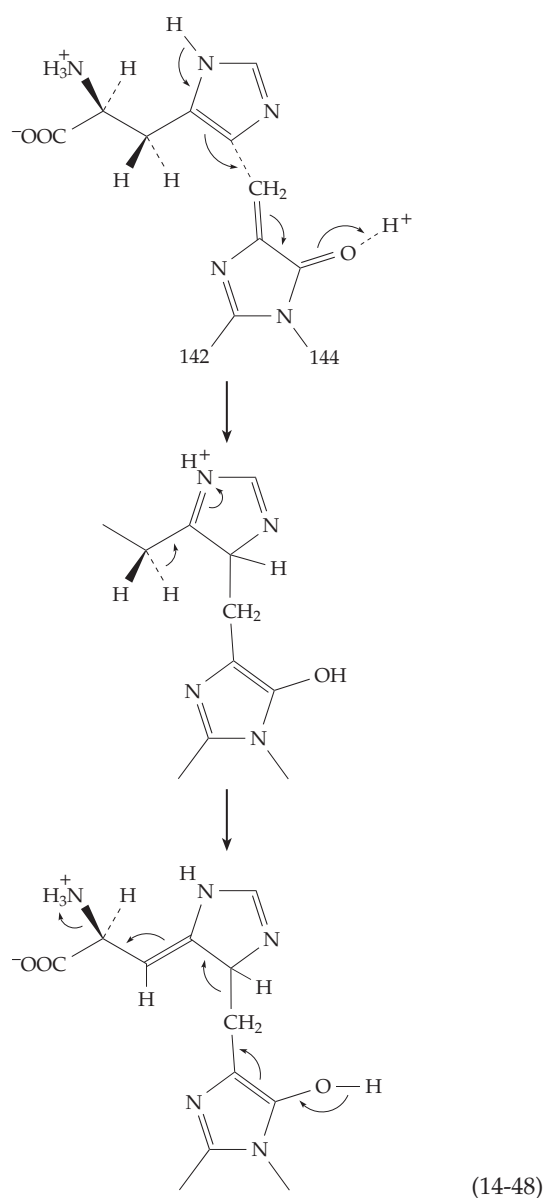
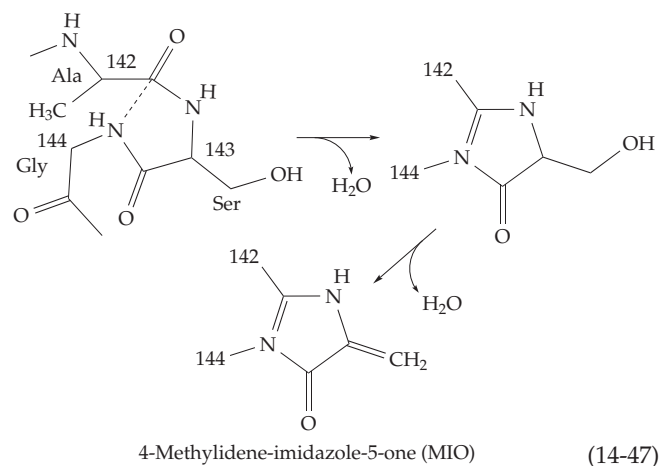
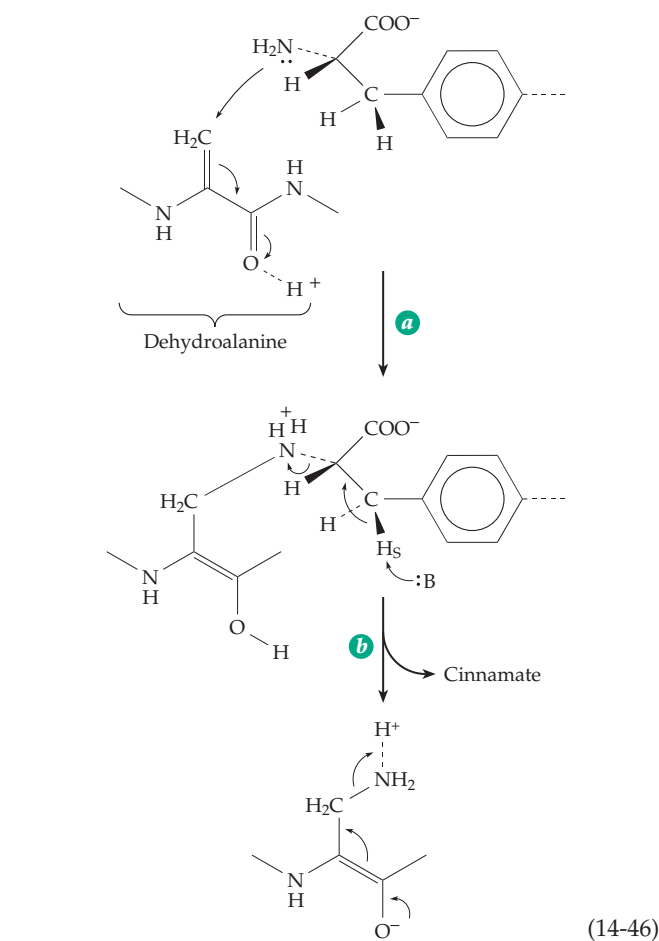
^e Bell, R. R., and Haskell, B. E. (1971) *Arch. Biochem. Biophys.* **147**, 588–601

^f Pascal, T. A., Gaull, G. E., Beratis, N. G., Gillam, B. M., Tallan, H. H., and Hirschhorn, K. (1975) *Science* **190**, 1209–1211

^g Bender, D. A. (1999) *Br. J. Nutr.* **81**, 7–20

was inactive.^{288,289} These results support the formation of dehydroalanine. One proposed mechanism of action of the enzymes involved addition of the substrate amino group to the C = C bond of the dehydroalanine followed by elimination (Eq. 14-46). Dissociation of the C – H bond in this step would be assisted by the electron-accepting properties of the adjacent aromatic ring. However, the proposed chemistry was not convincing. Noting that a 5-nitro substituent on the histidine greatly enhances the rate of reaction, Langer

et al. proposed that the histidine reacts with the electrophilic carbon in the pyruvoyl center as in Eq. 14-47.^{290,291} However, this chemistry, too, was unprecedented in enzymology. Determination of the three-dimensional structure of histidine ammonia-lyase²⁹² led to the discovery of a new prosthetic group and a solution to the problem. Two dehydration steps, (Eq. 14-47) convert an Ala-Gly-Ser sequence within the protein into **4-methylidene-imidazole-5-one (MIO)**, a modified dehydroalanine with enhanced electron-accepting properties. The proposed mechanism of action is portrayed in Eq. 14-48.



References

1. Moodie, S. L., Mitchell, J. B. O., and Thornton, J. M. (1996) *J. Mol. Biol.* **263**, 486–500
2. Lipmann, F. (1945) *J. Biol. Chem.* **160**, 173–190
3. Roskoski, R. N., Jr. (1987) *Trends Biochem. Sci.* **12**, 136–138
- 3a. Bentley, R. (2000) *Trends Biochem. Sci.* **25**, 302–305
4. Martin, D. P., and Drucehammer, D. G. (1992) *J. Am. Chem. Soc.* **114**, 7287–7288
5. Baraniak, J., and Frey, P. A. (1988) *J. Am. Chem. Soc.* **110**, 4059–4060
6. Richter, D., and Hilz, H. (1979) *Trends Biochem. Sci.* **1**, N123–N124
7. Ohlrogge, J. B., Jaworski, J. G., and Post-Beittenmiller, D. (1993) in *Lipid Metabolism in Plants* (Moore, T. S., Jr., ed), pp. 3–32, CRC Press, Boca Raton, Florida
8. Pavela-Vrancic, M., Pfeifer, E., Schröder, W., von Döhren, H., and Kleinkauf, H. (1994) *J. Biol. Chem.* **269**, 14962–14966
9. Haese, A., Pieper, R., von Ostrowski, T., and Zocher, R. (1994) *J. Mol. Biol.* **243**, 116–122
10. Pieper, R., Ebert-Khosla, S., Cane, D., and Khosla, C. (1996) *Biochemistry* **35**, 2054–2060
11. Cortes, J., Wiesmann, K. E. H., Roberts, G. A., Brown, M. J. B., Staunton, J., and Leadlay, P. F. (1995) *Science* **268**, 1487–1489
12. Brambl, R., and Plesofsky-Vig, N. (1986) *Proc. Natl. Acad. Sci. U.S.A.* **83**, 3644–3648
13. Berg, M., Hilbi, H., and Dimroth, P. (1996) *Biochemistry* **35**, 4689–4696
14. Dimroth, P. (1987) *Microbiol. Rev.* **51**, 320–340
- 14a. Hoenke, S., Wild, M. R., and Dimroth, P. (2000) *Biochemistry* **39**, 13223–13232
15. Jackowski, S., and Rock, C. O. (1984) *J. Biol. Chem.* **259**, 1891–1895
- 15a. McAllister, K. A., Peery, R. B., Meier, T. I., Fischl, A. S., and Zhao, G. (2000) *J. Biol. Chem.* **275**, 30864–30872
- 15b. Chirgadze, N. Y., Briggs, S. L., McAllister, K. A., Fischl, A. S., and Zhao, G. (2000) *EMBO J.* **19**, 5281–5287
16. Whitty, A., Fierke, C. A., and Jencks, W. P. (1995) *Biochemistry* **34**, 11678–11689
17. Fierke, C. A., and Jencks, W. P. (1986) *J. Biol. Chem.* **261**, 7603–7606
18. Langer, B. W., Jr., and Gyorgy, P. (1968) in *The Vitamins*, 2nd ed., Vol. 2 (Sebrell, W. H., Jr., and Harris, R. S., eds), pp. 294–322, Academic Press, New York
19. Bentley, R. (1985) *Trends Biochem. Sci.* **10**, 51–56
20. DeTitta, G. T., Edmonds, J. W., Stallings, W., and Donohue, J. (1976) *J. Am. Chem. Soc.* **98**, 1920–1926
- 20a. Garton, S. D., Temple, C. A., Dhawan, I. K., Barber, M. J., and Rajagopalan, K. V. (2000) *J. Biol. Chem.* **275**, 6798–6805
- 20b. McIver, L., Baxter, R. L., and Campopiano, D. J. (2000) *J. Biol. Chem.* **275**, 13888–13894
21. Wright, L. D., Cresson, E. L., Skeggs, H. R., Peck, R. L., Wolf, D. E., Wood, T. R., Valent, J., and Folkers, K. (1951) *Science* **114**, 635–636
22. Knowles, J. R. (1989) *Ann. Rev. Biochem.* **58**, 195–221
23. López-Casillas, F., Bai, D.-H., Luo, X., Kong, I. S., Hermanson, M. A., and Kim, K.-H. (1988) *Proc. Natl. Acad. Sci. U.S.A.* **85**, 5784–5788
24. Browner, M. F., Taroni, F., Sztul, E., and Rosenberg, L. E. (1989) *J. Biol. Chem.* **264**, 12680–12685
25. Samols, D., Thornton, C. G., Murtif, V. L., Kumar, G. K., Haase, F. C., and Wood, H. G. (1988) *J. Biol. Chem.* **263**, 6461–6464
26. León-Del-Río, A., Leclerc, D., Akerman, B., Wakamatsu, N., and Gravel, R. A. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 4626–4630
27. Ahmad, P. M., and Ahmad, F. (1991) *FASEB J.* **5**, 2482–2485
28. Wang, X., Wurtele, E. S., Keller, G., McKean, A. L., and Nikolau, B. J. (1994) *J. Biol. Chem.* **269**, 11760–11769
- 28a. McKean, A. L., Ke, J., Song, J., Che, P., Achenbach, S., Nikolau, B. J., and Wurtele, E. S. (2000) *J. Biol. Chem.* **275**, 5582–5590
29. Weaver, L. M., Lebrun, L., Wurtele, E. S., and Nikolau, B. J. (1995) *Plant Physiol.* **107**, 1013–1014
30. Al-Feel, W., Chirala, S. S., and Wakil, S. J. (1992) *Proc. Natl. Acad. Sci. U.S.A.* **89**, 4534–4538
31. Lamzin, V. S., Dauter, Z., Popov, V. O., Harutyunyan, E. H., and Wilson, K. S. (1994) *J. Mol. Biol.* **236**, 759–785
32. Roessler, P. G., and Ohlrogge, J. B. (1993) *J. Biol. Chem.* **268**, 19254–19259
- 32a. Abu-Elheiga, L., Almaraz-Ortega, D. B., Baldini, A., and Wakil, S. J. (1997) *J. Biol. Chem.* **272**, 10669–10677
- 32b. Abu-Elheiga, L., Brinkley, W. R., Zhong, L., Chirala, S. S., Woldegiorgis, G., and Wakil, S. J. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 1444–1449
- 32c. Nikolskaya, T., Zagnitko, O., Tevzadze, G., Haselkorn, R., and Gornicki, P. (1999) *Proc. Natl. Acad. Sci. U.S.A.* **96**, 14647–14651
- 32d. Schulte, W., Töpfer, R., Stracke, R., Schell, J., and Martini, N. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 3465–3470
- 32e. Jitrapakdee, S., Booker, G. W., Cassady, A. I., and Wallace, J. C. (1997) *J. Biol. Chem.* **272**, 20522–20530
33. Lim, F., Morris, C. P., Occhiodoro, F., and Wallace, J. C. (1988) *J. Biol. Chem.* **263**, 11493–11497
34. Kalousek, F., Darigo, M. D., and Rosenberg, L. E. (1980) *J. Biol. Chem.* **255**, 60–65
35. Leon – Del-Río, A., and Gravel, R. A. (1994) *J. Biol. Chem.* **269**, 22964–22968
36. Sasaki, Y., Hakamada, K., Suama, Y., Nagano, Y., Furusawa, I., and Matsuno, R. (1993) *J. Biol. Chem.* **268**, 25118–25123
37. Gornicki, P., Podkowinski, J., Scappino, L. A., DiMaio, J., Ward, E., and Haselkorn, R. (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 6860–6864
38. Nenortas, E., and Beckett, D. (1996) *J. Biol. Chem.* **271**, 7559–7567
39. Waldrop, G. L., Rayment, I., and Holden, H. M. (1994) *Biochemistry* **33**, 10249–10256
- 39a. Blanchard, C. Z., Lee, Y. M., Frantom, P. A., and Waldrop, G. L. (1999) *Biochemistry* **38**, 3393–3400
40. Li, S.-J., and Cronan, J. E., Jr. (1992) *J. Biol. Chem.* **267**, 855–863
41. Li, S.-J., and Cronan, J. E. J. (1992) *J. Biol. Chem.* **267**, 16841–16847
42. Ha, J., Daniel, S., Broyles, S. S., and Kim, K.-H. (1994) *J. Biol. Chem.* **269**, 22162–22168
43. Wood, H. G., Harmon, F. R., Wuhr, B., Hubner, K., and Lynen, F. (1980) *J. Biol. Chem.* **255**, 7397–7409
44. Shenoy, B. C., and Wood, H. G. (1988) *FASEB J.* **2**, 2396–2401
45. Wilson, K. P., Shewchuk, L. M., Brennan, R. G., Otsuka, A. J., and Matthews, B. W. (1992) *Proc. Natl. Acad. Sci. U.S.A.* **89**, 9257–9261
46. Wolf, B. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 2 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 3151–3177, McGraw-Hill, New York
47. Taroni, F., and Rosenberg, L. E. (1991) *J. Biol. Chem.* **266**, 13267–13271
- 47a. Reche, P., and Perham, R. N. (1999) *EMBO J.* **18**, 2673–2682
48. Tanaka, K. (1981) *N. Engl. J. Med.* **304**, 839–840
49. Craft, D. V., Goss, N. H., Chandramouli, N., and Wood, H. G. (1985) *Biochemistry* **24**, 2471–2476
50. Chauhan, J., and Dakshinamurti, K. (1986) *J. Biol. Chem.* **261**, 4268–4275
51. Alberts, A. W., and Vagelos, P. R. (1972) in *The Enzymes*, 3rd ed., Vol. 6 (Boyer, P. D., ed), pp. 37–82, Academic Press, New York
52. Kumar, G. K., Beegen, H., and Wood, H. G. (1988) *Biochemistry* **27**, 5972–5978
53. Lynen, F., Knappe, J., Lorch, E., Jutting, G., and Ringelmann, E. (1959) *Angew. Chem. Int. Ed. Engl.* **71**, 481–486
54. Wood, H. G. (1979) *Trends Biochem. Sci.* **4**, N300–N302
55. Polakis, S. E., Guchhait, R. B., and Lane, M. D. (1972) *J. Biol. Chem.* **247**, 1235–1337
- 55a. Chapman-Smith, A., Forbes, B. E., Wallace, J. C., and Cronan, J. E., Jr. (1997) *J. Biol. Chem.* **272**, 26017–26022
- 55b. Reddy, D. V., Shenoy, B. C., Carey, P. R., and Sönnichsen, F. D. (2000) *Biochemistry* **39**, 2509–2516
56. Tipton, P. A., and Cleland, W. W. (1988) *J. Am. Chem. Soc.* **110**, 5866–5869
57. Tipton, P. A., and Cleland, W. W. (1988) *Biochemistry* **27**, 4317–4325
58. Ogita, T., and Knowles, J. R. (1988) *Biochemistry* **27**, 8028–8033
- 58a. Branson, J. P., and Attwood, P. V. (2000) *Biochemistry* **39**, 7480–7491
59. Perrin, C. L., and Dwyer, T. J. (1987) *J. Am. Chem. Soc.* **109**, 5163–5167
60. Kluger, R., Davis, P. P., and Adawadkar, P. D. (1979) *J. Am. Chem. Soc.* **101**, 5995–6000
61. King, S. W., Natarajan, R., Bembi, R., and Fife, T. H. (1992) *J. Am. Chem. Soc.* **114**, 10715–10721
62. Kluger, R., and Taylor, S. D. (1991) *J. Am. Chem. Soc.* **113**, 996–1001
63. Taylor, S. D., and Kluger, R. (1993) *J. Am. Chem. Soc.* **115**, 867–871
64. Hoving, H., Crysell, B., and Leadbay, P. F. (1985) *Biochemistry* **24**, 6163–6169
65. O'Keefe, S. J., and Knowles, J. R. (1986) *Biochemistry* **25**, 6077–6084
66. Kuo, D. J., and Rose, I. A. (1993) *J. Am. Chem. Soc.* **115**, 387–390
67. Stubbe, J., Fish, S., and Abeles, R. H. (1980) *J. Biol. Chem.* **255**, 236–242
68. Reed, G. H., and Scrutton, M. G. (1974) *J. Biol. Chem.* **249**, 6156–6162
- 68a. Werneburg, B. G., and Ash, D. E. (1997) *Biochemistry* **36**, 14392–14402
69. DeTitta, G. T., Blessing, R. H., Moss, G. R., King, H. F., Sukumaran, D. K., and Roskwital-ski, R. L. (1994) *J. Am. Chem. Soc.* **116**, 6485–6493
70. Legge, G. B., Branson, J. P., and Attwood, P. V. (1996) *Biochemistry* **35**, 3849–3856
71. Mabrouk, G. M., Helmy, I. M., Thampy, K. G., and Wakil, S. J. (1990) *J. Biol. Chem.* **265**, 6330–6338
72. Kim, K.-H., López-Casillas, F., Bai, D. H., Luo, X., and Pape, M. E. (1989) *FASEB J.* **3**, 2250–2256
73. Harwood, J. L. (1988) *Trends Biochem. Sci.* **13**, 330–331
74. LauBermaier, E., Schwarz, E., Oesterheld, D., Reinke, H., Beyreuther, K., and Dimroth, P. (1989) *J. Biol. Chem.* **264**, 14710–14715
- 74a. Jockel, P., Schmid, M., Choinowski, T., and Dimroth, P. (2000) *Biochemistry* **39**, 4320–4326
75. Laussermaier, E., Schwarz, E., Oesterheld, D., Reinke, H., Beyreuther, K., and Dimroth, P. (1989) *J. Biol. Chem.* **264**, 14710–14715
76. Huder, J. B., and Dimroth, P. (1993) *J. Biol. Chem.* **268**, 24564–24571

References

77. Bendrat, K., and Buckel, W. (1993) *Eur. J. Biochem.* **211**, 697–702
78. Ahmad, F., Lygre, D. G., Jacobson, B. E., and Wood, H. G. (1972) *J. Biol. Chem.* **247**, 6299–6305
79. Metzler, D. E. (1960) in *The Enzymes*, 2nd ed., Vol. 2 (Boyer, P. D., Lardy, H., and Myrback, K., eds), pp. 295–337, Academic Press, New York
80. Schellenberger, A., and Schowen, R. L., eds. (1988) *Thiamin Pyrophosphate Biochemistry*, CRC Press, Boca Raton, Florida (2 Vols.)
81. Krampitz, L. O. (1970) *Thiamin Diphosphate and Its Catalytic Functions*, Dekker, New York
82. Gubler, C. J., ed. (1976) *Thiamine*, Wiley, New York
83. Gallo, A., Mieryl, J. J., and Sable, H. Z. (1978) *Bioorg. Chem.* **4**, 147–177
- 83a. Jordan, F., Li, H., and Brown, A. (1999) *Biochemistry* **38**, 6369–6373
84. Kluger, R. (1987) *Chem. Rev.* **87**, 863–876
85. Sable, H. Z., ed., and Gubler, C. J., ed. (1982) *Ann. N.Y. Acad. Sci.* **378**
86. Karlson, P. (1984) *Trends Biochem. Sci.* **9**, 536–537
87. Panijpan, B. (1979) *J. Chem. Educ.* **56**, 805–806
88. Suchy, J., Mieryl, J. J., Bantle, G., and Sable, H. Z. (1972) *J. Biol. Chem.* **247**, 5905–5912
89. Williams, R. R., and Spies, T. D. (1938) *Vitamin B₁ and Its Use in Medicine*, Macmillan, New York
90. Kluger, R., Lam, J. F., Pezacki, J. P., and Yang, C.-M. (1995) *J. Am. Chem. Soc.* **117**, 11383–11389
91. Windheuser, J. J., and Higuchi, T. (1962) *J. Pharm. Sci.* **51**, 354–364
92. Costello, C. A., Kelleher, N. L., Abe, M., McLafferty, F. W., and Begley, T. P. (1996) *J. Biol. Chem.* **271**, 3445–3452
- 92a. Campobasso, N., Costello, C. A., Kinsland, C., Begley, T. P., and Ealick, S. E. (1998) *Biochemistry* **37**, 15981–15989
- 92b. Webb, E., and Downs, D. (1997) *J. Biol. Chem.* **272**, 15702–15707
- 92c. Nosaka, K., Onozuka, M., Nishino, H., Nishimura, H., Kawasaki, Y., and Ueyama, H. (1999) *J. Biol. Chem.* **274**, 34129–34133
93. Mizuhara, S., and Handler, P. (1954) *J. Am. Chem. Soc.* **76**, 571–573
94. Breslow, R. (1958) *J. Am. Chem. Soc.* **80**, 3719–3726
95. Hopmann, R. F. W., Brugnoni, G. P., and Fol, B. (1982) *J. Am. Chem. Soc.* **104**, 1341–1344
96. Washabaugh, M. W., and Jencks, W. P. (1989) *J. Am. Chem. Soc.* **111**, 683–692
97. Breslow, R., and McNelis, E. (1959) *J. Am. Chem. Soc.* **81**, 3080–3082
98. Kluger, R., Chin, J., and Smyth, T. (1981) *J. Am. Chem. Soc.* **103**, 884–888
99. Kluger, R., and Smyth, T. (1981) *J. Am. Chem. Soc.* **103**, 1216–1218
100. Kluger, R., Karimian, K., Gish, G., Pangborn, W. A., and DeTitta, G. T. (1987) *J. Am. Chem. Soc.* **109**, 618–620
101. Crosby, J., Stone, R., and Lienhard, G. E. (1970) *J. Am. Chem. Soc.* **92**, 2891–2900
102. Shin, W., Oh, D.-G., Chae, C.-H., and Yoon, T.-S. (1993) *J. Am. Chem. Soc.* **115**, 12238–12250
103. Lobell, M., and Crout, D. H. G. (1996) *J. Am. Chem. Soc.* **118**, 1867–1873
104. Gallo, A. A., and Sable, H. Z. (1974) *J. Biol. Chem.* **249**, 1382–1389
105. Schellenberger, A. (1967) *Angew. Chem. Int. Ed. Engl.* **6**, 1024–1035
106. Golbik, R., Neef, H., Hübner, G., König, S., Seliger, B., Meshalkina, L., Kochetov, G. A., and Schellenberger, A. (1991) *Bioorg. Chem.* **19**, 10–17
107. König, S., Schellenberger, A., Neef, H., and Schneider, G. (1994) *J. Biol. Chem.* **269**, 10879–10882
108. Jordan, F., and Mariam, Y. H. (1978) *J. Am. Chem. Soc.* **100**, 2534–2541
109. Metzler, D. E. (1977) *Biochemistry; The Chemical Reactions of Living Cells*, Academic Press, New York (440–441)
- 109a. Kern, D., Kern, G., Neef, H., Tittmann, K., Killenberg-Jabs, M., Wikner, C., Schneider, G., and Hübner, G. (1997) *Science* **275**, 67–70
110. Singleton, C. K., Wang, J. J.-L., Shan, L., and Martin, P. R. (1996) *Biochemistry* **35**, 15865–15869
111. Guo, F., Zhang, D., Kahyaoglu, A., Farid, R. S., and Jordan, F. (1998) *Biochemistry* **37**, 13379–13391
- 111a. Li, H., Furey, W., and Jordan, F. (1999) *Biochemistry* **38**, 9992–10003
112. Gish, G., Smyth, T., and Kluger, R. (1988) *J. Am. Chem. Soc.* **110**, 6230–6234
113. Annan, N., Paris, R., and Jordan, F. (1989) *J. Am. Chem. Soc.* **111**, 8895–8901
114. Alvarez, F. J., Ermer, J., Hübner, G., Schellenberger, A., and Schowen, R. L. (1991) *J. Am. Chem. Soc.* **113**, 8402–8409
115. Alvarez, F. J., Ermer, J., Hübner, G., Schellenberger, A., and Schowen, R. L. (1995) *J. Am. Chem. Soc.* **117**, 1678–1683
116. Sun, S., Duggleby, R. G., and Schowen, R. L. (1995) *J. Am. Chem. Soc.* **117**, 7317–7322
117. Harris, T. K., and Washabaugh, M. W. (1995) *Biochemistry* **34**, 13994–14000
118. Harris, T. K., and Washabaugh, M. W. (1995) *Biochemistry* **34**, 14001–14011
119. Muller, Y. A., and Schulz, G. E. (1993) *Science* **259**, 965–967
120. Muller, Y. A., Schumacher, G., Rudolph, R., and Schulz, G. E. (1994) *J. Mol. Biol.* **237**, 315–335
121. Dyda, F., Furey, W., Swaminathan, S., Sax, M., Farrenkopf, B., and Jordan, F. (1993) *Biochemistry* **32**, 6165–6170
122. Arjunan, P., Umland, T., Dyda, R., Swaminathan, S., Furey, W., Sax, M., Farrenkopf, B., Gao, Y., Zhang, D., and Jordan, F. (1996) *J. Mol. Biol.* **256**, 590–600
- 122a. Dobritsch, D., König, S., Schneider, G., and Lu, G. (1998) *J. Biol. Chem.* **273**, 20196–20204
- 122b. Chang, A. K., Nixon, P. F., and Duggleby, R. G. (2000) *Biochemistry* **39**, 9430–9437
- 122c. Li, H., and Jordan, F. (1999) *Biochemistry* **38**, 10004–10012
123. Wikner, C., Meshalkina, L., Nilsson, U., Nikkola, M., Lindqvist, Y., Sundström, M., and Schneider, G. (1994) *J. Biol. Chem.* **269**, 32144–32150
124. Nikkola, M., Lindqvist, Y., and Schneider, G. (1994) *J. Mol. Biol.* **238**, 387–404
- 124a. Hasson, M. S., Muscate, A., McLeish, M. J., Polovnikova, L. S., Gerlt, J. A., Kenyon, G. L., Petsko, G. A., and Ringe, D. (1998) *Biochemistry* **37**, 9918–9930
125. Killenberg-Jabs, M., König, S., Eberhardt, I., Hohmann, S., and Hübner, G. (1997) *Biochemistry* **36**, 1900–1905
126. Koga, J., Adachi, T., and Hidaka, H. (1992) *J. Biol. Chem.* **267**, 15823–15828
127. Hasson, M. S., Muscate, A., Henahan, G. T. M., Guidinger, P. F., Petsko, G. A., Ringe, D., and Kenyon, G. L. (1995) *Protein Sci.* **4**, 955–959
128. Ibdah, M., Bar-Ilan, A., Livnah, O., Schloss, J. V., Barak, Z., and Chipman, D. M. (1996) *Biochemistry* **35**, 16282–16291
129. Bernasconi, P., Woodworth, A. R., Rosen, B. A., Subramanian, M. V., and Siehl, D. L. (1995) *J. Biol. Chem.* **270**, 17381–17385
130. Ott, K.-H., Kwagh, J.-G., Stockton, G. W., Sidorov, V., and Kakefuda, G. (1996) *J. Mol. Biol.* **263**, 359–368
131. Stivers, J. T., and Washabaugh, M. W. (1993) *Biochemistry* **32**, 13472–13482
132. Chang, Y.-Y., Wang, A.-Y., and Cronan, J. E., Jr. (1993) *J. Biol. Chem.* **268**, 3911–3919
- 132a. Kuzuyama, T., Takahashi, S., Takagi, M., and Seto, H. (2000) *J. Biol. Chem.* **275**, 19928–19932
- 132b. Wikner, C., Nilsson, U., Meshalkina, L., Udekwi, C., Lindqvist, Y., and Schneider, G. (1997) *Biochemistry* **36**, 15643–15649
133. Chuang, L. F., and Collins, E. B. (1968) *J. Bacteriol.* **95**, 2083–2089
134. Gruys, K. J., Datta, A., and Frey, P. A. (1989) *Biochemistry* **28**, 9071–9080
135. Flournoy, D. S., and Frey, P. A. (1989) *Biochemistry* **28**, 9594–9602
- 135a. Tittmann, K., Proske, D., Spinka, M., Ghisla, S., Rudolph, R., Hübner, G., and Kern, G. (1998) *J. Biol. Chem.* **273**, 12929–12934
136. Matsuda, T., and Cooper, J. R. (1983) *Biochemistry* **22**, 2209–2213
137. Barchi, R. L., and Viale, R. O. (1976) *J. Biol. Chem.* **251**, 193–197
138. Nishino, K., Itokawa, Y., Nishino, N., Piros, K., and Cooper, J. R. (1983) *J. Biol. Chem.* **258**, 1871–1878
139. Braunstein, A. E., and Kritzmman, M. G. (1937) *Enzymologia* **2**, 751–752
140. Braunstein, A. E., and Kritzmman, M. G. (1937) *Biokhim.* **2**, 242–262, 859–874
141. Braunstein, A. E. (1973) in *The Enzymes*, 3rd ed., Vol. 9 (Boyer, P. D., ed), pp. 379–481, Academic Press, New York
142. Christen, P., and Metzler, D. E., eds. (1984) *Transaminases*, Wiley, New York
143. Torchinsky, Y. (1986) in *Vitamin B₆, Pyridoxal Phosphate: Chemical, Biochemical and Medical Aspects*, Vol. 1A (Dolphin, R., Poulson, R., and Avramovic, O., eds), pp. 169–221, Wiley, New York
144. Snell, E. E. (1944) *J. Biol. Chem.* **154**, 313–314
145. Snell, E. E. (1986) in *Vitamin B₆, Pyridoxal Phosphate: Chemical, Biochemical and Medical Aspects*, Vol. 1A (Dolphin, D., Poulson, R., and Avramovic, O., eds), pp. 1–12, Wiley, New York
146. Gunsalus, I. C., Bellamy, W. D., and Umbreit, W. W. (1944) *J. Biol. Chem.* **155**, 685–686
147. Snell, E. E. (1981) in *Methods in Vitamin B₆ Nutrition* (Leklem, J. E., and Reynolds, R. D., eds), pp. 1–19, Plenum, New York
148. Metzler, D. E., Ikawa, M., and Snell, E. E. (1954) *J. Am. Chem. Soc.* **76**, 648–652
149. Kallen, R. G., Korpela, T., Martell, A. E., Matsushima, Y., Metzler, C. M., Metzler, D. E., Morozov, Y. V., Ralston, I. M., Savin, F. A., Torchinsky, Y. M., and Ueno, H. (1984) in *Transaminases* (Christen, P., and Metzler, D. E., eds), pp. 37–106, Wiley, New York
150. Leussing, D. L. (1986) in *Vitamin B₆, Pyridoxal Phosphate: Chemical, Biochemical and Medical Aspects*, Vol. 1A (Dolphin, D., Poulson, R., and Avramovic, O., eds), pp. 69–115, Wiley, New York
151. Braunstein, A. E., and Kritzmman, M. G. (1952) *Dokl. Akad. Nauk. SSSR* **85**, 1115–1118
152. Braunstein, A. E., and Kritzmman, M. G. (1952) *Biokhim.* **18**, 393–411
153. Shaw, J. P., Petsko, G. A., and Ringe, D. (1997) *Biochemistry* **36**, 1329–1342
- 153a. Stamper, C. G. F., Morollo, A. A., and Ringe, D. (1998) *Biochemistry* **37**, 10438–10445
- 153b. Morollo, A. A., Petsko, G. A., and Ringe, D. (1999) *Biochemistry* **38**, 3293–3301
- 153c. Sun, S., and Toney, M. D. (1999) *Biochemistry* **38**, 4058–4065
154. Aswad, D. W., and Johnson, B. A. (1987) *Trends Biochem. Sci.* **12**, 155–158
- 154a. Watababe, A., Kurokawa, Y., Yoshimura, T., Kurihara, T., Soda, K., and Esaki, N. (1999) *J. Biol. Chem.* **274**, 4189–4194

References

155. Hoffmann, K., Schneider-Scherzer, E., Kleinkauf, H., and Zocher, R. (1994) *J. Biol. Chem.* **269**, 12710–12714
- 155a. Cheng, Y.-Q., and Walton, J. D. (2000) *J. Biol. Chem.* **275**, 4906–4911
156. Schnell, M. J., Cooper, O. B., and Snyder, S. H. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 2013–2018
157. Yamauchi, T., Choi, S.-Y., Okada, H., Yohda, M., Kumagai, H., Esaki, N., and Soda, K. (1992) *J. Biol. Chem.* **267**, 18361–18364
158. Yohda, M., Endo, I., Abe, Y., Ohta, T., Iida, T., Maruyama, T., and Kagawa, Y. (1996) *J. Biol. Chem.* **271**, 22017–22021
159. Li, N., and Mattoo, A. K. (1994) *J. Biol. Chem.* **269**, 6908–6917
160. Li, N., Jiang, X. N., Cai, G. P., and Yang, S. F. (1996) *J. Biol. Chem.* **271**, 25738–25741
161. Hohenester, E., White, M. F., Kirsch, J. F., and Jansonius, J. N. (1994) *J. Mol. Biol.* **243**, 947–949
- 161a. Capitani, G., Hohenester, E., Feng, L., Storici, P., Kirsch, J. F., and Jansonius, J. N. (1999) *J. Mol. Biol.* **294**, 745–756
162. Ramalingam, K., Lee, K.-M., Woodard, R. W., Blecker, A. B., and Kende, H. (1985) *Proc. Natl. Acad. Sci. U.S.A.* **82**, 7820–7824
163. Rhee, S., Silva, M. M., Hyde, C. C., Rogers, P. H., Metzler, C. M., Metzler, D. E., and Arnone, A. (1997) *J. Biol. Chem.* **272**, 17293–17302
164. Malashkevich, V. N., Strokopytov, B. V., Borisov, V. V., Dauter, Z., Wilson, K. S., and Torchinsky, Y. M. (1995) *J. Mol. Biol.* **247**, 111–124
165. McPhalen, C. A., Vincent, M. G., and Jansonius, J. N. (1992) *J. Mol. Biol.* **225**, 495–517
166. Kamitori, S., Okamoto, A., Hirotsu, K., Higuchi, T., Kuramitsu, S., Kagamiyama, H., Matsuura, Y., and Katsube, Y. (1990) *J. Biochem.* **108**, 175–184
167. Okamoto, A., Higuchi, T., Hirotsu, K., Kuramitsu, S., and Kagamiyama, H. (1994) *J. Biochem.* **116**, 95–107
- 167a. Jeffery, C. J., Barry, T., Doonan, S., Petsko, G. A., and Ringe, D. (1998) *Protein Sci.* **7**, 1380–1387
168. Uchida, C., Funai, T., Oda, T., Ohbayashi, K., and Ichiyama, A. (1994) *J. Biol. Chem.* **269**, 8846–8856
- 168a. Matsui, I., Matsui, E., Sakai, Y., Kikuchi, H., Kawarabayashi, Y., Ura, H., Kawaguchi, S.-i., Kuramitsu, S., and Harata, K. (2000) *J. Biol. Chem.* **275**, 4871–4879
- 168b. Okada, K., Hirotsu, K., Sato, M., Hayashi, H., and Kagamiyama, H. (1997) *J. Biochem.* **121**, 637–641
- 168c. Davoodi, J., Drown, P. M., Bledsoe, R. K., Wallin, R., Reinhart, G. D., and Hutson, S. M. (1998) *J. Biol. Chem.* **273**, 4982–4989
- 168d. van Ophem, P. W., Peisach, D., Erickson, S. D., Soda, K., Ringe, D., and Manning, J. M. (1999) *Biochemistry* **38**, 1323–1331
- 168e. Hester, G., Stark, W., Moser, M., Kallen, J., Markovic-Housley, Z., and Jansonius, J. N. (1999) *J. Mol. Biol.* **286**, 829–850
- 168f. Storici, P., Capitani, G., Müller, R., Schirmer, T., and Jansonius, J. N. (1999) *J. Mol. Biol.* **285**, 297–309
- 168g. Shah, S. A., Shen, B. W., and Brünger, A. T. (1997) *Structure* **5**, 1067–1075
- 168h. Ledwidge, R., and Blanchard, J. S. (1999) *Biochemistry* **38**, 3019–3024
- 168i. Smith, M. A., King, P. J., and Grimm, B. (1998) *Biochemistry* **37**, 319–329
- 168j. Käck, H., Sandmark, J., Gibson, K., Schneider, G., and Lindqvist, Y. (1999) *J. Mol. Biol.* **291**, 857–876
- 168k. Contestabile, R., Angelaccio, S., Maytum, R., Bossa, F., and John, R. A. (2000) *J. Biol. Chem.* **275**, 3879–3886
169. Kintanar, A., Metzler, C. M., Metzler, D. E., and Scott, R. D. (1991) *J. Biol. Chem.* **266**, 17222–17229
170. Miles, E. W. (1986) in *Vitamin B₆, Pyridoxal Phosphate: Chemical, Biochemical and Medical Aspects*, Vol. 1B (Dolphin, D., Poulson, R., and Avramovic, O., eds), pp. 253–310, Wiley, New York
171. Marceau, M., Lewis, S. D., Kojiro, C. L., and Shafer, J. A. (1989) *J. Biol. Chem.* **264**, 2753–2757
172. Ogawa, H., Gomi, T., Konishi, K., Date, T., Nakashima, H., Nose, K., Matsuda, Y., Peraino, C., Pilot, H. C., and Fujioka, M. (1989) *J. Biol. Chem.* **264**, 15818–15823
173. Obmolova, G., Teplakov, A., Harutyunyan, E., Wahler, G., and Schnackerz, K. D. (1990) *J. Mol. Biol.* **214**, 641–642
- 173a. Ogawa, H., Takusagawa, F., Wakaki, K., Kishi, H., Eskandarian, M. R., Kobayashi, M., Date, T., Huh, N.-H., and Pitot, H. C. (1999) *J. Biol. Chem.* **274**, 12855–12860
174. Snell, E. E. (1975) *Adv. Enzymol.* **42**, 287–333
175. Metzler, C. M., Viswanath, R., and Metzler, D. E. (1991) *J. Biol. Chem.* **266**, 9374–9381
176. Sloan, M. J., and Phillips, R. S. (1996) *Biochemistry* **35**, 16165–16173
- 176a. Isupov, M. N., Antson, A. A., Dodson, E. J., Dodson, G. G., Dementieva, I. S., Zakomirdina, L. N., Wilson, K. S., Dauter, Z., Lebedev, A. A., and Harutyunyan, E. H. (1998) *J. Mol. Biol.* **276**, 603–623
- 176b. Ikushiro, H., Hayashi, H., Kawata, Y., and Kagamiyama, H. (1998) *Biochemistry* **37**, 3043–3052
- 176c. Phillips, R. S., Sundararaju, B., and Faleev, N. G. (2000) *J. Am. Chem. Soc.* **122**, 1008–1014
177. Antson, A. A., Demidkina, T. V., Gollnick, P., Dauter, Z., Von Tersch, R. L., Long, J., Berezhnoy, S. N., Phillips, R. S., Harutyunyan, E. H., and Wilson, K. S. (1993) *Biochemistry* **32**, 4195–4206
- 177a. Sundararaju, B., Antson, A. A., Phillips, R. S., Demidkina, T. V., Barbolina, M. V., Gollnick, P., Dodson, G. G., and Wilson, K. S. (1997) *Biochemistry* **36**, 6502–6510
178. Chen, H. Y., Demidkina, T. V., and Phillips, R. S. (1995) *Biochemistry* **34**, 12276–12283
- 178a. Sundararaju, B., Chen, H., Shilcutt, S., and Phillips, R. S. (2000) *Biochemistry* **39**, 8546–8555
179. Block, E. (1985) *Sci. Am.* **252**(Mar), 114–119
180. Block, E., Gillies, J. Z., Gillies, C. W., Bazzi, A. A., Putman, D., Revelle, L. K., Wang, D., and Zhang, X. (1996) *J. Am. Chem. Soc.* **118**, 7492–7501
181. Gentry-Weeks, C. R., Spokes, J., and Thompson, J. (1995) *J. Biol. Chem.* **270**, 7695–7702
182. Clausen, T., Huber, R., Laber, B., Pohlenz, H.-D., and Messerschmidt, A. (1996) *J. Mol. Biol.* **262**, 202–224
183. Nagasawa, T., Tanizawa, K., Satoda, T., and Yamada, H. (1988) *J. Biol. Chem.* **263**, 958–964
184. Hyde, C. C., Ahmed, S. A., Padlan, E. A., Miles, E. W., and Davies, D. R. (1988) *J. Biol. Chem.* **263**, 17857–17871
- 184a. Rhee, S., Miles, E. W., and Davies, D. R. (1998) *J. Biol. Chem.* **273**, 8553–8555
- 184b. Weyand, M., and Schlichting, I. (1999) *Biochemistry* **38**, 16469–16480
- 184c. Sachpatzidis, A., Dealwis, C., Lubetsky, J. B., Liang, P.-H., Anderson, K. S., and Lolis, E. (1999) *Biochemistry* **38**, 12665–12674
- 184d. Rhee, S., Miles, E. W., Mozzarelli, A., and Davies, D. R. (1998) *Biochemistry* **37**, 10653–10659
185. Rege, V. D., Kredich, N. M., Tai, C.-H., Karsten, W. E., Schnackerz, K. D., and Cook, P. F. (1996) *Biochemistry* **35**, 13485–13493
- 185a. Burkhard, P., Tai, C.-H., Jansonius, J. N., and Cook, P. F. (2000) *J. Mol. Biol.* **303**, 279–286
186. Cook, P. F., Tai, C.-H., Hwang, C.-C., Woehl, E. U., Dunn, M. F., and Schnackerz, K. D. (1996) *J. Biol. Chem.* **271**, 25842–25849
- 186a. Benci, S., Vaccari, S., Mozzarelli, A., and Cook, P. F. (1997) *Biochemistry* **36**, 15419–15427
187. Kery, V., Bukovska, G., and Kraus, J. P. (1994) *J. Biol. Chem.* **269**, 25283–25288
188. Peracchi, A., Bettati, S., Mozzarelli, A., Rossi, G. L., Miles, E. W., and Dunn, M. F. (1996) *Biochemistry* **35**, 1872–1880
- 188a. Jhee, K.-H., McPhie, P., and Miles, E. W. (2000) *J. Biol. Chem.* **275**, 11541–11544
- 188b. Taoka, S., Widjaja, L., and Banerjee, R. (1999) *Biochemistry* **38**, 13155–13161
- 188c. Ojha, S., Hwang, J., Kabil, O., Penner-Hahn, J. E., and Banerjee, R. (2000) *Biochemistry* **39**, 10542–10547
189. Palcic, M. M., and Floss, H. G. (1986) in *Vitamin B₆, Pyridoxal Phosphate: Chemical, Biochemical and Medical Aspects*, Vol. 1A (Dolphin, D., Poulson, R., and Avramovic, O., eds), pp. 25–68, Wiley, New York
190. Pan, P., Woehl, E., and Dunn, M. F. (1997) *Trends Biochem. Sci.* **22**, 22–27
191. Sukhareva, B. S. (1986) in *Vitamin B₆, Pyridoxal Phosphate: Chemical, Biochemical and Medical Aspects*, Vol. 1B (Dolphin, D., Poulson, R., and Avramovic, O., eds), pp. 325–353, Wiley, New York
192. Sandmeier, E., Hale, T. I., and Christen, P. (1994) *Eur. J. Biochem.* **221**, 997–1002
193. Nathan, B., Hsu, C.-C., Bao, J., Wu, R., and Wu, J.-Y. (1994) *J. Biol. Chem.* **269**, 7249–7254
194. Porter, T. G., and Martin, D. L. (1988) *Biochim. Biophys. Acta.* **874**, 235–244
195. Bu, D.-F., Erlander, M. G., Hitz, B. C., Tillakaratne, N. J. K., Kaufman, D. L., Wagner-McPherson, C. B., Evans, G. A., and Tobin, A. J. (1992) *Proc. Natl. Acad. Sci. U.S.A.* **89**, 2115–2119
196. Dirx, R., Jr., Thomas, A., Li, L., Lernmark, Å., Sherwin, R. S., DeCamilli, P., and Solimena, M. (1995) *J. Biol. Chem.* **270**, 2241–2246
- 196a. Qu, K., Martin, D. L., and Lawrence, C. E. (1998) *Protein Sci.* **7**, 1092–1105
- 196b. Kanaani, J., Lissin, D., Kash, S. F., and Baekkeskov, S. (1999) *J. Biol. Chem.* **274**, 37200–37209
197. Malashkevich, V. N., Filippini, P., Sauder, U., Dominici, P., Jansonius, J. N., and Voltattorni, C. B. (1992) *J. Mol. Biol.* **224**, 1167–1170
198. Dominici, P., Filippini, P., Schinnina, M. E., Barra, D., and Voltattorni, C. B. (1990) *Ann. N.Y. Acad. Sci.* **585**, 162–171
199. Reith, J., Benkelfat, C., Sherwin, A., Yasuhara, Y., Kuwabara, H., Andermann, F., Bachneff, S., Cumming, P., Diks, M., Dyve, S. E., Etinne, P., Evans, A. C., Lal, S., Shevell, M., Savard, G., Wong, D. F., Chouinard, G., and Gjedde, A. (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 11651–11654
- 199a. Hayashi, H., Tsukiyama, F., Ishii, S., Mizuguchi, H., and Kagamiyama, H. (1999) *Biochemistry* **38**, 15615–15622
- 199b. Nishino, J., Hayashi, H., Ishii, S., and Kagamiyama, H. (1997) *J. Biochem.* **121**, 604–611
200. Bhattacharjee, M. K., and Snell, E. E. (1990) *J. Biol. Chem.* **265**, 6664–6668
201. Chudomelka, P. J., Ramaley, R. F., and Murrin, L. C. (1990) *Neurochemical Research* **15**, 17–24
202. Yatsunami, K., Tsuchikawa, M., Kamada, M., Hori, K., and Higuchi, T. (1995) *J. Biol. Chem.* **270**, 30813–30817

References

203. Osterman, A. L., Kinch, L. N., Grishin, N. V., and Phillips, M. A. (1995) *J. Biol. Chem.* **270**, 11797–11802
204. Momany, C., Ernst, S., Ghosh, R., Chang, N.-L., and Hackert, M. L. (1995) *J. Mol. Biol.* **252**, 643–655
205. Osterman, A. L., Brooks, H. B., Jackson, L., Abbott, J. J., and Phillips, M. A. (1999) *Biochemistry* **38**, 11814–11826
206. Viguera, E., Trelles, O., Urdiales, J. L., Matés, J. M., and Sánchez-Jiménez, F. (1994) *Trends Biochem. Sci.* **19**, 318–319
- 206a. Almrud, J. J., Oliveira, M. A., Kern, A. D., Grishin, N. V., Phillips, M. A., and Hackert, M. L. (2000) *J. Mol. Biol.* **295**, 7–16
- 206b. Jackson, L. K., Brooks, H. B., Osterman, A. L., Goldsmith, E. J., and Phillips, M. A. (2000) *Biochemistry* **39**, 11247–11257
- 206c. Grishin, N. V., Osterman, A. L., Brooks, H. B., Phillips, M. A., and Goldsmith, E. J. (1999) *Biochemistry* **38**, 15174–15184
207. Kume, A., Koyata, H., Sakakibara, T., Ishiguro, Y., Kure, S., and Hiraga, K. (1991) *J. Biol. Chem.* **266**, 3323–3329
208. Pasternack, L. B., Laude, D. A., Jr., and Appling, D. R. (1992) *Biochemistry* **31**, 8713–8719
209. Kennard, O., and Salisbury, S. A. (1993) *J. Biol. Chem.* **268**, 10701–10704
- 209a. Okamura-Ikeda, K., Fujiwara, K., and Motokawa, Y. (1999) *J. Biol. Chem.* **274**, 17471–17477
- 209b. Guilhaudis, L., Simorre, J.-P., Blackledge, M., Marion, D., Gans, P., Neuburger, M., and Douce, R. (2000) *Biochemistry* **39**, 4259–4266
210. Akhtar, M., Stevenson, D. E., and Gani, D. (1990) *Biochemistry* **29**, 7648–7660
- 210a. Zhou, X., and Toney, M. D. (1999) *Biochemistry* **38**, 311–320
- 210b. Malashkevich, V. N., Strop, P., Keller, J. W., Jansonius, J. N., and Toney, M. D. (1999) *J. Mol. Biol.* **294**, 193–200
211. Scarsdale, J. N., Kazanina, G., Radaev, S., Schirch, V., and Wright, H. T. (1999) *Biochemistry* **38**, 8347–8358
212. Stover, P. J., Chen, L. H., Suh, J. R., Stover, D. M., Keyomarsi, K., and Shane, B. (1997) *J. Biol. Chem.* **272**, 1842–1848
- 212a. Szebenyi, D. M. E., Liu, X., Kriksunov, I. A., Stover, P. J., and Thiel, D. J. (2000) *Biochemistry* **39**, 13313–13323
213. Webb, H. K., and Matthews, R. G. (1995) *J. Biol. Chem.* **270**, 17204–17209
- 213a. Jagath, J. R., Sharma, B., Appaji Rao, N., and Savithri, H. S. (1997) *J. Biol. Chem.* **272**, 24355–24362
- 213b. Scarsdale, J. N., Radaev, S., Kazanina, G., Schirch, V., and Wright, H. T. (2000) *J. Mol. Biol.* **296**, 155–168
214. Tong, H., and Davis, L. (1994) *J. Biol. Chem.* **269**, 4057–4064
215. Tong, H., and Davis, L. (1995) *Biochemistry* **34**, 3362–3367
216. Cotter, P. D., Baumann, M., and Bishop, D. F. (1992) *Proc. Natl. Acad. Sci. U.S.A.* **89**, 4028–4032
217. Ferreira, G. C., Neame, P. J., and Dailey, H. A. (1993) *Protein Sci.* **2**, 1959–1965
218. Cox, T. C., Bottomley, S. S., Wiley, J. S., Bawden, M. J., Matthews, C. S., and May, B. K. (1994) *N. Engl. J. Med.* **330**, 675–679
- 218a. Hunter, G. A., and Ferreira, G. C. (1999) *J. Biol. Chem.* **274**, 12222–12228
- 218b. Hunter, G. A., and Ferreira, G. C. (1999) *Biochemistry* **38**, 3711–3718
219. Williams, R. D., Nixon, D. W., and Merrill, A. H., Jr. (1984) *Cancer Research* **44**, 1918–1923
220. Webster, S. P., Alexeev, D., Campopiano, D. J., Watt, R. M., Alexeeva, M., Sawyer, L., and Baxter, R. L. (2000) *Biochemistry* **39**, 516–528
- 220a. Steegborn, C., Messerschmidt, A., Laber, B., Streber, W., Huber, R., and Clausen, T. (1999) *J. Mol. Biol.* **290**, 983–996
- 220b. Steegborn, C., Clausen, T., Sondernmann, P., Jacob, U., Worbs, M., Marinkovic, S., Huber, R., and Wahl, M. C. (1999) *J. Biol. Chem.* **274**, 12675–12684
- 220c. Laber, B., Gerbling, K.-P., Harde, C., Neff, K.-H., Nordhoff, E., and Pohlenz, H.-D. (1994) *Biochemistry* **33**, 3413–3423
221. Tate, S. E., and Meister, A. (1971) *Adv. Enzymol.* **35**, 503–543
222. Mihara, H., Kurihara, T., Watanabe, T., Yoshimura, T., and Esaki, N. (2000) *J. Biol. Chem.* **275**, 6195–6200
223. Esaki, N., Seraneeprakarn, V., Tanaka, H., and Soda, K. (1988) *J. Bacteriol.* **170**, 751–756
- 223a. Mihara, H., Maeda, M., Fujii, T., Kurihara, T., Hata, Y., and Esaki, N. (1999) *J. Biol. Chem.* **274**, 14768–14772
- 223b. Fujii, T., Maeda, M., Mihara, H., Kurihara, T., Esaki, N., and Hata, Y. (2000) *Biochemistry* **39**, 1263–1273
224. Zheng, L., and Dean, D. R. (1994) *J. Biol. Chem.* **269**, 18723–18726
225. Phillips, R. S., and Dua, R. K. (1991) *J. Am. Chem. Soc.* **113**, 7385–7388
226. Dua, R. K., Taylor, E. W., and Phillips, R. S. (1993) *J. Am. Chem. Soc.* **115**, 1264–1270
227. Krebs, E. G., and Fischer, E. H. (1964) *Vitam. Horm. (N. Y.)* **22**, 399–410
228. Rubenstein, P. A., and Strominger, J. L. (1974) *J. Biol. Chem.* **249**, 3776–3781
229. Johnson, D. A., Gassner, G. T., Bandarian, V., Ruzicka, F. J., Ballou, D. P., Reed, G. H., and Liu, H.-w. (1996) *Biochemistry* **35**, 15846–15856
230. Gonzalez-Porqué, P. (1986) in *Vitamin B₆, Pyridoxal Phosphate: Chemical, Biochemical and Medical Aspects*, Vol. 1B (Dolphin, D., Poulson, R., and Avramovic, O., eds), pp. 391–419, Wiley, New York
231. Bailey, G. B., Chotamangsa, O., and Vuttivej, K. (1970) *Biochemistry* **9**, 3243–3248
232. Toney, M. D., Hohenester, E., Cowan, S. W., and Jansonius, J. N. (1993) *Science* **261**, 756–759
233. Dunathan, H. C. (1971) *Adv. Enzymol.* **35**, 79–134
234. Tobler, H. P., Gehring, H., and Christen, P. (1987) *J. Biol. Chem.* **262**, 8985–8989
235. Sugio, S., Petsko, G. A., Manning, J. M., Soda, K., and Ringe, D. (1995) *Biochemistry* **34**, 9661–9669
236. Syke, G. E., Potts, R., and Floss, H. G. (1974) *J. Am. Chem. Soc.* **96**, 1593–1595
237. Floss, H. G., and Vederas, J. C. (1982) in *New Comprehensive Biochemistry Stereochemistry*, Vol. 3 (Tamm, C., ed), pp. 161–199, Elsevier, Amsterdam
238. Palcic, M. M., and Floss, H. G. (1986) in *Vitamin B₆, Pyridoxal Phosphate: Chemical, Biochemical and Medical Aspects*, Vol. 1A (Dolphin, D., Poulson, R., and Avramovic, O., eds), pp. 25–68, Wiley, New York
239. Metzler, D. E., Harris, C. M., Johnson, R. J., Siano, D. B., and Thomson, J. A. (1973) *Biochemistry* **12**, 5377–5392
240. Metzler, C. M., Cahill, A. E., and Metzler, D. E. (1980) *J. Am. Chem. Soc.* **102**, 6075–6082
241. Schirch, L., and Jenkins, W. T. (1964) *J. Biol. Chem.* **239**, 3801–3807
242. Metzler, C. M., Harris, A. G., and Metzler, D. E. (1988) *Biochemistry* **27**, 4923–4933
243. Schnackerz, K. D. (1986) in *Vitamin B₆, Pyridoxal Phosphate: Chemical, Biochemical and Medical Aspects*, Vol. 1A (Dolphin, D., Poulson, R., and Avramovic, O., eds), pp. 245–264, Wiley, New York
- 243a. Mehta, P. K., and Christen, P. (1998) in *Advances in Enzymology and Related Areas of Molecular Biology*, Vol. 74 (Purich, D. L., ed), Wiley, New York (pp. 129–184)
- 243b. Jansonius, J. N. (1998) *Current Opinion in Structural Biology* **8**, 759–769
244. Watanabe, N., Yonaha, K., Sakabe, K., Sakabe, N., Aibara, S., and Morita, Y. (1991) in *Enzymes Dependent on Pyridoxal Phosphate and Other Carbonyl Compounds as Cofactors* (Fukui, T., Kagamiyama, H., Soda, K., and Wada, H., eds), pp. 121–124, Pergamon Press, Oxford
245. Huston, S. M., Bledsoe, R. K., Hall, T. R., and Dawson, P. A. (1995) *J. Biol. Chem.* **270**, 30344–30352
246. Ivanov, V. I., and Karpeisky, M. Y. (1969) *Adv. Enzymol.* **32**, 21–53
247. Arnone, A., Christen, P., Jansonius, J. N., and Metzler, D. E. (1985) in *Transaminases* (Christen, P., and Metzler, D. E., eds), pp. 326–370, Wiley, New York
248. Rhee, S., Silva, M. M., Hyde, C. C., Rogers, P. H., Metzler, C. M., Metzler, D. E., and Arnone, A. (1997) *J. Biol. Chem.* **272**, 17293–17302
249. Metzler, C. M., Mitra, J., Metzler, D. E., Makinen, M. W., Hyde, C. C., Rogers, P., and Arnone, A. (1988) *J. Mol. Biol.* **203**, 197–220
250. Malashkevich, V. N., Toney, M. D., and Jansonius, J. N. (1993) *Biochemistry* **32**, 13451–13462
251. von Stosch, A. G. (1996) *Biochemistry* **35**, 15260–15268
252. Metzler, C. M., Metzler, D. E., Kintanar, A., Scott, R. D., and Marceau, M. (1991) *Biochem. Biophys. Res. Commun.* **178**, 385–392
253. Fukui, T., Kagamiyama, H., Soda, K., and Wada, H., eds. (1991) *Enzymes Dependent on Pyridoxal Phosphate and Other Carbonyl Compounds as Cofactors*, Pergamon Press, Oxford
254. Marino, G., Sannia, G., and Bossa, F., eds. (1994) *Biochemistry of Vitamin B₆ and PQQ*, Birkhäuser, Basel
- 254a. Iriarte, A., Kagan, H. M., and Martinez-Carrion, M., eds. (2000) *Biochemistry and Molecular Biology of Vitamin B₆ and PQQ-dependent Proteins*, Birkhäuser Verlag, Basel
255. Relimpio, A., Iriarte, A., Chlebowski, J. F., and Martinez-Carrion, M. (1981) *J. Biol. Chem.* **256**, 4478–4488
256. Gloss, L. M., and Kirsch, J. F. (1995) *Biochemistry* **34**, 12323–12332
257. Hwang, C.-C., Woehl, E. U., Minter, D. E., Dunn, M. F., and Cook, P. F. (1996) *Biochemistry* **35**, 6358–6365
258. Pan, P., and Dunn, M. F. (1996) *Biochemistry* **35**, 5002–5013
259. Anderson, K. S., Kim, A. Y., Quillen, J. M., Sayers, E., Yang, X.-J., and Miles, E. W. (1995) *J. Biol. Chem.* **270**, 29936–29944
260. Schnackerz, K. D., Tai, C.-H., Simmons, J. W., III, Jacobson, T. M., Rao, G. S. J., and Cook, P. F. (1995) *Biochemistry* **34**, 12152–12160
261. Hayashi, H., and Kagamiyama, H. (1995) *Biochemistry* **34**, 9413–9423
262. Mattingly, M. E., Mattingly, J. R., Jr., and Martinez-Carrion, M. (1982) *J. Biol. Chem.* **257**, 8872–8878
263. Metzler, D. E. (1997) *Methods Enzymol.* **280**, 30–40
264. Higaki, T., Tanase, S., Nagashima, F., Morino, Y., Scott, A. I., Williams, H. J., and Stolowich, N. J. (1991) *Biochemistry* **30**, 2519–2526
265. Goldberg, J. M., and Kirsch, J. F. (1996) *Biochemistry* **35**, 5280–5291
266. Recsei, P. A., and Snell, E. E. (1984) *Ann. Rev. Biochem.* **53**, 357–387
267. van Peolje, P. D., and Snell, E. E. (1990) *Ann. Rev. Biochem.* **59**, 29–59

References

268. Pishko, E. J., and Robertus, J. D. (1993) *Biochemistry* **32**, 4943–4948
269. Gallagher, T., Snell, E. E., and Hackert, M. L. (1989) *J. Biol. Chem.* **264**, 12737–12743
270. Recsei, P. A., Huynh, Q. K., and Snell, E. E. (1983) *Proc. Natl. Acad. Sci. U.S.A.* **80**, 973–977
271. Van Poelje, P. D., and Snell, E. E. (1988) *Proc. Natl. Acad. Sci. U.S.A.* **85**, 8449–8453
272. Huynh, Q. H., and Snell, E. E. (1986) *J. Biol. Chem.* **261**, 1521–1524
273. Shirahata, A., and Pegg, A. E. (1986) *J. Biol. Chem.* **261**, 13833–13837
274. Stanley, B. A., Shantz, L. M., and Pegg, A. E. (1994) *J. Biol. Chem.* **269**, 7901–7907
- 274a. Xiong, H., and Pegg, A. E. (1999) *J. Biol. Chem.* **274**, 35059–35066
275. Clancey, C. J., Chang, S.-C., and Dowhan, W. (1993) *J. Biol. Chem.* **268**, 24580–24590
276. Satre, M., and Kennedy, E. P. (1978) *J. Biol. Chem.* **253**, 479–483
277. Williamson, J. M., and Brown, G. M. (1979) *J. Biol. Chem.* **254**, 8074–8082
278. Yang, H., and Abeles, R. H. (1987) *Biochemistry* **26**, 4076–4081
279. Seto, B., and Stadtman, T. C. (1976) *J. Biol. Chem.* **251**, 2435–2439
280. Seto, B. (1978) *J. Biol. Chem.* **253**, 4525–4529
281. Hodgins, D. S., and Abeles, R. H. (1969) *Arch. Biochem. Biophys.* **130**, 274–285
282. Arkowitz, R. A., and Abeles, R. H. (1989) *Biochemistry* **28**, 4639–4644
283. Stadtman, T. C., and Davis, J. N. (1991) *J. Biol. Chem.* **266**, 22147–22153
- 283a. Kabisch, U. C., Gräntzdörffer, A., Schierhorn, A., Rücknagel, K. P., Andreesen, J. R., and Pich, A. (1999) *J. Biol. Chem.* **274**, 8445–8454
284. Furuta, T., Takahashi, H., Shibasaki, H., and Kasuya, Y. (1992) *J. Biol. Chem.* **267**, 12600–12605
285. Levy, H. L., Taylor, R. G., and McInnes, R. R. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 1 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 1107–1123, McGraw-Hill, New York
286. Havir, E. A., and Hanson, K. R. (1985) *Biochemistry* **24**, 2959–2967
287. Consevage, M. W., and Phillips, A. T. (1985) *Biochemistry* **24**, 301–308
288. Langer, M., Reck, G., Reed, J., and Rétey, J. (1994) *Biochemistry* **33**, 6462–6467
289. Langer, M., Lieber, A., and Rétey, J. (1994) *Biochemistry* **33**, 14034–14038
290. Langer, M., Pauling, A., and Rétey, J. (1995) *Angew. Chem. Int. Ed. Engl.* **34**, 1464–1465
291. Schuster, B., and Rétey, J. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 8433–8437
292. Schwede, T. F., Rétey, J., and Schulz, G. E. (1999) *Biochemistry* **38**, 5355–5361

Study Questions

- Discuss the role of biotin in metabolism and the chemical mechanism of its action. Illustrate with examples.
- Biotin has been shown to be an essential component of a bacterial oxaloacetate *decarboxylase* that pumps two sodium ions out of a cell for each oxaloacetate molecule decarboxylated. Propose a chemical mechanism for the functioning of biotin and also any ideas that you may have for the operation of the sodium pump.
- Write out the equation by which **acetyl-CoA** and **pyruvate** can be converted into **L-glutamate**, which can be observed in living animals by ^{13}C NMR. If $[2-^{13}\text{C}]$ sodium acetate were injected into the animal what labeling pattern could be anticipated in the L-glutamate?
- Compare mechanisms and coenzyme requirements for biological decarboxylation of the following three groups of compounds:
 - β -oxoacids
 - α -oxoacids
 - α -amino acids
- Illustrate, using structural equations, the chemical mechanisms of the following biochemical reactions.
 - Pyruvate $\rightarrow$ acetaldehyde + CO_2
 - 2-Pyruvate $\rightarrow$ α -acetolactate + CO_2
 - Pyruvate + NAD^+ + CoA-SH $\rightarrow$ acetyl-CoA + NADH + H^+ + CO_2
 - Fructose 6-*P* + glyceraldehyde-3-*P* $\rightarrow$ xylulose-5-*P* + erythrose-4-*P*
- 3-Fluoropyruvate is converted quantitatively by pyruvate decarboxylase from wheat germ into acetate, fluoride (F^-), and carbon dioxide. Propose a reaction mechanism. See Gish, G., Smyth, T., and Kluger, R. (1988) *J. Am. Chem. Soc.* **110**, 6230-6234.
- In *E. coli* **L-cysteine** is formed from L-serine and the sulfide ion S^{2-} in a reaction that also requires acetyl-CoA and is catalyzed by the consecutive action of an acyl transferase and cysteine synthase. Outline the mechanism of this conversion indicating participation of any essential coenzymes.
- Illustrate, using structural equations, the chemical mechanisms of the following biochemical reactions.
 - L-Glutamate + oxaloacetate $\rightarrow$ 2-oxoglutarate + L-aspartate
 - L-Serine $\rightarrow$ pyruvate + NH_4^+
 - L-Serine + indole (from cleavage of indole-3-glycerol phosphate) $\rightarrow$ L-tryptophan
 - L-Serine + tetrahydrofolate $\rightarrow$ glycine + N^5 , N^{10} -methylene-tetrahydrofolate
 - L-Selenocysteine $\rightarrow$ L-alanine + Se^0
- Threonine is formed by *E. coli* from homoserine via the intermediate γ -phosphohomoserine. Write out an abbreviated reaction sequence for its conversion to L-threonine by the action of threonine synthase.
- Write out a plausible step-by-step mechanism by which 1-aminocyclopropane-1-carboxylate synthase (ACC synthase) of plant tissues can form ACC from S-adenosylmethionine. This reaction requires a specific cofactor
- Tissues of the mammalian central nervous system contain a pyridoxal phosphate-dependent glutamate decarboxylase that catalyzes conversion of Glu to γ -aminobutyrate (GABA), an inhibitory synaptic transmitter. GABA is degraded by transimination with α -oxoglutarate as the acceptor to yield succinic semialdehyde, which then is oxidized to succinate by an NAD-linked dehydrogenase.
 - Show how these reactions can operate as a shunt pathway that allows the citric acid cycle to function without the enzymes α -oxoglutarate dehydrogenase and succinate thiokinase.
 - Is the shunt more or less efficient than the normal cycle from the standpoint of energy recovery? Explain.



Biological oxidation-reduction reactions rely upon many organic coenzymes and transition metal ions. Left, a molecule of the hydrogen carrier pyrroloquinoline quinone (PQQ, Fig. 15-23) is seen at the bottom of a 7-bladed β propeller of a bacterial methanol dehydrogenase. In adrenodoxin reductase (right) the reducing power of NADPH is passed to FAD and then to the small redox protein adrenodoxin. In this stereoscopic view the pyridine ring of the oxidized coenzyme NADP⁺ and the tricyclic flavin ring of FAD are seen stacked against each other in the center; the adenylate ends stretching toward the top and bottom of the complex. From Ziegler and Schulz, (2000) *Biochemistry* 39, 10986-10995.

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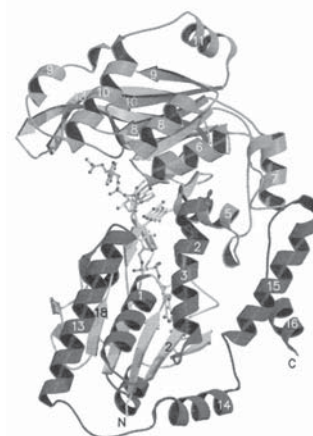
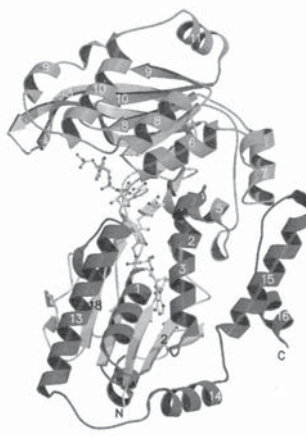
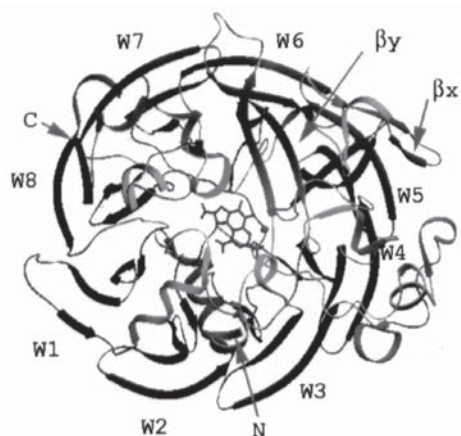
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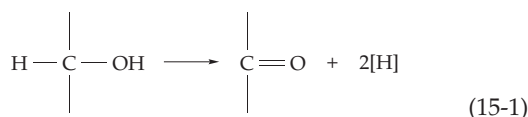
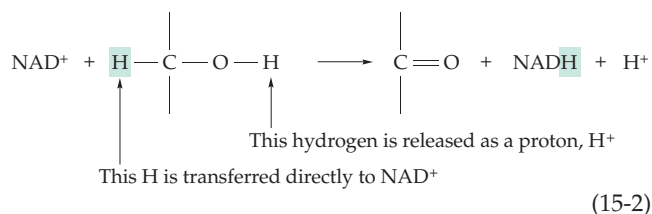
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Coenzymes of Oxidation–Reduction Reactions

15



The **dehydrogenation** of an alcohol to a ketone or aldehyde (Eq. 15-1) is one of the most frequent biological oxidation reactions. Although the hydrogen atoms removed from the substrate are often indicated simply as 2[H], it was recognized early in the twentieth century that they are actually transferred to hydrogen-carrying coenzymes such as NAD⁺, NADP⁺, FAD, and **riboflavin**



5'-phosphate (FMN). This chapter deals with these coenzymes and also with a number of other organic oxidation–reduction coenzymes. They may be considered either as carriers of hydrogen or as carriers of electrons ($\text{H} = \text{H}^+ + \text{e}^-$) in metabolic reactions.

When NAD⁺ becomes reduced by dehydrogenation of an alcohol, one of the hydrogen atoms removed from the alcohol becomes firmly attached to the NAD⁺, converting it to NADH. The other is released as a proton (Eq. 15-2). Study of ²H-labeled alcohols and their oxidation by NAD⁺ has shown that *dehydrogenases catalyze direct transfer to NAD⁺ of the hydrogen that is attached to carbon in the alcohol.* There is never any exchange of this hydrogen atom with protons of the medium. At the same time, the hydrogen attached to the oxygen of the alcohol is released into the medium as H⁺:

The foregoing observations suggested that these biological dehydrogenations may be viewed as removal of a hydride ion (H[−]) together with a proton (H⁺) rather than as removal of two hydrogen atoms. NAD⁺ and NADP⁺ are regarded as hydride ion-accepting coenzymes. However, it has been impossible to establish conclusively that the hydrogen atom and electron are transferred simultaneously as H[−]. Transfer of the hydrogen atom to or from these coenzymes may conceivably be followed by or preceded by transfer of an electron. The situation is even less clear for the flavin coenzymes FAD and riboflavin phosphate for which intermediate **free radical** oxidation states are known to exist. However, regardless of the actual mechanism, it is convenient to classify most hydrogen transfer reactions of metabolism as if they occurred by transfer of a hydride ion. *The hydride ion can be regarded as a nucleophile which can add to double bonds or can be eliminated from substrates in reactions of types that we have already considered.* Some of these reactions are listed in Table 15-1.

Why are there *four* major hydrogen transfer coenzymes, NAD⁺, NADP⁺, FAD, and riboflavin phosphate (FMN), instead of just one? Part of the answer is that the **reduced pyridine nucleotides** NADPH and NADH are more powerful reducing agents than are reduced **flavins** (Table 6-7). Conversely, flavin coenzymes are more powerful oxidizing agents than are

NAD⁺ and NADP⁺. This difference reflects the chemical difference between the vitamins **riboflavin** and **nicotinamide** which form the oxidation–reduction centers of the coenzymes. Another difference is that NAD⁺ and NADP⁺ tend to be present in free forms within cells, diffusing from a site on one enzyme to a site on another. These coenzymes are *sometimes* tightly bound but flavin coenzymes are *usually* firmly bound to proteins, fixed, and unable to move. Thus, they

tend to accept hydrogen atoms from one substrate and to pass them to a second substrate while attached to a single enzyme.

The oxidation–reduction potential of a pyridine nucleotide coenzyme system is determined by the standard redox potential for the free coenzyme (Table 6-8) together with the ratio of concentrations of oxidized to reduced coenzyme ([NAD⁺] / [NADH], Eq. 6-64). If these concentrations are known, a redox

TABLE 15-1
Some Biochemical Hydrogen Transfer Reactions^a

Reaction	Example (oxidant)
A Dehydrogenation of an alcohol $\begin{array}{c} \\ \text{HC} - \text{OH} \\ \end{array} \xrightarrow{\text{H}^+} \text{H}^- + \begin{array}{c} \diagup \quad \diagdown \\ \text{C} = \text{O} \end{array}$	Alcohol dehydrogenase (NAD ⁺)
B Dehydrogenation of an amine $\begin{array}{c} \\ \text{HC} - \text{NH}_2 \\ \end{array} \xrightarrow{\text{H}^+} \text{H}^- + \begin{array}{c} \diagup \quad \diagdown \\ \text{C} = \text{NH} \end{array}$ $\begin{array}{c} \text{H}_2\text{O} \\ \downarrow \\ \text{NH}_4^+ \\ \diagup \quad \diagdown \\ \text{C} = \text{O} \end{array}$	Amino acid dehydrogenases, amine oxidases (NAD ⁺ or flavin)
C Dehydrogenation of adduct of thiol and aldehyde $\text{R} - \text{SH} + \begin{array}{c} \text{O} \\ \\ \text{---C---H} \end{array} \rightleftharpoons \begin{array}{c} \text{OH} \\ \\ \text{---C---SR} \\ \\ \text{H} \end{array}$ $\xrightarrow{\text{H}^+} \text{H}^- + \begin{array}{c} \text{O} \\ \\ \text{---C---SR} \end{array}$	Glyceraldehyde 3-phosphate dehydrogenase (NAD ⁺ or NADP ⁺)
D Dehydrogenation of acyl-CoA, acyl-ACP, or carboxylic acid $\begin{array}{c} \text{H} \\ \\ \text{R} - \text{C} - \text{H} \\ \quad \diagup \\ \text{C} - \text{C} = \text{O} \\ \quad \\ \text{H} \quad \text{H} \end{array} \xrightarrow{\text{H}^+} \text{H}^- + \begin{array}{c} \text{H} \\ \\ \text{R} - \text{C} = \text{C} - \text{C} = \text{O} \\ \quad \diagup \\ \text{H} \quad \text{H} \end{array}$ (Y = —S—CoA or —OH)	Acyl-CoA dehydrogenases (Flavin) Succinic dehydrogenase (Flavin) Opposite: enoyl reductase (NADPH)
E Reduction of desmosterol to cholesterol by NADPH $\text{H}^- + \begin{array}{c} \text{H} \\ \\ \text{R} - \text{C} = \text{C} \end{array} \xrightarrow{\text{H}^+} \begin{array}{c} \text{H} \\ \\ \text{R} - \text{CH} - \text{CH} \end{array}$	Reduction of desmosterol to cholesterol by NADPH

^a Reaction type 9 A,B of Table on the inside cover at the end of the book: A hypothetical hydride ion H[−] is transferred from the substrate to a coenzyme of suitable reduction potential such as NAD⁺, NADP⁺, FAD, or riboflavin 5'-phosphate. The reverse of hydrogenation is shown for E. Many of the reactions are reversible and often go spontaneously in the reverse direction from that shown here.

potential can be defined for the NAD^+ system within a cell. This potential may vary in different parts of the cell because of differences in the $[\text{NAD}^+]/[\text{NADH}]$ ratio, but within a given region of the cell it is constant. On the other hand, the redox potentials of flavoproteins vary. Since the flavin coenzymes are not dissociable, two flavoproteins may operate at very different potentials even when they are physically close together.

Why are there two pyridine nucleotides, NAD^+ and NADP^+ , differing only in the presence or absence of an extra phosphate group? One important answer is that they are members of two different oxidation–reduction systems, both based on nicotinamide but functionally independent. The experimentally measured ratio $[\text{NAD}^+]/[\text{NADH}]$ is much higher than the ratio $[\text{NADP}^+]/[\text{NADPH}]$. Thus, these two coenzyme systems also can operate within a cell at different redox potentials. A related generalization that holds much of the time is that *NAD^+ is usually involved in pathways of catabolism, where it functions as an oxidant, while NADPH is more often used as a reducing agent in biosynthetic processes.* See Chapter 17, Section I for further discussion.

A. Pyridine Nucleotide Coenzymes and Dehydrogenases

In 1897, Buchner discovered that “yeast juice,” prepared by grinding yeast with sand and filtering, catalyzed fermentation of sugar. This was a major discovery which excited the interest of many other biochemists.¹ Among them were Harden and Young, who, in 1904, showed that Buchner’s cell-free yeast juice lost its ability to ferment glucose to alcohol and carbon dioxide when it was dialyzed. Apparently, fermentation depended upon a low-molecular-weight substance that passed out through the pores of the dialysis membrane. Fermentation could be restored by adding back to the yeast juice either the concentrated dialysate or boiled yeast juice (in which the enzyme proteins had been destroyed). The heat-stable material, which Harden and Young called **cozymase**, was eventually found to be a mixture of inorganic phosphate ions, thiamin diphosphate, and NAD^+ . However, characterization of NAD^+ was not accomplished until 1935.

Pure NADP^+ was isolated from red blood cells in 1934 by Otto Warburg and W. Christian, who had been studying the oxidation of glucose 6-phosphate by erythrocytes.^{1a} They demonstrated a requirement for a dialyzable coenzyme which they characterized and named **triphosphopyridine nucleotide** (TPN^+ , but now officially NADP^+ ; Fig. 15-1). Thus, even before its recognition as an important vitamin in human nutrition, nicotinamide was identified as a component of NADP^+ .

Warburg and Christian recognized the relationship of NADP^+ and NAD^+ (then called DPN^+) and proposed that both of these compounds act as hydrogen carriers through alternate reduction and oxidation of the pyridine ring. They showed that the coenzymes could be reduced either enzymatically or with sodium dithionite $\text{Na}_2\text{S}_2\text{O}_4$.



The reduced coenzymes NADH and NADPH were characterized by a new light-absorption band at 340 nm. This is not present in the oxidized forms, which absorb maximally at 260 nm (Fig. 15-2). The reduced forms are stable in air, but their reoxidation was found

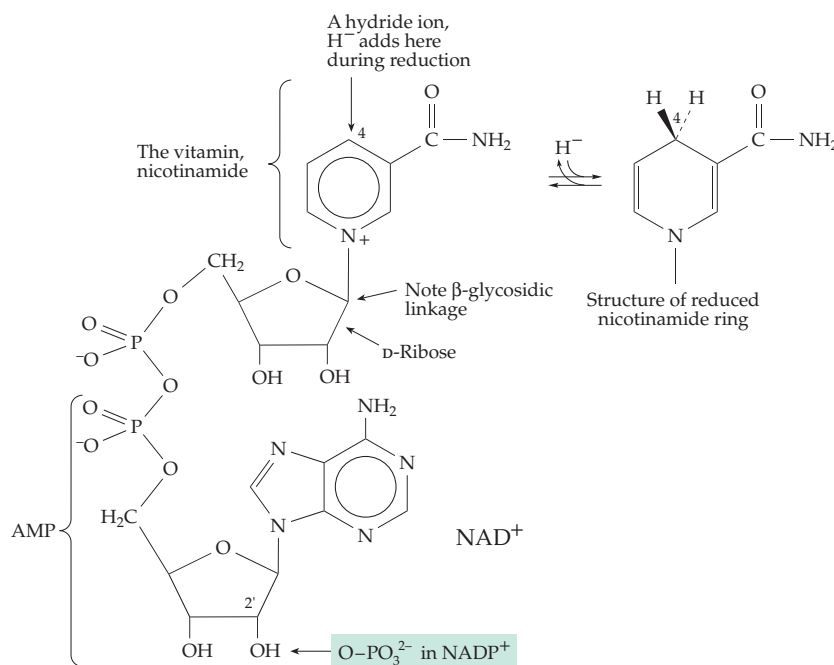


Figure 15-1 The hydrogen-carrying coenzymes NAD^+ (nicotinamide adenine dinucleotide) and NADP^+ (nicotinamide adenine dinucleotide phosphate). We use the abbreviations NAD^+ and NADP^+ , even though the net charge on the entire molecule at pH 7 is negative because of the charges on oxygen atoms of the phospho groups.

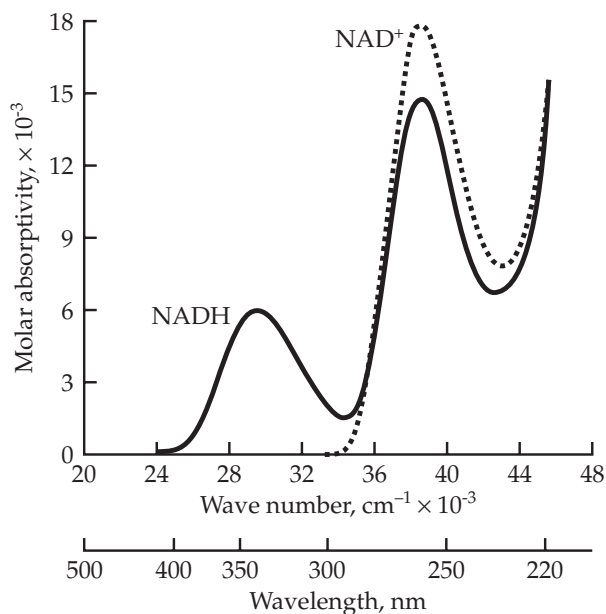


Figure 15-2 Absorption spectra of NAD^+ and NADH . Spectra of NADP^+ and NADPH are nearly the same as these. The difference in absorbance between oxidized and reduced forms at 340 nm is the basis for what is probably the single most often used spectral measurement in biochemistry. Reduction of NAD^+ or NADP^+ or oxidation of NADH or NADPH is measured by changes in absorbance at 340 nm in many methods of enzyme assay. If a pyridine nucleotide is not a reactant for the enzyme being studied, a **coupled assay** is often possible. For example, the rate of enzymatic formation of ATP in a process can be measured by adding to the reaction mixture the following enzymes and substrates: hexokinase + glucose + glucose-6-phosphate dehydrogenase + NADP^+ . As ATP is formed, it phosphorylates glucose via the action of hexokinase. NADP^+ then oxidizes the glucose 6-phosphate that is formed with production of NADPH , whose rate of appearance is monitored at 340 nm.

to be catalyzed by certain yellow enzymes which were later identified as flavoproteins.

1. Three-Dimensional Structures of Dehydrogenases

Most NAD^+ - or NADP^+ -dependent dehydrogenases are dimers or trimers of 20- to 40-kDa subunits. Among them are some of the first enzymes for which complete structures were determined by X-ray diffraction methods. The structure of the 329-residue per subunit muscle (M_4) isoenzyme of **lactate dehydrogenases** (see Chapter 11) from the dogfish was determined to 0.25 nm resolution by Rossmann and associates in 1971.^{2–4} More recently, structures have been determined for mammalian muscle and heart type (H_4) isoenzymes,⁵ for the testicular (C_4) isoenzyme from the

mouse,⁶ and for bacterial lactate dehydrogenases.^{6–8} In all of these the polypeptide is folded nearly identically. The structures of the homologous cytosolic and mitochondrial isoenzymes of **malate dehydrogenase** are also similar,^{9–11} as are those of the bacterial enzyme.^{12,13} All of these proteins consist of two structural domains and the NAD^+ is bound to the nucleotide-binding domain in a similar manner as is shown in Fig. 2-13 for glyceraldehyde phosphate dehydrogenase. The coenzyme-binding domains of the dehydrogenases of known structures all have this nearly constant structural feature (often called the Rossmann fold) consisting of a six-stranded parallel sheet together with several α helical coils¹⁴ (Figs. 2-13 and 2-27).

The coenzyme molecule curls around one end of the nucleotide-binding domains in a “C” conformation with the nicotinamide ring lying in a pocket (Figs. 2-13, 15-3). Even before the crystal structure of lactate dehydrogenase was known, the lack of pH dependence of coenzyme binding from pH 5 to 10 together with observed inactivation by butanedione suggested that the pyrophosphate group of NAD^+ binds to a guanidinium group of an arginine residue. This was identified by X-ray diffraction studies as Arg 101. This ion pairing interaction, as well as the hydrogen bond between Asp 53 and the 2' oxygen atom of a ribose ring (Fig. 15-3), is present in all of the lactate and malate dehydrogenases.

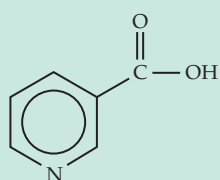
The adenine ring of the coenzyme is bound in a hydrophobic pocket with its amino group pointed out into the solvent. A second structural domain holds additional catalytic groups needed to form the active site.

2. Stereospecificity and Mechanism

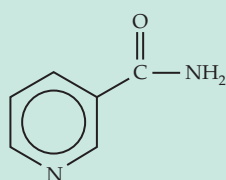
When NAD^+ is reduced in $^2\text{H}_2\text{O}$ by dithionite (Eq. 15-3) an atom of ^2H is introduced into the reduced pyridine. Chemical degradation showed the ^2H to be present at the 4 position of the ring para to the nitrogen atom¹⁵ (see Fig. 15-1). As shown by Westheimer and associates, during enzymatic reduction of NAD^+ by deuterium-containing ethanol, $\text{CH}_3\text{--C}^2\text{H}_2\text{OH}$, one of the ^2H atoms is transferred into the NADH formed, thus establishing the *direct transfer of a hydrogen atom*.^{16,17} When the NAD^2H formed in this way is reoxidized enzymatically with acetaldehyde, with regeneration of NAD^+ and ethanol, the ^2H is completely removed.

This was one of the first recognized examples of the ability of an enzyme to choose between two identical atoms at a *pro*-chiral center (Chapter 9). The two sides of the nicotinamide ring of NAD were designated A and B and the two hydrogen atoms at the 4 position of NADH as H_A (now known as *pro*-R) and H_B (*pro*-S). Alcohol dehydrogenase always removes the *pro*-R hydrogen. Malate, isocitrate, lactate, and D-glycerate dehydrogenases select the same hydrogen. However,

BOX 15-A NICOTINIC ACID AND NICOTINAMIDE



Nicotinic acid



Nicotinamide

Nicotinic acid was prepared in 1867 by oxidation of nicotine. Although it was later isolated by Funk and independently by Suzuki in 1911–1912 from yeast and rice polishings, it was not recognized as a vitamin. Its biological significance was established in 1935 when nicotinamide was identified as a component of NAD^+ by von Euler and associates and of NADP^+ by Warburg and Christian.^a Both forms of the vitamin are stable, colorless compounds highly soluble in water.

In 1937, Elvehjem and coworkers demonstrated that nicotinic acid cured canine “blacktongue.” In the same year it was found to cure human **pellagra**, a terrible disease characterized by weakness, indigestion, and loss of appetite followed by dermatitis,

diarrhea, mental disorders, and eventual death. At that time pellagra was common in the United States, especially in the south. The U.S. Public Health Service estimated that during 1912–1916 there were 100,000 victims and 10,000 deaths a year.^{b,c}

The daily requirement for an adult is about 7.5 mg. The amount is decreased by the presence in the diet of tryptophan, which can be converted partially to nicotinic acid (Chapter 25).^{d,e} Tryptophan is about 1/60 as active as nicotinic acid itself. The one-time prevalence of pellagra in the southern United States was a direct consequence of a diet high in maize whose proteins have an unusually low tryptophan content.

^a Schlenk, F. (1984) *Trends Biochem. Sci.* **9**, 286–288

^b Wagner, A. F., and Folkers, K. (1964) *Vitamins and Coenzymes*, Wiley (Interscience), New York (p. 73)

^c Rosenkrantz, B. G. (1974) *Science* **183**, 949–950

^d Teply, L. J. (1993) *FASEB J.* **7**, 1300

^e Sauberlich, H. E. (1987) in *Pyridine Nucleotide Coenzymes: Chemical, Biochemical and Medical Aspects*, Vol. B (Dolphin, D., Avramović, O., and Poulson, R., eds), pp. 599–626, Wiley (Interscience), New York

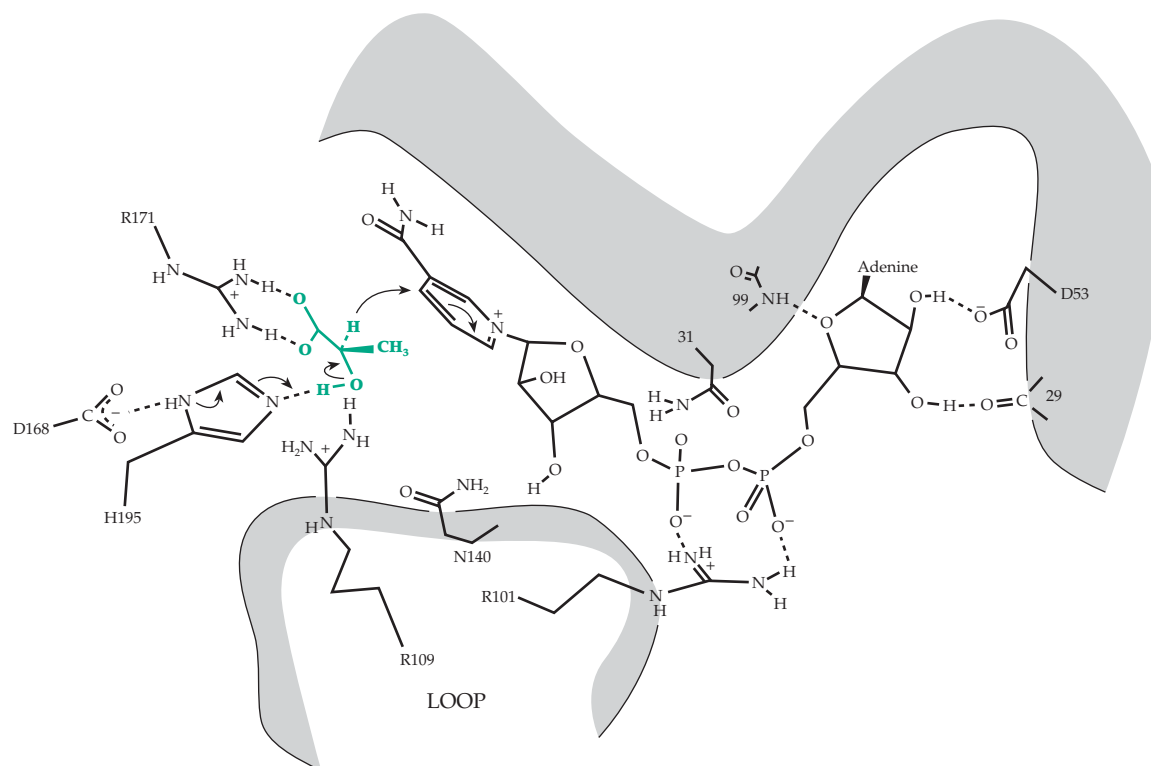
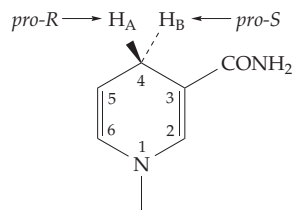


Figure 15-3 Diagrammatic structure of NAD^+ and L-lactate bound into the active site of lactate dehydrogenase. See Eventhoff *et al.*⁵

dehydrogenases acting on glucose 6-phosphate, glutamate, 6-phosphogluconate, and 3-phosphoglyceraldehyde remove the *pro-S* hydrogen. By 1979 this stereospecificity had been determined for 127 dehydrogenases, about half of which were found A-specific and half B-specific.¹⁸ Isotopically labeled substrates



have usually been used for this purpose but a simple NMR method has been devised.^{19,20}

How complete is the stereospecificity? Does the enzyme sometimes make a mistake? *Lactate dehydrogenase displays nearly absolute specificity*, transferring a proton into the “incorrect” *pro-S* position no more than once in 5×10^7 catalytic cycles.²¹ This suggests a difference in transition-state energies of about 40 kJ/mol for the two isomers. X-ray structural studies suggested that a major factor determining this stereoselectivity might be the location of hydrogen-bonding groups that hold the $-\text{CONH}_2$ group of the nicotinamide. If these are located in such a way that the nicotinamide ring has a *syn* orientation with respect to the ribose ring to which it is attached (Fig. 15-4), the A face of the coenzyme will be against the substrate undergoing oxidation or reduction. If the ring has an *anti* conformation, the B side will be against the substrate.^{18,22} However, the explanation cannot be this simple. High specificity is still retained when NADH is modified, for example, by replacement of the $-\text{CONH}_2$ on the pyridine ring with $-\text{COCH}_3$ or $-\text{CHO}$ or by use of α -NADH, in which the ribose–nicotinamide linkage is α instead of β as in normal NADH.²³

An intriguing idea is that the enzyme selectively stabilizes one of the boat conformations of NADH or NADPH. According to stereoelectronic principles the axially oriented hydrogen in such a boat structure (*pro-R* in Fig. 15-4) will be the most readily transferred. On the basis of this principle, together with the idea that enzymes adjust the Gibbs energies of intermediate states to achieve optimum catalytic rates, the following prediction was made by Benner and associates:²⁴ *The thermodynamically most easily reduced carbonyl compounds will react by enzymatic transfer of the hydride ion from the pro-R position of NADH. Conversely, the most difficultly reduced carbonyl compounds will receive the hydride ion from the pro-S position of NADH.* The proposal has been controversial and it has been argued that evolutionary relationships play a dominant role in determining stereoselectivity.²³ Theoretical computations suggest that the boatlike puckering of the ring is

flexible, raising some doubts about the rigidity of the reactants in the active site.^{12,25} Nevertheless, it is likely that the coenzyme as well as bound substrates must assume very specific conformations before the enzyme is able to move to the transition state for the reaction. A great deal of effort has been expended in trying to understand other factors that may explain the high stereospecificity of dehydrogenases.^{12,22,26–28}

When a hydrogen atom is transferred by an enzyme from the 4 position of NADH or NADPH to an aldehyde or ketone to form an alcohol, the placement of the hydrogen atom on the alcohol is also stereospecific. Thus, alcohol dehydrogenase acting on NAD^2H converts acetaldehyde to (*R*)-mono- $[\text{}^2\text{H}]$ ethanol (Eq. 9-73). Pyruvate is reduced by lactate dehydrogenase to L-lactate, and so on. Even mutations that disrupt the binding of the carboxylate of lactate, e.g., substitution of arginine 171 (see Fig. 15-3) by tryptophan or phenylalanine and introduction of a new arginine on the other side of the active site, do not alter the L-stereospecificity. However, the specificity for lactate is lost.²⁹

One step or two-step transfer? Another major question about dehydrogenases is whether the hydrogen atom that is transferred moves as a hydride ion, as is generally accepted, or as a hydrogen atom with separate transfer of an electron and with an intermediate NAD or NADPH free radical. In one study para-substituted benzaldehydes were reduced with NADH and NAD^2H using yeast alcohol dehydrogenase as a catalyst.³⁰ This permitted the application of the Hammett equation (Box 6-C) to the rate data. For a series of benzaldehydes for which σ^+ varied widely, a value

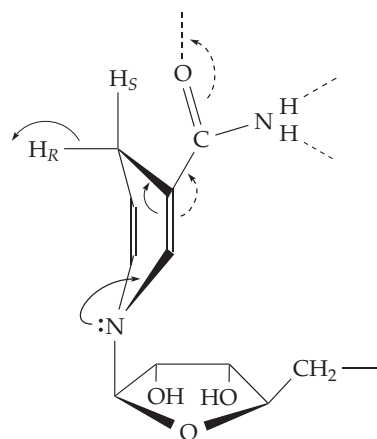


Figure 15-4 The nicotinamide ring of NADH in a *syn* boat conformation suitable for transfer of an axially oriented *pro-R* hydrogen atom from its A face as H^- . The flow of electrons is shown by the solid arrows. The dashed arrows indicate competing resonance which favors planarity of the ring and opposes the H^- transfer. Hydrogen bonds from the protein to the carboxamide group (dashed lines) affect both this tendency and the conformation of the nucleotide.

of $\rho = +2.2$ was observed for the rate constant with both NADH and NAD²H. Thus, electron-accepting substituents in the para position hasten the reaction. While the significance of this observation is not immediately obvious, the relatively low value of ρ is probably incompatible with a mechanism that requires complete transfer of a single electron from NADH to acetaldehyde in the first step. A primary isotope effect on the rates was $k_{\text{H}}/k_{\text{D}} = 3.6$, indicating that the C–H bond in NADH is broken in the rate-limiting step. The fact that the isotope effect is the same for all of the substituted benzaldehydes also argued in favor of a hydride ion transfer.

Studies of the kinetics of nonenzymatic model reactions of NADH with quinones^{31,32} have also been interpreted to favor a single-step hydride ion transfer. Application of Marcus theory (Chapter 9, Section D,4) to data from model systems also supports the hydride transfer mechanism.^{33,34} Quantum mechanical tunneling may be involved in enzymatic transfer of protons, hydride ions, and electrons.^{35–36} Tunneling is often recognized by unusually large primary or secondary kinetic isotope effects. According to semi-classical theory (Chapter 12), the maximum effects for deuterium and tritium are given by the following ratios:

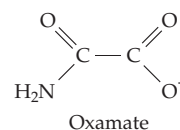
$$(k_{2\text{H}}/k_{1\text{H}}) = (k_{3\text{H}}/k_{2\text{H}})^{3.3} \approx 7.$$

Higher ratios, which suggest tunneling, are frequently observed for dehydrogenases.^{36a,b} Tunneling is apparently coupled to fluctuations in motion within the enzyme-substrate complex. Study of effects of pressure on reactions provides a new approach that can aid interpretation.^{36c,36d}

Coenzyme and substrate analogs. The structures of enzyme•NAD⁺•substrate complexes (Fig. 15-3) may be studied by X-ray crystallography under certain conditions or can be inferred from those of various stable enzyme-inhibitor complexes or from enzyme reconstituted with NAD⁺ that has been covalently

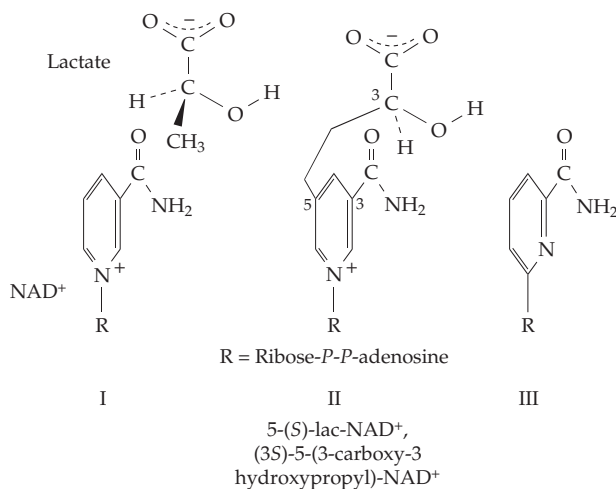
linked to the substrate. In the following diagram,²⁷ NAD⁺ (I) is shown with L-lactate lying next to its A face, ready to transfer a hydride ion to the *pro-R* position in NADPH. Also shown (II) is 5-(*S*)-lac-NAD⁺, with the covalently linked lactate portion in nearly the same position as in diagram I. This NAD derivative was used to obtain the first 0.27-nm structure with a bound substrate-like molecule in lactate dehydrogenase.³⁷ Since then the structures of many complexes with a variety of dehydrogenases have been studied. In coenzyme analog III the ring is bound to ribose with a C–C bond and it lacks the positive charge on the nicotinamide ring in NAD⁺. It does not react with substrate. However, it binds to the coenzyme site of alcohol dehydrogenase and forms with ethanol a ternary complex whose structure has been solved.³⁸

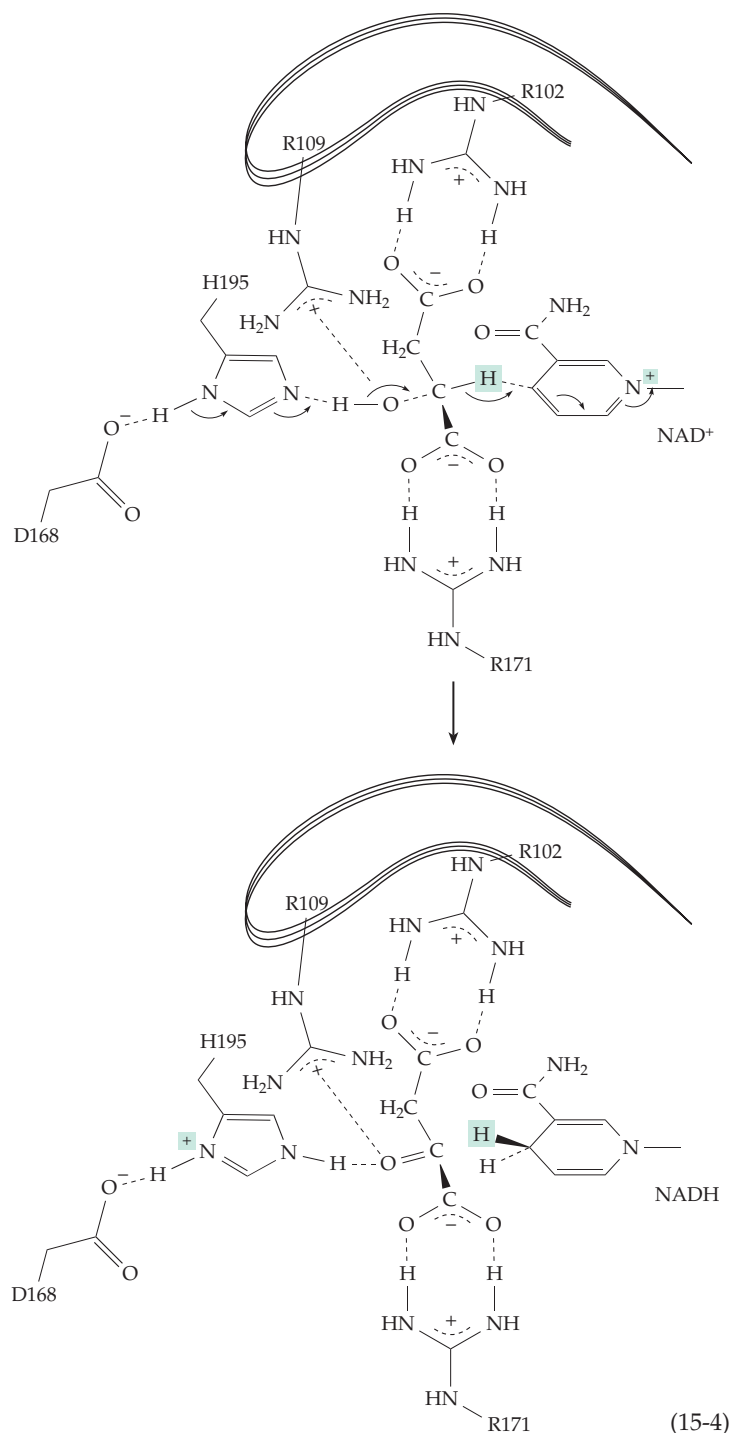
A related approach is to study complexes formed with normal NAD⁺ but with an unreactive second substrate. An example is oxamate, which binds well to lactate dehydrogenase to form stable ternary complexes for which equilibrium isotope effects have been studied.³⁹



In the structure of the lactate dehydrogenase active site shown in Fig. 15-3, the lactate carboxylate ion is held and neutralized by the guanidinium group of Arg 171, and the imidazole group of His 195 is in position to serve as a general base catalyst to abstract a proton from the hydroxyl group of the substrate (Eq. 15-4). This imidazole group is also hydrogen bonded to the carboxylate of Asp 168 as in the “charge-relay” system of the serine proteases (Fig. 12-10). The same features are present in the active site of malate dehydrogenases and have been shown essential by study of various mutant forms.^{11,40–42} The His:Asp pair of the dehydrogenases is not part of the nucleotide-binding domain but is present in the second structural domain, the “catalytic domain.” This is another feature reminiscent of the serine proteases. A bacterial D-glycerate dehydrogenase also has a similar structure and the same catalytic groups. However, the placement of the catalytic histidine and the arginine that binds the α -carboxylate group of the substrate is reversed, allowing the enzyme to act on the D-isomer.⁴³

Conformational changes during dehydrogenase action. Dehydrogenases bind coenzyme and substrate in an ordered sequence. The coenzyme binds first, then the oxidizable or reducible substrate. The binding of the coenzyme to lactate dehydrogenase is accompanied by a conformational change by which a loop, involving residues 98–120 and including one helix,





folds over the coenzyme like a lid.^{41,44} This conformational change must occur during each catalytic cycle, just as in the previously discussed cases of citrate synthase (Chapter 13) and aspartate aminotransferase (Chapter 14). One effect of folding of the loop is to bring the side chain of Arg 109 into close proximity to His 195 and to the OH group of the bound lactate (Eq. 15-4). The closing of the loop also forces the positively charged nicotinamide ring more deeply into a relatively nonpolar pocket. This may induce a move-

ment of the positive charge toward the 4' carbon of the ring, assisting in the transfer of the hydride ion (Eq. 15-4).⁴¹ At the end of this transfer both Arg 109 and His 196 are positively charged and electrostatic repulsion between them may help to move the loop and release the products. If the reaction proceeds in the opposite direction the presence of two positive charges will assist in the hydride ion transfer. The importance of Arg 109 is demonstrated by the fact that a mutant with glutamine in place of Arg 109 has a value of k_{cat} only 1/400 that of native enzyme.⁴⁵

The loop closure also causes significant changes in the Raman spectra of the bound NAD^+ , especially in vibrational modes that involve the carboxamide group of the nicotinamide ring. These have been interpreted as indicating an increased rigidity of binding of the coenzyme in the closed conformation.⁴⁶ As with many other enzymes acting on polar substrates, a characteristic of the pretransition state complex appears to be formation of a complex with a network of hydrogen bonds extending into the protein, exclusion of most water molecules, and tight packing of protein groups around the substrate. It is also significant that an overall net electrical charge on the ES complexes must be correct for tight binding of substrates to occur for lactate or malate dehydrogenase.⁴⁴ Positive and negative charges are balanced except for one excess positive charge which may be needed for catalysis.

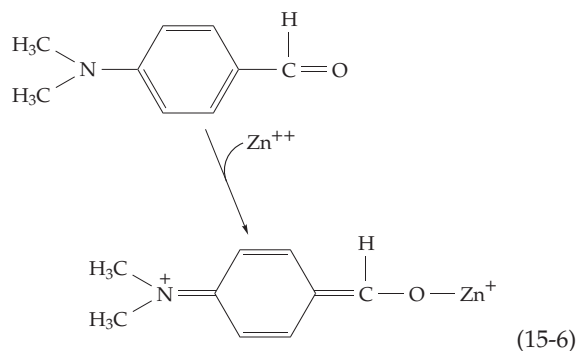
Zinc-containing alcohol dehydrogenases. Liver alcohol dehydrogenase is a relatively nonspecific enzyme that oxidizes ethanol and many other alcohols. The much studied horse liver enzyme^{47–60} is a dimer of 374-residue subunits, each of which contains a “catalytic” Zn^{2+} ion deeply buried in a crevice between the nucleotide-binding and catalytic domains. The enzyme also contains a “structural” Zn^{2+} ion⁴⁸ that is bound by four sulfur atoms from cysteine side chains but does not represent a conserved feature of all alcohol dehydrogenases. The catalytic Zn^{2+} is ligated

by sulfur atoms from cysteines 46 and 174 and by a nitrogen atom of the imidazole group of His 67 (Fig. 15-5). In the free enzyme a water molecule is thought to occupy the 4th coordination position and its dissociation to form the Zn^+-HO^- complex (Eq. 15-5) may account for a pK_a of 9.2 in the free enzyme and of 7.6 in the NAD^+ complex.⁴⁸ The apparent pK_a drops further to about 6.4 in the presence of the substrate



ethanol. The assignment of this pK_a value has been controversial.^{48,49} Histidine 51 (Fig. 15-5), Glu 68 and Asp 49 have side chains close to the zinc and, as we have seen (Chapter 7), macroscopic pK_a values of proteins can be *shared* by two or more closely placed groups.

Substrate binding also induces a conformational change in this enzyme. When both coenzymes and substrate bind the “closed” conformation of the enzyme is formed by a rotation of the catalytic domains of the two subunits relative to the coenzyme-binding domains.^{50–51a} Structures of ternary complexes with inhibitors and with substrates have also been established. For example, liver alcohol dehydrogenase was crystallized as the enzyme•NAD⁺•*p*-bromobenzyl alcohol complex with saturating concentrations of substrates in an equilibrium mixture^{51b} and studied at low resolution. Transient kinetic studies or direct spectroscopic determinations led to the conclusion that the internal equilibrium ($E\cdot NAD^+ \cdot \text{alcohol} = E\cdot NADH \cdot \text{aldehyde}$) favors the NAD⁺•alcohol complex.⁵² Subsequently, the complex was studied at higher resolution, and the basic structural features were confirmed with a



structure of the enzyme complexed with NAD⁺ and 2,3,4,5,6-pentafluorobenzyl alcohol.⁵³ From the crystal structures of the NAD⁺•*p*-bromobenzyl alcohol complex and the previously mentioned complex with ethanol and analog III it appears that the oxygen of the alcohol substrate coordinates with the Zn²⁺, displacing the bound water. Binding of the chromophoric aldehyde 4-*trans*-(*N,N*-dimethylamino) cinnamaldehyde shifts the wavelength of maximum absorbance by 66 nm,

suggesting that Zn²⁺ binds directly to the oxygen of this ligand (Eq. 15-6).⁵⁴ Resonance-enhanced Raman spectroscopy (Chapter 23) of the complex of dimethylaminobenzaldehyde with alcohol dehydrogenase also supports an intermediate in which the Zn²⁺ becomes coordinated directly with the substrate oxygen (Eq. 15-6).⁵⁵ Rapid scanning spectrophotometry of complexes with another chromophoric substrate, 3-hydroxy-4-nitrobenzyl alcohol, also suggested that the alcohol first formed a complex with an undissociated alcoholic –OH group (Eq. 15-7, step *a*) and then lost a proton to form a zinc alcoholate complex (step *b*)⁵⁶ which could react by hydride ion transfer (step *c*).

Eklund *et al.* suggested that the side chains of Ser 48 and His 51 act as a proton relay system to remove the proton from the alcohol, in step *b* of Eq. 15-7, leaving the transient zinc-bound alcoholate ion, which can then transfer a hydride ion to NAD⁺, in step *c*.⁵² The shaded hydrogen atom leaves as H⁺. The role of His 51 as a base is supported by studies of the inactivation of the horse liver enzyme by diethyl pyrocarbonate⁵⁷ and by directed mutation of yeast and liver enzymes. When His 51 was substituted by Gln the pK_a of 7 was abolished and the activity was decreased ten-fold.⁵⁸

The functioning of zinc ions in enzymes has been controversial and other mechanisms have been proposed. Makinen *et al.* suggested a transient pentacoordinate Zn²⁺ complex on the basis of EPR measurements on en-

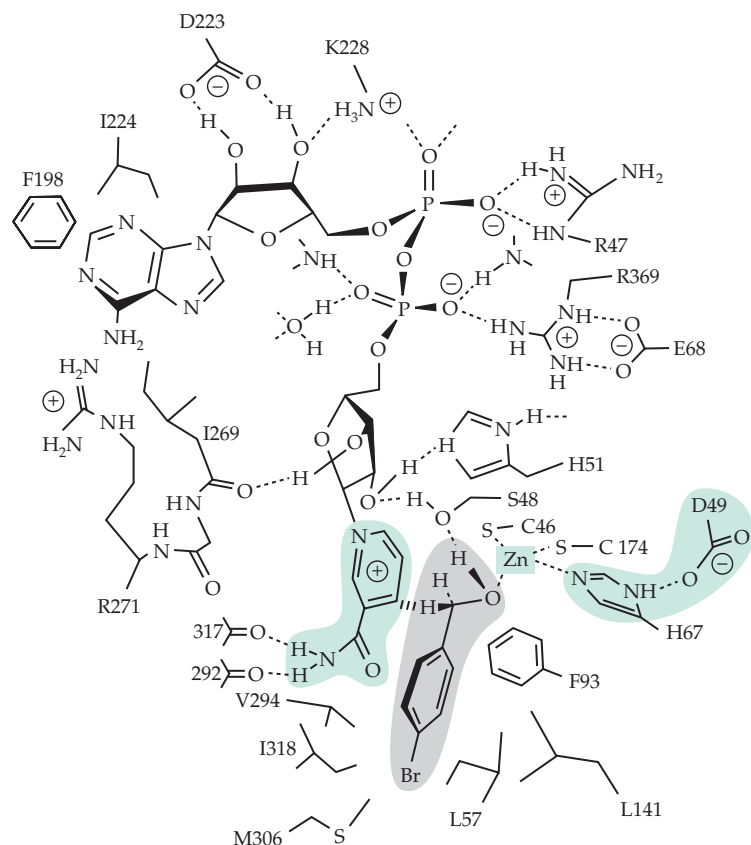
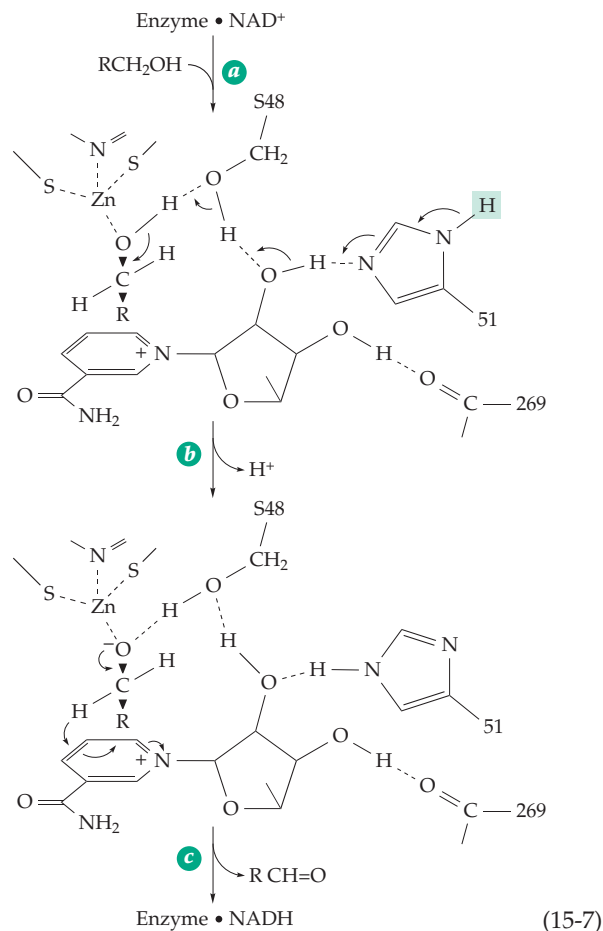


Figure 15-5 Structure of the complex of horse liver alcohol dehydrogenase with NAD⁺ and the slow substrate *p*-bromobenzyl alcohol. The zinc atom and the nicotinamide ring of the bound NAD⁺ are shaded. Adjacent to them is the bound substrate. Courtesy of Bryce Plapp. Based on Ramaswamy *et al.*⁵³



zyme containing Co²⁺. Such a complex, in which the side chain of nearby Glu 68 would participate, would allow the coordinated water molecule to act as the base in deprotonation of the alcohol.⁵⁹ Molecular dynamics calculations indicate that Glu 68 can coordinate the zinc ion in this fashion, but Ryde suggests that its function may be to assist the exchange of ligands, i.e., the release of an alcohol or aldehyde product.⁶⁰ A variety of kinetic and spectroscopic studies have provided additional information that makes alcohol dehydrogenase one of the most investigated of all enzymes.

Liver alcohol dehydrogenase is important to the metabolism of ethanol by drinkers. Human beings exhibit small individual differences in their rates of alcohol metabolism which may reflect the fact that there are several isoenzymes and a number of genetic variants whose distribution differs from one person to the next as well as among tissues.^{61–64} Inhibition of these isoenzymes by *uncompetitive* inhibitors, discussed in Chapter 9, is a goal in treatment of poisoning by methanol or ethylene glycol. Inhibition of the dehydrogenases slows the two-step oxidation of these substrates to toxic carboxylic acids.⁶⁵

Yeast contains two cytosolic alcohol dehydrogenase isoenzymes.⁶⁶ Alcohol dehydrogenase I, present in large amounts in cells undergoing fermentation,

functions to reduce acetaldehyde in the fermentation process. Alcohol dehydrogenase II is synthesized by cells growing on such carbon sources as ethanol itself and needing to oxidize it to obtain energy. A third isoenzyme is present in mitochondria. An alcohol dehydrogenase isoenzyme of green plants is induced by anaerobic stress such as flooding. It permits ethanolic fermentation to provide energy temporarily to submerged roots and other tissues.⁶⁷ Some bacteria contain an NADP⁺-dependent, Zn²⁺-containing alcohol dehydrogenase.^{67a}

Other alcohol dehydrogenases and aldo-keto reductases. The oxidation of an alcohol to a carbonyl compound and the reverse reaction of reduction of a carbonyl group are found in so many metabolic pathways that numerous specialized dehydrogenases exist. A large group of “short-chain” dehydrogenases and reductases had at least 57 known members by 1995.^{68–69a} Their structures and functions are variable. Most appear to be single-domain proteins with a large β sheet, a nucleotide-binding pocket, and a conserved pair of residues: Tyr 152 and Lys 156. These may function in a manner similar to that of the His-Asp pair in lactate dehydrogenase. A possible role of a cysteine side chain has been suggested for another member of this group, 3-hydroxyisobutyrate dehydrogenase, an enzyme of valine catabolism.⁷⁰

Dehydrogenases often act primarily to reduce a carbonyl compound rather than to dehydrogenate an alcohol. These enzymes may still be called dehydrogenases. For example, in the lactic acid fermentation lactate is formed by reduction of pyruvate but we still call the enzyme lactate dehydrogenase. In our bodies this enzyme functions in both directions. However, some enzymes that act mainly in the direction of reduction are called reductases. An example is **aldose reductase**, a member of a family of **aldo-keto reductases**^{71–73} which have (α/β)₈-barrel structures.^{74–76}

The normal physiological function of aldose reductase is uncertain but it can cause a problem in diabetic persons by reducing glucose to sorbitol (glucitol), ribose to ribitol, etc. The resulting sugar alcohols accumulate in the lens and are thought to promote cataract formation and may also be involved in the severe damage to retinas and kidneys that occurs in diabetes mellitus. Inhibitors of aldose reductase delay development of these complications in animals but the compounds tested are too toxic for human use.⁷⁶

The active sites of aldo-keto reductases contain an essential tyrosine (Tyr 48) with a low pK_a value. The nearby His 110, Asp 43, and Lys 77 may also participate in catalysis.^{76,77} The kinetics are unusual. Both NAD⁺ and NADH are bound tightly and the overall rate of reduction of a substrate is limited by the rate of dissociation of NAD⁺.⁷⁸ Citrate is a natural uncompetitive inhibitor of aldose reductase.⁷⁹

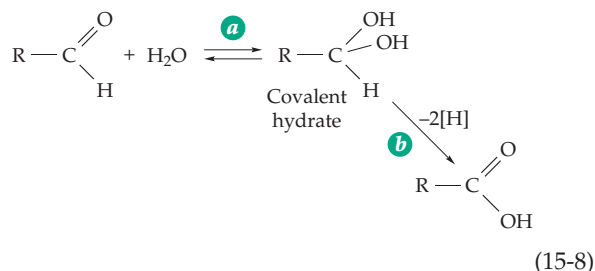
3. Dehydrogenation of Amino Acids and Amines

The dehydrogenation of an amine or the reverse reaction, the reduction of a Schiff base (Table 15-1, reaction type B), is another important pyridine nucleotide-dependent process. **Glutamate dehydrogenase**, a large oligomeric protein whose subunits contain 450 or more residues, is the best known enzyme catalyzing this reaction. An intermediate Schiff base of 2-oxo-glutarate and NH_3 is a presumed intermediate.^{80,81} Similar reactions are catalyzed by dehydrogenases for alanine,^{82,83} leucine,⁸⁴ phenylalanine,⁸⁵ and other amino acids.^{83,86,87}

4. Glyceraldehyde-3-Phosphate Dehydrogenase and the Generation of ATP in Fermentation Reactions

The oxidation of an aldehyde to a carboxylic acid, a highly exergonic process, often proceeds through a thioester intermediate whose cleavage can then be coupled to synthesis of ATP. This sequence is of central importance to the energy metabolism of cells (Chapters 10 and 17) and is shown in Fig. 15-6.

The best known enzyme catalyzing the first step of this reaction sequence is glyceraldehyde 3-phosphate dehydrogenase which functions in the glycolytic sequence (steps *a* and *b* of Fig. 10-3). It is present in both prokaryotes and eukaryotes as a tetramer of identical 36- to 37-kDa subunits. Three-dimensional structures have been determined for enzyme from several species, including lobster,^{47,88} *E. coli*,⁸⁹ the thermophilic bacterium *Bacillus stearothermophilus* (Fig. 2-13),^{90-90a} and trypanosomes.⁹¹ Recall that aldehydes are in equilibrium with their covalent hydrates (Eq. 15-8, step *a*). Dehydrogenation of such a hydrate yields an acid (Eq. 15-8, step *b*) but such a mechanism offers no possibility of conserving the energy available from the reaction. However, during catalysis by glyceraldehyde-phosphate dehydrogenase the SH group of Cys 149, in the first step (step *a*, Fig. 15-6), adds to the substrate carbonyl group to form an adduct, a thiohemiacetal. This adduct is oxidized by NAD^+ to a thioester, an *S*-acyl enzyme (step *b*), which is then cleaved by the same enzyme through a displacement on carbon by an oxygen atom of P_i (phosphorolysis; step *c*). The sulfhydryl group of the enzyme is released simultaneously and the product, the acyl phosphate **1,3-bisphosphoglycerate**, is formed. The imidazole group of His 176 may catalyze both steps *a* and *b*.^{89,92} A separate enzyme then transfers the phospho group from the 1 position of 1,3-bisphosphoglycerate to ADP to form ATP and 3-phosphoglycerate (step *d*). The overall sequence of Fig. 15-6 is the synthesis of one mole of ATP coupled to the oxidation of an aldehyde



to a carboxylic acid and the conversion of NAD^+ to NADH.

In green plants and in some bacteria an NADP^+ -dependent cytoplasmic glyceraldehyde 3-phosphate dehydrogenase *does not* use inorganic phosphate to form an acyl phosphate intermediate but gives 3-phosphoglycerate with a free carboxylate as in Eq. 15-8.^{93,93a} Because it doesn't couple ATP cleavage to the dehydrogenation, it drives the $[\text{NADPH}] / [\text{NADP}^+]$ ratio to a high value favorable to biosynthetic processes (see Chapter 17).

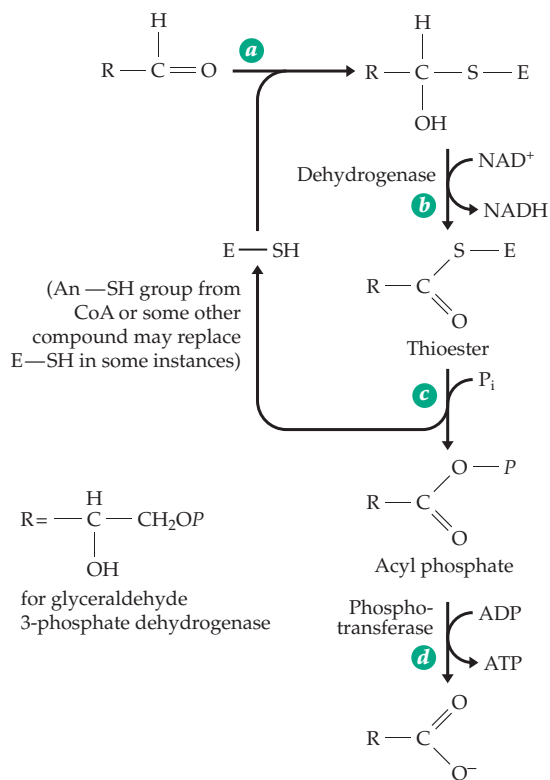
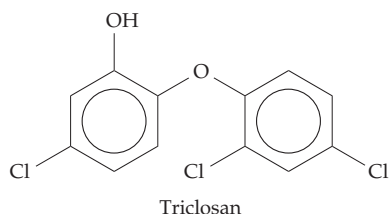
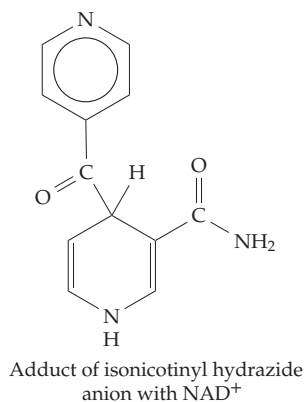


Figure 15-6 Generation of ATP coupled to oxidation of an aldehyde to a carboxylic acid. The most important known example of this sequence is the oxidation of glyceraldehyde 3-phosphate to 3-phosphoglycerate (Fig. 10-3, steps *a* and *b*). Other important sequences for “substrate-level” phosphorylation are shown in Eq. 14-23 and in Fig. 15-16.

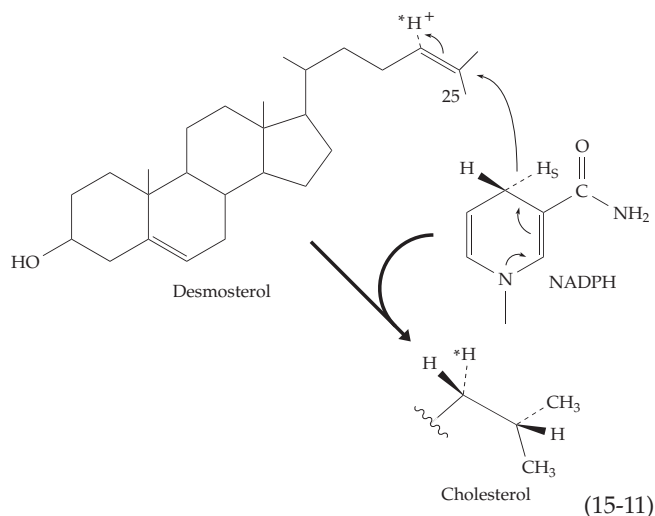
induced hemolytic anemia in human beings. About 400 variant forms of this enzyme are known.¹⁰⁵ Like the sickle cell trait (Box 7-B) some mutant forms of glucose-6-phosphate dehydrogenase appear to confer resistance to malaria.

6. Reduction of Carbon–Carbon Double Bonds

Neither NADP^+ nor NAD^+ is a strong enough oxidant to carry out the dehydrogenation of an acyl-CoA (reaction type D of Table 15-1). However, NADPH or NADH can participate in the opposite reaction. Thus, NADPH transfers a hydride ion to the β -carbon of an unsaturated acyl group during the biosynthesis of fatty acids (Chapter 17) and during elongation of shorter fatty acids (Chapter 21).^{106–107b} A discovery of medical importance is that isonicotinyl hydrazide (INH), the most widely used antituberculosis drug, forms an adduct (of an INH anion or radical) with NAD^+ of long-chain **enoyl-acyl carrier protein reductase** (enoyl-ACP reductase).^{133a,b} This enzyme utilizes NADH in reduction of a $\text{C}=\text{C}$ double bond during synthesis of mycolic acids. The same enzyme is blocked by **triclosan**, an antibacterial compound used widely in household products such as antiseptic soaps, toothpastes, cosmetics, fabrics, and toys.^{133c–e}



Less frequently NADPH is used to reduce an *isolated double bond*. An example is the hydrogenation of **desmosterol** by NADPH (Eq. 15-11), the final step in one of the pathways of biosynthesis of cholesterol (Fig. 22-7). In this and in other reactions of the same

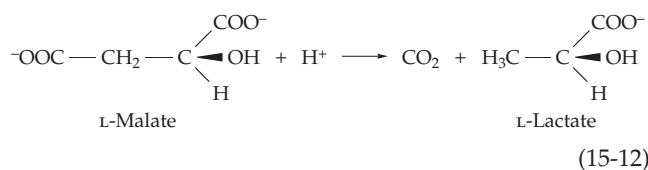


type, hydrogen transfer has been shown to be from the *pro-S* position in NADPH directly to C-25 of the sterol. The proton introduced from the medium (designated by the asterisks in Eq. 15-11) enters *trans* to the H^- ion from NADPH. The proton always adds to the more electron-rich terminus of the double bond, i.e., it follows the Markovnikov rule. This result suggests that protonation of the double bond may precede H^- transfer.¹⁰⁸

Additional pyridine nucleotide-dependent dehydrogenases include **glutathione reductase** (Figs. 15-10, 15-12), **dihydrofolate reductase** (Fig. 15-19), isocitrate dehydrogenase, ***sn*-glycerol-3-phosphate dehydrogenase** (Chapter 21), **L-3-hydroxyacyl-CoA dehydrogenase** (Chapter 21), **retinol dehydrogenase** (Chapter 23), and a bacterial quinone oxidoreductase.¹⁰⁹ Some of these also contain a flavin coenzyme.

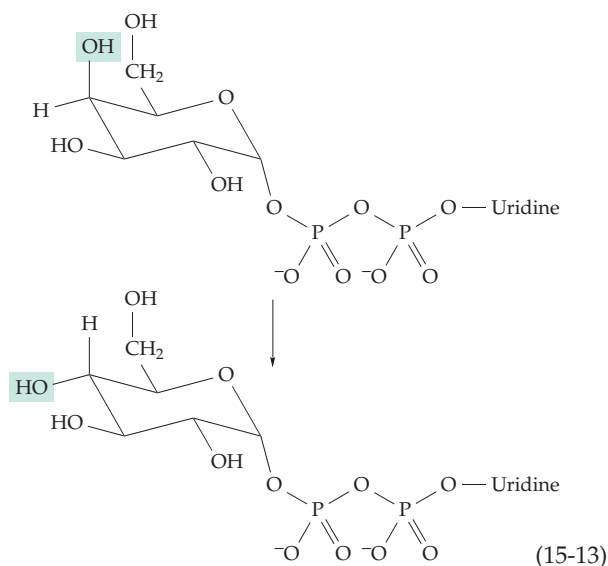
7. Transient Carbonyl Groups in Catalysis

Some enzymes contain bound NAD^+ which oxidizes a substrate alcohol to facilitate a reaction step and is then regenerated. For example, the **malolactic enzyme** found in some lactic acid bacteria and also in *Ascaris* decarboxylates L-malate to lactate (Eq. 15-12). This reaction is similar to those of isocitrate dehydrogenase,^{110–112} 6-phosphogluconate dehydrogenase,¹¹³ and the malic enzyme (Eq. 13-45)¹¹⁴ which utilize free NAD^+ to first dehydrogenate the substrate to a bound oxoacid whose β carbonyl group facilitates decarboxylation. Likewise, the bound NAD^+ of the malolactic



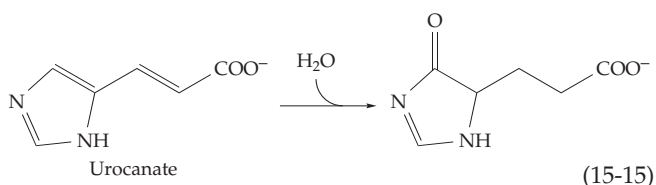
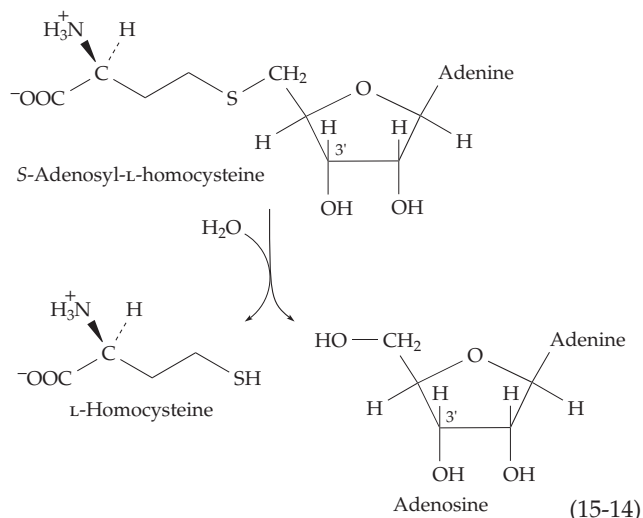
enzyme apparently dehydrogenates the malate to a bound oxaloacetate which is decarboxylated to pyruvate.¹¹⁵ The latter remains in the active site and is reduced by the bound NADH to lactate which is released from the enzyme.

Another reason for introducing a carbonyl group is to form a symmetric intermediate in a reaction that inverts the configuration about a chiral center. An example is **UDP-galactose 4-epimerase**, an enzyme that converts UDP-galactose to UDP-glucose (Eq. 15-13; Chapter 20) and is essential in the metabolism of galactose in our bodies. The enzyme contains bound NAD⁺ and forms a transient 4-oxo intermediate and bound NADH. Rotation of the intermediate allows nonstereospecific reduction by the NADH, leading to epimerization.^{116–117a}



The enzyme is a member of the short-chain dehydrogenase group with a catalytic Tyr–Lys pair in the active site.¹¹⁸ Another way that formation of an oxo group can assist in epimerization of a sugar is through enolization with nonstereospecific return of a proton to the intermediate enediol. A third possible mechanism of epimerization is through aldol cleavage followed by aldol condensation, with inversion of configuration. In each case the initial creation of an oxo group by dehydrogenation is essential.

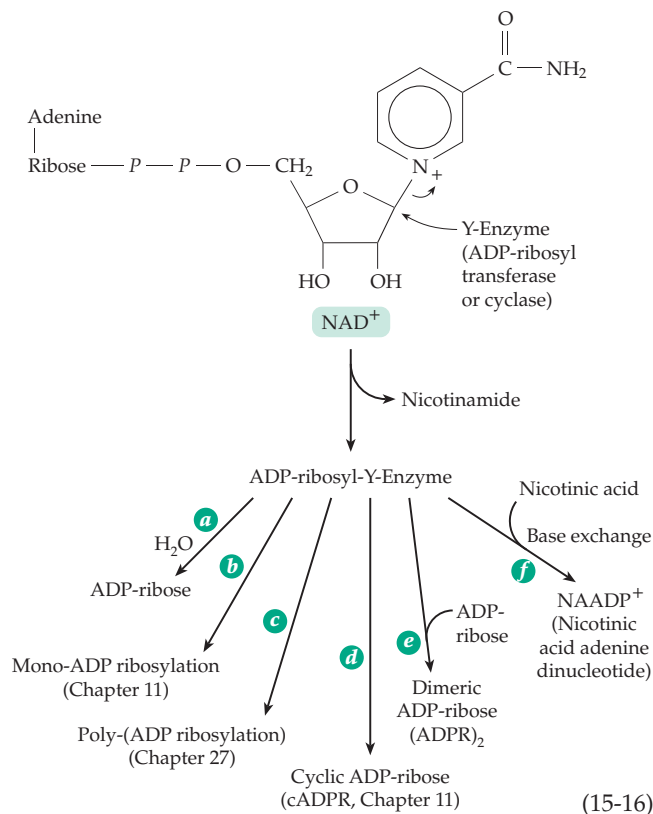
Bound NAD⁺ is also present in **S-adenosylhomocysteine hydrolase**,^{119,120} which catalyzes the irreversible reaction of Eq. 15-14. Transient oxidation at the 3 position of the ribose ring facilitates the reaction. The reader can doubtless deduce the function that has been established for the bound NAD⁺ in this enzyme. However, the role of NAD in the **urocanase** reaction (Eq. 15-15) is puzzling. This reaction, which is the second step in the catabolism of histidine, following Eq. 14-44, appears simple. However, there is no obvious



mechanism and no obvious role for NAD⁺. See Frey for a discussion.¹¹⁷

8. ADP Ribosylation and Related Reactions of NAD⁺ and NADP⁺

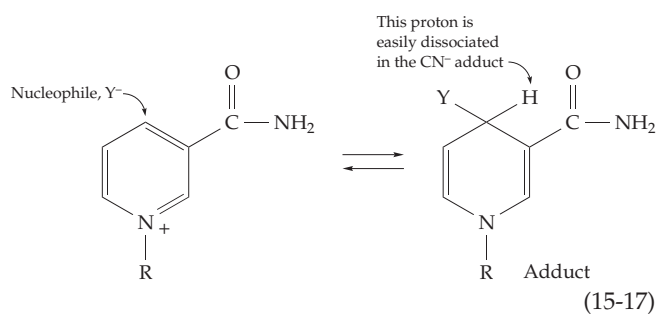
The linkage of nicotinamide to ribose in NAD⁺ and NADP⁺ is easily broken by nucleophilic attack on C-1 of ribose. In Chapter 11 enzyme-catalyzed ADP ribosylation, which can be shown as in Eq. 15-16, is discussed briefly. The nucleophilic group –Y from an ADP-ribosyltransferase carries the ADP-ribosyl group which can then be transferred by a second displacement onto a suitable nucleophilic acceptor group.^{121,122} Hydrolysis (Eq. 15-16, step *a*) gives free ADP-ribose.¹²³ Other known products of enzymatic action are indicated in steps *c–f*. Poly-(ADP ribosylation) is discussed in Chapter 27. The structure of cyclic ADP-ribose (cADPR)¹²⁴ is shown in Chapter 11, Section E.2. The acceptor nucleophile is N-1 of the adenine ring which is made more nucleophilic by electron donation from the amino group. A similar reaction with ADP ribose (Eq. 15-16, step *e*) produces a dimeric ADP-ribose (ADPR)₂¹²⁵ while reaction with free nicotinic acid (step *f*) yields, in an overall base exchange, nicotinic acid adenine dinucleotide (NAADP⁺).^{126,127} Some of these compounds, e.g., cADPR, NAADP⁺, and (ADPR)₂, are involved in signaling with calcium ions.¹²⁴



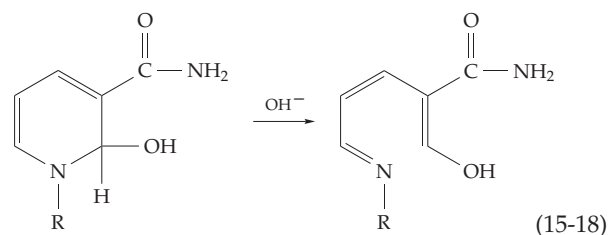
9. The Varied Chemistry of the Pyridine Nucleotides

Despite the apparent simplicity of their structures, the chemistry of the nicotinamide ring in NAD⁺ and NADP⁺ is surprisingly complex.^{128,129} NAD⁺ is extremely unstable in basic solutions, whereas NADH is just as unstable in slightly acidic media. These properties, together with the ability of NAD⁺ to undergo condensation reactions with other compounds, have sometimes caused serious errors in interpretation of experiments and may be of significance to biological function.

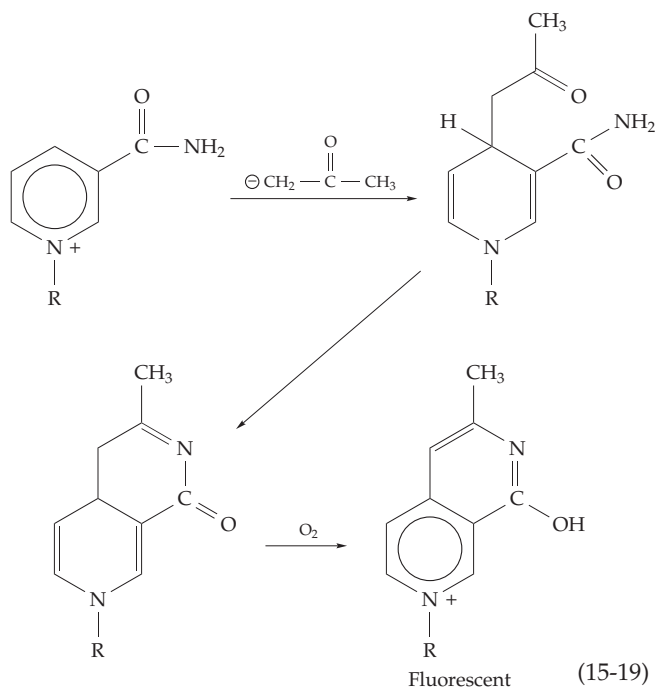
Addition to NAD⁺ and NADP⁺. Many nucleophilic reagents add reversibly at the para (or 4) position (Eq. 15-17) to form adducts having structures resembling those of the reduced coenzymes. Formation of



the cyanide adduct, whose absorption maximum is at 327 nm, has been used to introduce deuterium into the para position of the pyridine nucleotides. In the adduct, the hydrogen adjacent to the highly polarized C — N bond is easily dissociated as a proton. Thiolate anions and bisulfite also add. Dithionite ion, S₂O₄²⁻, can lose SO₂ and acquire a proton to form the sulfoxylate ion HSO₂⁻ which also adds to the 4 position of the NAD⁺ ring.¹³⁰ The resulting adduct is unstable and loses SO₂ to give NADH + H⁺. Addition can also occur at the two ortho positions. The adducts of HO⁻ to the 4 position of NAD⁺ are stable but those to the 2 position undergo ring opening (Eq. 15-18) in base-catalyzed reactions which are followed by further degradation.^{128,131}

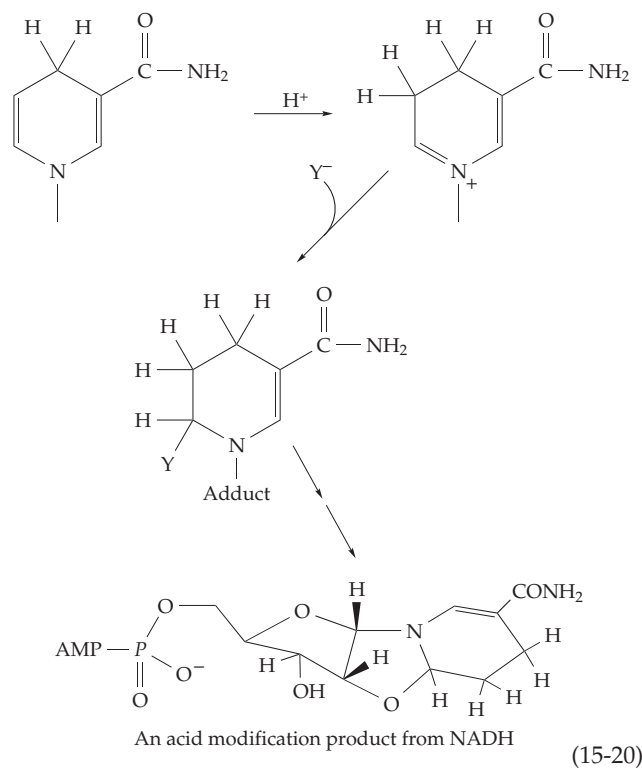


Another base-catalyzed reaction is the addition of enolate anions derived from ketones to the 4 position of the pyridine nucleotides (Eq. 15-19). The adducts undergo ring closure and in the presence of oxygen are converted slowly to fluorescent materials. While forming the basis for a useful analytical method for determination of NAD⁺ (using 2-butanone), these reactions also have created a troublesome enzyme inhibitor from traces of acetone present in commercial NADH.¹³²



The reactions of Eq. 15-19 occur nonenzymatically only under the influence of strong base but dehydrogenases often catalyze similar condensations relatively rapidly and reversibly. Pyruvate inhibits lactate dehydrogenase, 2-oxoglutarate inhibits glutamate dehydrogenase, and ketones inhibit a short-chain alcohol dehydrogenase in this manner.^{133,69a}

Modification of NADH in acid. Reduced pyridine nucleotides are destroyed rapidly in dilute HCl and more slowly at pH 7 in reactions catalyzed by buffer acids.^{128,131,134} Apparently the reduced nicotinamide ring is first protonated at C-5, after which a nucleophile Y^- adds at the 6 position (Eq. 15-20). The nucleophile may be OH^- , and the adduct may undergo further reactions. For example, water may add to the other double bond and the ring may open on either side of the nitrogen. The glycosidic linkage can be isomerized from β to α or can be hydrolytically cleaved. The early steps in the modification reaction are partially reversible, but the overall sequence is irreversible. One of the products, which has been characterized by crystal structure determination, is shown in Eq. 15-20.^{128,135} It can arise if the group Y of Eq. 15-20 is the C-2' hydroxyl of the ribose ring and if the configuration of the glycosidic linkage is inverted (anomerized).

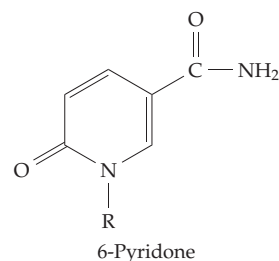


The foregoing reactions have attracted interest because glyceraldehyde-3-phosphate dehydrogenase, in a side reaction, converts NADH to a substance referred to as NADH-X which has been shown to be

the 6(R) adduct of Eq. 15-20, where Y is $-OH$. In an ATP-dependent reaction an enzyme from yeast reconverts NADH-X to NADH.¹³⁶

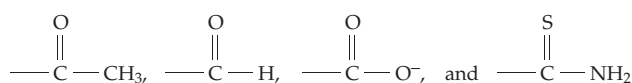
Mercury (II) ions can add in place of H^+ in the first step of Eq. 15-20 and subsequent reactions similar to those promoted by acid can occur.¹³⁷

Other reactions of pyridine nucleotides. Alkaline hexacyanoferrate (III) oxidizes NAD^+ and $NADP^+$ to 2,4-, and 6-pyridones. The 6-pyridone of *N*-methyl-nicotinamide is a well-known excretion product of nicotinic acid in mammals. Reoxidation of NADH and NADPH to NAD^+ and $NADP^+$ can be accomplished with hexacyanoferrate (III), quinones, and riboflavin



but not by H_2O_2 or O_2 . However, O_2 does react at neutral pH with uptake of a proton to form a peroxide derivative of NADH.¹³⁸ When heated in 0.1 N alkali at $100^\circ C$ for 5 min, NAD^+ is hydrolyzed to nicotinamide and adenosine-diphosphate-ribose.

Treatment of NAD^+ with nitrous acid deaminates the adenine ring. The resulting deamino NAD^+ as well as synthetic analogs containing the following groups in place of the carboxamide have been used



widely in enzyme studies. In fact, almost every part of the coenzyme molecule has been varied systematically and the effects on the chemical and enzymatic properties have been investigated.¹³⁹⁻¹⁴¹ "Caged" NAD^+ and $NADP^+$ have also been made.¹⁴² These compounds do not react as substrates until they are released ("uncaged") by photolytic action of a laser beam (see Chapter 23).

B. The Flavin Coenzymes

Flavin adenine diphosphate (FAD, flavin adenine dinucleotide) and **riboflavin 5'-monophosphate (FMN**, flavin mononucleotide), whose structures are shown in Fig. 15-7, are perhaps the most versatile of all

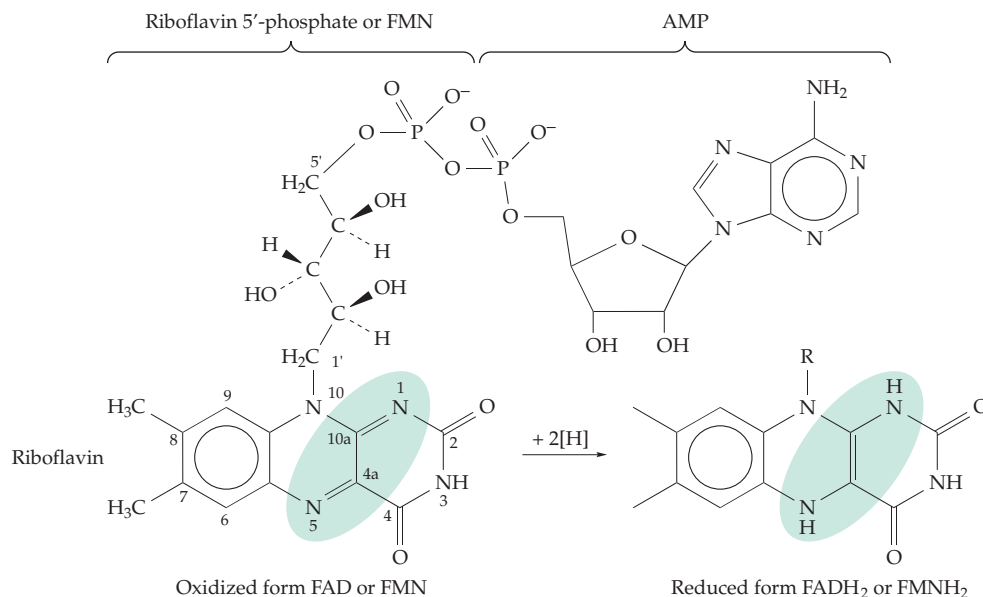


Figure 15-7 The flavin coenzymes flavin adenine dinucleotide (FAD) and riboflavin 5'-phosphate (FMN). Dotted lines enclose the region that is altered upon reduction.

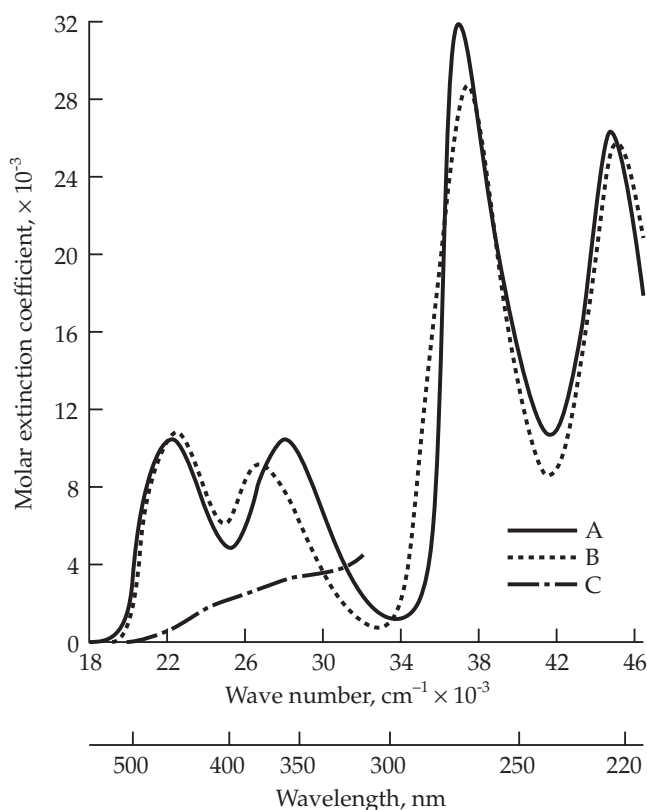


Figure 15-8 Absorption spectrum of neutral, uncharged riboflavin (A), the riboflavin anion (B), and reduced to the dihydro form (Fig. 15-7) by the action of light in the presence of EDTA (C). A solution of 1.1×10^{-4} M riboflavin containing 0.01 M EDTA was placed 11.5 cm from a 40-W incandescent lamp for 30 min.

the oxidation coenzymes. The name flavin adenine dinucleotide is not entirely appropriate because the D-ribityl group is not linked to the riboflavin in a glycosidic linkage. Hemmerich suggested that FAD be called flavin adenine diphosphate.¹⁴³

The attention of biochemists was first attracted to flavins as a result of their color and fluorescence. The study of spectral properties of flavins (Fig. 15-8) has been of importance in understanding these coenzymes. The biochemical role of the flavin coenzymes was first recognized through studies of the "old yellow enzyme"^{144,145} which was shown by Theorell to contain riboflavin 5'-phosphate. By 1938, FAD was recognized as the coenzyme of a different yellow protein, **D-amino acid oxidase** of kidney tissue. Like the pyridine nucleotides, the new flavin coenzymes were reduced by dithionite to nearly colorless dihydro forms (Figs. 15-7 and 15-8) revealing the chemical basis for their function as hydrogen carriers.

Flavins are also among the natural light receptors and display an interesting and much studied photochemistry (Fig. 3-5).¹⁴³ Flavins may function in some photoresponses of plants, and they serve as light emitters in bacterial bioluminescence (Chapter 23).^{146,147}

Three facts account for the need of cells for both the flavin and pyridine nucleotide coenzymes: (1) Flavins are usually stronger oxidizing agents than is NAD⁺. This property fits them for a role in the electron transport chains of mitochondria where a sequence of increasingly more powerful oxidants is needed and makes them ideal oxidants in a variety of other dehydrogenations. (2) Flavins can be reduced either by one- or two-electron processes. This enables them to participate in oxidation reactions involving free radicals and in reactions with metal ions. (3) Reduced flavins

are “autooxidizable,” i.e., they can be reoxidized directly and rapidly by O_2 , a property shared with relatively few other organic substances. For example, NADH and NADPH are not spontaneously reoxidized by oxygen. Autooxidizability allows flavins of some enzymes to pass electrons directly to O_2 and also provides a basis for the functioning of flavins in hydroxylation reactions.

1. Flavoproteins and Their Reduction Potentials

Flavin coenzymes are usually bound tightly to proteins and cycle between reduced and oxidized states while attached to the same protein molecule. In a free unbound coenzyme the redox potential is determined by the structures of the oxidized and reduced forms of the couple. Both riboflavin and the pyridine nucleotides contain aromatic ring systems that are stabilized by resonance. Part of this resonance stabilization is lost upon reduction. The value of E° depends in part upon the varying amounts of resonance in the oxidized and reduced forms. The structures of the coenzymes have apparently evolved to provide values of E° appropriate for their biological functions.

The relative strengths of binding of oxidized and reduced flavin coenzymes to a protein also have strong effects upon the reduction potential of the coenzyme.¹⁴⁸ If the oxidized form is bound weakly, but the reduced form is bound tightly, a bound flavin will have a greater tendency to stay in the reduced form than it did when free. The reduction potential E° will be less negative than it is for the free flavin-dihydroflavin couple. On the other hand, if the oxidized form of the flavin is bound more tightly by the protein than is the reduced form, E° will be more negative and the flavoenzyme will be a less powerful oxidizing agent. In fact, the values of E° at pH 7 for flavoproteins span a remarkably wide range from -0.49 to $+0.19$ V. The state of protonation of the reduced flavin when bound to the enzyme will also have a major effect on the oxidation–reduction potential. For example, acyl-CoA dehydrogenases are thought to form an anionic species of reduced FAD ($FADH^-$) which is tightly bound to the protein.¹⁴⁹

Every flavoprotein accepts electrons from the substrate that it oxidizes and passes these electrons on to another substrate, an oxidant. In the following sections we will consider for several enzymes how the electrons may get into the flavin from the oxidizable substrate and how they may flow out of the flavin into the final electron acceptor.

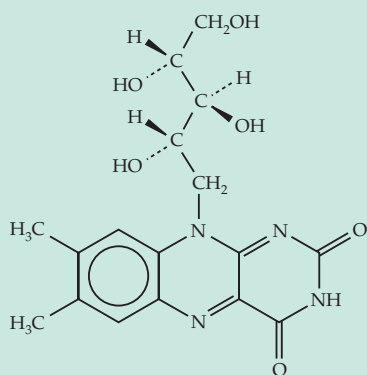
2. Typical Dehydrogenation Reactions Catalyzed by Flavoproteins

The functions of flavoprotein enzymes are numerous and diversified.^{151–153a} A few of them are shown in Table 15-2 and are classified there as follows: (A) oxidation of hemiacetals to lactones, (B) oxidation of alcohols to aldehydes or ketones, (C) oxidation of amines to imines, (D) oxidation of carbonyl compounds or carboxylic acids to α,β -unsaturated compounds, (E) oxidation of NADH and NADPH in electron transport chains, and (F) oxidation of dithiols to disulfides or the reverse reaction. Three-dimensional structures are known for enzymes of each of these types.

Reactions of types A–C could equally well be catalyzed by pyridine nucleotide-requiring dehydrogenases. Recall that D-glucose-6-phosphate dehydrogenase uses $NADP^+$ as the oxidant (Eq. 15-10). The first product is the lactone which is hydrolyzed to 6-phosphogluconic acid. The similar reaction of free glucose (Table 15-2, reaction type A) is catalyzed by fungal **glucose oxidase**, a 580-residue FAD-containing enzyme.^{154–155} A bacterial cholesterol oxidase has a similar structure.¹⁵⁶ The important plant enzyme **glycolate oxidase** is a dimer containing riboflavin 5'-phosphate.^{157–159} It catalyzes a reaction of type B, which plays an important role in photorespiration (Chapter 23). **Amino acid oxidases** (reaction type C) are well-known. The peroxisomal D-amino acid oxidase from kidney was the source from which Warburg first isolated FAD and has been the subject of much investigation of mechanism and structure.^{153,160–163} Many snake venoms contain an active 140-kDa L-amino-acid oxidase which contains FAD. Flavin-dependent **amine oxidases**, important in the human body, catalyze the related reaction with primary, secondary, or tertiary amines and in which a carboxyl group need not be present.^{164–166a} Reduced flavin produced by all of these oxidases is reoxidized with molecular oxygen and hydrogen peroxide is the product. Nature has chosen to forego the use of an electron transport chain (Fig. 10-5), giving up the possible gain of ATP in favor of simplicity and a more direct reaction with oxygen. In some cases there is specific value to the organism in forming H_2O_2 (see Chapter 18).

In contrast to the flavin oxidases, flavin dehydrogenases pass electrons to carriers within electron transport chains and the flavin does *not* react with O_2 . Examples include a bacterial **trimethylamine dehydrogenase** (Fig. 15-9) which contains an iron–sulfur cluster that serves as the immediate electron acceptor^{167–169} and yeast **flavocytochrome b_2** , a lactate dehydrogenase that passes electrons to a built-in heme group which can then pass the electrons to an external acceptor, another heme in cytochrome *c*.^{170–173} Like glycolate oxidase, these enzymes bind their flavin coenzyme at the ends of 8-stranded $\alpha\beta$ barrels similar

BOX 15-B RIBOFLAVIN



The bright orange-yellow color and brilliant greenish fluorescence of riboflavin first attracted the attention of chemists. Blyth isolated the vitamin from whey in 1879 and others later obtained the same fluorescent, yellow compound from eggs, muscle, and urine. All of these substances, referred to as **flavins** because of their yellow color, were eventually recognized as identical. The structure of riboflavin was established in 1933 by R. Kuhn and associates, who had isolated 30 mg of the pure material from 30 kg of dried albumin from 10,000 eggs. The intense fluorescence assisted in the final stages of purification. The vitamin was synthesized in 1935 by P. Karrer.^a

Riboflavin, a yellow solid, has a low solubility of ~100 mg/l at 25°C. Three crystalline forms are known. One of these, the “readily soluble form,” is ten times more soluble than the others and can be used to prepare metastable solutions of higher concentration. One crystalline form is platelike and occurs naturally in the tapetum (Box 13-C) of the nocturnal lemur.

Discovery of the role of riboflavin in biological oxidation was an outgrowth of biochemists’ interest in respiration. In the 1920s Warburg provided evidence that oxygen reacted with an iron-containing respiration catalyst and it was shown that the dye **methylene blue** could often substitute for oxygen as an oxidant (Box 18-A). Oxidation of glucose 6-phosphate by methylene blue within red blood cells required both a “ferment” (enzyme) and a “coferment,” later identified as NADP⁺. A yellow protein, isolated from yeast, was found to have the remarkable property of being decolorized by the reducing system of glucose 6-phosphate plus the protein and coferment from red blood cells.

Warburg and Christian showed that the color of this **old yellow enzyme** came from a flavin and proposed that its cyclic reduction and reoxidation played a role in cellular oxidation. When NADP⁺ was isolated the proposal was extended to encompass a **respiratory chain**. The two hydrogen carriers NADP⁺ and flavin would work in sequence to link dehydrogenation of glucose to the iron-containing catalyst that interacted with oxygen. While we still do not know the physiological function of the old yellow enzyme,^b the concept of respiratory chain was correct.

Human beings require about 1.5 mg of riboflavin per day. Because of its wide distribution in food, a deficiency, which affects skin and eyes, is rarely seen. Riboflavin is produced commercially in large quantities by fungi such as *Eremothecium asbyii* which, apparently because of some metabolic anomaly, produce the vitamin in such copious amounts that it crystallizes in the culture medium.

When taken up by the body, riboflavin is converted into its coenzyme forms (Chapter 25) and any excess is quickly excreted in the urine. Urine also contains smaller amounts of metabolites. The ribityl group may be cut by the action of intestinal bacteria acting on riboflavin before it is absorbed. The resulting 10-hydroxyethyl flavin may sometimes be a major urinary product.^{c,d} The related 10-formylmethyl flavin is also excreted,^c as are small amounts of 7 α - and 8 α -hydroxyriboflavins, apparently formed in the body by hydroxylation. These may be degraded farther to the 7 α - and 8 α -carboxylic acids of lumichrome (riboflavin from which the ribityl side chain is totally missing).^e A riboflavin glucoside has also been found in rat urine.^f

The choroid layer of the eye (behind the retina) in many animals contains a high concentration of free riboflavin. Cats’ eyes also contain a large amount of 7 α -hydroxyriboflavin (neko flavin), as do their livers.^g Neko flavin is also present in human blood.^h Hen egg white contains a 219-residue riboflavin-binding protein whose functions are thought to be storage of the vitamin and delivery to the developing embryo.^{i-k} Most of the riboflavin in human blood is bound to proteins such as albumin and immunoglobulins. However, during pregnancy a riboflavin-binding protein similar to that of the chicken appears, apparently to carry riboflavin to the fetus.^j

Riboflavin is stable to heat but is extremely sensitive to light, a fact of some nutritional significance. Do not leave bottled milk in the sunshine (see Fig. 15-8)! Many products of photolysis are formed (Fig. 3-5). Among them is lumichrome.

^a Yagi, K. (1990) in *Flavins and Flavoproteins* (Curti, B., Ronchi, S., and Zanetti, G., eds), pp. 3–16, Walter de Gruyter, Berlin

^b Kohli, R. M., and Massey, V. (1998) *J. Biol. Chem.* **273**, 32763–32770

^c Owen, E. C., West, D. W., and Coates, M. E. (1970) *Br. J. Nutr.* **24**, 259–267

^d Roughead, Z. K., and McCormick, D. B. (1991) *European Journal of Clinical Nutrition* **45**, 299–307

^e Ohkawa, H., Ohishi, N., and Yagi, K. (1983) *J. Biol. Chem.* **258**, 5623–5628

^f Ohkawa, H., Ohishi, N., and Yagi, K. (1983) *J. Nutr. Sci. Vitaminol.* **29**, 515–522

^g Matsui, K., and Kasai, S. (1996) *J. Biochem.* **119**, 441–447

^h Zemleni, J., Galloway, J. R., and McCormick, D. B. (1995) *Int. J. Vitamins Nutr. Res.* **66**, 151–157

ⁱ Matsui, K., Sugimoto, K., and Kasai, S. (1982) *J. Biochem.* **91**, 469–475

^j Miura, R., Tojo, H., Fujii, S., Yamano, T., Miyake, Y., (1984) *J. Biochem.* **96**, 197–206

^k Monaco, H. L. (1997) *EMBO J.* **16**, 1475–1483

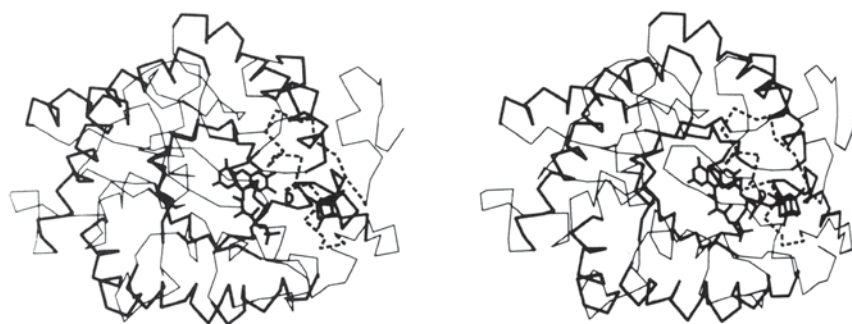


Figure 15-9 Stereoscopic view of the large domain (residues 1–383) of trimethylamine dehydrogenase from a methylotrophic bacterium. The helices and β strands of the $(\alpha\beta)_8$ barrel are drawn in heavy lines as are the FMN (center) and the Fe_4S_4 iron–sulfur cluster at the lower right edge. The α/β loop to which it is bound is drawn with dashed lines. The 733-residue protein also contains two other structural domains. From Lim *et al.*¹⁵⁰ Courtesy of F. S. Mathews.

to that of triose phosphate isomerase (Fig. 2-28). Flavocytochrome b_2 has an additional domain which carries the bound heme. Two additional domains of trimethylamine dehydrogenase have a topology resembling that of the FAD- and NADH-binding domains of glutathione reductase shown in Fig. 15-10. A bacterial **mandelate dehydrogenase** is structurally and mechanistically closely related to the glycolate oxidase family.^{173a}

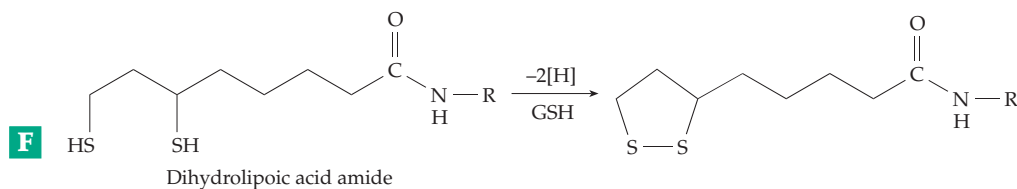
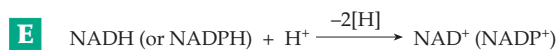
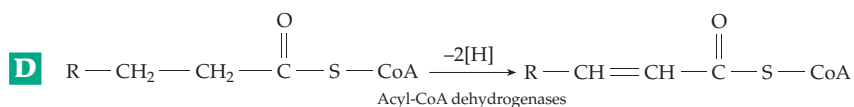
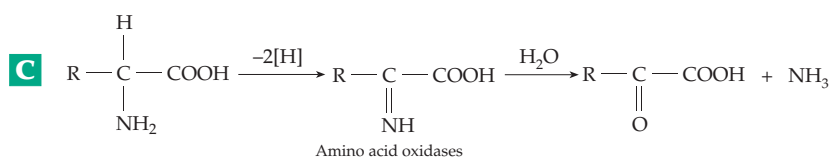
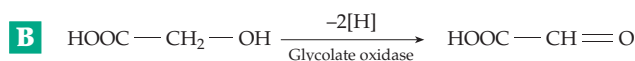
Reaction type D of Table 15-2, the dehydrogenation of an acyl-coenzyme A (CoA), could not be accomplished by a pyridine nucleotide system because the reduction

potential (E° , pH 7 = -0.32 V) is inappropriate. The more powerfully oxidizing flavin system is needed. (However, the reverse reaction, hydrogenation of a C=C bond, is often carried out biologically with a reduced pyridine nucleotide.) Dehydrogenation reactions of this type are important in the energy metabolism of aerobic cells. For example, the first oxidative step in the β oxidation of fatty acids (Fig. 17-1) is the α,β dehydrogenation of fatty acyl-CoA derivatives. The *pro-R* hydrogen atoms are removed from both the α - and β -carbon atoms to create the double bond (Table 15-2, type C).

TABLE 15-2
Some Dehydrogenation Reactions Catalyzed by Flavoproteins^a



(See Eq. 17-12 for structures in a closely related reaction.)

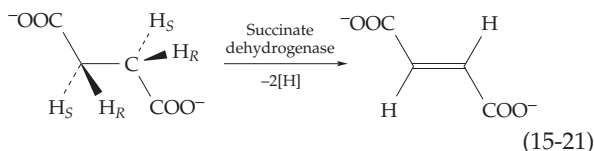


^a These are shown as removal of two H atoms [H] and may occur by transfer of H^- , $\text{H}^+ + \text{e}^-$, or $2\text{H}^+ + 2\text{e}^-$. They represent reaction type 9C of the table inside the back cover.

Figure 15-10 The three-dimensional structure of glutathione reductase. Bound FAD is shown. NAD⁺ binds to a separate domain below the FAD. The two cysteine residues forming the reducible disulfide loop are indicated by dots. From Thieme *et al.*¹⁸²



Animal mitochondria contain several different **acyl-CoA dehydrogenases** with differing preferences for chain length or branching pattern.^{174–177} A related reaction that occurs in the citric acid cycle is dehydrogenation of succinate to fumarate by **succinate dehydrogenase** (Eq. 15-21).^{177a} The dehydrogenation also involves *trans* removal of one of the two hydrogens, one *pro-S* hydrogen and one *pro-R*.¹⁷⁸ The enzyme has a large 621-residue flavoprotein subunit and a smaller 27-kDa iron–sulfur protein subunit.^{179–181}

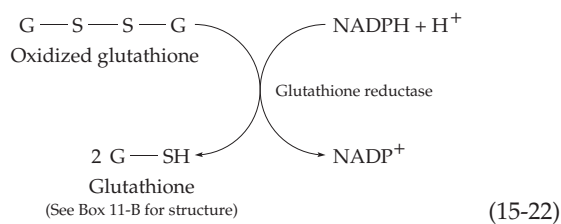


Neither the acyl-CoA dehydrogenases nor succinate dehydrogenase react with O₂. Acyl-CoA dehydrogenases pass the electrons removed from substrates to another flavoprotein, a soluble electron transferring flavoprotein (p. 794), which carries the electrons to an iron–sulfur protein embedded in the inner mitochondrial membrane where they enter the electron-transport chain. Succinate dehydrogenase as well as NADH dehydrogenase (Table 15-2, reaction E)¹⁸³ are embedded in the same membrane and also pass their electrons to iron–sulfur clusters and eventually to oxygen through the electron transport chain of the mitochondria (Chapter 18). **Fumarate reductase**¹⁸⁴ has properties similar to those of succinate dehydrogenase but catalyzes the opposite reaction in “anaerobic respiration” (Chapter 18),¹⁸⁴ as do similar reductases of bacteria¹⁸⁵ and of some eukaryotes.¹⁸⁶

Dihydrolipoyl dehydrogenase (lipoamide dehydrogenase), **glutathione reductase**, and human **thioredoxin reductase**^{187–190} belong to a subclass

of flavoproteins that act on dithiols or disulfides.

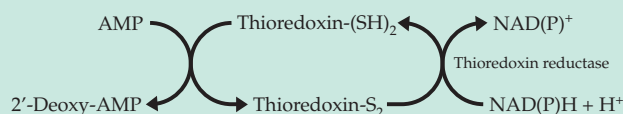
The reaction catalyzed by the first of these is illustrated in Table 15-2 (reaction type F). The other two enzymes usually promote the reverse type of reaction, the reduction of a disulfide to two SH groups by NADPH (Eq. 15-22). Glutathione reductase splits its substrate into two halves while reduction of the small 12-kDa protein **thioredoxin** (Box 15-C) simply opens a loop in its peptide chain. The reduction of lipoic acid opens the small disulfide-containing 5-membered ring in that molecule. Each of these flavoproteins also contains within its structure a reducible disulfide group that participates in catalysis.



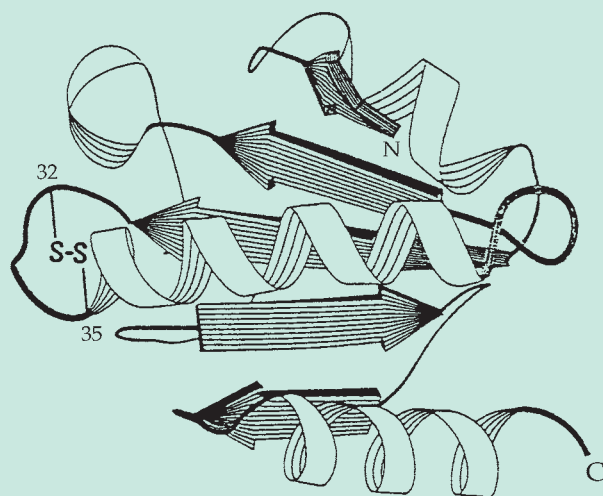
Each 50-kDa chain of the dimeric glutathione reductase is organized into three structural domains (Fig. 15-10).^{191–193} Two of the domains each contain a nucleotide-binding motif resembling those of the NAD⁺-dependent dehydrogenases. One of these domains binds NADPH and the other FAD. The latter domain also contains the reducible disulfide which is formed from Cys 58 and Cys 63. It serves as an intermediate hydrogen carrier which can in turn reduce oxidized glutathione. A **trypanothione reductase** from trypanosomes and related flagellated protozoa has a similar structure and acts on trypanothione, which replaces glutathione in these organisms.^{188,194–196} Because it is unique to trypanosomes, this enzyme is a target for design of drugs against these organisms

BOX 15-C THIOREDOXIN AND GLUTAREDOXIN

The small proteins thioredoxin and glutaredoxin are present in relatively high concentrations in bacteria, plants, and animals. For example, thioredoxin has a concentration of 15 μM in *E. coli*. Both proteins were discovered by their role as reducing agents in conversion of the ribonucleotides AMP, GMP, CMP, and UMP to the corresponding 2-deoxyribonucleotides which are needed for synthesis of DNA.^{a,b}



The redox group in thioredoxin is a **disulfide loop** located on a protrusion at one end of the molecule. In the 108-residue *E. coli* thioredoxin, as well as in the 105-residue human thioredoxin,^c it is formed by cysteines 32 and 35 which are present in the conserved sequence CGPC. The $-\text{SH}$ groups of these two interacting cysteines have pK_a values of ~ 6.9 and 7.5 , the former belonging predominantly to the more exposed Cys 32.^d The buried Asp 26 carboxyl group, which may be a proton donor to Cys 35 during reduction of the disulfide form,^e forms a salt bridge with the Lys 57 $-\text{NH}_3^+$ and has a high pK_a of ~ 7.4 .^{e-g}



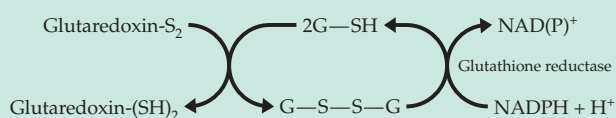
Oxidized *E. coli* thioredoxin. From Langsetmo *et al.*^f

The pK_a assignments, which have been controversial, are discussed in Chapter 7. This disulfide loop is reduced by NADPH through the action of the flavoprotein enzyme **thioredoxin reductase**.

The resulting thiol pair of the reduced thioredoxin is the reductant used for ribonucleotide reductase (Chapter 16). The standard redox potential $E^{\circ'}$ of *E. coli* thioredoxin is -0.27 V , appropriately low for coupling to the NADPH / NADP⁺ system.

Reduced thioredoxin has a variety of functions.^h It is the reductant for conversion of methionine sulf-oxide to methionine in bacteria, for reduction of sulfate in yeast, and for additional specific enzymatic reactions.^{b,h} However, its major function may be to reduce disulfide linkages in various proteins.ⁱ Several photosynthetic enzymes are activated by reduction of disulfide linkages via photosynthetically generated reduced ferredoxin and thioredoxin^{j,k} (Chapter 23). Reduced thioredoxin may also play a similar role in nonphotosynthetic cells. It may reduce mixed disulfides such as those formed between glutathione and proteins (Box 11-B).^{l-n} Thioredoxin may participate in regulation of the level of nitric oxide (NO) in tissues^m and it is needed in the assembly of filamentous bacteriophages.^{b,o} For reasons that are not clear, thioredoxin is also an essential subunit for a virus-induced DNA polymerase formed in *E. coli* following infection by bacteriophage T7 (see Chapter 27).

It was a surprise to discover that a mutant of *E. coli* lacking thioredoxin can still reduce ribonucleotides. In the mutant cells thioredoxin is replaced by glutaredoxin, whose active site disulfide linkage is reduced by glutathione rather than directly by NADPH. Oxidized glutathione is, in turn, reduced by NADPH and glutathione reductase. Thus, the end result is the same with respect to ribonucleotide reduction.



However, the two proteins have significantly different specificities and functions. The disulfide loop in glutaredoxin, whose eukaryotic forms are often called **thioltransferases**,^p has the sequence CPYC. Although glutaredoxins are weaker reductants of mixed disulfides of proteins with glutathione than are thioredoxins,^{q-s} they are more specific.

Both thioredoxin and glutaredoxin are members of a larger group of thiol:disulfide oxidoreductases which are found in all known organisms. In *E. coli* there are one thioredoxin, three different glutaredoxins,^{h,t} and the periplasmic protein disulfide

BOX 15-C (continued)

isomerases DsbA and DsbC, which were discussed in Chapter 10.^{u-w} Similar enzymes are found in the endoplasmic reticulum of eukaryotic cells.^{u,x} Like thioredoxins, they contain disulfide loops which contain the sequences CGHC and CGYC as well as CPHC. Thioredoxin reductase itself also can keep cytoplasmic bacterial proteins reduced.^y Redox potentials vary within this family of proteins:^{n,s,z,aa}

	E° , pH 7
Thioredoxins	−.27 V
Glutaredoxins	−.20 to −.23 V
Protein disulfide isomerase	−.18 V
DsbA	−.09 to −.12 V

These differences are correlated with differing functions. The intracellular reduced thioredoxins are thermodynamically the best reductants of disulfide linkages in proteins and they help keep intracellular proteins reduced. Glutaredoxin can be reduced efficiently by reduced glutathione or by NADPH and glutathione reductase and can, in turn, reduce cysteine and the oxidized form of vitamin C, **dehydroascorbic acid** (Box 18-D).^{bb,cc} The periplasmic bacterial proteins DsbA and DsbC have the highest redox potentials and an unusually low first pK_a for the dithiol pair in their thioredoxinlike domains.^{aa} The basis for these properties has been hard to understand but is consistent with their role in assisting formation of disulfide bridges in extracellular proteins and with the role of the related protein disulfide isomerase in the ER.^{u,dd-ff} These disulfide exchanges are nucleophilic displacement reactions (Chapter 12).

^a Holmgren, A. (1981) *Trends Biochem. Sci.* **6**, 26–29

^b Holmgren, A. (1989) *J. Biol. Chem.* **264**, 13963–13966

^c Forman-Kay, J. D., Clore, G. M., Wingfield, P. T., and Gronenborn, A. M. (1991) *Biochemistry* **30**, 2685–2698

^d Dyson, H. J., Jeng, M.-F., Tennant, L. L., Slaby, I., Lindell, M., Cui, D.-S., Kuprin, S., and Holmgren, A. (1997) *Biochemistry* **36**, 2622–2636

^e Jeng, M.-F., and Dyson, H. J. (1996) *Biochemistry* **35**, 1–6

^f Langsetmo, K., Fuchs, J. A., and Woodward, C. (1991) *Biochemistry* **30**, 7603–7609

^g Ladbury, J. E., Wynn, R., Hellinga, H. W., and Sturtevant, J. M. (1993) *Biochemistry* **32**, 7526–7530

^h Gvakharia, B. O., Hanson, E., Koonin, E. K., and Mathews, C. K. (1996) *J. Biol. Chem.* **271**, 15307–15310

ⁱ Thomas, J. A., Poland, B., and Honzatko, R. (1995) *Arch. Biochem. Biophys.* **319**, 1–9

^j Capitani, G., Markovic-Housley, Z., DelVal, G., Morris, M., Jansonius, J. N., and Schürmann, P. (2000) *J. Mol. Biol.* **302**, 135–154

^k Buchanan, B. B., Schürmann, P., Decottignies, P., and Lozano, R. M. (1994) *Arch. Biochem. Biophys.* **314**, 257–260

^l Wynn, R., Cocco, M. J., and Richrds, F. M. (1995) *Biochemistry* **34**, 11807–11813

^m Nikitovic, D., and Holmgren, A. (1996) *J. Biol. Chem.* **271**, 19180–19185

ⁿ Prinz, W. A., Åslund, F., Holmgren, A., and Beckwith, J. (1997) *J. Biol. Chem.* **272**, 15661–15667

^o Russel, M., and Model, P. (1985) *Proc. Natl. Acad. Sci. U.S.A.* **82**, 29–33

^p Srinivasan, U., Mieyal, P. A., and Mieyal, J. J. (1997) *Biochemistry* **36**, 3199–3206

^q Katti, S. K., Robbins, A. H., Yang, Y., and Wells, W. W. (1995) *Protein Sci.* **4**, 1998–2005

^r Gravina, S. A., and Mieyal, J. J. (1993) *Biochemistry* **32**, 336–3376

^s Nikkola, M., Gleason, F. K., and Eklund, H. (1993) *J. Biol. Chem.* **268**, 3845–3849

^t Åslund, F., Nordstrand, K., Berndt, K. D., Nikkola, M., Bergman, T., Ponstingl, H., Jörnval, H., Otting, G., and Holmgren, A. (1996) *J. Biol. Chem.* **271**, 6736–6745

^u Freedman, R. B., Hirst, T. R., and Tuite, M. F. (1994) *Trends Biochem. Sci.* **19**, 331–335

^v Rietsch, A., Belin, D., Martin, N., and Beckwith, J. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 13048–13053

^w Darby, N. J., and Creighton, T. E. (1995) *Biochemistry* **34**, 16770–16780

^x Kanaya, E., Anaguchi, H., and Kikuchi, M. (1994) *J. Biol. Chem.* **269**, 4273–4278

^y Derman, A. I., Prinz, W. A., Belin, D., and Beckwith, J. (1993) *Science* **262**, 1744–1747

^z Chivers, P. T., Prehoda, K. E., and Raines, R. T. (1997) *Biochemistry* **36**, 4061–4066

^{aa} Jacobi, A., Huber-Wunderlich, M., Hennecke, J., and Glockshuber, R. (1997) *J. Biol. Chem.* **272**, 21692–21699

^{bb} Wells, W. W., Xu, D. P., Yang, Y., and Rocque, P. A. (1990) *J. Biol. Chem.* **265**, 15361–15364

^{cc} Bischoff, R., Lepage, P., Jaquinod, M., Cauet, G., Acker-Klein, M., Clesse, D., Laporte, M., Bayol, A., Van Dorselaer, A., and Roitsch, C. (1993) *Biochemistry* **32**, 725–734

^{dd} Hwang, C., Sinskey, A. J., and Lodish, H. F. (1992) *Science* **257**, 1496–1502

^{ee} Ruoppolo, M., Freedman, R. B., Pucci, P., and Marino, G. (1996) *Biochemistry* **35**, 13636–13646

^{ff} Couprie, J., Vinci, F., Dugave, C., Quéméneur, E., and Mourtiez, M. (2000) *Biochemistry* **39**, 6732–6742

which cause such terrible diseases as African sleeping sickness and Chagas disease.^{195,197}

Another flavoprotein constructed on the glutathione reductase pattern is the bacterial plasmid-encoded **mercuric reductase** which reduces the highly toxic Hg^{2+} to volatile elemental mercury, Hg^0 . A reducible

disulfide loop corresponding to that in glutathione reductase is present in this enzyme but there is also a second pair of cysteines nearby. All of these may participate in binding and reduction of Hg^{2+} .^{198–199a}